Presentation-Relative Structural Complexity: Quantitative Limits for General Intelligent Computation

Guillem Duran Ballester
guillem@fragile.tech

Abstract

Can the capabilities of an arbitrary intelligent system be bounded without fixing its architecture? We develop a presentation-relative theory that replaces algorithm-specific analysis with a saturated structural profile: the strongest target-relevant structure that any resource-bounded computation can construct from the public observables, constructors, terminal data, and cost model of a task. Universal accountability theorems convert this profile into quantitative performance bounds that hold simultaneously for every admissible algorithm, including architectures and runtime-generated representations not anticipated in advance.

The framework unifies represented geometry, abstraction, learned features, information acquisition, memory, Bayesian inference, control, reinforcement learning, and attention within a common cost-, provenance-, temporal-, source-, and noise-aware ledger. It separates prospective sources from retrospective explanations and charges generated representations at their complete saturated cost. Source-resolved Rademacher, sequential Rademacher, PAC–Bayes, and effective-dimension bounds keep statistical capacity, structural ancestry, noise contraction, construction cost, and deployment attached to the same computation. The resulting statistical, computational, and information-theoretic noise thresholds quantify transitions among learnable, resource-limited, and information-limited regimes.

Universal quantification over the declared resource class gives precise architecture-independent bounds on learning, reasoning, control, and target-directed performance for increasingly general intelligent systems, including artificial general intelligence. As a proof-producing stress test, the LPN specialization isolates one compact residual-ancestry certification target and proves that a deterministically accepted finite-to-asymptotic certificate discharges it. The instantiated certificate analysis is not executed here, so the current LPN hardness consequences remain conditional on this internal certificate; verifying it would make them unconditional relative to the declared presentation.

Universal capability bound. For every declared task presentation $\mathfrak{M}$, scale n, and resource budget B, the framework has the schematic form

$$\mathrm{Performance}_{\mathfrak{M},n}(B) \;\leq\; \mathrm{Overhead}(B,n)\; \mathrm{SaturatedProfile}_{\mathfrak{M},n}(\Psi(B,n)).$$

Here the left-hand side is optimized over every admissible algorithm in the declared resource class. The right-hand side depends only on the task presentation and the cost-filtered closure of all structure derivable from it. Consequently, the bound applies to arbitrary architectures without enumerating them or anticipating how they represent, search, learn, remember, attend, or control. Advice, target information unavailable through the declared presentation, structures above the transformed budget $\Psi(B,n)$, and enriched interfaces are assigned to distinct accounting classes rather than included in the saturated profile without charge.

Contents

Part I
Task Spaces and Observable Structure

I.0 Introduction: Static Task Presentations

Part contract. *Inputs:* only the declared data of a task presentation. *Output:* the static observable and target-relevance structures used in Parts II and III. *First pass:* the examples and audit procedures may be skimmed after the definitions they illustrate.

Most performance analyses begin with a chosen computational method. They inspect its states, representations, and transitions, then derive a bound for that method:

$$\text{choose an algorithm} \longrightarrow \text{analyze its computation} \longrightarrow \text{bound its performance.}$$

Such a bound is tied to the analyzed design. A different architecture, representation, search rule, or controller requires a new analysis unless the argument identifies an invariant shared by the entire admissible resource class.

This paper begins instead with the information and operations supplied by the task. Across theoretical computer science, optimization, machine learning, and control, these data comprise a state space, a public observable interface, admissible constructors, resource accounting, and terminal criteria distinguishing success from failure. The analysis follows the reverse order:

$$\text{specify the public task presentation} \longrightarrow \text{close all affordable structure}$$
$$\longrightarrow \text{measure the saturated target-relevant profile} \longrightarrow \text{bound every admissible computation.}$$

The resulting bound is architecture-independent within the declared presentation and resource class. Its defining closure property is summarized in the following budget-level statement.

Saturation in one paragraph. At budget B, saturation closes the public presentation under every admissible oracle-free constructor and represented computation whose complete derivation is admissible at B; it does not select a preferred algorithm. If an algorithm generates a quotient, feature map, potential, kernel, controller, value representation, or target-aligned observable at runtime, the complete generated record re-enters the same closure with its derivation and admissible budget set. Under scalar cost accounting, it first appears at its least saturated representation cost and remains outside the budget-B slice while that cost exceeds B. For a partially ordered resource budget, it contributes precisely at the budgets in its represented admissible budget set. The resulting profile is therefore intrinsic to the presentation and budget, rather than to the method that discovered the representation.

The exact eight-state calculation in Example III.16.7 instantiates this closure at a finite budget: it lists the entire affordable quotient slice, computes every conditional-algebra increment, and turns their conserved charge into a bound for every admissible transition rule in the example.

Temporal source typing. Fix a surviving pre-action history h_t. After the run, let a represented terminal evaluator label the next action when the completed trajectory succeeds and output $\perp$ otherwise. On a successful run this label identifies the first action on the successful continuation. If two legal completions of h_t produce different labels, no function of the available history can equal that terminal label on both completions. Once its terminal evaluator and complete derivation are charged, the label is an accounting record for the achieved performance; its retrospective representation is not a source for the action selected at h_t.
A representation of the same denotation enters the prospective source profile at this use precisely when it independently factors through the pre-action history, has no later outcome, verifier flag, terminal record, or oracle answer as an ancestor, is evaluated and charged before the action, and is an actual ancestor of the selector, transition, or target-aligned represented rule whose target progress it is invoked to explain:

$$\boxed{\text{prospective source } \neq \text{ retrospective explanation.}}$$

The display distinguishes represented derivations at a claimed use, rather than extensional denotations. One denotation may have both a retrospective terminal representation and an independent prefix representation; each is classified and charged through its own derivation. Later verifier outcomes, exact first-hit labels, completed value functions, and other terminal postprocessings remain in the accounting profile under a retrospective representation when that representation and its complete derivation are charged. An independent prefix derivation re-enters the saturated closure with its complete ancestry and cost. Exact first-hit accounting preserves the hitting law already generated, whereas prospective selector accountability charges improvement over the declared neutral enumeration baseline only to the pre-hit sampler and its temporally admissible ancestry. Definitions III.21.1 and III.23.3, Corollaries III.23.1 and III.23.2, and Theorems III.21.3 and V.6.4 formalize this proof strategy.

I.0.1 What this paper introduces

Stated in classical terms, the manuscript develops a single quantitative theory around three objects: a canonical decomposition of sub-Markov generators, a cost filtration on an algebra of observables, and the hitting-time analysis of a distinguished target set. No familiarity with the internal terminology of the framework is needed to read this subsection; the remainder of the paper makes each ingredient precise.

The presented object. A computational or learning problem is recorded as a measurable state space together with a distinguished unital algebra of bounded observables, a slot for state-to-state structure, and a pair of terminal events (success and failure). This is the common measurable core of decision problems, optimization landscapes, statistical models, and controlled systems; it fixes what is public about a problem before any algorithm is chosen.

The generator decomposition. Any search procedure, algorithm, or stochastic dynamics acting on such a space and respecting a stated resource budget induces a positive sub-Markov operator, hence, by the positive maximum principle, a generator. The first structural theorem is an exhaustive decomposition of every such generator into five components with distinct classical meanings—a reversible (Dirichlet-form) part, a drift part directed by a declared potential, a lumping part in the sense of Markov-chain lumpability (quotients of the state space), a state-augmentation part (lifts to enlarged carriers), and a killing/boundary part—plus a residual term that provably carries no structure attributable to the five listed sources. Normal forms are proved for each component, and the decomposition is stable under composition and under valid lifts.

The cost filtration. The observable algebra carries a filtration indexed by a nonnegative cost: an object lies at level B when it admits a public derivation of total cost at most B from the declared data. The value of this filtration at the working budget—the *saturated* cost—measures how much structure is legally available to any procedure operating within that budget. All later upper and lower bounds are stated against this filtration rather than against a machine model.

Hitting times and hardness. Complexity is then measured by the discounted hitting functional of the target set, equivalently by the target-projected resolvent of the killed generator. The central inequality bounds this functional by the structure extractable at the saturated budget: each unit of discounted target mass is charged, via an exact first-hit decomposition, to a specific affordable component of the generator. Hardness statements take the form "the saturated profile is negligible, hence the hitting functional is negligible," and are presentation-relative: they quantify over every procedure representable within the declared budget, not over a fixed algorithm.

Consequences. Because the saturated filtration is a reproducing-kernel object, the same ledger yields effective-dimension and two-sided generalization bounds for kernel and neural-network learners, including under label noise and dependent (mixing or adaptive) sampling. For controlled Markov systems and partially observed models, the decomposition of the selected generator produces a coordinatewise certificate covering committors, reach–avoid probabilities, Bellman error, and learned policies. Softmax attention is identified with a Gibbs-measure selector whose normalization satisfies exact Kullback–Leibler and Fisher–Rao identities, so trained attention layers are audited by the same accounting. A meta-complexity part inverts the picture: strict Pareto frontiers record the least resource vectors at which a certificate of prescribed quality exists, are exact generalized inverses of the saturated profiles, and specialize to circuit and universal-program minimization. Finally, the framework is instantiated on Learning Parity with Noise: after all geometric, quotient, and provenance sources are exhausted, the surviving coordinate of the saturated profile is isolated explicitly. At fixed positive noise, every materialized located spectral block has exponentially small alignment, while compact non-lumpable residual-ancestry records form one internal certification target. The paper proves a deterministic finite-to-asymptotic verification theorem: exact source coverage, certified capacity and attenuation, outward-rounded error bounds, symbolic uniform scale laws, a crossover check, and a finite-prefix check together imply the required bound $(1-\eta)^n$. The present manuscript does not execute that final certificate analysis. Consequently, its fixed-noise hardness theorem and the resulting $P \neq NP$ consequence remain conditional on acceptance of this explicit proof certificate. An accepted certificate would discharge the sole remaining certification target internally; no external LPN or complexity-theoretic hardness assumption is used.

Uniformity over new high-n structure. For an accepted proof-producing package, exact finite saturation enumerates the complete budgeted structural slice at every size in the audited prefix, not the output of one selected optimizer. Its tail certificate quantifies over the complete qualified source cell, and its recurrence cover assigns every admissible represented transition in that cell to a verified transition map. A different high-n regime must therefore enter through a represented change of recurrence grammar, acquisition interface, reuse rule, precision, or locator. Such a change has a complete construction and deployment cost; the crossing-price certificate excludes it whenever that cost exceeds the declared ceiling (Definitions V.15.15 and V.15.17, Theorem V.15.18, and Corollary V.15.19). Every representation constructed by a future admissible algorithm remains in the same saturated derivability closure, so the verified source-cell and crossing bounds apply to it without naming its architecture in advance.

Representative learning consequences. The target-qualified prospective source selected by a complete learning record is first assigned to one of the 4960 exact structural ancestry cells $\alpha \in \mathfrak{A}_{\mathrm{learn}}$, indexed by its master mode and state–coefficient provenance. For losses in $[-1, 1]$, the nonempty cell-restricted output or loss classes satisfy the source-resolved Rademacher union bound

$$\widehat{\mathfrak{R}}_S\left(\bigcup_{\alpha\in\mathfrak{A}_{\mathrm{learn}}} \mathcal{H}_\alpha(B,m)\right) \le \max_{\alpha\in\mathfrak{A}_{\mathrm{learn}}} \widehat{\mathfrak{R}}_S(\mathcal{H}_\alpha(B,m)) + \sqrt{\frac{2\log|\mathfrak{A}_{\mathrm{learn}}|}{m}}.$$

For a cell-mixture prior $P = \sum_\alpha w_\alpha P_\alpha$ and posterior $Q = \sum_\alpha q_\alpha Q_\alpha$ with matching conditional cells, the PAC–Bayes complexity decomposes exactly as

$$\mathrm{KL}(Q\|P) = \mathrm{KL}(q\|w) + \sum_\alpha q_\alpha \,\mathrm{KL}(Q_\alpha\|P_\alpha).$$

Thus the familiar inequalities are applied after structural ancestry has been resolved: source selection is paid by the finite-cell Rademacher term or by the cell-selection divergence $\mathrm{KL}(q\|w)$. Every complete learning bound assembled from these cellwise estimates retains the sample size m, confidence $1-\delta$, resource budget B, exact source cell α, noise contraction, representation and optimization cost, decoder or policy deployment, and a represented same-law, same-unit conversion to the target performance quantity. See Definition IV.13.2 and Theorem IV.13.3.

Three non-interchangeable noise thresholds.

threshold	quantitative meaning
Statistical	For a clean separation margin Δ, exact identification is certified when $2\Gamma_m^{\mathrm{Rad}}(\mathcal{H},\delta) + \varepsilon_{\mathrm{opt}} < (1-2\eta)\Delta$. This is a sufficient $1-\delta$ sample-and-margin certificate; it does not certify that the empirical minimizer is affordable.
Computational	$\eta^{\mathrm{sat}}_{\mathrm{comp}}(n,m;\sigma,\alpha) = \sup\{0 \le \eta < \frac{1}{2} : \mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta,n}}(\sigma(n)) \ge \alpha\}$. The supremum is zero when the affordable region is empty. When the declared ceiling is closed under represented noise degradation, it is the right endpoint of the noise region in which target performance α remains affordable under the complete saturated ceiling $\sigma(n)$.
Information	Every estimator, without a computational restriction, needs $m(1-H_2(\eta)) \ge \alpha\log_2 N - 1$ to recover a uniform target among N possibilities with probability at least α. Violation of this inequality rules out that recovery level regardless of computational resources.

Put $\rho = 1-2\eta$. Clean target margins and correlations are attenuated by ρ, so recovering a clean quantity from its noisy counterpart introduces the inverse correction ρ^{-1}; quadratic error bounds carry ρ^{-2}. The distinct forward coefficient $\vartheta(\eta) = \rho^2 = (1-2\eta)^2$ is the admissible strong-data-processing coefficient that upper-bounds source information transmitted through a binary symmetric reveal. The computation-unrestricted information threshold itself uses the channel capacity term $1 - H_2(\eta)$; these coordinates are compared on the same experiment but do not replace one another. See Definition IV.14.2 and Theorems IV.14.4 and IV.16.8.

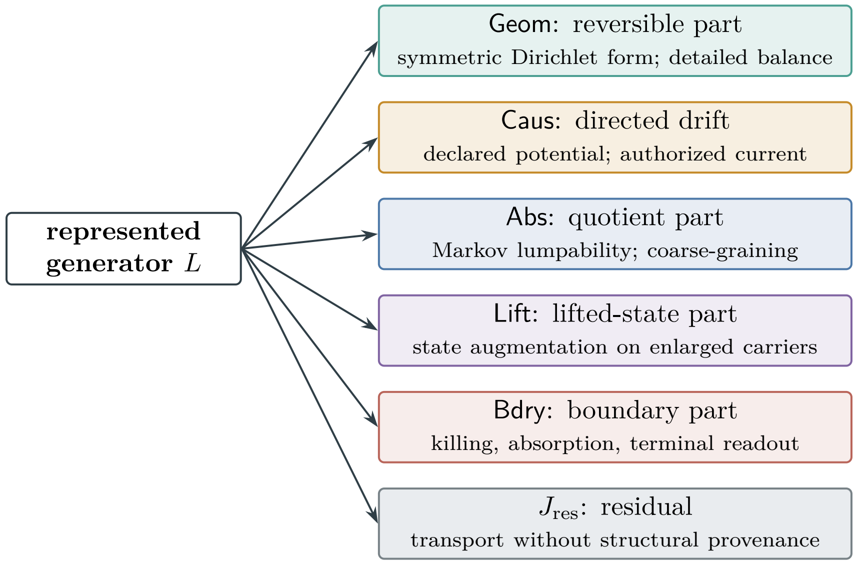

Figure 1: Dictionary for the exhaustive generator decomposition (Theorem II.7.1). Every budget-admissible positive sub-Markov operator splits into the five structural components and a residual; the small print gives the classical counterpart of each component. The intrinsic sources of reusable structure are Geom and Abs; the channels Caus, Lift, and Bdry orient, reorganize, and read out that structure, and J_{res} is transport that provably admits no such attribution.

Figure 1 summarizes the generator decomposition together with its classical reading; the internal channel names used throughout the paper appear in the left column.

The labels in Fig. 1 name typed framework objects, not only their classical operator cores. A row of Table 1 applies to a complete represented record; the classical ingredient in its middle column supplies an algebraic coordinate, while the final column states the additional conditions under which that coordinate has framework status.

Part I formalizes this structure independently of any solution method. A task presentation consists of complete states, observables, a declared slot for state-to-state structure, and terminal data. Search procedures, stochastic processes, generators, transition kernels, algorithms, and complexity bounds are introduced only in subsequent parts. The symbol Dyn therefore denotes a static relation slot in this part; its dynamical interpretation begins in Part II.

The static spine is

$$\text{states} \longrightarrow \text{observables} \longrightarrow \text{observable event algebra} \longrightarrow \text{terminal sectors} \longrightarrow \text{the presented task.}$$

Framework term	Classical ingredient	Additional framework meaning
Geom	reversible or symmetric operator; Dirichlet form	represented intrinsic geometry with public provenance, qualification, complete representation cost, and target-visibility accounting
Caus	directed current or drift	authorized target-oriented use of an exposed source, using the represented projection of the actual selected current under a declared authority envelope
Abs	quotient, sufficient statistic, or lumping	represented abstraction with explicit fibers, terminal compatibility, qualification, temporal availability, and saturated representation cost
Lift	state augmentation	represented enlargement with a charged interface that preserves target fractions and inherits the provenance and qualification of its base source
Bdry	killing, absorption, or terminal value	represented terminal readout that evaluates previously exposed structure and creates no independent interior transport source
J_{res}	orthogonal remainder	post-saturation transport remaining after all affordable channel-qualified target-relevant representatives have been reclassified and charged; it carries no prospective structural source charge

Table 1: What the channel names mean. The middle column records the classical operator ingredient; the final column records the additional typed requirements imposed by the framework. The precise gates are the budgeted representative condition of Definition II.4.1, the authority condition of Definition II.4.3, the quotient and lift conditions of Definitions II.5.1 and II.5.2, the boundary readout condition of Definition II.3.4, and the saturated residual condition of Corollary II.8.5.

I.0.2 Architecture-Independent Capability Analysis

Definition I.0.1 (General intelligent computation relative to a presentation)**.** Fix a presented task family, a declared resource envelope B, and declared observation, memory, action, and oracle interfaces. A *general intelligent computation* in this model is an arbitrary family-uniform, presentation-relative represented computation

$$\mathcal{A}_n \in \mathsf{Alg}_{n,\le B}$$

that interacts with the task only through those interfaces. Its internal architecture, state organization, update rule, learned representations, and control strategy are unrestricted within the declared represented-computation model. What is fixed is the public task presentation, the information available through the declared interfaces, and the complete resource account, including code, initialization, construction, peak memory, updates, oracle use, total work, and deployment. No unpresented observation or uncharged interface is part of the admissible class. This is the architecture-independent computational class used throughout the paper; see Definitions III.24.1 and II.5.3 and Convention III.5.10.

For a task-family functional $\mathfrak{M}_n$, define the corresponding budgeted capability envelope by

$$\mathrm{Cap}_{\mathfrak{M},n}(B) := \mathrm{Succ}_{\mathfrak{M},n}(B) = \sup_{\mathcal{A}\in\mathsf{Alg}_{n,\le B}} \mathfrak{M}_n(I \mapsto \alpha_I(\mathcal{A}))\,.$$

The universal accountability theorem then gives the architecture-independent bound

$$\mathrm{Cap}_{\mathfrak{M},n}(B) \le eH_B(n)\, m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n))\,, \tag{1}$$

where H_B and Ψ are derived from the same declared resource ledger rather than chosen for a particular algorithm (Theorem III.24.3). The right-hand side saturates over all affordable represented structure, including features, memories, abstractions, policies, and runtime-generated representations. It therefore applies simultaneously to every architecture in the class of Definition I.0.1.

Different specializations of this envelope provide:

- capability bounds over task families and performance functionals;
- learning and inference bounds that retain noise contraction, statistical capacity, and complete represented cost (Theorem IV.13.3);
- limits on adaptive information acquisition (Theorem V.9.7);
- explicit charges for memory, representation construction, and deployment (Definitions III.14.1 and II.5.3);
- controlled target-arrival and reach–avoid bounds (Theorems V.1.2 and V.12.3);
- Bayesian, compact-belief, and attention-based acquisition limits (Theorems V.11.22, VI.2.2 and VI.2.4); and
- a common comparison of statistical, computational, and information-theoretic bottlenecks (Theorems IV.14.4 and IV.16.8).

Thus the AGI interpretation is a quantified modeling statement: the internal architecture is universally quantified while the presentation, access model, and resource envelope remain explicit. The framework supplies mathematical language and universal quantitative machinery for analyzing general intelligent computation; it does not construct an AGI system or assert that the engineering problem of AGI has been solved.

I.0.3 Contributions

The contributions form three levels. The first consists of the framework's new mathematical objects. The second contains the theorems made possible by those objects. The third records the established analytic ingredients used inside the proofs.

A. New foundational objects.

- *Task presentations as machine-independent computational objects.* A task-space signature separates complete states, public observables, state-to-state structure, admissible constructors, resource accounting, and terminal data (Definitions I.1.1 and I.2.1).
- *Budgeted derivability and saturated represented closure.* Budgeted derivability closes the declared presentation under oracle-free constructors and represented computations, while saturated representation cost records the least complete budget of each derived object (Definitions III.14.1 and III.1.3).
- *Typed structural channels.* The channels $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$, and J_{res} carry cost, provenance, qualification, target-use, and residual semantics in addition to their operator-theoretic coordinates (Definitions II.4.1 to II.4.2).
- *Affordable conditional-algebra frontiers.* An affordable frontier records the represented backward slice, its nested conditional algebras, complete carrier cost, and contextual target charges (Definition III.16.1).
- *Prospective source and retrospective accounting types.* The framework distinguishes structure available when a transition is chosen from terminal-compatible records constructed after the trajectory is complete (Definition III.21.1 and Corollary III.23.1).
- *Algorithm-independent saturated capability profiles.* The master and channelwise profiles optimize over every qualified represented structure generated within budget, including structures constructed at runtime (Definitions III.25.1 and III.14.1).
- *Structural meta-frontiers.* Strict Pareto frontiers are generalized inverses of saturated profiles over complete resource vectors and retain their temporal and source types (Definitions VII.2.1 and VII.1.2).
- *Source-resolved saturated learning ledgers.* Learning records retain the exact structural source cell, statistical capacity, information, noise, selection, dynamics, and deployment costs (Definition IV.13.2).

B. New theorems enabled by these objects.

- *Exhaustive channels and exact structural-charge conservation.* Every represented positive sub-Markov operator has the exhaustive channel decomposition and explicit normal forms, while every affordable frontier satisfies an exact Pythagorean charge identity (Theorems II.5.10, II.7.1, II.7.2, II.9.5, II.10.5 and III.16.6).
- *Universal algorithm accountability.* The quantitative backward cut and exact first-hit ledger bound every budget-admissible computation by the corresponding saturated profile (Theorems III.15.6, III.21.3 and III.24.3 and Lemma III.20.2).
- *No-retrospective-source and no-uncharged-value principles.* Completed first-hit labels enter prospective profiles only through an independent pre-action representation at complete cost, and represented value cannot amplify prospective target gain without a charged source (Corollary III.23.2 and Theorem V.6.4).
- *Source-cell exhaustion.* The five-family provenance normal form and the exact state–coefficient classification exhaust the prospective source cells and their meta-frontiers (Theorems III.2.2,

III.3.4 and VII.3.2).

- *Source-resolved Rademacher and PAC–Bayes compilation.* The learning ledger gives cellwise Rademacher union and PAC–Bayes mixture bounds, together with operator, effective-dimension, label-noise, and dependent-sampling consequences (Theorems IV.3.3, IV.3.12 and IV.13.3).
- *Controlled-system, attention, and compact-belief compilation.* Selected generators, Bellman and reach–avoid quantities, attention records, and compact latent beliefs compile into the same source- and cost-resolved capability ledger (Theorems V.1.2, V.11.22, V.12.3, VI.1.3, VI.2.2 and VI.2.4).
- *Explicit statistical, computational, and information noise thresholds.* The source-resolved ledger separates the sample, represented-computation, and information limits and compares their phase transitions (Theorems IV.14.4 and IV.16.8).
- *Proof-producing finite-to-asymptotic certificate machinery.* For the LPN application, exact quotient and geometric exhaustion isolates one residual-ancestry certification target. Exact enumeration, deterministic verification, symbolic inequalities, outward-rounded interval bounds, finite-prefix checks, and uniform scale propagation form a sufficient certificate whose deterministic acceptance discharges that target. The paper proves this discharge implication and specifies every verification field; it does not execute the final instantiated certificate analysis (Definition V.15.15, Theorems VIII.3.4 and VIII.20.6, Lemma VIII.3.1, and Corollary V.15.14).

C. Standard ingredients used inside the proofs.

- Markov and sub-Markov kernels, generators, and killed resolvents;
- Dirichlet forms, reversible operators, and Hodge projection;
- conditional expectation and projection Pythagoras;
- lumpability, quotient operators, and state augmentation;
- Rademacher complexity, PAC–Bayes inequalities, effective dimension, and concentration bounds;
- Bellman equations, Bayesian identities, and Gibbs selectors; and
- interval arithmetic, finite enumeration, and induction.

These established ingredients serve inside a presentation-relative architecture that assigns complete cost, provenance, temporal availability, source type, and deployment semantics before compiling them into universal capability bounds.

I.0.4 How to read this paper

The manuscript has a three-part structural spine followed by learning, controlled-systems, attention, structural meta-complexity, and LPN parts. The beginning of each part states its inputs, output, and a suggested first pass. The complete organization is as follows.

Part I: presented structure.

The task-space signature fixes the state, observable, relation, and terminal interfaces. The part then constructs the observable algebra, formalizes static target relevance, and closes with admissible presentation equivalence. Its output is the complete static object on which all later dynamics act.

Part II: admissible dynamics.

Budget-admissible movement is written in generator form and decomposed into its structural channels. The channel-exhaustiveness theorem proves that the list is complete, while the operational embedding places ordinary computations on the same transcript carrier. Its output is an algorithm-independent channel decomposition with the exhaustive channel normal forms of Theorems II.5.10, II.7.2, II.9.5 and II.10.5 and explicit resource provenance.

Part III: saturated extraction cost.
The killed resolvent measures target arrival. The combined extraction functional then assembles the saturated ledger, quantitative backward cut, and structural charging identity. Exact first-hit accounting culminates in the universal accountability theorem. The saturated cost invariant and the presentation-extension tests complete the framework core.

Part IV: learning.
The saturated structural objects are read as kernels and RKHS features. This part derives effective-dimension and two-sided learning bounds, treats learned neural kernels, transports these capacity certificates through symmetric label corruption, develops computable evaluation and deployment certificates, and extends structural Rademacher and PAC–Bayes control to independent episodes, mixing trajectories, and predictable adaptive paths. It ends with the noise-indexed statistical, computational, and information threshold comparison. It uses Parts I–III. Its statistical results feed the quantitative LPN phase comparison, while the structural LPN reduction uses Parts I–III directly.

Part V: controlled dynamical systems.
Represented policies select cost-tagged generators on finite or clocked carriers. This part combines the existing reversible geometry, authorized Caus current, committor accounting, reach–avoid capacity, Bellman contraction, the audited functional-inequality profile, and dependent-data learning certificates in the universal dynamical-systems theorem and the master learned-control certificate. Continuous and stochastic models enter through the represented Lift transfer. The optimal structural Bayesian section also defines compact latent beliefs, their information and sufficiency coordinates, and the exact joint acquisition–encoding ceiling in Theorems V.11.22 and V.11.24. The trained-model chapters give complementary white-box and black-box certificates: Section V.15 audits a complete training record, while Section V.16 audits an oracle and represented test law at finite query cost.

Part VI: attention.
Attention rows are treated as represented Gibbs selectors. Their score geometry, directed current, fibers, values, masks, and residual transport are compiled into the existing source-resolved ledger in Theorem VI.1.3; Theorem VI.2.2 gives the target-gain envelope. The log-likelihood specialization identifies attention with the compact latent Bayesian update, and the joint theorem equates optimal compact latent attention with optimal acquisition on its saturated transfer class (Corollary VI.1.2 and Theorem VI.2.4).

Part VII: structural meta-complexity.
The complete prospective records are indexed by the Pareto cost required to cross a target-gain threshold. The strict structural meta-frontier is the exact generalized inverse of the saturated profile in Theorem VII.1.3. The part separates representation, extraction, decision, search, certification, and prospective deployment; recovers circuit and program minimization as typed specializations; and derives the meta-accountability implication in Theorem VII.2.2.

Part VIII: LPN.
The public LPN presentation is inserted into the framework. Its noise–resource phase diagram compares the polynomial, statistical, and information thresholds. Public verification is separated from extraction; the constant-noise spectral ledger evaluates materialized located spectra and the exact residual-parity fiber gate; the geometric and five abstraction-provenance sources are core-refined; and the resulting saturated-profile reduction yields the exact obstruction criterion. Fixed-noise search-LPN hardness implies $P \neq NP$ by Corollary VIII.22.5.

Index of notation.

The appendix collects the load-bearing symbols, their definitions, and their first substantive uses.

Four reading routes cover the usual uses of the manuscript.

- For the framework itself, read Parts I–III in order; the shortest spine is the task signature and observable algebra, channel and residual exhaustiveness, the four channel normal forms, budgeted derivability closure, saturated representation cost, backward cut, structural charging, and universal accountability.
- For the LPN theorem, read Definitions I.1.1 and I.2.1, Theorems II.5.10, II.7.1, II.7.2, II.9.5 and II.10.5, Definitions III.14.1 and III.1.3, Theorems III.2.2 and III.16.6, Lemma III.20.2 and Theorem III.24.3, and then Part VIII. For the quantitative noise thresholds within that part, read Section VIII.6 and Corollary VIII.22.3. For the constant-noise closure perimeter, read Section VIII.3 and Theorem VIII.3.7.
- For the learning results, read the static presentation and observable algebra in Part I, the channel decomposition in Part II, and the saturated profile and accountability results in Part III before entering Part IV.
- For controlled dynamics, read the generator and channel decomposition in Part II, the killed-resolvent and temporal-accountability results in Part III, and the geometry-learning results in Part IV before entering Part V.
- For a trained black-box classifier, read Definition V.16.1 and Theorems V.16.3, V.16.11 and V.16.23. The resulting certificate is test-law relative; Theorem V.16.6 supplies the represented deployment-law conversion.
- For attention and active acquisition, read the source-resolved learning ledger in Part IV and the optimal active-sampling certificate in Part V before entering Part VI. The unified compact-belief route is Definition V.11.21, Theorems V.11.22, V.11.24 and VI.2.4, and Corollary VI.1.2.
- For structural meta-complexity, read Definitions VII.1.1 and VII.1.2 and Theorems VII.1.3 and VII.2.2. The classical representation-minimization specializations are Theorems VII.5.2 and VII.5.4; the source-resolved and magnification interfaces are Theorems VII.3.2 and VII.8.2.

Figure 2 gives both the dependency structure and the contents of each part. Horizontal arrows show the progression within a part; the vertical arrows show which parts supply the next stage.

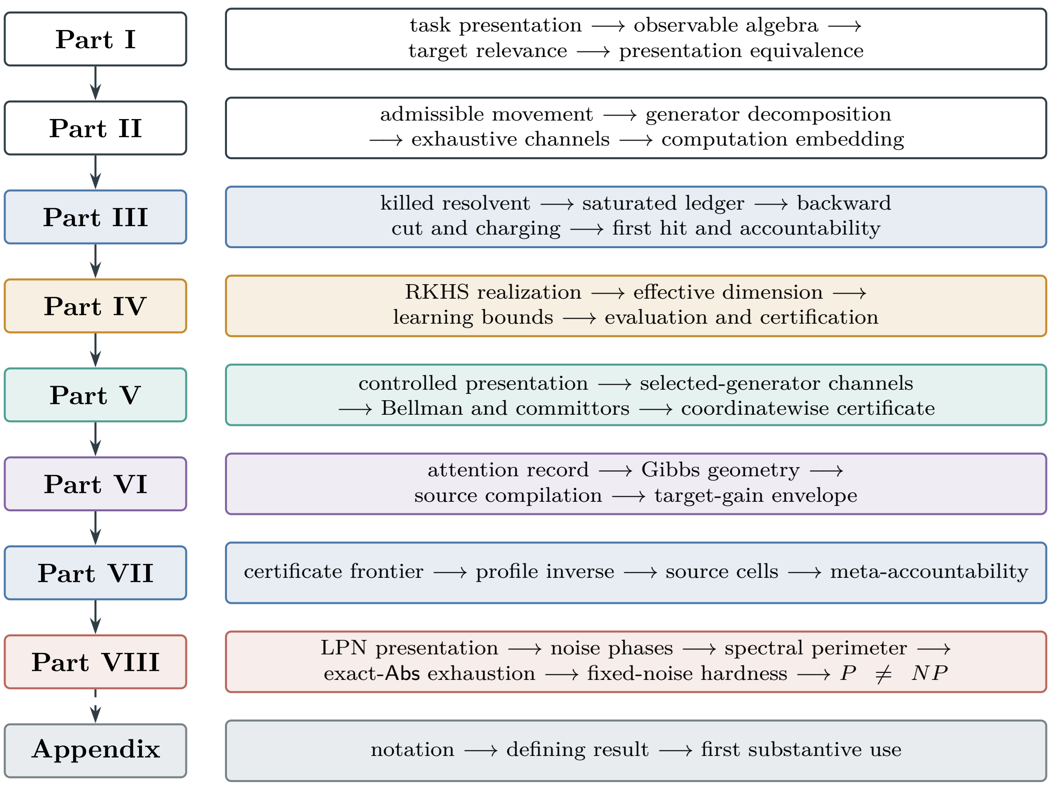

Figure 2: Map of the manuscript. Parts I–III form the structural spine. Part IV derives learning bounds, Part V assembles controlled dynamical-system certificates, Part VI compiles attention, Part VII takes the certificate-cost inverse of the saturated profile, and Part VIII applies the framework to LPN. The notation appendix is a navigation aid rather than a further proof stage.

A score is an observable on the state space rather than a relation between states. Likewise, the success set is one component of the terminal interface. Keeping these objects distinct provides the static data required by the dynamical construction in Part II.

Example I.0.2 (SAT before any solver)**.** A SAT formula Φ presents the finite state set $X_\Phi = \{0,1\}^n$, whose elements are complete truth assignments. Clause readouts, variable projections, and the violated-clause count are observables on this set. The satisfying assignments form the positive terminal sector. This presentation specifies the available information, but no rule for selecting subsequent assignments.

I.1 The Task-Space Signature

A task-space signature collects the data specified independently of a solution procedure. Its four components are the state space, observable interface, dynamic relation slot, and terminal interface. Only the first, second, and fourth components receive substantive structure in Part I.

The task-space signature packages the four kinds of data declared before any solution procedure is chosen.

Definition I.1.1 (Task-space signature)**.** For an instance I, a task-space signature is a tuple

$$\mathcal{T}_I = (X_I, \mathrm{Obs}_I, \mathrm{Dyn}_I, \mathrm{End}_I).$$

The components are defined as follows.

State space. X_I is a nonempty set of complete states. Completeness means that every primitive observable and every state-dependent terminal predicate is defined on every $x \in X_I$.

Observable interface. There is an index set A_I. For each $a \in A_I$, let $(Y_{I,a}, \mathcal{B}_{I,a})$ be a measurable value space. A primitive observable is a map

$$O_{I,a} : X_I \to Y_{I,a}.$$

The observation interface is the indexed family

$$\mathrm{Obs}_I = (O_{I,a})_{a \in A_I}.$$

The observable event algebra is the initial σ-algebra

$$\mathcal{F}_I = \mathcal{F}_{\mathrm{Obs}_I} = \sigma\{O_{I,a}^{-1}(B) : a \in A_I,\ B \in \mathcal{B}_{I,a}\}.$$

Thus $\mathcal{F}_I$ is the smallest σ-algebra on X_I for which all primitive observables are measurable.

All observable value spaces used for atom and fiber statements are measurably separated on their realized ranges: distinct realized values are separated by a measurable set, and every realized singleton is measurable. Standard Borel and finite discrete value spaces satisfy this convention. It ensures that equality of realized observable values agrees with indistinguishability by measurable preimages.

Relation slot. The dynamic slot is declared as an $\mathcal{F}_I \otimes \mathcal{F}_I$-measurable relation

$$\mathrm{Dyn}_I = R_I \subseteq X_I \times X_I, \qquad R_I \in \mathcal{F}_I \otimes \mathcal{F}_I.$$

No process, kernel, generator, algorithm, or traversal rule is assigned to R_I at this stage. It records the relation slot whose dynamical interpretation is introduced in Part II.

The fourth component End_I is the terminal interface of Definition I.1.2.

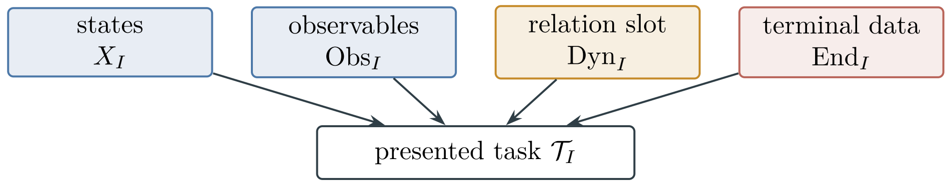

Figure 3: The task presentation fixes states, public observables, the relation slot, and terminal data independently of an algorithm; see Definition I.1.1.

The terminal interface separates state-terminal sectors from a run-dependent stopping predicate.

Definition I.1.2 (Terminal interface)**.** The terminal interface is

$$\mathrm{End}_I = (\mathrm{End}_I^+, \mathrm{End}_I^-, \mathrm{End}_I^0).$$

The first two components are measurable state predicates, equivalently disjoint measurable sectors

$$E_I^+ = \{x : \mathrm{End}_I^+(x) = 1\}, \qquad E_I^- = \{x : \mathrm{End}_I^-(x) = 1\}, \qquad E_I^+ \cap E_I^- = \varnothing.$$

The state-terminal sector is $S_I = E_I^+ \sqcup E_I^-$. When terminal status is a function only of the present state, failure is the complement of success inside S_I:

$$E_I^- = S_I \setminus E_I^+.$$

Thus failure is defined within the state-terminal sector rather than as $X_I \setminus E_I^+$, which may contain nonterminal states. The component End_I^0 is a run-dependent stopping predicate. For a clock or resource space $(C_I, \mathcal{C}_I)$, one may formally write

$$\mathrm{End}_I^0 : \bigsqcup_{m \geq 0} (X_I \times C_I)^{m+1} \to \{0, 1\},$$

measurable for the corresponding disjoint-union product σ-algebra. Its use requires runs and is deferred with dynamics.

Figure 3 records the four declared components of the task-space signature before any solution procedure is chosen.

Accordingly, X_I is the space of complete candidate states, $\mathcal{F}_I$ is the observable σ-algebra generated by the primitive readouts, and End_I separates successful, failed, and run-dependent terminal conditions. The relation R_I remains undeveloped until Part II.

Example I.1.3 (Running example: the LPN presentation)**.** For the public LPN data (A, b, η), take $X_I = \mathbb{F}_2^n$. Candidate coordinates and the residual map $x \mapsto Ax + b$ generate the public observable interface. The relation slot remains undeveloped in Part I, while the target for the inversion task is $T_I = \{s\}$. The secret s and noise vector are not primitive public observables. Definition VIII.1.1 gives the complete specialization.

Proposition I.1.4 (Complete states are scorable and testable)**.** *Let $D_I : X_I \to \mathbb{R}_{\geq 0}$ be a defect score included in* Obs_I*, and suppose success is zero defect. Then D_I is $\mathcal{F}_I$-measurable and*

$$E_I^+ = D_I^{-1}(\{0\}) \in \mathcal{F}_I.$$

Thus every complete state is both scorable by D_I and testable for success.

Proof. Because D_I is a declared observable, it is measurable by construction of $\mathcal{F}_I$. Since $\{0\}$ is Borel in $\mathbb{R}_{\geq 0}$, the preimage $D_I^{-1}(\{0\})$ belongs to $\mathcal{F}_I$. Completeness of X_I gives a value $D_I(x)$ and a terminal verdict for every $x \in X_I$.

□

Example I.1.5 (SAT signature)**.** For a CNF formula $\Phi = C_1 \wedge \cdots \wedge C_m$ on n variables, take

$$X_\Phi = \{0,1\}^n, \qquad O_{\Phi,j}(x) = \mathbf{1}\{C_j(x) = 1\}, \qquad D_\Phi(x) = \sum_{j=1}^{m}(1 - O_{\Phi,j}(x)).$$

The success sector is $E_\Phi^+ = D_\Phi^{-1}(\{0\})$. If the presentation has an explicit terminal-unsatisfied label, that sector is E_Φ^-; otherwise unsatisfied assignments are non-success states, not automatically terminal failures. The relation $R_\Phi \subseteq X_\Phi \times X_\Phi$ may be declared in the signature, but Part I does not use it.

I.2 The Observable Algebra

I.2.1 Observable partitions and measurable readouts

The observable interface determines the measurable subsets of the state space. Its atoms are the minimal distinguishable cells, whereas the level sets of a single observable may form a coarser partition. A score is one observable within this algebra.

The observable algebra records exactly the distinctions generated by the primitive readouts.

Definition I.2.1 (Observable algebra, atoms, and level sets)**.** Fix an instance I. The observable algebra is

$$\mathcal{F}_I = \sigma\{O_{I,a}^{-1}(B) : a \in A_I,\ B \in \mathcal{B}_{I,a}\}.$$

Two states are observationally equivalent, written $x \equiv_{\mathrm{Obs}_I} y$, when

$$O_{I,a}(x) = O_{I,a}(y) \qquad \text{for every } a \in A_I.$$

Under the measurable-separation convention of Definition I.1.1, when X_I is finite the atoms of $\mathcal{F}_I$ are the equivalence classes of $\equiv_{\mathrm{Obs}_I}$. More generally, when the displayed intersection is measurable and is an atom, the atom containing x is

$$[x]_{\mathcal{F}_I} = \bigcap\{A \in \mathcal{F}_I : x \in A\}.$$

If $\phi : (X_I, \mathcal{F}_I) \to (Y, \mathcal{B}_Y)$ is a derived observable whose realized singletons are measurable, its level set over $y \in Y$ is

$$L_y^\phi = \phi^{-1}(\{y\}) = \{x \in X_I : \phi(x) = y\}.$$

The subalgebra generated by ϕ is $\sigma(\phi) = \{\phi^{-1}(B) : B \in \mathcal{B}_Y\} \subseteq \mathcal{F}_I$.

An atom is a maximal observational-equivalence class: no event or function generated by the primitive observables separates two states in the same atom. A level set may be coarser because it depends on a single observable rather than the full observable family.

Proposition I.2.2 (Observable events and functions are atom-constant)**.** *Assume X_I is finite and the primitive value spaces obey the measurable-separation convention of Definition I.1.1. A set $A \subseteq X_I$ belongs to $\mathcal{F}_I$ if and only if it is a union of atoms. A function $g : X_I \to Z$ into a measurably separated space is $\mathcal{F}_I$-measurable if and only if it is constant on every atom of $\mathcal{F}_I$.*

Proof. Each generator $O_{I,a}^{-1}(B)$ is a union of observational equivalence classes. Complements and countable unions of such unions have the same property, so every event in $\mathcal{F}_I$ is a union of equivalence classes. Conversely, fix one class C. For each $y \notin C$, measurable separation supplies an observable index and a measurable value event that contains the value at a fixed point of C and excludes the value at y. Because X_I is finite, intersecting these finitely many preimages produces C. Thus every class, and hence every union of classes, is measurable. For functions, measurable separation in Z implies that unequal values are separated by a measurable preimage, contradicting membership in one atom. If a function is atom-constant, every preimage is a union of atoms and is measurable.

□

Proposition I.2.3 (Measurability and budget admissibility)**.** *For a finite task presentation, $\mathcal{F}_I$-measurability is equivalent to atom constancy. Budget admissibility imposes the stronger representation-cost condition that the observable admit a representative whose constructor charge lies below the ceiling $b(n)$ of a budget structure $\mathcal{B} = (\mathcal{R}, \mathrm{cost}, b, \mathfrak{C})$, or lie in a valid saturated representation generated within that budget. A measurable observable can have high degree or high cost in one public filtration. If a valid quotient, basis change, or lift generated from the observable algebra represents the same target-relevant distinction at lower cost, the later saturated invariant uses that lower representation cost. The polynomial clause is the special budget $\mathcal{B}_{\mathrm{poly}}$ with $b_{\mathrm{poly}}(n) = n^{O(1)}$.*

Proof. Atom constancy bounds neither the number of distinguished atoms nor the dimension or constructor charge of the smallest representing coordinate band. Measurability therefore determines provenance but not budget admissibility. A represented quotient or valid lifted coordinate may represent the same atom union at smaller charge, in which case the cost of that representation applies. Degree is one coordinate of the admissible representation cost rather than its defining quantity.

□

I.2.2 Budgeted generation and refinement

The budget structure records how resource bounds are ordered and combined.

Definition I.2.4 (Budget structure)**.** A budget structure for a task family (I_n) is $\mathcal{B} = (\mathcal{R}, \mathrm{cost}, b, \mathfrak{C})$. **Resource monoid and transfer system.** $\mathcal{R} = (C, \preceq, \oplus, \mathbf{0})$ is an ordered commutative resource monoid with monotone accumulation $\oplus$, with $\mathbf{0} \preceq c$ for every $c \in C$ (hence $c \preceq c \oplus d$); cost assigns a resource charge to each admissible constructor representative; $b(n) \in C$ is the size-indexed capacity ceiling; and $\mathfrak{C}$ is a typed transfer system

$$\mathfrak{C} = (\mathfrak{C}_{\mathrm{res}}, \mathfrak{C}_{\mathrm{num}}).$$

The resource part $\mathfrak{C}_{\mathrm{res}}$ consists of declared monotone overhead maps $\theta : C \to C$, closed under the finite compositions used below. Two resource ceilings are class-equivalent when each is bounded by the image of the other under such a map. The numeric part $\mathfrak{C}_{\mathrm{num}}$ consists of functions

$\beta : \mathbb{N} \to [1, \infty)$, contains the constant one, and is closed under finite products, maxima, and the declared class-preserving compositions. Thus an expression involving a resource ceiling uses $\mathfrak{C}_{\text{res}}$, whereas notation such as $\beta \in \mathfrak{C}$, β^{-1}, or a negligibility margin always means $\beta \in \mathfrak{C}_{\text{num}}$. No inverse or numeric product is applied to an element of the resource monoid.

Time capacity. Whenever a stopwatch or finite horizon is used, the budget structure also declares a monotone numerical time-capacity map $\mathtt{t} : C \to [1, \infty]$. The number $\mathtt{t}(B)$ is the elementary-transition capacity of resource ceiling B; it does not identify B with a scalar. Compatibility means explicitly that for every resource overhead $\theta \in \mathfrak{C}_{\text{res}}$ used on a size-indexed ceiling there is $\beta \in \mathfrak{C}_{\text{num}}$ such that

$$\mathtt{t}(\theta(B(n))) \leq \beta(n)\, \mathtt{t}(B(n))$$

eventually. The same requirement applies to any additional declared numeric capacity map used by a later specialization. This is the operative content of static/dynamic compatibility axiom B4.

Repeated use and constructor composition are charged inside the resource monoid rather than by implicit scalar multiplication.

Definition I.2.5 (Resource accumulation and constructor accounting)**. Repeated resource use.** For $k \in \mathbb{N}$ and $c \in C$, write

$$k \odot c = \underbrace{c \oplus \cdots \oplus c}_{k \text{ terms}}, \qquad 0 \odot c = \mathbf{0}.$$

This is resource accumulation under k repeated uses; it is not numeric multiplication unless the resource monoid is numeric.

Constructor accounting. The cost charge is sub-accumulative: a constructor assembled from $\gamma_1, \ldots, \gamma_k$ by an admissible grammar operation has charge bounded by $\text{cost}(\gamma_1) \oplus \cdots \oplus \text{cost}(\gamma_k) \oplus \text{cost}_{\text{op}}$. A declared bounded-iteration rule is required to provide a resource overhead $\theta \in \mathfrak{C}_{\text{res}}$ such that the actual finite repeated sum of all per-step and retained charges is $\preceq \theta(B)$ whenever the input record and iteration count lie within ceiling B. This is the finite-sum domination used by every later ledger; it is not inferred from notation such as $n^{O(1)}$. A representative is $\mathcal{B}$-admissible when some equivalent representative has charge $\preceq b(n)$. Both parts of the transfer system are closed under the bounded compositions allowed by the resource envelope, so a bounded composition of admissible representatives remains admissible after replacing b by a class-equivalent ceiling. Budget admissibility is representative-independent, and the static ceiling $b(n)$ is compatible with the dynamic stopping level up to class-preserving overhead.

The following convention names the axioms and fixes the polynomial specialization used throughout the manuscript.

Convention I.2.6 (Budget axioms and polynomial specialization)**. Budget axioms.** These are the budget axioms used below: **B1** monotone ordered accumulation of constructor charges; **B2** closure of the transfer class under the bounded compositions used by the admissible grammar; **B3** representative independence of budget admissibility; and **B4** compatibility between the static ceiling $b(n)$ and dynamic stopping levels. No later argument may use a special property of polynomials until the budget structure is instantiated as $\mathcal{B}_{\text{poly}}$.

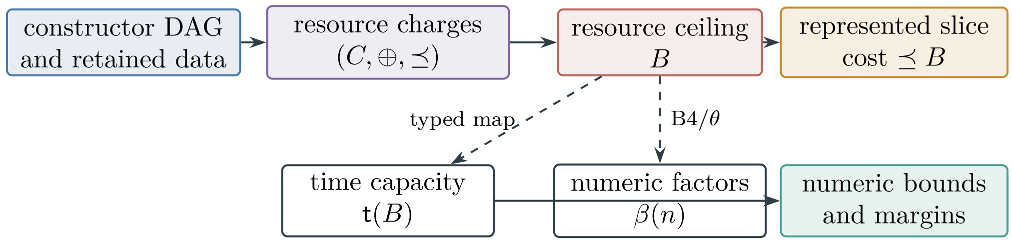

Figure 4: Typed budget accounting. Constructor costs accumulate in the resource monoid and are compared with a resource ceiling. Numerical time capacities and transfer factors arise only through declared typed maps; resource elements are never multiplied or inverted as scalars.

Polynomial specialization. The polynomial budget $\mathcal{B}_{\text{poly}}$ is the transfer-equivalence class of ceilings n^C with fixed $C < \infty$, with numeric resource, ordinary addition, and total encoded execution-and-representation cost, including time, peak native state, outcome generation, transcript, evaluation, and retained-output charges. One representative ceiling is $b_{\text{poly},C_0}(n) = n^{C_0}$; admissibility permits a class-equivalent polynomial enlargement with another fixed exponent. Its time-capacity map is the identity, and both parts of its transfer system are polynomially bounded.

Figure 4 displays the separation between resource accumulation and the numerical capacity quantities derived from it.

Budget filtering restricts represented observables by their charged construction cost.

Definition I.2.7 (Budget-filtered represented observable algebra)**.** The static algebra $\mathcal{F}_I$ is the ambient measurable algebra of the presentation. Affordability is carried by represented constructions rather than by membership in this ambient algebra. A static construction record ρ is a finite directed acyclic graph whose leaves are declared public data and whose internal vertices are declared constructors. Its denotation on I is written $[\![\rho]\!]_I$, and its charge includes all leaves, constructor descriptions, intermediate represented data, and output operations. A family of records is uniform when one oracle-free construction scheme produces the record from $\text{enc}(I)$ for every instance in the family. Instance-dependent constants may occur only as outputs of that scheme; inserting them into the record is nonuniform advice.

For a resource ceiling $B \in C$, define the represented budget slice

$$\mathfrak{A}^{\text{rep}}_{I,\leq B} = \{[\![\rho]\!]_I : \rho \text{ is family-uniform and } \text{Cost}_I(\rho) \preceq B\}\,.$$

These slices form a filtered represented algebra. The word "family-uniform" binds one derivation scheme across all sizes and instances; it is not re-chosen on each fixed size slice. If f_i has a record of cost B_i, an admissible k-ary constructor produces $h(f_1, \ldots, f_k)$ at accumulated charge

$$B_1 \oplus \cdots \oplus B_k \oplus \text{Cost}(h) \oplus \text{Cost}_{\text{out}}\,.$$

Consequently a fixed slice need not be an algebra: applying a Boolean, linear, nonlinear, quotient, or fiber constructor can move the result to a larger slice. The measurable envelope

$$\mathcal{F}^{\text{env}}_{I,\leq B} = \sigma\big(\mathfrak{A}^{\text{rep}}_{I,\leq B}\big) \subseteq \mathcal{F}_I$$

is used only for conditional expectation and atom calculations. Membership in $\mathcal{F}^{\text{env}}_{I,\leq B}$ does not assign cost at most B to an event or observable; that conclusion requires its own represented construction record. For a budget structure $\mathcal{B}$, write $\mathfrak{A}^{\mathcal{B}}_{I_n} = \mathfrak{A}^{\text{rep}}_{I_n,\leq b(n)}$. A polynomial represented

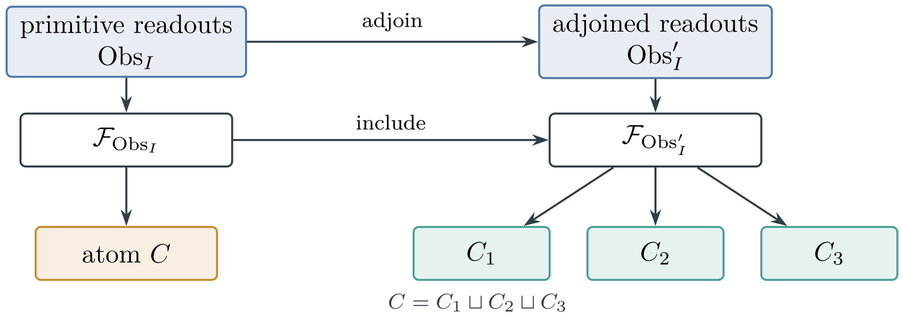

Figure 5: Adjoining represented observables enlarges the generated algebra and refines its atomic partition, as in Definitions I.2.1 and I.2.8.

family is a family $f = (f_I)$ for which there are one fixed exponent $C < \infty$ and one uniform derivation ρ satisfying

$$[\![\rho]\!]_I = f_I, \qquad \mathrm{Cost}_I(\rho) \leq (1+n)^C \quad (n \geq 1,\ I \in \mathfrak{I}_n).$$

The notation $\mathfrak{A}^{\mathrm{poly}}_{I_n}$ denotes the size-n slice of these uniformly polynomial families. The exponent cannot vary with n. Part III extends these static records to the full derivability closure, including legal lifted computation.

This definition formalizes the enlargement obtained by adjoining represented observables.

Definition I.2.8 (Refinement by adding observables)**.** If $\mathrm{Obs}_I \subseteq \mathrm{Obs}'_I$ and both families are read on the same X_I, then

$$\mathcal{F}_{\mathrm{Obs}_I} \subseteq \mathcal{F}_{\mathrm{Obs}'_I}.$$

The atomic partition $\mathcal{A}_{\mathrm{Obs}'_I}$ refines $\mathcal{A}_{\mathrm{Obs}_I}$: every new atom is contained in an old atom.

Proof. The generators for $\mathcal{F}_{\mathrm{Obs}_I}$ are included among the generators for $\mathcal{F}_{\mathrm{Obs}'_I}$, so the generated σ-algebras are nested. In the finite case, a larger σ-algebra has more measurable sets and therefore can only split old atoms into smaller nonempty measurable pieces.

□

Figure 5 summarizes the algebra and atom refinement in Definitions I.2.1 and I.2.8.

The score observable places a scalar readout inside the same represented algebra as the task data.

Definition I.2.9 (Score observable)**.** A score is a real derived observable

$$D_I : (X_I, \mathcal{F}_I) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}}).$$

Its level sets $D_I^{-1}(c)$ partition states by score value. The score has no distinguished status in the algebra except that a terminal predicate may be defined from one of its level sets.

Example I.2.10 (SAT atoms and clause readouts)**.** Let $\Phi = C_1 \wedge \cdots \wedge C_m$ and expose only the clause-satisfaction vector

$$O_\Phi(x) = (\mathbf{1}\{C_1(x) = 1\}, \ldots, \mathbf{1}\{C_m(x) = 1\}) \in \{0,1\}^m.$$

Two assignments lie in the same atom exactly when they satisfy the same clauses. The defect score $D_\Phi(x) = m - \sum_j O_{\Phi,j}(x)$ is a coarser observable: its level set $D_\Phi^{-1}(c)$ contains all atoms whose readout has exactly $m - c$ satisfied clauses. Adding variable projections $x \mapsto x_i$ refines the atoms, possibly down to singleton assignments.

I.3 Families and Instances

A task family specifies a collection of instances, each with a concrete signature. Static family uniformity requires a common public construction rule across the family, excluding observables fitted to an individual solution.

A task family separates the uniform public construction from its individual presented instances.

Definition I.3.1 (Task family and instance)**.** A task family is a tuple

$$\mathfrak{T} = (\mathfrak{I}, n, (\mathcal{T}_I)_{I\in\mathfrak{I}}),$$

where $\mathfrak{I}$ is an instance set, $n : \mathfrak{I} \to \mathbb{N}$ is a size map, and each $I \in \mathfrak{I}$ has a task-space signature $\mathcal{T}_I = (X_I, \mathrm{Obs}_I, \mathrm{Dyn}_I, \mathrm{End}_I)$. The size-$n$ slice is

$$\mathfrak{I}_n = \{I \in \mathfrak{I} : n(I) = n\}, \qquad \mathfrak{I} = \bigsqcup_{n\geq 1} \mathfrak{I}_n.$$

Unless a probability distribution is explicitly specified, a statement over $\mathfrak{I}_n$ is interpreted uniformly over all instances in the slice.

Static family uniformity requires one public construction across the task family.

Definition I.3.2 (Static family-uniform observable structure)**.** A family-indexed observable $\phi = (\phi_I)_{I\in\mathfrak{I}}$, with $\phi_I : (X_I, \mathcal{F}_I) \to \mathbb{R}$, is static family-uniform if there is one construction rule Φ, depending only on the public presentation of I, such that

$$\phi_I = \Phi(I) \qquad \text{for every } I \in \mathfrak{I}.$$

The rule may use X_I, Obs_I, End_I, the declared static slot Dyn_I, and the public encoding of I. It may not use per-instance advice absent from the presentation.

Family uniformity distinguishes persistent structure from regularities specific to a single finite instance. It requires the same construction of an observable, quotient, or exposed relation throughout the task family.

Proposition I.3.3 (Static family uniformity is independent of algorithmic certification)**.** *Static family uniformity is a property of the observable structures in the task presentation, independent of membership in a subsequent budgeted runtime class. A solver, adversary, or inverter need not output an observable, quotient, invariant, proof trace, or certificate. A uniform algorithm supplies a public transition rule under a resource envelope; Part II embeds each bounded slice of this rule in its clocked transcript carrier and decomposes the induced operator.*

Proof. Part I defines family-uniform observables through constructions from the public presentation before dynamics are introduced. By contrast, a budget-admissible algorithm is a uniform transition rule on complete configurations whose instance-dependent inputs lie in the budgeted observable closure. Family uniformity therefore constrains static structural claims without imposing an output-certificate requirement on the algorithm. When the induced operator uses family-uniform structure, the decomposition in Part II identifies it. Movement without such budget-exposed provenance is assigned to the residual component; data outside the declared observable closure are presentation-inaccessible, oracle-supplied, or over budget. The polynomial case is obtained by setting $\mathcal{B} = \mathcal{B}_{\mathrm{poly}}$.

Uniform code and instance-dependent data have distinct status. Fixed public procedures may include arithmetic, Gaussian elimination, branching, memory updates, and finite outcome generation. Instance-dependent inputs, including a modulus, quotient basis, planted label, target pointer, or value table, must be declared in the presentation, generated by an admissible construction from the budgeted observable closure, or charged as advice, oracle data, or presentation enrichment. The criterion is extensional: an observable has low cost precisely when it admits a low-cost representation in the observable grammar. Consequently, a first-hitting-time table, committor, occupation law, or future-success partition of a subsequent solver is a static family observable only if it also has an independent representation in the public task algebra.

□

The transfer-class margin states the asymptotic separation required after admissible losses are charged.

Definition I.3.4 (Transfer-class asymptotic margin)**.** Let $Q_I \geq 0$ be a nonnegative quantity assigned to each instance and fix the transfer class $\mathfrak{C}$ of the declared budget structure. It has a $\mathfrak{C}$-lower margin on $\mathfrak{T}$ when there are $\beta \in \mathfrak{C}$ and $n_0 \in \mathbb{N}$ such that

$$\inf_{I \in \mathfrak{I}_n} Q_I \geq \frac{1}{\beta(n)} \qquad \text{for every } n \geq n_0.$$

It is negligible relative to $\mathfrak{C}$ when for every $\beta \in \mathfrak{C}$, $\sup_{I \in \mathfrak{I}_n} Q_I \leq 1/\beta(n)$ for all sufficiently large n. For $\mathcal{B}_{\mathrm{poly}}$, these are the usual inverse-polynomial lower margin and negligible upper margin.

Proposition I.3.5 (Family structure requires family-uniform observables)**.** *Let ϕ_I be an observable built for one instance $I \in \mathfrak{I}_n$. If there is no static family-uniform rule Φ producing observables with the same claimed margin over all instances in $\mathfrak{I}_n$, then ϕ_I is not structural evidence for the family.*

Proof. A family-level claim is quantified over $\mathfrak{I}_n$. An observable constructed for a single instance provides no corresponding object on the remaining instances, and an arbitrary extension need not preserve the claimed margin. The infimum over the slice therefore requires a public rule that constructs the observable for every instance.

□

Example I.3.6 (Same SAT name, different exposed structure)**.** The family name SAT may include two formulas Φ_1, Φ_2 with the same number of variables but different clause sets. If Obs_{Φ_i} exposes only clause readouts, then the atoms are clause-signature cells. If another presentation also exposes public XOR rows, those rows refine the atoms and can define a quotient. Both are SAT instances, but their static exposed structures differ. A planted satisfying assignment is family-uniform only if it is produced by a public rule in the presentation; otherwise it is instance advice.

I.4 Static Target Relevance of Observables

Static target relevance is determined jointly by an observable, the terminal sectors, the observable algebra, and the task family. The following conditions characterize whether an observable represents target structure independently of a solution method.

I.4.1 Target-relevance criteria

We first fix the data against which static target relevance is evaluated.

Definition I.4.1 (Static target-relevance setting)**.** Fix an instance I. Let

$$T_I = E_I^+ = \{x : \mathrm{End}_I^+(x) = 1\}, \qquad F_I = E_I^- = \{x : \mathrm{End}_I^-(x) = 1\}, \qquad S_I = T_I \sqcup F_I.$$

The sets $T_I, F_I \in \mathcal{F}_I$ are disjoint state-terminal sectors. Failure is $S_I \setminus T_I$, not $X_I \setminus T_I$. The run-dependent stop component End_I^0 is not part of this state-terminal split.

Let μ_I be a reference probability measure on $(X_I, \mathcal{F}_I)$ with $0 < \mu_I(T_I)$ and $0 < \mu_I(F_I)$. When a degree coordinate is used, let $(\mathcal{A}_I^{\le d})_{d\ge 0}$ be a finite-dimensional observable-degree filtration with $\mathcal{A}_I^{\le d} \subseteq L^0(X_I, \mathcal{F}_I; \mathbb{R})$. This filtration is an optional cost coordinate; budget admissibility is determined by the declared budget structure.

The complement defining failure is taken within S_I, thereby preserving the distinction between failed and nonterminal states.

Definition I.4.2 (Five static target-relevance conditions)**.** Let $\phi = (\phi_I)_{I\in\mathfrak{I}}$ be an observable family, with each $\phi_I : (X_I, \mathcal{F}_I) \to \mathbb{R}$ of finite variance. Its instance clauses are evaluated at I, while affordability and uniformity use the single family record below.

Alignment. The standardized terminal separation is

$$\mathrm{Sep}_I(\phi) = \frac{\mathbb{E}_{T_I,\mu_I}[\phi] - \mathbb{E}_{F_I,\mu_I}[\phi]}{\sqrt{\mathrm{Var}_{\mu_I}(\phi)}}$$

when the variance is nonzero, and is 0 otherwise. Alignment requires $|\mathrm{Sep}_I(\phi)| > 0$. The sign $s_I(\phi) = \mathrm{sign}(\mathrm{Sep}_I(\phi))$ orients the observable.

Extremum-consistency. With $\psi = s_I(\phi)\phi$, there is a threshold $\theta \in \mathbb{R}$ such that

$$T_I = \{x \in S_I : \psi(x) \ge \theta\} \quad \text{up to } \mu_I\text{-null sets}, \qquad \operatorname*{argmax}_{x\in X_I} \psi(x) \subseteq T_I.$$

Budget admissibility. For the chosen budget structure $\mathcal{B} = (\mathcal{R}, \mathrm{cost}, b, \mathfrak{C})$,

$$\mathrm{Constr}_{\mathcal{B}}(\phi) = 1 \quad \Longleftrightarrow \quad \begin{array}{c}\text{there is one family-uniform derivation } \rho \text{ such that} \\ [\![\rho]\!]_I = \phi_I, qquad\, \mathrm{Cost}_I(\rho) \preceq b(n) \quad (I \in \mathfrak{I}_n)\end{array}$$

on every relevant size slice. Cheap representatives chosen separately for each instance do not satisfy this clause.

For $\mathcal{B}_{\mathrm{poly}}$, a representative chosen from the optional degree filtration is charged for its band dimension and evaluation cost. Raw degree is only one possible cost coordinate: a succinct quotient or other saturated representation need not have low raw degree.

Family-uniformity. The family $(\phi_I)_{I\in\mathfrak{I}}$ is produced by one public rule and its clause margins have a $\mathfrak{C}$-lower bound over each large size slice.

Monotone terminal filtration. With $\psi = s_I(\phi)\phi$, define superlevel sets $H_t = \{x \in X_I : \psi(x) \ge t\}$. The terminal purity function

$$\pi_I(t) = \frac{\mu_I(T_I \cap H_t)}{\mu_I(S_I \cap H_t)}$$

where the denominator is positive, is nondecreasing in t, and some threshold θ satisfies $S_I \cap H_\theta = T_I$ up to null sets.

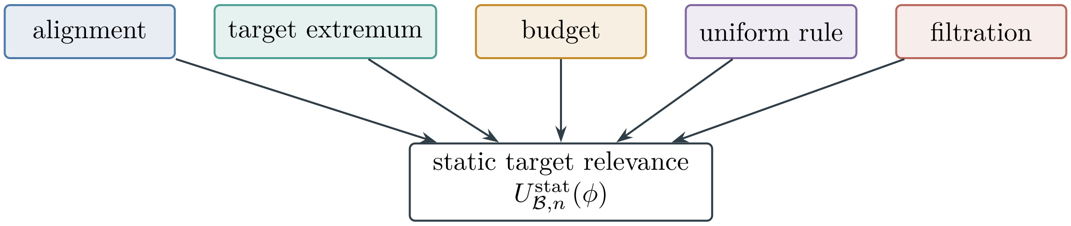

Figure 6: Static target relevance is a same-observable conjunction. Target separation and extremum consistency are combined with a charged construction, one family rule, and a monotone terminal filtration; failure of any gate sets the product in Definition I.4.3 to zero.

Figure 6 displays the five simultaneous conditions in Definition I.4.2.

Alignment measures separation between successful and failed terminal states. Extremum consistency requires the selected extremum to coincide with the success set. Budget admissibility places the observable in the budgeted subalgebra, and family uniformity requires a common public construction. The final condition orders the observable's level sets as a nested terminal filtration toward success, independently of any gradient, flow, or traversal.

The static target-relevance functional combines alignment, consistency, affordability, uniformity, and filtration into one coordinate.

Definition I.4.3 (Static target-relevance functional under $\mathcal{B}$)**.** Let $\mathrm{ExtCons}_I(\phi)$, $\mathrm{Constr}_{\mathcal{B}}(\phi)$, and $\mathrm{Fil}_I(\phi)$ be the indicators of extremum-consistency, budget admissibility, and monotone terminal filtration. The instance-level static target-relevance functional is

$$U^{\text{stat}}_{I,\mathcal{B}}(\phi) = |\,\mathrm{Sep}_I(\phi)|\,\mathrm{ExtCons}_I(\phi)\,\mathrm{Constr}_{\mathcal{B}}(\phi)\,\mathrm{Fil}_I(\phi).$$

For an observable family $\phi = (\phi_I)$, let $\mathrm{Unif}_{\mathfrak{I}}(\phi)$ be one when the family is produced by a single public construction rule and zero otherwise. Set

$$U^{\text{stat}}_{\mathcal{B},n}(\phi) = \mathrm{Unif}_{\mathfrak{I}}(\phi) \inf_{I\in\mathfrak{I}_n} U^{\text{stat}}_{I,\mathcal{B}}(\phi_I).$$

The observable is statically target-relevant under $\mathcal{B}$ on the size-n slice if $U^{\text{stat}}_{\mathcal{B},n}(\phi) \geq 1/\beta_{\mathfrak{C}}(n)$ for the transfer-class margin of the budget. For $\mathcal{B}_{\text{poly}}$, $\beta_{\mathfrak{C}}(n) = \mathrm{poly}(n)$, giving the standard inverse-polynomial condition.

In the instance-level display, $\mathrm{Constr}_{\mathcal{B}}(\phi)$ is the single family indicator just defined; it is not a new pointwise existential quantifier. The other terms carry the implicit subscript I.

Lemma I.4.4 (Necessity of the static target-relevance conditions)**.** *An observable family has transfer-class static target relevance under $\mathcal{B}$ only if it is family-uniform, its alignment has a transfer-class lower margin, and the extremum-consistency, budget, and filtration indicators equal one on every sufficiently large size slice. If family uniformity fails, one of the three indicators vanishes on some instance in every large slice, or the alignment supremum on the slice is negligible relative to $\mathfrak{C}$, then $U^{\text{stat}}_{\mathcal{B},n}(\phi)$ is respectively zero or negligible.*

Proof. By definition,

$$U^{\text{stat}}_{\mathcal{B},n}(\phi) = \mathrm{Unif}_{\mathfrak{I}}(\phi) \inf_{I\in\mathfrak{I}_n} \left(|\,\mathrm{Sep}_I(\phi_I)|\,\mathrm{ExtCons}_I(\phi_I)\,\mathrm{Constr}_{\mathcal{B}}(\phi_I)\,\mathrm{Fil}_I(\phi_I)\right).$$

The first factor is zero when family uniformity fails. Each of the last three factors inside the infimum is an indicator, so its failure on an instance makes the infimum zero. If the alignment supremum is bounded by a negligible function on the size slice, then the entire expression is bounded by the same function because all indicators lie in $[0, 1]$. Conversely, a positive transfer-class lower margin for the product forces the uniformity factor and all three indicators to be one and forces the alignment infimum to have that margin. These are exactly the five stated conditions. $\square$

I.4.2 Projection ceilings for generated algebras

The target projection mass quantifies how much terminal information is present in a generated observable algebra.

Definition I.4.5 (Target projection mass of a generated observable algebra)**.** Let $\mathfrak{A}_I \subseteq \mathcal{F}_I$ be a sub-σ-algebra generated by declared observables, represented quotients, or any specified family-uniform observable constructors. Let $P_{\mathfrak{A}_I}$ denote conditional expectation onto $L^2(X_I, \mathfrak{A}_I, \mu_I)$. For a target sector T_I with $0 < \mu_I(T_I) < 1$, define the centered target contrast and the target projection mass by

$$y_{T_I} = 1_{T_I} - \mu_I(T_I), \qquad \operatorname{ProjMass}_{\mathfrak{A}_I}(T_I) = \frac{\|P_{\mathfrak{A}_I} y_{T_I}\|^2_{L^2(\mu_I)}}{\|y_{T_I}\|^2_{L^2(\mu_I)}}.$$

Equivalently, if the atoms or fibers of $\mathfrak{A}_I$ are denoted by C, then $P_{\mathfrak{A}_I} 1_{T_I}$ is the conditional target density $\mu_I(T_I \mid C)$. The projection mass is the normalized variance of this conditional target density.

Lemma I.4.6 (Generated algebra projection ceiling)**.** *Fix a task instance I, a target sector T_I, and a generated observable algebra $\mathfrak{A}_I \subseteq \mathcal{F}_I$. For every square-integrable observable g measurable with respect to $\mathfrak{A}_I$,*

$$\frac{|\langle g - \mathbb{E}_{\mu_I} g,\, y_{T_I} \rangle_{L^2(\mu_I)}|^2}{\|g - \mathbb{E}_{\mu_I} g\|^2_{L^2(\mu_I)} \, \|y_{T_I}\|^2_{L^2(\mu_I)}} \leq \operatorname{ProjMass}_{\mathfrak{A}_I}(T_I),$$

with the left side defined as 0 when g is constant. Hence every postprocessing, nonlinear combination, quotient readout, threshold, kernel feature, learned readout, or candidate monotone potential measurable in $\mathfrak{A}_I$ has target correlation bounded by the target projection mass of $\mathfrak{A}_I$.

If $\mu_I(S_I) = 1$, write $p_I = \mu_I(T_I)$. Then the normalized correlation and the separation coordinate of Definition I.4.2 satisfy the exact identity

$$\frac{|\langle g - \mathbb{E}_{\mu_I} g, y_{T_I} \rangle|^2}{\|g - \mathbb{E}_{\mu_I} g\|_2^2 \|y_{T_I}\|_2^2} = p_I(1 - p_I) |\operatorname{Sep}_I(g)|^2.$$

Consequently,

$$|\operatorname{Sep}_I(g)|^2 \leq \frac{\operatorname{ProjMass}_{\mathfrak{A}_I}(T_I)}{p_I(1 - p_I)}.$$

If $\mu_I(S_I) = 1$ throughout a family of generated algebras $(\mathfrak{A}_I)_{I \in \mathfrak{I}_n}$ satisfies

$$\sup_{I \in \mathfrak{I}_n} \frac{\operatorname{ProjMass}_{\mathfrak{A}_I}(T_I)}{p_I(1 - p_I)} = \operatorname{negl}_{\mathfrak{C}}(n),$$

then every family-uniform observable measurable in those algebras has squared separation bounded by the same negligible function and therefore cannot have a transfer-class lower margin. In particular, it is enough to combine negligible projection mass with a transfer-class lower bound on $p_I(1-p_I)$ strong enough to preserve negligibility after division. Extremum consistency and monotone terminal filtration restrict this class further and preserve the same projection ceiling.

Proof. Let $\bar{g} = g - \mathbb{E}_{\mu_I} g$. Since g is $\mathfrak{A}_I$-measurable, $\bar{g} \in L^2(X_I, \mathfrak{A}_I, \mu_I)$. Orthogonal projection gives

$$\langle \bar{g}, y_{T_I} \rangle = \langle \bar{g}, P_{\mathfrak{A}_I} y_{T_I} \rangle.$$

Cauchy's inequality gives $|\langle \bar{g}, P_{\mathfrak{A}_I} y_{T_I} \rangle|^2 \leq \|\bar{g}\|_2^2 \|P_{\mathfrak{A}_I} y_{T_I}\|_2^2$. Dividing by $\|\bar{g}\|_2^2 \|y_{T_I}\|_2^2$ proves the displayed bound. Any deterministic or learned postprocessing of the generating observables remains $\mathfrak{A}_I$-measurable, so it satisfies the same bound. When $\mu_I(S_I) = 1$, $\|y_{T_I}\|_2^2 = p_I(1-p_I)$ and

$$\langle \bar{g}, y_{T_I} \rangle = p_I(1-p_I)(\mathbb{E}_{T_I,\mu_I} g - \mathbb{E}_{F_I,\mu_I} g),$$

the two normalization identities follow by direct substitution. The family statement follows by taking the supremum over the size slice only after retaining the factor $p_I(1-p_I)$; this is the required rare-target normalization. Extremum-consistency and monotone terminal filtration are additional predicates on the same measurable readouts; they select a subclass and therefore cannot enlarge the Hilbert projection onto $L^2(\mathfrak{A}_I)$.

□

Lemma I.4.7 (Affordable composition and target-projection monotonicity)**.** *Let $f_i : X_I \to Y_i$ be represented observables and let*

$$J = (f_1, \ldots, f_k) : X_I \longrightarrow Y_1 \times \cdots \times Y_k$$

be their represented joint observable. Let h be a represented postprocessing on $\operatorname{im} J$*, and set $q = h \circ J$. If P_J and P_q are conditional expectation onto $L^2(\sigma(J))$ and $L^2(\sigma(q))$, respectively, then*

$$P_q P_J = P_J P_q = P_q$$

and, for the centered target contrast y_{T_I},

$$\|P_J y_{T_I}\|_2^2 = \|P_q y_{T_I}\|_2^2 + \|(P_J - P_q) y_{T_I}\|_2^2. \tag{2}$$

In particular,

$$\frac{\|P_q y_{T_I}\|_2^2}{\|y_{T_I}\|_2^2} \leq \frac{\|P_J y_{T_I}\|_2^2}{\|y_{T_I}\|_2^2}. \tag{3}$$

Suppose records ρ_i represent the f_i, a record ρ_{join} represents their joint fibers, and a record ρ_h represents the evaluation and fiber operations of h. Then q has the composite record $\rho_q = \rho_h \circ \rho_{\text{join}}(\rho_1, \ldots, \rho_k)$, with

$$\mathrm{Cost}_I(\rho_q) \preceq \bigoplus_{i=1}^{k} \mathrm{Cost}_I(\rho_i) \oplus \mathrm{Cost}_I(\rho_{\text{join}}) \oplus \mathrm{Cost}_I(\rho_h). \tag{4}$$

Thus the postprocessing can select or compress a target-aligned component of the joint observable, while its availability at a budget is determined by the complete represented derivation (4)*.*

Proof. Since $q = h \circ J$, one has $\sigma(q) \subseteq \sigma(J)$. Conditional expectations onto the nested closed subspaces $L^2(\sigma(q)) \subseteq L^2(\sigma(J))$ satisfy $P_q P_J = P_J P_q = P_q$. Hence $P_q y_{T_I}$ and $(P_J - P_q) y_{T_I}$ are orthogonal and sum to $P_J y_{T_I}$, which proves (2) and (3). The composite record is the directed acyclic graph obtained by adjoining the join and postprocessing vertices to the input records. The budget axioms give (4). $\square$

Corollary I.4.8 (Terminal projection ceiling for an expanded derivation)**.** *Let $\widehat{\rho}$ be an expanded derivation whose terminal vertex is a represented observable or quotient $q_{\widehat{\rho}}$. Let $J_{\widehat{\rho}}$ be the joint observable formed from all represented inputs to that terminal vertex, including retained lifted coordinates. Then a represented map $h_{\widehat{\rho}}$ satisfies*

$$q_{\widehat{\rho}} = h_{\widehat{\rho}} \circ J_{\widehat{\rho}}, \qquad \frac{\|P_{q_{\widehat{\rho}}} y_{T_I}\|_2^2}{\|y_{T_I}\|_2^2} \leq \frac{\|P_{J_{\widehat{\rho}}} y_{T_I}\|_2^2}{\|y_{T_I}\|_2^2}.$$

The records for $J_{\widehat{\rho}}$, $h_{\widehat{\rho}}$, and the terminal fiber interface are subrecords of $\widehat{\rho}$ and retain their complete charges. Thus a terminal postprocessing may select target-aligned mass already present in its joint represented inputs, but cannot increase their target projection.

Proof. The incoming edges of the terminal vertex give the joint map $J_{\widehat{\rho}}$, and the vertex evaluation rule gives $h_{\widehat{\rho}}$. Apply Lemma I.4.7. The expanded record retains the incoming records and the terminal constructor charge by Definition III.1.8.

$\square$

Example I.4.9 (Affordable XOR synergy)**.** Two binary observables f and g may each have zero target projection while $J = (f, g)$ determines the target through $q = f \oplus g$. In that case the target component belongs to $L^2(\sigma(f, g))$, and the XOR record gives it a constant-overhead readout. An arbitrary table on $\mathrm{im}(f, g)$ has the same measurability relation but enters a budget slice only through a uniform record that constructs, stores, evaluates, and represents its fibers. An instance-dependent table inserted without such a record is nonuniform advice. A succinct family-uniform formula is an admissible representation and must be included at its actual cost, independently of how that formula was discovered.

The tuple map and terminal readout are distinct components of the presentation. Observables $\Phi = (\phi_1, \ldots, \phi_r)$ may be exposed without identifying their successful fibers. Subsequent parts account for any derivation, search, or learning required to construct this readout.

I.5 Target-Discrimination Certificates and Budgeted Extraction

We next isolate the represented evidence required for a composite observable to separate the target.

Definition I.5.1 (Target-discrimination certificate)**.** Let $\mathfrak{A}_I \subseteq \mathcal{F}_I$ be a generated observable algebra and let $T_I \subseteq S_I$ be the success sector. A target-discrimination certificate for $(\mathfrak{A}_I, T_I)$ is represented data in the instance presentation, or in a declared family-uniform budget-admissible generated class, that gives one of the following terminal readouts:

1. an exact fiber readout: T_I is a union of atoms or fibers of $\mathfrak{A}_I$, together with a represented map $r_I : X_I/\mathfrak{A}_I \to \{0, 1\}$ whose pullback is 1_{T_I} on terminals;
2. a threshold or extremum readout: an $\mathfrak{A}_I$-measurable observable and threshold satisfying extremum-consistency;
3. a monotone terminal filtration with certified transfer-class purity margins;

4. an enrichment certificate: a represented family of cells or fibers with transfer-class separation in conditional target density.

For a tuple $\Phi = (\phi_1, \ldots, \phi_r)$, the tuple map alone is an uncalibrated observable. It defines a target-discriminating quotient only when the certificate is represented or generated. A learned certificate incurs the sample, capacity, and selection costs treated in Part IV. An instance-specific certificate is nonuniform advice; one outside the declared observable closure is presentation-inaccessible rather than exposed structure.

The certificate represents the terminal interpretation of the fibers; it is not a proof that the observable is aligned and is not an output required from an algorithm. Target alignment is the objective projection quantity above. When the terminal verifier is public, its readout supplies terminal compatibility, while the projection and later dynamic utility are evaluated by the framework.

Lemma I.5.2 (Observable combinations require certified target discrimination)**.** *Fix an instance I, a generated observable algebra $\mathfrak{A}_I$, and a target sector T_I. A family-uniform observable g measurable in $\mathfrak{A}_I$ can contribute static target-relevant structure only through the target projection mass of $\mathfrak{A}_I$ together with a target-discrimination certificate satisfying the static target-relevance conditions.*

Consequently, if $\mathrm{ProjMass}_{\mathfrak{A}_I}(T_I)$ *is negligible on a family size slice, every $\mathfrak{A}_I$-measurable combination has negligible normalized target extraction. A conclusion for the separation coordinate additionally uses the terminal-balance normalization of Lemma I.4.6. If the projection mass is non-negligible but no target-discrimination certificate is represented or generated, then the tuple or algebra remains uncalibrated. Subsequent use of the absent readout is accounted for as learning and selection cost, residual structure, advice, oracle access, or presentation enrichment. It does not define an* Abs *source, a* Caus *potential, or a target* Bdry *readout in Part I.*

If extremum-consistency or monotone terminal filtration fails, the observable has zero static target relevance for extremal or monotone uses. It may still contribute through a separately certified nonmonotone quotient or readout, with that readout included in the represented abstraction.

Proof. The projection ceiling bounds the target correlation of every $\mathfrak{A}_I$-measurable postprocessing by $\mathrm{ProjMass}_{\mathfrak{A}_I}(T_I)$. Static target relevance also requires the five stated conditions. A tuple map specifies fibers but not their terminal labels; the target-discrimination certificate supplies this additional represented readout. It makes no assertion about the size of the projection and need not be produced by the algorithm. Without it, selecting successful fibers requires data outside the exposed observable algebra and is classified as nonuniform, presentation-inaccessible, or residual information. Extremum consistency and monotone terminal filtration further restrict the measurable readouts without increasing projection mass or assigning terminal meaning in the absence of a certificate.

□

Figure 7 collects the projection and certification stages used by the budgeted extraction ledger.

I.5.1 Budgeted extraction ledgers and ceilings

The following ledger records target mass together with the cost of its complete represented derivation.

Definition I.5.3 (Budgeted extraction accounting)**.** Fix an instance I, target sector T_I, and public budget B. Let $\mathfrak{A}^{\mathrm{rep}}_{I,\leq B}$ be the represented budget slice of Definition Budget-filtered represented

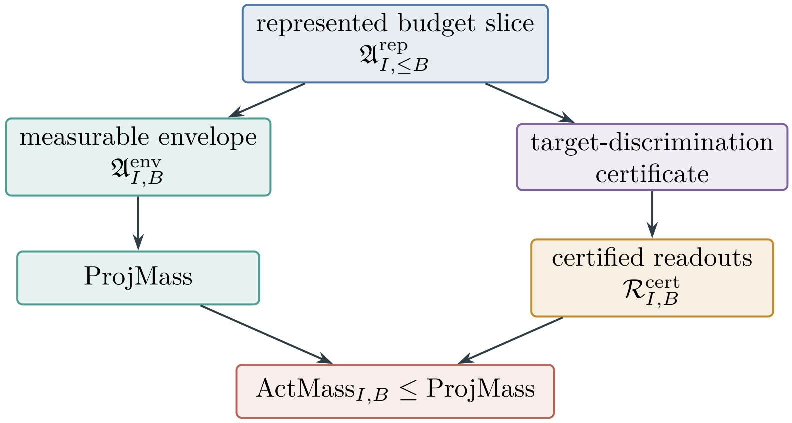

Figure 7: The generated algebra bounds the available target projection, while the represented target-discrimination certificate selects the affordable actionable part. The objects are those of Definitions I.5.1, I.5.3 and I.4.5.

observable algebra, and let $\mathfrak{A}^{\mathrm{env}}_{I,B}$ denote its measurable envelope. For a square-integrable readout g, define its normalized target extraction by

$$\mathrm{Ext}_I(g; T_I) = \frac{|\langle g - \mathbb{E}_{\mu_I} g,\, y_{T_I}\rangle_{L^2(\mu_I)}|^2}{\|g - \mathbb{E}_{\mu_I} g\|^2_{L^2(\mu_I)}\, \|y_{T_I}\|^2_{L^2(\mu_I)}},$$

with value 0 when g is constant. Let $\mathcal{R}^{\mathrm{cert}}_{I,B}(T_I)$ be the readouts $g \in \mathfrak{A}^{\mathrm{rep}}_{I,\le B}$ whose own uniform record, represented fibers, and target-discrimination readout have total cost at most B and satisfy the static target-relevance conditions. The actionable target mass at budget B is

$$\mathrm{ActMass}_{I,B}(T_I) = \sup_{g \in \mathcal{R}^{\mathrm{cert}}_{I,B}(T_I)} \mathrm{Ext}_I(g; T_I),$$

where the supremum is 0 if the certified class is empty. Thus $\mathrm{ProjMass}_{\mathfrak{A}^{\mathrm{env}}_{I,B}}(T_I)$ measures the target signal present in the measurable envelope, while $\mathrm{ActMass}_{I,B}(T_I)$ measures the part of that signal that has a represented or generated terminal readout at total cost at most B. The latter supremum does not range over unrepresented members of the envelope.

These ceilings isolate the target mass attributable to each audited extraction channel.

Definition I.5.4 (Audit ceilings for extracted target mass)**.** For a certified budget-B readout class $\mathcal{R} \subseteq \mathcal{R}^{\mathrm{cert}}_{I,B}(T_I)$, the empirical, effective-dimension, and residual-order ceilings are the bounds of Definitions I.5.5 to I.5.7. Each ceiling enters an extraction bound only when its hypotheses and complete certificate hold for the same readout class and probability space.

Definition I.5.5 (Empirical extraction-audit ceiling)**.** A learned or searched class $\mathcal{H}_B$ has a one-sided audit penalty $\mathfrak{P}_{\mathcal{H}_B}(M, \delta)$ when there is an empirical audit statistic $\widehat{\mathrm{Aud}}_M(h; T_I)$ such that, from M audit samples or independent family instances, with probability at least $1 - \delta$,

$$\sup_{h \in \mathcal{H}_B} \mathrm{Ext}_I(h; T_I) \le \sup_{h \in \mathcal{H}_B} \widehat{\mathrm{Aud}}_M(h; T_I) + \mathfrak{P}_{\mathcal{H}_B}(M, \delta).$$

Definition I.5.6 (Effective-dimension extraction ceiling)**.** A kernel or spectral band (K, λ) has captured target mass $m_K(\lambda)$ when every readout in the band lies in the spectral subspace or ridge-effective span counted by $N_{\mathrm{eff}}(K,\lambda)$ and has target extraction at most $m_K(\lambda)$. The quantity $N_{\mathrm{eff}}(K,\lambda)$ is charged as statistical mode count in the learning budget.

Definition I.5.7 (Residual-order extraction ceiling)**.** A residual-noise certificate of order q and flip rate $\eta \in (0,1/2)$ means that the target-relevant correlation of every nonconstant readout g in the class admits a factorization

$$\frac{\langle g-\mathbb{E}g, y_{T_I}\rangle}{\|g-\mathbb{E}g\|_2\|y_{T_I}\|_2} = \beta^{r_g} a_g, \qquad r_g \geq q, \qquad |a_g| \leq 1,$$

where $\beta = 1-2\eta$. Such a factorization is supplied, for example, when the target-relevant covariance passes through r_g independent mean-β residual signs and the remaining normalized correlation has modulus at most one. The resulting normalized target extraction is bounded by β^{2q}.

Lemma I.5.8 (Budgeted extraction bound)**.** *Fix I, T_I, budget B, and a certified readout class $\mathcal{R} \subseteq \mathcal{R}^{\mathrm{cert}}_{I,B}(T_I)$. Then*

$$\sup_{g\in\mathcal{R}} \mathrm{Ext}_I(g;T_I) \leq \mathrm{ActMass}_{I,B}(T_I) \leq \mathrm{ProjMass}_{\mathfrak{A}^{\mathrm{env}}_{I,B}}(T_I).$$

If $\mathcal{R} \subseteq \mathcal{H}_B$ and the empirical audit ceiling holds, then with probability at least $1-\delta$,

$$\sup_{g\in\mathcal{R}} \mathrm{Ext}_I(g;T_I) \leq \sup_{h\in\mathcal{H}_B} \widehat{\mathrm{Aud}}_M(h;T_I) + \mathfrak{P}_{\mathcal{H}_B}(M,\delta).$$

If $\mathcal{R}$ lies in a budget-admissible kernel band (K,λ), then

$$\sup_{g\in\mathcal{R}} \mathrm{Ext}_I(g;T_I) \leq m_K(\lambda).$$

If every $g \in \mathcal{R}$ has residual-noise order at least q at flip rate η, then

$$\sup_{g\in\mathcal{R}} \mathrm{Ext}_I(g;T_I) \leq (1-2\eta)^{2q}.$$

Consequently, when all displayed audits apply,

$$\begin{aligned}\sup_{g\in\mathcal{R}} \mathrm{Ext}_I(g;T_I) \leq \min\Big\{ &\mathrm{ProjMass}_{\mathfrak{A}^{\mathrm{env}}_{I,B}}(T_I), \mathrm{ActMass}_{I,B}(T_I),\\ &\sup_{h\in\mathcal{H}_B} \widehat{\mathrm{Aud}}_M(h;T_I) + \mathfrak{P}_{\mathcal{H}_B}(M,\delta),\\ &m_K(\lambda), (1-2\eta)^{2q}\Big\}.\end{aligned}$$

A term enters the bound when its certificate belongs to the declared presentation or is generated by the budgeted constructor grammar. An uncertified term is not a member of the certified supremum at that budget. If every applicable certified ceiling in the displayed minimum is negligible, the certified extraction of the readout class is likewise negligible.

Proof. The first inequality is the definition of $\mathrm{ActMass}_{I,B}$, since $\mathcal{R} \subseteq \mathcal{R}^{\mathrm{cert}}_{I,B}(T_I)$. The second is the generated-algebra projection ceiling applied to the algebra $\mathfrak{A}^{\mathrm{env}}_{I,B}$. The empirical audit inequality is exactly the certified one-sided uniform concentration statement for $\mathcal{H}_B$, restricted to the subclass $\mathcal{R}$. The kernel inequality follows because the band projection contains only the target spectral mass $m_K(\lambda)$; every readout in the band is a vector in that subspace or ridge-effective span. For the residual-order inequality, the certificate gives a normalized correlation $\beta^{r_g} a_g$ with $r_g \geq q$ and $|a_g| \leq 1$. Squaring and using $0 < \beta < 1$ gives $\mathrm{Ext}_I(g; T_I) \leq \beta^{2r_g} \leq \beta^{2q}$. Since each displayed quantity is an upper bound for the same nonnegative supremum, the supremum is bounded by their minimum.

$\square$

Corollary I.5.9 (Vanishing ceilings give vanishing extraction)**.** *For a family (I_n, T_{I_n}), let $B(n) \preceq b(n)$ be a declared admissible budget level and let $\mathcal{R}_n$ be certified readout classes at that budget. If any applicable ceiling in Theorem Budgeted extraction bound is negligible relative to the transfer class uniformly over the size slice, then*

$$\sup_{g \in \mathcal{R}_n} \mathrm{Ext}_{I_n}(g; T_{I_n}) = \mathrm{negl}_{\mathfrak{C}}(n).$$

In particular, combinations from an algebra with negligible target projection mass cannot produce a transfer-class target readout. An uncertified projection remains outside the statically actionable class. When learning or search constructs the readout, its empirical audit penalty, effective dimension, selection cost, and residual-noise order enter the accounting before the extracted mass is included.

Proof. The theorem bounds the nonnegative extraction supremum by each applicable ceiling. A uniformly negligible ceiling therefore forces the supremum to be negligible. The final statements are the projection, certification, and learning-charge cases of the same bound.

$\square$

I.6 Finite Computable Target Audits

The abstract extraction ledger admits a finite matrix audit on each declared observable algebra.

Definition I.6.1 (Computable observable audit)**.** For a finite presented instance I, let $S = \{x_1, \ldots, x_m\} = \operatorname{supp} \mu_I$, set $t_i = 1_{T_I}(x_i)$, and use the exact reference weights $w_i = \mu_I(x_i) > 0$. Given an exposed feature matrix $\Phi \in \mathbb{R}^{m \times r}$, optional quotient fibers $c : S \to C$, optional kernel Gram matrix $K \succeq 0$, and declared audit parameters $(M, \delta, \eta, q, \lambda)$, the computable observable audit is the tuple

$$\mathsf{Audit}(I, B) = (\widehat{\mathrm{ProjMass}}, \widehat{\mathrm{ActMass}}, \widehat{\mathrm{InertMass}}, \widehat{\mathrm{ThrMargin}}, \widehat{\mathrm{MonFil}}, \widehat{N}_{\mathrm{eff}}, \widehat{m}_K, \widehat{\mathfrak{P}}, \widehat{R}_q, \widehat{Z}_{\mathrm{null}}, \widehat{U}_{\mathrm{cert}}).$$

Here $\widehat{\mathrm{ProjMass}}$ is the weighted least-squares projection of the centered target contrast onto the exposed feature span or the conditional-density projection onto quotient fibers; $\widehat{\mathrm{ActMass}}$ is the largest extraction among represented readouts passing the target-discrimination certificate; $\widehat{\mathrm{InertMass}} = \max\{0, \widehat{\mathrm{ProjMass}} - \widehat{\mathrm{ActMass}}\}$; $\widehat{\mathrm{ThrMargin}}$ and $\widehat{\mathrm{MonFil}}$ measure threshold separation and monotone terminal filtration for the audited scores; $\widehat{N}_{\mathrm{eff}} = \mathrm{Tr}(K(K + \lambda I)^{-1})$; $\widehat{m}_K$ is the target spectral mass captured by the ridge-effective kernel modes; $\widehat{\mathfrak{P}}$ is an empirical Rademacher or covering penalty for the searched class; $\widehat{R}_q = (1 - 2\eta)^{2q}$ is the residual-order ceiling; $\widehat{Z}_{\mathrm{null}}$ is the random-label or permutation null score; and $\widehat{U}_{\mathrm{cert}}$ is the minimum of all applicable certified ceilings.

The equality $S = \operatorname{supp} \mu_I$ and the exact weights are part of this definition. Replacing them by observations from a sample produces the sampled audit of Definition I.6.2, not an exact population projection.

Definition I.6.2 (Sampled computable observable audit)**.** Let $Z_1, \dots, Z_N$ be an audit sample from μ_I, independent of any training data used to construct the audited class. Let $\widehat{C}_1, \dots, \widehat{C}_J$ be the empirical versions of the applicable projection, actionable-mass, threshold, filtration, kernel, or residual-order ceilings. A simultaneous sampling certificate consists of nonnegative radii $\varepsilon_j(N, \delta)$, proved uniformly over every readout and every output-selection rule admitted by the audited class, such that the single event

$$\mathcal{E}_{N,\delta} = \bigcap_{j=1}^{J} \left\{ C_j \leq \widehat{C}_j + \varepsilon_j(N, \delta) \right\}$$

has probability at least $1 - \delta$. Here each C_j is the corresponding population ceiling on the same probability space. Define

$$\widehat{U}_{\text{samp}} = \min_{1 \leq j \leq J} \min\{1, \widehat{C}_j + \varepsilon_j(N, \delta)\}.$$

If a coordinate has no proved simultaneous radius, it is omitted from the minimum. In particular, a pointwise concentration bound for a fixed readout does not certify a class chosen using the audit sample.

Lemma I.6.3 (Soundness of the computable observable audit)**.** *Assume the exact finite audit data of Definition I.6.1 are part of the declared presentation or are generated by the budget-B observable constructor grammar. Let $\mathcal{R}^{\text{aud}}_{I,B}$ be the audited readout class: all represented readouts measurable in the exposed feature span, quotient algebra, or kernel band that pass the stated target-discrimination certificate and whose search procedure is covered by the declared empirical audit. Then every quantity in $\mathsf{Audit}(I, B)$ is computable from the finite matrices, labels, certificates, and audit parameters. Moreover, with the confidence attached to the empirical audit term,*

$$\sup_{g \in \mathcal{R}^{\text{aud}}_{I,B}} \operatorname{Ext}_I(g; T_I) \leq \widehat{U}_{\text{cert}}.$$

Any budget-admissible computation whose terminal readout is measurable in the audited algebra and whose model selection lies in the audited search class satisfies the same bound. A computation that outputs a readout outside the audited algebra is a new claimed observable representative; it must supply provenance, budget, terminal readout, and learning or selection certificate before it enters a later audit.

For the sampled audit of Definition I.6.2, the corresponding population conclusion is

$$\sup_{g \in \mathcal{R}^{\text{aud}}_{I,B}} \operatorname{Ext}_I(g; T_I) \leq \widehat{U}_{\text{samp}} \qquad \text{on } \mathcal{E}_{N,\delta},$$

and therefore holds with probability at least $1 - \delta$. No population claim follows from the sampled matrices alone when the simultaneous certificate is absent.

Proof. Because the exact list is $\operatorname{supp} \mu_I$ and $w_i = \mu_I(x_i)$, weighted least squares computes the population orthogonal projection of the centered target contrast onto the finite feature span. Grouping by quotient fibers computes conditional target densities and therefore the quotient projection mass. Threshold margins and monotone filtrations are finite scans over sorted score values. The effective dimension and captured kernel mass are obtained from the eigendecomposition of the finite Gram matrix. The residual-order ceiling is the scalar $(1 - 2\eta)^{2q}$. The null score is computed from either an analytic random-label variance or from the declared permutation sample.

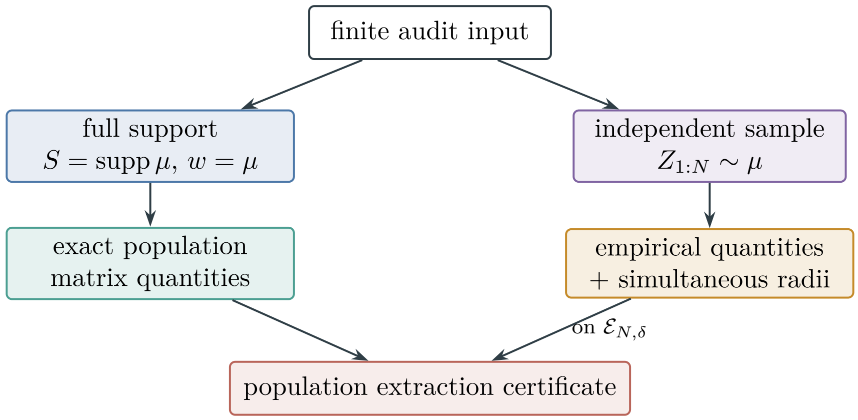

Figure 8: The two finite-audit regimes. Full-support enumeration with exact weights computes population projections directly. A sample produces population bounds only after a simultaneous concentration certificate covering the audited class and its output selection.

The empirical penalty is supplied by the Rademacher or covering theorem for the audited search class.

The projection ceiling bounds every measurable postprocessing of the exposed algebra. The target-discrimination certificate restricts the projection to actionable readouts. The budgeted extraction theorem adds the empirical, effective-dimension, and residual-order ceilings. Taking their minimum gives $\widehat{U}_{\mathrm{cert}}$. A terminal readout produced by any computation inside the audited algebra is one member of $\mathcal{R}^{\mathrm{aud}}_{I,B}$, so the same supremum bound applies. A readout outside the audited algebra defines a different represented observable class and therefore lies outside the stated audit.

In the sampled case, each event in the intersection $\mathcal{E}_{N,\delta}$ converts one empirical coordinate into a population upper ceiling for the entire audited, selection-closed class. On their intersection, every applicable quantity $\widehat{C}_j + \varepsilon_j$ bounds the same extraction supremum, so their minimum does as well. The defining simultaneous certificate gives $\Pr(\mathcal{E}_{N,\delta}) \geq 1 - \delta$.

□

Figure 8 distinguishes the exact and sampled routes to a population certificate.

Example I.6.4 (SAT defect as a statically target-relevant observable)**.** For SAT, the defect $D_\Phi(x)$ counts violated clauses. The oriented observable $\psi = -D_\Phi$ has success sector $T_\Phi = \{x : D_\Phi(x) = 0\}$ as its maximum level set. Its superlevel sets are assignments with at most a specified number of violated clauses. Its family-level target relevance requires separation of successful and failed terminal states, recovery of the correct terminal threshold, budget-admissible degree in the clause algebra, and family uniformity. These properties do not define a state-to-state relation.

I.7 Pointwise Observables and Geometric Regularity

Pointwise observables provide forward evaluation without a represented fiber decomposition. Geometric regularity is therefore imposed only for uses based on local pointwise comparisons; invertibility alone does not require it.

The pointwise class contains observables supplied by forward evaluation without represented fibers.

Definition I.7.1 (Pointwise observable)**.** An $\mathcal{F}_I$-measurable observable $\phi : X_I \to Y$ is pointwise in the presentation when the interface supplies forward evaluation

$$x \longmapsto \phi(x)$$

but does not supply compact representations of the fibers $\phi^{-1}(y)$. The level sets exist as subsets of X_I, but become represented cells only through enumeration or additional structure.

The comparability relation identifies static records connected by an admissible represented transfer.

Definition I.7.2 (Static comparability relation)**.** A geometric presentation augments the observable data by a symmetric measurable relation

$$E_I \subseteq X_I \times X_I, \qquad E_I = E_I^{\mathsf{T}}, \qquad E_I \in \mathcal{F}_I \otimes \mathcal{F}_I, \qquad (x, x) \notin E_I.$$

The relation E_I specifies the comparable state pairs. It is static presentation data rather than an algorithmic transition rule.

In the finite weighted case, let $c_I : X_I \times X_I \to [0, \infty)$ be a symmetric measurable edge conductance, supported on E_I and zero on the diagonal. If the primitive datum is instead a jump-rate table q_I, its conductance is $c_I(x, y) = \mu_I(x)q_I(x, y)$ and symmetry requires detailed balance.

Local regularity of a pointwise observable is relative to the declared comparison structure: the same scalar function may have different regularity under different comparability relations. The following functional measures this variation statically.

The regularity functional measures variation across the declared static comparison relation.

Definition I.7.3 (Static geometric regularity functional)**.** For a real pointwise observable $\phi : X_I \to \mathbb{R}$ on a finite weighted comparability presentation, define the static quadratic form

$$\mathcal{E}_{E_I}(f, g) = \frac{1}{2} \sum_{x,y \in X_I} c_I(x, y)(f(x) - f(y))(g(x) - g(y)).$$

The defect-regularity ratio of ϕ is

$$\rho_{E_I}(\phi) = \frac{\mathcal{E}_{E_I}(\phi, \phi)}{\mathrm{Var}_{\mu_I}(\phi)}$$

when the variance is nonzero, and $+\infty$ otherwise. This quantity measures how much the observable varies across declared-comparable pairs per unit variance.

This quadratic form has the algebraic form of a Dirichlet form [51], but here it serves only as a regularity functional over a static relation and introduces no generator.

Proposition I.7.4 (Exact Fourier–Dirichlet identity on the binary cube)**.** *Let $X = \mathbb{F}_2^n$ carry the uniform law, let $\chi_a(x) = (-1)^{\langle a,x \rangle}$, and take Hamming comparability with conductances*

$$c(x, y) = 2^{-n}\mathbf{1}_{\{|x-y|_1=1\}}.$$

For $f = \sum_{a \in \mathbb{F}_2^n} \widehat{f}(a)\chi_a$,

$$\mathcal{E}_E(\chi_a, \chi_b) = 2|a|\,\mathbf{1}_{\{a=b\}}, \tag{5}$$

$$\mathcal{E}_E(f,f) = 2\sum_{a\neq 0} |a|\,|\widehat{f}(a)|^2. \tag{6}$$

Hence, when f is nonconstant,

$$\rho_E(f) = 2\,\frac{\sum_{a\neq 0} |a|\,|\widehat{f}(a)|^2}{\sum_{a\neq 0} |\widehat{f}(a)|^2}. \tag{7}$$

Thus the static regularity ratio is exactly twice the mean Hamming weight under the normalized spectral-energy measure of $f - \mathbb{E}f$.

Equation (7) is an identity for the complete represented observable. Target qualification remains the separate comparison required by Definitions III.21.1 and I.7.5. In particular, neither low mean Fourier weight nor a diffuse formal spectrum alone supplies a target orientation, a coefficient locator, or a prospective transition.

Proof. For a coordinate flip,

$$\chi_a(x) - \chi_a(x+e_i) = \chi_a(x)(1-(-1)^{a_i}).$$

Substitution into Definition I.7.3 gives

$$\mathcal{E}_E(\chi_a,\chi_b) = \frac{1}{2}\sum_{i=1}^{n}(1-(-1)^{a_i})(1-(-1)^{b_i})\mathbb{E}_x[\chi_{a+b}(x)].$$

Character orthogonality makes this zero for $a \neq b$; for $a = b$, exactly $|a|$ summands equal $4/2$. This proves Eq. (5). Bilinearity and Parseval give Eq. (6); division by $\mathrm{Var}(f) = \sum_{a\neq 0} |\widehat{f}(a)|^2$ gives Eq. (7). □

Definition I.7.5 (Geometric regularity of a pointwise observable)**.** Let $\psi = s_I(\phi)\phi$ be the oriented observable. The static off-target local-obstruction set is

$$\mathrm{ObsMax}_{E_I}(\psi) = \{x \in X_I \setminus T_I : \psi(y) \leq \psi(x) \text{ for every } y \text{ with } (x,y) \in E_I\}.$$

The pointwise geometric regularity factor is

$$\mathrm{Reg}_I^{\mathrm{Geom}}(\phi) = \mathbf{1}\{\mathrm{ObsMax}_{E_I}(\psi) = \varnothing\} \cdot \frac{1}{1+\rho_{E_I}(\psi)}.$$

The observable is geometrically regular when this factor has a transfer-class lower margin over the family.

Proposition I.7.6 (Non-identifiability of fiber quotients from pointwise access)**.** *If ϕ is pointwise, two evaluations decide the pairwise predicate $x \sim_\phi y \iff \phi(x) = \phi(y)$. Pointwise access alone does not supply a represented quotient object: a uniform cell encoding, construction of the represented fiber indexed by a value, and the declared finite Boolean fiber operations require their own constructor record. Such a quotient is available precisely when the presentation supplies that record or the budgeted grammar derives it.*

Proof. Pointwise access returns $\phi(x)$, and hence decides equality of the values at two supplied states. The definition of a fiber-represented observable requires more: a uniform representation of the cell associated with a realized value, together with its construction and manipulation costs. These operations are not constructors in the pointwise interface. They enter the represented algebra only through an additional declared or derived record, at which point Definition I.8.1 applies.

□

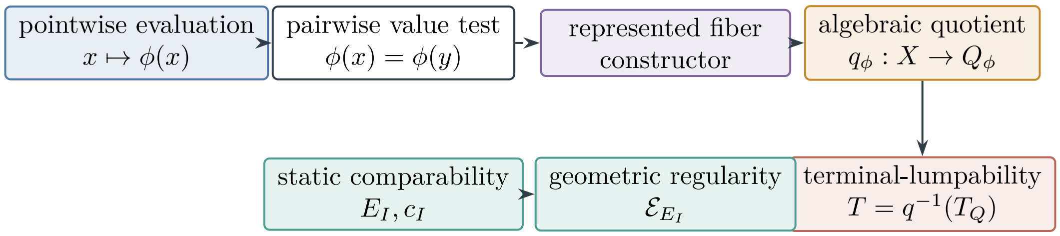

Figure 9: Observable interfaces have distinct types. Pointwise evaluation supplies value comparisons; a represented quotient additionally requires a charged fiber constructor. Terminal lumpability is a property of that quotient, whereas comparability and geometric regularity form an independent static interface.

Example I.7.7 (SAT defect over Hamming comparability)**.** For SAT, take E_Φ to relate assignments at Hamming distance one [37]. This is a static comparability relation on $\{0,1\}^n$. The defect score D_Φ is pointwise if the presentation lets us evaluate the number of violated clauses at a given assignment but does not give compact descriptions of all assignments with the same defect. The ratio

$$\rho_{E_\Phi}(D_\Phi) = \frac{\frac{1}{2}\sum_{x,y} c_\Phi(x,y)(D_\Phi(x) - D_\Phi(y))^2}{\mathrm{Var}_{\mu_\Phi}(D_\Phi)}$$

is a static measurement of defect variation across one-bit comparable assignments, independent of any bit-flip transition rule.

I.8 Fiber-Represented Observables, Lumpability, and Algebraic Quotients

A fiber-represented observable supplements pointwise evaluation with a representation of its level sets, thereby inducing a static quotient of the state space. Compatibility of this quotient with a search process is treated separately in Part II.

Fiber representation adds an affordable description of an observable's level sets to its pointwise evaluation rule.

Definition I.8.1 (Fiber-represented observable)**.** An observable $\phi : X_I \to Y$ is fiber-represented in the task presentation when the presentation supplies compact descriptions of every nonempty fiber

$$\phi^{-1}(y) = \{x \in X_I : \phi(x) = y\}.$$

Compact representation requires a uniform value-indexed fiber constructor, fiber membership, construction of the compact fiber description, and finite Boolean operations on represented fibers to have constructor charge below the declared budget ceiling. It does not require selecting or finding an element of a nonempty fiber. Under the polynomial budget, this is the standard polynomial bound on description length and manipulation cost.

Figure 9 separates pointwise evaluation, represented fibers, algebraic quotients, and the independent geometric comparability interface.

A pointwise observable supplies state values, whereas a fiber-represented observable also represents the associated fibers. This additional data names measurable cells without specifying transitions between them.

The induced algebraic quotient names the represented fibers without assigning dynamics to them.

Definition I.8.2 (Algebraic quotient induced by a fiber-represented observable)**.** Let ϕ be fiber-represented. Define an equivalence relation

$$x \sim_\phi y \quad \iff \quad \phi(x) = \phi(y).$$

The quotient set is

$$Q_\phi = X_I/{\sim_\phi}, \qquad q_\phi : X_I \to Q_\phi, \qquad q_\phi(x) = [x]_\phi.$$

Because the fibers are represented, the quotient cells $q_\phi^{-1}(q)$ are represented observable objects.

The terminology below is adapted from finite-state Markov-chain lumpability [45].

Definition I.8.3 (Static lumpability)**.** A represented quotient $q : X_I \to Q$ is statically terminal-lumpable when the terminal sectors are unions of quotient cells:

$$T_I = q^{-1}(T_Q), \qquad F_I = q^{-1}(F_Q)$$

for some disjoint subsets $T_Q, F_Q \subseteq Q$. Equivalently, if $q(x) = q(y)$, then x and y have the same success/failure terminal status whenever either is state-terminal.

Static lumpability requires each quotient cell to have consistent terminal status. It is a compatibility condition between the quotient partition and terminal structure; dynamical lumpability is introduced in Part II.

Proposition I.8.4 (Quotient events are exactly unions of represented fibers)**.** *Let $\phi : X_I \to Y$ be fiber-represented with quotient map q_ϕ. In the finite case, assume the realized singletons of Y are measurable as in Definition I.1.1. A set $A \subseteq X_I$ is measurable with respect to $\sigma(\phi)$ if and only if it is a union of quotient cells.*

Proof. The atoms of $\sigma(\phi)$ are the nonempty fibers $\phi^{-1}(y)$, which are exactly the quotient cells. By the atom characterization for finite observable algebras, the measurable sets in $\sigma(\phi)$ are precisely unions of those atoms.

□

Example I.8.5 (SAT parity quotient)**.** Suppose a SAT presentation declares a public parity row $\ell(x) = b \cdot x \pmod 2$. The observable $\ell : X_\Phi \to \mathbb{F}_2$ is fiber-represented when the affine fibers $\ell^{-1}(0)$ and $\ell^{-1}(1)$ are represented by the public linear equation. The quotient $X_\Phi/{\sim_\ell}$ has two cells. It is statically terminal-lumpable only if the satisfying assignments and terminal failures are unions of those parity cells. If a parity cell mixes satisfying and terminal-failing assignments, the parity quotient is not a valid static success quotient.

I.9 Observable Versus Hidden Structure

A relation may hold on an ambient instance without being generated by its presentation. The distinction is measurable: the observable algebra separates exposed structure from information outside the primitive presentation.

Exposure distinguishes relations generated by the presentation from relations available only in the ambient instance.

Definition I.9.1 (Exposed and hidden structure)**.** A set $A \subseteq X_I$ is exposed when $A \in \mathcal{F}_I$. A k-ary relation $R \subseteq X_I^k$ is exposed when

$$R \in \mathcal{F}_I^{\otimes k}.$$

A function $g : X_I \to Z$ is exposed when it is $\mathcal{F}_I$-measurable. A structure is hidden relative to the presentation when it may be true in an ambient description of the instance but is not exposed in this sense.

This definition concerns whether the declared observation interface already generates the structure. Formally hidden structure may hold in the ambient instance while remaining presentation-inaccessible. Part II classifies movement based on such structure as residual relative to the public presentation unless a budget-admissible computation generates it from exposed observables in a lifted state space.

The subsequent source hierarchy distinguishes exposed local geometry, defined by the comparability relation and regularity functional, from represented abstraction, defined by fibers and quotient cells. The directed/drift channel Caus orients these sources, the lifted-state channel Lift reorganizes exposed or generated coordinates, and the boundary/killing channel Bdry evaluates terminal observables. Each operation therefore acts on geometry or abstraction already represented in the observable algebra or generated in an admissible lifted configuration.

Proposition I.9.2 (Finite membership test for exposed relations)**.** *Assume X_I is finite with atomic partition $\mathcal{A}_I$. A relation $R \subseteq X_I^k$ is exposed if and only if its indicator is constant on every product atom $A_1 \times \cdots \times A_k$ with $A_j \in \mathcal{A}_I$.*

Proof. The product σ-algebra $\mathcal{F}_I^{\otimes k}$ is generated by rectangles whose sides are observable events. In the finite case its atoms are the products of atoms of $\mathcal{F}_I$. By the atom characterization, a subset of X_I^k is measurable in the product algebra exactly when it is a union of these product atoms. This is equivalent to constancy of its indicator on each product atom. □

Example I.9.3 (Hidden modulus)**.** Let $X = \{0,1\}^n$ and suppose the ambient instance has a residue relation $x \equiv_M y$ for a modulus M. If M and the residue map are not generated by Obs_I, then the relation

$$R_M = \{(x,y) : x \equiv_M y\}$$

is hidden precisely when $R_M \notin \mathcal{F}_I \otimes \mathcal{F}_I$. In a finite presentation this means that some product atom contains both residue-equivalent and non-residue-equivalent pairs.

Example I.9.4 (Hidden planted assignment in SAT)**.** A random planted SAT generator may choose a satisfying assignment $x^\star$ and then sample clauses consistent with it. If the final task presentation exposes only the clauses, the singleton $\{x^\star\}$ is exposed only when it is a union of atoms of the clause-readout algebra. If another assignment has the same public readout but is not $x^\star$, the planted label is hidden even though it was used by the external generator that produced the instance.

I.10 Scores and State-to-State Relations

A score is a function on states, whereas a state-to-state relation is a subset of $X_I \times X_I$. These data have different mathematical types.

Score coloring records the state partition induced by a scalar score.

Definition I.10.1 (Score coloring)**.** A score observable is a measurable map $D_I : (X_I, \mathcal{F}_I) \to (\mathbb{R}, \mathcal{B}_\mathbb{R})$. It induces the coloring

$$\kappa_D : X_I \to \mathbb{R}, \qquad \kappa_D(x) = D_I(x),$$

and the level-set equivalence relation $x \sim_D y \iff D_I(x) = D_I(y)$. It does not by itself specify any relation of admissible pairs between different score levels.

A monotone transformation changes score values while preserving their level-set partition. The score alone canonically induces only equality of score values, not a relation between distinct fibers.

Proposition I.10.2 (Non-identifiability of transition relations from scores)**.** *Let X be a set with $|X| \geq 2$, and let $D : X \to \mathbb{R}$. The data of D determine the partition into fibers $D^{-1}(c)$. They do not determine a distinguished relation $R \subseteq X \times X$ between distinct fibers.*

Proof. Every construction from D is invariant under score-preserving permutations of X. It can test equality and order of score values, but cannot distinguish two state pairs with the same ordered pair of scores. For distinct fibers A and B, both the empty and complete relations on $A \times B$ are compatible with D. The score therefore does not determine a distinguished state-to-state relation. □

Example I.10.3 (SAT defect partitions assignments)**.** The SAT defect $D_\Phi(x)$ partitions assignments by their numbers of violated clauses; the zero-defect fiber is the success set. This partition specifies neither state pairs nor a bit-flip, clause-repair, or quotient-comparison rule. Such relations require additional structure.

I.11 Examples of Static Task Presentations

The SAT and subset-sum examples illustrate state spaces, observable readouts, atoms, level sets, and the distinction between exposed and hidden relations.

I.11.1 SAT as a Task Space

SAT provides the first example [23]. A formula $\Phi = C_1 \wedge \cdots \wedge C_m$ on n variables has state space $X_\Phi = \{0,1\}^n$. Clause predicates and the defect score are observables. The success sector is the zero-defect level set.

The SAT presentation instantiates the static signature with assignments, clause readouts, and the satisfying sector.

Definition I.11.1 (SAT task presentation)**.** For a CNF formula Φ, define

$$X_\Phi = \{0,1\}^n, \qquad O_{\Phi,j}(x) = \mathbf{1}\{C_j(x) = 1\}, \qquad D_\Phi(x) = \#\{j : C_j(x) = 0\}.$$

The observable algebra is generated by the declared clause readouts and any additional public observables. The positive terminal sector is

$$T_\Phi = D_\Phi^{-1}(\{0\}).$$

I.11.2 Subset Sum and Exposed Arithmetic

Subset sum provides a second instance of the same static construction [43]. The state set can be the binary inclusion cube $X = \{0,1\}^n$. The public input consists of weights and a target,

$$a_1, \ldots, a_n, \qquad t.$$

The natural observable is the partial-sum defect

$$D(x) = \left| \sum_{i=1}^{n} a_i x_i - t \right|.$$

If the presentation also exposes residues modulo a public modulus, then residue classes are observable quotient cells. If the modulus is not in the observable algebra, the same residue relation may be true in an ambient description but hidden from the task presentation.

Example I.11.2 (Observable and hidden residue structure)**.** Let $r_M(x) = \sum_i a_i x_i \pmod M$. If M is public and r_M is declared or derived from the observables, then the residue fibers $r_M^{-1}(b)$ are exposed. If M is absent from Obs_I, then the relation $r_M(x) = r_M(y)$ is hidden unless it is still measurable in $\mathcal{F}_I \otimes \mathcal{F}_I$.

The same pattern appears in SAT. Public XOR clauses expose affine fibers. A planted assignment or secret parity used only by an instance generator is hidden unless the final presentation declares it.

I.12 Admissible Presentation Equivalence

Because the invariant is presentation-relative, changing the observable interface may change its value. Presentations that encode the same observable information at comparable representation cost must, nevertheless, yield equivalent structural conclusions.

Admissible equivalence identifies presentations that preserve observable information and its charged representation cost.

Definition I.12.1 (Admissible presentation equivalence)**.** Two task-family presentations $\mathcal{T} = (X, \mathcal{F}, T, E, \mathcal{R}_{\le d})$ and $\mathcal{T}' = (X', \mathcal{F}', T', E', \mathcal{R}'_{\le d})$ are admissibly equivalent with degree distortions $a(d)$ and $b(d)$ when there are actual mutually inverse measurable bijections $\phi : X \to X'$ and $\psi : X' \to X$. Their pointwise pullbacks

$$\Phi^* : L^\infty(X', \mathcal{F}') \to L^\infty(X, \mathcal{F}), \qquad \Psi^* : L^\infty(X, \mathcal{F}) \to L^\infty(X', \mathcal{F}')$$

are inverse Boolean-algebra isomorphisms on the represented observable events and observables, including their values on null states, and satisfy the following conditions. The induced maps on equivalence classes in L^∞ are denoted by the same symbols.

Terminals. $\Phi^* 1_{T'} = 1_T$ and $\Psi^* 1_T = 1_{T'}$, and the same holds for all named terminal sectors.

Measures and Hilbert structures. The pullbacks extend to unitary maps between the declared state and edge Hilbert spaces. In particular, the reference measure is transported exactly and the edge inner products used for geometric, Hodge, quotient, lift, and boundary projections are preserved. A merely bounded nonconstant Radon–Nikodym density gives equivalent norms but need not preserve a correlation, angle, zero projection, or orthogonal channel. Such a change is not an admissible equivalence for the exact invariance theorem unless separate cross-pairing and projection-comparison estimates are declared and charged.

Static structure. Observable comparability relations, represented fibers, quotient cells, and static terminal-lumpability are transported by Φ^* and Ψ^*.

Cost filtration. Every represented observable, quotient, comparability relation, or static constructor admissible for $\mathcal{B}'$ pulls back to one admissible for a class-equivalent enlargement of $\mathcal{B}$, and conversely. The distortion functions preserve the transfer class: polynomial-dimensional bands map to polynomial-dimensional bands only in the special case $\mathcal{B} = \mathcal{B}_{\mathrm{poly}}$.

Lemma I.12.2 (Static invariance under admissible equivalence)**.** *Under admissible presentation equivalence, exposed versus hidden structure, terminal membership, represented quotient structure, static terminal-lumpability, comparability geometry, and the five static target-relevance conditions are preserved up to the stated budget/cost distortion and with their Hilbert-space projection and correlation values transported exactly.*

Proof. A pointwise Boolean-algebra isomorphism sends atoms to atoms, measurable sets to measurable sets, observable values to the corresponding values, and nonmeasurable distinctions to nonmeasurable distinctions relative to the transported presentation. In particular, pointwise maximizers and essential maximizers are both transported exactly; no change on a null state is discarded. Terminal preservation gives the same success and failure sectors. Transport of represented fibers sends quotient cells to quotient cells, so unions of terminal cells and static lumpability are invariant. Transport of comparability relations and the unitary edge transport send the static Dirichlet quadratic form to the corresponding transported quadratic form. Unitarity preserves normalized projections and cross-pairings. Finally, the cost-filtration condition sends each budget-admissible static observable or quotient on one side to an admissible representative on the other after the declared transfer-class distortion. Thus the static target-relevance conditions are preserved under the declared budget distortion.

□

A presentation containing an additional quotient, modulus, planted label, or target pointer changes the observable algebra and is therefore inequivalent. A recoding of the same observable algebra with comparable constructor cost is equivalent, and the subsequent invariants agree up to the prescribed cost distortion.

I.13 Summary of the Static Task Presentation

Part I represents each problem instance as the static object

$$\mathcal{T}_I = (X_I, \mathrm{Obs}_I, \mathrm{Dyn}_I, \mathrm{End}_I),$$

where X_I, Obs_I, and End_I carry the developed structure and Dyn_I remains a static relation slot. The observable algebra, its atoms and level sets, and its pointwise and represented-fiber observables determine the exposed static structure. Target projection mass bounds the target signal obtainable from any generated subalgebra, while a target-discrimination certificate assigns terminal meaning to its fibers. Budgeted extraction accounting then bounds the actionable mass after certification, learning, spectral, and residual-noise audits. For finite presentations, the computable audit evaluates these quantities from matrices, labels, quotient fibers, kernels, and declared audit parameters. The intrinsic sources used in Part II are exposed local geometry and represented abstraction; the directed/drift, lifted-state, and boundary/killing channels operate on these sources.

Dependency trail. The static interface is fixed by Definitions I.3.2, I.3.4, I.2.8 and I.2.9. Target relevance and its audits use Definitions I.5.1 to I.5.4. The geometric and quotient vocabulary is Definitions I.7.1 to I.8.3 and I.7.5. The signature and observable-algebra consequences are Propositions I.2.2 to I.3.5. Static target relevance is developed in Lemmas I.5.2 to I.4.4 and I.4.6 and Corollaries I.4.8 and I.5.9. The pointwise and quotient distinctions are Propositions I.9.2, I.10.2, I.8.4 and I.7.6. Presentation invariance and the resulting static closure are Lemmas I.13.1 and I.12.2. The running static examples are collected in Examples I.0.2 to I.7.7, I.4.9 and I.2.10.

Lemma I.13.1 (Static closure of Part I)**.** *Every object developed in Part I is determined by the state set, observable algebra, terminal sectors, family presentation, represented fibers, target-discrimination certificates, budgeted extraction accounts, and static comparability relations. The target signal available to any generated subalgebra is bounded by the Hilbert projection of the target contrast onto that subalgebra, while terminal relevance requires a represented or generated target-discrimination certificate. Certified extraction is further bounded by the applicable empirical, effective-dimension, and residual-order audits. The only intrinsic reusable sources for later structured movement are the local geometric source from exposed comparability and the abstraction source from represented fibers. None requires a run, transition kernel, generator, search process, or algorithm.*

Proof. The task signature is defined from sets, maps, measurable structures, and terminal predicates. Atoms, level sets, refinements, exposed relations, hidden relations, quotient cells, static lumpability, score partitions, target projection masses, target-discrimination certificates, and budgeted extraction accounts are all set-theoretic or measurable constructions from those data. The comparability relation E_I is declared as a symmetric measurable relation, and the regularity functional is a quadratic form on functions over that relation. No definition quantifies over runs or assigns probabilities to state-to-state evolution. Therefore the development is closed at the static level. $\square$

The forward correspondences are now named and ready for Part II:

Static object in Part I	Later role in Part II
declared slot Dyn_I	relation slot for search and state-to-state dynamics
comparability relation E_I	intrinsic local geometry to be activated dynamically as Geom

Static object in Part I	Later role in Part II
regularity functional $\mathcal{E}_{E_I}$	quantitative form of the intrinsic local geometry
fiber-represented observable ϕ	intrinsic represented abstraction to be activated as Abs
static terminal-lumpability	compatibility condition for the later quotient/lumping channel
terminal sectors End_I	boundary/value readout through Bdry
score observable D_I	state partition without a state-to-state transition relation
target projection mass of a generated algebra	ceiling on target signal available to any combination of its observables
target-discrimination certificate	equips observable fibers with a terminal-relevant quotient readout; without it, the tuple remains uncalibrated
budgeted extraction accounting	quantitative upper bound on certified extraction after projection, certification, learning, spectral, and residual-noise audits
computable observable audit	finite matrix implementation of the accounting for a declared exposed algebra and audit class

Example I.13.2 (SAT static closure)**.** For SAT, Part I gives the assignment cube, clause observables, defect score, satisfying sector, observable atoms, level sets, possible exposed parity quotients, possible hidden planted labels, and the static Hamming comparability relation when declared. The selection of subsequent assignments belongs to the dynamics of Part II.

Part II
Budget-Admissible Dynamics and Structural Decomposition

II.0 Introduction: Budget-Admissible Dynamics

Part contract. *Inputs:* the task presentation, observable algebra, terminal sectors, comparability data, and represented quotients from Part I. *Output:* the exhaustive structural-channel decomposition in Theorem II.7.1, the explicit channel normal forms in Theorems II.5.10, II.7.2, II.9.5 and II.10.5, and their stability under represented composition. *First pass:* the post-exhaustion comparison band and worked examples may be skimmed.

Part II equips the static task presentation with dynamics and develops a decomposition of every budget-admissible mass-transport operator.

Part I defined the presented task

$$\mathcal{T}_I = (X_I, \mathrm{Obs}_I, \mathrm{Dyn}_I, \mathrm{End}_I),$$

while leaving Dyn_I without an assigned process, kernel, generator, traversal rule, or algorithm. The state space, observable algebra, terminal sectors, static comparability relation, and represented quotients now provide the data for the dynamic construction.

The proposed algebraic decomposition contains five structural channel slots and one residual component; at a fixed budget, only charged represented projections are exposed:

$$\mathsf{Geom}, \qquad \mathsf{Caus}, \qquad \mathsf{Abs}, \qquad \mathsf{Lift}, \qquad \mathsf{Bdry}, \qquad J_{\mathrm{res}},$$

Every budget-admissible represented computation without oracle access induces such an operator on its full observable transcript carrier. The results below establish exhaustiveness of the decomposition.

The geometric/reversible channel Geom and the quotient/lumping channel Abs represent the intrinsic sources: exposed local comparability and represented abstraction. The directed/drift channel Caus orients these sources, the lifted-state channel Lift reorganizes their coordinates, and the boundary/killing channel Bdry evaluates terminal observables. The orthogonal remainder is denoted by J_{res}.

The directed channel is relative to a declared authority envelope. A problem may permit an affordable control to select the skew part of its generator, may impose that part as dynamic data, or may allow a hybrid of imposed and controlled motion. Subsequent projections and lifts transport the selected operator and its current; they do not replace them by a more favorable target-aligned flow.

For a declared budget structure $\mathcal{B} = (\mathcal{R}, \mathrm{cost}, b, \mathfrak{C})$, admissibility is filtered by the capacity functional $b(n)$. Instance-dependent information is restricted to the budgeted saturated observable closure generated by the presented observable algebra, public parameters, admissible lifted constructors, and finite-horizon computations whose transcripts have charge below $b(n)$. Relations, quotients, potentials, lift interfaces, boundary values, transition data, trajectory statistics, and finite-horizon target-contact observables generated in this way enter the closure at their saturated computation cost. Objects outside the closure are associated with an enriched presentation, oracle or advice data, over-budget recovery, or the residual component. The polynomial model is the special case $\mathcal{B}_{\mathrm{poly}}$ with $b_{\mathrm{poly}}(n) = n^{O(1)}$.

The analysis proceeds through five closure results. The positive maximum principle yields nonnegative transition and killing rates. A fixed local relation then separates local symmetric, local directed, nonlocal, and boundary terms. Hodge and quotient projections resolve these terms into the five structural channels and the residual component. Any family-uniform budget-admissible reduction, quotient, or target alignment is reclassified from the post-exhaustion residual band into the corresponding structural channel. Part III treats the saturated representation cost of these components. Finally, finite composition and valid lifting preserve the classification.

The principal decomposition results are stated for finite state spaces, including the SAT and subset-sum examples and the finite transcript carriers of resource-bounded computations. Part III gives a conditional sectorial-form and carrier interface for continuum realizations; specific applications still require their own domain, closure, capacity, and regularity hypotheses. Arrival times, resolvents, saturated cost thresholds, and quantitative bounds are deferred to Part III.

II.1 Budget-Admissible Movement and the Search Operator

A frontier is a nonnegative measure on the state space of Part I. In the finite setting it is a vector in $\mathbb{R}^{X_I}_{\geq 0}$ representing the mass currently assigned to each state before channel decomposition.

The frontier formalizes the current mass distribution on which a search operator acts.

Definition II.1.1 (Frontier and search operator)**.** Fix an instance I with finite state space X_I and observable algebra $\mathcal{F}_I$. Let $\mathcal{M}(X_I)$ be the real vector space of signed measures on X_I, and let $\mathcal{M}_+(X_I)$ be its positive cone. A frontier is an element $\mu \in \mathcal{M}_+(X_I)$.

A discrete search operator is a linear map $P : \mathcal{M}(X_I) \to \mathcal{M}(X_I)$ such that $P\mathcal{M}_+(X_I) \subseteq \mathcal{M}_+(X_I)$. With row-vector frontiers this means

$$(\mu P)(y) = \sum_{x \in X_I} \mu(x) P_{xy}, \qquad P_{xy} \geq 0.$$

The dual operator on observable functions $f \in \mathbb{R}^{X_I}$ is

$$(Kf)(x) = \sum_{y \in X_I} P_{xy} f(y),$$

and it is positive in the sense that

$$f \geq 0 \quad \Longrightarrow \quad Kf \geq 0.$$

A continuous-time search operator is a linear operator $L : \mathbb{R}^{X_I} \to \mathbb{R}^{X_I}$ whose semigroup is positive and sub-Markov:

$$e^{tL} f \geq 0 \quad (f \geq 0), \qquad e^{tL}\mathbf{1} \leq \mathbf{1} \quad (t \geq 0).$$

It is Markov when equality holds in the second relation. On a finite state space this condition is equivalent to the positive maximum principle below. Evolution is written on observables as $\partial_t u_t = L u_t$, or dually on frontiers as $\partial_t \mu_t = L^* \mu_t$.

Budget admissibility is inherited from the closure defined in Part I. A dynamic edge, kernel, or generator may use only distinctions exposed at the declared budget. The available finite data are generated by the observable algebra, its product algebra on edges, and the coordinates of an admissible lifted configuration; after lifting, the same condition is imposed with respect to the lifted observable algebra.

When a budget-admissible computation generates additional data from this closure, including through finite-horizon iteration and postprocessing, those data enter the lifted observable coordinates at their accumulated cost and the decomposition is applied on the lifted space. The polynomial case sets $\mathcal{B} = \mathcal{B}_{\text{poly}}$. Data outside the closure and any admissible budget-generated lift are presentation-inaccessible, oracle-supplied, or over budget. Movement without exposed structural provenance is assigned to the residual component.

Budget-admissible movement restricts transition rules to the declared saturated closure and its charged lifts.

Definition II.1.2 (Budget-admissible movement)**.** Let $\mathcal{A}_I$ be the atomic partition of the finite observable algebra $\mathcal{F}_I$. A kernel $P = (P_{xy})$ is budget-admissible at the declared budget when it is Markov or sub-Markov, when its positive support is contained in an exposed relation represented within the budgeted saturated closure,

$$\operatorname{supp}_{\text{off}} P := \{(x, y) : x \neq y,\ P_{xy} > 0\} \subseteq R_I, \qquad R_I \in \mathcal{F}_I \otimes \mathcal{F}_I.$$

Diagonal holding probabilities are unrestricted by this support condition; equivalently, $\operatorname{supp} P \subseteq R_I \cup \Delta_{X_I}$. The rate table must also be product-measurable and represented at charged cost: if x, x' lie in the same atom of $\mathcal{F}_I$ and y, y' lie in the same atom, then $P_{xy} = P_{x'y'}$ whenever both pairs are in the same product atom. For a generator with rates $q(x, y)$, the same condition is imposed on q. Equivalently, $q : X_I \times X_I \to [0, \infty)$ is $\mathcal{F}_I \otimes \mathcal{F}_I$-measurable and its representation lies inside the active budgeted closure.

This definition extends the exposed-versus-hidden distinction of Part I to dynamics. A hidden modulus, planted assignment, secret parity, or target pointer may exist in the ambient instance, but it supports a budget-admissible transition only when generated from exposed data in an admissible lifted configuration. Otherwise its use is oracle-supplied, presentation-inaccessible, over budget, or residual relative to the public presentation.

II.2 The Generator Form and the Positive Maximum Principle

A budget-admissible infinitesimal frontier update preserves nonnegativity. Dually, the positive maximum principle requires the generator to be nonpositive at a nonnegative maximum. On finite state spaces this condition determines the matrix form [24, 61].

The positive maximum principle fixes the finite-generator condition that preserves nonnegative frontier evolution.

Definition II.2.1 (Positive maximum principle)**.** A linear operator L on observable functions satisfies the positive maximum principle when

$$f(x_0) = \max_{x \in X_I} f(x) \geq 0 \quad \Longrightarrow \quad (Lf)(x_0) \leq 0.$$

This is the local infinitesimal form of positive sub-Markov preservation for the frontier semigroup.

Lemma II.2.2 (Finite generator form)**.** *Let X be finite and let L be a linear operator on functions $f : X \to \mathbb{R}$. If L satisfies the positive maximum principle, then there are nonnegative rates $q(x, y) \geq 0$ for $x \neq y$ and a nonnegative killing rate $\kappa(x) \geq 0$ such that*

$$(Lf)(x) = \sum_{y \neq x} q(x,y)(f(y) - f(x)) - \kappa(x)f(x).$$

Conversely, every operator of this form satisfies the positive maximum principle.

Proof. Write $L_{xy} = (L\mathbf{1}_y)(x)$, so $(Lf)(x) = \sum_y L_{xy}f(y)$. Fix $x \neq y$. Let $f(y) = -1$ and $f(z) = 0$ for $z \neq y$. Then $f(x) = 0 = \max_z f(z)$, so the positive maximum principle at x gives $(Lf)(x) = -L_{xy} \leq 0$. Hence $L_{xy} \geq 0$. Put $q(x,y) = L_{xy}$ for $x \neq y$.

Apply the principle to the constant function 1. Since every point is a nonnegative maximum, $(L1)(x) \leq 0$. Define $\kappa(x) = -(L1)(x) \geq 0$. Then

$$L_{xx} = -\sum_{y \neq x} q(x,y) - \kappa(x),$$

because $(L1)(x) = L_{xx} + \sum_{y \neq x} L_{xy}$. This gives the displayed representation. Conversely, if f has a nonnegative maximum at x, then $f(y) - f(x) \leq 0$ for every y, and $-\kappa(x)f(x) \leq 0$. Thus $(Lf)(x) \leq 0$.

□

Thus every positive finite-state generator consists of off-diagonal transition rates and diagonal loss. With a represented terminal observable space, diagonal loss is the zero-boundary-value specialization of boundary readout. These terms exhaust the matrix entries.

II.3 The Fixed-Space Decomposition

The finite generator theorem yields a rate table, while the static comparability relation E_I from Part I identifies local state pairs. Relative to this declared relation, every rate has a unique local or nonlocal location.

The positive weight m is a reference measure used to express local rates as conductance and current. Any positive weight provides algebraic coordinates for the same finite rate table. Quantitative statements, however, use only the admissible reference measures of Definition II.3.1. A task-dependent weight supplied by the presentation or a public lifted initialization is charged at that level; otherwise it is presentation-inaccessible, oracle-supplied, or over budget. Changing an admissible weight changes the conductance-current coordinates but not the underlying transport operator.

Reference-measure legality determines which conductance and current coordinates may enter quantitative certificates.

Definition II.3.1 (Reference measure legality)**.** A base reference measure is admissible when it is part of the task presentation or is transported from an admissibly equivalent presentation by its unitary measure transport. A lifted reference measure on Ω is admissible when it is fixed by the lifted presentation before the measured operator is used. Concretely, it is one of the following: a public product gauge on finitely encoded lifted coordinates, including clocked transcript coordinates; the public initialization lift $\Lambda\mu$; or a measure absolutely continuous with respect to one of these public gauges whose density is $\mathcal{G}_{\mathrm{obs}}$-measurable, generated from the public presentation and lifted initialization data, and bounded above and below by transfer-class factors on the family. For $\mathcal{B}_{\mathrm{poly}}$, these are polynomial factors.

A measure used only for a static integral may have smaller support. A measure used to realize L on L^2, define an orthogonal Hodge or channel projection, or compress to a quotient must have

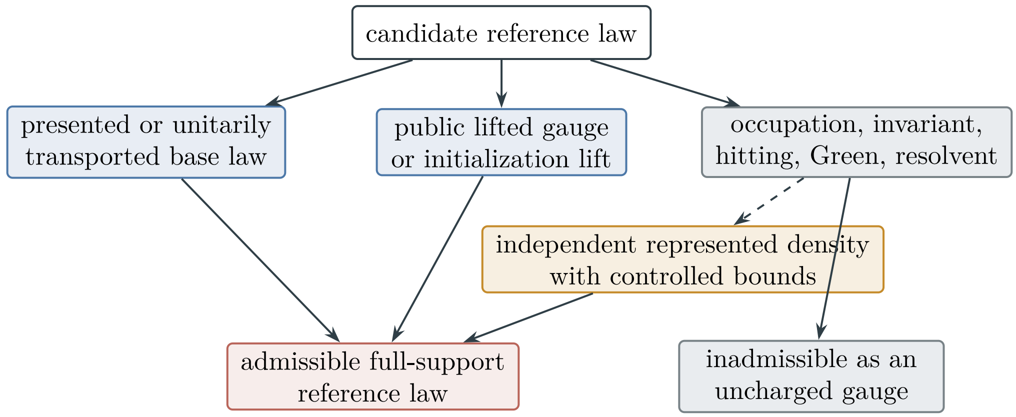

Figure 10: Reference-measure legality. Presented and public lifted gauges are admissible subject to the stated support conditions. A trajectory-defined law enters only through an independent charged representation with controlled density; its operator-dependent definition alone is not a reference gauge.

full support on the represented carrier. Equivalently, one may first restrict the carrier to its support, provided that support is invariant under the operator and all quotient and terminal maps. Without full support or this invariant-support reduction, a transition into a null state need not define an operator on L^2 equivalence classes.

A public product gauge may be used for the exact clocked-transcript and first-hit accounting of an arbitrary finite computation. It supports a structural Lift comparison only when the separate projection and target-fraction conditions of Definition II.5.2 hold. Thus the configuration gauge does not create a target-mass transfer identity.

Trajectory-defined laws, including occupation laws $\sum_{t\le p(n)} \mu_0 P^t$, killed or hitting-conditioned occupation laws, and measures obtained from an invariant, Green, or resolvent problem for the measured operator, are admissible only when they also possess an independent representation satisfying the preceding conditions. Their dynamic definition alone does not provide a reference gauge for the presentation.

Figure 10 summarizes the admissible routes to a reference law and the additional representation gate for trajectory-defined laws.

Proposition II.3.2 (Exclusion of operator-dependent trajectory measures)**.** *Let P or L be the induced operator whose channels are being decomposed. A reference law constructed from the trajectory, occupation, hitting-conditioned, invariant, Green, or resolvent behavior of that same operator is inadmissible as an uncharged lifted reference measure in the channel decomposition or in the Part III quantitative projections.*

Proof. Such a law is operator-dependent because it is defined from P, L, or their killed or conditioned dynamics rather than from the presented observable algebra, its budgeted closure, and the public initialization lift. It also differs from a gauge change: a gauge modifies L^2 coordinates by a controlled observable density, whereas an occupation law may concentrate on a trajectory or reachable band and alter the Hilbert space and channel projections. Finally, it is not an initialization lift, since in general $\pi_* \sum_{t\le p(n)} \mu_0 P^t$ is the time-occupation law of the projected run rather than the base frontier μ required by $\pi_*\Lambda\mu = \mu$. The law therefore supplies dynamical information rather than reference coordinates. An independent public representation with controlled density restores admissibility at the cost of that representation.

□

Consequently, a clock, program-counter, or depth coordinate aligned with the target under an admissible public lifted reference law belongs to the directed/drift or geometric/reversible channel. Alignment arising only under the solver's trajectory or occupation law is supplied by an inadmissible reference measure.

The fixed-space split decomposes one represented rate table into symmetric, skew, killing, and residual terms.

Definition II.3.3 (Fixed-space rate split)**.** Let L have rates $q(x,y)$ and killing $\kappa(x)$. Fix a symmetric local relation $E_I \subseteq X_I \times X_I \setminus \Delta$ and a positive reference weight $m(x) > 0$. Put $q_E(x,y) = q(x,y)\mathbf{1}_{E_I}(x,y)$ and write the local oriented edge mass

$$r_E(x,y) = m(x)q_E(x,y).$$

For $x \neq y$, split the local edge mass into symmetric conductance and antisymmetric current:

$$c_E(x,y) = \frac{1}{2}\big(r_E(x,y) + r_E(y,x)\big), \qquad j_E(x,y) = \frac{1}{2}\big(r_E(x,y) - r_E(y,x)\big).$$

The four fixed-space operators, in zero-boundary reduction, are

$$\begin{aligned}
(L^{\mathsf{Sym}} f)(x) &= \frac{1}{m(x)} \sum_{y:(x,y)\in E_I} c_E(x,y)(f(y) - f(x)),\\
(L^{\mathsf{Anti}} f)(x) &= \frac{1}{m(x)} \sum_{y:(x,y)\in E_I} j_E(x,y)(f(y) - f(x)),\\
(L^{\mathsf{Jump}} f)(x) &= \sum_{y\neq x:(x,y)\notin E_I} q(x,y)(f(y) - f(x)),\\
(L^{\mathsf{Bdry}} f)(x) &= -\kappa(x)f(x).
\end{aligned}$$

Here $c_E(x,y) = c_E(y,x) \geq 0$ and $j_E(x,y) = -j_E(y,x)$. The antisymmetric term is a signed current operator; it is not itself required to be a positive generator. The displayed boundary term is the special case in which removed mass has boundary value zero.

The boundary-value extension records the destination and value of mass removed from the interior carrier.

Definition II.3.4 (Boundary-value channel extension)**.** For terminal readout, let ∂X_I be a finite boundary/value space, let $b(x,\zeta) \geq 0$ be the rate of transfer from $x \in X_I$ to $\zeta \in \partial X_I$, and put $\kappa(x) = \sum_\zeta b(x,\zeta)$. For an interior observable $f : X_I \to \mathbb{R}$ and a boundary readout $v : \partial X_I \to \mathbb{R}$, the same boundary channel acts on the extended observable pair by

$$(L^{\mathsf{Bdry}}(f,v))(x) = \sum_{\zeta\in\partial X_I} b(x,\zeta)(v(\zeta) - f(x)) = \kappa(x)(Bv(x) - f(x)),$$

where $Bv(x) = \kappa(x)^{-1}\sum_\zeta b(x,\zeta)v(\zeta)$ when $\kappa(x) > 0$. Setting $v \equiv 0$ gives the zero-value killing term $-\kappa(x)f(x)$. Thus killing/removing support is the zero-value specialization of boundary value processing.

A boundary value is budget-admissible as Bdry when the terminal observable and value readout are declared by the presentation or generated from budget-admissible observable structure. A committor or hitting-time table of an induced kernel becomes a boundary readout when it admits an observable representation, including one produced by a budget-admissible finite-horizon computation. An uncomputed resolvent or infinite-horizon hitting process does not constitute an admissible boundary constructor.

Lemma II.3.5 (Fixed-space split). *For a finite budget-admissible generator, the exposed local relation E_I, and a positive reference weight m used only to coordinatize local edge mass, the decomposition*

$$L = L^{\mathsf{Sym}} + L^{\mathsf{Anti}} + L^{\mathsf{Jump}} + L^{\mathsf{Bdry}}$$

is canonical relative to these coordinates. The four terms are the local symmetric, local directed, nonlocal, and boundary/killing components. Dependence on m is confined to the conductance-current coordinates.

Proof. The generator form determines all off-diagonal rates and diagonal loss. The relation E_I partitions off-diagonal pairs into local and nonlocal pairs. On local pairs, the ordered edge mass $r_E(x, y)$ has the unique decomposition $r_E = c_E + j_E$ into its symmetric and antisymmetric parts. Substituting $q_E(x, y) = m(x)^{-1}(c_E(x, y) + j_E(x, y))$ into the local part of L gives $L^{\mathsf{Sym}} + L^{\mathsf{Anti}}$. The nonlocal rates give L^{Jump}. The diagonal loss, equivalently transfer to an observable boundary value space, gives L^{Bdry}. The supports are disjoint, and the symmetric/antisymmetric split is unique in the chosen edge-mass coordinates. Since m only defines those coordinates, it leaves the underlying movement channels unchanged.

□

Proposition II.3.6 (Exhaustiveness of the local rate decomposition). *Let A be any linear operator satisfying the positive maximum principle whose off-diagonal support is contained in the observable local relation E_I. Then A has only local pair rates and diagonal loss, equivalently zero-value boundary readout. After the symmetric/current split above, no third local term remains.*

Proof. Apply the finite generator theorem to A. Its off-diagonal coefficients are nonnegative rates $a(x, y) \geq 0$, and the support hypothesis gives $a(x, y) = 0$ when $(x, y) \notin E_I$. Its diagonal part is exactly $-\sum_{y \neq x} a(x, y) - \kappa_A(x)$ with $\kappa_A(x) \geq 0$. Thus every local positive generator is already a sum of pair-difference terms and boundary loss. If terminal values are carried, that loss is the boundary value term $\kappa_A(Bv - f)$; if the terminal value is zero, it is killing. Splitting the local edge mass into $c_E + j_E$ accounts for all local pair terms, so a further local primitive would have zero coefficients in every matrix entry.

□

This algebraic split locates the rates but does not yet identify their structural provenance.

II.4 From the Algebraic Split to Structural Channels

The fixed-space decomposition separates the rate table algebraically. The structural decomposition further classifies each component by its observable provenance.

The relevant subspaces are generated by the observable algebra, its budgeted saturated closure, and admissible coordinates in a lifted configuration. A potential, quotient, or lifted interface outside this closure can enter the operator only through an admissible lifted derivation. Otherwise it is presentation-inaccessible, oracle-supplied, over budget, or part of an enriched presentation. Its operator term is retained in the budget-filtered residual ledger, but it has no affordable target-aligned source status for the original presentation.

Syntactic part	Location	Structural channel
Sym	paired local rates inside E_I	Geom (geometric/reversible)
local and nonlocal skew flow	oriented imbalance	Caus (directed/drift) $+J_{\mathrm{res}}$
nonlocal jump	outside E_I	Abs (quotient/lumping) +Lift (lifted-state) when supplied by observable quotient or valid lifted-interface data; otherwise J_{res}
Bdry	boundary readout, value processing, zero-value killing	Bdry (boundary/killing)

Provenance determines the nonlocal classification. A directed component belongs to Caus when it is the gradient of a represented potential constructed from local geometry, quotient coordinates, or lifted-state coordinates. A nonlocal transition has algebraic quotient or lift provenance when represented by an observable quotient or valid lifted interface, respectively. The quotient projection contributes a qualified Abs source only through Definition III.15.1; the lift retains the qualification of its represented base source. The remainder is residual transport. These are properties of the presented operator, independent of whether a representation has been discovered.

II.4.1 Projected channels and causal authority

We first define the represented channel projections and the authority envelope for their directed component.

Definition II.4.1 (Budgeted channel representatives)**.** The representative spaces for Caus, Abs, valid Lift, and Bdry consist of objects in the presented observable algebra, the public initialization lift, and the budgeted saturated closure of lifted observable coordinates. At a declared budget this includes budget-admissible finite-horizon computations performed in the lifted configuration and charged at their saturated computation cost. Stopped dynamics, powers, occupations, Green operators, resolvents, committors, first-hitting maps of the same operator, and ambient relations enter these representative spaces when the denoted finite object is generated by a budget-admissible computation from the closure, or otherwise has a budget-admissible saturated representative at the claimed budget. A solver-indexed future-hitting observable such as

$$D_L(\omega) = \inf\{t \geq 0 : P^t\omega \in T\}$$

for a deterministic transition rule, or the analogous stopped hitting distribution for a randomized kernel, has the cost of an admissible representation rather than cost zero by definition. A computed finite-horizon version belongs to the budgeted saturated closure at the cost of its simulation. An uncomputed future-hitting functional is a semantic property of the induced operator rather than a representative. Thus the criterion is extensional and cost-sensitive: any budget-admissible saturated representation of the same denotation is sufficient.

The Hodge split separates the directed-flow components used by the operator ledger.

Definition II.4.2 (Hodge split of directed flow)**.** Fix the represented generator rates $q(x,y)$ and a legal full-support reference weight m. On every ordered off-diagonal pair put

$$r(x,y) = m(x)q(x,y), \qquad j(x,y) = \frac{1}{2}\big(r(x,y) - r(y,x)\big).$$

Let $\vec{\mathcal{E}}$ contain one declared orientation of every unordered pair on which j may be nonzero, including local and nonlocal pairs, and define the actual skew-current vector by $F_J(e) = j(x,y)$ for $e = (x,y)$. Fix a represented positive edge metric $w = (w_e)_{e\in\vec{\mathcal{E}}}$ and set $\mathcal{H}_{\mathrm{edge}} = \mathbb{R}^{\vec{\mathcal{E}}}$ with

$$\langle a,b\rangle_w = \sum_{e\in\vec{\mathcal{E}}} w_e a_e b_e.$$

The orientation, reference weight, and edge metric are declared as part of the channel-coordinate record before the projection is evaluated. Their representations and all metric arithmetic are charged. Thus the projection below is canonical relative to a named represented metric; an arbitrary post-hoc choice of w is not an exposed Caus certificate.

Let $\mathcal{P}_{\mathrm{str}} \le \mathbb{R}^{X_I}$ be the observable finite-dimensional space of exposed structural potentials: geometric coordinates, quotient coordinates from represented fibers, and lifted interface coordinates satisfying Definition Budgeted channel representatives. Define the gradient map

$$G : \mathcal{P}_{\mathrm{str}} \to \mathcal{H}_{\mathrm{edge}}, \qquad (G\Phi)(x,y) = \Phi(y) - \Phi(x).$$

Let $\mathcal{C}_{\mathrm{str}} = \operatorname{im} G$. Since $\mathcal{H}_{\mathrm{edge}}$ is finite dimensional, the orthogonal projection $\Pi_{\mathcal{C}_{\mathrm{str}}}$ is well-defined. This is the finite-dimensional Hodge projection used below [30, 42]. Define

$$F_{\mathrm{Caus}} = \Pi_{\mathcal{C}_{\mathrm{str}}} F_J, \qquad F_{\mathrm{res}} = F_J - F_{\mathrm{Caus}}.$$

The corresponding skew matrices are denoted J_{Caus} and $J_{\mathrm{res}}^{\mathrm{skew}}$. This unique orthogonal decomposition separates the structural-gradient projection from the directed flow orthogonal to every exposed structural gradient. A Poisson solution, committor, first-hitting-time function, or other future functional enlarges the potential space only when it has an admissible representation. A finite-horizon potential produced by budget-admissible computation is a lifted observable charged at its saturated computation cost.

The displayed projection is an unconditional finite-dimensional algebraic projection relative to the declared potential subspace and edge metric. At a budget B, it is called an exposed Caus projection only when one represented record of cost $\preceq B$ supplies a finite basis for $\mathcal{P}_{\mathrm{str}}$, its gradients, the orientation, reference weight, edge metric, and the projection coefficients, or a uniform procedure that computes them, with all arithmetic and output costs charged. The set of objects having individual cost at most B is not silently replaced by its free linear span. An algebraic gradient component lacking this record remains in the budget-filtered residual ledger, although the unfiltered orthogonal identity remains valid.

The authority envelope specifies which causal currents the presentation permits the certificate to use.

Definition II.4.3 (Caus authority envelope)**.** Let a realized budget-admissible generator on $L^2(X_I, \mu_I)$ have the operator decomposition

$$L = L^{\mathsf{s}} + A + L^{\mathsf{b}}, \qquad (L^{\mathsf{s}})^* = L^{\mathsf{s}}, \qquad A^* = -A,$$

where the diagonal killed or boundary contribution may be included in L^{s} when convenient. A Caus authority envelope at budget B is an affine represented family

$$\mathfrak{A}^{\mathsf{Caus}}_{I,\le B} = A^{\mathrm{imp}}_I + \mathcal{A}^{\mathrm{ctrl}}_{I,\le B}.$$

The imposed operator A^{imp}_I is part of the declared dynamic realization. Each $A^{\mathrm{ctrl}}_I \in \mathcal{A}^{\mathrm{ctrl}}_{I,\le B}$ is a skew represented control constructed from the public presentation with its saturated cost charged. A choice is admissible only when the total operator

$$L^{\mathsf{s}} + A^{\mathrm{imp}}_I + A^{\mathrm{ctrl}}_I + L^{\mathsf{b}}$$

satisfies the positive maximum principle, the declared support relation, and the terminal conditions. The three authority regimes are

regime	A^{imp}_I	$\mathcal{A}^{\mathrm{ctrl}}_{I,\le B}$
free	0	all budget-admissible represented controls,
imposed	A^{imp}_I	$\{0\}$,
hybrid	A^{imp}_I	a declared budget-admissible control family.

Thus free authority removes an external restriction on the directed operator; it does not remove the derivation or selection cost of the chosen control.

The current record distinguishes authorized operator transport from auxiliary geometric descent.

Definition II.4.4 (Authorized operator current and geometric descent current)**.** Fix a generator selected from $\mathfrak{A}^{\mathsf{Caus}}_{I,\le B}$. Its authorized Caus operator current is the structural-gradient projection of the actual edge current induced by the selected skew operator $A^{\mathrm{imp}}_I + A^{\mathrm{ctrl}}_I$.

Separately, a represented symmetric conductance c_Φ and a represented target-oriented potential D define the auxiliary geometric descent current

$$J^{\mathrm{grad}}_{\Phi,D}(x,z) = -c_\Phi(x,z)\big(D(z) - D(x)\big),$$

It may orient a Geom drift or descent certificate when the same full generator supplies the stated local-consistency, drift, or carrier comparison. It is not the Caus operator component of that generator unless the certificate includes an explicit represented identity equating it with the structural-gradient projection of the actual selected current.

The potential is target-normalized: $D \ge 0$ and $D = 0$ on the positive terminal sector. A provenance record contains the authority envelope, selected control, the actual-current identity or auxiliary geometric-current identity as appropriate, every quotient or coordinate transport, and the comparison gate. All entries belong to one charged saturated derivation.

Lemma II.4.5 (Caus authority and no substitution)**.** *For a fixed dynamic realization and authority envelope, the exposed Caus component is the Hodge projection of the actual edge current of the selected skew operator. A quotient, coordinate change, or valid lift may transport this current through a represented dynamic interface, but may not replace it by another target-aligned current. An auxiliary geometric descent current may be used only in its Geom certificate role, unless an explicit represented equality identifies it with the actual operator current. A different represented* *current contributes only as a newly charged control in the free or hybrid envelope, or as residual directed motion relative to the imposed envelope.*

Proof. The Hodge projection in Definition II.4.2 is unique for the selected generator. Its input edge current is fixed by the imposed operator and the selected represented control. The auxiliary geometric current is fixed by its represented conductance and target-normalized potential, but a same-generator comparison does not make it equal to the skew current. A legal transport applies the represented interface to the appropriate data and retains their derivation provenance. A different operator current is not the image of the selected generator. It must consequently be supplied as a new control, if the authority envelope permits it, or remains in the orthogonal residual ledger. This exhausts the alternatives in the budgeted channel representative space. □

Theorem II.4.6 (Reversible geometric carrier and exhaustive directed classification)**.** *Fix a finite presented task, a budget B, and a represented generator selected from the authority envelope of Definition II.4.3. Write its existing operator split as*

$$L = L^{\mathsf{s}} + A + L^{\mathsf{b}}, \qquad (L^{\mathsf{s}})^* = L^{\mathsf{s}}, \qquad A^* = -A. \tag{8}$$

The local conductance represented by L^{s} is the Geom *carrier. In particular, the geometric carrier is reversible with respect to the declared reference measure. Let F_A be the actual edge current of A, let $\mathcal{C}_{\mathrm{str}} = \operatorname{im} G$ be the exposed structural gradient space of Definition II.4.2, and put*

$$F_A = \Pi_{\mathcal{C}_{\mathrm{str}}} F_A + (I - \Pi_{\mathcal{C}_{\mathrm{str}}}) F_A. \tag{9}$$

The first summand is the unique exposed Caus *current of the selected generator whenever its complete projection and authority record has cost at most B. The second summand is the directed J_{res} coordinate. For every exposed structural potential $D \in \mathcal{P}_{\mathrm{str}}$, one has the exact identities*

$$\langle F_A, -GD \rangle_w = \langle \Pi_{\mathcal{C}_{\mathrm{str}}} F_A, -GD \rangle_w \,, \tag{10}$$

$$\big\langle (I - \Pi_{\mathcal{C}_{\mathrm{str}}}) F_A, -GD \big\rangle_w = 0. \tag{11}$$

Consequently, every target-oriented use of the selected directed current that is visible at budget B is evaluated through the existing authorized Caus *alignment factor of the reversible* Geom *carrier. A directed term whose charged projection record is unavailable at budget B, or whose current is orthogonal to the exposed structural gradient space, remains in the budget-filtered J_{res} coordinate. If a later affordable represented record supplies a structural potential, projection, quotient, or lift that qualifies the term, saturated reclassification assigns it to the corresponding existing channel at the complete cost of that record.*

Thus the channel decomposition contains no independent nonreversible geometric source. Nonreversibility of the full selected generator is carried by the authorized Caus *current and its directed residual complement. An auxiliary geometric descent current remains a derived use of the reversible carrier unless the represented equality required by Definition II.4.4 identifies it with the actual selected operator current.*

The general-resolvent stability factor $S^{\mathrm{nr}}_{q,T}(\lambda)$ of Definition III.11.7 does not add a source channel. It is a charged target-transfer comparison for a quotient of the full selected Geom + Caus *generator. Its active geometric slot retains the reversible carrier, and its causal slot retains the transported current and authority record prescribed by the master certificate. Failure of that*

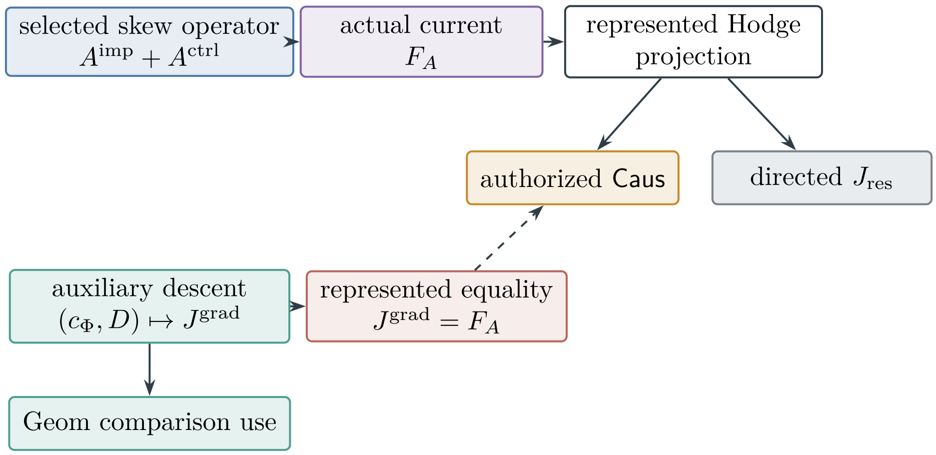

Figure 11: Causal-authority routing. The Caus current is the represented Hodge projection of the actual selected operator current. An auxiliary geometric descent current remains a Geom comparison unless a charged equality record identifies it with that actual current.

provenance sets the existing alignment factor to zero; it does not create a nonreversible Geom *alternative.*

Proof. Self-adjointness of L^{s} in the declared $L^2(X_I, \mu_I)$ realization is detailed balance for its off-diagonal conductance table. By the fixed-space and channel decompositions in Lemma II.3.5 and Theorem II.7.1, this symmetric local table is exactly the Geom coordinate, whereas the actual skew current of A is the input to the Hodge projection. Orthogonal projection onto $\mathcal{C}_{\mathrm{str}}$ gives Eq. (9); uniqueness follows from the orthogonal direct sum

$$\mathcal{H}_{\mathrm{edge}} = \mathcal{C}_{\mathrm{str}} \oplus \mathcal{C}_{\mathrm{str}}^{\perp}.$$

For $D \in \mathcal{P}_{\mathrm{str}}$, the vector GD belongs to $\mathcal{C}_{\mathrm{str}}$. The second summand of Eq. (9) is orthogonal to that space, which proves Eq. (11); adding this zero term proves Eq. (10). The charged-projection criterion and the assignment of an unavailable projection to filtered residual are part of Definition II.4.2. The authority and same-current requirements are Definition II.4.4 and Lemma II.4.5. Therefore every represented directed pairing used in a geometric certificate is already the authorized Caus factor, while the orthogonal or unrepresented remainder has no exposed structural-gradient pairing at budget B. Saturated reclassification is Proposition II.8.3 and Lemma II.8.4.

Finally, Definition III.11.7 defines $S^{\mathrm{nr}}_{q,T}(\lambda)$ as a scalar comparison of the quotient and full-process target functionals. The causal-provenance and dynamic-stability slots of Definition III.18.1 retain, respectively, the actual selected current and that comparison factor. Proposition II.5.9 prevents either derived slot from becoming an additional primitive source. Hence a non-self-adjoint full-generator comparison changes neither the reversible meaning of Geom nor the exhaustive Caus/J_{res} classification of its skew current.

□

Figure 11 shows the no-substitution routing enforced by the authority envelope.

The target sector does not enter this projection. Target alignment without an exposed structural gradient is classified as an oracle-supplied edge, a presentation-inaccessible edge, or residual directed motion.

II.5 Quotient, Lifted-State, and Operational Channels

The nonlocal structural term now separates into represented quotient motion and valid lifted-state transport.

Definition II.5.1 (Structured nonlocal quotient and lift channels)**.** Let $\mathcal{H}_{\mathrm{nl}}$ be the finite-dimensional Hilbert space of all nonlocal off-diagonal operator tables remaining after the declared Geom and Caus projections, with the same weighted edge inner-product convention. Its elements are signed operator components; an orthogonal projection need not itself be a positive generator. An observable represented quotient $q : X_I \to Q$ contributes an abstraction subspace $\mathcal{S}_{\mathrm{Abs}}(q) \leq \mathcal{H}_{\mathrm{nl}}$ spanned by represented nonlocal operator tables whose quotient-constant subspace is invariant and whose restriction descends to a quotient operator. With lift $U : L^2(Q) \to L^2(X_I)$, exact descent means

$$L_X U = U L_Q.$$

The abstraction subspace is the span of these observable quotient subspaces, including symmetric or directed nonlocal tables after the earlier projections. An observable lifted interface contributes a subspace $\mathcal{S}_{\mathrm{Lift}} \leq \mathcal{H}_{\mathrm{nl}}$ generated by nonlocal moves on lifted configuration coordinates: memory, stack state, proof fragments, subproblem interfaces, or product factors. To make the structured split unique, set

$$\mathcal{S}_{\mathrm{struct}} = \mathcal{S}_{\mathrm{Abs}} \oplus (\mathcal{S}_{\mathrm{Lift}} \cap \mathcal{S}_{\mathrm{Abs}}^{\perp}).$$

Projection onto $\mathcal{S}_{\mathrm{Abs}}$ defines the quotient/lumping channel coordinate Abs at the algebraic-ledger stage, while projection onto $\mathcal{S}_{\mathrm{Lift}} \cap \mathcal{S}_{\mathrm{Abs}}^{\perp}$ defines the lifted-state component Lift. The remaining nonlocal edge mass has neither quotient nor lift provenance and is included in J_{res}. A single observed transition does not determine symmetry of the full rate table.

The algebraic channel coordinate records quotient provenance. A represented fiber partition contributes to the target-bearing Abs structural source profile precisely when its complete record satisfies the quotient-utility gate of Definition III.15.1, the uniform saturated-cost condition of Definition III.15.2, and, for prospective source use, the temporal condition of Definition III.21.1. Exact quotient records used only to preserve a completed hitting law remain in the accounting profile of Definition III.23.3. Thus quotient provenance, accounting admissibility, and qualified structural-source status are separate stages of the same ledger.

Likewise, the spans and orthogonal projections in this definition are unconditional algebraic coordinates. Their budget-B exposed versions require one charged represented record containing finite bases for the quotient and lift table spaces, the represented interfaces, and the projection procedure and output. Individual low-cost quotient or lift records do not make their unrestricted linear span or its orthogonal projection free. Any algebraic portion without such a record is retained in $J_{\mathrm{res},B}$ for the filtered accounting.

Validity identifies the lifts that preserve the declared observable and target interfaces.

Definition II.5.2 (Valid Lift)**.** Write $\mathcal{F}_{\mathrm{obs}} = \mathcal{F}_I$ for the observable algebra of the presented task. Let $T_I \in \mathcal{F}_{\mathrm{obs}}$ be any target or terminal sector named by the task. A finite lift is valid when it consists of a finite lifted state space Ω, a lifted observable algebra $\mathcal{G}_{\mathrm{obs}}$, a measurable projection $\pi : \Omega \to X_I$, and a positive lifting map $\Lambda : \mathcal{M}(X_I) \to \mathcal{M}(\Omega)$. The map Λ is linear, sends $\mathcal{M}_+(X_I)$ into $\mathcal{M}_+(\Omega)$, and satisfies

$$\pi^{-1}\mathcal{F}_{\mathrm{obs}} \subseteq \mathcal{G}_{\mathrm{obs}}, \qquad \pi_*\Lambda\mu = \mu \quad \text{for every } \mu \in \mathcal{M}_+(X_I).$$

In particular, Λ preserves total mass and is a Markov kernel from base states to lifted fibers. Linearity is part of Valid Lift; it is what permits the fiber disintegration and composition identities below.

The structural-Lift target sector is $T_\Omega = \pi^{-1}(T_I)$. If the instance presentation or a budget-generated computation specifies a different terminal sector, that sector is a legal computation interface but is a structural Valid-Lift target only when the same target-fraction and average identities below are verified separately. For the pullback target, every nonzero frontier satisfies the target-fraction identity

$$\frac{(\Lambda\mu)(T_\Omega)}{(\Lambda\mu)(\Omega)} = \frac{\mu(T_I)}{\mu(X_I)}.$$

Equivalently, for every bounded $\mathcal{F}_{\mathrm{obs}}$-measurable target-relevant observable h, the lifted average of $h \circ \pi$ under $\Lambda\mu$ equals the original average of h under μ. A budget-controlled lift, including the polynomial-budget lifts used in later examples, must satisfy these measurability and fraction-preservation conditions. The lifted-state channel may therefore transform, organize, and expose information in the observable algebra or explicitly charged lifted coordinates while preserving target mass.

II.5.1 Resource envelopes and operational transcripts

Resource envelopes place the quotient and lift constructions on the complete operational transcript carrier.

Definition II.5.3 (Budget-compatible resource envelope)**.** For a task family (I_n) and budget structure $\mathcal{B} = (\mathcal{R}, \mathrm{cost}, b, \mathfrak{C})$, a budget-compatible resource envelope bounds update steps, native internal-state capacity, live work bits or cells, stack depth, stored tables, learned records, query transcripts, preprocessing output, outcome records, processor registers, and total parallel work. Data within the envelope are generated by a uniform rule without oracle access and are stored, queried, or updated with accumulated charge $\preceq b(n)$. Remaining data are advice, oracle-supplied, or over budget. For $\mathcal{B}_{\mathrm{poly}}$, this recovers the usual polynomial resource envelope.

Parallelism is charged by total represented work. A parallel configuration records the finite list of active branch states, branch weights or scheduler state, and shared memory/interface records. The envelope bounds the total number of represented branch states and cells; unrecorded exponential branching is therefore inadmissible. An internal operational state is charged by its native capacity and update rule, rather than by the cardinality of an extensional coordinate expansion. Thus this clause charges actual replicated work and does not replace one represented internal state by a list of hypothetical branches.

This definition charges an operational transition rule as a represented object of the presentation.

Definition II.5.4 (Represented operational transition rule)**.** A represented operational transition rule consists of a complete native operational state z, a finite represented outcome interface $A(z)$, a subprobability law $p(\cdot \mid z)$ on that interface, a represented successor map $(z, a) \mapsto z \cdot a$, and finite represented public-readout and terminal-record interfaces. Completeness means that the conditional law of the next native state and every declared record is determined by the current native state. The native transition law is

$$P(z, B) = \sum_{a \in A(z)} p(a \mid z)\, \mathbf{1}_B(z \cdot a).$$

The native state contains every internal coordinate needed by the update semantics, together with the finite control, work, and history records used by later transitions. It need not be a member of a finite enumerated set. The represented initializer produces z_0; any nondeterministic initialization is incorporated into the first finite outcome interface. Starting from that initializer, a finite outcome prefix $h = (a_1, \ldots, a_t)$ determines the current native state recursively by

$$z(\varnothing) = z_0, \qquad z(ha) = z(h) \cdot a.$$

The observable algebra contains the finite outcome and terminal records and the declared public effects of $z(h)$; semantic internal coordinates are not automatically adjoined as free readouts. For a finite horizon, the represented records

$$\xi_t = (t, a_1, \ldots, a_t, v_0, \ldots, v_t),$$

where each v_j is the declared finite public terminal/readout record at time j, form a finite padded transcript carrier. Execution keeps only the current native state and appends the represented records. It does not retain the sequence $z_0, \ldots, z_t$. Representation cost charges the update implementation, precision, outcome and terminal records, accumulated transcript and total work, together with the peak native-state capacity. It does not require an extensional native-state vector, transition table, or enumeration of native states. The represented row or effect at an isolated prefix h may reconstruct $z(h)$ by replaying that prefix through the successor map, using one live native state and charging at most t updates. Sequential execution already carries $z(h)$ and needs no such reconstruction. Neither evaluation retains the intermediate native states. The definition uses only the conditional transition law on complete native states and makes no classification by implementation substrate.

The full transcript representation places every stored operational field on one charged carrier.

Definition II.5.5 (Full operational transcript representation)**.** Let A_n be a represented operational transition rule under a finite resource envelope $\mathcal{B}$, and let H be the step bound supplied by that envelope. Its full operational transcript representation is the clocked transcript carrier $\Omega_{A,n}^{[H]}$ of padded outcome, public-readout, and terminal records from Definition II.5.4. A legal transcript prefix h determines the current complete native state $z(h)$ by the represented initializer and successor recursion. The live state may contain the candidate state, finite control, work or memory contents, head positions, stacks, preprocessed data, stored tables, learned records, priority queues, active parallel branch states, scheduler state, branch weights, and output registers. These are charged through the peak native-state capacity and their update operations; their successive values are not copied into the finite carrier. Any outcome, query, public-history, or terminal record retained by the computation is instead appended to the transcript and charged there. Malformed or unused syntactic transcripts are completed by a fixed inert failure transition, so the transcript update is total without changing the legal computation.

The representation ledger distinguishes sequential execution from isolated carrier access. Sequential execution carries the current native state. Evaluating a kernel row or declared native-state effect from an isolated transcript prefix replays that prefix through the outcome successor interface and charges every repeated update, while retaining only one native state.

The lifted observable algebra $\mathcal{G}_{A,n}^{[H]}$ is generated by the transcript coordinates and declared represented effects on the current native state $z(h)$. The complete native state determines the update but is exposed only through these declared effects. The transition representation itself requires neither a target nor a task projection. When the rule is coupled to a presented task, this algebra also contains the pullback of the base observable algebra, and the task interface adjoins an exposed total readout $\pi_A : \Omega_{A,n}^{[H]} \to X_n$, a public initialization, and a terminal verifier. In that case the lifted target may be $T_{A,n} = \pi_A^{-1}(T_n)$, with any additional terminal-output convention exposed in $\mathcal{G}_{A,n}^{[H]}$. The initialization lift Λ_A attaches the public initialization distribution over auxiliary records to a base frontier.

When these task-interface data are present, the transcript representation is additionally a valid structural Lift when $\pi_A^{-1}\mathcal{F}_n \subseteq \mathcal{G}_{A,n}^{[H]}$, $\pi_{A*}\Lambda_A\mu = \mu$, and the target-fraction identity of Definition II.5.2 holds. Preprocessing is part of the initialized native state and is charged to $\mathcal{B}$. Finite outcomes are either recorded in the transcript or marginalized into its Markov kernel. Persistent adaptive memory, tables, and actual parallel branches remain in the current native state unless a declared represented effect exposes them; retained public records are transcript coordinates.

The finite transcript representation exists without the displayed valid-Lift identities. Those identities are required only when the auxiliary coordinates are used as a structural Lift source or when a target-fraction comparison is transported to the base task. The Markov embedding and exact first-hit construction act directly on $\Omega_{A,n}^{[H]}$ and require no structural-Lift classification.

II.5.2 Admissibility and transcript completeness

The following conditions identify valid lifted interfaces and establish that resource-bounded computations have complete transcript representations.

Definition II.5.6 (Structural admissibility conditions for Lift)**.** A lifted component is a budget-admissible Lift contribution at saturated public budget B precisely when the task presentation and the induced generator on the full represented carrier satisfy all of the following objective conditions:

Generation. The lifted coordinates are generated by the public observable algebra and the uniform transition rule without oracle access within the resource envelope.

Measurability. The base observables pull back into the lifted algebra: $\pi^{-1}\mathcal{F}_{\mathrm{obs}} \subseteq \mathcal{G}_{\mathrm{obs}}$.

Projection. The lift preserves the base frontier under projection: $\pi_*\Lambda\mu = \mu$ for every base frontier under consideration.

Reference law. Every lifted L^2 reference measure used to prove cost, regularity, projection, or reach satisfies Definition Reference measure legality. In particular, occupation, hitting-conditioned, invariant, Green, and resolvent measures of the induced operator are not admissible reference laws unless they have an independent public representation with controlled density.

Target-fraction preservation. For every named target or terminal sector T, the lifted target is the pullback $\pi^{-1}T$ unless an alternative terminal convention is itself exposed, and the identity $(\Lambda\mu)(\pi^{-1}T)/(\Lambda\mu)(\Omega) = \mu(T)/\mu(X)$ holds.

Cost. The description length, represented state dimension, lifted interfaces, and comparison data used by the charge have saturated public cost at most B.

These conditions are predicates of the presented or lifted operator, rather than outputs required from an algorithm, inverter, or adversary. When they hold at budget B, the Lift component exists independently of whether its representation has been identified. When they fail, the component is

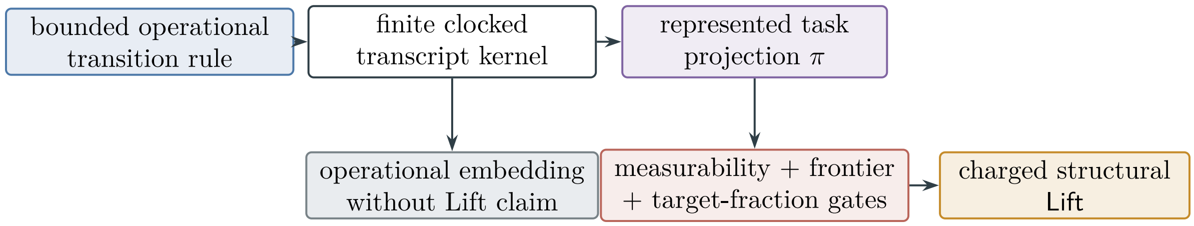

Figure 12: Operational embedding and structural lifting are distinct. Every bounded represented transition rule has a finite clocked transcript kernel. It becomes a structural Lift only after a represented task projection, measurability, frontier-preservation, target-fraction, reference-law, and cost gates are verified.

not exposed at budget B and is classified as another represented channel, presentation-inaccessible or oracle-supplied data, presentation enrichment, or J_{res}, according to the provenance theorem.

Figure 12 separates unconditional operational embedding from the additional gates required for a structural Lift certificate.

Proposition II.5.7 (Channel admissibility conditions are operator properties)**.** *The channel conditions used by later bounds, including quotient lumpability, valid-lift conditions, target-fraction preservation, saturated-cost bounds, boundary readout, and any mixing or regularity comparison, are objective predicates of the presented task object, the lifted configuration, and the induced operator. They are not membership conditions for a runtime class and are not obligations imposed on the algorithm, inverter, or adversary.*

Absent a proof of a condition, the corresponding quantitative theorem cannot assign the component a budget-admissible charge. The induced operator nevertheless remains in the Part II decomposition. At a fixed budget, a component without the required representative is classified as another represented channel, presentation-inaccessible or oracle-supplied information, presentation enrichment, or J_{res}.

Proof. The generator form, fixed-space split, Hodge projection, quotient projection, valid-lift projection, and boundary split apply directly to the induced operator. The admissibility conditions enter only when a subsequent quantitative theorem assigns a cost to a component. Their truth is independent of the availability of a proof. Without a proof from the presentation and induced operator, that quantitative application is unavailable, while the structural decomposition remains valid. Verification is therefore an application hypothesis rather than a condition for budget-admissible execution.

□

Proposition II.5.8 (Full transcript representation of resource-bounded computations)**.** *Every represented operational transition rule under a finite resource envelope has a full finite transcript representation once every external input on which its transition depends is included in the initialized native state or in the represented interface. Every datum used by a future transition is either retained in the current native state, recorded in the outcome/terminal transcript, marginalized into the induced transcript kernel, or supplied as an external input. The ledger charges the current native state at peak capacity and the transcript at its accumulated represented size; it does not store or charge a list of intermediate native states. Relative to a fixed presented task, an external instance-dependent datum not declared or generated within budget is advice, oracle-supplied data,* *or over-budget information; this affects affordability for that presentation, not existence of the transition representation. When the projection and initialization also satisfy Definition II.5.2, this representation is a valid structural* Lift.

Proof. The resource envelope bounds the horizon and each finite outcome, public-readout, and terminal-record alphabet. Their padded transcript carrier is therefore finite. For a legal prefix h, the initializer and successor recursion determine $z(h)$, and the represented conditional outcome law appends one record. Extend this rule by the declared inert failure update on malformed transcripts. This gives a Markov or sub-Markov kernel on the finite transcript carrier without enumerating the native-state space. Preprocessing belongs to the initialized native state. Adaptive memory, learned records, tables, replicated branches, and scheduler state are updated inside the single live native state and are charged through its peak capacity; declared public records are appended to the transcript. A transition depending on an unrecorded datum is a transition system with an external input and is represented after that input is adjoined to its represented interface. For a fixed task presentation, using such a datum without declaring or deriving it is advice, oracle information, or data over budget for $\mathcal{B}$. This proves the unconditional finite transcript representation. The task projection and target-fraction identities are additional Valid Lift conditions invoked only for the corresponding structural channel. □

Proposition II.5.9 (The directed, lifted-state, and boundary channels are derived)**.** *The reusable sources of observable structure are exposed local geometry and represented abstraction. The channels* Caus*,* Lift*, and* Bdry *process these sources. More precisely,* Caus *is the gradient projection of directed flow onto potentials represented in the observable or lifted algebra;* Lift *is valid only when the lifted coordinates are observable or explicitly charged and the target fraction is preserved by projection; and* Bdry *processes exposed terminal observables into boundary value functions represented by the observable grammar. Removing support is the zero-boundary-value case. A committor or hitting-time value of the induced generator is counted as* Bdry *when it is represented by exposed terminal data together with already budget-admissible Geom/Abs/Caus/Lift structure, or when a finite-horizon approximation of the same value is generated by budget-admissible computation and charged as a lifted observable.*

Proof. For Caus, the definition is $F_{\mathrm{Caus}} = \Pi_{\mathcal{C}_{\mathrm{str}}} F_J$ with $\mathcal{C}_{\mathrm{str}} = \operatorname{im} G$. If the observable structural potential space is zero, then $\mathcal{C}_{\mathrm{str}} = 0$ and the causal component is zero. A directed edge that points at the target without such a potential is residual or oracle-supplied rather than a directed/drift structural component.

For Lift, validity gives $\pi_* \Lambda\mu = \mu$ and $T_\Omega = \pi^{-1}(T_I)$. Therefore $(\Lambda\mu)(T_\Omega) = \mu(T_I)$ and $(\Lambda\mu)(\Omega) = \mu(X_I)$. The displayed target-fraction identity follows. Any lifted coordinate not generated by the observable algebra must be generated as part of a valid lifted observable state before use. Otherwise it is presentation-inaccessible, oracle-supplied, or over budget, and its associated movement remains residual relative to the public presentation.

For Bdry, the general boundary form is $(L^{\mathsf{Bdry}}(f, v))(x) = \kappa(x)(Bv(x) - f(x))$. With an interior transport operator A from the other channels, the associated boundary value function V is equivalently characterized by the finite Dirichlet value equation

$$(AV)(x) + \kappa(x)(Bv(x) - V(x)) = 0 \qquad (x \in X_I).$$

Thus Bdry maps observable terminal readouts to interior value functions, with support removal given by the special case $Bv \equiv 0$. Because this channel has no interior off-diagonal transport edge, it evaluates or removes mass according to exposed terminal data and budget-admissible transport structure. The map $L \mapsto (\lambda I - L_N)^{-1} k_T$, or the corresponding infinite-horizon hitting table of the same induced kernel, requires an admissible representation to enter the boundary channel. A finite-horizon table generated by budget-admissible simulation is charged at its saturated computation cost. These three derived channels therefore act only on exposed geometry, represented abstraction, or explicitly charged lifted data.

□

Theorem II.5.10 (Exhaustive boundary normal form)**.** *Fix the represented boundary space ∂X_I, transfer rates $b(x,\zeta) \geq 0$, and boundary value input v from Definition II.3.4. Put $\kappa(x) = \sum_{\zeta \in \partial X_I} b(x,\zeta)$, and define the two boundary blocks*

$$(L_{\mathrm{kill}}^{\mathsf{Bdry}}(f,v))(x) = -\kappa(x) f(x), \tag{12}$$

$$(L_{\mathrm{val}}^{\mathsf{Bdry}}(f,v))(x) = \sum_{\zeta \in \partial X_I} b(x,\zeta) v(\zeta) = \kappa(x) B v(x). \tag{13}$$

Every finite boundary component has the unique block decomposition

$$L^{\mathsf{Bdry}} = L_{\mathrm{kill}}^{\mathsf{Bdry}} + L_{\mathrm{val}}^{\mathsf{Bdry}}. \tag{14}$$

The value block has the unique boundary-atom expansion

$$L_{\mathrm{val}}^{\mathsf{Bdry}} = \sum_{\zeta \in \partial X_I} b_\zeta \, \mathrm{ev}_\zeta, \qquad b_\zeta(x) = b(x,\zeta), \qquad \mathrm{ev}_\zeta(v) = v(\zeta). \tag{15}$$

Thus the algebraic Bdry *channel has exactly two blocks: homogeneous boundary loss and inhomogeneous represented boundary-value readout. Killing and support removal are the specialization $v = 0$. Terminal indicators, support colorings, committors, hitting-time values, and other scalar terminal tables belong to the value block precisely when the same denotation is a declared boundary observable or has a budget-admissible represented derivation. A finite-horizon value generated by an admissible computation is in the latter provenance class at its saturated computation cost.*

At budget B, the complete record consists of the boundary space, the rate vectors b_ζ, the represented value input, and the evaluation procedure. A record exposed at cost at most B is charged to Bdry*. The filtered-residual convention of Theorem II.7.2 retains a structurally typed boundary block whose complete record is unavailable at that budget. The channel contains no interior off-diagonal transport block.*

Proof. The finite generator form assigns every diagonal loss not balanced by an interior off-diagonal rate to transfer into the represented boundary space. For the extended input pair (f,v), the boundary term is

$$\sum_{\zeta \in \partial X_I} b(x,\zeta)\big(v(\zeta) - f(x)\big) = -\kappa(x) f(x) + \sum_{\zeta \in \partial X_I} b(x,\zeta) v(\zeta),$$

which proves (14). Restriction to inputs $(f,0)$ recovers $L_{\mathrm{kill}}^{\mathsf{Bdry}}$, while restriction to $(0,v)$ recovers $L_{\mathrm{val}}^{\mathsf{Bdry}}$; hence the two-block decomposition is unique. The coordinate evaluations $\{\mathrm{ev}_\zeta\}_{\zeta \in \partial X_I}$ form the canonical basis of $(\mathbb{R}^{\partial X_I})^*$. The coefficients of the value block in that basis are exactly the rate vectors b_ζ, proving (15) and its uniqueness.

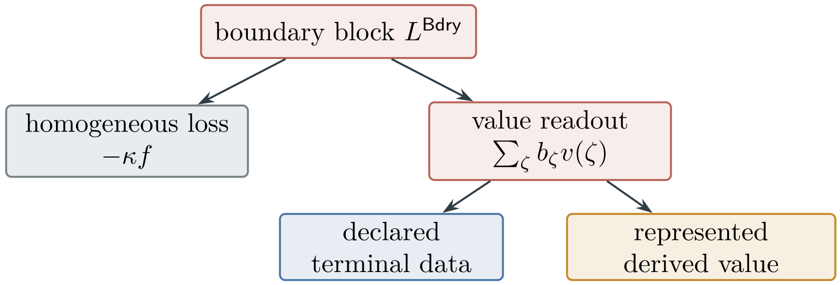

Figure 13: The exhaustive boundary normal form of Theorem II.5.10: homogeneous killing and inhomogeneous value readout are the two algebraic blocks; the value is declared or supplied by a charged represented derivation.

Proposition II.5.9 identifies a represented terminal observable or a budget-generated finite-horizon value as the admissible input to this block. The charged projection convention gives the budget statement. Both blocks map from the interior/boundary input pair to the interior value coordinate, and neither introduces an interior off-diagonal rate. Therefore the two blocks exhaust Bdry.

□

Figure 13 displays the algebraic and provenance layers of Theorem II.5.10.

II.6 Partial Lumpability and Full-Operator Accounting

Partial lumpability is measured on the restricted block while its contribution remains charged to the full represented operator.

Definition II.6.1 (Partial lumpability defect)**.** Let $q : X_I \to Q$ be a represented quotient. Let $U : \mathbb{R}^Q \to \mathbb{R}^{X_I}$ be the lift $(Ug)(x) = g(q(x))$, and let $P : \mathbb{R}^{X_I} \to \mathbb{R}^Q$ be a declared projection with $PU = I_Q$. For a finite generator L_X, the canonical quotient operator and defect are

$$L_Q = PL_XU, \qquad E_q = L_XU - UL_Q = (I - UP)L_XU.$$

The quotient is exactly lumpable when $E_q = 0$. It is partially or approximately lumpable when $E_q \neq 0$, with the size of E_q measured relative to the quotient and within-fiber dynamics. This is the standard distinction between exact and approximate aggregation of a finite Markov chain [17, 45]. The quotient operator L_Q is a valid quotient generator only when it satisfies the finite generator form on Q; otherwise the part failing that form is counted as defect rather than as Abs.

Proposition II.6.2 (Full-operator accounting of partial lumpability)**.** *A partially lumpable quotient decomposes the action of L_X on quotient observables, but that restricted identity is not by itself a decomposition of the full operator. With $\Pi = UP$, the exact full-domain identity is*

$$L_X = UL_QP + E_qP + L_X(I - \Pi). \tag{16}$$

The channel projections are applied to all three terms. The represented quotient-descending portion of UL_QP that lies in $\mathcal{S}_{\mathrm{Abs}}$ is an Abs *contribution; the remaining local, causal, lifted, boundary, and residual portions of the complete identity are assigned by the same channel decomposition.*

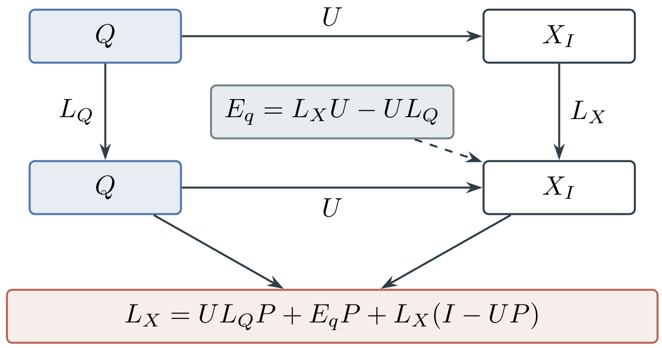

Figure 14: Exact lumpability is the commuting case $E_q = 0$. Partial lumpability retains the defect and the within-fiber complement in the full operator identity of Proposition II.6.2.

Proof. The restricted identity

$$L_X U = U L_Q + E_q$$

is an algebraic decomposition of the image of quotient observables. Because $PU = I_Q$, $\Pi = UP$ is idempotent, and

$$L_X = L_X \Pi + L_X(I - \Pi) = U L_Q P + E_q P + L_X(I - \Pi),$$

which proves (16). The block $U L_Q P$ is a full-domain quotient-intertwined block; it is not automatically an entire channel component. Apply the declared Hilbert projections to all three full-domain terms. This is a linear-algebraic decomposition and does not assert that $E_q P$, $L_X(I - \Pi)$, or any projected piece is a positive Markov generator. A piece with a represented geometric, causal, quotient, lift, or boundary witness is charged to that channel, and the orthogonal complement is charged to J_{res}. Any hitting or resolvent statement still uses the full positive generator L_X. $\square$

Figure 14 displays the intertwining defect and the full-domain identity of Definition II.6.1 and Proposition II.6.2.

The residual analysis later treats iteration of approximate quotients across a residual degree threshold. The structural conclusion used here is that approximate intertwining supplies the full identity (16), whose three terms are covered by the same channel projections.

Accordingly, the comparability relation and regularity functional induce the geometric/reversible channel, represented fibers and static lumpability induce the quotient/lumping channel, level-set filtrations and exposed potentials induce the directed/drift channel, lifted task states induce the lifted-state channel, and terminal readouts induce the boundary/killing channel. Nonlocal transitions without these provenance witnesses belong to J_{res}.

II.7 Exhaustiveness of the Structural Decomposition

The generator form reduces the operator to transition rates and diagonal loss; after adjoining the boundary-value space, the latter is boundary readout with killing as its zero-value specialization. The fixed-space decomposition separates local symmetric, local directed, nonlocal, and boundary terms. The structural projections below exhaust these components.

II.7.1 Exhaustive channel decomposition

The first result gives the exact operator identity and its budget-filtered interpretation.

Theorem II.7.1 (Component exhaustiveness)**.** *Every finite search generator satisfying the positive maximum principle has, relative to any declared finite channel subspaces on $(X_I, \mathcal{F}_I)$ or on a full lifted observable configuration, the unconditional algebraic decomposition*

$$L = L^{\mathsf{Geom}} + L^{\mathsf{Caus}} + L^{\mathsf{Abs}} + L^{\mathsf{Lift}} + L^{\mathsf{Bdry}} + L^{J_{\mathrm{res}}}$$

in the direct-sum operator Hilbert space determined by the fixed-space, Hodge, quotient, lift, and boundary projections. In particular,

$$\|L\|^2_{\mathscr{H}_{\mathrm{ch}}} = \sum_{j\in\{\mathsf{Geom},\mathsf{Caus},\mathsf{Abs},\mathsf{Lift},\mathsf{Bdry}\}} \|L^j\|^2_{\mathscr{H}_{\mathrm{ch}}} + \|L^{J_{\mathrm{res}}}\|^2_{\mathscr{H}_{\mathrm{ch}}}, \tag{17}$$

where $\|\cdot\|_{\mathscr{H}_{\mathrm{ch}}}$ denotes the direct-sum Hilbert norm used to define the channel projections. It is not the induced operator norm.

At budget B, a displayed component is an exposed structural channel only when its subspace, projection, and output have the charged representation required above. Algebraic portions lacking that record are included in the filtered residual $L^{J_{\mathrm{res}},B}$. This budget convention changes no full-operator identity or Pythagorean identity for the corresponding declared algebraic subspaces.

Here the local relation, structural potentials, quotient subspaces, and valid lift subspaces are exactly those generated by the observable algebra of the presented or lifted task. The reference weight m only fixes conductance/current coordinates. The exposed structural components are the geometric/reversible Geom*, directed/drift* Caus*, quotient/lumping* Abs*, lifted-state* Lift*, and boundary/killing* Bdry *channels. The term $L^{J_{\mathrm{res}}}$ is the orthogonal remainder left after the declared projections. It includes directed local cycle/current components orthogonal to the represented gradient space as well as unclassified nonlocal transport. These terms exhaust the algebraic operator decomposition together with the five structural projections; their budget-exposed versions use the charged projection convention above.*

Proof. By the finite generator theorem, the operator consists only of nonnegative off-diagonal rates and nonnegative diagonal loss. In zero-boundary reduction this loss is killing. With terminal values represented, it is the boundary value operator $\kappa(Bv - f)$. This is Bdry.

Fix the exposed local relation E_I and reference weight m. The off-diagonal rate table splits uniquely into local symmetric, local antisymmetric, and nonlocal entries in these coordinates. Local symmetric conductance is Geom. The weight m provides conductance-current coordinates for the same rate table and does not alter transitions or target data.

Apply the declared Hodge projection to the directed current. This gives the gradient component Caus and an orthogonal directed remainder, which may contain local cycle current and is assigned to J_{res}. After setting aside that remainder, the remaining nonlocal table lies in the finite Hilbert space $\mathcal{H}_{\mathrm{nl}}$. Orthogonal projection onto the observable quotient subspace $\mathcal{S}_{\mathrm{Abs}}$, then onto the valid observable lifted subspace $\mathcal{S}_{\mathrm{Lift}} \cap \mathcal{S}_{\mathrm{Abs}}^{\perp}$, gives Abs and Lift. Partial lumpability gives no additional case: the full identity (16) is projected by this same component calculus, and no restricted block is silently identified with the full Abs component. Any remaining nonlocal component is also assigned to J_{res}. These projections establish the algebraic identity. Restricting exposed terms to charged projection records and moving every unrepresented projected denotation into the filtered residual gives the budget-indexed statement. Hence the five structural projections and their residual complement exhaust the rate table.

□

The residual complement has its own canonical normal form within the same projection calculus.

Theorem II.7.2 (Exhaustive residual normal form)**.** *Fix the data and projection order of Definitions II.5.1 to II.3.3, and let L be a finite search generator covered by Theorem II.7.1. Let F_J be the complete directed-flow vector passed to the Hodge projection. Define the directed residual coordinate $L_{\mathrm{dir}}^{J_{\mathrm{res}}}$ to be the operator represented by*

$$F_{\mathrm{res}} = (I - \Pi_{\mathcal{C}_{\mathrm{str}}})F_J, \tag{18}$$

After setting aside $L_{\mathrm{dir}}^{J_{\mathrm{res}}}$, let $L_{\mathrm{nl}} \in \mathcal{H}_{\mathrm{nl}}$ be the nonlocal table remaining after the declared Geom *and* Caus *projections, in the projection order of Theorem II.7.1. Define the nonlocal residual coordinate by*

$$L_{\mathrm{nl}}^{J_{\mathrm{res}}} = (I - \Pi_{\mathcal{S}_{\mathrm{struct}}})L_{\mathrm{nl}}, \qquad \mathcal{S}_{\mathrm{struct}} = \mathcal{S}_{\mathrm{Abs}} \oplus (\mathcal{S}_{\mathrm{Lift}} \cap \mathcal{S}_{\mathrm{Abs}}^{\perp}). \tag{19}$$

Then the algebraic residual has the unique orthogonal decomposition

$$L^{J_{\mathrm{res}}} = L_{\mathrm{dir}}^{J_{\mathrm{res}}} + L_{\mathrm{nl}}^{J_{\mathrm{res}}}, \qquad \langle L_{\mathrm{dir}}^{J_{\mathrm{res}}}, L_{\mathrm{nl}}^{J_{\mathrm{res}}} \rangle_{\mathscr{H}_{\mathrm{ch}}} = 0. \tag{20}$$

Consequently,

$$\|L^{J_{\mathrm{res}}}\|_{\mathscr{H}_{\mathrm{ch}}}^2 = \|L_{\mathrm{dir}}^{J_{\mathrm{res}}}\|_{\mathscr{H}_{\mathrm{ch}}}^2 + \|L_{\mathrm{nl}}^{J_{\mathrm{res}}}\|_{\mathscr{H}_{\mathrm{ch}}}^2. \tag{21}$$

The first summand is precisely the directed flow orthogonal to every exposed structural gradient. It includes cycle current and, when $\mathcal{P}_{\mathrm{str}}$ is a proper subspace of the full potential space, the orthogonal components of gradients whose potentials have no represented structural realization. The second summand is precisely the remaining nonlocal table orthogonal to both the represented quotient subspace and the orthogonalized valid-lift subspace. These two summands exhaust the unfiltered residual.

At a budget B, every term retained in the filtered residual $L^{J_{\mathrm{res}},B}$ has exactly one of the following ledger provenances:

(i) *the directed algebraic remainder $L_{\mathrm{dir}}^{J_{\mathrm{res}}}$;*
(ii) *the nonlocal algebraic remainder $L_{\mathrm{nl}}^{J_{\mathrm{res}}}$;*
(iii) *an algebraic component in one of the five ordered channel slots* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$ *whose subspace, projection procedure, or projected output lacks the complete charged representation required at budget B.*

The third case retains the unique algebraic channel tag supplied by the ordered orthogonal decomposition; it is a budget-ledger assignment rather than an additional algebraic summand. Whenever its complete charged record is exposed at a budget, the charged projection convention assigns the component to its algebraic channel, and orthogonality removes it from the filtered residual. Every partial-lumpability defect and within-fiber complement is passed through the same decomposition by Proposition II.6.2. Thus partial quotients, finite composition, lifted transcripts, memory, and adaptive branching create no further residual class.

Proof. By Lemma II.3.5, the fixed-space rate table is the unique sum of local symmetric, local directed, nonlocal, and boundary terms. The local symmetric term is Geom, and the boundary term is Bdry. Proposition II.3.6 excludes any further local primitive.

Apply the Hodge projection of Definition II.4.2 to the complete directed-flow vector. Orthogonal projection gives

$$F_J = \Pi_{\mathcal{C}_{\mathrm{str}}} F_J + (I - \Pi_{\mathcal{C}_{\mathrm{str}}}) F_J.$$

The first term is the Caus coordinate and the second is (18). After this directed remainder is set aside, Definition II.5.1 places the remaining nonlocal table in $\mathcal{H}_{\mathrm{nl}}$. Its structural subspace is the orthogonal sum

$$\mathcal{S}_{\mathrm{Abs}} \oplus (\mathcal{S}_{\mathrm{Lift}} \cap \mathcal{S}_{\mathrm{Abs}}^{\perp}).$$

Projection onto the two displayed summands gives Abs and Lift, respectively. Projection onto their orthogonal complement gives exactly (19). The fixed-space blocks and the successive structural projections are the direct-sum coordinates of $\mathscr{H}_{\mathrm{ch}}$. Hence the two residual coordinates are orthogonal and give (20) and (21). The symmetric and boundary blocks have already been exhausted, so no third algebraic residual coordinate remains.

The charged projection convention in Theorem II.7.1 and Definitions II.5.1 and II.4.2 retains in $L^{J_{\mathrm{res}},B}$ every algebraic projected denotation whose complete projection record is unavailable at budget B. The unfiltered ordered orthogonal decomposition supplies its unique channel tag, proving the third ledger case without replacing the budgeted objects by a free linear span. The charged projection convention removes every channel-qualified term once its complete representation record is exposed. Finally, Proposition II.6.2 sends all three terms of the full partial-lumpability identity back through these same projections. By Proposition II.5.8, a resource-bounded operational rule is represented on its complete finite lifted transcript carrier. Theorem II.7.1 applies the identical calculation on that carrier, including memory, branching, and retained outcomes. This exhausts the algebraic and budget-filtered alternatives.

□

Corollary II.7.3 (Completeness of the residual provenance normal form)**.** *For every presentation-relative finite generator on its complete represented carrier, each admitted residual operator term is either Hodge-orthogonal directed flow or quotient/lift-orthogonal nonlocal transport. At a fixed budget it may instead be a structurally typed algebraic component retained by the filtered-residual convention until its complete charged record is exposed. These alternatives exhaust the residual provenance of the admitted generator.*

Proof. Apply Theorem II.7.2 on the complete represented carrier supplied by Proposition II.5.8.

□

A transition rule using data outside the admitted presentation is assigned by Proposition II.11.7 to the corresponding enriched presentation. Target arrival produced by arbitrary interaction of the two algebraic residual coordinates with the five channel coordinates is included in the full-process bounds of Proposition III.31.3 and Corollary III.31.4.

Figure 15 displays the structural classification given by Theorem II.7.1, together with the Abs provenance classification of Theorem III.2.2. Figure 16 displays the exhaustive residual normal form of Theorem II.7.2.

Example II.7.4 (Running example: the LPN transcript channels)**.** Apply Theorem II.7.1 to any represented clocked LPN transcript generator built from the presentation in Example I.1.3. Its positive operator decomposes into $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$, and J_{res} on that transcript carrier.

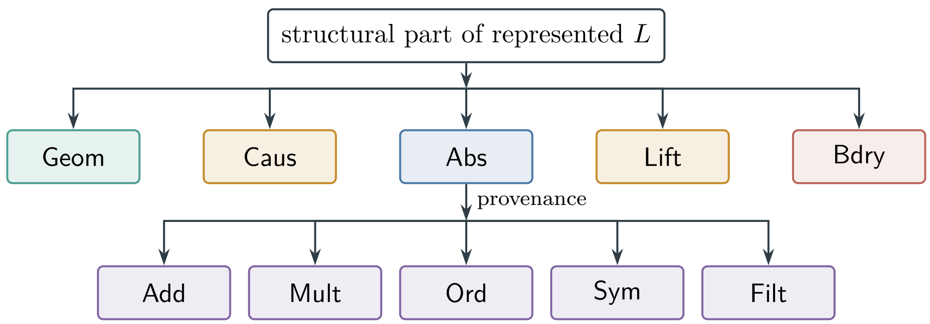

Figure 15: The five structural channels of Theorem II.7.1. The Abs channel has the five exhaustive, possibly overlapping provenance families of Theorem III.2.2. The residual complement is classified separately in Fig. 16.

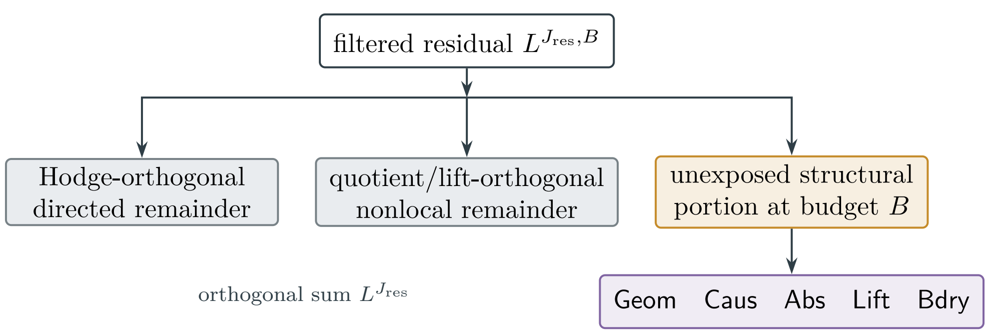

Figure 16: Exhaustive residual classification from Theorem II.7.2. The two gray leaves form the orthogonal unfiltered residual. The budget-filtered ledger also retains a structurally typed portion until its complete charged projection record is exposed.

This is the complete channel ledger; the LPN source analysis in Part VIII determines which target-aligned source terms remain.

Remark II.7.5 (Operator components and executable dynamics). The decomposition is an exact operator identity. An orthogonal component or a partial sum of components need not preserve positivity and need not be a Markov generator. Hitting probabilities, resolvents, and first-arrival laws are therefore evaluated for the full positive generator L. Component terms provide structural accounting; they are used as independent dynamics only when positivity and the finite generator form are verified separately. In particular, $L^{J_{\mathrm{res}}}$ is an orthogonal remainder, not an executable residual process by default.

Proposition II.7.6 (Channel decomposition is invariant under admissible presentation equivalence)**.** *Let two presentations be admissibly equivalent in the sense of Part I, and suppose the equivalence transports the dynamic data: admissible edge algebras, local relations, represented quotient subspaces, valid lift interfaces, terminal readouts, and the state and edge Hilbert structures by the unitary pullbacks required in Definition I.12.1. If L' is the transported generator of L, then each component transports to the corresponding component:*

$$L^{\mathsf{Geom}} \leftrightarrow L'^{\mathsf{Geom}}, \quad L^{\mathsf{Caus}} \leftrightarrow L'^{\mathsf{Caus}}, \quad L^{\mathsf{Abs}} \leftrightarrow L'^{\mathsf{Abs}},$$
$$L^{\mathsf{Lift}} \leftrightarrow L'^{\mathsf{Lift}}, \quad L^{\mathsf{Bdry}} \leftrightarrow L'^{\mathsf{Bdry}}, \quad L^{J_{\mathrm{res}}} \leftrightarrow L'^{J_{\mathrm{res}}}.$$

Thus an admissible recoding preserves each structural channel and may change only the coordinate cost of its transported representation. A bounded-density but nonunitary reweighting is not covered by this exact componentwise assertion.

Proof. The generator form is invariant under conjugating a finite Markov generator by a positive observable-algebra isomorphism: off-diagonal rates remain rates and diagonal loss remains diagonal loss. The transported local relation sends local symmetric conductance to local symmetric conductance and local oriented current to local oriented current. The Hodge projection is the orthogonal projection onto the transported structural-gradient subspace, so causal and orthogonal residual parts correspond. Represented quotient subspaces and valid lift subspaces are transported by hypothesis, so the Abs and Lift orthogonal projections correspond. Terminal readouts are transported by terminal preservation, so Bdry corresponds. The residual is the orthogonal complement left after these transported subspaces are removed, so it corresponds as well.

□

Corollary II.7.7 (Completeness of the structural channel classification)**.** *A budget-admissible operator obtains observable target-relevant structure through the five exposed structural channels. The intrinsic sources are exposed local geometry* Geom *and represented abstraction* Abs*. The other exposed channels are uses of those sources:* Caus *orients exposed potentials, valid* Lift *reorganizes observable or generated coordinates while preserving target fractions, and* Bdry *reads terminal observables as value functions. A budget-admissible method without oracle access either represents one of these objects on its clocked transcript carrier or through a declared native-state effect, or contributes to the residual operator component, which includes local cycle current and unclassified nonlocal transport. After saturated reclassification, this residual is an operator remainder and contains no affordable target-aligned source.*

Proof. Lemma II.8.2 partitions every target-relevant edge effect, after the full transcript and declared native-state effects are included, into public structural data, lifted/generated data, and residual operator transport. The public structural part is then decomposed by Theorem II.7.1

into the five exposed channels. The classification of generative sources identifies the only intrinsic reusable sources among those channels as valid Geom, exact Abs, and their coupled quasi-lumpable case. Caus, Lift, and Bdry retain the source they orient, reorganize, or read in their complete carrier-cost record. Their conditional target charge is transferred to that source only when a represented comparison proves the transfer inequality. By definition, every effect outside these projections belongs to the orthogonal residual J_{res}, completing the classification.

□

The same conclusion admits an information-provenance formulation. Public-algebra factorization, channel decomposition, partial-lumpability defect accounting, and finite-composition closure exhaust the origins of target-relevant information. Later low-degree and enumeration results quantify residual correlation but are not required for this structural partition.

II.8 Information Provenance and the Residual Band

Having decomposed the operator, we classify the represented origin of each state-dependent edge rule.

Definition II.8.1 (Information provenance of an edge rule)**.** Fix the finite observable algebra $\mathcal{F}_I$ with atom map $\alpha : X_I \to \mathcal{A}_I$. A provenance witness for an edge rule $q : X_I \times X_I \to [0, \infty)$ is a finite list of auxiliary variables $D = (D_1, \dots, D_r)$ such that q is measurable with respect to the finite algebra generated by $(\alpha(x), \alpha(y), D)$. The witness is a mathematical object associated with the presentation or induced operator, independently of its algorithmic discovery. Relative to the task presentation, each auxiliary variable has one of three provenances:

Public. It is generated by $\mathcal{F}_I$, or by the declared product algebra $\mathcal{F}_I \otimes \mathcal{F}_I$ for edge data.

Lifted/generated. It is generated beyond the base observable algebra by a budget-admissible computation without oracle access from the public presentation and is present as a coordinate in the lifted observable configuration.

Residual. After the public and lifted/generated data have been included, the edge flow lies outside the observable structural subspaces. It is recorded as J_{res}. This includes directed local cycle current orthogonal to every exposed structural gradient, as well as observed nonlocal jumps whose full symmetric/reversible rate table is not part of the observable structure. This provenance tag records operator transport only. Any affordable represented target-aligned use is reclassified and charged before the post-saturation residual is formed.

Lemma II.8.2 (Exhaustion of information provenance)**.** *Let L be a finite budget-admissible generator on the full clocked transcript carrier of a budget-admissible operational rule without oracle access. Assume that the transcript contains every retained finite outcome, query, public-history, and terminal record, and that every declared effect of the current native state is represented and charged. Then every target-relevant edge effect has one of the following provenances:*

$$\begin{gathered}\textit{public structural} \quad \sqcup \quad \textit{lifted/generated data} \quad \sqcup \quad \textit{residual operator transport,}\\ \textit{where the residual includes local cycle current and unclassified nonlocal transport.}\end{gathered}$$

After budget-generated data are lifted into the observable state, the combined represented public and lifted/generated part is decomposed by the five exposed channel projections

$$\mathsf{Geom}, \quad \mathsf{Caus}, \quad \mathsf{Abs}, \quad \mathsf{Lift}, \quad \mathsf{Bdry}.$$

The term J_{res} records the movement left after these projections and has the exhaustive directed/nonlocal normal form of Theorem II.7.2.

Proof. **Step 1: public factorization.** A finite function or relation is $\mathcal{F}_I$-measurable precisely when it is constant on atoms. Hence any admissible rate table $q(x, y)$ factors through the product atom map $(x, y) \mapsto (\alpha(x), \alpha(y))$ whenever that table is public-$\mathcal{F}_I \otimes \mathcal{F}_I$-measurable. This accounts for the public part. Tables depending on generated coordinates are handled after adjoining those coordinates in Step 3.

Step 2: structural partition. The positive maximum principle gives rates and diagonal loss. The fixed-space split partitions the rates into local symmetric, local directed, and nonlocal parts, and records the diagonal loss as boundary readout. The Hodge projection partitions directed flow into the structural-gradient subspace and its residual. The quotient/valid-lift projections extract the nonlocal flow with represented abstraction or target-fraction-preserving lift provenance. A partially lumpable quotient supplies the full-domain identity $L = UL_QP + E_qP + L(I - UP)$; all three terms are passed through the channel projections, and only the quotient-descending projection of the first term is Abs. The boundary component processes terminal observables into value functions; if the terminal value is zero, it is ordinary killing. Every remaining directed or nonlocal edge effect has no observable structural provenance and is assigned to J_{res}.

Step 3: lifted/generated data become observable coordinates. If an edge rule depends on data outside $\mathcal{F}_I$, then those data are either generated by a budget-admissible computation from public history or are presentation-inaccessible, oracle-supplied, or over budget. In the generated case replace X_I by the lifted space $\Omega = X_I \times D$, where D is the finite generated transcript or declared-readout component. The datum is now a coordinate observable on Ω, so Step 2 applies on the lifted space. If D contains target information that is neither primitive public data nor generated by the charged derivation just described, then its use is presentation-inaccessible, oracle-supplied, nonuniform, or over budget. Target information actually derived from public data by that budget-admissible computation remains in the lifted/generated case and is charged at the cost of the derivation.

Step 4: conditional outcomes give positive mixing. At a complete operational state, the induced kernel is the nonnegative sum of its represented outcome transitions. The positive cone and the finite structural subspaces are closed under finite nonnegative sums. Mixing typed edges therefore preserves the channel subspaces and their residual projection.

Step 5: finite composition is lifted state evolution. A finite computation $g_{t+1} = F_t(g_t, \phi_1, \ldots, \phi_k, D, \xi_t)$ is represented by its outcome/terminal transcript together with the current native field g_t, whose peak capacity is charged. Intermediate native fields are not automatically observable coordinates. If the final field supplies an exposed budget-admissible structural potential, only the authorized Hodge projection of the actual selected current is a Caus operator component; an auxiliary potential may instead support a Geom descent certificate. If the final field supplies represented level sets, they are a candidate quotient. They enter Abs only after terminal compatibility, exact descent, and nonzero utility are verified; otherwise the partial-lumpability and residual calculus applies. If the computation uses a family, stack, table, or interface, that organization is Lift only when the lift is observable, charged, and target-fraction preserving. A stopped-future functional produced by budget-admissible finite-horizon computation is charged as such a lifted observable when it satisfies the channel conditions; an uncomputed semantic future of the same operator has no such representation. If the computation supplies none of these structural coordinates, or if the proposed lift enriches the target fraction without being charged as operator information, its nonlocal movement is projected to J_{res} or classified as inadmissible oracle data. These alternatives exhaust the finite transcript coordinates and declared native-state effects.

□

Proposition II.8.3 (Reclassification of family-uniform residual alignment)**.** *Let (I_n) be a task family and let $J_{\mathrm{res},I}$ be the residual component of a budget-admissible generator after the five exposed structural projections. Suppose a family-uniform budget-admissible relation, potential, quotient, lifted interface, terminal readout, or finite-horizon trajectory statistic is generated from the public presentations, including through presentation-admissible oracle evaluators, and is verified to satisfy the defining predicate of one of the five channel subspaces. Then its projection onto that subspace is reclassified into one of* Geom*,* Caus*,* Abs*,* Lift*, or* Bdry*. An uncomputed committor, first-hitting-time partition, occupation law, or stopped-future functional of the residual operator is a witness only when it also has a budget-admissible representation, including one computed at a budget-admissible finite horizon. Consequently a component that remains J_{res} has no family-uniform budget-admissible one-step structural representative. If the full process has positive excess over the declared neutral baseline, Theorem III.21.3 exposes the represented pre-hit sampler carrying that excess. At saturated cost, an affordable target-aligned source in its ancestry is reclassified into an exposed channel; otherwise the purported source is over budget, nonuniform advice, presentation-inaccessible, or an inadmissible oracle enrichment. The completed first-hit quotient remains an accounting record of the full process.*

Proof. By Part I's family-uniformity clause, a target-aligned construction that works across the family by one public rule is an observable family invariant, provided the construction has a budgeted channel representative in the sense of Definition Budgeted channel representatives. If the rule is produced by a budget-admissible computation, after expanding every presentation-admissible oracle call by its represented evaluator, its records are coordinates in the lifted observable configuration. Thus the aligned edge predicate, potential, quotient, lifted interface, or terminal readout is measurable in the presented or lifted observable algebra. The additional hypothesis that it satisfies a channel predicate, rather than measurability alone, makes it a structural representative.

The information-provenance theorem applies on that lifted observable space. An aligned edge effect with a verified budgeted channel representative has public or lifted/generated provenance, so the channel exhaustiveness theorem assigns it to the appropriate exposed component: local geometry, causal gradient, represented abstraction, valid lift, or boundary readout. The residual operator's own committor, first-hitting-time partition, occupation law, or stopped-future functional requires an admissible representation. If the same finite object is generated by budget-admissible computation and its induced operator effect satisfies a channel predicate, that structural projection is reclassified. Any orthogonal remainder stays residual. If no budget-admissible channel-qualified representative exists, any semantic target pairing has no admissible source representation at that budget. The operator term remains one of the two algebraic coordinates in Theorem II.7.2; it supplies no affordable target-aligned source in the saturated ledger. The prospective hazard identity accounts for neutral contact and exposes every affordable excess, while the completed first-hit quotient records the resulting arrival retrospectively.

□

Lemma II.8.4 (No represented structural channel witness remains residual)**.** *After the five exposed structural projections and the family-uniform reclassification step have been performed, J_{res} contains no family-uniform budget-admissible target-aligned* channel representative *lying in one of the five projected structural subspaces while remaining in J_{res}. Equivalently, if a represented edge effect, quotient, potential, structural Valid Lift, or boundary readout satisfies the defining predicate of an exposed channel, its orthogonal projection to that channel is removed from the residual.*

The algebraic projection and the saturated affordability statement are distinct. A semantic pairing of the algebraic remainder with the target has no algorithmic force at budget B without a family-uniform admissible representation. Once such a representation is generated affordably, the family-uniform reclassification and post-exhaustion irreducibility results assign its target-aligned component to an exposed channel. Thus the saturated residual contains no affordable target-aligned source. Neutral target contact is retained in the declared enumeration baseline, and every affordable excess is carried by the prospective sampler of Theorem III.21.3.

Proof. Let $S_1, \ldots, S_5$ be the five declared closed channel subspaces in the finite operator Hilbert space, orthogonalized in the order fixed by the channel definition, and let R be their orthogonal complement. The residual is $\Pi_R L$. If a represented structural witness gives a component $K \in S_j$, then $K \perp R$, so its S_j-projection is part of the exposed channel and cannot simultaneously be a nonzero component of $\Pi_R L$. This is precisely the asserted reclassification. This proves the algebraic assertion. The affordability assertion follows by applying Proposition II.8.3 and Lemma II.8.8 to every family-uniform represented target-aligned use at its least saturated cost. A denotation lacking such a representation is inactive at the declared budget.

□

Corollary II.8.5 (No post-saturation residual source alternative)**.** *Fix a budget B and form the saturated closure before taking the residual coordinate. Every family-uniform, budget-admissible represented mechanism invoked to explain target progress passes through the following exhaustive ordered accounting:*

(i) *every represented projection satisfying an exposed channel predicate is assigned to* Geom*,* Caus*,* Abs*,* Lift*, or* Bdry *at its complete saturated cost;*

(ii) *a* Caus*, valid-*Lift*, or* Bdry *use inherits the contextual charge of the represented* Geom*,* Abs*, or coupled source that it orients, transports, or reads;*

(iii) *the post-projection orthogonal component lies in J_{res} and supplies no prospective structural source charge.*

This ordered accounting is exhaustive. In particular, an affordable represented target-aligned use of an operator component cannot remain a residual source: its least-cost channel-qualified representative is removed from J_{res} and charged before the post-saturation residual is formed. Neutral target contact belongs to the declared enumeration baseline. Every affordable excess over that baseline is carried by the represented pre-hit selector and its charged source ancestry. A completed first-hit record remains an accounting record and creates no additional source alternative.

Proof. Channel exhaustiveness and the full residual normal form are Theorems II.7.1 and II.7.2. The no-creation identity for the three derived channels is Proposition II.5.9. Apply Proposition II.8.3 at the least saturated cost of every affordable represented target-aligned use. Its structural projection is assigned to the corresponding exposed channel, while the orthogonal complement remains one of the two residual operator coordinates. Lemma II.8.4 then gives zero prospective source status to that post-reclassification remainder. Finally, Theorem III.21.3 and Corollary III.23.1 assign neutral contact to the baseline, every nonbaseline term to a represented pre-hit selector, and the completed first-hit record to accounting. These operations cover every represented mechanism in the saturated closure.

□

Proposition II.8.6 (Distinction between provenance and degree)**.** *The Part II decomposition assigns each target-relevant edge effect a provenance and a transport type. Budget admissibility at* *degree d additionally requires a valid saturated representation of degree or effective cost at most d in the Part I filtration or in a lifted observable configuration generated from it.*

Consequently, $\mathcal{F}_I \otimes \mathcal{F}_I$-measurability records only extensional observability. Exposed provenance and placement in a budget-admissible cost band require a uniform represented derivation. Valid Lift may lower, preserve, or raise the raw coordinate degree of a target-relevant distinction; Part III charges the distinction to the least saturated representation budget at which it is represented. Below that budget the effect is not an exposed channel for the filtered problem. For a declared budget structure, a distinction requiring recovery, representation, or application above $b(n)$ is residual from the budgeted observable presentation, presentation-inaccessible, oracle-supplied, or structure for an enriched presentation. In the polynomial instance, this is the usual superpolynomial-recovery statement.

Proof. The information-provenance theorem is a finite represented decomposition: public data, uniformly lifted or generated data, and residual transport exhaust the possible sources of edge information. Degree is a separate filtration on represented observables and generated interfaces. Since a finite measurable function may distinguish many atoms, ambient measurability gives neither provenance nor a dimension bound. Conversely, a generated quotient or lifted interface can give a compact representation of a high raw-degree coordinate. Thus the channel assignment is exact at the level of source, while budget admissibility is determined later by the minimal saturated representation cost of the assigned source.

□

II.8.1 Post-Exhaustion Band: Irreducibility, Enumeration Complexity, and Composition Closure

The residual band records represented operator effects remaining after all declared structural projections.

Definition II.8.7 (Post-exhaustion residual band)**.** Let (I_n) be a task family equipped with the Part I observable filtration and the Part III saturated public cost budget B. For a budget-admissible generator L_n, first remove all components with exposed geometry, exposed structural gradients, represented quotients, valid lifted interfaces, and boundary readouts. Also remove the exact quotient part of every partially lumpable quotient and send its defect back through the same channel decomposition. The remaining component is $L_n^{J_{\mathrm{res}}}$.

The residual cost band above B, denoted $\mathcal{R}_{n,>B}(L)$, is the part of $L_n^{J_{\mathrm{res}}}$ whose target-aligned structural factor or terminal readout has no saturated admissible representation of public cost at most B: it does not factor through a base observable band, through a represented quotient, or through a valid lifted coordinate whose charged saturated cost is at most B. The raw transition implementation may be represented; what is absent below B is an admissible target-aligned structural representative. This band is defined after accounting for quotient, lifted-state, and boundary components. Membership is determined by saturated representation cost rather than measurability alone.

Lemma II.8.8 (Properties of the post-exhaustion residual band)**.** *Fix a budget-compatible resource envelope and a budgeted task family. After the five exposed structural channels have been removed, $\mathcal{R}_{n,>B}(L)$ has the following three properties.*

Irreducibility. *It admits no family-uniform budget-admissible observable reduction, including one using presentation-admissible oracle evaluators, that produces target alignment at the declared threshold. Any such reduction is a represented quotient, a structural potential, a valid lifted* *coordinate, a finite-horizon trajectory statistic generated by budget-admissible computation, or a boundary readout, and is therefore reclassified into* Abs*,* Caus*,* Lift*, or* Bdry *before the residual band is formed. A first-hitting partition or stopped-future value of the residual operator enters this classification precisely when it has a budget-admissible observable representative.*

Direct-enumeration lower bound. *Let $B_{n,d}$ be an independent degree-d coordinate band of dimension $N_{n,d}$. A direct coordinate scan exhausts $B_{n,d}$ only if the inspected coordinates span $B_{n,d}$; hence it requires at least $N_{n,d}$ coordinate tests. In the Boolean Walsh filtration, $N_{n,d} = \binom{n}{d}$ for the exact degree-d band. If $\binom{n}{d}$ is superpolynomial, no polynomial-cardinality direct scan exhausts that band.*

Closure under valid composition. *A finite composition of classified channel components and valid lifts either exposes a quotient, potential, lifted interface, or terminal value function and is charged to one of the five structural channels, or it leaves residual operator movement—nonlocal transport or directed cycles—in J_{res}. Target information outside the budgeted saturated observable closure is presentation-inaccessible, oracle-supplied, over budget, or part of an enriched presentation.*

Proof. For irreducibility, suppose a residual subcomponent above a cost level B has a family-uniform budget-admissible observable reduction r_n, after expanding every presentation-admissible oracle call by its represented evaluator, with target alignment at the declared threshold and with saturated representation cost at most B. If r_n is a map $X_n \to Q_n$ whose fibers and quotient operations are generated and manipulated within the declared budget, then it is a represented observable-quotient candidate once its fibers are generated by the public rule. Its exact intertwined projection has algebraic Abs provenance. It contributes to the Abs structural source profile only through the qualification conditions of Definitions III.15.1 and III.15.2. If the quotient is only partial, the identity $L_X U = U L_Q + E_q$ separates the exact Abs part from a defect that is decomposed again. If the rule produces a potential, it lies in the budgeted structural-gradient space and is Caus. If it exposes memory, tables, products, proof objects, interfaces, or finite-horizon trajectory statistics as represented output or retained public records, those readouts are part of the transcript observable algebra. Internal work state remains charged at peak native capacity and is not a free observable. The exposed readouts are Lift only when the valid-lift conditions and target-fraction preservation hold. If it produces terminal values, it is Bdry. The residual operator's own first-hitting partition, committor, or stopped-future table is included when generated by budget-admissible finite-horizon computation or represented by another admissible observable. These are precisely the cases in the information-provenance theorem and the family-uniform residual-reclassification proposition. Thus a family-uniform aligned reduction of cost at most d cannot remain in $\mathcal{R}_{n,>d}(L)$. If the least saturated representation cost is larger than d, the effect is not a channel below d; at that filtered level it remains in the residual band, is presentation-inaccessible, oracle-supplied, or enriched-presentation data, or is charged only when the budget is raised to its represented cost.

For direct enumeration, let $W \le B_{n,d}$ be the span of all coordinates inspected by the scan. Each coordinate test adds at most one dimension, so $\dim W \le |S_n|$, where S_n is the inspected set. If $|S_n| < N_{n,d} = \dim B_{n,d}$, then $W \ne B_{n,d}$, and there exists a nonzero coordinate direction in $B_{n,d} \cap W^{\perp}$ unseen by the scan. Therefore exhaustive scanning requires $|S_n| \ge N_{n,d}$. For Boolean Walsh characters the exact degree-d band consists of one independent character for each d-subset of variables, so $N_{n,d} = \binom{n}{d}$. Superpolynomial $\binom{n}{d}$ therefore exceeds every polynomial-cardinality scan. This establishes a dimension lower bound for direct enumeration.

For composition closure, use the finite insertion identity and interface expansion proved in the composition-closure lemma. Those identities decompose every composed edge into the same typed pieces: local conductance, directed structural gradient, quotient or valid-lift jump, boundary readout, and residual transport. If the composition creates a family-uniform aligned coordinate, the first paragraph reclassifies it into one of the five structural channels. If it creates no such coordinate, it remains residual. Hence valid composition remains within the five structural channels and preserves irreducibility of the post-exhaustion residual band.

□

II.9 Represented Quotient Geometry and Geom/Abs Sources

The residual-band analysis leaves the represented fiber algebra and the conditions for a geometric source certificate.

Lemma II.9.1 (Fiber algebra of a represented quotient)**.** *Let $q : X_I \to Q$ be a represented quotient, and let $Q_q \subseteq Q$ be its set of nonempty fibers. Its quotient-observable algebra is*

$$\mathscr{A}(q) := \{a \circ q : a : Q_q \to \mathbb{R}\} \subseteq \mathbb{R}^{X_I},$$

with pointwise addition and multiplication. If $k = |Q_q|$, then

$$\mathscr{A}(q) \cong \mathbb{R}^k$$

as a unital commutative real algebra. Its primitive idempotents are the indicators of the nonempty quotient fibers. In particular, $\mathscr{A}(q)$ is reduced, and both its nilradical and its Jacobson radical are zero.

Consequently, the abstract isomorphism type of the pointwise quotient algebra records the number of quotient cells but not the represented derivation that produced them. Construction provenance is carried by the quotient's uniform derivation record, represented fiber operations, and saturated cost rather than by an Artinian decomposition of $\mathscr{A}(q)$.

Proof. Enumerate the nonempty fibers as $C_i = q^{-1}(u_i)$, $1 \le i \le k$. The map

$$\mathscr{A}(q) \longrightarrow \mathbb{R}^k, \qquad a \circ q \longmapsto (a(u_1), \ldots, a(u_k)),$$

is a well-defined unital algebra isomorphism. Under this map the fiber indicators $\mathbf{1}_{C_i}$ are the standard coordinate idempotents. If $f^r = 0$ for some $r \ge 1$, every coordinate of the image of f has zero r-th power in $\mathbb{R}$, and hence $f = 0$. Thus the algebra is reduced and its nilradical is zero. A finite product of fields also has zero Jacobson radical. The remaining assertions follow. The displayed isomorphism depends only on the fiber partition; the derivation and its cost are additional represented data.

□

Lemma II.9.2 (Characterization of transfer-class geometric certificates)**.** *A pointwise observable $\phi : X_I \to \mathbb{R}$, together with its declared conductance and comparison data, is a transfer-class* Geom *certificate precisely when the following five conditions hold:*

1. ***Represented provenance.*** *ϕ is generated in the budgeted observable or lifted closure at the stated budget.*
2. ***Exposed locality.*** *The differences $\phi(y) - \phi(x)$ used by the source are supported on the exposed comparability relation E_I.*
3. ***Symmetric conductance.*** *The local conductance $c_E(x, y) = c_E(y, x)$ is exposed, so ϕ defines the symmetric Dirichlet form $\sum_{(x,y)\in E_I} c_E(x, y)(\phi(y) - \phi(x))^2$.*

4. ***Positive killed gap.*** *The killed local form has a represented positive Poincare/Cheeger gap toward the terminal sector at the stated budget [47, 51].*
5. ***Target reach.*** *The level-set algebra of ϕ has non-negligible projection on the centered target contrast.*

Conditions (1) and (2) specify provenance and locality. Condition (3) identifies the symmetric local source and separates Geom *from* Caus*. Conditions (4) and (5) impose the rate and alignment requirements.*

Proof. The symmetric local part of a positive generator is represented by an exposed conductance on E_I and pointwise potential differences. A source outside the represented closure lacks budget-admissible provenance; one using unexposed comparisons is outside the local Geom channel. Nonsymmetric orientation belongs to Caus. Without a positive killed-gap comparison the record does not supply the required transfer-class rate certificate; without target reach it does not supply the required alignment certificate. Conversely, conditions (1)–(3) construct the represented symmetric local channel, while conditions (4) and (5) provide exactly the rate and alignment margins required by the definition of a transfer-class certificate. This proves the characterization of certificate records. It does not assert that every symmetric local generator has a positive killed gap or non-negligible target reach.

□

II.9.1 The five Geom/Abs source configurations

The final phase combines exact quotient structure with ambient or quotient geometry through the compensated certificate.

Definition II.9.3 (Geometrically compensated quotient)**.** Let $q : \Omega \to Q$ be a represented terminal-compatible quotient. Write

$$U : H_Q \longrightarrow H_\Omega, \qquad P : H_\Omega \longrightarrow H_Q, \qquad PU = I_Q,$$

and let $L_Q = PL_\Omega U$. The lumpability defect is the operator

$$E_q = L_\Omega U - UL_Q : H_Q \longrightarrow H_\Omega.$$

A geometrically compensated quotient consists of q, represented Dirichlet data for the quotient modes and the within-fiber modes, and a charged comparison certificate showing that the coupling E_q is small relative to both forms. In the reversible case the form comparison is controlled by the Schur-complement parameter $\theta_\lambda(q) < 1$ of Definition III.12.1; target visibility additionally uses the target-functional margin $\vartheta^T_\lambda(q) < 1$ of Corollary III.12.3. In the general case it uses the target-functional normalized resolvent loss $\Delta^T_\lambda(q) < 1$ of Definition III.11.7. The reciprocal of the resulting positive stability factor $S_q(\lambda)$ must lie in the declared transfer class.

When the quotient is used for directed target progress, the same certificate also contains a represented potential and its Caus current. Their quantitative alignment and drift margin are those of Definition III.12.8 and Lemma III.12.10. Thus geometric compensation controls an operator between quotient and ambient spaces; it does not identify that operator with a single edge gradient.

The source classification enumerates the represented configurations available to the two structural channels.

Theorem II.9.4 (Classification of represented Geom/Abs source configurations)**.** *For any presented finite task family, every dynamically qualified source record in the declared channel calculus whose reusable target-bearing data are represented by* Geom*,* Abs*, or their composition has one of the following five configurations of the compensated quotient–geometry form:*

1. *represented ambient* Geom*, oriented by an authority-compatible* Caus *current and represented potential when it is used for directed progress;*
2. *exact represented* Abs*, corresponding to a null geometry slot;*
3. *represented* Geom *on an exact* Abs *quotient, with authority-compatible* Caus *providing target orientation;*
4. *represented* Geom *on a quasi-lumpable* Abs *quotient whose defect satisfies a charged reversible Schur-complement compensation bound, again with* Caus *providing target orientation under the declared authority envelope; and*
5. *represented* Geom *on a quasi-lumpable* Abs *quotient whose defect satisfies a charged general resolvent compensation bound.*

The last three cases combine the two source coordinates; they do not define an additional coordinate. Within this calculus the remaining channels are derived uses: Caus *supplies only the orientation authorized by the selected imposed/control generator,* Lift *reorganizes represented coordinates, and* Bdry *supplies a terminal readout. A quotient whose defect has no transfer-class compensation certificate supplies no quotient-to-ambient comparison at that budget. Its defect may instead be resolved by refining the quotient, representing a different geometry, or retaining the unclassified operator part in J_{res}. The theorem's domain is the represented, budget-admissible channel-certificate class of Definition II.9.3 and Lemma II.9.2; affordability is the charged derivation condition of that class.*

Proof. Component exhaustiveness leaves local symmetric, local oriented, nonlocal quotient, lifted, boundary, and residual pieces. The local symmetric piece is Geom, characterized by Lemma II.9.2. The represented quotient piece is Abs. Its pointwise fiber algebra is described by Lemma II.9.1, while its construction provenance is the resource-filtered derivation established in Lemma III.15.4. If no nonidentity quotient is used, identity support gives the first case. A nonidentity exact quotient with a null geometry slot gives the second case. If a represented geometry is constructed on a quotient, the intertwining defect $E_q = L_\Omega U - U L_Q$ separates the exact and quasi-lumpable cases. Exact intertwining gives the third case. For nonzero defect, the comparison is valid precisely when the defect is controlled relative to the quotient and within-fiber forms, as quantified in Section III.11. The reversible Schur-complement and general resolvent records give the fourth and fifth cases. The transformed-gradient decomposition then places the target-directed part in Caus, and Lemma II.4.5 restricts it to an authorized current of the selected generator. Valid Lift and Bdry remain charged uses of these sources, while every unrepresented remainder lies in J_{res}. These alternatives exhaust the dynamically qualified represented Geom/Abs source records specified in the statement.

□

Theorem II.9.5 (Exhaustive geometric normal form)**.** *Fix the symmetric local relation E_I, reference weight m, and conductance c_E of Definition II.3.3. Let*

$$\mathcal{E}_I^{\mathrm{un}} = \{\{x, y\} : x \neq y,\ (x, y) \in E_I\}$$

be the associated unordered edge set. For $e = \{x, y\}$, define the weighted edge Laplacian

$$(\Delta_e^{(m)} f)(z) = \begin{cases} m(x)^{-1}(f(y) - f(x)), & z = x, \\ m(y)^{-1}(f(x) - f(y)), & z = y, \\ 0, & z \notin \{x, y\}. \end{cases}$$

Then every algebraic geometric component has the unique edge-conductance normal form

$$L^{\mathsf{Geom}} = \sum_{e \in \mathcal{E}_I^{\mathrm{un}}} c_e \Delta_e^{(m)}, \qquad c_e = c_E(x, y) = c_E(y, x) \geq 0. \tag{22}$$

A target-bearing transfer-class certificate for this component satisfies the five necessary and sufficient conditions of Lemma II.9.2. After composing its represented quotient maps and retaining the complete derivation cost, its active geometry has one or more of the following exhaustive provenance configurations:

(i) *ambient represented geometry, including ambient geometry on a complete valid lifted carrier, with identity quotient;*
(ii) *represented geometry on an exact* Abs *quotient;*
(iii) *represented geometry on a quasi-lumpable* Abs *quotient with a charged reversible Schur-complement compensation record;*
(iv) *represented geometry on a quasi-lumpable* Abs *quotient with a charged general resolvent-compensation record.*

The last two provenance families may overlap when both represented compensation records are available. Selecting and charging one record gives the disjoint certificate tag used by Theorem III.18.9. Authorized Caus *orientation, valid* Lift*, and* Bdry *readout retain this geometric source and its carrier cost; they introduce no additional geometric configuration. At a budget B, a geometric record is exposed precisely when its relation, conductance, potential or comparison data, target-reach data, projection procedure, and output are represented at total saturated cost at most B. Otherwise its algebraic Geom tag is retained in the budget-filtered residual ledger of Theorem II.7.2.*

Proof. The symmetric local block in Lemma II.3.5 is

$$(L^{\mathsf{Sym}} f)(x) = \frac{1}{m(x)} \sum_{y:(x,y) \in E_I} c_E(x, y)(f(y) - f(x)).$$

Grouping the two orientations of each unordered pair gives (22). Conversely, the off-diagonal matrix entries recover $c_e = m(x)(L^{\mathsf{Geom}})_{xy} = m(y)(L^{\mathsf{Geom}})_{yx}$, so the coefficients and the edge sum are unique. By Proposition II.3.6 and Theorem II.7.1, every symmetric local operator term is in this block and no further local geometric primitive remains.

Lemma II.9.2 gives the necessary and sufficient represented-provenance, locality, conductance, killed-gap, and target-reach conditions. Apply Theorem II.9.4 to any record satisfying them. Its five source configurations consist of ambient Geom, null-geometry exact Abs, Geom on exact Abs, and the reversible and general compensated quasi-lumpable cases. Removing the null-geometry configuration, which has zero active Geom slot, leaves exactly the four configurations in the statement. The derived-channel proposition Proposition II.5.9 and the exact Lift identity Lemma II.10.3 preserve the source assignment under orientation, lifting, and terminal readout. The charged projection convention proves the final budget clause.

□

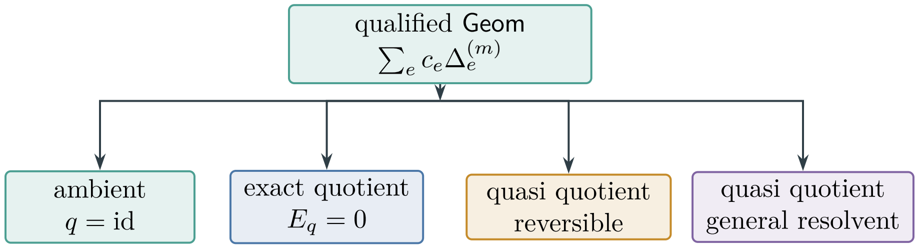

Figure 17: The four exhaustive, possibly overlapping active-Geom provenance configurations of Theorem II.9.5. Every configuration uses the same unique local edge-conductance normal form.

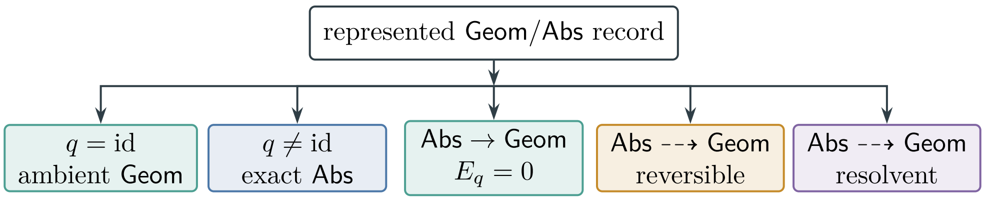

Figure 18: The compensated quotient–geometry certificate has five alternative configurations, classified by Theorem II.9.4.

Figure 17 displays the four active-geometry configurations of Theorem II.9.5.

Figure 18 compresses the five source configurations into their quotient and geometry slots.

Consequently, the five exposed structural channels together with J_{res} provide the complete source decomposition used in the subsequent quantitative analysis.

II.10 Stability Under Composition and Valid Lifting

Algorithms may augment the state space with stacks, products, proof trees, dynamic-programming tables, learned records, or subproblem interfaces. When this augmentation defines a valid lift of the observable task, the resulting generator remains within the same infinitesimal decomposition.

Proposition II.10.1 (Defect calculus for composed quotients)**.** *Let*

$$X \xrightarrow{q_1} Q_1 \xrightarrow{q_2} Q_2$$

be represented quotients with lifts U_1, U_2, projections P_1, P_2, and $P_i U_i = I$. Starting from $L_0 = L_X$, define

$$L_1 = P_1 L_0 U_1, \qquad L_2 = P_2 L_1 U_2,$$

and

$$E_1 = L_0 U_1 - U_1 L_1, \qquad E_2 = L_1 U_2 - U_2 L_2.$$

For $\sharp \in \{\mathsf{s}, \mathsf{a}\}$, let $L_i^\sharp$ be obtained by compressing the corresponding component at each stage, and set

$$E_1^\sharp = L_0^\sharp U_1 - U_1 L_1^\sharp, \qquad E_2^\sharp = L_1^\sharp U_2 - U_2 L_2^\sharp.$$

The composite lift is $U_{12} = U_1 U_2$, and its defect satisfies the exact identity

$$E_{q_2 \circ q_1} = E_1 U_2 + U_1 E_2. \tag{23}$$

Componentwise,

$$E^{\sharp}_{q_2 \circ q_1} = E^{\sharp}_1 U_2 + U_1 E^{\sharp}_2, \qquad \sharp \in \{\mathsf{s}, \mathsf{a}\}. \tag{24}$$

Consequently,

$$\|E_{q_2 \circ q_1}\| \le \|E_1\| \, \|U_2\| + \|U_1\| \, \|E_2\|.$$

For conditional-expectation projections the lifts are isometries, and the right-hand side is $\|E_1\| + \|E_2\|$. Exact lumpability therefore composes. The same estimate holds separately for the symmetric and causal defects. The saturated cost of the composite includes both quotient derivations, both represented-fiber interfaces, and the composition constructor.

Proof. Insert the two defect identities:

$$\begin{aligned} L_0 U_1 U_2 - U_1 U_2 L_2 &= (L_0 U_1 - U_1 L_1) U_2 + U_1 (L_1 U_2 - U_2 L_2) \\ &= E_1 U_2 + U_1 E_2. \end{aligned}$$

The norm estimate follows from the triangle inequality and submultiplicativity. The cost statement is the join-and-composition clause of the budgeted derivability closure. Replacing each L_i by L_i^{s} or L_i^{a} gives (24) by the same calculation.

□

The lift closure collects the represented observables generated by valid lift interfaces.

Definition II.10.2 (Lift closure)**.** Let $\mathcal{G}_0$ be the class of budget-admissible fixed-space generators decomposed into the five exposed structural channels plus residual accounting. The valid lift closure is the least class containing $\mathcal{G}_0$ and closed under finite direct sums, tensor products, product semigroups, bounded iteration, split/base/merge decompositions, and finite interface contraction along observable interfaces, subject at every stage to the valid-lift conditions $\pi^{-1}\mathcal{F}_{\mathrm{obs}} \subseteq \mathcal{G}_{\mathrm{obs}}$, $\pi_* \Lambda \mu = \mu$, and target-fraction preservation.

Lemma II.10.3 (Exact Lift identity)**.** *Let $\pi : \widetilde{X} \to X$ be a valid lift with lifted initialization $\Lambda\mu$. Set the lifted measure $\widetilde{\mu} := \Lambda\mu$, and let the target be $T \subseteq X$. Then*

$$\widetilde{\mu}(\pi^{-1}T) = \mu(T), \qquad \int_{\widetilde{X}} h \circ \pi \, d\widetilde{\mu} = \int_X h \, d\mu$$

for every base-measurable target-relevant observable h. These identities compose exactly under finite products, staged lifts, bounded iteration, and finite interface contractions that satisfy the valid-lift conditions. Hence Lift has zero target-fraction defect: it reorganizes represented source mass but does not create a new target source.

Proof. The identity $\pi_* \widetilde{\mu} = \mu$ is the valid-lift projection condition. Applying the definition of pushforward to 1_T gives $\widetilde{\mu}(\pi^{-1}T) = \mu(T)$. Applying it to a bounded base observable h gives the average identity. For a composition of valid lifts, pushforwards compose; induction gives the same identities for the composed projection. Interface contraction is valid only when the contracted interface has an observable positive representative preserving the same projection identities.

Therefore finite lifted composition preserves target fractions exactly and carries provenance back to the represented base source.

□

Lemma II.10.4 (Fiber disintegration of a finite valid lift). *Let $(\Omega, \mathcal{G}_{\mathrm{obs}}, \pi, \Lambda)$ be a finite valid lift of X_I in the sense of Definition II.5.2. For each $x \in X_I$, define the positive measure*

$$\kappa_x = \Lambda\delta_x.$$

Then κ_x is a probability measure supported on the fiber $\pi^{-1}(x)$, and every positive frontier has the unique fiber-kernel normal form

$$\Lambda\mu = \sum_{x \in X_I} \mu(x)\kappa_x, \qquad \operatorname{supp}\kappa_x \subseteq \pi^{-1}(x). \tag{25}$$

Let $U : \mathbb{R}^{X_I} \to \mathbb{R}^{\Omega}$ be pullback and define

$$(P_\kappa h)(x) = \sum_{\omega \in \pi^{-1}(x)} \kappa_x(\omega)h(\omega).$$

Then $P_\kappa U = I_{X_I}$.

Proof. The valid-lift identity gives $\pi_*\Lambda\delta_x = \delta_x$. Positivity implies that $\Lambda\delta_x$ vanishes outside $\pi^{-1}(x)$, and preservation of total mass gives $\kappa_x(\Omega) = 1$. Linearity of Λ on the point-mass basis gives (25), and evaluation on each δ_x proves uniqueness. Finally,

$$(P_\kappa U g)(x) = \sum_{\omega \in \pi^{-1}(x)} \kappa_x(\omega)g(\pi(\omega)) = g(x),$$

so $P_\kappa U = I_{X_I}$.

□

Theorem II.10.5 (Exhaustive finite-Lift normal form). *Let $(\Omega, \mathcal{G}_{\mathrm{obs}}, \pi, \Lambda)$ be a finite valid lift of X_I, and let κ_x, U, and P_κ be the unique fiber data of Lemma II.10.4.*

For every finite lifted generator L_Ω, put

$$L_X = P_\kappa L_\Omega U, \qquad E_\pi = L_\Omega U - U L_X, \qquad \Pi_\pi = U P_\kappa.$$

The lifted operator has the exact full-domain decomposition

$$L_\Omega = U L_X P_\kappa + E_\pi P_\kappa + L_\Omega(I - \Pi_\pi). \tag{26}$$

Apply the ordered channel projections to all three terms. The represented quotient-descending projection of $U L_X P_\kappa$ is Abs. *After the declared* Geom, Caus, *and* Abs *projections, the represented fiber/interface projection*

$$L^{\mathsf{Lift}} = \Pi_{\mathcal{S}_{\mathrm{Lift}} \cap \mathcal{S}_{\mathrm{Abs}}^{\perp}} L_{\mathrm{nl}} \tag{27}$$

is the complete Lift *component. Every part of $E_\pi P_\kappa + L_\Omega(I - \Pi_\pi)$ outside this subspace is assigned by the same* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Bdry}, J_{\mathrm{res}}$ *decomposition. Consequently every finite structural Lift has exactly the following normal-form data:*

(i) *the unique family of target-neutral fiber laws $(\kappa_x)_{x \in X_I}$;*

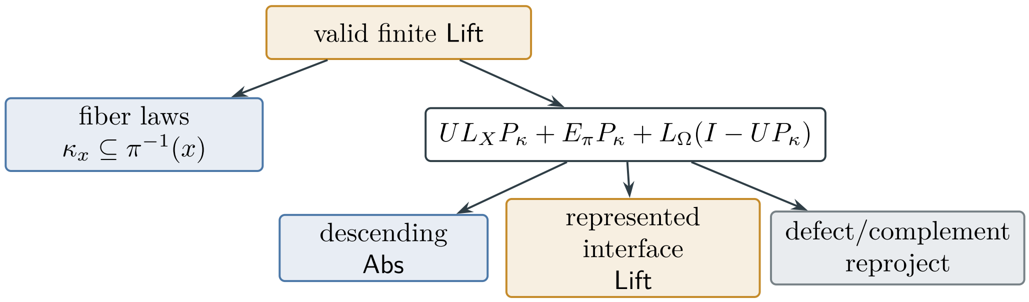

Figure 19: The exhaustive finite-Lift normal form of Theorem II.10.5. A valid initialization is a unique family of fiber laws; the lifted operator separates into its descending quotient block, represented Lift interface, and recursively decomposed defect/complement.

(ii) *the represented lifted interface and its orthogonalized* Lift*-channel projection* (27)*;*
(iii) *the quotient defect and within-fiber complement of* (26)*, recursively assigned by the existing channel decomposition.*

Finite products, staged lifts, bounded iteration, split/base/merge constructions, and represented interface contractions produce new fiber kernels of the same form and preserve the exact target-fraction identity. At budget B, the fiber laws, interface, projection, update rule, and output are included in the charged record. A structurally typed Lift whose complete record is unavailable at B is retained by the filtered-residual convention of Theorem II.7.2.

Proof. By Lemma II.10.4, $P_\kappa U = I_{X_I}$, so $\Pi_\pi = UP_\kappa$ is idempotent. Therefore

$$\begin{aligned} L_\Omega &= L_\Omega \Pi_\pi + L_\Omega(I - \Pi_\pi) \\ &= L_\Omega U P_\kappa + L_\Omega(I - \Pi_\pi) \\ &= UL_XP_\kappa + E_\pi P_\kappa + L_\Omega(I - \Pi_\pi), \end{aligned}$$

which is (26). This is Proposition II.6.2 for the represented projection π. The structured nonlocal split of Definition II.5.1 first projects onto $\mathcal{S}_{\mathrm{Abs}}$ and then onto $\mathcal{S}_{\mathrm{Lift}} \cap \mathcal{S}_{\mathrm{Abs}}^{\perp}$, proving (27). Component exhaustiveness assigns the orthogonal remainder, the defect, and the within-fiber complement without adding another Lift class.

For every named target $T \subseteq X_I$, fiber support gives

$$\kappa_x(\pi^{-1}T) = \mathbf{1}_T(x), \qquad (\Lambda\mu)(\pi^{-1}T) = \mu(T).$$

This is the exact identity of Lemma II.10.3. Pushforwards and fiber kernels compose under the constructors in Definition II.10.2, so the same normal form and target identity persist under finite valid composition. The structural Lift gate Definition II.5.6 supplies the represented-generation, reference-law, and cost conditions, and the charged projection convention gives the budget statement.

□

Figure 19 displays the fiber and operator layers of Theorem II.10.5.

Lemma II.10.6 (Closure of the channel decomposition under composition)**.** *If the observable target-directed structure of each factor lies in the five exposed structural channels and every lift is valid, then the same holds for the composed generator. Nonlocal movement without observable* *structural provenance remains residual unless the composition generates an exposed quotient or valid-lift coordinate in the enlarged observable space.*

Proof. For product spaces, infinitesimal composition has the insertion form

$$L_{X_1 \otimes \cdots \otimes X_k} = \sum_{i=1}^{k} I_{X_1} \otimes \cdots \otimes L_{X_i} \otimes \cdots \otimes I_{X_k}.$$

Each summand is the same generator type acting on one factor and identity on the others. Local conductance remains local conductance on its factor, directed current remains directed current, quotient structure remains quotient structure on a represented macro-coordinate, valid lifted organization is recorded as Lift, boundary readout remains boundary readout, and non-structural nonlocal movement remains residual. The projection maps of valid lifts compose, so the identities $\pi_* \Lambda\mu = \mu$ and $(\Lambda\mu)(\pi^{-1}T_I)/(\Lambda\mu)(\Omega) = \mu(T_I)/\mu(X_I)$ are preserved through finite products and finite staged lifts.

For a finite interface $X = A \sqcup I$, eliminating the interface through a declared positive interface inverse produces

$$L_{\mathrm{eff}}(\lambda) = L_{AA} + L_{AI}(\lambda I - L_{II})^{-1} L_{IA}.$$

The positive inverse and the resulting Schur-complement matrix are treated as one represented interface-contraction constructor, with the inverse, precision, evaluation, and output table fully charged. A finite interface does not make its Neumann/path expansion finite when cycles are present. After the contraction is formed, apply the finite channel decomposition directly to its resulting rate table: local conductance, directed current, quotient or valid lifted jump, boundary readout, and residual operator transport are projected in the usual way. An interface containing an undeclared solver, presentation-inaccessible target pointer, or target-enriching lift that violates fraction preservation introduces oracle-supplied or over-budget information. Otherwise composition remains within the same component decomposition.

□

Corollary II.10.7 (Target-mass preservation under valid lifting)**.** *A valid Lift composes, exposes, routes, and extracts structure represented in its factors or observable interfaces while preserving target-relevant mass fractions under projection. Target enrichment outside those factors is presentation-inaccessible, oracle-supplied, over budget, or residual relative to the public presentation.*

Proof. For a valid lift $\pi : \Omega \to X_I$, the lifted target is $T_\Omega = \pi^{-1}(T_I)$ and the initialization lift satisfies $\pi_* \Lambda\mu = \mu$. Therefore $(\Lambda\mu)(T_\Omega) = \mu(T_I)$ and, for every base-measurable observable h, the lifted average of $h \circ \pi$ equals the base average of h. These identities persist through finite compositions of valid lifted interfaces. Generated coordinates may route represented Geom and Abs sources, but target fractions and target-aligned averages remain those of the pre-lift source. A lift changing the target fraction is invalid for the original presentation.

□

II.11 Operational Representation: Computations as Markov Kernels

II.11.1 Finite transcript kernels

The operator formulation applies to represented computations through their Markov representation. Under the resource envelope of Definition Budget-compatible resource envelope, the clocked outcome/terminal transcript carrier of Definition Full operational transcript representation is finite. Each transcript prefix determines the current native state, and one operational step defines a Markov or sub-Markov kernel on the finite transcript carrier.

The finite-envelope statement below is a budget-slice theorem, not a finiteness axiom for computation. Corollary II.11.3 represents an arbitrary unbounded or nonhalting DTM as one countable process; its finite kernels are exact stopped restrictions used for finite-budget accounting.

Lemma II.11.1 (Operational computation embedding)**.** *Fix an input instance and a finite resource envelope with horizon H. Any represented operational computation embeds as a finite Markov or sub-Markov kernel $P^{[H]}$ on the clocked transcript carrier $\Omega^{[H]}$ of Definition II.5.5; its legal transcript prefixes form an invariant subset from the public initialization. For any scalar $\rho > 0$,*

$$L^{[H]} = \rho(P^{[H]} - I)$$

is a positive continuous-time generator on $\Omega^{[H]}$. It is a represented generator when ρ is represented and charged; the canonical choice $\rho = 1$ is always available. The embedding depends only on the complete operational transition law. Its execution ledger charges the accumulated finite transcript and one peak native state; it does not enumerate or store the intervening native states.

Proof. The bounded horizon and padded finite record alphabets form a finite represented transcript carrier. For a legal prefix h, let $z(h)$ be the current native state obtained by the initializer and successor recursion. The transition from h appends an outcome a and its declared terminal/readout record with conditional weight $p(a \mid z(h))$. During execution the update is applied to the single live state $z(h)$; earlier native states are neither retained nor exposed. An isolated row evaluation reconstructs $z(h)$ by the same prefix recursion and charges its repeated updates; this affects the representation ledger, not the finite-carrier identity. Extend malformed and unused transcripts by the fixed inert failure rule from Definition II.5.5. This gives a total Markov or sub-Markov kernel $P^{[H]}$ without enumerating the native-state space, the reachable transcript subset, or an extensional transition table.

The matrix $L^{[H]} = \rho(P^{[H]} - I)$ has off-diagonal entries $L^{[H]}_{\xi\xi'} = \rho P^{[H]}_{\xi\xi'} \geq 0$ for $\xi \neq \xi'$. Its generator-form rates and killing are

$$q(\xi, \xi') = \rho P^{[H]}_{\xi\xi'} \quad (\xi \neq \xi'), \qquad \kappa(\xi) = \rho\Big(1 - \sum_{\xi'} P^{[H]}_{\xi\xi'}\Big) \geq 0.$$

Then $L^{[H]}_{\xi\xi} = \rho(P^{[H]}_{\xi\xi} - 1) = -\sum_{\xi' \neq \xi} q(\xi, \xi') - \kappa(\xi)$. Hence $L^{[H]}$ satisfies the generator form and the positive maximum principle. The continuous-time process generated by $\rho(P^{[H]} - I)$ performs the same clocked transcript transition chain at Poisson-distributed jump counts, so it represents the same finite-horizon step relation without adding a movement primitive.

□

Figure 20 shows the finite clocked carrier used to represent an arbitrary computation at horizon H.

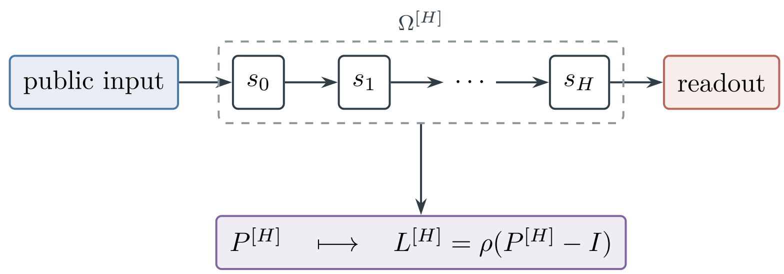

Figure 20: A bounded computation becomes a represented clocked transcript kernel and hence a positive generator by Lemma II.11.1.

II.11.2 Countable processes and finite stopping

The countable-process record connects an unbounded description to finite charged stopping filtrations.

Definition II.11.2 (Countably represented process and finite stopping filtration)**.** A countably represented process is a tuple

$$\mathfrak{P} = (\Omega, \mathcal{P}(\Omega), P, \mu_0),$$

where Ω is a countable set of uniformly finite-encoded states, P is a Markov or sub-Markov kernel specified by one represented transition rule, and μ_0 is a represented initial law. Put

$$R_H(\mu_0) = \bigcup_{t=0}^{H} \operatorname{supp}(\mu_0 P^t).$$

The process is *finitely stoppable* when every $R_H(\mu_0)$ is finite and, uniformly from the codes of P, μ_0, and H, the finite disjoint-time carrier

$$\Omega^{[H]} = \left(\bigsqcup_{t=0}^{H} R_t(\mu_0) \times \{t\} \right) \sqcup \{\dagger_H\}$$

and its stopped kernel are represented. For $t < H$, the transition from (x, t) is the pushforward of $P(x, \cdot)$ under $y \mapsto (y, t+1)$; states at time H enter the absorbing timeout state $\dagger_H$. A sub-Markov deficit is sent to the same state. The sets R_H are increasing. The clocked kernels need not be nested, but their path laws agree with the original process on every cylinder event determined before their deadlines. The countable tuple together with all of these stopped kernels is its *filtered process object*.

This definition is a representation layer, not a finiteness assumption on the underlying computation. Finite-state profile and decomposition theorems may be applied to the stopped kernels; a direct infinite-state analytic theorem requires its own operator-domain hypotheses.

Corollary II.11.3 (All Turing machines are represented)**.** *Let M be an arbitrary deterministic Turing machine, with no halting, time, space, reversibility, locality, lumpability, or regularity assumption beyond its finite transition table. Let Ω_M be the countable set of all finite configurations of M, with the discrete sigma-algebra. Extend the transition map F_M by fixing halting configurations, and define*

$$P_M(c, A) = \mathbf{1}_A(F_M(c)).$$

For every finite input I, with initial configuration c_I,

$$\mathfrak{P}_{M,I} = (\Omega_M, \mathcal{P}(\Omega_M), P_M, \delta_{c_I})$$

is a single countably represented, finitely stoppable process object. It represents the complete run, whether or not that run halts. Its finite stopped slice at horizon H reproduces the first H Turing transitions exactly and is a deterministic finite Markov kernel whose generator is the point-mass specialization of Lemma II.11.1.

Let $G_{M,I,H}$ instead denote the partial, unclocked transition graph containing the configurations and transition edges encountered before time H, without replacing its boundary edge by timeout. Then $G_{M,I,H} \subseteq G_{M,I,H+1}$, and their union is the complete reachable computation graph of M on I. Every finite prefix and every halting run is represented exactly in a sufficiently large slice. Taking the same exhaustion over all finite input encodings (or all finite starting configurations) recovers the ordinary countable configuration graph of M. The clocked timeout kernels themselves need not be nested, and no result below assumes that they are.

Consequently every DTM, including every unbounded or nonhalting DTM, is represented by the framework as one countable filtered object. No structural Valid-Lift, target, geometric, reversibility, or lumpability hypothesis is used for this representation. Point-mass kernels are a proper subclass of the permitted stochastic and sub-Markov kernels; continuous-time kernels and general represented positive generators add further operator objects. Thus the represented operator class strictly contains deterministic Turing evolution.

Proof. Every Turing configuration has a finite encoding, so Ω_M is countable. The finite transition table computes $F_M(c)$ uniformly from c, hence P_M is a represented point-mass kernel. From the point-mass initial law, at most one configuration is reached at each time, so $|R_H(\delta_{c_I})| \leq H + 1$. Definition II.11.2 therefore supplies a finite stopped kernel for every H, and its transitions before the deadline are exactly those of M.

Removing the artificial deadline edge gives the partial graph $G_{M,I,H}$. Increasing H only appends configurations and Turing-transition edges, so these partial graphs are nested. Their union contains exactly every configuration and edge reached after some finite number of steps on I. Every ordinary configuration has a finite encoding, and ranging also over finite starting configurations recovers the full countable Turing configuration graph. This proves both the finite-slice representation and its exhaustion statement.

Finally, on any represented carrier containing two states, a kernel having a row with two positive transition probabilities is permitted by the framework but is not a deterministic point-mass kernel. Hence the operator inclusion is proper; sub-Markov killing and general represented positive generators provide further permitted cases.

□

Remark II.11.4 (Operator domain does not restrict Turing representation). On $\ell^\infty(\Omega_M)$,

$$(P_M f)(c) = f(F_M(c)), \qquad L_M = P_M - I$$

defines, respectively, a positive contraction P_M and a bounded generator L_M with $\|L_M\| \leq 2$. Thus the complete countable computation is itself an operator object without an additional analytic assumption. The finite-dimensional L^2 projections, Hodge decompositions, spectral gaps, Schur complements, and matrix resolvents developed below apply unconditionally to every finite stopped slice. Applying a specifically Hilbert-space or spectral statement directly on the countable carrier requires a declared reference measure and the usual domain, closure, and limit hypotheses. Those hypotheses govern that analytic extension; they are not conditions for representing the Turing machine.

The operational representation retains every finite outcome, query, public-history, and terminal record in the transcript carrier. Memory, stacks, learned databases, priority queues, preprocessing state, actual parallel branches, and decomposition work remain in the live native state and are charged at peak capacity; a declared represented effect exposes any of them that is used as an observable. The channel decomposition applies to the induced transcript kernel rather than only to candidate assignments.

II.11.3 Extraction and structural admissibility

Proposition II.11.5 separates extraction of the represented operator from certification of its structure.

Proposition II.11.5 (Operator extraction is independent of algorithmic certification)**.** *Budget-admissible membership is a statement about a uniform operational transition rule without oracle access under the declared resource envelope. In the polynomial instance, the procedure supplies its code, native state bound, and resource bound. No quotient, invariant, proof of correctness, or explanation of target alignment is required. Once the transition rule and a finite horizon are fixed, the outcome/terminal transcript determines a Markov kernel $P^{[H]}$ and hence a generator $L^{[H]} = \rho(P^{[H]} - I)$. The channel decomposition is then applied to $L^{[H]}$. Any structural witness used in later bounds is a budget-admissible observable-structural representative of the induced operator, not part of the algorithm's membership in the declared budgeted class. A future-hitting statistic generated by budget-admissible finite-horizon computation is counted at that cost; an unrepresented semantic future of the operator is not a channel representative. In the polynomial instance, this is membership in the declared uniform polynomial operational class.*

Proof. A represented operational transition rule gives the conditional outcome law and native successor. On the finite clocked transcript carrier this yields the kernel $P^{[H]}$ in the embedding theorem. The finite generator theorem and channel decomposition depend on $L^{[H]}$ and the observable algebra of that carrier, independently of an explanatory annotation of the code. After embedding, every family-uniform target-aligned structure has a provenance component. It is task structure at the declared budget precisely when represented in the budgeted saturated observable closure below the ceiling. Otherwise it belongs to the residual component, presentation-inaccessible or oracle-supplied information, over-budget recovery, or an enriched presentation containing the missing datum. The polynomial case is the instance $\mathcal{B}_{\text{poly}}$.

□

Proposition II.11.6 (Operational embedding and structural budget admissibility)**.** *Every uniform resource-bounded computation without oracle access determines a finite generator on each bounded clocked transcript carrier. This embedding does not assign zero structural representation cost to the generator, transition graph, committor, first-hitting partition, or hitting-time function. A finite-horizon object generated by budget-admissible simulation is instead a lifted observable charged at the saturated cost of that computation.*

A target-relevant component of the induced generator becomes budget-admissible structure when the component is represented as one of the existing channel objects at the relevant saturated public *cost: a declared conductance or potential for* Geom/Caus*, represented lumpable fibers for* Abs*, a target-fraction-preserving valid interface for* Lift*, or an observable terminal value readout for* Bdry*. A budget-admissible finite-horizon simulation is one way to produce such a representative, and the resulting denotation is counted at the cost of that simulation. Naming an uncomputed infinite future, occupation law, or resolvent of the same generator does not define a constructor. A denotation available only through such an operation is outside the budget-admissible algebra of the original task unless the same object has a budget-admissible saturated representative.*

Proof. The embedding theorem records the complete transition rule in the finite transcript kernel $P^{[H]}$ and generator $L^{[H]}$. This construction uses only the next-step dynamics and the resource envelope, after the admissible instance-dependent data have been fixed by the budgeted saturated observable closure. It charges one peak native state and the accumulated finite transcript, rather than a sequence of native states. Saturated representation cost is the minimal description cost in that observable structural grammar for the component extracted from $L^{[H]}$, including the cost of any budget-admissible finite-horizon simulation used to generate it. Part III records provenance, execution, and structural representation as distinct readings inside the same budgeted closure. The embedding theorem establishes the induced operator and leaves the representation costs of associated denotations unchanged. A committor, first-hitting partition, or hitting-time table obtained by solving a global boundary problem for $L^{[H]}$ is an operator property. The same finite object is a budget-admissible datum exactly when generated by budget-admissible computation or otherwise represented by a budget-admissible quotient, potential, geometry, lifted interface, or boundary readout.

□

Proposition II.11.7 (Oracle accounting)**.** *An oracle or advice mechanism is external to the presented task. If an edge or observable supplied by such a mechanism is already $\mathcal{F}_I \otimes \mathcal{F}_I$-measurable and structurally explained, it belongs to one of the exposed structural channels. Extra target information outside the presentation is oracle-supplied and therefore outside the original budgeted presentation. Directed motion without exposed structural provenance lies in J_{res}.*

Proof. Admissibility is relative to the primitive presentation and its budgeted saturated closure. If an oracle-supplied relation or edge table is already measurable in the public product algebra and has a represented Geom, Caus, Abs, Lift, or Bdry provenance, the Part II projections place it in that channel. If the oracle supplies data not generated by the presentation, the transition rule uses oracle access and is therefore outside the budget-admissible rules for the original task; it is an oracle/advice-relative rule or a changed signature. Finally, if the supplied motion has no exposed structural provenance after the projections, the residual complement assigns it to J_{res}. The information-provenance theorem exhausts these alternatives.

□

Thus every presentation-relative budget-admissible algorithm without oracle access induces a budget-admissible generator on each finite clocked transcript carrier. Presentation-relative means that the instance-dependent data read by the transition rule lie in the budgeted saturated observable closure of the original task, and that any finite-horizon trajectory statistic used as structure is generated by budget-admissible computation and charged there. Every such generator decomposes into five algebraic structural channel slots and a residual. Their charged projections are the exposed channels. Remaining directed cycle current and nonlocal movement lacking exposed provenance are assigned to J_{res}. The decomposition derives provenance from the induced operator without requiring an algorithmic proof witness. Part III determines the saturated public cost at which each extracted component appears.

II.12 Operator-Level Universality

The decomposition is a property of operators on the observable algebra. An algorithm enters the analysis through the generator induced on its clocked transcript carrier.

Proposition II.12.1 (Universality of the operator decomposition)**.** *Universality follows from object exhaustiveness: every budget-admissible positive generator has exactly five algebraic channel slots and a residual. Only charged represented projections are exposed at a budget; remaining directed cycle current and nonlocal movement lacking observable structural provenance are recorded as J_{res}. Every resource-bounded represented computation without oracle access embeds as such a generator on each finite clocked transcript carrier. For a declared budget structure, budget-admissible means generated from the budgeted saturated observable closure of the presented task. Therefore every algorithmic use of budget-admissible task structure is represented by the channel decomposition. In particular, an algorithm uses the intrinsic problem geometry supplied by* Geom *and* Abs*, through causal orientation, valid lifting, and boundary/value readout. Structure outside the budgeted saturated closure lies outside the admissible data of the original presentation.*

Proof. By the operational embedding theorem, every resource-bounded represented computation without oracle access becomes a positive finite generator on its clocked outcome/terminal transcript carrier. The ledger separately charges the live internal state and memory at peak native capacity. The channel exhaustiveness theorem applies to every such generator and decomposes its movement into the five algebraic structural slots plus its residual component. At a budget, only charged represented projections are exposed. The classification of generative sources identifies Geom and Abs as the intrinsic sources, with exact and compensated quotient geometry as their joint modes. The channels Caus, Lift, and Bdry retain the sources they orient, reorganize, or read in the complete carrier ledger; target charge moves between contextual extensions only by a represented transfer. Because the embedding includes every retained admissible instance-dependent record and every declared native-state effect in the observable space, every induced effect either changes the primitive signature, uses oracle or advice information, requires over-budget recovery, or appears in this decomposition. Universality therefore concerns the operator class rather than an enumeration of algorithms.

□

Once the observable algebra, budget-admissibility condition, positivity principle, and clocked transcript carrier are fixed, every algorithm in the stated class induces an operator already covered by the decomposition. A budget-admissible method without oracle access may change the exposed transcript/readout space, generate an exposed edge relation, compute a finite-horizon trajectory statistic, or select a different kernel using admissible instance data. Each case still produces a positive generator on the budgeted observable algebra. Intrinsic problem structure appears through Geom or Abs, its derived uses through Caus, valid Lift, or Bdry, and the remainder through J_{res}. A semantic value function on the configuration graph is budget-admissible only when it has an admissible representation, including one produced by finite-horizon computation from the observable closure.

II.13 Examples: Dynamic Channels for SAT and Subset Sum

The static SAT and subset-sum presentations from Part I now illustrate the dynamic channel decomposition.

II.13.1 SAT

For a CNF formula Φ, the state space is $X_\Phi = \{0,1\}^n$. A one-bit Hamming relation is an exposed local relation when the presentation declares bit flips. Symmetric one-bit rates belong to the geometric/reversible channel Geom. A directed local imbalance belongs to Caus when it descends an exposed structural potential, such as a computable clause-defect or propagation potential. If the directed imbalance has no such provenance, it is J_{res}.

XOR-SAT supplies the clean abstraction witness. Public parity equations define a represented quotient

$$\pi(x) = Ax, \qquad \pi^{-1}(b) = \{x : Ax = b\}.$$

A jump or walk that descends to the quotient and satisfies the intertwining relation $L_X U = U L_Q$ belongs to Abs. Row reduction operates on the represented quotient.

Random planted 3-SAT illustrates the opposite case. The satisfying states exist, but a planted assignment outside the final formula's observable closure does not define a budget-admissible operator edge. A directed search path without a represented potential or quotient coordinate contributes to J_{res}.

II.13.2 Subset Sum

For subset sum, states are inclusion vectors $x \in \{0,1\}^n$, and the public defect observable is

$$D(x) = \left| \sum_i a_i x_i - t \right|.$$

One-bit inclusion moves belong to the geometric/reversible channel when declared local. A public residue map

$$r_M(x) = \sum_i a_i x_i \pmod{M}$$

gives a quotient/lumping channel when the modulus and residue fibers are exposed by the task presentation. A hidden modulus may hold in the ambient arithmetic, but it is budget-inadmissible unless made observable by the presentation or generated within budget in the lifted configuration. Otherwise it is presentation-inaccessible, oracle-supplied, over budget, or residual relative to the public presentation.

II.14 Summary of the Dynamic Decomposition

Part II represents dynamics as a search operator on frontiers. Under positive sub-Markov evolution, equivalently the positive maximum principle in finite dimension, this operator has generator form; under budget admissibility it is measurable with respect to the observable algebra and represented in the active budgeted closure. Relative to the local relation and structural projections, it decomposes as

$$\mathsf{Geom}, \qquad \mathsf{Caus}, \qquad \mathsf{Abs}, \qquad \mathsf{Lift}, \qquad \mathsf{Bdry}, \qquad J_{\mathrm{res}}.$$

Dependency trail. The dynamic objects and structural subspaces are Definitions II.1.1 to II.3.3 and II.5.6. Their information, residual, coupled-source, and lifting records are Definitions II.8.1 to II.9.3 and II.8.7. Generator and fixed-space accounting are established in Propositions II.3.2 and II.3.6 and Lemma II.3.5. The structural gates and complete lifted carrier are treated in Propositions II.6.2 and II.7.6 to II.5.9. The provenance and residual consequences are Theorem II.7.2, Corollaries II.7.3 and II.7.7, Propositions II.8.3 and II.8.6, and Lemmas II.8.4 and II.8.8. Composition, lifting, and the operational bridge are summarized by Lemmas II.14.1 and II.10.6, Corollary II.10.7, and Propositions II.12.1 and II.11.6. The interpretive qualifications concerning channel components and the countable DTM carrier are recorded in Remarks II.11.4 and II.7.5.

Lemma II.14.1 (Dynamical closure of Part II)**.** *Every budget-admissible positive search operator, including every represented computation without oracle access on its finite clocked transcript carrier, has exactly five algebraic structural channel slots:* Geom*,* Caus*,* Abs*,* Lift*, and* Bdry*. After those five projections are removed, the remaining movement is the residual component* J_{res}*. At a budget* B*, a slot is exposed only through a charged represented projection record; an unrepresented algebraic portion is retained in the filtered residual. The intrinsic sources of reusable problem geometry are* Geom *and* Abs*. Exact quotient structure is* Abs*; partial quotient structure is exact* Abs *plus a defect operator that is decomposed again. A quotient-level geometric comparison transfers to the ambient process only through the quantitative compensation criterion of Part III. The channels* Caus*, valid* Lift*, and* Bdry *orient, reorganize, or read out those intrinsic sources. The* Caus *projection is the one authorized by the selected imposed/control generator, and legal lifts preserve its current identity. Any local cycle current orthogonal to the exposed gradient space, or nonlocal movement without quotient, valid lift, or boundary-readout provenance, is residual transport. Composition and valid lifting preserve this classification.*

Proof. The positive maximum principle gives rates and diagonal loss. The fixed-space split separates local symmetric, local directed, and nonlocal rates, together with boundary processing. The structural projections identify local symmetric rates as geometric/reversible, directed flow as a structural gradient plus residual, and nonlocal structure as quotient/lumping or lifted-state when the observable algebra and target-fraction preservation supply that provenance, partial-lumpability defects by recursive component decomposition, and terminal observables as boundary value functions. Every remaining directed or nonlocal operator portion is J_{res} residual accounting. Its directed and nonlocal coordinates are exhausted by Theorem II.7.2. The budget-filtered assignment uses the charged projection convention of Theorems II.7.1 and II.7.2. The Hodge projection is taken from the selected imposed/control generator, so its uniqueness also gives the no-substitution property. The operational embedding theorem places algorithms into this same generator class, and the composition lemma prevents lifted constructions from adding another structural primitive. Hence the displayed decomposition is complete. □

Structured movement is generated by exposed geometry and represented abstraction. The directed/drift channel orients exposed potentials, the lifted-state channel reorganizes observable coordinates while preserving target fractions, and the boundary/killing channel maps exposed terminal observables to value functions, with termination or absorption as the zero-value case. Nonlocal motion without one of these provenance witnesses belongs to J_{res}. In particular, a restart without declared reversible structure is an ordered nonlocal jump in the residual component.

Part III introduces the quantitative arrival-time analysis and the saturated public cost at which each exposed structural channel becomes budget-admissible. Its structural quantities are defined on the five exposed channels, with J_{res} treated separately as the component without exposed structural provenance.

Part III

Spectral Complexity and Saturated Extraction Cost

III.0 Introduction: Discounted Hitting Functionals and Spectral Cost

Part contract. *Inputs:* the static presentation of Part I and the channel decomposition of Part II. *Output:* the saturated structural profile, exact charging identities, first-hit accounting, and universal accountability used in Parts IV and VIII. *First pass:* the Walsh specialization, continuum interface, and examples may be skimmed after the general profile results.

Part I specifies the static task, and Part II classifies its budget-admissible dynamics. This part introduces the corresponding quantitative theory of target-hitting times.

Part I defined the presented object

$$\mathcal{T}_I = (X_I, \mathrm{Obs}_I, \mathrm{Dyn}_I, \mathrm{End}_I)$$

and developed the observable algebra, terminal sectors, pointwise and represented-fiber observables, measurable and presentation-inaccessible structure, and static comparability. Part II proves that every budget-admissible evolution decomposes into five structural components and a residual term:

$$\mathsf{Geom}, \qquad \mathsf{Caus}, \qquad \mathsf{Abs}, \qquad \mathsf{Lift}, \qquad \mathsf{Bdry}, \qquad J_{\mathrm{res}}.$$

These components have distinct analytic roles. The intrinsic structure is encoded by the geometric or reversible component Geom and the quotient or lumping component Abs. The directed or drift component Caus, the lifted-state component Lift, and the boundary or killing component Bdry orient, reorganize, and evaluate that intrinsic structure. In particular, Bdry consists of terminal or support indicators and boundary values; it has no off-diagonal support and therefore induces no transport by itself. The residual term records transport without a representative in the exposed structural decomposition.

For a budget-admissible operator, the central quantity is the discounted mass reaching the positive terminal sector. The analysis is organized by the comparison

$$\boxed{\text{complexity} = \text{arrival time} = \text{target resolvent} \leq \text{saturated extraction at budget } B}$$

and its associated invariant is the least saturated public cost at which an exposed structural component produces non-negligible target transport.

The interaction of Geom and Abs is expressed by one compensated quotient–geometry certificate. Its five parameter regimes are ambient geometry, exact quotient structure with a null geometry slot, geometry on an exact quotient, reversible compensation of a nonzero quotient defect, and general resolvent compensation of a nonzero quotient defect. The identity quotient gives pure ambient Geom, while zero defect and a null slot give pure Abs. Caus supplies the represented descending current whenever the geometry slot is active. The saturated profile charges the complete certificate, including its quotient fibers, geometry, potential, comparison window, and defect loss.

III.1 Global Notation and the Saturated Closure

We begin by fixing the common finite-state notation and the budgeted closure in which all subsequent certificates are represented.

Definition III.1.1 (Global notation for Part III)**.** A family is indexed by a size parameter n. The finite state space is X_n, the reference measure is μ_n, and the positive terminal sector is

$$T_n = E^+_{I_n} \subseteq X_n, \qquad N_n = X_n \setminus T_n.$$

The centered target contrast is

$$y_n = \mathbf{1}_{T_n} - \mu_n(T_n), \qquad \bar{y}_n = \frac{y_n}{\|y_n\|_{L^2(\mu_n)}} \quad \text{when } 0 < \mu_n(T_n) < 1.$$

The ambient resource model is a budget structure $\mathcal{B} = (\mathcal{R}, \mathrm{cost}, b, \mathfrak{C})$: a resource space, a constructor cost, a size-indexed capacity functional $b(n)$, and a transfer class. A numeric margin has transfer-class size when it is bounded below by $1/\beta(n)$ for some $\beta \in \mathfrak{C}$, and it is negligible relative to $\mathfrak{C}$ when it is bounded above by $1/\beta(n)$ for every $\beta \in \mathfrak{C}$ eventually. For $\mathcal{B}_{\mathrm{poly}}$, these specialize to inverse-polynomial and negligible. A representative lies within $\mathcal{B}$ when its saturated description, construction, fiber-use, lift, comparison, boundary-readout, and finite-postprocessing charges lie below $b(n)$. The optional degree symbol d denotes one cost coordinate, often the log-cost level $d = \lceil \log_2 B \rceil$ for a numeric ceiling B, not raw Walsh degree.

The polynomial model is the instance $\mathcal{B}_{\mathrm{poly}}$ with $C = \mathbb{N}$, $\preceq = \leq$, $\oplus = +$, and $b_{\mathrm{poly}}(n) = n^{C_0}$, with the polynomial transfer class taking the union over constants C_0.

The dense Boolean Walsh band is one charged coordinate inside this profile. If r is raw Walsh degree, the dense scan/projection cost is

$$\mathrm{Cost}_W(n, r) = \dim V^W_{n,r} = \sum_{k \leq r} \binom{n}{k}.$$

Thus, under $\mathcal{B}_{\mathrm{poly}}$, a dense Walsh representation lies within budget precisely at fixed raw r for each fixed exponent in the polynomial budget instance, while a log-cost parameter satisfies the polynomial budget when $d = O(\log n)$. Under another budget, the admissible Walsh cutoff is the largest r with $\mathrm{Cost}_W(n, r) \preceq b(n)$. These are different coordinates in the same saturated cost profile.

The saturated represented closure collects all observables obtainable within the declared budget.

Definition III.1.2 (Budgeted saturated represented closure)**.** For a presented task family $(I)_{I \in \mathfrak{I}_n}$, let $\mathcal{R}^{\mathrm{sat}}_{n,\leq B}$ be the denotations produced by one family-uniform oracle-free represented construction of saturated public cost at most B. The available constructors include declared public observables, represented quotients and fibers, admissible public basis changes, budget-generated clocked transcript carriers and their declared readouts, structural Valid-Lift interfaces, boundary readouts, forward observable transport data, and finite represented postprocessings. Every denotation retains its complete construction record.

The ambient measurable envelope

$$\mathcal{F}^{\mathrm{env}}_{I,\leq B} = \sigma\left\{ R_I : R \in \mathcal{R}^{\mathrm{sat}}_{n,\leq B} \right\}$$

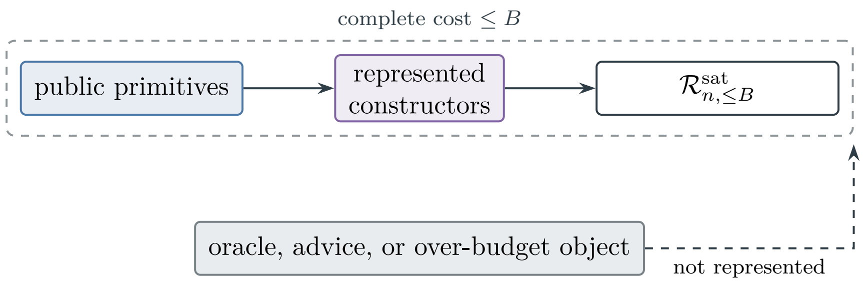

Figure 21: The saturated closure contains exactly the uniformly represented objects whose complete construction records fit the declared budget of Definition III.1.2.

supports conditional expectations and finite atom calculations. It is not the budget slice: an element of this envelope belongs to $\mathcal{R}^{\text{sat}}_{n,\le B}$ only when it has its own family-uniform construction record of total cost at most B. Constructor application therefore satisfies a filtered closure rule, with the input, operation, output, fiber, and postprocessing costs added to determine the output slice.

For a budget structure $\mathcal{B} = (\mathcal{R}, \text{cost}, b, \mathfrak{C})$, the within-budget represented closure is $\mathcal{R}^{\mathcal{B}}_n = \mathcal{R}^{\text{sat}}_{n,\le b(n)}$. It is the source of legal instance-dependent data for computation under $\mathcal{B}$. Legal finite-horizon iteration extends this closure only through a represented computation whose transcript, lifted coordinates, precision, and postprocessing are included in its accumulated charge. An ambient relation or semantic future object requiring over-budget construction, oracle access, nonuniform advice, an uncomputed resolvent, or an undeclared primitive has no within-budget representative for the original presentation.

The polynomial represented closure consists of uniform families $R = (R_I)$ for which one fixed exponent $C < \infty$ and one uniform derivation ρ satisfy

$$[\![\rho]\!]_I = R_I, \qquad \text{Cost}_I(\rho) \le (1+n)^C \quad (n \ge 1,\ I \in \mathfrak{I}_n).$$

Write $\mathcal{R}^{\text{poly}}_n$ for the size-n slice of these families. The exponent is uniform in n and in the instance.

Figure 21 distinguishes the represented budget slice from the larger ambient collection of semantic objects.

The derivability closure records the constructors and provenance available at a fixed budget.

Definition III.1.3 (Budgeted derivability closure: represented grammar)**.** The budgeted derivability closure consists of the represented grammar below, the uniform budget filtration of Definition III.1.4, and the admissibility criterion of Definition III.1.5. The present definition fixes the grammar and therefore retains the public label for the combined construction.

For the same presentation, budget structure, and resource envelope, let $\mathsf{Der}_{\mathfrak{I}}$ be the smallest class of family-uniform derivation schemes generated by the following clauses. For $\rho \in \mathsf{Der}_{\mathfrak{I}}$, write $[\![\rho]\!]_I$ for its denotation on I and $\text{Cost}_I(\rho)$ for its complete charged execution and representation cost.

(i) Local coding. The derivation coding is locally finite: the description alphabet is finite, the charged cost dominates encoded description length, and there are only finitely many uniform derivation descriptions of cost at most B for every finite B. Encoded coefficients, tables, and precision parameters contribute their full description lengths. This is a convention on the resource coding, independent of the presented task family.

(ii) Primitive records. Declared public observables, public parameters, terminal predicates, budget-generated transcript coordinates, finite outcome-generation interfaces, and the constructor signatures declared by the presentation are leaves of a derivation, with their public description and access costs.

(iii) Constructors. Derivations are closed under the admissible oracle-free constructors of Parts I–II: finite postprocessing, represented joins, quotient and fiber formation, public basis changes, valid lifts with $\pi_*\Lambda\mu = \mu$ and target-fraction preservation, boundary readouts through already derived structure, conductance and potential formation, forward observable transport data, channel projection data, and the finite comparison data required by the discounted hitting estimates. The charged cost of a constructor output is the sum of input costs plus the declared constructor charge for its arity, output description length, evaluation rule, represented fiber operations, lifted-interface size, retained coefficients or tables, and displayed comparison constants. For $\mathcal{B}_{\mathrm{poly}}$, admissible constructor records have polynomially bounded total charge.

(iv) Computation closure. If one uniform oracle-free procedure has all instance-dependent inputs represented by prior derivations and halts within the declared budget-compatible resource envelope, its finite output has the composite derivation obtained by adjoining the represented outcome/terminal transcript and terminal output record. Its full padded transcript carrier, initializer, operational update, terminal interface, and any public full-support product gauge are legal computation records independently of the structural Valid-Lift identities. The charged cost includes input access, time, peak native-state capacity, outcome generation, accumulated transcript, retained data, output representation, and postprocessing. This clause includes finite-horizon trajectory statistics, first-hit quotients, truncated value functions, represented propagation components, and target-contact statistics computed by legal simulation on the clocked transcript carrier with the current native state charged at peak capacity.

(v) Evaluator normalization. Every finite state-dependent primitive and evaluator in an oracle-free derivation has an explicit charged table or a uniform charged implementation over a fixed finite operational basis of encoded coordinate access, affine and product Boolean gates, comparisons and selection, represented group-action instructions, and clocked state or memory updates. The implementation record is part of the derivation. State-dependent access supplied without either representation belongs to the oracle-extended derivability closure. This clause normalizes an evaluator already present in a derivation; it does not add a semantic function merely because the operational basis is functionally complete. An explicit table retains its full encoded length, storage, and access cost. A compact circuit or other factorization is admitted only through its own represented derivation and is charged at its actual description, evaluation, storage, and retained-state cost.

Definition III.1.4 (Uniform derivations and budget filtration)**.** Fix the presentation, resource envelope, and represented grammar of Definition III.1.3.

(vi) Uniform provenance. The program producing the derivation is fixed across the family. Instance-dependent strings, coefficients, tables, or fiber labels must be generated from $\mathrm{enc}(I)$ by prior vertices of the same derivation. Changing the program or inserting a string separately for each instance is nonuniform advice rather than a public derivation. A fixed succinct uniform formula remains admissible at its actual description and execution cost; the framework does not charge the process by which that formula was discovered.

(vii) Minimality and budget slices. No other derivations belong to $\mathsf{Der}_{\mathfrak{J}}$. For a denoted family $R = (R_I)$, define its upward-closed set of admissible uniform budgets by

$$\mathfrak{B}_{\mathrm{sat},n}(R) := \{B : \ \exists \rho \in \mathsf{Der}_{\mathfrak{J}} \quad [\![\rho]\!]_I = R_I, \quad \mathrm{Cost}_I(\rho) \preceq B \quad (I \in \mathfrak{I}_n)\} .$$

The primary budget slice is therefore

$$\mathcal{D}_{n,\leq B} := \{R : B \in \mathfrak{B}_{\mathrm{sat},n}(R)\}.$$

When the resource order is partial, the saturated cost of R is the Pareto-minimal frontier of $\mathfrak{B}_{\mathrm{sat},n}(R)$, not a putative least vector. When a declared admissible scalar cost $\kappa : C \to [0, \infty]$ has been fixed, define

$$\mathrm{Cost}^{\kappa}_{\mathrm{sat},n}(R) = \inf_{\rho:\, [\![\rho]\!]_I = R_I \ \ \forall I \in \mathfrak{I}_n} \ \sup_{I \in \mathfrak{I}_n} \kappa(\mathrm{Cost}_I(\rho)) ,$$

and omit the superscript only in scalar-cost statements. Thus no infimum or supremum in the resource monoid is used to define a budget slice. The saturated represented class used below is identified with this closure:

$$\mathcal{R}^{\mathrm{sat}}_{n,\leq B} = \mathcal{D}_{n,\leq B}.$$

Here admissibility of κ means that it is monotone, dominates encoded description length up to a fixed class-preserving factor, and has finite description sublevels. These conditions prevent a scalarization from discarding an unbounded charged coordinate. A budget-indexed expression such as $\mathrm{Cost}_{\mathrm{sat}}(R) \preceq B$ always means $B \in \mathfrak{B}_{\mathrm{sat},n}(R)$, witnessed by one actual uniform derivation. The typography $\leq B$ is used only after a numeric scalarization has been declared.

Definition III.1.5 (Derivation admissibility criterion)**.** Use the grammar and budget filtration of Definitions III.1.3 and III.1.4.

(viii) Admissibility criterion. An object is admissible within the resource envelope precisely when it is the denotation of one uniform derivation whose cost is charged to that envelope. Ambient measurability and membership in a measurable envelope do not supply such a derivation. A datum without one is not admissible for the presentation under $\mathcal{B}$; its use requires presentation-inaccessible information, nonuniform advice, oracle access, over-budget recovery, or an enlarged primitive signature.

Lemma III.1.6 (Description length is a charged resource)**.** *Fix the locally finite derivation coding of Definition III.1.3 and an admissible scalarization κ from Definition III.1.4. There is a coding constant $c_{\mathrm{desc}} \geq 1$, independent of the presented family and size slice, such that every represented derivation ρ satisfies*

$$|\mathrm{enc}(\rho)| \leq c_{\mathrm{desc}} \left(1 + \sup_{I \in \mathfrak{I}_n} \kappa(\mathrm{Cost}_I(\rho)) \right) . \tag{28}$$

Consequently, a derivation below a polynomial scalar ceiling has polynomial encoded description length. A single family-uniform derivation has one fixed finite program code across all sizes. A size-dependent table or coefficient list is admissible only when that fixed program constructs it from the public presentation and its output, construction, storage, and evaluation costs fit the current ceiling. Inserting such data separately for each instance is nonuniform advice and belongs to an enriched presentation.

Proof. Local coding in Definition III.1.3 makes encoded description length a coordinate dominated by complete cost. Admissibility of κ in Definition III.1.4 preserves that domination up to one fixed class-preserving coding factor, which gives Eq. (28). The same two definitions charge generated tables, retained coefficients, output length, storage, and evaluation. Their uniform-provenance clause requires every instance-dependent string to be produced by an earlier vertex of the same fixed derivation. A separately inserted string therefore has no derivation in the original presentation and is classified by Definition III.1.5 and Proposition II.11.7. □

Theorem III.1.7 (Complete charging of explicit and implicit spectral aggregation)**.** *Let I be a presented instance, let X_I be a finite abelian carrier, and let $h_I : X_I \to \mathbb{C}$ be the denotation of a represented family-uniform derivation ρ_h. An orthogonal expansion*

$$h_I = \sum_{\gamma \in \widehat{X}_I} \widehat{h}_I(\gamma)\chi_\gamma$$

is an analytic identity. It creates no derivation vertex, coefficient oracle, locator, or prospective source. At every declared charged ceiling, the following alternatives exhaust the represented uses of this expansion.

(i) *If a derivation materializes a finite set $L_I \subseteq \widehat{X}_I$ of indices or coefficients, the complete record contains the rule constructing L_I, every coefficient evaluator, the represented output, its storage and precision, and every subsequent access. Under an admissible scalarization, the output and evaluation coordinates of its complete cost dominate $|L_I|$ up to the fixed coding constant.*

(ii) *If h_I is evaluated without materializing its formal expansion, then ρ_h is the compression record. Its circuit, factorization, recurrence, tensor rule, shared subexpressions, generated parameters, memory, precision, and evaluation cost are charged exactly as represented. The cardinality of the formal spectral support is not charged term by term unless those terms are individually constructed or accessed.*

(iii) *If a compact evaluator is used prospectively, the complete derivation additionally contains the represented comparison, index-selection, aggregation, potential, transition, attention, or active-acquisition rule that converts its output into the claimed target-directed use. Every instance-dependent field of that rule is generated from the declared public presentation by an ancestral derivation. A selection predicate involving a nonpublic target is unavailable unless an independent represented derivation supplies the same comparison, in which case that derivation and its complete cost are retained.*

(iv) *The complete prospective record in item* (iii) *is assigned by the existing structural gates: qualified nonidentity fibers enter an exact or compensated quotient mode; identity-supported qualified transport enters the ambient* $\mathsf{Geom}/\mathsf{Caus}$ *mode; and a readout with no represented authorized intermediate use is terminal accounting. The implicit spectrum is not an additional source modality.*

Consequently, an exponentially supported formal spectrum has two distinct budget interpretations. An explicitly accessed expansion pays its materialized cardinality. A compact implicit aggregation pays the complete cost of the structural compression rule and of its prospective consumer. Neither representation supplies free coefficient location or free target alignment.

Proof. Evaluator normalization in Definition III.1.3 admits either a charged table or a charged uniform implementation and retains the implementation record. Local coding and Lemma III.1.6 charge every materialized index, coefficient, output coordinate, precision field, and stored table. This proves item (i). The same evaluator-normalization clause charges a compact circuit or

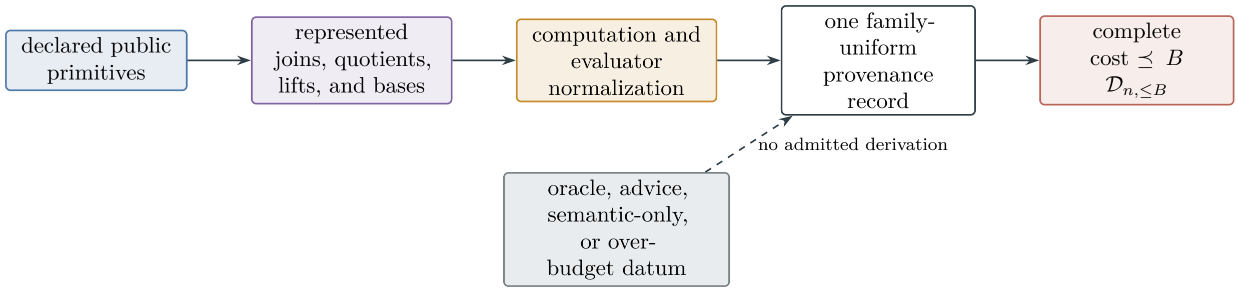

Figure 22: The derivability filtration is generated from declared primitives by represented constructors and charged computations. Family uniformity and the complete resource record are independent gates: semantic measurability without both gates does not place a denotation in the budget slice.

factorization at its actual representation and execution cost, rather than at the support size of an analytic basis expansion; this proves item (ii).

Computation closure and the dependency principle retain every ancestor of a prospective use. Uniform provenance requires all instance-dependent fields to be generated by the same family-uniform derivation. Temporal availability and target qualification are Definition III.21.1 and Theorem III.21.5; they prove item (iii). Finally, Theorem III.18.9 and Corollaries III.22.1 and III.3.7 give the exhaustive assignment in item (iv). □

Figure 22 records the distinction between a semantic object and a denotation admitted by the charged derivation grammar.

III.1.1 Resource-filtered derivation normal form

The derivation normal form exposes every constructor and retained datum in a budget-admissible representation.

Definition III.1.8 (Expanded resource-filtered derivation normal form)**.** Let $\rho \in \mathsf{Der}_{\mathfrak{J}}$. Its expanded normal form $\mathrm{Exp}(\rho)$ is the same uniform typed derivation graph with every computation-closure vertex displayed through the finite outcome/terminal transcript already charged by Definition III.1.3. Every vertex has one of the following presentation-independent forms:

1. a declared primitive or public parameter;
2. a finite represented composition, join, quotient/fiber constructor, or public change of coordinates;
3. a budget-generated transcript coordinate, elementary update, finite outcome-interface update, retained terminal record, or a separately certified structural Valid-Lift interface;
4. a represented conductance, potential, forward-transport, channel-projection, or comparison-data constructor; or
5. a boundary or terminal readout through previously derived data.

All instance-dependent inputs to a vertex occur at earlier vertices of the same graph. The normal-form cost is inherited from the original record:

$$\mathrm{Cost}_I^{\mathrm{nf}}(\mathrm{Exp}(\rho)) := \mathrm{Cost}_I(\rho).$$

Displaying a transcript therefore does not charge its data a second time. Let $\mathsf{Der}_{\mathfrak{J}}^{\mathrm{nf}}$ be the family of expanded records obtained in this way.

Lemma III.1.9 (Expanded derivation equivalence). *Expansion and contraction preserve uniform denotation and total saturated cost. More precisely, for every $\rho \in \mathsf{Der}_{\mathfrak{J}}$,*

$$[\![\mathrm{Exp}(\rho)]\!]_I = [\![\rho]\!]_I, \qquad \mathrm{Cost}_I^{\mathrm{nf}}(\mathrm{Exp}(\rho)) = \mathrm{Cost}_I(\rho)$$

on every instance. Conversely, contracting the displayed elementary transcript vertices of any $\widehat{\rho} \in \mathsf{Der}_{\mathfrak{J}}^{\mathrm{nf}}$ produces a derivation in $\mathsf{Der}_{\mathfrak{J}}$ with the same denotation and cost. Hence the budget slices of the two derivation systems have identical denotations.

Proof. Proceed by structural induction on the derivation graph. Primitive vertices agree. Each constructor vertex retains its inputs, output, evaluation rule, represented interface, and declared charge, so the induction hypothesis passes through every constructor clause. For a computation-closure vertex, Definition III.1.3 already includes the finite outcome/terminal transcript, peak native and work-state capacity, retained data, and postprocessing in the record and its charge. Expansion displays these existing entries without changing the induced map or adding a charge. Contraction reverses this bookkeeping operation. The uniform program and every instance-dependent dependency are unchanged, which preserves family uniformity and proves both directions. □

III.2 Five-Family Provenance for Represented Evaluators

Every normalized state-dependent evaluator instruction receives exhaustive constructor provenance. Exact quotient and active geometric records are the two structural applications used below.

Definition III.2.1 (Five-family evaluator provenance). Let

$$\mathfrak{J}^5_{\mathsf{Abs}} = \{\mathsf{Add}, \mathsf{Mult}, \mathsf{Ord}, \mathsf{Sym}, \mathsf{Filt}\}.$$

The five represented provenance families for state-dependent evaluator instructions are as follows.

1. Add contains encoded coordinate access, affine and additive combinations, characters, module operations, parity maps, and additive convolution.
2. Mult contains Boolean conjunction, products, composition gates, polynomial and ring operations, equality tests, and represented fiber formation.
3. Ord contains comparisons, branching and selection, weights, metric comparisons, quadratic comparisons, and ordered reduction rules.
4. Sym contains represented group actions, orbit maps, equivariant operations, averaging operators, and isotypic projections. The authenticated action record represents the action on the active public object and verifies equivariance of the state-dependent primitive interface used by the orbit or isotypic constructor. A represented permutation without this equivariance record retains its operational implementation provenance but does not acquire a Sym tag.
5. Filt contains clocks, memory updates, recurrence, bounded iteration, refinement, outcome histories, transcript coordinates, flags, and kernel towers. Nilpotent update rules are filtration provenance in this sense. This tag records the operational carrier and does not, by itself, authenticate an independent generative source. Its target charge is governed by filtration provenance charging below.

The predicates may overlap. A symmetry or order instruction may retain its additive and multiplicative implementation tags. Pure tuple, relabeling, and interface instructions expose their state-dependent arguments and inherit the union of their argument provenance. The same opcode assignment applies to the represented derivation of an instance-dependent coefficient when that coefficient is used by a structural source record. Public constants independent of both the represented state and the presented instance remain untagged coefficients. Thus a coefficient computed from public instance data retains the provenance and complete cost of its computation; declaring its output state-independent does not erase that ancestry.

For a normalized state-dependent vertex, or for a normalized instance-dependent coefficient vertex used by a structural source, write

$$\mathrm{Prov}_5(v) \subseteq \mathfrak{J}^5_{\mathsf{Abs}}$$

for its nonempty set of authenticated opcode tags. For a pure interface vertex, $\mathrm{Prov}_5(v)$ is the union of the tags of its exposed state-dependent arguments.

For an expanded derivation $\widehat{\rho}$, its five-family expansion $\mathrm{Exp}_5(\widehat{\rho})$ replaces every finite state-dependent evaluator by the charged operational implementation provided by evaluator normalization. Boolean gates use the affine and product basis over $\mathbb{F}_2$; finite-precision arithmetic, comparison, lookup, and memory access use their charged Boolean implementations. Represented symmetry instructions retain their authenticated action data. Bounded stateful composition retains its clock, work state, outcome record, and transcript under Filt.

The five-family normal form exhausts the represented provenance of structural source evaluators.

Theorem III.2.2 (Budgeted five-family evaluator-provenance normal form)**.** *For every finite task presentation under the budgeted derivability closure and every oracle-free derivation $\rho \in \mathsf{Der}_{\mathfrak{J}}$, the record $\mathrm{Exp}_5(\mathrm{Exp}(\rho))$ is a uniform derivation with the same denotation. Its cost is at most the class-preserving overhead fixed by the encoded operational model:*

$$\mathrm{Cost}_I(\mathrm{Exp}_5(\mathrm{Exp}(\rho))) \preceq \Psi_5(\mathrm{Cost}_I(\rho))$$

on every instance.

Every state-dependent instruction in this normal form has provenance in at least one member of $\mathfrak{J}^5_{\mathsf{Abs}}$, or is a pure interface exposing earlier tagged arguments. In particular, if the terminal denotation is a represented quotient q, its backward slice admits a represented joint frontier J_5 and a charged postprocessing h_5 such that

$$q = h_5 \circ J_5, \qquad \sigma(q) \subseteq \sigma(J_5).$$

Consequently, for every centered target contrast y,

$$\|P_q y\|_2^2 \le \|P_{J_5} y\|_2^2.$$

Every coordinate of J_5 is a declared public state-dependent primitive or the output of a prior charged represented constructor. Thus the quotient visibility is carried by the public derivation ancestry. The represented joint frontier retains the complete state-dependent and instance-coefficient ancestry with every applicable provenance tag.

The same charged normalization applies to budget-generated transcript and exact first-hit quotients already present in the saturated execution class. It neither constructs a new evaluator nor certifies target alignment.

Proof. Fix the represented derivation ρ. Evaluator normalization supplies a charged finite implementation of each state-dependent primitive and evaluator occurring in that derivation. Affine

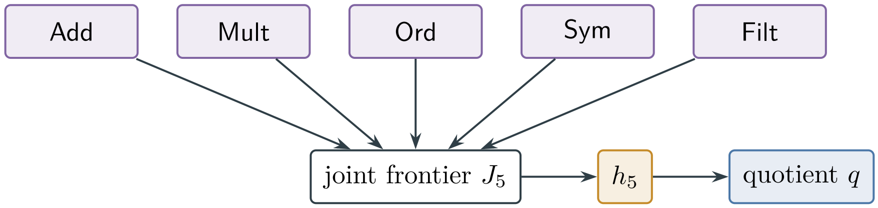

Figure 23: Every exact quotient factors through a charged joint frontier whose state-dependent instructions have the provenance classified by Theorem III.2.2.

gates and multiplication over $\mathbb{F}_2$ form a complete Boolean gate basis because conjunction is multiplication and negation is addition of the constant one. This is a coding fact about the implementation of ρ, not an admission rule for other Boolean maps. The finite encoded arithmetic, comparison, selection, or table lookup retained by ρ compiles to the basis with its complete implementation cost; in particular, an explicit table retains its encoded length and storage cost. The resulting gates have Add, Mult, or Ord provenance. Authenticated group-action instructions also carry Sym, and every stateful execution record carries Filt. The encoded implementation and evaluator-normalization clauses give the class-preserving cost map Ψ_5.

Expand interface instructions until their state-dependent arguments are exposed. The terminal backward slice is a finite derivation DAG, so its tagged state-dependent vertices form a finite represented joint frontier. Topological contraction of the remaining downstream record gives h_5 and the factorization $q = h_5 \circ J_5$. The sigma-algebra inclusion follows. The target-projection inequality is the affordable composition and target-projection theorem. Computation closure and the exact first-hit construction provide derivation records of the same form for finite-horizon algorithms.

□

Figure 23 shows the task-independent charged provenance frontier retained by a budgeted exact quotient evaluator.

III.2.1 Exhaustive source provenance for valid geometry

The geometric normal form classifies the carrier on which a valid geometry acts. The normalized derivation DAG classifies every represented source from which its active geometric data are generated.

Definition III.2.3 (Active Geom ancestry, charged cut, and provenance cells)**.** Let

$$\mathfrak{S}^4_{\mathsf{Geom}} = \{\mathsf{X}, \mathsf{Q}_{\mathrm{ex}}, \mathsf{Q}_{\mathrm{rev}}, \mathsf{Q}_{\mathrm{nr}}\}$$

denote the four active-geometry support tags of Theorem II.9.5: ambient identity support, exact quotient support, reversible compensated quotient support, and general-resolvent compensated quotient support.

Let $\mathscr{C}$ be a dynamically qualified represented source record in one of the four configurations of Theorem II.9.5, with active geometry and complete saturated cost at most B. Its *active Geom record* is the represented tuple containing:

(i) the normalized kernel, selected generator, reference gauge, and initialization in $\mathfrak{N}$;
(ii) the exposed relation and conductance of $\mathfrak{G}$;

(iii) every potential, local-difference evaluator, target-reach evaluator, killed-gap or mixing comparison, regularity comparison, and same-generator local-consistency record used by its visibility;

(iv) the selected current and its complete authority record when directed progress is invoked; and

(v) for nonidentity support, the quotient, represented fibers, quotient dynamics, defect ledger, target-functional comparison, and selected exact, reversible, or general stability record.

(vi) every represented qualification indicator used by the source. Such an indicator is either identically one for an already-qualified record or is retained as an active pre-hit gate together with its complete represented derivation and evaluation cost.

Write this complete tuple as $\mathrm{Act}_{\mathsf{Geom}}(\mathscr{C})$. Its entries are conjunctive validity data: a qualified geometry must pass every applicable gate. The classification below does not replace this joint record by any single gate.

First expand every presentation-admissible oracle interface into its charged family-uniform represented evaluator. Then apply $\mathrm{Exp}_5(\mathrm{Exp}(\rho_{\mathscr{C}}))$ to one complete derivation of this tuple. Adjoin a pure tuple-interface vertex whose arguments are precisely the active fields above, and take the backward slice of that vertex. After pure interfaces expose their arguments, let $\mathcal{A}_{\mathsf{Geom},5}(\mathscr{C})$ be the encoded-order list of every tagged non-interface vertex in this finite backward slice: every state-dependent vertex and every instance-dependent coefficient vertex whose value is an ancestor of an active field or gate. Thus the list contains both primitive leaves and the tagged internal constructors that combine them. A public constant independent of the instance is omitted. This is the *complete active Geom ancestral set.*

Let $t_{\mathscr{C}}$ be the adjoined active-tuple interface vertex and set

$$\Gamma_{\mathsf{Geom},5}(\mathscr{C}) = (t_{\mathscr{C}}), \qquad J_{\mathsf{Geom},5}(\mathscr{C}) = \mathrm{Act}_{\mathsf{Geom}}(\mathscr{C}).$$

This singleton is an antichain and meets every path terminating at the active-tuple vertex. It is the active-tuple analogue of the permitted singleton terminal cut in Definition III.15.5. The cut supplies factorization and cost accounting; the complete ancestral set supplies provenance. Keeping these two objects separate prevents an internal constructor from disappearing from the signature while preserving the antichain and separation properties required of a cut.

The *five-family source signature* of $\mathscr{C}$ is

$$\mathrm{Prov}_{\mathsf{Geom},5}(\mathscr{C}) = \bigcup_{v \in \mathcal{A}_{\mathsf{Geom},5}(\mathscr{C})} \mathrm{Prov}_5(v) \subseteq \mathfrak{I}^5_{\mathsf{Abs}}.$$

It is nonempty. Mixed source derivations retain every applicable tag.

For $\sigma \in \mathfrak{S}^4_{\mathsf{Geom}}$ and $c \in \mathfrak{I}^5_{\mathsf{Abs}}$, let

$$G^{\mathrm{src,sat}}_{\sigma,c,n}(B)$$

be the restriction of the existing prospective Geom profile to records whose selected master support tag is σ and whose source signature contains c, with supremum 0 for an empty class. These quantities are provenance restrictions of $G^{\mathrm{src,sat}}_n$; they introduce no new structural channel.

Let

$$\mathfrak{P}_{\mathsf{Geom},5} = 2^{\mathfrak{I}^5_{\mathsf{Abs}}} \setminus \{\varnothing\}$$

be the thirty-one nonempty exact provenance signatures. For $S \in \mathfrak{P}_{\mathsf{Geom},5}$, let

$$G^{\mathrm{src,sat},=}_{\sigma,S,n}(B)$$

be the restriction to records with selected support tag σ and canonical source signature exactly S. Exact-signature restrictions are pairwise disjoint. Their union is the complete normalized prospective Geom record family. The superscript $=$ distinguishes exact signatures from the overlapping single-tag restrictions above.

Theorem III.2.4 (Exhaustive five-family source classification of valid Geom)**.** *For every presented finite task family, every temporally admissible, dynamically qualified Geom source, including one using presentation-admissible oracle interfaces, has:*

(i) *exactly one selected support tag $\sigma \in \mathfrak{S}^4_{\mathsf{Geom}}$;*

(ii) *a nonempty source signature*

$$\mathrm{Prov}_{\mathsf{Geom},5}(\mathscr{C}) \subseteq \{\mathsf{Add}, \mathsf{Mult}, \mathsf{Ord}, \mathsf{Sym}, \mathsf{Filt}\};$$

(iii) *a charged factorization of its complete active tuple through its joint source frontier,*

$$\mathrm{Act}_{\mathsf{Geom}}(\mathscr{C}) = h_{\mathscr{C}} \circ J_{\mathsf{Geom},5}(\mathscr{C}), \tag{29}$$

where the downstream record $h_{\mathscr{C}}$ retains public constants independent of the instance, all active gate evaluations, and every comparison listed in Definition III.2.3; and

(iv) *a class-preserving complete-cost bound*

$$\mathrm{Cost}_I\big(\rho_{\Gamma_{\mathsf{Geom},5}(\mathscr{C})}\big) \preceq \Psi_{\mathsf{Geom},5}(B), \tag{30}$$

where $\Psi_{\mathsf{Geom},5}$ is the five-family expansion cost together with the charged represented join interface.

Consequently,

$$G^{\mathrm{src,sat}}_n(B) \le \max_{\substack{\sigma\in\mathfrak{S}^4_{\mathsf{Geom}}\\ c\in\mathfrak{J}^5_{\mathsf{Abs}}}} G^{\mathrm{src,sat}}_{\sigma,c,n}\big(\Psi_{\mathsf{Geom},5}(B)\big) \le G^{\mathrm{src,sat}}_n\big(\Psi_{\mathsf{Geom},5}(B)\big). \tag{31}$$

Thus every nonnegligible valid Geom source at a budget in the transfer class produces a nonnegligible record in at least one of the twenty support–provenance restrictions

$$\mathfrak{S}^4_{\mathsf{Geom}} \times \mathfrak{J}^5_{\mathsf{Abs}}.$$

The support coordinate is unique. The provenance restrictions may overlap, because a mixed derivation retains every applicable five-family tag. Negligibility of these twenty provenance-restricted envelopes throughout a class-preserving budget closure implies negligibility of the complete prospective Geom profile.

Authorized Caus *orientation does not add a source cell: its current and authority comparison occur in the active tuple and retain the source signature of that tuple. Valid* Lift *and* Bdry *records retain the same source assignment. A represented record failing the Geom qualification gates contributes zero to every cell, while its post-saturation gradient-orthogonal complement belongs to J_{res}.*

Proof. Theorem II.9.5 gives the four active-geometry support configurations. Selecting one exact, reversible, or general comparison record gives the disjoint master support tag of Theorem III.18.9; hence σ is unique.

Theorem III.33.2 expands every class-preserving oracle interface into its represented evaluator without changing the presented task family. Evaluator normalization and Lemma III.1.9 and Theorem III.2.2 expand every finite state-dependent evaluator in the complete active record without changing its denotation. Each exposed state-dependent vertex has at least one tag in $\mathfrak{J}^5_{\mathsf{Abs}}$.

The active tuple contains a state-dependent potential, conductance, kernel, current, quotient, or comparison whenever its visibility is nonzero, so its source signature is nonempty.

The active backward slice is a finite derivation DAG. Its terminal active-tuple vertex is a singleton separating antichain, and its represented output is $J_{\mathsf{Geom},5}(\mathscr{C}) = \mathrm{Act}_{\mathsf{Geom}}(\mathscr{C})$. Taking $h_{\mathscr{C}}$ to be the represented identity on that tuple proves Eq. (29). The complete ancestral set $\mathcal{A}_{\mathsf{Geom},5}(\mathscr{C})$, rather than the cut alone, retains the construction of every tagged internal vertex for provenance. The ancestral records and the downstream active tuple are subrecords of the original expanded derivation. Only the represented join interface adds a constructor charge. Evaluator normalization and computation closure therefore give Eq. (30), with a class-preserving $\Psi_{\mathsf{Geom},5}$. This is the same singleton-terminal cost argument as in Theorem III.15.6, applied to the represented active tuple rather than a quotient-map output.

Every qualified prospective Geom record of cost at most B is consequently present, with unchanged visibility, in at least one restricted cell at the enlarged cost. Taking its support tag and any member of its nonempty source signature proves the first inequality in Eq. (31). Each restricted envelope is a subfamily of the complete Geom profile at the same enlarged budget, proving the second inequality. The index set has $4 \cdot 5 = 20$ elements, so the negligibility conclusion follows from closure of the transfer class under finite maxima.

Finally, Proposition II.5.9 and Lemmas II.10.3 and II.4.5 preserve the source assignment under authorized orientation, valid lifting, and boundary readout. Component exhaustiveness and the residual normal form Theorems II.7.1 and II.7.2 assign the remaining gradient-orthogonal component to J_{res}.

□

Theorem III.2.5 (Canonical exact-signature decomposition of valid Geom)**.** *For every presented finite task family and saturated budget B, the canonical coefficient-complete ancestral set of Definition III.2.3 assigns every temporally admissible, dynamically qualified Geom source to exactly one cell*

$$(\sigma, S) \in \mathfrak{S}^4_{\mathsf{Geom}} \times \mathfrak{P}_{\mathsf{Geom},5}.$$

The support tag σ records ambient identity, exact quotient, reversible compensated quotient, or general-resolvent compensated quotient support. The exact signature S records all opcode families occurring in the active state-dependent inputs and in the represented derivations of the instance-dependent coefficients used by the active tuple. In particular, computing a center, weight, ordering, conductance, kernel parameter, current parameter, or authority comparison from public instance data cannot be hidden in an untagged coefficient.

At the class-preserving normalized ceiling $\widehat{B} = \Psi_{\mathsf{Geom},5}(B)$,

$$G^{\mathrm{src,sat}}_n(B) \le \max_{\substack{\sigma \in \mathfrak{S}^4_{\mathsf{Geom}} \\ S \in \mathfrak{P}_{\mathsf{Geom},5}}} G^{\mathrm{src,sat},=}_{\sigma,S,n}(\widehat{B}) \le G^{\mathrm{src,sat}}_n(\widehat{B}). \tag{32}$$

For the normalized record family at a fixed ceiling, the first comparison is an equality. Thus the Geom source family is indexed by $4(2^5 - 1) = 124$ support–provenance normal forms, with empty cells assigned value zero and at most 124 cells nonempty. Authorized Caus *is the gradient projection inside the selected Geom cell; its gradient-orthogonal complement is J_{res} and is absent from all 124 cells.*

Proof. The four support alternatives are disjoint by Theorems II.9.5 and III.18.9. Evaluator normalization assigns a nonempty subset of $\mathfrak{I}^5_{\mathsf{Abs}}$ to every tagged non-interface vertex of the coefficient-complete active ancestral DAG. Taking their union gives one and only one exact signature S. The active backward slice includes the represented derivation of every instance-dependent coefficient used in an active field or gate, so contraction leaves only instance-independent constants untagged. This proves existence and uniqueness of (σ, S).

Expansion preserves denotation and has cost at most $\widehat{B}$ by Lemma III.1.9, Theorem III.2.2, and Eq. (30). The exact-signature cells are disjoint and cover the normalized record family. A supremum over a finite disjoint union equals the maximum of the cell suprema, proving Eq. (32). The final current assignment is Proposition II.5.9 and Theorem II.7.2.

□

Theorem III.2.6 (No unclassified Geom source remains after quotient assignment)**.** *Fix a presented finite task family and a saturated ceiling B, and put $\widehat{B} = \Psi_{\mathsf{Geom},5}(B)$. At the normalized ceiling, the complete prospective Geom profile is the exact finite maximum*

$$G_n^{\mathrm{src,sat}}(\widehat{B}) = \max_{\substack{\sigma \in \mathfrak{S}^4_{\mathsf{Geom}} \\ S \in \mathfrak{P}_{\mathsf{Geom},5}}} G^{\mathrm{src,sat,=}}_{\sigma,S,n}(\widehat{B}). \tag{33}$$

Consequently its quotient-free remainder is exactly

$$G_{X,n}^{\mathrm{src,sat}}(\widehat{B}) = \max_{S \in \mathfrak{P}_{\mathsf{Geom},5}} G^{\mathrm{src,sat,=}}_{\mathsf{X},S,n}(\widehat{B}). \tag{34}$$

Every record on the right-hand side has identity support, a nonempty exact signature drawn from

$$\{\mathsf{Add}, \mathsf{Mult}, \mathsf{Ord}, \mathsf{Sym}, \mathsf{Filt}\},$$

and one coefficient-complete represented ancestral DAG containing the potential, geometry, transition rule, current, authority comparison, target readout, every instance-dependent coefficient, and their complete costs. Authorized Caus*, valid* Lift*, and* Bdry *operations inherit this same source cell and create no additional one. The gradient-orthogonal component is J_{res} and is absent from the prospective Geom family.*

Thus a represented computation contributes prospective geometric structure only through a quotient-supported exact or compensated record, or through one of the ambient identity-support cells in Eq. (34)*. There is no additional algorithmic, coefficient, mixed-operation, orientation, spectral, or postprocessing Geom source outside this decomposition.*

Proof. Theorem III.18.9 gives the four pairwise disjoint support tags: ambient identity support, exact quotient support, reversible compensated quotient support, and general-resolvent compensated quotient support. Theorem III.2.5 assigns every normalized qualified record exactly one nonempty coefficient-complete five-family signature S, including the derivations of all instance-dependent coefficients. Their product indexing gives the disjoint record classification whose profile form is Eq. (33). Restricting that classification to the ambient tag X gives Eq. (34).

Source inheritance under authorized orientation, lifting, and boundary readout is Proposition II.5.9 and Lemmas II.10.3 and II.4.5. The orthogonal assignment is Theorem II.7.2. These results exhaust support, coefficient provenance, temporal source status, and derived-channel status, which proves the final assertion.

□

III.3 State–Coefficient Indexing of the Saturated Source Profile

The master source normal form admits an exact finite reindexing by the provenance already present in each complete represented record. The indices below add neither a source channel nor an admissibility predicate or numerical visibility factor.

III.3.1 State–coefficient source coordinates

Definition III.3.1 (State–coefficient provenance and restricted source-profile notation)**.** This definition fixes the ancestral signatures and their profile cells. The same-record interaction coordinate attached to those cells is defined separately in Definition III.3.2.

Let $\mathscr{C}$ be a coefficient-complete normalized record of one of two temporal types: a prospective source in one of the five source configurations of Theorem II.9.4, subsequently formalized as the five master families in Definition III.18.4, or a completed terminal-compatible exact quotient retained only for accounting. Its *complete ancestral set* (with the historical notation) $\mathcal{A}_{\mathrm{src},5}(\mathscr{C})$ is defined as follows.

(i) For a pure exact-Abs source record or a completed exact-quotient accounting record, expand every presentation-admissible oracle interface and then apply $\mathrm{Exp}_5 \circ \mathrm{Exp}$ to the complete backward slice of the terminal quotient. After pure interfaces expose their arguments, define $\mathcal{A}_{\mathrm{src},5}(\mathscr{C})$ to be the encoded-order list of every tagged non-interface vertex in the complete ancestral DAG of that quotient, separated into state-dependent vertices and instance-dependent coefficient vertices.

(ii) For any of the four active-Geom records, set

$$\mathcal{A}_{\mathrm{src},5}(\mathscr{C}) = \mathcal{A}_{\mathsf{Geom},5}(\mathscr{C}),$$

with $\mathcal{A}_{\mathsf{Geom},5}$ as in Definition III.2.3.

The same syntactic ancestral-set construction applies to a completed terminal-compatible exact quotient retained only for accounting. In that use the notation $\mathcal{A}_{\mathrm{src},5}$ is historical: it records normalized ancestry and does not assign prospective-source status. The record's source/accounting temporal type is retained separately throughout.

In both cases, an instance-independent public constant is omitted, whereas the complete represented derivation of every instance-dependent coefficient is retained. The encoded order and the applicable master family make this ancestral set canonical for the normalized record. The applicable terminal singleton—the quotient-map vertex in the pure exact-Abs mode and the active-tuple vertex in an active-Geom mode—is the associated canonical charged cut. The terminal cut is used for factorization, whereas the complete ancestral set is used for provenance.

Split $\mathcal{A}_{\mathrm{src},5}(\mathscr{C})$ into its represented state-dependent vertices $\mathcal{A}^{\mathrm{st}}_{\mathrm{src},5}(\mathscr{C})$ and its represented instance-dependent coefficient vertices $\mathcal{A}^{\mathrm{cf}}_{\mathrm{src},5}(\mathscr{C})$. Define

$$S_{\mathrm{st}}(\mathscr{C}) = \bigcup_{v \in \mathcal{A}^{\mathrm{st}}_{\mathrm{src},5}(\mathscr{C})} \mathrm{Prov}_5(v), \tag{35}$$

$$S_{\mathrm{cf}}(\mathscr{C}) = \bigcup_{v \in \mathcal{A}^{\mathrm{cf}}_{\mathrm{src},5}(\mathscr{C})} \mathrm{Prov}_5(v). \tag{36}$$

The state signature is nonempty because a prospective structural-source record contains a state-dependent quotient or active geometric field, and that tagged vertex belongs to the complete ancestral DAG. The coefficient signature may be empty. In the active-Geom modes their union is the exact signature of Theorem III.2.5. For

$$\varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}, \qquad S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}},$$

write

$$G^{\mathrm{src,sat,=}}_{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}},n}(B)$$

for the restriction to records having precisely these two signatures and support tag σ. When a family functional $\mathfrak{M}_n$ is declared, write

$$G^{\mathrm{src,sat,=}}_{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}},\mathfrak{M},n}(B)$$

for the supremum of $\mathfrak{M}_n(I \mapsto Q_I V_{\mathscr{C},I})$ over the same refined Geom cell, with the qualification gate included in the complete source record as specified below.

Definition III.3.2 (State–coefficient structural-interaction coordinate)**.** Fix a budget B, a nonempty state signature $S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}$, a coefficient signature $S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}$, and a monotone sublinear positively homogeneous family functional $\mathfrak{M}_n$. The following symbol abbreviates the restriction of the existing prospective saturated source profile to one exact signature pair:

$$\begin{aligned} &\mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}(B) \\ &\quad := \sup_{(\rho,\mathscr{C},Q)} \mathfrak{M}_n\big(I \longmapsto Q_I V_{\mathscr{C},I}\big)\,. \end{aligned} \tag{37}$$

Here $\mathscr{C}$ ranges over the five temporally admissible prospective source modes, $Q_I \in \{0,1\}$ is its represented qualification indicator, and ρ is the complete family-uniform represented derivation of the qualified pair $(\mathscr{C}, Q)$, of saturated cost at most B. Either $Q \equiv 1$, or Q, its complete pre-hit derivation, storage, and evaluation are an active gate of $\mathscr{C}$. In the latter case every tagged ancestor of Q is adjoined to $\mathcal{A}_{\mathrm{src},5}(\mathscr{C})$ before the exact signatures are formed. Thus no qualification predicate is external to the charged source record. The complete normalized ancestral DAG of the qualified pair has exact state–coefficient signature $(S_{\mathrm{st}}, S_{\mathrm{cf}})$, and $V_{\mathscr{C},I}$ is the complete source visibility, including every dynamic, target-utility, compatibility, regularity, and temporal-source gate applicable to its master mode. The record ρ contains the derivation, storage, and evaluation of every instance-dependent coefficient used by that complete source. The supremum is zero when this represented class is empty. Thus StructAct is profile-restriction notation, not a new structural coordinate.

This coordinate retains the complete same-record source product. A public target verifier or target-discriminating readout alone therefore contributes no structural activity: it enters this coordinate only as part of a prospective source record that passes all applicable gates. A coefficient supplied without a family-uniform derivation belongs to advice or to an enlarged presentation. An over-budget coefficient is absent from the budget slice.

Theorem III.3.3 (Exact state–coefficient decomposition and coefficient accountability for Geom)**.** *For every presented finite task family and saturated budget B, every temporally admissible dynamically qualified Geom source has exactly one structural index*

$$(\sigma, S_{\mathrm{st}}, S_{\mathrm{cf}}), \qquad \sigma \in \mathfrak{S}^4_{\mathsf{Geom}}, \qquad \varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}, \qquad S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}.$$

At the coefficient-complete normalized ceiling $\widehat{B} = \Psi_{\mathsf{Geom},5}(B)$,

$$G^{\mathrm{src,sat}}_n(B) \le \max_{\substack{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}} \\ S_{\mathrm{st}}\neq\varnothing}} G^{\mathrm{src,sat,=}}_{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}},n}(\widehat{B}) \le G^{\mathrm{src,sat}}_n(\widehat{B}). \tag{38}$$

Let $G^{\mathrm{src,sat}}_{n,\mathrm{Norm}(B)}(\widehat{B})$ *denote the supremum over the exact image of the cost-* B *family under coefficient-complete normalization; a semicolon* $\mathrm{Norm}(B)$ *on a cell denotes the same restriction. Disjointness gives the literal equality*

$$G^{\mathrm{src,sat}}_{n,\mathrm{Norm}(B)}(\widehat{B}) = \max_{\substack{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}}\\ S_{\mathrm{st}}\neq\varnothing}} G^{\mathrm{src,sat,=}}_{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}},n;\mathrm{Norm}(B)}(\widehat{B}).$$

The inequalities in Eq. (38) *compare this exact normalized image with the enclosing ambient budget slices. Hence the normalized Geom family is indexed by*

$$4(2^5-1)2^5 = 3968$$

support–state-provenance–coefficient-provenance cells, with every empty cell assigned value zero; therefore at most 3968 cells are nonempty. Its coarser 124-cell decomposition is recovered exactly from

$$G^{\mathrm{src,sat,=}}_{\sigma,S,n}(B) = \max_{\substack{\varnothing\neq S_{\mathrm{st}}\subseteq\mathfrak{I}^5_{\mathsf{Abs}}\\ S_{\mathrm{cf}}\subseteq\mathfrak{I}^5_{\mathsf{Abs}}\\ S_{\mathrm{st}}\cup S_{\mathrm{cf}}=S}} G^{\mathrm{src,sat,=}}_{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}},n}(B). \tag{39}$$

Let $\mathfrak{M}_n$ *evaluate the same family of records. There is a class-preserving normalization overhead* Ψ_D *such that every refined cell satisfies*

$$G^{\mathrm{src,sat,=}}_{\sigma,S_{\mathrm{st}},S_{\mathrm{cf}},\mathfrak{M},n}(B) \leq \mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}(\Psi_D(B,n)). \tag{40}$$

Thus an instance-dependent coefficient enters a Geom source only through a represented affordable interaction inside the complete qualified source record. Its provenance, qualification event, full gate product, and complete cost remain in the right-hand coordinate.

Proof. The coefficient-complete ancestral set of Definition III.2.3 gives a disjoint partition of its vertices into state-dependent and instance-dependent coefficient vertices. Taking the union of the authenticated five-family tags in each part gives Eqs. (35) and (36). Both sets are uniquely determined by the normalized backward slice, and the state signature is nonempty. Together with the unique support tag from Theorem III.2.5, this proves the existence and uniqueness of the displayed structural index.

Expansion preserves denotation and has cost at most $\widehat{B}$. The refined indexed cells are disjoint and cover the normalized record family; an empty indexed cell has supremum zero. Hence a supremum over the family is the maximum of their suprema. This proves Eq. (38). For a fixed coarse signature S, the refined cells with $S_{\mathrm{st}} \cup S_{\mathrm{cf}} = S$ are again a finite disjoint cover, which proves Eq. (39). There are $2^5 - 1$ choices for the nonempty state signature and 2^5 choices for the coefficient signature; multiplication by the four support tags gives 3968 cells.

Let $\mathscr{C} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K})$ be a record in one refined cell and let Q_I be its qualification indicator. The normalized complete source record, including every active gate and every ancestral coefficient derivation, is one of the records in Eq. (37) at the class-preserving normalization overhead $\Psi_D(B,n)$. Its exact state–coefficient signature is unchanged. Apply the monotone family functional and take the supremum over the refined Geom cell to obtain Eq. (40).

□

III.3.2 Universal source decomposition

Theorem III.3.4 (Universal state–coefficient decomposition of affordable structural sources)**.** *Let*

$$\mathfrak{S}^5_{\mathrm{src}} = \{\mathsf{A}, \mathsf{X}, \mathsf{Q}_{\mathrm{ex}}, \mathsf{Q}_{\mathrm{rev}}, \mathsf{Q}_{\mathrm{nr}}\}$$

denote, respectively, pure exact Abs*, ambient* Geom*,* Geom *on an exact quotient, reversible compensated quotient–geometry, and general-resolvent compensated quotient–geometry. These are the five certificate families of Definition III.18.4.*

There are class-preserving normalization and complete-source overheads $\Psi_{\mathrm{src},5}$, Ψ_{read}, and Ψ_{all}, fixed by the encoded operational model. The first expands admissible oracle interfaces, performs the five-family evaluator normalization, and constructs the canonical ancestral set and its charged cut. The second retains the complete qualified source product at the normalized ceiling, and the third is a common class-preserving majorant of the finite compositions used below. Normalize every temporally admissible prospective structural source at $\widehat{B} = \Psi_{\mathrm{src},5}(B)$. For $\chi \in \mathfrak{S}^5_{\mathrm{src}}$ and $\varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}}$ and $S_{\mathrm{cf}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}}$, let

$$\mathfrak{U}^{\mathrm{src,sat,=}}_{\chi,S_{\mathrm{st}},S_{\mathrm{cf}},n}(\widehat{B})$$

be the restriction of the master source coordinate χ to records with the exact state–coefficient signature of Definition III.3.1. When a family functional $\mathfrak{M}_n$ is declared, write $\mathfrak{U}^{\mathrm{src,sat,=}}_{\chi,S_{\mathrm{st}},S_{\mathrm{cf}},\mathfrak{M},n}(B)$ for the supremum of $\mathfrak{M}_n(I \mapsto Q_I V_{\mathscr{C},I})$ over that same cell, where the qualification gate belongs to the complete charged pair as in Definition III.3.2, and use $\mathfrak{U}^{\mathrm{src,sat}}_{\mathfrak{M},n}$ and $I^{\mathrm{src,sat}}_{\mathfrak{M},n}$ for the corresponding master and combined evaluations. Then

$$\mathfrak{U}^{\mathrm{src,sat}}_{n}(B) \le \max_{\substack{\chi \in \mathfrak{S}^5_{\mathrm{src}} \\ \varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}} \\ S_{\mathrm{cf}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}}}} \mathfrak{U}^{\mathrm{src,sat,=}}_{\chi,S_{\mathrm{st}},S_{\mathrm{cf}},n}(\widehat{B}) \le \mathfrak{U}^{\mathrm{src,sat}}_{n}(\widehat{B}). \tag{41}$$

Let $\mathfrak{U}^{\mathrm{src,sat}}_{n,\mathrm{Norm}(B)}(\widehat{B})$ denote the supremum over the exact normalized image of the cost- B record family; a semicolon $\mathrm{Norm}(B)$ on a cell denotes the same restriction. The cell partition gives

$$\mathfrak{U}^{\mathrm{src,sat}}_{n,\mathrm{Norm}(B)}(\widehat{B}) = \max_{\substack{\chi, S_{\mathrm{st}}, S_{\mathrm{cf}} \\ S_{\mathrm{st}} \neq \varnothing}} \mathfrak{U}^{\mathrm{src,sat,=}}_{\chi,S_{\mathrm{st}},S_{\mathrm{cf}},n;\mathrm{Norm}(B)}(\widehat{B}).$$

The inequalities in Eq. (41) *compare this exact image with its enclosing ambient slices. Thus that normalized prospective source family is indexed by*

$$5(2^5 - 1)2^5 = 4960$$

source-mode–state-provenance–coefficient-provenance cells, with empty cells assigned value zero; hence at most 4960 cells are nonempty.

For every monotone sublinear positively homogeneous family functional $\mathfrak{M}_n$, every cell satisfies

$$\mathfrak{U}^{\mathrm{src,sat,=}}_{\chi,S_{\mathrm{st}},S_{\mathrm{cf}},\mathfrak{M},n}(B) \le \mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}\big(\Psi_{\mathrm{read}}(B,n)\big). \tag{42}$$

Consequently,

$$\mathfrak{U}^{\mathrm{src,sat}}_{\mathfrak{M},n}(B) \le \max_{\substack{\varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}} \\ S_{\mathrm{cf}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}}}} \mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}\big(\Psi_{\mathrm{all}}(B,n)\big), \tag{43}$$

$$I^{\mathrm{src,sat}}_{\mathfrak{M},n}(B) \le 2 \max_{\substack{\varnothing \ne S_{\mathrm{st}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}} \\ S_{\mathrm{cf}} \subseteq \mathfrak{I}^5_{\mathsf{Abs}}}} \mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}(\Psi_{\mathrm{all}}(B,n)). \tag{44}$$

Caus, *valid* Lift, *and* Bdry *retain the cell of the source record that they use. Their represented currents, lifted coordinates, and boundary readouts remain in its complete derivation. The post-saturation* J_{res} *coordinate contains no affordable target-bearing source and therefore contributes no cell to Eq.* (41)*.*

Proof. The five master certificate families form an exhaustive disjoint partition by Theorems II.9.4 and III.18.9 and the selected certificate record in Definition III.18.4. In particular, the classification is made by the displayed source carrier and stability record, not by the numerical value or eventual success of the certificate. In the pure exact-Abs family, Theorem III.33.2 first expands every presentation-admissible oracle interface, and the coefficient-complete evaluator expansion of Theorem III.2.2 then applies to the target-bearing quotient readout. The complete-ancestry expansion of Theorem III.3.3 applies to the other four families. The encoded backward slice and the state–coefficient type of each retained vertex are fixed by Definition III.3.1. Hence every normalized record has one unique state signature and one unique coefficient signature. The terminal quotient in the pure exact-Abs mode, and the complete active tuple in each active-Geom mode, contains a state-dependent source vertex. By Theorem III.2.2, that state-dependent vertex has at least one five-family tag, so $S_{\mathrm{st}} \ne \varnothing$. Intersecting the five master families with these disjoint signature pairs, and assigning value zero to every empty intersection, proves Eq. (41). The number of cells is the product of five source modes, $2^5 - 1$ nonempty state signatures, and 2^5 coefficient signatures. The oracle expansion, evaluator normalization, backward slice, and represented join have the declared class-preserving combined overhead $\Psi_{\mathrm{src},5}$.

For the quantitative domination, normalize a record in a fixed refined cell and retain its qualification indicator and complete source visibility. The resulting record belongs to the class defining Eq. (37) with the same exact state–coefficient signature. Application of $\mathfrak{M}_n$ and the supremum over the fixed refined cell prove Eq. (42). Taking the finite maximum and using the common overhead majorant Ψ_{all} gives Eq. (43). The source-profile comparison Eq. (184) gives Eq. (44).

Finally, Proposition II.5.9 and Lemma II.10.3 preserve source assignment under orientation, valid lifting, and boundary readout. The residual assertion is Theorem II.7.2 and Lemma II.8.4. □

III.3.3 Boolean-complete mixed ancestry

Theorem III.3.5 (Affine/product evaluator normalization and charge conservation in mixed ancestral cells)**.** *Fix a presented family with finite binary encodings and a saturated budget B. Consider a normalized prospective source record whose active state-dependent fields or instance-dependent coefficients are evaluated from the declared binary interfaces by a finite represented Boolean circuit. Evaluator normalization gives the same denotation at cost at most $\Psi_5(B)$ using affine gates and product gates over $\mathbb{F}_2$. The complete ancestral set of Definition III.3.1 assigns every active affine gate the* Add *tag and every active product gate the* Mult *tag.*

Consequently, the exact state–coefficient classification has the following structural alternatives.

(i) *A record whose complete active ancestry uses only affine gates and has no additional authenticated opcode lies in a pure-*Add *signature cell. If an authenticated non-Boolean opcode is also present, its tag remains and the full record lies in the corresponding mixed cell.*

(ii) *A record with an active product gate lies in a mixed signature cell containing* Mult*. If its declared state interface has* Add *provenance, then*

$$\mathsf{Add} \in S_{\mathrm{st}}, \qquad \mathsf{Mult} \in S_{\mathrm{st}} \cup S_{\mathrm{cf}}.$$

(iii) *Coordinate access, addition of the constant one, and multiplication implement XOR, negation, and conjunction. They therefore normalize the gate syntax of each finite Boolean circuit already carried by the represented record. The union is taken over the exact cells occupied by those budgeted records. Expressivity of the affine/product basis does not establish membership of an unrepresented map in a budgeted cell. Authenticated comparison, symmetry, and stateful execution retain their* Ord*,* Sym*, and* Filt *tags in addition to the Boolean implementation tags.*

For every exact signature pair, its numerical coordinate remains the complete same-record quantity

$$\mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}(B) = \sup_{(\rho,\mathscr{C},Q)} \mathfrak{M}_n(I \longmapsto Q_I V_{\mathscr{C},I})$$

from Eq. (37)*. Zero incremental contextual charge of an internal deterministic constructor preserves this complete coordinate by attributing the target projection to its charged joint ancestry; the qualification gate, source utility, and all geometric or quotient gates remain in* $Q_I V_{\mathscr{C},I}$*. Thus the mixed cells give the exhaustive operational provenance of the represented Boolean evaluator. Product gates record implementation and cost; they are not independent structural sources. The numerical value is the complete qualified source product in the existing master mode, and independent source attribution is performed by the charge-bearing core of Definition III.5.2. In particular, the gate identities neither construct a circuit nor place its denotation in the budget slice. Membership still requires the complete uniform derivation and cost prescribed by Definitions III.1.2 and III.1.3.*

Proof. By Lemma III.1.9 and Theorem III.2.2, evaluator normalization preserves denotation and retains the charged operational implementation. Over $\mathbb{F}_2$,

$$u \,\mathrm{XOR}\, v = u + v, \qquad \mathrm{NOT} u = u + 1, \qquad u \,\mathrm{AND}\, v = uv.$$

These identities show that the affine/product basis normalizes the already represented finite Boolean circuit. The first two gates have Add provenance and the third has Mult provenance. Because $\mathcal{A}_{\mathrm{src},5}(\mathscr{C})$ contains every tagged non-interface vertex in the complete normalized ancestral DAG, an active product gate and every additional authenticated opcode remain in the exact signature. By Definitions III.1.2 and III.1.3, the circuit occupies the budget slice only through the represented derivation from which normalization started, with its complete cost. This proves the three structural alternatives.

The final display is the definition in Eq. (37). The deterministic no-creation identity of Lemma III.17.1 attributes no incremental conditional projection to a postprocessing once its complete input algebra is exposed. The exact contextual decomposition of Corollary III.16.11 retains the earlier joint load, and the source definition retains all applicable gates in $Q_I V_{\mathscr{C},I}$. Hence incremental charge and the complete qualified source value obey the two distinct identities stated in the theorem. This proves the charge-conservation assertion.

□

III.3.4 Exhaustion and terminal-readout consequences

Corollary III.3.6 (Finite universal exhaustion criterion for affordable structural sources)**.** *Fix a transfer class, a monotone sublinear positively homogeneous family functional $\mathfrak{M}_n$, and a budget sequence B_n. Suppose a bound ε_n satisfies*

$$\mathrm{CoreAct}^{\mathrm{sat}}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}},n}(\Psi_{\mathrm{all}}(B_n,n)) \le \varepsilon_n$$

throughout the finite charge-bearing core index set, with empty cells assigned value zero. Then

$$\mathfrak{U}^{\mathrm{src,sat}}_{\mathfrak{M},n}(B_n) \le \varepsilon_n,$$
$$I^{\mathrm{src,sat}}_{\mathfrak{M},n}(B_n) \le 2\varepsilon_n.$$

In particular, a transfer-negligible common bound exhausts the affordable target-aligned prospective source algebra.

Proof. Apply Eq. (55) and then Eqs. (43) and (44). The finite maximum is over charge-bearing core pairs; operational decorators remain in the complete records defining those coordinates.

□

Corollary III.3.7 (Separation of terminal verification from prospective structural activity)**.** *Let $v_I : X_I \to \{0,1\}$ be a represented terminal verifier, and suppose that $\sigma(v_I)$ has nonzero target projection. The readout v_I alone contributes zero to every structural-interaction coordinate in Eq.* (37)*. If a complete temporally admissible source record $\mathscr{C}$ uses v_I, then its contribution is*

$$Q_I V_{\mathscr{C},I},$$

with every applicable source gate and the complete ancestral derivation of v_I retained in the unique state–coefficient cell of $\mathscr{C}$. In particular, target projection of a terminal verifier cannot replace dynamic utility, compatibility, authorized orientation, target reach, regularity, or temporal-source admissibility.

Proof. The defining supremum in Eq. (37) ranges over complete prospective source records and evaluates their qualified visibility, rather than the projection mass of an isolated readout. Hence an isolated verifier is outside that class. When it is an ancestor of a complete source record, Definition III.3.1 retains every tagged vertex in the normalized ancestral DAG of the qualified pair, including every nonconstant qualification gate, and the same definition evaluates the full product $Q_I V_{\mathscr{C},I}$. These are exactly the two asserted cases.

□

Corollary III.3.8 (An extremal readout requires represented intermediate transport)**.** *Let $D_I : X_I \to \mathbb{R}$ be a represented observable whose selected minimum or maximum contains T_I. Extremum consistency supplies a terminal-discrimination readout. It supplies prospective ambient* Geom *activity precisely through a complete temporally admissible record $\mathscr{C} = (\mathfrak{N}, \Phi, D_I, J, \mathfrak{K})$ that retains, before the transition in which it is used, the represented transition rule, its same-generator authority comparison, its authorized* Caus *alignment, its target reach, and its regularity comparison. The contribution is the complete qualified visibility $Q_I V_{\mathscr{C},I}$.*

If no such record is present at the charged ceiling, D_I is a terminal verifier and contributes zero to the prospective source profile. If the record is present, failure of any required gate sets its prospective value to zero, while a quantitative regularity loss is retained in the regularity factor of $V_{\mathscr{C},I}$. Thus knowledge that the target is an extremum cannot be substituted for represented control of the intermediate transitions leading to that extremum.

Proof. Extremum consistency is one of the terminal-readout alternatives in Definition I.5.1. By Corollary III.3.7, an isolated readout has zero prospective structural activity. When D_I belongs to a complete ambient record, Definitions III.14.9 and III.12.9 retain the transition, authority, alignment, reach, regularity, temporal, and cost gates in the same qualified visibility. The empty-record and failed-gate cases follow from the defining supremum and product conventions.

□

III.4 Quantitative Normal Forms for Exact Geom Cells

The universal decomposition specializes to the following same-record evaluations on the active-geometry branch.

Corollary III.4.1 (Cellwise quantitative form of the exact Geom decomposition)**.** *At the normalized ceiling of Theorem III.2.5, the complete prospective Geom profile satisfies the exact finite identity*

$$G_n^{\mathrm{src,sat}}(\widehat{B}) = \max_{S\in\mathfrak{P}_{\mathsf{Geom},5}} \max\Big\{G^{\mathrm{src,sat,=}}_{\mathsf{X},S,n}(\widehat{B}), G^{\mathrm{src,sat,=}}_{Q,\mathrm{ex},S,n}(\widehat{B}), G^{\mathrm{src,sat,=}}_{Q,\mathrm{rev},S,n}(\widehat{B}), G^{\mathrm{src,sat,=}}_{Q,\mathrm{nr},S,n}(\widehat{B})\Big\}. \tag{45}$$

For every exact signature S, the three quotient-supported coordinates obey

$$G^{\mathrm{src,sat,=}}_{Q,\mathrm{ex},S,n}(\widehat{B}) \le m_n^{\mathrm{src,sat}}(\widehat{B}), \tag{46}$$

$$G^{\mathrm{src,sat,=}}_{Q,\mathrm{rev},S,n}(\widehat{B}) \le G^{\mathrm{src,sat}}_{Q,\mathrm{rev},n}(\widehat{B}), \tag{47}$$

$$G^{\mathrm{src,sat,=}}_{Q,\mathrm{nr},S,n}(\widehat{B}) \le G^{\mathrm{src,sat}}_{Q,\mathrm{nr},n}(\widehat{B}). \tag{48}$$

The ambient cell has the exact same-record evaluation

$$G^{\mathrm{src,sat,=}}_{\mathsf{X},S,n}(\widehat{B}) = \sup_{\mathrm{Prov}_{\mathsf{Geom},5}(\mathfrak{g})=S} M_\Phi C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D) R_T(D) \frac{1}{1+\rho_\Phi(D)}, \tag{49}$$

where the supremum is over the temporally admissible normalized records at that ceiling and is zero when the cell is empty. Consequently it is bounded by each of the three corresponding cell-restricted suprema of $C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D)$, $R_T(D)$, and $(1+\rho_\Phi(D))^{-1}$.

Thus a family-specific Geom proof is complete precisely when it supplies a bound for every coordinate appearing in Eq. (45)*. A provenance label is not an additional source coordinate: zero-charge or inherited constructors may be contracted only through a theorem preserving the complete same-record visibility in Eq.* (49)*.*

Proof. The disjoint-union assertion of Theorem III.2.5 gives Eq. (45). The exact-quotient comparison is Theorem III.19.1; the other two support bounds are inclusions of exact-signature subfamilies into their master compensated coordinates. Restricting Eq. (190) to one exact signature gives Eq. (49), and the factorwise bounds follow because all displayed factors lie in $[0, 1]$.

□

Theorem III.4.2 (Reversible carriers and authenticated symmetry provenance)**.** *Let $\mathscr{C}$ be a dynamically qualified represented Geom source and let $\mathcal{A}_{\mathsf{Geom},5}(\mathscr{C})$ be its complete active ancestral set from Definition III.2.3. The conductance identity*

$$c_\Phi(x,z) = c_\Phi(z,x)$$

certifies the reversible Dirichlet carrier of $\mathscr{C}$. It contributes no Sym *provenance tag by itself.*

The source signature of $\mathscr{C}$ contains Sym *precisely when its complete active ancestral set contains a state-dependent symmetry constructor with the authenticated action and equivariance record required by Definition III.2.1. A data-dependent center, ordering, relabeling, potential, conductance, or current lacking that authenticated action retains the provenance of its represented construction and its complete cost, and receives no* Sym *tag.*

Suppose every authenticated public action occurring in the complete active ancestral set acts trivially on each state-dependent primitive interface used by the record. Then its orbit maps and averaging operations are identity-equivalent on the active carrier, and every isotypic output is either inherited from a differently tagged state-dependent argument or is state-independent. Hence the record contains no independent Sym*-derived Geom source and produces no nonidentity* Sym*-derived orbit quotient. Mixed records retain and are charged through their remaining provenance tags.*

Proof. Symmetry of c_Φ is the defining condition of the static Dirichlet form in Definition III.14.8; it is an edge-reversal identity and contains no group-action datum. The Sym tag is assigned by Definition III.2.1 only to a represented group action, orbit map, equivariant operation, averaging operator, or isotypic projection whose action on the active public object is authenticated. The complete active ancestral set in Definition III.2.3 exposes every state-dependent ancestor of the active tuple. These definitions prove the first two assertions and retain the construction cost through Eq. (30).

When every authenticated action is the identity on the active primitive interfaces, each orbit is a singleton and group averaging acts as the identity on state-dependent inputs. A nontrivial isotypic component then has no state-dependent input from the authenticated action; any remaining state dependence is inherited from another exposed argument and retains that argument's provenance tag. Pure interfaces preserve the union of those tags by Definition III.2.1. Thus the symmetry constructors add neither an independent active Geom datum nor a nonidentity orbit partition. The mixed-provenance conclusion follows from Theorem III.2.4.

□

III.5 Operational Provenance and Charge-Bearing Structural Cores

The exact provenance signature records the authenticated opcodes in the complete active ancestry. Source attribution is a separate operation. Constructors certified to create no new contextual charge retain their operational provenance and complete cost, while their source credit remains attached to the represented arguments from which they are derived.

III.5.1 Inherited decorators and charge-bearing cores

Definition III.5.1 (Certified inherited decorator)**.** Fix a coefficient-complete normalized record $\mathscr{C}$ to which the ancestral construction of Definition III.3.1 applies. It may be a prospective source record or a completed exact-quotient accounting record; its temporal type is unchanged. Write its complete ancestral set as $\mathcal{A}_{\mathrm{src},5}(\mathscr{C})$, and use the fixed encoded topological order of this finite DAG.

A tagged vertex $v \in \mathcal{A}_{\mathrm{src},5}(\mathscr{C})$ is a *certified inherited decorator* when its retained record supplies one of the following family-uniform certificates.

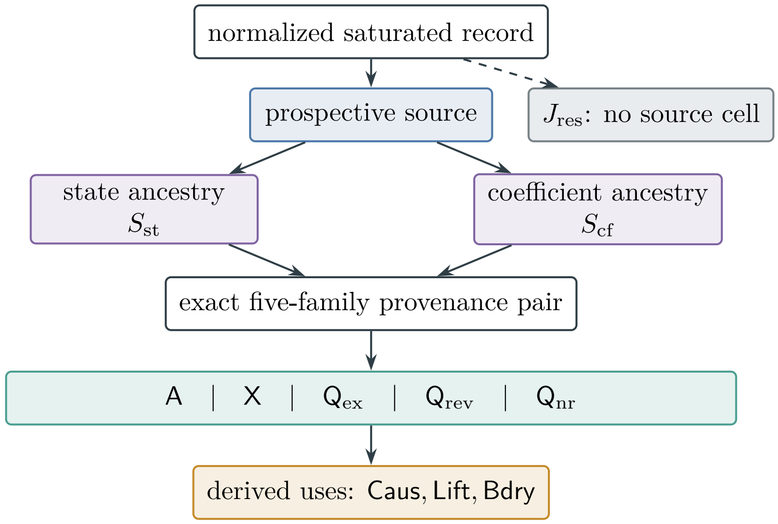

Figure 24: Universal state–coefficient classification of affordable prospective structural sources inside the saturated ledger. The five master modes are crossed with the two exact provenance signatures. Causal orientation, valid lifting, and boundary readout inherit their source cell; the residual coordinate is the post-saturation non-source complement.

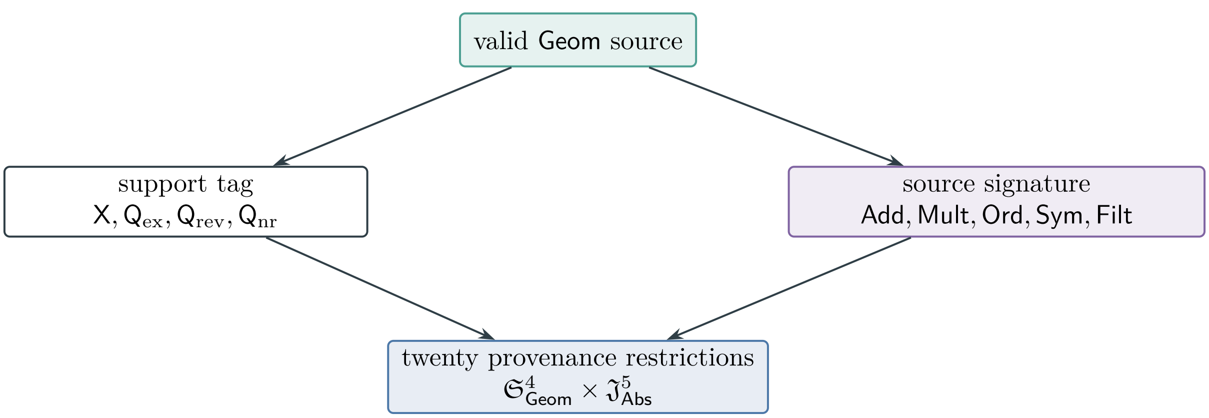

Figure 25: Every valid geometric source has one of four support configurations and a nonempty five-family source signature. The resulting twenty cells are an exhaustive finite cover; mixed-provenance cells may overlap. This is Theorem III.2.4.

(i) A set P of earlier active arguments and a represented evaluator h_v satisfy

$$f_v = h_v \circ J_P,$$

and v has no primitive public read or independently authenticated structural source outside those arguments. This includes the deterministic constructors covered by Lemmas III.17.1 and I.4.7.

(ii) The vertex is a filtration, transcript, or valid-Lift coordinate whose target contribution is assigned to an earlier represented base record by Lemmas III.17.2 and II.10.3. Its charge-parent set is the complete retained set of those base-record arguments.

(iii) The vertex has authenticated Sym provenance and its represented action is certified to act trivially on each active primitive interface used by the record, as in Theorem III.4.2. Its charge parents are the nontrivial active arguments from which its output is inherited.

A certificate is either syntactic, as in the displayed represented factorization, or a represented structural certificate whose description, verifier, and verification cost belong to the complete record. A semantic assertion without a retained certificate does not make a vertex a decorator. For each applicable clause, order its retained certified parent sets first by inclusion and then lexicographically in the fixed encoded vertex order, and select the lexicographically first inclusion-minimal set. The collection is finite because the ancestral DAG is finite. Define $\mathrm{Par}_{\mathrm{src}}(v)$ as the union of the selected sets over the applicable clauses. This rule also applies when several factorizations or structural certificates satisfy one clause. Each selected parent precedes v, so the selection is canonical and charge inheritance is acyclic. The subscript src is retained as notation and does not retype a completed accounting record. Write

$$\mathrm{Dec}_{\mathrm{inh}}(\mathscr{C}) \subseteq \mathcal{A}_{\mathrm{src},5}(\mathscr{C})$$

for the resulting canonical decorator set.

Definition III.5.2 (Charge-bearing core signature)**.** For $v \in \mathcal{A}_{\mathrm{src},5}(\mathscr{C})$, define its anchor set recursively in the fixed topological order by

$$\mathrm{Anch}_{\mathscr{C}}(v) = \begin{cases} \{v\}, & v \notin \mathrm{Dec}_{\mathrm{inh}}(\mathscr{C}), \\ \displaystyle\bigcup_{u \in \mathrm{Par}_{\mathrm{src}}(v)} \mathrm{Anch}_{\mathscr{C}}(u), & v \in \mathrm{Dec}_{\mathrm{inh}}(\mathscr{C}). \end{cases}$$

An empty charge-parent set gives an empty anchor set. The *charge-bearing structural core* is

$$\mathcal{K}_{\mathrm{src},5}(\mathscr{C}) = \bigcup_{v \in \mathcal{A}_{\mathrm{src},5}(\mathscr{C})} \mathrm{Anch}_{\mathscr{C}}(v).$$

Split it according to the state–coefficient type of its retained vertices and define

$$K_{\mathrm{st}}(\mathscr{C}) = \bigcup_{v \in \mathcal{K}_{\mathrm{src},5}(\mathscr{C}) \cap \mathcal{A}^{\mathrm{st}}_{\mathrm{src},5}(\mathscr{C})} \mathrm{Prov}_5(v), \tag{50}$$

$$K_{\mathrm{cf}}(\mathscr{C}) = \bigcup_{v \in \mathcal{K}_{\mathrm{src},5}(\mathscr{C}) \cap \mathcal{A}^{\mathrm{cf}}_{\mathrm{src},5}(\mathscr{C})} \mathrm{Prov}_5(v). \tag{51}$$

The existing pair

$$\mathrm{OpSig}(\mathscr{C}) := \big(S_{\mathrm{st}}(\mathscr{C}), S_{\mathrm{cf}}(\mathscr{C})\big)$$

is the *exact operational signature*; the pair

$$\mathrm{CoreSig}(\mathscr{C}) := \big(K_{\mathrm{st}}(\mathscr{C}), K_{\mathrm{cf}}(\mathscr{C})\big)$$

is the *charge-bearing core signature.*

At a coefficient-complete normalized ceiling B, define

$$\begin{aligned}&\mathrm{CoreAct}^{\mathrm{sat},=}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}}|S_{\mathrm{st}},S_{\mathrm{cf}},n}(B)\\ &:= \sup\left\{\mathfrak{M}_n(I \longmapsto Q_I V_{\mathscr{C},I}) : \begin{array}{l}(\rho, \mathscr{C}, Q) \text{ is a complete normalized}\\ \quad\text{prospective source record,}\\ \mathrm{Cost}_I(\rho) \preceq B,\\ \mathrm{OpSig}(\mathscr{C}) = (S_{\mathrm{st}}, S_{\mathrm{cf}}),\\ \mathrm{CoreSig}(\mathscr{C}) = (K_{\mathrm{st}}, K_{\mathrm{cf}})\end{array}\right\},\end{aligned} \tag{52}$$

with value zero for an empty class. The record, qualification gate, complete visibility, and complete cost in Eq. (52) are the original ones. Core formation is charge attribution in the structural ledger and does not contract the represented derivation.

Finally, put

$$\mathrm{CoreAct}^{\mathrm{sat}}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}},n}(B) = \max_{\substack{S_{\mathrm{st}}\supseteq K_{\mathrm{st}}\\ S_{\mathrm{cf}}\supseteq K_{\mathrm{cf}}}} \mathrm{CoreAct}^{\mathrm{sat},=}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}}|S_{\mathrm{st}},S_{\mathrm{cf}},n}(B). \tag{53}$$

Theorem III.5.3 (Exact separation of operational provenance and charge-bearing cores)**.** *Fix a coefficient-complete normalized record $\mathscr{C}$ in the domain of Definition III.3.1: either a prospective source or a completed terminal-compatible exact quotient retained for accounting. Its operational and core signatures are canonical and satisfy*

$$K_{\mathrm{st}}(\mathscr{C}) \subseteq S_{\mathrm{st}}(\mathscr{C}), \qquad K_{\mathrm{cf}}(\mathscr{C}) \subseteq S_{\mathrm{cf}}(\mathscr{C}).$$

For the prospective-source specialization, each exact operational cell has the finite refinement

$$\mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}(B) = \max_{\substack{K_{\mathrm{st}}\subseteq S_{\mathrm{st}}\\ K_{\mathrm{cf}}\subseteq S_{\mathrm{cf}}}} \mathrm{CoreAct}^{\mathrm{sat},=}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}}|S_{\mathrm{st}},S_{\mathrm{cf}},n}(B). \tag{54}$$

Consequently,

$$\max_{\substack{S_{\mathrm{st}}\neq\varnothing\\ S_{\mathrm{cf}}\subseteq\mathfrak{I}^5_{\mathsf{Abs}}}} \mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M},S_{\mathrm{st}},S_{\mathrm{cf}},n}(B) = \max_{K_{\mathrm{st}},K_{\mathrm{cf}}} \mathrm{CoreAct}^{\mathrm{sat}}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}},n}(B). \tag{55}$$

These are exact prospective-profile reindexing identities. The same complete derivation cost $\mathrm{Cost}_I(\rho)$ *and the same qualified visibility* $Q_I V_{\mathscr{C},I}$ *occur on both sides. A comparison, clock, stored transcript, valid-Lift auxiliary, or certified trivial symmetry can enlarge the operational signature and the complete cost, but it creates no independent charge-bearing core index.*

For a terminal-compatible exact quotient of either temporal type, view its complete normalized record as $\mathscr{C}_{\widehat{\rho}}$ *and let* $(v_1, \ldots, v_k)$ *be its canonical five-family contextual frontier. The exact structural-charge identity refines to*

$$V_I(q_{\widehat{\rho}}) = \sum_{\substack{1\le\ell\le k\\ v_\ell\notin\mathrm{Dec}_{\mathrm{inh}}(\mathscr{C}_{\widehat{\rho}})}} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma_5,\ell}). \tag{56}$$

Thus each nonzero contextual charge is carried by a core anchor; inherited decorators retain their operational records and receive no independent summand.

A reduced core signature is not a restricted operational grammar. In particular,

$$\mathrm{CoreSig}(\mathscr{C}) = (\{\mathsf{Add}\}, \{\mathsf{Add}\})$$

does not restrict the complete evaluator to its retained core anchors. The complete operational record and its cost remain present while qualified visibility is charged to those anchors. Gate expressivity supplies no independent structural credit, and deterministic Boolean constructors cannot make a failed qualification gate nonzero.

Proof. The complete normalized ancestral DAG is finite and has a fixed encoded topological order. Each charge parent in Definition III.5.1 precedes its decorated vertex. Therefore the anchor recursion terminates and is canonical. Each anchor is a vertex of the original complete ancestral set, which proves the two signature inclusions.

Fix an exact operational signature pair in the prospective-source profile. Core formation assigns each record in that exact cell one core pair. The finitely many core-pair restrictions are disjoint and cover the exact cell. A supremum over this finite disjoint union is the maximum of the restricted suprema, proving Eq. (54). Taking the maximum over the operational signatures and regrouping the same records by their core signatures proves Eq. (55).

No represented vertex is removed, contracted, or re-evaluated. The original derivation, qualification gate, active source tuple, and operational decorators remain in the record defining Eq. (52). Hence the complete cost and $Q_I V_{\mathscr{C},I}$ are unchanged.

Consider the canonical contextual frontier of a terminal-compatible exact quotient. In case (i) of Definition III.5.1, the distinguished output is measurable with respect to the joint algebra of its earlier charge parents. Adjoining it leaves that algebra unchanged, so Lemmas III.17.1 and I.4.7 gives zero contextual charge. The filtration and valid-Lift cases have zero incremental charge, with their target contribution assigned to the retained base record, by Lemmas III.17.2 and II.10.3. For a certified trivial symmetry, every orbit operation is identity-equivalent or state-independent by Theorem III.4.2; adjoining its output likewise leaves the earlier target-bearing algebra unchanged.

The contextual summands whose distinguished vertices are inherited decorators are therefore zero. Removing those zero summands from Eq. (105) proves Eq. (56). Their costs remain in the complete carrier ledger.

Finally, core formation records charge attribution rather than evaluator contraction. By Theorem III.3.5, deterministic Mult- and Ord-tagged postprocessings in an already represented finite Boolean evaluator carry zero incremental contextual charge after their joint inputs are exposed. Their complete implementation cost remains in the record, and their complete qualified source value remains $Q_I V_{\mathscr{C},I}$, proving the final assertion.

□

III.5.2 Budget-faithful Boolean normalization

Theorem III.5.4 (Budget-faithful Boolean normalization)**.** *Let $(\rho, \mathscr{C}, Q)$ be a complete family-uniform represented prospective source record for a finite binary presentation, after expansion of each presentation-admissible oracle interface, and suppose*

$$\mathrm{Cost}_I(\rho) \preceq B \qquad (I \in \mathfrak{I}_n).$$

Apply $\mathrm{Exp}_5 \circ \mathrm{Exp}$ *to the complete derivation of the qualified source pair, rather than only to its terminal readout. Denote the resulting record by* $(\widehat{\rho}, \widehat{\mathscr{C}}, \widehat{Q})$*. Then*

$$[\![\widehat{\rho}]\!]_I = [\![\rho]\!]_I, \qquad \widehat{Q}_I = Q_I, \qquad V_{\widehat{\mathscr{C}},I} = V_{\mathscr{C},I}, \qquad \mathrm{Cost}_I(\widehat{\rho}) \preceq \Psi_5(B). \tag{57}$$

The cost on the right charges the complete normalized pair: its gates, instance-dependent coefficients, tables, work state, storage, qualification evaluator, and retained authentication records.

At the level of the single saturated budget slice, normalization has the exact denotational image

$$\left\{ \left([\![\mathrm{Exp}_5(\mathrm{Exp}(\rho))]\!]_I\right)_{I \in \mathfrak{I}_n} : \begin{array}{l} \rho \in \mathsf{Der}_{\mathfrak{I}}, \\ \mathrm{Cost}_I(\rho) \preceq B \quad (I \in \mathfrak{I}_n) \end{array} \right\} = \mathcal{R}^{\mathrm{sat}}_{n,\le B}. \tag{58}$$

The normalized representatives on the left have cost at most the class-preserving ceiling $\Psi_5(B)$*. This set is precisely the normalized denotational image of the original budget-*B *slice and may be a proper subset of* $\mathcal{R}^{\mathrm{sat}}_{n,\le\Psi_5(B)}$*.*

The implementation–source and exact-charge consequences of this same normalization are stated separately in Lemma III.5.5 and Corollary III.5.6. Together the three statements constitute the budget-faithful normalization result carried by the public label above.

Lemma III.5.5 (Boolean implementation–source separation)**.** *Each active affine or product gate that is an internal deterministic vertex of the normalized implementation of the already represented Boolean map is a certified inherited decorator. Its retained direct-input set supplies an earlier certified parent set and its gate evaluator supplies the factorization in Definition III.5.1; the canonical source-parent set is then selected by the rule in that definition. Hence*

$$\mathrm{Anch}_{\widehat{\mathscr{C}}}(v) = \bigcup_{u \in \mathrm{Par}_{\mathrm{src}}(v)} \mathrm{Anch}_{\widehat{\mathscr{C}}}(u) \tag{59}$$

for each such gate v*. The gate remains in* $\mathrm{OpSig}(\widehat{\mathscr{C}})$*, with its complete cost, but it supplies no independent core anchor. An independently authenticated primitive source retains its own core tag; the assertion concerns only the deterministic implementation gates.*

A product-free normalized Boolean evaluator is affine over $\mathbb{F}_2$*. If an active product gate occurs, then*

$$\mathsf{Mult} \in S_{\mathrm{st}}(\widehat{\mathscr{C}}) \cup S_{\mathrm{cf}}(\widehat{\mathscr{C}}),$$

and the Mult *tag records implementation provenance. Source contribution remains the complete qualified value* $Q_I V_{\mathscr{C},I}$*;* CoreSig *then attributes that value to its charge-bearing anchors.*

Let

$$(S_{\mathrm{st}}, S_{\mathrm{cf}}) = \mathrm{OpSig}(\widehat{\mathscr{C}}), \qquad (K_{\mathrm{st}}, K_{\mathrm{cf}}) = \mathrm{CoreSig}(\widehat{\mathscr{C}}).$$

For a declared monotone sublinear positively homogeneous family functional $\mathfrak{M}_n$*, the normalized record satisfies the same-record bound*

$$\begin{aligned} \mathfrak{M}_n(I \longmapsto Q_I V_{\mathscr{C},I}) &= \mathfrak{M}_n\Big(I \longmapsto \widehat{Q}_I V_{\widehat{\mathscr{C}},I}\Big) \\ &\le \mathrm{CoreAct}^{\mathrm{sat},=}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}}|S_{\mathrm{st}},S_{\mathrm{cf}},n}\big(\Psi_5(B)\big). \end{aligned} \tag{60}$$

Profile membership requires one fully charged representation. A table realization retains its complete encoded-table cost; a compact circuit retains the complete description, access, evaluation, storage, retained-data, and output cost of its represented DAG. A charged conversion may supply a second representation at the conversion's complete cost. Normalization preserves denotation and complete

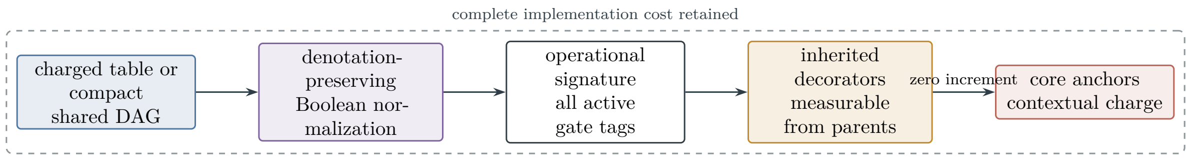

Figure 26: Boolean normalization records every implementation gate in the operational signature. A deterministic gate already measurable from its charged parents is an inherited decorator: its representation cost remains, but it creates no independent contextual source charge.

ancestry accounting; it does not infer an expanded formula, truth table, or compact DAG from semantic existence. Thus a compact DAG is charged at its actual complete shared-circuit cost even when a fully expanded formula is large, while semantic existence alone supplies no record in the budget slice.

Corollary III.5.6 (Exact core charge after Boolean normalization)**.** *If the terminal record is a terminal-compatible exact quotient, let $(v_1, \ldots, v_k)$ be its canonical five-family contextual frontier. The internal implementation gates contribute no contextual-charge summand, and*

$$V_I(q_{\widehat{\rho}}) = \sum_{\substack{1 \le \ell \le k \\ v_\ell \notin \mathrm{Dec}_{\mathrm{inh}}(\widehat{\mathscr{C}})}} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma_5,\ell}). \tag{61}$$

The total charge identity is invariant under affine/product evaluator normalization.

Proof of Theorem III.5.4, Lemma III.5.5, and Corollary III.5.6. Apply Lemma III.1.9 and Theorem III.2.2 to the represented tuple containing the complete source record and its qualification indicator. Componentwise denotation preservation fixes the source data, every qualification comparison, and every factor in its visibility. The monotone transfer map in Definition I.2.4 gives the cost comparison in Eq. (57).

For the inclusion from left to right in Eq. (58), denotation preservation replaces the normalized record by its originating cost-B derivation. Conversely, the definitions in Definitions III.1.2 and III.1.3 give each member of $\mathcal{R}^{\mathrm{sat}}_{n,\le B}$ a complete uniform derivation ρ of cost at most B; normalizing that witness places the same denotation on the left. The cost comparison from B to $\Psi_5(B)$ is the one-way enlargement in Theorem III.2.2.

An internal affine or product gate is a represented deterministic function of its retained direct inputs and has no primitive read outside those inputs. It therefore satisfies clause (i) of Definition III.5.1, and the recursive definition in Definition III.5.2 gives Eq. (59). The Boolean gate identities in Theorem III.3.5 show that absence of a product gate leaves only affine operations. The complete ancestral-set definition retains every active product gate in the operational signature. Core formation removes no vertex or cost and assigns the same qualified visibility to the unique exact operational/core cell. Membership in that cell proves Eq. (60).

Finally, internal deterministic gates have zero incremental contextual charge by Lemmas III.17.1 and I.4.7. Substituting the certified decorator set in the exact core-charge identity Eq. (56) proves Eq. (61).

□

Figure 26 separates implementation provenance from the anchors that receive contextual source charge.

III.5.3 Derivability filtration and cost frontiers

Lemma III.5.7 (Well-posedness of the derivability filtration)**.** *The classes $\mathcal{D}_{n,\leq B}$ are nested in B, and consequently $m_n^{\mathrm{sat}}(B)$, $G_n^{\mathrm{sat}}(B)$, and $I_n^{\mathrm{sat}}(B)$ are nondecreasing in B. Below any chosen finite-cost witness, every represented object has a Pareto-minimal description cost. If the resource order is total and well ordered (in particular, for a scalar discrete cost), this is a least cost.*

Proof. If $B_1 \preceq B_2$, every derivation of cost at most B_1 is also a derivation of cost at most B_2, so $\mathcal{D}_{n,\leq B_1} \subseteq \mathcal{D}_{n,\leq B_2}$. At fixed n, each extraction coordinate is a supremum of nonnegative visibility terms over its corresponding budget slice. The inclusion of slices therefore proves monotonicity without any finiteness or attainment argument. By the local-finiteness coding convention, the nonempty set of descriptions of a represented object having cost at most any one of its finite-cost descriptions is finite. Its finite set of costs has a minimal element in the resource partial order, which proves the Pareto-minimal assertion. Under a total well order that element can be chosen least. All budget-slice arguments use an actual witness of cost $\preceq B$ and therefore require neither scalarization nor a global least element.

□

Remark III.5.8 (Cost-frontier convention). When the resource monoid is only partially ordered, subsequent phrases such as "least saturated cost" refer to the upward-closed admissible budget set, represented by its Pareto-minimal frontier. They do not assert a least vector among incomparable resources. Statements using a single number, logarithmic degree, infimum, or equality of threshold costs are made after the declared scalar cost or monotone scalarization has been fixed. The constructor-charge theorems themselves use only an actual witness with cost $\preceq B$.

Remark III.5.9 (Derivability of committors and finite-horizon approximations). For a killed generator L_N, the exact discounted committor $u_{\lambda,L} = (\lambda I - L_N)^{-1} k_T$ is legal under $\mathcal{B}$ exactly when it has a derivation in $\mathcal{D}_{n,\leq b(n)}$. Thus the resolvent must be represented by an admissible derivation. A finite-horizon truncation such as $\sum_{j\leq M} a_j P_N^j k_T$ is admissible when the one-step rule, coefficients, horizon, and terminal edge gate are derived data and the finite computation lies within the resource envelope. Its saturated computation cost is then charged explicitly. In the polynomial instance this condition is $b(n) = n^C$ for some fixed C.

Convention III.5.10 (Budgeted computation is presentation-relative)**.** Throughout this part, a budget-admissible oracle-free computation is relative to $\mathcal{R}_n^{\mathcal{B}}$, the budgeted saturated represented closure of the presented task. The algorithm may have any uniform oracle-free code allowed by the resource model, but every instance-dependent datum read by that code must be declared public or generated by a prior vertex of the same represented derivation and its legal lifted computation. This includes initialization data, finite outcome-generation interfaces, lifted coordinates, auxiliary tables, predicates, basis changes, quotient/fiber operations, and update maps. Any generated datum becomes part of the full lifted observable configuration and retains its derivation cost when used as structural data. Measurability in $\mathcal{F}^{\mathrm{env}}_{I,\leq b(n)}$ alone does not authorize a read or assign a budget.

If a transition rule uses a datum outside $\mathcal{R}_n^{\mathcal{B}}$, such as a withheld modulus, trapdoor, target table, oracle answer, nonuniform advice string, or relation recoverable only above the declared budget, then the resulting algorithm is not budget-admissible and oracle-free for the presented task. Such a rule uses oracle or advice data, performs over-budget recovery, or enlarges the primitive signature. This convention is the computational counterpart of Part I's distinction between exposed and presentation-inaccessible structure after imposing the saturated cost filtration.

The complexity theorems below quantify over the presentation-relative generators embedded by Part II at the declared budget. Transfer-class target arrival by such a generator forces a saturated source in the observable accounting at the same budget, up to the transfer-class distortion. Structurality follows from the Part II channel decomposition and the Part III cost filtration. For $\mathcal{B}_{\mathrm{poly}}$, this convention recovers ordinary presentation-relative polynomial computation and inverse-polynomial target-arrival thresholds inside $\mathcal{R}_n^{\mathcal{B}_{\mathrm{poly}}}$.

III.6 Encoding Adequacy for Budgeted Presentations and the Polynomial Instance

The encoded presentation determines the input model and supplies all public instance-dependent data. The budgeted saturated observable closure is the closure of these data under the admissible budgeted oracle-free constructions.

Budget completeness connects the abstract presentation to a uniform finite input encoding with comparable costs.

Definition III.6.1 (Budget-complete encoding)**.** An encoding $\mathrm{enc}(I)$ of a task-family presentation is complete for the declared budget structure $\mathcal{B}$ when the following simulations are uniformly charged by the resource monoid and stay below $b(n)$ whenever the represented derivation lies within the declared budget. For $\mathcal{B}_{\mathrm{poly}}$, this is the usual polynomial-size encoding with polynomial-time simulations.

Decoding. The primitive public task data are decoded from $\mathrm{enc}(I)$ within the declared decoding budget: state-size data, primitive observables, terminal predicates, declared public parameters, static comparability data, and the declared dynamic/lift constructor signatures.

Constructor simulation. Every admissible constructor used to generate $\mathcal{R}_n^{\mathcal{B}}$ has a uniform implementation whose overhead is charged in $\mathcal{B}$ from $\mathrm{enc}(I)$ and the current lifted configuration.

Evaluator normalization. Every state-dependent primitive, lookup rule, and finite evaluator supplied by the encoding has either an explicit charged table or a uniform charged implementation over a fixed finite operational basis. Finite-precision arithmetic and Boolean evaluators are compiled to this basis with class-preserving overhead. An explicit table retains its encoded length and lookup cost. A state-dependent access rule supplied without either representation is an oracle interface in the oracle-extended presentation.

Transition representation. Every uniform oracle-free operational transition rule whose transcript charge is $\preceq b(n)$ is represented by its code, initializer, successor, clock or budget counter, peak native-state capacity, finite outcome/terminal transcript, and declared native-state effects. The one-step kernel on the clocked transcript carrier produced by that rule is therefore a legal forward observable transport datum. In the polynomial instance, this includes exactly the uniform operational transition rules whose native resources are polynomially bounded on $\mathrm{enc}(I)$.

Input closure. A datum not decoded from $\mathrm{enc}(I)$, not generated by the admissible constructors, and not present in the legal lifted configuration is outside the budgeted observable algebra of this presentation. Supplying it changes the primitive signature, gives advice, or gives oracle access.

Encoding adequacy connects the operational representation to the declared computation model.

Theorem III.6.2 (Encoding adequacy)**.** *Fix a budget-complete encoding of the task family. A represented operational transition rule is presentation-relative and lies within $\mathcal{B}$ exactly when its instance-dependent data have records in $\mathcal{R}_n^{\mathcal{B}}$, its lifted transcript is legally represented, and its accumulated charge is $\preceq b(n)$. Consequently the budget-admissible algorithms for the encoded* *presentation are exactly the budget-admissible generators embedded by Part II. Setting $\mathcal{B} = \mathcal{B}_{\mathrm{poly}}$ recovers the uniform polynomial-resource equivalence inside $\mathcal{R}_n^{\mathcal{B}_{\mathrm{poly}}}$.*

Proof. Let M be a uniform operational transition rule whose transcript charge lies within $\mathcal{B}$. At any represented step, its single live native state contains the input-dependent working data, finite control, work state, budget counter, and memory records. The finite clocked carrier contains the outcome/terminal transcript; declared effects expose only the required native-state readouts. The current native state determines the represented conditional outcome law and successor. By transition representation in the definition of budget-complete encoding, this one-step rule is a legal forward observable transport datum generated from $\mathcal{R}_n^{\mathcal{B}}$ and the lifted configuration. Hence M is a presentation-relative budget-admissible transition rule, and Part II embeds each bounded slice as $L_M^{[H]} = \rho(P_M^{[H]} - I)$. For $\mathcal{B}_{\mathrm{poly}}$, this is the native polynomial-resource simulation declared by the budget structure.

Conversely, let A be a presentation-relative budget-admissible transition rule. Its transition is built from declared public data, legal lifted coordinates, and admissible constructors in $\mathcal{R}_n^{\mathcal{B}}$, all within the declared resource envelope. By decoding and constructor simulation, the encoded model can decode the presentation, execute each constructor and lifted-coordinate update, and realize the same one-step operational transition law with charged overhead still below $b(n)$. Thus A is a within-budget uniform transition rule for the encoded presentation.

The equivalence concerns transition rules and their admissible instance-dependent input data. A semantic function of the induced future path has zero structural cost only when it has an explicit representative. In particular, a committor, first-hitting partition, occupation law, or resolvent value of the same induced generator contributes to m^{sat} or G^{sat} when the same denotation has a budget-admissible saturated representative as a quotient, potential, geometry, lifted interface, boundary readout, or admissible within-budget finite-horizon computation. Derivability accounting requires a construction of the semantic object; an admissible computation from $\mathcal{R}_n^{\mathcal{B}}$ is counted at its saturated cost.

□

Lemma III.6.3 (Budgeted generation gives budgeted representation)**.** *Fix a presentation-complete encoding and a budget-compatible resource envelope. If a datum, relation, finite-horizon trajectory statistic, first-hit quotient, target-contact statistic, boundary value, or postprocessed observable is produced by a uniform oracle-free computation from $\mathcal{R}_n^{\mathcal{B}}$ and the legal clocked transcript representation, with peak native-state and transcript charge below $b(n)$, then its denotation has a budget-admissible saturated representative whose cost lies within budget: the public code for the computation, the horizon, the lifted coordinates it reads, and the finite postprocessing rule. Conversely, if every budget-admissible representative of such a denotation has cost above the declared ceiling, then the denotation has not been produced as legal budgeted data for the original presentation. Using it in a purported budget-admissible transition rule therefore requires oracle or advice data, an enlarged primitive signature, or over-budget recovery. Consequently every uniform public target alignment admissible for the presented task belongs to the budgeted saturated observable closure.*

Proof. The forward implication is the definition of $\mathcal{R}_n^{\mathcal{B}}$ together with the constructor-simulation clause of presentation completeness. An admissible within-budget finite-horizon computation has a finite lifted transcript whose charge is below $b(n)$, reads only legal lifted coordinates and declared public data, and applies only budget-admissible oracle-free postprocessings. This transcript description is a budget-admissible representative, with saturated cost bounded by the declared ceiling. For the converse, suppose such a denotation were produced by a presentation-relative within-budget computation from the legal closure. The first part would give a within-budget budget-admissible representative, contradicting the assumed lower bound above $b(n)$ on all budget-admissible representatives. Hence any use of the denotation requires data outside the budgeted observable algebra, an enlarged primitive signature, oracle or advice access, or computation exceeding the declared budget.

□

All L^2 projections, normalized target correlations, reach ratios, regularity ratios, and extraction quantities in this part use the presentation reference measure on the base space and a Part II-legitimate reference measure on any lifted configuration space. A lifted reference law may be the canonical public full-support product gauge on a padded finite configuration encoding, the public initialization lift when it satisfies the support condition, or a publicly represented transfer-class-bounded density change of one of these gauges. The product gauge is sufficient for exact configuration and first-hit accounting but supplies no structural Valid-Lift projection or target-fraction identity. Occupation measures, hitting-conditioned laws, invariant laws, Green measures, and resolvent measures obtained from the same measured operator qualify as reference measures for these quantities only when they also have an independent admissible representation with controlled density. Accordingly, a clock or program-counter coordinate aligned under a valid public lifted law is exposed structure, whereas alignment arising only after replacing that law by the algorithm's trajectory distribution is not exposed by the presentation.

The function y_n is the centered target indicator. Low saturated cost means that this indicator admits a representation in a low-complexity admissible observable class. A large saturated hardness cost means that every observable class within the declared budget has negligible target contrast. Raw Walsh degree is one coordinate in this saturated cost profile.

This part develops the discounted hitting functional, the target-projected killed resolvent, the structural operator ledger with full-process first-hit accounting, the Walsh/spectral bound, the combined extraction functional, the saturated cost invariant B^* and its log-cost coordinate d^*, and the admissibility dichotomy. It proves hardness for presentation-relative budget-admissible oracle-free generators in regimes where the exposed channel representatives and comparison data have been bounded: random targets, removed structure, obstructed placement against the declared geometry, and proven saturated-cost lower-bound constructions. These comparison data enter only the analysis of a budget-admissible computation. A uniform oracle-free procedure supplies only its transition rule; every instance-dependent datum used by that rule must have a derivation in $\mathcal{R}_n^{\mathcal{B}}$, be a declared public parameter, or lie in the legal lifted closure. Part II embeds that rule as a generator, and the present part estimates the exposed projections of that generator. The problem-specific input supplied by an application is an explicitly proved saturated structural negligibility bound for its complete derivation-indexed quotient and geometric families. For a fixed cryptographic instance whose gapping structure is admissible, Part III reduces hardness to the statement that B^* lies above the declared budget; in the polynomial instance this is the usual superpolynomial, equivalently superlogarithmic, lower bound in log-cost notation. Part IV supplies the complementary quantitative learning analysis, and Part VIII records a separate application of the framework.

Admissibility is a computational provenance condition relative to the budgeted saturated observable closure of the instance presentation. Part II proves that every presentation-relative budget-admissible oracle-free computation, represented on its full observable configuration, induces a generator whose target-relevant transport is decomposed into exposed structural components and a residual term. The exposed structural components are Geom, Caus, Abs, Lift, and Bdry.

Among them, the geometric/reversible component Geom and the quotient/lumping component Abs provide the intrinsic structure. The directed/drift component Caus, the lifted-state component Lift, and the boundary/killing component Bdry respectively orient exposed edge differences, reorganize valid lifted coordinates, and apply terminal or support values to an already represented structure. The scalar Bdry data mark, kill, weight, or evaluate states and affect transport only through a represented Geom, Abs, or Lift edge relation whose differences are oriented by Caus. The residual term J_{res} is residual accounting: it is the part of the movement with no target-relevant representative in $\mathcal{R}_n^{\mathcal{B}}$ after the five exposed structural projections have been applied. A proposed budget-admissible method using an instance-dependent relation outside $\mathcal{R}_n^{\mathcal{B}}$ requires presentation-inaccessible information: oracle or advice access, an over-budget recovery procedure, or an enlarged primitive signature. If the relation is generated inside $\mathcal{R}_n^{\mathcal{B}}$, it appears as intrinsic $\mathsf{Geom}/\mathsf{Abs}$ structure or as a valid use of it after lifting. The full target arrival of this decomposition is bounded by the uniform saturated profile after applying the exact first-hit construction; no separate residual-coupling bound is inferred here.

Example III.6.4 (SAT after the structural decomposition)**.** For SAT, Part I gave the assignment space, clause observables, defect score, success sector, atoms, level sets, and possible exposed quotients. Part II classified every budget-admissible transport rule on that space. The present analysis concerns the hitting time of the satisfying sector. Its Laplace transform is expressed by the target-projected killed resolvent of a budget-admissible operator and bounded by the saturated public cost at which the observable algebra exposes either a quotient or a regular aimed geometry.

III.7 Arrival Time as the Complexity Measure

Fix the positive terminal sector defined in Part I. Throughout this section, write

$$T = T_n = E^{+}_{I_n}, \qquad N = N_n = X_n \setminus T.$$

The non-target sector N contains both nonterminal states and terminal failures. Thus a terminal failure remains terminal even though it belongs to N.

Killed dynamics isolate the non-target evolution and the flux entering the target sector.

Definition III.7.1 (Killed dynamics at the target)**.** Let X_n be finite. An admissible continuous-time operator L is written in row-frontier convention, so a row probability vector evolves by

$$p_t = p_0 e^{tL}.$$

Assume $L_{xy} \geq 0$ for $x \neq y$ and $\sum_y L_{xy} = 0$ after adjoining an absorbing cemetery state if needed. The killed non-target operator is the restriction

$$L_N = (L_{xy})_{x,y \in N}, \qquad K_N = -L_N.$$

The target killing rate is

$$k_T(x) = \sum_{z \in T} L_{xz}, \qquad x \in N.$$

When the killed chain is stable, every eigenvalue of K_N has positive real part. In the reversible case K_N is self-adjoint in $L^2(\mu_N)$, and the killed-target gap is

$$\gamma_T(L) = \lambda_{\min}(K_N).$$

Killing removes from the non-target equation all mass that reaches T. The vector k_T gives the rate of transition from each non-target state into the target sector. These are the standard killed-chain and hitting-time conventions [61].

The arrival functional measures the target flux generated by the killed process.

Definition III.7.2 (Arrival functional)**.** Let μ_0 be any initial probability measure, put $a_0 = \mu_0(T)$, and let $\mu_{0,N}$ be its restriction to N. Let τ_T be the first hitting time of T, with $\tau_T = 0$ on an initially successful state, for the absorbed process generated by L. The arrival functional is

$$\Phi_L(t) = \mathbb{P}_{\mu_0}(\tau_T \le t).$$

The killed frontier on N is $q_t = \mu_{0,N} e^{tL_N}$. Its survival mass and hit flux are

$$\mathbb{P}_{\mu_0}(\tau_T > t) = q_t \mathbf{1}_N, \qquad \Phi_L'(t) = q_t k_T = \mu_{0,N} e^{tL_N} k_T$$

whenever Φ_L is differentiable.

The resulting complexity parameter is $\Phi_L(\mathtt{t}(b(n)))$, evaluated at the numerical time capacity of the declared resource ceiling.

Definition III.7.3 (Discounted hitting functional)**.** For discount rate $\lambda > 0$, define

$$S_T(\lambda; L, \mu_0) = \int_0^\infty e^{-\lambda t}\, d\Phi_L(t) = \mathbb{E}_{\mu_0}\left[e^{-\lambda \tau_T} \mathbf{1}_{\{\tau_T < \infty\}}\right].$$

A budget ceiling $b(n)$ of finite time capacity is read at discount $\lambda = \mathtt{t}(b(n))^{-1} > 0$. The functional is within budget when $S_T(\mathtt{t}(b(n))^{-1}; L, \mu_0)$ has the required transfer-class margin. For $\mathcal{B}_{\mathrm{poly}}$, this is the usual inverse-polynomial reading at $b(n) = n^{O(1)}$. For an infinite time capacity, use the finite-slice readings or a separately justified $\lambda \downarrow 0$ limit; the resolvent definition itself always assumes $\lambda > 0$.

The exponential discount weights target hits occurring before the budget more heavily than those occurring substantially later. It thereby reduces the hitting-time distribution to a scalar functional that is comparable across operators.

Proposition III.7.4 (Within-budget arrival bounds the discounted hitting functional)**.** *For every budget $B > 0$,*

$$S_T(1/B; L, \mu_0) \ge e^{-1}\, \mathbb{P}_{\mu_0}(\tau_T \le B).$$

Proof. On the event $\{\tau_T \le B\}$, the factor $e^{-\tau_T/B}$ is at least e^{-1}. Therefore

$$\mathbb{E}[e^{-\tau_T/B} \mathbf{1}_{\{\tau_T < \infty\}}] \ge \mathbb{E}[e^{-\tau_T/B} \mathbf{1}_{\{\tau_T \le B\}}] \ge e^{-1} \mathbb{P}(\tau_T \le B).$$

□

Example III.7.5 (Discounted hitting for SAT)**.** For a SAT instance, T is the set of satisfying assignments and N is the set of all other assignments together with any non-success terminal states in the presentation. The quantity $\Phi_L(t)$ is the probability that the budget-admissible operator has hit a satisfying assignment by time t. The defect score assigns a scalar value to each assignment, and the hitting functional measures arrival at zero defect.

III.8 Killed Resolvent Representation

Part II introduced budget-admissible generators as transport operators. The Laplace transform of their target-hitting flux is a killed resolvent [44, 61].

The target-projected resolvent is the Laplace-domain representation of discounted target arrival.

Definition III.8.1 (Target-projected resolvent)**.** Let L_N, $K_N = -L_N$, and k_T be as in Definition Killed dynamics at the target. For $\lambda > 0$, the target-projected resolvent is

$$A_T(\lambda; L, \mu_0) = \mu_0(T) + \mu_{0,N}(\lambda I_N - L_N)^{-1}k_T = \mu_0(T) + \mu_{0,N}(\lambda I_N + K_N)^{-1}k_T.$$

Here $(\lambda I_N - L_N)^{-1}$ integrates all discounted non-target histories, while k_T restricts to histories terminating upon entry into the target. The terminal edge gate identifies the final crossing, whereas the killed resolvent accounts for the entire discounted history preceding that crossing.

The full-history transport record exposes the complete represented path contribution preceding target contact.

Definition III.8.2 (Full-history target-contact transport)**.** Write the off-diagonal rates of L as $q_L(x, z)$. The terminal-contact edge gate is

$$W_T^L(x, z) = q_L(x, z)\mathbf{1}_T(z), \qquad k_T^L(x) = \sum_z W_T^L(x, z).$$

For an initial frontier μ_0 and discount $\lambda > 0$, the full-history target-contact transport is the nonnegative measure on target-entering edges

$$\mathsf{H}^T_{\lambda,L,\mu_0}(x, z) = \int_0^\infty e^{-\lambda t}\,(\mu_{0,N}e^{tL_N})(x)\,W_T^L(x, z)\,dt.$$

Its total mass is the noninitial part of the target-projected resolvent:

$$\sum_{x\in N}\sum_{z\in T}\mathsf{H}^T_{\lambda,L,\mu_0}(x, z) = \mu_{0,N}(\lambda I_N - L_N)^{-1}k_T^L = A_T(\lambda; L, \mu_0) - \mu_0(T).$$

The measure $\mathsf{H}^T_{\lambda,L,\mu_0}$ is an operator-level hitting quantity rather than a representation constructor. It separates the killed propagation e^{tL_N} from the terminal edge gate W_T^L. The Part II decomposition applies to the generator driving the propagation, and the displayed formula integrates the corresponding component projections.

Lemma III.8.3 (Killed-resolvent identity for the discounted hitting functional)**.** *For every $\lambda > 0$,*

$$A_T(\lambda; L, \mu_0) = S_T(\lambda; L, \mu_0).$$

This identity is the standard Laplace-transform representation of a killed semigroup resolvent [44, 61].

Proof. The killed frontier on N is $\mu_{0,N}e^{tL_N}$, so the instantaneous post-initial hit flux is $\mu_{0,N}e^{tL_N}k_T$. The atom at $\tau_T = 0$ has mass $a_0 = \mu_0(T)$. Hence

$$S_T(\lambda) = a_0 + \int_0^\infty e^{-\lambda t}\mu_{0,N}e^{tL_N}k_T\,dt = a_0 + \mu_{0,N}\left(\int_0^\infty e^{-t(\lambda I_N - L_N)}\,dt\right)k_T.$$

Because L_N is a finite sub-Markov generator, its spectrum lies in the closed left half-plane. Hence $\lambda I_N - L_N$ is invertible for every $\lambda > 0$, the exponentially discounted matrix integral

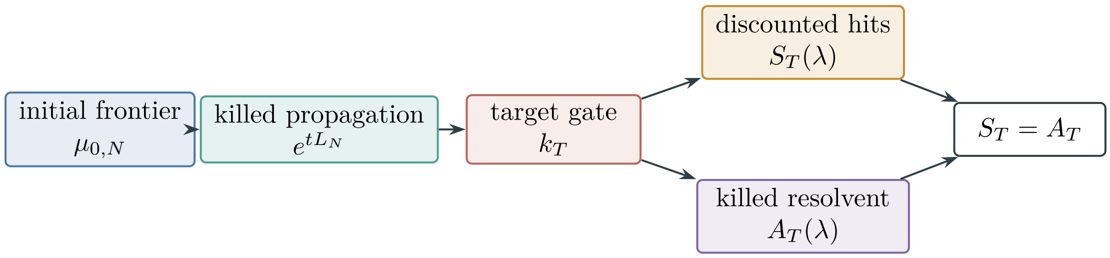

Figure 27: Discounted target arrival and the target-projected killed resolvent are the same scalar functional by Lemma III.8.3.

converges, and it equals $(\lambda I_N - L_N)^{-1}$. This includes an absorbing non-target failure state, for which 0 may belong to the spectrum of L_N.

$\square$

Figure 27 summarizes the exact identity in Lemma III.8.3.

This comparison record contains the charged data needed to transfer an operator estimate to target hitting.

Definition III.8.4 (Budgeted hitting-functional comparison data)**.** Budgeted hitting-functional comparison data for a budget-admissible operator L under a declared budget structure $\mathcal{B}$, with operative ceiling $b(n)$, are finite exposed data whose charge is $\preceq b(n)$ and that verify one of the following admissible arrival readings for the already fixed operator. The first reading is available only when $\mathtt{t}(b(n)) < \infty$:

$$A_T(\mathtt{t}(b(n))^{-1}; L, \mu_0) \succeq \beta_{\mathfrak{C}}(n)^{-1},$$

$$\Phi_L(h(n)) \succeq \beta_{\mathfrak{C}}(n)^{-1} \quad \text{for some } 1 \le h(n) \le \mathtt{t}(b(n)),$$

$$\gamma_T(L) \succeq \beta_{\mathfrak{C}}(n)^{-1} \quad \text{together with exposed resolvent/survival comparison data.}$$

When $\mathtt{t}(b(n)) = \infty$, one instead uses finite stopping slices, or a separately proved $\lambda \downarrow 0$ limit; the resolvent expression is never evaluated at $\lambda = 0$.

The comparison data may be a direct target-resolvent computation, a killed-gap/Dirichlet comparison, or an exact quotient comparison. Its ingredients must be generated by the observable algebra and the Part II decomposition into exposed structural channels plus residual accounting. These data belong to the analysis of the fixed operator and are not algorithmic inputs or conditions for membership in the declared budgeted operational class. They assign a budgeted hitting bound to an exposed component of L.

Hitting-functional comparison data are analytic objects. Budget-admissible membership is uniform execution under the declared resource envelope using only $\mathcal{R}_n^{\mathcal{B}}$ and its legal lifted closure. After that execution rule is embedded as a generator, the killed-resolvent identity identifies the operator quantity that must be large. If the large quantity is carried by exposed structure in the budgeted closure, the corresponding projection of the generator supplies the comparison data. If it is not carried by exposed structure, the remaining operator term belongs to J_{res}. Any target arrival of the complete process is then charged through the exact first-hit quotient rather than through a standalone residual estimate.

Convention III.8.5 (Analytic role of comparison data)**.** Throughout this part, hitting-functional comparison data establish a quantitative bound for an already fixed operator. The structural component, saturated cost, membership in the declared budgeted operational class, and existence of quotient or geometric structure are properties of the presented task and its induced operator. An oracle-free procedure may output only a candidate solution; its transition rule still determines L. If no comparison data have been established, the corresponding quantitative estimate remains unavailable, while the underlying structure is unchanged.

For example, a public linear quotient makes XOR-type instances low-cost abstraction instances as a structural fact. Their membership in polynomial time is realized by Gaussian elimination, which implements structure already present in the observable presentation.

Proposition III.8.6 (Budget-admissible procedures are covered before comparison data enter)**.** *Let $A = (A_n)$ be a uniform oracle-free procedure for the presented task family whose transcript and instance-dependent data lie within budget under $\mathcal{B}$, in the presentation-relative sense of Convention Budgeted computation is presentation-relative. For each n, record the clocked finite outcome/terminal transcript of A_n and charge the current memory, tables, work state, and output state at peak native capacity. Its one-step execution rule defines a Markov kernel $P^{[H]}_{A,n}$ on this transcript carrier and a generator $L^{[H]}_{A,n} = \rho(P^{[H]}_{A,n} - I)$. This construction depends only on the transition rule of A; the procedure need not output comparison data, a quotient, an invariant, or a structural decomposition.*

Proof. The resource envelope makes the clocked transcript carrier finite. The code and represented finite outcome interface of A_n, together with public data generated from $\mathcal{R}^{\mathcal{B}}_n$, determine the conditional transcript transition law, hence determine $P^{[H]}_{A,n}$. Part II applies the exhaustive decomposition to $L^{[H]}_{A,n}$. If A has non-negligible target arrival at the declared margin, the killed-resolvent identity gives a non-negligible target-resolvent quantity for this induced generator. Comparison data, when used, charge that operator quantity to an exposed structural projection. If no exposed projection carries it, the contribution remains J_{res} residual accounting, or it uses presentation-inaccessible or oracle-dependent information outside the budgeted saturated observable closure. A program whose transition rule depends on such outside information is analyzed instead as an oracle/advice computation, an over-budget recovery procedure, or a computation after changing the primitive signature so that the datum is public. Thus coverage of presentation-relative A is independent of any comparison data.

□

Lemma III.8.7 (Quantitative arrival–resolvent comparison and conditional gap route)**.** *For every $B, c > 0$, the arrival distribution and discounted hitting functional satisfy*

$$e^{-1}\Phi_L(B) \leq S_T(B^{-1}; L, \mu_0) \leq \Phi_L(cB) + e^{-c}.$$

Consequently $\Phi_L(B) \geq \varepsilon$ implies $S_T(B^{-1}) \geq e^{-1}\varepsilon$, while $S_T(B^{-1}) \geq \varepsilon$ implies $\Phi_L(cB) \geq \varepsilon/2$ for $c = \log(2/\varepsilon)$. A killed-gap estimate is a third proof route only when its declared comparison data also bound the relevant survival, initial-overlap, and target-flux terms so as to imply one of these two quantities. Under transfer-class bounds on those comparison factors and on c, the three proof routes yield the same transfer-class non-negligibility conclusion.

Proof. The first inequality is Proposition Within-budget arrival bounds the discounted hitting functional. For the second, split according to $\{\tau_T \leq cB\}$:

$$
\begin{aligned}
S_T(B^{-1}) &= \mathbb{E}\Big[e^{-\tau_T/B}\mathbf{1}_{\{\tau_T<\infty\}}\Big] \\
&\le \mathbb{P}(\tau_T \le cB) + e^{-c}\mathbb{P}(cB < \tau_T < \infty) \\
&\le \Phi_L(cB) + e^{-c}.
\end{aligned}
$$

The two consequences follow by substitution. The resolvent reading equals S_T by Theorem Killed-resolvent identity for the discounted hitting functional. A gap alone contains no information about initial overlap or flux into T; hence the gap route is valid only with the additional comparison data stated in the theorem. Those data imply an arrival or resolvent bound, after which the displayed inequalities apply.

□

Example III.8.8 (SAT resolvent)**.** For SAT, $k_T(x)$ records the rate at which the operator sends a non-satisfying assignment into a satisfying assignment. The expression $\mu_{0,N}(\lambda I - L_N)^{-1}k_T$ sums all discounted non-satisfying histories that end by entering a satisfying assignment. A polynomial SAT channel supports this quantity, or an equivalent killed-gap or quotient reading, through the exposed clause structure, declared geometry, quotient, lift, or boundary data.

III.9 Structural Operator Ledger and Full-Process Hitting

III.9.1 Channel coordinates and full-process accounting

Part II classifies the full generator in an operator Hilbert space. Probabilistic hitting functionals remain attached to the full positive generator unless a component is separately verified to lie in the Markov-generator cone.

The channel coordinates place every structural component on the common full-process operator ledger.

Definition III.9.1 (Channel operator coordinates)**.** Let a budget-admissible generator decompose as

$$L = L_{\mathsf{Geom}} + L_{\mathsf{Caus}} + L_{\mathsf{Abs}} + L_{\mathsf{Lift}} + L_{\mathsf{Bdry}} + L_J.$$

For a component ξ, define its operator energy by

$$\mathcal{E}_\xi(L) = \|L_\xi\|^2_{\mathscr{H}_{\mathrm{ch}}}.$$

The Pythagorean identity (17) gives $\|L\|^2_{\mathscr{H}_{\mathrm{ch}}} = \sum_\xi \mathcal{E}_\xi(L)$; no Pythagorean claim is made for the induced operator norm. A quantity of the form $A_T(\lambda; L - L_\xi, \mu_0)$ is a hitting functional only when $L - L_\xi$ separately satisfies the finite generator form.

These coordinates measure the exact operator decomposition without assigning standalone transition probabilities to its summands.

Proposition III.9.2 (Roles of the five structural components and the residual term)**.** *For finite components with verified hitting-functional data, their analytic roles are inherited from the Part II decomposition. The term J_{res} is the residual component rather than an exposed structural component.*

component	*role in $A_T(\lambda)$*
Geom	*intrinsic local geometry: conductance and Poincaré/Cheeger scale*
Caus	*orientation/use of an exposed potential built from intrinsic structure*
Abs	*intrinsic represented quotient abstraction*
Lift	*valid re-expression of already charged intrinsic structure*
Bdry	*terminal/support coloring or boundary value; no movement by itself*
J_{res}	*orthogonal operator remainder; full-process first-hit accounting*

The table records operator provenance. It does not identify contextual target charges: every Lift/Bdry occurrence remains in the complete carrier, and a smaller-source charge bound requires a represented contextual transfer.

Proof. The fixed-space and component decompositions of Part II identify the origin of each term. Local symmetric conductance belongs to the geometric/reversible component Geom and contributes through the killed Poincaré or Cheeger gap. Represented nonlocal jumps constant on quotient cells have algebraic quotient/lumping provenance. When the quotient passes Definition III.15.1, its qualified Abs coordinate reduces the dimension of the killed problem. Directed exact one-form components constitute the directed/drift component Caus, given by the monotone use of edge differences of an exposed potential built from the observable geometry or abstraction. Lift is functorial composition or enlargement, so its target-directed part is supported on the structure present before lifting at the level of operator provenance. Its contextual charge is nevertheless retained until a represented transfer bounds it. Boundary terms color, kill, weight, or read terminal states, with terminal removal as the zero-value case; they have no off-diagonal transition support, so they affect movement only through differences across an already represented Geom/Abs/Lift edge structure that is then oriented by Caus. The residual term is orthogonal to the declared structural potentials and quotient/lift subspaces. The killed-resolvent identity of Section III.8 is applied to the full positive generator. The exact first-hit quotient then supplies the retrospective accounting-profile identity for target arrival; its source status is governed separately by Definition III.21.1. □

Lemma III.9.3 (Structural ledger and saturated accountability)**.** *After the Part II decomposition, transfer-class target advantage is assigned to the intrinsic problem structure: exposed local geometry* (Geom) *and represented quotient structure* (Abs)*. Directed descent, valid lifted-state constructions, and boundary readout respectively orient, reorganize, and evaluate these components. Boundary readout consists of scalar state data and has no independent transition mechanism. The residual component J_{res} is the orthogonal remainder in the operator ledger; no standalone Markov or baseline interpretation is assigned to it. For every affordable full generator, all target arrival—including effects arising from interactions among the six operator coordinates—is bounded by the saturated accounting quotient profile at the cost of the corresponding exact first-hit construction. Prospective source attribution is instead governed by Theorem III.21.3.*

Proof. The intrinsic local component is Geom: it supplies the observable comparison relation, conductance, and killed gap. The intrinsic quotient component is Abs: it supplies represented compression compatible with the terminal structure. Caus uses these sources by orienting edge differences of an exposed potential; Lift composes or enlarges existing sources and retains that structure in its complete carrier; Bdry supplies terminal/support colorings and boundary values on states. The conditional charges of Lift and Bdry transfer to a smaller source only through a represented comparison. The residual term is only an orthogonal operator coordinate; it need not be a positive generator, and no hitting probability is attributed to it in isolation. Instead, apply the killed resolvent to the full positive generator and construct its exact first-hit quotient. That quotient is a uniform represented exact-quotient accounting record whose cost includes the entire computation, including all interactions with J_{res}. The universal saturated-profile theorem therefore charges the resulting target arrival as an audit identity without a componentwise positivity assumption. By Corollary III.23.1, this completed record does not supply prospective source attribution; that attribution uses the neutral baseline and selector advantage in Theorem III.21.3. The roles of Lift and Bdry in the structural ledger remain as stated: lifting is charged to its represented construction, while boundary data supplies no movement by itself.

□

III.9.2 Transformed gradients and directed drift

The transformed-gradient space fixes the represented directions eligible for geometric comparison.

Definition III.9.4 (Admissible transformed-gradient space)**.** Fix a saturated budget B and a presentation-relative generator on its full lifted configuration space. Let $E_{\mathrm{tr},\leq B}$ be the represented transformed edge, quotient, or lifted comparability structure generated at budget B from intrinsic $\mathsf{Geom}/\mathsf{Abs}$ data and valid Lift operations. Let $\mathcal{V}_{\mathrm{tr},\leq B}$ be the represented scalar state-coloring space on that structure: pointwise Geom potentials, quotient coordinates, lifted coordinates, and Bdry terminal/support colorings or boundary values, all generated from the same budgeted observable algebra. The admissible information flow is

$$
\begin{aligned}
\mathcal{R}_n^{\mathcal{B}} &\xrightarrow{\mathsf{Geom}/\mathsf{Abs}} \text{represented edge or quotient structure}\\
&\xrightarrow{\mathsf{Lift}} \text{valid lifted comparability structure}\\
&\qquad\text{and}\\
&\mathsf{Bdry} \text{ supplies scalar terminal/support values on it.}
\end{aligned}
$$

The boundary/killing component Bdry is scalar: it marks, kills, weights, or evaluates states and terminal sectors. It contributes no off-diagonal edge and no transport operator. Transport appears only after $E_{\mathrm{tr},\leq B}$ supplies represented comparable pairs and a scalar $V \in \mathcal{V}_{\mathrm{tr},\leq B}$ supplies differences across those pairs. With represented nonnegative edge weights $a_B(x,z)$ supported on $E_{\mathrm{tr},\leq B}$, the associated transformed-gradient edge space is

$$
\nabla_{E_{\mathrm{tr},\leq B}} \mathcal{V}_{\mathrm{tr},\leq B} = \{(x,z) \mapsto a_B(x,z)(V(z) - V(x)) : V \in \mathcal{V}_{\mathrm{tr},\leq B}\}.
$$

The edge inner product is the one supplied by the Part II Hodge split on the represented current support. A directed edge current is a Caus contribution at budget B exactly through its orthogonal projection onto $\nabla_{E_{\mathrm{tr},\leq B}} \mathcal{V}_{\mathrm{tr},\leq B}$. This definition states that the directed/drift component Caus follows gradients over represented geometric, quotient, or lifted-state structure; Bdry may assign the state values defining the gradient but contributes no transition by itself.

Figure 28 shows how the transformed-gradient space separates authorized direction from the orthogonal operator remainder.

Lemma III.9.5 (Directed drift has transformed-gradient normal form)**.** *Let L_n be any presentation-relative uniform oracle-free budget-admissible generator, represented on its finite Part II carrier (the clocked transcript carrier for an operational computation). Let J_{dir} be any directed or antisymmetric*

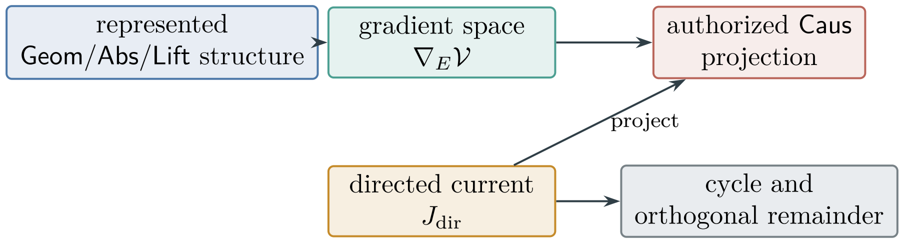

Figure 28: The Hodge projection of Definition III.9.4 and Lemma III.9.5. Only the component of a represented current lying in the transformed-gradient space receives Caus status; the complementary components remain in their cycle or residual coordinates.

target-relevant current appearing in the killed full-history accounting after the symmetric Geom and exact Abs quotient parts have been separated. For each saturated budget B there is an orthogonal decomposition

$$J_{\mathrm{dir}} = J_{\mathsf{Caus},B} + J_{\mathrm{cyc},B} + J^{\mathrm{dir}}_{\mathrm{res},B},$$

where $J_{\mathsf{Caus},B}$ lies in $\nabla_{E_{\mathrm{tr},\le B}}\mathcal{V}_{\mathrm{tr},\le B}$, $J_{\mathrm{cyc},B}$ is the divergence-free cycle component relative to the represented transformed edge structure, and $J^{\mathrm{dir}}_{\mathrm{res},B}$ is orthogonal to every transformed gradient generated at budget B. The first component is a Caus channel. Its complete carrier charges the Geom/Abs data, every Lift that supplies the represented edge structure, and every Bdry scalar coloring or boundary value used in the potential. This is cost accounting, not an automatic transfer of target charge between extensions. The cycle component contributes to target arrival only through represented terminal contact across the same edge structure; a Bdry readout may record or kill a terminal state, but it does not move mass. Its contextual charge remains explicit unless a represented comparison transfers that charge to the source it reads. Any transfer-class target excess left in $J^{\mathrm{dir}}_{\mathrm{res},B}$ is either represented by a budget-admissible saturated observable and therefore reclassified by Part II into Geom/Caus/Abs/Lift/Bdry, or has no such one-step representative at budget B. In the latter case it remains an orthogonal operator remainder; full-process arrival is charged at the cost of the first-hit derivation. Presentation-inaccessible or oracle-dependent data, nonuniform advice, and over-budget recovery are inadmissible for the original presentation.

Consequently, a represented directed source at budget B consists of a construction from the budgeted represented algebra of a represented edge/quotient/lift geometry together with an admissible scalar coloring whose edge differences point to the target. If no such transformed gradient or represented quotient is available at that budget, the directed operator contribution remains outside the represented source coordinates. The universal profile theorem accounts for any success of the complete computation at its enlarged finite-horizon cost.

Proof. Work on the finite edge Hilbert space of the directed current support, with the positive edge weights used in the Part II Hodge split. By definition, $E_{\mathrm{tr},\le B}$ is a represented edge/quotient/lift support and $\mathcal{V}_{\mathrm{tr},\le B}$ is a finite-dimensional scalar state-coloring space on it. Therefore $\nabla_{E_{\mathrm{tr},\le B}}\mathcal{V}_{\mathrm{tr},\le B}$ is a closed finite-dimensional subspace of the edge space. Let $J_{\mathsf{Caus},B}$ be the orthogonal projection of J_{dir} onto this transformed-gradient subspace. This is exactly the structural-gradient component in Part II's Hodge split, with the edge support enlarged only by admissible Abs and Lift transformations and with Bdry entering only as scalar terminal/support values already present in the saturated closure. Hence it is Caus, and its carrier cost includes the represented edge structure, scalar potential, and Geom/Abs data that produced them. This provenance statement does not transfer contextual target charge.

Inside the orthogonal complement, take the finite Hodge decomposition relative to the represented transformed graph with boundary: the divergence-free part is $J_{\mathrm{cyc},B}$, and the remaining part is $J^{\mathrm{dir}}_{\mathrm{res},B}$. A divergence-free cycle has no net boundary flux in the represented quotient/geometry. A represented Bdry coloring can record, stop, or weight terminal states, but since it has no off-diagonal support it cannot turn a cycle into motion. If terminal contact is accountable, it is because the same represented edge structure supplies terminal-crossing differences or terminal-contact edges already charged in Geom/Abs/Lift and then read by Bdry. Without such represented terminal contact, the cyclic part has no accountable target descent.

It remains to classify $J^{\mathrm{dir}}_{\mathrm{res},B}$. By construction it is orthogonal to every admissible transformed gradient at budget B and has no represented quotient or boundary readout already removed. If its target excess is represented by an admissible observable relation, potential, quotient, lifted interface, scalar boundary readout, represented propagation component, or legal finite-horizon trajectory statistic, Part II's no-uniform-residual-correlation theorem reclassifies that representative into one of the five structural components. The reclassified contribution is then counted in m_n^{sat} or G_n^{sat} at its least saturated carrier cost. Lift and Bdry remain represented contextual extensions; moving their target charge to another source requires a charged comparison satisfying the contextual-transfer inequality. If no such representative exists, the directed target excess is not legal budget-admissible structure for the original presentation; it is an orthogonal residual at this budget, or it uses presentation-inaccessible or oracle-dependent data, nonuniform advice, over-budget recovery, or data outside the budgeted represented algebra. These alternatives exhaust the Part II component theorem and the budgeted derivability closure. Any target arrival of their full composition is subject to Theorem III.24.3.

□

Example III.9.6 (Structural components for SAT)**.** A one-bit Hamming graph provides the geometric/reversible component for SAT. Unit propagation or implication closure provides a directed/drift component when the implications are exposed and target-aligned. Public XOR clauses provide a quotient/lumping component because the syndrome quotient collapses assignments into represented affine cells. Adding auxiliary variables is Lift. Detecting $D_\Phi = 0$ is boundary/killing data. Random restarts without structural alignment belong to the residual term J_{res}.

III.10 The Walsh/Spectral Bound

The spectral bound restricts a low-degree comparison vector to the low-degree projection of the target. Consequently, low-degree target mass bounds the discounted hitting estimate obtainable from degree-d structure.

The degree projection measures how much target contrast lies in the declared low-degree observable band.

Definition III.10.1 (Degree projection and spectral mass)**.** On $X = \{0,1\}^n$ with uniform measure, let

$$\chi_S(x) = (-1)^{\sum_{i\in S} x_i}, \qquad S \subseteq \{1,\dots,n\},$$

be the Walsh characters [62, 78]. For a centered target contrast $y \in L^2(X)$, write

$$y = \sum_{S\subseteq[n]} \widehat{y}(S)\chi_S, \qquad \widehat{y}(S) = \mathbb{E}_x[y(x)\chi_S(x)].$$

The degree-$\leq d$ projection and spectral mass are

$$P_d y = \sum_{|S|\leq d} \widehat{y}(S)\chi_S, \qquad m(d) = \frac{\|P_d y\|^2_{L^2(\mu)}}{\|y\|^2_{L^2(\mu)}} = \frac{\sum_{|S|\leq d} |\widehat{y}(S)|^2}{\sum_S |\widehat{y}(S)|^2}.$$

For a general finite observable algebra, P_d denotes orthogonal projection onto the chosen degree-$\leq d$ observable subspace $V_d \subseteq L^2(X_I, \mu_I)$. The invariant later optimizes over admissible public basis changes.

The quantity $m(d)$ is the fraction of the target contrast contained in the observable subspace of degree at most d. If $m(d)$ is negligible, then degree-d observables have asymptotically equal means on the target and its complement.

The comparison vector supplies the represented low-degree direction used in the Walsh bound.

Definition III.10.2 (Degree-d target comparison vector)**.** A degree-d target comparison vector for a structural component consists of an exposed unit vector $b \in V_d$, a transfer-class comparison factor $C_{\mathrm{comp}}(n)$, and a verified discounted hitting strength $\Gamma_T(L)$ such that the comparison proves

$$0 \leq \Gamma_T(L) \leq C_{\mathrm{comp}}(n) \left| \langle \bar{y}, b \rangle_{L^2(\mu)} \right|^2 .$$

Here $\Gamma_T(L)$ may be the verified killed gap, a verified target-resolvent lower-bound strength, or the equivalent verified arrival strength from Section III.8. If no such exposed b and verification exist, then no degree-d budgeted hitting contribution has been charged to that structural component.

This condition is required when applying the theorem. A presentation-relative budget-admissible algorithm supplies a uniform transition rule whose instance-dependent data are legal for the presented task. To prove a budgeted hitting contribution at raw observable degree d in this Walsh-bound subcase, the argument must exhibit an exposed low-degree statistic in the induced generator and the corresponding inequality. Without that proof, no raw degree-d contribution has been established in the application; the objective saturated cost of the component is unchanged.

The accounted degree condition ties a low-degree operator contribution to an exposed statistic and its charged derivation.

Definition III.10.3 (Degree-d accounted budget-admissible operator)**.** A budget-admissible killed operator is accounted at effective degree at most d when there is a proof that every target-directed structural component of L is generated by observables in V_d, after any charged public basis change, and every claimed budgeted hitting contribution has a degree-d target comparison vector. This is an analytic classification of the operator rather than a membership condition for the algorithm that induced it.

Lemma III.10.4 (Walsh/spectral comparison-vector bound)**.** *For every degree-d accounted budget-admissible operator L,*

$$\Gamma_T(L) \leq C_{\mathrm{comp}}(n) m(d), \qquad C_{\mathrm{comp}} \in \mathfrak{C}.$$

In particular, a transfer-class verified hitting estimate at degree d requires transfer-class target mass in the degree-$\leq d$ observable band. In the polynomial instance this is the usual inverse-polynomial statement.

Proof. Let $b \in V_d$ be the exposed unit vector in the comparison. Since P_d is orthogonal projection onto V_d,

$$\langle \bar{y}, b\rangle = \langle P_d \bar{y}, b\rangle.$$

By Cauchy-Schwarz and $\|b\| \le 1$,

$$|\langle \bar{y}, b\rangle|^2 \le \|P_d \bar{y}\|^2 = \frac{\|P_d y\|^2}{\|y\|^2} = m(d).$$

Substituting this inequality into the verified comparison inequality gives the displayed bound. $\square$

Remark III.10.5 (Basis robustness). The relevant degree is the degree in the public observable algebra after admissible public basis changes and represented quotients are charged. A raw-coordinate Walsh calculation is a diagnostic; the invariant uses the basis that the presentation exposes.

Example III.10.6 (XOR-SAT and basis dependence). An XOR-SAT instance may have a target contrast that occupies degree three in raw coordinate Walsh characters when one equation involves three variables. The public syndrome observable represents that same equation as one syndrome coordinate. In the public syndrome basis, the target-facing proof vector is degree one, so the admissible invariant reads the problem through the syndrome quotient.

III.11 Quantitative Geom–Abs Interaction: Quotient Defect and Local Compensation

Part I defined static lumpability as compatibility between represented quotient cells and terminal structure. Part II expresses quotient dynamics through intertwining. The present section measures how accurately a represented quotient describes the killed dynamics, and when local geometry controls the modes omitted by that quotient.

III.11.1 Quotient defects and exact resolvent identities

We first compare the ambient and quotient generators through their represented lift, projection, and lumpability defect.

Definition III.11.1 (Quotient lift, projection, and quotient generator)**.** Let $q : X_I \to Q$ be a represented quotient of saturated cost at most B. Let $H = L^2(X_I, \mu_I)$ and $H_Q = L^2(Q, \nu_Q)$, where $\nu_Q = q_*\mu_I$. The lift and projection are

$$U_q : H_Q \to H, \qquad (U_q f)(x) = f(q(x)),$$
$$P_q : H \to H_Q, \qquad (P_q g)(a) = \mathbb{E}_{\mu_I}[g(X) \mid q(X) = a].$$

For a killed generator L on X_I, the quotient generator induced by q is

$$L_Q = P_q L U_q.$$

The lift turns a quotient observable into a state observable by making it constant on every quotient cell. The projection averages a state observable over each cell. With the displayed conditional-expectation projection, U_q is an isometry and $P_q = U_q^*$. The quotient generator is the compression of the original generator to the quotient subspace.

The lumpability defect measures the failure of quotient lifting to intertwine the full and compressed generators.

Definition III.11.2 (Lumpability defect)**.** The lumpability defect of q is the operator

$$E_q = LU_q - U_q L_Q : H_Q \to H.$$

Exact lumpability means $E_q = 0$, equivalently

$$LU_q = U_q L_Q.$$

Proposition III.11.3 (Unit transfer for a canonical exact quotient)**.** *Let $q : X \to Q$ be a finite terminal-compatible quotient on a finite carrier equipped with a full-support reference law μ, equip Q with the canonical law $\nu_Q = q_*\mu$, and let $U_q f = f \circ q$. Let L and L_Q be Markov or sub-Markov generators satisfying*

$$LU_q = U_q L_Q.$$

Suppose that $T = q^{-1}(T_Q)$, that the target sectors are absorbing, and that the quotient initialization is $\bar{\mu}_0 = q_*\mu_0$. Then U_q is isometric, $P_q = U_q^*$, and

$$e^{tL} U_q = U_q e^{tL_Q} \qquad (t \geq 0). \tag{62}$$

The full and quotient target-hitting laws, and hence their discounted target functionals, agree exactly:

$$\Pr_{\mu_0}\{\tau_T \leq t\} = \Pr_{\bar{\mu}_0}\{\tau_{T_Q} \leq t\}, \qquad A_T(\lambda; L, \mu_0) = A_T^Q(\lambda; L_Q, \bar{\mu}_0). \tag{63}$$

Consequently the exact-quotient comparison and normalization factor is one for the canonical record:

$$C_q = 1. \tag{64}$$

Representation, simulation, and fiber-evaluation overhead contributes to saturated cost and does not introduce a numerical transfer loss.

Proof. By the definition of the pushforward law,

$$\|U_q f\|^2_{L^2(\mu)} = \int_X |f \circ q|^2 \, d\mu = \int_Q |f|^2 \, d\nu_Q = \|f\|^2_{L^2(\nu_Q)}.$$

Thus $U_q^* U_q = I$, and the conditional-expectation formula in Definition III.11.1 gives $P_q = U_q^*$. Exact intertwining passes to every power of the generators and therefore to their exponential series, proving (62). Terminal compatibility gives $\mathbf{1}_T = U_q \mathbf{1}_{T_Q}$. Since both sectors are absorbing, their occupation probabilities at time t equal their first-hit probabilities by time t. Hence

$$
\begin{aligned}
\Pr_{\mu_0}\{\tau_T \le t\} &= \mu_0 e^{tL}\mathbf{1}_T \\
&= \mu_0 e^{tL} U_q \mathbf{1}_{T_Q} \\
&= \bar{\mu}_0 e^{tL_Q}\mathbf{1}_{T_Q} \\
&= \Pr_{\bar{\mu}_0}\{\tau_{T_Q} \le t\}.
\end{aligned}
$$

Taking the Laplace–Stieltjes transform proves the functional identity. All norm, initialization, dynamic, and terminal comparisons used by the canonical exact-quotient record are equalities, which proves (64).

□

Exact and approximate lumpability in this finite-state setting agree with the usual Markov-chain aggregation notions [17, 45]. The defect is an operator $H_Q \to H$; geometric compensation must therefore control its action on a space of quotient observables.

The component defect record resolves lumpability failure into symmetric and causal parts.

Definition III.11.4 (Symmetric and causal quotient defects)**.** Let the killed generator selected from a Caus authority envelope be

$$
L = S + A, \qquad S^* = S, \qquad A^* = -A,
$$

where diagonal killing is included in S. Define the compressed operators

$$
S_Q = P_q S U_q, \qquad A_Q = P_q A U_q,
$$

and the component defects

$$
E_q^{\mathsf{s}} = SU_q - U_q S_Q, \qquad E_q^{\mathsf{a}} = AU_q - U_q A_Q.
$$

Then

$$
L_Q = S_Q + A_Q, \qquad E_q = E_q^{\mathsf{s}} + E_q^{\mathsf{a}}. \tag{65}
$$

The first term is the symmetric or geometric compatibility defect. The second is the causal transport defect of the selected imposed/control operator. A reversible realization has $A = 0$ and hence $E_q^{\mathsf{a}} = 0$. An imposed nonreversible realization must retain $A_Q = P_q A U_q$; choosing a different quotient skew operator changes the dynamic realization and is not a quotient comparison for the original authority envelope.

Proposition III.11.5 (Orthogonality of the compressed authority split)**.** *With the conditional-expectation projection $P_q = U_q^*$, one has*

$$
S_Q^* = S_Q, \qquad A_Q^* = -A_Q, \qquad \operatorname{Re}\operatorname{tr}(S_Q^* A_Q) = 0.
$$

Thus quotient compression preserves the reversible/nonreversible operator split, and the two compressed components are orthogonal in the real Hilbert–Schmidt inner product. Exact full lumpability is equivalent to

$$
E_q^{\mathsf{s}} = -E_q^{\mathsf{a}};
$$

componentwise causal compatibility is the stronger condition $E_q^{\mathsf{s}} = E_q^{\mathsf{a}} = 0$. Every authority-preserving certificate records the component defects even when their sum cancels.

Proof. Since $P_q = U_q^*$,

$$S_Q^* = U_q^* S^* U_q = S_Q, \qquad A_Q^* = U_q^* A^* U_q = -A_Q.$$

Moreover,

$$\overline{\operatorname{tr}(S_Q^* A_Q)} = \operatorname{tr}(A_Q^* S_Q) = -\operatorname{tr}(A_Q S_Q) = -\operatorname{tr}(S_Q A_Q),$$

where cyclicity of the finite-dimensional trace is used in the last equality. Hence $\operatorname{tr}(S_Q^* A_Q)$ is purely imaginary and its real part is zero. Equation (65) gives the remaining statements. Recording the two terms prevents an exact full intertwining caused by cancellation from being used as evidence that the imposed causal and symmetric components descend separately.

□

Proposition III.11.6 (Exact resolvent defect identity)**.** *For every $\lambda > 0$ in the common resolvent set of L and L_Q,*

$$(\lambda I - L)^{-1} U_q - U_q(\lambda I - L_Q)^{-1} = (\lambda I - L)^{-1} E_q (\lambda I - L_Q)^{-1}.$$

Consequently,

$$\left\|(\lambda I - L)^{-1} U_q - U_q(\lambda I - L_Q)^{-1}\right\| \le \|(\lambda I - L)^{-1}\| \, \|E_q\| \, \|(\lambda I - L_Q)^{-1}\|.$$

The displayed equality is the resolvent identity in perturbation form [44].

Proof. From the definition of E_q,

$$(\lambda I - L)U_q - U_q(\lambda I - L_Q) = -E_q.$$

Multiplying on the left by $(\lambda I - L)^{-1}$ and on the right by $(\lambda I - L_Q)^{-1}$ gives the identity. The norm bound is submultiplicativity.

□

The normalized defect measures the target-relevant loss in a nonreversible quotient resolvent comparison.

Definition III.11.7 (Normalized nonreversible resolvent defect)**.** Let $W_Q \le H_Q$ and $W \le H$ be represented comparison windows, with orthogonal projections Π_Q and Π_W. Set

$$b_\lambda(q) = \|\Pi_W U_q (\lambda I - L_Q)^{-1} \Pi_Q\|,$$

and

$$d_\lambda(q) = \|\Pi_W (\lambda I - L)^{-1} E_q (\lambda I - L_Q)^{-1} \Pi_Q\|.$$

The normalized resolvent defect is

$$\Delta_\lambda(q; W_Q, W) = \begin{cases} d_\lambda(q)/b_\lambda(q), & b_\lambda(q) > 0, \\ 0, & b_\lambda(q) = d_\lambda(q) = 0, \\ \infty, & b_\lambda(q) = 0 < d_\lambda(q). \end{cases}$$

For a target-arrival comparison, the certificate must additionally specify a represented functional $\ell_T \in W^*$ and input $v_T \in W_Q$ corresponding to its initial-law and terminal-flux reading. Define

$$b_\lambda^T(q) = \left|\ell_T\Big(\Pi_W U_q(\lambda I - L_Q)^{-1}\Pi_Q v_T\Big)\right|,$$
$$d_\lambda^T(q) = \left|\ell_T\Big(\Pi_W(\lambda I - L)^{-1}E_q(\lambda I - L_Q)^{-1}\Pi_Q v_T\Big)\right|,$$

and define $\Delta_\lambda^T = d_\lambda^T/b_\lambda^T$ with the same zero-denominator conventions as above. Its target-transfer stability factor is

$$S_{q,T}^{\mathrm{nr}}(\lambda) = \max\{0, 1 - \Delta_\lambda^T(q;\ell_T, v_T)\}.$$

All windows, resolvent bounds, and retained operator data used to evaluate these numbers belong to the charged comparison certificate. In particular, Δ_λ or Δ_λ^T is admissible at budget B only when these data have a derivation of cost at most B; the exact resolvents are not free constructors.

Corollary III.11.8 (General quotient reconstruction bound)**.** *On the represented windows of Definition III.11.7,*

$$\begin{aligned}&\|\Pi_W((\lambda I - L)^{-1}U_q - U_q(\lambda I - L_Q)^{-1})\Pi_Q\|\\ &\qquad\le \Delta_\lambda(q; W_Q, W)b_\lambda(q).\end{aligned}$$

Consequently,

$$\|\Pi_W(\lambda I - L)^{-1}U_q\Pi_Q\| \le (1 + \Delta_\lambda(q; W_Q, W))b_\lambda(q),$$

and, when $\Delta_\lambda < 1$, the same norm is at least $(1 - \Delta_\lambda)b_\lambda(q)$. Thus quotient resolvent estimates transfer to the full process with the explicit additive loss $d_\lambda(q)$. A lower transfer certificate is nontrivial precisely when $\Delta_\lambda < 1$, with loss $(1-\Delta_\lambda)^{-1}$; the upper reconstruction bound has factor $1+\Delta_\lambda$. For a general nonreversible generator this statement is the available conclusion; a killed-gap comparison requires additional self-adjoint or sector hypotheses.

For the represented target functional in Definition III.11.7, the same identity and the reverse triangle inequality also give

$$\left|\ell_T\Big(\Pi_W(\lambda I - L)^{-1}U_q\Pi_Q v_T\Big)\right| \ge (1 - \Delta_\lambda^T(q;\ell_T, v_T))b_\lambda^T(q)$$

whenever $\Delta_\lambda^T < 1$. This scalar inequality, rather than the operator-norm reconstruction bound alone, is the lower-transfer record used in a target-visibility certificate.

Proof. Project the identity in Proposition III.11.6 onto W_Q and W, then apply the definitions of d_λ and Δ_λ. The zero-denominator conventions cover the two degenerate cases. Applying the represented functional to the same identity gives a scalar quotient term plus an error term of modulus d_λ^T; reverse triangle and $d_\lambda^T = \Delta_\lambda^T b_\lambda^T$ prove the final inequality. □

III.12 Reversible Defect Compensation

In the reversible setting, Schur-complement control converts the defect bound into quotient-to-ambient resolvent estimates.

III.12.1 Compensated defects and Schur complements

Definition III.12.1 (Reversible compensated defect)**.** Let $N = X \setminus T$, let $H_N = L^2(N, \mu|_N)$, and suppose the killed operator $K = -L_N$ is self-adjoint and positive semidefinite. Let $U : H_Q \to H_N$ be the isometric lift of a terminal-compatible quotient, and set $\Pi = UU^*$. Relative to $H_N = \operatorname{ran} U \oplus \operatorname{ran} U^\perp$, define

$$A = U^* K U, \qquad B = (I - \Pi)KU, \qquad D = (I - \Pi)K(I - \Pi)|_{\operatorname{ran} U^\perp}.$$

Then

$$K = \begin{pmatrix} A & B^* \\ B & D \end{pmatrix}.$$

For $\lambda > 0$, the reversible compensated-defect parameter is

$$\theta_\lambda(q) = \left\| (D + \lambda I)^{-1/2} B (A + \lambda I)^{-1/2} \right\|^2. \tag{66}$$

The associated form-conditioning factor is

$$S_q^{\mathrm{rev}}(\lambda) = \max\{0, 1 - \theta_\lambda(q)\}.$$

This factor controls the Loewner comparison in the theorem below. It is not, by itself, a lower bound for an arbitrary cross pairing $\langle u, (K + \lambda I)^{-1} v \rangle$.

The operator A measures quotient geometry, D measures within-fiber geometry, and B is the coupling between them. Since $L_Q = -A$ and $E_q = -B$, with the latter embedded in $\operatorname{ran} U^\perp$, exact lumpability is equivalent to $B = 0$.

Lemma III.12.2 (Schur-complement compensation theorem)**.** *Under Definition III.12.1,*

$$U^*(K + \lambda I)^{-1} U = \Big(A + \lambda I - B^*(D + \lambda I)^{-1} B \Big)^{-1}. \tag{67}$$

If $\theta_\lambda(q) < 1$, then

$$(A + \lambda I)^{-1} \preceq U^*(K + \lambda I)^{-1} U \preceq \frac{1}{1 - \theta_\lambda(q)} (A + \lambda I)^{-1}. \tag{68}$$

In particular, with $R_q = (A + \lambda I)^{-1}$ and $R^q_{\mathrm{full}} = U^*(K + \lambda I)^{-1} U$,

$$0 \preceq R^q_{\mathrm{full}} - R_q \preceq \frac{\theta_\lambda(q)}{1 - \theta_\lambda(q)} R_q.$$

Consequently, for $u, v \in H_Q$,

$$|\langle u, (R^q_{\mathrm{full}} - R_q) v \rangle| \le \frac{\theta_\lambda(q)}{1 - \theta_\lambda(q)} \langle u, R_q u \rangle^{1/2} \langle v, R_q v \rangle^{1/2}.$$

Proof. The upper-left block of the inverse of $K + \lambda I$ is the inverse of its Schur complement, which gives (67). Let $X = A + \lambda I$ and $Y = D + \lambda I$. By (66),

$$0 \preceq B^* Y^{-1} B \preceq \theta_\lambda(q) X.$$

Hence

$$(1-\theta_\lambda(q))X \preceq X - B^*Y^{-1}B \preceq X.$$

Both sides are positive definite when $\theta_\lambda(q) < 1$. Inverting reverses Loewner order and proves (68). The operator difference is positive and bounded above by $\theta_\lambda(q)(1-\theta_\lambda(q))^{-1}R_q$. Cauchy–Schwarz for its quadratic form gives the final estimate.

□

III.12.2 Target-functional and killed-gap transfer

Corollary III.12.3 (Target-functional reversible transfer)**.** *With R_q and R^q_{full} as above, fix represented target-comparison vectors $u_T, v_T \in H_Q$, and put*

$$b^T_{\lambda,\mathrm{rev}} = |\langle u_T, R_q v_T\rangle|, \qquad d^T_{\lambda,\mathrm{rev}} = |\langle u_T, (R^q_{\mathrm{full}} - R_q)v_T\rangle|.$$

Define $\vartheta^T_\lambda = d^T_{\lambda,\mathrm{rev}}/b^T_{\lambda,\mathrm{rev}}$ with the zero-denominator conventions of Definition III.11.7. If $\vartheta^T_\lambda < 1$, then

$$|\langle u_T, R^q_{\mathrm{full}} v_T\rangle| \geq (1-\vartheta^T_\lambda)b^T_{\lambda,\mathrm{rev}}.$$

Moreover, the Schur-complement theorem gives the computable estimate

$$d^T_{\lambda,\mathrm{rev}} \leq \frac{\theta_\lambda(q)}{1-\theta_\lambda(q)}\langle u_T, R_q u_T\rangle^{1/2}\langle v_T, R_q v_T\rangle^{1/2}.$$

Proof. The first assertion is the reverse triangle inequality. The second is the final bilinear estimate in Lemma III.12.2.

□

Corollary III.12.4 (Two-scale killed-gap bound)**.** *Suppose, in addition, that*

$$A \succeq \gamma_Q I, \qquad D \succeq \gamma_\perp I, \qquad \|B\| \leq \varepsilon,$$

where $\gamma_Q, \gamma_\perp > 0$. Then

$$\theta_\lambda(q) \leq \frac{\varepsilon^2}{(\gamma_Q+\lambda)(\gamma_\perp+\lambda)},$$

and

$$K \succeq \gamma_- I, \qquad \gamma_- = \frac{\gamma_Q + \gamma_\perp - \sqrt{(\gamma_Q-\gamma_\perp)^2 + 4\varepsilon^2}}{2}.$$

In particular, $\gamma_- > 0$ precisely when $\varepsilon^2 < \gamma_Q\gamma_\perp$. Thus a non-lumpable quotient is controlled only when the coupling is small relative to both the quotient gap and the within-fiber geometric gap.

Proof. For $f = Uu + v$, with $v \perp \operatorname{ran} U$,

$$\langle f, Kf\rangle \geq \gamma_Q\|u\|^2 + \gamma_\perp\|v\|^2 - 2\varepsilon\|u\|\|v\|.$$

The least eigenvalue of the resulting two-dimensional quadratic form is γ_-. Its determinant is $\gamma_Q\gamma_\perp - \varepsilon^2$, which gives the positivity criterion. The bound on θ_λ follows from $\|(D+\lambda I)^{-1/2}\| \leq (\gamma_\perp+\lambda)^{-1/2}$, $\|(A+\lambda I)^{-1/2}\| \leq (\gamma_Q+\lambda)^{-1/2}$, and $\|B\| \leq \varepsilon$.

□

Corollary III.12.5 (Represented local geometry compensates the defect)**.** *Let A_{Φ_Q} and $D_{\Phi_\perp}$ be the operators of represented quotient and within-fiber Dirichlet forms. Suppose that, on the certified windows,*

$$A + \lambda I \succeq (1-\delta_Q)(A_{\Phi_Q} + \lambda I), \qquad D + \lambda I \succeq (1-\delta_\perp)(D_{\Phi_\perp} + \lambda I),$$

where $0 \le \delta_Q, \delta_\perp < 1$, and that

$$A_{\Phi_Q} \succeq g_Q I, \qquad D_{\Phi_\perp} \succeq g_\perp I, \qquad \|B\| \le \varepsilon.$$

Then

$$\theta_\lambda(q) \le \frac{\varepsilon^2}{(1-\delta_Q)(1-\delta_\perp)(g_Q+\lambda)(g_\perp+\lambda)}. \tag{69}$$

If the right-hand side is smaller than one, the represented geometries yield a valid Schur-complement compensation certificate. Its transfer loss is at most the reciprocal of one minus the right-hand side.

Proof. The form comparisons imply

$$\|(A+\lambda I)^{-1/2}\| \le [(1-\delta_Q)(g_Q+\lambda)]^{-1/2}$$

and the analogous estimate for $D + \lambda I$. Insert these two bounds and $\|B\| \le \varepsilon$ into (66). The Schur-complement loss then follows from Lemma III.12.2.

□

III.12.3 Coercive regularity and ambient consequences

Theorem III.12.6 (Coercive regularity forces reversible quotient compensation)**.** *Let L be a represented reversible killed generator on the nonterminal space H_N, put $K = -L_N$, and suppose that the same represented comparison record certifies*

$$0 < \gamma I \preceq K \preceq \Lambda I \qquad (0 < \gamma \le \Lambda). \tag{70}$$

Let $q \neq \mathrm{id}$ be a represented terminal-compatible quotient for this generator, with isometric lift U, orthogonal projection $\Pi = UU^*$, and represented block data A, B, D from Definition III.12.1. For every represented discount $\lambda > 0$, set

$$r_{\gamma,\Lambda,\lambda} = \frac{\gamma+\lambda}{\Lambda+\lambda}.$$

Then its existing reversible compensated-defect parameter satisfies

$$\sqrt{\theta_\lambda(q)} \le 1 - r_{\gamma,\Lambda,\lambda}, \tag{71}$$

$$S_q^{\mathrm{rev}}(\lambda) = 1 - \theta_\lambda(q) \ge 2r_{\gamma,\Lambda,\lambda} - r_{\gamma,\Lambda,\lambda}^2 \ge r_{\gamma,\Lambda,\lambda}. \tag{72}$$

In particular, $E_q = 0$ gives an exact quotient record. If $E_q \neq 0$, then q has a valid reversible compensated record and

$$(S_q^{\mathrm{rev}}(\lambda))^{-1} \le \frac{\Lambda+\lambda}{\gamma+\lambda}. \tag{73}$$

Consequently, whenever the quotient, its compressed and within-fiber block data, the comparison in Eq. (70)*, and the quotient target-functional record have total saturated cost at most B, and the right side of Eq.* (73) *belongs to the declared transfer class, every dynamically qualified target-visible use of this coarsening is classified by the existing exact or reversible compensated nonidentity quotient coordinate. It cannot remain in the identity-supported ambient coordinate.*

Proof. Relative to $H_N = \operatorname{ran} U \oplus \operatorname{ran} U^{\perp}$, write

$$K + \lambda I = \begin{pmatrix} X & B^* \\ B & Y \end{pmatrix}, \qquad X = A + \lambda I, \qquad Y = D + \lambda I.$$

Both X and Y are positive definite. Define

$$Z = Y^{-1/2} B X^{-1/2}, \qquad \mathcal{D} = \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix}.$$

Block normalization gives the exact identity

$$\mathcal{D}^{-1/2}(K + \lambda I)\mathcal{D}^{-1/2} = \begin{pmatrix} I & Z^* \\ Z & I \end{pmatrix}. \tag{74}$$

The least eigenvalue of the self-adjoint operator on the right is $1 - \|Z\|$. Indeed, the singular-value decomposition of Z reduces each nontrivial invariant block to $\left(\begin{smallmatrix} 1 & \sigma \\ \sigma & 1 \end{smallmatrix}\right)$, whose eigenvalues are $1 \pm \sigma$; any unmatched directions have eigenvalue one. By Definition III.12.1, $\|Z\| = \sqrt{\theta_\lambda(q)}$.

The coercive comparison gives $K + \lambda I \succeq (\gamma + \lambda)I$. The two diagonal blocks are compressions of $K + \lambda I$, so

$$\mathcal{D} \preceq (\Lambda + \lambda)I.$$

For every vector w, therefore,

$$\begin{aligned} \left\langle w, \mathcal{D}^{-1/2}(K + \lambda I)\mathcal{D}^{-1/2} w \right\rangle &\geq (\gamma + \lambda)\|\mathcal{D}^{-1/2} w\|^2 \\ &\geq \frac{\gamma + \lambda}{\Lambda + \lambda}\|w\|^2. \end{aligned}$$

Taking the least eigenvalue in Eq. (74) yields

$$1 - \sqrt{\theta_\lambda(q)} \geq r_{\gamma,\Lambda,\lambda},$$

which proves Eq. (71). Since $0 < r_{\gamma,\Lambda,\lambda} \leq 1$,

$$1 - \theta_\lambda(q) \geq 1 - (1 - r_{\gamma,\Lambda,\lambda})^2 = 2r_{\gamma,\Lambda,\lambda} - r_{\gamma,\Lambda,\lambda}^2 \geq r_{\gamma,\Lambda,\lambda}.$$

This is Eqs. (72) and (73). The identity $E_q = -B$ from Definition III.12.1 separates the exact case from the nonzero-defect case. In the latter case the strict inequality $\theta_\lambda(q) < 1$, the transfer bound, and the represented target-functional comparison give precisely the reversible dynamic-stability record of Definition III.18.1. The master normal form in Theorem III.18.9 then assigns the record to the exact or reversible compensated nonidentity quotient family at its complete saturated cost.

□

Corollary III.12.7 (Strict meaning of ambient geometry after quotient exhaustion)**.** *Fix a saturated budget B, and suppose that the exact and reversible compensated nonidentity quotient envelopes are bounded by α_n as in Eq. (195). Let $q \neq \mathrm{id}$ be a represented terminal-compatible coarsening of a reversible geometric record satisfying Eq. (70), with $(\Lambda+\lambda)/(\gamma+\lambda)$ in the transfer class. Assume that the quotient and every compressed, within-fiber, utility, target-functional, and active-geometry datum in Definition III.18.1 have total saturated cost at most B. Then*

$$\overline{A}_q\, M_{\mathfrak{G}_q}\, C_{\mathfrak{G}_q}\, \frac{R_{\mathfrak{G}_q}}{1+\rho_{\mathfrak{G}_q}} \leq \alpha_n\, \frac{\Lambda+\lambda}{\gamma+\lambda}. \tag{75}$$

Put

$$\beta_n(q) = \min\left\{1, \alpha_n \frac{\Lambda+\lambda}{\gamma+\lambda}\right\}. \tag{76}$$

For arbitrary thresholds $a, c, r, g \in (0,1]$ satisfying

$$acrg > \beta_n(q), \tag{77}$$

the same represented coarsening cannot satisfy all five conditions

$$M_{\mathfrak{G}_q} = 1, \qquad \overline{A}_q > a, \qquad C_{\mathfrak{G}_q} > c, \qquad R_{\mathfrak{G}_q} > r, \qquad \frac{1}{1+\rho_{\mathfrak{G}_q}} > g. \tag{78}$$

In particular, every such coarsening satisfies the symmetric obstruction

$$M_{\mathfrak{G}_q} = 0 \quad \text{or} \quad \min\left\{\overline{A}_q, C_{\mathfrak{G}_q}, R_{\mathfrak{G}_q}, \frac{1}{1+\rho_{\mathfrak{G}_q}}\right\} \leq \beta_n(q)^{1/4}. \tag{79}$$

Thus, after quotient exhaustion, an ambient geometry can remain above scale $\alpha_n(\Lambda+\lambda)/(\gamma+\lambda)$ only when every affordable nonidentity spectral or level-set coarsening loses that scale in a represented quotient-utility, target-reach, descended-geometry, or target-functional gate. Coercive regularity itself cannot be the failed compensation gate. This is the quantitative meaning of quotient-incompressible ambient geometry.

Proof. If $E_q = 0$, the master stability factor is one. If $E_q \neq 0$, Theorem III.12.6 gives

$$S_q^{\mathrm{rev}}(\lambda) \geq \frac{\gamma+\lambda}{\Lambda+\lambda}.$$

In either case the master normal form places the coarsening in the exact or reversible compensated nonidentity quotient envelope. By Eq. (124), its visibility is

$$V_{\mathscr{C}} = \overline{A}_q S_q(\lambda) M_{\mathfrak{G}_q} C_{\mathfrak{G}_q} \frac{R_{\mathfrak{G}_q}}{1+\rho_{\mathfrak{G}_q}} \leq \alpha_n.$$

Dividing by the displayed stability lower bound proves Eq. (75). Hence any coarsening that retains a larger unconditioned geometric product would contradict the nonidentity quotient-envelope bound.

Every factor in Eq. (75) lies in $[0,1]$, and the mixing factor is binary. If Eq. (78) held under Eq. (77), their product would exceed $\beta_n(q)$, contradicting Eqs. (75) and (76). Taking $a = c = r = g = \beta_n(q)^{1/4}$ gives Eq. (79).

□

Figure 29 summarizes the three admissible routes from an ambient generator to a dynamically qualified quotient certificate.

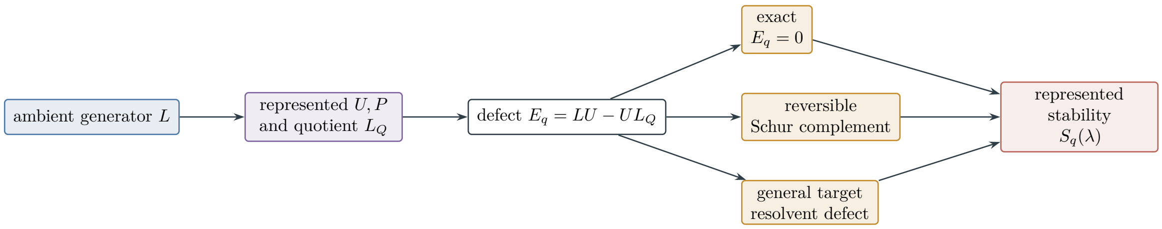

Figure 29: A quotient is dynamically qualified through a charged route: exact intertwining, reversible Schur-complement compensation, or a general target-functional resolvent estimate. All routes terminate in the same stability factor and use the same represented ambient generator.

III.12.4 Causal alignment under the authority envelope

The directed contribution is quantified by the alignment of the selected authorized current with the represented potential.

Definition III.12.8 (Quantitative Caus alignment)**.** Let $\mathcal{H}_E$ be the represented weighted edge space used by the Hodge decomposition, let $J \in \mathcal{H}_E$ be a represented directed current, and let V be a represented scalar potential. Define

$$\mathrm{Align}_{\mathsf{Caus}}(J, V) = \begin{cases} \dfrac{(\langle J, -\nabla V\rangle_E)_+^2}{\|J\|_E^2 \|\nabla V\|_E^2}, & \|J\|_E \|\nabla V\|_E > 0, \\ 0, & \text{otherwise,} \end{cases}$$

where $a_+ = \max\{a, 0\}$. This number lies in $[0, 1]$. It is one for the canonical descending current $J = -\nabla V$, and it vanishes when the represented current has no descending component along V. The construction cost of V, the edge structure, the weights, and the current is included in the saturated certificate.

This alignment factor retains only causal descent supported by the declared authority record.

Definition III.12.9 (Authority-compatible Caus alignment)**.** Let $\mathfrak{K}$ be a causal provenance record for a target-normalized potential V. If $\mathfrak{K}$ verifies that its current $J_{\mathfrak{K}}$ is authorized by Definition II.4.4 and is transported through every quotient, coordinate change, and lift in the certificate, set

$$\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, V) = \mathrm{Align}_{\mathsf{Caus}}(J_{\mathfrak{K}}, V).$$

Set the value to zero when the current identity, target polarity, or dynamic transport record is absent. The authority-compatible local factor is

$$C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(V) = \chi^{\mathrm{auth}}_{\Phi}(V)\, \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, V),$$

where $\chi^{\mathrm{auth}}_{\Phi}(V) = 1$ only when the represented local-consistency, drift, or carrier comparison is evaluated for the same full generator that authorizes $\mathfrak{K}$. More precisely, the record must contain and verify a displayed quantitative implication for that generator—for example a drift margin $-LV \geq a > 0$ on the nonterminal sector, a resolvent/form comparison with declared constants, or the authorized carrier inequality (81). The margin and every comparison loss must lie in the declared transfer class. Mere construction of V, or the formal identity $J = -\nabla_{\Phi} V$, does not set the indicator to one. This factor lies in $[0, 1]$.

Lemma III.12.10 (Caus drift survives a controlled quotient defect)**.** *Let L be the generator of a finite Markov process with absorbing target T, and let $q : X \to Q$ be a represented terminal-compatible quotient. Assume that L is selected from a declared Caus authority envelope, that $L_Q = P_q L U_q$, and that the current paired with V has a causal provenance record $\mathfrak{K}$ transported from the same selected generator. Let $V : Q \to [0, \infty)$ vanish on the quotient target. Suppose that, on every nonterminal quotient state,*

$$-L_Q V \geq a > 0,$$

and that the pointwise defect action satisfies

$$\|E_q V\|_{\infty,N} \leq \delta < a.$$

Then the lifted potential satisfies

$$-L(UV) \geq a - \delta \qquad \text{on } N = X \setminus T. \tag{80}$$

If τ_T is the first hitting time of T, then

$$\mathbb{E}_x \tau_T \leq \frac{(UV)(x)}{a-\delta}, \qquad \mathbb{E}_{\mu_0} \tau_T \leq \frac{\mathbb{E}_{\mu_0}[UV]}{a-\delta}.$$

The potential gives a Caus *certificate when $C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(V) > 0$. Exact lumpability is the case $\delta = 0$; geometric compensation is the case in which represented local regularity makes the positive margin $a - \delta$ affordable and verifiable. A current with no authority-compatible transport record cannot be substituted into this conclusion.*

Proof. The defect identity gives

$$L(UV) = U(L_Q V) + E_q V.$$

The two hypotheses therefore imply (80). Dynkin's formula for the stopped process gives

$$\mathbb{E}_x[(UV)(X_{t\wedge\tau_T})] = (UV)(x) + \mathbb{E}_x \int_0^{t\wedge\tau_T} L(UV)(X_s)\, ds.$$

The left-hand side is nonnegative, so $(a-\delta)\mathbb{E}_x[t \wedge \tau_T] \leq (UV)(x)$. Monotone convergence as $t \to \infty$ proves the first hitting-time bound; integration against μ_0 proves the second [61]. $\square$

III.13 Sectorial and Continuum Interfaces

The sectorial interface transports the same causal-authority record to the conditional continuum realization.

Definition III.13.1 (Sectorial Caus-authority interface)**.** Let $H = L^2(X, \mu)$, let $\mathcal{D} \subseteq H$ be a dense common form domain, and let

$$\mathcal{E} = \mathcal{E}^{\mathsf{s}} + \mathcal{E}^{\mathsf{a},\mathrm{imp}} + \mathcal{E}^{\mathsf{a},\mathrm{ctrl}} + \mathcal{E}^{\mathsf{b}}$$

be a represented closed sectorial Markov form on $\mathcal{D}$. Here $\mathcal{E}^{\mathsf{s}}$ is symmetric and nonnegative, $\mathcal{E}^{\mathsf{a},\mathrm{imp}}$ is the imposed antisymmetric form, $\mathcal{E}^{\mathsf{a},\mathrm{ctrl}} \in \mathfrak{E}^{\mathsf{a},\mathrm{ctrl}}_{I,\leq B}$ belongs to a declared affordable control family, and $\mathcal{E}^{\mathsf{b}}$ is the symmetric nonnegative boundary or killing form. The selected antisymmetric form must satisfy a represented sector estimate

$$\left|\mathcal{E}^{\mathsf{a},\mathrm{imp}}(u,v)+\mathcal{E}^{\mathsf{a},\mathrm{ctrl}}(u,v)\right|\le K_{\mathrm{sec}}\,\mathcal{E}^{\mathsf{s}}_{1}(u,u)^{1/2}\mathcal{E}^{\mathsf{s}}_{1}(v,v)^{1/2},\qquad \mathcal{E}^{\mathsf{s}}_{1}(u,u)=\mathcal{E}^{\mathsf{s}}(u,u)+\|u\|_2^2.$$

The associated generator uses the sign convention $\mathcal{E}(u,v)=-\langle Lu,v\rangle$ on its operator domain.

A form-compatible quotient has an isometric pullback $U_q : H_Q \to H$ with $U_q\mathcal{D}_Q \subseteq \mathcal{D}$, adjoint projection $P_q = U_q^*$, and compressed forms

$$\mathcal{E}^{\sharp}_{Q}(u,v)=\mathcal{E}^{\sharp}(U_q u,U_q v),\qquad \sharp\in\{\mathsf{s},\mathsf{a},\mathsf{b}\}.$$

For $u\in\mathcal{D}_Q$ and $v\in\mathcal{D}$ with $P_q v\in\mathcal{D}_Q$, define the form defects

$$\langle\mathfrak{E}^{\sharp}_{q}u,v\rangle=\mathcal{E}^{\sharp}(U_q u,v)-\mathcal{E}^{\sharp}_{Q}(u,P_q v),\qquad \sharp\in\{\mathsf{s},\mathsf{a}\}.$$

They are elements of the dual form space $\mathcal{D}^*$. The antisymmetric defect is the continuum causal transport defect.

A represented carrier interface consists of functionals $\mathcal{K},\mathfrak{C},\Phi,R$ on a common admissible profile domain such that, along every declared admissible profile $t\mapsto V(t)$,

$$\frac{d}{dt}\mathcal{K}(V(t))\le-\mathfrak{C}(V(t))+\Phi(V(t))+R(V(t)),\qquad \mathfrak{C}(V(t))\ge 0. \tag{81}$$

A quotient, rescaling, compactification, or coordinate change is legal for this interface when it transports the imposed and controlled forms by the displayed compression rule, preserves the target and declared capacity, and descends (81). Every discrepancy is recorded in $\mathfrak{E}^{\mathsf{s}}_{q}$, $\mathfrak{E}^{\mathsf{a}}_{q}$, or the represented remainder R, with its derivation and comparison cost charged.

Lemma III.13.2 (Conditional continuum realization of Caus authority)**.** *Assume Definition III.13.1. The closed form determines a sectorial generator whose antisymmetric component has the Caus authority envelope*

$$\mathcal{E}^{\mathsf{a},\mathrm{imp}}+\mathfrak{E}^{\mathsf{a},\mathrm{ctrl}}_{I,\le B}.$$

Every legal form-compatible quotient or coordinate transformation therefore produces the authority-compatible operator record required by the master quotient–geometry construction: its symmetric geometry is the compression of $\mathcal{E}^{\mathsf{s}}$, its imposed Caus component is the compression of $\mathcal{E}^{\mathsf{a},\mathrm{imp}}$, and its component defects are $\mathfrak{E}^{\mathsf{s}}_{q}$ and $\mathfrak{E}^{\mathsf{a}}_{q}$. A different transformed antisymmetric form is admissible only as a declared control or as a new dynamic presentation.

In a finite, Galerkin, finite-element, or other represented finite-rank realization, this form record supplies the dynamic and authority slots of a master certificate. It becomes a master certificate of Definition III.18.1 only when the target-normalized potential, reach, utility, and required comparison estimates are also represented. In the continuum it supplies the corresponding closed-form slots under the same condition. Effective-resistance, sector, commutator, hypocoercive, positive- commutator, or flux/dissipation estimates may supply its comparison record when their domains, closures, constants, and transport maps are represented. The theorem transfers provenance and compatibility; the compactness, capacity, rigidity, and realization hypotheses of a specific continuum problem remain hypotheses of its application theorem.

Proof. The representation theorem for closed sectorial forms gives the generator on the common form domain [44]. The symmetric and antisymmetric parts are fixed by polarization, and the sector bound makes the selected antisymmetric form a form-bounded perturbation of $\mathcal{E}_1^{\mathsf{s}}$. Compression by U_q gives the three quotient forms in the definition. Subtracting their pullbacks from the ambient forms gives the displayed dual-form defects, so the imposed antisymmetric part cannot change without a declared control or a change of presentation.

In a finite-rank realization, the forms are represented matrices and their component defects agree with Definition III.11.4. When the remaining current and target-potential data are represented, their causal gate is precisely Definition III.12.9. In the continuum, the corresponding operator identities hold in the form/dual-form pairing. Any additional routing or carrier theorem is usable precisely when its represented comparison data control these defects and descend through the same interface. This proves the conditional bridge.

□

Corollary III.13.3 (Authorized flux–dissipation polarity)**.** *Suppose $t \mapsto \mathcal{K}(V(t))$ is absolutely continuous, the terms below are integrable on finite time intervals, and a legal carrier interface satisfies, for some $\beta \geq 0$,*

$$\Phi_+(V) \leq \beta\, \mathfrak{C}(V)$$

on its reduced represented profile domain, with no positive-flux mode in $\ker \mathfrak{C}$*. Then*

$$\mathcal{K}(V(T)) + (1-\beta) \int_0^T \mathfrak{C}(V(t))\, dt \leq \mathcal{K}(V(0)) + \int_0^T R(V(t))\, dt.$$

For $\beta < 1$, the imposed dynamics are draining with a quantitative margin. For $\beta = 1$, the comparison leaves the represented critical carrier sector to an equality or rigidity analysis. For $\beta > 1$, this carrier supplies no draining conclusion. These alternatives are invariant under every legal authority-preserving transformation.

Proof. Substitute $\Phi \leq \Phi_+ \leq \beta\mathfrak{C}$ into (81) and integrate. Legality of the transformation makes the descended carrier identity identical up to the recorded remainder, so the same estimate holds after quotienting or changing coordinates.

□

Remark III.13.4 (Imposed dissipation and transport)*.* For a dissipative evolution, viscosity or friction belongs to the declared symmetric form. Conservative transport belongs to the imposed antisymmetric form when it is skew under the declared pairing and boundary conditions; any remaining symmetric or residual part is retained in its corresponding ledger. Constraints enter through their represented boundary, lift, or remainder interfaces. A legal transformation may reveal a target-aligned quotient or potential, but it preserves the sign and provenance of the carrier identity. In particular, a dissipative term cannot be replaced by a favorable directed current.

III.13.1 Local geometric form comparison

We next bound the resolvent effect of a represented perturbation of the local Dirichlet geometry.

Lemma III.13.5 (Local Dirichlet-form error for the geometric component)**.** *The comparison uses only represented objects already defined: the Part II* Geom *component of a budget-admissible generator and a declared geometry Φ already eligible for $G_n^{\mathrm{sat}}(B)$ at its saturated public cost. The scalar below is charged auxiliary comparison data. It does not define a structural component,* *invariant, constructor, or admissible object, but it may be used at budget B only when the common form domain, the compression to W, and a derivation or proof of the displayed bound are represented within the declared comparison budget.*

Let $N = X \setminus T$ be the nonterminal sector and let $H_N = L^2(N, \mu|_N)$. For a budget-admissible killed generator L, let L^{Geom} be its Part II geometric component restricted to N. This is the nonnegative paired local-rate component of the fixed-space split; it is not the symmetric part of an arbitrary nonreversible generator. Write its Dirichlet form as

$$\mathcal{E}_L^{\mathsf{Geom}}(f,g) = \frac{1}{2} \sum_{x,z \in N} c_L^{\mathsf{Geom}}(x,z)(f(x)-f(z))(g(x)-g(z))\mu(x)\mu(z).$$

Let $\Phi = (E_\Phi, c_\Phi)$ be a declared geometry with Dirichlet form $\mathcal{E}_\Phi$. Let $W \leq H_N$ be a finite target-oriented observable subspace, typically $W = \mathcal{M}(D)|_N$. For $\lambda > 0$, set the relative form error

$$\delta_{\lambda,W}(L,\Phi) = \sup_{0 \neq f \in W} \frac{|\mathcal{E}_L^{\mathsf{Geom}}(f,f) - \mathcal{E}_\Phi(f,f)|}{\mathcal{E}_\Phi(f,f) + \lambda \|f\|_{H_N}^2}.$$

The denominator is strictly positive on $W \setminus \{0\}$. This number records the relative form error on the verified target-oriented window. It contributes no separate term to $I_n^{\mathrm{sat}}(B)$; the geometric/reversible data remain the declared geometries, potentials, defects, and value readouts in $G_n^{\mathrm{sat}}(B)$ at their least saturated public cost.

Proof. The Part II fixed-space split represents the geometric component by a symmetric nonnegative conductance c_L^{Geom} on the exposed local relation, so the displayed Dirichlet form is a finite symmetric positive semidefinite form on H_N. The declared geometry Φ gives another such form on the same finite space. For $\lambda > 0$, the denominator $\mathcal{E}_\Phi(f,f) + \lambda\|f\|^2$ is positive for every nonzero $f \in W$; hence the supremum defining $\delta_{\lambda,W}$ is finite. The lemma only introduces this scalar comparison between two already represented forms. Since no new observable, edge relation, quotient, or readout is introduced, the scalar is auxiliary comparison data and makes no separate contribution to the saturated invariant. Its evaluation or proof cost nevertheless belongs to the comparison-data charge specified in the statement.

□

Proposition III.13.6 (Geometric Dirichlet-form comparison of resolvents)**.** *Let A_L^W and A_Φ^W be the self-adjoint positive operators on W representing the compressed forms $\mathcal{E}_L^{\mathsf{Geom}}|_W$ and $\mathcal{E}_\Phi|_W$. If $\delta = \delta_{\lambda,W}(L,\Phi) < 1$, then, as quadratic forms on W,*

$$(1-\delta)(A_\Phi^W + \lambda I) \preceq A_L^W + \lambda I \preceq (1+\delta)(A_\Phi^W + \lambda I).$$

Consequently,

$$(A_L^W + \lambda I)^{-1} \preceq \frac{1}{1-\delta}(A_\Phi^W + \lambda I)^{-1}.$$

Thus a comparison proof for the already-declared geometry Φ transfers to the actual geometric component of L with loss at most $(1-\delta)^{-1}$ on the verified window. The transfer is usable at the declared budget when this loss lies in the transfer class. For $\mathcal{B}_{\mathrm{poly}}$, this is the condition $1 - \delta \geq 1/\operatorname{poly}(n)$. When the declared transfer condition fails, the altered conductance cannot borrow the comparison proof for Φ; it must be represented as its own Geom datum and charged to $G_n^{\mathrm{sat}}(B)$ at its least saturated public cost, or it remains outside the exposed within-budget structure.

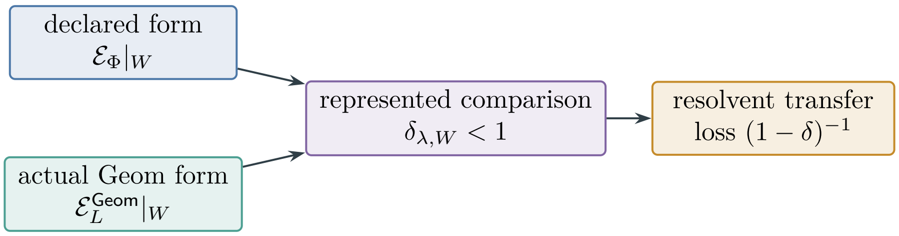

Figure 30: A declared geometry controls the actual geometric resolvent only on the represented target-oriented window W, with the explicit relative-form loss from Lemma III.13.5 and Proposition III.13.6. The comparison introduces no additional structural channel.

Proof. For every $f \in W$, the definition of δ gives

$$|\langle f, (A_L^W - A_\Phi^W)f\rangle| \le \delta\langle f, (A_\Phi^W + \lambda I)f\rangle.$$

Adding $\langle f, (A_\Phi^W + \lambda I)f\rangle$ to the middle term gives the two form inequalities. Since W is finite dimensional and $\lambda > 0$, both shifted operators are strictly positive. Inverting positive operators reverses Loewner order, which gives the displayed resolvent comparison. □

Figure 30 summarizes the local form comparison used in Proposition III.13.6.

Corollary III.13.7 (Form comparison caps nonlinear geometric gain)**.** *Let $y_W = \Pi_W y$. Under the hypotheses of Proposition III.13.6,*

$$\frac{\langle y_W, (A_L^W + \lambda I)^{-1} y_W\rangle}{\|y\|^2} \le \frac{1}{1 - \delta_{\lambda,W}(L,\Phi)} \frac{\langle y_W, (A_\Phi^W + \lambda I)^{-1} y_W\rangle}{\|y\|^2}.$$

If, in addition, represented hitting-functional comparison data bound the full killed target resolvent by these compressed quadratic forms, then Section III.8 converts the resulting full resolvent estimate into the corresponding arrival estimate, with the same form-comparison loss and the already declared transfer-class constants.

Proof. Apply the resolvent inequality from Proposition Geometric Dirichlet-form comparison of resolvents to the vector y_W and divide by $\|y\|^2$. The compressed quadratic-form inequality alone does not identify a hitting probability. Under the additional comparison hypothesis in the statement, it bounds the relevant full killed resolvent; Lemma III.8.7 then gives the asserted arrival comparison. □

III.13.2 Nonlinear closure and the quotient spectral split

The final phase classifies nonquotient nonlinear constructions and records the spectral split induced by a represented quotient.

Proposition III.13.8 (Nonquotient nonlinear constructions are geometric or residual)**.** *Let a presentation-relative uniform oracle-free budget-admissible construction introduce a nonlinear observable, feature map, score, metric, or update rule inside the lifted configuration space. Any finite observable postprocessing, coarsening, thresholding, or relabeling of that object is itself another admissible feature map at the corresponding saturated cost. If such a postprocessing* *satisfies the represented-fiber, terminal-event, exact-compatibility, quotient-utility, and nontrivial-public-structure clauses of Definition III.15.1, it is a qualified* Abs *quotient and is counted by* $m_n^{\mathrm{sat}}(B)$ *at the least saturated budget containing its complete uniform record. A prospective use is further restricted by Definition III.21.1. After all qualified quotient postprocessings have been removed, every remaining target-relevant exposed contribution of the construction is accounted for by one of the following alternatives.*

1. *It induces a pointwise geometry, defect observable, directed potential, or scalar terminal/support coloring of least saturated representation cost* k*. Then its Geom/Caus contribution is one of the terms counted by* $G(k)$*. A Bdry coloring or readout remains in the complete carrier; its conditional target charge is bounded by the represented source only when a contextual transfer inequality is supplied.*
2. *It claims to improve an existing declared geometry* Φ *only through its actual geometric conductance. Then the comparison with* Φ *is valid on a target-oriented window* W *only with the form-comparison loss* $(1-\delta_{\lambda,W})^{-1}$ *from Corollary III.13.7. If this loss is not in the declared transfer class, no budgeted hitting comparison transfers from* Φ*; the altered conductance must be counted as its own geometry at every budget in its admissible budget set or treated by the remaining alternatives.*
3. *It has no exposed Geom/Caus/Abs/Lift/Bdry provenance in the observable or lifted algebra. Then, after the Part II projections, its contribution is the residual term* J_{res}*, presentation-inaccessible or oracle-dependent information, nonuniform advice, or over-budget representation work.*

Hence the decomposition contains no additional nonlinear component between valid quotient structure and geometry. A nonlinear construction either enforces exact lumpability and is charged to $m_n^{\mathrm{sat}}(B)$ *at the budget representing its quotient, or it supplies exposed geometric/causal data together with possible scalar boundary colorings and is charged to* $G_n^{\mathrm{sat}}(B)$ *exactly when* $B \in \mathfrak{B}_{\mathrm{sat},n}(R_{\mathsf{Geom}})$*, or it is residual, inadmissible, or outside the declared budget.*

Here every boundary or lifted occurrence remains in the complete carrier. Its separate contextual charge is dominated by the displayed Geom/Caus source only when a represented transfer comparison is part of the record.

Proof. Embed the computation as a finite Markov generator on its bounded clocked transcript carrier. Part II exhaustiveness decomposes the induced movement into Geom, Caus, Abs, valid Lift, Bdry, and J_{res}. The preliminary closure under finite observable postprocessing removes every valid Abs quotient derivable from the nonlinear object. Valid Lift creates no primitive target channel and is retained with the structure it lifts; Bdry supplies scalar terminal/support colorings or boundary values and is retained with the source that made those values exposed. This is carrier-cost and provenance accounting; their contextual target charges transfer only through a represented comparison. Therefore any remaining exposed target-relevant movement is Geom or Caus, represented by a conductance, pointwise observable, directed potential, or edge difference generated from the lifted observable algebra; Bdry can only color or read the states on which that represented movement is evaluated. By definition of saturated representation degree, that datum enters $G_n^{\mathrm{sat}}(B)$ exactly for budgets $B \in \mathfrak{B}_{\mathrm{sat},n}(R_{\mathsf{Geom}})$ when its stated admissibility conditions hold. If the construction is presented only as a perturbation of an existing geometry, the preceding form-comparison lemma is exactly the finite-dimensional condition under which the existing geometric comparison proof transfers. If the condition fails, the perturbation does not yield an uncharged geometric improvement; it must be represented and counted as its own geometry or else remains outside the exposed within-budget structure. The residual and presentation-inaccessible cases are the remaining alternatives in Part II's decomposition.

□

Proposition III.13.9 (Quotient projection gives the spectral split)**.** *A degree-d quotient represents the component $P_d y$ of the centered target contrast. In the Walsh basis, the orthogonal complement has squared mass*

$$\|(I - P_d)y\|^2 = \sum_{|S|>d} |\widehat{y}(S)|^2.$$

The represented quotient captures the normalized mass

$$\frac{\|P_d y\|^2}{\|y\|^2} = m(d).$$

Proof. The represented degree-d quotient subspace is the range of the orthogonal projection P_d. Hence $P_d y$ is the component of y represented by quotient observables of degree at most d, while $(I - P_d)y$ is orthogonal to every such observable. Parseval's identity in the Walsh basis gives the displayed tail formula. Dividing by $\|y\|^2$ gives the normalized represented mass.

□

Example III.13.10 (Clause buckets and lumpability)**.** Grouping SAT assignments by equal violated-clause count gives represented score buckets when the score is available. Exact lumpability requires more: all assignments in one bucket must have the same quotient-visible transition behavior and the same terminal status after projection. Assignments with the same defect can have different clause patterns and different target access under a budget-admissible operator. The defect operator E_q measures this within-bucket mismatch.

III.14 Combined Geometric and Quotient Extraction Functional

The spectral bound handles represented quotient extraction. Pointwise geometric extraction has a separate regularity condition. The combined functional records the target-relevant information available under a saturated public cost budget. The primary variable is a budget B; the notation d is used only as a cost level, for example $d = \lceil \log_2 B \rceil$. Neither B nor d is raw Walsh degree or mere measurability.

III.14.1 Saturated representation cost and the three-layer ledger

We first place every represented structural object and its execution record on one saturated resource ledger.

Definition III.14.1 (Saturated budget-admissible representation cost)**.** Fix a budget structure $\mathcal{B} = (\mathcal{R}, \mathrm{cost}, b, \mathfrak{C})$ and a cost level $B \preceq b(n)$. The saturated budget-admissible representation class $\mathcal{R}^{\mathrm{sat}}_{n,\leq B}$ consists of all target-relevant observable representatives generated from the presented observable algebra, declared public parameters, and legal lifted closure by the declared oracle-free constructors with public cost at most B. At the declared ceiling these representatives constitute $\mathcal{R}^{\mathcal{B}}_n$; for the polynomial instance $\mathcal{B}_{\mathrm{poly}}$, the union over polynomial ceilings recovers $\mathcal{R}^{\mathcal{B}_{\mathrm{poly}}}_n$, not the entire ambient measurable envelope. When the representative includes a forward transition rule, its uniform code is allowed, but every instance-dependent datum read by that rule must lie in the same cost-filtered budgeted closure. These representatives include pointwise observables, geometries, potentials, represented quotients, valid lifted interfaces, boundary readouts, forward observable transport kernels, and generator components after Part II channel projection. The charged public cost includes description length, effective dimension, dense-basis scan cost, represented fiber operations, charged public basis-change data, lifted memory/interface data, comparison data, terminal readout data, and finite postprocessings used as observable structure.

The constructors describe the resulting fibers, potentials, conductances, forward transition rules, interfaces, terminal value readouts, finite-horizon trajectory statistics, or component-projected generator components as observable objects. Executing a named solver provides no constructor for its exact future hitting behavior, first-hitting partition, committor, occupation law, or resolvent unless that semantic object is represented explicitly. A finite-horizon computation generated from budget-admissible data is instead charged at its saturated computation cost. In the general resource order, the saturated cost of a target-relevant component R is the admissible-budget set and its Pareto frontier from Definition III.1.3. After a monotone scalar cost κ has been fixed, its scalar saturated cost is

$$\mathrm{Cost}^{\kappa}_{\mathrm{sat},n}(R) = \inf_{\rho\colon [\![\rho]\!]_I = R_I \ \ \forall I \in \mathfrak{I}_n} \ \sup_{I \in \mathfrak{I}_n} \kappa(\mathrm{Cost}_I(\rho))\,.$$

We omit κ after fixing that scalarization. The optional log-cost degree is $\deg_{\mathrm{sat},n}(R) = \lceil \log_2(1 + \mathrm{Cost}_{\mathrm{sat},n}(R)) \rceil$. Thus $\mathcal{R}^{\mathrm{sat}}_{n,\le d}$ means $\mathcal{R}^{\mathrm{sat}}_{n,\le 2^d}$ when log-cost notation is used. This is an objective infimum over the saturated class and is attained under the locally finite discrete scalar-cost convention. It does not depend on whether the representative has been discovered or whether optimality has been proved.

The budget frontier, or the scalar cost after scalarization, is the budget-admissible representation cost. A component may have high raw Walsh degree and low saturated cost when a valid quotient, basis change, lift, forward transport, or legal finite-horizon computation generated from the budgeted saturated observable closure represents it compactly. Broad measurability in the finite state space is insufficient: a set-theoretic description or an exact semantic future value may exist without any budgeted saturated representative. Conversely, if a denoted object is produced by a presentation-relative within-budget computation from $\mathcal{R}_n^{\mathcal{B}}$ and the legal lifted closure, that computation is a budget-admissible representative at its saturated computation cost by Lemma Budgeted generation gives budgeted representation. Therefore every admissible budgeted datum belongs to the budgeted saturated closure. Broad measurability gives an ambient set-theoretic description, whereas membership in this closure gives admissible budgeted structure.

The three ledgers separate displayed, represented, and saturated resource accounting.

Definition III.14.2 (Three layers of cost accounting)**.** Every object used in the hitting-time bounds is evaluated through three accounting layers, all evaluated relative to the same source of legal instance-dependent data: $\mathcal{R}_n^{\mathcal{B}}$.

Provenance. The provenance account records whether the object is generated from $\mathcal{R}_n^{\mathcal{B}}$, declared public parameters, public lifted coordinates, and oracle-free constructor rules. For transition rules, this account records whether all instance-dependent data read by the uniform code have that provenance. It determines whether the object belongs to the budgeted observable algebra of the original presentation or instead requires presentation-inaccessible information, nonuniform advice, oracle access, over-budget recovery, or an enlarged primitive signature.

Execution. The execution account records the resources needed by a named transition rule, simulation, evaluator, or update procedure under the budget-compatible resource envelope, after the provenance account has fixed the admissible input data. This account determines whether a program or induced kernel is executable within the declared budget for the presented task.

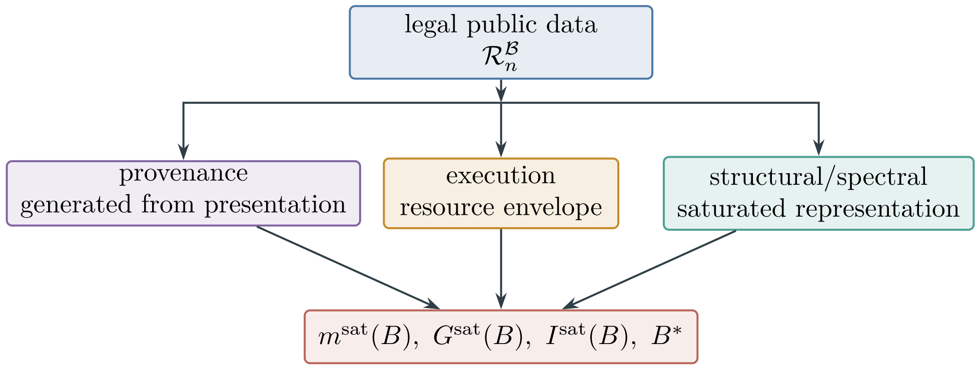

Figure 31: Provenance, execution, and structural/spectral representation are three accounts of the same legal data source and resource ledger, as defined in Definition III.14.2.

Budget-compatible execution relative to an uncharged modulus, trapdoor, target table, over-budget recovered relation, or advice string is budget-compatible execution for a different input object, not for the original presentation.

Structural/spectral representation. The structural account records the least saturated public cost B, or equivalently the least log-cost level d, at which the denoted object itself is represented in $\mathcal{R}^{\mathrm{sat}}_{n,\le B}$. This account controls $m^{\mathrm{sat}}(B)$, $G^{\mathrm{sat}}(B)$, $I^{\mathrm{sat}}(B)$, and B^*.

These accounts are distinct aspects of one budgeted closure rather than independent admissibility criteria. The underlying data source is $\mathcal{R}_n^{\mathcal{B}}$ and its legal lifted closure. Budget-compatible execution does not assign zero saturated structural/spectral cost to every semantic object attached to the execution. A finite-horizon object generated by an admissible within-budget simulation is legal budgeted data and is charged at the saturated cost of that computation; an uncomputed semantic future object is not. The execution account is evaluated only after inadmissible information has been removed or charged: a fast update rule using data outside the budgeted saturated observable closure is not admissible for the original task, even if it becomes efficient after enlarging the primitive signature. Finite charged composition of budget-admissible updates preserves legality when all leaves and constructors remain inside $\mathcal{R}_n^{\mathcal{B}}$. If the composition produces a quotient, potential, geometry, valid lifted interface, boundary readout, forward observable transport component, or finite-horizon trajectory statistic by an admissible within-budget computation, Lemma Budgeted generation gives budgeted representation places that denotation in the budgeted saturated closure at its computation cost. If it produces only an uncomputed stopped-future or exact resolvent semantic object, it is not an admissible budgeted datum.

Figure 31 displays the three accounts of Definition III.14.2 as views of one budgeted closure.

Remark III.14.3 (Structural projection of forward transport). A forward observable transport kernel is included in $\mathcal{R}^{\mathrm{sat}}_{n,\le B}$ so that an algorithm-induced transition rule is part of the saturated accounting. The extraction functional has no additional transport summand; it evaluates such a kernel by applying the Part II channel projection to the represented generator that drives the killed propagation and to the terminal edge gate W_T^L. The full-history hitting functional then integrates those projected histories as in Definition Full-history target-contact transport. A represented quotient propagation contributes to $m_n^{\mathrm{sat}}(B)$; a represented geometric or causal propagation contributes to $G_n^{\mathrm{sat}}(B)$; valid Lift and Bdry retain their full carrier fields and contextual charge unless a represented transfer bounds that charge by the source they reorganize or read; and the post-projection remainder is J_{res}. This projection is component accounting on the represented transition rule together with the exposed terminal edge gate. The projection itself neither computes a resolvent or committor nor changes the reference measure, and it does not treat the uncomputed stopped future as a representative. Every target-contact-bearing represented history is counted through the existing Geom/Caus or Abs coordinate, while uncomputed stopped-future semantics are handled by derivability accounting below.

This definition charges a committor according to the derivation that makes it available.

Definition III.14.4 (Committor derivability accounting)**.** For an induced killed generator L, the discounted committor or target value function

$$u_{\lambda,L} = (\lambda I - L_N)^{-1} k_T$$

is a semantic function on the lifted configuration graph [59]. It is counted in $\mathcal{R}^{\mathrm{sat}}_{n,\le B}$, and hence in $G^{\mathrm{sat}}_n(B)$ or boundary accounting, precisely when the same function has a representation of cost at most B by the observable structural grammar: an exposed potential or geometry, a represented quotient, a valid lifted interface, a terminal value readout, or a legal finite-horizon computation built from budget-admissible data. The operation $L \mapsto (\lambda I - L_N)^{-1} k_T$ becomes a budget-admissible representation only through an admissible constructor. A finite-horizon first-hit quotient, hit indicator, or truncated value produced by simulating the represented one-step rule for a within-budget horizon is a budget-admissible representative at its saturated computation cost. Thus the derivability closure contains computed future observables and requires an explicit derivation for each semantic object.

Lemma III.14.5 (Truncated-Neumann derivability accounting)**.** *Let P_N be a killed one-step kernel or L_N a killed generator induced by a presentation-relative oracle-free budget-admissible rule, and let k_T be the exposed terminal edge gate. For any coefficients $a_0, \dots, a_M$ and horizon $M = M(n)$ generated within the declared budget, set*

$$u_M = \sum_{j=0}^{M} a_j P_N^j k_T.$$

This expression has legal budgeted provenance exactly when P_N, k_T, and the coefficients are generated inside the budgeted saturated observable closure. A truncated finite-horizon expression that is actually generated by an admissible within-budget simulation is itself a budget-admissible representative at its saturated computation cost. It contributes at budget B precisely when that cost and its quotient or Geom/Caus/Bdry admissibility conditions place the denoted function u_M in $\mathcal{R}^{\mathrm{sat}}_{n,\le B}$. If such a representative exists, its target contribution is already included in $m^{\mathrm{sat}}_n(B)$ or $G^{\mathrm{sat}}_n(B)$. If no such representative exists at budget B, then the expression defines no within-budget structural component, although it may enter at a larger budget.

Consequently, a truncated Neumann approximation to

$$(\lambda I - L_N)^{-1} k_T \quad \textit{or} \quad (\lambda + \rho)^{-1} \sum_{j=0}^{M} \left(\frac{\rho}{\lambda + \rho} \right)^j P_N^j k_T$$

remains subject to the saturated filtration. Either the approximant is generated as a legal finite-horizon observable representative and is counted by $I^{\mathrm{sat}}_n(B)$ at its saturated computation cost, or it is only an uncomputed stopped-future denotation and has no low-cost observable representation. The theorem concerns the approximant as a quotient, potential, or boundary value; the represented generator, its Part II component projections, and the full-history hitting accounting remain part of the saturated accounting.

Proof. The expression u_M is obtained by finite composition and finite linear postprocessing of budget-admissible data. If M, the one-step rule, terminal edge gate, and coefficients lie in the budgeted saturated closure and their accumulated simulation charge lies within budget, then this computation is a budget-admissible representative of u_M at its saturated computation cost. By Definition Saturated budget-admissible representation cost, the cost charged to a denoted object is the infimum over all budget-admissible representatives of that denotation. Hence u_M belongs to $\mathcal{R}^{\mathrm{sat}}_{n,\leq B}$ exactly when its least budget-admissible representative has cost at most B; if the explicit finite computation lies within budget, the declared budget contains it; in the polynomial instance this becomes the usual fixed-exponent statement. If the representative is a quotient, its projection kernel is one of the kernels optimized in $m^{\mathrm{sat}}_n(B)$. A represented potential or geometry is optimized in $G^{\mathrm{sat}}_n(B)$. If the object is a terminal value readout or valid lifted interface over such data, its complete carrier is retained in the structural profile; it is dominated by the smaller Geom/Caus datum only through a represented contextual transfer. For budgets below its actual saturated cost, u_M is not yet counted. The full exact resolvent or infinite stopped-future value is different: without a budget-admissible finite representative or other observable representation, semantic notation for that exact future is not a low-cost observable.

□

Lemma III.14.6 (Budgeted full-history composition bound)**.** *Fix a task family and a budget-compatible resource envelope $B(n)$. Let L_n be the transcript generator of a presentation-relative uniform oracle-free computation, including the clock and every retained outcome, branch, public-history, and terminal record, with internal state, memory, and tables charged at peak native capacity. Suppose the computation has horizon at most $H_B(n)$, and let α_I be its probability of reaching the public target verifier within that horizon. Let $\Psi(B,n)$ be a uniform saturated-cost bound for the canonical first-hit quotient furnished by Lemma III.20.2. Then*

$$\alpha_I \leq eH_B(n)m^{\mathrm{sat}}_I(\Psi(B,n)). \tag{82}$$

The inequality applies to every finite composition of represented updates, postprocessings, outcome branches, valid lifted coordinates, boundary readouts, and all five channel components together with J_{res}. It does not require a componentwise resolvent expansion, a positive partial sum of channel projections, or a residual baseline assumption. Consequently, a non-negligible full-history target arrival forces non-negligible represented quotient visibility at the charged enlarged saturated budget.

Proof. Execute the operational rule once while retaining its finite outcome and terminal transcript, then replay only the terminal flags with the deadline coordinate as in Lemma III.20.2. The transcript law is exactly the law of the original computation. The execution ledger charges one peak native state, including all memory, branching, postprocessing, and component interaction, together with the accumulated transcript. The lemma constructs a uniform exactly lumpable quotient of saturated cost at most $\Psi(B,n)$ and gives

$$\alpha_I \leq eH_B(n)V_I(q_{\mathcal{A}}).$$

Since $q_{\mathcal{A}}$ is one of the quotient families in the definition of $m^{\mathrm{sat}}_I(\Psi(B,n))$, this is (82). The argument is applied before any channel expansion, so it includes J_{res} and all cancellations automatically.

□

Remark III.14.7 (Committor cases). If the presentation exposes a low-cost committor, target value observable, or hitting-time partition, then the problem has a low-cost progress observable and $G_n^{\mathrm{sat}}(B)$ or $m_n^{\mathrm{sat}}(B)$ is nonzero at that budget. If the budgeted saturated observable closure derives such a value by an admissible within-budget finite-horizon computation or by other allowed constructors, the same conclusion holds, with the value charged at its saturated computation cost. If the only description is semantic, namely "the infinite future of this induced kernel" or "the exact resolvent for this induced kernel" with no legal finite representative at the budget, then the value exists extensionally but has no admissible budgeted representation. This distinguishes semantic existence from a charged observable representation.

III.14.2 Ambient geometric extraction

The ambient coordinate combines represented Dirichlet geometry, authorized directed alignment, and target reach.

Definition III.14.8 (Static Dirichlet geometry)**.** A budget-B pointwise geometry is a pair $\Phi = (E_\Phi, c_\Phi)$, where $E_\Phi \subseteq X \times X$ is a symmetric declared comparability relation and $c_\Phi(x,z) = c_\Phi(z,x) \geq 0$ is an exposed conductance represented with saturated cost at most B. Its static Dirichlet form is

$$\mathcal{E}_\Phi(f,g) = \frac{1}{2} \sum_{x,z \in X} c_\Phi(x,z)(f(x) - f(z))(g(x) - g(z))\mu(x)\mu(z).$$

This is a quadratic form measuring variation across declared comparable states. Here it serves as a regularity functional for hitting-time estimates; Part II associates such forms with generators.

The ambient extraction coordinate measures target-visible geometry and authorized causal alignment.

Definition III.14.9 (Ambient Geom/Caus extraction coordinate)**.** Fix a Caus authority envelope $\mathfrak{A}$. Let $\mathcal{G}_{n,\leq B}^{X,\mathfrak{A}}$ be the family of represented records $(\mathfrak{N}, \Phi, D, J, \mathfrak{K})$ of total saturated cost at most B, where $\mathfrak{N}$ is the charged normalized-kernel record of Lemma III.17.5, Φ is a pointwise geometry, $D : X \to [0,\infty)$ is a progress observable vanishing on the positive terminal sector, J is a directed current on the represented edge space of Φ, and $\mathfrak{K}$ proves that J is authorized by a generator selected from $\mathfrak{A}$. In particular, the canonical current $J = -\nabla_\Phi D$ is admissible only when it is formed from the same represented conductance and potential and the same full generator supplies its local or carrier comparison, with the complete construction cost charged. Each record family is produced by one derivation scheme fixed across all sizes, and its cost bound is uniform on $\mathfrak{I}_n$, exactly as in Definition III.15.2. For $\mathrm{Var}_\mu(D) > 0$, define

$$\rho_\Phi(D) = \frac{\mathcal{E}_\Phi(D,D)}{\mathrm{Var}_\mu(D)}.$$

Let $M_\Phi \in \{0,1\}$ be the mixing indicator, equal to one when the declared geometry has a Poincare or conductance lower bound whose reciprocal lies in the declared transfer class. In the $\mathcal{B}_{\mathrm{poly}}$ instance this is the usual inverse-polynomial scale. Let $\chi_\Phi^{\mathrm{auth}}(D) \in \{0,1\}$ be the local consistency gate, equal to one when the positive-defect landscape has no represented local obstruction away from T on the declared comparison window for the same selected generator. Define the quantitative Caus factor

$$C_{\Phi,\mathfrak{K}}^{\mathrm{auth}}(D) = \chi_\Phi^{\mathrm{auth}}(D)\, \mathrm{Align}_{\mathsf{Caus}}^{\mathrm{auth}}(\mathfrak{K}, D).$$

Let $\mathcal{M}(D)$ be the closed span of monotone functions of D oriented toward $D = 0$, and define the target reach of D by

$$R_T(D) = \frac{\|\Pi_{\mathcal{M}(D)} y\|^2}{\|y\|^2}.$$

The conditioned ambient Geom/Caus visibility contribution is

$$V^X_{\Phi,D,J,\mathfrak{K}} := M_\Phi C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D) \frac{R_T(D)}{1+\rho_\Phi(D)}.$$

The authority-relative ambient geometric extraction coordinate is

$$G^{\mathrm{sat}}_{X,n,\mathfrak{A}}(B) = \sup_{(\mathfrak{N},\Phi,D,J,\mathfrak{K})\in\mathcal{G}^{X,\mathfrak{A}}_{n,\le B}} V^X_{\Phi,D,J,\mathfrak{K}}.$$

Because the free envelope ranges over every budget-admissible represented selected skew operator, one has $\mathcal{C}^{\mathrm{all}}_{QG}(B) = \mathcal{C}^{\mathrm{free}}_{QG}(B)$. Write $G^{\mathrm{sat}}_{X,n}(B)$.

The reach $R_T(D)$ measures alignment of the defect with the target, $\rho_\Phi(D)$ measures its regularity across declared comparable states, and $C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D)$ measures the descending fraction of the authorized represented current. The mixing and consistency gates impose the stated local comparison conditions for the selected generator. The corresponding target-arrival conclusion follows from an applicable arrival theorem, such as Lemma III.12.10 when a pointwise drift margin is available.

III.15 Quotient Extraction and the Derivation Normal Form

We now define exact quotient visibility and expose its complete represented derivation.

Definition III.15.1 (Part I quotient-utility condition)**.** Fix a represented-fiber quotient Q, an induced budget-admissible operator L, a legal initialization μ_0, and a hitting-time discount λ. The quotient-utility factor $A_Q = A_Q(\lambda; L, \mu_0)$ is the Part I fiber-represented-observable utility condition, transported to the comparison window. It is a number in $[0, 1]$, and it is set to zero unless all of the following structural conditions hold.

1. **Represented fibers.** Q is a fiber-represented observable in the Part I sense: fiber membership, construction of the compact fiber description, and finite Boolean operations on represented fibers have saturated cost at most the current budget. No point selection from a fiber is required.
2. **Terminal quotient event.** The target and terminal/failure sectors used by the hitting functional are unions of quotient cells.
3. **Exact dynamic compatibility.** On the target window, the induced dynamics satisfy the exact intertwining $L_X U = U L_Q$. A quotient with nonzero defect contributes only through the compensated Geom/Caus certificate of Definition III.18.1; its defect loss is not absorbed into the pure Abs coordinate.
4. **Quotient utility.** The represented quotient dynamics, represented fiber operations, or quotient-level observables have strictly positive discounted target functional from the legal initialization: $A^Q_T(\lambda; L_Q, \bar{\mu}_0) > 0$.
5. **Nontrivial public structure.** The quotient is generated from the public observable algebra and, after empty fibers are removed, its partition strictly refines the verifier partition:

$$\sigma(\mathbf{1}_T) \subsetneq \sigma(q).$$

Equivalently, at least one represented nonterminal quotient distinction is not determined by $\mathbf{1}_T$. A large target-containing cell may contribute visibility only through the normalized projection below; it need not include a selector or sampling procedure.

When these conditions hold, let $A_T^Q(\lambda; L_Q, \bar{\mu}_0)$ be the target-projected quotient hitting functional from the pushed-forward initialization $\bar{\mu}_0$, and let $C_Q \geq 1$ be the product of the charged exact-quotient comparison and normalization factors. We set

$$A_Q(\lambda; L, \mu_0) = \min\{1, \lambda C_Q^{-1} A_T^Q(\lambda; L_Q, \bar{\mu}_0)\}.$$

For the canonical pushforward measure and initialization of a terminal-compatible exact quotient, Proposition III.11.3 gives $C_Q = 1$. A nonunit factor is retained only when the quotient record explicitly uses a different represented gauge or an additional comparison map.

Thus A_Q is the normalized structural utility of a represented quotient on the active comparison window. It records exposed quotient alignment independently of any target solver.

The quotient extraction coordinate measures the target-visible value of an exact represented quotient.

Definition III.15.2 (Quotient extraction coordinate)**.** Let $\mathfrak{I}_n$ be a size-n task slice. Define $\mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\leq B}$ to be the families $\mathbf{Q} = (\mathscr{R}_I)_{I\in\mathfrak{I}_n}$ produced by one uniform exactly lumpable represented-fiber quotient derivation of saturated cost at most B uniformly on the slice. The family is the restriction to size n of one derivation scheme $\rho \in \mathsf{Der}_{\mathfrak{I}}$ fixed across all sizes and instances; the program is not re-chosen for each n. Here a carrier record

$$\mathscr{R}_I = (\Omega_I, \mu_{\Omega,I}, T_{\Omega,I}, \mu_{0,\Omega,I}, \mathfrak{N}_{\Omega,I}, q_I, Q_I)$$

contains a finite represented carrier, a legal public full-support reference law, an exposed terminal sector, a legal initialization, the charged normalization record $\mathfrak{N}_{\Omega,I} = (\widetilde{L}_{\Omega,I}, \zeta_I, c_I, P_{\Omega,I}, L_{\Omega,I}, \lambda_I)$, with $L_{\Omega,I} = P_{\Omega,I} - I$, and the represented exact quotient. The utility factor is evaluated using this same $L_{\Omega,I}$ and λ_I. Thus normalization belongs to the exact-profile cost from the outset; it is not added later as a cost-free extension. The carrier may be the base task carrier or any budget-generated clocked transcript representation produced by the computation-closure clause. Its encoding, initialization, update, target verifier, reference gauge, quotient fibers, and comparison data are all charged in the same derivation. Such a computation carrier need not satisfy the Valid-Lift projection or target-fraction identities. Those identities are required only if the record separately claims a structural Lift contribution or transports a comparison back to a base task. This uniformity excludes a separate carrier, quotient program, or fiber table for each instance. For $\mathbf{Q} \in \mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\leq B}$, let P_{Q_I} be orthogonal projection in $L^2(\Omega_I, \mu_{\Omega,I})$ onto the quotient-observable subspace on instance I, let

$$y_{\Omega,I} = \mathbf{1}_{T_{\Omega,I}} - \mu_{\Omega,I}(T_{\Omega,I}),$$

and let $A_{Q_I} \in [0, 1]$ be the Part I quotient-utility factor from Definition Part I quotient-utility condition, evaluated on the current comparison window. The conditioned quotient visibility contribution at I is

$$V_I(\mathbf{Q}) := A_{Q_I} \frac{\|P_{Q_I} y_{\Omega,I}\|^2}{\|y_{\Omega,I}\|^2},$$

with value zero when the carrier target has reference mass zero or one. The base-carrier case has $\Omega_I = X_I$ and $y_{\Omega,I} = y_I$.

When the instance and uniform family are fixed, write this quantity as V_Q.

The instance-level abstraction extraction coordinate is

$$m_I^{\mathrm{sat}}(B) = \sup_{\mathbf{Q} \in \mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n, \leq B}} V_I(\mathbf{Q}), \qquad \sup \varnothing := 0.$$

For a distinguished instance sequence (I_n), retain the shorthand $\mathcal{Q}^{\mathrm{sat}}_{n,\leq B}$ for the corresponding family of quotients and write $m_n^{\mathrm{sat}}(B) = m_{I_n}^{\mathrm{sat}}(B)$. In log-cost notation, $m_n(d) = m_n^{\mathrm{sat}}(2^d)$.

The exact quotient normal form expands a represented derivation into the record used by the profile.

Definition III.15.3 (Derivation-induced exact quotient normal form)**.** An exact quotient normal-form record is an expanded derivation $\widehat{\rho} \in \mathsf{Der}^{\mathrm{nf}}_{\mathfrak{I}}$ whose terminal vertex denotes a finite map

$$q_{\widehat{\rho},I} : \Omega_I \longrightarrow Q_{\widehat{\rho},I}$$

on the base space or on a budget-generated clocked transcript carrier. The latter is legal for quotient accounting by computation closure and does not have to be a structural Valid Lift. The record contains the carrier encoding, legal reference gauge, initialization, exposed terminal sector, and the same charged normalized-kernel record required by Definition III.15.2 in addition to the data below. The record may use a finite public label set containing unused labels; an unused label has the represented false fiber predicate, while every nonempty fiber has the representation required below. The record contains the quotient evaluation rule, represented nonempty fibers and their finite Boolean operations, the terminal cell sets, the induced quotient operator, and the comparison and normalization data entering $A_{q_{\widehat{\rho},I}}$. These data are typed attachments to the terminal quotient interface and are copied into every backward-cut record, whether they are stored as separate vertices or as fields of the terminal vertex. The fiber map and its induced operator satisfy exact intertwining. Its profile contribution is evaluated by Definition III.15.1, which assigns $A_{q_{\widehat{\rho},I}} = 0$ unless the remaining terminal, utility, and nontriviality conditions hold.

Let $\mathsf{NFQ}_{\mathfrak{I}_n,\leq B}$ be the family of such uniform normal-form records whose actual recorded witness cost satisfies $\mathrm{Cost}_I^{\mathrm{nf}}(\widehat{\rho}) \preceq B$ for every $I \in \mathfrak{I}_n$.

Thus membership bounds the chosen record itself, not merely the cost of another record with the same denotation. Local finiteness makes a Pareto-minimal normal-form witness available below any chosen finite-cost witness. The budgeted statements below use the actual normal-form witness of cost $\preceq B$, not a globally least element. Let $\mathcal{Q}^{\mathrm{nf}}_{\mathfrak{I}_n,\leq B}$ be their quotient denotations, and write $\mathsf{NFQ}_{\mathfrak{I}_n} := \bigcup_B \mathsf{NFQ}_{\mathfrak{I}_n,\leq B}$.

Lemma III.15.4 (Affordable exact quotient derivation normal form)**.** *For every presented finite task family and every saturated budget B,*

$$\mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\leq B} = \mathcal{Q}^{\mathrm{nf}}_{\mathfrak{I}_n,\leq B}. \tag{83}$$

Consequently, the instance-level quotient profile has the exact derivation-indexed representation

$$y_{\widehat{\rho},I} := \mathbf{1}_{T_{\Omega_{\widehat{\rho},I}}} - \mu_{\Omega_{\widehat{\rho},I}}(T_{\Omega_{\widehat{\rho},I}}),$$

where the corresponding summand is defined to be zero when $\|y_{\widehat{\rho},I}\|_2 = 0$, as in Definition III.15.2.

$$m_I^{\mathrm{sat}}(B) = \sup_{\widehat{\rho} \in \mathsf{NFQ}_{\mathfrak{I}_n, \le B}} A_{q_{\widehat{\rho},I}} \frac{\|P_{q_{\widehat{\rho},I}} y_{\widehat{\rho},I}\|_2^2}{\|y_{\widehat{\rho},I}\|_2^2}, \tag{84}$$

where $\sup \varnothing = 0$. *Thus the exact* Abs *profile ranges over the complete resource-filtered derivation calculus, rather than over a semantic classification of the resulting pointwise fiber algebras.*

Proof. Let $\mathbf{Q} \in \mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n, \le B}$. By Definition III.15.2, one uniform derivation of cost at most B produces its quotient map, fibers, dynamic intertwining, and utility data. Lemma III.1.9 expands that record without changing its denotation or cost, so $\mathbf{Q} \in \mathcal{Q}^{\mathrm{nf}}_{\mathfrak{I}_n, \le B}$. Conversely, contraction of a normal-form quotient record gives an admissible derivation with identical quotient data and cost. This proves (83). Substitution of the identical indexing family into Definition III.15.2 gives (84). □

III.15.1 The charged backward cut

The backward cut isolates the first joint frontier carrying all state-dependent information used by the terminal quotient.

Definition III.15.5 (Charged backward cut and joint-source load)**.** Let $\widehat{\rho} \in \mathsf{NFQ}_{\mathfrak{I}_n, \le B}$ be an expanded exact-quotient record, and let $\mathsf{G}_{\widehat{\rho}}$ be the backward slice of its terminal quotient-map vertex. A *charged backward cut* is an ordered finite antichain

$$\Gamma = (v_1, \ldots, v_k)$$

in $\mathsf{G}_{\widehat{\rho}}$ that separates every represented state-dependent input of the terminal slice from the terminal vertex. The singleton containing the terminal quotient-map vertex is permitted as a cut and separates these paths at their common terminal endpoint. Every state-dependent retained lifted coordinate or derived datum read below the cut is included among the cut outputs. State-independent public parameters may remain as represented coefficients in the downstream record.

The record carries a unique finite carrier $(\Omega_I, \mu_{\Omega,I}, T_{\Omega,I})$. Throughout this backward-cut and structural-charge subsection, write

$$y_I := y_{\Omega,I} = \mathbf{1}_{T_{\Omega,I}} - \mu_{\Omega,I}(T_{\Omega,I}),$$

and take every projection in $L^2(\Omega_I, \mu_{\Omega,I})$. This local shorthand includes the base carrier as a special case and does not replace a computation-carrier contrast by the base-task contrast.

If $f_{v,I}$ denotes the output of a cut vertex, write

$$J_{\Gamma,I} = (f_{v_1,I}, \ldots, f_{v_k,I}).$$

The cut record consists of the union of the ancestral records of the vertices in Γ, their represented joint-fiber interface, and the complete downstream subrecord terminating in $q_{\widehat{\rho},I}$. In particular, the downstream record retains the quotient operator, terminal cells, fiber predicates, comparison data, and the utility factor $A_{q_{\widehat{\rho},I}}$.

For a represented observable F, set

$$\mathsf{p}_I(F) := \frac{\|P_F y_I\|_2^2}{\|y_I\|_2^2},$$

and define the joint-source load carried across Γ by

$$\mathrm{Load}_I(\widehat{\rho},\Gamma) := A_{q_{\widehat{\rho},I}}\, \mathsf{p}_I(J_{\Gamma,I}). \tag{85}$$

The factor $A_{q_{\widehat{\rho},I}}$ belongs to the complete terminal record. Thus a public terminal discriminator contributes projection mass only with the dynamic utility supplied by the represented transport and quotient record.

The backward factorization exposes the charged sources on which a terminal quotient depends.

Theorem III.15.6 (Quantitative backward factorization through a charged joint frontier)**.** *Let $\widehat{\rho}$ and Γ be as in Definition III.15.5. Topological contraction of the downstream subrecord gives one represented family-uniform map $h_{\widehat{\rho},\Gamma,I}$ such that*

$$q_{\widehat{\rho},I} = h_{\widehat{\rho},\Gamma,I} \circ J_{\Gamma,I}. \tag{86}$$

If materializing the cut tuple requires a separate interface, the complete cut record has cost

$$\mathrm{Cost}_{I'}(\widehat{\rho}_\Gamma) \preceq B \oplus \mathrm{Cost}_{\mathrm{join}}(\Gamma) =: \Psi_{\mathrm{cut}}(B,\Gamma), \qquad I' \in \mathfrak{I}_n. \tag{87}$$

When the tuple interface is already retained in $\widehat{\rho}$, no additional charge is incurred. The map Ψ_{cut} is class preserving whenever the cut arity and represented join interface lie in the declared transfer class; in particular, it is polynomial for polynomial-cost expanded records.

For every instance,

$$\mathsf{p}_I(J_{\Gamma,I}) = \mathsf{p}_I(q_{\widehat{\rho},I}) + \frac{\|(P_{J_{\Gamma,I}} - P_{q_{\widehat{\rho},I}})y_I\|_2^2}{\|y_I\|_2^2}, \tag{88}$$

$$V_I(q_{\widehat{\rho}}) \le \mathrm{Load}_I(\widehat{\rho},\Gamma). \tag{89}$$

Moreover, put $J_{\Gamma,\ell} = (f_{v_1},\dots,f_{v_\ell})$ for $\ell \ge 1$, let $J_{\Gamma,0}$ be the constant observable, and write $P_{J_{\Gamma,0,I}} = P_{\Gamma,0}$ for projection onto constants. Define

$$\Delta_{I,\ell}(\widehat{\rho},\Gamma) := \frac{\|(P_{J_{\Gamma,\ell,I}} - P_{J_{\Gamma,\ell-1,I}})y_I\|_2^2}{\|y_I\|_2^2}. \tag{90}$$

Then

$$\mathsf{p}_I(J_{\Gamma,I}) = \sum_{\ell=1}^{k} \Delta_{I,\ell}(\widehat{\rho},\Gamma), \qquad V_I(q_{\widehat{\rho}}) \le A_{q_{\widehat{\rho},I}} \sum_{\ell=1}^{k} \Delta_{I,\ell}(\widehat{\rho},\Gamma). \tag{91}$$

Consequently, when $k \ge 1$, some conditional join extension $(J_{\Gamma,\ell-1}, f_{v_\ell})$ satisfies

$$A_{q_{\widehat{\rho},I}} \Delta_{I,\ell}(\widehat{\rho},\Gamma) \ge \frac{V_I(q_{\widehat{\rho}})}{k}. \tag{92}$$

The load-bearing object in (92) *is the charged conditional join, including its preceding context and its complete downstream terminal record. No conclusion about one cut coordinate in isolation is required.*

Proof. The terminal backward slice is a finite directed acyclic graph. Treat the outputs at Γ as inputs and evaluate the remaining vertices in topological order. The separating property ensures that every state-dependent input used by this evaluation occurs in $J_{\Gamma,I}$; represented public coefficients remain in the same downstream record. This produces $h_{\widehat{\rho},\Gamma,I}$ and proves (86). The ancestral records, the downstream evaluator, and its terminal interface are subrecords of $\widehat{\rho}$. Only a newly materialized tuple interface adds a constructor charge, which proves (87). In particular, a formula, table, recurrence, branching rule, or fiber selector used below the cut remains in the downstream subrecord with its construction, storage, evaluation, and fiber-operation costs.

Equation (86) gives $\sigma(q_{\widehat{\rho},I}) \subseteq \sigma(J_{\Gamma,I})$. Apply Lemma I.4.7 to obtain (88). Since $0 \leq A_{q_{\widehat{\rho},I}} \leq 1$, multiplication by this same terminal utility factor gives

$$V_I(q_{\widehat{\rho}}) = A_{q_{\widehat{\rho},I}} \mathsf{p}_I(q_{\widehat{\rho},I}) \leq A_{q_{\widehat{\rho},I}} \mathsf{p}_I(J_{\Gamma,I}),$$

which is (89). Finally, the spaces generated by $J_{\Gamma,0}, J_{\Gamma,1}, \dots, J_{\Gamma,k}$ are nested. Successive conditional-expectation differences are orthogonal, and $P_{\Gamma,0} y_I = 0$ because y_I is centered. Pythagoras therefore gives the first identity in (91); the second follows from (89). At least one of the k nonnegative summands is at least their average, which proves (92).

□

Remark III.15.7 (Projection ancestry and dynamic source transfer). Theorem III.15.6 is an unconditional factorization and cost-accounting result. The joint cut observable need not be lumpable, terminal compatible, or dynamically qualified. For an exact terminal-compatible quotient whose terminal sector is a union of quotient cells, the centered terminal contrast is already quotient measurable; in that case $\mathsf{p}_I(q_{\widehat{\rho},I}) = 1$, and the nontrivial quantity is the utility factor $A_{q_{\widehat{\rho},I}}$. Transferring that factor to a proper ancestral source requires the represented dynamic comparison in (93). This distinction also permits joint effects: two cut coordinates may have zero individual increments before conditioning on their joint context.

The compressed-source cover packages the finitely many source envelopes needed for a budgeted audit.

Definition III.15.8 (Quantitative compressed-source cover)**.** Let $\mathfrak{C}_{\mathrm{src}}$ be a finite set of represented source signatures and, for each $c \in \mathfrak{C}_{\mathrm{src}}$, let $\mathcal{S}_I^{(c)}(B)$ be a family of dynamically qualified represented source certificates of saturated cost at most B. Such a certificate is the evaluation of one family-uniform derivation record containing a terminal target-aligned object and all dynamic, terminal, comparison, and normalization data that define its recorded visibility $V_I(R)$. In particular, an exact-quotient record is a source certificate with

$$V_I(R) = A_{q_{R,I}} \mathsf{p}_I(q_{R,I}).$$

Write

$$s_I^{(c)}(B) := \sup_{R \in \mathcal{S}_I^{(c)}(B)} V_I(R), \qquad \sup \varnothing := 0.$$

A quantitative compressed-source cover at budget B consists of a class-preserving cost enlargement Ψ_{src}, a factor K_B in the transfer class, an error $\delta_B \geq 0$, and one family-uniform syntactic rule with the following quantifier order. For every $\widehat{\rho} \in \mathsf{NFQ}_{\mathfrak{I}_n, \leq B}$, the rule selects, independently of the instance, a charged backward cut Γ, a represented witness of a signature $c \in \mathfrak{C}_{\mathrm{src}}$, a uniform source record R, and a represented comparison record. Their single transfer carrier is

$$\mathfrak{T}_{\widehat{\rho}} = (\widehat{\rho}^{\Gamma}, R, \mathrm{Cmp}_{\widehat{\rho},\Gamma,R}),$$

and satisfies, for every $I \in \mathfrak{I}_n$,

$$\mathrm{Cost}_I(\mathfrak{T}_{\widehat{\rho}}) \preceq \Psi_{\mathrm{src}}(B), \qquad R_I \in \mathcal{S}_I^{(c)}(\Psi_{\mathrm{src}}(B)).$$

This carrier includes the materialized joint cut, the source, and the comparison verifier. Whenever $A_{q_{\widehat{\rho},I}} > 0$, the same uniformly selected records satisfy

$$\mathrm{Load}_I(\widehat{\rho}, \Gamma) \leq K_B V_I(R) + \delta_B. \tag{93}$$

In particular, (93) is a dynamic source-transfer condition: projection monotonicity alone supplies (89), while lumpability, target utility, and comparison loss for R are verified by the transfer certificate. The selection therefore has quantifier order $\forall \widehat{\rho}\, \exists (\Gamma, c, R, \mathrm{Cmp})\, \forall I$, rather than permitting an instance-dependent witness. Records with $A_{q_{\widehat{\rho},I}} = 0$ contribute zero to the exact-quotient profile and require no source witness.

The finite-envelope theorem bounds all charged backward sources by one family-uniform cover.

Theorem III.15.9 (Finite compressed-source envelope)**.** *Under Definition III.15.8,*

$$m_I^{\mathrm{sat}}(B) \leq K_B \max_{c \in \mathfrak{C}_{\mathrm{src}}} s_I^{(c)}(\Psi_{\mathrm{src}}(B)) + \delta_B. \tag{94}$$

If K_B lies in the transfer class, δ_B is negligible, and every source envelope on the right is negligible, then $m_I^{\mathrm{sat}}(B)$ is negligible.

Theorem III.16.6 below provides unconditional constructor-level accounting without replacing a contextual extension by a smaller source. The present hypothesis is needed precisely when such an extension is compressed to an independently dynamically qualified certificate.

Proof. Fix $\widehat{\rho}$. If $A_{q_{\widehat{\rho},I}} = 0$, then $V_I(q_{\widehat{\rho}}) = 0$ and the asserted upper bound is immediate. Assume $A_{q_{\widehat{\rho},I}} > 0$. Theorem III.15.6 and the source transfer give

$$V_I(q_{\widehat{\rho}}) \leq \mathrm{Load}_I(\widehat{\rho}, \Gamma) \leq K_B V_I(R) + \delta_B \leq K_B \max_{c \in \mathfrak{C}_{\mathrm{src}}} s_I^{(c)}(\Psi_{\mathrm{src}}(B)) + \delta_B.$$

Taking the supremum over $\widehat{\rho} \in \mathsf{NFQ}_{\mathfrak{I}_n, \leq B}$ and applying Lemma III.15.4 proves (94). The negligibility assertion follows from closure of the transfer class under finite maxima and multiplication by K_B.

□

III.16 Affordable Frontiers and Structural Charging

Having exposed the backward cut, we decompose its visibility into contextual increments along an affordable conditional-algebra frontier.

The construction proceeds through the carrier in Section III.16.1, the exact charge identity in Section III.16.2, the constructor covers in Section III.16.3, filtration provenance in Section III.17, and lossless uniformization in Section III.17.1. The corresponding load-bearing results are Definition III.16.1, Theorem III.16.6, Corollary III.16.10, and Lemmas III.17.2 and III.17.5.

III.16.1 Frontier records and complete carrier cost

Definition III.16.1 (Affordable conditional-algebra frontier)**.** An affordable frontier consists of the factorization below, the complete carrier/cost record of Definition III.16.2, and the contextual charges of Definition III.16.3. The present definition retains the public label for this combined object.

(i) Frontier factorization. Let $\widehat{\rho} \in \mathsf{NFQ}_{\mathfrak{I}_n,\le B}$. An *affordable conditional-algebra frontier* for $\widehat{\rho}$ is a family-uniform expanded record $\widehat{\rho}^{\Gamma}$ with an ordered finite represented interface

$$\Gamma = (v_1, \ldots, v_k), \qquad k \ge 0,$$

whose entries are actual state-dependent vertices in the terminal backward slice. The entries may be comparable in the derivation order. The record contains represented join vertices

$$J_{\Gamma,0} = *, \qquad J_{\Gamma,\ell} = (J_{\Gamma,\ell-1}, f_{v_\ell}), \qquad 1 \le \ell \le k,$$

and one represented downstream contraction satisfying

$$q_{\widehat{\rho},I} = h_{\widehat{\rho},\Gamma,I} \circ J_{\Gamma,k,I} \tag{95}$$

on every instance. Pure tuple, relabeling, lift-interface, comparison, boundary, and terminal-readout vertices remain in the downstream record whenever their state-dependent arguments occur at earlier represented vertices. Those arguments pass through the interface, while public codes and state-independent parameters remain represented coefficients of $h_{\widehat{\rho},\Gamma}$. A declared state-dependent primitive whose argument is not separately exposed may instead occur as a frontier entry; a boundary or terminal primitive is then witnessed by an explicit boundary or terminal constructor class and cannot be reassigned to a generative class without a represented transfer. Frontier entries are strictly ancestral to the terminal quotient for a proper internal cut. The terminal quotient vertex itself is always permitted as the sole catch-all frontier entry, as already required by Definition III.15.5; its complete ancestral derivation and downstream identity interface remain in the carrier. The charged opcode is its actual primitive, boundary, quotient, postprocessing, or five-family evaluator schema. Thus the catch-all case preserves every affordability field rather than treating a derived terminal map as a free primitive. When $k = 0$, the factorization requires the quotient to be constant; such a record has zero target charge under the nontriviality gate.

Definition III.16.2 (Complete carrier and cost allocation for an affordable frontier)**.** Let $(\widehat{\rho}, \Gamma)$ be an affordable conditional-algebra frontier in the sense of Definition III.16.1.

(ii) Complete carrier. The carrier $\widehat{\rho}^{\Gamma}$ retains the complete original record, all mixed ancestors of the frontier, every join interface, all later frontier entries, and the full downstream quotient operator, fibers, dynamics, terminal sectors, comparison data, normalization, and utility factor. Its uniform charged cost ceiling $D(\widehat{\rho}, \Gamma)$ is, by definition, a represented resource vector that dominates the sum of the allocated carrier fields below. In particular,

$$\mathrm{Cost}_{I'}(\widehat{\rho}^{\Gamma}) \preceq D(\widehat{\rho}, \Gamma) \qquad (I' \in \mathfrak{I}_n). \tag{96}$$

(iii) Cost allocation. The complete carrier-cost record consists of charged upper-bound fields for description, evaluation, storage, finite outcome generation, precision, fibers, interfaces, comparisons, and retained global execution resources. Each field is assigned once to the opcode schema of its recorded origin; nonlocal execution or comparison fields are assigned once to a distinguished carrier-overhead schema. Shared fields are not duplicated. For each schema σ, let

$$B_{\sigma,I'}(\widehat{\rho},\Gamma)$$

be the resource aggregation of the upper-bound fields assigned to σ on instance I'. This is an allocation of the bounds already recorded in the charged carrier, not a second charge on the displayed trace. Sub-accumulation of represented cost gives the complete affordability ledger

$$\mathrm{Cost}_{I'}(\widehat{\rho}^{\Gamma}) \preceq \Lambda_{I'}(\widehat{\rho},\Gamma) := \bigoplus_{\sigma} B_{\sigma,I'}(\widehat{\rho},\Gamma) \preceq D(\widehat{\rho},\Gamma). \tag{97}$$

Definition III.16.3 (Contextual extension and structural charge)**.** Let $(\widehat{\rho},\Gamma)$ carry the complete carrier and cost allocation of Definition III.16.2.

(iv) Contextual charge. The ℓ-th *contextual source extension* is the complete carrier with the inclusion

$$\sigma(J_{\Gamma,\ell-1,I}) \subseteq \sigma(J_{\Gamma,\ell,I})$$

distinguished; denote it by $E_{\widehat{\rho},\Gamma,\ell}$. Its cost is the full carrier cost (96), rather than the marginal cost of v_ℓ. Its target-aligned structural charge is

$$\mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}) := A_{q_{\widehat{\rho},I}} \frac{\|(P_{J_{\Gamma,\ell,I}} - P_{J_{\Gamma,\ell-1,I}})y_I\|_2^2}{\|y_I\|_2^2}. \tag{98}$$

The complete affordable structural-charge accounting datum is

$$\mathbf{SC}_I(E_{\widehat{\rho},\Gamma,\ell}) := \Big(\mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}), D(\widehat{\rho},\Gamma), \big(B_{\sigma,I}(\widehat{\rho},\Gamma)\big)_{\sigma}\Big). \tag{99}$$

This tuple is metamathematical bookkeeping evaluated at I; it is not an additional represented observable and is not charged as a new derivation vertex.

Thus the charged object is the conditional algebra extension together with its complete preceding context and downstream dynamics. The coordinate f_{v_ℓ} in isolation is not the charged source, and no visibility estimate may replace the carrier cost by its marginal gate cost.

This cover assigns each contextual charge to a finite affordable constructor schema.

Definition III.16.4 (Finite affordable constructor-charge cover)**.** A *finite affordable constructor-charge cover* consists of:

1. a single family-uniform syntactic rule $\mathfrak{F}$, fixed independently of the size, budget, instance, and target values, that assigns every exact-quotient normal-form record a frontier realization $\widehat{\rho}^{\Gamma} = \mathfrak{F}(\widehat{\rho})$; the bounds below are required on records with positive utility;
2. class-preserving majorants $\Psi_{\mathrm{alg}}(B)$ and $K_{\mathrm{cut}}(B)$ such that

$$D(\widehat{\rho},\Gamma) \preceq \Psi_{\mathrm{alg}}(B), \qquad |\Gamma| \le K_{\mathrm{cut}}(B); \tag{100}$$

3. a finite set $\mathfrak{J}_{\mathrm{con}}$, fixed independently of n and B, of represented constructor-source predicates W_c, where $\mathsf{W}_c(E_{\widehat{\rho},\Gamma,\ell})$ is certified by the actual opcode of the distinguished vertex v_ℓ, or by a represented structural certificate specifically comparing the conditional extension $\sigma(J_{\Gamma,\ell-1}) \subseteq \sigma(J_{\Gamma,\ell})$ with a class-c source. An unrelated opcode elsewhere in the carrier is not such a witness.

Every contextual extension produced by $\mathfrak{F}$ satisfies at least one predicate W_c. Predicates may overlap, and a mixed derivation may carry charges from several constructor families. No priority rule assigns the complete derivation to one family. Under the five-family provenance normal form, evaluator instructions carry one or more predicates in $\mathfrak{J}^5_{\mathsf{Abs}}$. Boundary and terminal interfaces retain their state-dependent arguments in the frontier and their readout records in the downstream carrier. All other opcodes in the retained upstream and downstream carrier remain in the cost ledger but do not acquire the extension's target charge merely by co-occurrence.

A predicate W_c is not auxiliary metadata. It is a family-uniform decidable relation on the encoded distinguished extension. Reading its authenticated opcode is syntactic parsing. If membership instead uses a semantic structural certificate, the certificate description, its verifier, and the cost of verification occur in the ledger (97). If the claimed comparison cannot be verified from those charged fields within the carrier budget, W_c is false.

For $c \in \mathfrak{J}_{\mathrm{con}}$, define the aggregate contextual charge envelope

$$\mathcal{C}^{(c)}_{I,\mathfrak{F}}(D) := \sup_{\substack{\widehat{\rho}\in\mathsf{NFQ}_{\mathfrak{J}_n} \\ A_{q_{\widehat{\rho},I}}>0,\ D(\widehat{\rho},\Gamma)\preceq D \\ \widehat{\rho}^{\Gamma}=\mathfrak{F}(\widehat{\rho})}} \sum_{\substack{1\le\ell\le|\Gamma| \\ \mathsf{W}_c(E_{\widehat{\rho},\Gamma,\ell})}} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}), \qquad \sup\varnothing := 0, \tag{101}$$

and the individual-extension envelope

$$\mathfrak{c}^{(c)}_{I,\mathfrak{F}}(D) := \sup_{\substack{\widehat{\rho}\in\mathsf{NFQ}_{\mathfrak{J}_n},\ A_{q_{\widehat{\rho},I}}>0 \\ D(\widehat{\rho},\Gamma)\preceq D,\ \widehat{\rho}^{\Gamma}=\mathfrak{F}(\widehat{\rho}) \\ 1\le\ell\le|\Gamma|,\ \mathsf{W}_c(E_{\widehat{\rho},\Gamma,\ell})}} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}), \qquad \sup\varnothing := 0. \tag{102}$$

Remark III.16.5 (Structural witnesses do not assign charge). The quantity $\mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell})$ is fixed by the nested affordable algebras and by the terminal utility of the complete quotient before any witness relation W_c is evaluated. A relation W_c only certifies the represented constructor provenance of an already charged conditional extension so that its charge can be included in an envelope. It neither creates, reallocates, nor discounts that charge. Removing all class names leaves the exact charge-conservation identity unchanged; an overlapping witness cover only repeats nonnegative summands in an upper bound. In every case the affordability argument uses the complete carrier ledger, not the cost of the witness name or of the distinguished opcode alone.

The charging theorem identifies terminal quotient visibility with its affordable contextual source charges.

III.16.2 Exact structural-charge identity

Theorem III.16.6 (Exact affordable-algebra structural charging). *Let $\widehat{\rho}^{\Gamma}$ be an affordable conditional-algebra frontier. Then*

$$\sum_{\ell=1}^{|\Gamma|} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}) = V_I(q_{\widehat{\rho}}) + A_{q_{\widehat{\rho},I}} \frac{\|(P_{J_{\Gamma,|\Gamma|,I}} - P_{q_{\widehat{\rho},I}})y_I\|_2^2}{\|y_I\|_2^2}, \tag{103}$$

$$V_I(q_{\widehat{\rho}}) \le \sum_{\ell=1}^{|\Gamma|} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}). \tag{104}$$

For every positive-utility exact quotient in the saturated profile, terminal compatibility gives $P_{q_{\widehat{\rho},I}} y_I = y_I$. In that case the remainder in (103) *is zero and*

$$V_I(q_{\widehat{\rho}}) = \sum_{\ell=1}^{|\Gamma|} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}). \tag{105}$$

Under Definition III.16.4, the complete exact-quotient profile satisfies

$$\begin{aligned} m_I^{\mathrm{sat}}(B) &\le \sum_{c \in \mathfrak{J}_{\mathrm{con}}} \mathcal{C}_{I,\mathfrak{F}}^{(c)}(\Psi_{\mathrm{alg}}(B)) \\ &\le |\mathfrak{J}_{\mathrm{con}}| \max_{c \in \mathfrak{J}_{\mathrm{con}}} \mathcal{C}_{I,\mathfrak{F}}^{(c)}(\Psi_{\mathrm{alg}}(B)), \end{aligned} \tag{106}$$

and also

$$m_I^{\mathrm{sat}}(B) \le K_{\mathrm{cut}}(B) \max_{c \in \mathfrak{J}_{\mathrm{con}}} \mathfrak{c}_{I,\mathfrak{F}}^{(c)}(\Psi_{\mathrm{alg}}(B)). \tag{107}$$

Consequently, negligible aggregate constructor-charge envelopes at every budget in the transfer class imply negligible saturated exact-quotient visibility. The same conclusion follows from negligible individual-extension envelopes when K_{cut} belongs to the transfer class.

Proof. The represented factorization (95) gives $\sigma(q_{\widehat{\rho},I}) \subseteq \sigma(J_{\Gamma,|\Gamma|,I})$. Projection Pythagoras therefore gives

$$\mathsf{p}_I(J_{\Gamma,|\Gamma|,I}) = \mathsf{p}_I(q_{\widehat{\rho},I}) + \frac{\|(P_{J_{\Gamma,|\Gamma|,I}} - P_{q_{\widehat{\rho},I}})y_I\|_2^2}{\|y_I\|_2^2}.$$

The represented join algebras form a nested chain beginning with the constants. Successive conditional-expectation differences are orthogonal, so

$$\mathsf{p}_I(J_{\Gamma,|\Gamma|,I}) = \sum_{\ell=1}^{|\Gamma|} \frac{\|(P_{J_{\Gamma,\ell,I}} - P_{J_{\Gamma,\ell-1,I}})y_I\|_2^2}{\|y_I\|_2^2}.$$

Multiplication by the retained terminal utility factor proves (103) and (104). If the quotient is terminal compatible, then y_I is quotient measurable. Hence $P_q y_I = y_I$, and the factorization also gives $P_{J_{\Gamma,|\Gamma|}} y_I = y_I$. The remainder vanishes, proving (105).

Fix a positive-utility record of cost at most B and use the frontier selected by $\mathfrak{F}$. Every extension is covered by at least one actual constructor-source predicate. Nonnegativity gives

$$V_I(q_{\widehat{\rho}}) \le \sum_{c \in \mathfrak{J}_{\mathrm{con}}} \sum_{\ell:\, \mathsf{W}_c(E_{\widehat{\rho},\Gamma,\ell})} \mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}).$$

The carrier has cost at most $\Psi_{\mathrm{alg}}(B)$, so each inner sum is bounded by its aggregate envelope. Taking the supremum over $\widehat{\rho}$ proves (106). Alternatively, each of at most $K_{\mathrm{cut}}(B)$ extensions is bounded by the largest individual-extension envelope, which proves (107). Closure of the transfer class under finite sums, maxima, and multiplication by K_{cut} proves the final assertions.

□

Example III.16.7 (An exact eight-state affordable algebra)**.** We give a complete finite calculation of the objects in Definition III.16.3 and Theorem III.16.6. The carrier is

$$\Omega = \{0,1\}^3, \qquad x = (c,u,v), \qquad \mu_\Omega = \mathrm{Unif}(\Omega),$$

with target and failure sectors

$$T = \{111\}, \qquad F = \{100, 101, 110\}.$$

Thus the states with $c = 1$ are terminal; 111 is success and the other three are failure. The public state observables are the three bit reads c, u, v, and the terminal verifier is $t(c,u,v) = cuv = \mathbf{1}_T(c,u,v)$. The legal initialization is

$$\mu_0 = \mathrm{Unif}\{000, 001, 010\}.$$

The single toy presentation declares exactly three admissible one-step transition records, indexed by $p \in \{0, \frac{1}{2}, 1\}$. They are

$$\begin{array}{c|cccc} x & 000 & 001 & 010 & 011 \\ \hline K_p(x,\cdot) & \delta_{100} & \delta_{101} & p\delta_{111} + (1-p)\delta_{110} & \delta_{110}, \end{array} \qquad K_p(x,\cdot) = \delta_x \quad (c=1). \tag{108}$$

The finite code for p is part of this transition record. Every terminal state reachable from the support of μ_0 is reached after exactly one transition.

Grammar, budget, and complete slice. Fix the following finite constructor grammar. It may read any of c, u, v, form ordered tuples of distinct coordinate reads by binary joins, apply the downstream verifier t, and attach one of the three records K_p together with the displayed initialization, reference law, and discount normalization. The verifier is a downstream terminal operation: the grammar does not feed t back into an upstream tuple constructor. The declared scalar cost ledger for the complete three-coordinate quotient record is

charged field	cost
primitive reads c, u, v	3
two nontrivial tuple joins	2
target and terminal readout	1
kernel/clock, initialization, reference law, and discount record	2
represented fibers and exact-intertwining verification	1
complete carrier cost B_0	9

These are declared representation costs, not operation counts inferred after the calculation. In particular, the finite p-code is included in the transition row, as is the one-step clock needed for the displayed hitting functional. Every contextual extension below retains the complete carrier and therefore has cost $D = 9$, rather than the marginal unit cost of the newly distinguished read. At budget B_0, the grammar admits no iteration, additional clock extension, feedback of the verifier into a tuple, or other quotient constructor; each omitted opcode is assigned declared cost greater than B_0.

For each of the three transition records, the affordable quotient-output slice at $B_0 = 9$, modulo equality of the induced partitions, consists of the constant map, the verifier partition, the coordinate reads and their proper coordinate tuples, and the full tuple

$$q = (c, u, v) : \Omega \longrightarrow \{0, 1\}^3.$$

This list exhausts the affordable derivation slice. Every derivation attaches one of the three K_p records to a constant, a verifier output, a coordinate read, or an ordered binary join of distinct coordinate reads. Different orders for the same coordinate subset induce the same partition and hence the same projection and terminal-compatibility test. The constant fails the nontriviality condition. The verifier partition has zero qualified utility: although it separates T, its zero-cell combines F with nonterminal states and hence does not represent the failure sector. Every proper coordinate tuple also fails terminal compatibility. Indeed, for the three two-coordinate tuples, respectively,

$$(c, u) : 110 \sim 111, \qquad (c, v) : 101 \sim 111, \qquad (u, v) : 011 \sim 111,$$

and deleting another coordinate cannot repair the failure. The full tuple q is the identity partition, so its fibers are represented and both terminal sectors are unions of fibers. If $\bar{K}_p$ denotes the induced quotient kernel, then $K_p U_q = U_q \bar{K}_p$ holds exactly. The quotient strictly refines the verifier partition. Write q_p for the complete quotient record obtained by attaching K_p to this map. Consequently q_p is the sole potentially positive-utility quotient record for each p, with positive utility at $p = \frac{1}{2}, 1$; at $p = 0$ the same exact quotient map has utility zero.

Backward cut and conditional algebras. Choose the ordered backward cut

$$\Gamma = (c, u, v), \qquad J_0 = *, \quad J_1 = c, \quad J_2 = (c, u), \quad J_3 = (c, u, v) = q.$$

This cut separates every state-dependent path from the public reads to the terminal quotient. The kernel, normalization, terminal cells, fiber records, and exact-intertwining check remain in the downstream carrier. In the state order $000, 001, 010, 011, 100, 101, 110, 111$, the centered target contrast and its successive conditional expectations are

$$\begin{aligned} y = \mathbf{1}_T - \mu_\Omega(T) &= \frac{1}{8}(-1, -1, -1, -1, -1, -1, -1, 7), \qquad (109)\\ P_{J_1} y &= \frac{1}{8}(-1, -1, -1, -1, 1, 1, 1, 1),\\ P_{J_2} y &= \frac{1}{8}(-1, -1, -1, -1, -1, -1, 3, 3),\\ P_{J_3} y &= y. \end{aligned}$$

With the $L^2(\mu_\Omega)$ norm,

$$\|y\|_2^2 = \frac{7}{64}, \qquad \|P_{J_1} y\|_2^2 = \frac{1}{64}, \qquad \|P_{J_2} y\|_2^2 = \frac{3}{64}, \tag{110}$$

and orthogonality of the conditional increments gives

$$\|(P_{J_1} - P_{J_0})y\|_2^2 = \frac{1}{64}, \quad \|(P_{J_2} - P_{J_1})y\|_2^2 = \frac{2}{64}, \quad \|(P_{J_3} - P_{J_2})y\|_2^2 = \frac{4}{64}. \tag{111}$$

The normalized projection increments are therefore $1/7, 2/7, 4/7$, whose sum is one.

Utility, charges, and the rule-uniform bound. For an arbitrary admissible K_p, the one-step success probability is

$$\alpha_p = \Pr_{\mu_0}\{\tau_T \le 1\} = \frac{p}{3}. \tag{112}$$

Take $\lambda = 1$. Since success can occur only on the first transition, the uniformized target functional in (146) is $A_T^q = \alpha_p/(1+\lambda) = p/6$. The quotient uses the canonical pushforward comparison, so $C_q = 1$, and hence

$$A_q = \min\{1, \lambda C_q^{-1} A_T^q\} = \frac{p}{6}, \qquad V(q_p) = A_q \frac{\|P_q y\|_2^2}{\|y\|_2^2} = \frac{p}{6}.$$

Writing E_ℓ for the complete contextual carrier with $J_{\ell-1} \subset J_\ell$ distinguished, the three charges are exactly

$$\begin{aligned} \mathrm{ch}(E_1) &= \frac{p}{6}\frac{1}{7} = \frac{p}{42}, \\ \mathrm{ch}(E_2) &= \frac{p}{6}\frac{2}{7} = \frac{p}{21}, \\ \mathrm{ch}(E_3) &= \frac{p}{6}\frac{4}{7} = \frac{2p}{21}. \end{aligned} \tag{113}$$

Thus the theorem's remainder vanishes and its charge-conservation identity is visible numerically:

$$\sum_{\ell=1}^{3} \mathrm{ch}(E_\ell) = \frac{p}{42} + \frac{p}{21} + \frac{2p}{21} = \frac{p}{6} = V(q_p). \tag{114}$$

Let m^{sat} denote the saturated profile of this single toy presentation. Its supremum ranges over the three admissible K_p records. Because the exhaustive affordable slice has no other positive-utility quotient,

$$m^{\mathrm{sat}}(B_0) = \max_{p \in \{0, \frac{1}{2}, 1\}} V(q_p) = \frac{1}{6}. \tag{115}$$

Combining (112) and (114) yields, simultaneously for every admissible transition rule K_p, the exact class bound

$$\Pr_{\mu_0}\{\tau_T \le 1\} = 2V(q_p) = 2\sum_{\ell=1}^{3} \mathrm{ch}(E_\ell) \le 2m^{\mathrm{sat}}(B_0) = \frac{1}{3}, \tag{116}$$

with equality for K_1. By comparison, applying the generic universal accountability estimate (175) with $H = 1$ and a declared first-hit compilation ceiling $B_{\mathcal{A}} = B_0$ gives the valid but coarser bound $\alpha_p \le e/6$. The sharper 1/3 above comes from evaluating the entire finite affordable algebra and its exact conditional charges, rather than replacing the one-step discount by the universal e^{-1} lower bound.

Figure 32 records the charge identity in Theorem III.16.6.

Remark III.16.8 (Order dependence and invariant total charge). The individual quantities $\mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell})$ depend on the family-uniform frontier and on its encoded order, just as conditional information increments do. No individual increment is claimed to be an intrinsic contribution of v_ℓ in isolation. What is invariant for a terminal-compatible exact quotient is their sum (105). All envelope statements therefore fix $\mathfrak{F}$ and its order before the instance is evaluated.

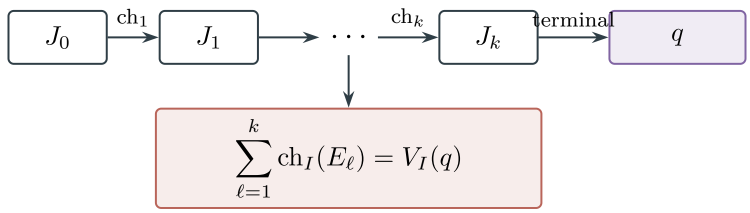

Figure 32: For a terminal-compatible exact quotient, successive contextual charges telescope to its visibility by Theorem III.16.6.

Corollary III.16.9 (Contextual structural-charge transfer)**.** *Assume a finite affordable constructor-charge cover at budget B. Suppose a family-uniform rule assigns every contextual extension $E_{\widehat{\rho},\Gamma,\ell}$ a signature $c_\ell \in \mathfrak{C}_{\mathrm{src}}$, a dynamically qualified source record $R_\ell \in \mathcal{S}_I^{(c_\ell)}(\Psi_{\mathrm{tr}}(B))$, and a represented comparison such that the complete transfer carrier*

$$(\widehat{\rho}^\Gamma, R_\ell, \mathrm{Cmp}_{E_\ell,R_\ell})$$

has cost at most $\Psi_{\mathrm{tr}}(B)$ uniformly over the task slice and

$$\mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}) \le K'_B V_I(R_\ell) + \delta'_B \tag{117}$$

for every instance. Then

$$m_I^{\mathrm{sat}}(B) \le K_{\mathrm{cut}}(B) K'_B \max_{c \in \mathfrak{C}_{\mathrm{src}}} s_I^{(c)}(\Psi_{\mathrm{tr}}(B)) + K_{\mathrm{cut}}(B)\delta'_B. \tag{118}$$

Thus an incremental transfer preserves negligibility when K_{cut}, K'_B, and Ψ_{tr} lie in the declared transfer class and δ'_B is negligible.

Proof. Use (104), apply (117) to each of at most $K_{\mathrm{cut}}(B)$ extensions, and take the supremum over actual-cost records in $\mathsf{NFQ}_{\mathfrak{I}_n,\le B}$. This proves (118). The final claim follows from transfer-class closure.

□

III.16.3 Canonical constructor covers

Corollary III.16.10 (Canonical leaf charge cover under succinct interfaces)**.** *Fix a presented finite task family with the derivation coding of Definition III.1.3. Assume its declared resource envelope supplies class-preserving majorants $L_{\mathrm{tr}}(B)$ and $J_{\mathrm{tuple}}(B,k)$ with the following two properties: an expanded record of actual witness cost at most B has at most $L_{\mathrm{tr}}(B)$ encoded vertices, and a nested represented tuple of at most k vertices from that record has complete description, materialization, storage, evaluation, joint-fiber-membership, and finite Boolean fiber-operation cost at most $J_{\mathrm{tuple}}(B,k)$. These are the succinct-interface hypotheses.*

Let π_{op} be a fixed finite schema map on the normal-form opcodes declared by the presentation; opcode parameters and public coefficients are data, not new schemas. For an exact-quotient record $\widehat{\rho}$, list in encoded order all state-dependent leaves of its terminal backward slice and call the list $\Gamma_{\mathrm{syn}}(\widehat{\rho})$. Public constants and state-independent parameters remain coefficients of the downstream contraction. Define

$$\mathsf{W}_\sigma(E_{\widehat{\rho},\Gamma,\ell}) \iff \pi_{\mathrm{op}}(\mathrm{op}(v_\ell)) = \sigma.$$

Thus the predicate is anchored at the distinguished algebra extension. An internal or terminal opcode on the downstream path remains charged in the carrier ledger but does not receive the leaf's target charge. Then $\widehat{\rho}^{\Gamma_{\mathrm{syn}}} = \mathfrak{F}_{\mathrm{syn}}(\widehat{\rho})$ and the anchored predicates satisfy Definition III.16.4, with

$$|\Gamma_{\mathrm{syn}}(\widehat{\rho})| \le K_{\mathrm{syn}}(B) := L_{\mathrm{tr}}(B), \qquad D(\widehat{\rho}, \Gamma_{\mathrm{syn}}) \preceq \Psi_{\mathrm{syn}}(B) := B \oplus J_{\mathrm{tuple}}(B, L_{\mathrm{tr}}(B))$$

for every record of cost at most B. The standard finite operational-history configuration model satisfies these hypotheses when the charged trace dominates its encoded length and tuples are represented by references to their charged component records; in that model both majorants are polynomial. Consequently,

$$m_I^{\mathrm{sat}}(B) \le \sum_{\sigma \in \mathrm{im}(\pi_{\mathrm{op}})} \mathcal{C}^{(\sigma)}_{I,\mathfrak{F}_{\mathrm{syn}}}(\Psi_{\mathrm{syn}}(B)). \tag{119}$$

Any application-specific frontier may instead select outputs of actual derived structural constructors, together with the remaining leaves needed for factorization. A coarser semantic list is valid only through anchored represented certificates for those selected extensions. No assignment of a complete record to one terminal class is involved.

Proof. Every state-dependent input to a vertex of an expanded record is an earlier vertex of the same finite DAG. Evaluation of the retained DAG in topological order therefore gives a represented contraction of the terminal quotient through the joint observable of its state-dependent leaves, proving (95). Every distinguished leaf has exactly one opcode schema under π_{op}, so the anchored predicates cover every extension. A primitive state-dependent terminal quotient is itself such a leaf and is admitted by the affordable conditional-algebra frontier; its actual schema supplies its source designation.

The first succinct-interface hypothesis bounds the number of leaves. The second bounds the added nested tuple, including its representation and evaluation; retaining the original record contributes at most B. These are precisely the displayed majorants. The complete DAG remains in every carrier, so every upstream and downstream operation is still present in (97). Theorem III.16.6 now gives (119).

□

Corollary III.16.11 (Exhaustive five-family envelope for the saturated exact-quotient profile)**.** *Apply the five-family expansion of the universal five-family Abs provenance normal form to every $\widehat{\rho} \in \mathsf{NFQ}_{\mathfrak{I}_n, \le B}$. In encoded topological order, let Γ_5 contain the state-dependent typed vertices in the backward slice of the terminal quotient map after pure interfaces have exposed their arguments. Include the terminal quotient vertex. Let $\mathfrak{F}_5$ be this family-uniform frontier rule and put*

$$\mathsf{W}_c(E_{\widehat{\rho},\Gamma_5,\ell}) \iff c \in \mathrm{Prov}_5(v_\ell), \qquad c \in \mathfrak{I}^5_{\mathsf{Abs}}.$$

Then $\mathfrak{F}_5$ is a finite affordable constructor-charge cover. There are class-preserving majorants Ψ_5^{alg} and K_5, fixed by the encoded operational model, such that

$$m_I^{\mathrm{sat}}(B) \le \sum_{c \in \mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(\Psi_5^{\mathrm{alg}}(B)), \tag{120}$$

$$m_I^{\mathrm{sat}}(B) \le K_5(B) \max_{c \in \mathfrak{I}^5_{\mathsf{Abs}}} \mathfrak{c}^{(c)}_{I,\mathfrak{F}_5}(\Psi_5^{\mathrm{alg}}(B)). \tag{121}$$

For every positive-utility terminal-compatible exact quotient, its visibility is exactly the sum of the contextual charges on this frontier, and every summand belongs to at least one of the five families. Consequently, negligibility of the five aggregate envelopes, or of the five individual envelopes with K_5 in the transfer class, implies negligibility of the complete saturated exact-quotient accounting profile. Restricting the same frontier to temporally admissible records gives the source profile of Definition III.23.3. Exact first-hit quotients belong to the accounting frontier and enter the source frontier only through an independent pre-hit representation.

Proof. The five-family normal form gives a finite typed backward slice with class-preserving expansion cost. The encoded topological order is family uniform. Since the terminal quotient is included, $q_{\widehat{\rho}}$ factors through the represented join of Γ_5. Every selected state-dependent vertex has at least one authenticated five-family tag, and interface vertices have exposed their tagged arguments. The expanded-record length and represented tuple-interface bounds give K_5 and Ψ_5^{alg}.

Apply the exact affordable-algebra structural-charging theorem with $\mathfrak{J}_{\mathrm{con}} = \mathfrak{J}^5_{\mathsf{Abs}}$. Equations (120) and (121) are its aggregate and individual bounds. Terminal compatibility gives the exact charge identity. The exact first-hit lemma supplies a normal-form accounting quotient for every represented finite-horizon computation. Temporal source admissibility is applied separately as in Corollary III.23.1.

□

III.17 No-Creation and Filtration Provenance

Lemma III.17.1 (Derived evaluator constructors carry zero contextual charge)**.** *Let $E_{\widehat{\rho},\Gamma_5,\ell}$ be a contextual extension in the five-family frontier, and suppose its distinguished vertex v_ℓ is the output of a represented* Mult *or* Ord *constructor, or a* Filt*-tagged deterministic constructor whose state-dependent arguments occur earlier in the frontier. Then*

$$\mathrm{ch}_I(E_{\widehat{\rho},\Gamma_5,\ell}) = 0.$$

This applies to conjunction, products, composition gates, polynomial and ring operations, equality tests, represented fiber formation, comparisons, branching and selection, weights, metric and quadratic comparisons, and ordered reductions. The same conclusion applies to a Filt*-tagged clock, stored copy, memory update, recurrence step, or history coordinate that is a represented deterministic function of earlier state-dependent arguments.*

Proof. Every state-dependent argument of v_ℓ is an earlier vertex in the encoded topological order defining Γ_5; state-independent public data remain coefficients. Hence the observable at v_ℓ is a represented postprocessing of $J_{\Gamma_5,\ell-1,I}$. The affordable composition and target-projection theorem gives

$$\sigma(v_\ell) \subseteq \sigma(J_{\Gamma_5,\ell-1,I}).$$

Therefore adjoining v_ℓ does not change the represented joint algebra:

$$\sigma(J_{\Gamma_5,\ell,I}) = \sigma(J_{\Gamma_5,\ell-1,I}).$$

The two conditional-expectation projections in (98) are equal, so the contextual charge is zero.

□

Lemma III.17.2 (Filtration provenance charging). *Let $E_{\widehat{\rho},\Gamma_5,\ell}$ be a contextual extension in the five-family frontier whose distinguished vertex has* Filt *provenance. Retain the complete carrier, including the generating update, verifier, lifted interfaces, terminal records, and all earlier state-dependent arguments. Then filtration provenance supplies no independent generative source. More precisely:*

1. *If the distinguished observable is a represented deterministic function of earlier frontier coordinates, then*

$$\mathrm{ch}_I(E_{\widehat{\rho},\Gamma_5,\ell}) = 0.$$

2. *If it is an auxiliary coordinate of a valid Lift, the represented base projection occurs earlier in the retained interface and the lifted target contrast is its pullback. The auxiliary coordinate therefore adds zero contextual target charge; the target contribution remains attached to the represented base record.*
3. *For a budget-generated outcome or terminal transcript, apply the component-exhaustiveness projection to its retained generating update and terminal edge gate. Normalize its represented evaluator into the five-family backward slice of Theorem III.2.2. Independent outcome randomness is a valid-Lift auxiliary coordinate and adds zero target charge by the preceding case. The update, verifier, stored flags, terminal map, and every subsequent first-hit postprocessing are represented deterministic functions of earlier frontier coordinates and add zero contextual charge by the first case. Hence the exact charge identity of Corollary III.16.11 assigns every nonzero contextual charge to an earlier authenticated vertex of the completed derivation. If that vertex is invoked prospectively, the temporal-source and represented-transfer conditions of Definition III.21.1 and Corollary III.16.9 classify and compare its source record. Without those additional conditions, the charge remains completed accounting. The post-saturation J_{res} term contains no represented structural source by Proposition II.8.3 and Lemma II.8.4.*

Thus a clock, memory record, stored flag, outcome history, recurrence, or first-hit record can retain Filt *carrier provenance but cannot receive an independent source credit from that tag.*

Proof. In the first case, all state-dependent arguments of the distinguished vertex occur in $J_{\Gamma_5,\ell-1,I}$. The derived constructor zero-charge lemma gives the asserted equality.

For a valid Lift $\pi : \widetilde{X} \to X$, the target is $\widetilde{T} = \pi^{-1}(T)$ and target-fraction preservation gives

$$\mathbf{1}_{\widetilde{T}} - \widetilde{\mu}(\widetilde{T}) = \big(\mathbf{1}_T - \mu(T)\big) \circ \pi.$$

The represented Lift interface exposes π and its base state-dependent arguments before any auxiliary coordinate. Hence the lifted target contrast is already measurable with respect to the preceding joint algebra. Adjoining an auxiliary Lift coordinate leaves its conditional expectation unchanged, so its contextual charge is zero. This is the conditional-algebra form of the exact Lift identity.

It remains to consider an operational outcome or terminal record. The budgeted derivability closure retains its initializer, generating update, verifier, outcome interface, transcript, and terminal map at their accumulated cost. Theorem III.2.2 expands this complete record into a charged topological derivation. Independent random coordinates are valid-Lift auxiliaries, so the second case gives zero incremental target charge. After those coordinates and the earlier state have been exposed, the update, verifier flag, stored transcript coordinate, and terminal map are deterministic represented postprocessings. The first case therefore gives zero incremental charge at each such vertex. The same conclusion applies to the first-hit map, which is a deterministic postprocessing of the stored flags and deadline.

The exact telescoping identity in Corollary III.16.11 leaves every nonzero charge on an earlier authenticated vertex of the retained derivation. When the vertex is used prospectively, the component-exhaustiveness theorem, Theorem II.9.4, and the temporal and transfer conditions in Definition III.21.1 and Corollary III.16.9 classify its source and supply the applicable comparison. Otherwise the telescoping identity remains an accounting identity. In either case, Proposition II.8.3 and Lemma II.8.4 exclude a represented source from the post-saturation residual. Thus no operation in the filtration chain assigns a new generative source to the Filt tag. This proves the three alternatives.

□

Remark III.17.3 (A verifier and first-hit quotient do not supply a selector). Let V_I be a represented public verifier and let $q_{\mathcal{A}}$ be the exact first-hit quotient of a represented computation. The quotient map is an affordable function of the generated terminal flags and deadline, so

$$\sigma(q_{\mathcal{A}}) \subseteq \sigma(s_0, \ldots, s_H, r).$$

The affordable composition theorem therefore assigns it no target information beyond that generated transcript. Represented fibers provide uniform membership predicates and finite Boolean fiber operations. They do not provide a map selecting a successful transcript or candidate from a success fiber. Such a selector is a separate represented derivation and is charged at its own saturated cost; without one, the quotient cannot generate a verifier-accepting candidate.

The quotient utility retains the actual success probability α_I through $V_{q_{\mathcal{A}}} \geq \alpha_I/(eH)$. Consequently the first-hit construction records success already produced by the computation and does not create it. A terminal test using a latent target rather than a represented public verifier belongs to oracle or advice accounting.

Remark III.17.4 (Dynamic transfer from contextual charge). The structural charge $\mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell})$ retains the terminal utility of the complete computation; it does not assert that the frontier coordinate is independently lumpable. A comparison with a smaller dynamically qualified source requires a represented inequality

$$\mathrm{ch}_I(E_{\widehat{\rho},\Gamma,\ell}) \leq K_c(D)V_I(R_E) + \delta_c(D),$$

with the full comparison carrier charged, followed by Corollary III.16.9. Under the hypotheses of Lemma III.20.2, every finite-horizon uniform represented computation supplies an exact quotient record to which the structural charging theorem applies.

III.17.1 Lossless uniformization

Lemma III.17.5 (Lossless charged uniformization). *Let $\widetilde{L}$ be any represented finite Markov or sub-Markov generator, let $\zeta > 0$ be represented and charged, and write its generator-form data as $q(x, y)$ and $\kappa(x)$. Choose and charge a represented scalar c satisfying*

$$c \geq \zeta, \qquad c \geq \max_x \left\{ \sum_{y \neq x} q(x, y) + \kappa(x) \right\}.$$

Then

$$P = I + c^{-1}\widetilde{L}, \qquad L = P - I = c^{-1}\widetilde{L}, \qquad \lambda = \zeta/c$$

defines a represented Markov or sub-Markov kernel P, a normalized generator $L = P - I$, and a discount $0 < \lambda \le 1$. For every absorbing target T and every initial law μ_0,

$$A_T(\zeta; \widetilde{L}, \mu_0) = A_T(\lambda; L, \mu_0). \tag{122}$$

The normalization changes no state, target, quotient fiber, or off-diagonal transition support. It may add holding steps, as every uniformization does, but preserves the target-hitting functional exactly. Its scalar, rate-table arithmetic, normalized kernel, discount conversion, and every transported quotient, defect, geometry, and authority record are included in the saturated cost. Such a scalar exists analytically on every finite carrier. It is a represented datum when the carrier has a represented finite enumeration, when the generator presentation supplies an exit-rate majorant, or when the majorant is included explicitly in the certificate. Its construction and verification are charged. This is not a restriction on DTM representation: for a discrete kernel generator $\widetilde{L} = P - I$, one may take $c = \max\{1, \zeta\}$, and the canonical algorithmic reading uses $\zeta \le 1$ and $c = 1$.

Proof. For $x \neq y$, $P_{xy} = q(x, y)/c \ge 0$, while

$$P_{xx} = 1 - \frac{\sum_{y \neq x} q(x, y) + \kappa(x)}{c} \ge 0, \qquad \sum_y P_{xy} = 1 - \frac{\kappa(x)}{c} \in [0, 1].$$

Thus P is sub-Markov, and it is Markov when $\kappa = 0$. Let $\widetilde{L}_N$ and $\widetilde{k}_T$ be the killed operator and target flux for $\widetilde{L}$. The normalized data are $L_N = c^{-1}\widetilde{L}_N$ and $k_T = c^{-1}\widetilde{k}_T$. Therefore

$$(\lambda I - L_N)^{-1} k_T = \left(c^{-1}(\zeta I - \widetilde{L}_N)\right)^{-1} c^{-1}\widetilde{k}_T = (\zeta I - \widetilde{L}_N)^{-1}\widetilde{k}_T.$$

Adding the unchanged initial target atom proves (122). Positive scalar rescaling also preserves the generator's off-diagonal zero pattern, absorbing sectors, and every exact intertwining identity. All other transported numerical data are recomputed for L and retained in the charged record. □

III.18 The Master Compensated Quotient–Geometry Certificate

The master certificate assembles the exact quotient, geometric, causal, and defect-compensation records on the common saturated ledger.

The construction proceeds through the certificate data in Section III.18.1, the saturated profiles in Section III.18.2, and the authority-restricted normal form in Section III.18.3. The profile definition and its authority controls are Definition III.18.4, Lemma III.18.8, Theorem III.18.9, and Corollary III.18.10.

III.18.1 Certificate data and visibility

Definition III.18.1 (Master compensated quotient–geometry certificate)**.** The master certificate consists of the structural record and slot typing in parts (i)–(iv), the dynamic qualification and quotient utility of Definition III.18.2, and the visibility and charge rule of Definition III.18.3. The present definition retains both established public labels for the combined object.

A master compensated quotient–geometry certificate of total saturated cost at most B is a tuple

$$\mathscr{C} = (\mathfrak{N}, q, \mathfrak{G}_q, \mathfrak{S}, \mathfrak{K}_q),$$

(i) Uniformization and budget. Here

$$\mathfrak{N} = (\widetilde{L}, \zeta, c, P, L, \lambda)$$

is the lossless charged uniformization record of Lemma III.17.5. Thus every master record contains a represented normalized kernel form $L = P - I$, has $0 < \lambda \le 1$, and preserves the original selected process's discounted target functional by (122). All quotient, geometry, stability, and causal slots below refer to this same normalized generator L. If an authority envelope was declared for $\widetilde{L}$, the record must also transport its actual current by the fixed factor c^{-1}; normalization cannot select a different current or control.

Membership at budget B includes one chosen family-uniform derivation of the complete tuple with actual cost $\preceq B$; it is not defined through an infimum in the resource partial order.

(ii) Quotient and geometry. The map $q : X \to Q$ is a represented terminal-compatible quotient, with the identity quotient allowed. A nonidentity support quotient is admitted only when the same record satisfies the represented-fiber, terminal-event, positive quotient-utility, and nontrivial-public-structure conditions of Definition III.15.1. In the exact-support case its dynamic condition is exact intertwining. In a nonzero-defect case that condition is replaced, and only replaced, by the charged stability and target-functional comparison in part (v) below. Thus the support map is an affordable target-qualified quotient record; a represented fiber map that fails these existing gates is not admitted merely because a geometry was written on its image. The geometry slot $\mathfrak{G}_q$ is either active or null. An active slot is the represented tuple

$$\mathfrak{G}_q = (\Phi_Q, \Phi_\perp, D_Q, J_Q),$$

where Φ_Q is a Dirichlet geometry on Q, $D_Q : Q \to \mathbb{R}$ is a progress observable, J_Q is a quotient-edge current, and $\Phi_\perp$ is the within-fiber geometry used by a nonzero-defect stability record. In an exact active certificate $\Phi_\perp$ may be empty. The potential is target-normalized:

$$D_Q \ge 0, \qquad D_Q = 0 \quad \text{on the positive quotient terminal sector.}$$

(iii) Causal provenance. The causal slot $\mathfrak{K}_q$ is a provenance record in the sense of Definition II.4.4. For an active geometry slot it verifies that $J_Q = J_{\mathfrak{K}_q}$, records the selected authority envelope and generator, declares whether J_Q is the transported structural-gradient projection of the actual selected skew current or the auxiliary geometric current of (Φ_Q, D_Q), and transports that same current through every quotient, coordinate change, and lift used by the certificate. In the first case it is a Caus operator component. In the second it only orients the Geom descent comparison and is not identified with the operator's Caus component unless an explicit represented equality is included. An active certificate without the appropriate provenance record has zero alignment factor.

(iv) Null geometry. The null slot, denoted $\mathfrak{G}_q = \varnothing$, is permitted precisely when $E_q = 0$ and q is an exact quotient record of Definition III.15.2; the identity quotient is allowed. Thus a null slot records pure exact quotient accounting and does not assert any geometric comparison; in this case $\mathfrak{K}_q = \varnothing$. An identity quotient with an active geometry slot instead belongs to the ambient Geom branch. These two records are disjoint because the slot is part of the certificate.

Definition III.18.2 (Dynamic qualification and utility of a master certificate)**.** Let $\mathscr{C} = (\mathfrak{N}, q, \mathfrak{G}_q, \mathfrak{S}, \mathfrak{K}_q)$ have the structural slots of Definition III.18.1.

(v) Dynamic stability. The record $\mathfrak{S}$ is one of the following dynamic certificates:

$$S_q(\lambda) = \begin{cases} 1, & E_q = 0, \\ 1 - \vartheta_\lambda^T(q), & E_q \neq 0 \text{ and the reversible hypotheses hold with } \vartheta_\lambda^T(q) < 1, \\ \max\{0, 1 - \Delta_\lambda(q)\}, & E_q \neq 0 \text{ and a general resolvent certificate is used,} \\ 0, & \text{otherwise.} \end{cases}$$

In the second line $\vartheta_\lambda^T(q)$ is the represented target-functional relative loss of Corollary III.12.3; its Schur-complement bound, target vectors, and quotient-arrival identification are part of $\mathfrak{S}$. In the third line $\Delta_\lambda(q)$ abbreviates the represented target-functional defect $\Delta_\lambda^T(q; \ell_T, v_T)$ from Definition III.11.7. The windows, functional, and input are part of $\mathfrak{S}$ and must realize the same quotient-arrival scalar used in $\overline{A}_q$. A nonzero-defect certificate is dynamically qualified only when $S_q(\lambda) > 0$ and $S_q(\lambda)^{-1}$ belongs to the declared transfer class. In particular, the null geometry slot cannot qualify a nonzero defect. The stability record is evaluated for the full selected generator. Its component ledger contains E_q^{s} and E_q^{a} from Definition III.11.4; the reversible line is available only when the selected skew operator vanishes. The discount scale is part of the resource record: a certificate used at resource bound B must have λ^{-1} in the corresponding time-transfer class. Finally, dynamic qualification requires the represented target vectors and terminal-flux functional to identify the actual full-process discounted target functional. Thus the reversible and general records both certify

$$A_T(\lambda; L, \mu_0) \geq S_q(\lambda) A_T^Q(\lambda; L_Q, \bar{\mu}_0). \tag{123}$$

This is the scalar target-functional conclusion of Corollary III.12.3 or Corollary III.11.8, respectively. An operator-norm comparison without the displayed target-functional identification is not a dynamically qualified nonzero-defect record.

(vi) Quotient utility. For a nonidentity quotient, let

$$\overline{A}_q = \min\{1, \lambda \overline{C}_q^{-1} A_T^Q(\lambda; L_Q, \bar{\mu}_0)\},$$

where $\overline{C}_q$ contains the represented-fiber, terminal, normalization, and quotient-level comparison factors, but not the dynamic defect loss already recorded by S_q. For an identity quotient with an active geometry slot set $\overline{A}_{\mathrm{id}} = 1$. For an identity quotient with a null slot set $\overline{A}_{\mathrm{id}} = A_{\mathrm{id}}$. Thus every exact null-slot quotient, including the identity, satisfies $\overline{A}_q = A_q$. For a nonidentity exact-support record, the comparison data are the same data used by Definition III.15.1; hence $\overline{C}_q = C_q$ and $\overline{A}_q = A_q$. In a nonzero-defect record, $\overline{A}_q$ retains those same nondynamic qualification factors, while the defect loss occurs exactly once in S_q.

Definition III.18.3 (Master-certificate reach, visibility, and charge)**.** Let $\mathscr{C}$ be a structurally typed and dynamically qualified master certificate in the sense of Definitions III.18.1 and III.18.2.

(vii) Target reach. For an active geometry slot, let $\mathcal{M}_Q(D_Q)$ be the monotone target-oriented span on Q. When $\|y\|_2 = 0$, set every reach factor and $V_{\mathscr{C}}$ to zero. Otherwise, when $\mathrm{Var}_{\nu_Q}(D_Q) > 0$, define

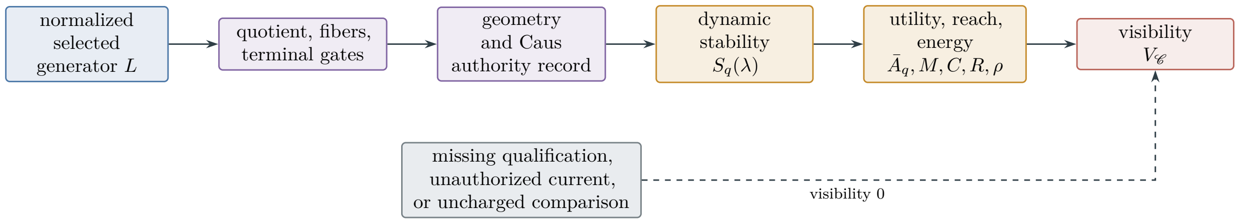

Figure 33: The master certificate is a dependency DAG, not a menu of independent estimates. Quotient utility, represented geometry, authorized orientation, dynamic transfer, and target reach must occur in one affordable record. A missing gate cannot be supplied from another certificate.

$$R_T^q(D_Q) = \frac{\|\Pi_{U\mathcal{M}_Q(D_Q)} y\|^2}{\|y\|^2}, \qquad \rho_{\Phi_Q}(D_Q) = \frac{\mathcal{E}_{\Phi_Q}(D_Q, D_Q)}{\mathrm{Var}_{\nu_Q}(D_Q)}.$$

When $\mathrm{Var}_{\nu_Q}(D_Q) = 0$, set the active-slot reach factor and $\rho_{\Phi_Q}(D_Q)$ to zero. The resulting certificate visibility is zero.

Write the four geometric factors associated with an active slot as

$$M_{\mathfrak{G}_q} = M_{\Phi_Q}, \qquad C_{\mathfrak{G}_q} = C^{\mathrm{auth}}_{\Phi_Q, \mathfrak{K}_q}(D_Q), \qquad R_{\mathfrak{G}_q} = R_T^q(D_Q), \qquad \rho_{\mathfrak{G}_q} = \rho_{\Phi_Q}(D_Q).$$

For the null slot use the conventions

$$M_\varnothing = C_\varnothing = 1, \qquad \rho_\varnothing = 0, \qquad R_\varnothing = \frac{\|P_q^{\mathrm{amb}} y\|^2}{\|y\|^2},$$

where $P_q^{\mathrm{amb}} = U_q P_q$ is the orthogonal projection onto the quotient-observable subspace of $L^2(X, \mu)$. The certificate visibility is the single expression

$$V_{\mathscr{C}} = \overline{A}_q S_q(\lambda) M_{\mathfrak{G}_q} C_{\mathfrak{G}_q} \frac{R_{\mathfrak{G}_q}}{1 + \rho_{\mathfrak{G}_q}}. \tag{124}$$

(viii) Complete charge. Every quotient, fiber predicate, conductance, potential, current, authority envelope, selected control, transport map, component defect, comparison window, defect estimate, and normalization factor in this expression is charged in the single derivation of $\mathscr{C}$. A current that is target-aligned but not authorized by that derivation cannot be inserted into the certificate. A null geometry slot contributes no additional cost, so its value is exactly

$$V_{\mathscr{C}} = A_q \frac{\|P_q^{\mathrm{amb}} y\|^2}{\|y\|^2} = V_q.$$

Figure 33 displays the same-record gates whose product defines master visibility.

III.18.2 Saturated master profiles

The master profile takes the budgeted supremum over the compensated certificate families.

Definition III.18.4 (Master saturated quotient–geometry profile)**.** The master profile comprises the certificate partition below, the numerical coordinates of Definition III.18.5, and the authority restriction of Definition III.18.6. The present definition retains the established public labels for the combined profile.

Fix a task family and a size slice $\mathfrak{I}_n$. In this definition a symbol $\mathscr{C}$ denotes a family $(\mathscr{C}_I)_{I\in\mathfrak{I}_n}$ produced by one derivation scheme $\rho \in \mathsf{Der}_{\mathfrak{I}}$ fixed across all sizes and instances. Every displayed predicate holds pointwise, and $\mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B$ means that the same derivation has that cost bound for every $I \in \mathfrak{I}_n$. For a distinguished instance sequence (I_n), write $V_{\mathscr{C}} = V_{\mathscr{C}_{I_n}}$; an instance-level or family-functional profile replaces this evaluation by the corresponding declared functional. Per-instance re-selection of the certificate program is not permitted.

(i) Certificate families. Partition the master certificates into the following five disjoint families, according to the displayed data in the certificate record:

$$\begin{aligned}
\mathcal{C}_X(B) &= \{\mathscr{C} : q = \mathrm{id},\ \mathfrak{G}_q \neq \varnothing,\ \mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B\},\\
\mathcal{C}_A(B) &= \{\mathscr{C} : E_q = 0,\ \mathfrak{G}_q = \varnothing,\ \mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B\},\\
\mathcal{C}_{Q,\mathrm{ex}}(B) &= \{\mathscr{C} : q \neq \mathrm{id},\ E_q = 0,\ \mathfrak{G}_q \neq \varnothing,\ \mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B\},\\
\mathcal{C}_{Q,\mathrm{rev}}(B) &= \{\mathscr{C} : q \neq \mathrm{id},\ E_q \neq 0,\ A = 0,\ S_q(\lambda) = 1 - \vartheta_\lambda^T(q) > 0,\ S_q(\lambda)^{-1} \in \mathfrak{C},\ \mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B\},\\
\mathcal{C}_{Q,\mathrm{nr}}(B) &= \{\mathscr{C} : q \neq \mathrm{id},\ E_q \neq 0,\ S_q(\lambda) = 1 - \Delta_\lambda(q) > 0,\ S_q(\lambda)^{-1} \in \mathfrak{C},\ \mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B\}.
\end{aligned}$$

The fourth and fifth families are disjoint as certificate records: they are indexed by the chosen reversible Schur-complement record or general resolvent record, respectively. A reversible operator may be analyzed with either record, but each charged certificate has one declared stability record.

Definition III.18.5 (Master-profile coordinates)**.** Fix the task family, size slice, budget, and five certificate families of Definition III.18.4.

(ii) Profile coordinates. Define

$$\begin{aligned}
G^{\mathrm{sat}}_{A,n}(B) &= \sup_{\mathscr{C}\in\mathcal{C}_A(B)} V_{\mathscr{C}},\\
G^{\mathrm{sat}}_{X,n}(B) &= \sup_{\mathscr{C}\in\mathcal{C}_X(B)} V_{\mathscr{C}},\\
G^{\mathrm{sat}}_{Q,\mathrm{ex},n}(B) &= \sup_{\mathscr{C}\in\mathcal{C}_{Q,\mathrm{ex}}(B)} V_{\mathscr{C}},\\
G^{\mathrm{sat}}_{Q,\mathrm{rev},n}(B) &= \sup_{\mathscr{C}\in\mathcal{C}_{Q,\mathrm{rev}}(B)} V_{\mathscr{C}},\\
G^{\mathrm{sat}}_{Q,\mathrm{nr},n}(B) &= \sup_{\mathscr{C}\in\mathcal{C}_{Q,\mathrm{nr}}(B)} V_{\mathscr{C}},
\end{aligned}$$

with $\sup\varnothing = 0$. Set

$$G^{\mathrm{sat}}_{Q,\mathrm{ql},n}(B) = \max\left\{G^{\mathrm{sat}}_{Q,\mathrm{rev},n}(B), G^{\mathrm{sat}}_{Q,\mathrm{nr},n}(B)\right\},$$

and retain the active-geometry profile

$$G^{\mathrm{sat}}_{n}(B) = \max\left\{G^{\mathrm{sat}}_{X,n}(B), G^{\mathrm{sat}}_{Q,\mathrm{ex},n}(B), G^{\mathrm{sat}}_{Q,\mathrm{ql},n}(B)\right\}. \tag{125}$$

For a declared monotone family functional $\mathfrak{M}_n$, replace $V_{\mathscr{C}}$ in each of the five displayed suprema by $\mathfrak{M}_n(I \mapsto V_{\mathscr{C},I})$ and write

$$G^{\mathrm{sat}}_{A,\mathfrak{M},n},\quad G^{\mathrm{sat}}_{X,\mathfrak{M},n},\quad G^{\mathrm{sat}}_{Q,\mathrm{ex},\mathfrak{M},n},\quad G^{\mathrm{sat}}_{Q,\mathrm{rev},\mathfrak{M},n},\quad G^{\mathrm{sat}}_{Q,\mathrm{nr},\mathfrak{M},n}.$$

Restricting the same record families to temporally admissible prospective sources gives the corresponding superscript $\mathrm{src},\mathrm{sat}$. This is family evaluation of the existing five coordinates, not an additional source profile.

Definition III.18.6 (Authority-relative master profile)**.** Fix the task family, size slice, budget, and master-profile coordinates of Definition III.18.5.

(iii) Authority restriction. Let $\mathcal{C}^{\mathrm{all}}_{QG}(B)$ be the union of the five certificate families. For a declared Caus authority envelope $\mathfrak{A}$, let

$$\mathcal{C}^{\mathfrak{A}}_{QG}(B) = \left\{ \mathscr{C} \in \mathcal{C}^{\mathrm{all}}_{QG}(B) : \begin{array}{l} \text{the selected generator belongs to } \mathfrak{A}, \\ \text{and } \mathfrak{K}_q \text{ is its transported authority record} \end{array} \right\}.$$

For a null geometry slot, the second condition is replaced by $\mathfrak{K}_q = \varnothing$; its quotient and stability data must still be computed for a generator selected from $\mathfrak{A}$. The authority-relative master profile is

$$\mathfrak{U}^{\mathrm{sat}}_{n,\mathfrak{A}}(B) = \sup_{\mathscr{C} \in \mathcal{C}^{\mathfrak{A}}_{QG}(B)} V_{\mathscr{C}}. \tag{126}$$

For the free authority envelope, write

$$\mathcal{C}_{QG}(B) = \mathcal{C}^{\mathrm{free}}_{QG}(B)$$

and

$$\mathfrak{U}^{\mathrm{sat}}_{n}(B) = \mathfrak{U}^{\mathrm{sat}}_{n,\mathrm{free}}(B) = \sup_{\mathscr{C} \in \mathcal{C}_{QG}(B)} V_{\mathscr{C}}. \tag{127}$$

(iv) Regime interpretation. The imposed and hybrid profiles are restrictions of the same universal master-certificate family, not additional structural regimes.

The first active term agrees with the ambient definition above. The five parameter regimes are pure ambient Geom/Caus, pure exact Abs, Geom/Caus on exact Abs, reversible geometrically compensated quasi-Abs, and general-resolvent geometrically compensated quasi-Abs. In log-cost notation, $G_n(d) = G^{\mathrm{sat}}_n(2^d)$ and $\mathfrak{U}_n(d) = \mathfrak{U}^{\mathrm{sat}}_n(2^d)$.

Example III.18.7 (Running example: an LPN parity quotient)**.** Continue Example II.7.4. A represented parity map $q_u(x) = \langle u, x \rangle$ is counted by the exact quotient profile only when its public derivation, fibers, induced dynamics, terminal data, and utility record have total saturated cost at most B. Its contribution is exactly $V_I(q_u)$ from Definition III.15.2; no separate semantic quotient is added to the profile.

III.18.3 Authority restriction and master normal form

Lemma III.18.8 (Authority monotonicity of the master profile)**.** *Let $\mathfrak{A}_1$ and $\mathfrak{A}_2$ have the same symmetric generator, boundary conditions, and terminal sector. Suppose every selected total skew operator legal under $\mathfrak{A}_1$, together with its provenance, has an identical representation of no greater cost under $\mathfrak{A}_2$. Equivalently, as resource-filtered represented families,*

$$\mathfrak{A}^{\mathsf{Caus}}_{1,I,\le B} \subseteq \mathfrak{A}^{\mathsf{Caus}}_{2,I,\le B}.$$

Then

$$\mathfrak{U}^{\mathrm{sat}}_{n,\mathfrak{A}_1}(B) \le \mathfrak{U}^{\mathrm{sat}}_{n,\mathfrak{A}_2}(B).$$

The imposed envelope is contained in every hybrid envelope whose control family contains zero. Such a hybrid envelope is contained in the free envelope whenever each of its total skew operators is budget-admissibly represented there. Under these conditions, the imposed profile is bounded by the hybrid profile, which is bounded by the free profile.

Proof. Every selected generator and causal provenance record legal under $\mathfrak{A}_1$ is legal under $\mathfrak{A}_2$, with the same derivation and cost. Hence $\mathcal{C}_{QG}^{\mathfrak{A}_1}(B) \subseteq \mathcal{C}_{QG}^{\mathfrak{A}_2}(B)$. The inequality follows by taking suprema. The final assertion is the free/imposed/hybrid specialization of this inclusion. □

The master normal form places every affordable quotient–geometry record in the common certificate space.

Theorem III.18.9 (Affordable quotient–geometry master normal form)**.** *Fix a declared Caus authority envelope. Every dynamically qualified, authority-preserving Geom/Abs structural certificate counted by the normalized source profiles above, and hence already carrying the charged record $\mathfrak{N}$, has, after composing its represented quotients and retaining the full accumulated cost, a master certificate of Definition III.18.1 in $\mathcal{C}_{QG}^{\mathfrak{A}}(B)$. The intersections of the five families in Definition III.18.4 with $\mathcal{C}_{QG}^{\mathfrak{A}}(B)$ form an exhaustive disjoint partition of these authority-relative certificate records. Under the free authority envelope, moreover,*

$$\mathfrak{U}_n^{\mathrm{sat}}(B) = \max\left\{m_n^{\mathrm{sat}}(B), G_{X,n}^{\mathrm{sat}}(B), G_{Q,\mathrm{ex},n}^{\mathrm{sat}}(B), G_{Q,\mathrm{rev},n}^{\mathrm{sat}}(B), G_{Q,\mathrm{nr},n}^{\mathrm{sat}}(B)\right\}. \tag{128}$$

A directed use of an active geometry contributes only through its authorized Caus *projection and the factor* $\mathrm{Align}_{\mathsf{Caus}}^{\mathrm{auth}}$*. Its current is the transported current of the same selected generator; a different target-aligned current is a new control, a new dynamic presentation, or a residual term. Valid Lift and Bdry add no additional intrinsic regime.*

A non-lumpable quotient without a transfer-class stability factor is not a dynamically qualified quotient comparison. It can enter the profile only after a represented quotient refinement, a separately charged geometry, or a compensated certificate satisfying Definition III.18.1.

Proof. Choose any represented derivation of the certificate having cost at most B, and proceed through its constructor DAG, carrying the selected generator and its authority record at every node. Primitive conductances and potentials have identity support and give $q = \mathrm{id}$ with an active geometry slot. Their directed use is admitted only when the full selected generator supplies the local or carrier comparison for the canonical geometric current. A represented quotient constructor gives $q \neq \mathrm{id}$. Nested quotient constructors collapse to one represented quotient; Proposition II.10.1 gives its exact defect and its symmetric and causal components, and charges every intermediate fiber interface. If the composite defect vanishes, the geometry slot is null precisely when the record uses only the quotient extraction coordinate; this is pure exact Abs. The component ledger is retained even when $E_q^{\mathsf{s}} + E_q^{\mathsf{a}} = 0$, so exactness by cancellation does not erase the selected causal operator. An active slot gives Geom/Caus on exact Abs. If the defect is nonzero, dynamic qualification requires an active slot and a transfer-class stability record for the full generator. The reversible Schur-complement record requires $A = 0$; otherwise the general resolvent record gives the last family. When $A = 0$, either comparison may be selected and charged.

Finite joins, postprocessing, and computation closure may produce a new potential or quotient, but the output and the entire construction trace then form a new charged derivation to which the same alternatives apply. The transformed-gradient normal form assigns every represented directed use to Caus, while Lemma II.4.5 restricts that use to the Hodge projection of the actual selected current. An auxiliary canonical geometric current may orient the Geom certificate but is not substituted for the operator current without an explicit represented identity. Exact Lift preserves the source measure, target fraction, and current identity, while Bdry supplies only scalar terminal data. Induction over the derivation DAG therefore yields the authority-relative five-case partition.

It remains to identify the null-slot profile. Erasing a null slot changes neither the quotient derivation nor its cost, and the null-slot conventions in Definition III.18.1 give $V_{\mathscr{C}} = V_q$. Conversely, every exact quotient record counted by $m_n^{\mathrm{sat}}(B)$ already contains the same charged normalization record and becomes a master certificate by the canonical null-slot annotation, without adding a constructor or output datum. Hence $G_{A,n}^{\mathrm{sat}}(B) = m_n^{\mathrm{sat}}(B)$. Taking suprema over the disjoint partition proves (128).

□

Corollary III.18.10 (No causal substitution in master normal form)**.** *Expansion into, and contraction from, the authority-relative master normal form preserve the selected total skew operator and the identity of every authorized current. Consequently, $\mathfrak{U}_{n,\mathfrak{A}}^{\mathrm{sat}}(B)$ optimizes only legal transforms of the imposed dynamics and declared controls. A different target-aligned current can affect this profile only after it is admitted and charged as a control in $\mathfrak{A}$; otherwise its authority-compatible factor is zero and it remains in the residual ledger.*

Proof. The expansion in Theorem III.18.9 copies the generator and current provenance through every constructor. Contraction composes the recorded transports and therefore recovers the same selected operator and current. The final assertion is Lemma II.4.5 applied to the resulting record.

□

III.19 Visibility Domination and First-Hit Accounting

The reduction proceeds through exact-geometry domination in Section III.19.1, represented path lifts and compensated success in Section III.19.2, the two-source reductions in Section III.19.3, the canonical exact first-hit quotient in Section III.20, and pointwise computation accountability in Section III.23.1. The resulting profile reductions are Corollaries III.19.4 to III.19.6.

III.19.1 Exact quotient-geometry domination

Theorems III.19.1 and III.19.3 and Lemma III.20.2 reduce compensated dynamic success to exact quotient visibility through a represented first-hit construction.

Theorem III.19.1 (Exact quotient geometry is visibility-dominated)**.** *For every budget B,*

$$G_{Q,\mathrm{ex},n}^{\mathrm{sat}}(B) \le m_n^{\mathrm{sat}}(B). \tag{129}$$

The comparison preserves temporal typing and family evaluation. In particular, fix a monotone family functional $\mathfrak{M}_n$. In the following display the two symbols are the $\mathfrak{M}_n$-evaluated restrictions of the existing prospective exact-support and exact-quotient profiles; they introduce no additional profile coordinate:

$$G_{Q,\mathrm{ex},\mathfrak{M},n}^{\mathrm{src,sat}}(B) \le m_{\mathfrak{M},n}^{\mathrm{src,sat}}(B). \tag{130}$$

Consequently, the master profile has the reduced representation

$$\mathfrak{U}_n^{\mathrm{sat}}(B) = \max\left\{m_n^{\mathrm{sat}}(B), G_{X,n}^{\mathrm{sat}}(B), G_{Q,\mathrm{rev},n}^{\mathrm{sat}}(B), G_{Q,\mathrm{nr},n}^{\mathrm{sat}}(B)\right\}$$

$$= \max\left\{m_n^{\mathrm{sat}}(B), G_{X,n}^{\mathrm{sat}}(B), G_{Q,\mathrm{ql},n}^{\mathrm{sat}}(B)\right\}. \tag{131}$$

Thus geometry on an exact quotient remains an operational search mode, but it introduces no larger target-visibility term than the exact quotient from which it is constructed.

Proof. Let $\mathscr{C} \in \mathcal{C}_{Q,\mathrm{ex}}(B)$. Its nonidentity support quotient is itself a qualified exact-quotient record by Definition III.18.1: it retains the same represented-fiber, terminal-event, positive-utility, nontrivial-public-structure, normalization, and cost data. Since $E_q = 0$, one has $S_q(\lambda) = 1$, $\overline{C}_q = C_q$, and $\overline{A}_q = A_q$. Moreover,

$$U_q \mathcal{M}_Q(D_Q) \subseteq \operatorname{ran} U_q = \operatorname{ran} P_q^{\mathrm{amb}}.$$

Orthogonal projection onto a subspace cannot have greater norm than projection onto a containing subspace, so

$$R_T^q(D_Q) \le \frac{\|P_q^{\mathrm{amb}} y\|^2}{\|y\|^2}.$$

The mixing indicator and the authority-compatible local factor lie in $[0,1]$ by Definition III.12.9, and $\rho_{\Phi_Q}(D_Q) \ge 0$. Substitution in (124) therefore gives

$$V_{\mathscr{C}} \le A_q \frac{\|P_q^{\mathrm{amb}} y\|^2}{\|y\|^2} = V_q.$$

The qualified exact quotient record, including the gate data just listed, is a subrecord of $\mathscr{C}$, hence its saturated cost is at most B. Taking the supremum proves (129). If $\mathscr{C}$ is a prospective source, its quotient subrecord, nondynamic Abs gates, and the same qualification indicator are available before the transition for which the geometry is invoked. Thus the pointwise comparison preserves temporal source admissibility. Applying a monotone $\mathfrak{M}_n$ and taking the restricted supremum proves (130). Combining the first inequality with (128) and the definition of $G_{Q,\mathrm{ql},n}^{\mathrm{sat}}$ proves (131).

□

III.19.2 Represented path lifts and compensated success

Lemma III.19.2 (Represented finite-horizon path lift). *Let P be a represented finite Markov or sub-Markov kernel on X, let $T \subseteq X$ be absorbing, and let μ_0 be a represented initial law. Assume that the declared public full-support gauge gives T mass strictly between zero and one. Adjoin a cemetery state in the sub-Markov case, with its represented deficit weight. For every integer $H \ge 1$, there is a finite represented outcome/terminal transcript carrier and an exact first-hit quotient $q_{P,H}$ such that*

$$\Pr_{\mu_0}\{\tau_T \le H\} \le eH\, V_{q_{P,H}}. \tag{132}$$

The carrier record has cost bounded by a monotone ledger $\Psi_{\mathrm{path}}(\mathrm{Cost}(P), H, n)$ consisting of H executions of the represented transition interface and retention of their realized outcomes, $H+1$ target tests, peak live-state capacity, the transcript coordinates, the initialization, the target test, the public full-support product gauge, the exact quotient fibers, and the deadline interface. No extensional transition table, rationality, dyadic-probability, reversibility, or Valid-Lift hypothesis is required.

Proof. On $\widehat{X} = X \sqcup \{\partial\}$, complete a sub-Markov kernel by sending missing mass to the absorbing cemetery state ∂. Choose a represented finite-outcome realization of $\widehat{P}$ and μ_0. Such a realization is always available on finite X: the initial outcome may be x_0 with law μ_0, and the next outcome at x may be the successor y with law $\widehat{P}(x, y)$. Write a_0 for the initialization outcome and a_{t+1} for the outcome of transition t. The recursion of Definition II.5.4 determines

$$x_0 = x(a_0), \qquad x_{t+1} = x_t \cdot a_{t+1}.$$

For each t, record the target flag $s_t = \mathbf{1}_T(x_t)$. Let Ξ_H be the finite padded syntactic carrier of records

$$\xi = (a_0, a_1, \dots, a_H; s_0, s_1, \dots, s_H),$$

and put $\Omega_H = \Xi_H \times \{0, \dots, H\}$. Its legal initialization at pointer $k = 0$ is obtained by sequentially executing the initializer and transition interface once, retaining only the outcomes and target flags. Equivalently, its law is the analytic identity

$$\mathbb{P}_{\mu_0}(\xi) = p_{\text{init}}(a_0) \prod_{t=0}^{H-1} p(a_{t+1} \mid x_t(\xi)) \prod_{t=0}^{H} \mathbf{1}\{s_t = \mathbf{1}_T(x_t(\xi))\}.$$

Equip the entire syntactic transcript carrier with a public full-support product gauge fixed by the padded record encoding and independently of the legal transcript law. On every syntactic record, whether or not it has positive legal probability, use the deterministic replay update

$$F_H(\xi, k) = \begin{cases} (\xi, k), & s_k = 1 \text{ or } k = H, \\ (\xi, k+1), & s_k = 0 \text{ and } k < H. \end{cases}$$

Under the legal initialization, this update has exactly the first-hit law of the first H transitions of P. Preparation uses one live state x_t at a time. Replay reads only the stored flags and never reconstructs or stores $(x_0, \dots, x_H)$. The displayed product is an analytic identity for the generated transcript law; its individual weights need not be evaluated or materialized by a separate constructor. Let P_H be the point-mass kernel induced by F_H and set $L_H = P_H - I$. The path quotient carries the canonical exact-profile normalization record

$$\mathfrak{N}_H = (L_H, 1/H, 1, P_H, L_H, 1/H),$$

whose entries are already charged by the path update and deadline ledger.

Adjoin the remaining deadline $r = H - k$ and define

$$q_{P,H}(\xi, k) = (r, \min\{0 \le j \le r : s_{k+j} = 1\}),$$

with the second coordinate equal to ∞ when there is no hit. Its fibers are finite Boolean combinations of the represented stored terminal flags. The deterministic shift sends (r, j) to $(r-1, j-1)$ for $j \ge 1$, sends (r, ∞) to $(r-1, \infty)$, and makes the success and deadline cells absorbing. Thus the quotient intertwines exactly. With the canonical pushforward quotient gauge, Proposition III.11.3 gives unit transfer. The calculation in (146)–(148), which does not require $\mu_0(T) = 0$, now gives

$$V_{q_{P,H}} \ge \frac{\Pr_{\mu_0}\{\tau_T \le H\}}{eH}.$$

Sequential operational generation of the transcript, its retained outcome and terminal records, one peak native state, the cemetery interface, and the flag scan give the stated repeated-use cost ledger. Empty or full reference targets are excluded from this lemma because their centered contrast is zero. In any theorem whose left-hand structural visibility is then zero, the desired upper bound is immediate; an algorithmic full-target task is handled by the canonical fresh start phase in Lemma III.20.2.

□

The first-hit theorem charges complete target arrival through the represented dynamic certificate.

Theorem III.19.3 (First-hit accounting for dynamic compensation)**.** *Let $\mathscr{C} = (\mathfrak{N}, q, \mathfrak{G}_q, \mathfrak{S}, \mathfrak{K}_q)$ be a dynamically qualified nonzero-defect master certificate for a finite presented process. Its charged record $\mathfrak{N} = (\widetilde{L}, \zeta, c, P, L, \lambda)$ supplies $L = P - I$ and $0 < \lambda \leq 1$ by Lemma III.17.5. Let τ_T be the first target-hitting time of the full selected process, measured in uniformized transitions, and put*

$$\alpha_H = \Pr_{\mu_0}\{\tau_T \leq H\}.$$

For every integer $H \geq 1$, let $q_{\mathscr{C},H}$ be the exact first-hit quotient of the represented finite-horizon transcript carrier, including the quotient, geometry, within-fiber data, selected current, comparison record, path-weight rule, and remaining deadline. Then

$$V_{\mathscr{C}} \leq \lambda\big(\alpha_H + (1+\lambda)^{-H}\big) \leq e\lambda H V_{q_{\mathscr{C},H}} + \lambda(1+\lambda)^{-H}. \tag{133}$$

In particular, suppose a budget family and an integer $H_B \geq 1$ have uniform bounds

$$\mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B, \qquad 1 \geq \lambda \geq \lambda_B > 0$$

and every resulting first-hit quotient has saturated cost at most $\Psi(B, H_B, n)$. Then

$$G^{\mathrm{sat}}_{Q,\mathrm{ql},n}(B) \leq eH_B m^{\mathrm{sat}}_n(\Psi(B, H_B, n)) + (1+\lambda_B)^{-H_B}. \tag{134}$$

The displayed finite-budget inequality is available whenever the stated uniform λ_B is part of the budget slice. In the polynomial specialization set $B = b_{\mathrm{poly}}$. For a fixed numerical transfer margin n^{-k}, let

$$H_k = \lceil n^{k+3} \rceil, \qquad B^{\mathrm{hit}}_k = \Psi(b_{\mathrm{poly}}, H_k, n)$$

be the class-preserving path-lift ceiling supplied by the declared transfer system. No profile-wide lower bound on λ is needed: if $m^{\mathrm{sat}}_n(B^{\mathrm{hit}}_k)$ is negligible at these named path-lift ceilings, then the reversible and general-resolvent compensated quasi-lumpable envelopes at the single declared ceiling b_{poly} are negligible.

Proof. If the carrier target has reference mass zero or one, then $V_{\mathscr{C}} = 0$ by Definition III.18.1, and every asserted upper bound is immediate. Assume henceforth that its public full-support gauge gives the target mass strictly between zero and one.

All active-geometry factors in (124) lie in $[0, 1]$, and $\overline{C}_q \geq 1$. Therefore

$$V_{\mathscr{C}} \leq \overline{A}_q S_q(\lambda) \leq \lambda S_q(\lambda) A^Q_T(\lambda; L_Q, \bar{\mu}_0).$$

Dynamic qualification includes the target-functional comparison (123); hence

$$V_{\mathscr{C}} \le \lambda A_T(\lambda; L, \mu_0). \tag{135}$$

Uniformization identifies the last target functional with the discounted first-hit transform

$$A_T(\lambda; L, \mu_0) = \sum_{j\ge 0} \Pr_{\mu_0}\{\tau_T = j\}(1+\lambda)^{-j}.$$

Splitting this sum at H and using $(1+\lambda)^{-j} \le (1+\lambda)^{-H}$ for $j > H$ gives

$$A_T(\lambda; L, \mu_0) \le \alpha_H + (1+\lambda)^{-H}.$$

The uniformized kernel is represented because every datum in $\mathscr{C}$, its selected dynamics, and the uniformization is part of the charged constructor record. Lemma III.19.2 therefore constructs $q_{\mathscr{C},H}$ on the represented transcript carrier, with all of these data and the deadline included in its cost, and yields

$$\alpha_H \le eHV_{q_{\mathscr{C},H}}.$$

Combining the preceding three inequalities proves (133). The quotient $q_{\mathscr{C},H}$ is exact and has cost at most $\Psi(B, H, n)$, so its visibility is bounded by the exact saturated profile at that enlarged budget. Taking the supremum over both nonzero-defect master families, using $\lambda \le 1$, and observing that $(1+\lambda)^{-H_B} \le (1+\lambda_B)^{-H_B}$ proves (134).

For completeness, the polynomial conclusion does not require a uniform λ_B. Fix the numerical transfer margin n^{-k}. Every certificate satisfies $V_{\mathscr{C}} \le \lambda$ by (135). Split the budget slice into

$$\lambda < n^{-(k+2)} \qquad \text{and} \qquad \lambda \ge n^{-(k+2)}.$$

The first subfamily has visibility at most $n^{-(k+2)}$. For the second take $H_k = \lceil n^{k+3} \rceil$. Since $0 < \lambda \le 1$,

$$(1+\lambda)^{-H_k} \le \exp(-\lambda H_k/2) \le e^{-n/2}$$

for all sufficiently large n. The represented path lift at horizon H_k has cost at most the named class-preserving ceiling $B_k^{\mathrm{hit}} = \Psi(b_{\mathrm{poly}}, H_k, n)$. Hence this subfamily is bounded by

$$eH_k\, m_n^{\mathrm{sat}}(B_k^{\mathrm{hit}}) + e^{-n/2},$$

which is negligible by the stated bound at that path-lift ceiling. Closure of the numerical transfer class over its fixed margins gives negligibility of the full quasi-lumpable envelope. This argument keeps the one declared polynomial ceiling and its charged path-lift enlargements; it does not replace them by a quantifier over polynomial budgets.

□

III.19.3 Two-source and provenance reductions

Corollary III.19.4 (Two-source reduction of the affordable master profile)**.** *Under the uniform envelope hypotheses of Theorem III.19.3,*

$$\mathfrak{U}_n^{\mathrm{sat}}(B) \le \max\left\{G_{X,n}^{\mathrm{sat}}(B), m_n^{\mathrm{sat}}(B), eH_B m_n^{\mathrm{sat}}(\Psi(B, H_B, n)) + (1+\lambda_B)^{-H_B}\right\}. \tag{136}$$

Hence, at $B = b_{\mathrm{poly}}$, negligible ambient Geom at the declared ceiling, negligible exact-quotient visibility there, and the path-lift bounds $m_n^{\mathrm{sat}}(B_k^{\mathrm{hit}}) = \mathrm{negl}(n)$ from Theorem III.19.3 imply a negligible full master profile. Exact quotient geometry is absorbed at the declared ceiling; nonzero-defect compensated geometry is absorbed through its charged exact first-hit quotient at the named class-preserving enlargement.

Proof. Use the five-case identity (128). Theorem III.19.1 bounds geometry on an exact quotient by $m_n^{\mathrm{sat}}(B)$, while (134) bounds both compensated nonzero-defect families. Taking the maximum gives (136). The polynomial conclusion follows from the final assertion of Theorem III.19.3.

□

Corollary III.19.5 (Quantitative accounting of composite alignment). *Let R be a dynamically qualified target-aligned object obtained by an authority-preserving uniform derivation from public observables, including a represented finite join, product-containing postprocessing, quotient, geometry, potential, or lifted computation. If its declared authority envelope is $\mathfrak{A}$, let B_R be any admissible uniform budget for its chosen represented derivation, equivalently*

$$B_R \in \mathfrak{B}_{\mathrm{sat},n}(R).$$

then its conditioned structural visibility satisfies

$$\mathrm{Vis}_I(R) \le \mathfrak{U}_{n,\mathfrak{A}}^{\mathrm{sat}}(B_R) \le \mathfrak{U}_n^{\mathrm{sat}}(B_R).$$

More generally, for every family $\mathscr{R}_B$ of such derivations with total saturated cost at most B,

$$\sup_{R\in\mathscr{R}_B} \mathrm{Vis}_I(R) \le \mathfrak{U}_{n,\mathfrak{A}}^{\mathrm{sat}}(B) \le \mathfrak{U}_n^{\mathrm{sat}}(B),$$

when the derivations in $\mathscr{R}_B$ share $\mathfrak{A}$.

Thus a join of observables may expose target alignment, but the joined fiber map, the combining rule, and the dynamic comparison enter at their complete derivation cost. An extensional transformation with no derivation at budget B contributes no term to the budget-B profile.

Proof. Theorem III.18.9 assigns the declared derivation of R to one of the five master-certificate families. Its conditioned visibility is therefore a term optimized by $\mathfrak{U}_{n,\mathfrak{A}}^{\mathrm{sat}}(B_R)$. Authority monotonicity bounds this by the free profile. Taking the supremum proves the family statement. The final assertion is the minimality clause of the budgeted derivability closure.

□

Corollary III.19.6 (Compatibility with quotient derivation normal form). *The quotient supporting each of the last four regimes is the terminal output of an expanded quotient derivation in the sense of Definition III.15.3. Adding a represented conductance, potential, current, or stability comparison extends that derivation and retains the same quotient fibers. It changes the dynamic certificate but does not supply a second source of quotient provenance. The extension must also retain the selected generator and causal-current provenance; otherwise it changes the authority envelope or has zero authority-compatible causal factor. A quotient refinement is a new terminal fiber map and enters at the saturated cost of its complete expanded derivation.*

Proof. Lemma III.1.9 puts the represented quotient and every attached dynamic datum in one cost-preserving expanded record. The fiber map is unchanged when the record is extended by quotient geometry or Caus data. The causal data are copied from the selected generator by Lemma II.4.5.

The exact and nonzero-defect cases are then precisely the alternatives in Theorem III.18.9. A refinement changes the terminal map and therefore has its own quotient record and cost.

□

III.20 Canonical Exact First-Hit Quotient

Definition III.20.1 (Canonical normalized replay carrier and first-hit quotient)**.** Let $\mathcal{A}_n$ be a presentation-relative uniform represented operational computation with running time at most $H_0 = H_0(n) \geq 0$. Assume that its one-step update, peak native and work state, outcome/terminal transcript, and public terminal verifier have costs bounded by the declared resource envelope.

(i) Start normalization. Apply the canonical start-state normalization: adjoin a fresh public phase bit whose start value is outside the success event. The first transition realizes the original initialization, sets the phase bit to active, and thereafter runs the original update; the terminal verifier is enabled only in the active phase. Write

$$H := H_0 + 1$$

for the normalized horizon. Original success by time H_0, including success of the original initialization, equals normalized first-hit success by time H. The augmentation costs one transition, one phase bit, the represented initializer, and constant description overhead. Thus support outside success is a normalization available to every finite-horizon computation, not an assumption on the procedure.

(ii) Charged transcript carrier. Pass to the full operational transcript representation and execute the represented transition rule sequentially for the normalized horizon. At each step retain its finite outcome record and the public terminal record produced by the verifier. Denote the resulting padded transcript by

$$\xi = (a_1, \dots, a_H; s_0, \dots, s_H),$$

where $s_t \in \{0, 1\}$ is the success flag at time t. The execution keeps one current native state and does not store its intermediate values. Adjoin a replay pointer, represented equivalently by a remaining-deadline coordinate

$$r \in \{0, \dots, H\}.$$

Write the resulting replay configurations as $(\xi, r) \in \Omega_I$; the current transcript position is $H - r$. Pad every finite record coordinate to its declared resource bound and extend the update inertly on unused syntactic encodings. Equip this finite transcript product space with the canonical public full-support product gauge $\widetilde{\mu}_I$, fixed by the configuration encoding before the update is analyzed. This is an admissible configuration reference measure by Definition II.3.1; it requires no structural-Lift or target-fraction hypothesis.

(iii) Replay dynamics. The augmented replay update reads s_{H-r}, decrements r, and advances the pointer at each nonabsorbing step. It never invokes the native successor or the verifier. Its one-step update is the deterministic map

$$F_I : \Omega_I \longrightarrow \Omega_I.$$

Explicitly,

$$F_I(\xi, r) = \begin{cases} (\xi, r), & s_{H-r} = 1 \text{ or } r = 0, \\ (\xi, r-1), & s_{H-r} = 0 \text{ and } r \geq 1. \end{cases}$$

Success and deadline expiration are made absorbing. A non-successful early halt has terminal flags equal to zero thereafter, so replay continues inertly until it enters the failure cell at $r = 0$. Let

$$\widetilde{T}_I = \{(\xi, r) : s_{H-r} = 1\}$$

be the represented replay success event. Under the legal transcript law this event is exactly the public verifier event of the original computation. The fresh start flag is zero, and whenever the syntactic success sector is nonempty the full-support product gauge gives

$$0 < \widetilde{\mu}_I(\widetilde{T}_I) < 1.$$

If the syntactic success sector is empty, the success probability is zero and the asserted bound is immediate. Thus no nondegeneracy assumption is imposed on the computation.

(iv) First-hit quotient. For a transcript with remaining deadline r, define

$$\tau_r(\xi) := \min\{0 \leq j \leq r : F_I^j(\xi, r) \in \widetilde{T}_I\},$$

with $\tau_r(\xi) = \infty$ if the displayed set is empty, and set

$$q_{\mathcal{A}}(\xi, r) := (r, \tau_r(\xi)).$$

Its quotient state space is

$$Q_{\mathcal{A}} = \{(r, j) : 0 \leq j \leq r \leq H\} \cup \{(r, \infty) : 0 \leq r \leq H\}.$$

Lemma III.20.2 (Exact budget-admissible first-hit quotient)**.** *Let $\mathcal{A}_n$, H, Ω_I, F_I, $\widetilde{T}_I$, and $q_{\mathcal{A}}$ be the represented objects constructed under the hypotheses of Definition III.20.1.*

(v) Exactness, cost, and visibility. *Then $q_{\mathcal{A}} : \Omega_I \to Q_{\mathcal{A}}$ is a budget-admissible, terminal-compatible, exactly lumpable represented quotient, with saturated cost bounded by* (140) *below. As I varies, these maps are produced by one uniform quotient derivation from the represented rule $\mathcal{A}_n$, the public verifier, and the declared horizon. Write $\mu_{0,I}$ for the induced legal transcript law at replay position $r = H$. If*

$$\alpha_I := \Pr_{\mu_{0,I}}\{\tau_H < \infty\},$$

then its visibility in the sense of Definition Quotient extraction coordinate satisfies

$$V_{q_{\mathcal{A}}} \geq \frac{\alpha_I}{eH}. \tag{137}$$

For this canonical quotient the comparison and normalization factor in Definition III.15.1 is exactly one: the quotient law is the pushforward of the lifted reference law, the quotient lift is isometric, the dynamics intertwine exactly, and the terminal functional is preserved. Consequently, inverse-polynomial success of a polynomial-resource computation produces inverse-polynomial visibility at polynomial saturated cost without an additional comparison hypothesis.

Proof. Step 1: replay preserves the operational law.

Let z_0 be the normalized fresh native start state. For a transcript prefix $a_{1:t}$, define the single current native state recursively by

$$z_t = z(a_{1:t}), \qquad z_{t+1} = z_t \cdot a_{t+1}.$$

Writing ver_I for the public verifier, the sequentially generated transcript has law

$$\Pr\{\xi\} = \prod_{t=0}^{H-1} p(a_{t+1} \mid z_t) \prod_{t=0}^{H} \mathbf{1}\{s_t = \mathrm{ver}_I(z_t)\},$$

with any subprobability deficit represented by the declared failure outcome. This identity follows inductively from the defining conditional outcome law at the current native state. Hence preparation followed by deterministic flag replay has exactly the original terminal and first-hit law. No transition table or intermediate native-state list is materialized; preparation applies the represented update to one live state and appends only its outcome and terminal records.

Step 2: represented fibers and their cost.

For $0 \le j \le r$, the fiber over (r, j) is

$$q_{\mathcal{A}}^{-1}(r,j) = \Big\{(\xi, r) : F_I^i(\xi, r) \notin \widetilde{T}_I \;\; (0 \le i < j), \quad F_I^j(\xi, r) \in \widetilde{T}_I\Big\}. \tag{138}$$

The no-hit fiber is

$$q_{\mathcal{A}}^{-1}(r,\infty) = \Big\{(\xi, r) : F_I^i(\xi, r) \notin \widetilde{T}_I \;\; (0 \le i \le r)\Big\}. \tag{139}$$

Given a label (r, j), its fiber predicate is constructed uniformly from the represented transcript interface and the deadline. Preparing the transcript uses at most H applications of the represented one-step update and $H+1$ applications of the public verifier. Membership reads at most $H+1$ stored terminal flags and invokes neither operation. Consequently,

$$\begin{aligned} \mathrm{Cost}_{\mathrm{sat},n}(q_{\mathcal{A}}) \preceq H \odot c_{\mathrm{step}} \oplus (H+1) \odot c_T \oplus c_{\mathrm{native}}^{\mathrm{peak}} \oplus c_{\mathrm{transcript}} \oplus c_{\mathrm{init}} \\ \oplus c_{\mathrm{input}} \oplus c_{\mathrm{gauge}} \oplus c_{\mathrm{interface}} \oplus c_{\mathrm{desc}} \oplus c_{\mathrm{clock}}. \end{aligned} \tag{140}$$

where the terms are uniform size-slice majorants for the represented update, verifier, peak native/work state, accumulated outcome/terminal transcript, initializer, public-input access, reference-gauge description, task interface, procedure description, and deadline coordinate in the declared resource monoid. In the numeric polynomial instance this specializes to

$$O\Big(H(c_{\mathrm{step}} + c_T) + c_{\mathrm{native}}^{\mathrm{peak}} + c_{\mathrm{transcript}} + c_{\mathrm{init}} + c_{\mathrm{input}} + c_{\mathrm{gauge}} + c_{\mathrm{interface}} + |\operatorname{desc}(\mathcal{A}_n)| + \log H\Big).$$

Thus the quotient and its fiber predicates belong to the budgeted derivability closure. Here a fiber representative is a compact public fiber predicate or description on which finite Boolean operations can be performed; no representative element of a nonempty fiber is required.

Step 3: exact quotient dynamics.

The remaining-deadline coordinate makes the quotient transition exact. For $r \ge 1$,

$$(r, j) \longmapsto (r-1, j-1) \qquad (1 \le j \le r), \tag{141}$$

and

$$(r,\infty) \longmapsto (r-1,\infty). \tag{142}$$

The cells $(r,0)$ are absorbing success cells, and $(0,\infty)$ is the absorbing timeout/failure cell. Let P be the transition kernel induced by F_I, normalize the algorithm embedding as $L_{\Omega_I} = P - I$, and let U denote the lift of quotient observables. The target and failure sectors are unions of quotient cells, and the lifted update intertwines exactly with the quotient update:

$$L_{\Omega_I} U = U L_Q. \tag{143}$$

Equip the quotient with $\nu_Q = (q_{\mathcal{A}})_* \widetilde{\mu}_I$ and the pushed-forward initialization. The target is the pullback of the success labels, and both success and timeout sectors are absorbing. Proposition III.11.3, applied to (143), therefore identifies the full and quotient hitting laws exactly and gives

$$C_{q_{\mathcal{A}}} = 1. \tag{144}$$

Its quotient lumpability defect is zero, and all construction overhead is already included in (140).

Step 4: nontriviality of the quotient record.

If $\alpha_I = 0$, inequality (137) is immediate. Suppose henceforth that $\alpha_I > 0$. Since the normalized initial law is concentrated on the fresh start phase outside $\widetilde{T}_I$, some finite first-hit fiber with $j \geq 1$ has positive initial mass. Together with the deadline labels, this fiber strictly refines the two-cell verifier partition. Thus the quotient satisfies the nontrivial-public-structure condition, while its positive first-hit mass supplies nonzero quotient utility.

Step 5: discounted arrival and visibility.

Let

$$y_I = \mathbf{1}_{\widetilde{T}_I} - \widetilde{\mu}_I(\widetilde{T}_I).$$

Since $\widetilde{T}_I$ is a union of the cells $(r,0)$, $P^{\mathrm{amb}}_{q_{\mathcal{A}}} y_I = y_I$, where $P^{\mathrm{amb}}_{q_{\mathcal{A}}} = U_{q_{\mathcal{A}}} P_{q_{\mathcal{A}}}$, and hence

$$\frac{\|P^{\mathrm{amb}}_{q_{\mathcal{A}}} y_I\|_2^2}{\|y_I\|_2^2} = 1. \tag{145}$$

Conditional on requiring j discrete transitions to hit the target, uniformization for $L_{\Omega_I} = P - I$ gives the discount factor $(1+\lambda)^{-j}$ [41]. Consequently,

$$A_T^Q(\lambda) = \sum_{j=0}^{H} \Pr_{\mu_{0,I}}\{\tau_H = j\}(1+\lambda)^{-j}. \tag{146}$$

Taking $\lambda = 1/H$ gives

$$A_T^Q(1/H) \geq (1+1/H)^{-H} \alpha_I \geq e^{-1} \alpha_I. \tag{147}$$

The exact-profile normalization record is the canonical charged tuple

$$\mathfrak{N}_{\mathcal{A}} = (L_{\Omega_I}, 1/H, 1, P, L_{\Omega_I}, 1/H).$$

Its kernel, generator, discount, and unit majorant are already present in the transition and clock ledger of (140); no rate-table enumeration or additional normalization is used.

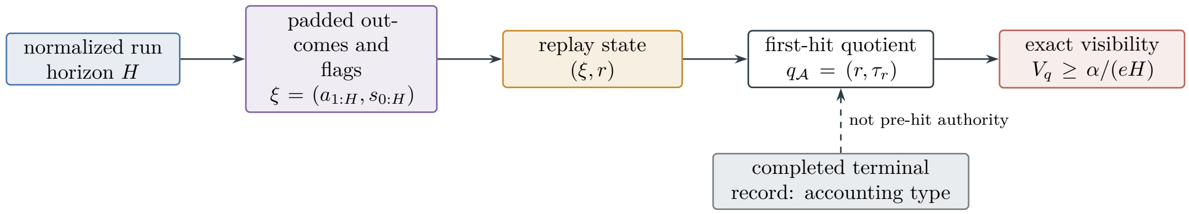

Figure 34: The canonical replay retains the charged transcript and advances only a public deadline pointer. The quotient (r, τ_r) is exactly lumpable and converts success probability to retrospective visibility. Its temporal type remains accounting unless a separate pre-hit source certificate is supplied.

By Definition Part I quotient-utility condition and exact lumpability,

$$A_{q_{\mathcal{A}}} = \min\left\{1, \lambda A_T^Q(\lambda)\right\} \geq \frac{\alpha_I}{eH}. \tag{148}$$

Combining (145) and (148) yields

$$V_{q_{\mathcal{A}}} = A_{q_{\mathcal{A}}} \frac{\|P_{q_{\mathcal{A}}}^{\mathrm{amb}} y_I\|_2^2}{\|y_I\|_2^2} \geq \frac{\alpha_I}{eH},$$

which proves (137).

□

Figure 34 shows how a completed operational run becomes an exact accounting quotient without being retyped as a prospective source.

III.21 Temporal Source Admissibility and Prospective Accounting

The structural ledger distinguishes information available in time to influence a search from labels constructed from its completed terminal record. The distinction is defined on the represented transcript carrier and is independent of the task family.

III.21.1 Pre-hit source and selector records

Definition III.21.1 (Pre-hit filtration and temporally admissible source)**.** Let $\mathcal{A}$ be a represented finite-horizon computation with normalized horizon H, full clocked transcript carrier $\Omega_{\mathcal{A}}^{[H]}$, and public verifier flags $s_0, \ldots, s_H$. The fresh-start normalization of Lemma III.20.2 gives $s_0 = 0$. Put

$$\tau = \inf\{t \in \{1, \ldots, H\} : s_t = 1\}, \qquad \inf \varnothing = \infty.$$

For $0 \leq t < H$, let

$$\mathrm{pref}_t : \Omega_{\mathcal{A}}^{[H]} \longrightarrow \Omega_{\mathcal{A}}^{[t]}$$

be the represented map retaining the legal transcript through time t. The complete pre-hit record $\mathcal{H}_t$ contains the public presentation, the represented effects of the current native state, the outcomes and internal randomness revealed by time t, the candidates generated by time t, the flags $s_0, \ldots, s_t$, and the answers of presentation-admissible oracle calls completed by time t. It contains no later outcome, verifier flag, terminal record, or oracle answer.

A represented record R is *available at time t* when a represented prefix evaluator R_t satisfies

$$R(\omega) = R_t(\mathrm{pref}_t\,\omega) \tag{149}$$

for every pair consisting of a legal prefix in $\{\tau > t\}$ and a legal completion ω of that prefix.

A record is a *temporally admissible structural source* for the transition after time t when:

(i) it is available at time t;
(ii) its complete represented derivation and evaluation cost are charged before that transition;
(iii) its derivation has no later verifier flag $s_{t+1}, \ldots, s_H$, later terminal record, or later oracle answer as an ancestor; and
(iv) it is an ancestor in the represented derivation DAG of the transition, selector, potential, quotient comparison, target-discrimination readout, or cell-ordering rule whose target progress it is invoked to explain.

Earlier negative verifier flags may occur in $\mathcal{H}_t$. Their effect is included in the remaining-search-space baseline of Definition III.21.2; the flags receive no independent structural-source charge. A represented record that fails the four conditions remains available for terminal or accounting purposes but is not a source for an earlier transition.

Temporal admissibility is representation-sensitive. If one denotation has a retrospective representation and an independent prefix representation, source admissibility is established only by the latter, with its complete cost and provenance. An inadmissible oracle that supplies a later answer changes the presented family by Theorem III.33.2; it does not make the answer pre-hit data for the original presentation.

Definition III.21.2 (Represented pre-hit selector and neutral baseline)**.** Use the charged clock refinement in which each transition performs at most one public verifier test. A transition that performs $k > 1$ tests is expanded into k represented microsteps, and a transition that performs no test selects a null symbol $\perp$. The refined horizon counts these microsteps and remains within the class-preserving execution enlargement.

On $\{\tau > t\}$, let $\mathcal{C}_t(\mathcal{H}_t)$ be the represented finite carrier of eligible next-test cells, candidates, and, when required, the null symbol. Extend the public verifier V_I to this carrier by $V_I(\perp) = 0$.

A represented algorithmic selector is a represented next-test sampler whose conditional law is the probability kernel

$$K_t(\,\cdot \mid \mathcal{H}_t)$$

on $\mathcal{C}_t(\mathcal{H}_t)$. Its sampler, random-bit use, and transition implementation are temporally admissible in the sense of Definition III.21.1; the kernel table need not be materialized. A deterministic selector is the point-mass case. Let

$$\nu_t(\,\cdot \mid \mathcal{H}_t)$$

be a declared search-neutral baseline kernel on the same eligible set. Examples include uniform enumeration, weighted enumeration, and another explicitly declared target-neutral testing rule.

For either kernel $M_t \in \{K_t, \nu_t\}$, write

$$M_t(V_I = 1 \mid \mathcal{H}_t) := \sum_{C \in \mathcal{C}_t(\mathcal{H}_t)} M_t(C \mid \mathcal{H}_t) V_I(C).$$

This is analytic notation for the acceptance probability of the next test. It does not enumerate the carrier or construct the set of accepting candidates.

Define the step-t target-alignment advantage by

$$\mathrm{Adv}_t(K_t;\nu_t) = \mathbb{E}\Big[\mathbf{1}_{\{\tau>t\}}\left(K_t(V_I = 1 \mid \mathcal{H}_t) - \nu_t(V_I = 1 \mid \mathcal{H}_t)\right)\Big]. \tag{150}$$

The expectation evaluates the represented sampler analytically; it gives the computation no accepting candidate or verifier-positive fiber.

For a saturated budget B, define the prospective selector profile

$$\mathrm{Sel}_n^{\mathrm{sat}}(B) = \sup\left\{ \left(\mathrm{Adv}_t(K_t;\nu_t)\right)_+ : \begin{array}{l} 0 \le t < H; \\ K_t \text{ is family-uniform;} \\ K_t \text{ is temporally admissible;} \\ \text{complete prefix and transition cost is at most } B; \\ \nu_t \text{ is the baseline} \end{array} \right\}, \tag{151}$$

with value 0 when the sampler class is empty. The baseline is an analytic comparison rule and need not be executed by the computation. Instance and task-family subscripts are added when more than one presentation is under discussion.

III.21.2 Prospective selector accountability

Theorem III.21.3 (Prospective selector accountability and no-free target-aligned rearrangement)**.** *Let $\mathcal{A}$ be a presentation-relative uniform represented finite-horizon computation, including computations expressed through presentation-admissible oracle interfaces. Apply the one-test clock refinement of Definition III.21.2. Let K_t be its next-test selector after time t, let ν_t be the declared neutral baseline, and put*

$$\mathrm{Enum}_\nu(\mathcal{A}, H) = \sum_{t=0}^{H-1} \mathbb{E}\Big[\mathbf{1}_{\{\tau>t\}}\nu_t(V_I = 1 \mid \mathcal{H}_t)\Big].$$

Then

$$\Pr\{\tau \le H\} = \mathrm{Enum}_\nu(\mathcal{A}, H) + \sum_{t=0}^{H-1} \mathrm{Adv}_t(K_t;\nu_t). \tag{152}$$

If every temporally admissible represented sampler within the class-preserving enlarged budget $\Psi(B,n)$ has advantage at most $\delta_n \ge 0$, then every budget-B computation satisfies

$$\Pr\{\tau \le H\} \le \mathrm{Enum}_\nu(\mathcal{A}, H) + H\,\mathrm{Sel}_n^{\mathrm{sat}}(\Psi(B,n)) \le \mathrm{Enum}_\nu(\mathcal{A}, H) + H\delta_n. \tag{153}$$

Conversely, if

$$\Pr\{\tau \le H\} > \mathrm{Enum}_\nu(\mathcal{A}, H) + \varepsilon,$$

then some $0 \le t < H$ satisfies

$$\mathrm{Adv}_t(K_t;\nu_t) > \frac{\varepsilon}{H}. \tag{154}$$

Thus every improvement over the declared enumeration baseline exposes a temporally admissible represented target-aligned sampler. The responsible record is its pre-hit sampler and transition implementation. Any ranking, potential, or quotient readout invoked as its structural source must occur in that sampler's pre-hit ancestry. A label formed after target contact is not a source for the transitions preceding that contact.

Proof. The events $\{\tau = t+1\}$, $0 \le t < H$, are pairwise disjoint. Conditional on $\mathcal{H}_t$ and $\{\tau > t\}$, the next test is sampled from $K_t(\cdot \mid \mathcal{H}_t)$. The one-test normalization and the analytic acceptance-mass notation give the conditional identity

$$\Pr\{\tau = t+1 \mid \mathcal{H}_t, \tau > t\} = K_t(V_I = 1 \mid \mathcal{H}_t).$$

Therefore

$$\Pr\{\tau = t+1\} = \mathbb{E}\Big[\mathbf{1}_{\{\tau > t\}} K_t(V_I = 1 \mid \mathcal{H}_t)\Big].$$

Summing over t and adding and subtracting $\nu_t(V_I = 1 \mid \mathcal{H}_t)$ in each summand gives (152).

The represented sampler inducing K_t is an ancestor of the transition that makes the corresponding test and is evaluated before that transition. Encoding adequacy, computation closure, and the explicit microstep refinement place the sampler, its random-bit use, and its transition implementation in the saturated derivability closure at the class-preserving enlarged budget $\Psi(B,n)$. No probability table or accepting set is added to that record. The profile definition bounds each advantage by $\mathrm{Sel}_n^{\mathrm{sat}}(\Psi(B,n))$; the assumed uniform bound then bounds the profile by δ_n. Summing at most H terms proves (153).

Finally, if a sum of H real numbers exceeds ε, at least one summand exceeds ε/H. Apply this observation to the advantage sum in (152). The extracted record is the represented pre-hit sampler inducing K_t, proving (154).

□

III.21.3 Structural charging of selector success

Theorem III.21.4 (Prospective selector accounting through five-family quotient charge)**.** *Fix a represented task family, the five-family frontier rule $\mathfrak{F}_5$ of Corollary III.16.11, and a saturated budget B with finite time capacity. Let $H_B(n)$ be the largest refined horizon admitted by B. Let $\Psi_{\mathrm{hit}}(B,n)$ be the monotone ceiling obtained by substituting the resource majorants in B into the explicit first-hit ledger* (140)*, and set*

$$D_B^{\mathrm{sel}}(n) := \Psi_5^{\mathrm{alg}}(\Psi_{\mathrm{hit}}(B,n)). \tag{155}$$

For every presented instance I, write $\mathrm{Sel}_{I,n}^{\mathrm{sat}}(B)$ for the pointwise profile obtained from (151)*. Every represented selector K_t counted by this profile satisfies*

$$\big(\mathrm{Adv}_t(K_t;\nu_t)\big)_+ \le eH_B(n) \sum_{c\in\mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D_B^{\mathrm{sel}}(n)). \tag{156}$$

Consequently,

$$\mathrm{Sel}_{I,n}^{\mathrm{sat}}(B) \le eH_B(n) \sum_{c\in\mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D_B^{\mathrm{sel}}(n)), \tag{157}$$

Applying any monotone positively homogeneous family functional to the pointwise inequality gives the corresponding size-slice statement. Every budget-B computation also satisfies the whole-transcript bound

$$\Pr\{\tau \le H_B\} \le eH_B(n) \sum_{c\in\mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D_B^{\mathrm{sel}}(n)). \tag{158}$$

The right-hand sides are contextual-charge envelopes of completed exact first-hit accounting quotients. They retain the complete pre-hit sampler and all later terminal records, but their numerical value is accounting visibility. Matching provenance tags or zero incremental charge of terminal postprocessing does not transfer that value to a prospective source profile.

A prospective-source conclusion follows when the complete record also contains, for every nonzero frontier term, the represented pre-hit comparison required by Remark III.17.4 and Corollary III.16.9. Under that additional same-record comparison, the corresponding source envelopes replace the accounting envelopes in the three displayed bounds. Without it, Eqs. (156) *to* (158) *are exact accounting estimates only. In either interpretation, negligibility of the displayed right-hand side, after multiplication by the admitted horizon factor, implies the corresponding negligibility conclusion.*

Proof. Step 1: reduce selector advantage to next-hit mass.

Fix a selector occurrence at time t, and let C_{t+1} be the candidate sampled from $K_t(\cdot \mid \mathcal{H}_t)$. Its unconditional next-hit mass is

$$h_t := \mathbb{E}\Big[\mathbf{1}_{\{\tau>t\}} K_t(V_I = 1 \mid \mathcal{H}_t)\Big] = \Pr\{\tau = t+1\}. \tag{159}$$

Put

$$b_t := \mathbb{E}\Big[\mathbf{1}_{\{\tau>t\}} \nu_t(V_I = 1 \mid \mathcal{H}_t)\Big].$$

By (150), $\mathrm{Adv}_t(K_t;\nu_t) = h_t - b_t$, and $b_t \geq 0$. Hence

$$\big(\mathrm{Adv}_t(K_t;\nu_t)\big)_+ \leq h_t. \tag{160}$$

When $h_t = 0$, (156) follows immediately. Assume henceforth that $h_t > 0$.

Step 2: construct the truncated first-hit quotient.

Form the represented truncated computation $\mathcal{A}^{\langle t\rangle}$ by retaining the original initialization and transitions through the next test, declaring success only at that final test, and then stopping. Its terminal flag is the represented Boolean observable

$$z_{t+1} = \mathbf{1}_{\{s_0=\cdots=s_t=0\}} V_I(C_{t+1}).$$

The definition of the selector profile charges the complete derivation of $\mathcal{H}_t$, the sampler, and the next transition. The survival gate, terminal flag, and stopping instruction are deterministic filtration postprocessings of retained fields. Thus $\mathcal{A}^{\langle t\rangle}$ lies in the same execution class, has horizon $t+1 \leq H_B(n)$, and its exact first-hit quotient $q_{\mathcal{A}^{\langle t\rangle}}$ has saturated cost at most $\Psi_{\mathrm{hit}}(B,n)$ by Lemma III.20.2 and Definition III.24.1. Its success probability is exactly h_t.

Apply Lemma III.20.2 to the truncated computation. The quotient is terminal compatible and satisfies

$$h_t \leq e(t+1) V_I(q_{\mathcal{A}^{\langle t\rangle}}) \leq e H_B(n) V_I(q_{\mathcal{A}^{\langle t\rangle}})\,. \tag{161}$$

Step 3: expand its exact structural charge.

Expand this quotient by the family-uniform five-family frontier $\mathfrak{F}_5$. The composed cost bound is precisely (155). Terminal compatibility and the exact structural-charge identity give

$$V_I(q_{\mathcal{A}^{\langle t\rangle}}) = \sum_{\ell=1}^{|\Gamma_5|} \mathrm{ch}_I\Big(E_{\mathcal{A}^{\langle t\rangle},\Gamma_5,\ell}\Big)\,. \tag{162}$$

Every extension in Γ_5 has at least one authenticated provenance tag in $\mathfrak{J}^5_{\mathsf{Abs}}$. Since the charges are nonnegative,

$$\begin{aligned} V_I(q_{\mathcal{A}\langle t\rangle}) &\le \sum_{c\in\mathfrak{J}^5_{\mathsf{Abs}}} \sum_{\ell:\,\mathsf{W}_c(E_{\mathcal{A}\langle t\rangle,\Gamma_5,\ell})} \mathrm{ch}_I\Big(E_{\mathcal{A}\langle t\rangle,\Gamma_5,\ell}\Big) \\ &\le \sum_{c\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_B(n)). \end{aligned} \tag{163}$$

The last inequality is the defining supremum (101), applied at the composed carrier cost. Combining (160), (161), and (163) proves (156). Taking the supremum over the selector records in (151) proves (157).

Step 4: retain the provenance scope.

Independent outcome randomness is a valid-Lift auxiliary coordinate and has zero incremental target charge. Once its state-dependent arguments are present, every update, stored flag, survival gate, terminal flag, stopping instruction, and first-hit postprocessing is deterministic and has zero incremental charge by Lemmas III.17.1 and III.17.2. This locates the contextual charge inside the complete accounting DAG. It does not prove that a charge-bearing ancestor was available before the selector occurrence with a numerically dominating prospective visibility. That latter conclusion requires the represented comparison of Remark III.17.4 and Corollary III.16.9.

Step 5: obtain the whole-computation bound.

For the whole computation, the assertion is immediate when $\Pr\{\tau \le H_B\} = 0$. Otherwise apply the same argument once to its canonical first-hit quotient $q_{\mathcal{A}}$, rather than separately to its H_B hazards. The quotient then has positive utility. The first-hit visibility bound, exact structural-charge identity, and five-family cover give

$$\Pr\{\tau \le H_B\} \le eH_B(n)V_I(q_{\mathcal{A}}) \le eH_B(n) \sum_{c\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_B(n)).$$

The same exact quotient expansion proves (158) without an additional horizon summation. By Corollary III.23.1, the completed quotient remains an accounting record. A prospective interpretation of a frontier term requires its separate represented pre-hit comparison, as stated in the theorem.

□

An accounting application instantiates Theorem III.21.4 by estimating the fixed five contextual-charge envelopes on the right-hand side. A source-exhaustion application additionally supplies the represented transfer comparison in the theorem. The supremum in each envelope is already taken over the declared saturated budget slice.

III.21.4 Transition-rule provenance

Theorem III.21.5 (No-free succinct transition rule and active-source routing)**.** *Let $\mathcal{A}$ be a family-uniform represented finite-horizon computation of cost at most B, and let $K_t(\cdot \mid \mathcal{H}_t)$ be the kernel used for its transition after time t. Evaluator normalization and computation closure produce one represented derivation ρ_{K_t} of that kernel. Let $\widehat{B}$ be a common class-preserving majorant of its encoding, evaluator normalization, expanded-derivation, and active-source join overheads. At this ceiling the following properties hold.*

(i) *The inputs of ρ_{K_t} are the declared public presentation, the represented pre-hit record $\mathcal{H}_t$, and the random coordinates revealed by time t. Every state-dependent intermediate value and every instance-dependent coefficient used by the kernel occurs in its normalized ancestral DAG. A table occurring in the representation is charged by its encoded length, storage, lookup, and evaluation costs.*

(ii) *For each dynamically qualified Geom/Abs source record $\mathscr{C}$ invoked to explain the transition, join ρ_{K_t} with its canonical source derivation. In the pure exact-*Abs *mode this is the complete backward slice of the terminal quotient. In each active-Geom mode it is the represented derivation of $\mathrm{Act}_{\mathsf{Geom}}(\mathscr{C})$, containing the potential, local comparisons, selected generator, current, authority comparison, qualification record, and every applicable Geom gate. These are exactly the two cases of Definition III.3.1. A succinct evaluator may generate different transitions at exponentially many states, but those differences and the source record used to explain them remain in the corresponding charged joint derivation and its exact five-family signature.*

(iii) *If ρ_{K_t} is unavailable from the prefix at time t, it is absent from the prospective selector profile. If it is available and has positive selector advantage, that advantage is counted by Theorem III.21.3. Structural credit requires the same-record source comparison of Corollary III.16.9; a terminal or retrospective readout supplies accounting only.*

(iv) *Every dynamically qualified structural use of the complete active tuple belongs to exactly one support mode:*

$$\mathsf{A}, \qquad \mathsf{Q}_{\mathrm{ex}}, \qquad \mathsf{Q}_{\mathrm{rev}}, \qquad \mathsf{Q}_{\mathrm{nr}}, \qquad \mathsf{X}.$$

The mode A *is pure exact* Abs *with a null geometry slot. The next three modes retain a represented nonidentity quotient together with the applicable exact or compensated Geom data. The ambient mode* X *has identity support and retains the complete full-carrier Geom record. Its directed contribution is the authorized* Caus *projection of the selected current; the gradient-orthogonal component is J_{res} and has zero authorized target alignment.*

Consequently, a family-uniform rule with positive advantage over its declared neutral selector has no unindexed policy-table source. Its prospective structural contribution is carried by a represented exact or compensated quotient record, by a complete identity-supported ambient Geom record, or by no qualified structural source; neutral baseline contact remains in the enumeration term of Theorem III.21.3. Instance-dependent transition data outside the declared public derivability closure constitute charged advice, an oracle extension, or a change of task presentation.

Proof. Encoding adequacy and budgeted generation give a represented derivation of every transition used by $\mathcal{A}$; see Lemma III.6.3. Evaluator normalization, expanded-derivation equivalence, and affordable composition retain the complete implementation and its accumulated cost: Lemmas I.4.7 and III.1.9. In particular, a lookup table is an encoded input to that implementation and is charged at its represented size. When a source record $\mathscr{C}$ is invoked, the represented join constructor combines the kernel record with the terminal quotient in mode A, or with $\mathrm{Act}_{\mathsf{Geom}}(\mathscr{C})$ in one of the four active-Geom modes. The composition-cost identity (4) charges both derivations and the join. Budget compatibility and the common normalization overhead Ψ_{all} of Theorem III.3.4 provide the declared class-preserving majorant $\widehat{B}$. Expanding every interface and taking the applicable terminal-quotient or active-tuple backward slice gives exactly the two-case canonical ancestry and coefficient-complete signature of Definition III.3.1; its active-Geom specialization is Definition III.2.3. This proves (i) and (ii).

Temporal availability is exactly Definition III.21.1. The selector hazard identity and extraction statement are Theorem III.21.3; the required numerical source transfer is Corollary III.16.9.

Completed terminal flags and first-hit records retain their accounting status by Corollary III.23.1. This proves (iii).

The five master modes and their disjoint exhaustiveness are Theorem III.18.9. The four active-Geom support tags are refined by Theorem III.2.6. The active current split and the absence of an independent residual source are Proposition II.5.9 and Theorem II.7.2. In an active-Geom record, the represented active potential D satisfies $\nabla D \in \mathcal{C}_{\mathrm{str}}$, and therefore

$$\langle (I - \Pi_{\mathcal{C}_{\mathrm{str}}}) F_J, -\nabla D \rangle_E = 0.$$

The residual current consequently has zero value in $\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}$ by Definitions III.12.8 and III.12.9. This proves (iv) and the conclusion.

□

Theorem III.21.6 (Adaptive potential accounting on the represented transcript)**.** *Let $Z_0, \ldots, Z_H$ be a represented finite-horizon transcript and let $D_0, \ldots, D_{H-1}$ be real represented pre-hit potentials, where D_t is available before the transition from Z_t to Z_{t+1}. Then, pathwise,*

$$\sum_{t=0}^{H-1} \big(D_t(Z_{t+1}) - D_t(Z_t)\big) = D_{H-1}(Z_H) - D_0(Z_0) + \sum_{t=1}^{H-1} \big(D_{t-1}(Z_t) - D_t(Z_t)\big). \tag{164}$$

For a fixed potential $D_t = D$, the update sum vanishes and

$$\sum_{t=0}^{H-1} \big(D(Z_{t+1}) - D(Z_t)\big) = D(Z_H) - D(Z_0). \tag{165}$$

Equivalently, if time, history, learned state, and current parameters are included in the represented clocked carrier and one fixed lifted potential $\widetilde{D}$ is evaluated on that carrier, its increments telescope as in Eq. (165)*. The construction and update of every field of $\widetilde{D}$ remain in its ancestral record.*

The identities supply no target comparison by themselves. A target-directed interpretation of either endpoint term or any update term requires that term to occur in a temporally admissible qualified source record. Its complete prospective value is then evaluated by the existing target projection and the exact or compensated quotient, ambient $\mathsf{Geom}/\mathsf{Caus}$*, or terminal gates. For a singleton target on $\mathbb{F}_2^n$, the target indicator has the full expansion*

$$\mathbf{1}_{\{s\}}(x) = 2^{-n} \sum_{a \in \mathbb{F}_2^n} \chi_a(x + s), \tag{166}$$

so target-directed endpoint progress is not, in general, one distinguished Fourier coefficient of D.

Proof. Expanding the left side of Eq. (164), group the two occurrences at each intermediate state Z_t. The uncancelled intermediate term is $D_{t-1}(Z_t) - D_t(Z_t)$, which proves the identity. The fixed-potential case follows by cancellation. Computation closure and Lemma IV.0.2 retain the clock, history, learned state, parameter updates, potential evaluations, and their complete costs. Temporal and structural assignments are Definition III.21.1 and Theorems III.18.9 and III.21.5. Finally, character orthogonality gives Eq. (166). □

III.22 Full-Carrier Controls and Core-Refined Accounting

Corollary III.22.1 (Prospective routing of a represented nonlinear readout)**.** *In the setting of Theorem III.21.5, let $\mathscr{C}$ be a represented candidate source record and let $D_t = F_t \circ J_t$ be a represented nonlinear readout generated from its pre-hit active frontier, with complete derivation retained in the normalized joint transition–source record. Its structural use has the following exhaustive routing.*

(i) *If D_t is not an ancestor of the canonical prospective terminal field of $\mathscr{C}$—the terminal quotient in mode A, or a field of $\mathrm{Act}_{\mathsf{Geom}}(\mathscr{C})$ in an active-Geom mode—then it is inactive in the prospective contribution of $\mathscr{C}$. In particular, it contributes nothing through $\mathscr{C}$. Any prospective use by a transition, selector, potential, quotient comparison, cell-ordering rule, or same-generator comparison belongs to the separate complete source record in whose canonical ancestry it occurs. The present readout may remain a terminal or accounting record.*

(ii) *Suppose its represented fiber map q_{D_t} is nonidentity and target-compatible. Exact fiber dynamics with a null geometry slot place the record in A; exact fiber dynamics with an active geometry slot place it in Q_{ex}. A represented nonzero defect with a valid reversible or general-resolvent stability record places it in $\mathsf{Q}_{\mathrm{rev}}$ or Q_{nr}. Coercive regularity supplies the reversible compensation bound whenever the quotient, block data, target comparison, and coercive comparison occur in the same affordable record. A fiber map that lacks target compatibility or a required dynamic comparison receives no quotient-supported source value.*

(iii) *Every structural use not admitted by the quotient-supported case is evaluated through the complete active transition tuple. In particular, if the transition uses full-state fields not determined by q_{D_t}, or uses D_t as a potential without admitting its fibers as a dynamically qualified quotient, those fields and their derivations remain in the active frontier. A qualified identity-support record belongs to X and is evaluated by the complete same-record ambient gate product. The scalar readout D_t receives no separate source value.*

(iv) *In every support mode, only the authorized gradient projection of the selected current contributes directed structure. Absence of a required same-generator comparison, target alignment, target reach, or mixing gate sets the corresponding ambient product to zero. A represented quantitative regularity loss remains in the factor $1/(1+\rho_\Phi(D_t))$. The gradient-orthogonal component is J_{res}.*

Thus polynomial evaluation of a nonlinear scalar readout supplies neither quotient dynamics nor target-oriented transport by itself. Dynamically sufficient fiber structure is charged to the exact or compensated quotient modes. When the complete master record has identity support, full-state transport is charged to the ambient mode with every active field and gate retained. When another nonidentity quotient is active, the complete tuple is charged to the exact or compensated support mode selected by Theorem III.18.9.

Proof. Part (i) is clause (iv) of Definition III.21.1; the extremal-readout specialization is Corollary III.3.8. For part (ii), apply the support alternatives in Theorem III.18.9; the coercive specialization is Theorem III.12.6. Failure of target compatibility or dynamic qualification excludes the readout from the quotient-supported certificate class by Theorem III.18.9. Failure of fiber sufficiency, or use of the scalar as a full-state potential, preserves every additional active argument by Definition III.2.3; identity support is the ambient mode of Theorem II.9.5. This proves part (iii). Part (iv) follows from Definitions III.12.9 and III.14.9 and Theorem II.7.2. These cases are exhaustive by Theorem III.21.5. □

Corollary III.22.2 (Quotient-free full-carrier control normal form)**.** *Work at one saturated ceiling B and suppose that the four quotient-supported prospective master coordinates vanish there:*

$$G^{\mathrm{src,sat}}_{A,n}(B) = G^{\mathrm{src,sat}}_{Q,\mathrm{ex},n}(B) = G^{\mathrm{src,sat}}_{Q,\mathrm{rev},n}(B) = G^{\mathrm{src,sat}}_{Q,\mathrm{nr},n}(B) = 0. \tag{167}$$

Every positive dynamically qualified prospective contribution of a represented transition rule then has the ambient support mode X. For such a record $\mathscr{C}$, adjoining the complete pre-hit record to the state presents its active tuple as a represented full-carrier controlled process with:

(i) *state equal to the represented pre-hit record $\mathcal{H}_t$, selected generator and kernel $K_t(\cdot \mid \mathcal{H}_t)$, and complete transition ancestry retained at the charged ceiling;*

(ii) *geometry given by the represented relation, conductance, potential, local comparisons, reach comparison, mixing comparison, and regularity comparison in $\mathrm{Act}_{\mathsf{Geom}}(\mathscr{C})$; and*

(iii) *direction given by the authorized Caus projection of the current selected from that same active record.*

Its conditioned directed visibility is exactly the already-defined ambient record contribution

$$V^{X}_{\Phi,D,J,\mathfrak{K}} = M_\Phi C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D)\frac{R_T(D)}{1+\rho_\Phi(D)}. \tag{168}$$

Thus the usual control or reinforcement-learning description introduces no additional structural channel: the represented kernel is the transition rule, the selected generator is the control authority, the potential is the progress observable, and its authorized gradient current is the Caus component. Likewise, smooth, spectral, hierarchical, multiscale, and self-similar descriptions introduce no additional channel. Such a description contributes precisely through the represented Geom fields and comparisons that it supplies in the active tuple, with its complete derivation cost retained. Recursive reuse may shorten that derivation; it does not remove its public ancestors or any active Geom gate.

Consequently, after quotient exhaustion, a prospective structural contribution is positive precisely through a represented full-carrier ambient record for which every factor in (168) *is positive. Failure of temporal availability excludes the record from the prospective profile. Failure of authority, same-generator consistency, target alignment, target reach, or mixing sets its qualified contribution to zero. Quantitative irregularity is retained in $1/(1+\rho_\Phi(D))$. The remaining gradient-orthogonal current belongs to J_{res}.*

Proof. The support alternatives in Theorem III.21.5 are exhaustive. Let $\mathscr{C}$ have positive prospective visibility. If its support were A, Q_{ex}, $\mathsf{Q}_{\mathrm{rev}}$, or Q_{nr}, its visibility would be bounded by the corresponding nonnegative coordinate in (167), a contradiction. Hence $\mathscr{C}$ has support X. The active fields in (i)–(iii) are exactly the ambient fields of Definition III.2.3. A history-dependent finite-horizon kernel becomes a state-dependent kernel when the represented pre-hit record is used as the state; this is the clocked transcript representation already retained by Definition III.21.1.

The value of the qualified ambient record is (168) by Definition III.14.9. The same-generator authority and directional projection are those of Definition III.12.9. Evaluator normalization and no-creation retain the ancestry and cost of every recursive or multiscale implementation by Lemmas I.4.7 and III.1.9. The exhaustive geometric normal form assigns every represented relation and conductance to the same Geom coordinate, independently of the descriptive vocabulary used for it; see Theorem II.9.5. Finally, Theorem II.7.2 assigns the orthogonal current to J_{res}. These statements prove the corollary.

□

Figure 35 displays the support and source routing of the preceding theorem and corollary.

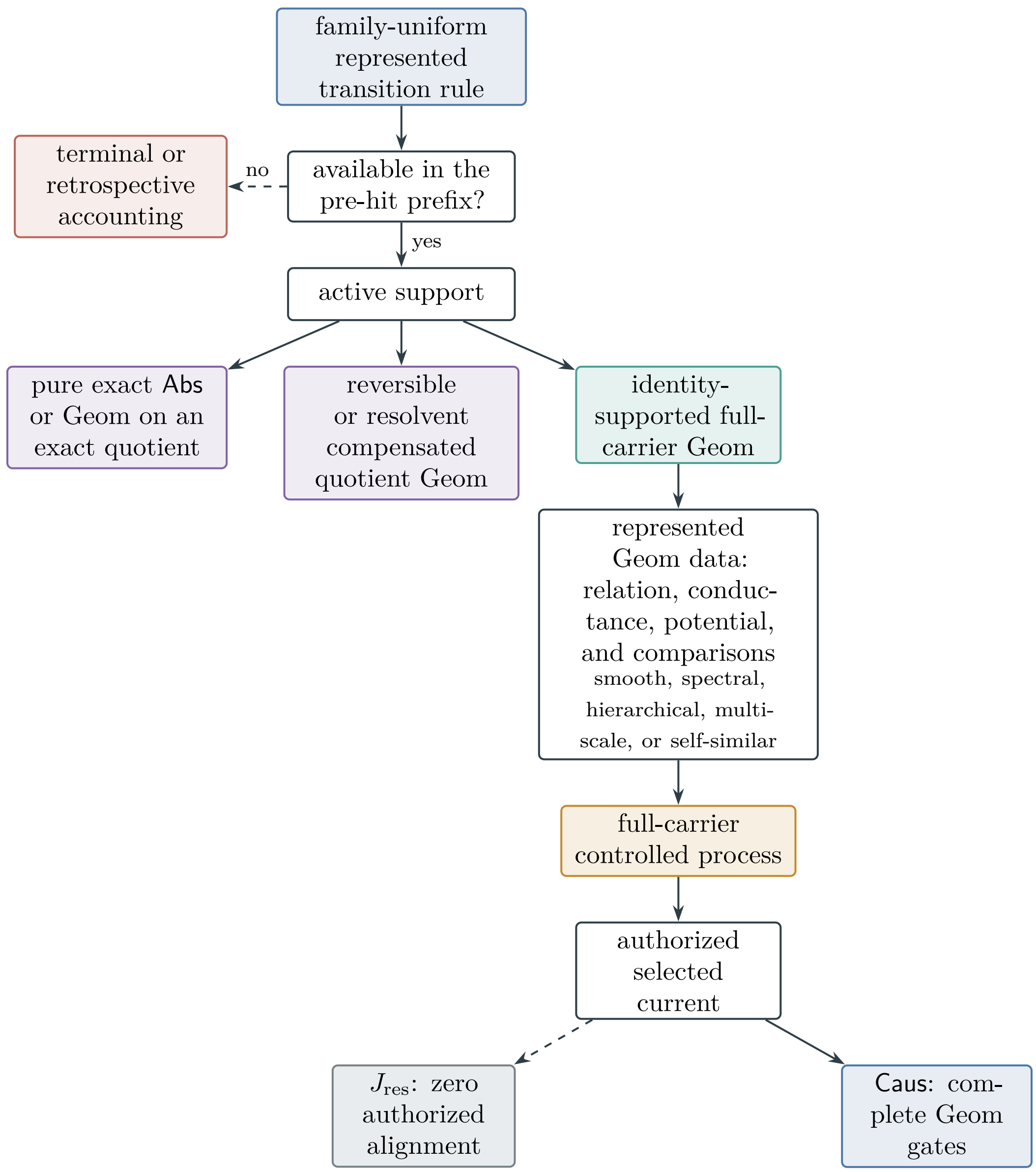

Figure 35: Exhaustive routing of a succinct transition rule. Succinct representation exposes one charged derivation rather than a state-indexed policy table. The exact branch contains pure exact Abs and Geom on an exact quotient; the compensated branch contains reversible and general-resolvent quotient–Geom. The remaining support is identity-supported ambient Geom. In the ambient branch, control and reinforcement-learning terminology describes the represented kernel and selected generator; smooth, spectral, hierarchical, multiscale, or self-similar terminology describes the represented Geom data. These are not additional source channels. Only the authorized Caus component of the selected ambient current contributes directed structure; the remaining current is J_{res}.

Definition III.22.3 (Core-refined first-hit accounting profile)**.** Fix a saturated ceiling B, a finite horizon bound H_B, and a monotone sublinear positively homogeneous family functional $\mathfrak{M}_n$. Apply the coefficient-complete five-family normalization to the exact first-hit record of a represented computation in the budget slice. For exact operational and charge-bearing core signatures, define

$$\begin{aligned}&\mathrm{HitCore}^{\mathrm{sat},=}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}}|S_{\mathrm{st}},S_{\mathrm{cf}},n}(B)\\ &\quad := \sup \mathfrak{M}_n(I \longmapsto V_I(q_{\mathcal{A}}))\,,\end{aligned} \tag{169}$$

where the supremum ranges over the normalized first-hit records whose complete carrier cost is at most B, whose exact operational signature is $(S_{\mathrm{st}}, S_{\mathrm{cf}})$, and whose charge-bearing core signature is $(K_{\mathrm{st}}, K_{\mathrm{cf}})$. An empty restriction has value zero. Replacing $\mathfrak{M}_n$ by evaluation at one presented instance I defines the pointwise notation $\mathrm{HitCore}^{\mathrm{sat}}_{I,n}$. Put

$$\mathrm{HitCore}^{\mathrm{sat}}_{\mathfrak{M},n}(B) = \max_{\substack{K_{\mathrm{st}},K_{\mathrm{cf}}\\ S_{\mathrm{st}}\supseteq K_{\mathrm{st}}\\ S_{\mathrm{cf}}\supseteq K_{\mathrm{cf}}}} \mathrm{HitCore}^{\mathrm{sat},=}_{\mathfrak{M},K_{\mathrm{st}},K_{\mathrm{cf}}|S_{\mathrm{st}},S_{\mathrm{cf}},n}(B). \tag{170}$$

This is an accounting refinement of the existing exact first-hit records. It is not a new structural source and does not assert that an indexed cell is nonempty. A record occurs in the supremum only through its represented constructor, complete carrier, qualification data, and actual saturated cost. In particular, a spectral datum occurs in this profile only when its constructor and every certificate used to authenticate it occur in that record at cost at most B.

Theorem III.22.4 (Core-refined prospective accountability)**.** *Let D^{hit}_B be a common class-preserving ceiling for the microstep refinement, exact first-hit construction, five-family normalization, and core-signature construction of a budget-B record. Then*

$$\mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n}(B) \le eH_B\ \mathrm{HitCore}^{\mathrm{sat}}_{\mathfrak{M},n}(D^{\mathrm{hit}}_B), \tag{171}$$

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le eH_B\ \mathrm{HitCore}^{\mathrm{sat}}_{\mathfrak{M},n}(D^{\mathrm{hit}}_B). \tag{172}$$

For each cell in Eq. (169)*, its visibility has the exact charge representation*

$$V_I(q_{\mathcal{A}}) = \sum_{\substack{1\le \ell\le |\Gamma_5|\\ v_\ell \notin \mathrm{Dec}_{\mathrm{inh}}(\mathscr{C}_{\mathcal{A}})}} \mathrm{ch}_I(E_{\mathcal{A},\Gamma_5,\ell}). \tag{173}$$

Thus the exact operational cell, core cell, cost, and contextual charge of a successful first-hit record are retained simultaneously. The theorem makes no transfer from this accounting visibility to a smaller prospective source coordinate. Such a transfer is available only under the represented comparison required by Corollary III.16.9.

Proof. The coefficient-complete normalized ancestral record has one exact operational signature and, by Theorem III.5.3, one charge-bearing core signature. The finitely many restrictions in Eq. (170) are disjoint and exhaust the normalized first-hit accounting family. No record is introduced by taking this maximum.

For the success bound, apply the exact first-hit inequality of Lemma III.20.2 to the represented budget-B transcript. Its normalized first-hit record occurs in exactly one cell at the ceiling D^{hit}_B, whence

$$\Pr\{\tau \le H_B\} \le eH_B V_I(q_{\mathcal{A}}) \le eH_B\,\mathrm{HitCore}^{\mathrm{sat}}_{I,n}(D^{\mathrm{hit}}_B).$$

Apply the family functional and then take the saturated supremum to obtain Eq. (172). For a selector occurrence, use the truncated record ending at its next represented verifier call, as in the proof of Theorem III.21.4. Its positive advantage is at most its next-hit mass, and the same first-hit inequality places that mass in one core-refined cell. Taking the saturated supremum gives Eq. (171).

Finally, terminal compatibility and Eq. (56) give Eq. (173). The accounting record remains retrospective unless it has a separate pre-hit representation, by Corollary III.23.1. Therefore no source-profile comparison follows merely from the core label; the last assertion is exactly the transfer requirement of Remark III.17.4 and Corollary III.16.9.

□

Corollary III.22.5 (Uniform enumeration in a target-permutation-neutral cell family)**.** *Assume that initially there are N represented cells, exactly one of which is target-containing, and that after every all-negative pre-hit transcript the target cell is conditionally uniform over the remaining untested cells. Assume that one verifier call tests at most one cell, and let ν_t be uniform on the remaining untested cells.*

Every represented sampler has zero expected advantage over ν_t. Every computation making $H \le N$ distinct tests satisfies

$$\Pr\{\tau \le H\} = \frac{H}{N}. \tag{174}$$

With repeated tests or early stopping, H/N is an upper bound. More generally, success exceeding H/N implies that the pre-hit record breaks target-permutation neutrality. The represented datum, score, quotient readout, potential, or sampler implementation producing that distinction is target-aligned structure and is charged at its complete saturated cost.

Proof. After t distinct failures, the target is conditionally uniform among the $N-t$ remaining cells. Hence, for every probability kernel K_t,

$$\mathbb{E}[K_t(J \mid \mathcal{H}_t) \mid \mathcal{H}_t, \tau > t] = \frac{1}{N-t}\sum_{C\in\mathcal{C}_t} K_t(C \mid \mathcal{H}_t) = \frac{1}{N-t}.$$

The same identity holds for the uniform baseline, so the expected advantage is zero. The survival recursion is

$$\Pr\{\tau > t+1 \mid \tau > t\} = 1 - \frac{1}{N-t} = \frac{N-t-1}{N-t}.$$

Multiplying for $t = 0, \dots, H-1$ yields $\Pr\{\tau > H\} = (N-H)/N$, proving (174). More generally, expose the random ordered list of distinct cells tested before stopping. Conditional target-permutation neutrality makes each set of d distinct tested cells contain the target with probability d/N. Repetition and early stopping give $d \le H$, so their success probability is at most H/N.

□

III.23 Retrospective Quotients and Source Profiles

Corollary III.23.1 (Retrospective first-hit quotients are accounting records)**.** *Let $q_{\mathcal{A}}$ be the exact first-hit quotient of Lemma III.20.2. Its coordinates are represented by postprocessing the completed terminal flags and remaining deadline:*

$$q_{\mathcal{A}} = F(s_0, \dots, s_H, r).$$

For an earlier transition after time $t < H$, this representation depends on later flags and fails temporal source admissibility whenever two legal continuations of the same surviving prefix have different later first-hit times. In that case $q_{\mathcal{A}}$ does not factor through pref_t *and cannot influence the selector used after time t.*

The quotient remains a valid saturated accounting record: it preserves the success law already generated by the computation. It enters a prospective source profile only through an independent temporally admissible prefix representation, charged at that representation's complete cost. Executing later transitions and verifier calls to construct the label retains those operations in the completed derivation and does not move the label before them in the derivation order.

Consequently, a retrospective first-hit record cannot close a source exhaustion cycle

$$\textit{residual success} \longrightarrow q_{\mathcal{A}} \longrightarrow \textit{claimed pre-hit source.}$$

The charged source ancestry of the completed record is the five-family frontier of Theorem III.2.2 and Corollary III.16.11; the first-hit postprocessing itself contributes zero incremental charge by Lemma III.17.2.

Proof. If two legal completions agree through time t but have different future first-hit times, their images under $q_{\mathcal{A}}$ differ. No function of their common prefix can equal $q_{\mathcal{A}}$ on both completions, so (149) fails.

An independent prefix evaluator is assessed through its own represented derivation. An evaluator that obtains its result by executing later updates and verifier calls is downstream of those operations. Treating the resulting completed label as their source reverses the derivation order. By Theorem III.2.2, the completed quotient factors through the charged joint frontier of its derivation, and Lemmas III.17.2 and I.4.7 assign no new target projection or contextual charge to the terminal postprocessing. This proves the asserted accounting/source distinction.

□

Corollary III.23.2 (No retrospective first-hit source loophole)**.** *An exact first-hit denotation cannot serve as an uncharged structural source against a saturated-profile exhaustion theorem. Every represented use of that denotation falls into one of the following two cases.*

(i) *The representation fails temporal source admissibility at the claimed use. In particular, its derivation may have a later transition outcome, verifier flag, or terminal record as an ancestor, as in the completed flag construction of Lemma III.20.2. It may remain accounting-admissible but is source-inadmissible for that use by Definition III.21.1 and Corollary III.23.1.*

(ii) *The same denotation has an independent pre-hit representation. Then that representation, its full derivation cost, and its complete ancestry belong to the saturated closure. The source is classified by Theorems II.9.4 and III.2.2 and is charged by Corollary III.16.11.*

The same denotation may possess representations of both kinds; each is charged according to its own derivation. Thus defining an observable extensionally by the first-hit time supplies no third case. Without a represented evaluator it is unavailable in the presented resource class; with a represented evaluator it falls into case (i) *or* (ii) *at each claimed use.*

Proof. Every use of an observable in the budgeted closure has a represented derivation by Definition III.1.3 and Theorem III.6.2. Fix the transition for which source credit is claimed. If the derivation has a later transition outcome, verifier flag, or terminal record as an ancestor, it fails the ancestor and prefix-factorization conditions of Definition III.21.1, giving case (i). If it has no such ancestor and is available before the transition, its prefix evaluator is the independent representation in case (ii). A record that is not available before the transition falls under case (i) by the same definition. These alternatives exhaust the represented derivation at the claimed use. The five-family normal form and its exact contextual-charge identity classify and charge every case-(ii) record.

□

Definition III.23.3 (Source and accounting profiles)**.** Let $\mathcal{Q}^{\mathrm{src}}_{\mathfrak{I}_n,\le B}$ be the subfamily of $\mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\le B}$ whose target-aligned quotient/readout record is temporally admissible for the transition it is invoked to explain. A record invoked at several transitions satisfies the condition at each claimed use; a primitive public record available before execution satisfies it at time 0. Set

$$m_I^{\mathrm{src,sat}}(B) = \sup_{\mathbf{Q}\in\mathcal{Q}^{\mathrm{src}}_{\mathfrak{I}_n,\le B}} V_I(\mathbf{Q}).$$

Define $G_I^{\mathrm{src,sat}}(B)$ by imposing the same temporal condition on target-aligned potentials, geometries, rankings, current comparisons, and readouts. For a distinguished instance sequence (I_n), put

$$I_n^{\mathrm{src,sat}}(B) = m_{I_n}^{\mathrm{src,sat}}(B) + G_{I_n}^{\mathrm{src,sat}}(B).$$

The superscript $\mathrm{src,sat}$ on $G_X, G_{Q,\mathrm{ex}}, G_{Q,\mathrm{rev}}, G_{Q,\mathrm{nr}}$, and $\mathfrak{U}$ denotes the same restriction of their defining certificate families to temporally admissible records. Their algebraic maximum and two-coordinate identities are unchanged under this common restriction.

When the instance sequence is fixed, write $m_n^{\mathrm{src,sat}}$ and $G_n^{\mathrm{src,sat}}$ for the first two terms.

The *source profiles* range only over temporally admissible pre-hit quotients, potentials, geometries, rankings, and target-discrimination readouts. Prospective sampler implementations are measured by $\mathrm{Sel}_n^{\mathrm{sat}}$ in Definition III.21.2. The *accounting profiles* may also contain finite-horizon terminal records, completed transcripts, truncated future values, and exact first-hit quotients. Thus

$$m_I^{\mathrm{src,sat}}(B) \le m_I^{\mathrm{sat}}(B).$$

An accounting record may certify the hitting law but supplies no causal explanation of an earlier transition without a separate temporally admissible representation. Universal hardness therefore uses Theorem III.21.3: a negligible prospective selector profile together with a negligible or explicitly bounded enumeration baseline implies negligible target arrival. The exact first-hit quotient remains an accounting identity.

III.23.1 Pointwise computation accountability

Corollary III.23.4 (Pointwise algorithm-to-accounting-profile bound)**.** *Let $B_{\mathcal{A}}(n)$ denote the right-hand side of* (140)*, including its fixed constructor overhead. For every instance I,*

$$\alpha_I(\mathcal{A}_n) \le eH(n)m_I^{\mathrm{sat}}(B_{\mathcal{A}}(n)). \tag{175}$$

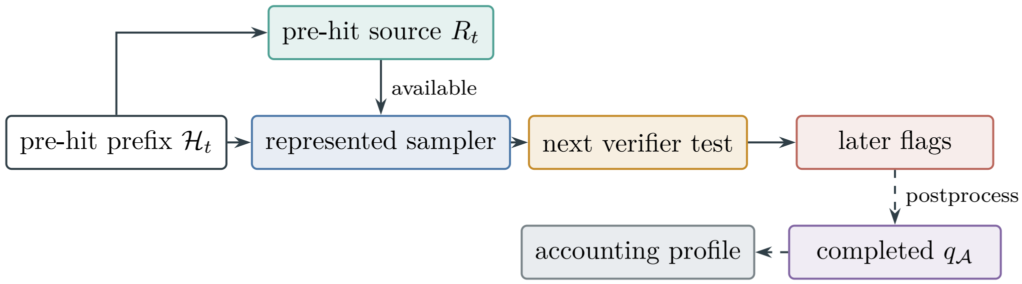

Figure 36: Temporal typing of the structural ledger. A source used by the next transition factors through the pre-hit prefix and is charged before the test, as required by Definition III.21.1. Later verifier flags generate the exact first-hit quotient only in the accounting branch, as proved in Corollary III.23.1.

This is an exact accounting implication within the declared uniform presentation-relative resource envelope, including presentation-admissible oracle interfaces. It is not a source-exhaustion implication: the retrospective first-hit quotient belongs to the accounting profile by Corollary III.23.1.

Proof. The uniform family $q_{\mathcal{A}}$ belongs to $\mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\le B_{\mathcal{A}}}$. Hence $V_I(q_{\mathcal{A}}) \le m_I^{\mathrm{sat}}(B_{\mathcal{A}})$. Rearranging (137) proves the result. $\square$

Corollary III.23.5 (Authority-relative computation accountability)**.** *Assume, in addition, that the full lifted generator of $\mathcal{A}_n$ is selected from a declared Caus authority envelope $\mathfrak{A}$, and that every dynamic interface in its represented derivation preserves that selection. Then, pointwise in the presented instance,*

$$\alpha_I(\mathcal{A}_n) \le eH(n)\mathfrak{U}^{\mathrm{sat}}_{n,\mathfrak{A}}(B_{\mathcal{A}}(n)). \tag{176}$$

Thus the generic clocked-transcript audit remains valid when a mathematical object imposes its dynamics. The exact first-hit quotient is an authority-relative accounting record and is not a prospective source unless it also has the prefix representation required by Definition III.21.1.

Proof. Lemma III.20.2 constructs an exact quotient for the actual lifted generator. With a null geometry and causal slot, this is a pure-Abs master certificate. The additional hypothesis puts that certificate in $\mathcal{C}^{\mathfrak{A}}_{QG}(B_{\mathcal{A}})$. Hence $V_I(q_{\mathcal{A}}) \le \mathfrak{U}^{\mathrm{sat}}_{n,\mathfrak{A}}(B_{\mathcal{A}})$, and rearranging the first-hit visibility inequality proves the claim.

$\square$

III.24 The Uniform Saturated Profile and DTM Accountability

Finally, we aggregate the pointwise bounds uniformly over an instance family and over every represented computation at the declared budget.

The family-level conclusion is organized into uniform profiles in Section III.24.1, universal accountability in Section III.24.2, finite-slice and residual accounting in Section III.24.3, and the combined profile in Section III.25. The family-level formulas are Corollaries III.24.2 and III.24.5, Theorem III.24.3, Lemma III.27.5, and Proposition III.27.6.

III.24.1 Uniform visibility and success profiles

Definition III.24.1 (Uniform saturated accounting visibility and success profiles)**.** Let $\mathfrak{M}_n$ be a monotone positively homogeneous functional on nonnegative functions on $\mathfrak{I}_n$. The principal examples are expectation under a declared instance distribution, supremum over a size slice, and infimum over a size slice. Define the uniform quotient visibility profile by

$$m^{\mathrm{sat}}_{\mathfrak{M},n}(B) = \sup_{\mathbf{Q}\in\mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\le B}} \mathfrak{M}_n(I \mapsto V_I(\mathbf{Q}))\,. \tag{177}$$

In the temporal typing of Definition III.23.3, this original saturated profile is the family-level accounting profile.

Let $\mathsf{Alg}_{n,\le B}$ be the uniform presentation-relative represented computations, including computations using presentation-admissible oracle interfaces, whose code, initialization, peak native state and memory, updates, outcome/terminal transcript, and total work lie in the resource envelope B. Their task-facing representation includes the canonical fresh start phase of Lemma III.20.2; this changes neither their original output law nor their original success probability. Their optimal affordable success profile is

$$\mathrm{Succ}_{\mathfrak{M},n}(B) = \sup_{\mathcal{A}\in\mathsf{Alg}_{n,\le B}} \mathfrak{M}_n(I \mapsto \alpha_I(\mathcal{A}))\,. \tag{178}$$

The envelope supplies these bounds rather than assuming them. When $\mathsf{t}(B) < \infty$, set

$$H_B(n) = 1 + \lfloor \mathsf{t}(B) \rfloor$$

for the normalized horizon, and let $\Psi_{\mathrm{hit}}(B,n)$ be the monotone resource ceiling obtained by substituting the componentwise majorants declared by B into the explicit ledger (140). Equivalently, one may use the coarser repeated-monoid bound that charges every per-step field by B, every retained field once by B, and adds the public verifier, gauge, interface, and clock charges. Axioms B1–B4 make this a uniform ceiling for every member of $\mathsf{Alg}_{n,\le B}$. Write $\Psi(B,n) = \Psi_{\mathrm{hit}}(B,n)$. Thus "envelope comparison" is notation for a bound derived from the declared resource ledger, not an additional computational or structural hypothesis. In the polynomial resource envelope, H_B and $\Psi(B,n)$ are polynomial after a fixed exponent enlargement.

Corollary III.24.2 (Uniform derivation-indexed quotient profile)**.** *For every family functional $\mathfrak{M}_n$ and saturated budget B,*

$$m^{\mathrm{sat}}_{\mathfrak{M},n}(B) = \sup_{\widehat{\rho}\in\mathsf{NFQ}_{\mathfrak{I}_n,\le B}} \mathfrak{M}_n\left(I \longmapsto A_{q_{\widehat{\rho},I}} \frac{\|P_{q_{\widehat{\rho},I}} y_{\widehat{\rho},I}\|_2^2}{\|y_{\widehat{\rho},I}\|_2^2} \right). \tag{179}$$

The right-hand side ranges over one problem-independent constructor grammar. A task family enters only through its declared primitives, reference measures, terminal data, and the denotations of those constructors.

Proof. The indexing families in $\mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\le B}$ and $\mathcal{Q}^{\mathrm{nf}}_{\mathfrak{I}_n,\le B}$ agree by Lemma III.15.4. Substitute their common normal-form records into (177).

□

Universal accountability bounds any admissible computation by the saturated structural profile at its charged budget.

III.24.2 Universal saturated-profile accountability

Theorem III.24.3 (Universal saturated accounting-profile bound)**.** *For every task family, family functional $\mathfrak{M}_n$, and budget ceiling B with finite time capacity $\mathsf{t}(B) < \infty$, with H_B and Ψ derived from the declared resource ledger as above,*

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \leq eH_B(n) m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)). \tag{180}$$

Consequently, for every target success scale $\varepsilon(n) > 0$,

$$m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)) < \frac{\varepsilon(n)}{eH_B(n)} \quad \Longrightarrow \quad \mathrm{Succ}_{\mathfrak{M},n}(B) < \varepsilon(n). \tag{181}$$

The implication is an exact audit statement. Source-exhaustion arguments use Theorem III.21.3 rather than substituting a retrospective first-hit quotient into the source profile.

Proof. Fix $\mathcal{A} \in \mathsf{Alg}_{n,\leq B}$. The pointwise bound and the uniform envelope estimates give

$$\alpha_I(\mathcal{A}) \leq eH_B(n)V_I(q_{\mathcal{A}})$$

for every I. Monotonicity and positive homogeneity of $\mathfrak{M}_n$ therefore imply

$$\mathfrak{M}_n(\alpha_I(\mathcal{A})) \leq eH_B(n)\mathfrak{M}_n(V_I(q_{\mathcal{A}})).$$

The first-hit maps form one uniform represented quotient family of cost at most $\Psi(B,n)$, so the last functional is bounded by $m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n))$. Taking the supremum over $\mathcal{A}$ proves (180); (181) is its contrapositive. □

Corollary III.24.4 (Uniform prospective-selector hardness criterion)**.** *Fix a size slice on which every budget-B computation has refined horizon at most $H_B(n)$. Suppose that, pointwise on that slice,*

$$\mathrm{Enum}_\nu(\mathcal{A}, H_B) \leq \beta_n$$

for every such computation, and every temporally admissible represented sampler of saturated cost at most $\Psi(B,n)$ satisfies

$$\mathrm{Adv}_t(K_t; \nu_t) \leq \delta_n.$$

Then every computation satisfies

$$\Pr\{\tau \leq H_B\} \leq \beta_n + H_B(n)\delta_n. \tag{182}$$

Consequently, for every monotone positively homogeneous family functional $\mathfrak{M}_n$,

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \leq \mathfrak{M}_n(\mathbf{1})\left(\beta_n + H_B(n)\delta_n\right).$$

If β_n and $H_B\delta_n$ are negligible, every budget-B computation has negligible target-arrival probability.

Proof. Apply Eq. (153) pointwise and use the two uniform hypotheses. Monotonicity and positive homogeneity of $\mathfrak{M}_n$ give the family-functional bound.

□

The prospective hardness map and retrospective audit are summarized in Figure 37.

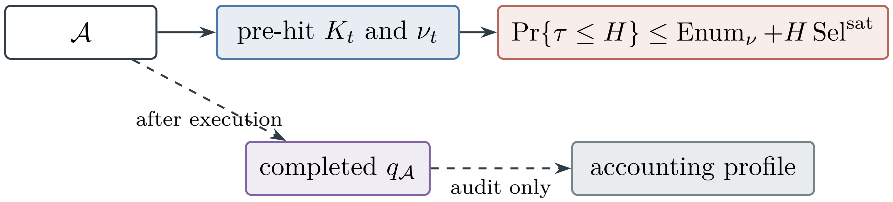

Figure 37: Prospective target advantage is charged through the pre-hit selector in Theorem III.21.3. The completed first-hit quotient remains the retrospective accounting record of Corollary III.23.1.

III.24.3 Finite-slice and residual accountability

Corollary III.24.5 (Finite-slice accountability for an arbitrary DTM)**.** *Let M be any deterministic Turing machine and I any finite input. Fix also one exposed target verifier whose program is uniform on the input-size slices. For every finite horizon H_0, the clocked representation in Corollary II.11.3, followed by the canonical start normalization, has a finite resource ceiling $B_{M,H_0}(n)$, uniform over every input of encoded length n, and an exact first-hit quotient satisfying*

$$\Pr\{M(I) \textit{ reaches the exposed target by time } H_0\} \le e(H_0+1)\, m_I^{\mathrm{sat}}(\Psi_{\mathrm{hit}}(B_{M,H_0}(n(I)), n(I)))\,,$$

where $n(I)$ is the encoded input size.

No running-time, halting, reversibility, geometry, lumpability, or Valid-Lift assumption is made on M. If M is unbounded or nonhalting, these inequalities hold for every finite slice. Its eventual target-arrival probability is the monotone limit of the finite-horizon probabilities. Passing a spectral, resolvent, or budget estimate through that limit is a separate analytic question and is not part of the DTM representation theorem.

Proof. Use the finite clocked point-mass kernel from Corollary II.11.3 and apply Lemma III.20.2. For fixed M, an input of length n can expose or alter at most $O_M(H_0)$ additional tape cells in H_0 steps. The padded configuration width, transition cost, and fixed verifier cost therefore have one worst-case majorant depending only on M, n, H_0, uniformly over the size slice. The explicit first-hit ledger gives the displayed ceiling $B_{M,H_0}(n)$. Monotone convergence of the events $\{\tau_T \le H_0\}$ gives the final assertion.

□

Remark III.24.6 (Saturated accountability of the residual)*.* Apply Theorem Component exhaustiveness to the full lifted generator of $\mathcal{A}$. Its five structural components and the two residual coordinates classified by Theorem II.7.2 may interact through cancellation, memory, iteration, and adaptive branching. Theorem III.24.3 evaluates their combined target arrival through the single exact first-hit accounting quotient. Thus a successful use of dynamics initially assigned to J_{res} produces a represented accounting record at the charged finite-horizon cost. It produces a prospective Abs source only when an independent pre-hit representation satisfies Definition III.21.1. No positivity assumption on component partial sums is needed; the prospective lower-bound direction uses the explicit baseline in Theorem III.21.3.

The projection $P_Q y$ is the target mass visible to the represented quotient, and A_Q is the Part I quotient-utility factor. Their product defines structural visibility. A target-correlated quotient need not provide a budget-admissible selector; its correlation remains part of the visibility invariant.

Any budget-instantiated extraction consequence is supplied by the applicable arrival theorem for the declared class rather than by the definition of $m_n^{\mathrm{sat}}(B)$.

The supremum over $\mathcal{Q}^{\mathrm{sat}}_{\mathfrak{I}_n,\le B}$ includes every nonlinear or runtime-generated projection U produced by one uniform represented derivation and satisfying the stated conditions. If such a projection is generated without oracle access from the presented instance, has represented fibers with saturated cost at most B, is exactly lumpable on the target window, and has non-negligible Part I quotient utility factor A_Q, then it is one of the null-geometry certificates counted by $\mathfrak{U}_n^{\mathrm{sat}}(B)$. If it is pointwise geometric or causal rather than represented-fiber structure, its conditioned target visibility is counted by the identity-quotient branch of the same master profile at every budget in its admissible budget set. If it only perturbs an already declared geometry, the Dirichlet form comparison of Section III.11 is the condition under which the Part III geometric comparison proof transfers; otherwise the perturbation must be counted as its own geometry at every budget in its admissible budget set. High raw Walsh degree does not exclude a compact valid representation. Conversely, $\mathcal{F}_I$-measurability does not imply within-budget cost or target utility: a measurable target distinction is charged to $\mathfrak{U}_n^{\mathrm{sat}}(B)$ only when $B \in \mathfrak{B}_{\mathrm{sat},n}(R)$ and its Part I quotient-utility factor is nonzero. If it is not counted by $\mathfrak{U}_n^{\mathrm{sat}}(B)$ at any declared budget level, then it is not exposed within-budget structure for the original presentation: it is presentation-inaccessible, oracle-dependent, over-budget to construct, or residual from the public presentation. Thus a nonlinear change of variables can reclassify residual transport as an Abs component only by supplying an exposed represented target-relevant quotient, and it can turn residual motion into a Geom/Caus component only by supplying exposed geometric or potential data; either contribution is charged in $\mathfrak{U}_n^{\mathrm{sat}}(B)$ exactly at the budgets in its admissible budget set.

III.25 Combined Extraction and Profile Decomposition

The combined extraction functional joins the exact-quotient and active-geometry profiles at one budget.

Definition III.25.1 (Combined extraction functional)**.** The untyped quantities in this definition are the accounting profiles of Definition III.23.3. The budget-B accounting extraction functional is

$$I_n^{\mathrm{sat}}(B) = m_n^{\mathrm{sat}}(B) + G_n^{\mathrm{sat}}(B).$$

The master-certificate definition gives the exact identity

$$\mathfrak{U}_n^{\mathrm{sat}}(B) = \max\{m_n^{\mathrm{sat}}(B), G_n^{\mathrm{sat}}(B)\}, \qquad \mathfrak{U}_n^{\mathrm{sat}}(B) \le I_n^{\mathrm{sat}}(B) \le 2\mathfrak{U}_n^{\mathrm{sat}}(B). \tag{183}$$

Thus the two-coordinate sum is retained for channelwise accounting. Its prospective restriction satisfies

$$\mathfrak{U}_n^{\mathrm{src,sat}}(B) = \max\{m_n^{\mathrm{src,sat}}(B), G_n^{\mathrm{src,sat}}(B)\}, \qquad \mathfrak{U}_n^{\mathrm{src,sat}}(B) \le I_n^{\mathrm{src,sat}}(B) \le 2\mathfrak{U}_n^{\mathrm{src,sat}}(B). \tag{184}$$

In log-cost notation $I_n(d) = I_n^{\mathrm{sat}}(2^d)$. When the surrounding text writes $m(d)$, $G(d)$, or $I(d)$, the variable d is this saturated cost level, not raw Walsh degree.

For a family functional $\mathfrak{M}_n$, define $G^{\mathrm{sat}}_{\mathfrak{M},n}(B)$ by taking the analogous supremum over family-uniform represented Geom/Caus data. Define

$$\mathfrak{U}^{\mathrm{sat}}_{\mathfrak{M},n}(B) = \max\left\{m^{\mathrm{sat}}_{\mathfrak{M},n}(B), G^{\mathrm{sat}}_{\mathfrak{M},n}(B)\right\},$$

and set

$$I^{\mathrm{sat}}_{\mathfrak{M},n}(B) = m^{\mathrm{sat}}_{\mathfrak{M},n}(B) + G^{\mathrm{sat}}_{\mathfrak{M},n}(B).$$

The retrospective audit bound therefore has the channel-compatible form

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le eH_B\, m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)) \le eH_B\, \mathfrak{U}^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)) \le eH_B\, I^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)). \tag{185}$$

The first inequality is an accounting identity; prospective hardness uses Corollary III.24.4.

Theorem III.25.2 (Full cost-tagged saturated-profile decomposition)**.** *Fix a represented family, its resource algebra, and the budgeted derivability closure and saturated cost of Definitions III.14.1 and III.1.3. Apply the complete-carrier channel decomposition of Theorem II.7.1 to every represented operator record and assign each complete record its upward-closed admissible budget set and Pareto-minimal frontier, which exist by Lemma III.5.7. The resulting global cost-tagged ledger has the following exhaustive decomposition.*

(i) *Its prospective* Abs *coordinate is* $m^{\mathrm{src,sat}}_n(B)$*. Every contributing source quotient has exactly one of the five provenance tags in Theorem III.2.2 and satisfies Definition III.21.1. The accounting coordinate* $m^{\mathrm{sat}}_n(B)$ *may additionally contain completed records such as the exact first-hit quotient.*

(ii) *Its prospective* $\mathsf{Geom}/\mathsf{Caus}$ *coordinate is* $G^{\mathrm{src,sat}}_n(B)$*. Every contributing geometry has the edge-conductance form and one of the four represented configurations in Theorem II.9.5. Its complete active tuple factors through the canonical charged source frontier of Theorem III.2.4; hence its unique support tag and nonempty five-family provenance signature place it in the exhaustive finite cover* $\mathfrak{S}^4_{\mathsf{Geom}} \times \mathfrak{J}^5_{\mathsf{Abs}}$*.* Caus *is the authorized oriented projection of that same represented source and introduces no additional source class.*

(iii) *Every* Lift *record has the fiber-kernel and operator factorization of Theorem II.10.5, and every* Bdry *record has the killing/value factorization of Theorem II.5.10. These derived channels retain their complete carrier cost and transfer target visibility only through a represented contextual identity, as required by Proposition II.5.9.*

(iv) *After all represented channel-qualified denotations have been assigned their admissible budget sets, the post-saturation residual is exactly*

$$L^{J_{\mathrm{res}},\mathrm{sat}} = L^{J_{\mathrm{res}}}_{\mathrm{dir}} + L^{J_{\mathrm{res}}}_{\mathrm{nl}}$$

from Theorem II.7.2. In particular, non-structural *means orthogonal to the represented structural channel subspaces, not merely absent from the current execution record. A family-uniform represented structural witness cannot remain in this post-saturation residual by Proposition II.8.3 and Lemma II.8.4.*

Refining the two prospective source coordinates in items (i) *and* (ii) *by their canonical complete ancestral sets gives exactly the source-mode–state-provenance–coefficient-provenance decomposition of Theorem III.3.4. Thus the finite 4960-cell source classification is a refinement of this full saturated ledger, while* Caus*,* Lift*, and* Bdry *inherit those cells and* J_{res} *contributes none.*

Corollary III.25.3 (Prospective saturated-profile identities)**.** *Consequently the intrinsic prospective source profile is precisely*

$$I^{\mathrm{src,sat}}_n(B) = m^{\mathrm{src,sat}}_n(B) + G^{\mathrm{src,sat}}_n(B), \tag{186}$$

and its master certificate satisfies

$$\mathfrak{U}_n^{\mathrm{src,sat}}(B) = \max\Big\{m_n^{\mathrm{src,sat}}(B), G_{X,n}^{\mathrm{src,sat}}(B), G_{Q,\mathrm{ex},n}^{\mathrm{src,sat}}(B), G_{Q,\mathrm{rev},n}^{\mathrm{src,sat}}(B), G_{Q,\mathrm{nr},n}^{\mathrm{src,sat}}(B)\Big\}. \tag{187}$$

A structural record R for which $B \notin \mathfrak{B}_{\mathrm{sat},n}(R)$ remains tagged to its structural channel in the global ledger and is inactive at budget B. The execution-level filtered residual $L^{J_{\mathrm{res}},B}$ may retain that same block below its exposure cost solely as the budget-bookkeeping case of Theorem II.7.2; it is not part of $L^{J_{\mathrm{res}},\mathrm{sat}}$. Thus saturation removes every represented structural component from J_{res} without treating an over-budget record as affordable.

Corollary III.25.4 (Budgeted computation success and source-charge bounds)**.** *For every budget-B computation $\mathcal{A}$, prospective selector accountability gives*

$$\Pr\{\tau_{\mathcal{A}} \le H_B\} \le \mathrm{Enum}_\nu(\mathcal{A}, H_B) + H_B \,\mathrm{Sel}_n^{\mathrm{sat}}(\Psi(B,n)). \tag{188}$$

The quantitative five-family accounting-charge form is

$$\Pr\{\tau_{\mathcal{A}} \le H_B\} \le eH_B(n) \sum_{c \in \mathfrak{J}_{\mathsf{Abs}}^5} \mathcal{C}_{I,\mathfrak{F}_5}^{(c)}\Big(\Psi_5^{\mathrm{alg}}(\Psi_{\mathrm{hit}}(B,n))\Big). \tag{189}$$

The displayed contextual envelopes account for the completed first-hit record at its saturated cost. A prospective-source interpretation of a frontier term requires the represented comparison in Corollary III.16.9. Lift *and* Bdry *propagate represented sources, while the post-saturation J_{res} component supplies no additional structural coordinate.*

Proof of Theorem III.25.2 and Corollaries III.25.3 and III.25.4. Theorem II.7.1 gives the exhaustive ordered carrier split. The normal forms Theorems II.5.10, II.7.2, II.9.5, II.10.5, III.2.2 and III.2.4 classify the five non-causal blocks, and Proposition II.5.9 identifies Caus, Lift, and Bdry as derived uses of represented source data. Budget-set assignment is well posed by Lemma III.5.7. Therefore a represented structural denotation is charged to its structural channel exactly at budgets belonging to its upward-closed admissible budget set. The reclassification and residual-correlation results Proposition II.8.3 and Lemma II.8.4 then leave in the post-saturation residual exactly the two orthogonal remainders of Theorem II.7.2. This proves the ledger decomposition and (186).

Theorem III.18.9 restricted by temporal source admissibility gives the five branches in (187); taking their maximum is the definition of the prospective master certificate. The two-sided equivalence (184) then identifies that maximum with the two-coordinate sum up to the displayed constant factor. Finally, Theorem III.21.3 gives (188), while Theorem III.21.4 gives the direct five-envelope accounting estimate (189). No residual structural source remains after least-cost saturation, and Corollary III.23.1 places the completed first-hit record in the separate accounting layer.

□

Figure 38 combines the complete classification of every channel in one cost-tagged tree.

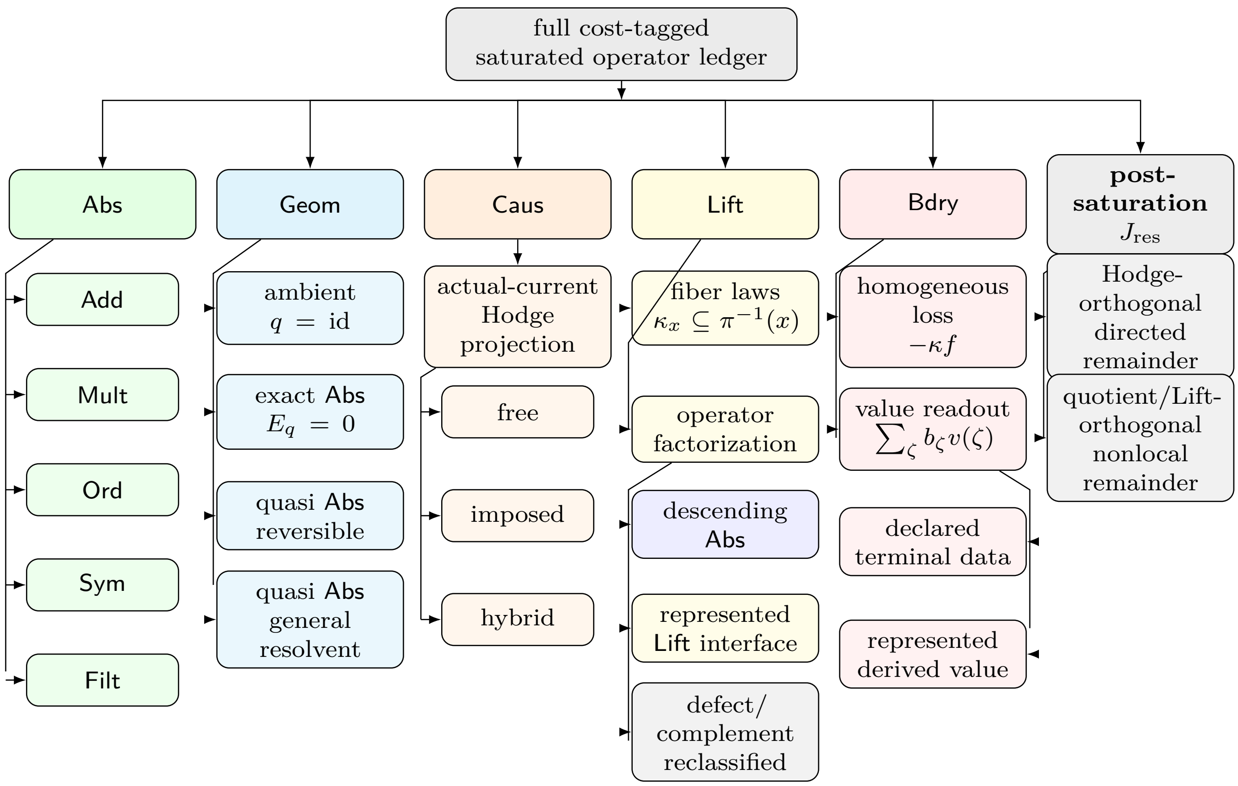

Figure 38: Exhaustive classification of the full cost-tagged saturated operator ledger. The Abs column applies Theorem III.2.2; the Geom column applies Theorem II.9.5, and each of its four support forms carries the nonempty five-family source signature classified by Theorem III.2.4; the Caus column applies Definition II.4.3; the Lift column applies Theorem II.10.5; and the Bdry column applies Theorem II.5.10. After every represented structural denotation has been charged at its least saturated cost, J_{res} consists exactly of the two non-structural orthogonal remainders in Theorem II.7.2.

III.26 Ambient-Geometry Accountability

Theorem III.26.1 (Saturated ambient-geometry accountability after quotient exhaustion)**.** *Fix a represented task family, the free Caus authority envelope, and a saturated budget B. Restrict the ambient record family $\mathcal{G}_{n,\le B}^{X,\mathrm{free}}$ of Definition III.14.9 to temporally admissible prospective source records. Then*

$$G_{X,n}^{\mathrm{src,sat}}(B) = \sup_{(\mathfrak{N},\Phi,D,J,\mathfrak{K})} M_\Phi C_{\Phi,\mathfrak{K}}^{\mathrm{auth}}(D) \frac{R_T(D)}{1+\rho_\Phi(D)}, \tag{190}$$

where the supremum is over that single restricted family and has value zero when the family is empty. The same zero convention applies to the three factorwise suprema below. Every factor belongs to $[0,1]$*. Consequently,*

$$\begin{aligned} G_{X,n}^{\mathrm{src,sat}}(B) \le \min\Bigg\{ &\sup_{(\mathfrak{N},\Phi,D,J,\mathfrak{K})} C_{\Phi,\mathfrak{K}}^{\mathrm{auth}}(D), \quad \sup_{(\mathfrak{N},\Phi,D,J,\mathfrak{K})} R_T(D), \\ &\sup_{(\mathfrak{N},\Phi,D,J,\mathfrak{K})} \frac{1}{1+\rho_\Phi(D)} \Bigg\}. \end{aligned} \tag{191}$$

More sharply, for every $0<\varepsilon<1$,

$$G_{X,n}^{\mathrm{src,sat}}(B) > \varepsilon \tag{192}$$

holds only if one family-uniform represented record of complete saturated cost at most B satisfies simultaneously

$$M_\Phi = 1, \qquad C_{\Phi,\mathfrak{K}}^{\mathrm{auth}}(D) > \varepsilon, \qquad R_T(D) > \varepsilon, \qquad \rho_\Phi(D) < \varepsilon^{-1} - 1. \tag{193}$$

Conversely, if no record in the saturated ambient family satisfies all four inequalities in (193)*, then*

$$G_{X,n}^{\mathrm{src,sat}}(B) \le \varepsilon. \tag{194}$$

The witness in (193) *has identity quotient support. Its geometry therefore acts on the full presented state space, and every step used by its Dirichlet form lies in the exposed comparability relation E_Φ. The condition $M_\Phi = 1$ supplies the represented transfer-class Poincare or conductance comparison; positivity of $C_{\Phi,\mathfrak{K}}^{\mathrm{auth}}(D)$ supplies the local-consistency gate and an authorized descending current for the same selected generator; $R_T(D) > \varepsilon$ supplies target reach; and the last inequality is the exact regularity bound appearing in the ambient visibility coordinate.*

Now suppose that, at the same budget, the four prospective nonambient coordinates satisfy

$$\max\Big\{ m_n^{\mathrm{src,sat}}(B), G_{Q,\mathrm{ex},n}^{\mathrm{src,sat}}(B), G_{Q,\mathrm{rev},n}^{\mathrm{src,sat}}(B), G_{Q,\mathrm{nr},n}^{\mathrm{src,sat}}(B) \Big\} \le \alpha_n \qquad (\alpha_n \ge 0). \tag{195}$$

Then the full prospective source profiles obey

$$\mathfrak{U}_n^{\mathrm{src,sat}}(B) \le \max\Big\{ G_{X,n}^{\mathrm{src,sat}}(B), \alpha_n \Big\}, \tag{196}$$

$$I_n^{\mathrm{src,sat}}(B) \le 2\max\Big\{ G_{X,n}^{\mathrm{src,sat}}(B), \alpha_n \Big\}. \tag{197}$$

Thus quotient exhaustion leaves exactly one possible affordable structural source: an identity-supported ambient record satisfying the joint local-gap, authorized-descent, target-reach, and regularity conditions above. A represented transformation that preserves identity support is already in the supremum in (190). *A transformation that creates nonidentity fibers is charged to one of the four coordinates in* (195).

Proof. Temporal source restriction changes only the admissible record family in the supremum of Definition III.14.9. The visibility of each remaining identity-supported record is, by definition,

$$V^X_{\Phi,D,J,\mathfrak{K}} = M_\Phi C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D) R_T(D) \frac{1}{1+\rho_\Phi(D)}.$$

Taking the supremum proves (190). The mixing indicator is binary, the authority-compatible alignment and target-reach factors lie in $[0,1]$, and $\rho_\Phi(D) \geq 0$. Hence the product is bounded by each of its three nonbinary factors. Taking suprema over the same record family proves (191).

Assume (192). By the defining property of a supremum, some record in the restricted family has visibility greater than ε. Its mixing indicator cannot be zero. Since the remaining three factors lie in $[0,1]$, their product can exceed ε only when each factor exceeds ε. The regularity inequality is equivalent to

$$\frac{1}{1+\rho_\Phi(D)} > \varepsilon \quad \Longleftrightarrow \quad \rho_\Phi(D) < \varepsilon^{-1} - 1.$$

This proves (193). Conversely, if no record satisfies all four conditions, then every record has mixing indicator zero or at least one of the remaining factors at most ε. Its visibility is therefore at most ε. Taking the supremum proves (194).

The interpretation of the four conditions follows directly from Lemma II.9.2 and Definitions III.12.9 and III.14.9. In particular, the authority-compatible factor is positive only when the local comparison and current provenance belong to the same selected generator. By Lemma II.4.5, a separately chosen target-oriented current cannot be inserted into this record. The master normal form of Theorem III.18.9 charges every represented change of variables, join, potential, conductance, and current. It assigns the result to the identity-supported ambient family precisely when the terminal quotient is the identity; otherwise it assigns the record to an exact or compensated nonidentity quotient family.

Finally, apply the prospective five-branch identity (187). Under (195), four entries in that maximum are at most α_n, leaving only $G^{\mathrm{src,sat}}_{X,n}(B)$. This proves (196). The prospective combined-profile comparison (184) gives (197).

□

Theorem III.26.2 (Represented-current admission and no semantic target support)**.** *Fix a presented finite task family, a saturated budget B, and the prospective ambient family of Theorem III.26.1. A current contributes to $G^{\mathrm{src,sat}}_{X,n}(B)$ only as the current entry of one complete record*

$$\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K}) \in \mathcal{G}^{X,\mathrm{free}}_{n,\leq B}$$

whose represented derivation satisfies all of the following conditions. In this record, $J = J_{\mathfrak{K}}$: the current entry is the current identified and authenticated by the authority record $\mathfrak{K}$.

(i) *the normalized kernel, edge relation, conductance, potential, selected current, and authority comparison are derived from the declared public presentation and presentation-admissible oracle evaluators;*

(ii) *their complete derivation and evaluation cost is at most B;*
(iii) *the record is available before the transition whose progress it is invoked to explain; and*
(iv) *$\mathfrak{K}$ authenticates the current of the selected generator, or authenticates an auxiliary geometric current together with the same-generator local, drift, resolvent, or carrier comparison required by Definitions II.4.4 and III.12.9.*

If any condition fails, that denotation is absent from the prospective ambient record family. Equivalently, under the zero-extension convention for incomplete records, a missing current identity, target polarity, dynamic transport, or same-generator comparison gives

$$\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) = 0, \qquad V^X_{\Phi,D,J,\mathfrak{K}} = 0. \tag{198}$$

Let $S_I \subseteq E_\Phi$ be a represented edge sector such that $\nabla_\Phi D$ is supported on S_I. For every admitted record with nonzero current,

$$V^X_{\Phi,D,J,\mathfrak{K}} \leq \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) \leq \frac{\|\mathbf{1}_{S_I} J_{\mathfrak{K}}\|_E^2}{\|J_{\mathfrak{K}}\|_E^2} \leq 1. \tag{199}$$

The ratio in Eq. (199) *is an analytic evaluation of an already admitted current. It neither constructs the current nor admits an extensionally specified current into the saturated profile. Evaluating the predicate $\mathbf{1}_{S_I}(x, z)$ likewise does not construct a normalized kernel or a sampler supported on S_I; such a kernel contributes only with its represented normalization, sampling procedure, random-bit use, selected-generator identity, and complete cost.*

For a randomized family-uniform derivation, let $Q_{I,\omega}$ be the indicator that its output record satisfies the four admission conditions and all remaining Geom qualification gates on instance I and seed ω. The qualified-record convention gives the pointwise bounds

$$0 \leq Q_{I,\omega} V^X_{\Phi,D,J,\mathfrak{K}} \leq Q_{I,\omega} \frac{\|\mathbf{1}_{S_I} J_{\mathfrak{K}}\|_E^2}{\|J_{\mathfrak{K}}\|_E^2} \leq Q_{I,\omega}, \tag{200}$$

with the middle ratio set to zero for a zero current. Consequently, for every monotone positively homogeneous family functional $\mathfrak{M}_n$,

$$\mathfrak{M}_n\Big(I \longmapsto \mathbb{E}_\omega[Q_{I,\omega} V^X_{\Phi,D,J,\mathfrak{K}}]\Big) \leq \mathfrak{M}_n\Big(I \longmapsto \Pr_\omega\{Q_{I,\omega} = 1\}\Big). \tag{201}$$

If qualification entails a represented success event $\mathcal{E}_{I,\omega}$, the right-hand side is at most the same family functional applied to $\Pr_\omega(\mathcal{E}_{I,\omega})$. Thus a current constructed from a candidate target, decoded center, learned ranking, or selector is charged through the complete derivation of that record and through the probability that the resulting record is target-qualified.

An oracle or advice string that supplies the current without a represented evaluator in the declared presentation changes the presented family as in Theorem III.33.2. A current produced from later verifier flags is an accounting record rather than a prospective source by Definition III.21.1.

Proof. The ambient profile in Eq. (190) is a supremum over complete represented records in $\mathcal{G}^{X,\mathrm{free}}_{n,\leq B}$. Membership in that family gives conditions (i) and (ii) and identifies the current entry with the current authenticated by $\mathfrak{K}$, while Definition III.21.1 gives condition (iii). The current and comparison clauses in Definitions II.4.4 and III.12.9 give condition (iv) and set the authorized alignment to zero when its record is missing. The complete visibility product then proves Eq. (198).

Because $\nabla_\Phi D$ is supported on S_I,

$$\langle J_{\mathfrak{K}}, -\nabla_\Phi D\rangle_E = \langle \mathbf{1}_{S_I} J_{\mathfrak{K}}, -\nabla_\Phi D\rangle_E.$$

For nonzero $J_{\mathfrak{K}}$ and nonzero $\nabla_\Phi D$, Definitions III.12.8 and III.12.9 identify the authorized alignment with the normalized positive pairing below. Cauchy–Schwarz on the represented weighted edge space gives

$$\begin{aligned}\operatorname{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) &= \frac{\left(\langle \mathbf{1}_{S_I} J_{\mathfrak{K}}, -\nabla_\Phi D\rangle_E\right)_+^2}{\|J_{\mathfrak{K}}\|_E^2 \|\nabla_\Phi D\|_E^2} \\ &\le \frac{\|\mathbf{1}_{S_I} J_{\mathfrak{K}}\|_E^2 \|\nabla_\Phi D\|_E^2}{\|J_{\mathfrak{K}}\|_E^2 \|\nabla_\Phi D\|_E^2} = \frac{\|\mathbf{1}_{S_I} J_{\mathfrak{K}}\|_E^2}{\|J_{\mathfrak{K}}\|_E^2}.\end{aligned}$$

If $\nabla_\Phi D = 0$, the alignment is zero by definition and the same inequality holds. This proves the second inequality in Eq. (199); the first follows because every other visibility factor lies in $[0, 1]$. Restriction of a weighted norm to an edge subset gives the last inequality.

The budgeted derivability closure contains outputs of represented constructors, not arbitrary extensional tables. The no-substitution result Lemma II.4.5 prevents replacement of the selected current by another target-oriented vector, and Lemma I.4.7 retains the complete ancestry of every represented postprocessing. A Boolean sector predicate supplies an evaluator only; normalization and sampling require the additional represented operations listed in the theorem. This proves the admission and predicate statements.

Finally, multiply the pointwise inequalities by the qualification indicator. Every visibility and support fraction lies in $[0, 1]$, which proves Eq. (200). In particular, pointwise in the instance,

$$\mathbb{E}_\omega[Q_{I,\omega} V^X_{\Phi,D,J,\mathfrak{K}}] \le \mathbb{E}_\omega[Q_{I,\omega}] = \Pr_\omega\{Q_{I,\omega} = 1\}.$$

Monotonicity of $\mathfrak{M}_n$ proves Eq. (201); positive homogeneity is the declared profile convention and is preserved. The success-event conclusion is monotonicity under $Q_{I,\omega} \le \mathbf{1}_{\mathcal{E}_{I,\omega}}$. Oracle separation and temporal classification are the two cited framework results.

□

III.27 Authorized Perturbations and Obstruction Criteria

Definition III.27.1 (Authorized ambient perturbation witness)**.** Let

$$\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K}) \in \mathcal{G}^{X,\mathrm{free}}_{n,\le B}$$

be a temporally admissible prospective ambient record of Definition III.14.9, with identity quotient support, and write

$$\mathfrak{N} = (\widetilde{L}, \zeta, c, P, L, \lambda)$$

for its charged uniformization record. Its *authorized ambient perturbation kernel* is the edge-restricted sample-and-hold kernel

$$\begin{aligned} P_\Phi(x, z) &= \mathbf{1}_{\{(x,z)\in E_\Phi\}} P(x, z), \qquad\qquad z \ne x, \\ P_\Phi(x, x) &= P(x, x) + \sum_{\substack{z\ne x \\ (x,z)\notin E_\Phi}} P(x, z). \end{aligned} \tag{202}$$

The killing mass of P is unchanged, so P_Φ is sub-Markov, and it is Markov whenever P is Markov. Operationally, one samples one P-transition and replaces an outcome outside E_Φ by a holding step. The construction uses one represented kernel step and one represented edge-membership test and is therefore charged within the same class-preserving derivation.

The *authorized ambient perturbation data* are the existing represented data together with this derived kernel:

$$\mathrm{Pert}(\mathfrak{g}) = \big(P, P_\Phi, E_\Phi, c_\Phi, D, J_{\mathfrak{K}}, \mathfrak{K}\big).$$

The tuple retains the complete derivation and cost of $\mathfrak{g}$. The normalized kernel P is the selected public transition kernel; P_Φ is its canonical exposed local perturbation; E_Φ and c_Φ are the geometric comparison data; and $J_{\mathfrak{K}}$ is either the structural-gradient projection of the actual selected current or the auxiliary geometric current accompanied by the represented comparison required in Definitions II.4.4 and III.12.9.

Define the represented descending-edge predicate

$$E_\downarrow(\mathfrak{g}) = \big\{(x,z) \in E_\Phi : J_{\mathfrak{K}}(x,z)\big(D(x) - D(z)\big) > 0\big\}\,.$$

For $0 < \varepsilon < 1$, the perturbation witness is *ε-effective* when

$$M_\Phi = 1, \qquad \chi_\Phi^{\mathrm{auth}}(D) = 1, \qquad \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) > \varepsilon, \qquad R_T(D) > \varepsilon, \qquad \frac{1}{1+\rho_\Phi(D)} > \varepsilon. \tag{203}$$

This is a predicate on the existing ambient certificate record. It defines neither a new structural channel nor an additional profile coordinate. Raw displacement in a chosen presentation coordinate is not part of the predicate: locality means support on the represented relation E_Φ, with the selected-kernel comparison and its cost retained.

Theorem III.27.2 (Perturbation characterization and obstruction after quotient exhaustion)**.** *Fix a represented task family, saturated budget B, and $0 < \varepsilon < 1$. Every dynamically qualified, temporally admissible identity-supported Geom/Caus source is an authorized ambient perturbation witness in the sense of Definition III.27.1. Conversely, an authorized ambient perturbation record satisfying the five certificate conditions of Lemma II.9.2 is precisely an identity-supported ambient source record in $\mathcal{G}^{X,\mathrm{free}}_{n,\le B}$.*

If

$$V^X_{\Phi,D,J,\mathfrak{K}} > \varepsilon, \tag{204}$$

then $\mathrm{Pert}(\mathfrak{g})$ *is ε-effective. Its authorized current satisfies*

$$\langle J_{\mathfrak{K}}, -\nabla_\Phi D\rangle_E > \sqrt{\varepsilon}\,\|J_{\mathfrak{K}}\|_E \|\nabla_\Phi D\|_E, \tag{205}$$

and its descending edges carry at least the same positive pairing:

$$\Big\langle \mathbf{1}_{E_\downarrow(\mathfrak{g})} J_{\mathfrak{K}}, -\nabla_\Phi D\Big\rangle_E \ge \langle J_{\mathfrak{K}}, -\nabla_\Phi D\rangle_E\,. \tag{206}$$

In particular, the perturbation is represented before use; its counted geometric current and potential differences are supported on exposed comparisons; and its descent is authorized by the same selected generator that supplies the local-consistency, drift, resolvent, or carrier comparison.

Conversely, every ε-effective perturbation witness satisfies

$$V^X_{\Phi,D,J,\mathfrak{K}} > \varepsilon^3. \tag{207}$$

More generally, let $0 \le \delta < 1$. If every temporally admissible ambient record of cost at most B satisfies at least one of

$$\begin{gathered} M_\Phi = 0, \qquad \chi_\Phi^{\mathrm{auth}}(D) = 0, \qquad \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) \le \delta, \\ R_T(D) \le \delta, \qquad \frac{1}{1+\rho_\Phi(D)} \le \delta, \end{gathered} \tag{208}$$

then

$$G^{\mathrm{src,sat}}_{X,n}(B) \le \delta. \tag{209}$$

If, at the same budget, the nonambient quotient envelope in Eq. (195) *is at most* α_n*, then*

$$\mathfrak{U}^{\mathrm{src,sat}}_{n}(B) \le \max\{\alpha_n, \delta\}, \tag{210}$$

$$I^{\mathrm{src,sat}}_{n}(B) \le 2\max\{\alpha_n, \delta\}. \tag{211}$$

Thus quotient exhaustion and the perturbation obstruction cover jointly bound the complete prospective structural profile. A represented transformation with nonidentity composite fibers is assigned to the exact or compensated quotient envelope. A transformation with identity support, including one with large displacement in raw coordinates, remains in the ambient family and must satisfy Eq. (203) *to contribute above the corresponding scale.*

Proof. The ambient branch in Theorems II.9.5 and III.18.9 has identity quotient support and an active geometry slot. Its charged normalized kernel is $\mathfrak{N}$, its local relation and conductance are Φ, and its current provenance is $\mathfrak{K}$. These are exactly the entries of $\mathrm{Pert}(\mathfrak{g})$. The converse follows from the necessary-and-sufficient certificate conditions in Lemma II.9.2, together with the authority condition of Definition III.12.9. This proves the record characterization.

Assume (204). By Definition III.14.9,

$$V^{X}_{\Phi,D,J,\mathfrak{K}} = M_\Phi\, \chi^{\mathrm{auth}}_{\Phi}(D)\ \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D)\, R_T(D)\, \frac{1}{1+\rho_\Phi(D)}.$$

The mixing and consistency gates are binary and the other three factors lie in $[0,1]$. A product greater than ε forces both binary gates to equal one and each remaining factor to exceed ε. Hence $\mathrm{Pert}(\mathfrak{g})$ is ε-effective.

With the authority gate equal to one, Definitions III.12.8 and III.12.9 give

$$\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) = \frac{\left(\langle J_{\mathfrak{K}}, -\nabla_\Phi D\rangle_E\right)_+^2}{\|J_{\mathfrak{K}}\|_E^2 \|\nabla_\Phi D\|_E^2}.$$

The strict lower bound by ε proves (205). On every edge outside $E_{\downarrow}(\mathfrak{g})$, the summand in the edge pairing is nonpositive. Removing those summands can only increase the pairing, proving (206). Temporal availability, represented support, complete cost, and same-generator authority are the defining properties of Definitions III.21.1, II.4.4 and III.12.9.

Conversely, under (203), the two binary gates equal one and the alignment, reach, and regularity factors each exceed ε. Their product is greater than ε^3, proving (207).

Now assume (208). Every ambient visibility has a zero binary factor or one of its three quantitative factors is at most δ. Since all remaining factors lie in $[0,1]$, every record has visibility at most δ. Taking the supremum proves (209). Finally, Eqs. (196) and (197) with $G^{\mathrm{src,sat}}_{X,n}(B) \le \delta$ gives Eqs. (210) and (211).

□

Corollary III.27.3 (Polynomial perturbation obstruction criterion)**.** *Let* $B = B(n)$ *be a budget schedule in the polynomial resource envelope, and suppose the nonambient quotient envelope is negligible at* $B(n)$*. The complete prospective source profile is negligible whenever, for every fixed*

$k \geq 1$ and all sufficiently large n, every temporally admissible identity-supported perturbation record of cost at most $B(n)$ satisfies at least one of

$$\begin{gathered} M_\Phi = 0, \qquad \chi_\Phi^{\mathrm{auth}}(D) = 0, \qquad \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) \leq n^{-k}, \\ R_T(D) \leq n^{-k}, \qquad \frac{1}{1+\rho_\Phi(D)} \leq n^{-k}. \end{gathered} \tag{212}$$

Equivalently, a nonnegligible prospective source profile produces one family-uniform affordable perturbation record for which all five displayed obstructions fail at an inverse-polynomial scale. The record includes the selected kernel, exposed perturbation relation, potential, current, authority proof, and complete represented derivation.

Proof. Apply Theorem III.27.2 with $\delta = n^{-k}$. The obstruction cover gives $G^{\mathrm{src,sat}}_{X,n}(B(n)) \leq n^{-k}$. Since k is arbitrary, the ambient profile is negligible. The quotient envelope is negligible by hypothesis, so Eqs. (210) and (211) proves negligibility of the master and combined prospective profiles. The equivalent formulation is the contrapositive, using Eq. (204).

□

Corollary III.27.4 (Profile-level criterion for absence of an affordable ambient geometry)**.** *Let $B = B(n)$ be a declared budget schedule. The ambient prospective coordinate $G^{\mathrm{src,sat}}_{X,n}(B(n))$ is negligible if and only if, for every fixed $k \geq 1$ and all sufficiently large n, the saturated ambient family contains no temporally admissible record satisfying*

$$M_\Phi = 1, \qquad C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D) > n^{-k}, \qquad R_T(D) > n^{-k}, \qquad \frac{1}{1+\rho_\Phi(D)} > n^{-k}. \tag{213}$$

If the quotient-branch envelope in (195) *is also negligible, then the complete prospective source profile is negligible. This is one saturated profile evaluation: its quantifier ranges over the single family-uniform derivation class fixed by the presentation and budget, rather than over a separately enumerated collection of algorithms.*

Proof. Suppose first that the ambient coordinate is negligible. Fix k. For all sufficiently large n, negligibility gives

$$G^{\mathrm{src,sat}}_{X,n}(B(n)) \leq n^{-(3k+1)}.$$

A record satisfying (213) would have visibility strictly greater than n^{-3k}, a contradiction.

Conversely, assume the displayed witness is absent for every fixed k. For a chosen k and all sufficiently large n, every ambient record has mixing indicator zero or at least one of the remaining three factors at most n^{-k}. Its visibility is at most n^{-k}, and so is the supremum. Since k is arbitrary, the ambient coordinate is negligible. The final assertion follows from Eqs. (196) and (197).

□

For a same-carrier statistical presentation, Lemma IV.12.1 bounds this complete ambient supremum by the saturated learned projection envelope at the class-preserving enlarged cost. The represented-readout and selector refinements are Proposition IV.12.3 and Theorem IV.12.4.

Lemma III.27.5 (Combined geometric and quotient structural comparison bound)**.** *Let $\mathcal{S}_B$ be a class of budget-admissible represented structural certificates. Suppose that each $R \in \mathcal{S}_B$ comes with charged comparison data and two nonnegative visibility terms V_R^{Abs} and V_R^{Geom} satisfying*

$$\Gamma_T(R) \le C_{\mathcal{B}}(n)\big(V_R^{\mathsf{Abs}} + V_R^{\mathsf{Geom}}\big), \qquad V_R^{\mathsf{Abs}} \le m_n^{\mathrm{sat}}(B), \qquad V_R^{\mathsf{Geom}} \le G_n^{\mathrm{sat}}(B).$$

Define $\Gamma_T^{\mathrm{str}}(B) = \sup_{R\in\mathcal{S}_B}\Gamma_T(R)$, *with value zero when* $\mathcal{S}_B$ *is empty. Then*

$$\Gamma_T^{\mathrm{str}}(B) \le C_{\mathcal{B}}(n) I_n^{\mathrm{sat}}(B), \qquad I_n^{\mathrm{sat}}(B) = m_n^{\mathrm{sat}}(B) + G_n^{\mathrm{sat}}(B).$$

Equivalently,

$$\Gamma_T^{\mathrm{str}}(B) \le 2C_{\mathcal{B}}(n)\mathfrak{U}_n^{\mathrm{sat}}(B).$$

Here $C_{\mathcal{B}}$ *is the transfer-class comparison factor supplied by the declared structural comparison data; for* $\mathcal{B}_{\mathrm{poly}}$ *it is polynomial. This structural comparison bounds target contact; the construction of a target-reaching procedure requires the corresponding arrival theorem.*

Proof. For every $R \in \mathcal{S}_B$, the three assumed inequalities give

$$\Gamma_T(R) \le C_{\mathcal{B}}(n)\big(m_n^{\mathrm{sat}}(B) + G_n^{\mathrm{sat}}(B)\big).$$

Taking the supremum over R proves the first displayed bound. Since $a + b \le 2\max\{a, b\}$ for $a, b \ge 0$, the second follows. The theorem therefore records exactly what the charged comparison data prove; the Part II decomposition by itself does not supply the first assumed inequality. Full-process target arrival, including interaction with J_{res}, is controlled instead by (185).

□

Proposition III.27.6 (Structural extraction identifies a component)**.** *If a term in the supremum defining* $\mathfrak{U}_n^{\mathrm{sat}}(B)$ *has transfer-class size, then the corresponding admissible quotient–geometry certificate is an exposed target-aligned component at saturated cost at most* B*. This proposition identifies the exposed component; target-arrival consequences are supplied by the applicable arrival theorem rather than by the visibility definition.*

Proof. For a null geometry slot, a transfer-class-size term is $V_q = A_q\|P_q^{\mathrm{amb}}y\|^2/\|y\|^2$, so the exact quotient subspace contains target mass at the declared threshold after imposing the Part I utility condition. For an active slot, a transfer-class-size term in (124) supplies represented target reach, controlled regularity, and quantitative descending-current alignment. A non-lumpable quotient additionally has an explicit transfer-class stability factor. These are structural visibility properties. A pointwise drift certificate gives the hitting-time conclusion of Lemma III.12.10; other arrival conclusions require the comparison theorem declared by the certificate.

□

Example III.27.7 (SAT extraction)**.** For XOR-SAT, $m_n^{\mathrm{sat}}(B)$ is inverse-polynomial at the polynomial budget needed to represent the public syndrome quotient, so the Abs term in $I_n^{\mathrm{sat}}(B)$ is non-negligible. For a random planted-looking SAT instance with no exposed quotient and a defect landscape whose Hamming geometry has off-target obstructions, the corresponding raw-basis and Hamming-geometry fragment profiles are negligible. The full $m_n^{\mathrm{sat}}(B)$ and $G_n^{\mathrm{sat}}(B)$ have that conclusion only after a proved uniform cover of the complete saturated kernel and geometry families.

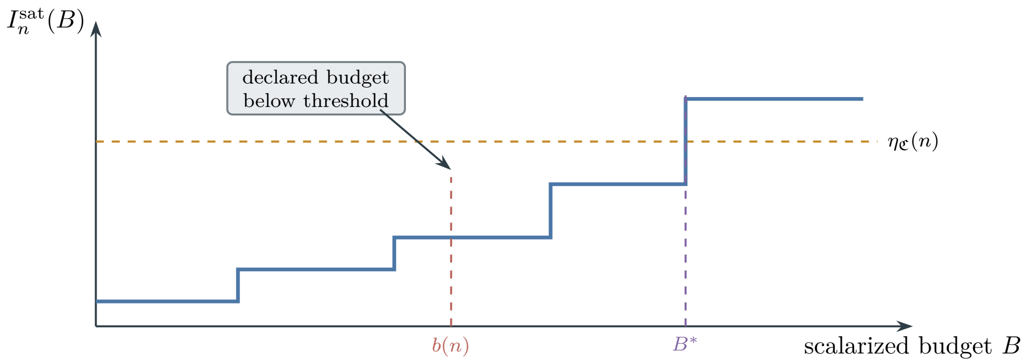

Figure 39: Schematic hard-side reading of the saturated cost invariant: B^* is the infimum budget at which the monotone saturated profile reaches $\eta_{\mathfrak{C}}(n)$, while the displayed ceiling satisfies $b(n) < B^*$. The diagram asserts neither continuity nor general attainment.

III.28 The Saturated Cost Invariant

The central object is the first saturated public cost budget at which the combined extraction functional reaches the declared transfer-class threshold. The log-cost degree d^* is a convenient coordinate for this budget; it is not raw Walsh degree.

The saturated cost invariant is the first budget at which the combined profile reaches the declared threshold.

Definition III.28.1 (Saturated cost invariant)**.** Fix a structural-visibility threshold $\eta_{\mathfrak{C}}(n) \succeq 1/\beta_{\mathfrak{C}}(n)$ and one admissible scalarization $\kappa : C \to [0, \infty]$ from Definition III.1.3. For a scalar $s \geq 0$, interpret $I^{\mathrm{sat}}_{n,\kappa}(s)$ as the supremum over complete represented records having $\kappa(\mathrm{Cost}) \leq s$. In the polynomial instance this is n^{-c}. Define the first extraction budget

$$B^*_\eta(n) = \inf\{s \geq 0 : I^{\mathrm{sat}}_{n,\kappa}(s) \geq \eta_{\mathfrak{C}}(n)\}.$$

If the set is empty, set $B^*_\eta(n) = \infty$. The corresponding log-cost invariant is

$$d^*_\eta(n) = \left\lceil \log_2(1 + B^*_\eta(n)) \right\rceil.$$

The notation $B^*(n)$ or $d^*(n)$ denotes this threshold up to class-equivalent changes of $\eta_{\mathfrak{C}}$. All infima, logarithms, and numerical comparisons involving B^* or d^* in this section use this same scalarization. The underlying resource-valued invariant remains the Pareto frontier of admissible ceilings.

Figure 39 gives the threshold reading of Definition III.28.1 in the hard-side regime.

The invariant B^* is the first saturated public budget at which the observable closure contains either a target-aligned quotient or a regular target-oriented geometry. It is minimized over all admissible representations generated from the presentation. Below this budget, every structural representation beneath the declared ceiling has negligible target contrast at the transfer-class threshold. Raw Walsh degree alone does not determine B^*: a valid low-cost quotient, lift, or forward transport may represent the same target distinction, whereas measurability alone provides no low-cost representation.

Proposition III.28.2 (The saturated cost invariant is intrinsic to the problem presentation)**.** *The invariants B^* and d^* depend only on the family of task presentations, the admissible dynamic slot activated by Part II, and the saturated cost filtration generated from the observable algebra. They are independent of the choice of method, program, solver, or learning algorithm.*

Proof. The quantities $m_n^{\mathrm{sat}}(B)$ and $G_n^{\mathrm{sat}}(B)$ are suprema over all observable structures in the saturated admissible representation class at public cost at most B. This class is generated by the presentation and the oracle-free constructor rules: public basis changes, represented quotients, valid lifts, exposed potentials, pointwise geometries, scalar boundary readouts, and forward observable transport/generator components after Part II projection. The class is fixed before choosing a particular solver. Part II proves that every algorithm-induced movement lies in Geom, Caus, Abs, Lift, Bdry, or J_{res}. The target-amplifying exposed contributions charged in $I_n^{\mathrm{sat}}(B)$ are the two intrinsic coordinates: Geom/Caus, which measures exposed local geometry and its oriented use, and Abs, which measures represented quotient abstraction. Lift is retained with the intrinsic structure it lifts, and Bdry is retained with the scalar readout and the structure it colors. Their contextual target charges are not reassigned without a represented comparison. The label J_{res} supplies no additional structural coordinate: a verified channel projection is reclassified at its charged cost, while the full-process target effect is accounted for by the exact first-hit quotient. Therefore minimizing the budget over the charged exposed coordinates is a property of the presented problem and its admissible structural closure.

□

Lemma III.28.3 (Saturated cost is invariant under admissible presentation equivalence)**.** *Let two task-family presentations $\mathcal{T}$ and $\mathcal{T}'$ be admissibly equivalent in the sense of Part I, and assume the equivalence transports the Part II dynamic constructors: budget-admissible kernels, represented quotients, valid lifts, boundary readouts, the Geom/Caus/Abs/Lift/Bdry projections. Let $a(d)$ and $b(d)$ be log-cost distortions for saturated public representation cost. Then*

$$I_{\mathcal{T}}(d) \le I_{\mathcal{T}'}(b(d)), \qquad I_{\mathcal{T}'}(d) \le I_{\mathcal{T}}(a(d)).$$

Consequently the threshold degrees are comparable up to the same distortion:

$$d^*_{\mathcal{T}',\eta} \le b(d^*_{\mathcal{T},\eta}), \qquad d^*_{\mathcal{T},\eta} \le a(d^*_{\mathcal{T}',\eta}).$$

If a and b preserve budget admissibility, then largeness of d^* beyond every within-budget band is invariant under the presentation equivalence.

Proof. Take any structural visibility term counted by $I_{\mathcal{T}}(d)$, with d in log-cost notation. If it is a Geom/Caus term, the presentation equivalence transports the pointwise observable, comparability relation, potential, current, mixing and consistency gates, Caus authority envelope and current provenance, Caus alignment, quotient stability record, scalar boundary coloring/readout, and structural reach datum to $\mathcal{T}'$ with log-cost at most $b(d)$. If it is an Abs term, the equivalence transports the represented fibers, terminal compatibility, lumpability condition, and charged quotient structural data to a represented quotient of log-cost at most $b(d)$. Valid Lift and Bdry terms are transported because their target-fraction and terminal-readout conditions are part of the transported dynamic constructor data. The unitary transport in Definition I.12.1 preserves the visibility diagnostic exactly. Taking the supremum gives $I_{\mathcal{T}}(d) \le I_{\mathcal{T}'}(b(d))$. The reverse inequality is identical with $a(d)$. The displayed threshold inequalities follow directly from the definition of d^*_η. Since budget admissibility is preserved by hypothesis, escaping every within-budget band is also preserved.

$\square$

Proposition III.28.4 (Closure under generated representations)**.** *The saturated class used in B^* is the cost-filtered closure of the presented observable algebra under the admissible oracle-free constructors fixed by the task object; its budget-filtered part is $\mathcal{R}_n^{\mathcal{B}}$. If a budget-admissible uniform rule generates a target-aligning quotient, potential, geometry, valid lift, boundary readout, or forward observable transport component, then the generated object belongs to this saturated class at the least saturated public cost of its representation. If that cost is within the declared budget and the relevant conditions hold, then $I_n^{\mathrm{sat}}(B)$ reaches the declared transfer-class margin at that budget. If B^* lies above the declared ceiling, up to the allowed transfer-class overheads, no such within-budget admissible representation exists in the budgeted saturated closure.*

Proof. The constructor grammar is fixed by the task presentation, the valid-lift conditions, the quotient conditions, and the resource envelope. A runtime-generated object produced by an oracle-free uniform rule has a finite lifted configuration trace, so Part II records it as lifted/generated observable data. If it supplies represented fibers, an exposed potential, a valid lifted interface, a scalar boundary value/readout, or a forward observable transport component after the component projection, it is one of the objects over which $m_n^{\mathrm{sat}}(B)$ or $G_n^{\mathrm{sat}}(B)$ takes its supremum, with B equal to the least saturated public representation cost. Thus the invariant ranges over the fixed admissible closure, including representations outside a raw Walsh cover. A lower bound for the cost of every target-aligned object in this closure is a structural representation-cost theorem over the admissible constructor class.

$\square$

Proposition III.28.5 (Saturated structural cost can be tested by budget)**.** *Fix n, a saturated public budget B, and the saturated admissible constructor grammar of the presentation. For every admissible quotient $Q \in \mathcal{Q}_{n,\le B}^{\mathrm{sat}}$, set the visibility diagnostic kernel $K_Q^{\mathrm{vis}} = A_Q P_Q$. For every admissible active-slot master certificate $\mathscr{C} = (\mathfrak{N}, q, \mathfrak{G}_q, \mathfrak{S}, \mathfrak{K}_q)$, where $\mathfrak{G}_q = (\Phi_Q, \Phi_\perp, D_Q, J_Q)$, set*

$$K_{\mathscr{C}}^{\mathrm{vis}} = \overline{A}_q S_q(\lambda) M_{\Phi_Q} C^{\mathrm{auth}}_{\Phi_Q,\mathfrak{K}_q}(D_Q) \frac{1}{1+\rho_{\Phi_Q}(D_Q)} \Pi_{U\mathcal{M}_Q(D_Q)}.$$

Then the structural coordinates are exactly the suprema of the corresponding normalized visibility forms:

$$m_n^{\mathrm{sat}}(B) = \sup_{Q \in \mathcal{Q}_{n,\le B}^{\mathrm{sat}}} \frac{\langle y, K_Q^{\mathrm{vis}} y\rangle}{\|y\|^2},$$

$$G_n^{\mathrm{sat}}(B) = \sup_{\mathscr{C}:\mathrm{Cost}_{\mathrm{sat}}(\mathscr{C}) \preceq B} \frac{\langle y, K_{\mathscr{C}}^{\mathrm{vis}} y\rangle}{\|y\|^2}.$$

Consequently $B^* > B$ is proved at size n by bounding the explicit visibility-kernel family on the target contrast for every budget-admissible representative at budget B. If the budget-admissible visibility-kernel family has a finite enumeration, a semialgebraic description, or another exact parametrization, this is an ordinary finite-dimensional linear-algebra or optimization problem.

Proof. For quotients, $\langle y, K_Q^{\mathrm{vis}} y\rangle/\|y\|^2$ is the normalized visibility diagnostic of the represented quotient subspace after applying the Abs utility condition. For Geom/Caus, $\langle y, K_{\mathscr{C}}^{\mathrm{vis}} y\rangle/\|y\|^2$ is the normalized target-oriented certificate visibility after the quotient support, stability, mixing, authority-compatible Caus alignment, consistency, and regularity conditions. These are precisely the terms used in the definitions of $m_n^{\mathrm{sat}}(B)$ and $G_n^{\mathrm{sat}}(B)$. Taking suprema over the saturated constructor class gives the displayed identities.

$\square$

Raw Walsh spectra and algebraic-immunity diagnostics can rule out target visibility in a declared raw basis or in a covered family of public bases. A bound for the full set of executable extraction procedures requires the tested kernel family to cover the saturated structural closure. A large-B^* theorem for the present visibility invariant bounds the visibility diagnostics above, budget by budget, over all budget-admissible saturated quotients, valid lifts, nonlinear represented kernels, and Geom/Caus data. For declared fragments, source-to-witness implications give additional consequences. The hardness direction uses the accountability implication from target arrival to counted visibility.

III.28.1 Universal Derivability Accountability

The next two statements give the unconditional lower-bound direction: successful admissible computation forces visibility in the uniform saturated profile. They follow from the budgeted derivability closure and the clocked-transcript first-hit quotient.

Proposition III.28.6 (Forced derivable witness)**.** *Let A_n be a presentation-relative uniform oracle-free algorithm running inside a budget-compatible resource envelope. If its target arrival probability is $\alpha(n)$, let $h(n)$ be its normalized horizon, including the fresh start phase, and let $B_{\mathrm{hit}}(n)$ be the explicit first-hit ceiling from* (140)*. Then*

$$I_n^{\mathrm{sat}}(B_{\mathrm{hit}}(n)) \geq m_n^{\mathrm{sat}}(B_{\mathrm{hit}}(n)) \geq \frac{\alpha(n)}{eh(n)}.$$

In particular, successful budget-admissible arrival produces a charged exact first-hit quotient in the derivability closure with no additional numerical transfer factor. If α has a numerical transfer-class lower margin and $h(n) \leq \gamma(n)$ for some $\gamma \in \mathfrak{C}_{\mathrm{num}}$, the displayed exact inequality gives the corresponding numerical transfer-class margin. A finite but otherwise unrestricted declared time capacity does not imply this last closure property. The polynomial wording is recovered by setting $\mathcal{B} = \mathcal{B}_{\mathrm{poly}}$, where the horizon is itself polynomially bounded.

Proof. Embed the run of A_n on its clocked Part II transcript carrier. The finite control, work tapes, and memory remain in the current native state and are charged at peak capacity; the carrier records the clock and finite outcome/terminal transcript. The one-step transition is a legal represented kernel by Encoding Adequacy. Let $h(n)$ be the charged stopping horizon supplied by the budget-compatible resource envelope after start normalization. Lemma III.20.2 constructs an exact represented first-hit quotient q_A whose saturated cost is at most B_{hit} and whose visibility satisfies

$$V_{q_A} \geq \frac{\alpha(n)}{eh(n)}.$$

Since $V_{q_A} \leq m_n^{\mathrm{sat}}(B_{\mathrm{hit}}) \leq I_n^{\mathrm{sat}}(B_{\mathrm{hit}})$, the claimed exact lower bound follows. For the optional transfer-margin conclusion, division by $h \leq \gamma$ preserves the numerical transfer class by its declared multiplicative closure. B4 alone is not used for that conclusion.

$\square$

Lemma III.28.7 (Full derivability profile criterion)**.** *Fix a task family and one budget envelope of finite time capacity. At its declared ceiling B,*

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le eH_B(n) m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)).$$

Therefore a saturated-profile bound at the named enlarged ceiling controls the complete success profile of the declared operational class. Under $\mathcal{B}_{\mathrm{poly}}$, set $B = b_{\mathrm{poly}}$. The concrete implication is

$$m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(b_{\mathrm{poly}}, n)) = \mathrm{negl}(n) \quad \Longrightarrow \quad \mathrm{Succ}_{\mathfrak{M},n}(b_{\mathrm{poly}}) = \mathrm{negl}(n),$$

with the polynomial horizon factor already included in the declared numerical transfer system.

The converse requires an additional executable source-to-arrival theorem. A visible quotient or geometry need not by itself supply a selector, sampler, descent rule, or decoder.

Proof. This is Theorem III.24.3. Its proof uses the uniform exact first-hit quotient and therefore applies to the complete positive process without a residual-baseline hypothesis. The final sentence records the logical direction of the result: the proof constructs visibility from an algorithm, not an algorithm from arbitrary visible structure.

□

Remark III.28.8 (Scope of the characterization)*.* The theorem reduces the declared-class success profile to the full uniform saturated profile at one charged enlarged ceiling. A large B^* supplies a sufficient lower-bound certificate. Without a source-to-arrival theorem, a small B^* records visible structure but does not assert the existence of a solver.

III.28.2 Alignment Complexity and the Saturated Budget

Alignment complexity records the least budget supporting the required target-oriented comparison.

Definition III.28.9 (Budgeted alignment complexity)**.** Fix the same admissible scalarization κ used for B^*. Let $\mathrm{KT}(\mathscr{R} \mid \mathrm{enc}(I_n))$ be the least scalarized charge of one family-uniform program that, from $\mathrm{enc}(I)$, constructs a *complete* exact-quotient or master quotient–geometry certificate $\mathscr{R}_I$ for every instance in the family. The record includes its carrier, reference law, initialization, generator, quotient fibers, terminal sectors, utility, geometry, authority/current provenance, defect and stability data, normalization, and every comparison used by its visibility. The charge includes the program, running resources, and every generated instance-dependent datum; a cheap terminal evaluator with an expensive carrier is therefore not cheap alignment. For a threshold $\zeta > 0$, define

$$\begin{aligned}\kappa_{\mathrm{align},\zeta}(n) = \min\bigl\{\mathrm{KT}(\mathscr{R} \mid \mathrm{enc}(I_n)) ::; \mathscr{R} &\text{ is a complete exact-quotient or master certificate} \\ &\text{with visibility at least } \zeta(n)\bigr\}.\end{aligned}$$

If no such record exists, set $\kappa_{\mathrm{align},\zeta}(n) = \infty$. Write κ_{align} only when the threshold is displayed or fixed in context.

Lemma III.28.10 (Saturated budget and alignment complexity are class-equivalent)**.** *There are monotone scalar overhead maps Π_1, Π_2, determined by the constructor simulation overheads of the presentation and the transfer class, such that*

$$\kappa_{\mathrm{align},\eta/2}(n) \le \Pi_1(B^*_\eta(n)), \qquad B^*_\eta(n) \le \Pi_2(\kappa_{\mathrm{align},\eta}(n)).$$

Thus $B^*(n)$ lies above the declared budget if and only if the corresponding alignment complexity does, up to the transfer-class overheads and the class-equivalent constant change from η to $\eta/2$. For $\mathcal{B}_{\mathrm{poly}}$, this is the usual polynomial/superpolynomial equivalence. The first inequality is read with an attaining witness when the coding is locally finite as in Lemma III.5.7; without attainment it holds after replacing $B^*(n)$ by any class-equivalent budget $B_+(n) > B^*(n)$ for which $I_n^{\mathrm{sat}}(B_+(n)) \geq \eta_{\mathfrak{C}}(n)$.

Proof. Under local finiteness, if $B_\eta^*(n) = B$, choose witnesses for $I_n^{\mathrm{sat}}(B) = m_n^{\mathrm{sat}}(B) + G_n^{\mathrm{sat}}(B) \geq \eta$. At least one of the two complete records has visibility at least $\eta/2$; choose it and put $\widetilde{B} = B$. Without local finiteness, fix $B_+ > B^*$ satisfying the condition in the statement, choose the witness at B_+, and put $\widetilde{B} = B_+$. Its derivation is a finite sequence of primitive reads, admissible constructors, and legal computations of total cost at most $\widetilde{B}$. Budget completeness simulates each step with class-preserving overhead, so the resulting complete target-aligned record has conditional charged complexity at most $\Pi_1(\widetilde{B})$. This is $\Pi_1(B^*)$ in the attained case and the stated B_+-replacement otherwise.

Conversely, if $\kappa_{\mathrm{align},\eta}(n) = K$, fix a charged program of scalarized total cost at most K constructing the complete certificate from $\mathrm{enc}(I_n)$. By Encoding Adequacy and the computation-closure clause of Definition Budgeted derivability closure, this program is a budget-admissible represented record of scalarized cost at most $\Pi_2(K)$. Its visibility is at least η, so $B_\eta^*(n) \leq \Pi_2(K)$. □

III.28.3 Reversible Spectrally Compressed Fragment

The reversible fragment isolates the represented spectral subfamily used by the compressed audit.

Definition III.28.11 (Reversible spectrally compressed fragment)**.** Fix a declared monotone numerical spectral-capacity map $\mathsf{s} : C \to [0, \infty]$, subject to the B4 compatibility condition of Definition I.2.4. Let $\mathsf{Rev}_{\mathrm{gate}}$ denote the class of budget-admissible killed generators L such that $K_N = -L_N$ is self-adjoint and positive semidefinite in $L^2(\mu_N)$, has spectrum in $[0, \Lambda]$ with $\Lambda \leq \mathsf{s}(b(n))$, is represented in the saturated closure, starts from $d\mu_0 = f_0\, d\mu_N$ with $\|f_0\|_2$ controlled by the declared transfer class, and whose induced reversible geometry and boundary readouts satisfy the Geom/Caus/Bdry conditions with transfer-class constants. For a depth bound $q(n)$, let $\mathcal{D}^{(q)}$ be the subclosure allowing at most $q(n)$ dependent applications of represented kernels or maps, with breadth and postprocessing allowed exactly when their total charge remains budget-admissible. In the polynomial instance this recovers the usual polynomial spectral, norm, breadth, and comparison bounds.

Lemma III.28.12 (Reversible spectral compression)**.** *Let $K_N \succeq 0$ be self-adjoint with spectrum in $[0, \Lambda]$. For every $\lambda > 0$ and $0 < \varepsilon < 1$, there is an explicit real polynomial p_q of degree*

$$q = O\Big((1 + \sqrt{\Lambda/\lambda})\, \log(2/\varepsilon)\Big)$$

such that $\sup_{s \in [0,\Lambda]} |(\lambda + s)^{-1} - p_q(s)| \leq \varepsilon/\lambda$.

Proof. Map $[0, \Lambda]$ affinely to $[-1, 1]$. The function $s \mapsto (\lambda + s)^{-1}$ is analytic on a Bernstein ellipse whose parameter $\rho > 1$ satisfies $\log \rho \geq c(1 + \sqrt{\Lambda/\lambda})^{-1}$ for an absolute constant $c > 0$. Truncating its Chebyshev expansion at degree q therefore gives uniform error

$$O\Bigg(\lambda^{-1} \exp\Bigg\{-\frac{cq}{1 + \sqrt{\Lambda/\lambda}}\Bigg\}\Bigg).$$

Choosing $q = C(1+\sqrt{\Lambda/\lambda})\log(2/\varepsilon)$ with C large proves the claim [74].

□

Lemma III.28.13 (Reversible compressed-fragment hardness)**.** *Fix $\lambda \geq 1/\beta_{\mathfrak{C}}(n)$, $\Lambda \leq \mathsf{s}(b(n))$, and a tolerance $\varepsilon_{\mathfrak{C}}(n)$ negligible relative to $\mathfrak{C}$. Let $q(n) = O((1+\sqrt{\Lambda/\lambda})\log(2/\varepsilon_{\mathfrak{C}}(n)))$, with total approximation charge lying within $\mathcal{B}$. Assume a charged spectral-to-visibility comparison: for every represented $h = p(K_N)k_T$ in this fragment there is a certificate $\mathscr{C}_h \in \mathcal{D}^{(q)}_{\leq b(n)}$ such that*

$$|\langle f_0, h\rangle| \leq C_{\mathrm{bridge}}(n)V_{\mathscr{C}_h} + \varepsilon_{\mathrm{bridge}}(n),$$

where C_{bridge} lies in the transfer class and $\varepsilon_{\mathrm{bridge}}$ is negligible. If $I^{(q)}_n(b(n)) = \mathrm{negl}_{\mathfrak{C}}(n)$, then every $L \in \mathsf{Rev}_{\mathrm{gate}}$ has negligible discounted target arrival at discount λ. For $\mathcal{B}_{\mathrm{poly}}$ and inverse-polynomial thresholds, this recovers the usual $O((1+\sqrt{\Lambda/\lambda})\log n)$ degree choice.

Proof. Assume, contrapositively, that $A_T(\lambda; L, \mu_0) = \langle f_0, (\lambda + K_N)^{-1}k_T\rangle$ has transfer-class size. Choose $\varepsilon_{\mathfrak{C}}$ negligible relative to $\mathfrak{C}$ after absorbing the transfer-class losses in $\|f_0\|_2$, $\|k_T\|_2$, and λ^{-1}. By Lemma Reversible spectral compression, the observable

$$h = p_q(K_N)k_T$$

is within negligible pairing error of the exact resolvent value. Since K_N, k_T, and the coefficients of p_q are represented, $h \in \mathcal{D}^{(q)}_{\leq b(n)}$. The defining conditions of $\mathsf{Rev}_{\mathrm{gate}}$ make this a represented reversible geometry and boundary readout. The approximation estimate makes $|\langle f_0, h\rangle|$ transfer-class non-negligible. The assumed bridge then gives

$$V_{\mathscr{C}_h} \geq \frac{|\langle f_0, h\rangle| - \varepsilon_{\mathrm{bridge}}(n)}{C_{\mathrm{bridge}}(n)},$$

which is transfer-class non-negligible. Hence $I^{(q)}_n(b(n)) \geq V_{\mathscr{C}_h}$ is non-negligible, contradicting the hypothesis. Without the displayed bridge, polynomial approximation of the resolvent supplies an observable pairing but does not by itself prove visibility or target reach.

□

Example III.28.14 (Saturated cost for SAT presentations)**.** XOR-SAT with public syndrome observables has polynomial B^*, hence logarithmic d^* in log-cost notation, because the public syndrome quotient is represented compactly even when raw Walsh degree is high. A random 3-SAT presentation without an exposed quotient and without a regular aimed defect geometry has large saturated cost in the random-target model. A planted SAT instance has low saturated cost when the presentation exposes a low-cost component that recovers the planted structure.

III.29 Nonlinear Quotients and the Kernel Normal Form

III.29.1 Quotient kernels and represented nonlinear features

A nonlinear projection used by a computation is accounted for by the same saturated invariant B^*. In a finite Boolean task space, every represented nonlinear feature map determines a finite feature subspace and hence a weighted projection operator. For Abs, the counted object is the quotient projection associated to represented fibers. A feature Gram matrix is a coordinate description of the same quotient after projection onto the represented feature span. If the nonlinear construction instead supplies a pointwise geometry, potential, metric, score, or scalar terminal/support coloring, it is counted in the Geom/Caus coordinate, while a Bdry readout remains a full-cost contextual extension over the represented structure. Its target charge is transferred to that structure only by a represented comparison. If it only modifies a declared geometry, the Dirichlet form comparison of Section III.11 is the perturbation bound that determines whether the geometric comparison transfers. If the feature map induces an exactly lumpable quotient, its projection has algebraic quotient provenance. It is counted in $m_n^{\mathrm{sat}}(B)$ at its least saturated public cost precisely when its complete record satisfies Definitions III.15.1 and III.15.2. A prospective source use also satisfies Definition III.21.1. A nonzero-defect quotient enters only through the supported Geom/Caus coordinate when its compensation record is dynamically qualified. This is the kernel-trick form of the invariant.

The nonlinear quotient kernel is the coordinate-free projection determined by an affordable represented feature partition.

Definition III.29.1 (Budget-admissible nonlinear quotient kernel)**.** Let $(X, \mathcal{F}, \mu, T)$ be a finite presented task. A budget-admissible nonlinear quotient representation at saturated public cost at most B is a finite feature map or projection rule U generated by the observable algebra, valid lifted coordinates, and resource-envelope operations at cost at most B. The cost includes the feature description, quotient labels, comparison data, charged public basis changes, lifted memory, retained preprocessing tables, and finite outcome/terminal transcript coordinates whenever those objects are used as observable structure. Nonuniform advice and oracle data are inadmissible coordinates unless they are charged to the presentation; otherwise they are excluded from the uniform oracle-free class. Observable finite postprocessings $h \circ U$, including coarsenings and threshold labels, are treated as feature maps at their own saturated cost. The equality fibers of the chosen U define a quotient

$$q_U : X \to Q_U.$$

It is admissible as an Abs quotient when its fibers are represented in the Part I fiber-represented-observable sense, the target and terminal sectors are quotient events, the quotient is exactly lumpable on the target window, and the quotient has nonzero quotient-utility factor on that window. Approximate lumpability is handled separately by Definition III.18.1: it contributes to the supported Geom/Caus profile only with a represented transfer-class stability factor, and every remaining defect is decomposed again by Part II. Let $\mathcal{H}_U \leq L^2(X, \mu)$ be the subspace of functions constant on the fibers of q_U, and let P_U be the weighted orthogonal projection onto $\mathcal{H}_U$. The associated quotient-visibility kernel is the positive self-adjoint idempotent

$$K_U = P_U, \qquad \mathrm{Vis}_T(U) = \frac{\langle y, K_U y \rangle_{L^2(\mu)}}{\|y\|^2_{L^2(\mu)}}.$$

This projection kernel supplies the quotient visibility. The extraction contribution also includes the quotient-utility factor A_U, which records the represented level-set control and quotient target utility. Different feature coordinates, Gram matrices, or nonlinear formulas with the same represented fibers give the same P_U. The degree of U is the least saturated representation cost of the induced quotient projection K_U, independently of discovery time or the proof of optimality, together with its charged utility data relative to the observable algebra.

Figure 40 records the structural routing of the nonlinear feature map in Definition III.29.1.

Proposition III.29.2 (Kernel-trick normal form)**.** *Every admissible nonlinear quotient representation U at cost at most d has quotient visibility contribution*

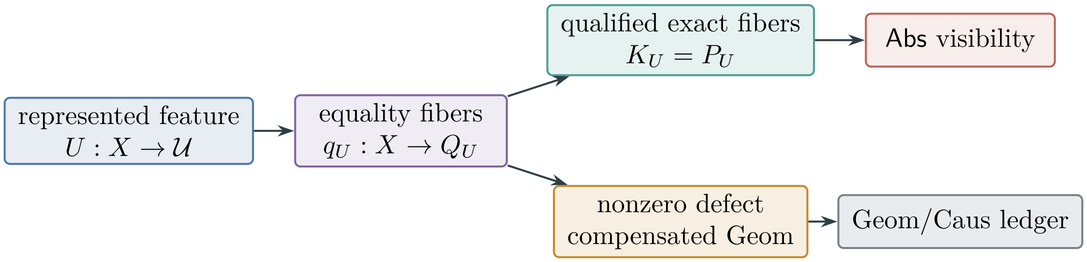

Figure 40: A nonlinear formula is classified through its represented structural effect. Exact target-compatible fibers give the quotient projection kernel of Proposition III.29.2; a nonzero lumpability defect remains in the compensated geometric ledger with its stability data.

$$V_U = A_U \frac{\langle y, K_U y\rangle}{\|y\|^2},$$

where $A_U \in [0,1]$ is the same Part I quotient-utility factor used in Definition III.15.2. Thus $m_n^{\mathrm{sat}}(B) \geq V_U$ when U is budget-admissible. The visibility form depends only on the induced quotient subspace and its utility data, independently of the coordinates, feature basis, Gram representation, discovery procedure, or algorithmic construction of U.

Proof. Since X is finite, the feature map U has finite image and its equality fibers form a finite partition of X. The quotient-observable space is the closed finite-dimensional subspace of $L^2(X,\mu)$ consisting of functions constant on those fibers. Weighted orthogonal projection onto that subspace is unique; different feature coordinates for the same fibers produce the same projection P_U. A non-idempotent feature kernel contributes to Abs only through this projection onto the represented quotient subspace, multiplied by the Part I quotient-utility factor; otherwise its target-relevant part is classified by the Geom/Caus/Bdry/residual alternatives. If the quotient-utility conditions hold, q_U is one of the quotients in $\mathcal{Q}^{\mathrm{sat}}_{\leq B}$. The definition of $m_n^{\mathrm{sat}}(B)$ is the supremum of the conditioned quotient visibility over these represented target-relevant quotient subspaces, hence the displayed inequality. □

Lemma III.29.3 (Nonlinear kernel test is algorithm-independent)**.** *Fix the presented observable task family, the budget-compatible resource envelope, the admissibility rules of Part II, and the structural visibility threshold $\eta(n)$. The invariant B^* is a property of this observable task object. If the generator induced by any presentation-relative uniform oracle-free budget-admissible algorithm contains a target-relevant nonlinear projection that induces a target-relevant* Abs *quotient at saturated public cost at most B, then $m_n^{\mathrm{sat}}(B)$ counts the induced quotient visibility kernel K_U through its conditioned projection mass. If the same nonlinear construction instead induces a pointwise potential, defect, score, metric, conductance, or geometry, then $G_n^{\mathrm{sat}}(B)$ counts its conditioned Geom/Caus visibility exactly at budgets in its admissible budget set. If it is only approximately lumpable, it contributes through $G^{\mathrm{sat}}_{Q,\mathrm{ql},n}(B)$ precisely when the represented data supply the compensated certificate of Section III.11. If it only approximately matches an existing geometry, Section III.11 supplies the Dirichlet form comparison; without a transfer-class comparison loss, the construction must be counted as its own geometry at budgets in its admissible budget set. If it fails the valid quotient, lift, or boundary conditions, then its exact admissible part is decomposed again by Part II, while any target information outside the observable/lifted algebra is* *presentation-inaccessible or oracle-dependent, and any remaining nonlocal transport belongs to the residual term J_{res}.*

Consequently, for every budget B,

$$B^*_\eta(n) > B \iff I^{\mathrm{sat}}_n(B) = m^{\mathrm{sat}}_n(B) + G^{\mathrm{sat}}_n(B) < \eta(n)$$

for the chosen structural visibility threshold. In kernel language this says that the quotient visibility kernels K^{vis}_Q and the supported Geom/Caus visibility kernels $K^{\mathrm{vis}}_{\mathscr{C}}$ from Proposition III.28.5 have total conditioned target visibility below $\eta(n)$. The kernel notation is the normal form of the already-defined accounting $I^{\mathrm{sat}}_n(B) = m^{\mathrm{sat}}_n(B) + G^{\mathrm{sat}}_n(B)$. This identity concerns the represented structural profile. For a budget-B algorithm, full-process target arrival is related to that profile by the exact first-hit quotient at the enlarged budget $\Psi(B, n)$, as in Theorem III.24.3.

Proof. Part II embeds every presentation-relative uniform oracle-free budget-admissible algorithm as a finite Markov generator on each bounded clocked transcript carrier. Component exhaustiveness then places every target-relevant part of that generator in Geom, Caus, Abs, valid Lift, Bdry, or J_{res}. Valid Lift and Bdry retain their full represented carrier cost and contextual charge. A target-charge transfer to an intrinsic Geom/Caus or Abs source requires a represented comparison with its declared loss and residual term. By the kernel-trick normal form, every nonlinear quotient source is represented by a unique quotient projection kernel and is counted in $m^{\mathrm{sat}}_n(B)$ exactly when $B \in \mathfrak{B}_{\mathrm{sat},n}(R_Q)$. Pointwise geometric or causal sources are counted in $G^{\mathrm{sat}}_n(B)$. Approximate quotients enter only after the Section III.11 quotient defect estimate is applied; the controlled quotient part is counted at budgets in its admissible budget set and the defect is decomposed by the same Part II channels. Nonquotient nonlinear geometric modifications enter only through the Section III.11 form comparison: a small relative form error transfers an existing geometric comparison proof with the displayed loss, while a large form error yields no uncharged improvement and must be represented as its own Geom/Caus datum or reclassified as residual or external information. A family-uniform residual target correlation whose budget-admissible representative satisfies one of the five structural-channel predicates cannot remain residual by Part II's no-uniform-residual-correlation theorem. Mere measurability or correlation is not enough. The residual class is defined after every represented structural-channel projection in the current saturated closure has been removed. At budget B, the reclassified contribution is counted precisely when $B \in \mathfrak{B}_{\mathrm{sat},n}(R)$; otherwise it is absent from that budget slice. If the representative is generated by an admissible within-budget finite-horizon computation, Lemma Budgeted generation gives budgeted representation puts that cost in the declared budget account. If no budgeted representative exists, a transition rule using the datum is not presentation-relative budget-admissible for the original task; it uses data outside the budgeted observable algebra, including presentation-inaccessible or oracle-dependent data, nonuniform advice, over-budget recovery, or an enlarged primitive signature. The induced operator's exact first-hitting partition, committor, occupation law, or stopped-future map is represented only by a budget-admissible saturated representative; a finite-horizon version actually generated by an admissible within-budget computation is counted. Thus $m^{\mathrm{sat}}_n(B) + G^{\mathrm{sat}}_n(B)$ exhausts the represented quotient and Geom/Caus structural denotations at budget B. This is not a componentwise bound on the effect of J_{res}; the complete target-arrival probability is charged through the first-hit quotient at $\Psi(B, n)$. The saturated classes are nested in B, hence $I^{\mathrm{sat}}_n(B)$ is nondecreasing. Since $B^*_\eta(n)$ is the least budget with $I^{\mathrm{sat}}_n(B) \geq \eta_{\mathfrak{C}}(n)$, the displayed equivalence follows. Under an admissible presentation equivalence, the Lemma III.28.3 transports the same statement with the stated cost and normalization distortions.

□

The saturated kernel test is therefore intrinsic to the fixed presentation and resource envelope: algorithms matter only through the budget-admissible projections, kernels, scalar readouts, and value functions their induced generators instantiate over represented structure. Comparison data identify the objective component term in the existing accounting and need not be produced by an algorithm or adversary. Raw Walsh spectra, nonlinearity, and algebraic-immunity diagnostics can support this test only when they bound the full saturated accounting $I_n^{\mathrm{sat}}(B) = m_n^{\mathrm{sat}}(B) + G_n^{\mathrm{sat}}(B)$, including nonlinear represented quotient projections.

III.29.2 Universal nonlinear accounting

The universal accounting record collects the quotient, geometric, and residual treatment of a represented nonlinear construction.

Definition III.29.4 (Universal observable structural accounting)**.** Fix a task family, a budget-compatible resource envelope, a saturated public budget family $\mathcal{B}_n$, and the admissible presentation-relative oracle-free operator class obtained from the Part II representation theorem. For every induced generator L_n, the task-level structural accounting is the following object-level decomposition.

First apply the Part II component decomposition on the induced finite carrier as an operator identity. For a represented computation this is its clocked transcript carrier. Discounted hitting quantities are evaluated on the full positive killed propagation, with the exposed terminal edge gate applied as in Definition Full-history target-contact transport. Represented quotient components contribute to $m_n^{\mathrm{sat}}(B)$ when B belongs to their admissible budget sets; pointwise geometric or causal components contribute to $G_n^{\mathrm{sat}}(B)$ under the same membership criterion. Valid Lift and Bdry are retained as full-cost contextual extensions. Bdry contributes no off-diagonal movement, but neither its charge nor a Lift charge is reassigned to a represented Geom/Caus or Abs source without a represented contextual transfer inequality. A committor, first-hitting partition, occupation law, or stopped-future value of L_n contributes when the same denotation has a budget-admissible representative in $\mathcal{R}_{n,\leq B}^{\mathrm{sat}}$, including a representative generated by an admissible within-budget finite-horizon computation. A semantic object contributes only through such a representation. The orthogonal remainder J_{res} receives no standalone hitting probability unless it is separately verified to be a positive kernel. Target arrival of the complete process is retained retrospectively in its exact represented first-hit accounting quotient. Prospective structural credit is assigned only under Definition III.21.1, and non-enumerative target arrival is charged by Theorem III.21.3.

Proposition III.29.5 (Structural accounting after operational embedding)**.** *The operational embedding maps the declared presentation-relative uniform oracle-free budget class into the clocked-transcript generator class. The accounting above acts on this image and therefore requires no separately supplied quotient, comparison datum, or invariant. A represented program enters through its induced generator L_A; once L_A is formed, the same admissible component projections, full-process first-hit accounting, and saturated representation costs apply. A rule that becomes budget-admissible only after adding unpresented data is not in this operator class for the original task.*

Proof. Part II embeds every budget-bounded oracle-free algorithm whose instance-dependent data are generated from $\mathcal{R}_n^{\mathcal{B}}$ and its legal lifted closure as a finite Markov generator on each bounded clocked transcript carrier. The generator form, fixed-space split, Hodge projection, quotient projection, valid-lift projection, and boundary split are finite algebraic operations on that generator and on the fixed observable grammar. Derivability accounting decides which semantic future values have saturated representatives. Every generator in the class and every finite-horizon object generated by an admissible within-budget computation remains included. Therefore two algorithms inducing the same generator receive the same accounting, and a semantic value function that lacks a representation in the budgeted observable grammar has no low-cost representation associated with a named algorithm alone.

□

Lemma III.29.6 (Accounting-profile bound at the declared ceiling)**.** *At the declared ceiling B, the operational image satisfies*

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \leq eH_B(n)m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)).$$

Hence a negligible right-hand profile at that named enlargement gives a negligible target-contact profile after the declared numerical transfer. For $\mathcal{B}_{\mathrm{poly}}$, this is the single substitution $B = b_{\mathrm{poly}}$, with enlarged ceiling $\Psi(b_{\mathrm{poly}}, n)$.

Proof. Apply (180). The exact first-hit quotient supplies the required exact-quotient accounting visibility term for the complete computation. This implication assumes negligibility of the accounting profile itself; source-profile hardness is instead the prospective criterion of Corollary III.24.4.

□

Remark III.29.7 (Role of the fragment criterion)*.* The visibility invariant measures represented target visibility. The first-hit construction supplies target-contact accountability for every uniform finite-horizon computation. Fragment theorems may give sharper constants or separate Geom/Caus estimates, but the universal lower-bound direction no longer depends on a fragment-specific residual argument.

III.30 Charged Noisy Combination and Semantic Cancellation

The saturated ledger distinguishes cancellation in the semantic label of a derived observable from the noisy public primitives used to construct it. The distinction is intrinsic to the represented derivation and therefore applies independently of a particular elimination schedule.

Definition III.30.1 (Charged binary noisy-combination record)**.** Let G be a finite binary vector space with uniform latent state $\vartheta \in G$. A binary noisy-character presentation consists of public records

$$Y_i = \chi_{\gamma_i}(\vartheta)\xi_i, \qquad 1 \leq i \leq q,$$

where $\gamma_i \in \widehat{G}$, the signs ξ_i are independent of ϑ, and

$$\mathbb{E}\xi_i = \rho_i \in [-1, 1].$$

A *charged binary noisy-combination record* contains a represented matrix

$$\Lambda = (\lambda_1^\mathsf{T}, \ldots, \lambda_K^\mathsf{T}) \in \mathbb{F}_2^{K\times q}$$

and the derived characters

$$Y_{\lambda_j} := \prod_{i:(\lambda_j)_i=1} Y_i.$$

For a nonzero row λ, define its semantic label, provenance support, and attenuation by

$$\Gamma(\lambda) := \sum_{i=1}^{q} \lambda_i \gamma_i, \qquad \operatorname{Prov}(\lambda) := \{i : \lambda_i = 1\},$$
$$w(\lambda) := |\operatorname{Prov}(\lambda)|, \qquad a(\lambda) := \prod_{i \in \operatorname{Prov}(\lambda)} \rho_i. \tag{214}$$

For the whole materialized record put

$$d_i(\Lambda) := |\{j : (\lambda_j)_i = 1\}|, \qquad D_2(\Lambda) := \sum_{i=1}^{q} d_i(\Lambda)^2. \tag{215}$$

For coefficients $c = (c_1, \ldots, c_K)$, also put

$$D_c(\Lambda) := \sum_{i=1}^{q} \left(\sum_{j:(\lambda_j)_i = 1} |c_j| \right)^2. \tag{216}$$

The complete record contains the primitive acquisition interface, the rule constructing Λ, every row locator, the shared combination DAG, the materialized outputs, retained storage, the comparison or decoder, and the prospective transition using the result. Its cost is the resource sum of those represented fields. A shared vertex is charged once in the DAG; every output, stored reference, and evaluation using that vertex remains charged.

Semantic cancellation means that $\Gamma(\lambda)$ has smaller represented support in a declared basis of $\widehat{G}$. It does not alter $\operatorname{Prov}(\lambda)$, $w(\lambda)$, or the construction record.

Theorem III.30.2 (Exact transport and complete charging for noisy combinations)**.** *For the record of Definition III.30.1 and every $v \in \widehat{G}$,*

$$\mathbb{E}_{\vartheta,\xi}[Y_\lambda \chi_v(\vartheta)] = \mathbf{1}_{\{\Gamma(\lambda)=v\}} a(\lambda). \tag{217}$$

For two rows, write

$$a(\lambda_j + \lambda_l) = \prod_{i:(\lambda_j+\lambda_l)_i = 1} \rho_i.$$

Their exact conditional covariance over the primitive noise is

$$\operatorname{Cov}_\xi \left(Y_{\lambda_j}, Y_{\lambda_l} \mid \vartheta \right) = \chi_{\Gamma(\lambda_j+\lambda_l)}(\vartheta) \left(a(\lambda_j + \lambda_l) - a(\lambda_j) a(\lambda_l) \right). \tag{218}$$

Consequently, for a materialized located block

$$H = \sum_{j=1}^{K} c_j Y_{\lambda_j}, \qquad \sum_{j=1}^{K} |c_j|^2 \le 1,$$

one has

$$\begin{aligned} |\mathbb{E}[H \chi_v(\vartheta)]| &\le \left(\sum_{j:\Gamma(\lambda_j)=v} |a(\lambda_j)|^2 \right)^{1/2} \\ &\le \sqrt{L_v} \max_{j:\Gamma(\lambda_j)=v} |a(\lambda_j)|, \end{aligned} \tag{219}$$

where $L_v = |\{j : \Gamma(\lambda_j) = v\}|$. If $|\rho_i| \le \rho < 1$ and every aligned row has provenance weight at least q_0, then

$$|\mathbb{E}[H\chi_v(\vartheta)]| \le \sqrt{L_v}\,\rho^{q_0}. \tag{220}$$

Moreover, conditional on ϑ, for every $t > 0$,

$$\Pr_{\xi}\left\{\sum_{j=1}^{K} c_j\big(Y_{\lambda_j} - \mathbb{E}_\xi[Y_{\lambda_j} \mid \vartheta]\big) \ge t \;\middle|\; \vartheta\right\} \le \exp\left[-\frac{t^2}{2D_c(\Lambda)}\right]. \tag{221}$$

Every fan-in-two combination DAG producing one row of provenance weight w contains at least $w-1$ binary combination vertices in the ancestral sub-DAG of that output. For several outputs, the complete construction cost dominates the number of distinct vertices in the union of their ancestral sub-DAGs. Thus sharing is charged at its actual represented cost and does not erase provenance weight or attenuation.

Finally, any represented postprocessing of the complete materialized tuple is measurable in the tuple algebra and obeys the target-projection monotonicity of Lemma I.4.7. A prospective use of that postprocessing retains the complete combination record. Its structural assignment is the exact or compensated quotient-supported mode when it has qualified nonidentity fibers, ambient Geom/Caus *when it has identity support and qualified transport, or terminal accounting when it supplies no authorized intermediate transport.*

Proof. Expanding the product gives

$$Y_\lambda = \chi_{\Gamma(\lambda)}(\vartheta) \prod_{i:\lambda_i=1} \xi_i.$$

Orthogonality of characters under the uniform law on G, followed by independence of the ξ_i, proves Eq. (217). Multiplying two derived characters replaces their provenance rows by $\lambda_j + \lambda_l$; subtracting the product of their conditional means proves Eq. (218). Applying Cauchy–Schwarz to the aligned summands proves Eq. (219); the uniform reliability and weight hypotheses give Eq. (220).

Changing the primitive sign ξ_i changes the centered weighted sum by at most

$$2\sum_{j:(\lambda_j)_i=1} |c_j|.$$

The bounded-difference inequality with the squared sensitivities summed over i gives Eq. (221).

A fan-in-two DAG with g combination vertices can join at most $g+1$ distinct primitive leaves into one output. Hence an output depending on w distinct leaves requires $g \ge w-1$. Taking the union of ancestral sub-DAGs counts a shared vertex once and proves the multi-output statement. The cost assertion is then Definition III.1.3 and Lemmas IV.0.2 and III.1.6. Projection monotonicity is Lemma I.4.7; the exhaustive support and temporal assignments are Theorem III.18.9, Definition III.21.1, and Corollary III.3.7. □

Definition III.30.3 (Saturated located-combination profile)**.** At a charged ceiling B, define

$$\mathrm{Comb}^{\mathrm{loc,sat}}_{\mathfrak{M},n}(B) := \sup_R \mathfrak{M}_n\left(I \longmapsto \sup_{v\ne 0} |\mathbb{E}_I[H_R\chi_v(\vartheta)]|\right), \tag{222}$$

where R ranges over family-uniform, temporally admissible records of Definition III.30.1 whose complete cost is at most B, and H_R is a normalized materialized located block in that record. The supremum is zero when the class is empty.

This is a source-resolved subprofile of the existing saturated profile. It does not add a source modality. Quotient, Geom/Caus, Lift, learning, and terminal gates remain those of the complete record.

Corollary III.30.4 (Capacity–attenuation closure at one charged ceiling)**.** *Suppose every record contributing to* $\mathrm{Comb}^{\mathrm{loc,sat}}_{\mathfrak{M},n}(B_n)$ *materializes at most* L_n *target-aligned rows, each of attenuation at most* a_n*, outside a presentation event* $\mathcal{E}_n$ *satisfying* $\mathfrak{M}_n(\mathbf{1}_{\mathcal{E}_n}) \le \varepsilon_n$*. Then*

$$\mathrm{Comb}^{\mathrm{loc,sat}}_{\mathfrak{M},n}(B_n) \le \min\{1, \sqrt{L_n} a_n + \sqrt{L_n}\varepsilon_n\}. \tag{223}$$

In particular, polynomial L_n *and exponentially decreasing* a_n, ε_n *give a negligible saturated located-combination profile.*

Proof. On the complement of the exceptional event apply Eq. (219); on the exceptional event use the same inequality with unit attenuation. Apply the monotone sublinear family functional and the union bound. Truncation at one gives the displayed formula. □

Corollary III.30.5 (Prospective accountability of a noisy-combination use)**.** *Let a represented pre-hit selector use a complete record from Definition III.30.1 at ceiling* B*. Its positive step advantage obeys the existing saturated structural-charge inequality*

$$(\mathrm{Adv}_t(K_t; \nu_t))_+ \le e H_B(n) \sum_{c \in \mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_B(n)). \tag{224}$$

The complete combination record is an ancestor of every nonzero term used by the selector. Its materialized located-character contribution is bounded by Eq. (219)*; any further contribution is the already-defined exact-*Abs*, compensated, ambient-*Geom/Caus*, Lift, or terminal coordinate selected by the complete record's qualification gates. Thus combination, repeated cancellation, and reordering introduce no additional source coordinate.*

Proof. Apply Theorems III.21.4 and III.21.5 to the selector's complete derivation. Complete ancestry and the located bound are Lemma IV.0.2 and Theorem III.30.2. The structural assignment is Theorem III.18.9. □

III.31 The Trace-Support Admissibility Dichotomy

Part I separated extensional observability from affordable represented derivability. The present section applies that distinction to operators and records a residual baseline diagnostic for components that are separately verified to be nonnegative.

Trace-support admissibility distinguishes represented structural support from support visible only in an induced trajectory.

Definition III.31.1 (Trace-support admissibility)**.** For a finite operator L, the off-diagonal trace support is

$$\mathrm{supp}_{\mathrm{tr}}(L) = \{(x, z) : x \neq z,\ L_{xz} \neq 0\}.$$

The operator is trace-support admissible at budget B when its support relation and every quotient cell, potential level set, conductance, or boundary set used to define its nonzero rates have uniform represented derivations of total saturated cost at most B. Such relations are necessarily measurable in the ambient envelope $\mathcal{F}_I \otimes \mathcal{F}_I$, but measurability alone is not an admissibility certificate.

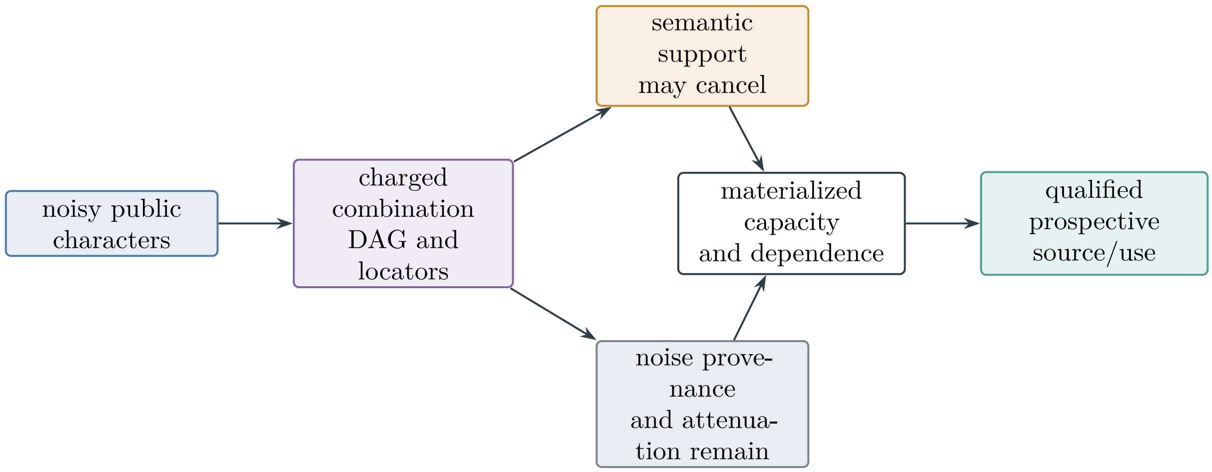

Figure 41: Charged noisy combination separates semantic cancellation from noise provenance. Cancellation may simplify the derived character, while the primitive leaves, attenuation, shared DAG, located outputs, dependence, and prospective use remain in one saturated record.

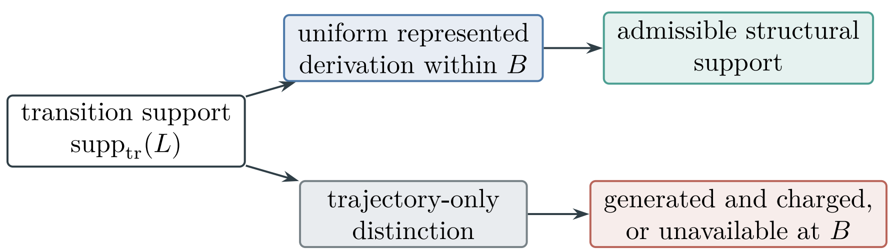

Figure 42: Trace support is structural at budget B precisely when its distinctions have the represented derivation required by Definition III.31.1. Measurability or occurrence in a trajectory alone supplies no admissibility certificate.

Figure 42 distinguishes a represented support relation from a relation visible only after an induced trajectory is known.

An admissible operator uses only distinctions generated affordably from the presentation. A jump rule whose cells have no uniform represented derivation within the declared budget defines no budget-admissible structural component for the presented problem.

Non-admissibility means that the relation is unavailable as structural data at the declared budget. An ambient transition table may still contain such a relation. If a presentation-relative budget-admissible oracle-free computation can generate and use the relation from the presented instance, then the generated relation appears in the lifted observable configuration and Part II classifies its movement into the exposed components. If it cannot be so generated, then any transition table that still points along it has no exposed provenance; after the exposed projections it is J_{res} rather than a structural component. This operator classification alone gives no probabilistic baseline bound: the orthogonal remainder need not be nonnegative, and its interaction with the other components may affect the full process. Quantitative target arrival is evaluated on the full positive generator and charged through Theorem III.24.3.

The residual baseline records a target-coupling bound only when the residual operator is separately positive.

Definition III.31.2 (Residual baseline target coupling)**.** Let J be a residual component that has separately been verified to be a nonnegative transition kernel or rate table from $N = X \setminus T$ to X. With reference law $\mu_N = \mu(\cdot \mid N)$, its target hit rate is

$$\mathrm{Hit}_T(J) = \sum_{x \in N} \mu_N(x) \sum_{z \in T} J(x,z).$$

The baseline target coupling at the same total residual activity is

$$\mathrm{Base}_T(J) = \mu(T) \sum_{x \in N} \mu_N(x) \sum_{z \in X} J(x,z).$$

The residual excess is

$$\mathrm{Exc}_T(J) = \big(\mathrm{Hit}_T(J) - \mathrm{Base}_T(J)\big)_+.$$

These quantities are defined only under the stated positivity hypothesis. They do not apply automatically to the orthogonal component $L^{J_{\mathrm{res}}}$ of Theorem II.7.1, and their definition implies no bound on $\mathrm{Exc}_T(J)$.

The baseline is a comparison statistic for a separately executable kernel. For the full positive process, the pre-hit hazard identity of Theorem III.21.3 separates neutral target contact from represented prospective alignment. The completed first-hit quotient remains an exact audit of the same process.

Proposition III.31.3 (Residual interaction is charged by saturation)**.** *Let $L_{\mathcal{A}}$ be the full positive generator induced by a uniform presentation-relative algorithm, and decompose it into the five structural channels and $L_{\mathcal{A}}^{J_{\mathrm{res}}}$. For a declared horizon H, choose a declared pre-hit neutral selector ν_t. Every target-arrival probability α_I of the full process satisfies*

$$\alpha_I \le \mathrm{Enum}_\nu(\mathcal{A}, H) + H\,\mathrm{Sel}_n^{\mathrm{sat}}(B_{\mathcal{A}}), \tag{225}$$

where $B_{\mathcal{A}}$ is the complete finite-horizon derivation cost after the class-preserving clock refinement. The bound includes every effect of $L_{\mathcal{A}}^{J_{\mathrm{res}}}$, including its interaction with the other components. Independently, the retrospective audit satisfies

$$\alpha_I \le eH\, m_I^{\mathrm{sat}}(\mathrm{Cost}_{\mathrm{sat}}(q_{\mathcal{A}}))$$

by Lemma III.20.2. If the transition rule uses instance-dependent data with no uniform represented derivation, then it is not a presentation-relative algorithm in the declared resource class.

Proof. Apply the channel decomposition only as the exact operator identity of Theorem II.7.1; the transition implementing the pre-hit sampler belongs to the full positive update before any component expansion. The exact identity (152), followed by the definition of the prospective selector profile, gives (225). Applying Lemma III.20.2 after execution gives the separate accounting inequality. The derivability definition supplies the final provenance statement.

□

Corollary III.31.4 (No residual bypass of prospective accounting)**.** *For a uniform task family, including presentation-admissible oracle interfaces, suppose every budget-B computation has*

$\mathrm{Enum}_\nu(\mathcal{A}, H_B) \le \beta_n$*, and every temporally admissible selector at the enlarged cost has advantage at most* δ_n*. Then the full-process success probability, including every interaction with* J_{res}*, satisfies*

$$\Pr\{\tau \le H_B\} \le \beta_n + H_B(n)\delta_n.$$

Hence residual interaction supplies no bypass around the pre-hit selector ledger. Negligibility of the prospective selector profile and of the declared neutral baseline implies negligible full-process success. The orthogonal residual itself need not satisfy a componentwise probabilistic baseline.

Proof. Apply Corollary III.24.4 and Proposition III.31.3 to the full positive process.

□

This dichotomy separates gaps within the fixed presentation from gaps created by inadmissible extensions.

Definition III.31.5 (Admissible-gap and inadmissible-gap cases)**.** An instance is in the inadmissible-gap case at budget B when every operator with transfer-class verified discounted hitting strength uses at least one quotient cell, potential level set, conductance relation, or trace-support relation lacking a uniform represented derivation at that budget. It is in the admissible-gap case when some transfer-class verified hitting component is trace-support admissible at budget B.

Lemma III.31.6 (Inadmissible-gap saturated-profile criterion)**.** *For an instance family in the inadmissible-gap case, every uniform budget-*B *algorithm still satisfies*

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le eH_B(n)m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)).$$

Thus negligibility of the saturated profile at the enlarged budget implies negligibility of every budget-admissible discounted target arrival. Trace-support inadmissibility alone is a provenance statement; it does not imply a probabilistic baseline estimate for J_{res}*.*

Proof. Apply Theorem III.24.3. The first-hit quotient is generated from the complete computation, so the conclusion does not depend on whether its one-step transition initially lies in an exposed channel or in the orthogonal residual. The negligibility statement follows from (181).

□

Lemma III.31.7 (Admissible gaps are governed by saturated cost)**.** *In the admissible-gap case, structured movement is governed by the growth of* B^**. A within-budget admissible gapping quotient or regular aimed geometry puts* B^* *below the declared ceiling; if every budget-admissible gapping structure has saturated cost above* $b(n)$*, then* B^* *is large relative to the declared budget. In the polynomial instance this is the usual polynomial versus superpolynomial distinction.*

Proof. When a gapping operator is budget-admissible, all of the structure it uses has a represented derivation from the public or lifted observable presentation. Part II's exhaustiveness theorem places that structure in Geom/Caus, Abs, or Lift. Ambient measurability does not determine budget admissibility. The least saturated public budget at which the component contributes extraction at the declared threshold is exactly the definition of B^*. The hardness criterion of Section III.28 gives the conclusion.

□

Example III.31.8 (Withheld modulus)**.** In subset sum, a public residue map $x \mapsto \sum_i a_i x_i \pmod{q}$ can define an admissible quotient. If the modulus q is withheld and no uniform represented construction derives the residue cells within the declared budget, then the quotient that would gap the target is inadmissible at that budget, regardless of its extensional measurability. A rule that uses the withheld modulus leaves the budgeted observable algebra of the original presentation; it becomes admissible only after enlarging the primitive signature so that q is public, or through oracle or advice access relative to the original presentation. It is therefore not a budget-admissible oracle-free computation for the original task object.

III.32 Structural Negligibility Certificates for Budget-Admissible Generators

III.32.1 The negligibility certificate

The following regimes yield hardness results once the instance presentation and Part II budget-admissibility theorem are fixed. For a presentation-relative uniform computation, including one using presentation-admissible oracle interfaces, the clocked-transcript embedding produces a generator that is decomposed by Part II and evaluated by the full saturated accounting profile. Transition tables depending on data outside the budgeted saturated observable closure use presentation-inaccessible or oracle-dependent data, require over-budget recovery, or require changing the primitive signature so that the data are public; transition tables whose target contact has exposed provenance are counted through the corresponding budget-admissible structure. For a fixed cryptographic instance, the analysis reduces hardness to B^* lying above the declared budget; in the polynomial instance this is the usual superpolynomial cost statement. The required application theorem must bound the single master quotient–geometry profile over its complete resource-filtered derivation family. Part IV supplies the complementary quantitative learning analysis, and Part VIII is a separate specialization.

The structural negligibility certificate bounds the single full saturated master profile. Universal accountability then applies directly.

Definition III.32.1 (Structural negligibility certificate for a family)**.** Let $\{I_n\}$ be a presentation-complete task family. A structural negligibility certificate under the budget structure $\mathcal{B}$ is the family-specific bound

$$\mathfrak{U}_n^{\mathrm{sat}}(\Psi(b(n), n)) := \sup_{\mathscr{C} \in \mathcal{C}_{QG}(\Psi(b(n),n))} V_{\mathscr{C}} = \mathrm{negl}_{\mathfrak{C}}(n);$$

The supremum ranges over every family-uniform accounting master certificate generated by the resource-filtered derivation calculus. The bound may be proved by separate estimates on the five parameter regimes, but completeness is imposed on their union. For the exact Abs coordinate, the exhaustive five-family envelope supplies the exhaustive additive, multiplicative, order, symmetry, and filtration provenance cover, including mixed contextual charges. The component estimates and channel normal forms are family-specific applications of the general framework. Completed first-hit records are included in the accounting profile, while Corollary III.23.2 prevents their retrospective representation from being used as an independent source.

Structural negligibility yields presentation-relative hardness through universal accountability.

Theorem III.32.2 (Structural negligibility implies presentation-relative hardness)**.** *If $\{I_n\}$ has a structural negligibility certificate under $\mathcal{B}$, then*

$$\mathfrak{U}_n^{\mathrm{sat}}(\Psi(b(n),n)) = \mathrm{negl}_{\mathfrak{C}}(n), \qquad I_n^{\mathrm{sat}}(\Psi(b(n),n)) \le 2\mathfrak{U}_n^{\mathrm{sat}}(\Psi(b(n),n)) = \mathrm{negl}_{\mathfrak{C}}(n).$$

Moreover, every presentation-relative uniform generator lying within $\mathcal{B}$, including computations using presentation-admissible oracle interfaces, has negligible target arrival.

Proof. The master normal-form theorem places every dynamically qualified accounting Geom/Abs certificate at the enlarged budget in $\mathcal{C}_{QG}(\Psi(b(n),n))$. The assumed supremum therefore bounds all five parameter regimes simultaneously. Equation (183) gives $I_n^{\mathrm{sat}} \le 2\mathfrak{U}_n^{\mathrm{sat}}$, proving the displayed negligibility statement. Part II's exact-Lift identity and non-generativity hierarchy determine structural provenance but do not erase contextual charge. The master certificate retains every Caus, Lift, and Bdry field at full carrier cost; a smaller-source transfer is used only when its represented comparison is part of that certificate.

Encoding adequacy and Lemma Budgeted generation gives budgeted representation place any uniform computation whose transcript lies within budget inside the same saturated budget accounting. The universal saturated-profile accountability theorem Theorem III.24.3 bounds every target-arrival probability by the full exact-quotient accounting profile times the declared horizon factor. Transfer-class closure preserves negligibility. The exact first-hit quotient is the audit witness in that implication; by Corollary III.23.2, its retrospective representation supplies no independent source, and any independent pre-hit representation is already included in the assumed saturated supremum.

□

III.32.2 Certificate regimes

The first regime treats target placement that is random relative to the represented structural family.

Lemma III.32.3 (Random target regime)**.** *Put $N = 2^n$, and let $y \in \{-1,1\}^N$ have independent Rademacher coordinates. Let $\mathcal{K}_{n,\le B}^{\mathrm{sat}}$ be a family of positive-semidefinite contractions that is independent of y and contains the quotient and Geom/Caus visibility kernels under consideration. Suppose it has an operator-norm ε-net of cardinality M_B. Define*

$$r_1(B) = \sup_{K \in \mathcal{K}_{n,\le B}^{\mathrm{sat}}} \frac{\operatorname{tr} K}{N}, \qquad r_2(B) = \sup_{K \in \mathcal{K}_{n,\le B}^{\mathrm{sat}}} \frac{\|K\|_{\mathrm{F}}}{N}.$$

There is an absolute constant C such that, with probability at least $1-\delta$,

$$\sup_{K \in \mathcal{K}_{n,\le B}^{\mathrm{sat}}} \frac{\langle y, Ky\rangle}{\|y\|_2^2} \le r_1(B) + C\left[r_2(B)\sqrt{\log \frac{2M_B}{\delta}} + \frac{\log(2M_B/\delta)}{N}\right] + \varepsilon. \tag{226}$$

If the right-hand side is $\mathrm{negl}_{\mathfrak{C}}(n)$, then each structural visibility coordinate indexed by this kernel family is negligible and $I_n^{\mathrm{sat}}(B)$ is at most twice the right-hand side. The same conclusion holds for a random fixed-density target contrast only after supplying the analogous sampling-without-replacement quadratic-form estimate. In particular, cover entropy alone is insufficient: the normalized trace and Frobenius terms must also be small. Any subexponential-budget lower bound is conditional on these quantities being negligible uniformly throughout that budget range.

Proof. For a fixed real symmetric K, the Hanson–Wright inequality gives, for every $t > 0$,

$$\mathbb{P}\Big\{y^{\mathsf{T}}Ky - \operatorname{tr} K > C\big(\|K\|_{\mathrm{F}}\sqrt{t} + \|K\|_{\mathrm{op}}t\big)\Big\} \le 2e^{-t}$$

for an absolute C [71]. Apply this estimate to every member K_0 of the declared net with $t = \log(2M_B/\delta)$, use $\|K_0\|_{\mathrm{op}} \le 1$, and take a union bound. For an arbitrary K choose a net point K_0 with $\|K - K_0\|_{\mathrm{op}} \le \varepsilon$. Since $\|y\|_2^2 = N$,

$$\frac{|y^{\mathsf{T}}(K - K_0)y|}{N} \le \|K - K_0\|_{\mathrm{op}} \le \varepsilon.$$

Dividing the simultaneous fixed-net bounds by N and taking the supremum proves (226). If both the Abs and Geom/Caus coordinates use kernels in this family, each is bounded by the same right-hand side; their sum is therefore bounded by twice that quantity.

□

Lemma III.32.4 (Presentation-inaccessible structure regime)**.** *If every transfer-class gapping structure requires data absent from the declared represented closure $\mathcal{R}_n^{\mathcal{B}}$, then no budget-admissible operator in the class has a transfer-class gapping component.*

Proof. This is the provenance half of Theorem Inadmissible-gap saturated-profile criterion. A datum without a uniform represented derivation cannot be used by a presentation-relative operator at the declared budget. No ambient measurability or degree estimate supplies the missing derivation.

□

Lemma III.32.5 (Adversarial placement against the declared geometry)**.** *Suppose the only budget-admissible non-residual exposed structural component below saturated budget B is a pointwise geometry. Assume either that every such record fails its consistency condition, or that there is one negligible function $\nu_B(n)$ such that, uniformly over all budget-admissible geometry records at that budget,*

$$\sup_{(\mathfrak{N},\Phi,D,J,\mathfrak{K})\in\mathcal{G}_{n,\le B}^{X,\mathrm{free}}} R_T(D) \le \nu_B(n).$$

Then $G_n^{\mathrm{sat}}(B) = \mathrm{negl}_{\mathfrak{C}}(n)$. If also $m_{\mathfrak{M},n}^{\mathrm{sat}}(\Psi(B,n))$ is negligible, every budget-B algorithm has target arrival negligible after the explicit horizon factor eH_B.

Proof. Failure of the consistency condition sets the corresponding term in $G_n^{\mathrm{sat}}(B)$ to zero. In the second case the single uniform bound controls the numerator $R_T(D)$ for every remaining record. Since $(1+\rho_\Phi(D))^{-1} \le 1$ and the mixing indicator is at most one, the supremum defining $G_n^{\mathrm{sat}}(B)$ is negligible. If the uniform saturated quotient profile is also negligible at the enlarged first-hit budget, Theorem III.24.3 bounds the full target-arrival probability. This last step is applied to the complete positive process and therefore includes residual interactions.

□

Lemma III.32.6 (Saturated-cost lower-bound regime)**.** *Assume a family has a proven saturated-cost threshold $B_0(n)$: there is one negligible function $\nu(n)$ such that for its centered target contrast,*

$$\sup_{B<B_0(n)} \sup_{K\in\mathcal{K}_{n,\le B}^{\mathrm{sat}}} \frac{\langle y, Ky\rangle}{\|y\|^2} \le \nu(n).$$

Then

$$I_n^{\mathrm{sat}}(B) = \mathrm{negl}_{\mathfrak{C}}(n) \quad (B < B_0(n)), \qquad B^* \geq B_0(n).$$

In the raw Boolean direct-coordinate subcase, if a raw Walsh support threshold is $r_0(n)$, then exhaustive coordinate scanning below that threshold has cost $\mathrm{Cost}_W(n, r_0) = \sum_{k<r_0(n)} \binom{n}{k}$. This is one charged coordinate inside $B_0(n)$, rather than the complete budget-admissibility definition.

Proof. By Proposition III.28.5, the quotient and Geom/Caus structural visibility coordinates are bounded above by the displayed target quadratic forms over the budget-admissible saturated visibility kernels. The hypothesis therefore gives $I_n^{\mathrm{sat}}(B) = \mathrm{negl}_{\mathfrak{C}}(n)$ for every $B < B_0(n)$. The definition of B^* implies $B^* \geq B_0$ up to the chosen transfer-class threshold; for $\mathcal{B}_{\mathrm{poly}}$ this is the inverse-polynomial threshold. In the raw Boolean Walsh subcase, the dense-band dimension below raw degree r_0 is $\sum_{k<r_0} \binom{n}{k}$, which is the displayed direct-scan cost.

□

Remark III.32.7 (Saturated-cost threshold versus raw degree). The hypothesis is a lower support theorem for the saturated budget-admissible kernel family. A raw Walsh, nonlinearity, or algebraic-immunity statement supplies this hypothesis only when it controls every budget-admissible represented quotient, public basis change, valid lift, and geometric/causal kernel in the saturated class. Otherwise it remains a diagnostic for one coordinate system and does not establish that B^* lies above the declared budget; in $\mathcal{B}_{\mathrm{poly}}$, this is the usual superpolynomial lower-bound statement. Any application claiming such a threshold must bound the complete derivation-indexed quotient family and all supported geometric modes. Learning fragments provide quantitative diagnostics for the subfamilies they cover.

III.33 Complexity Barriers as Presentation-Extension Tests

III.33.1 Oracle extensions and presentation identity

The invariant measures spectral transport relative to a task presentation. The primitive signature is part of the task object, so an oracle extension must be interpreted relative to that signature. An oracle already generated by the saturated closure is a conservative interface. An oracle not generated by that closure enlarges the public presentation and defines a different presented task. Classical relativization, natural-proofs, and algebrization results test claims that remain valid under specified presentation extensions; they do not identify oracle access as an unclassified source of computation.

Figure 43 gives the common presentation-relative form of the three barrier tests discussed in this section.

The oracle-extended closure records precisely the presentation obtained by adjoining a charged query interface.

Definition III.33.1 (Oracle-extended derivability closure)**.** For a presented task $\mathcal{T}$ and an oracle O, let $\mathcal{T}^O$ be the presentation obtained by adjoining the query interface O, and let $\mathcal{D}_n^O$ be Definition Budgeted derivability closure with one extra primitive: oracle-query operations $\omega \mapsto O(g(\omega))$ for represented queries g, charged once per call plus the cost of deriving g. All constructors, minimality, computation-closure, and structural-provenance clauses are unchanged. This query charge prices access to the declared interface; it does not charge construction of O. Consequently $\mathcal{T}^O$ is the same effective task as $\mathcal{T}$ only when the oracle interface has a class-preserving represented derivation from the original presentation.

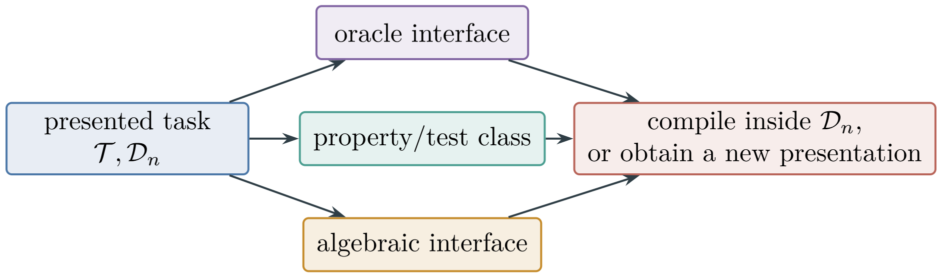

Figure 43: Relativization, natural-proofs, and algebrization are evaluated as tests of a specified extension. A represented class-preserving interface compiles into the original closure; an inadmissible interface changes the presented derivation system, as formalized in Theorem III.33.2.

This theorem distinguishes admissible oracle access from an oracle extension that changes the presentation.

Theorem III.33.2 (Oracle admissibility and presentation separation)**.** *Suppose the oracle evaluator $x \mapsto O(x)$ has one family-uniform represented derivation from $\mathcal{T}$, including its retained data and evaluation rule. If substituting this derivation for every oracle call has class-preserving accumulated cost $\Psi_O(B,n)$, then every record in $\mathcal{D}^O_{n,\le B}$ compiles to a record in $\mathcal{D}_{n,\le \Psi_O(B,n)}$ with the same denotation. In particular,*

$$I^{\mathrm{sat}}_{\mathcal{T}^O,n}(B) \le I^{\mathrm{sat}}_{\mathcal{T},n}(\Psi_O(B,n)).$$

The extension is then conservative for the declared resource class. If no such derivation exists at the declared budget, then

$$\mathcal{T}^O \ne \mathcal{T}, \qquad \mathcal{D}^O_n \ne \mathcal{D}_n \quad \text{as presented derivation systems.}$$

Indeed, the oracle evaluator is a primitive of $\mathcal{T}^O$ and is not an admissible primitive of $\mathcal{T}$. Every target-aligned oracle output is classified and charged as a source in the saturated profile of $\mathcal{T}^O$. An instance-dependent oracle table not produced by one family-uniform rule is nonuniform advice and likewise defines a different presented family.

Proof. Replace each oracle-query vertex by the represented evaluator for O. Closure under composition preserves the denotation, and the complete substituted derivation has cost at most $\Psi_O(B,n)$. Applying the substitution to every saturated certificate proves the profile inequality. In the absence of a represented evaluator, the Oracle accounting proposition places the output outside the original derivability closure. Once declared primitive, the Component exhaustiveness theorem and the Universal five-family evaluator provenance normal-form theorem classify its state-dependent contribution in the enlarged presentation.

□

Figure 44 shows the two cases of Theorem III.33.2.

Corollary III.33.3 (Presentation-coarse classifications cannot separate P from NP)**.** *Let $\mathfrak{C}$ be a classification of presented problem families that assigns the same class to $\mathcal{T}$ and $\mathcal{T}^O$ whenever O is inadmissible for $\mathcal{T}$. For every presented NP search or decision family $\mathcal{T}$, adjoin an oracle $O_{\mathcal{T}}$ that returns a valid witness or the verified decision bit. Then $\mathcal{T}^{O_{\mathcal{T}}}$ has a uniform polynomial solver in its enriched presentation. Therefore $\mathfrak{C}$ cannot be both sound and complete for polynomial-time*

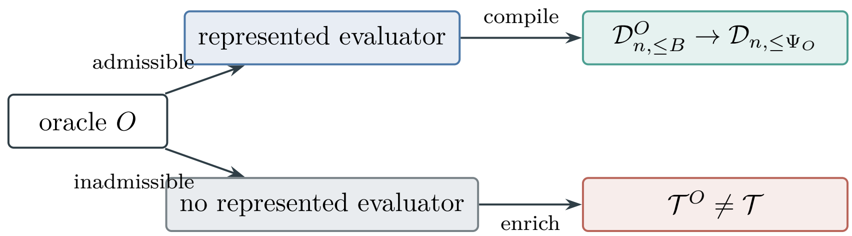

Figure 44: An admissible oracle compiles into the original closure; an inadmissible oracle defines an enriched presentation, as formalized in Theorem III.33.2.

solvability on presented NP families unless $P = NP$. Equivalently, a classification that identifies a family with all of its inadmissible oracle enrichments cannot distinguish the original P-versus-NP question.

Proof. The witness oracle supplies a polynomial query, verification, and readout procedure in $\mathcal{T}^{O_\mathcal{T}}$. If it is admissible for $\mathcal{T}$, the Oracle admissibility and presentation separation theorem compiles that procedure into a polynomial solver for $\mathcal{T}$. If it is inadmissible, the same theorem gives $\mathcal{T}^{O_\mathcal{T}} \neq \mathcal{T}$, while the assumed classification assigns them the same value. Soundness and completeness would then classify every presented NP family as polynomial-time solvable. Hence such a classifier implies $P = NP$; absent that collapse, it cannot distinguish the two complexity classes.

□

III.33.2 Relativization, natural proofs, and algebrization

Relativized accountability applies the same derivability ledger after a declared oracle extension.

Lemma III.33.4 (Universal derivability accountability relativizes)**.** *For every fixed oracle presentation $\mathcal{T}^O$, the Encoding adequacy theorem, the Forced derivable witness proposition, and the Full derivability lower-bound criterion hold verbatim over $\mathcal{D}^O_n$. Hence negligibility of the saturated profile of $\mathcal{T}^O$ at the charged enlarged budget rules out inverse-polynomial target arrival by uniform polynomial computations in that oracle presentation. The profile is recomputed after adjoining O; the theorem asserts no identity between the profiles of $\mathcal{T}$ and $\mathcal{T}^O$.*

Proof. An oracle call is a declared primitive constructor of $\mathcal{T}^O$. Encoding Adequacy simulates ordinary and oracle steps one by one. The forced-witness proof applies the legal clocked-transcript embedding and first-hit quotient to the induced transition. The algorithm-to-profile inequality therefore holds inside the enlarged presentation. The Oracle admissibility and presentation separation theorem determines whether this is a conservative interface for $\mathcal{T}$ or a distinct task presentation.

□

Corollary III.33.5 (Presentation-relative relativization test)**.** *If, for every oracle A, both the Universal saturated-profile accountability theorem and a negligible-profile proof applied to the standard presentation of an NP^A-complete family, they would imply $P^A \neq NP^A$ for every oracle A. The Baker–Gill–Solovay theorem excludes that extension-invariant conclusion [6]. In the presentation-relative framework, the failure occurs in the profile premise: an arbitrary oracle may add a target-aligned primitive, including a solver or answer table, and thereby make the saturated* *profile non-negligible. Conservative oracle extensions preserve the original conclusion through the Oracle admissibility and presentation separation theorem; nonconservative extensions define different presented tasks and require their own structural analysis.*

Proof. Assume the displayed accountability theorem and its negligible-profile premise held for the standard presentation of an NP^A-complete family for every oracle A. The profile-to-arrival implication would then rule out every polynomial-time oracle decider for that family and yield $P^A \neq NP^A$ for every A. Baker–Gill–Solovay provide an oracle for which $P^A = NP^A$, a contradiction. Therefore the extension-uniform conclusion can fail only because at least one required premise is not preserved. In this framework the presentation-separation theorem locates that failure in the recomputed profile when the oracle contributes a target-aligned primitive; for a conservative extension its simulation hypothesis instead preserves the original profile conclusion. $\square$

Lemma III.33.6 (Presentation-relative natural-proofs test)**.** *The predicate $\Phi_n(\mathcal{T})$ that a presented task has superpolynomial full-closure B^* is a property of the complete presentation, not of an unadorned Boolean truth table. Its value may change when advice, a circuit family, or another target-aligned primitive is adjoined. The Random target regime theorem proves largeness for the declared random sparse presentation under its cover-entropy hypothesis. The Razborov–Rudich barrier applies only to a derived Boolean-function property that is simultaneously polynomial-time constructive from the truth table, large, and useful against the stated nonuniform circuit class [69]. A fixed-presentation proof of saturated-profile negligibility does not acquire those three properties from the generic framework.*

Proof. Largeness is the conclusion of the Random target regime theorem for the specified presented kernel family. Usefulness in the present framework is relative to the declared uniform saturated closure. Passing to P/poly adjoins nonuniform circuit data and changes the computational authority being classified. If one additionally constructs a polynomial-time truth-table test whose acceptance is large and excludes that nonuniform circuit class, the resulting truth-table property satisfies the hypotheses of Razborov–Rudich. In the absence of that additional construction, the theorem remains a presentation-relative structural statement, and the natural-proofs hypotheses are not present. $\square$

The algebraic-oracle closure records the presentation obtained by adjoining algebraic oracle primitives.

Definition III.33.7 (Algebraic-oracle closure)**.** For an algebraic oracle $\widetilde{O}$, such as a low-degree extension or interpolation operation, define the enriched presentation $\mathcal{T}^{\widetilde{O}}$ by adjoining evaluations of $\widetilde{O}$ on represented queries as primitive operations and leaving the remaining closure rules unchanged. The extension is conservative precisely when these evaluations have a class-preserving represented derivation in $\mathcal{T}$.

Lemma III.33.8 (Algebraic-oracle accountability and presentation separation)**.** *The Encoding adequacy theorem, the Forced derivable witness proposition, and the Full derivability lower-bound criterion hold verbatim for the presentation $\mathcal{T}^{\widetilde{O}}$. Therefore the accountability direction algebrizes in the Aaronson–Wigderson sense [1]. Its conclusion is evaluated using the saturated profile of the enriched presentation. When $\widetilde{O}$ has a class-preserving represented evaluator in $\mathcal{T}$, the extension compiles conservatively into $\mathcal{T}$. Otherwise $\mathcal{T}^{\widetilde{O}} \neq \mathcal{T}$, and every structural source introduced by* *the algebraic oracle is charged in the enriched profile. Thus algebrization preserves accountability inside each presentation; it does not preserve a family-specific profile-negligibility premise across nonconservative algebraic extensions.*

Proof. The Encoding adequacy theorem and the Exact budget-admissible first-hit quotient lemma operate on the finite update induced by the declared primitive operations. An algebraic-oracle evaluation is one such operation in $\mathcal{T}^{\widetilde{O}}$, so the algorithm-to-profile implication holds in that presentation. The profile premise must then be proved for the enriched saturated closure. Applying Oracle admissibility and presentation separation theorem to the evaluator $\widetilde{O}$ gives the two stated cases.

□

Proposition III.33.9 (Barrier interpretation for fixed presentations)**.** *A full-closure lower bound for a fixed presentation follows from a family-specific proof that its saturated profile is negligible at the declared budget. No additional barrier hypothesis is required for that fixed-presentation implication. Classical relativization, natural-proofs, and algebrization barriers constrain stronger statements that discard the presentation or require the same structural premise to survive nonconservative changes of computational authority. A specified extension is handled by Oracle admissibility and presentation separation theorem: conservative interfaces compile into the original closure, while nonconservative interfaces define a different family whose new sources require their own analysis.*

Proof. The generic accountability implication is evaluated in whichever presentation supplies the computation. A family-specific lower bound proves its profile premise in the declared closure. A conservative extension preserves that premise through the compilation inequality. A nonconservative extension adds primitive information and changes the family, so the original premise is not a statement about the enriched task. The classical barriers arise precisely when these distinct presentations are identified or when one demands one premise uniformly across all of them. The presentation-coarse classification corollary shows why such identification cannot classify polynomial solvability without collapsing P and NP.

□

Remark III.33.10 (Scope). The barrier results distinguish a fixed-presentation theorem from a claim invariant under changes of computational authority. The LPN specialization of this scope statement is indexed in Section VIII.23.

III.34 Examples: Discounted Hitting for SAT, Subset Sum, and Degree Scans

The examples below are static spectral tests of the hitting-time analysis. They compute how much target contrast is visible at low degree in selected observable bases and how much is exposed by public comparison data generated from the SAT presentation, without access to a target oracle.

Example III.34.1 (XOR-SAT)**.** For public XOR-SAT, the syndrome map is a represented observable with affine fibers. Projecting onto the public syndrome quotient captures the satisfying sector exactly. A raw-coordinate Walsh diagnostic may place the same contrast in higher coordinate degrees when equations involve many variables, but the public quotient is the budget-admissible basis and controls the invariant.

Example III.34.2 (Random 3-SAT model)**.** In the random-target approximation to random 3-SAT, raw low-degree Walsh projections of the centered success contrast are negligible with high probability for the declared basis family. If the clause-defect geometry also fails the regular target-orientation condition, then the tested raw low-degree and clause-geometry fragment profiles are negligible at polynomial saturated budgets. Negligibility of the full $m_n^{\mathrm{sat}}(B)$ or $G_n^{\mathrm{sat}}(B)$ additionally requires a uniform exhaustive cover of their saturated certificate families. Any operator portion outside the tested exposed fragments remains in the residual ledger, while its full target effect is accounted for by the exact first-hit quotient.

Example III.34.3 (Subset sum with public or withheld modulus)**.** If the residue map $x \mapsto \sum_i a_i x_i \pmod{q}$ is public, the residue quotient is budget-admissible when its representation fits the declared ceiling; in $\mathcal{B}_{\mathrm{poly}}$ this can make B^* polynomial. If the modulus and residue map are withheld, the same quotient is not generated in the declared budgeted saturated observable closure of the original presentation. Its quotient is then budget-inadmissible rather than an admissible high-cost representation.

Example III.34.4 (XOR, Gaussian elimination, and public closures)**.** Budget-admissible computations whose raw Walsh support is high are handled by admissible public basis changes and generated observables; in the polynomial instance these are the ordinary polynomial computations. Gaussian elimination on XOR-SAT computes the public syndrome quotient, so its target signal is charged to $m_n^{\mathrm{sat}}(B)$ in the syndrome basis rather than left in J_{res}. The implication graph for 2-SAT and forward chaining for Horn-SAT are generated from exposed clauses and are charged as Caus over public graph/closure geometry [5, 27, 72], with Bdry only marking terminal satisfied states. A public weighted-sum or residue map for subset-sum gives an Abs quotient; the same map, when absent from the presentation and not generated within the declared budget from exposed data, is presentation-inaccessible and budget-inadmissible rather than residual structure.

The raw Walsh scan records the degree statistic used by the finite diagnostic.

Definition III.34.5 (Raw Walsh degree scan)**.** For the dense raw Walsh diagnostic, the empirical raw-degree increment is

$$\Delta I(d) = \widehat{I}(d) - \widehat{I}(d-1), \qquad \widehat{I}(d) = \widehat{m}(d) + \widehat{G}(d).$$

For a sample of M states and a raw degree band of dense scan cost $N_d = \sum_{k \le d} \binom{n}{k}$, Walsh coefficient estimates concentrate at scale

$$O\left(\sqrt{\frac{N_d + \log(1/\delta)}{M}}\right).$$

The scan certifies flatness only for this raw-coordinate cost model, up to degrees whose dense band dimension lies within budget and whose sampling floor is below the claimed signal. It is a diagnostic coordinate inside the saturated profile rather than the complete budget-admissibility definition.

III.35 Summary: Hardness as Saturated Extraction Cost

The Part II structural decomposition yields a quantitative theory of target-arrival time. Its Laplace transform is the target-projected killed resolvent. For self-adjoint killed dynamics satisfying the stated comparison hypotheses, the resolvent, killed-target gap, and arrival functional are comparable on the declared window. General nonreversible dynamics retain the exact normalized resolvent-defect bound. The resulting estimate depends on the extraction available within the saturated public budget. These hypotheses classify the induced operator and need not be supplied by the algorithm that induces it.

Dependency trail. The global encoding, resolvent, and channel coordinates are Definitions III.1.1 to III.9.1, III.8.4 and III.9.4. The spectral and saturated ledgers are Definitions III.10.1 to III.14.4. The later cost, kernel, admissibility, and oracle coordinates are Definitions III.29.1 to III.31.2, III.29.4 to III.34.5, III.33.7, III.28.9 and III.28.11. Resolvent and channel timing are provided by Proposition III.8.6 and Lemmas III.9.3 to III.9.5. The geometric, causal, and defect comparisons are Propositions III.11.5, III.13.8 and III.13.9, Corollaries III.13.3 to III.12.5, and Lemmas III.13.2 and III.13.5. The saturated-history and source-envelope results are Lemmas III.14.5 and III.14.6, Theorems III.15.9 and III.26.1, and Corollaries III.27.4 and III.19.6. The cost-invariant, nonlinear-kernel, and universal-accounting consequences are Propositions III.28.2, III.29.2, III.28.4 and III.29.5 and Lemmas III.29.3, III.29.6, III.31.7, III.28.10 and III.28.13. The certificate regimes and presentation-extension tests are Lemmas III.32.4 to III.33.6 and III.33.8, Corollary III.33.5, and Proposition III.33.9. Conventions and qualifications used in these arguments are indexed by Remarks III.14.3 to III.33.10. The corresponding worked examples are Examples III.34.1 to III.31.8, III.13.10 and III.28.14.

The summary theorem collects the accountability and structural-negligibility consequences used by the application parts.

Theorem III.35.1 (Saturated-profile hardness summary)**.** *Fix a task family, its declared budget ceiling B, and a family functional $\mathfrak{M}_n$. Then*

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le eH_B(n) m^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)) \le eH_B(n) I^{\mathrm{sat}}_{\mathfrak{M},n}(\Psi(B,n)).$$

Thus negligibility of the saturated profile at the displayed class-preserving enlargement implies negligible target arrival for the declared operational class. For $\mathcal{B}_{\mathrm{poly}}$, the statement is evaluated at $B = b_{\mathrm{poly}}$ and $\Psi(b_{\mathrm{poly}}, n)$; no additional polynomial-budget quantifier is introduced.

Proof. Apply Theorem III.24.3 and then $m^{\mathrm{sat}}_{\mathfrak{M},n} \le I^{\mathrm{sat}}_{\mathfrak{M},n}$. The first-hit proof operates on the full positive generator before channel decomposition, so the result includes all five channels, their interactions, and J_{res} without a componentwise positivity assumption.

□

The principal results of this part are the budgeted derivability closure, the arrival functional, the discounted hitting functional, the target-projected killed resolvent, the budgeted comparison-data equivalence, the exact structural operator ledger, the full-process first-hit accountability theorem, the directed-drift transformed-gradient normal form, the budget-parametrized full-history composition bound, and the spectral comparison-vector bound with declared transfer-class comparison factor, the normalized nonreversible defect bound, the reversible Schur-complement compensation theorem, the Caus drift-margin theorem, the five-regime master quotient–geometry normal form and exact-quotient domination theorem, the extraction functional $I^{\mathrm{sat}}_n(B) = m^{\mathrm{sat}}_n(B) + G^{\mathrm{sat}}_n(B)$, the saturated ambient-geometry accountability and inverse-polynomial witness criterion after quotient exhaustion, the universal five-family Abs provenance normal form and its exhaustive contextual-charge envelope, the intrinsic cost invariant B^* and log-cost coordinate d^*, the universal derivability accountability theorem, the alignment-complexity sandwich, the reversible compressed-fragment theorem, the admissibility dichotomy, and hardness in the exposed-structure regimes.

For a fixed transfer-class-gap cryptographic instance under the polynomial budget, the results give the reduction

$$\begin{gathered}B^* \text{ is superpolynomial in the saturated represented algebra} \\ \Longrightarrow \\ \text{hardness against presentation-relative movement.}\end{gathered}$$

The full closure gives an unconditional presentation-relative lower-bound criterion at a chosen budget: if B^* lies above the declared ceiling, up to the allowed transfer-class overheads, target arrival is absent in the legal derivability closure at that budget. Fragment and family-specific theorems supply sufficient conditions by proving that the relevant saturated subclosure has no transfer-class target-aligned representative. Part IV uses the same $I_n^{\mathrm{sat}}(B)$ invariant as a learning-capacity bound. Part VIII applies the structural decomposition to the LPN charged-envelope criterion, while Part IV gives the learning-capacity reading of the same invariant.

Example III.35.2 (SAT final reading)**.** The discounted hitting functional for SAT depends on its presentation. XOR-SAT with public parity rows has polynomial B^*, hence logarithmic d^* in log-cost notation. A random-target SAT model has superpolynomial B^* with high probability. A subset-sum-derived SAT instance with a withheld modulus is budget-inadmissible. The invariant is attached to the presented problem rather than to the name SAT alone.

Part IV
Learning and Computability Bounds

IV.0 Introduction

Part contract. *Inputs:* the saturated cost and extraction profiles from Part III. *Output:* operator, kernel, neural-tangent-kernel, and certified generalization bounds. *First pass:* the computational evaluation procedures may be skimmed.

Parts I–III define the task presentation, characterize budget-admissible dynamics, and introduce the saturated cost invariant. This part develops the corresponding statistical theory. Effective dimension measures the number of statistically accessible modes, alignment measures their orientation relative to the target, and the source residual measures target mass outside the selected spectral subspace.

Part II shows that every budget-admissible oracle-free computation on its finite clocked transcript carrier decomposes into the five exposed structural channels

$$\mathsf{Geom}, \qquad \mathsf{Caus}, \qquad \mathsf{Abs}, \qquad \mathsf{Lift}, \qquad \mathsf{Bdry},$$

and the residual term J_{res}. For a task family whose exposed structural components satisfy the stipulated cost conditions, Part III shows that target arrival at the declared transfer-class margin requires saturated extraction at the same margin,

$$I_n^{\mathrm{sat}}(B) = m_n^{\mathrm{sat}}(B) + G_n^{\mathrm{sat}}(B)$$

under the declared budget structure $\mathcal{B}$, with ceiling $B = b(n)$. This accounting includes public basis changes, represented quotients, runtime-generated observables, admissible finite-horizon statistics, boundary readouts, and valid lifts. A learning algorithm is itself a budget-admissible oracle-free computation on the presented data; ordinary polynomial-time learning corresponds to $\mathcal{B}_{\mathrm{poly}}$. Consequently, generalization within this formalism requires a fixed or learned observable algebra that captures target mass at a statistical cost within the same saturated represented closure. The choice $\mathcal{B} = \mathcal{B}_{\mathrm{poly}}$ yields the polynomial-time specialization.

We consider three settings.

- For fixed features, the observable algebra is an RKHS.
- For learned features, a neural tangent kernel describes the evolving tangent-feature space.
- After training, the spectrum of the learned kernel determines an empirical effective degree.

The analysis uses standard results on RKHSs and Mercer eigendecompositions, the representer theorem, effective-dimension bounds for kernel ridge regression, kernel–target alignment, neural tangent kernels, and spectral-margin bounds for neural networks [3, 9, 18, 25, 40, 46, 58].

Definition IV.0.1 (Imported objects from Parts I–III)**.** Part IV uses the following objects with the meanings fixed earlier.

Part I. A task presentation has state space X_I, observable algebra $\mathcal{F}_I$, positive terminal sector E_I^+, terminal-failure sector E_I^- when declared, static family-uniform observable constructions, pointwise observables, represented-fiber observables, and the exposed/presentation-inaccessible distinction. The five static admissibility criteria are terminal alignment, extremum or threshold consistency, cost within the declared budget structure, family-uniformity, and monotone terminal filtration.

Part II. A budget-admissible oracle-free computation becomes a positive generator on each finite clocked transcript carrier. Component exhaustiveness decomposes every budget-admissible generator into Geom, Caus, Abs, Lift, Bdry, plus J_{res} residual accounting. The term J_{res} contains no additional exposed structural source.

Part III. Computation is presentation-relative to a declared budget structure $\mathcal{B}$, inside the budgeted saturated represented closure $\mathcal{R}_n^{\mathcal{B}}$, equivalently the derivability filtration $\mathcal{D}_{n,\leq b(n)}$. The extraction functional is $I_n^{\mathrm{sat}}(B) = m_n^{\mathrm{sat}}(B) + G_n^{\mathrm{sat}}(B)$. The term $m_n^{\mathrm{sat}}(B)$ is the supremum over represented-fiber quotient sources after admissible public basis changes, valid lifted/generated coordinates, terminal compatibility, and exact-lumpability conditions. The term $G_n^{\mathrm{sat}}(B)$ is the maximum of ambient Geom/Caus, Geom/Caus on an exact quotient, and Geom/Caus on a compensated quasi-lumpable quotient. The saturated cost invariant B^* is the first public budget at which $I_n^{\mathrm{sat}}(B)$ reaches the declared transfer-class margin; its log-cost coordinate is $d^* = \lceil \log_2 B^* \rceil$ when the resource is numeric. Raw Walsh degree is one charged diagnostic coordinate; the budget criterion is defined by the complete saturated cost.

Lemma IV.0.2 (Statistical interpretation of saturated extraction)**.** *Each learning object considered below consists of a Part III saturated representative together with a statistical sampling procedure. A fixed kernel, learned kernel, empirical NTK, feature quotient, margin bound, or post-training spectral profile contributes to structural learning when its instance-dependent construction belongs to $\mathcal{D}_{n,\leq B}$ for the original presentation and its readout satisfies the terminal conditions of Part I and the channel admissibility conditions of Part II. The statistical quantities $\mathrm{Tr}(A^{-1})$, $N_{\mathrm{eff}}(\lambda)$, $\log \mathcal{N}$, source residual, margin, and concentration errors measure the sample cost of estimating or selecting these representatives. They introduce no additional structural channel beyond $I_n^{\mathrm{sat}}(B)$.*

Proof. By the encoding-adequacy result of Part III, a budget-admissible learner on a budget-complete encoding defines a presentation-relative generator on its clocked training transcript carrier. The polynomial specialization recovers uniform polynomial-time learners. Part II decomposes the resulting target-relevant dynamics into Geom, Caus, Abs, Lift, Bdry, and residual accounting. If the learner constructs an operator, kernel, quotient, spectral projector, feature map, or terminal readout within the declared budget, the budgeted-generation lemma of Part III supplies a saturated representative and assigns its derivation cost. Otherwise the object requires oracle or advice data, an over-budget construction, or a different primitive signature. Standard learning bounds then quantify the estimation, capacity, and selection costs of the admissible representative. Hence every included generalization term measures a Part III source; all remaining terms are residual or inadmissible at the declared budget.

□

Source-resolved convention for Part IV. Every hypothesis class, operator, kernel, prior, posterior, learned feature, decoder, and deployed selector below is restricted by its exact source index when a target-qualified structural use is made. The complete quantitative compilation is Definition IV.13.2 and Theorem IV.13.3. Thus each statistical bound retains B, m, δ, its source cell, and its complete construction cost. A noisy target-dependent reveal additionally retains its source-assigned channel coefficient and noise parameter. Generic population–empirical deviations remain in risk units; they enter a target-alignment, recovery, selector, or arrival bound only through a represented same-law, same-unit converter. This convention applies to every theorem in this part, including fixed and learned kernels, NTK feature learning, effective dimension, PAC–Bayes, dependent samples, channel learning, active querying, and the complete learning decomposition.

IV.1 The Observable Algebra Is a Reproducing-Kernel Hilbert Space

The observable algebra of Part I acquires a statistical structure when its functions are embedded in a Hilbert space. A feature map specifies the observable representation, and its kernel records inner products between represented states.

The feature observable embeds represented task functions in an RKHS and induces their kernel.

Definition IV.1.1 (Feature observable and induced kernel)**.** Let $(X, \mathcal{F}, \mu)$ be a task presentation and let $\Phi : X \to \mathcal{H}$ be a measurable feature observable into a Hilbert space. The induced kernel is

$$k_\Phi(x, z) = \langle \Phi(x), \Phi(z) \rangle_{\mathcal{H}}.$$

The RKHS $\mathcal{H}_k$ is the closure of finite linear combinations of kernel sections $k_x(\cdot) = k(x, \cdot)$ with inner product determined by $\langle k_x, k_z \rangle_{\mathcal{H}_k} = k(x, z)$ [3].

The Mercer decomposition expresses the represented kernel in its observable spectral coordinates.

Definition IV.1.2 (Mercer eigendecomposition)**.** Assume k is square-integrable and positive definite. Its integral operator is

$$(T_k f)(x) = \int_X k(x, z) f(z) \, d\mu(z).$$

Let (η_j, φ_j) denote its Mercer eigenpairs [58], $\eta_1 \geq \eta_2 \geq \cdots \geq 0$, a function $f = \sum_j \widehat{f}_j \varphi_j$ has RKHS norm

$$\|f\|^2_{\mathcal{H}_k} = \sum_{j:\eta_j > 0} \frac{\widehat{f}_j^2}{\eta_j}.$$

The eigenfunctions of T_k provide the observable basis for kernel learning, in the same way that Part III uses Walsh characters on the Boolean cube and Laplacian eigenfunctions on continuous spaces.

Proposition IV.1.3 (RKHS norm as Dirichlet regularity)**.** *Let $L_W \geq 0$ be the self-adjoint generator of an exposed geometric Dirichlet form from Part III, with eigenpairs (λ_j, ψ_j). Define the resolvent kernel by spectral weights*

$$k_W(x, z) = \sum_j \frac{1}{1 + \lambda_j} \psi_j(x) \psi_j(z).$$

Then for $f = \sum_j a_j \psi_j$,

$$\|f\|^2_{\mathcal{H}_{k_W}} = \sum_j (1 + \lambda_j) a_j^2 = \|f\|^2_{L^2(\mu)} + \mathcal{E}_W(f, f).$$

Proof. The RKHS norm divides each squared coefficient by the Mercer eigenvalue. Here the eigenvalue is $\eta_j = (1 + \lambda_j)^{-1}$. Hence $\|f\|^2_{\mathcal{H}_{k_W}} = \sum_j a_j^2 / \eta_j = \sum_j (1 + \lambda_j) a_j^2$. The spectral theorem for the Dirichlet form gives $\mathcal{E}_W(f, f) = \sum_j \lambda_j a_j^2$.

□

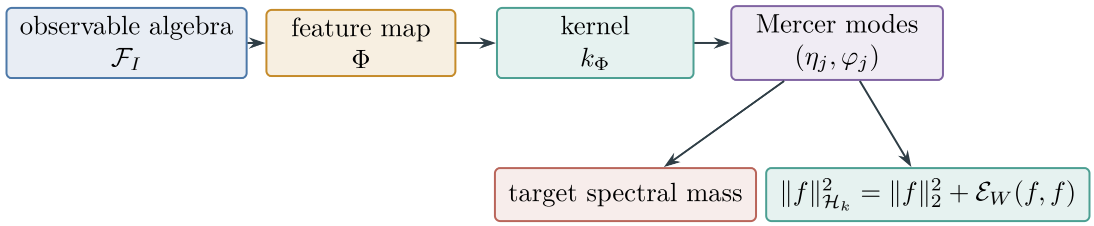

Figure 45: A represented feature observable selects a kernel and hence a Mercer observable basis. Target alignment is spectral mass in that basis, while the resolvent-kernel RKHS norm is the represented Dirichlet regularity of Proposition IV.1.3.

Corollary IV.1.4 (Kernel formulation of the observable criteria)**.** *The static criteria of Part I have the following kernel formulation. Terminal alignment is the label mass carried by Mercer modes. Extremum or threshold consistency requires kernel scores to agree with the terminal interface. Effective dimension supplies the statistical complexity measure. Family uniformity requires a public kernel rule for the entire family, and monotone terminal filtration requires stability of the induced score bands under terminal readout. RKHS/Dirichlet regularity is the Part III geometric regularity coordinate expressed in the Mercer basis, rather than an additional static criterion.*

Proof. Part I evaluates an observable through target projection, extremum or threshold consistency, affine cost, family-uniform construction, and terminal filtration. A kernel selects a feature map and therefore a closed observable span. The Mercer eigendecomposition identifies target projection with label mass in the kernel eigenmodes. Threshold consistency requires the score formed from these modes to induce the same terminal readout as the target interface. Effective dimension counts the modes available within the statistical and computational budget. Family uniformity requires the kernel rule to be generated from the presentation uniformly over the family, while monotone terminal filtration corresponds to the nested spectral or score bands induced by the kernel readout. Finally, Proposition IV.1.3 identifies the RKHS norm with the Dirichlet energy associated with $I + L_W$, namely the Geom regularity coordinate of Part III.

□

Figure 45 summarizes the Part I–Part IV bridge in Proposition IV.1.3 and Corollary IV.1.4.

Admissibility requires the learning kernel and its construction to remain inside the charged presentation.

Definition IV.1.5 (Admissible learning kernel)**.** A fixed kernel k_I is admissible for an instance I when its feature rule, kernel evaluations, and required Gram or integral-operator approximations have derivations in $\mathcal{D}_{n,\le B}$ from $(X_I, \mathcal{F}_I, E_I^+, E_I^-)$ by a family-uniform oracle-free rule. A learned kernel is admissible when a budget-admissible oracle-free training computation produces it from the observed sample and labels, and the terminal kernel is a represented terminal output on the clocked training transcript carrier, with its finite outcome-generation interface, precision, and postprocessing charged in the same saturated budget accounting. The polynomial case is the instance $\mathcal{B}_{\mathrm{poly}}$.

Observed labels are part of the learning sample. Unobserved labels, target membership outside the sample, planted certificates, and relations not generated from the budgeted saturated observable closure are not available to the learner. If a spectral projector or feature quotient produced by a budget-admissible kernel has total cost within $\mathcal{B}$ and satisfies the represented-fiber, terminal-compatibility, and lumpability conditions of Definitions III.15.1 and III.15.2, its captured mass is a candidate contribution to $m_n^{\mathrm{sat}}(B)$ at the charged cost. If it is pointwise Geom/Caus structure, it contributes through $G_n^{\mathrm{sat}}(B)$. Otherwise the same empirical correlation is presentation-inaccessible, requires an oracle, exceeds the discovery or application budget, changes the primitive signature, or remains residual accounting.

IV.2 Saturated Cost and Effective Dimension

Part III accounts for public basis changes, represented quotients, generated coordinates, admissible finite-horizon statistics, boundary readouts, and valid lifts in the saturated public budget. In the RKHS setting, ridge regularization assigns each Mercer mode a continuous statistical weight.

Effective dimension measures the statistically active Mercer modes at the chosen regularization scale.

Definition IV.2.1 (Effective dimension)**.** For a kernel integral operator T_k with eigenvalues $\eta_j \geq 0$ and ridge scale $\lambda > 0$, define

$$N_{\mathrm{eff}}(\lambda) = \mathrm{Tr}\big(T_k(T_k + \lambda I)^{-1}\big) = \sum_j \frac{\eta_j}{\eta_j + \lambda}.$$

For an empirical Gram matrix $K \in \mathbb{R}^{m\times m}$, the corresponding quantity is

$$\widehat{N}_{\mathrm{eff}}(\lambda) = \mathrm{Tr}\big(K(K + \lambda m I)^{-1}\big),$$

equivalently $\sum_j \widehat{\eta}_j/(\widehat{\eta}_j + \lambda)$ for the eigenvalues of K/m.

Proposition IV.2.2 (Effective dimension as statistical mode count)**.** *The function $N_{\mathrm{eff}}(\lambda)$ interpolates between the rank of the kernel and zero. If $\eta_j \asymp j^{-2\beta}$ with $\beta > 1/2$, then $N_{\mathrm{eff}}(\lambda) \asymp \lambda^{-1/(2\beta)}$. On a compact q-dimensional geometric domain, Weyl's law for a Laplacian kernel gives $N_{\mathrm{eff}}(\lambda)$ of the same order as a spectral cutoff count $C_q\Lambda^{q/2}$ under the resolvent relation $\eta = (1+\Lambda)^{-1}$ [79]. Thus N_{eff} is the RKHS realization of the statistical mode count in the saturated budget. The provenance and representation conditions of Part III remain separate requirements.*

Proof. The summand $\eta_j/(\eta_j + \lambda)$ is close to one for modes with $\eta_j \gg \lambda$ and close to zero for modes with $\eta_j \ll \lambda$. Thus the trace is a smoothed count of modes above the learnability threshold. Polynomial eigenvalue decay gives the displayed inverse relation by solving $j^{-2\beta} \approx \lambda$. Weyl's law gives the continuous-domain count for Laplacian eigenvalues below Λ. The definition of a budget-admissible kernel supplies the independent provenance condition required for interpreting this count structurally.

□

Budget-feasible bands identify which spectral target mass is available to the represented learner.

Definition IV.2.3 (Budget-feasible spectral bands and candidate capture)**.** A budget-admissible kernel band is budget-feasible at ridge scale λ when its total saturated learning charge does not exceed $b(n)$. For a fixed kernel this budget contains the derivation and representation cost of k, the Gram or operator approximation cost, the numerical precision, and $N_{\mathrm{eff}}(\lambda)$. For a learned kernel it also contains the training transcript and the operator-selection cost. For a centered target field $y = \sum_j b_j\varphi_j$, its kernel spectral capture at a hard cutoff is

$$m_k(\tau) = \frac{\sum_{\eta_j \geq \tau} b_j^2}{\sum_j b_j^2},$$

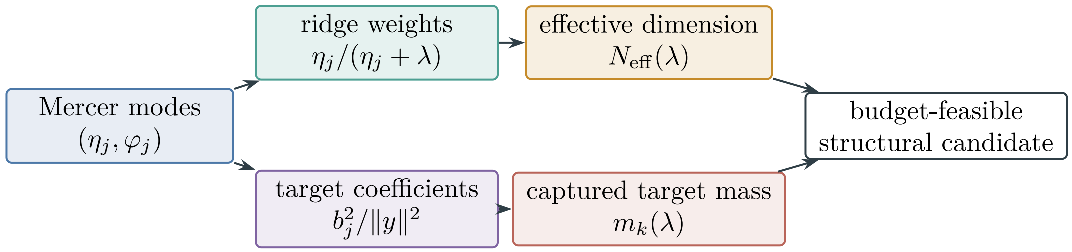

Figure 46: Effective dimension counts ridge-active modes, whereas captured mass measures their alignment with the target. A kernel band contributes to the saturated profile only through the budget and admissibility conditions of Definition IV.2.3.

and its ridge-smoothed capture is

$$m_k(\lambda) = \frac{\sum_j \left(\frac{\eta_j}{\eta_j+\lambda}\right)^2 b_j^2}{\sum_j b_j^2}.$$

The quantity m_k is a kernel-specific candidate capture curve. It contributes to $m_n^{\mathrm{sat}}(B)$ only when the kernel, spectral band, and induced quotient or lift satisfy the admissibility, represented-fiber, terminal-compatibility, and lumpability conditions within total budget B. It contributes to $G_n^{\mathrm{sat}}(B)$ only when the same data define an admissible pointwise Geom/Caus source. Otherwise m_k remains an empirical correlation statistic rather than counted structural extraction.

Figure 46 separates statistical mode count from target-aligned spectral capture in Definitions IV.2.1 and IV.2.3.

This construction is the continuous RKHS analogue of the Walsh-to-Laplacian passage in Part III. A spectral cutoff or ridge scale is a coordinate of the saturated public budget, and N_{eff} gives the corresponding smoothed mode count. Generator–observable alignment is an additional requirement: the dominant kernel modes must carry target mass. A low-dimensional kernel whose eigenspaces are misaligned with the target remains statistically uninformative even when it lies within budget.

IV.3 Two-Sided Learning Bounds

IV.3.1 Structural Operator Bounds Before the Kernel Refinement

The RKHS effective-dimension bound refines the operator-level statistical bound associated with the saturated budget of Part III. The operator formulation separates three contributions that persist in the kernel and neural-network results: complexity relative to a fixed budget-admissible operator, target–operator alignment, and the cost of operator selection.

The structural energy ball fixes the represented function class used by the operator-level generalization bound.

Definition IV.3.1 (Structural energy ball)**.** On an empirical task space $S = \{x_1, \ldots, x_m\}$, let $A \succeq \eta I$ be a budget-admissible symmetric structural operator. For $f \in \mathbb{R}^m$, set $\mathcal{E}_A(f) = f^T A f$ and

$$\mathcal{F}_A(R) = \{f \in \mathbb{R}^m : f^T A f \le R^2\}.$$

Lemma IV.3.2 (Exact Rademacher complexity of an energy ball)**.** *For any sign vector $\sigma \in \{\pm 1\}^m$,*

$$\sup_{f \in \mathcal{F}_A(R)} \sigma^T f = R\sqrt{\sigma^T A^{-1} \sigma}.$$

Consequently the empirical Rademacher complexity satisfies

$$\mathrm{Rad}_S^A(R) = \frac{R}{m} \mathbb{E}_\sigma \sqrt{\sigma^T A^{-1} \sigma} \le \frac{R}{m} \sqrt{\mathrm{Tr}(A^{-1})}.$$

Proof. Lagrange multipliers give the optimizer $f = RA^{-1}\sigma / \sqrt{\sigma^T A^{-1} \sigma}$. Averaging the support function over signs gives the equality. Jensen's inequality and $\mathbb{E}_\sigma \sigma\sigma^T = I$ give the trace bound. □

The fixed-operator bound controls generalization after the represented kernel has been selected.

Theorem IV.3.3 (Fixed-operator generalization bound)**.** *For a loss taking values in* $[0, 1]$ *that is L_ℓ-Lipschitz in its prediction argument, and for a fixed budget-admissible operator A, with probability at least* $1 - \delta$*, uniformly over $h \in \mathcal{F}_A(R)$,*

$$L(h) \le L_S(h) + \frac{2L_\ell R}{m} \sqrt{\mathrm{Tr}(A^{-1})} + 3\sqrt{\frac{\log(2/\delta)}{2m}} + \varepsilon_{\mathrm{task}}.$$

The bound is specific to the declared operator; optimization over the budget-admissible operator class is a separate model-selection problem. Here $\varepsilon_{\mathrm{task}} \ge 0$ is the fixed approximation or task residual declared by the benchmark; it is zero when the population and empirical risks are compared without such an additional residual. For a target-qualified deployment, restrict A and its output class to their exact source cell α. The displayed radius is then $\Gamma^{\mathrm{learn}}_{R,\alpha}(B, m, \boldsymbol{\eta}, \delta)$ in Definition IV.13.2; conversion to target gain is governed by Eq. (281)*.*

Proof. Apply the standard uniform generalization inequality for an L_ℓ-Lipschitz loss [8, 48]: with probability at least $1-\delta$, uniformly over a class $\mathcal{F}$, $L(h) \le L_S(h) + 2L_\ell \widehat{\mathfrak{R}}_S(\mathcal{F}) + 3\sqrt{\log(2/\delta)/(2m)}$. For $\mathcal{F}_A(R) = \{h : h^T A h \le R^2\}$, the ellipsoid Rademacher lemma gives $\widehat{\mathfrak{R}}_S(\mathcal{F}_A(R)) \le Rm^{-1}\sqrt{\mathrm{Tr}(A^{-1})}$. Substitution gives the displayed bound. The term $\varepsilon_{\mathrm{task}}$ is the fixed approximation or task residual already present in the declared benchmark, so it adds linearly. □

These capacities compare total task complexity with the target-aligned represented component.

Definition IV.3.4 (Task capacity and structural capacity)**.** For a target field y, define the fixed-operator capacity score

$$\mathrm{TaskCap}_A(y) = \frac{1}{m} \sqrt{(y^T A y)\, \mathrm{Tr}(A^{-1})}.$$

The identity fallback and modal gap score are

$$\mathrm{GapCap}_A(h) = \frac{1}{m} \sqrt{(h^T A h)\, \mathrm{Tr}(A^{-1})}, \qquad \mathrm{GapCap}_I(h) = \frac{\|h\|_2}{\sqrt{m}}.$$

The resulting bound is the minimum of the modal and identity bounds. The identity operator provides a reference value for detecting operator misalignment.

For a parameterized budget-admissible class $\Theta \ni \theta \mapsto A_\theta$, define

$$\mathrm{StructCap}(y;\Theta) = \min_{\theta\in\Theta}\left[\mathrm{TaskCap}_{A_\theta}(y) + c\sqrt{\frac{\log\mathcal{N}(\Theta,\rho)}{m}}\right].$$

The first term measures target complexity after fixing the operator, whereas the second accounts for data-dependent operator selection.

The random-label score supplies the null calibration for empirical target alignment.

Definition IV.3.5 (Random-label null score)**.** For a structural operator A and target field y, define

$$Z_A(y) = \frac{\mathrm{Tr}(A) - y^T A y}{2\sqrt{\sum_{i<j} A_{ij}^2 + \eta}}.$$

This statistic is the standardized reduction of $y^T A y$ relative to the random-sign null mean. A large positive value indicates operator–target alignment, whereas a value near zero is consistent with the null scale.

Lemma IV.3.6 (Learned-operator generalization bound)**.** *Let $\mathcal{H}(R) = \bigcup_{\theta\in\Theta}\mathcal{F}_{A_\theta}(R)$. Suppose $A_\theta \succeq \eta I$ uniformly, $\theta \mapsto A_\theta^{-1}$ is* Lip*-Lipschitz in operator norm, and $\mathcal{N}(\Theta,\rho)$ is a finite ρ-covering number. Then*

$$\mathrm{Rad}_S^{\mathcal{H}}(R) \le \inf_{\rho>0}\left[\frac{R}{m}\sqrt{\max_\theta \mathrm{Tr}(A_\theta^{-1})} + R\sqrt{\frac{\mathrm{Lip}\,\rho}{m}} + \frac{R}{m\sqrt{\eta}}\sqrt{2\log\mathcal{N}(\Theta,\rho)}\right].$$

Consequently, for a $[0,1]$-valued L_ℓ-Lipschitz loss, the generalization gap over $\mathcal{H}(R)$ has the same confidence form as the fixed-operator theorem, with twice L_ℓ times the displayed quantity plus the sampling tail and declared task residual.

Proof. Fix $\rho > 0$ and choose a ρ-net Θ_ρ. For each net point, the ellipsoid lemma bounds the empirical Rademacher support function by $Rm^{-1}\sqrt{\mathrm{Tr}(A_\theta^{-1})}$. The Lipschitz condition and $|\sqrt{a}-\sqrt{b}| \le \sqrt{|a-b|}$ give, for every sign vector σ and every θ within distance ρ of a net point θ_0,

$$\frac{R}{m}\left|\sqrt{\sigma^\top A_\theta^{-1}\sigma} - \sqrt{\sigma^\top A_{\theta_0}^{-1}\sigma}\right| \le R\sqrt{\frac{\mathrm{Lip}\rho}{m}},$$

because $\|\sigma\|_2^2 = m$. Moreover $A_\theta \succeq \eta I$ implies $\sup_{f\in\mathcal{F}_{A_\theta}(R)}\|f\|_2 \le R/\sqrt{\eta}$. The finite-union Rademacher lemma therefore contributes $R\sqrt{2\log|\Theta_\rho|}/(m\sqrt{\eta})$ [55]. Taking the infimum over ρ gives the displayed Rademacher bound, and the same symmetrization/contraction inequality used in the fixed-operator theorem gives the stated generalization form.

□

Figure 47 separates the fixed-operator capacity term from the cost of selecting an operator from data.

Corollary IV.3.7 (Rademacher bound for the Part I empirical audit)**.** *Let $\mathcal{H}_B$ be a budget-admissible readout class at budget B and let $a_h(Z) \in [0,1]$ be a bounded audit statistic such that $\mathrm{Ext}_I(h;T_I) \le \mathbb{E}a_h(Z)$ for every $h \in \mathcal{H}_B$. For an audit sample $Z_1,\ldots,Z_M$, write $\widehat{a}_M(h) = M^{-1}\sum_i a_h(Z_i)$. Then with probability at least $1-\delta$,*

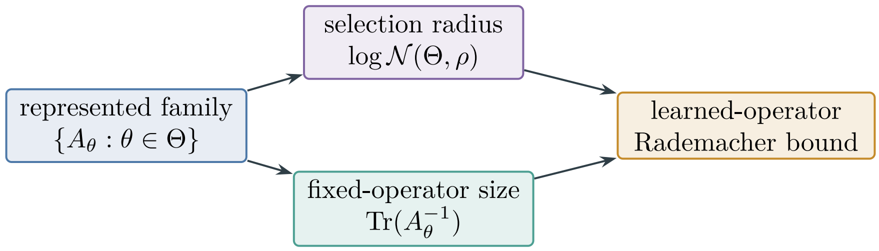

Figure 47: The bound in Lemma IV.3.6 charges two distinct quantities: capacity after fixing A_θ, and the represented choice among operators. The covering term prevents data-dependent geometry selection from being treated as free.

$$\sup_{h \in \mathcal{H}_B} \mathrm{Ext}_I(h; T_I) \le \sup_{h \in \mathcal{H}_B} \widehat{a}_M(h) + 2\,\mathrm{Rad}_M(\{a_h : h \in \mathcal{H}_B\}) + 3\sqrt{\frac{\log(2/\delta)}{2M}}.$$

Thus the Part I empirical audit penalty may be taken as

$$\mathfrak{P}_{\mathcal{H}_B}(M, \delta) = 2\,\mathrm{Rad}_M(\{a_h : h \in \mathcal{H}_B\}) + 3\sqrt{\frac{\log(2/\delta)}{2M}},$$

or by the learned-operator cover bound above when $\mathcal{H}_B = \bigcup_{\theta \in \Theta} \mathcal{F}_{A_\theta}(R)$. This inequality converts empirical optimization over observable combinations into a uniform upper bound on actionable extraction.

Proof. Apply the standard uniform Rademacher generalization inequality to the bounded audit class $\{a_h : h \in \mathcal{H}_B\}$. Since $\mathrm{Ext}_I(h; T_I) \le \mathbb{E}a_h(Z)$ by hypothesis for each h, the same upper bound holds for the extraction supremum. Substituting the fixed-operator or learned-operator Rademacher estimates gives the corresponding budgeted audit penalty.

□

Corollary IV.3.8 (Operator-selection conservation)**.** *Enlarging the operator class may reduce* $\mathrm{TaskCap}_{A_\theta}(y)$*, but it also increases the selection term* $\sqrt{\log \mathcal{N}(\Theta, \rho)/m}$ *unless the instance determines an aligned subfamily of small covering number. Universal approximation establishes representability of an aligned operator but does not control the cost of selecting it.*

Proof. The learned-operator theorem decomposes the cost of selecting an operator into a fixed-operator capacity term, a Lipschitz approximation term, and the covering term

$$\frac{R}{m\sqrt{\eta}}\sqrt{2 \log \mathcal{N}(\Theta, \rho)}.$$

Enlarging the operator family can reduce $\mathrm{TaskCap}_{A_\theta}(y)$ through minimization over θ. The same enlargement cannot decrease the covering number at a fixed resolution. It yields a net certificate improvement only when the task-capacity reduction exceeds the additional approximation and covering terms. Universal approximation controls approximation after a suitable θ is known; the theorem also charges the cost of selecting that θ.

□

Lemma IV.3.9 (Condition-number lower bound for operator capacity)**.** *Let $A_\theta \succ 0$ for every $\theta \in \Theta$, and set $\kappa = \sup_\theta \operatorname{cond}(A_\theta) < \infty$. Then, deterministically for every target vector y,*

$$\min_{\theta\in\Theta} \operatorname{TaskCap}_{A_\theta}(y) \geq \frac{\|y\|_2}{\sqrt{m\kappa}}.$$

For a sign field the right-hand side is $1/\sqrt{\kappa}$. For the centered indicator of a fixed-density target sector it is $\sqrt{\nu(T)(1-\nu(T))/\kappa}$. This estimate concerns the capacity functional only; it does not by itself give a statistical minimax lower bound or require a covering argument.

Proof. Let $\lambda_{\min}$ and $\lambda_{\max}$ be the extremal eigenvalues of A_θ. Then

$$y^\mathsf{T} A_\theta y \geq \lambda_{\min}\|y\|_2^2, \qquad \operatorname{tr}(A_\theta^{-1}) \geq \frac{m}{\lambda_{\max}}.$$

Substitution into the definition of TaskCap gives

$$\operatorname{TaskCap}_{A_\theta}(y) \geq \frac{\|y\|_2}{\sqrt{m}}\sqrt{\frac{\lambda_{\min}}{\lambda_{\max}}} \geq \frac{\|y\|_2}{\sqrt{m\kappa}}.$$

Taking the infimum over θ proves the first claim. The two specializations follow from $\|y\|_2^2 = m$ for signs and $\|y\|_2^2 = m\nu(T)(1-\nu(T))$ for a centered fixed-density indicator.

□

IV.3.2 Kernel Ridge Bounds at a Saturated Representative

For a fixed budget-admissible kernel, the learning bound is the statistical form of the saturated extraction bound in Part III. The effective dimension refines the trace/Rademacher estimate and attains the corresponding minimax rate. Derivability, admissibility, and alignment determine whether the kernel's spectral mass contributes to $I_n^{\mathrm{sat}}(B)$.

The kernel ridge estimator is the regularized learner evaluated by the effective-dimension bound.

Definition IV.3.10 (Kernel ridge estimator)**.** Given samples $(x_i, y_i)_{i=1}^m$ from $Y = f_*(X) + \xi$, with $\mathbb{E}[\xi \mid X] = 0$ and $\mathbb{E}[\xi^2 \mid X] \leq \sigma^2$, kernel ridge regression chooses

$$\widehat{f}_\lambda = \arg\min_{f\in\mathcal{H}_k} \frac{1}{m}\sum_{i=1}^m (f(x_i) - y_i)^2 + \lambda\|f\|_{\mathcal{H}_k}^2.$$

Empirical alignment measures the observed target mass carried by the learned kernel.

Definition IV.3.11 (Empirical kernel-target alignment)**.** Let $H = I - m^{-1}\mathbf{1}\mathbf{1}^T$, $K_c = HKH$, and $y_c = Hy$. The centered kernel-target alignment is

$$A_m(K, y) = \frac{\langle K_c, y_c y_c^T\rangle_F}{\|K_c\|_F\,\|y_c y_c^T\|_F}.$$

This is the centered kernel–target alignment of Cristianini et al. [25]. It measures the empirical association between the pairwise kernel geometry and label agreement. The statistic contains no information about unobserved labels and is used only to select among kernels satisfying the preceding provenance conditions.

The kernel-ridge theorem transfers a fixed-kernel oracle certificate into the structural ledger without changing its probability event or units.

Theorem IV.3.12 (Transfer of a certified effective-dimension bound). *Fix a kernel k, ridge scale λ, regression function $f_* \in L^2(\mu)$, and confidence event $\mathcal{E}_{k,\lambda}$ with $\Pr(\mathcal{E}_{k,\lambda}) \geq 1 - \delta$. A* fixed-kernel oracle certificate *consists of represented constants $C_{\text{var}}, C_{\text{bias}} < \infty$ and a proof, on that same event, of*

$$\|\widehat{f}_\lambda - f_*\|^2_{L^2(\mu)} \leq C_{\text{var}} \frac{\sigma^2 N_{\text{eff}}(\lambda)}{m} + C_{\text{bias}} B_k(\lambda; f_*). \tag{227}$$

The event may be supplied by any declared operator-concentration theorem, but its hypotheses, constants, confidence allocation, and covariance/noise normalization belong to the complete record. Then the saturated fixed-kernel risk coordinate obeys, on $\mathcal{E}_{k,\lambda}$,

$$\|\widehat{f}_\lambda - f_*\|^2_{L^2(\mu)} \leq C_{\text{var}} \frac{\sigma^2 N_{\text{eff}}(\lambda)}{m} + C_{\text{bias}} B_k(\lambda; f_*),$$

where $B_k(\lambda; f_*) = \|(I - T_k(T_k + \lambda I)^{-1}) f_*\|^2_{L^2(\mu)}$ is the source residual. If a named source/eigenvalue class also supplies a matching minimax lower certificate with constants $c_{\text{low}}, C_{\text{high}} > 0$, optimizing the displayed upper certificate is rate-matched only for that declared class and those constants.

Proof. Equation (227) is already a population bound on the named event and in the named norm. Entering the represented kernel, event, and constants into the saturated ledger changes neither side, so the displayed structural coordinate follows verbatim. Optimization over λ is permitted only after the record charges that selection or the certificate is simultaneous in λ. A matching lower certificate, when separately supplied for the same source/eigenvalue class, gives the stated rate comparison. No unnamed concentration or minimax theorem is used by this transfer step.

□

Corollary IV.3.13 (Source-condition learning rate). *Assume $f_* = T_k^r g$ with $0 < r \leq 1$, source radius $\|g\|_{L^2} \leq R_s$, and Mercer eigenvalues $\eta_j \asymp j^{-2\beta}$ with $\beta > 1/2$. Assume also that* (227) *holds at the selected λ_m, or simultaneously over the represented candidate grid, with C_{var} and C_{bias} bounded independently of m. The selection of λ_m must be predeclared, independently certified, or charged in the complete record. Then the source residual satisfies $B_k(\lambda; f_*) \leq C_r R_s^2 \lambda^{2r}$, while $N_{\text{eff}}(\lambda) \asymp \lambda^{-1/(2\beta)}$. On the oracle-certificate event, optimizing the fixed-kernel risk gives*

$$\lambda_m \asymp m^{-1/(2r+1/(2\beta))}, \qquad \|\widehat{f}_{\lambda_m} - f_*\|^2_{L^2} \lesssim m^{-2r/(2r+1/(2\beta))} = m^{-4\beta r/(4\beta r+1)},$$

with an implicit constant determined by the declared spectral, source, C_{var}, and C_{bias} constants. This establishes the quantitative correspondence between extraction degree and the source condition: greater smoothness relative to a budget-admissible kernel yields a smaller effective degree and a faster rate.

Proof. Write the target in the Mercer basis. The source condition $f_* = T_k^r g$ gives coefficients $a_j = \eta_j^r g_j$, so the ridge bias is bounded by $C_r R_s^2 \lambda^{2r}$. The eigenvalue law $\eta_j \asymp j^{-2\beta}$ gives $N_{\text{eff}}(\lambda) = \sum_j \eta_j/(\eta_j + \lambda) \asymp \lambda^{-1/(2\beta)}$. The certified kernel-ridge inequality therefore gives, on its named event, risk of order $\lambda^{2r} + m^{-1}\lambda^{-1/(2\beta)}$. Equating the two terms yields $\lambda_m \asymp m^{-1/(2r+1/(2\beta))}$, and substituting this value gives $m^{-2r/(2r+1/(2\beta))} = m^{-4\beta r/(4\beta r+1)}$.

□

Corollary IV.3.14 (Alignment-calibrated kernel-selection diagnostic)**.** *For kernel selection, define the empirical diagnostic*

$$\mathcal{B}_\lambda(K,y) = \frac{\sigma^2 N_{\mathrm{eff}}(\lambda)}{m\,\max\{A_m(K,y),a_0\}} + (1-m_k(\lambda))\|y_c\|_m^2 + \sigma^2,$$

where $a_0 > 0$ is a declared minimum alignment floor. When K is derivable under the declared budget structure, $A_m(K,y)$ is bounded below, and $m_k(\lambda)$ is close to one at the declared total learning budget, the score records a within-budget candidate for captured structure. It is not an excess-risk upper bound unless a separate theorem derives the alignment denominator and source term in this form. Small A_m does not certify Part III extraction, and alignment obtained only above the declared total budget is inadmissible for the original presentation.

Proof. The effective-dimension, captured-mass, and noise entries are measurable diagnostics motivated by the kernel-ridge decomposition. The alignment denominator is not present in the kernel-ridge oracle inequality; it records the separate Part III requirement that captured spectral mass be oriented toward the target rather than concentrated in irrelevant low modes. The floor a_0 keeps the certificate finite when the empirical alignment is null. If the kernel is derivable under the declared budget structure and the alignment and captured mass conditions hold, each term in $\mathcal{B}_\lambda$ is a charged statistical or structural diagnostic. A risk or minimax conclusion requires the explicit hypotheses of Lemma IV.3.15.

□

Lemma IV.3.15 (Kernel minimax lower bound)**.** *Fix a budget-reachable kernel class $\mathcal{K}$ whose data-dependent selection cost lies in the declared transfer class. Assume a uniform minimax lower-bound theorem for the declared regression class: for each fixed $k \in \mathcal{K}$, every admissible estimator has excess risk at least*

$$c_0 \min\left\{\frac{\sigma^2 N_{\mathrm{eff},k}(\lambda)}{m} + B_k(\lambda; f_*),\ 1\right\}$$

for the relevant choice of λ, with the same constant $c_0 > 0$. Suppose every reachable kernel either has $N_{\mathrm{eff},k}(\lambda)/m \geq c_1 > 0$, or has $B_k(\lambda; f_*) \geq c_1 > 0$, uniformly over its budget-feasible ridge scales. Suppose also that a declared finite-net or information-theoretic selection lower bound makes this fixed-kernel lower bound uniform over the data-dependent choice of k. Then no estimator in the stated class learns the target to vanishing excess risk.

Proof. For each fixed reachable kernel and budget-feasible ridge scale, one of the two alternatives makes the assumed minimax lower bound at least c_0c_1 (capped by c_0). The assumed selection lower bound makes this conclusion simultaneous for the learner's data-dependent kernel choice. Hence the infimum excess risk over the declared learner class is bounded away from zero. Alignment or empirical capture alone would not imply this conclusion; the uniform source-residual/capacity and selection hypotheses are essential.

□

Corollary IV.3.16 (Learning lower bound from family-specific structural negligibility)**.** *Fix a presented task family and a budget-reachable kernel or learned-kernel class $\mathcal{K}$. Suppose a later family-specific argument proves that every representative within budget in $\mathcal{K}$ has negligible counted structural capture in $I_n^{\mathrm{sat}}(B)$, unless its effective dimension or selection cost exceeds the declared budget. Assume additionally a proved capture-to-source bridge for this kernel class: negligible*

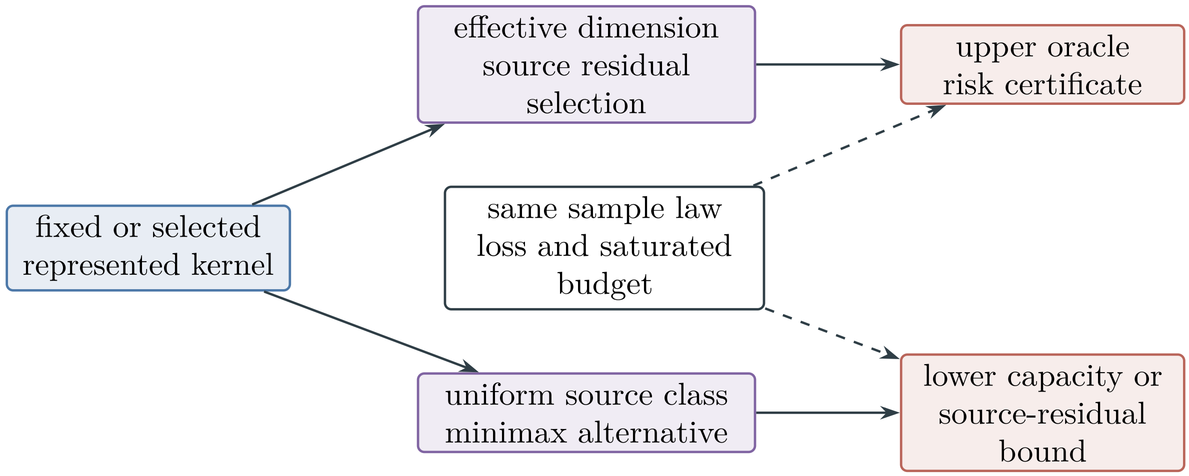

Figure 48: Two-sided learning control requires common scope. Effective dimension and source residual give an upper oracle inequality, whereas a uniform source-class or minimax argument gives the hard side. Both are interpretable together only after their kernel family, population law, loss, and saturated budget have been matched.

counted capture implies a source residual bounded below by a fixed positive constant. Then no budget-admissible learner whose fixed or learned kernel lies in $\mathcal{K}$ and satisfies the hypotheses of the kernel minimax and selection theorem learns the target to vanishing excess risk.

Proof. Condition on the budget-admissible terminal kernel or on a net element of the learned-kernel class. If its effective dimension exceeds the declared statistical budget, the capacity term in the kernel minimax lower bound cannot vanish within the available sample resources. If its effective dimension lies within budget, the family-specific structural bound says that the kernel has negligible counted capture, and the assumed capture-to-source bridge bounds the source residual below. The learned-operator selection hypothesis accounts for data-dependent selection. Thus budget-admissible learning would imply a structural representative within budget that the family-specific bound excludes.

□

Under the source-class, concentration, and minimax hypotheses stated above, effective dimension provides matching-order upper and lower bounds. Alignment determines whether the selected spectral subspace is target-relevant: a low-capacity kernel misaligned with the target does not contribute to $I_n^{\mathrm{sat}}(B)$, while a highly aligned kernel whose derivation, evaluation, selection, or effective dimension exceeds the declared budget is inadmissible for the original presentation.

Figure 48 aligns the upper and lower learning certificates on the same represented kernel family and budget.

IV.4 The Kernel Trick as the Lift Channel

The Lift channel of Part II permits an admissible enlargement of the observable configuration while preserving the original source of target information. Kernel methods give the corresponding statistical construction.

Lemma IV.4.1 (Kernel trick as composition-bounded Lift). *Let $k(x,z) = \langle \Psi(x), \Psi(z) \rangle$ be an admissible feature lift, possibly infinite-dimensional, whose kernel evaluations are budget-admissible public computations from the presentation or from the learned lifted configuration. For any empirical loss depending only on the sample evaluations and any regularizer that is a strictly increasing function of the RKHS norm, every empirical minimizer has the form*

$$\widehat{f}(x) = \sum_{i=1}^{m} \alpha_i k(x_i, x).$$

Thus the computational lift is confined to the sample span, while $N_{\text{eff}}(\lambda)$, rather than the ambient feature dimension, controls statistical complexity. The lift contributes to the structural count only when the feature map preserves the observable/terminal interface required by Part II's valid-lift definition.

Proof. This is the representer theorem [46]. Decompose any candidate $f \in \mathcal{H}_k$ into $f = f_S + f_\perp$, where f_S lies in the span of $\{k_{x_i}\}_{i=1}^m$ and $f_\perp$ is orthogonal to that span. Since $f_\perp(x_i) = \langle f_\perp, k_{x_i} \rangle = 0$, it does not change the empirical loss, while it increases the RKHS norm unless it is zero. Hence the minimizer lies in the sample span. The KRR bound in Section IV.3 then charges the lifted span by the effective dimension of its spectrum.

□

Corollary IV.4.2 (Budget lower bound for feature lifts). *Assume the uniform source-class, fixed-kernel minimax, and data-dependent selection hypotheses of Lemma IV.3.15. If every admissible aligned kernel has $N_{\text{eff}}(\lambda)/m \geq c_1 > 0$ at every budget-feasible scale, or has the alternative source-residual lower bound in that theorem, then samples within the budget cannot learn the declared source class to vanishing excess risk. The conclusion applies to infinite-coordinate feature lifts because it concerns effective dimension and source residual, not ambient feature dimension.*

Proof. Apply Lemma IV.3.15. Its uniform fixed-kernel bound supplies the capacity/source-residual alternative, and its selection hypothesis makes that alternative simultaneous for the data-dependent terminal kernel. The representer theorem identifies the empirical sample span but is not itself used as a minimax lower bound. Infinite ambient dimension does not alter the stated uniform effective-dimension and source-class hypotheses.

□

The kernel trick is therefore an instance of valid Lift whenever the kernel is publicly generated and terminal-compatible. Composition may access a high- or infinite-dimensional feature representation, but the target information available within budget is precisely the counted spectral mass at the corresponding effective dimension. The remaining mass is a source residual or part of J_{res}, rather than an additional source.

IV.5 Neural Networks and Learned Kernels

IV.5.1 Kernel evolution during training

A budget-admissible oracle-free neural-network training procedure constructs represented terminal outputs and readouts on its clocked training transcript carrier; ordinary polynomial-time training is the $\mathcal{B}_{\text{poly}}$ specialization. The empirical neural tangent kernel (NTK) is the Gram kernel of the tangent features used by gradient descent.

The empirical neural tangent kernel records inner products of the network's tangent features during training.

Definition IV.5.1 (Empirical neural tangent kernel)**.** For a network $f(x;\theta)$, the empirical NTK at parameter value θ is

$$\Theta_\theta(x,z) = \langle \nabla_\theta f(x;\theta), \nabla_\theta f(z;\theta) \rangle .$$

The Gram matrix Θ_t generally evolves during training and is approximately constant in the lazy-training regime [40].

The movement trajectory tracks how training changes the learned kernel and its target alignment.

Definition IV.5.2 (Kernel movement and alignment trajectory)**.** On a sample, define

$$\delta_t = \frac{\|\Theta_t - \Theta_0\|_F}{\|\Theta_0\|_F}, \qquad A_t = A_m(\Theta_t, y).$$

Small δ_t together with nearly constant A_t is an empirical diagnostic consistent with lazy training, but is not by itself a prediction-linearization theorem. Order-one kernel movement accompanied by increasing alignment is evidence of feature learning.

Lemma IV.5.3 (Lazy-training regime)**.** *Assume the complete network record contains a verified population prediction comparison*

$$\|\widehat{f}_{\mathrm{NN}} - \widehat{f}_{\Theta_0}\|_{L^2(\mu)} \le \varepsilon_{\mathrm{lazy}},$$

where $\widehat{f}_{\Theta_0}$ is the fixed-kernel flow or ridge predictor. Assume on the same event that Θ_0 carries the certified fixed-kernel bound (227)*, with constants C_{var} and C_{bias}. Then*

$$\|\widehat{f}_{\mathrm{NN}} - f_*\|^2_{L^2(\mu)} \le 2C_{\mathrm{var}} \frac{\sigma^2 N_{\mathrm{eff}}(\Theta_0, \lambda)}{m} + 2C_{\mathrm{bias}} B_{\Theta_0}(\lambda; f_*) + 2\varepsilon^2_{\mathrm{lazy}}.$$

The empirical diagnostic $\|\Theta_t - \Theta_0\|_F = o(\|\Theta_0\|_F)$ alone does not imply these hypotheses; in particular, it supplies neither spectral-scale operator control nor a population generalization theorem [4, 22, 40, 49].

Proof. By hypothesis, the network predictor is within $\varepsilon_{\mathrm{lazy}}$ in L^2 of the fixed-kernel predictor. Thus

$$\|\widehat{f}_{\mathrm{NN}} - f_*\|^2_{L^2(\mu)} \le 2\|\widehat{f}_{\Theta_0} - f_*\|^2_{L^2(\mu)} + 2\varepsilon^2_{\mathrm{lazy}}.$$

Substitution of (227) gives the stated bound. No conclusion is drawn from Frobenius kernel movement alone.

□

IV.5.2 Feature learning and selection cost

The feature-learning bound charges movement of the learned operator together with its selection complexity.

Lemma IV.5.4 (Feature-learning regime)**.** *Suppose budget-admissible oracle-free training selects a represented terminal kernel Θ_T from an admissibly parameterized feature-map class $\mathcal{K}_\Theta$ with covering number $N_\rho = \mathcal{N}(\mathcal{K}_\Theta, \rho) < \infty$. The terminal kernel, training transcript, precision, and selection cover are charged as a Part III saturated representative. Let $\mathcal{K}_\rho$ be the represented ρ-net. Assume that one simultaneous event $\mathcal{E}_\rho$, of probability at least $1-\delta$, carries for every $K \in \mathcal{K}_\rho$ the fixed-kernel certificate*

$$\mathrm{ExcessRisk}(K) \leq C_{\mathrm{var}} \frac{\sigma^2 N_{\mathrm{eff}}(K,\lambda)}{m} + C_{\mathrm{bias}} B_K(\lambda; f_*) + r_{\mathrm{sel}}. \tag{228}$$

Here $C_{\mathrm{var}}, C_{\mathrm{bias}}$, the confidence split, and the represented radius r_{sel} are part of the record. For an independent audit sample of size m_a and a loss in an interval of length L_{loss}, one admissible simultaneous empirical-to-population radius is

$$r_{\mathrm{sel}} = L_{\mathrm{loss}} \sqrt{\frac{\log(2N_\rho/\delta)}{2m_a}}.$$

Assume further that the fixed-kernel right side and the corresponding KRR prediction risk are jointly L_{sel}-Lipschitz in the declared kernel operator metric. Let represented nonnegative $\varepsilon_{\mathrm{est}}$ and $\varepsilon_{\mathrm{opt}}$ bound kernel estimation and optimization error in the same excess-risk units. Then, on $\mathcal{E}_\rho$,

$$\mathrm{ExcessRisk} \leq C_{\mathrm{var}} \frac{\sigma^2 N_{\mathrm{eff}}(\Theta_T,\lambda)}{m} + C_{\mathrm{bias}} B_{\Theta_T}(\lambda; f_*) + r_{\mathrm{sel}} + L_{\mathrm{sel}}\rho + \varepsilon_{\mathrm{est}} + \varepsilon_{\mathrm{opt}}.$$

The first two terms are the learned kernel's certified capacity and source residual. The radius and stability term charge selection from the finite cover. Any target-relevant movement not represented by the terminal kernel, valid lifts, or terminal readouts remains residual accounting rather than an independent source.

Proof. Choose $K_T^\rho \in \mathcal{K}_\rho$ within distance ρ of Θ_T. On $\mathcal{E}_\rho$, apply (228) to K_T^ρ. Joint Lipschitz stability transports its right side and prediction risk to Θ_T at total cost at most $L_{\mathrm{sel}}\rho$; the two named algorithmic errors add in the same units. For the displayed independent audit option, Hoeffding's inequality gives failure probability at most $2\exp(-2m_a r^2/L_{\mathrm{loss}}^2)$ for one net point. Substituting the displayed r_{sel} and taking a union bound over N_ρ points gives failure probability at most δ.

Part II supplies the operator embedding on the clocked training transcript carrier. Part III includes the terminal kernel, transcript, cover resolution, precision, and readout in the saturated derivability accounting before the learned kernel can contribute to extraction.

□

Figure 49 organizes the two represented kernel training regimes of Lemmas IV.5.3 and IV.5.4.

Lemma IV.5.5 (Observable-success accountability and residual audit)**.** *Let $\{I_n\}$ be a presented task family with exposed target sectors $T_I \subseteq X_I$ and declared budget structure $\mathcal{B}$. Let $\mathcal{A}$ be a uniform oracle-free represented computation whose operational transcript, lifted coordinates, precision, and output map have cost within $\mathcal{B}$, with finite normalized horizon $H_B(n)$. If*

$$\Pr[\mathcal{A}(I) \in T_I] \geq \Gamma(n),$$

then its clocked transcript representation induces the exact first-hit record

$$q_{\mathcal{A}} \in \mathcal{Q}^{\mathrm{sat}}_{n,\leq B_{\mathrm{hit}}}, \qquad B_{\mathrm{hit}} = \Psi_{\mathrm{hit}}(b(n), n),$$

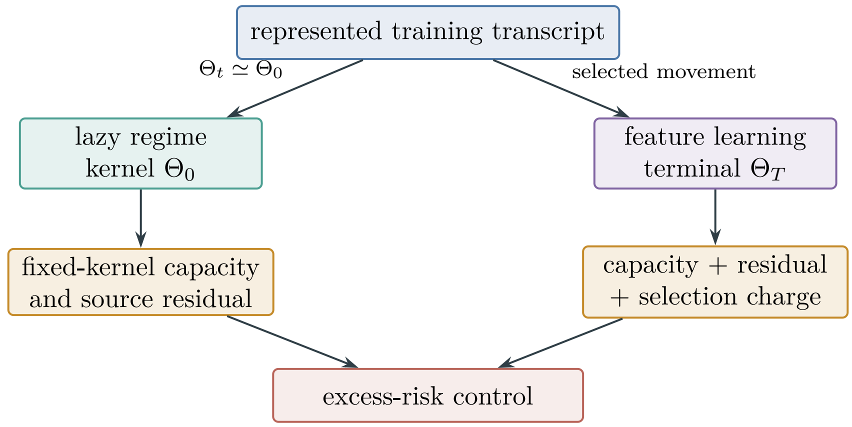

Figure 49: Under the hypotheses of Lemmas IV.5.3 and IV.5.4, lazy training is controlled by the initial kernel, whereas feature learning uses the terminal kernel and pays its represented selection charge.

and, pointwise in the instance,

$$V_I(q_{\mathcal{A}}) \geq \Gamma_{\text{hit}}(n) := \frac{\Gamma(n)}{eH_B(n)}. \tag{229}$$

This counted exact-quotient witness is unconditional within the declared finite resource envelope. In the polynomial instance, $B_{\text{hit}} = n^{O_C(1)}$ for a time-n^C algorithm. Separately, for every residual audit class $\mathcal{H}_B$ generated from the same lifted transcript and represented at budget B_{hit}, put

$$\mathcal{P}_{\mathcal{H}_B}(M, \delta, \rho) = c\left(\sqrt{\frac{N_{\mathcal{H}_B} + \log \mathcal{N}(\mathcal{H}_B, \rho) + \log(1/\delta)}{M}} + \rho\right),$$

where $N_{\mathcal{H}_B}$ is the effective mode count of the audited residual span, M is the number of audit samples or independent family instances, and ρ is the operator-net resolution. For the following audit assertion, assume the independent-sample, bounded-statistic, and Lipschitz/cover hypotheses of Corollary IV.3.7, together with a proved class bound

$$\mathfrak{R}_M(\mathcal{H}_B) \leq c\left(\sqrt{\frac{N_{\mathcal{H}_B} + \log \mathcal{N}(\mathcal{H}_B, \rho)}{M}} + \rho\right).$$

With probability at least $1 - \delta$ over the audit sample, every statistic in that declared class has population audited capture at most its empirical audited capture plus $\mathcal{P}_{\mathcal{H}_B}(M, \delta, \rho)$. Thus an audit resolves effects at the forced-witness scale when $\mathcal{P}_{\mathcal{H}_B} = o(\Gamma_{\text{hit}})$. If the penalty is not smaller than that scale, this audit is inconclusive; failure of a sufficient Rademacher bound is not an indistinguishability or hardness theorem.

A terminal observable outside $\mathcal{D}_{n, \leq B_{\text{hit}}}$ is inadmissible for the original presentation. It requires oracle or advice data, a changed primitive signature, or discovery, representation, precision, or application cost above the declared budget.

Proof. Lemma III.20.2 applies to the complete configuration record and gives both the cost $B_{\text{hit}} = \Psi_{\text{hit}}(b(n), n)$ and (229). This step already includes every exposed channel, residual interaction, memory effect, and terminal readout; no componentwise transfer premise is used.

For a separately declared residual audit class, the explicitly assumed Rademacher estimate and Corollary IV.3.7 give the stated uniform empirical-to-population inequality. Comparing its penalty to Γ_{hit} gives the audit-resolution statement and no converse lower bound.

The final assertion follows from the admissibility dichotomy of Part III. A terminal object absent from the derivability filtration of the original presentation is not a budget-admissible representative: it belongs to an oracle or advice extension, changes the primitive signature, or requires an over-budget construction.

□

Corollary IV.5.6 (Parameter-cover selection cost)**.** *If the parameter domain has dimension p, radius R_Θ, and the map $\theta \mapsto \Theta_\theta$ is L_K-Lipschitz in empirical operator norm, then*

$$\log \mathcal{N}(\mathcal{K}_\Theta, \rho) \le p \log\left(1 + \frac{2R_\Theta L_K}{\rho}\right).$$

For neural networks, L_K may be upper-bounded by layer Lipschitz or spectral-norm controls. For the independent bounded-loss audit option in Lemma IV.5.4, substitution into its simultaneous radius gives the explicit certificate

$$r_{\mathrm{sel}} \le L_{\mathrm{loss}} \sqrt{\frac{\log(2/\delta) + p\log(1 + 2R_\Theta L_K/\rho)}{2m_a}}.$$

Proof. A radius-R_Θ subset of $\mathbb{R}^p$ has a Euclidean ϵ-net of cardinality at most $(1 + 2R_\Theta/\epsilon)^p$. If $\theta \mapsto \Theta_\theta$ is L_K-Lipschitz in empirical operator norm, an $\epsilon = \rho/L_K$ parameter net maps to a ρ-net of terminal kernels. Taking logarithms gives the displayed covering bound. Substituting it into the exact radius in Lemma IV.5.4 gives the second display.

□

Corollary IV.5.7 (Feature-learning capacity tradeoff)**.** *Under the hypotheses of Lemma IV.5.4, write*

$$B_\Theta(\lambda) = B_\Theta(\lambda; f_*).$$

Assume the initial kernel carries the same fixed-kernel certificate constants $C_{\mathrm{var}}, C_{\mathrm{bias}}$ and no corresponding selection charges. The terminal-kernel upper certificate is no larger than the initial-kernel upper certificate whenever

$$C_{\mathrm{var}} \frac{\sigma^2}{m} \left[N_{\mathrm{eff}}(\Theta_0, \lambda) - N_{\mathrm{eff}}(\Theta_T, \lambda)\right] + C_{\mathrm{bias}} \left[B_{\Theta_0}(\lambda) - B_{\Theta_T}(\lambda)\right] \ge r_{\mathrm{sel}} + L_{\mathrm{sel}}\rho + \varepsilon_{\mathrm{est}} + \varepsilon_{\mathrm{opt}}.$$

When the source terms agree and the other errors are negligible, this reduces to

$$N_{\mathrm{eff}}(\Theta_0, \lambda) - N_{\mathrm{eff}}(\Theta_T, \lambda) \ge \frac{m\, r_{\mathrm{sel}}}{C_{\mathrm{var}}\sigma^2}.$$

The alignment-adjusted quantity

$$\Delta\left(\frac{N_{\mathrm{eff}}}{A_m}\right) = \frac{N_{\mathrm{eff}}(\Theta_0, \lambda)}{A_m(\Theta_0, y)} - \frac{N_{\mathrm{eff}}(\Theta_T, \lambda)}{A_m(\Theta_T, y)}$$

remains the structural diagnostic of Corollary IV.3.14; it is not substituted into the risk inequality.

Proof. Subtract the terminal-kernel certificate in Lemma IV.5.4 from the initial fixed-kernel certificate. The resulting difference is exactly the left side of the displayed condition, while the terminal-only charges form its right side. The capacity-only specialization sets the source difference and all charges except r_{sel} to zero. No alignment denominator appears in either oracle inequality. □

IV.5.3 Composite capacity control

The composite bound reports the stronger of the learned-kernel and network-capacity controls.

Lemma IV.5.8 (Composite neural-network bound)**.** *Assume the hypotheses of Lemma IV.5.4 for the trained network's terminal kernel. Let $\mathcal{E}_{\mathrm{ker}}$ be its declared simultaneous event, of probability at least $1-\delta_{\mathrm{ker}}$, on which the same declared population risk $R(f)$ is bounded by*

$$\mathcal{B}_{\mathrm{ker}} = C_{\mathrm{var}}\frac{\sigma^2 N_{\mathrm{eff}}(\Theta_T,\lambda)}{m} + C_{\mathrm{bias}}B_{\Theta_T}(\lambda; f_*) + r_{\mathrm{sel}} + L_{\mathrm{sel}}\rho + \varepsilon_{\mathrm{est}} + \varepsilon_{\mathrm{opt}}.$$

Independently, suppose a named represented margin theorem supplies a finite quantity $\mathcal{B}_{\mathrm{margin}}$ and an event $\mathcal{E}_{\mathrm{margin}}$, of probability at least $1-\delta_{\mathrm{margin}}$, such that

$$R(f) \le \mathcal{B}_{\mathrm{margin}} \quad \text{on } \mathcal{E}_{\mathrm{margin}}.$$

That certificate must state its margin convention, loss conversion, norm normalization, architectural hypotheses, constants, and confidence term; for example it may instantiate a fully stated spectral-margin theorem [9]. It must cover the actual data-dependent trained weights, rather than only a fixed network chosen before the sample.

Then, with probability at least $1-\delta_{\mathrm{ker}}-\delta_{\mathrm{margin}}$,

$$R(f) \le \min\{\mathcal{B}_{\mathrm{ker}}, \mathcal{B}_{\mathrm{margin}}\}.$$

The minimum is permitted because both quantities bound the same population risk on one simultaneous event. Terms already included in one certificate are not added again to the other.

Proof. On $\mathcal{E}_{\mathrm{ker}} \cap \mathcal{E}_{\mathrm{margin}}$, both displayed inequalities hold for the same random trained network and the same population risk. Hence their minimum is an upper bound. The union bound gives $\Pr(\mathcal{E}_{\mathrm{ker}} \cap \mathcal{E}_{\mathrm{margin}}) \ge 1-\delta_{\mathrm{ker}}-\delta_{\mathrm{margin}}$. □

IV.6 Estimating the Learned Effective Degree

The spectrum of the learned kernel provides a post-training estimate of the effective degree captured by the model.

Definition IV.6.1 (Label spectral profile and effective degree)**.** Let $K=\sum_{i=1}^m s_i u_i u_i^T$ be the centered learned Gram matrix, with $s_1 \ge \cdots \ge s_m \ge 0$. For centered labels y_c, define

$$\beta_i = \frac{\langle y_c, u_i\rangle^2}{\|y_c\|^2}, \qquad \sum_i \beta_i = 1.$$

The cumulative label mass in the top k modes is $M(k)=\sum_{i\le k}\beta_i$. The effective degree at tolerance ε is

$$d_{\text{eff}}(\varepsilon) = \min\{k : M(k) \geq 1 - \varepsilon\}.$$

The smooth participation form is $d_{\text{part}} = (\sum_i \beta_i)^2 / \sum_i \beta_i^2$.

Lemma IV.6.2 (Spectral upper bound conditional on profile concentration)**.** *Let $M_{\text{pop}}(k)$ and $M_{\text{emp}}(k)$ be cumulative population and empirical label-mass profiles in corresponding ordered spectral bands. Assume that an independently established concentration event of probability at least $1 - \delta$ gives*

$$\sup_k |M_{\text{pop}}(k) - M_{\text{emp}}(k)| \leq \Delta_m.$$

Then, on that event,

$$d_{\text{eff}}^{\text{pop}}(\varepsilon + \Delta_m) \leq \widehat{d}_{\text{eff}}(\varepsilon),$$

For bounded kernels, a concrete value of Δ_m may be supplied by an operator-concentration and Davis–Kahan estimate for hard eigengap-separated bands, or by a Lipschitz functional-calculus estimate for declared smoothed bands.

Proof. Put $k = \widehat{d}_{\text{eff}}(\varepsilon)$. By definition, $M_{\text{emp}}(k) \geq 1 - \varepsilon$. The assumed event gives $M_{\text{pop}}(k) \geq 1 - \varepsilon - \Delta_m$, so the smallest population prefix containing mass $1 - (\varepsilon + \Delta_m)$ has size at most k. This is the displayed inequality. Matrix Bernstein and Davis–Kahan are standard ways to verify the stated event when their boundedness and eigengap hypotheses hold [26, 75].

□

Lemma IV.6.3 (Generalization certifies captured mass)**.** *Assume the centered target is normalized so that $\|f_*\|^2_{L^2(\mu)} = 1$, and measure squared-loss test error r and noise level σ^2 in these units. Assume an independent test bound gives $R(\widehat{f}) \leq r$, and assume a separately proved lower approximation inequality*

$$R(\widehat{f}) \geq 1 - m_{\Theta_T}(\lambda) - C\frac{\sigma^2 N_{\text{eff}}(\lambda)}{m} - \sigma^2 - o(1).$$

Then the ridge-feasible band captured at least

$$m_{\Theta_T}(\lambda) \geq 1 - r - C\frac{\sigma^2 N_{\text{eff}}(\lambda)}{m} - \sigma^2 - o(1)$$

of the normalized target mass, whenever the kernel-bound hypotheses of Section IV.5 apply.

Proof. Combine the assumed upper and lower bounds:

$$1 - m_{\Theta_T}(\lambda) - C\frac{\sigma^2 N_{\text{eff}}(\lambda)}{m} - \sigma^2 - o(1) \leq r.$$

Rearranging proves the claim. The lower approximation inequality is essential; an upper kernel-risk bound alone cannot be inverted to infer captured mass.

□

Lemma IV.6.4 (Population–empirical prefix-index sandwich)**.** *Let $M_{\text{pop}}(k)$ and $M_{\text{emp}}(k)$ be the population and empirical cumulative target masses in the first k kernel modes, and suppose*

$$\sup_k |M_{\text{pop}}(k) - M_{\text{emp}}(k)| \leq \Delta_m.$$

For $a \in [0,1]$, let $k_{\mathrm{pop}}(a)$ and $k_{\mathrm{emp}}(a)$ be the smallest prefix indices whose respective cumulative masses are at least a, using the endpoint conventions $k(a) = 0$ for $a \le 0$ and $k(a) = \infty$ for $a > 1$. Then

$$k_{\mathrm{emp}}(a - \Delta_m) \le k_{\mathrm{pop}}(a) \le k_{\mathrm{emp}}(a + \Delta_m).$$

When Lemma IV.6.3 supplies the mass level $q = 1 - r - C\sigma^2 N_{\mathrm{eff}}(\lambda)/m - \sigma^2 - o(1)$, set $a = q$ to obtain empirical lower and upper indices for the population prefix needed to contain that mass.

Proof. If $M_{\mathrm{pop}}(k) \ge a$, then $M_{\mathrm{emp}}(k) \ge a - \Delta_m$; applying this at $k = k_{\mathrm{pop}}(a)$ gives the left inequality. Conversely, $M_{\mathrm{emp}}(k) \ge a + \Delta_m$ implies $M_{\mathrm{pop}}(k) \ge a$; applying this at $k = k_{\mathrm{emp}}(a + \Delta_m)$ gives the right inequality.

□

Remark IV.6.5 (Scope of finite-sample degree error bars). Bounded kernels and sub-Gaussian labels alone do not determine a universal finite-sample radius for a data-dependent spectral cutoff. A quantitative degree certificate must name, on one simultaneous event,

(i) an operator estimate $\|\widehat{T} - T\|_{\mathrm{op}} \le \varepsilon_T(m, \delta_T)$;
(ii) either a population eigengap $g_\tau > 2\varepsilon_T$ at every hard cutoff used by the procedure, or a displayed Lipschitz constant for a smoothed spectral filter;
(iii) a label-moment estimate in the same population law and norm;
(iv) simultaneous confidence allocation over every selected cutoff, regularization level, architecture, and subsampling rule.

For a hard cutoff, Davis–Kahan converts the first item to projector error at most $2\varepsilon_T/g_\tau$. For a smoothed filter φ, an applicable functional-calculus theorem instead contributes its displayed operator-Lipschitz constant times ε_T. Either term must be added to the separately certified label-moment radius before Lemma IV.6.4 is applied. Effective-dimension and participation-ratio estimates likewise require, respectively, a stated trace/resolvent perturbation bound and a certified positive denominator. DKW, bootstrap, Nyström, and delta-method conclusions are available only through those named hypotheses; this remark asserts no distribution-free rate for them [26, 29, 54].

Independent held-out error is separate. For a $[0,1]$-valued loss and M models fixed before the held-out sample, Hoeffding and a union bound give the explicit simultaneous radius

$$\sqrt{\frac{\log(2M/\delta)}{2m_{\mathrm{test}}}}.$$

Proof. Bounded kernels give operator-norm concentration of the empirical Gram/integral operator by matrix Bernstein [75]. For a fixed spectral band, applying the spectral projector to sub-Gaussian labels gives the stated square-root concentration; hard cutoffs add the Davis–Kahan projector perturbation term [26], while smoothed projectors are Lipschitz and avoid eigengap denominators. Uniform cutoff control follows by applying a DKW-style envelope to the cumulative smoothed spectral distribution, or by a union bound over the finite empirical spectrum. The map $K \mapsto \mathrm{Tr}(K(K + \lambda m I)^{-1})$ is Lipschitz on the positive cone away from the numerical floor with constant controlled by λ^{-1}. The participation ratio is a smooth function of the first two spectral moments, so the delta method gives its rate. Independent test error is an average of bounded or sub-exponential losses, controlled by Hoeffding or Bernstein.

□

Sweeping architectures gives the problem-level degree relative to the model class:

$$d_{\text{eff}}^{\text{problem}}(\varepsilon) = \min_{\text{admissible architectures}} d_{\text{eff}}^{\text{captured}}(\varepsilon).$$

If every admissible architecture has low alignment or requires high effective degree before attaining an error level within budget, the architecture sweep provides an empirical lower-bound diagnostic for that model class.

IV.7 Computational Evaluation Procedure

The preceding quantities can be evaluated from empirical operators and Gram matrices as follows.

- **Initialization analysis.** Compute Θ_0, $N_{\text{eff}}(\lambda)$, $A_m(\Theta_0, y)$, and the label spectral profile before full training.
- **Training-regime classification.** Track $\delta_t = \|\Theta_t - \Theta_0\|_F / \|\Theta_0\|_F$ and A_t. Approximately constant values characterize lazy training, whereas increasing alignment indicates feature learning.
- **Post-training feature-learning bound.** Compare N_{eff}/A at initialization and convergence, then subtract the selection penalty $\sqrt{\log \mathcal{N}/m}$.
- **Composite bound.** Report the minimum of the learned-kernel and spectral-margin bounds.
- **Model-class lower-bound diagnostic.** Compare admissible architectures. If every learned kernel within budget has low alignment or requires high effective degree before capture, the target has no counted low-degree structure in that model class.
- **Fixed-operator analysis.** For any declared operator A, compute $\text{Tr}(A^{-1})$, $y^T A y$, $\text{TaskCap}_A(y)$, and the fallback identity score. This determines whether the operator yields a structural reduction relative to the identity reference or only enlarges the hypothesis class.
- **Random-label null comparison.** For an operator family with a polynomial net, compare $y^T A y$ against the sign-null mean $\text{Tr}(A)$ and variance $2\sum_{i \neq j} A_{ij}^2$. A target whose score stays at the null scale across the net has no counted operator-level discount in that class.
- **Finite-sample observable analysis.** From the exposed feature matrix, quotient fibers, labels, kernel Gram matrix, and audit parameters, compute Part I's $\widehat{\text{ProjMass}}$, $\widehat{\text{ActMass}}$, inert mass, threshold margin, monotone-filtration score, N_{eff}, captured kernel mass, Rademacher penalty, residual-order ceiling, null score, and certified upper bound. These computations implement the budgeted extraction accounting on finite data.

IV.8 Structural Generalization Bounds

This section extends the kernel, NTK, effective-dimension, and feature-learning analysis. It quantifies statistical capacity, structural KL divergence, effective-resistance defect, coefficient selection, defect-geometry profiles, and uncertainty from learning the geometry. Each quantity is computed from a represented operator, unlabeled geometry, labeled sample, or declared model family and is included in the same saturated accounting. These statistical bounds do not replace a family-specific structural lower bound such as Part VIII.

Whenever the required objects are represented, we use the defect-restricted, harmonic-reduced formulation. Effective resistance is the Poincaré component of the more general defect-geometry profile. The global spectral-gap and isotropic-prior expressions serve as comparison bounds when the reduced profile cannot be represented.

IV.8.1 Defect-restricted PAC–Bayes control

We first restrict the PAC–Bayes comparison to the represented defect profile selected by the structural ledger.

Definition IV.8.1 (Defect-restricted structural PAC–Bayes divergence)**.** Let $\mathbf{M}_{\mathcal{D}} \succeq 0$ be the domain Dirichlet operator computed from unlabeled input geometry. Let H be the represented coupled hypothesis subspace after the Abs coordinates and harmonic Geom slack are fixed, and let Π_E project to the represented lumpability-defect directions. Let $R_{E,H} \subseteq H \cap \operatorname{ran} \Pi_E$ be the represented retained subspace, and let $P_{E,H}$ be its orthogonal projection. Define the reduced structural precision, as an operator on $R_{E,H}$, by

$$\mathbf{M}_{E,H} = P_{E,H}\mathbf{M}_{\mathcal{D}}P_{E,H}|_{R_{E,H}} .$$

Put $R_+ = \operatorname{ran} \mathbf{M}_{E,H}$. The structural prior is the proper Gaussian measure on R_+

$$P^{\mathsf{str}}_{E,H} = \mathcal{N}_{R_+}(0, \mathbf{M}^+_{E,H}).$$

There is no improper flat factor on null or harmonic directions. If a model retains such directions, it must assign them a separate proper, label-independent prior and include their KL contribution. For a Gaussian posterior $Q = \mathcal{N}_{R_+}(\widehat{w}, \Sigma_Q)$, with Σ_Q positive definite on R_+, define

$$\mathrm{KL}^{\mathsf{str}}_{E,H}(Q) = \frac{1}{2}\left[\widehat{w}^T\mathbf{M}_{E,H}\widehat{w} + \operatorname{tr}(\mathbf{M}_{E,H}\Sigma_Q) - \log\det{}_+(\mathbf{M}_{E,H}\Sigma_Q) - d_{E,H}\right],$$

where $d_{E,H} = \operatorname{rank}(\mathbf{M}_{E,H})$ and $\det_+$ is the pseudo-determinant on the positive spectral subspace. The mean-dependent part is

$$\frac{1}{2}\widehat{w}^T\mathbf{M}_{E,H}\widehat{w}.$$

For graph or quotient defects this mean term is the effective-resistance energy $\frac{1}{2}\langle E_q, \mathcal{L}^+_W E_q\rangle$ after the represented identification between coefficient directions and defect currents.

Lemma IV.8.2 (PAC–Bayes bound with a defect-restricted prior)**.** *Assume the loss is bounded in* $[0, 1]$ *and the structural prior* $P^{\mathsf{str}}_{E,H}$ *is fixed independently of the labeled sample, for example from a fixed public geometry or an independent unlabeled sample. Then for every posterior* Q *on* R_+*, with probability at least* $1 - \delta$ *over the labeled sample,*

$$R(Q) \le \widehat{R}_S(Q) + \sqrt{\frac{\mathrm{KL}^{\mathsf{str}}_{E,H}(Q) + \log(2\sqrt{m}/\delta)}{2(m-1)}}.$$

If the defect and harmonic projections are unavailable in the representation, the same bound may be applied on the positive spectral range of $\mathbf{M}_{\mathcal{D}}$ *with a proper Gaussian prior there, together with proper priors for any retained null directions. No ordering between the reduced and unreduced KL terms is asserted without the factorization hypotheses of Proposition IV.8.3. When source cells are selected or mixed, the prior and posterior use the cell-mixture decomposition in Eq.* (280)*; the cell-selection divergence and every source-dependent prior construction remain charged.*

Proof. By hypothesis, the prior is fixed independently of the labeled sample to which the inequality is applied. It is therefore an admissible PAC–Bayes prior. McAllester's PAC-Bayes inequality for bounded losses gives the displayed inequality for any fixed legal prior [56, 57]. Substituting $P^{\mathsf{str}}_{E,H}$ gives the KL in the definition above. The Gaussian KL formula on a positive semidefinite

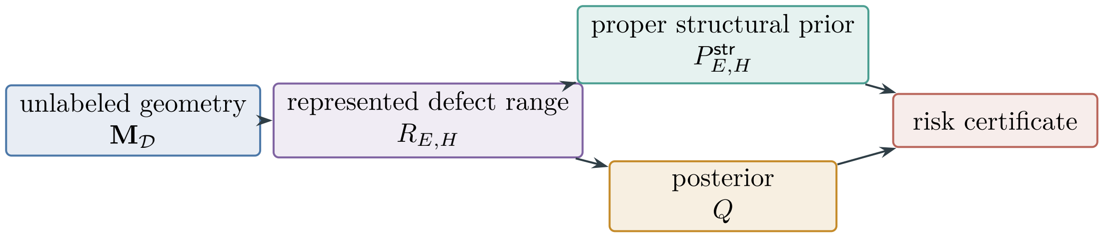

Figure 50: The defect-restricted construction of Definition IV.8.1 and Lemma IV.8.2. The prior is formed on the represented positive defect range; retained null directions require their own proper prior and KL charge.

precision is the standard finite-dimensional formula restricted to R_+; it is a KL between two proper probability measures on the same finite-dimensional space. The mean-dependent term is $\frac{1}{2}\widehat{w}^T \mathbf{M}_{E,H}\widehat{w}$. When the represented coordinates encode a defect current E_q routed through a graph Laplacian $\mathcal{L}_W$, solving the compensation equation $\mathcal{L}_W\phi = E_q$ gives energy $\langle E_q, \mathcal{L}_W^+ E_q\rangle$, so the KL divergence consists of the effective-resistance term together with covariance and normalization terms. An unreduced proper prior is obtained by choosing the full represented positive spectral subspace. It is a valid alternative, but is not automatically weaker in KL.

□

Figure 50 shows how the represented defect geometry enters the PAC–Bayes certificate without adding an uncharged prior direction.

Proposition IV.8.3 (KL comparison under reducing factorization)**.** *Suppose an orthogonal decomposition $R_{\text{full}} = R_{E,H} \oplus R_{\text{rem}}$ reduces $\mathbf{M}_{\mathcal{D}}$, so the full proper Gaussian prior factors as $P_{\text{full}} = P_{E,H} \otimes P_{\text{rem}}$. Suppose also that the full posterior factors as $Q_{\text{full}} = Q_{E,H} \otimes Q_{\text{rem}}$. Then*

$$\mathrm{KL}(Q_{E,H} \| P_{E,H}) \le \mathrm{KL}(Q_{\text{full}} \| P_{\text{full}}),$$

with equality exactly when $\mathrm{KL}(Q_{\text{rem}} \| P_{\text{rem}}) = 0$. For a graph defect, the exact penalty is

$$\mathcal{R}_W(E_q) = \langle E_q, \mathcal{L}_W^+ E_q\rangle = \sum_{k\ge 2} \lambda_k^{-1} \langle E_q, v_k\rangle^2.$$

The global-gap expression $\|E_q\|^2/\lambda_2$ is the comparison upper bound

$$\mathcal{R}_W(E_q) \le \frac{\|E_q\|^2}{\lambda_2};$$

and is exact only when the defect lies in the slowest active eigendirection. Isotropic PAC–Bayes is recovered when the domain precision is isotropic on the represented subspace and the defect/harmonic restrictions are trivial.

Proof. The KL chain rule for the assumed product measures gives

$$\mathrm{KL}(Q_{\text{full}} \| P_{\text{full}}) = \mathrm{KL}(Q_{E,H} \| P_{E,H}) + \mathrm{KL}(Q_{\text{rem}} \| P_{\text{rem}}).$$

Nonnegativity of KL proves the comparison and its equality condition. Without the reducing and factorization hypotheses, compression of a precision matrix need not order Gaussian KL divergences. Diagonalizing $\mathcal{L}_W = \sum_k \lambda_k v_k v_k^T$ gives the displayed effective-resistance identity. Since

$\lambda_k \geq \lambda_2$ for $k \geq 2$, the global-gap inequality follows, with equality exactly when all defect mass is on the λ_2 eigenspace. If $\mathbf{M}_{\mathcal{D}} = \sigma^{-2} I$ and no directions are removed, the structural KL is the ordinary isotropic Gaussian KL.

□

IV.8.2 Selection of structural loss coefficients

The next bounds choose the loss coefficients from the same certified structural quantities.

Lemma IV.8.4 (Selection of loss coefficients)**.** *Let Λ be a finite, declared set of admissible coefficient vectors. For each $\lambda \in \Lambda$, let the training computation produce a posterior Q_λ from a represented loss*

$$\widehat{R}_S(\theta) + \sum_{j=1}^{r} \lambda_j C_j(\theta),$$

where each C_j is a computable certificate term, for example $\mathcal{R}_W(E_q)$, effective dimension, harmonic capacity, or connectivity. With probability at least $1-\delta$, simultaneously for all $\lambda \in \Lambda$,

$$R(Q_\lambda) \leq \widehat{R}_S(Q_\lambda) + \sqrt{\frac{\mathrm{KL}^{\mathsf{str}}_{E,H}(Q_\lambda) + \log(2\sqrt{m}|\Lambda|/\delta)}{2(m-1)}}.$$

Thus the coefficient vector minimizing the simultaneous bound is

$$\lambda^* \in \operatorname*{argmin}_{\lambda \in \Lambda} \left[\widehat{R}_S(Q_\lambda) + \sqrt{\frac{\mathrm{KL}^{\mathsf{str}}_{E,H}(Q_\lambda) + \log(2\sqrt{m}|\Lambda|/\delta)}{2(m-1)}} \right].$$

For continuous coefficient domains, the admissible implementation is a finite ε-net or certified optimizer whose transcript gives a finite cover and a stability error. The cover size and stability error are charged in the saturated budget.

Proof. Apply the structural PAC–Bayes theorem to each fixed $\lambda \in \Lambda$ with confidence $\delta/|\Lambda|$, then use the union bound. The resulting event holds for all candidate posteriors at once. On that event, minimizing the right-hand side yields the smallest upper bound among the declared candidates. The same labeled sample may be used to choose λ^* because the factor $|\Lambda|$ accounts for this finite model-selection step. For a continuous domain, replacing it by a net adds $\log \mathcal{N}(\Lambda, \varepsilon)$ and the certified Lipschitz stability error of the objective. Hence the coefficient-selection cost is explicit in the bound.

□

Corollary IV.8.5 (Two computable coefficient-selection bounds)**.** *Two complementary forms apply to different estimators.*

PAC-Bayes posterior form. *For bounded losses and a posterior Q_λ, use the theorem's square-root bound with $\mathrm{KL}^{\mathsf{str}}_{E,H}$. This gives the stochastic-posterior bound.*

Deterministic effective-resistance form. *For a deterministic coupled model, suppose an independently established excess-risk theorem gives, simultaneously over the declared finite coefficient set, the following right-hand side. Then use*

$$\mathrm{Cert}_{\mathrm{ER}}(\theta_\lambda) = \widehat{R}_S(\theta_\lambda) + c_{\mathrm{loss}}\, \mathcal{R}_W(E_q(\theta_\lambda)) + C\sqrt{\frac{N^{\mathsf{Abs}}_{\mathrm{eff}}(\theta_\lambda) + N^{\mathsf{harm}}_{\mathrm{eff}} + \log(|\Lambda|/\delta)}{m}},$$

where c_{loss} is the declared Lipschitz/Poincare conversion constant for the loss geometry. The selected coefficient minimizes this displayed bound over Λ. This form retains the exact pseudoinverse penalty $\mathcal{R}_W(E_q)$. Replacing it by $\|E_q\|^2/\lambda_2$ gives the weaker global-gap bound.

Proof. The PAC–Bayes form is Lemma IV.8.4. The deterministic conclusion is conditional on the excess-risk inequality stated in the corollary; it is not a consequence of the PAC–Bayes theorem. Once that inequality has been proved for each fixed coefficient vector, a union bound over the finite set Λ supplies the displayed logarithmic term and permits minimization on the simultaneous event. Since $\mathcal{R}_W(E_q)$ is computed by solving $\mathcal{L}_W z = E_q$ on the orthogonal complement of the harmonic kernel, it is computable for finite represented graphs and is the smallest graph-energy penalty compatible with the defect current. The global-gap expression follows by replacing every active eigenvalue by λ_2, hence it can only increase the penalty unless the defect is worst-mode aligned.

□

IV.8.3 Active querying from resistance certificates

Submodularity turns the represented resistance certificate into a greedy querying guarantee.

Lemma IV.8.6 (Greedy querying under a submodular resistance certificate)**.** *For defect/current E_q and domain Laplacian $\mathcal{L}_\mathcal{D}$, the value of querying a point x is the certified reduction*

$$\Delta(x) = \mathcal{C}(S) - \mathcal{C}(S \cup \{x\}) = \langle E_q, \mathcal{L}^+_{\mathcal{D},S} E_q \rangle - \langle E_q, \mathcal{L}^+_{\mathcal{D},S\cup\{x\}} E_q \rangle.$$

Assume, for the declared grounding model and admissible query family, that $F(S) = \mathcal{C}(\varnothing) - \mathcal{C}(S)$ is nonnegative, monotone, and submodular. Then greedy querying achieves the standard $1 - 1/e$ approximation to the optimal batch of the same size, with stopping criterion $\max_x \Delta(x) \le c_{\text{query}}$.

Proof. Grounding a queried point adds a boundary condition to the domain Laplacian and lowers the effective resistance seen by the defect current. Rayleigh monotonicity gives monotonicity for the standard grounding order. Submodularity is an explicit additional hypothesis: it is not implied by Rayleigh monotonicity for an arbitrary query operation or current functional. Under that hypothesis, the Nemhauser–Wolsey greedy theorem gives the batch approximation [60]. The stopping rule is the value-of-information inequality comparing the best marginal certificate drop with query cost.

□

IV.8.4 Intrinsic harmonic dimension and saturation

The intrinsic harmonic dimension identifies the point at which a fixed geometry exhausts its represented target projection.

Definition IV.8.7 (Intrinsic harmonic dimension, ceiling, and coverage gap)**.** Fix a represented unlabeled data geometry $\mathcal{L}_\mathcal{D}$, self-adjoint and nonnegative on the empirical or population task space, with orthonormal eigenbasis (v_k). Let y_T be the centered target contrast or centered regression field, normalized only when explicitly stated. For a declared structural threshold τ above the residual/noise floor, define

$$\mathcal{V}^* = \operatorname{span}\{v_k : |\langle v_k, y_T \rangle|^2/\|y_T\|^2 \ge \tau\}, \qquad d^* = \dim \mathcal{V}^*, \qquad \mathcal{C} = \frac{\|P_{\mathcal{V}^*} y_T\|^2}{\|y_T\|^2}.$$

The number d^* is the intrinsic harmonic dimension of the target relative to the declared geometry, and $\mathcal{C}$ is the structural information ceiling. For a represented model/readout subspace $\mathcal{R}$, define the certified structural extraction and coverage gap by

$$\mathcal{E}_{\mathrm{sig}}(\mathcal{R}) = \frac{\|P_{\mathcal{R}} P_{\mathcal{V}^*} y_T\|^2}{\|y_T\|^2}, \qquad \mathrm{Gap}(\mathcal{R}) = \mathcal{C} - \mathcal{E}_{\mathrm{sig}}(\mathcal{R}) = \frac{\|(I - P_{\mathcal{R}}) P_{\mathcal{V}^*} y_T\|^2}{\|y_T\|^2}.$$

The equality $\mathrm{Gap}(\mathcal{R}) = 0$ is equivalent only to target-vector coverage,

$$P_{\mathcal{R}} P_{\mathcal{V}^*} y_T = P_{\mathcal{V}^*} y_T.$$

Full structural-subspace coverage is the stronger identity $P_{\mathcal{R}} P_{\mathcal{V}^*} = P_{\mathcal{V}^*}$, and it implies zero gap. These quantities are computable from a finite represented operator by eigensolve, projection, and a declared threshold or spectral gap.

Lemma IV.8.8 (Projection bound for a fixed geometry). *For the declared geometry $\mathcal{L}_{\mathcal{D}}$ and threshold τ, every budget-admissible learner has certified structural extraction at most $\mathcal{C}$. Target projection outside $\mathcal{V}^*$ is classified as residual or noise mass by this criterion.*

Proof. The target decomposes orthogonally as $y_T = P_{\mathcal{V}^*} y_T + (I - P_{\mathcal{V}^*}) y_T$. By definition, the first term is the whole target-bearing component above the declared structural threshold, and the second term lies below that threshold. Part II and Part III classify target arrival outside the represented structural component as residual accounting, oracle/advice information, a changed presentation, or a new saturated derivation. Therefore the mass counted by this geometry's structural criterion is $P_{\mathcal{V}^*} y_T$. For any represented readout subspace $\mathcal{R}$, projection cannot increase norm, so $\mathcal{E}_{\mathrm{sig}}(\mathcal{R}) \le \|P_{\mathcal{V}^*} y_T\|^2 / \|y_T\|^2 = \mathcal{C}$. Raw projection onto $(I - P_{\mathcal{V}^*}) y_T$ may improve empirical fit, but by construction it is below the structural threshold and is charged as residual/noise unless a different geometry or source certificate reclassifies it.

□

Lemma IV.8.9 (Saturation at the intrinsic harmonic dimension). *Let $\mathcal{R}_K$ be a represented model/readout subspace with* $\dim \mathcal{R}_K \le K$*. The certified gap is exactly*

$$\mathcal{C} - \mathcal{E}_{\mathrm{sig}}(\mathcal{R}_K) = \frac{\|(I - P_{\mathcal{R}_K}) P_{\mathcal{V}^*} y_T\|^2}{\|y_T\|^2}.$$

Consequently, if $K \ge d^*$ and $\mathcal{R}_K$ covers $\mathcal{V}^*$, then $\mathcal{E}_{\mathrm{sig}}(\mathcal{R}_K) = \mathcal{C}$. By Lemma IV.8.8, no budget-admissible learner for the same declared geometry has larger certified structural extraction. Thus a covering model of capacity at least d^* is optimal within the declared geometry. If $K > d^*$, any additional raw fit beyond $\mathcal{C}$ lies in the residual component and is an overfitting/noise term for this certificate.

Proof. The displayed identity is the Pythagorean projection identity applied to the structural target component $P_{\mathcal{V}^*} y_T$. If $P_{\mathcal{R}_K} \mathcal{V}^* = \mathcal{V}^*$, then $P_{\mathcal{R}_K} P_{\mathcal{V}^*} y_T = P_{\mathcal{V}^*} y_T$, so $\mathcal{E}_{\mathrm{sig}} = \mathcal{C}$. Lemma IV.8.8 proves that $\mathcal{C}$ upper-bounds every budget-admissible learner's certified structural extraction for the same geometry, so equality is global optimality within that declared presentation. Capacity below d^* cannot cover a d^*-dimensional structural subspace. Capacity above d^* can only add directions orthogonal to $\mathcal{V}^*$; their target overlap is residual by definition and enters the PAC-Bayes/effective-resistance certificate as capacity or residual cost, not as new certified signal.

□

Corollary IV.8.10 (Coverage criterion for model-family optimality)**.** *Fix a metric family and a represented learning rule whose reachable readout spans include a subspace $\mathcal{R}$ of dimension d^* with $P_{\mathcal{R}}P_{\mathcal{V}^*} = P_{\mathcal{V}^*}$. Then choosing capacity $K^* = d^*$, training or solving the terminal reach problem inside that family, and verifying this projection identity proves optimality within the declared geometry: no budget-admissible learner for the same declared geometry and task presentation has a better certified structural extraction. If the gap is positive, its value is the exact structural shortfall and quantifies the additional coverage required from the model family.*

Proof. The saturation theorem shows that a covering subspace at capacity d^* reaches $\mathcal{C}$. The ceiling theorem shows that $\mathcal{C}$ cannot be exceeded by any budget-admissible learner in the declared geometry. The displayed projection identity verifies full subspace coverage and hence zero gap; conversely, zero gap alone verifies only coverage of the single target vector. Otherwise the Pythagorean identity gives the exact missing target mass. Thus, once the model family is chosen rich enough to contain a covering subspace, the remaining optimization problem has a certificate-verifiable optimum; the irreducible choice is the model/metric family that makes coverage possible.

□

IV.9 Certification of Learned Geometries

We now separate label-independent geometry certification from the subsequent propagation of its bound to a selected model family.

Lemma IV.9.1 (Validity of a label-independent geometry certificate)**.** *Let $\{\mathcal{L}_{\mathcal{D}}(\theta) : \theta \in \Theta\}$ be a finite-coverable metric family computed from unlabeled data. Learning θ by maximizing the label criterion $\mathcal{E}(\theta; y)$ does not produce a label-independent geometry certificate or a label-independent PAC–Bayes prior: sufficient metric/readout capacity can place arbitrary random labels in the leading modes. Such a procedure may still be analyzed as label-dependent learned-operator selection, with its selection complexity and generalization error charged. To obtain the label-independent certificate used here, select the learned geometry through a label-independent self-consistency criterion, for example*

$$\theta^* \in \operatorname*{argmin}_{\theta \in \Theta} \left[\mathcal{R}_\theta(E_{\mathrm{self}}(\theta)) + C\sqrt{\frac{N_{\mathrm{eff}}^{\mathrm{metric}}(\theta) + \log \mathcal{N}(\Theta, \rho) + \log(1/\delta)}{m_u}} \right],$$

where m_u is the unlabeled sample size,

$$E_{\mathrm{self}}(\theta) = L_{\mathrm{data}}(\theta)U_\theta - U_\theta L_{Q,\theta}, \qquad \mathcal{R}_\theta(E) = \langle E, \mathcal{L}_{\mathcal{D}}(\theta)^+ E \rangle.$$

The labels are used only after θ^* and the geometry capacity are fixed by unlabeled data. The subsequent label-dependent reach, intrinsic harmonic dimension, and PAC–Bayes prior under θ^* are then based on a label-independent geometry selection.

Proof. If θ is optimized against $\mathcal{E}(\theta; y)$, the metric family can use the labels to reorder the spectrum. At capacity approaching the sample size, the label vector lies in the readout span for any labeling, including random labels, so the certificate becomes circular if presented as label-independent; it must instead be charged as learned-operator selection. In contrast, the displayed self-defect objective depends only on unlabeled samples and the metric family. The finite cover term charges data-dependent selection over Θ. Since θ^* is independent of the labels, PAC-Bayes priors and spectral subspaces computed from $\mathcal{L}_{\mathcal{D}}(\theta^*)$ satisfy the required prior-independence condition.

Capping model capacity by the intrinsic dimension read from the unlabeled geometry prevents the label-dependent readout from expanding until it certifies arbitrary labels. Thus the geometry is selected independently of the labels, and the subsequent label reach is evaluated within that fixed geometry.

□

Lemma IV.9.2 (Two-level geometry certification and propagation to the model certificate)**.** *Split or resample the unlabeled data used to learn geometry. Let $\epsilon_{\mathrm{metric}}$ satisfy the explicit high-probability operator estimate*

$$\|\mathcal{L}_{\mathcal{D}}^{\mathrm{emp}}(\theta^*) - \mathcal{L}_{\mathcal{D}}^{\mathrm{pop}}(\theta^*)\|_{\mathrm{op}} \le \epsilon_{\mathrm{metric}},$$

obtained from held-out unlabeled data or another declared concentration theorem. If the structural eigengap separating $\mathcal{V}^*$ from the residual spectrum is $g_* > 0$, assume $2\epsilon_{\mathrm{metric}} < g_*$ and define the empirical cluster through the corresponding isolated spectral window. Then the population and empirical structural projectors differ by

$$\|P_{\mathcal{V}^*}^{\mathrm{pop}} - P_{\mathcal{V}^*}^{\mathrm{emp}}\|_{\mathrm{op}} \le \frac{2\,\epsilon_{\mathrm{metric}}}{g_*}.$$

For every model-certificate coordinate Q used after geometry selection, the complete record must also name a represented constant $L_Q < \infty$ such that

$$|Q(P) - Q(P')| \le L_Q \|P - P'\|_{\mathrm{op}}$$

on the admitted projector class. Such a coordinate satisfies

$$|Q(P_{\mathcal{V}^*}^{\mathrm{pop}}) - Q(P_{\mathcal{V}^*}^{\mathrm{emp}})| \le \frac{2L_Q\epsilon_{\mathrm{metric}}}{g_*}.$$

For target energy $Q(P) = \|Py_T\|_2^2$, one may take $L_Q = \|y_T\|_2^2$.

Proof. The spectral-window hypothesis permits Davis–Kahan projector perturbation to be applied to the same isolated cluster, giving $\|P_{\mathrm{pop}} - P_{\mathrm{emp}}\|_{\mathrm{op}} \le 2\|\Delta\mathcal{L}\|_{\mathrm{op}}/g_*$. The declared Lipschitz inequality for Q, followed by substitution of this projector bound, proves the coordinatewise transfer inequality. For target energy, orthogonality gives $\|Py_T\|_2^2 = \langle y_T, Py_T\rangle$; hence $|Q(P) - Q(P')| \le \|y_T\|_2^2\|P - P'\|_{\mathrm{op}}$.

□

Lemma IV.9.3 (Finite-net termination within a declared geometry family)**.** *Fix a represented metric family Θ, a precision $\rho > 0$, and a budget B. If the budget-admissible part of Θ has a finite ρ-net, then exhaustive certificate evaluation over that net terminates after at most $\mathcal{N}(\Theta, \rho)$ evaluations. If the certificate objective is L_Θ-Lipschitz in the covering metric, the best net value differs from the infimum over the covered family by at most $L_\Theta\rho$.*

This is a termination statement only for the fixed represented family and precision. It neither proves that recursive enlargement of metric families terminates nor that the chosen family contains an intrinsic population geometry.

Proof. By hypothesis the net is finite, so evaluating its members requires at most its cardinality many evaluations. For any θ in the covered family choose a net point θ_0 at distance at most ρ. Lipschitz continuity gives $|F(\theta) - F(\theta_0)| \le L_\Theta\rho$. Taking infima proves the approximation statement. No assertion about families outside the declared cover follows from this argument.

□

Definition IV.9.4 (Certified geometry evaluation protocol)**.** Combining Lemmas IV.9.1 to IV.8.4 and IV.8.6 yields the following procedure.

1. Declare a finite-coverable metric/model family Θ.
2. Learn θ^* from unlabeled data by minimizing self-defect/effective-resistance plus metric capacity, independently of the labels.
3. Certify metric generalization on held-out unlabeled data and carry the perturbation term forward.
4. Compute $\mathcal{V}^*$, d^*, and $\mathcal{C}$ under the certified geometry.
5. Choose a model/readout family of capacity $K = d^*$ or the smallest capacity whose represented span covers $\mathcal{V}^*$.
6. Verify the projection identity $P_{\mathcal{R}} P_{\mathcal{V}^*} = P_{\mathcal{V}^*}$ when full subspace coverage is claimed, and compute the target-vector gap $\mathcal{C} - \mathcal{E}_{\text{sig}}$. A zero target-vector gap, up to metric and sampling errors, proves saturation for this target; a positive gap quantifies its missing structural mass.
7. Select regularization coefficients using the structural PAC–Bayes or deterministic effective-resistance bound above.

The protocol reports full coverage only when every displayed certificate is finite on one simultaneous event and the projection identity is verified. Otherwise it reports the measured target-vector gap and does not infer which unverified condition caused it.

Theorem IV.9.5 (Conditional completeness of the geometry evaluation protocol)**.** *Assume the protocol of Definition IV.9.4 returns, on one event $\mathcal{E}$, a valid represented geometry, its intrinsic target subspace $\mathcal{V}^*$, and a represented readout space $\mathcal{R}$ satisfying*

$$P_{\mathcal{R}} P_{\mathcal{V}^*} = P_{\mathcal{V}^*}.$$

Let y be the target vector and define the geometry-accessible target energy by $\mathcal{C} = \|P_{\mathcal{V}^*} y\|_2^2$. Then the readout captures all of that energy:

$$\|P_{\mathcal{R}} P_{\mathcal{V}^*} y\|_2^2 = \mathcal{C}.$$

If the certified metric, projector, and empirical target-energy errors sum to $\varepsilon_{\text{cert}}$ in the comparison norm named by the record, assume the record also supplies a represented constant $L_{\text{cert}} < \infty$ for which the reported gap functional is L_{cert}-Lipschitz in that norm on the admitted certificate class. The reported target-vector gap is then at most

$$L_{\text{cert}} \varepsilon_{\text{cert}}.$$

Generalization and regularization claims remain exactly those supplied by the separately invoked PAC–Bayes, resistance, and model-family certificates; no claim is made for geometries or readouts outside the declared family.

Proof. Apply the verified projection identity to y: $P_{\mathcal{R}} P_{\mathcal{V}^*} y = P_{\mathcal{V}^*} y$, and take squared norms. The exact target-vector gap is zero. The triangle inequality bounds the distance between the exact and reported certificate tuples by $\varepsilon_{\text{cert}}$; the declared Lipschitz inequality gives the displayed bound. The final sentence is a scope statement: the cited learning theorems apply on $\mathcal{E}$ with their own hypotheses and additive errors.

□

Figure 51 records the independence and simultaneity requirements of the certified geometry protocol.

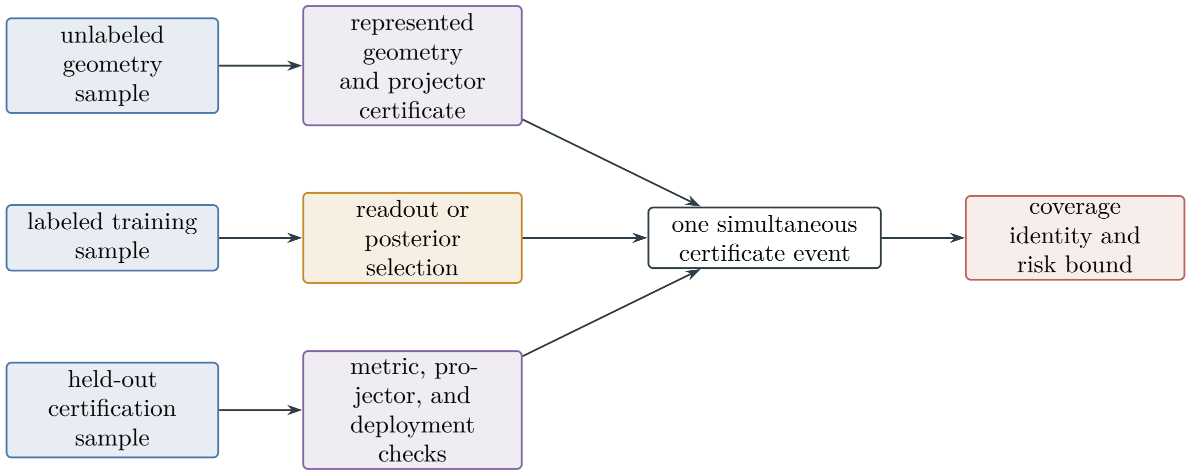

Figure 51: A learned geometry may enter a population-risk certificate only through an independence split or a uniform theorem covering its selection. Geometry construction, labeled readout fitting, and held-out propagation checks join only on one simultaneous event before coverage or deployment is claimed.

IV.9.1 Defect-Geometry Inequality Profiles

The following constants belong to the represented defect geometry, not to an ambient smoothness class. The learned routing form $\mathcal{L}_D$ acts on the lumpability defect E_q. Effective resistance determines whether the signed defect admits finite-energy routing. The full profile additionally measures entropy dissipation, gradient contraction, target capacity, and stability under distribution shift.

A certified defect profile is

$$\Pi_D = (\mathcal{L}_D, E_q, \rho_D, \lambda_D, \alpha^D_{\mathrm{MLSI}}, \kappa_D, \mathrm{Cap}_D, \Phi_D, \alpha^D_T),$$

with every entry audited on the same active defect support and the same quotient defect used to form $\mathcal{R}_D(E_q) = \langle E_q, \mathcal{L}_D^+ E_q\rangle$. The constants are lower or upper certified bounds in the direction used by the theorem: lower bounds for rates such as α^D_{MLSI}, κ_D, transport constants, and target capacities; upper bounds for residual energies, entropy defects, and boundary-risk terms.

Lemma IV.9.6 (Selection among defect-profile bounds)**.** *Fix a represented quotient, a sample, and a label-independent or independently evaluated admissible defect geometry. For each component $s \in S$, let $C_s \geq 0$ denote the single true contribution in the risk units of the surrounding theorem. Suppose a finite audited family $\mathcal{J}_s$ and nonnegative random bounds $B_{s,j} = \widehat{C}_{s,j} + r_{s,j}$ satisfy, on one event $\mathcal{E}_s$ of probability at least $1 - \delta_s$,*

$$C_s \leq B_{s,j} \qquad \textit{simultaneously for every } j \in \mathcal{J}_s.$$

Then, on $\mathcal{E} = \bigcap_{s\in S} \mathcal{E}_s$,

$$\mathcal{C}^{\mathrm{prof}}_s(\Pi_D) = \min_{j\in\mathcal{J}_s} B_{s,j} \quad \textit{satisfies} \quad C_s \leq \mathcal{C}^{\mathrm{prof}}_s(\Pi_D), \qquad \Pr(\mathcal{E}) \geq 1 - \sum_{s\in S} \delta_s.$$

The radii $r_{s,j}$ include every finite-family, sample-split, or uniform selection penalty required to establish the simultaneous event. They may be zero for deterministic analytic bounds. Candidates for different components are never minimized against one another.

Proof. On $\mathcal{E}_s$, the same scalar C_s is no larger than every $B_{s,j}$, hence it is no larger than their minimum. The probability claim is the union bound over s.

$\square$

Remark IV.9.7 (Typed defect-profile candidates). The routing component always contains the Poincaré/effective-resistance bound

$$\mathcal{C}_{\mathrm{route,Poincare}}(\Pi_D) = c_R \mathcal{R}_D(E_q),$$

with c_R the conversion constant of the surrounding risk theorem. The remaining inequalities apply only to the components they bound:

- **MLSI/entropy.** With the declared normalization, a defect MLSI such as $\mathrm{Ent}^{\mathcal{B}}_{\mu_D}(\rho_D) \leq c_{\mathrm{MLSI}} \alpha_D^{-1} \mathcal{E}_D(\rho_D, \log \rho_D)$ supplies a concentration or entropy-production bound for the actual residual density [15, 36].
- **Bakry–Emery.** A curvature certificate $\Gamma_{2,D}(f) \geq \kappa_D \Gamma_D(f)$, with $\kappa_D > 0$, supplies a propagation/contraction bound, for example $\Gamma_D(P_t f) \leq e^{-2\kappa_D t} P_t \Gamma_D(f)$ under the declared reversible semigroup calculus, or a static curvature scale after multiplication by the theorem's conversion constant [7].
- **Capacity/Cheeger.** Target access, boundary access, and death avoidance are capacity components. Target access may use inverse target capacity; death avoidance uses polar excision when $\mathrm{Cap}_D(\Sigma^-) = 0$, or a monotone boundary-risk/death-capacity upper charge otherwise.
- **Transport entropy.** Deployment, replay, or off-policy shift is a distribution-shift component, for example $\mathrm{Lip}_D(\ell)\sqrt{\mathrm{KL}(\nu\|\mu)/\alpha_T^D}$ [14, 65].

Logically implied inequalities are counted once. If a Bakry–Émery condition implies the MLSI or transport constant in the declared setting, the induced constant is substituted into the corresponding component.

Corollary IV.9.8 (Substitution of simultaneous defect-profile certificates)**.** *Suppose a represented risk theorem has already established, on the event $\mathcal{E}$ of Lemma IV.9.6,*

$$R(h) \leq \widehat{R}_S(h) + \sum_{s \in S} C_s + E_{\mathrm{op}} + E_{\mathrm{src}},$$

where all terms are in the same risk units. Then, on that event,

$$R(h) \leq \widehat{R}_S(h) + \sum_{s \in S} \mathcal{C}_s^{\mathrm{prof}}(\Pi_D) + E_{\mathrm{op}} + E_{\mathrm{src}}.$$

The routing coordinate occurs once in the index set S; any candidate that already contains its resistance contribution is treated as one complete bound $B_{s,j}$, rather than added to it again.

Proof. Substitute $C_s \leq \mathcal{C}_s^{\mathrm{prof}}(\Pi_D)$ from Lemma IV.9.6 term by term.

$\square$

Training losses alone do not establish the stated profile bounds. If the profile constants are selected using the labeled risk-evaluation data, the theorem requires an independent sample split, a label-independent selection rule, or a uniform concentration result over the declared finite family. Analytically established constants require only the finite-family selection term.

The Bakry–Émery condition provides a contraction estimate for the defect-induced Caus gradient and therefore describes active residual dissipation. An MLSI term remains distinct in optimization because a minibatch curvature loss evaluates only a finite function class, whereas the inequality is universal over the admissible domain. The training objective may therefore contain both terms:

$$\mathcal{L}_{\mathrm{geom}} = \lambda_R \mathcal{R}_D(E_q) + \lambda_{\Gamma_2}\mathcal{L}_{\Gamma_2} + \lambda_{\mathrm{MLSI}}\mathcal{L}_{\mathrm{MLSI}} + \mathcal{L}_{\mathrm{norm}}.$$

Here $\mathcal{L}_{\Gamma_2}$ targets defect-gradient contraction and $\mathcal{L}_{\mathrm{MLSI}}$ is a direct entropy witness for the actual residual density. The theorem uses the constants established after training rather than the inclusion of these terms in the objective.

Persistent large E_q after admissible resistance, MLSI, and curvature optimization indicates that the quotient has identified states that must remain distinct. In this case the admissible modification is to refine q, rather than to absorb the defect into $\mathcal{L}_D$.

IV.10 Learning Under Symmetric Label Corruption

The preceding Rademacher, effective-dimension, and PAC–Bayes bounds admit a common noise correction for binary labels. Symmetric label corruption attenuates the clean prediction advantage by the factor $1 - 2\eta$. The statistical capacity and represented model-selection costs remain those already defined in this part.

Figure 52 summarizes this transport from corrupted sample risk to clean prediction risk.

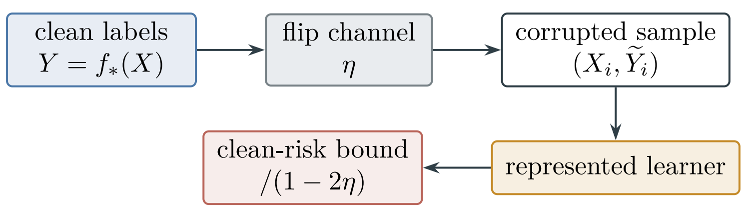

Figure 52: Symmetric corruption attenuates clean-label advantage by $1 - 2\eta$. The represented learner pays the existing Rademacher, effective-dimension, PAC–Bayes, and operator-selection terms; dividing the resulting corrupted-risk certificate by the attenuation factor gives the clean-risk bound.

IV.10.1 Risk transport through a binary flip channel

The following definition covers homogeneous symmetric noise and its instance-dependent Massart extension.

Definition IV.10.1 (Binary label-corruption channel)**.** Let $X \sim \mu$, let the clean binary target be $Y = f_*(X) \in \{-1, 1\}$, and let

$$\widetilde{Y} = Y\Xi, \qquad \Pr\{\Xi = -1 \mid X = x\} = \eta(x).$$

Assume

$$0 \le \eta(x) \le \bar{\eta} < \frac{1}{2}, \qquad \bar{\rho} := 1 - 2\bar{\eta} > 0.$$

For a binary predictor $h : X \to \{-1, 1\}$, write

$$R_0(h) = \Pr\{h(X) \neq Y\}, \qquad R_{\mathrm{corr}}(h) = \Pr\{h(X) \neq \widetilde{Y}\}.$$

In the homogeneous case $\eta(x) \equiv \eta$, put $\rho = 1 - 2\eta$ and write R_η for the corrupted-label risk.

More generally, a bounded loss $\ell : [-1, 1] \times \{-1, 1\} \to [0, 1]$ is *flip-symmetric* when

$$\ell(z, 1) + \ell(z, -1) = 1 \qquad (-1 \leq z \leq 1).$$

Its clean and corrupted risks are denoted by $R_0^\ell(f)$ and $R_{\mathrm{corr}}^\ell(f)$. The binary zero–one loss and the clipped linear loss $\ell(z, y) = (1 - yz)/2$ satisfy this identity.

Lemma IV.10.2 (Exact transport of clean risk through label corruption)**.** *Under Definition IV.10.1, every binary predictor satisfies*

$$R_{\mathrm{corr}}(h) - R_{\mathrm{corr}}(f_*) = \mathbb{E}\Big[(1 - 2\eta(X))\mathbf{1}_{\{h(X) \neq f_*(X)\}}\Big] \geq \bar{\rho} R_0(h). \tag{230}$$

For homogeneous noise, the identity becomes

$$R_\eta(h) = \eta + \rho R_0(h). \tag{231}$$

For every flip-symmetric loss,

$$R_{\mathrm{corr}}^\ell(f) = \mathbb{E}[\eta(X) + (1 - 2\eta(X))\ell(f(X), Y)]\,. \tag{232}$$

In particular, homogeneous corruption gives

$$R_\eta^\ell(f) = \eta + \rho R_0^\ell(f).$$

Proof. Condition on $X = x$. If $h(x) = f_*(x)$, the corrupted label disagrees with $h(x)$ with probability $\eta(x)$. If $h(x) \neq f_*(x)$, it disagrees with $h(x)$ with probability $1 - \eta(x)$. Subtracting the conditional risk of f_* gives

$$(1 - 2\eta(x))\mathbf{1}_{\{h(x) \neq f_*(x)\}}.$$

Integration proves (230); the lower bound follows from $1 - 2\eta(x) \geq \bar{\rho}$. Constant $\eta(x) = \eta$ gives (231). For a flip-symmetric loss,

$$\begin{aligned}
\mathbb{E}[\ell(f(X), \widetilde{Y}) \mid X, Y] &= (1 - \eta(X))\ell(f(X), Y) + \eta(X)\ell(f(X), -Y) \\
&= \eta(X) + (1 - 2\eta(X))\ell(f(X), Y),
\end{aligned}$$

which proves (232).

□

IV.10.2 Rademacher capacity and the tolerable-noise region

The exact transport identity converts the existing generalization radius into a clean-label denoising guarantee.

Theorem IV.10.3 (Rademacher bound under label corruption)**.** *Let $\mathcal{H}$ be a represented class of binary predictors and let $S = ((X_i, \widetilde{Y}_i))_{i=1}^m$ be an i.i.d. corrupted sample from Definition IV.10.1, where $m \geq 1$ and $0 < \delta < 1$. Let*

$$\mathcal{L}_{\mathcal{H}} = \Big\{(x, \widetilde{y}) \longmapsto \mathbf{1}_{\{h(x) \neq \widetilde{y}\}} : h \in \mathcal{H}\Big\}$$

and define

$$\Gamma_m^{\mathrm{Rad}}(\mathcal{H}, \delta) := 2\widehat{\mathfrak{R}}_S(\mathcal{L}_{\mathcal{H}}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}. \tag{233}$$

With probability at least $1 - \delta$, simultaneously for every $h \in \mathcal{H}$,

$$\Big|R_{\mathrm{corr}}(h) - \widehat{R}_{\mathrm{corr},S}(h)\Big| \leq \Gamma_m^{\mathrm{Rad}}(\mathcal{H}, \delta). \tag{234}$$

Assume $f_* \in \mathcal{H}$, and let $\widehat{h}$ be an $\varepsilon_{\mathrm{opt}}$-empirical minimizer:

$$\widehat{R}_{\mathrm{corr},S}(\widehat{h}) \leq \inf_{h \in \mathcal{H}} \widehat{R}_{\mathrm{corr},S}(h) + \varepsilon_{\mathrm{opt}}.$$

On the same event,

$$R_0(\widehat{h}) \leq \frac{2\Gamma_m^{\mathrm{Rad}}(\mathcal{H}, \delta) + \varepsilon_{\mathrm{opt}}}{\bar{\rho}}. \tag{235}$$

Under homogeneous noise, every fixed $h \in \mathcal{H}$ also satisfies

$$R_0(h) \leq \left[\frac{\widehat{R}_{\eta,S}(h) - \eta + \Gamma_m^{\mathrm{Rad}}(\mathcal{H}, \delta)}{\rho}\right]_{[0,1]}, \tag{236}$$

where $[u]_{[0,1]} = \min\{1, \max\{0, u\}\}$.

For a target-qualified use, apply these inequalities to each exact source cell of $\mathcal{H}$. Source selection is charged by Eq. (279), and every corrupted reveal retains the coefficient $\vartheta(\eta) = (1 - 2\eta)^2$ in Eq. (275). The inverse factor ρ^{-1} in clean-risk correction and the forward information contraction $\vartheta(\eta)$ remain separate coordinates.

Proof. The standard two-sided empirical Rademacher inequality for a $[0, 1]$-valued loss class gives (234); it is the two-sided form of the bound used in Theorem IV.3.3 and Corollary IV.3.7. On this event,

$$\begin{aligned}
R_{\mathrm{corr}}(\widehat{h}) &\leq \widehat{R}_{\mathrm{corr},S}(\widehat{h}) + \Gamma_m^{\mathrm{Rad}} \\
&\leq \widehat{R}_{\mathrm{corr},S}(f_*) + \varepsilon_{\mathrm{opt}} + \Gamma_m^{\mathrm{Rad}} \\
&\leq R_{\mathrm{corr}}(f_*) + \varepsilon_{\mathrm{opt}} + 2\Gamma_m^{\mathrm{Rad}}.
\end{aligned}$$

Apply (230) and divide by $\bar{\rho}$ to obtain (235). In the homogeneous case, combine the upper half of (234) with (231), then clip the resulting risk bound to $[0, 1]$. $\square$

Corollary IV.10.4 (Capacity-controlled noise tolerance)**.** *In the setting of Theorem IV.10.3, a sufficient condition for clean risk at most $\varepsilon > 0$ is*

$$2\Gamma_m^{\mathrm{Rad}}(\mathcal{H},\delta) + \varepsilon_{\mathrm{opt}} \le (1-2\bar{\eta})\varepsilon. \tag{237}$$

Equivalently,

$$\bar{\eta} \le \frac{1}{2} - \frac{\Gamma_m^{\mathrm{Rad}}(\mathcal{H},\delta) + \varepsilon_{\mathrm{opt}}/2}{\varepsilon}.$$

The same conclusion holds for a represented score class, a flip-symmetric loss, and an $\varepsilon_{\mathrm{opt}}$-empirical minimizer whenever the class contains a comparator f_* with $R_0^\ell(f_*) = 0$. For a fixed structural energy ball $\mathcal{F}_A(R)$, an L_ℓ-Lipschitz loss, and clipped scores in $[-1,1]$, the contraction principle and Lemma IV.3.2 then permit the explicit radius

$$\Gamma_{A,m}(R,\delta) = \frac{2L_\ell R}{m}\sqrt{\mathrm{Tr}(A^{-1})} + 3\sqrt{\frac{\log(2/\delta)}{2m}}. \tag{238}$$

For a learned operator class, the expression in Lemma IV.3.6 replaces the fixed-operator Rademacher term. Thus denoising without sample memorization occurs whenever the fixed or selected operator radius and $\varepsilon_{\mathrm{opt}}$ are $o(1-2\bar{\eta})$; operator selection retains the covering cost already present in that theorem.

Proof. The first assertion is (235) with its right-hand side bounded by ε. Solving the resulting inequality for $\bar{\eta}$ gives the second display. If $R_0^\ell(f_*) = 0$, (232) gives $R_{\mathrm{corr}}^\ell(f_*) = \mathbb{E}\eta(X)$ and

$$R_{\mathrm{corr}}^\ell(f) - R_{\mathrm{corr}}^\ell(f_*) \ge \bar{\rho} R_0^\ell(f).$$

The same empirical-minimizer argument therefore proves the loss version. For an L_ℓ-Lipschitz loss, Rademacher contraction bounds the loss-class complexity by L_ℓ times the score-class complexity. Lemma IV.3.2 supplies $Rm^{-1}\sqrt{\mathrm{Tr}(A^{-1})}$. The learned-operator substitution follows identically from Lemma IV.3.6.

$\square$

IV.10.3 Structural PAC–Bayes correction

The same attenuation factor converts the structural PAC–Bayes certificate into a clean-risk certificate.

Theorem IV.10.5 (PAC–Bayes bound under symmetric label noise)**.** *Assume homogeneous corruption at a known rate $0 \le \eta < 1/2$, a bounded flip-symmetric loss, a sample size $m \ge 2$, a confidence level $0 < \delta < 1$, and a represented prior P fixed independently of the labeled sample. With probability at least $1-\delta$, simultaneously for every posterior $Q \ll P$,*

$$R_0^\ell(Q) \le \left[\frac{\widehat{R}_{\eta,S}^\ell(Q) - \eta + \sqrt{\dfrac{\mathrm{KL}(Q\|P) + \log(2\sqrt{m}/\delta)}{2(m-1)}}}{1-2\eta}\right]_{[0,1]}. \tag{239}$$

For the structural prior $P = P_{E,H}^{\mathsf{str}}$ of Definition IV.8.1, the divergence in (239) is exactly $\mathrm{KL}_{E,H}^{\mathsf{str}}(Q)$. Thus the noise correction preserves the defect-restricted structural complexity term of Lemma IV.8.2.

For a declared finite coefficient family Λ, the bound holds simultaneously after replacing the logarithmic term by $\log(2\sqrt{m}|\Lambda|/\delta)$, exactly as in Lemma IV.8.4. Minimizing this simultaneous right-hand side gives the certified noise-corrected coefficient choice.

Proof. The McAllester inequality used in the proof of Lemma IV.8.2 applies to any represented prior fixed independently of the labeled sample; choosing $P = P^{\mathsf{str}}_{E,H}$ recovers that theorem's structural specialization. The prior-independence requirement is unchanged because the flip channel modifies the labels rather than the public or independently learned geometry. The homogeneous flip-symmetric identity in Lemma IV.10.2 gives $R^{\ell}_{\eta}(Q) = \eta + (1-2\eta)R^{\ell}_{0}(Q)$. Subtract η, divide by $1-2\eta$, and clip to $[0,1]$. The simultaneous coefficient form is the union-bound argument of Lemma IV.8.4.

□

IV.10.4 Effective dimension after label debiasing

For squared-loss prediction, symmetric flips scale the regression function without changing its source subspace.

Corollary IV.10.6 (Debiased kernel ridge bound for binary labels)**.** *Let $f_* : X \to \{-1,1\}$, let $\widetilde{Y} = f_*(X)\Xi$ under homogeneous noise, and put $\rho = 1-2\eta$. Then*

$$\mathbb{E}[\widetilde{Y} \mid X] = \rho f_*(X).$$

Let $\widehat{g}_\lambda$ be the kernel-ridge estimator trained on $(X_i, \widetilde{Y}_i)_{i=1}^m$, and define the debiased predictor $\widehat{f}_\lambda = \widehat{g}_\lambda/\rho$. Suppose a fixed-kernel oracle certificate (227) holds on an event $\mathcal{E}_{\eta,k,\lambda}$ for the regression target $g_* = \rho f_*$, with constants $C_{\mathrm{var}}, C_{\mathrm{bias}}$ and with v_η^2 a valid conditional sub-Gaussian variance proxy for $\widetilde{Y} - \rho f_*(X)$. Then, on the same event,

$$\|\widehat{f}_\lambda - f_*\|^2_{L^2(\mu)} \le C_{\mathrm{var}} \frac{v_\eta^2 N_{\mathrm{eff}}(\lambda)}{\rho^2 m} + C_{\mathrm{bias}} B_k(\lambda; f_*). \tag{240}$$

Since the centered flip variable is bounded, one may always take a universal bounded sub-Gaussian proxy; its exact conditional variance is $1-\rho^2 = 4\eta(1-\eta)$. Moreover,

$$\Pr\{\mathrm{sign}(\widehat{f}_\lambda(X)) \ne f_*(X)\} \le \|\widehat{f}_\lambda - f_*\|^2_{L^2(\mu)}.$$

Consequently a sufficient effective-dimension condition for vanishing denoising error is

$$C_{\mathrm{var}} \frac{v_\eta^2 N_{\mathrm{eff}}(\lambda)}{(1-2\eta)^2 m} + C_{\mathrm{bias}} B_k(\lambda; f_*) = o(1).$$

Proof. Conditional expectation gives $\mathbb{E}[\widetilde{Y} \mid X] = f_*(X)\mathbb{E}\Xi = \rho f_*(X)$. Apply the named fixed-kernel certificate to $g_* = \rho f_*$:

$$\|\widehat{g}_\lambda - \rho f_*\|_2^2 \le C_{\mathrm{var}} \frac{v_\eta^2 N_{\mathrm{eff}}(\lambda)}{m} + C_{\mathrm{bias}} B_k(\lambda; \rho f_*).$$

The ridge source residual is quadratic, so $B_k(\lambda; \rho f_*) = \rho^2 B_k(\lambda; f_*)$. Divide the inequality by ρ^2 to obtain (240). If the sign of $\widehat{f}_\lambda$ is incorrect, its distance from the binary value $f_*(X)$ is at least one. The classification inequality follows by Markov's inequality applied pointwise to the squared error.

□

IV.11 Saturated Statistical Attacks and Population Alignment

The statistical bounds above apply simultaneously to the full represented budget slice. The following definition fixes that class before the training sample is drawn.

Definition IV.11.1 (Budget-saturated statistical readout class)**.** Fix a presented binary prediction family, a sample length m, and a budget B. Let $\mathcal{H}^{\mathrm{sat,stat}}_{n,B,m}$ be the sample-independent union of all scores

$$h_{\rho,S,\omega} : X_I \longrightarrow [-1,1]$$

obtained as follows. The record ρ is one family-uniform derivation in $\mathsf{Der}_{\mathfrak{J}}$; S ranges over every legal encoded training sample of length m; ω ranges over every represented internal random seed; and the complete cost of training, retained state, output representation, and score evaluation is at most B. Every score retains its complete derivation record. The union over legal S and ω is taken before the realized sample is selected, so the class is a fixed statistical class. A data-dependent kernel, feature map, posterior, regularization choice, or terminal postprocessing belongs to this class only through its complete represented construction.

Let $T_I = \{x : f_*(x) = 1\}$. Define the saturated learned projection envelope

$$M^{\mathrm{sat,stat}}_{I,n}(B,m) = \sup_{h\in\mathcal{H}^{\mathrm{sat,stat}}_{n,B,m}} \mathrm{ProjMass}_{\sigma(h)}(T_I), \tag{241}$$

with supremum zero when the class is empty. The algebra $\sigma(h)$ is generated by the represented score itself. It is distinct from the ambient measurable envelope of all public data.

For the clipped linear loss

$$\ell_h(x,\widetilde{y}) = \frac{1-\widetilde{y}h(x)}{2},$$

let $\Gamma^{\mathrm{sat,stat}}_{n,B,m,\delta}$ denote the radius in (233) for the loss class generated by $\mathcal{H}^{\mathrm{sat,stat}}_{n,B,m}$. A learned-operator cover may instead be inserted through Lemma IV.3.6.

Theorem IV.11.2 (Uniform accountability of budget-saturated statistical attacks)**.** *Assume homogeneous symmetric label corruption at rate $0 \le \eta < 1/2$, and put $\rho = 1-2\eta$. For $h : X_I \to [-1,1]$, write*

$$c_0(h) = \mathbb{E}[h(X)f_*(X)], \qquad \widehat{c}_{\eta,S}(h) = \frac{1}{m}\sum_{i=1}^{m} h(X_i)\widetilde{Y}_i.$$

With probability at least $1-\delta$, simultaneously for every family-uniform represented statistical computation of complete cost at most B, every represented random seed, and its output $h \in \mathcal{H}^{\mathrm{sat,stat}}_{n,B,m}$,

$$|\widehat{c}_{\eta,S}(h) - \rho c_0(h)| \le 2\Gamma^{\mathrm{sat,stat}}_{n,B,m,\delta}. \tag{242}$$

Consequently,

$$|c_0(h)| \le \frac{|\widehat{c}_{\eta,S}(h)| + 2\Gamma^{\mathrm{sat,stat}}_{n,B,m,\delta}}{1-2\eta}. \tag{243}$$

If the clean binary target is balanced, then every such output also satisfies

$$|c_0(h)|^2 \le \mathrm{Var}_{\mu_I}(h)\ \mathrm{ProjMass}_{\sigma(h)}(T_I) \le M^{\mathrm{sat,stat}}_{I,n}(B,m). \tag{244}$$

Hence, on the same event,

$$\sup_{\mathcal{A},\omega} |c_0(h_{\mathcal{A},S,\omega})| \le \min\left\{ \sqrt{M^{\mathrm{sat,stat}}_{I,n}(B,m)}, \frac{\sup_{h\in\mathcal{H}^{\mathrm{sat,stat}}_{n,B,m}} |\widehat{c}_{\eta,S}(h)| + 2\Gamma^{\mathrm{sat,stat}}_{n,B,m,\delta}}{1-2\eta} \right\}. \tag{245}$$

For sign-valued outputs in the balanced setting, the same projection envelope gives the universal approximation floor

$$\inf_{\mathcal{A},\omega} R_0(h_{\mathcal{A},S,\omega}) \ge \frac{1-\sqrt{M^{\mathrm{sat,stat}}_{I,n}(B,m)}}{2}. \tag{246}$$

Every represented use of such an output as a target selector, ranking, potential, quotient readout, or transition rule retains the same derivation record and adds its represented use cost. The quantities $c_0(h)$ and $\mathrm{ProjMass}_{\sigma(h)}(T_I)$ are analytic population audits of that affordable score. A downstream use enters the Geom/Abs *prospective source ledger only when the same complete pre-hit record also satisfies the represented channel predicate, target-discrimination comparison, temporal admissibility, and cost gates of Lemma IV.0.2 and Theorem III.25.2. Otherwise the audit remains a statistical-accounting coordinate and target arrival is controlled by the prospective-selector ledger. Statistical estimation introduces no additional structural channel.*

Proof. The sample-independent hull in Definition IV.11.1 contains the output of every represented budget-B learner for every realized sample and every represented seed. Apply the simultaneous score-class form of Theorem IV.10.3 and Corollary IV.10.4 to the clipped linear losses. Since

$$R^{\ell}_{\eta}(h) = \frac{1-\rho c_0(h)}{2}, \qquad \widehat{R}^{\ell}_{\eta,S}(h) = \frac{1-\widehat{c}_{\eta,S}(h)}{2},$$

the uniform risk deviation is exactly (242). Division by $\rho > 0$ gives (243). The event is uniform over the whole output class, so data-dependent selection of the learner, operator, score, or postprocessing preserves the inequality.

For the projection statement, balance gives $\mu_I(T_I) = 1/2$ and $y_{T_I} = f_*/2$. If h is nonconstant, the normalized correlation in Lemma I.4.6, applied to $\mathfrak{A}_I = \sigma(h)$, is

$$\frac{|\langle h - \mathbb{E}h, y_{T_I}\rangle|^2}{\|h-\mathbb{E}h\|_2^2 \|y_{T_I}\|_2^2} = \frac{|c_0(h)|^2}{\mathrm{Var}_{\mu_I}(h)}.$$

Thus the first inequality in (244) follows from that ceiling. It also holds for constant h, because balance gives $c_0(h) = 0$. The score range implies $\mathrm{Var}(h) \le 1$, and taking the saturated supremum gives the second inequality. Combining the projection and statistical bounds and taking the supremum over computations and seeds proves (245). For a sign-valued output, $R_0(h) = (1-c_0(h))/2$; applying $|c_0(h)| \le \sqrt{M^{\mathrm{sat,stat}}_{I,n}(B,m)}$ proves (246).

Finally, Lemma IV.0.2 places the learner and every represented downstream use in the saturated derivability closure. Theorem III.25.2 assigns every target-bearing structural use to the existing source ledger at its complete cost.

□

Corollary IV.11.3 (Target alignment forces saturated learned structure)**.** *In the balanced setting of Theorem IV.11.2, if a budget-B statistical computation produces an output with $|c_0(h)| > \varepsilon$, then*

$$M_{I,n}^{\mathrm{sat,stat}}(B, m) > \varepsilon^2. \tag{247}$$

On the simultaneous event of that theorem it also satisfies

$$|\widehat{c}_{\eta,S}(h)| > (1 - 2\eta)\varepsilon - 2\Gamma_{n,B,m,\delta}^{\mathrm{sat,stat}}. \tag{248}$$

Thus a clean alignment of size ε requires a represented learned score whose generated algebra has projection mass greater than ε^2, while its statistical visibility above the uniform selection radius is attenuated by $1 - 2\eta$.

Proof. The first conclusion is the contrapositive of (244). Rearranging (242) proves the second. □

Theorem IV.11.4 (Two-level provenance and uniform noise transport for learned readouts)**.** *Adopt the balanced setting and notation of Definition IV.11.1 and Theorem IV.11.2, and put $\rho = 1 - 2\eta$. For every realized sample S and every represented output h, define the scalar transport residual*

$$r_{\eta,S}(h) := \widehat{c}_{\eta,S}(h) - \rho c_0(h). \tag{249}$$

With probability at least $1 - \delta$, simultaneously over the complete budget-saturated statistical class,

$$|r_{\eta,S}(h)| \le 2\Gamma_{n,B,m,\delta}^{\mathrm{sat,stat}}, \tag{250}$$

$$|c_0(h)| \le \sqrt{M_{I,n}^{\mathrm{sat,stat}}(B, m)}, \tag{251}$$

$$|\widehat{c}_{\eta,S}(h)| \le \rho\sqrt{M_{I,n}^{\mathrm{sat,stat}}(B, m)} + 2\Gamma_{n,B,m,\delta}^{\mathrm{sat,stat}}. \tag{252}$$

Conversely, an output satisfying $|c_0(h)| > \varepsilon$ witnesses

$$M_{I,n}^{\mathrm{sat,stat}}(B, m) > \varepsilon^2, \qquad |\widehat{c}_{\eta,S}(h)| > \rho\varepsilon - 2\Gamma_{n,B,m,\delta}^{\mathrm{sat,stat}}. \tag{253}$$

These estimates have the following provenance interpretation.

(i) *A family-uniform learner is one represented derivation rule. Its output may depend on the presented instance, the realized sample, and a represented random seed; family uniformity does not require those outputs to be identical.*

(ii) *Every retained sample table, empirical prediction vector, posterior, optimization transcript, and memorized lookup record is a represented finite coordinate. Its construction, storage, and evaluation costs remain in the lifted computation record. Its population projection is an analytic audit. It is charged to the existing* Geom/Abs *prospective source ledger only when the represented use satisfies the same-record channel predicate, target comparison, temporal, and cost gates; otherwise it remains statistical accounting. The orthogonal remainder is assigned to J_{res} by the exhaustive residual normal form. Thus memorization creates neither an uncharged population certificate nor a sixth structural source.*

(iii) *If one family-uniform affordable rule has nonnegligible population alignment, then* (253) *records its nonnegligible analytic target projection. Any represented use of that alignment as a selector, ranking, potential, quotient readout, or transition rule retains the learner's full ancestry. It is classified as a prospective source only after its relevant represented channel predicate and pre-hit comparison are verified. If it induces positive selector advantage, that advantage is bounded by the existing five contextual-charge envelopes through Theorem III.21.4; conversion of those contextual charges to source values additionally uses Corollary III.16.9.*

(iv) *A target-bearing table supplied separately for each instance without one family-uniform represented derivation is nonuniform advice, not an output of $\mathcal{H}^{\mathrm{sat,stat}}_{n,B,m}$. An over-budget construction is likewise absent from the budget slice.*

Hence family-uniform learning may discover affordable structure already derivable from the presentation, but sample-dependent fitting does not create an additional provenance channel.

Proof. Equation (250) is (242) after the abbreviation (249). The projection estimate (244) gives (251). The triangle inequality then gives

$$|\widehat{c}_{\eta,S}(h)| \le \rho|c_0(h)| + |r_{\eta,S}(h)|,$$

which proves (252). The first conclusion in (253) is the contrapositive of (244); the second follows by applying the reverse triangle inequality to (242).

It remains to verify the provenance statements. By Definition IV.11.1, the union over legal samples and represented seeds is formed from one family-uniform derivation record and includes the complete cost of training, retained state, output representation, and evaluation. Consequently every empirical table and memorized value used by the learner is a charged coordinate on its finite lifted carrier. Apply Lemma IV.0.2 and Theorem III.25.2 to every represented population use that satisfies their channel predicates and temporal comparisons. Its qualified structural projection is assigned to the existing channels, and Theorem II.7.2 assigns the remaining orthogonal component to J_{res}. If that remainder acquires a family-uniform affordable target-aligned channel representative, it is reclassified by Proposition II.8.3 and Lemma II.8.4; therefore it cannot remain simultaneously residual and an affordable structural source.

Finally, a downstream selector is a represented use of the learned score and retains the derivation as an ancestor. Its positive advantage is bounded by Theorem III.21.4. The family-uniformity and budget exclusions are precisely the membership requirements in Definition IV.11.1. This proves all four provenance assertions.

□

Figure 53 depicts the population-structural and empirical-residual decomposition of a learned readout.

Corollary IV.11.5 (Residual-noise-order specialization)**.** *In the setting of Theorem IV.11.4, suppose the certified readout class has residual-noise order at least q in the sense of Definition I.5.4. Then, on the same simultaneous event, every h in that certified class satisfies*

$$|c_0(h)| \le (1-2\eta)^q, \qquad |\widehat{c}_{\eta,S}(h)| \le (1-2\eta)^{q+1} + 2\Gamma^{\mathrm{sat,stat}}_{n,B,m,\delta}. \tag{254}$$

Proof. The residual-noise certificate and Lemma I.5.8 give $\mathrm{Ext}_I(h;T_I) \le \rho^{2q}$. The normalization calculation in the proof of Theorem IV.11.2 gives

$$|c_0(h)|^2 \le \mathrm{Var}_{\mu_I}(h)\,\mathrm{Ext}_I(h;T_I).$$

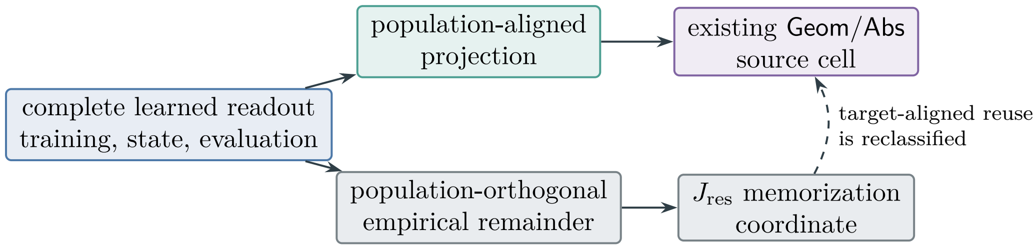

Figure 53: Learned-readout provenance from Theorem IV.11.4. Population-aligned use keeps its complete structural ancestry; the orthogonal sample-dependent component is memorization in J_{res}. An affordable target-aligned reuse returns to the existing source ledger.

Since $h \in [-1,1]$, its variance is at most one. Taking square roots proves the first inequality in (254). Substitution into $|\widehat{c}_{\eta,S}(h)| \le \rho|c_0(h)| + 2\Gamma$ proves the second.

□

IV.11.1 Noise-indexed population alignment and learning accountability

The sample-independent hull in Definition IV.11.1 is the correct class for a simultaneous concentration event. Computational recovery, however, is evaluated under the realized sample and the algorithm's represented randomness. The following profiles retain that quantifier order.

Definition IV.11.6 (Average-seed population alignment and generalizing projection profiles)**.** Fix a balanced presented binary prediction family, sample length m, budget B, and a monotone positively homogeneous family functional $\mathfrak{M}_n$. Let the supremum below range over one family-uniform represented learner $\mathcal{A}$ whose complete training, retained-state, output, fresh-input evaluation, and any invoked decoder cost is at most B. For instance I, let S have the declared training-sample law, let ω have the learner's represented seed law, and write

$$h_{I,S,\omega} = h_{\mathcal{A},I,S,\omega}, \qquad c_{0,I}(h) = \mathbb{E}_{X\sim\mu_I}[h(X)f_{*,I}(X)].$$

Define

$$\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n}(B,m) = \sup_{\mathcal{A}} \mathfrak{M}_n\Big(I \longmapsto \mathbb{E}_{S,\omega}|c_{0,I}(h_{I,S,\omega})|^2\Big), \tag{255}$$

$$\mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n}(B,m) = \sup_{\mathcal{A}} \mathfrak{M}_n\Big(I \longmapsto \mathbb{E}_{S,\omega}\Big[\mathrm{Var}_{\mu_I}(h_{I,S,\omega})\,\mathrm{ProjMass}_{\sigma(h_{I,S,\omega})}(T_I)\Big]\Big), \tag{256}$$

with supremum zero when the learner class is empty. These are evaluations of the existing saturated statistical class, not additional structural channels. The expectation over (S,ω) occurs inside the family functional and before the supremum over the one uniform learner rule. In particular, a seed or sample on which the learner happens to output the target readout contributes according to its probability under the declared execution law.

For homogeneous flip noise, append a subscript η to either profile. For a scalarized budget sublevel s, append κ and restrict the same supremum to complete scalarized cost at most s.

Theorem IV.11.7 (Population alignment, generalization, and memorization separation)**.** *In the setting of Definition IV.11.6,*

$$\operatorname{Align}^{\text{sat,stat}}_{\mathfrak{M},n}(B,m) \le \operatorname{GenProj}^{\text{sat,stat}}_{\mathfrak{M},n}(B,m) \le \mathfrak{M}_n\Big(I \longmapsto M^{\text{sat,stat}}_{I,n}(B,m)\Big). \tag{257}$$

On the simultaneous event of Theorem IV.11.2, every represented learner and every realized sample satisfy

$$\begin{aligned}\mathbb{E}_\omega |c_{0,I}(h_{I,S,\omega})|^2 \le \mathbb{E}_\omega \min\Bigg\{ &\operatorname{Var}_{\mu_I}(h_{I,S,\omega}) \operatorname{ProjMass}_{\sigma(h_{I,S,\omega})}(T_I),\\ &\left(\frac{|\widehat{c}_{\eta,S}(h_{I,S,\omega})| + 2\Gamma^{\text{sat,stat}}_{n,B,m,\delta}}{1-2\eta}\right)^2\Bigg\}.\end{aligned} \tag{258}$$

Writing E_I for this simultaneous event, its probability bound and $|c_{0,I}(h)| \le 1$ also give the execution-averaged certificate

$$\begin{aligned}\mathbb{E}_{S,\omega} |c_{0,I}(h_{I,S,\omega})|^2 \le \delta + \mathbb{E}_S\Bigg[\mathbf{1}_{E_I}\, \mathbb{E}_\omega \min\Bigg\{ &\operatorname{Var}_{\mu_I}(h_{I,S,\omega}) \operatorname{ProjMass}_{\sigma(h_{I,S,\omega})}(T_I),\\ &\left(\frac{|\widehat{c}_{\eta,S}(h_{I,S,\omega})| + 2\Gamma^{\text{sat,stat}}_{n,B,m,\delta}}{1-2\eta}\right)^2\Bigg\}\Bigg].\end{aligned} \tag{259}$$

Suppose a represented decoder D applied to the retained output has the following target-alignment margin: whenever $D(h_{I,S,\omega})$ is the correct target, $|c_{0,I}(h_{I,S,\omega})| \ge \gamma$, for some represented $0 < \gamma \le 1$. Then its recovery probability obeys

$$\mathfrak{M}_n\left(I \longmapsto \Pr_{S,\omega}\{D(h_{I,S,\omega}) \in T_I\}\right) \le \gamma^{-2} \operatorname{Align}^{\text{sat,stat}}_{\mathfrak{M},n}(B,m). \tag{260}$$

Every sample table, optimization transcript, empirical prediction vector, and seed-dependent lookup record retains its complete represented storage and evaluation cost. Its population projection is the term measured by Eq. (256)*. The orthogonal population component is assigned to J_{res} by Theorem II.7.2. If a family-uniform affordable use of that component has positive population target alignment and satisfies the relevant represented channel predicate and pre-hit comparison, its projection is reclassified into the existing* Geom/Abs *source ledger at its complete cost. Without those gates it remains an analytic population audit. Thus instance-specific memorization contributes no population alignment unless the represented learner generalizes it under the declared family law.*

Proof. For every fixed (I,S,ω), the projection calculation in Theorem IV.11.2 gives

$$|c_{0,I}(h)|^2 \le \operatorname{Var}_{\mu_I}(h) \operatorname{ProjMass}_{\sigma(h)}(T_I).$$

Take expectation over (S,ω), apply the monotonicity of $\mathfrak{M}_n$, and then take the supremum over the same uniform learner rules. This proves the first inequality in (257). Each realized output belongs to $\mathcal{H}^{\text{sat,stat}}_{n,B,m}$, its variance is at most one, and its projection mass is at most $M^{\text{sat,stat}}_{I,n}(B,m)$. The second inequality follows.

On the simultaneous event, Eq. (243) and the pointwise projection bound are both valid for every represented seed. Squaring the first, taking the minimum of the two pointwise ceilings, and averaging over ω proves (258). On E_I^c, the score range and the binary target give $|c_{0,I}(h)|^2 \le 1$. Splitting the sample expectation over E_I and E_I^c, and using $\Pr(E_I^c) \le \delta$, proves (259).

The decoder-margin condition gives the pointwise inequality

$$\mathbf{1}_{\{D(h_{I,S,\omega})\in T_I\}} \le \gamma^{-2}|c_{0,I}(h_{I,S,\omega})|^2.$$

Expectation, application of $\mathfrak{M}_n$, and the defining supremum in (255) prove (260).

Finally, the complete learner record belongs to the lifted computation carrier by Definition IV.11.1. Apply Theorems II.7.2 and III.25.2 to its population use. The represented structural projection is charged to the existing source channels, and its orthogonal complement is J_{res}. Reclassification of an affordable target-aligned residual representative is Proposition II.8.3. This proves the final assertion.

□

IV.12 Geometry-Aware Learning and Affordable Target Selection

The learning record and every geometric use of its output belong to one saturated derivation. The next lemma compares the target-reach factor of a learned potential with the generated-algebra projection already used by the statistical ledger.

Lemma IV.12.1 (Generated-algebra ceiling for a saturated geometric potential)**.** *Fix an instance I, target T_I, and a represented ambient record*

$$\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K}) \in \mathcal{G}^{X,\mathfrak{A}}_{n,\le B}$$

from Definition III.14.9. Define the bounded represented score

$$h_D = \frac{1-D}{1+D}. \tag{261}$$

Then $h_D : X_I \to (-1, 1]$, the maps $D \mapsto h_D$ and $h_D \mapsto D = (1 - h_D)/(1 + h_D)$ generate the same observable algebra, and

$$R_{T_I}(D) \le \mathrm{ProjMass}_{\sigma(D)}(T_I) = \mathrm{ProjMass}_{\sigma(h_D)}(T_I). \tag{262}$$

Consequently,

$$V^X_{\Phi,D,J,\mathfrak{K}} \le \frac{M_\Phi C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D)}{1+\rho_\Phi(D)} \mathrm{ProjMass}_{\sigma(h_D)}(T_I) \le \mathrm{ProjMass}_{\sigma(h_D)}(T_I). \tag{263}$$

If the presented task is a same-carrier statistical task and the complete record $\mathfrak{g}$ is generated from a length-m sample, then h_D belongs to $\mathcal{H}^{\mathrm{sat,stat}}_{n,\Psi(B,n),m}$. Let $G^{\mathrm{src,sat}}_{X,n,\mathrm{learn}(m)}(B)$ denote the restriction of the ambient source profile to precisely those complete same-carrier geometric records generated from a length-m sample under this presentation. Then

$$G^{\mathrm{src,sat}}_{X,n,\mathrm{learn}(m)}(B) \le \sup_{I\in\mathfrak{I}_n} M^{\mathrm{sat,stat}}_{I,n}(\Psi(B,n), m). \tag{264}$$

No conclusion for primitive, nonlearned, or differently sampled records in the unrestricted profile $G^{\mathrm{src,sat}}_{X,n}(B)$ follows from this statistical-hull inclusion.

For a distinguished instance sequence or a declared monotone family functional, the same conclusion holds with the corresponding evaluation on both sides, as in Definition III.18.4.

Proof. Every function of D, and therefore every member of the closed monotone span $\mathcal{M}(D)$, is $\sigma(D)$-measurable. Thus

$$\mathcal{M}(D) \subseteq L^2(X_I, \sigma(D), \mu_I).$$

Orthogonal projection onto a closed subspace cannot have larger norm than orthogonal projection onto a containing closed subspace. Applying this to the centered target contrast y_{T_I} gives

$$\|\Pi_{\mathcal{M}(D)} y_{T_I}\|_2 \leq \|P_{\sigma(D)} y_{T_I}\|_2.$$

Division by $\|y_{T_I}\|_2$ proves the first inequality in (262). The transform in (261) is one-to-one on $[0, \infty)$, with the displayed inverse on its image. Hence $\sigma(h_D) = \sigma(D)$. The forward transform is a finite represented postprocessing, and its complete evaluation cost is included in $\Psi(B, n)$. This proves the equality in (262).

Insert that ceiling into the visibility definition in Definition III.14.9. Since M_Φ, $C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D)$, and $(1 + \rho_\Phi(D))^{-1}$ lie in $[0, 1]$, both inequalities in (263) follow.

For a same-carrier statistical presentation, the derivation producing $\mathfrak{g}$ already contains its sample, retained state, represented random seed, potential, and evaluation rule. Appending the bounded transform therefore produces one member of $\mathcal{H}^{\mathrm{sat,stat}}_{n,\Psi(B,n),m}$ by Definition IV.11.1 and Lemma IV.0.2. Its projection mass is bounded by the defining supremum $M^{\mathrm{sat,stat}}_{I,n}(\Psi(B, n), m)$. Taking the saturated supremum over the same family-uniform geometric derivations proves (264). The other evaluation conventions follow by pointwise application followed by the declared monotone functional.

□

Corollary IV.12.2 (Qualified learned-current accountability)**.** *Adopt the same-carrier statistical setting of Lemma IV.12.1. Let $\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K})$ be one complete represented learned geometric record of saturated cost at most B, let $h_D = (1 - D)/(1 + D)$, and let S_I be a represented edge sector supporting $\nabla_\Phi D$. For its qualification indicator $Q_{I,S,\omega}$,*

$$Q_{I,S,\omega} V^X_{\Phi,D,J,\mathfrak{K}} \leq Q_{I,S,\omega} \min\left\{ \mathrm{ProjMass}_{\sigma(h_D)}(T_I), \frac{\|\mathbf{1}_{S_I} J_{\mathfrak{K}}\|_E^2}{\|J_{\mathfrak{K}}\|_E^2} \right\}, \tag{265}$$

where the current ratio is zero when $J_{\mathfrak{K}} = 0$.

Suppose qualification entails correctness of a represented decoder whose derivation and evaluation are included in the same cost-B record, and that decoder has the target-alignment margin γ of Theorem IV.11.7. Then every monotone sublinear positively homogeneous family functional satisfies

$$\begin{aligned} &\mathfrak{M}_n\Big(I \longmapsto \mathbb{E}_{S,\omega}[Q_{I,S,\omega} V^X_{\Phi,D,J,\mathfrak{K}}]\Big) \\ &\quad \leq \gamma^{-2} \, \mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n}(B, m) \leq \gamma^{-2} \, \mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n}(B, m) \\ &\quad \leq \gamma^{-2} \mathfrak{M}_n\Big(I \longmapsto M^{\mathrm{sat,stat}}_{I,n}(B, m)\Big). \end{aligned} \tag{266}$$

On the simultaneous noise-transport event, the same learned record also satisfies the realized empirical certificate Eq. (258)*, including the factor $1 - 2\eta$ and the complete saturated selection radius.*

Thus the admitted current, the potential generating its gradient component, the learned target projection, and the decoder qualification probability are bounded in one cost-preserving record and then evaluated through the single saturated family supremum.

Proof. Equation (263) bounds the visibility by $\mathrm{ProjMass}_{\sigma(h_D)}(T_I)$, while Eq. (199) bounds it by the current-energy fraction on S_I. Multiplication by the qualification indicator proves Eq. (265).

Qualification and the decoder hypothesis give

$$Q_{I,S,\omega} \le \mathbf{1}_{\{D_{\mathrm{dec}}(h_{I,S,\omega})\in T_I\}}.$$

Since visibility lies in $[0,1]$, Eq. (201) bounds the left side of Eq. (266) by the family-functional decoder success probability. Apply Eq. (260) and then Eq. (257). The realized noise-transport statement is Eq. (258).

□

The measurable projection ceiling becomes statistically testable through a represented readout. The following proposition applies the existing actionable-mass and noise-transport bounds to every such readout at once.

Proposition IV.12.3 (Noise-corrected actionable extraction for learned geometries)**.** *Adopt the balanced homogeneous-noise setting of Theorem IV.11.2, put $\rho = 1 - 2\eta > 0$, and fix a budget B. Let $g : X_I \to [-1,1]$ be any represented target-discrimination, monotone-potential, ranking, or quotient readout produced by a budget-B learned geometric record. Include its complete construction, retained training state, target-discrimination certificate, and evaluation cost, so that*

$$g \in \mathcal{R}^{\mathrm{cert}}_{I,\Psi(B,n)}(T_I) \cap \mathcal{H}^{\mathrm{sat,stat}}_{n,\Psi(B,n),m}.$$

With probability at least $1-\delta$, simultaneously for every such readout with positive variance,

$$\mathrm{Ext}_I(g;T_I) \le \min\left\{\begin{array}{l} \mathrm{ActMass}_{I,\Psi(B,n)}(T_I), \\ \mathrm{ProjMass}_{\sigma(g)}(T_I), \\ \dfrac{\left(|\widehat{c}_{\eta,S}(g)| + 2\Gamma^{\mathrm{sat,stat}}_{n,\Psi(B,n),m,\delta}\right)^2}{\rho^2\,\mathrm{Var}_{\mu_I}(g)} \end{array}\right\}. \tag{267}$$

For a constant readout the extraction is zero. If an empirical audit, effective-dimension ceiling, or certified residual-noise order also applies, the corresponding ceilings of Definition I.5.4 and Lemma I.5.8 enter the same minimum.

Proof. Membership in the certified class and Lemma I.5.8 give the actionable-mass ceiling. The generated-algebra projection ceiling Lemma I.4.6 gives the second ceiling. Balance implies $y_{T_I} = f_*/2$, and hence, for nonconstant g,

$$\mathrm{Ext}_I(g;T_I) = \frac{|c_{0,I}(g)|^2}{\mathrm{Var}_{\mu_I}(g)}. \tag{268}$$

The simultaneous transport inequality (243), applied at the enlarged complete cost, bounds

$$|c_{0,I}(g)| \le \frac{|\widehat{c}_{\eta,S}(g)| + 2\Gamma^{\mathrm{sat,stat}}_{n,\Psi(B,n),m,\delta}}{\rho}.$$

Squaring and substituting in (268) proves the third ceiling. All three hold on the same simultaneous event, so their minimum is valid. The additional audit ceilings are the remaining conclusions of Lemma I.5.8.

□

Theorem IV.12.4 (Saturated geometry–learning–selection accountability)**.** *Let a balanced binary-label task family have homogeneous noise $0 \le \eta < 1/2$, and let $\mathfrak{M}_n$ be a monotone sublinear positively homogeneous family functional. Equip the task with its represented public verifier and the one-test clock refinement of Definition III.21.2. Consider one family-uniform represented computation whose complete learner, retained state, learned geometry, target-discrimination readout, decoder, pre-hit sampler, verifier calls, and terminal output have saturated cost at most B. Let $H_B(n)$ be its largest admitted refined horizon. Suppose the represented decoder has target-alignment margin $0 < \gamma \le 1$ in the sense of Theorem IV.11.7. Then*

$$\mathfrak{M}_n\Big(I \longmapsto \Pr_{S,\omega}\{\textit{the represented decoder reaches } T_I\}\Big) \le \min\left\{\begin{matrix}\gamma^{-2}\,\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m), \\ eH_B(n)\displaystyle\sum_{c\in\mathfrak{J}^5_{\mathsf{Abs}}}\mathfrak{M}_n\Big(I \longmapsto \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_B(n))\Big)\end{matrix}\right\}. \tag{269}$$

The second entry is the whole-transcript first-hit accounting-charge bound of Theorem III.21.4. It becomes a prospective source envelope only when the same record also supplies the represented pre-hit comparison required by Corollary III.16.9.

The first entry admits both existing uniform ceilings

$$\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m) \le \mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m) \le \mathfrak{M}_n\Big(I \longmapsto M^{\mathrm{sat,stat}}_{I,n}(B,m)\Big), \tag{270}$$

and, on the simultaneous event of Theorem IV.11.2, the empirical certificate in Eq. (258)*. A same-carrier learned geometric source also satisfies Eq.* (264)*; a represented cross-carrier decoder is controlled by its displayed margin γ and the first entry of* (269)*.*

Every score evaluation and every target-aligned selection operation are charged in the same record. A score with an affordable evaluator enters the statistical class through a represented training and output derivation. A decoder or selector enters the budget slice when its own derivation, state, queries, and execution cost fit that slice. Empirical tables and memorized lookup state remain charged lifted coordinates. Their population-orthogonal component is J_{res}; an affordable population-aligned use is reclassified into the existing Geom/Abs *source ledger only after the relevant channel predicate and pre-hit comparison are verified.*

Proof. The decoder-margin conclusion (260), applied to the complete joint record, proves the first entry of (269). Its two projection ceilings are exactly (257), which proves (270). The realized-sample bound is (258).

For every fixed instance, the complete decoder and its verifier calls form a represented finite-horizon computation. Apply Theorem III.21.4 at budget B:

$$\Pr_{S,\omega}\{\text{the decoder reaches } T_I\} \le eH_B(n)\sum_{c\in\mathfrak{J}^5_{\mathsf{Abs}}}\mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_B(n)).$$

Apply $\mathfrak{M}_n$, use positive homogeneity and subadditivity, and obtain the second entry of (269). Both entries bound the same success quantity, so their minimum also bounds it. The source interpretation of the second entry, when used, is exactly the additional transfer in Corollary III.16.9.

By Definition IV.11.1, class membership includes training, retained state, output representation, and fresh-input evaluation. By Definitions III.21.1 and III.21.2, the selector record also includes every pre-hit state, sampler, random-bit use, and transition implementation. The budgeted derivability closure Definition III.1.3 charges their composition as one derivation. Thus inexpensive evaluation of a fixed score supplies no target-aligned selector without the represented selection derivation and its cost.

Finally, apply Theorems II.7.2 and IV.11.4. The population structural projection is charged to the existing source ledger, and the orthogonal empirical component is J_{res}. Its affordable target-aligned reuse is reclassified by Proposition II.8.3. This proves the provenance assertions.

□

IV.13 Complete Learned-Source Exhaustion and Noise Degradation

Corollary IV.13.1 (Complete structural exhaustion of learned sources)**.** *Adopt the setting of Theorem IV.12.4. Every temporally admissible, dynamically qualified source record produced or used by the complete learner–decoder–selector computation has, after the class-preserving normalization of Part III, exactly one index*

$$(\chi, S_{\mathrm{st}}, S_{\mathrm{cf}}), \qquad \chi \in \{\mathsf{A}, \mathsf{X}, \mathsf{Q}_{\mathrm{ex}}, \mathsf{Q}_{\mathrm{rev}}, \mathsf{Q}_{\mathrm{nr}}\},$$

with

$$\varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}, \qquad S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}.$$

Thus the complete learned source record lies in one of the 4,960 existing source-mode–state-provenance–coefficient-provenance cells of Theorem III.3.4. The index retains the learner, selected kernel or geometry, represented coefficients, decoder, sampler, qualification gates, and every downstream structural use at their complete joint cost.

Restricting the Part III supremum to learned prospective source records of complete cost at most B gives

$$\sup_{\substack{\mathscr{C} \text{ is a learned prospective source record} \\ \mathrm{Cost}(\mathscr{C}) \preceq B}} \mathfrak{M}_n(I \longmapsto Q_I V_{\mathscr{C},I}) \leq \max_{\substack{\varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}} \\ S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}}} \mathrm{StructAct}^{\mathrm{sat}}_{\mathfrak{M}, S_{\mathrm{st}}, S_{\mathrm{cf}}, n}(\Psi_{\mathrm{all}}(B, n)). \tag{271}$$

Every learned source therefore retains the complete numerical visibility of its existing master cell. In particular:

(i) *the pure exact-*Abs *mode* A *is evaluated by the prospective quotient profile;*

(ii) Q_{ex}*,* $\mathsf{Q}_{\mathrm{rev}}$*, and* Q_{nr} *retain, respectively, the exact, reversible-compensated, and general-resolvent quotient–Geom visibility, including utility, stability, mixing, authorized Caus alignment, target reach, and regularity;*

(iii) *the ambient mode* X *retains the exact same-record product*

$$Q_I V^X_{\Phi,D,J,\mathfrak{K}} = Q_I M_\Phi C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D) \frac{R_{T_I}(D)}{1 + \rho_\Phi(D)}. \tag{272}$$

For a same-carrier learned geometry, the generated-algebra ceiling gives the simultaneous refinement

$$Q_I V^X_{\Phi,D,J,\mathfrak{K}} \leq Q_I \min\left\{ M_\Phi C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D) \frac{R_{T_I}(D)}{1 + \rho_\Phi(D)}, \mathrm{ProjMass}_{\sigma(h_D)}(T_I) \right\}, \qquad h_D = \frac{1-D}{1+D}; \tag{273}$$

(iv) *authorized* Caus*, valid* Lift*, and* Bdry *uses inherit the same provenance cell, while their complete represented costs remain in the record. They inherit a numerical source value only through the represented contextual transfer required by that cell; provenance alone supplies no numerical transfer. The post-saturation* J_{res} *component contributes no prospective source cell.*

Consequently the generalized learning bounds use the full Part III structural exhaustion. The statistical alignment and generalization ceilings in Eq. (270) *and the structural source ceiling in Eq.* (271) *bound the same complete computation through their respective valid comparisons. The success estimate retains their minimum as in Eq.* (269)*; neither the statistical bound nor a provenance label replaces any Geom/Abs gate. The source-resolved noise, data, confidence, and complexity coordinates are compiled recordwise by Theorem IV.13.3; hence every learned source cell carries its own statistical candidate, ordered-reveal contraction, contextual charge, and deployment converter.*

Proof. The complete learning computation belongs to the budgeted derivability closure by Definition IV.11.1 and Lemma IV.0.2. Apply Theorem III.3.4 to each temporally admissible qualified source record in that derivation. This gives the unique master mode and exact state–coefficient provenance pair, with all instance-dependent coefficient derivations and qualification gates retained. Since the learned records form a subfamily of the records defining the existing saturated source profile, restriction of the supremum followed by Eq. (43) proves Eq. (271).

The mode assignments and their numerical visibility factors are Theorems III.18.9 and III.25.2. The exact cellwise quotient and ambient evaluations are Corollary III.4.1. In the ambient case, Definition III.14.9 gives Eq. (272), and Lemma IV.12.1 gives the second ceiling in Eq. (273). Provenance inheritance and the residual assignment are Proposition II.5.9 and Theorem II.7.2; numerical transfer uses Corollary III.16.9 when its hypotheses hold. Finally, Theorem IV.12.4 already takes the minimum of the valid statistical and structural estimates for the complete joint record. This proves the final assertion.

□

IV.13.1 Source-resolved quantitative learning ledger

Definition IV.13.2 (Complete source-resolved learning ledger)**.** Let R be a complete represented learning record of sample size m, confidence level $1-\delta$, horizon H, and saturated cost at most B. The record includes data acquisition, feature and kernel construction, optimization, retained state, output evaluation, every decoder or selector, and deployment. Normalize every temporally admissible target-qualified source in its backward slice by Theorem III.3.4. Its exact index is

$$\alpha = (\chi, S_{\mathrm{st}}, S_{\mathrm{cf}}), \qquad \chi \in \{\mathsf{A}, \mathsf{X}, \mathsf{Q}_{\mathrm{ex}}, \mathsf{Q}_{\mathrm{rev}}, \mathsf{Q}_{\mathrm{nr}}\},$$

with $\varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}$ and $S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}$. Write $\mathfrak{A}_{\mathrm{learn}}$ for this 4960-element index set.

For $a \in \mathfrak{J}^5_{\mathsf{Abs}}$, let

$$\mathcal{L}^{(a)}_{I,R} = \sum_{\ell\colon \mathsf{W}_a(E_{R,\ell})} \mathrm{ch}_I(E_{R,\ell}) \tag{274}$$

be the five-family contextual charge in the complete learner–deployment backward slice. Every selected likelihood, feature locator, coefficient, optimizer, belief coordinate, acquisition score, kernel, current, Lift map, boundary converter, and output rule remains in the carrier of this sum.

Enumerate the target-dependent reveals actually available before deployment as $Y_1, \ldots, Y_{L_R}$, use the canonical anchored assignment of Corollary III.16.10, and write a_j for the source tag of Y_j.

Define

$$J_{I,R}^{(a)} = \sum_{\substack{1\le j\le L_R\\ a_j=a}} \mathbb{E}\min\{c_j, \vartheta_j a_j^{\mathrm{info}}\}, \qquad J_{I,R}^{\mathrm{src}} = \sum_{a\in\mathfrak{I}^5_{\mathsf{Abs}}} J_{I,R}^{(a)}. \tag{275}$$

For a binary symmetric reveal of rate η_j, the admissible channel coefficient is $\vartheta_j = (1-2\eta_j)^2$. Other observation, reward, transition, and filter channels use their typed coefficients from Definition V.3.1. Deterministic postprocessing contributes zero information and retains its construction cost.

For a time $0 \le t < H$ and common ceiling D, let $\mathscr{R}_t^{\mathrm{learn}}(D)$ denote the complete family-uniform learning-record derivations deployed at time t with saturated cost at most D, and write R_I for the record produced on instance I. Define the saturated cellwise envelopes

$$\mathcal{L}_{\infty,t}^{(a),\mathrm{sat}}(D) = \sup_{R\in\mathscr{R}_t^{\mathrm{learn}}(D)} \sup_I \mathcal{L}_{I,R_I}^{(a)}, \tag{276}$$

$$J_{\infty,t}^{(a),\mathrm{sat}}(D) = \sup_{R\in\mathscr{R}_t^{\mathrm{learn}}(D)} \sup_I J_{I,R_I}^{(a)}. \tag{277}$$

The supremum uses one family-uniform represented record on the whole size slice; only its numerical evaluation is then maximized over presented instances. Thus these envelopes are pointwise ceilings and require no interchange of a square root with a family functional.

Let $\mathcal{H}_\alpha(B,m)$ be the output or loss class of complete represented records of saturated cost at most B whose selected prospective source has exact index α. On the same probability law and in the same loss units, let $\Gamma_{R,\alpha}^{\mathrm{learn}}(B,m,\boldsymbol{\eta},\delta)$ be the minimum of the applicable fixed-operator Rademacher, learned-operator, effective-dimension, structural PAC–Bayes, sequential-Rademacher, martingale PAC–Bayes, mixing-block, and finite-description candidates for that restricted class. Its value is $+\infty$ when no candidate applies.

Finally, let $U_R^{\mathrm{stat}\to\mathrm{tgt}}$ be the smallest bound obtained by a represented same-law, same-unit converter from one of these statistical loss bounds to the deployed target-contact advantage. Let $U_R^{\mathrm{dyn}\to\mathrm{tgt}}$ be the corresponding minimum of applicable capacity, Blackwell, Caus, Lift, observation, active-acquisition, and controlled-dynamics bounds. An unavailable target converter has value one. All objects are evaluated at one class-preserving common ceiling

$$D_B^{\mathrm{learn}} = \Psi_{\mathrm{learn}}(B,H,n,m,\boldsymbol{\eta},\delta), \tag{278}$$

which majorizes the complete record, normalized frontier, ordered reveals, confidence certificates, channel corrections, and every converter used in a displayed minimum. This ceiling changes no public constructor algebra.

Theorem IV.13.3 (Source-resolved compilation of saturated learning)**.** *Use the ledger of Definition IV.13.2. The following statements hold.*

(i) *The target-qualified prospective source records form the disjoint union of their exact cells $\alpha \in \mathfrak{A}_{\mathrm{learn}}$. Authorized* Caus*, valid* Lift*, and* Bdry *processing retain the exact cell of their earlier source. Their evaluation and deployment costs remain in D_B^{learn}. The post-saturation J_{res} component has zero target-qualified learning gain; an affordable aligned representative is reclassified before the residual is formed.*

(ii) *For losses in $[-1,1]$, the empirical Rademacher complexity of the complete exact-cell union satisfies*

$$\widehat{\mathfrak{R}}_S\left(\bigcup_{\alpha\in\mathfrak{A}_{\mathrm{learn}}} \mathcal{H}_\alpha(B,m)\right) \le \max_{\alpha\in\mathfrak{A}_{\mathrm{learn}}} \widehat{\mathfrak{R}}_S(\mathcal{H}_\alpha(B,m)) + \sqrt{\frac{2\log|\mathfrak{A}_{\mathrm{learn}}|}{m}}. \tag{279}$$

Empty cells are omitted. For a PAC–Bayes prior $P = \sum_\alpha w_\alpha P_\alpha$ *and posterior* $Q = \sum_\alpha q_\alpha Q_\alpha$ *with matching conditional cells,*

$$\mathrm{KL}(Q\|P) = \mathrm{KL}(q\|w) + \sum_\alpha q_\alpha \, \mathrm{KL}(Q_\alpha\|P_\alpha). \tag{280}$$

Thus source selection is paid either by the explicit finite-cell Rademacher term or by the PAC–Bayes cell-selection divergence. Effective-dimension and source-condition candidates are evaluated inside the selected exact cell; they are aggregated across cells only through a valid union, mixture, or represented selection bound.

(iii) *Let* K_R *be the represented deployed selector, let* ν_R *be its declared target-neutral baseline, and put* $g_R = (\mathrm{Adv}(K_R; \nu_R))_+$. *With* β_R^+ *the applicable decoupled neutral excess under the same pre-hit law,*

$$g_R \le \min\left\{1,\, U_R^{\mathrm{stat}\to\mathrm{tgt}},\, \beta_R^+ + \sqrt{\frac{\log 2}{2} J_{I,R}^{\mathrm{src}}},\, eH_B(n) \sum_{a\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{L}_{I,R}^{(a)},\, U_R^{\mathrm{dyn}\to\mathrm{tgt}}\right\}. \tag{281}$$

Each entry bounds the same survival-weighted target advantage under the same law. The minimum is taken for one complete record before any supremum.

(iv) *Let* $\mathscr{R}_t^{\mathrm{learn}}(D_B^{\mathrm{learn}})$ *be the complete family-uniform learning-record derivations deployed at time t, and denote the right side of Eq.* (281) *by* $U_{I,R,t}^{\mathrm{learn}}$. *Let* $\mathrm{Enum}_{I,\nu}^{\mathrm{sat,learn}}(B,H)$ *be the supremum of the declared neutral-contact sum over the same complete learning computations. Then*

$$\mathrm{Sel}_{I,n}^{\mathrm{sat,learn}}(B) \le \max_{0\le t<H} \sup_{R\in\mathscr{R}_t^{\mathrm{learn}}(D_B^{\mathrm{learn}})} U_{I,R_I,t}^{\mathrm{learn}}, \tag{282}$$

$$\mathrm{Succ}_{I,n}^{\mathrm{sat,learn}}(B) \le \mathrm{Enum}_{I,\nu}^{\mathrm{sat,learn}}(B,H) + \sum_{t=0}^{H-1} \sup_{R\in\mathscr{R}_t^{\mathrm{learn}}(D_B^{\mathrm{learn}})} U_{I,R_I,t}^{\mathrm{learn}}. \tag{283}$$

Both inequalities persist under a monotone family functional.

Accordingly every learning bound has four typed numerical layers: complete construction cost, source-resolved statistical capacity, source-resolved noise/information contraction, and a represented converter to the deployed target quantity. A finite value in one layer never substitutes for an unavailable value in another.

Proof. The exact-cell partition and channel inheritance in (i) are Theorems II.7.2 and III.3.4 and Proposition II.5.9. Every cell restriction keeps the complete represented record, so the common cost is Eq. (278).

For (ii), apply the standard exponential-moment proof of the Rademacher union bound to the finite collection of nonempty cell-restricted classes. Each empirical process is $1/\sqrt{m}$-sub-Gaussian for losses in $[-1,1]$; maximizing over at most $|\mathfrak{A}_{\mathrm{learn}}|$ classes gives the displayed selection term. The PAC–Bayes identity is the relative-entropy chain rule for the joint cell index and within-cell hypothesis. Restricting an operator or kernel to one cell leaves its existing effective-dimension and source-condition proof unchanged.

For (iii), the information candidate is the ordered-reveal decoupling inequality of Theorem V.9.7, with the chain rule split according to Eq. (275). The structural candidate is Theorem III.21.4 applied to the complete learner–deployment slice. The statistical and dynamical entries are admitted only through their represented target converters, so all entries bound the same scalar. Taking their minimum is therefore valid.

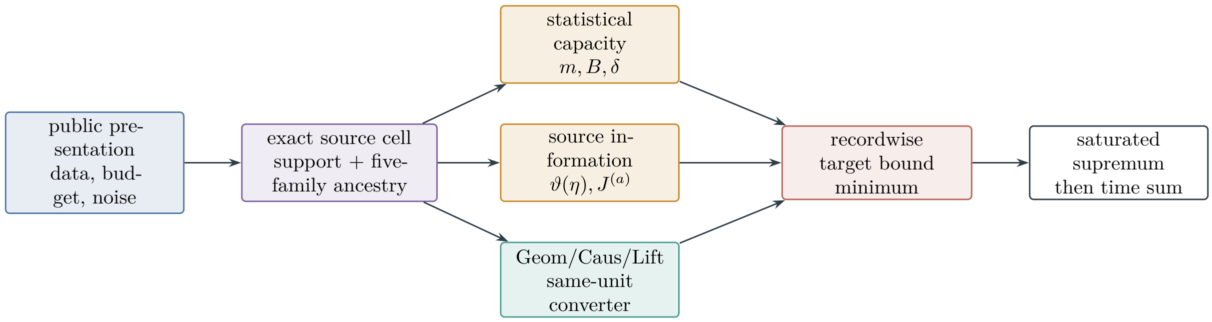

Figure 54: Source-resolved learning compilation. Each complete record keeps its exact structural ancestry, statistical capacity, source-assigned noise contraction, and dynamical converter. Comparable target bounds are minimized within the record before saturation, as required by Theorem IV.13.3.

Finally, take the record supremum only after the recordwise minimum and use Theorem III.21.3. This proves Eqs. (282) and (283). Monotonicity gives the family-functional form.

□

Corollary IV.13.4 (Noise–data–complexity learning consequence)**.** *At a common ceiling D_B^{learn}, suppose*

$$\mathcal{L}_{\infty,t}^{(a),\mathrm{sat}} \le \varepsilon_{a,t}(n, B, m, \boldsymbol{\eta}, \delta), \qquad J_{\infty,t}^{(a),\mathrm{sat}} \le \jmath_{a,t}(n, B, m, \boldsymbol{\eta}, \delta).$$

Then every complete saturated learning record satisfies

$$\mathrm{Succ}_{\mathfrak{M},n}^{\mathrm{learn}}(B) \le \mathrm{Enum}_{\mathfrak{M},\nu}(B, H) + \sum_{t=0}^{H-1} \min \left\{ \begin{matrix} 1,\ U_{\mathfrak{M},t}^{\mathrm{stat}\to\mathrm{tgt}},\ \beta_{\mathfrak{M},t}^{+} + \sqrt{\dfrac{\log 2}{2} \displaystyle\sum_a \jmath_{a,t}}, \\ eH_B \displaystyle\sum_a \varepsilon_{a,t},\ U_{\mathfrak{M},t}^{\mathrm{dyn}\to\mathrm{tgt}} \end{matrix} \right\}. \tag{284}$$

For a polynomial horizon, negligible neutral contact and negligible source-resolved inputs give negligible target arrival. The conclusion applies simultaneously to passive, active, structural Bayesian, sequential, and controlled learning because each is a restriction of the complete record class.

Proof. Apply Eq. (283) and substitute the two source-resolved ceilings before taking the time sum. The stated learning classes are restrictions of the same saturated record class, so no further union over learning paradigms is required.

□

Figure 55 summarizes the exact Part III classification inherited by every prospective learned source.

Corollary IV.13.5 (Finite-description radius for the complete saturated output class)**.** *Fix n, B, m, a scalarized locally finite derivation coding as in Definition III.1.3, and let*

$$N_{n,B,m}^{\mathrm{out}} = \sup_{I \in \mathfrak{I}_n} \left| \mathcal{H}_{n,B,m}^{\mathrm{sat,stat}}(I) \right|.$$

Set this value to $+\infty$ unless the fixed-budget output codebook is finite uniformly over $I \in \mathfrak{I}_n$. When it is finite, the clipped linear-loss class generated by the complete saturated outputs satisfies,

Learning coordinate	Statistical candidate	Source/noise ledger	Target deployment
Fixed or learned kernel, KRR, NTK	Rademacher, effective dimension, operator selection	exact cell, $\mathcal{L}^{(a)}$, $J^{(a)}$	represented decoder or selector margin
Structural PAC–Bayes	within-cell KL and cell-mixture KL	prior/posterior ancestry and channel SDPI	same-law posterior-to-action converter
Dependent or sequential learning	blocking, sequential Rademacher, martingale PAC–Bayes	ordered reveals with typed contraction	represented policy or next-test converter
Active and curiosity learning	observation resolution and learned-policy radius	acquisition information and five-family charge	next-contact, capacity, or master comparison
Controlled and Bayesian learning	model, Bellman, filter, and belief estimation	complete source cell with Caus/Lift inheritance	value, return, or arrival converter in matching units

Table 4: Application of the source-resolved ledger to the learning master bounds. Each row retains $B, m, \boldsymbol{\eta}, \delta$, and H when present. Statistical and structural quantities meet only in the final column through a represented converter.

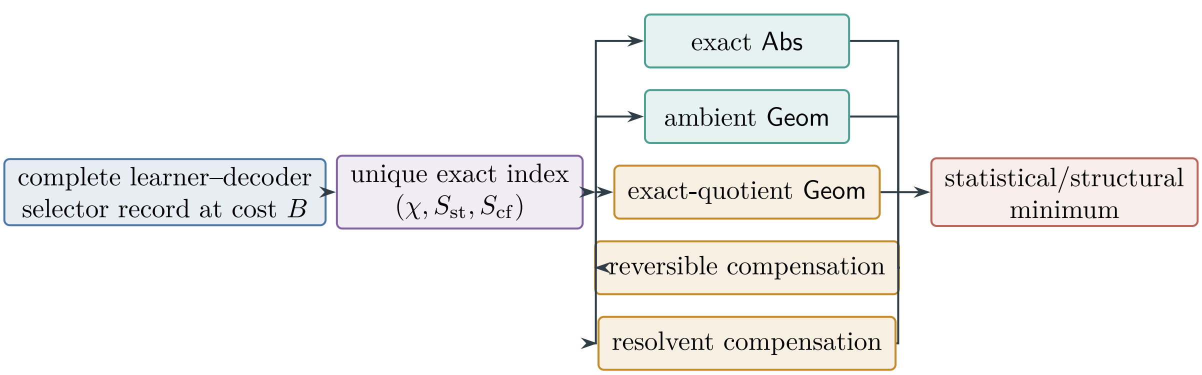

Figure 55: Complete structural exhaustion of learned sources. The complete learned record receives one of the 4,960 existing Part III indices from Corollary IV.13.1; its exact source visibility is combined recordwise with the applicable statistical, noise-information, and dynamical ceilings by Theorem IV.13.3.

$$\Gamma^{\mathrm{sat,stat}}_{n,B,m,\delta} \le 2\sqrt{\frac{2\log N^{\mathrm{out}}_{n,B,m}}{m}} + 3\sqrt{\frac{\log(2/\delta)}{2m}}. \tag{285}$$

The cardinality counts complete output records, including represented training state, selected geometry, retained coefficients and precision, decoder data, and every sample-dependent lookup value used at evaluation. When a declared coding estimate gives $\log N^{\mathrm{out}}_{n,B,m} \le L(B,n,m)$*, the first term in* (285) *is at most* $2\sqrt{2L(B,n,m)/m}$.

Proof. For each fixed instance, local coding in Definition III.1.3 uses a finite alphabet, charges every encoded coefficient, retained table, precision parameter, state value, and evaluation rule, and has finite scalar-cost sublevels. A uniform size-slice bound additionally requires the displayed supremum over instances to be finite; otherwise the corollary is applied pointwise in I, or its right side is $+\infty$.

Massart's finite-class lemma for a $[0,1]$-valued loss class gives

$$\widehat{\mathfrak{R}}_S(\mathcal{L}_{\mathcal{H}}) \le \sqrt{\frac{2\log N^{\mathrm{out}}_{n,B,m}}{m}}.$$

Substitution in the radius definition (233) proves (285). The final statement follows by monotonicity of the square root.

$\square$

Theorem IV.13.6 (Noise degradation and learning accountability)**.** *Let* $(\mathfrak{I}_{n,m,\eta})_{0\le\eta<1/2}$ *be a binary-label family under homogeneous flip noise, and let* $0 \le \eta_1 \le \eta_2 < 1/2$*. Put*

$$q(\eta_1,\eta_2) = \frac{\eta_2-\eta_1}{1-2\eta_1}.$$

Let $\psi_{\mathrm{flip}}(B,n,m)$ *charge the represented independent Bernoulli-*$q(\eta_1,\eta_2)$ *degradation of a length-*m *sample and every downstream retained use. Assume that* $\mathfrak{M}_n$ *is subadditive in addition to being monotone and positively homogeneous. Then*

$$\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_1}(\psi_{\mathrm{flip}}(B,n,m),m) \ge \mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_2}(B,m), \tag{286}$$

$$\mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_1}(\psi_{\mathrm{flip}}(B,n,m),m) \ge \mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_2}(B,m). \tag{287}$$

For a finite horizon H_B*, define the saturated neutral-contact profile*

$$\mathrm{Enum}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(B,H_B) = \sup_{\mathcal{A}:\mathrm{Cost}(\mathcal{A})\preceq B} \mathfrak{M}_n(I \longmapsto \mathrm{Enum}_\nu(\mathcal{A},I,H_B)), \tag{288}$$

where the same declared target-neutral baseline rule is used throughout the family. Write $\mathrm{Succ}_{\mathfrak{M},n,\eta}(B)$ *for Eq.* (178) *evaluated on* $\mathfrak{I}_{n,m,\eta}$*. Define*

$$\mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(B) = \sup_K \mathfrak{M}_n\Big(I \longmapsto \left(\mathrm{Adv}_t(K_{I,t};\nu_{I,t})\right)_+\Big), \tag{289}$$

where the supremum ranges over one family-uniform temporally admissible represented selector derivation, including its represented time rule, of complete saturated cost at most B*. Thus one selector program is used on the entire size slice; the supremum is not chosen separately for each instance. Every represented budget-*B *learning or search computation satisfies*

$$\mathrm{Succ}_{\mathfrak{M},n,\eta}(B) \le \mathrm{Enum}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(B, H_B) + H_B\, \mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(\Psi(B, n)). \tag{290}$$

For the complete learning subclass, the selector term in Eq. (290) *has the source-resolved evaluation*

$$\mathrm{Sel}^{\mathrm{sat,learn}}_{\mathfrak{M},n,\eta}(B) \le \max_{0\le t<H_B} \sup_{R\in\mathscr{R}^{\mathrm{learn}}_t(D^{\mathrm{learn}}_B)} \mathfrak{M}_n\Big(I \longmapsto U^{\mathrm{learn}}_{I,R_I,t}\Big), \tag{291}$$

where $U^{\mathrm{learn}}_{I,R,t}$ is the same-record minimum in Eq. (281)*. In particular, the binary noise factor $(1-2\eta)^2$, the five source charges, the sample size, confidence, complete construction cost, and every statistical-to-target or dynamics-to-target converter remain in the selector profile rather than being discarded after a risk estimate.*

Consequently, a negligible neutral-contact profile together with a negligible prospective selector profile gives negligible target arrival. If $\mathfrak{M}_n$ is expectation under the declared family law, then conversely, if one budget-B computation has family-averaged success exceeding its family-averaged neutral contact by ε, then its represented pre-hit selectors have total positive advantage exceeding ε, and at least one refined step has family-averaged positive advantage exceeding ε/H_B.

The profiles in Eqs. (286), (287) *and* (290) *range over the complete saturated budget class. Hence they quantify the noise dependence of all family-uniform represented learning rules at once; no algorithm-specific source family is introduced.*

Proof. An independent flip of rate $q(\eta_1, \eta_2)$ following a flip of rate η_1 has total flip rate η_2. Given any budget-B learner for the η_2 presentation, first apply this represented degradation to an η_1 sample and then run the same learner. The joint distribution of its retained output, represented seed, and every downstream evaluation is identical to the original η_2 execution. Its complete cost is at most $\psi_{\mathrm{flip}}(B, n, m)$. The two defining suprema therefore give (286) and (287).

For every fixed computation, the pointwise prospective hazard identity in Theorem III.21.3, followed by replacement of each signed advantage by its positive part, gives

$$\Pr_I\{\tau \le H_B\} \le \mathrm{Enum}_\nu(\mathcal{A}, I, H_B) + \sum_{t=0}^{H_B-1} \big(\mathrm{Adv}_t(K_{I,t}; \nu_{I,t})\big)_+.$$

Apply the monotone sublinear positively homogeneous functional $\mathfrak{M}_n$, and then take the supremum over the common budget class. For each fixed t, the transition of a uniform computation is one of the uniform selector derivations in (289); no pointwise choice of a different selector for each instance is made. Subadditivity contributes at most H_B selector terms, and the supremum of a sum is at most the sum of the suprema. This proves (290). For the linear expectation functional, the converse is the family-functional form of the exact identity (152): the sum of the positive parts of the step advantages dominates their signed sum, and one of at most H_B terms is at least their average.

Finally, Eq. (291) is Eq. (282) restricted to the noise-indexed family at the common ceiling. This restriction changes neither the public algebra nor the complete cost of a record.

□

Corollary IV.13.7 (Noise-indexed learning threshold criterion)**.** *Use the expectation functional $\mathfrak{M}_{n,m,\eta}$ and fixed scalarization κ of Definition IV.14.2. Assume that every computation counted by $B^*_{\mathrm{rec}}(n, m, \eta; \alpha)$ has, within the same complete scalar cost, a represented learner–decoder realization counted by Definition IV.11.6, and that every successful decoded output has alignment margin at least γ. For $0 < \theta \le 1$, define*

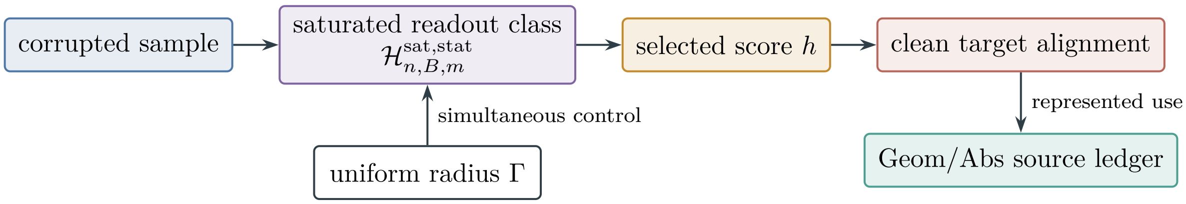

Figure 56: Every budget-admissible statistical attack selects one readout from the same saturated class. Uniform concentration controls sample-dependent selection. Empirical tables remain charged coordinates; on the population carrier, every target-bearing structural projection retains its complete Geom/Abs provenance and the orthogonal remainder is J_{res}.

$$B^*_{\mathrm{align}}(n, m, \eta; \theta) = \inf \left\{ s \geq 0 : \mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta,\kappa}(s, m) \geq \theta \right\}, \tag{292}$$

with value ∞ when the set is empty. Then

$$B^*_{\mathrm{align}}(n, m, \eta; \alpha\gamma^2) \leq B^*_{\mathrm{rec}}(n, m, \eta; \alpha). \tag{293}$$

Here B^*_{rec} is the noise-indexed saturated recovery cost of Definition IV.14.2. At a scalar budget s, the strict inequality

$$\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta,\kappa}(s, m) < \alpha\gamma^2$$

excludes recovery probability α for the complete represented learner class at that budget. Equations (286) *and* (287) *make the affordable alignment region downward closed under class-preserving flip degradation.*

Proof. If a scalar-cost-s computation recovers with probability at least α, the stated represented realization and Eq. (260) give

$$\mathrm{Align}^{\mathrm{sat,stat}}(s, m) \geq \alpha\gamma^2.$$

Taking the two infima proves (293). Its contrapositive gives the budgeted exclusion. Downward closure is Theorem IV.13.6.

□

Figure 56 displays the single uniform route from a represented learner to target alignment and structural charging.

IV.14 Noise-Indexed Recovery Budgets and the Three Threshold Scales

The statistical radius, the represented construction cost, and the information carried by the sample impose distinct thresholds. The following notation fixes their common binary-noise scale. For $0 \leq u \leq 1$, let

$$H_2(u) = -u \log_2 u - (1 - u) \log_2(1 - u), \tag{294}$$

with $0 \log_2 0 = 0$. For $0 < u, v < 1$, let

$$D_{\mathrm{Ber}}(u\|v) = u \log \frac{u}{v} + (1-u) \log \frac{1-u}{1-v}. \tag{295}$$

Thus D_{Ber} uses natural logarithms. The notation H_2^{-1} denotes the inverse of the increasing restriction $H_2 : [0, 1/2] \to [0, 1]$.

For integers $m \geq 1$ and $0 \leq k \leq m$, put

$$N_m(k) = \sum_{j=0}^{k} \binom{m}{j}, \qquad N_m(-1) = 0. \tag{296}$$

Lemma IV.14.1 (Binary Hamming-ball counting bounds)**.** *For $0 \leq k \leq m/2$, with $q = k/m$,*

$$\frac{2^{mH_2(q)}}{m+1} \leq \binom{m}{k} \leq N_m(k) \leq 2^{mH_2(q)}. \tag{297}$$

For every fixed integer $k \geq 1$, as $m \to \infty$,

$$N_m(k) = \frac{m^k}{k!} \left(1 + O_k(m^{-1})\right). \tag{298}$$

Proof. For $k = 0$, one has $\binom{m}{0} = N_m(0) = 2^{mH_2(0)} = 1$. Assume $1 \leq k \leq m/2$, so $0 < q \leq 1/2$. In the binomial law with success probability $q = k/m$, the value k is a mode. Since its $m+1$ point masses sum to one,

$$\binom{m}{k} q^k (1-q)^{m-k} \geq \frac{1}{m+1}.$$

Using $q^{-k}(1-q)^{-(m-k)} = 2^{mH_2(q)}$ gives the first inequality in Eq. (297). For $j \leq k$ and $q \leq 1/2$,

$$q^j (1-q)^{m-j} \geq q^k (1-q)^{m-k}.$$

Multiplying the sum defining $N_m(k)$ by the right-hand side and bounding it by the total binomial mass proves the upper type bound. Finally, $\binom{m}{k} = m^k/k! + O_k(m^{k-1})$, while the terms with $j < k$ are $O_k(m^{k-1})$. This proves Eq. (298).

$\square$

Definition IV.14.2 (Noise-indexed saturated recovery cost)**.** Let $(\mathfrak{I}_{n,m,\eta})_{0 \leq \eta < 1/2}$ be a family of presented binary-label tasks whose sample law is obtained by applying the homogeneous flip channel of Definition IV.10.1 at rate η. Let $\mathfrak{M}_{n,m,\eta}$ be expectation under the declared instance and sample law. Fix the scalarization κ used for the saturated cost invariant of Definition III.28.1, and write

$$\mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta},n}(s) = \sup_{\mathcal{A} \text{ represented } \kappa(\mathrm{Cost}(\mathcal{A})) \leq s} \mathfrak{M}_{n,m,\eta}(I \mapsto \alpha_I(\mathcal{A})). \tag{299}$$

This is Eq. (178) restricted by the scalar cost sublevel set of the already fixed scalarization. For a target recovery probability $0 < \alpha \leq 1$, define

$$B^*_{\mathrm{rec}}(n, m, \eta; \alpha) = \inf \left\{ s \geq 0 : \mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta},n}(s) \geq \alpha \right\}, \tag{300}$$

with value ∞ when the set is empty. For a declared scalar budget $\sigma(n)$, put

$$\eta_{\mathrm{comp}}^{\mathrm{sat}}(n,m;\sigma,\alpha)=\sup\left\{0\le\eta<\frac{1}{2}:\mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta},n}(\sigma(n))\ge\alpha\right\},\tag{301}$$

with supremum zero when the set is empty. This is the success-profile analogue of the threshold construction in Definition III.28.1. The success profile in Eq. (300) already takes the supremum over the complete budget-saturated class.

Proposition IV.14.3 (Degradation monotonicity of saturated success)**.** *Let $0\le\eta_1\le\eta_2<1/2$, and set*

$$q(\eta_1,\eta_2)=\frac{\eta_2-\eta_1}{1-2\eta_1}.\tag{302}$$

Appending an independent represented Bernoulli-$q(\eta_1,\eta_2)$ flip to an η_1-corrupted label produces an η_2-corrupted label. If $\psi_{\mathrm{flip}}(s,n,m)$ is a scalar ceiling for the complete class-preserving cost of this postprocessing after any computation of scalarized cost at most s, then

$$\mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta_1},n}(\psi_{\mathrm{flip}}(s,n,m))\ge\mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta_2},n}(s).\tag{303}$$

Consequently every budget ceiling closed under this class-preserving postprocessing has a downward-closed affordable-noise region. In particular, Eq. (301) *is its right endpoint.*

Proof. Let $Z_1\sim\mathrm{Bernoulli}(\eta_1)$ and $Z_q\sim\mathrm{Bernoulli}(q)$ be independent. Direct substitution of Eq. (302) gives

$$0\le q(\eta_1,\eta_2)<\frac{1}{2},$$

because $\eta_1\le\eta_2<1/2$, and

$$\Pr\{Z_1\oplus Z_q=1\}=\eta_1+q-2\eta_1q=\eta_2.$$

Apply any represented scalar-cost-s computation for the η_2 experiment after these additional flips. Its input and output laws are unchanged, while its complete cost is at most $\psi_{\mathrm{flip}}(s,n,m)$. Taking the supremum over the saturated budget slice proves Eq. (303). Closure of the declared ceiling under the same postprocessing proves downward closure.

□

Theorem IV.14.4 (Statistical, computational, and information thresholds)**.** *Let $\mathcal{T}$ be a finite parameter set of size $N\ge2$, let Θ be uniform on $\mathcal{T}$, and, conditional on $\Theta=\theta$, let the clean binary target be f_θ. Let $X_1,\dots,X_m$ be i.i.d. copies of X, independent of Θ. The observed labels are*

$$\widetilde{Y}_i=f_\Theta(X_i)\Xi_i,\qquad\Pr\{\Xi_i=-1\}=\eta,$$

where the flips are i.i.d. and independent of $(\Theta,X_1,\dots,X_m)$. Let $\mathcal{H}=\{f_\theta:\theta\in\mathcal{T}\}$, and assume the clean separation

$$\inf_{\substack{\theta,\theta'\in\mathcal{T}\\ \theta'\ne\theta}}\Pr\{f_{\theta'}(X)\ne f_\theta(X)\}\ge\Delta>0.\tag{304}$$

Then the following three statements hold.

(i) *Let* $\widehat{f} \in \mathcal{H}$ *be an* ε_{opt}*-empirical minimizer over* $\mathcal{H}$*, and let* $\Gamma_m^{\text{Rad}}(\mathcal{H}, \delta)$ *be the simultaneous radius in Eq. (233). With probability at least* $1 - \delta$*, exact identification is certified by*

$$2\Gamma_m^{\text{Rad}}(\mathcal{H}, \delta) + \varepsilon_{\text{opt}} < (1 - 2\eta)\Delta. \tag{305}$$

Equivalently, a sufficient statistical-noise threshold is

$$\eta < \frac{1}{2} - \frac{2\Gamma_m^{\text{Rad}}(\mathcal{H}, \delta) + \varepsilon_{\text{opt}}}{2\Delta}. \tag{306}$$

(ii) *Every estimator* $\widehat{\Theta} = \widehat{\Theta}(X^m, \widetilde{Y}^m)$*, without a computational restriction, satisfies*

$$\Pr\{\widehat{\Theta} = \Theta\} \le \min\left\{1, \frac{m(1 - H_2(\eta)) + 1}{\log_2 N}\right\}. \tag{307}$$

Hence success at least α *requires*

$$m(1 - H_2(\eta)) \ge \alpha \log_2 N - 1. \tag{308}$$

(iii) *If the right-hand side of Eq. (307) is smaller than* α*, then* $B^*_{\text{rec}}(n, m, \eta; \alpha) = \infty$*. If Eq. (305) holds and, for some* $s \ge 0$*, the represented empirical minimizer has scalarized complete cost at most* s*, then* $B^*_{\text{rec}}(n, m, \eta; 1 - \delta) \le s$*. Thus the saturated computational threshold lies between an explicitly represented achievable procedure and the information bound; statistical identification locates the sample-resolution threshold independently of the cost of constructing the empirical minimizer.*

Proof. By Eq. (235), on the simultaneous event,

$$R_0(\widehat{f}) \le \frac{2\Gamma_m^{\text{Rad}}(\mathcal{H}, \delta) + \varepsilon_{\text{opt}}}{1 - 2\eta}.$$

Under Eq. (305), this risk is strictly smaller than Δ. The separation Eq. (304) then forces $\widehat{f} = f_\Theta$, which proves (i).

For (ii), measure entropy in bits and write $S = (X^m, \widetilde{Y}^m)$. Since X^m is independent of Θ, the chain rule and the binary output alphabet give

$$\begin{aligned} I(\Theta; S) &= I(\Theta; \widetilde{Y}^m \mid X^m) \\ &\le \sum_{i=1}^m \Big(H(\widetilde{Y}_i \mid X^m, \widetilde{Y}^{i-1}) - H(\widetilde{Y}_i \mid \Theta, X^m, \widetilde{Y}^{i-1})\Big) \\ &\le m(1 - H_2(\eta)). \end{aligned}$$

Let $p = \Pr\{\widehat{\Theta} = \Theta\}$. Fano's entropy argument gives

$$H(\Theta \mid S) \le 1 + (1 - p)\log_2 N,$$

and therefore

$$p \log_2 N - 1 \le I(\Theta; S) \le m(1 - H_2(\eta)).$$

This proves Eq. (307) and Eq. (308). Part (iii) follows from the definition of B^*_{rec}: the information inequality ranges over all estimators, while a represented empirical minimizer is one member of the saturated budget slice.

□

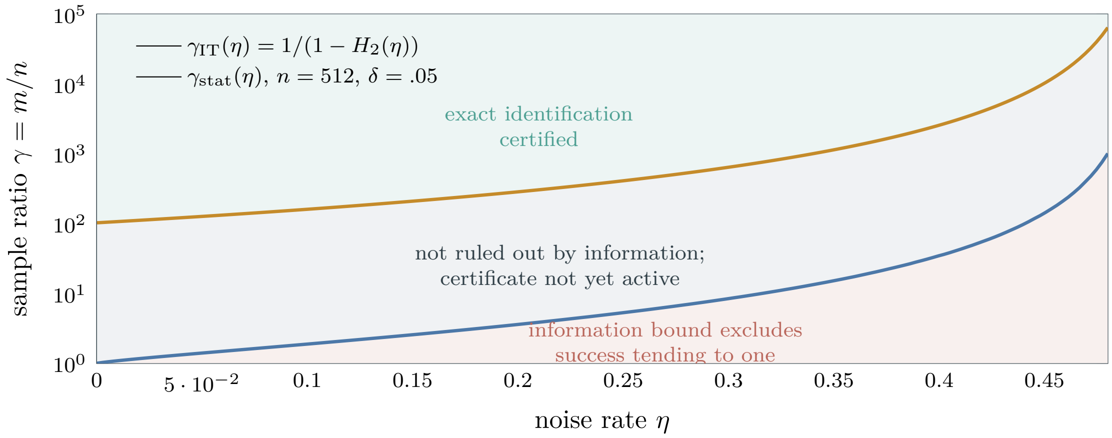

Figure 57: Information and statistical sample thresholds for $N = 2^n$ binary targets with clean separation 1/2. The lower curve is the asymptotic binary-symmetric-channel boundary $\gamma(1 - H_2(\eta)) = 1$ from Eq. (308) at success tending to one. The upper curve is the sufficient finite-class exact-identification condition Eq. (305), displayed at $n = 512$, $\delta = .05$, and zero optimization error.

For a class of 2^n binary targets with separation $\Delta = 1/2$, the information boundary and the finite-class Rademacher certificate can be drawn on the same sample-ratio scale. The latter is a sufficient certificate; the former bounds every estimator.

IV.15 Dependent Dynamical Samples and Sequential Structural Learning

The preceding statistical certificates use independent examples. A controlled system instead supplies independent episodes, dependent observations along one trajectory, or data selected predictably from an interaction history. The following results retain the same represented operator, source-projection, and complete-cost requirements while replacing independent symmetrization by the appropriate dynamical sampling certificate.

IV.15.1 Dynamical-sample certificates and blocking

Definition IV.15.1 (Represented dynamical-sample certificate)**.** Fix a budget-admissible represented loss class $\mathcal{L}_B = \{\ell_h : h \in \mathcal{H}_B\}$, with $0 \le \ell_h \le 1$. A *represented dynamical-sample certificate* is one of the following records, together with its complete construction, storage, and evaluation cost.

(i) An *independent-episode certificate* consists of independent episode records $E_1, \dots, E_M$ and represented normalized episode losses $\ell_h(E_i) \in [0, 1]$.

(ii) A *mixing-trajectory certificate* consists of a stationary represented process $(Z_t)_{t=1}^N$, its absolute-regularity coefficients

$$\beta(k) = \sup_{t \ge 1} \mathbb{E}\left[\|\mathcal{L}(Z_{t+k}, Z_{t+k+1}, \dots \mid \mathcal{F}_t) - \mathcal{L}(Z_{t+k}, Z_{t+k+1}, \dots)\|_{\mathrm{TV}}\right],$$

where $\mathcal{F}_t = \sigma(Z_1, \dots, Z_t)$, and represented losses $\ell_h(Z_t) \in [0, 1]$. A claimed coefficient is usable only when its upper bound is part of the certificate.

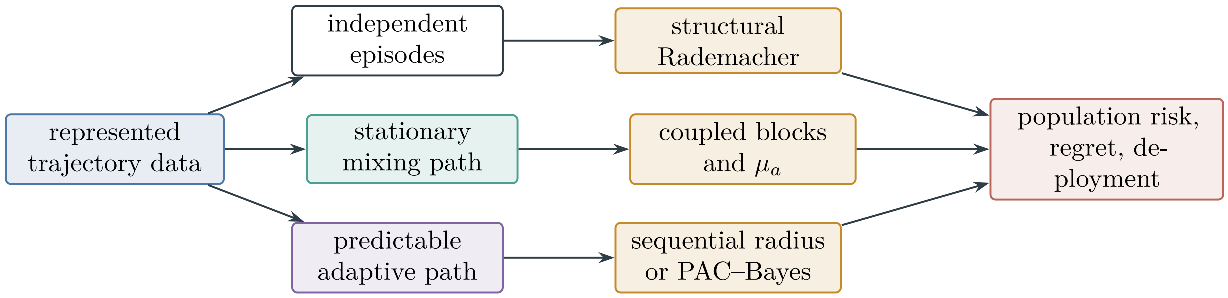

Figure 58: Each dynamical sampling regime has its own represented dependence certificate. Independent symmetrization, mixing blocks, and predictable-path complexity feed the same typed population-risk coordinates only through their proved conversion theorem.

(iii) A *predictable-path certificate* consists of a filtration $(\mathcal{F}_t)_{t=0}^T$, observations $Z_t \in \mathcal{F}_t$, represented predictable decisions, and losses $\ell_t(h) \in [0,1]$ such that $\ell_t(h)$ is $\mathcal{F}_t$-measurable. Its conditional risk is

$$r_t(h) = \mathbb{E}[\ell_t(h) \mid \mathcal{F}_{t-1}].$$

The hypothesis, operator, posterior, or policy used at time t is available from $\mathcal{F}_{t-1}$, in the temporal sense of Definition III.21.1.

For block length a in case (ii), put

$$\mu_a = \left\lfloor \frac{N}{2a} \right\rfloor, \qquad \delta_a = \delta - 2(\mu_a - 1)\beta(a).$$

The block length is certified at confidence $1-\delta$ when $\mu_a \geq 1$ and $\delta_a > 0$. All optimizations over a, over an operator family, or over a posterior are performed inside the same saturated derivation and retain their selection cost.

For a block $B = (Z_{(j-1)a+1}, \dots, Z_{ja})$, write $\bar{\ell}_h(B) = a^{-1} \sum_{Z_t \in B} \ell_h(Z_t)$. Let $\mathfrak{R}^{\mathrm{odd}}_{\mu_a}(a)$ and $\mathfrak{R}^{\mathrm{even}}_{\mu_a}(a)$ denote the expected Rademacher complexities of the block-average class on independent copies of the odd and even block laws, respectively.

Theorem IV.15.2 (Structural blocking bound for a mixing trajectory)**.** *Let case* (ii) *of Definition IV.15.1 hold and define*

$$R(h) = \mathbb{E}\ell_h(Z_1), \qquad \widehat{R}_N(h) = \frac{1}{N}\sum_{t=1}^N \ell_h(Z_t).$$

For every certified block length a, with probability at least $1-\delta$, simultaneously for all $h \in \mathcal{H}_B$,

$$\left| R(h) - \widehat{R}_N(h) \right| \leq 2 \max_{p \in \{\mathrm{odd},\mathrm{even}\}} \mathfrak{R}^p_{\mu_a}(a) + 3\sqrt{\frac{\log(4/\delta_a)}{2\mu_a}} + \frac{2a}{N}. \tag{309}$$

Consequently the right-hand side may be minimized over the represented set of certified block lengths. If the block-average evaluation vectors lie in a fixed structural energy ball $\mathcal{F}_A(R_A)$, the two Rademacher terms are bounded by

$$\mathfrak{R}^p_{\mu_a}(a) \leq \frac{R_A}{\mu_a}\sqrt{\mathrm{Tr}(A^{-1})},$$

with the block operator and its precision charged as in Definition IV.3.1.

Proof. Retain the first $2\mu_a a$ observations and divide them into alternating blocks of length a. The omitted suffix has length less than $2a$, so boundedness of the loss gives

$$\sup_h \left|\widehat{R}_N(h) - \widehat{R}_{2\mu_a a}(h)\right| \le \frac{2a}{N}.$$

Berbee coupling, applied separately to the odd and even block sequences, couples each sequence to μ_a independent blocks with total failure probability at most $(\mu_a - 1)\beta(a)$. The union of the two coupling failures therefore has probability at most $2(\mu_a - 1)\beta(a)$ [66].

Conditional on successful coupling, apply the two-sided independent-sample Rademacher inequality to each parity class at failure probability $\delta_a/2$. Stationarity makes the expectation of every block average equal to $R(h)$. The truncated empirical mean is the average of the odd and even block means, so its deviation is bounded by the larger of their two deviations. Adding the coupling and independent-sample failure probabilities gives δ and proves Eq. (309). The final display follows from Lemma IV.3.2 applied to the block evaluation vector. $\square$

The predictable regime is measured on history trees. For a sign sequence $\epsilon = (\epsilon_1, \dots, \epsilon_T)$, a depth-$T$ tree $\mathbf{z}$ has node $z_t(\epsilon_{1:t-1})$ at time t.

IV.15.2 Sequential structural complexity

Definition IV.15.3 (Sequential structural Rademacher complexity)**.** For a represented class $\mathcal{H}_B$ and a represented history tree $\mathbf{z}$, define

$$\mathfrak{R}_T^{\mathrm{seq}}(\mathcal{H}_B; \mathbf{z}) = \frac{1}{T}\mathbb{E}_\epsilon \sup_{h\in\mathcal{H}_B} \sum_{t=1}^{T} \epsilon_t h_t\big(z_t(\epsilon_{1:t-1})\big). \tag{310}$$

The saturated sequential complexity is the supremum over represented legal history trees at the declared budget. A path operator is a predictable tree $\mathbf{A}$ whose realization $A_\epsilon \succ 0$ acts on the length-T evaluation vector

$$h(\epsilon) = \big(h_t(z_t(\epsilon_{1:t-1}))\big)_{t=1}^{T}.$$

Its pathwise inverse-precision radius is

$$\mathfrak{q}_T(\mathbf{A}) = \frac{1}{T}\mathbb{E}_\epsilon \sqrt{\epsilon^\mathsf{T} A_\epsilon^{-1} \epsilon}. \tag{311}$$

The sequential structural energy ball of radius R_A consists of the represented trees satisfying $h(\epsilon)^\mathsf{T} A_\epsilon h(\epsilon) \le R_A^2$ on every path.

Lemma IV.15.4 (Pathwise support and fixed-operator sequential radius)**.** *For every path ϵ and every positive definite path operator,*

$$\sup_{f^\mathsf{T} A_\epsilon f \le R_A^2} \epsilon^\mathsf{T} f = R_A \sqrt{\epsilon^\mathsf{T} A_\epsilon^{-1} \epsilon}. \tag{312}$$

Hence a sequential structural energy ball satisfies

$$\mathfrak{R}_T^{\mathrm{seq}} \le R_A \mathfrak{q}_T(\mathbf{A}). \tag{313}$$

If $A_\epsilon = A$ is fixed over paths, then

$$\mathfrak{R}_T^{\mathrm{seq}} \le \frac{R_A}{T}\sqrt{\mathrm{Tr}(A^{-1})}. \tag{314}$$

Proof. Cauchy–Schwarz in the inner product generated by A_ϵ gives

$$\epsilon^\mathsf{T} f \le \sqrt{f^\mathsf{T} A_\epsilon f}\sqrt{\epsilon^\mathsf{T} A_\epsilon^{-1}\epsilon}.$$

Equality holds at $f = R_A A_\epsilon^{-1}\epsilon/\sqrt{\epsilon^\mathsf{T} A_\epsilon^{-1}\epsilon}$, proving Eq. (312). The represented class is contained in the pathwise ellipsoid, so expectation and division by T prove Eq. (313). For fixed A, Jensen's inequality and $\mathbb{E}_\epsilon \epsilon\epsilon^\mathsf{T} = I$ give

$$\mathbb{E}_\epsilon \sqrt{\epsilon^\mathsf{T} A^{-1}\epsilon} \le \sqrt{\operatorname{Tr}(A^{-1})},$$

which proves Eq. (314). □

IV.15.3 Learned-operator and PAC–Bayes control

Theorem IV.15.5 (Sequential learned-operator cover bound)**.** *Let $\{\mathbf{A}_\theta : \theta \in \Theta\}$ be a represented predictable operator family with $A_{\theta,\epsilon} \succeq \eta I$ on every path and*

$$\sup_\epsilon \|A_{\theta,\epsilon}^{-1} - A_{\theta',\epsilon}^{-1}\|_{\mathrm{op}} \le L_A d(\theta, \theta').$$

For the union of its radius-R_A sequential energy balls,

$$\mathfrak{R}_T^{\mathrm{seq}} \le \inf_{\rho>0} \left\{ R_A \sup_{\theta\in\Theta} \mathfrak{q}_T(\mathbf{A}_\theta) + R_A\sqrt{\frac{L_A\rho}{T}} + \frac{4R_A}{\sqrt{\eta}}\sqrt{\frac{\log \mathcal{N}(\Theta, d, \rho)}{T}} \right\}. \tag{315}$$

For fixed operators, the first term may be replaced by $R_A T^{-1}\sup_\theta \sqrt{\operatorname{Tr}(A_\theta^{-1})}$. The covering number, the construction of its net, and the selected operator remain part of the saturated cost.

Proof. Fix a ρ-net. On every path, the support identity and $|\sqrt{u} - \sqrt{v}| \le \sqrt{|u-v|}$ imply that replacing θ by a net point θ_0 changes the normalized support by at most

$$\frac{R_A}{T}\sqrt{\|A_{\theta,\epsilon}^{-1} - A_{\theta_0,\epsilon}^{-1}\|_{\mathrm{op}}\|\epsilon\|_2^2} \le R_A\sqrt{\frac{L_A\rho}{T}}.$$

For each net point, every coordinate of an evaluation vector has absolute value at most $R_A/\sqrt{\eta}$. The finite-union lemma for sequential Rademacher complexity therefore contributes at most $4R_A\eta^{-1/2}\sqrt{\log \mathcal{N}(\Theta, d, \rho)/T}$. Adding the fixed-net-point radius, the approximation term, and the finite-net term, then minimizing over ρ, proves Eq. (315). The trace specialization follows from Lemma IV.15.4. □

The existing structural prior also admits a predictable-path certificate. The next theorem uses the same proper prior and the same charged structural KL as Definition IV.8.1.

Theorem IV.15.6 (Martingale structural PAC–Bayes bound)**.** *Let case* (iii) *of Definition IV.15.1 hold. Let $P = P_{E,H}^{\mathsf{str}}$ be fixed before the evaluated innovations, or be constructed from public geometry or an independent sample, and let Q be any posterior with $Q \ll P$. In this theorem write*

$$\mathrm{KL}_{E,H}^{\mathsf{str}}(Q) := \mathrm{KL}(Q\|P);$$

for Gaussian Q, this is exactly the formula in Definition IV.8.1. For any $\lambda > 0$, with probability at least $1-\delta$, simultaneously for all such Q,

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}_{h\sim Q}\big[r_t(h) - \ell_t(h)\big] \le \frac{\mathrm{KL}_{E,H}^{\mathsf{str}}(Q) + \log(1/\delta)}{\lambda T} + \frac{\lambda}{8}. \tag{316}$$

More generally, suppose $|r_t(h) - \ell_t(h)| \le c$ *and put*

$$V_T(h) = \sum_{t=1}^{T} \mathbb{E}\left[(r_t(h) - \ell_t(h))^2 \mid \mathcal{F}_{t-1}\right].$$

For $0 < \lambda < 3/c$*, the predictable-variance refinement is*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}_Q[r_t(h) - \ell_t(h)] \le \frac{\mathrm{KL}^{\mathsf{str}}_{E,H}(Q) + \log(1/\delta)}{\lambda T} + \frac{\lambda\,\mathbb{E}_Q V_T(h)}{2T(1-\lambda c/3)}. \tag{317}$$

Any data-dependent choice of λ *is made over a represented countable grid and includes its union or prior weight in the logarithmic term.*

Proof. For fixed h, the increments $X_t(h) = r_t(h) - \ell_t(h)$ have conditional mean zero. Conditional Hoeffding's lemma gives

$$\mathbb{E}\left[\exp\left(\lambda X_t(h) - \lambda^2/8\right)\middle|\mathcal{F}_{t-1}\right] \le 1.$$

Iteration shows that the corresponding product is a supermartingale of mean at most one. Integrate it with respect to P, apply Markov's inequality, and then use the Donsker–Varadhan change-of-measure inequality

$$\mathbb{E}_Q F \le \mathrm{KL}(Q\|P) + \log \mathbb{E}_P e^F.$$

The resulting event is simultaneous in Q and yields Eq. (316) after division by λT. The KL is precisely $\mathrm{KL}^{\mathsf{str}}_{E,H}$ because the prior is the proper defect-restricted prior of Definition IV.8.1.

For the second assertion, the conditional Bernstein exponential inequality gives, for $0 < \lambda < 3/c$,

$$\mathbb{E}\exp\left\{\lambda \sum_t X_t(h) - \frac{\lambda^2 V_T(h)}{2(1-\lambda c/3)}\right\} \le 1.$$

The same integration, Markov, and change-of-measure steps prove Eq. (317). A weighted union bound over a represented grid proves the final assertion [73]. □

IV.15.4 Online regret and deployment

Theorem IV.15.7 (Sequential structural complexity and online regret)**.** *Let* $\mathcal{H}_B \subseteq [-1,1]^{\mathcal{Z}}$ *be represented and let* $\ell(\widehat{y}, y) \in [0,1]$ *be convex and* L_ℓ*-Lipschitz in* $\widehat{y}$*. The minimax expected regret against* $\mathcal{H}_B$ *over represented depth-*T *histories satisfies*

$$\mathcal{V}_T(\mathcal{H}_B) \le 2L_\ell T \sup_{\mathbf{z}} \mathfrak{R}^{\mathrm{seq}}_T(\mathcal{H}_B; \mathbf{z}). \tag{318}$$

If the minimax strategy is implemented through a represented finite tree, cover, or oracle in the declared presentation, its complete implementation cost is charged. Thus Lemma IV.15.4 and Theorem IV.15.5 give budget-admissible regret certificates exactly when that represented implementation lies within the budget.

Proof. Apply the minimax theorem at each history node and symmetrize the difference between the learner's cumulative loss and the best fixed $h \in \mathcal{H}_B$ by an independent tangent history. Sequential contraction for an L_ℓ-Lipschitz loss bounds the symmetrized loss tree by L_ℓ times the prediction tree, and the two tangent copies produce the factor two. The normalized prediction-tree supremum is Eq. (310); multiplication by T proves Eq. (318) [68]. The statistical minimax inequality specifies a strategy on the history tree. The framework admits it as a computation only through a represented implementation, whose tree, cover, or declared oracle and evaluation cost are then included by saturated no-creation and computation closure. □

Corollary IV.15.8 (Structural online-to-batch deployment)**.** *Suppose each observation is conditionally drawn from a common population law independently of $\mathcal{F}_{t-1}$, and an online learner produces predictable $h_1, \ldots, h_T$ with expected regret $\mathcal{R}_T$. If the loss is convex, the represented average $\bar{h}_T = T^{-1} \sum_{t=1}^{T} h_t$ satisfies*

$$\mathbb{E}R(\bar{h}_T) - \inf_{h \in \mathcal{H}_B} R(h) \leq \frac{\mathbb{E}\mathcal{R}_T}{T}. \tag{319}$$

Alternatively, choosing an index uniformly from $\{1, \ldots, T\}$ gives the same bound for the expected risk of the selected iterate without convexity. Every retained iterate, averaging operation, and deployment evaluator keeps its complete represented cost.

Proof. Independence of the next observation from the predictable iterate gives $\mathbb{E}\ell(h_t, Z_t) = \mathbb{E}R(h_t)$. Taking expectations in the regret inequality and dividing by T bounds the average of $R(h_t)$. Convexity gives $R(\bar{h}_T) \leq T^{-1} \sum_t R(h_t)$. Uniform selection has risk exactly that average. These two observations prove Eq. (319). □

The execution-law separation of population structure from retained sample records extends from independent examples to complete trajectories.

IV.15.5 Population alignment and source conditions

Theorem IV.15.9 (Trajectory population alignment and memorization separation)**.** *Let $\mathcal{A}$ be one family-uniform represented learner trained from any certificate in Definition IV.15.1, and let $h_{I,\mathsf{S},\omega}$ denote its retained output. Evaluate population alignment on a fresh episode or fresh legal continuation E', independent of the learner's private seed conditional on the declared public instance:*

$$c_I^{\mathrm{dyn}}(h) = \mathbb{E}\big[h(E') f_{*,I}(E') \mid I\big].$$

Then

$$|c_I^{\mathrm{dyn}}(h)|^2 \leq \mathrm{Var}(h(E') \mid I) \, \mathrm{ProjMass}_{\sigma(h)}(T_I), \tag{320}$$

with the target event interpreted on the fresh execution carrier. Averaging over the training trajectory and seed, and applying the monotone family functional, gives the same alignment–projection domination as Eq. (257)*.*

A table indexed only by training times, visited states, realized innovations, or one completed trajectory remains a charged lifted/transcript record. Its component orthogonal to the fresh-execution population projection lies in J_{res} by Theorem II.7.2. Any represented family-uniform use with nonzero fresh-execution target projection is charged to the existing qualified $\mathsf{Geom}/\mathsf{Abs}$ *source ledger at its complete training, storage, selection, and deployment cost. Thus trajectory memorization contributes to population alignment only through the represented rule that transfers it to fresh executions.*

Proof. On the fresh execution carrier, center $h(E')$ and the target indicator. Orthogonal projection of the centered target indicator onto $L^2(\sigma(h))$, followed by Cauchy–Schwarz, gives Eq. (320); it is the same Hilbert-space calculation used in Theorem IV.11.7. Expectation over the training record and learner seed preserves the pointwise inequality, and monotonicity of the family functional gives the averaged statement.

The complete training path is a represented lifted computation record by Theorem III.25.2. Applying Theorem II.7.2 to its fresh-execution use splits the represented population projection from its orthogonal component. The former retains its qualified $\mathsf{Geom/Abs}$ provenance; the latter is J_{res}. If an affordable family-uniform postprocessing of the latter has a nonzero fresh-execution projection, then Proposition II.8.3 reclassifies that represented postprocessing at its complete cost. This proves the provenance and memorization claims. □

Finally, the effective-dimension/source calculation depends on the sampling regime only through the certified variance scale.

Corollary IV.15.10 (Dynamical source-condition rate at certified effective sample size)**.** *Assume the source and eigenvalue hypotheses of Corollary IV.3.13. Let $m_{\mathrm{dyn}} > 0$ be a represented effective sample size for which the declared dynamical covariance and noise certificate proves the fixed-kernel oracle inequality*

$$\mathbb{E}\|\widehat{f}_\lambda - f_*\|^2_{L^2(\mu)} \le C\left(\frac{\sigma^2 N_{\mathrm{eff}}(\lambda)}{m_{\mathrm{dyn}}} + B_k(\lambda; f_*)\right). \tag{321}$$

For independent normalized episodes one may take $m_{\mathrm{dyn}} = M$. For a certified mixing block construction one may take $m_{\mathrm{dyn}} = \mu_a$ when the represented block-average feature and covariance operators satisfy the empirical-operator concentration event used to establish (321)*; its coupling tail and truncation error from Theorem IV.15.2 are retained separately. For a predictable martingale-noise certificate with variance proxy $0 < V_T < \infty$, the variance normalization supplies $m_{\mathrm{dyn}} = \sigma^2 T^2 / V_T$ whenever the represented empirical covariance concentration event used to establish* (321) *is certified.*

Then

$$\lambda_{\mathrm{dyn}} \asymp m_{\mathrm{dyn}}^{-1/(2r+1/(2\beta))}, \qquad \mathbb{E}\|\widehat{f}_{\lambda_{\mathrm{dyn}}} - f_*\|^2_{L^2} \lesssim m_{\mathrm{dyn}}^{-4\beta r/(4\beta r+1)}, \tag{322}$$

plus the displayed mixing coupling/truncation term or any represented operator-selection term. The effective sample size changes statistical resolution; it leaves the saturated derivation, target projection, and operator-selection charges unchanged.

Proof. The source condition gives $B_k(\lambda; f_*) \le C_r R_s^2 \lambda^{2r}$, and the eigenvalue law gives $N_{\mathrm{eff}}(\lambda) \asymp \lambda^{-1/(2\beta)}$. Substitution in Eq. (321) yields

$$C'\left(\lambda^{2r} + m_{\mathrm{dyn}}^{-1} \lambda^{-1/(2\beta)}\right).$$

Balancing the two terms proves Eq. (322). Independent episodes give the usual variance factor M^{-1}. Coupled blocks give μ_a independent block observations on the successful coupling event. For martingale noise, the normalized average has conditional variance at most V_T/T^2, which equals $\sigma^2/m_{\mathrm{dyn}}$ by definition. The stated empirical covariance event converts that scalar variance certificate into the kernel oracle inequality. The coupling, truncation, and selection terms arise outside the bias–variance balance and are therefore retained additively. □

IV.16 Dynamical Label Corruption at Saturated Computational Capacity

The binary corruption channel of Definition IV.10.1 extends to episode, mixing, and predictable-path records by conditioning at the time at which each label is revealed. The resulting clean-risk estimates retain both the sample size and the complete computational ceiling.

IV.16.1 Complete dynamical label-noise records

Definition IV.16.1 (Complete dynamical label-noise record)**.** Fix a problem size n, a represented computational ceiling $b(n)$, and m binary observations. A complete dynamical label-noise record consists of the following data.

(i) A filtration $(\mathcal{F}_t)_{t=0}^m$, represented inputs X_t, and clean labels $Y_t = f_{*,t}(X_t) \in \{-1, 1\}$. The rule $f_{*,t}$ is fixed by the presented task and the public pre-t record.

(ii) Predictable represented binary decision rules $h_t : \mathcal{X}_t \to \{-1, 1\}$. The random rule h_t, including every private random bit used to choose it, is $\mathcal{F}_{t-1}$-measurable; it is evaluated at X_t before Y_t or its corruption is revealed. Real-valued scores are treated separately through the flip-symmetric linear loss below.

(iii) Corrupted labels

$$\widetilde{Y}_t = Y_t \Xi_t, \qquad \Pr\{\Xi_t = -1 \mid \mathcal{F}_{t-1}, X_t\} = \eta_t(X_t),$$

with a represented rate certificate $0 \le \eta_t(X_t) \le \eta_+ < 1/2$. Put $\rho_{\min} = 1 - 2\eta_+ > 0$.

(iv) One dynamical-sample certificate from Definition IV.15.1, including every mixing, covariance, predictable-variance, operator, or history-tree constant used below.

(v) The clean-label semantics, corruption sampler, hypothesis or policy class, selected operator, empirical optimization, retained state, debiasing, deployment, comparison, confidence allocation, and numerical precision, all at complete saturated cost at most $b(n)$.

In the homogeneous case the record contains an exact represented $\eta_t \equiv \eta$, and $\rho = 1 - 2\eta$. A certified interval $\eta \in [\eta_-, \eta_+]$ permits the Massart conclusions with $\rho_{\min}$; an information upper bound uses the lower endpoint η_-. For a stationary mixing record, the admissible state-dependent extension has $\eta_t(X_t) = \eta(X_t)$. There are iid uniforms U_t, independent of the entire stationary clean process, such that Ξ_t is a measurable function of (X_t, Y_t, U_t). History-dependent rates use the predictable-path branch.

The record applies to binary target, failure, classification, committor-training, and binary policy-supervision labels. Reward, transition, and state-observation discrepancies retain the model-error coordinates of Definition V.13.1. The clean rule $f_{*,t}$ is an analytic task comparator. It is not an available pre-reveal source or policy input unless a separate represented derivation constructs it and pays its complete cost.

Theorem IV.16.2 (Conditional transport through dynamical label noise)**.** *Let h_t belong to a complete dynamical label-noise record and define*

$$r_{0,t}(h_t) = \mathbb{E}\Big[\mathbf{1}_{\{h_t(X_t) \neq Y_t\}} \mid \mathcal{F}_{t-1}\Big],$$
$$r_{\mathrm{corr},t}(h_t) = \mathbb{E}\Big[\mathbf{1}_{\{h_t(X_t) \neq \widetilde{Y}_t\}} \mid \mathcal{F}_{t-1}\Big].$$

Then

$$r_{\mathrm{corr},t}(h_t) - r_{\mathrm{corr},t}(f_{*,t}) = \mathbb{E}\Big[(1 - 2\eta_t(X_t))\mathbf{1}_{\{h_t(X_t) \neq f_{*,t}(X_t)\}} \mid \mathcal{F}_{t-1}\Big] \ge \rho_{\min} r_{0,t}(h_t). \tag{323}$$

Consequently,

$$\frac{1}{m}\sum_{t=1}^m r_{0,t}(h_t) \le \frac{1}{\rho_{\min} m}\sum_{t=1}^m \big[r_{\mathrm{corr},t}(h_t) - r_{\mathrm{corr},t}(f_{*,t})\big]. \tag{324}$$

For a bounded flip-symmetric loss,

$$\mathbb{E}[\ell(h_t(X_t), \widetilde{Y}_t) \mid \mathcal{F}_{t-1}, X_t, Y_t] = \eta_t(X_t) + (1 - 2\eta_t(X_t))\ell(h_t(X_t), Y_t). \tag{325}$$

Under homogeneous noise, Eqs. (323) *and* (324) *are equalities after replacing* $\rho_{\min}$ *by* ρ.

If the clean augmented process is stationary and absolutely regular with coefficients $\beta_{\text{clean}}(k)$, *and the stationary state-dependent corruption uses the independent iid innovations specified in Definition IV.16.1, then the observed process* $(X_t, \widetilde{Y}_t)_{t\in\mathbb{Z}}$ *satisfies*

$$\beta_{\text{corr}}(k) \le \beta_{\text{clean}}(k). \tag{326}$$

Proof. Condition on $\mathcal{F}_{t-1}, X_t$. Correct prediction incurs corrupted loss $\eta_t(X_t)$, while incorrect prediction incurs $1-\eta_t(X_t)$. Subtraction of the clean comparator's corrupted risk gives Eq. (323); summation proves Eq. (324). The identity $\ell(z,-y) = 1-\ell(z,y)$ gives Eq. (325).

For the mixing assertion, adjoin an independent innovation sequence to the clean stationary process. Independent products do not increase absolute regularity, and the corrupted observation is a measurable image of that product. Total-variation distance contracts under measurable maps, which proves Eq. (326).

□

IV.16.2 Saturated noise and sample profiles

Definition IV.16.3 (Saturated dynamical noise and sample profiles)**.** The complete profile consists of the affordable output and generalization coordinates below together with the corrupted-excess coordinate of Definition IV.16.4. A finite-codebook bound for the first coordinate is given separately in Proposition IV.16.5.

Let $\mathscr{A}^{\text{dyn}}_{n,b,m}(\eta_+)$ be the family of complete dynamical label-noise records of cost at most $b(n)$ and certified rates at most η_+. Let $\mathcal{H}^{\text{dyn,sat}}_{n,b,m}(\eta_+)$ be the sample-independent hull of all their complete outputs. Its 0–1 branch contains the binary decisions in Definition IV.16.1; its flip-symmetric linear-loss branch contains represented scores $h : \mathcal{X} \to [-1,1]$ and is never inserted into a mismatch indicator without a represented decision rule. On the fresh-execution carrier of Theorem IV.15.9, define

$$M^{\text{dyn,sat}}_{n,b,m}(\eta_+) = \sup_{\substack{I\in\mathfrak{I}_n \\ h\in\mathcal{H}^{\text{dyn,sat}}_{n,b,m}(I;\eta_+)}} \operatorname{ProjMass}_{\sigma(h)}(T_I), \tag{327}$$

with value zero when the indexed class is empty. The training sample and private seed range inside the fixed hull before the supremum is evaluated; the target projection is always measured on an independent fresh execution. For each $\mathcal{A} \in \mathscr{A}^{\text{dyn}}_{n,b,m}(\eta_+)$, let $\eta_{\mathcal{A},+}$ be its own certified rate ceiling, and let $\Gamma_{\mathcal{A}}(m,\delta)$ be the minimum of the represented two-sided upper bounds applicable to that record for the absolute population-minus-empirical corrupted-loss deviation. A minimum is allowed only after the candidates have been converted to the same population law and loss and placed on one declared simultaneous event, with confidence splitting over every data-dependent block length, operator, posterior, or model choice. Eligible candidates are the independent-episode radius, the mixing-block radius of Theorem IV.15.2, and the represented sequential or martingale comparisons of Theorems IV.15.5 and IV.15.6. The recordwise value is $+\infty$ when no candidate applies. Define

$$\Gamma^{\text{sat}}_{\text{dyn}}(n, b(n), m, \eta_+, \delta) = \sup_{\mathcal{A}\in\mathscr{A}^{\text{dyn}}_{n,b,m}(\eta_+)} \Gamma_{\mathcal{A}}(m,\delta). \tag{328}$$

Definition IV.16.4 (Saturated dynamical corrupted-excess profile)**.** Fix (n,b,m,η_+,δ) and the affordable record class and simultaneous generalization events of Definition IV.16.3.

For each record, define $\mathrm{Exc}_{\mathcal{A}}$ as the minimum of the following applicable upper bounds on its average corrupted excess relative to the analytic clean task rule, on that same event:

(a) $2\Gamma_{\mathcal{A}} + \varepsilon_{\mathrm{opt}} + \varepsilon_{\mathrm{app,corr}}$ for a represented empirical minimizer, where $\varepsilon_{\mathrm{app,corr}}$ is the corrupted-risk gap between the best admissible represented comparator and the analytic task rule;

(b) the normalized represented corrupted-loss regret plus its certified conditional-to-realized comparison for a predictable online learner, plus the chosen comparator's corrupted-risk gap to the task rule;

(c) the represented posterior-minus-comparator empirical excess plus both applicable martingale PAC–Bayes radii and that comparator's corrupted-risk gap to the task rule.

The comparator is used only for analysis unless it is independently constructed by the learner; every optimization error, prior, confidence split, and selection operation belongs to the complete record. The recordwise excess is $+\infty$ when none of the three candidates applies. Because each recordwise bound uses $\eta_{\mathcal{A},+}$, not the outer ceiling, enlarging η_+ only enlarges the nested affordable class. Thus the saturated profiles below are nondecreasing in η_+. The saturated corrupted-excess envelope is

$$\mathrm{Exc}^{\mathrm{sat}}_{\mathrm{corr}}(n, b(n), m, \eta_+, \delta) = \sup_{\mathcal{A} \in \mathscr{A}^{\mathrm{dyn}}_{n,b,m}(\eta_+)} \mathrm{Exc}_{\mathcal{A}}(m, \delta). \tag{329}$$

The supremum, rather than a cross-regime minimum, makes the bound uniform over every affordable sample-certificate type. Define

$$U^{\mathrm{clean}}_{\mathrm{noise}}(n, b(n), m, \eta_+, \delta) = \min\left\{1, \frac{\mathrm{Exc}^{\mathrm{sat}}_{\mathrm{corr}}(n, b(n), m, \eta_+, \delta)}{1 - 2\eta_+}\right\}. \tag{330}$$

IV.16.3 Finite-description generalization and clean-risk transport

Proposition IV.16.5 (Finite-description dynamical generalization radius)**.** *Work in the setting of Definition IV.16.3 at fixed $(n, b, m, \eta_+, \delta)$.*

Let

$$N^{\mathrm{dyn,out}}_{n,b,m}(\eta_+) = \sup_{I \in \mathfrak{I}_n} |\mathcal{H}^{\mathrm{dyn,sat}}_{n,b,m}(I; \eta_+)|,$$

and set this value to $+\infty$ unless the fixed-budget output codebook is finite. Assume the relevant loss takes values in $[0, 1]$, the codebook is fixed before the effective sample is revealed, and one represented sample certificate supplies a simultaneous comparison to m_{eff} independent normalized observations with additive error $\varepsilon_{\mathrm{dep}}$ and confidence δ_{eff}. Put

$$L_{\mathrm{out}}(n, b(n), m, \eta_+) = \log N^{\mathrm{dyn,out}}_{n,b,m}(\eta_+).$$

For any candidate with represented effective sample size m_{eff}, effective confidence δ_{eff}, and additive dependence error $\varepsilon_{\mathrm{dep}}$, the finite-description fallback is

$$\Gamma^{\mathrm{sat}}_{\mathrm{dyn}} \le 2\sqrt{\frac{2L_{\mathrm{out}}(n, b(n), m, \eta_+)}{m_{\mathrm{eff}}}} + 3\sqrt{\frac{\log(2/\delta_{\mathrm{eff}})}{2m_{\mathrm{eff}}}} + \varepsilon_{\mathrm{dep}}. \tag{331}$$

Here $m_{\mathrm{eff}} = m$ for independent normalized observations, $m_{\mathrm{eff}} = \mu_a$, $\delta_{\mathrm{eff}} = \delta_a/2$, and $\varepsilon_{\mathrm{dep}} = 2a/m$ for the certified mixing-block candidate, and the predictable candidate uses its declared history-tree or predictable-variance normalization. The displayed fallback applies when the effective-size and confidence quantities hold uniformly over the whole affordable class. Otherwise it is applied within each sample-certificate subclass and the resulting subclass bounds are combined by a supremum.

Proof. For a fixed finite codebook, Massart's finite-class lemma bounds its empirical Rademacher complexity by $\sqrt{2L_{\mathrm{out}}/m_{\mathrm{eff}}}$. The two-sided empirical Rademacher inequality for a $[0,1]$-valued loss class therefore gives the first two terms in Eq. (331) simultaneously over the codebook. The represented sample certificate transfers that independent comparison to the declared dynamical law and adds $\varepsilon_{\mathrm{dep}}$. Taking the supremum over record subclasses is valid only after their confidence events have been made simultaneous, which is exactly the final hypothesis.

□

Theorem IV.16.6 (Uniform dynamical clean-risk and alignment bounds)**.** *Every represented learner in $\mathcal{H}^{\mathrm{dyn,sat}}_{n,b,m}(\eta_+)$ whose corrupted excess is bounded by $\mathrm{Exc}^{\mathrm{sat}}_{\mathrm{corr}}$ satisfies, on the simultaneous event declared by its complete record,*

$$\frac{1}{m}\sum_{t=1}^{m} r_{0,t}(h_t) \le U^{\mathrm{clean}}_{\mathrm{noise}}(n, b(n), m, \eta_+, \delta). \tag{332}$$

For a homogeneous balanced target, a represented score $h : X \to [-1,1]$, and

$$c_0(h) = \mathbb{E}[h(X)f_*(X)], \qquad \widehat{c}_\eta(h) = \frac{1}{m}\sum_{t=1}^{m} h(X_t)\widetilde{Y}_t,$$

the same event gives

$$|c_0(h)| \le \min\left\{\sqrt{M^{\mathrm{dyn,sat}}_{n,b,m}(\eta)}, \frac{|\widehat{c}_\eta(h)| + 2\Gamma^{\mathrm{sat}}_{\mathrm{dyn}}(n, b(n), m, \eta, \delta)}{1-2\eta}\right\}, \tag{333}$$

where $M^{\mathrm{dyn,sat}}_{n,b,m}(\eta)$ is the fresh-execution generated-algebra projection envelope of Theorem IV.15.9.

For homogeneous noise and a dynamical kernel-ridge certificate with effective sample size m_{dyn}, let $\widehat{g}_\lambda$ be kernel ridge trained on the corrupted labels and define the represented debiased estimator $\widehat{f}_\lambda = \widehat{g}_\lambda/(1-2\eta)$. Then

$$\mathbb{E}\|\widehat{f}_\lambda - f_*\|^2_{L^2(\mu)} \le C\left(\frac{v_\eta^2 N_{\mathrm{eff}}(\lambda)}{(1-2\eta)^2 m_{\mathrm{dyn}}} + B_k(\lambda; f_*)\right) + \frac{\varepsilon_{\mathrm{dep}} + \varepsilon_{\mathrm{sel}}}{(1-2\eta)^2}. \tag{334}$$

The dependence and selection errors are the represented additive terms of Corollary IV.15.10.

Proof. The definition of the corrupted excess envelope and Eq. (324) give Eq. (332). In the homogeneous flip-symmetric linear loss,

$$R^\ell_\eta(h) = \frac{1-(1-2\eta)c_0(h)}{2}, \qquad \widehat{R}^\ell_\eta(h) = \frac{1-\widehat{c}_\eta(h)}{2}.$$

The simultaneous deviation event therefore gives the second entry of Eq. (333); the first is Eq. (320). For kernel ridge, apply Eq. (321) to the regression target $(1-2\eta)f_*$, define the displayed debiased estimator, divide the full oracle inequality by $(1-2\eta)^2$, and use quadratic homogeneity of the source residual. This proves Eq. (334).

□

IV.16.4 Dynamical threshold scales

Definition IV.16.7 (Dynamical sample, statistical-noise, and computational-noise thresholds)**.** For $0 < \varepsilon, \alpha \le 1$, define

$$m_{\text{noise}}^{\text{sat}}(n, b, \eta_+, \varepsilon, \delta) = \inf\left\{m_0 \ge 1 : \sup_{m' \ge m_0} \operatorname{Exc}_{\text{corr}}^{\text{sat}}(n, b(n), m', \eta_+, \delta) \le (1 - 2\eta_+)\varepsilon\right\}, \tag{335}$$

$$\eta_{\text{stat,dyn}}^{\text{sat}}(n, m; b, \varepsilon, \delta) = \sup\left\{0 \le \eta < \frac{1}{2} : \operatorname{Exc}_{\text{corr}}^{\text{sat}}(n, b(n), m, \eta, \delta) \le (1 - 2\eta)\varepsilon\right\}. \tag{336}$$

Infima of empty sets are $+\infty$, and suprema of empty sets are zero.

Let $\mathscr{A}_{n,b,m}^{\text{dyn,hom}}(\eta) \subseteq \mathscr{A}_{n,b,m}^{\text{dyn}}(\eta)$ be the complete records evaluated in the exact homogeneous experiment $\eta_t \equiv \eta$. Define

$$\operatorname{Succ}_{\text{dyn}}^{\text{sat}}(n, m, \eta; b) = \sup_{\mathcal{A} \in \mathscr{A}_{n,b,m}^{\text{dyn,hom}}(\eta)} \mathfrak{M}_{n,m,\eta}(\alpha_I(\mathcal{A})), \tag{337}$$

and define

$$\eta_{\text{comp,dyn}}^{\text{sat}}(n, m; b, \alpha) = \sup\left\{0 \le \eta < \frac{1}{2} : \operatorname{Succ}_{\text{dyn}}^{\text{sat}}(n, m, \eta; b) \ge \alpha\right\}. \tag{338}$$

For polynomial capacity write $b_C(n) = n^C$ and use the superscript (C) for the three profiles obtained by substituting b_C.

Theorem IV.16.8 (Dynamical statistical, computational, and information bounds)**.** *The sample threshold in Eq.* (335) *certifies clean average risk at most ε for every sample size at least that threshold. When the saturated numerator has a represented noise-independent value*

$$\operatorname{Exc}_{\text{corr}}^{\text{sat}}(n, b(n), m, \eta, \delta) = \overline{\operatorname{Exc}}_{\text{corr}}^{\text{sat}}(n, b(n), m, \delta) \qquad (0 \le \eta < 1/2),$$

$$\eta_{\text{stat,dyn}}^{\text{sat}} = \left[\frac{1}{2} - \frac{\overline{\operatorname{Exc}}_{\text{corr}}^{\text{sat}}(n, b(n), m, \delta)}{2\varepsilon}\right]_{[0,1/2]}. \tag{339}$$

The same statements hold after the polynomial substitution $b(n) = n^C$. Because the affordable record classes are nested in their certified rate ceiling and the recordwise numerators use their own rates, the statistical set in Eq. (336) *is downward closed.*

Suppose next that Θ is uniform on a finite target family $\mathcal{T}_n$ of size N_n. Let Z_0 be the initial public record, put $I_0 = I(\Theta; \mathsf{Z}_0)$, with mutual information measured in bits throughout this paragraph, and assume that a represented adaptive policy selects each query from Z_0 and the previous noisy labels. If the label returned at every query passes through an independent homogeneous binary symmetric channel of rate η, then the complete transcript Z_m satisfies

$$I(\Theta; \mathsf{Z}_m) \le I_0 + m(1 - H_2(\eta)). \tag{340}$$

Every estimator, irrespective of computational cost, therefore satisfies

$$\Pr\{\widehat{\Theta} = \Theta\} \le \min\left\{1, \frac{I_0 + m(1 - H_2(\eta)) + 1}{\log_2 N_n}\right\}. \tag{341}$$

Success at least α requires

$$m(1 - H_2(\eta)) \ge \alpha \log_2 N_n - I_0 - 1. \tag{342}$$

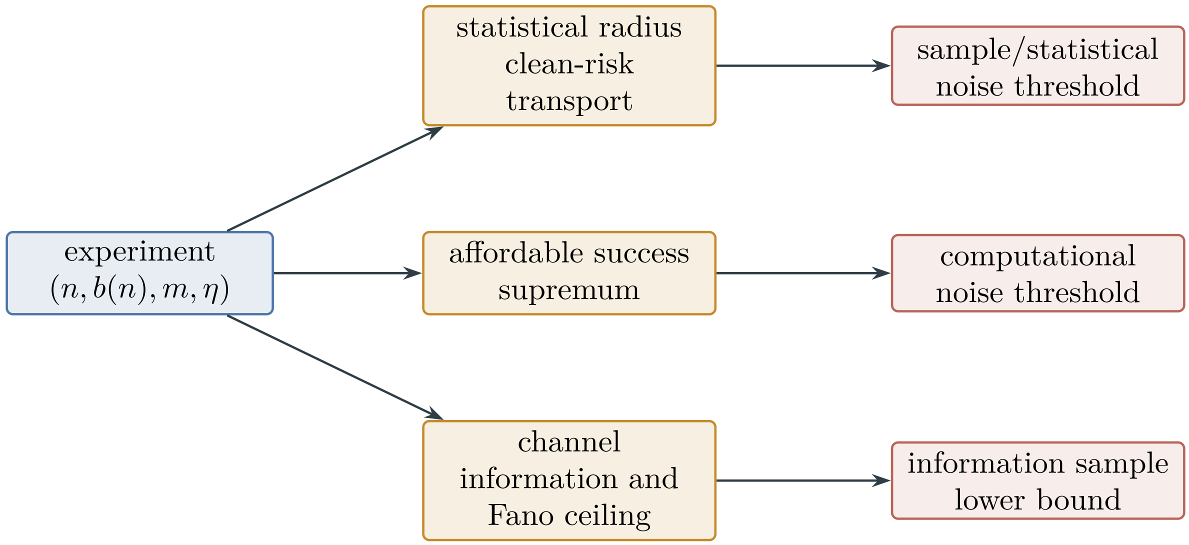

Figure 59: The three dynamical thresholds share the problem size, sample size, noise law, and computational ceiling, but they optimize different objects. A statistical excess radius, an affordable-success supremum, and a channel-information bound cannot replace one another; they may only be reported side by side after their experiment is matched.

For $N_n = 2^n$ and a target-independent public record, the right side is $\alpha n - 1$.

Finally, homogeneous noise degradation by the represented causal flip postprocessing of Proposition IV.14.3 gives

$$b^\sharp(n) = \psi_{\text{flip}}(b(n), n, m)$$

and

$$\mathrm{Succ}^{\text{sat}}_{\text{dyn}}(n, m, \eta_1; b^\sharp) \geq \mathrm{Succ}^{\text{sat}}_{\text{dyn}}(n, m, \eta_2; b) \quad (0 \leq \eta_1 \leq \eta_2 < 1/2).$$

Thus every ceiling closed under this postprocessing has a downward-closed computational-noise region with endpoint $\eta^{\text{sat}}_{\text{comp,dyn}}$.

Proof. The first statements are Eq. (330) and the definitions of the two thresholds. Solving the defining inequality for η proves Eq. (339). Nested record classes make the left side of the statistical condition nondecreasing in η, while $(1 - 2\eta)\varepsilon$ is decreasing; this proves downward closure.

For the information bound, first retain the contribution I_0 of the initial public record. Then condition successively on that record, the queries, actions, and previous noisy labels. The next query is generated from this conditioned record and contributes no additional conditional mutual information about Θ. The next binary label is one use of a memoryless binary symmetric channel, so its conditional mutual information is at most $1 - H_2(\eta)$ bits. The chain rule proves Eq. (340); Fano's entropy inequality proves Eqs. (341) and (342). The final assertion applies the same independent postprocessing coupling as Proposition IV.14.3 at each represented label read. □

Figure 59 separates the statistical, computational, and information coordinates evaluated on the same noisy experiment.

IV.17 Dynamical Model-Channel Corruption Beyond Binary Labels

Binary label flips are one instance of a represented observation channel. Reward reports, executed actions, state observations, and filter updates require different correction operators and different error norms. This section defines their common statistical record. The dynamical consequences of those typed errors are stated in Part V.

IV.17.1 Complete Typed Channel Records

Definition IV.17.1 (Complete dynamical model-channel noise record)**.** Fix a problem size n, a complete computational ceiling $b(n)$, and a sample of m logged transitions. Let $\mathcal{F}_t^{\text{obs}} \subseteq \mathcal{G}_t^{\text{lat}}$ be, respectively, the public learner filtration and the analytic latent filtration. A *complete dynamical model-channel noise record* consists of one or more of the following typed interfaces. When several interfaces occur in one system, the record also names the joint or composite object whose error is estimated; marginal certificates are not treated as a joint certificate.

(i) A reward record contains a clean conditional-mean comparator $r_{*,t} = \mathbb{E}[R_t \mid \mathcal{G}_{t-1}^{\text{lat}}, X_t, A_t]$, a public report $\widetilde{R}_t$, and represented predictable coefficients satisfying

$$\mathbb{E}[\widetilde{R}_t \mid \mathcal{F}_{t-1}^{\text{obs}}, Z_t] = c_t r_{*,t}^{\text{obs}}(Z_t) + d_t + u_t, \qquad |c_t| \geq c_{\min} > 0, \tag{343}$$

where $r_{*,t}^{\text{obs}}(Z_t) = \mathbb{E}[r_{*,t} \mid \mathcal{F}_{t-1}^{\text{obs}}, Z_t]$. The record gives the same-law bound $\|u\|_{L^2(d^b)} \leq \zeta_r$ and a bounded, sub-Gaussian, sub-exponential, or predictable-variance certificate for the centered reward innovation. The identity observation has $c_t = 1, d_t = u_t = 0$.
The record declares a probability law d^b on the disjoint union

$$\mathsf{S} = \bigsqcup_t \{t\} \times \mathsf{S}_t, \qquad s = (t, h_{t-1}^{\text{obs}}, z_t),$$

of pre-report predictor inputs. Thus the time index is part of the input. An $L^2(d^b)$ norm uses the S-marginal, whereas a reward loss uses the joint law $d^b(ds)Q_t(d\widetilde{r} \mid s)$. The functions $c, d, u, r_*^{\text{obs}}$ are measurable on S, the report and its conditional mean are square-integrable, and

$$\sigma(\mathcal{F}_{t-1}^{\text{obs}}, Z_t) \subseteq \sigma(\mathcal{G}_{t-1}^{\text{lat}}, X_t, A_t).$$

This inclusion and the tower property identify $r_{*,t}^{\text{obs}} = \mathbb{E}[R_t \mid \mathcal{F}_{t-1}^{\text{obs}}, Z_t]$.

(ii) An execution record contains intended actions A_t, executed actions A_t°, and a represented conditional channel

$$K_t^A(da^\circ \mid \mathcal{F}_{t-1}^{\text{obs}}, Z_t, X_t, A_t). \tag{344}$$

The physical analytic kernel may depend on latent X_t; a learner-evaluable version requires a represented public interface. If A_t° is not logged, only the composite conditional law of the observed next report given the declared public predictor input is potentially identifiable. A latent reward–transition composite additionally requires paired latent data or a represented reconstruction or stability theorem. No separate estimate of K_t^A is inferred.

(iii) An observation record contains a latent clean observation or state X_t, a public observation O_t, and a represented emission channel $K_t^O(do \mid X_t, A_{t-1}^\circ, \mathcal{F}_{t-1}^{\text{obs}})$. A clean-state comparison also contains paired latent data, a known channel with a represented correction, or the reconstruction certificate of Definition IV.17.3. No latent-state error is inferred from observation-law risk alone.

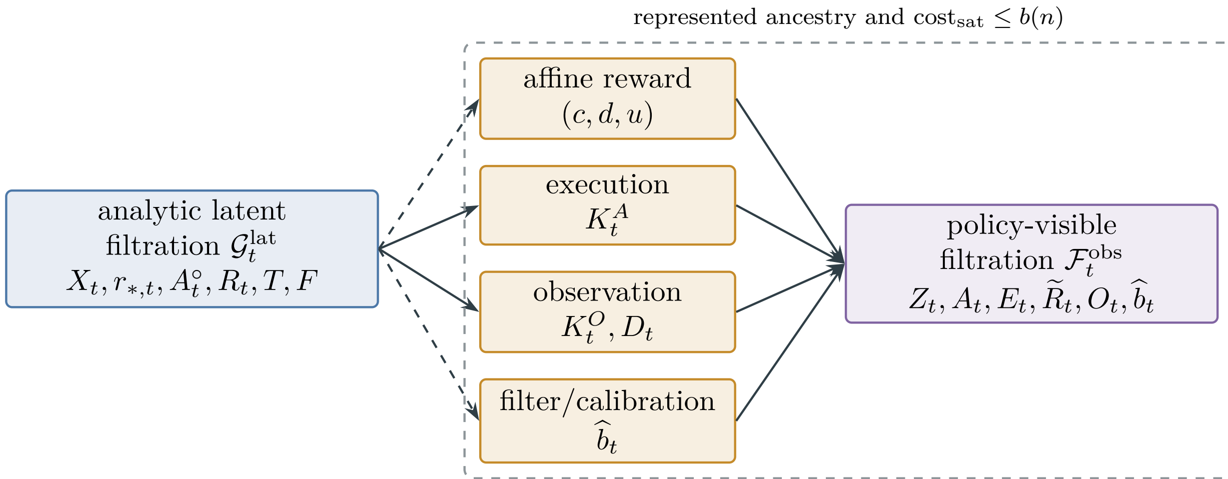

Figure 60: A complete dynamical channel record separates the analytic latent filtration from the policy-visible filtration. Reward, execution, observation, and filter interfaces retain their own norms and converters. Only represented public-measurable outputs may enter a prospective computation; latent comparators remain analytic.

(iv) A partial-observation or filter record contains either separately identified transition and emission channels or only their identifiable history kernel. It declares which object is estimated, the represented filter or memory update, its input and output norms, and a represented loss-to-filter-error calibration. A training loss without that calibration remains an empirical loss coordinate.

In every case, the record also contains a dynamical-sample certificate from Definition IV.15.1; the channel class, estimator, calibration or inverse, coverage event, behavior-to-deployment comparison, confidence allocation, numerical precision, retained state, and all data collection, training, selection, storage, and deployment costs. Their total saturated cost is at most $b(n)$. A requested norm whose coverage or calibration is absent receives the value $+\infty$. For square loss, the record either proves uniform bounds on both arguments $g(S)$ and the reported reward $\widetilde{R}$, and records the resulting loss normalization to $[0, 1]$, or supplies a separate represented unbounded-loss empirical-process or martingale theorem with its tail, variance or truncation, and selection penalties. The bounded dynamical-sample certificate is otherwise ineligible and the squared-loss excess certificate equals $+\infty$. The return-MGF calculation below is not, by itself, a uniform generalization theorem for a data-selected squared loss.

The latent variables and clean comparators in this definition are analytic objects. A policy may use only $\mathcal{F}_t^{\mathrm{obs}}$-measurable quantities. A channel is target-independent exogenous noise only when, conditional on the declared public pre-hit state, its innovation is generated by a target-free represented constructor and satisfies the declared conditional independence from the latent target. Information already present in that public state remains charged. State-, history-, learned-, or target-dependent channels are admissible only with their complete represented ancestry and the same-record source or contact route used by the claim; otherwise the proposed prospective use is ineligible. Mean-zero terminology alone supplies no target-neutrality or source exemption.

Figure 60 separates the analytic and policy-visible coordinates of this complete record.

IV.17.2 Reward-Channel Transport and Return Concentration

Theorem IV.17.2 (Affine reward-channel transport)**.** *Let a reward record satisfy Eq.* (343) *under its declared behavior-input law. For $\nu \in \{b, \pi\}$, let $\mathbb{P}^\nu$ be the declared joint law of $(S, R, \widetilde{R})$, let d^ν be its S-marginal, and fix the conditional-mean versions*

$$r^{\mathrm{obs}}_{*,\nu}(s) = \mathbb{E}_{\mathbb{P}^\nu}[R \mid S = s].$$

Assume $d^\pi \ll d^b$ whenever the deployment conclusion is requested. Under the behavior law put

$$g_*(s) = c(s) r^{\mathrm{obs}}_{*,b}(s) + d(s) + u(s).$$

Let $\mathcal{D}$ be the training and output-selection sigma-field. For every represented square-integrable predictor g measurable with respect to $\sigma(\mathcal{D}, S)$, require a fresh evaluation pair with the conditional law

$$\mathcal{L}(S, \widetilde{R} \mid \mathcal{D}) = d^b(ds) Q^b(d\widetilde{r} \mid s), \qquad \int \widetilde{r}\, Q^b(d\widetilde{r} \mid s) = g_*(s)$$

almost surely. Thus the evaluation input as well as its report innovation is fresh relative to selection; an in-sample identity requires a separate uniform empirical-process theorem. Then squared-loss excess against that fresh report satisfies, conditionally on $\mathcal{D}$ and hence also after averaging, the exact identity

$$\mathbb{E}\Big[(g(S) - \widetilde{R})^2 - (g_*(S) - \widetilde{R})^2 \,\Big|\, \mathcal{D}\Big] = \|g - g_*\|^2_{L^2(d^b)}. \tag{345}$$

Define the represented debiased predictor $\widehat{r} = (g - d)/c$, pointwise in the predictable coefficients. The displayed conclusion applies to exact coefficients satisfying Eq. (343)*; estimated coefficients require their own represented perturbation terms. If the left side of Eq.* (345) *is at most $E_r \geq 0$, then*

$$\|\widehat{r} - r^{\mathrm{obs}}_{*,b}\|_{L^2(d^b)} \leq \frac{\sqrt{E_r} + \zeta_r}{c_{\min}}, \tag{346}$$

$$\|\widehat{r} - r^{\mathrm{obs}}_{*,\pi}\|_{L^2(d^\pi)} \leq \frac{\sqrt{W_{\pi\leftarrow b}}(\sqrt{E_r} + \zeta_r)}{c_{\min}} + \Delta_{r,\pi\leftarrow b} \tag{347}$$

whenever the represented deployment comparison $d^\pi \leq W_{\pi\leftarrow b} d^b$ holds and the record certifies

$$\Delta_{r,\pi\leftarrow b} = \|r^{\mathrm{obs}}_{*,\pi} - r^{\mathrm{obs}}_{*,b}\|_{L^2(d^\pi)}.$$

This shift is zero only under a represented conditional-reward invariance statement, for example when the predictor input contains the full reward-relevant state and action.

For M independent episodes, let $(\mathcal{H}_{e,t})_{t=0}^H$ be a nested within-episode filtration such that $\mathcal{H}_{e,t}$ is pre-report, contains all earlier report errors, and $\widetilde{R}_{e,t} - R_{e,t}$ is $\mathcal{H}_{e,t+1}$-measurable. Define

$$B_{e,t} = \mathbb{E}[\widetilde{R}_{e,t} - R_{e,t} \mid \mathcal{H}_{e,t}],$$
$$\xi_{e,t} = \widetilde{R}_{e,t} - R_{e,t} - B_{e,t}.$$

Suppose $|B_{e,t}| \leq \beta_{r,t}$ almost surely and

$$\mathbb{E}[\exp(\lambda \xi_{e,t}) \mid \mathcal{H}_{e,t}] \leq \exp(\lambda^2 \sigma^2_{r,t}/2) \qquad (\lambda \in \mathbb{R}).$$

For fixed deterministic (hence $\mathcal{H}_{e,t}$-predictable) return weights w_t, with probability at least $1 - \delta$,

$$\left| \frac{1}{M} \sum_{e=1}^{M} \sum_{t=0}^{H-1} w_t (\widetilde{R}_{e,t} - R_{e,t}) \right| \leq \sum_{t=0}^{H-1} |w_t| \beta_{r,t} + \sqrt{\frac{2\log(2/\delta)}{M} \sum_{t=0}^{H-1} w_t^2 \sigma^2_{r,t}}. \tag{348}$$

The same iterated-MGF proof applies to any single globally ordered predictable sequence satisfying the displayed conditional MGF; independence is then unnecessary. A β-mixing replacement requires a separately proved bounded or truncated block certificate with its coupling and truncation terms. Neither mixing nor predictability alone authorizes substitution of a radius. A data-dependent predictor or weight choice uses a simultaneous class theorem rather than this fixed-choice display. At fixed horizon the logged transition count is $m_{\mathrm{tr}} = MH$; both M and m_{tr} remain in the complete cost record.

Proof. Put $\mathscr{X} = \sigma(\mathcal{D}, S)$. The conditional-law hypothesis gives $S \mid \mathcal{D} \sim d^b$ and $g_*(S) = \mathbb{E}[\widetilde{R} \mid \mathscr{X}]$. Hence, conditionally on $\mathcal{D}$,

$$\mathbb{E}[(g(S) - g_*(S))(g_*(S) - \widetilde{R}) \mid \mathcal{D}] = \mathbb{E}\Big[(g(S) - g_*(S))\mathbb{E}[g_*(S) - \widetilde{R} \mid \mathscr{X}]\Big|\mathcal{D}\Big] = 0.$$

The pointwise expansion

$$(g - \widetilde{R})^2 - (g_* - \widetilde{R})^2 = (g - g_*)^2 + 2(g - g_*)(g_* - \widetilde{R})$$

and $S \mid \mathcal{D} \sim d^b$ therefore prove Eq. (345). Moreover,

$$\widehat{r} - r^{\mathrm{obs}}_{*,b} = \frac{g - g_* + u}{c}.$$

Thus

$$\|\widehat{r} - r^{\mathrm{obs}}_{*,b}\|_{L^2(d^b)} \le c_{\min}^{-1}\big(\|g - g_*\|_{L^2(d^b)} + \|u\|_{L^2(d^b)}\big),$$

and the first bound follows from the excess identity. If $\rho_{\pi\leftarrow b} = dd^\pi/dd^b \le W_{\pi\leftarrow b}$, then for every measurable f,

$$\|f\|^2_{L^2(d^\pi)} = \int f^2 \rho_{\pi\leftarrow b}\, dd^b \le W_{\pi\leftarrow b}\|f\|^2_{L^2(d^b)}.$$

This bounds the behavior comparator under d^π. The triangle inequality with the certified conditional-reward shift proves Eq. (347).

For one episode, iterated conditional expectation and the conditional MGF hypothesis give

$$\mathbb{E}\exp\left(\lambda \sum_{t=0}^{H-1} w_t \xi_{e,t}\right) \le \exp\left(\frac{\lambda^2}{2} \sum_{t=0}^{H-1} w_t^2 \sigma_{r,t}^2\right).$$

Independence across episodes makes the MGF of the sum over e the product of these bounds. After division by M, the proxy is $M^{-1}\sum_t w_t^2 \sigma_{r,t}^2$. Applying Chernoff's bound to both signs gives the square-root term in Eq. (348). Finally,

$$\left|\frac{1}{M}\sum_{e,t} w_t B_{e,t}\right| \le \sum_t |w_t| \beta_{r,t},$$

which supplies its deterministic bias term. □

Figure 61 records the two distinct reward conversions: population debiasing and episode-return concentration.

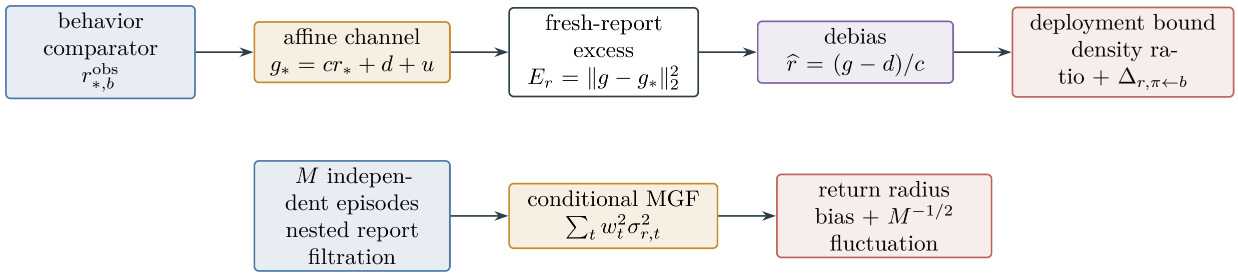

Figure 61: Affine reward correction is a same-law transport. Fresh-report orthogonality converts squared-loss excess to behavior-law L^2 error; density comparison and an explicit conditional-reward shift give deployment error. Episode-level return noise is controlled separately by its predictable bias and conditional-MGF radius.

IV.17.3 Observation Reconstruction and Identifiability

Definition IV.17.3 (Observation reconstruction and identifiability)**.** For a finite latent carrier X and observation carrier O, let $\mathcal{C}_{\mathrm{adm}}$ be the represented set of dynamic contexts on which the uniform conclusion is claimed, and fix $c = (t, h^{\mathrm{obs}}_{t-1}, a^{\circ}_{t-1})$. Let $\mathsf{O}_c(do \mid x)$ be the emission kernel and $D_c(dx' \mid o)$ a represented decoding kernel. Their affine pushforward operators are $\mathsf{O}_{c,*} : \mathcal{P}(X) \to \mathcal{P}(O)$ and $D_{c,*} : \mathcal{P}(O) \to \mathcal{P}(X)$. Their composition induces the Markov reconstruction kernel

$$K_{\mathrm{obs},c}(x, B) = (D_{c,*}\mathsf{O}_{c,*}\delta_x)(B) = \sum_{o\in O} \mathsf{O}_c(o \mid x) D_c(B \mid o), \qquad B \subseteq X.$$

We use $\|P - Q\|_{\mathrm{TV}} = \sup_B |P(B) - Q(B)| = \frac{1}{2}\|P - Q\|_1$. A uniform dynamic row-total-variation reconstruction certificate is

$$\varepsilon^{\mathrm{rec}}_{\mathrm{obs}} = \sup_{c\in\mathcal{C}_{\mathrm{adm}}} \sup_{x\in X} \|K_{\mathrm{obs},c}(x, \cdot) - \delta_x\|_{\mathrm{TV}}. \tag{349}$$

It gives, for every bounded latent function f,

$$\sup_{c\in\mathcal{C}_{\mathrm{adm}}} \|K_{\mathrm{obs},c} f - f\|_\infty \le \varepsilon^{\mathrm{rec}}_{\mathrm{obs}} \operatorname{Osc}(f). \tag{350}$$

An occupancy-averaged certificate must name its context occupancy and cannot be inserted into this sup-norm display. All context dependence and decoder selection remain in the same complete record.

Alternatively, for a declared joint law $\mu(dx)\mathsf{O}(do \mid x)$, write $\mu\mathsf{O}_*$ for its observation marginal and define, up to $\mu\mathsf{O}_*$-almost-sure equality, the conditional-expectation operator

$$(\mathsf{O}_\mu f)(o) = \mathbb{E}_\mu[f(X) \mid O = o].$$

A law-dependent stability certificate is a represented tuple $(\mathcal{D}_{\mathrm{obs}}, \kappa_{\mathrm{obs}}, \Delta_{\mathrm{obs}})$ such that, simultaneously for every $f \in \mathcal{D}_{\mathrm{obs}}$,

$$\|f\|_{L^2(\mu)} \le \kappa_{\mathrm{obs}} \|\mathsf{O}_\mu f\|_{L^2(\mu\mathsf{O}_*)} + \delta_{\mathrm{obs}}(f), \qquad 0 \le \delta_{\mathrm{obs}}(f) \le \Delta_{\mathrm{obs}}.$$

A learned clean-error conversion may use this inequality only after the same record proves that the estimator-minus-target difference belongs to $\mathcal{D}_{\mathrm{obs}}$ and that its observed error is exactly the displayed conditional expectation in the displayed law and norm. A deployment change in the

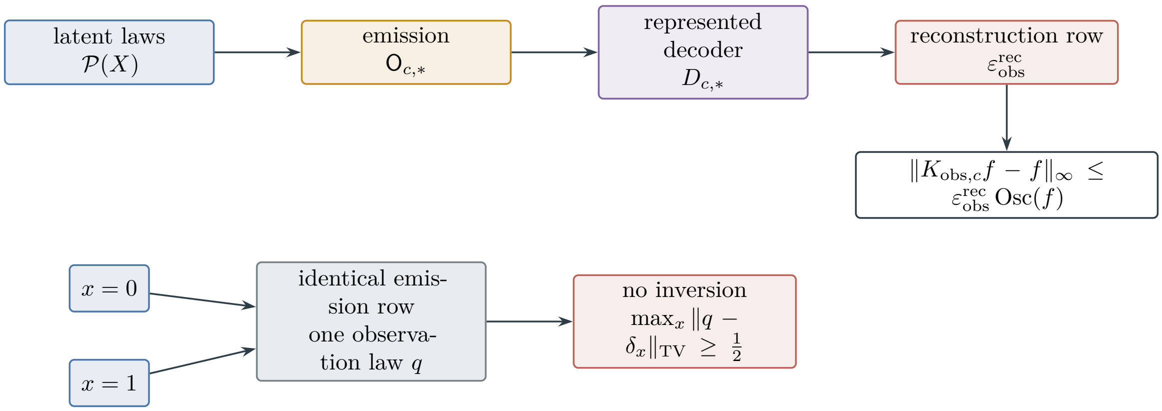

Figure 62: A represented decoder supplies a clean-state comparison through the composite reconstruction kernel. Row TV controls bounded latent functions by oscillation. Identical emission rows erase the binary latent distinction, forcing reconstruction error at least 1/2; observed-law risk alone gives no clean target-gate bound.

latent posterior or in O_μ requires a separate represented shift term. The decoder, operator, stability domain, constants, and construction costs belong to the same record. If neither branch applies, the clean latent-coordinate error is $+\infty$, although observed-law prediction risk may remain finite.

Proof of Eq. (350). For each c, x, the dual characterization of total variation gives

$$|K_{\mathrm{obs},c}f(x) - f(x)| \leq \|K_{\mathrm{obs},c}(x,\cdot) - \delta_x\|_{\mathrm{TV}} \operatorname{Osc}(f).$$

Taking the two suprema proves the claim.

□

Remark IV.17.4 (Observation noise is not invertible in general). Let $X = \{0,1\}$ and suppose the two latent states have identical emission rows. A single observation has the same law under the two states; under repeated conditionally memoryless observations of a fixed hidden state, with no other public kernel that distinguishes the states, induction gives the same law for every public transcript. For any decoder the two reconstruction rows therefore equal one law q. The triangle inequality gives

$$1 = \|\delta_0 - \delta_1\|_{\mathrm{TV}} \leq \|\delta_0 - q\|_{\mathrm{TV}} + \|q - \delta_1\|_{\mathrm{TV}},$$

so $\varepsilon^{\mathrm{rec}}_{\mathrm{obs}} \geq 1/2$. Thus emission entropy, sample size, and an observed-law generalization bound do not by themselves provide a clean-state or target-gate bound.

Figure 62 displays both the valid reconstruction route and the obstruction created by merged emission rows.

IV.17.4 Saturated Typed Errors and Entropy Continuity

Definition IV.17.5 (Saturated typed channel-error profiles). The fixed-law profile is defined below. Its uniform instance–law envelope and reward-square specialization are the separate coordinates in Definitions IV.17.6 and IV.17.7.

Let $q \in \{r, A, O, \mathrm{fil}, \mathrm{comp}\}$ denote reward, execution, observation, filter, or identifiable composite-kernel error, and fix a declared norm N. For each fixed presented instance I and data-generating law $\mathbb{P}_I$, a complete record $\mathcal{R}$ names the clean object, the returned object, their common population law, and the true discrepancy

$$e_{q,\mathsf{N}}(\mathcal{R}) \in [0, +\infty]$$

between them in N. For every candidate $j \in \mathcal{J}_{q,\mathsf{N}}(\mathcal{R})$, put

$$B_{\mathcal{R},j} = \widehat{e}_{q,\mathsf{N},j}(\mathcal{R}) + \Gamma_{q,\mathsf{N},j}(\mathcal{R}; m, \delta_j).$$

Any nonlinear converter is applied before this notation, so every $B_{\mathcal{R},j}$ has the same estimand, law, and norm as $e_{q,\mathsf{N}}(\mathcal{R})$.

Let $\mathscr{R}_{I,q,\mathsf{N}}(b(n))$ be the affordable complete record outputs defined on this fixed sample space and evaluated under this fixed law. Eligibility requires one pre-specified, output-selection-independent sample event $\mathcal{E}^{\mathrm{sat}}_{I,q,\mathsf{N}}(\delta)$, with $\mathbb{P}_I(\mathcal{E}^{\mathrm{sat}}_{I,q,\mathsf{N}}) \geq 1 - \delta$, on which

$$e_{q,\mathsf{N}}(\mathcal{R}) \leq B_{\mathcal{R},j} \quad \text{for every affordable data-dependent output } \mathcal{R} \text{ and every eligible } j. \tag{351}$$

A finite represented codebook may obtain this event from a pre-data global allocation $\sum_{(\mathcal{R},j)} \delta_{\mathcal{R},j} \leq \delta$. Otherwise it requires one uniform cover, PAC–Bayes, or sequential theorem over the affordable output hull. A per-record confidence allocation is not sufficient after the record is selected from data. Events belonging to different data-generating laws are never intersected. Define

$$\mathrm{Err}_{q,\mathsf{N}}(\mathcal{R}; m, \delta) = \min_{j \in \mathcal{J}_{q,\mathsf{N}}(\mathcal{R})} B_{\mathcal{R},j}, \tag{352}$$

where the minimum of an empty family is $+\infty$. The saturated profile is the recordwise minimum followed by the affordable supremum

$$\mathrm{Err}^{\mathrm{sat}}_{q,\mathsf{N};I,\mathbb{P}_I}(n, b(n), m, \delta) = \sup_{\mathcal{R} \in \mathscr{R}_{I,q,\mathsf{N}}(b(n))} \mathrm{Err}_{q,\mathsf{N}}(\mathcal{R}; m, \delta). \tag{353}$$

On $\mathcal{E}^{\mathrm{sat}}_{I,q,\mathsf{N}}(\delta)$,

$$\sup_{\mathcal{R} \in \mathscr{R}_{I,q,\mathsf{N}}(b(n))} e_{q,\mathsf{N}}(\mathcal{R}) \leq \mathrm{Err}^{\mathrm{sat}}_{q,\mathsf{N};I,\mathbb{P}_I}(n, b(n), m, \delta). \tag{354}$$

Indeed, Eq. (351) makes every $B_{\mathcal{R},j}$ an upper bound for the same number simultaneously, so that number is at most their minimum; taking the affordable supremum proves the display. The supremum of an empty record class is zero. An eligible record with an empty candidate family makes the unrestricted worst-record profile $+\infty$; finiteness is therefore a theorem only for a declared subclass in which every record has a finite converter. Profiles in distinct norms are not minimized or added. A represented norm, density, channel-inversion, filter-stability, Bellman, path-law, or Lift theorem must first convert them to one output coordinate. The polynomial-capacity specialization is $b_C(n) = n^C$.

Definition IV.17.6 (Uniform instance–law channel-error envelope)**.** Fix $(q, \mathsf{N}, n, b, m, \delta)$ and the fixed-law typed profiles of Definition IV.17.5.

When the presented instance and law are fixed, their subscripts are suppressed. If a deterministic number uniform over a represented family $\mathfrak{P}_n$ of instance–law pairs is required, its probability-space correct definition is the confidence envelope

$$\overline{\mathrm{Err}}^{\mathrm{sat}}_{q,\mathsf{N}}(n, b(n), m, \delta) = \inf\left\{ u : \inf_{(I,\mathbb{P}_I) \in \mathfrak{P}_n} \mathbb{P}_I\left(\sup_{\mathcal{R} \in \mathscr{R}_{I,q,\mathsf{N}}(b(n))} e_{q,\mathsf{N}}(\mathcal{R}) \leq u \right) \geq 1 - \delta \right\}.$$

A proposed finite value for this outer envelope requires a represented distribution-uniform theorem. It is not the pointwise supremum of random confidence bounds living on unrelated sample spaces.

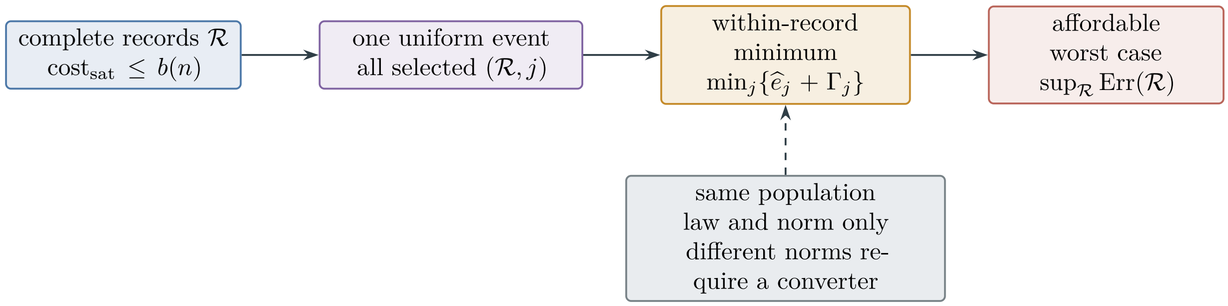

Figure 63: The typed saturated profile first selects the strongest simultaneous certificate within one complete record and then takes the affordable worst case over records. The order is $\sup_{\mathcal{R}} \min_j$, and only bounds for the same estimand, population law, and norm may compete.

Definition IV.17.7 (Reward-square specialization of the typed channel profile)**.** Fix the reward channel, behavior law, and complete record class appearing in Definition IV.17.5.

For reward square loss, the clean behavior-law squared-L^2 risk profile on the fixed $(I, \mathbb{P}_I)$ is

$$U^{\mathrm{sat}}_{r,L^2(d^b)^2;I,\mathbb{P}_I}(n, b(n), m, \delta) = \sup_{\mathcal{R} \in \mathscr{R}_{I,r,L^2(d^b)}(b(n))} \frac{\left(\sqrt{\mathrm{Exc}_r(\mathcal{R}; m, \delta)} + \zeta_r(\mathcal{R})\right)^2}{c_{\min}(\mathcal{R})^2}, \tag{355}$$

with the same complete record class as Eq. (353). A deployment or sup-norm version is obtained only through the record's represented comparison. The square in the subscript is literal: this coordinate bounds $\|\widehat{r} - r^{\mathrm{obs}}_{*,b}\|^2_{L^2(d^b)}$, not the unsquared norm. Here $\mathrm{Exc}_r(\mathcal{R}; m, \delta) \ge 0$ is the recordwise simultaneous upper confidence bound on the public squared-loss excess in Eq. (345): it is the minimum of the represented empirical excess-plus-deviation certificates in that same behavior law and has value $+\infty$ when no such certificate is present. It is not replaced by an RMSE, a clean risk, or an error in another norm without a represented converter. The empty record class has value zero; a missing excess certificate has value $+\infty$.

Figure 63 makes explicit the order of the two optimizations in a saturated channel profile.

Theorem IV.17.8 (Finite-alphabet channel estimation and entropy continuity)**.** *Let $0 < \delta < 1$. Let $K : \mathsf{U} \to \mathcal{P}(\mathsf{V})$ be a finite channel with $d_U = |\mathsf{U}| \ge 1$, $d_V = |\mathsf{V}| \ge 2$, and let $(U_i, V_i)_{i=1}^m$ be iid with joint law $p(u)K(v \mid u)$, where $p_* := \min_u p(u) > 0$. The affordable record contains a represented certified lower bound $0 < \underline{p}_* \le p_*$; the radius below is evaluated with $\underline{p}_*$. Let $\widehat{K}$ be the empirical row estimator, with an arbitrary row assigned when its count is zero. If*

$$m\underline{p}_* \ge 8 \log(2d_U/\delta),$$

then, with probability at least $1 - \delta$,

$$\sup_{u \in \mathsf{U}} \|\widehat{K}(u, \cdot) - K(u, \cdot)\|_{\mathrm{TV}} \le \tau_m := \min\left\{1, \sqrt{\frac{d_V \log 2 + \log(2d_U/\delta)}{m\underline{p}_*}}\right\}. \tag{356}$$

All logarithms in this concentration radius are natural. For a non-iid record, this theorem may be invoked only through a named sample certificate that supplies, on one event, a lower bound on every effective row count and a simultaneous row-TV inequality with a displayed radius. No iid multinomial radius is inferred from mixing or predictability alone. An unvisited row has *mathematical TV distance at most one under the arbitrary convention, but its framework certificate sentinel is* $+\infty$*: without coverage it is not an identified uniform-row converter. A data-dependent channel or alphabet choice additionally uses the global simultaneous event in Eq.* (351)*. Without a represented* $\underline{p}_*$*, the affordable uniform-row certificate is* $+\infty$*.*

Measure the following Shannon quantities in bits. For an input law ν*, put*

$$h_K(\nu) = H_\nu(V \mid U), \qquad C(K) = \sup_{\nu \in \mathcal{P}(\mathsf{U})} I_\nu(U;V).$$

Let

$$\mathfrak{f}_{d_V}(\tau) = \begin{cases} H_2(\tau) + \tau \log_2(d_V - 1), & 0 \le \tau \le 1 - d_V^{-1}, \\ \log_2 d_V, & 1 - d_V^{-1} < \tau \le 1. \end{cases}$$

On the event in Eq. (356)*,*

$$\sup_\nu |h_{\widehat{K}}(\nu) - h_K(\nu)| \le \mathfrak{f}_{d_V}(\tau_m), \tag{357}$$

$$|C(\widehat{K}) - C(K)| \le 2\mathfrak{f}_{d_V}(\tau_m). \tag{358}$$

Every estimated entropy or capacity retains the channel-estimation cost, coverage event, sample size m*, confidence, and complete ceiling* $b(n)$*.*

Proof. Let N_u be the number of observations with input u. Since $N_u \sim \mathrm{Bin}(m, p(u))$, the multiplicative Chernoff bound gives

$$\Pr\{N_u < mp(u)/2\} \le e^{-mp(u)/8} \le e^{-m\underline{p}_*/8}.$$

Consequently,

$$\Pr\{\min_u N_u < m\underline{p}_*/2\} \le d_U e^{-m\underline{p}_*/8} \le \delta/2.$$

Condition on the complementary count event. For one row, the multinomial Bretagnolle–Huber–Carol bound gives

$$\Pr\{\|\widehat{K}(u,\cdot) - K(u,\cdot)\|_1 > \epsilon \mid (N_{u'})_{u' \in \mathsf{U}}\} \le 2^{d_V} e^{-N_u \epsilon^2/2}.$$

If the square root defining τ_m is at least one, the TV claim is deterministic. Otherwise substitute $\epsilon = 2\tau_m$ and $N_u \ge m\underline{p}_*/2$ to obtain

$$\Pr\{\|\widehat{K}(u,\cdot) - K(u,\cdot)\|_{\mathrm{TV}} > \tau_m \mid (N_{u'})_{u' \in \mathsf{U}}\} \le 2^{d_V} e^{-m\underline{p}_* \tau_m^2}.$$

The union over u is at most

$$d_U 2^{d_V} \exp\{-d_V \log 2 - \log(2d_U/\delta)\} = \delta/2.$$

Combining the count and row-estimation failures proves Eq. (356).

To prove the required entropy continuity without an additional assumption, let P, Q be two laws on V at TV distance $\tau \le 1 - d_V^{-1}$, and take a maximal coupling (X, Y) with $\Pr\{X \ne Y\} = \tau$. For $E = \mathbf{1}_{\{X \ne Y\}}$,

$$H(X) - H(Y) = H(X \mid Y) - H(Y \mid X) \le H(X \mid Y) \le H_2(\tau) + \tau \log_2(d_V - 1).$$

The last step first reveals E; conditional on $E = 0$, $X = Y$, and conditional on $E = 1$ there are at most $d_V - 1$ possible values of X once Y is known. Interchanging X, Y controls the other sign.

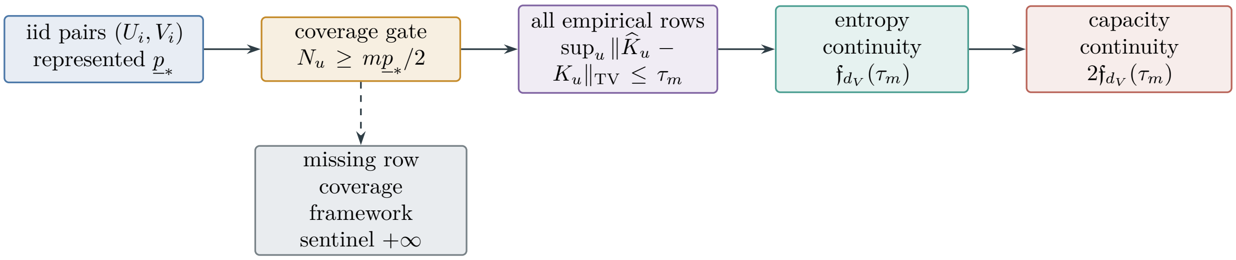

Figure 64: Uniform finite-channel estimation requires coverage of every input row. On the simultaneous coverage event, the common row-TV radius propagates to conditional entropy with loss $\mathfrak{f}_{d_V}(\tau_m)$ and to capacity with loss $2\mathfrak{f}_{d_V}(\tau_m)$.

For $\tau > 1 - d_V^{-1}$, the trivial bound $|H(P) - H(Q)| \leq \log_2 d_V$ applies. Hence, on the row-TV event, every row satisfies

$$|H(\widehat{K}(u,\cdot)) - H(K(u,\cdot))| \leq \mathfrak{f}_{d_V}(\tau_m).$$

Indeed, on $[0, 1 - d_V^{-1}]$,

$$\frac{d}{d\tau}\left[H_2(\tau) + \tau \log_2(d_V - 1)\right] = \log_2 \frac{(1-\tau)(d_V - 1)}{\tau} \geq 0,$$

and $\mathfrak{f}_{d_V}$ is constant thereafter. Thus replacing a row's actual TV distance by the common upper bound τ_m preserves the inequality. The definition of conditional entropy and averaging with $\nu(u)$ yield

$$|h_{\widehat{K}}(\nu) - h_K(\nu)| \leq \sum_u \nu(u)\mathfrak{f}_{d_V}(\tau_m) = \mathfrak{f}_{d_V}(\tau_m),$$

uniformly in ν, proving Eq. (357). Moreover,

$$\|\nu\widehat{K} - \nu K\|_{\mathrm{TV}} \leq \sum_u \nu(u)\|\widehat{K}(u,\cdot) - K(u,\cdot)\|_{\mathrm{TV}} \leq \tau_m.$$

Thus the output-entropy term and the conditional-entropy term in $I_\nu(U;V) = H_\nu(V) - H_\nu(V \mid U)$ each change by at most $\mathfrak{f}_{d_V}(\tau_m)$, so

$$\sup_\nu |I_\nu(\widehat{K}) - I_\nu(K)| \leq 2\mathfrak{f}_{d_V}(\tau_m).$$

Finally, $|\sup_\nu I_\nu(\widehat{K}) - \sup_\nu I_\nu(K)| \leq \sup_\nu |I_\nu(\widehat{K}) - I_\nu(K)|$, which proves Eq. (358).

□

Figure 64 summarizes the coverage, row-estimation, and information-continuity chain.

IV.18 Attainable Risk and the Complete Quantitative Learning Decomposition

The statistical envelopes above control estimation after a represented learning class has been selected. The saturated budget also determines which population risks are attainable by that class. Keeping these two quantities separate gives an exact approximation–estimation decomposition.

Definition IV.18.1 (Saturated attainable risk and approximation floor)**.** Fix an instance I, sample size m, channel parameter ζ, and budget B. Let $\mathscr{L}^{\mathrm{sat}}_{I,m,\zeta}(B)$ be the complete represented learning records whose data acquisition, selected hypothesis or posterior, retained state, optimization, evaluation, and deployment costs are at most B. For $R \in \mathscr{L}^{\mathrm{sat}}_{I,m,\zeta}(B)$, let h_R be its deployed readout and let $R_{0,I}(h_R) \in [0,1]$ be its clean population risk conditional on that readout. The expectation below averages the sample and private randomness of the record. Define

$$\mathcal{R}^{\mathrm{att,sat}}_{I,m,\zeta}(B) = \inf_{R\in\mathscr{L}^{\mathrm{sat}}_{I,m,\zeta}(B)} \mathbb{E}_R R_{0,I}(h_R), \tag{359}$$

$$\mathrm{App}^{\mathrm{sat}}_{I,m,\zeta}(B) = \mathcal{R}^{\mathrm{att,sat}}_{I,m,\zeta}(B) - R^*_{0,I}, \qquad R^*_{0,I} = \inf_h R_{0,I}(h). \tag{360}$$

The final infimum is an analytic population benchmark. It supplies no readout to a represented learner. Empty affordable classes have attainable risk one. A family-uniform version restricts the infimum to records produced by one represented derivation scheme on the declared family slice:

$$\mathcal{R}^{\mathrm{att,sat}}_{\mathfrak{M},n,m,\zeta}(B) = \inf_{R \text{ family-uniform}} \mathfrak{M}_{n,m,\zeta}\left(I \mapsto \mathbb{E}_R R_{0,I}(h_R)\right). \tag{361}$$

For a represented comparison class $\mathcal{H}_B$, put

$$\mathrm{App}_I(\mathcal{H}_B) = \inf_{h\in\mathcal{H}_B} R_{0,I}(h) - R^*_{0,I}. \tag{362}$$

The comparison class contains the construction and deployment cost of each member. Selection from the class is charged separately unless the statistical event is simultaneous over the complete class.

Theorem IV.18.2 (Complete quantitative learning decomposition)**.** *Let $\widehat{h} \in \mathcal{H}_B$ be the deployed output of a complete learning record. For every clean population loss,*

$$R_{0,I}(\widehat{h}) - R^*_{0,I} = \mathrm{App}_I(\mathcal{H}_B) + \left(R_{0,I}(\widehat{h}) - \inf_{h\in\mathcal{H}_B} R_{0,I}(h)\right). \tag{363}$$

Consequently the left side is at least $\mathrm{App}_I(\mathcal{H}_B)$.

Assume a binary flip-symmetric loss under homogeneous symmetric label noise $0 \le \eta < 1/2$, put $\rho = 1 - 2\eta$, and suppose one simultaneous event gives

$$\sup_{h\in\mathcal{H}_B} \left| R_{\eta,I}(h) - \widehat{R}_{\eta,S}(h) \right| \le \Gamma_I(B,m,\eta,\delta). \tag{364}$$

If the represented optimizer satisfies

$$\widehat{R}_{\eta,S}(\widehat{h}) \le \inf_{h\in\mathcal{H}_B} \widehat{R}_{\eta,S}(h) + \varepsilon_{\mathrm{opt}}, \tag{365}$$

then, on that event,

$$\mathrm{App}_I(\mathcal{H}_B) \le R_{0,I}(\widehat{h}) - R^*_{0,I} \le \mathrm{App}_I(\mathcal{H}_B) + \frac{2\Gamma_I(B,m,\eta,\delta) + \varepsilon_{\mathrm{opt}}}{1-2\eta}. \tag{366}$$

The radius may be the minimum of the Rademacher, structural PAC–Bayes, effective-dimension, finite-description, mixing-block, sequential, and martingale candidates already proved in this part, provided that every candidate controls the same corrupted population loss on the same simultaneous event. Behavior-to-deployment, observation reconstruction, and model-channel terms *enter additively only after their represented same-loss converters have been applied. For a source-selected class, the simultaneous radius includes the exact-cell selection term in Eq.* (279) *or the mixture divergence in Eq.* (280). *Each target-dependent reveal retains its source-assigned noise contraction in Eq.* (275). *If the deployed output is used for target selection, its success is additionally bounded by Eq.* (281)*; the risk sandwich itself remains in clean-risk units.*

Taking the infimum over all complete records of cost at most B gives the exact attainable-risk lower floor

$$\mathbb{E}_R R_{0,I}(h_R) - R^*_{0,I} \geq \mathrm{App}^{\mathrm{sat}}_{I,m,\eta}(B) \tag{367}$$

for every such record. Any record satisfying Eqs. (364) *and* (365) *supplies the corresponding upper certificate in Eq.* (366) *at its complete cost.*

Proof. Add and subtract $\inf_{h\in\mathcal{H}_B} R_{0,I}(h)$ to obtain Eq. (363). The second term is nonnegative by membership of $\widehat{h}$ in $\mathcal{H}_B$, which proves the lower bound.

Let h^*_B be an ϵ-minimizer of clean risk in $\mathcal{H}_B$. Homogeneous symmetric corruption gives the exact identity

$$R_{\eta,I}(h) - R_{\eta,I}(h^*_B) = \rho\big(R_{0,I}(h) - R_{0,I}(h^*_B)\big).$$

On the simultaneous event,

$$\begin{aligned} R_{\eta,I}(\widehat{h}) &\leq \widehat{R}_{\eta,S}(\widehat{h}) + \Gamma_I \\ &\leq \widehat{R}_{\eta,S}(h^*_B) + \varepsilon_{\mathrm{opt}} + \Gamma_I \\ &\leq R_{\eta,I}(h^*_B) + 2\Gamma_I + \varepsilon_{\mathrm{opt}}. \end{aligned}$$

Divide by ρ, let $\epsilon \downarrow 0$, and substitute the exact decomposition. This proves Eq. (366). The candidate-minimum rule is valid because all candidates bound the same quantity on one intersection event. The saturated lower floor follows directly from the infimum in Eq. (359).

□

Corollary IV.18.3 (Quantitative parameter limits)**.** *In the setting of Theorem IV.18.2, the clean case $\eta = 0$ removes the corruption factor. Increasing m can reduce the statistical radius but leaves the approximation floor unchanged. As $\eta \uparrow 1/2$, a corruption-transport certificate becomes neutral unless its numerator vanishes at the same rate. Increasing the saturated budget can only decrease* $\mathrm{App}^{\mathrm{sat}}_{I,m,\eta}(B)$*, because it enlarges the complete record class.*

Proof. The first three assertions follow from Eq. (366). Budget monotonicity follows by inclusion of the saturated cost sublevel sets in Eq. (359).

□

IV.19 Summary

This part interprets the saturated invariant in statistical terms.

Dependency trail. The imported objects, kernel spectrum, and admissible capacity sets are Definitions IV.0.1 to IV.2.3 and IV.1.5. The operator and neural coordinates are Definitions IV.3.1 to IV.5.2, IV.3.4, IV.3.5, IV.8.7 and IV.3.10. The RKHS and effective-dimension bridge is Lemma IV.0.2, Corollary IV.1.4, and Proposition IV.2.2. The operator and kernel bounds are Lemmas IV.4.1, IV.3.2, IV.3.6 and IV.3.9 and Corollaries IV.4.2, IV.3.8, IV.3.13 and IV.3.16. The

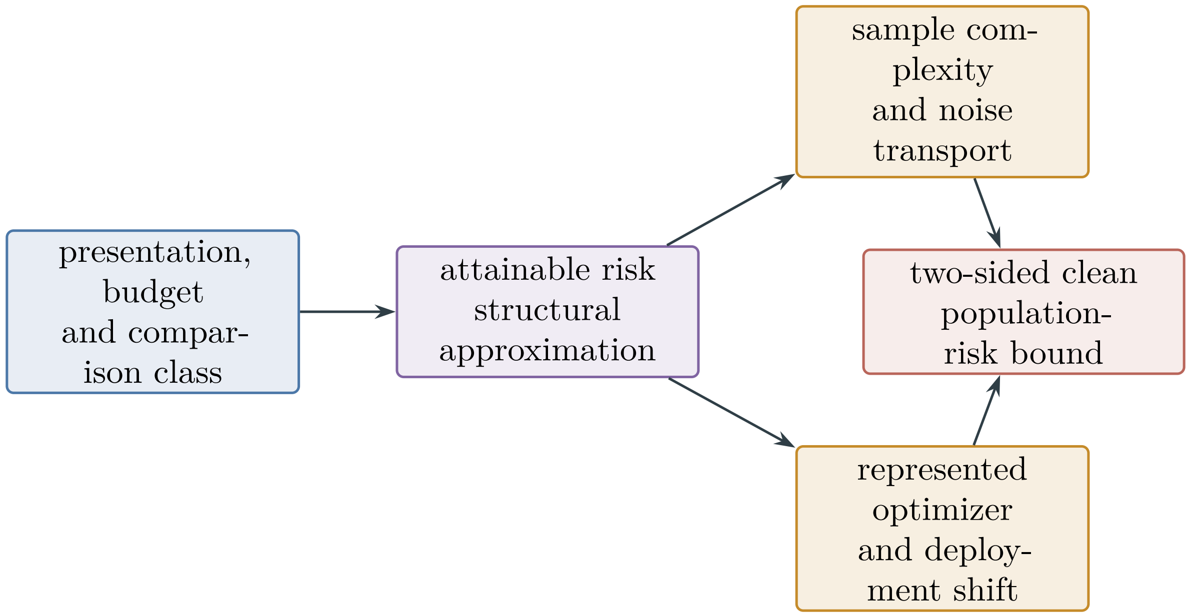

Figure 65: Complete quantitative learning decomposition. The saturated budget fixes the attainable population-risk floor. Statistical estimation, noise transport, represented optimization, and deployment comparison bound the excess above that floor. Exact source-cell selection and source-assigned noise contraction are supplied by Theorem IV.13.3.

neural-kernel and observable-success results are Lemmas IV.5.3, IV.5.5 and IV.5.8 and Corollaries IV.5.6 and IV.5.7. The symmetric-label-corruption results are Lemma IV.10.2, Theorems IV.10.3, IV.10.5 and IV.11.2, Corollaries IV.11.3, IV.10.4 and IV.10.6, and Definition IV.11.1. The empirical/population provenance split and its residual-noise-order specialization are Theorem IV.11.4 and Corollary IV.11.5. The execution-law population profiles, memorization separation, prospective learning accountability, and alignment-cost criterion are Definition IV.11.6, Theorems IV.11.7 and IV.13.6, and Corollary IV.13.7. The dependent-data extension, sequential capacity, martingale posterior certificate, and trajectory provenance split are Definitions IV.15.1 and IV.15.3, Theorems IV.15.2, IV.15.5 to IV.15.7 and IV.15.9, Lemma IV.15.4, and Corollaries IV.15.8 and IV.15.10. The uniform dynamical label-noise extension, including its computational, sample, statistical-noise, and information profiles, is Definitions IV.16.1, IV.16.3 and IV.16.7 and Theorems IV.16.2, IV.16.6 and IV.16.8. The reward, execution, observation, filter, and composite-channel statistical extension is Definitions IV.17.1, IV.17.3 and IV.17.5 and Theorems IV.17.2 and IV.17.8. The geometry-aware learning and affordable-selection bounds are Lemma IV.12.1, Proposition IV.12.3, Theorem IV.12.4, and Corollaries IV.13.1 and IV.13.5. The common source-resolved compilation that assigns every statistical, Bayesian, sequential, active, curiosity, and controlled-learning coordinate to its exact structural source, noise contraction, complete cost, and represented target converter is Definition IV.13.2, Theorem IV.13.3, and Corollary IV.13.4; its dependency diagram and audit table are Fig. 54 and Table 4. The attainable-risk profile and complete approximation–estimation decomposition are Definition IV.18.1, Theorem IV.18.2, and Corollary IV.18.3. The noise-indexed recovery cost, degradation law, and statistical–information threshold comparison are Definition IV.14.2, Proposition IV.14.3, and Theorem IV.14.4. The effective-degree and certification results are Lemmas IV.19.1 to IV.9.2, IV.6.4, IV.9.6 and IV.8.9, Remark IV.6.5, Corollaries IV.8.5 and IV.8.10, and Definition IV.9.4. The final SAT reading and the defect-geometry certificate profile are located at Example III.35.2 and Section IV.9.1.

Lemma IV.19.1 (Learning capacity summary)**.** *For a budget-admissible oracle-free learner on a presented task family, generalization requires target structure captured by an observable algebra within the declared budget. At the operator level, the fixed-operator complexity is* $\mathrm{Tr}(A^{-1})$*, target regularity is* $y^T Ay$*, and a learned operator incurs the selection term* $\sqrt{\log \mathcal{N}(\Theta, \rho)/m}$*. For a fixed kernel, the observable algebra is a budget-admissible RKHS. After declaring a monotone numerical statistical-capacity map* $\mathsf{n}_{\mathrm{stat}} : C \to [0, \infty]$*, the capacity condition is* $N_{\mathrm{eff}}(\lambda) \leq \mathsf{n}_{\mathrm{stat}}(b(n))$*, and captured structure is target-aligned spectral mass satisfying the provenance, representation, compatibility, and family-specific admissibility conditions of Parts I–III. For a neural network, the represented empirical tangent kernel defines the observable algebra. The lazy-training regime uses the initial kernel, whereas feature learning incurs the cost of selecting the terminal kernel. The observable-success theorem gives every successful finite-horizon computation an exact first-hit witness at scale* $\Gamma/(eH_B)$*. A separately declared Rademacher audit resolves residual statistics only when its sufficient error bound is smaller than that scale; otherwise it is inconclusive. The learned-kernel spectrum determines the effective degree, with the stated finite-sample errors. The structural PAC–Bayes bound uses the domain-geometry KL divergence as its complexity term and identifies effective resistance with the Poincaré component of the defect-geometry profile. Under homogeneous symmetric label corruption, exact risk transport rescales the Rademacher and PAC–Bayes clean-risk certificates by* $(1 - 2\eta)^{-1}$*. Debiased kernel ridge rescales its variance term by* $(1 - 2\eta)^{-2}$ *and preserves the source bias; the capacity and represented model-selection costs are unchanged. The saturated statistical-attack theorem applies these bounds simultaneously to every represented learner and its data-dependent output in one budget slice; target-bearing uses retain their full ancestry, and enter the prospective Geom/Abs ledger only after the same-record channel and temporal gates are verified. The average-seed alignment and generalizing-projection profiles evaluate the realized execution law rather than a worst seed in the sample-independent concentration hull. For dynamical samples, independent episodes retain the independent-sample certificate, stationary absolutely regular trajectories use the certified blocking bound, and adaptive paths use sequential structural Rademacher complexity or the martingale structural PAC–Bayes certificate. The pathwise inverse-precision radius and operator-cover term charge the geometry used along the history. Online-to-batch conversion preserves the represented deployment cost, and trajectory-indexed memorization contributes to a fresh execution only through its qualified population projection. Source-condition rates retain their exponent with the independent sample count replaced by the certified dynamical effective sample size. For binary labels on dynamical samples, conditional corruption transport adds the exact margin* $(1 - 2\eta_+)^{-1}$*, while the uniform excess envelope retains problem size* n*, computational ceiling* $b(n)$*, and sample size* m*. The associated sample, statistical-noise, and computational-noise thresholds specialize to polynomial capacity by* $b_C(n) = n^C$*. Adaptive information is bounded by the initial public information plus* $m(1 - H_2(\eta))$ *bits. General reward, execution, observation, filter, and composite-kernel corruption uses complete typed records. Affine reward transport gives a debiased behavior-law* L^2 *bound and, with a represented conditional-reward shift, a deployment bound. For fixed finite channels, the iid row radius depends on* $(m, \delta, d_U, d_V, \underline{p}_*)$*;* n *and* $b(n)$ *index the complete record and its computation. Observation-law risk alone converts to* $+\infty$ *clean-state error, whereas a represented noninvertible decoder may yield a finite but nonvanishing reconstruction floor, at least* $1/2$ *in the binary identical-row example. These quantities remain separate typed model-error coordinates. No model error, entropy, or channel-capacity coordinate enters* Geom/Caus/Lift/Bdry *by itself: target-aligned use requires the same complete record's model-to-Bellman, path, or contact converter, temporal gate, and charged ancestry. Theorem IV.12.4 bounds learned target reach by* *the generated-algebra projection envelope and charges the represented decoder and selector in the same derivation. Prospective learning accountability separates neutral contact from target-aligned selector advantage, while population-orthogonal memorization remains in J_{res}. The noise-indexed inverse success profile defines the exact budgeted computational endpoint, degradation makes its affordable region downward closed, and Fano information bounds supply an algorithm-independent upper endpoint. Ridge, structural-loss, and profile bounds are selected by minimizing a simultaneous bound over the declared finite set. The intrinsic-harmonic-dimension theorem gives the optimal capacity and coverage gap for the declared geometry. Once the model family covers the structural subspace, no budget-admissible learner has greater certified structural extraction relative to that geometry. Geometry learning is handled by a label-independent self-defect criterion, with the geometry-estimation error propagated to the model-level bound. Every learned prospective source retains the complete five-mode, state-provenance, and coefficient-provenance classification of Corollary IV.13.1. Thus learning introduces no coarser structural exception: its source belongs to one of the 4,960 existing Part III cells, with the exact quotient or ambient Geom visibility and every qualification gate evaluated on the same complete record. The attainable-risk profile adds the complementary optimization direction: it infimizes clean population risk over complete affordable learning records. The resulting oracle sandwich separates the structural approximation floor from estimation, homogeneous-noise transport, represented optimization, and deployment shift on one simultaneous event.*

Proof. Part II decomposes budget-admissible oracle-free dynamics into the exposed structural channels and residual accounting. Part III shows that non-negligible structured arrival requires non-negligible $I_n^{\mathrm{sat}}(b(n))$ at the declared budget. A generalizing predictor has non-negligible target correlation on unobserved states. The fixed-operator Rademacher theorem bounds this correlation through the trace term, and the learned-operator theorem includes the cost of data-dependent operator selection. The random-label theorem supplies the corresponding null lower bound for budget-coverable operator classes.

For a fixed budget-admissible kernel, the counted correlations are the Mercer modes of the RKHS after the quotient and lift admissibility conditions are imposed. The kernel ridge theorem in Section IV.3 gives matching effective-dimension, source-residual, and noise terms on the standard source classes, together with the optimized source-condition rates. The PAC–Bayes results in Section IV.8 establish the validity of a label-independent data-geometry prior and identify its KL divergence with Dirichlet/effective-resistance energy. The same section extends the Poincaré bound to defect-geometry profiles under verified MLSI, Bakry–Émery, capacity/Cheeger, or transport inequalities; includes coefficient and profile selection through finite-cover terms; proves saturation at the intrinsic harmonic dimension; and specifies a label-independent procedure for estimating the data geometry.

For neural networks, the fixed-kernel statement applies to the initial NTK in the lazy-training regime and to the represented terminal kernel, together with its covering cost, in the feature-learning regime. The observable-success theorem applies the same accounting to a successful budget-admissible computation: a represented target event is classified as source capture, residual capture controlled by Rademacher complexity, or a capacity barrier. Finally, the learned-kernel spectral profile gives the effective degree, while the test error lower-bounds the captured mass as in Section IV.6. The sample-independent saturated readout hull in Definition IV.11.1 contains every data-dependent output before the realized sample is selected. Theorem IV.11.2 therefore applies one simultaneous concentration event to all represented statistical attacks and uses Lemma I.4.6 to bound their balanced clean-target alignment by the learned projection envelope. Theorem IV.11.4 then keeps empirical memorization on its charged finite carrier, assigns only the orthogonal population remainder to J_{res}, and routes every represented target-bearing use through the existing source and selector accounting. Theorem IV.11.7 replaces the worst-seed projection envelope by the execution-law average needed for recovery, and Theorem IV.13.6 separates its neutral and prospective contributions before taking a noise threshold. Theorem IV.15.2 replaces independent symmetrization by coupled block symmetrization on an absolutely regular trajectory. On predictable paths, Lemma IV.15.4 and Theorem IV.15.5 express the sequential complexity through the same inverse structural precision and selection cover as the independent operator bounds, while Theorem IV.15.6 uses the existing proper defect-restricted prior in a conditional exponential supermartingale. Theorem IV.15.9 applies the existing population projection and residual normal form on a fresh execution carrier. These results establish the stated dependence of generalization on captured structure within the declared budget. Theorem IV.14.4 then places the represented statistical certificate, exact saturated computational crossing, and information converse on the same noise scale. The general model-channel extension follows by fresh-report conditional-mean orthogonality, represented decoder or stability comparisons, and finite-row multinomial concentration. On the single event $\mathcal{E}^{\mathrm{sat}}_{I,q,\mathsf{N}}$, every candidate bounds the same true discrepancy for every affordable data-selected record; hence the true record error is at most the candidate minimum, and taking the affordable supremum proves Eq. (354). These are precisely Theorems IV.17.2 and IV.17.8 and Definitions IV.17.3 and IV.17.5.

$\square$

Every operational certificate term is computed from represented finite data or supplied as a represented assumption with a coverage event. Analytic comparators and true coverage probabilities are not learner inputs. For an operator, they include $\mathrm{Tr}(A^{-1})$, $y^T Ay$, TaskCap, and the random-label null statistic. For a Gram matrix, they include N_{eff}, kernel–target alignment, candidate captured mass, source residual, and participation degree. A trained network additionally provides kernel movement, terminal-kernel improvement, the parameter-cover selection term, and a spectral-margin bound. Subject to the admissibility conditions, these quantities instantiate the $I_n^{\mathrm{sat}}(B)$ and B^* accounting for learnability, feature learning, and model-class lower bounds.

Standard references for the statistical pieces include Aronszajn's RKHS theory, Mercer's theorem, the representer theorem of Kimeldorf-Wahba and later RKHS treatments, Rademacher contraction and Dudley/covering bounds, Hanson-Wright concentration, Caponnetto-De Vito effective-dimension minimax rates for regularized least squares, Cristianini-Shawe-Taylor-Kandola kernel-target alignment, Davis-Kahan and matrix Bernstein spectral concentration, Jacot-Gabriel-Hongler neural tangent kernels, infinite-width NTK analyses such as Arora et al., and Bartlett-Foster-Telgarsky spectral-margin bounds [3, 4, 8, 9, 18, 25, 26, 28, 40, 46, 58, 71, 75].

Part V
Dynamical-Systems Geometry and Controlled Structural Flow

V.0 Introduction: Controlled Dynamics in the Saturated Framework

Part contract. *Inputs:* the represented task presentation and channel decomposition of Parts I and II, the saturated source and accountability theorems of Part III, and the learning certificates of Part IV. *Output:* a single framework-level geometry for represented controlled dynamical systems and coordinatewise bounds for arrival, reach–avoid, mixing, smoothing, concentration, hitting time, Bellman error, policy loss, learned control, and optimal structural active sampling. *First pass:* read Definitions V.1.1 and V.5.1 and Theorems V.1.2, V.5.8, V.6.2 and V.12.3. For affordable directed flow and lifted carriers, read Theorems V.7.2, V.7.9, V.7.12 and V.7.13. For binary target interfaces learned from corrupted labels, read Theorems V.7.7 and V.8.2. For reward, execution, observation, and stochastic partial-observation noise, read Theorems V.1.4, V.1.5, V.2.2, V.2.4 and V.3.2 and Definition V.4.1. For learned control, continue with Definition V.13.1 and Theorem V.14.2. For adaptive data acquisition and candidate transport, read Definition V.9.1 and Theorems V.9.5, V.9.8 and V.9.12. For the description-length axis, universal simulation, and its quantitative relation to the structural learner, read Definition V.17.2, Theorems V.17.6 and V.17.11, and Corollary V.17.22.

The preceding parts already contain the structural ingredients of a dynamical-system analysis: positive generators, reversible conductance, authorized directed current, represented quotients and lifts, boundary readouts, killed resolvents, temporal source admissibility, and learned geometry. This part assembles those ingredients for controlled dynamics. An action rule selects a represented generator; it creates no additional structural channel. Its geometry is evaluated by the same saturated $\mathsf{Geom}/\mathsf{Caus}/\mathsf{Abs}$ ledger used for search and learning.

The direct setting is a finite represented carrier. A deterministic system is a point kernel, a randomized system is a Markov kernel, and a history-dependent controller becomes Markovian on its clocked transcript carrier. Continuous and stochastic systems enter through a represented finite or clocked lift together with an explicit generator or form-comparison error. Thus every claimed bound has a finite charged record on which its constants can be evaluated.

Constructor exhaustion and numerical evaluation play distinct roles. The channel theorems classify every valid Caus and Lift record. The saturated action, aiming, capacity-amplification, and lift-access profiles introduced below evaluate uniform suprema over the affordable instances of those classes.

V.1 Represented Controlled Dynamical Systems

Definition V.1.1 (Controlled dynamical presentation)**.** Let I be a finite task presentation in the sense of Definition I.1.1. A controlled dynamical presentation over I consists of the following represented data:

(i) a finite action alphabet A_I and a nonempty legal-action set $\mathcal{A}_I(x) \subseteq A_I$ for every native state $x \in X_I$;

(ii) for every $a \in A_I$, a stochastic kernel $P_I^a(x,z)$ whose evaluator, retained coefficients, precision, and support are generated by the declared constructor grammar;

(iii) a represented initial law $\mu_{0,I}$, positive and negative terminal sectors T_I^+ and T_I^-, and, when a return criterion is used, a bounded represented reward $r_I : X_I \times A_I \to [-R_I, R_I]$;

(iv) the complete resource record for action testing, kernel evaluation, random-bit use, state update, reward and terminal readout, and storage.

A finite-horizon represented policy is a family of kernels

$$\pi_t(\cdot \mid \mathcal{H}_t) \quad (0 \le t < H)$$

supported on $\mathcal{A}_I(x_t)$, where $\mathcal{H}_t$ is the represented pre-transition history. The policy record contains one family-uniform derivation of these kernels and charges its evaluation, retained state, randomness, and selection operations. A stationary Markov policy has the special form $\pi(a \mid x)$.

For a stationary policy define

$$P_I^\pi(x,z) = \sum_{a \in \mathcal{A}_I(x)} \pi(a \mid x) P_I^a(x,z), \qquad L_I^\pi = \rho(P_I^\pi - I), \quad \rho > 0.$$

The scale ρ and its represented precision belong to the same resource record. The presentation is budget-admissible at B when all data used by the selected finite computation have complete saturated cost at most B.

Theorem V.1.2 (Clocked policy embedding and exhaustive channel decomposition)**.** *Fix a controlled dynamical presentation, a horizon H, and a represented policy of complete cost at most B. There is a represented clocked transcript carrier $\Omega_{I,\pi}^{[H]}$ and a time-homogeneous Markov kernel $P_{I,\pi}^{[H]}$ on that carrier whose projected native-state law is exactly the law generated by the original policy through time H. Its complete construction cost is at most a class-preserving enlarged ceiling $\Psi_{\mathrm{ctl}}(B,H,n)$.*

The normalized transcript generator has the exhaustive decomposition

$$L_{I,\pi}^{[H]} = L^{\mathsf{Geom}} + L^{\mathsf{Caus}} + L^{\mathsf{Abs}} + L^{\mathsf{Lift}} + L^{\mathsf{Bdry}} + L^{J_{\mathrm{res}}}. \tag{368}$$

Here L^{Geom} is the represented reversible carrier, L^{Caus} is the authorized Hodge projection of the selected actual current, L^{Abs} contains qualified represented quotient motion, L^{Lift} is the valid history/interface lift, L^{Bdry} contains terminal and reward readouts, and $L^{J_{\mathrm{res}}}$ is the post-exhaustion orthogonal remainder. The policy selection and every action-dependent coefficient remain ancestors of the corresponding component. Derived randomization, clocks, Bellman updates, stored values, and policy mixtures carry no independent structural source charge.

The same statement holds for every finite slice of an unbounded execution. Deterministic dynamics are included by taking each kernel to be a point mass.

Proof. The full record at time t contains the native state, clock, policy memory, revealed random bits, selected action, reward and terminal flags, and every retained auxiliary coordinate. Conditional on that record, one represented policy evaluation followed by one represented environment-kernel evaluation determines the next-record kernel. This is the finite transcript-kernel construction of Lemma II.11.1. Composition closure and the execution ledger give the ceiling $\Psi_{\mathrm{ctl}}(B,H,n)$, while projection to the native coordinate preserves the original finite-horizon law.

Apply Theorem II.7.1 to the positive transcript generator, then apply Theorems II.4.6, II.5.10, II.7.2 and II.10.5. The policy evaluator and selected action are coordinates of the valid transcript

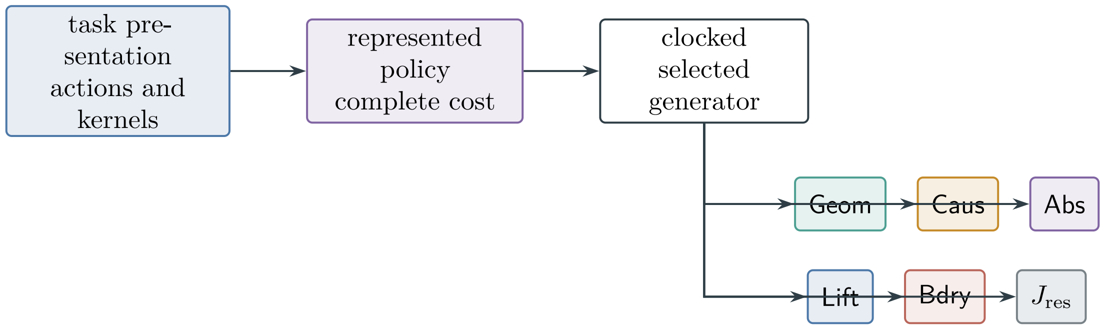

Figure 66: A represented policy selects one cost-tagged transcript generator. The existing exhaustive channel projection then classifies every component; control introduces no additional structural channel.

lift, so their derivations remain in the backward ancestry of every selected transition. No-creation under represented composition is Proposition II.5.9. A deterministic update is the stochastic kernel with one nonzero entry in each row. Applying the same finite construction at each horizon proves the final assertion.

□

Figure 66 records the dependency of the controlled generator on the public presentation and the selected policy.

V.1.1 Noisy execution and partially observed carriers

The selected generator is determined by the action that is executed, not by the action requested by the controller. Likewise, a partially observed controller acts on its public filtration, whereas target and failure events are evaluated on an analytic latent carrier. The following records make these distinctions explicit.

Definition V.1.3 (Represented action-execution channel)**.** Let $K^A(da^\circ \mid x, a, h)$ be a represented execution channel, where a is the intended action, a° is the executed action, and h is the public pre-transition history. For a stationary state policy, define the actual selected reward and kernel by

$$r^{\pi,K}(x) = \sum_a \pi(a \mid x) \sum_{a^\circ} K^A(a^\circ \mid x, a) r^{a^\circ}(x), \tag{369}$$

$$P^{\pi,K}(x, z) = \sum_a \pi(a \mid x) \sum_{a^\circ} K^A(a^\circ \mid x, a) P^{a^\circ}(x, z). \tag{370}$$

The history-dependent version is the corresponding clocked kernel on the transcript carrier. It includes the channel evaluator, random bits, execution feedback, and retained channel state.

The kernel in Eq. (370), rather than the intended kernel, supplies the invariant law, symmetric form, authorized current, Caus action, capacity, committor, and target-arrival law. A publicly fixed, target-independent channel has inherited randomization provenance. Every state-, history-, learned-, or target-dependent channel retains its complete ancestors and cost in the selected generator. If executed actions are unobserved, only the represented composite kernel is available for separate estimation.

Figure 67 shows where actuator noise enters the physical controlled law.

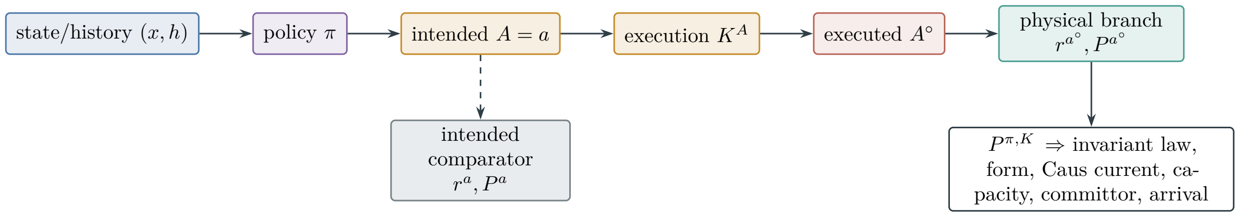

Figure 67: The physical controlled kernel is selected by the executed action. Averaging policy, execution channel, and physical transition produces $P^{\pi,K}$; this actual kernel, not the intended comparator, determines the invariant law, Dirichlet form, current, capacity, committor, and arrival.

Theorem V.1.4 (Execution-channel Bellman, path, and Caus comparison)**.** *Fix a discount $0<\beta<1$, a represented policy π, and a retained value class $\mathcal{V}_B$. Define the exact execution defect*

$$\varepsilon_A^{\mathrm{Bell}} = \sup_{v\in\mathcal{V}_B} \left\| \sum_a \pi(a\mid\cdot) \sum_{a^\circ} K^A(a^\circ\mid\cdot,a)\big[r^{a^\circ}+\beta P^{a^\circ}v - r^a - \beta P^a v\big] \right\|_\infty . \tag{371}$$

If the fixed points lie in $\mathcal{V}_B$, then

$$\|V^{\pi,K}-V^\pi\|_\infty \le \frac{\varepsilon_A^{\mathrm{Bell}}}{1-\beta}. \tag{372}$$

For a represented action metric d_A, suppose every retained $v\in\mathcal{V}_B$ satisfies

$$|r^{a^\circ}(x)-r^a(x)| + \beta|(P^{a^\circ}-P^a)v(x)| \le L_{A,v} d_A(a^\circ,a).$$

Then

$$\varepsilon_A^{\mathrm{Bell}} \le \sup_{v\in\mathcal{V}_B} L_{A,v} \sup_{x,a} \mathbb{E}_{K^A} d_A(A^\circ,a). \tag{373}$$

If the actual and intended reversible comparison forms share the same carrier and boundaries and satisfy the represented form comparison

$$0<a_A\le b_A<\infty, \qquad a_A\mathcal{E}_0(f,f)\le\mathcal{E}_K(f,f)\le b_A\mathcal{E}_0(f,f) \qquad (f|_T=1, f|_F=0),$$

then actuator noise changes condenser capacity only within

$$a_A\,\mathrm{Cap}_0(T,F) \le \mathrm{Cap}_K(T,F) \le b_A\,\mathrm{Cap}_0(T,F). \tag{374}$$

For a fixed state policy whose transition choice is independent of reward reports, reward-report noise does not alter this physical state-carrier capacity. For a history-dependent controller, reward reports can change later actions and therefore the actual clocked generator and its capacity. Observation noise likewise acts through the actual joint/filter carrier or a represented form/Lift comparison.

For a finite horizon, suppose the reference and actuator records have the same initial law and the same conditional transition, clean-reward, reward-report, observation, and feedback kernels once the executed action is fixed. Let a represented co-adapted coupling use identical downstream innovations at every common history for which intended and executed actions agree. Put $\tau_{\mathrm{mis}}=\inf\{t: A_t^\circ\neq A_t\}$, and suppose deterministic $q_t\in[0,1]$ satisfy, at every common history before mismatch,

$$\Pr\{A_t^\circ\neq A_t \mid \mathcal{H}_t^{\mathrm{cpl}}, \tau_{\mathrm{mis}}\ge t\} \le q_t.$$

Every common-path functional $F \in [0,1]$, including target-before-failure contact, satisfies

$$|\mathbb{E}_K F - \mathbb{E}_0 F| \le 1 - \prod_{t=0}^{H-1}(1-q_t) \le \sum_{t=0}^{H-1} q_t. \tag{375}$$

If the initial laws instead have TV distance at most $\varepsilon_{\text{init}}$, the first right side becomes $1-(1-\varepsilon_{\text{init}})\prod_{t<H}(1-q_t)$. For history-dependent hazards, the exact survival probability may instead be retained when their joint conditional mismatch law is represented; the union bound remains valid with the expected hazards. A discounted comparison uses Theorem V.7.12 or a represented resolvent identity; the finite-horizon sum is not inserted into a discounted slot.

Finally, suppose actual and reference authorized currents use the same mobility a_t, and the represented current difference $\Delta J_t = J_t^{\pi,K} - J_t^{\pi}$ has finite Caus action. Then

$$\left|\int_0^H [\mathcal{P}_{\mathsf{Caus}}(J_t^{\pi,K}, D_t) - \mathcal{P}_{\mathsf{Caus}}(J_t^{\pi}, D_t)]\,dt\right| \le \left(\int_0^H \mathcal{A}_{\mathsf{Caus}}(\Delta J_t; a_t)dt\right)^{1/2}\left(\int_0^H \mathcal{E}_{\text{aim}}(D_t; a_t)dt\right)^{1/2}. \tag{376}$$

This is an aiming-power comparison. Target contact still requires the same-record aiming-to-contact converter and its neutral baseline.

Proof. Write $\mathcal{T}_K v = r^{\pi,K} + \beta P^{\pi,K} v$ and $\mathcal{T}_0 v = r^{\pi} + \beta P^{\pi} v$. Since $\mathcal{T}_K$ is a β-contraction and $V^\pi \in \mathcal{V}_B$,

$$\begin{aligned}\|V^{\pi,K} - V^\pi\|_\infty &\le \|\mathcal{T}_K V^{\pi,K} - \mathcal{T}_K V^\pi\|_\infty + \|(\mathcal{T}_K - \mathcal{T}_0)V^\pi\|_\infty \\ &\le \beta\|V^{\pi,K} - V^\pi\|_\infty + \varepsilon_A^{\text{Bell}}.\end{aligned}$$

Rearrangement proves Eq. (372); only the reference fixed point is needed in the retained class. Averaging the pointwise metric inequality first under $K^A(\cdot \mid x, a)$ and then under $\pi(\cdot \mid x)$, followed by the two suprema, proves Eq. (373).

For the path comparison put $s_t = \Pr\{\tau_{\text{mis}} \ge t\}$. The conditional hazard hypothesis gives

$$s_{t+1} = \Pr\{\tau_{\text{mis}} \ge t, A_t^\circ = A_t\} \ge s_t(1-q_t).$$

Induction yields $s_H \ge \prod_{t<H}(1-q_t)$. The coupled full paths agree on $\{\tau_{\text{mis}} \ge H\}$, because every downstream innovation is then shared. Therefore

$$|\mathbb{E}_K F - \mathbb{E}_0 F| \le \Pr\{\tau_{\text{mis}} < H\} \le 1 - \prod_{t<H}(1-q_t) \le \sum_{t<H} q_t.$$

A maximal initial coupling gives the stated $1-(1-\varepsilon_{\text{init}})\prod_t(1-q_t)$ variant.

Let $\mathcal{A}_{T,F} = \{f : f|_T = 1, f|_F = 0\}$. By the variational definition,

$$\mathrm{Cap}_K(T,F) = \inf_{f \in \mathcal{A}_{T,F}} \mathcal{E}_K(f,f).$$

Taking the infimum of the lower form inequality and, separately, of the upper form inequality proves $a_A\,\mathrm{Cap}_0 \le \mathrm{Cap}_K \le b_A\,\mathrm{Cap}_0$.

Finally, linearity of the Caus edge pairing in its current gives

$$\mathcal{P}_{\mathsf{Caus}}(J_t^{\pi,K}, D_t) - \mathcal{P}_{\mathsf{Caus}}(J_t^{\pi}, D_t) = -\sum_e \Delta J_t(e)\nabla_e D_t.$$

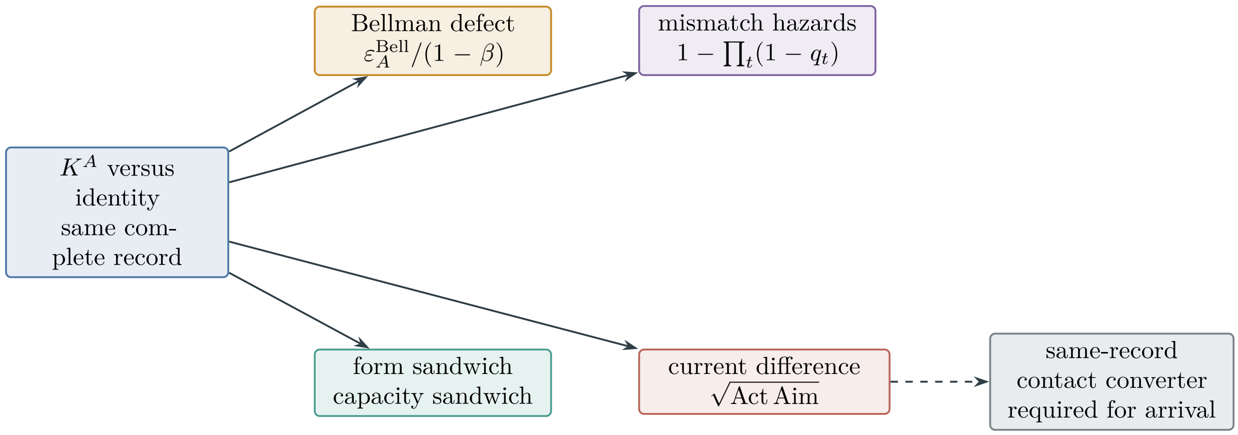

Figure 68: Execution noise has four typed consequences: Bellman defects control value error, form comparison controls capacity, first-mismatch coupling controls finite-horizon path events, and current perturbation controls aiming power. Only a same-record aiming-to-contact theorem converts the last branch to target arrival.

Indeed, $\mathcal{P}_{\mathsf{Caus}}(J,D) = -\sum_e J(e)\nabla_e D$; no mobility weight occurs in this pairing. Inserting $a_t(e)^{-1/2}a_t(e)^{1/2}$ edgewise and applying Cauchy–Schwarz bounds its absolute value by $\mathcal{A}_{\mathsf{Caus}}(\Delta J_t; a_t)^{1/2}\mathcal{E}_{\mathrm{aim}}(D_t; a_t)^{1/2}$; Cauchy–Schwarz in time proves Eq. (376). This calculation uses the same mobility and the same aiming field in both records and does not itself bound target contact.

□

Figure 68 separates the four numerical consequences of an execution-channel comparison.

Theorem V.1.5 (Closed-loop total-variation and relative-entropy comparison)**.** *Consider two complete closed-loop records on the same measurable finite-horizon latent–public path coordinates. Use the same within-step reveal order in both records, and let j range over initialization and all reward, action, observation, transition, and feedback kernels in that order. Suppose a represented common-history comparison gives either*

$$\sup_h \|Q_j^1(\cdot \mid h) - Q_j^0(\cdot \mid h)\|_{\mathrm{TV}} \le \delta_j, \qquad 0 \le \delta_j \le 1,$$

or

$$\sup_h \mathrm{KL}\Big(Q_j^1(\cdot \mid h)\|Q_j^0(\cdot \mid h)\Big) \le \kappa_j, \qquad 0 \le \kappa_j < \infty,$$

in nats. Then the full path laws satisfy, respectively,

$$\|\mathbb{P}^1 - \mathbb{P}^0\|_{\mathrm{TV}} \le 1 - \prod_j (1-\delta_j) \le \sum_j \delta_j, \tag{377}$$

$$\mathrm{KL}(\mathbb{P}^1\|\mathbb{P}^0) \le \sum_j \kappa_j, \qquad \|\mathbb{P}^1 - \mathbb{P}^0\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2}\sum_j \kappa_j}. \tag{378}$$

Consequently every $F \in [0,1]$ has expectation shift bounded by the applicable right side, clipped at one. The KL branch has value $+\infty$ when absolute continuity or a conditional KL certificate fails. A transport–entropy replacement requires a represented common path metric and reference law.

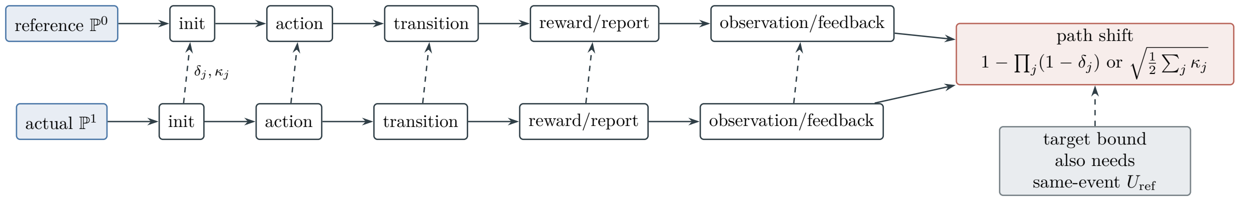

Figure 69: Closed-loop discrepancies accumulate in a common reveal order. Sequential coupling gives the TV product bound, while the relative-entropy chain rule gives a sum followed by Pinsker. Both are same-event shifts; a target upper bound additionally needs a represented reference-event bound.

These are path-law comparison errors, not structural source values. Write $\varepsilon_{\rm path}$ for the applicable clipped TV or Pinsker right side. If the same target-before-failure event has a represented reference upper bound $U_{\rm ref}$ under $\mathbb{P}^0$, then its probability under $\mathbb{P}^1$ is at most

$$\min\{1, U_{\rm ref} + \varepsilon_{\rm path}\}.$$

The complete-record target-contact bound is the minimum of this quantity and the eligible prospective selector, Caus-contact, capacity-contact, and Lift bounds for that same event. The raw comparison error alone is not a contact bound.

Proof. Enumerate the finite reveal slots as $j = 1, \dots, J$, including initialization. Conditional on agreement through slot $j-1$, a maximal coupling of the two conditional rows agrees at slot j with probability at least $1 - \delta_j$. Induction therefore gives

$$\Pr\{\text{all } J \text{ coordinates agree}\} \geq \prod_{j=1}^{J} (1 - \delta_j).$$

The coupling characterization of TV and $1 - \prod_j (1-\delta_j) \leq \sum_j \delta_j$ prove Eq. (377).

When the required absolute continuity holds, the exact chain rule is

$$\mathrm{KL}(\mathbb{P}^1 \| \mathbb{P}^0) = \sum_{j=1}^{J} \mathbb{E}_{\mathbb{P}^1} \mathrm{KL}\Big(Q_j^1(\cdot \mid H_{j-1}) \| Q_j^0(\cdot \mid H_{j-1})\Big).$$

The uniform common-history bound makes the right side at most $\sum_j \kappa_j$. Pinsker's inequality with natural-log KL gives $\|\mathbb{P}^1 - \mathbb{P}^0\|_{\rm TV} \leq \sqrt{\mathrm{KL}(\mathbb{P}^1 \| \mathbb{P}^0)/2}$, proving Eq. (378). Finally, for $0 \leq F \leq 1$, the layer-cake representation $F = \int_0^1 \mathbf{1}_{\{F > s\}} ds$ and the event definition of TV give $|\mathbb{E}_1 F - \mathbb{E}_0 F| \leq \|\mathbb{P}^1 - \mathbb{P}^0\|_{\rm TV}$.

□

Figure 69 compares TV and KL accumulation in the common reveal order.

V.2 Stochastic POMDP Carriers and Filter Transfer

Definition V.2.1 (Represented stochastic POMDP record)**.** A finite represented stochastic POMDP record has finite latent, action, reward-report, observation, feedback, and retained-memory alphabets. It contains a latent state carrier X, intended and executed action carriers A, A°, transition kernels $P_t(dx' \mid x, a^\circ, e, h)$, a clean reward kernel P^R, reward-report channel K^R, observation channel K^O, a joint execution-and-feedback channel $K^{A,E}$, initial law, and raw latent trigger sets $T^{\mathrm{raw}}, F^{\mathrm{raw}}$. These trigger sets may overlap; they are not themselves the terminal sectors of the task interface. The special case $P_t(dx' \mid x, a^\circ, e, h) = P_t^{a^\circ}(dx' \mid x)$ is the usual controlled Markov model. Its pre-action public history is

$$H_t^{\mathrm{obs}} = (O_0, A_0, E_0, \widetilde{R}_0, O_1, \dots, A_{t-1}, E_{t-1}, \widetilde{R}_{t-1}, O_t),$$

with $H_0^{\mathrm{obs}} = (O_0)$, where E_t denotes whatever execution feedback is actually revealed. The controller's random action kernel and memory update are measurable with respect to $\mathcal{F}_t^{\mathrm{obs}} = \sigma(H_t^{\mathrm{obs}}, \text{public seed})$. The analytic latent filtration $\mathcal{G}_t^{\mathrm{lat}}$ also contains $X_{0:t}$, clean rewards, executed actions, and unrevealed environment innovations, and $\mathcal{F}_t^{\mathrm{obs}} \subseteq \mathcal{G}_t^{\mathrm{lat}}$. The coordinate x includes every persistent latent environment or channel state on which a later row depends. A persistent state omitted from x is permitted only when the record proves that its conditional law is already a function of the retained clocked coordinates. This is the Markov adequacy requirement, not a notational convention.

For a horizon H, fix a represented tie map $\chi^{\mathrm{tie}}_{T^{\mathrm{raw}},F^{\mathrm{raw}}} : X \to \{\varnothing, \mathsf{T}, \mathsf{F}\}$ that returns $\varnothing$ off $T^{\mathrm{raw}} \cup F^{\mathrm{raw}}$, returns the unique winner label on $T^{\mathrm{raw}} \triangle F^{\mathrm{raw}}$, and returns the declared winner label on $T^{\mathrm{raw}} \cap F^{\mathrm{raw}}$. Augment the latent record by the absorbing first-arrival winner

$$W_0 = \chi^{\mathrm{tie}}_{T^{\mathrm{raw}},F^{\mathrm{raw}}}(X_0), \qquad W_{t+1} = \begin{cases} W_t, & W_t \neq \varnothing, \\ \chi^{\mathrm{tie}}_{T^{\mathrm{raw}},F^{\mathrm{raw}}}(X_{t+1}), & W_t = \varnothing. \end{cases}$$

and form the clocked carrier

$$\Omega^{[H]}_{\mathrm{POMDP}} = \{(t, x, h, \mathfrak{m}, w) : 0 \le t \le H,\ w \in \{\varnothing, \mathsf{T}, \mathsf{F}\}\}, \tag{379}$$

where $\mathfrak{m}$ is the represented controller/filter memory; the symbol m remains the training-sample size. The coordinate $\mathfrak{m}$ also contains every persistent public channel and policy-randomness state needed by a later row. With μ_0 the declared latent initial law, the initial clocked law is the pushforward of

$$\lambda_0(dx, do) = \mu_0(dx) K_0^O(do \mid x) \tag{380}$$

under $h = (o)$, the represented initial-memory constructor, clock zero, and $w = \chi^{\mathrm{tie}}_{T^{\mathrm{raw}},F^{\mathrm{raw}}}(x)$. Any initial channel state is included in x or $\mathfrak{m}$ before this pushforward. One transition is the represented composition

$$\begin{aligned} &\pi_t(da \mid h, \mathfrak{m}) K_t^{A,E}(da^\circ, de \mid x, a, h) P_t(dx' \mid x, a^\circ, e, h) \\ &\quad P_t^R(dr \mid x, a^\circ, e, x', h) K_t^R(d\widetilde{r} \mid r, x, a^\circ, e, x', h) \\ &\quad K_{t+1}^O(do' \mid x', a^\circ, e, r, \widetilde{r}, h). \end{aligned} \tag{381}$$

Every later factor is conditioned on the earlier outputs in this declared reveal order unless a represented conditional independence removes an argument. The product is followed by the deterministic history, memory, clock, and winner update. The A_t°-marginal of $K_t^{A,E}$ is the execution kernel K_t^A used by marginal execution comparisons. When A_t° and E_t are correlated, path-law and information comparisons use the joint kernel and do not multiply its two marginals. All kernels in the factorization may be replaced by one represented composite kernel when only that composite is identifiable. Continuous outcome spaces enter the finite framework only through a declared finite-precision discretization or continuum Lift with its represented approximation and target-interface error.

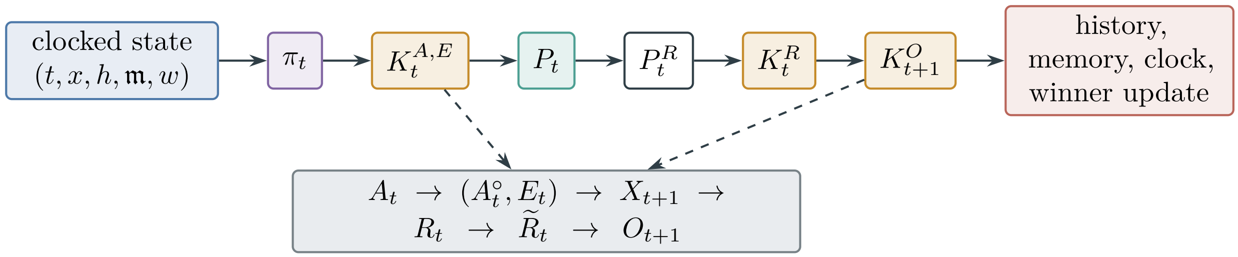

Figure 70: One POMDP step is a represented composition in a fixed reveal order: public policy, joint execution/feedback channel, latent transition, clean reward, reward report, observation, and deterministic record update. The product is Markov on the full clocked carrier even for a history-dependent controller.

The terminal interface on the augmented carrier is the disjoint pair

$$T_\Omega = \{(t, x, h, \mathfrak{m}, w) : w = \mathsf{T}\}, \qquad F_\Omega = \{(t, x, h, \mathfrak{m}, w) : w = \mathsf{F}\}. \tag{382}$$

Thus overlap is resolved before the task-space terminal interface is formed, and $T_\Omega \cap F_\Omega = \varnothing$.

The clean latent target and exact posterior used below are analytic comparators. They are not public policy inputs or prospective sources. A target gate, filter, belief, committor, or decoder becomes available to the controller only through a represented $\mathcal{F}_t^{\mathrm{obs}}$-measurable derivation with complete pre-hit cost. The complete record charges the channel and model descriptions, random bits, history or sufficient-memory representation, filtering, policy evaluation, target/failure gates, precision, sample size m, and every training and deployment operation under the ceiling $b(n)$.

Figure 70 expands one row of the clocked POMDP kernel in its declared reveal order.

Theorem V.2.2 (POMDP Markov embedding, source accounting, and target geometry)**.** *The kernel factorization in Eq.* (381) *defines a positive represented Markov kernel on* $\Omega^{[H]}_{\mathrm{POMDP}}$*. Its law projects exactly to the declared latent and public POMDP law through time* H*. At the class-preserving ceiling* $\Psi_{\mathrm{ctl}}(b(n), H, n)$*, its generator has the exhaustive decomposition of Eq.* (368)*. The actual executed kernel and observation-measurable policy remain ancestors of every selected current and transition.*

On the joint clocked carrier, the clean first-arrival event is the analytic boundary (T_Ω, F_Ω) *of Eq.* (382)*, with the chosen tie convention represented explicitly. Selector and first-arrival identities apply on this actual carrier under their temporal hypotheses. Native capacity, Caus action, and aiming apply only when their own represented reference-law, reversibility or comparison-form, authority, and boundary hypotheses hold; otherwise capacity is unavailable and probability entries retain their neutral value. A target-dependent capacity, aiming, or gate computation enters an affordable prospective minimum only when its boundary evaluator and target comparison are represented in the same temporally admissible record; otherwise it remains an external audit of the actual law. Projection to an unclocked latent state need not be a valid lift when histories with the same state choose different actions. A base-carrier form, capacity, or resolvent bound transfers only when the same record supplies the required lumpability, intertwining, form comparison, or clocked Lift defect. Valid-lift provenance by itself is not a numerical capacity comparison.*

Stochastic observations and rewards therefore add no structural channel. They alter the actual represented carrier, its public-filtration restriction, and the numerical constants evaluated by the existing channels.

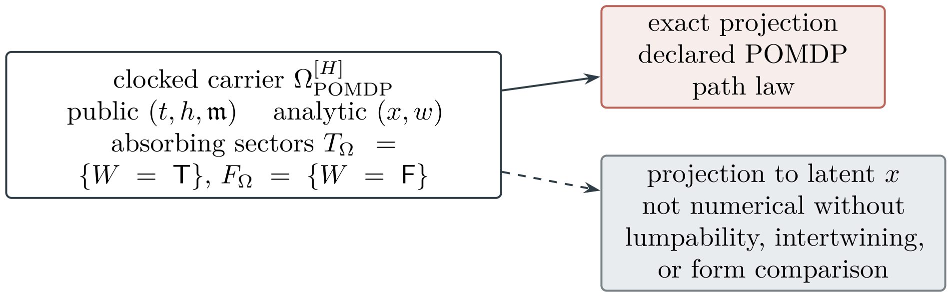

Figure 71: Winner augmentation makes first arrival an absorbing boundary event on the actual clocked carrier. Projection reproduces the declared POMDP path law, but projection to the unclocked latent state need not preserve dynamics: equal latent states can carry histories with different action mixtures.

Proof. Fix a clocked state. Sum the factors in Eq. (381) in reverse reveal order. Each factor is a nonnegative probability kernel, so each successive integral equals one; the deterministic history, memory, clock, and winner update is a point mass. The composed row is therefore nonnegative and has total mass one.

Equality of path laws follows on cylinder events by induction. At time zero, Eq. (380) followed by the deterministic memory and winner constructors is exactly the declared initial public–latent law, with $W_0 = \chi^{\mathrm{tie}}_{T^{\mathrm{raw}},F^{\mathrm{raw}}}(X_0)$. Suppose the projected cylinder law is correct through time t. Conditional on its terminal cylinder, the conditional law of $(A_t, A_t^\circ, E_t, X_{t+1}, R_t, \widetilde{R}_t, O_{t+1})$ is exactly the product in Eq. (381); the remaining coordinates are its deterministic record update. The tower property therefore gives the declared cylinder probability through time $t+1$. Projection proves the asserted equality through H.

Composition closure charges every factor, random bit, retained channel state, numerical precision, policy evaluation, and history or memory update. This gives the ceiling $\Psi_{\mathrm{ctl}}(b(n), H, n)$. Applying Theorem V.1.2 to this actual clocked kernel gives the exhaustive decomposition. Backward ancestry closure keeps the policy, execution, transition, reward, report, observation, and feedback factors attached to every selected transition and current.

The winner coordinate is absorbing. Hence $W = \mathsf{T}$ is exactly the event that the represented tie convention declared target as the first winner, and $W = \mathsf{F}$ is the analogous failure event. This proves the boundary identity. Selector, capacity, Caus, and Lift conclusions are then invoked separately only when the hypotheses listed in the theorem are present; the Markov embedding alone supplies none of their numerical comparisons.

Finally, two clocked states can share the same latent state x while their public histories induce different action mixtures. Their next-state rows then differ, so the projection to x is not lumpable in general. A numerical base-carrier transfer therefore requires the stated represented comparison. No-creation under represented composition is Proposition II.5.9.

$\square$

Figure 71 distinguishes exact path-law projection from an unjustified unclocked numerical transfer.

Figure 72 records the extra gates required before analytic target geometry enters a prospective minimum.

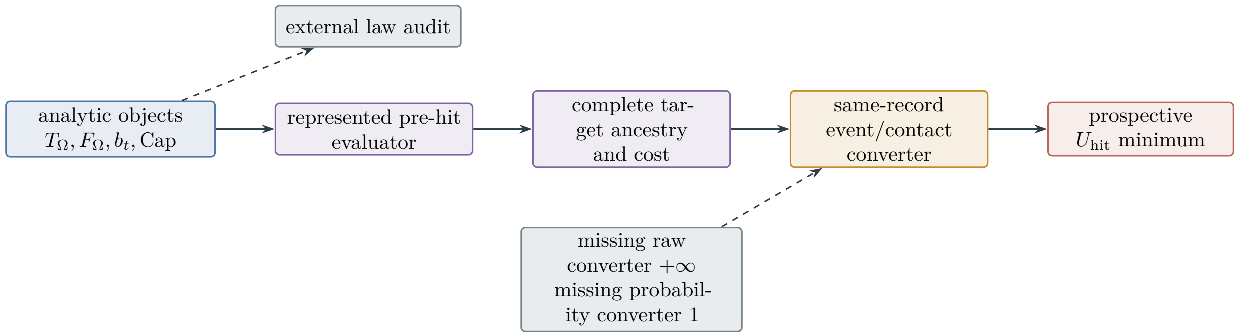

Figure 72: Analytic target geometry evaluates the actual law but is not an affordable prospective source. Entry into a probability minimum requires a represented pre-hit evaluator, complete target-dependent ancestry, and a same-record conversion to the declared event.

V.2.1 Approximate Filters and Transfer to Value and Arrival

Definition V.2.3 (Represented belief/filter comparison)**.** Let

$$b_t(h) = \mathcal{L}((X_t, W_t) \mid H_t^{\mathrm{obs}} = h)$$

be a version of the analytic exact belief on the augmented latent state. The record declares one common clocked carrier and measurable interfaces on which $b_t(h)$ and the represented $\widehat{b}_t(h)$ are compared for every admitted public history $h \in \mathcal{H}_t^{\mathrm{adm}}$. Put

$$e_t(h) = \|\widehat{b}_t(h) - b_t(h)\|_{\mathsf{B}}, \qquad \bar{e}_t = \sup_{h \in \mathcal{H}_t^{\mathrm{adm}}} e_t(h).$$

Here $\mathcal{H}_t^{\mathrm{adm}}$ contains the union of histories reachable under either compared clocked law. The record fixes common measurable versions of the two belief interfaces on that union. At a history having zero exact evidence but positive approximate probability, a finite comparison therefore requires a represented extension or a positive-evidence stability certificate; otherwise $e_t(h) = +\infty$. This convention prevents a regular conditional distribution on a null history from being chosen after seeing the desired bound. A represented filter $\widehat{b}_t$ is admissible only when its update uses public history and its model arithmetic, evidence normalization, retained precision, and deployment evaluation are included in the complete cost. It may update the augmented winner coordinate only when a represented pre-hit target/failure evaluator, or a learned public gate with its error interface, occurs in the filter's public ancestry. Otherwise the represented filter lives on the non-hit state/memory carrier and the exact winner remains analytic evaluation coordinates. In that non-hit branch, the b_t and $\widehat{b}_t$ used in the norm below denote the corresponding common-space state/memory marginal; the augmented belief notation and augmented norm are used only in the represented- gate branch. An approximate-filter certificate contains a declared norm and constants such that, for every admitted transition $h \mapsto h'$,

$$e_{t+1}(h') \le \alpha_t e_t(h) + \varepsilon_{\mathrm{fil},t}(h'), \qquad \bar{e}_0 \le \varepsilon_0, \tag{383}$$

where α_t is uniform over the admitted histories. With $\bar{\varepsilon}_{\mathrm{fil},t} = \sup_{h'} \varepsilon_{\mathrm{fil},t}(h')$, this implies $\bar{e}_{t+1} \le \alpha_t \bar{e}_t + \bar{\varepsilon}_{\mathrm{fil},t}$. Induction gives

$$\bar{e}_t \le \left(\prod_{j=0}^{t-1} \alpha_j\right) \varepsilon_0 + \sum_{s=0}^{t-1} \left(\prod_{j=s+1}^{t-1} \alpha_j\right) \bar{\varepsilon}_{\mathrm{fil},s}. \tag{384}$$

Empty products equal one. If $\alpha_t \le \alpha < 1$ and $\bar{\varepsilon}_{\mathrm{fil},t} \le \bar{\varepsilon}_{\mathrm{fil}}$, then

$$\bar{e}_t \le \alpha^t \varepsilon_0 + \frac{1-\alpha^t}{1-\alpha}\bar{\varepsilon}_{\mathrm{fil}}. \tag{385}$$

A Bayes update receives a finite α_t only from a represented evidence floor or a direct stability proof; normalization near zero evidence is not assumed Lipschitz.

Suppose the exact- and approximate-belief policies satisfy

$$\|\pi_t(\cdot \mid b_t(h)) - \pi_t(\cdot \mid \widehat{b}_t(h))\|_{\mathrm{TV}} \le L_{\pi,t} e_t(h). \tag{386}$$

Let $K_t^{\mathrm{ex}}(h, a; \cdot)$ and $\widehat{K}_t(h, a; \cdot)$ be the exact and represented approximate same-carrier clocked rows after action a. For $h = h(\omega)$, define the three rows

$$\begin{aligned} P_t(\omega, \cdot) &= \sum_a \pi_t(a \mid b_t(h)) K_t^{\mathrm{ex}}(h, a; \cdot), \\ P_t^{\mathrm{mid}}(\omega, \cdot) &= \sum_a \pi_t(a \mid \widehat{b}_t(h)) K_t^{\mathrm{ex}}(h, a; \cdot), \\ \widehat{P}_t(\omega, \cdot) &= \sum_a \pi_t(a \mid \widehat{b}_t(h)) \widehat{K}_t(h, a; \cdot), \end{aligned}$$

and the remaining model defect

$$\delta_{\mathrm{model},t}(h) = \|P_t^{\mathrm{mid}}(\omega, \cdot) - \widehat{P}_t(\omega, \cdot)\|_{\mathrm{TV}}.$$

Markov-kernel contraction applied to the first two rows gives

$$\|P_t(\omega, \cdot) - P_t^{\mathrm{mid}}(\omega, \cdot)\|_{\mathrm{TV}} \le \|\pi_t(\cdot \mid b_t(h)) - \pi_t(\cdot \mid \widehat{b}_t(h))\|_{\mathrm{TV}} \le L_{\pi,t} e_t(h).$$

The triangle inequality and the universal TV ceiling one therefore give

$$\|\widehat{P}_t(\omega, \cdot) - P_t(\omega, \cdot)\|_{\mathrm{TV}} \le \epsilon_t(\omega) := \min\{1, L_{\pi,t} e_t(h) + \delta_{\mathrm{model},t}(h)\}, \qquad \bar{\epsilon}_t := \sup_\omega \epsilon_t(\omega). \tag{387}$$

Thus $\delta_{\mathrm{model},t}$ is defined, not inferred from a differently typed model loss, and is required in that same row-total-variation norm. The record also contains the common-space comparisons

$$\|\widehat{\mu}_0 - \mu_0\|_{\mathrm{TV}} \le \varepsilon_{\mathrm{init}} \quad \text{and} \quad \mathbb{E}_{\mathbb{P}^{\widehat{b}}}\left|\widehat{\mathbf{1}}_{T,F,H}(\omega) - \mathbf{1}_{T,F,H}(\omega)\right| \le \varepsilon_{T,F,H}^{\mathrm{gate}}$$

for the initial law and target/failure event disagreement. The expectation is under the approximate-filter law on the declared common path carrier. Thus the event comparison below is the literal triangle inequality between $\mathbb{E}_{\mathbb{P}^b} \mathbf{1}_{T,F,H}$, $\mathbb{E}_{\mathbb{P}^{\widehat{b}}} \mathbf{1}_{T,F,H}$, and $\mathbb{E}_{\mathbb{P}^{\widehat{b}}} \widehat{\mathbf{1}}_{T,F,H}$. Missing probability interfaces take the neutral value one, whereas a requested raw model or filter error without its norm converter is $+\infty$. Analytic joint-carrier evaluation sets the gate comparison to zero only for that analytic probability; it does not expose the latent gate to the controller or create a prospective source. Without a filter-stability and policy/kernel converter, the filter-to-value and filter-to-hit errors are $+\infty$, while probability coordinates take their neutral value one.

Figure 73 displays the exact product–sum structure of the uniform history-wise filter certificate.

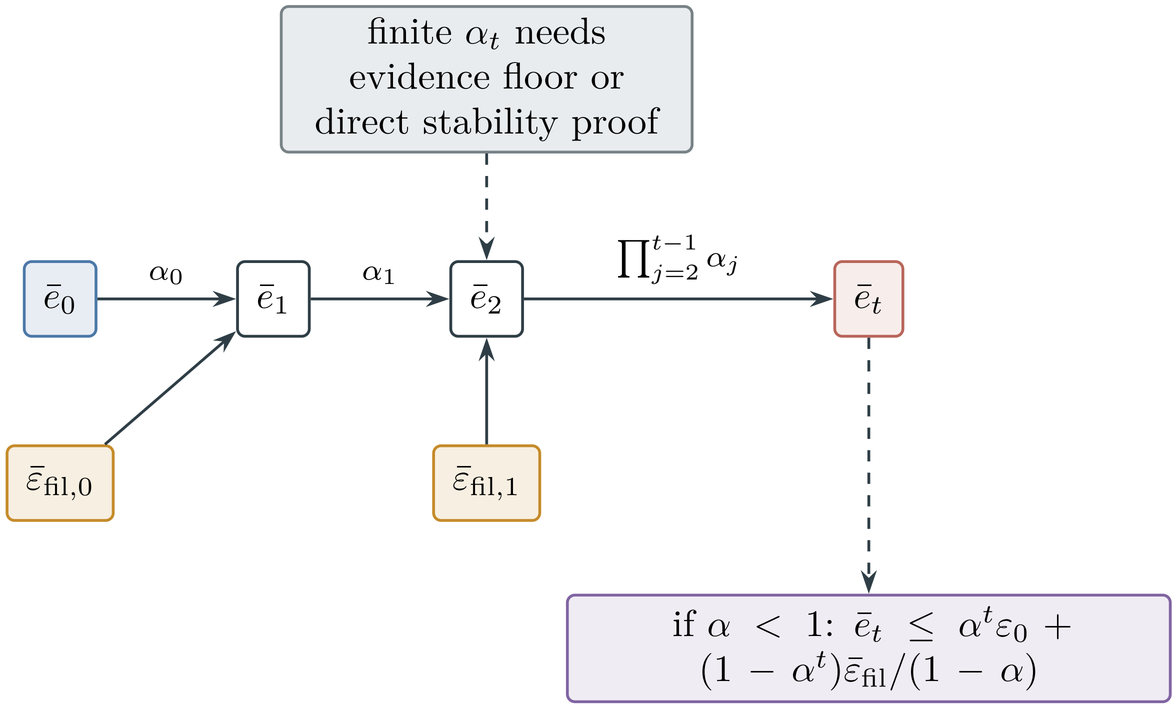

Figure 73: Filter error propagates by multiplicative stability and additive update defects. Iteration gives the exact product–sum bound; under a uniform contraction, initial error decays and accumulated defect approaches $\bar{\varepsilon}_{\mathrm{fil}}/(1-\alpha)$.

Theorem V.2.4 (Filter error to value and target-arrival error)**.** *Under Definition V.2.3, the exact-belief and represented-filter clocked laws satisfy the following bound whenever their initial-law total variation is at most $\varepsilon_{\mathrm{init}}$:*

$$p^{b}_{T,H} = \mathbb{E}_{\mathbb{P}^b} \mathbf{1}_{T,F,H}, \qquad p^{\widehat{b}}_{T,H} = \mathbb{E}_{\mathbb{P}^{\widehat{b}}} \widehat{\mathbf{1}}_{T,F,H}.$$

$$|p^{b}_{T,H} - p^{\widehat{b}}_{T,H}| \le \min\left\{1, \varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \bar{\epsilon}_t + \varepsilon^{\mathrm{gate}}_{T,F,H}\right\}. \tag{388}$$

Here the last term is zero for analytic evaluation on the joint carrier and is the represented target/failure interface error when the deployed record uses learned public gates. Discounted arrival uses the corresponding resolvent or fast-fiber comparison. A represented co-adapted coupling can replace $\sum_t \bar{\epsilon}_t$ by the sharper predictable hazard

$$\sum_{t=0}^{H-1} \mathbb{E}\Big[\mathbf{1}_{\{\tau_{\mathrm{mis}} \ge t\}} \epsilon_t(\Omega_t)\Big],$$

where $\tau_{\mathrm{mis}} = \inf\{t : \Omega^{b}_{t+1} \ne \Omega^{\widehat{b}}_{t+1}\}$ is the first divergent transition in that coupling; on $\{\tau_{\mathrm{mis}} \ge t\}$, the two pre-transition states at time t agree. An unconditioned average row error is not a valid substitute.

For the value branch, let Q^*_B be the exact action-value function of a declared restricted benchmark for which the Bellman and performance-difference identities are certified. Suppose

$$|Q^*_B(b,a) - Q^*_B(b',a)| \le L_Q \|b - b'\|_{\mathsf{B}}$$

for every retained action. Let the deployed approximate-filter action $\widehat{A}_t$ be ξ_t-greedy at $\widehat{b}_t$:

$$\max_a Q_B^*(\widehat{b}_t, a) - Q_B^*(\widehat{b}_t, \widehat{A}_t) \le \xi_t.$$

Then, with expectation under the deployed approximate-filter policy, the discounted benchmark loss is bounded by

$$V_B^*(b_0) - V^{\widehat{\pi}}(b_0) \le \sum_{t\ge 0} \beta^t \mathbb{E}_{\widehat{\pi}}[2L_Q e_t(H_t^{\text{obs}}) + \xi_t], \tag{389}$$

whenever the series and all comparisons are certified. Under exact greediness $\xi_t = 0$ and Eq. (385), the right side is at most

$$2L_Q \left[\frac{\varepsilon_0}{1-\beta\alpha} + \frac{\bar{\varepsilon}_{\text{fil}}}{1-\alpha}\left(\frac{1}{1-\beta} - \frac{1}{1-\beta\alpha}\right)\right]. \tag{390}$$

Proof. Let d_t be the TV distance between the two exact-gate clocked laws at time t. Markov-kernel contraction and Eq. (387) give

$$d_{t+1} \le \|\mu_t^b P_t - \mu_t^{\widehat{b}} P_t\|_{\text{TV}} + \|\mu_t^{\widehat{b}} P_t - \mu_t^{\widehat{b}} \widehat{P}_t\|_{\text{TV}} \le d_t + \bar{\epsilon}_t.$$

Since $d_0 \le \varepsilon_{\text{init}}$, induction yields $d_H \le \varepsilon_{\text{init}} + \sum_{t<H} \bar{\epsilon}_t$. The same recursion on path-prefix kernels bounds the TV distance of the full clocked path laws. Applying it to the exact target event and then using

$$\left|\mathbb{E}_{\mathbb{P}^{\widehat{b}}}(\mathbf{1}_{T,F,H} - \widehat{\mathbf{1}}_{T,F,H})\right| \le \varepsilon_{T,F,H}^{\text{gate}}$$

proves Eq. (388). Sequential maximal coupling makes this refinement explicit. First maximally couple the initial laws, whose mismatch probability is at most $\varepsilon_{\text{init}}$. Conditional on initial agreement and agreement through the state at time t, maximally couple the two rows at that shared state. Its conditional mismatch probability is at most $\epsilon_t(\Omega_t)$, so

$$\Pr\{\tau_{\text{mis}} = t\} \le \mathbb{E}\Big[\mathbf{1}_{\{\tau_{\text{mis}} \ge t\}} \epsilon_t(\Omega_t)\Big].$$

Summing the disjoint first-mismatch events and adding the initial and gate terms proves the stated survival-weighted refinement.

Fix a realized pair $b = b_t(H_t^{\text{obs}})$, $\widehat{b} = \widehat{b}_t(H_t^{\text{obs}})$. If $a^* \in \arg\max_a Q_B^*(b,a)$ and $\widehat{a} = \widehat{A}_t$, then

$$\begin{aligned} V_B^*(b) - Q_B^*(b, \widehat{a}) &\le |Q_B^*(b, a^*) - Q_B^*(\widehat{b}, a^*)| \\ &\quad + Q_B^*(\widehat{b}, a^*) - Q_B^*(\widehat{b}, \widehat{a}) \\ &\quad + |Q_B^*(\widehat{b}, \widehat{a}) - Q_B^*(b, \widehat{a})| \\ &\le 2L_Q e_t(H_t^{\text{obs}}) + \xi_t. \end{aligned}$$

The certified performance-difference identity is

$$V_B^*(b_0) - V^{\widehat{\pi}}(b_0) = \mathbb{E}_{\widehat{\pi}} \sum_{t\ge 0} \beta^t [V_B^*(b_t) - Q_B^*(b_t, \widehat{A}_t)].$$

Substitution proves Eq. (389). For exact greediness, insert the uniform bound $e_t(H_t^{\text{obs}}) \le \bar{e}_t$ from Eq. (385). The two sums are

$$\sum_{t\ge 0} (\beta\alpha)^t = \frac{1}{1-\beta\alpha}, \qquad \sum_{t\ge 0} \beta^t(1-\alpha^t) = \frac{1}{1-\beta} - \frac{1}{1-\beta\alpha},$$

which gives Eq. (390).

$\square$

Figure 74 shows the two distinct converters from history-wise filter error.

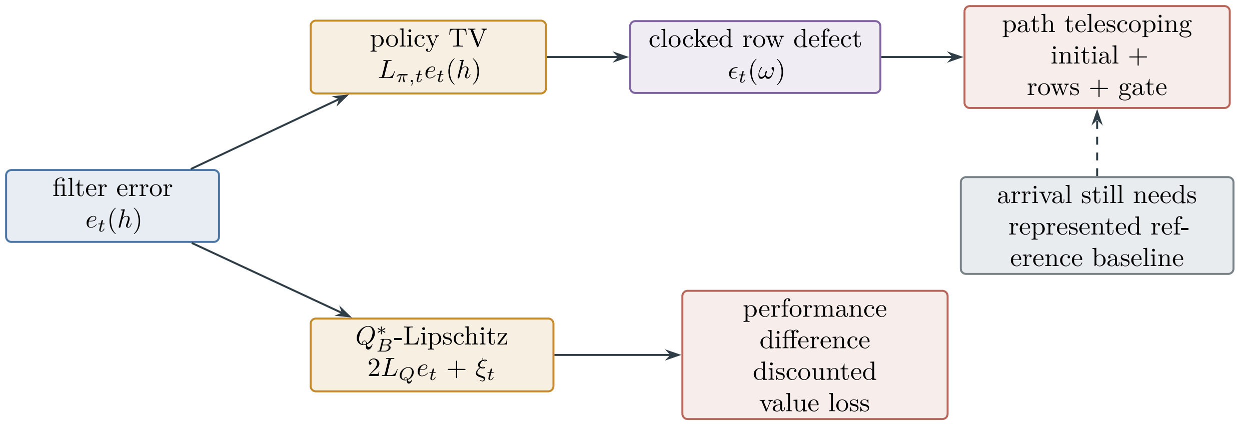

Figure 74: Filter error affects deployment only through explicit converters. Policy Lipschitzness gives a clocked-kernel row defect and hence a path-event shift. Exact benchmark action-value Lipschitzness plus the performance-difference identity gives value loss. The two output units are not interchanged.

V.3 Adaptive Channel Information and Belief Entropy

V.3.1 Conditional channel-information ledger

Definition V.3.1 (Conditional channel-information and entropy ledger)**.** Measure Shannon information in bits. For a represented channel $K : \mathsf{U} \to \mathcal{P}(\mathsf{V})$ and a declared nonempty represented class of input laws $\mathcal{Q}$, define

$$C_{\mathcal{Q}}(K) = \sup_{\nu \in \mathcal{Q}} I_\nu(U; V), \tag{391}$$

$$\vartheta_{\mathrm{KL}}(K; \mathcal{Q}) = \sup_{\substack{\nu,\nu' \in \mathcal{Q} \\ 0 < \mathrm{KL}(\nu \| \nu') < \infty}} \frac{\mathrm{KL}(\nu K \| \nu' K)}{\mathrm{KL}(\nu \| \nu')} \in [0, 1]. \tag{392}$$

The second display uses the same logarithm base in numerator and denominator. If the admissible-pair set in the second supremum is empty, its value is defined to be zero; with this convention $\vartheta_{\mathrm{KL}}(K; \mathcal{Q}) \in [0, 1]$ also for singleton input-law classes. For a restricted class $\mathcal{Q}$, an SDPI use additionally requires $\mathcal{Q}$ to contain, at each admitted public history, the conditional marginal law of U, every conditional law of U given the task index, and the mixtures used in the mutual-information decomposition. Taking $\mathcal{Q} = \mathcal{P}(\mathsf{U})$ is the unrestricted default. For a conditional history-dependent channel, the record contains predictable upper bounds $c_{q,t}$ and $\vartheta_{q,t}$ valid for every admitted public history and input law. Here q indexes the within-step reveal order of execution feedback, observation, reward report, and any other public output. A finite output alphabet gives the fallback $c_{q,t} \le \log_2 |\mathsf{V}_{q,t}|$. A continuous channel without a represented amplitude, moment, power, or input-law restriction has capacity $+\infty$.

For a finite-alphabet complete POMDP record, enumerate physical channel outputs in their actual reveal order. Bundle every initially sampled persistent channel state, policy seed, and other retained auxiliary random state into S_0^{ret}. If $\mathcal{F}_0^{\mathrm{clean}}$ is generated by the presentation, the clean initial task/latent state, and deterministic initialization data, set

$$\mathsf{H}_{\mathrm{init}}^{[H]} = H(S_0^{\mathrm{ret}} \mid \mathcal{F}_0^{\mathrm{clean}}). \tag{393}$$

This term is charged once. It is zero only when the retained auxiliary state is conditionally deterministic; otherwise the finite-alphabet fallback is $\log_2 |\mathsf{S}_0^{\mathrm{ret}}|$. It deliberately excludes clean initial task/latent uncertainty, which remains the prior-entropy or initial-information coordinate. A claimed total trajectory entropy must add that prior term; the present ledger does not silently identify injected randomness with total path entropy.

If $Z_{q,t}^{\mathrm{ch}}$ is one physical-channel output, $\mathcal{G}_{q,t}^{\mathrm{pre}}$ contains every earlier public and latent reveal and retained channel state, and $U_{q,t}$ is its declared clean channel input. With the initial slot separated, define the physical-channel noise entropy in bits by

$$\mathsf{H}_{\mathrm{ch}}^{[H]} = \sum_{t,q} H(Z_{q,t}^{\mathrm{ch}} \mid \mathcal{G}_{q,t}^{\mathrm{pre}}, U_{q,t}). \tag{394}$$

For the factorization in Eq. (381), this is the sum of the joint execution/feedback entropy $H(A_t^\circ, E_t \mid X_t, A_t, H_t^{\mathrm{obs}}, \text{channel state})$, the reward-report entropy conditional on the clean reward and every earlier reveal, and the observation entropy conditional on its clean latent input and every earlier reveal. The generic reveal-order definition prevents correlated outputs from being omitted or counted twice. Conditioning on a retained state does not erase its randomness: its initial entropy has already been charged once in Eq. (393). Let $\mathcal{G}_t^{A,\mathrm{pre}}$ be the full retained sigma-field immediately before (A_t°, E_t) is drawn, $\mathcal{G}_t^{\mathrm{tr,pre}} = \mathcal{G}_t^{A,\mathrm{pre}} \vee \sigma(A_t^\circ, E_t)$, and $\mathcal{G}_t^{R,\mathrm{pre}} = \mathcal{G}_t^{\mathrm{tr,pre}} \vee \sigma(X_{t+1})$. Let $\mathcal{F}_t^{\pi,\mathrm{pre}}$ contain the public history and every retained policy-randomness state immediately before the fresh action draw, but not that draw. The stochastic-transition, clean-reward, and policy-kernel entropies are stored separately:

$$\mathsf{H}_{\mathrm{env}}^{[H]} = \sum_{t=0}^{H-1} H(X_{t+1} \mid \mathcal{G}_t^{\mathrm{tr,pre}}), \tag{395}$$

$$\mathsf{H}_{R}^{[H]} = \sum_{t=0}^{H-1} H(R_t \mid \mathcal{G}_t^{R,\mathrm{pre}}), \tag{396}$$

$$H_{\pi}^{[H]} = \sum_{t=0}^{H-1} H(A_t \mid \mathcal{F}_t^{\pi,\mathrm{pre}}). \tag{397}$$

Conditioning transition entropy after the joint execution/feedback draw prevents actuator randomness from being counted again as environment randomness. The clean-reward row is separate from report-channel entropy; their chain-rule sum is used only when one combined environment coordinate is explicitly requested. Conditioning policy entropy on retained randomness prevents a persistent or global seed from being charged afresh at every time; its one-time charge is $\mathsf{H}_{\mathrm{init}}^{[H]}$. All Shannon entropies in these displays use base two. An absent or proved deterministic slot contributes zero. If an exact finite-alphabet upper bound is unavailable, the represented fallback is the sum of the applicable log-alphabet bounds, not zero. The quantity $\mathsf{H}_{\mathrm{ch}}^{[H]}$ measures randomization injected by the physical channels. It is distinct from the information $C_{\mathcal{Q}}(K)$ transmitted through them. For continuous channels, the record uses relative entropy to a declared reference or mutual information; unconstrained differential entropy is not used as a coordinate-invariant budget, and an unconstrained upper entry is $+\infty$.

Figure 75 keeps physical randomization and transmitted task information in separate coordinates.

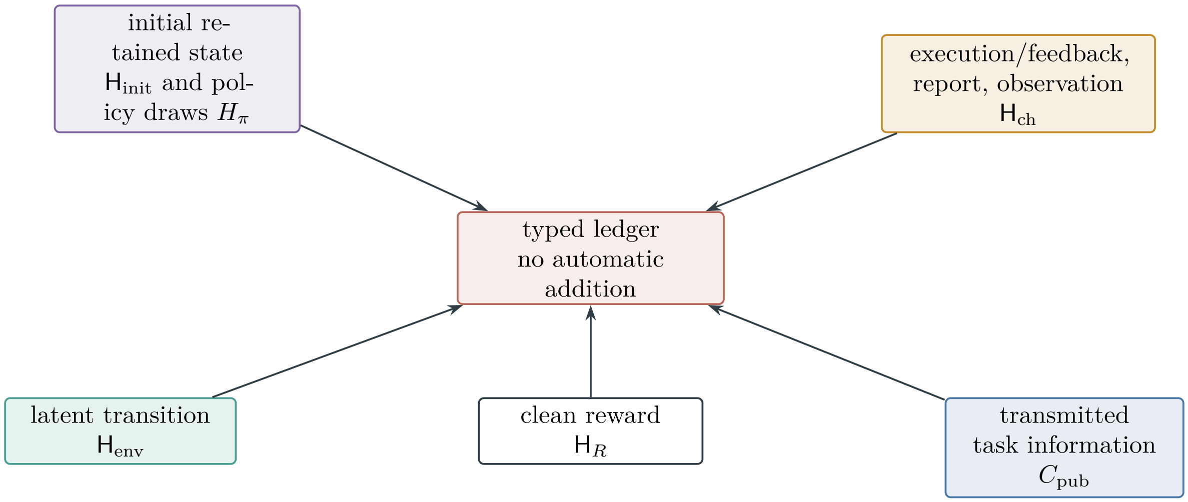

Figure 75: The physical entropy ledger charges initial retained randomness once and separates subsequent policy randomization, stochastic-transition entropy, clean-reward entropy, and channel-injected entropy. Public channel capacity is a fifth, non-entropic coordinate that bounds transmitted task information. The arrows denote typed registration, not the temporal order of a POMDP step.

V.3.2 Adaptive information and rate–distortion

Theorem V.3.2 (Adaptive multichannel information and rate–distortion bounds)**.** *Let Θ be a latent task index and let Z_0 be the initial public record. When Z_0 includes an estimator trained on the m logged samples, the complete record either keeps $I(\Theta;\mathsf{Z}_0)$ explicitly or factors $\mathsf{Z}_0 = (\mathsf{Z}_{\mathrm{prior}}, \mathsf{Z}_{\mathrm{train}})$ into the prior public presentation and m declared training reveal slots with capacities $c^{\mathrm{train}}_{q,s}$. At each deployment time and reveal slot, suppose the complete record contains a conditional Markov chain*

$$\Theta \longrightarrow U_{q,t} \longrightarrow Z_{q,t} \quad \text{conditional on the public past,} \tag{398}$$

and a capacity certificate $c_{q,t}$ from Definition V.3.1. The controller's intended action is generated from the public past and contributes zero conditional mutual information. Unrevealed target-independent actuator randomness also contributes zero. Execution feedback that depends on a latent state is a public observation channel and receives its own capacity slot. Every realized history-conditional input law must lie in the declared class $\mathcal{Q}$; the SDPI branch also uses the conditional and mixture closure specified in Definition V.3.1.

Let Z_H be the full public transcript, let $\mathsf{Z}^{\mathrm{dep}}_{1:H}$ denote its ordered deployment-reveal suffix so that $\mathsf{Z}_H = (\mathsf{Z}_0, \mathsf{Z}^{\mathrm{dep}}_{1:H})$, and abbreviate the represented deployment information budget by

$$C^{[H]}_{\mathrm{pub}} := \sum_{t,q} c_{q,t}.$$

Then

$$I(\Theta;\mathsf{Z}_H) \le I(\Theta;\mathsf{Z}_0) + \sum_{t,q} c_{q,t}, \tag{399}$$

$$H(\Theta \mid \mathsf{Z}_H) \ge H(\Theta \mid \mathsf{Z}_0) - \sum_{t,q} c_{q,t}. \tag{400}$$

If Θ is uniform on a family of size $N_n \geq 2$, let $I_0^{(m)}$ be a represented upper bound on $I(\Theta; \mathsf{Z}_0)$. The exact value is admissible; under the training factorization one may take

$$I(\Theta; \mathsf{Z}_0) \leq I_0^{(m)} := I_{\text{public}} + \sum_{s=1}^{m} \sum_{q} c_{q,s}^{\text{train}}, \qquad I_{\text{public}} \geq I(\Theta; \mathsf{Z}_{\text{prior}}). \tag{401}$$

Every estimator satisfies

$$\Pr\{\widehat{\Theta} = \Theta\} \leq \min\left\{1, \frac{I_0^{(m)} + \sum_{t,q} c_{q,t} + 1}{\log_2 N_n}\right\}. \tag{402}$$

Success at least α requires

$$\sum_{t,q} c_{q,t} \geq \alpha \log_2 N_n - I_0^{(m)} - 1. \tag{403}$$

More generally, let $d(\theta, \widehat{\theta})$ be a represented distortion and define the conditional rate–distortion function

$$R_{\Theta|\mathsf{Z}_0}(D) = \inf_{\substack{P_{\widehat{\Theta}|\Theta,\mathsf{Z}_0} \\ \mathbb{E}d(\Theta,\widehat{\Theta}) \leq D}} I(\Theta; \widehat{\Theta} \mid \mathsf{Z}_0). \tag{404}$$

Any transcript estimator with expected distortion at most D must satisfy

$$R_{\Theta|\mathsf{Z}_0}(D) \leq \sum_{t,q} c_{q,t}. \tag{405}$$

Equivalently, with the usual extended-real convention, define

$$D_{\min}(C) = \inf\{D \geq 0 : R_{\Theta|\mathsf{Z}_0}(D) \leq C\}. \tag{406}$$

Every transcript estimator obeys $\mathbb{E}d(\Theta, \widehat{\Theta}) \geq D_{\min}(C_{\text{pub}}^{[H]})$. Turning this distortion floor into an event- probability bound requires a represented distortion-to-event inequality. The rate–distortion branch is unavailable, rather than zero, when the source law, distortion, or conditional channel factorization is not represented. It is a converse and supplies no achievability claim. The computational ceiling $b(n)$ alone is not an information bound on $I_0^{(m)}$; the displayed m-dependence requires the represented training-channel factorization. If a target-dependent physical channel fails the conditional Markov factorization in Eq. (398), this capacity branch is unavailable; its dependence is not discarded as noise.

If the conditional channel also has an SDPI certificate, then each summand may be sharpened recordwise to

$$I(\Theta; Z_{q,t} \mid \textit{public past}) \leq \min\left\{c_{q,t}, \vartheta_{q,t} I(\Theta; U_{q,t} \mid \textit{public past})\right\}. \tag{407}$$

Proof. Write $\mathsf{Z}_{<q,t}$ for the public record immediately before slot (q, t). The chain rule gives

$$I(\Theta; \mathsf{Z}_H) = I(\Theta; \mathsf{Z}_0) + \sum_{t,q} I(\Theta; Z_{q,t} \mid \mathsf{Z}_{<q,t}).$$

For each admitted past h, conditional data processing and the capacity definition give

$$I(\Theta; Z_{q,t} \mid h) \leq I(U_{q,t}; Z_{q,t} \mid h) \leq c_{q,t}.$$

Integrating over h and summing proves Eq. (399). An intended action generated from the conditioned public past and fresh target-independent policy randomness has zero conditional mutual information and contributes no slot. Applying the identical argument to the declared training reveals proves Eq. (401). Subtracting the information bound in the identity $H(\Theta \mid Z) = H(\Theta) - I(\Theta; Z)$ proves Eq. (400).

Let $P_e = \Pr\{\widehat{\Theta} \neq \Theta\}$. For uniform Θ, Fano's entropy calculation is

$$H(\Theta \mid \widehat{\Theta}) \leq H_2(P_e) + P_e \log_2(N_n - 1) \leq 1 + P_e \log_2 N_n.$$

Because $\widehat{\Theta}$ is a transcript postprocessing, $I(\Theta; \widehat{\Theta}) \leq I(\Theta; \mathsf{Z}_H)$. Using $H(\Theta) = \log_2 N_n$ and rearranging gives $1 - P_e \leq (I(\Theta; \mathsf{Z}_H) + 1)/\log_2 N_n$, which proves Eq. (402). Requiring $1 - P_e \geq \alpha$ and rearranging proves Eq. (403).

Every transcript estimator induces a conditional test channel $P_{\widehat{\Theta}|\Theta,\mathsf{Z}_0}$. Conditional data processing and the deployment chain rule give

$$I(\Theta; \widehat{\Theta} \mid \mathsf{Z}_0) \leq I(\Theta; \mathsf{Z}^{\mathrm{dep}}_{1:H} \mid \mathsf{Z}_0) \leq C^{[H]}_{\mathrm{pub}}.$$

If its expected distortion is at most D, that test channel is feasible in Eq. (404); hence $R_{\Theta|\mathsf{Z}_0}(D) \leq C^{[H]}_{\mathrm{pub}}$. Its achieved distortion belongs to the set in Eq. (406) and is therefore at least the infimum of that set, proving the inverse floor.

For the SDPI branch, fix a public past h. The admitted-class closure allows the defining KL contraction to be applied to every conditional input law and its mixture:

$$\begin{aligned}
&\sum_{\theta} p(\theta \mid h)\, \mathrm{KL}\Big(P_{U|\theta,h}K \| P_{U|h}K\Big) \\
&\qquad \leq \vartheta_{q,t} \sum_{\theta} p(\theta \mid h)\, \mathrm{KL}\Big(P_{U|\theta,h} \| P_{U|h}\Big).
\end{aligned}$$

The two sides are respectively $I(\Theta; Z_{q,t} \mid h)$ and $\vartheta_{q,t} I(\Theta; U_{q,t} \mid h)$. Averaging over h and taking the minimum with the capacity bound proves Eq. (407).

□

V.3.3 Pre-Hit Sequential Information Contraction

Theorem V.3.3 (Pre-hit sequential SDPI and information contraction)**.** *Let Θ be a finite latent task index, let Z_0 be the initial public record, and fix a temporally admissible prefix ending before a represented selector is sampled. Enumerate every target-dependent public reveal in that prefix in its actual order as $Y_1, \ldots, Y_L$. The list includes observations, reward reports, execution feedback, verifier flags, and answers returned by declared oracle interfaces. Put $\mathscr{F}_0 = \sigma(\mathsf{Z}_0)$. Immediately before Y_j, let Q_j collect the represented query, action, disclosed policy randomness, and deterministic state updates performed since the preceding reveal, and set*

$$\mathscr{P}_j = \mathscr{F}_{j-1} \vee \sigma(Q_j), \qquad \mathscr{F}_j = \mathscr{P}_j \vee \sigma(Y_j).$$

Write H^{pre}_j for the represented pre-reveal history that generates $\mathscr{P}_j$. Assume

$$\Theta \perp Q_j \mid \mathscr{F}_{j-1}. \tag{408}$$

Every target-dependent coordinate in Q_j must instead occur as an earlier reveal slot.

For each j, suppose the complete record contains a clean channel input U_j, a history-conditional channel K_j, and the conditional Markov chain

$$\Theta \longrightarrow U_j \longrightarrow Y_j \quad \text{conditional on } \mathscr{P}_j. \tag{409}$$

Let $c_j(h)$ and $\vartheta_j(h)$ be, respectively, capacity and KL-SDPI certificates from Definition V.3.1, valid at every admitted pre-reveal history h for the required conditional input laws and their mixtures. Define

$$a_j(h) := I(\Theta; U_j \mid \mathscr{P}_j = h), \tag{410}$$

$$d_j := \mathbb{E}\Big[\min\{c_j(H_j^{\mathrm{pre}}), \vartheta_j(H_j^{\mathrm{pre}})a_j(H_j^{\mathrm{pre}})\}\Big]. \tag{411}$$

All information quantities in this theorem are measured in bits. Then

$$I(\Theta; \mathscr{F}_L \mid \mathsf{Z}_0) = \sum_{j=1}^{L} I(\Theta; Y_j \mid \mathscr{P}_j) \le \sum_{j=1}^{L} d_j, \tag{412}$$

$$I(\Theta; \mathscr{F}_L) \le I(\Theta; \mathsf{Z}_0) + \sum_{j=1}^{L} d_j. \tag{413}$$

In particular, adaptively selecting Q_j, U_j, or K_j from the represented past does not alter the bound; the certificate used at slot j is the one valid at the realized pre-reveal history.

Suppose, more specifically, that scalar bounds $c_j \in [0, +\infty]$ and $\vartheta_j \in [0,1]$ dominate the two history-dependent certificates at every admitted history. Let $h_j = H(\Theta \mid \mathscr{F}_j)$. Then

$$h_j \ge h_{j-1} - \min\{c_j, \vartheta_j h_{j-1}\} = \max\{h_{j-1} - c_j, (1-\vartheta_j)h_{j-1}\}. \tag{414}$$

Consequently, if $\underline{h}_0 \le H(\Theta \mid \mathsf{Z}_0)$ and

$$\underline{h}_j := \underline{h}_{j-1} - \min\{c_j, \vartheta_j \underline{h}_{j-1}\}, \tag{415}$$

then

$$H(\Theta \mid \mathscr{F}_L) \ge \underline{h}_L \ge \max\left\{\underline{h}_0 - \sum_{j=1}^{L} c_j,\ \underline{h}_0 \prod_{j=1}^{L}(1-\vartheta_j)\right\}. \tag{416}$$

These formulas permit $c_j = +\infty$ and $\vartheta_j = 1$; no small-increment hypothesis is used.

The complete prefix record retains the construction and evaluation cost of every Q_j and U_j, the selected channel and its state, the proof or certificate producing c_j and ϑ_j, every random bit, every memory update, and every returned answer. All of these are ancestors of the later selector and are charged before it according to Definition III.21.1. A presentation-admissible oracle answer occupies a channel slot with its declared provenance and cost. An oracle answer outside that presentation changes the presented family by Theorem III.33.2. Later verifier flags, terminal records, and future oracle answers are absent from $\mathscr{F}_L$.

Proof. Apply the conditional chain rule in the displayed reveal order:

$$\begin{aligned} I(\Theta; \mathscr{F}_L \mid \mathsf{Z}_0) &= \sum_{j=1}^{L} I(\Theta; Q_j, Y_j \mid \mathscr{F}_{j-1}) \\ &= \sum_{j=1}^{L} I(\Theta; Y_j \mid \mathscr{P}_j). \end{aligned}$$

The second equality is Eq. (408). Fix a pre-reveal history h. The admitted-class closure in Definition V.3.1 and the conditional Markov chain in Eq. (409) give

$$
\begin{aligned}
&I(\Theta; Y_j \mid \mathscr{P}_j = h) \le c_j(h),\\
&I(\Theta; Y_j \mid \mathscr{P}_j = h) \le \vartheta_j(h) I(\Theta; U_j \mid \mathscr{P}_j = h).
\end{aligned}
$$

Taking their minimum, averaging over the pre-reveal history, and summing proves Eq. (412). Adding $I(\Theta; \mathsf{Z}_0)$ proves Eq. (413).

Under uniform scalar certificates, the elementary mutual-information bound also gives

$$I(\Theta; U_j \mid \mathscr{P}_j) \le H(\Theta \mid \mathscr{P}_j) = H(\Theta \mid \mathscr{F}_{j-1}) = h_{j-1};$$

the middle equality again uses Eq. (408). Since

$$h_j = h_{j-1} - I(\Theta; Y_j \mid \mathscr{P}_j),$$

the capacity and SDPI bounds prove Eq. (414). The map $x \mapsto x - \min\{c_j, \vartheta_j x\}$ is nondecreasing on $[0, +\infty)$. Induction therefore proves $h_j \ge \underline{h}_j$. The same recursion separately implies $\underline{h}_j \ge \underline{h}_{j-1} - c_j$ and $\underline{h}_j \ge (1 - \vartheta_j)\underline{h}_{j-1}$. Iterating the two inequalities proves Eq. (416).

The final assertions follow from the definition of the complete pre-hit record and from the oracle-presentation separation theorem cited in the statement.

□

V.3.4 Posterior Concentration, Target Contact, and Selector Advantage

Corollary V.3.4 (Information contraction to posterior concentration, target contact, and selector advantage)**.** *Retain the record of Theorem V.3.3, and identify $\mathscr{F}_L$ with a complete pre-hit record $\mathcal{H}_t$. Let $K_t(\cdot \mid \mathcal{H}_t)$ be a temporally admissible next-test sampler as in Definition III.21.2; its fresh random bits are conditionally independent of Θ given $\mathcal{H}_t$. Let $V_\theta(c) \in \{0, 1\}$ be the represented verification relation for target index θ, and let C_{t+1} be the candidate sampled from K_t. Put*

$$p_t := \Pr\{\tau > t,\ V_\Theta(C_{t+1}) = 1\}. \tag{417}$$

The candidate carrier is common to all target indices, or is padded by a represented common carrier, so that $V_\theta(c)$ is measurable for every cross-pair (θ, c) appearing under the decoupled law below.

Let $\mathbb{P}_t$ be the joint law of $(\Theta, \mathcal{H}_t, C_{t+1})$. Define its target-decoupled reference by

$$\mathbb{Q}_t(d\theta, dh, dc) := P_\Theta(d\theta) P_{\mathcal{H}_t}(dh) K_t(dc \mid h), \tag{418}$$

and let

$$\beta_t := \mathbb{Q}_t\{\tau > t,\ V_\Theta(C_{t+1}) = 1\}. \tag{419}$$

If

$$J_t^{\uparrow} := I(\Theta; \mathsf{Z}_0) + \sum_{j=1}^{L} d_j, \tag{420}$$

then, with the lower-semicontinuous endpoint convention for D_{Ber} from Eq. (295),

$$D_{\mathrm{Ber}}(p_t \| \beta_t) \le (\log 2) J_t^{\uparrow}, \tag{421}$$

$$(p_t - \beta_t)_+ \le \sqrt{\frac{\log 2}{2} J_t^{\uparrow}}. \tag{422}$$

The same information budget controls the posterior:

$$\mathbb{E}\left[\left\| P_{\Theta|\mathcal{H}_t} - P_\Theta \right\|_{\mathrm{TV}}\right] \le \sqrt{\frac{\log 2}{2} J_t^{\uparrow}}. \tag{423}$$

Suppose Θ is uniform on $N \ge 2$ indices and the target sectors have represented overlap at most Λ_t, meaning

$$\sup_{h,c} \sum_{\theta=1}^{N} V_\theta(c) \le \Lambda_t. \tag{424}$$

Then $\beta_t \le \Lambda_t / N$, and hence

$$p_t \le \min\left\{1, \frac{\Lambda_t}{N} + \sqrt{\frac{\log 2}{2} J_t^{\uparrow}}\right\}. \tag{425}$$

For disjoint target sectors, $\Lambda_t = 1$. For any declared neutral baseline ν_t,

$$\left(\mathrm{Adv}_t(K_t; \nu_t)\right)_+ \le p_t, \tag{426}$$

so Eq. (425) *is also a prospective selector bound.*

There is an optional chi-square refinement requiring a separate represented certificate. If the complete record contains

$$\chi^2(\mathbb{P}_t \| \mathbb{Q}_t) := \int \left(\frac{d\mathbb{P}_t}{d\mathbb{Q}_t} - 1\right)^2 d\mathbb{Q}_t \le \Xi_t, \tag{427}$$

then

$$p_t \le \min\left\{1, \beta_t + \sqrt{\beta_t(1-\beta_t)\Xi_t}\right\}. \tag{428}$$

The derivation and provenance of Ξ_t are part of the same charged pre-hit record; no KL-to-chi-square conversion is inferred.

Proof. The sampler is a postprocessing of $\mathcal{H}_t$ using conditionally target-independent fresh randomness. Therefore

$$\mathrm{KL}(\mathbb{P}_t \| \mathbb{Q}_t) = (\log 2) I(\Theta; \mathcal{H}_t) \le (\log 2) J_t^{\uparrow}. \tag{429}$$

Applying the measurable map $(\theta, h, c) \mapsto \mathbf{1}_{\{\tau(h) > t\}} V_\theta(c)$ and using data processing for relative entropy proves Eq. (421). Pinsker's inequality in nats proves Eq. (422). Applying conditional Pinsker directly to $P_{\Theta|\mathcal{H}_t}$ and P_Θ, followed by Jensen's inequality, proves Eq. (423).

Under the decoupled law, Θ is independent of the history and the sampled candidate. Hence

$$\begin{aligned}\beta_t &= \mathbb{E}_{\mathbb{Q}_t}\left[\mathbf{1}_{\{\tau > t\}} \frac{1}{N} \sum_{\theta=1}^{N} V_\theta(C_{t+1})\right] \\ &\le \frac{\Lambda_t}{N}.\end{aligned}$$

Combining this estimate with Eq. (422) proves Eq. (425). The neutral selector has nonnegative next-hit mass. The definition Eq. (150) therefore gives $(\mathrm{Adv}_t)_+ \le p_t$, proving Eq. (426).

For the chi-square branch, write $L_t = d\mathbb{P}_t/d\mathbb{Q}_t$ and let E_t be the next-contact event. Since $\mathbb{E}_{\mathbb{Q}_t}(L_t - 1) = 0$, Cauchy–Schwarz gives

$$\begin{aligned} p_t - \beta_t &= \mathbb{E}_{\mathbb{Q}_t}[(L_t - 1)(\mathbf{1}_{E_t} - \beta_t)] \\ &\le \sqrt{\chi^2(\mathbb{P}_t \| \mathbb{Q}_t)}\sqrt{\beta_t(1-\beta_t)}, \end{aligned}$$

which proves Eq. (428).

□

Corollary V.3.5 (Cumulative first contact and full-transcript identification)**.** *Let Θ be uniform on $N \ge 2$ indices. For every $t = 0, \dots, H-1$, retain a complete prefix record satisfying Corollary V.3.4, with the quantities $J_t^{\uparrow}$ and Λ_t defined in Eqs.* (420) *and* (424)*. Then*

$$\Pr\{\tau \le H\} \le \min\left\{1, \sum_{t=0}^{H-1}\left(\frac{\Lambda_t}{N} + \sqrt{\frac{\log 2}{2} J_t^{\uparrow}}\right)\right\}. \tag{430}$$

If, in addition, the complete terminal record contains a represented decoder that identifies Θ on every verified hit, let $\mathsf{Z}_H^{\mathrm{full}}$ denote that terminal transcript. Suppose every target-dependent coordinate after Z_0, including terminal verification, decoder-selection, feedback, and oracle coordinates, is exposed as an ordered channel slot satisfying Theorem V.3.2. If $\mathcal{R}_H^{\mathrm{full}}$ is the resulting slot index set, with represented capacity certificates $c_{q,r}$, put

$$J_H^{\mathrm{full}} := I(\Theta; \mathsf{Z}_0) + \sum_{(q,r)\in\mathcal{R}_H^{\mathrm{full}}} c_{q,r}. \tag{431}$$

No terminal target-dependent coordinate is omitted from this sum; target-independent postprocessing and fresh target-independent randomness contribute zero conditional mutual information. Fano's inequality then gives the independent full-transcript bound

$$\Pr\{\tau \le H\} \le \min\left\{1, \frac{J_H^{\mathrm{full}} + 1}{\log_2 N}\right\}. \tag{432}$$

The recordwise minimum of Eqs. (430) *and* (432) *may be taken before saturation. Saturation then takes the supremum over complete records at the charged enlarged ceiling; each record retains its own channel, oracle, overlap, information, and decoder certificates.*

Proof. The first-hit events are disjoint, so summing their probabilities and using Eq. (425) proves Eq. (430). Under the identification gate, map a verified hit to its target index and assign an arbitrary index on failure. The decoder succeeds whenever a hit occurs. Its mutual information is at most that of $\mathsf{Z}_H^{\mathrm{full}}$. Applying Theorem V.3.2 to every slot in $\mathcal{R}_H^{\mathrm{full}}$ gives

$$I(\Theta; \mathsf{Z}_H^{\mathrm{full}}) \le J_H^{\mathrm{full}}.$$

The Fano calculation in Theorem V.3.2 proves Eq. (432). Every conversion was performed on one complete record, so taking its minimum before the affordable supremum has the order stated in the corollary.

□

Figure 76 shows where sample size and deployment horizon enter the information converse.

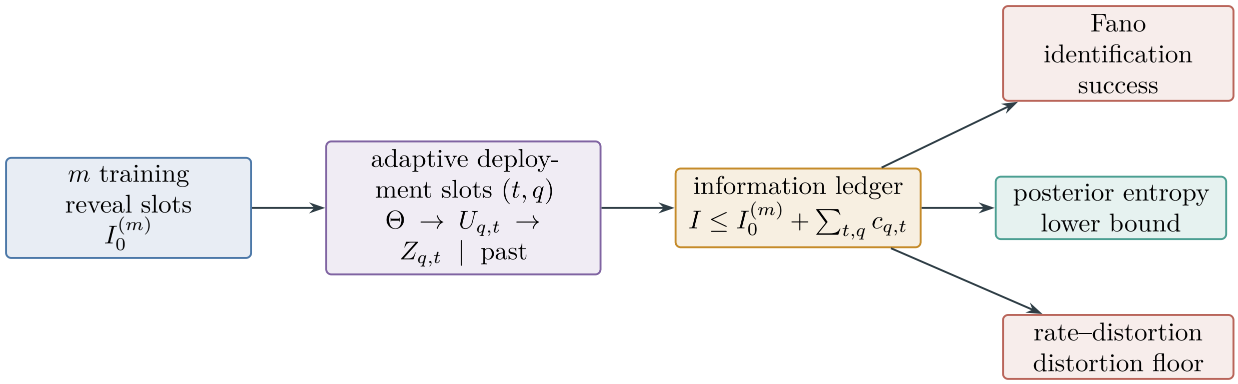

Figure 76: Adaptive information is charged reveal by reveal. Training and deployment budgets give $I(\Theta; \mathsf{Z}_H) \le I_0^{(m)} + \sum_{t,q} c_{q,t}$; posterior entropy, Fano identification, and rate–distortion are separate consequences of that same budget.

V.3.5 Explicit Channel-Information Budgets

Corollary V.3.6 (Explicit channel-information budgets)**.** *The following values are in bits per represented channel use.*

(i) *If $|\mathsf{V}| = d \ge 2$ and every admitted input law satisfies $0 \le h_- \le \log_2 d$ and $H(V \mid U) \ge h_-$, then*

$$C_{\mathcal{Q}}(K) \le \log_2 d - h_-. \tag{433}$$

(ii) *A q-ary symmetric channel that, conditional on an error, chooses each of the other $q-1$ symbols uniformly, with crossover probability $0 \le \epsilon \le (q-1)/q$, has*

$$C(K) = \log_2 q - H_2(\epsilon) - \epsilon \log_2(q-1). \tag{434}$$

An additive channel on a finite group G with independent noise N has $C(K) = \log_2 |G| - H(N)$. A q-ary erasure channel of erasure rate ϵ has $C(K) = (1-\epsilon)\log_2 q$.

(iii) *For an additive scalar Gaussian reporting channel with average input power at most P and independent noise variance $\sigma^2 > 0$,*

$$C_P(K) \le \frac{1}{2}\log_2(1 + P/\sigma^2). \tag{435}$$

Without the power constraint the capacity entry is $+\infty$.

Marginal memoryless formulas are not applied to correlated channel noise. Such a record uses conditional history-state capacities or a represented joint block-capacity certificate. The finite-alphabet estimates of Theorem IV.17.8 may be inserted after their continuity error is added.

Proof. For the first statement, $I(U; V) = H(V) - H(V \mid U) \le \log_2 d - h_-$.

For the q-ary symmetric channel, every row has entropy

$$H(V \mid U = u) = H_2(\epsilon) + \epsilon \log_2(q-1).$$

Consequently every input law has $I(U; V) \le \log_2 q - H_2(\epsilon) - \epsilon \log_2(q-1)$. Uniform input makes the output uniform and attains that upper bound. For the finite-group channel $V = U + N$, independence gives $H(V \mid U) = H(N)$, while $H(V) \le \log_2 |G|$; uniform U makes V uniform and proves equality. For the erasure channel, conditioning on the erasure indicator gives $I(U;V) = (1-\epsilon)H(U)$, maximized by uniform input.

For the scalar Gaussian channel $Y = X + N$, independence and the power bound give $\mathrm{Var}(Y) \le P + \sigma^2$. The maximum-entropy inequality therefore yields, in bits,

$$I(X;Y) = h(Y) - h(N) \le \frac{1}{2}\log_2 \frac{2\pi e(P+\sigma^2)}{2\pi e\sigma^2} = \frac{1}{2}\log_2(1 + P/\sigma^2).$$

For the ordinary unrestricted power-constrained input class, a centered Gaussian input attains equality; for a restricted represented input class, the displayed upper bound remains valid.

□

V.3.6 Belief entropy and target-gate uncertainty

Theorem V.3.7 (Belief-entropy balance and target-gate uncertainty)**.** *Assume finite latent alphabets, and let Z^{pub}_{t+1} collect the public observation, reward report, and execution feedback revealed between times t and $t+1$, with no additional public reveal omitted. Assume exactly*

$$\mathcal{F}^{\mathrm{obs}}_{t+1} = \mathcal{F}^{\mathrm{obs}}_t \vee \sigma(A_t, Z^{\mathrm{pub}}_{t+1}).$$

Assume the policy draw uses only the public pre-action record and target-independent policy randomness, so that $A_t \perp X_t \mid \mathcal{F}^{\mathrm{obs}}_t$. This includes the deterministic case in which the relevant policy seed is already in $\mathcal{F}^{\mathrm{obs}}_t$. Then

$$H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_{t+1}) = H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) - I(X_{t+1}; Z^{\mathrm{pub}}_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t). \tag{436}$$

The pre-observation uncertainty obeys

$$\begin{aligned} H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) \le{}& H(X_t \mid \mathcal{F}^{\mathrm{obs}}_t) + H(A^\circ_t, E_t \mid X_t, A_t, \mathcal{F}^{\mathrm{obs}}_t) \\ &+ H(X_{t+1} \mid X_t, A^\circ_t, E_t, \mathcal{F}^{\mathrm{obs}}_t, A_t). \end{aligned} \tag{437}$$

Write the last two terms as $h_{A,t}$ and $h_{P,t}$, respectively. Thus a represented information floor i^-_{t+1} for the mutual-information term gives the quantitative bounds

$$H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_{t+1}) \le H(X_t \mid \mathcal{F}^{\mathrm{obs}}_t) + h_{A,t} + h_{P,t} - i^-_{t+1}, \tag{438}$$

$$H(X_H \mid \mathcal{F}^{\mathrm{obs}}_H) \le \min\left\{\log_2|X|, H(X_0 \mid \mathcal{F}^{\mathrm{obs}}_0) + \sum_{t=0}^{H-1}(h_{A,t} + h_{P,t} - i^-_{t+1})\right\}. \tag{439}$$

The scalar right side may be clipped below at zero. In contrast, a channel-capacity upper bound c_{t+1} gives only

$$H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_{t+1}) \ge H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) - c_{t+1}. \tag{440}$$

An entropy upper bound cannot be obtained from a capacity upper bound alone. Here c_{t+1} must directly certify the displayed conditional mutual information, for example by summing the admitted conditional Markov-channel slots. Physical output entropy alone is not such a certificate.

For either augmented winner indicator $J_t \in \{\mathbf{1}_{\{W_t=T\}}, \mathbf{1}_{\{W_t=F\}}\}$, put $q_t = \Pr\{J_t = 1 \mid \mathcal{F}^{\mathrm{obs}}_t\}$ and let $p^{\mathrm{MAP}}_t = \mathbb{E}\min\{q_t, 1-q_t\}$ be the Bayes gate error. With binary entropy measured in bits,

$$H_2^{-1}\Big(H(J_t \mid \mathcal{F}^{\mathrm{obs}}_t)\Big) \le p^{\mathrm{MAP}}_t \le \frac{1}{2}H(J_t \mid \mathcal{F}^{\mathrm{obs}}_t), \tag{441}$$

where H_2^{-1} is the inverse on $[0,1/2]$. A represented learned gate adds its approximation and statistical errors to this analytic Bayes error through the deployment comparison of Definition V.8.1.

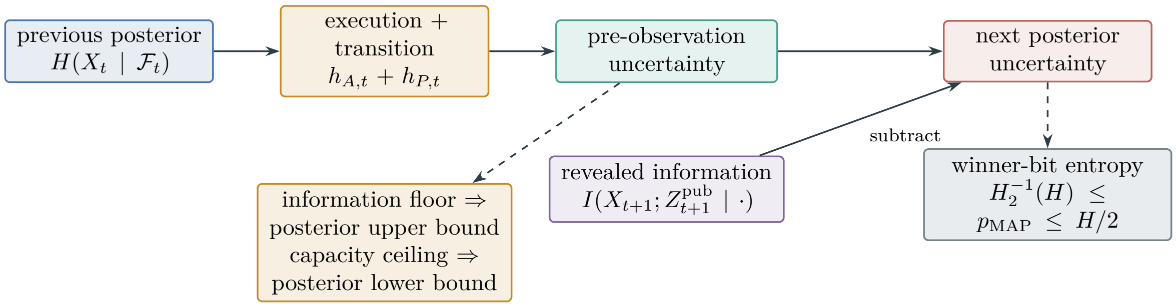

Figure 77: Belief entropy has an exact update: execution and transition uncertainty form the pre-observation reservoir, and public information removes mutual information from it. An information lower bound controls posterior uncertainty from above, whereas a capacity upper bound controls it only from below.

Proof. By the stated filtration identity,

$$I(X_{t+1}; Z^{\mathrm{pub}}_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) = H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) \\ - H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_{t+1}),$$

which rearranges to Eq. (436). For the injection bound, monotonicity under adjoining variables and the chain rule give

$$\begin{aligned} H(X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) &\le H(X_t, A^\circ_t, E_t, X_{t+1} \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) \\ &= H(X_t \mid \mathcal{F}^{\mathrm{obs}}_t) + H(A^\circ_t, E_t \mid X_t, A_t, \mathcal{F}^{\mathrm{obs}}_t) \\ &\quad + H(X_{t+1} \mid X_t, A^\circ_t, E_t, \mathcal{F}^{\mathrm{obs}}_t, A_t). \end{aligned}$$

In the chain-rule expansion, the policy conditional-independence assumption gives $H(X_t \mid \mathcal{F}^{\mathrm{obs}}_t, A_t) = H(X_t \mid \mathcal{F}^{\mathrm{obs}}_t)$; the last term is the entropy of the declared transition row by the transition Markov property. This proves Eq. (437). Substitution of i^-_{t+1} and induction prove Eqs. (438) and (439). Substitution of the upper information certificate instead proves Eq. (440).

Put $e(q) = \min\{q, 1-q\}$. Pointwise symmetry and the elementary chord bound give $H_2(q) = H_2(e(q)) \ge 2e(q)$. Taking expectations yields $p^{\mathrm{MAP}}_t \le \frac{1}{2} H(J_t \mid \mathcal{F}^{\mathrm{obs}}_t)$. Concavity gives

$$H(J_t \mid \mathcal{F}^{\mathrm{obs}}_t) = \mathbb{E} H_2(e(q_t)) \le H_2(\mathbb{E} e(q_t)) = H_2(p^{\mathrm{MAP}}_t).$$

Since H_2 is increasing on $[0, 1/2]$, inversion proves the lower bound in Eq. (441). □

Figure 77 emphasizes the opposite directions of an information floor and a capacity ceiling.

V.4 Typed Entropy, Affordable Success, and Blackwell Order

V.4.1 Typed entropy and affordable success

Definition V.4.1 (Typed entropy outputs and no-substitution rule)**.** One complete controlled record stores the entropy vector

$$\mathbf{H}_{\mathrm{ctl}} = (\mathsf{H}_{\mathrm{init}}, \mathsf{H}_{\mathrm{ch}}, \mathsf{H}_{\mathrm{env}}, \mathsf{H}_R, C_{\mathrm{pub}}, \mathrm{KL}_{\mathrm{path}}, \mathrm{Ent}_\mu, \mathrm{Act}_{\mathsf{Caus}}, \mathrm{KL}_{\mathrm{PB}}, \log N_{\mathrm{out}}, H_\pi). \tag{442}$$

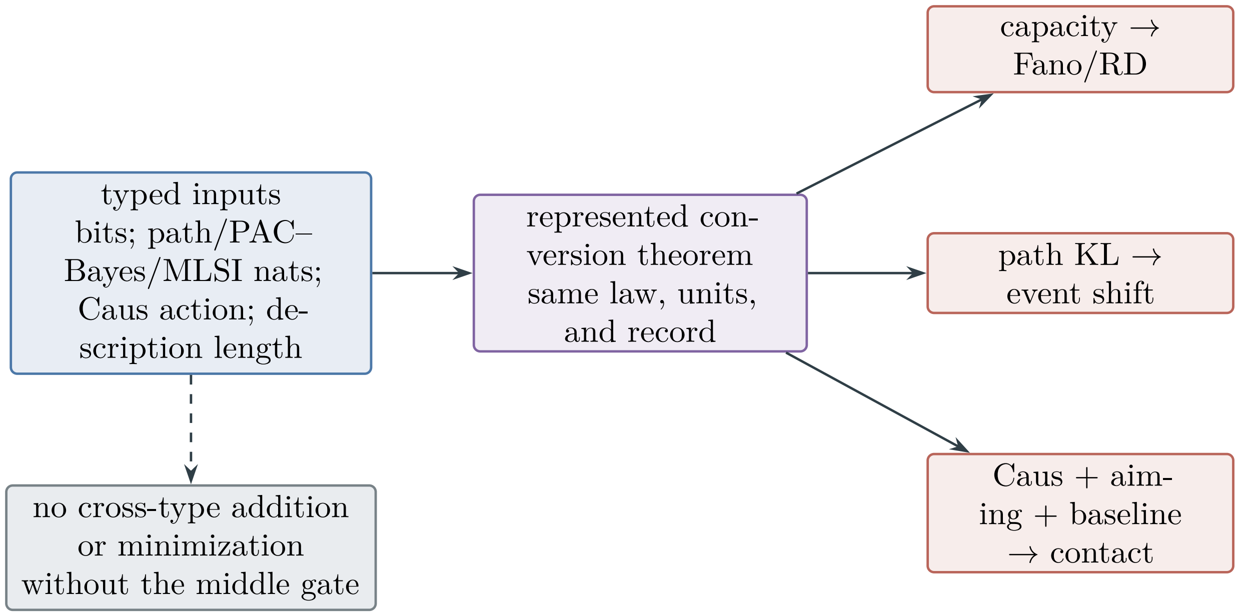

Figure 78: The controlled entropy vector is typed, not additive. Shannon bits, path-law and statistical KL, MLSI entropy, Caus action, and description length reach a common output only through a represented conversion theorem. In particular, Caus action reaches target contact only with aiming and a baseline.

The initial-retained-state, physical-channel, stochastic-transition, clean-reward, public-capacity, policy, and $\log_2 N_{\text{out}}$ entries are in bits. Path-law KL, PAC–Bayes KL, and MLSI relative entropy use nats unless an explicit conversion is applied. Caus action has edge-action units; it equals a thermodynamic entropy- production functional only under the logarithmic-mean flux hypotheses of the corresponding Caus theorem. It is not called entropy production merely because it is nonnegative. Conversion from nats to bits multiplies by $\log_2 e$.

Entries are compared or added only through a theorem with a common output. In particular:

(i) MLSI density decay does not bound observation-channel capacity without a represented channel/semigroup comparison;

(ii) channel or transition entropy does not bound reward, action, observation, or target distortion without a represented inverse, coupling, metric, or calibration;

(iii) Caus action enters target contact only through

$$\text{Hit}_{\text{contact}} \leq B^{\text{contact}} + K\sqrt{\text{Act}_{\mathsf{Caus}}\,\text{Aim}_{\mathsf{Caus}}} + \delta, \tag{443}$$

after the represented aiming-to-contact conversion; and

(iv) a Lift transports this numerical bound to another carrier only through a represented form, intertwining, clocked-defect, or resolvent comparison.

Missing aiming or contact converters give the probability-neutral value one; missing upper-error converters give $+\infty$, and missing lower-rate certificates give zero.

Figure 78 summarizes the no-substitution rule for the controlled entropy vector.

Definition V.4.2 (Affordable multichannel success profile)**.** Let $\mathscr{P}^{\text{ch}}_{n,b,m,H}$ be a represented class of complete POMDP or channelized-control records of cost at most $b(n)$, sample size m, and horizon

H. Fix a represented family on which Θ is uniform with size $N_n \geq 2$. For a target-identification task, define

$$\mathrm{Succ}_{\mathrm{ch}}^{\mathrm{sat}}(n, b(n), m, H) = \sup_{\mathcal{R} \in \mathscr{P}^{\mathrm{ch}}_{n,b,m,H}} \Pr_{\mathcal{R}}\{\widehat{\Theta} = \Theta\}. \tag{444}$$

For each record, let $U_{\mathrm{sel}}(\mathcal{R})$ be its prospective-selector probability bound or a qualified-source bound already converted to the same success probability by a represented same-record contact theorem; its value is one when that converter is absent. Let $U_{\mathrm{path}}(\mathcal{R})$ be the minimum of its same-event model, path-TV/KL, Caus-contact, capacity, and Lift bounds after conversion to the same success probability. In particular, every path-law candidate has the form

$$\min\{1, U_{\mathrm{ref}}(\mathcal{R}) + \varepsilon_{\mathrm{path}}(\mathcal{R})\},$$

where U_{ref} bounds the reference record's same target-success event and $\varepsilon_{\mathrm{path}}$ is the represented TV or Pinsker shift. Model, capacity, Caus, and Lift candidates are admitted only after their own baseline or contact term is included. A missing baseline or converter gives the value one. Finally, put

$$U_{\mathrm{info}}(\mathcal{R}) = \min\left\{1, \frac{I_0^{(m)}(\mathcal{R}) + C_{\mathrm{pub}}^{[H]}(\mathcal{R}) + 1}{\log_2 N_n}\right\}.$$

Recordwise unavailable probability bounds equal one. Then

$$\mathrm{Succ}_{\mathrm{ch}}^{\mathrm{sat}} \leq \min\left\{\sup_{\mathcal{R}} U_{\mathrm{sel}}(\mathcal{R}), \sup_{\mathcal{R}} U_{\mathrm{path}}(\mathcal{R}), \sup_{\mathcal{R}} U_{\mathrm{info}}(\mathcal{R})\right\}. \tag{445}$$

The selector term uses the complete target-aligned pre-hit ledger at $\Psi_{\mathrm{ctl}}(b(n), H, n)$; the information term is a complexity-free converse. For target hitting rather than explicit identification, the information term is eligible only when the represented hit/output event identifies Θ, for example through disjoint verified target sectors. The output-to-Θ decoder, sector verification, pre-hit or terminal admissibility, and their evaluation costs belong to the complete record. If uniformity, N_n, the decoder, either information budget, or the identification gate is unavailable, the information numerator is assigned $+\infty$ and the clipped probability candidate is one. For polynomial capacity substitute $b_C(n) = n^C$ everywhere. For a fixed clean model and channel tuple $\mathbf{K}$, write $\mathscr{P}^{\mathrm{ch}}_{n,b,m,H}(\mathbf{K})$ for the corresponding restricted complete-record class and $\mathrm{Succ}_{\mathrm{ch}}^{\mathrm{sat}}(n, b(n), m, H; \mathbf{K})$ for Eq. (444) with that class. This semicolon notation suppresses no channel, simulation, or garbling cost.

Justification of Eq. (445). Fix one complete record and write its success probability as $p_{\mathcal{R}}$. Prospective-selector accountability, including the complete channel, filter, policy, and deployment ancestry, gives $p_{\mathcal{R}} \leq U_{\mathrm{sel}}(\mathcal{R})$. Every admitted path candidate first applies the TV, KL, Caus, capacity, or Lift comparison and then adds its represented upper bound for the same reference event, so $p_{\mathcal{R}} \leq U_{\mathrm{path}}(\mathcal{R})$. Under the represented identification gate, Eq. (402) gives $p_{\mathcal{R}} \leq U_{\mathrm{info}}(\mathcal{R})$; without it the value one makes the inequality automatic. Therefore

$$p_{\mathcal{R}} \leq \min\{U_{\mathrm{sel}}(\mathcal{R}), U_{\mathrm{path}}(\mathcal{R}), U_{\mathrm{info}}(\mathcal{R})\}.$$

Taking the affordable supremum and using $\sup_{\mathcal{R}} \min_j U_j(\mathcal{R}) \leq \min_j \sup_{\mathcal{R}} U_j(\mathcal{R})$ proves the claim. □

Figure 79 shows why the bound is formed recordwise before saturation.

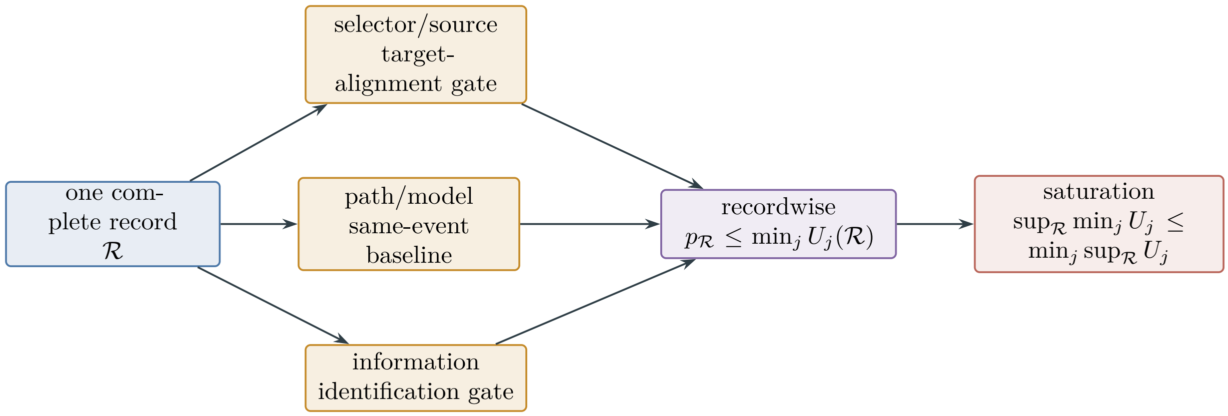

Figure 79: Affordable success is bounded recordwise before saturation. Selector, path/model, and information candidates enter only after their target-alignment, same-event-baseline, and identification gates are satisfied; unavailable probability candidates equal one.

V.4.2 Blackwell order for represented channels

Theorem V.4.3 (Multichannel degradation is a Blackwell partial order)**.** *Let two complete channel records have the same clean input, and include their initial and training public records in the following comparison. Let every public channel of the second be a represented causal garbling of the corresponding channel of the first:*

$$K^{(2)}_{q,t} = K^{(1)}_{q,t} G_{q,t}$$

in the declared history order. Suppose the garbling, its random bits, state, and precision enlarge the ceiling by the class-preserving map $b^\sharp = \psi_G(b, n, m, H)$. For an actuator channel, additionally assume that the full closed-loop factorization permits this garbling before the physical transition, so that the second executed kernel is a represented causal simulation from the first record. Equivalently, the entire second conditional path kernel, not only its reported actuator symbol, must be a causal postprocessing of the first. Then

$$\mathrm{Succ}^{\mathrm{sat}}_{\mathrm{ch}}(n, b^\sharp(n), m, H; \mathbf{K}^{(1)}) \ge \mathrm{Succ}^{\mathrm{sat}}_{\mathrm{ch}}(n, b(n), m, H; \mathbf{K}^{(2)}). \tag{446}$$

For the same compatible input-law class $\mathcal{Q}$ at every history, conditional capacity satisfies $C^{[H]}_{\mathrm{pub},2} \le C^{[H]}_{\mathrm{pub},1}$. More precisely, for each policy π_2 in the second record, let $\pi_{1\to 2}$ be its matched online simulation in the first record, and denote the resulting transcripts by Z^{2,π_2}_H and $\mathsf{Z}^{1,\pi_{1\to 2}}_H$. Then

$$I(\Theta; \mathsf{Z}^{2,\pi_2}_H) \le I(\Theta; \mathsf{Z}^{1,\pi_{1\to 2}}_H).$$

This inequality is not asserted for two independently chosen policies. For rate–distortion, retain the common degraded initial/training side information $\mathsf{Z}^{(2)}_0$ in both simulated experiments and use the same conditional source and distortion. The common inverse then satisfies

$$D_{\min}(C^{[H]}_{\mathrm{pub},2}) \ge D_{\min}(C^{[H]}_{\mathrm{pub},1}).$$

If one instead conditions each experiment on its different native $\mathsf{Z}^{(i)}_0$, no comparison of the two conditional rate–distortion functions is made without a separate side-information ordering theorem. No scalar "total-noise" threshold is declared for incomparable channel tuples. Downward-closed multichannel computational regions are defined only along a represented garbling chain.

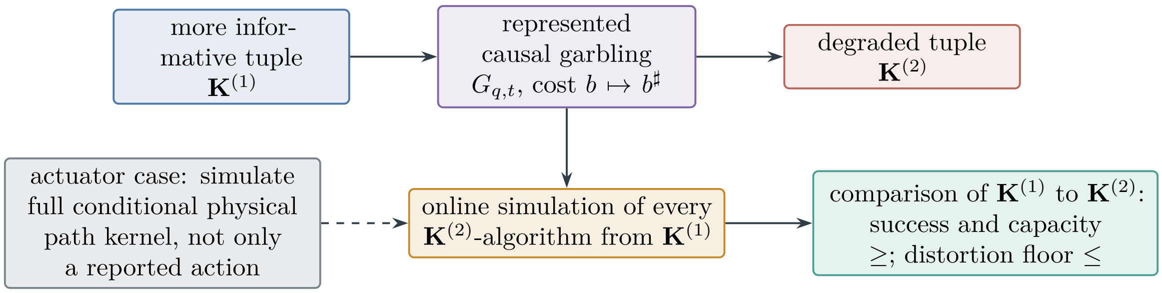

Figure 80: Represented causal garbling defines a partial, not scalar, degradation order. Online simulation proves success monotonicity and data processing proves information monotonicity. For actuator noise, the garbling must simulate the complete conditional physical path kernel before transition.

Proof. Fix an algorithm admitted for the second record. From the first initial and training records, apply the represented initial/training garblings. At every later reveal, condition on the already simulated second-record history, apply $G_{q,t}$ to the first-record output, and feed the result to the same policy, memory update, and estimator. Induction over reveal slots shows that the simulated transcript has exactly the second-record law. The actuator hypothesis ensures the induction includes the physical transition rather than merely an action report. Complete-cost closure charges every garbling and gives $b^\sharp$; taking the supremum over second-record algorithms proves Eq. (446).

Under the matched simulation, the second transcript is a causal postprocessing of the first transcript, including its initial and training coordinates. Data processing therefore gives the displayed matched-policy information inequality. Applying the same argument to each admitted history and compatible input law gives $C_2 \leq C_1$ slotwise and after summation. After both experiments retain the common degraded side information $\mathsf{Z}_0^{(2)}$, the conditional source, distortion, and rate–distortion function are literally the same. Its inverse floor is nonincreasing in the capacity argument; hence the smaller second deployment budget has the larger distortion floor. Distinct native side information would change the function itself, which is why the theorem makes no such unqualified comparison.

□

Figure 80 depicts the represented degradation order and its actuator guard.

V.4.3 Pre-hit decision value and sufficiency

Definition V.4.4 (Represented pre-hit target-decision value)**.** Fix a presented task, a time $0 \leq t < H$, its survival event $\{\tau > t\}$, and a complete pre-hit record $\mathcal{H}_t$. Let Θ denote the target parameter and let $\mathcal{A}_t(\mathcal{H}_t)$ be the represented carrier of decisions available after that record. A bounded target-decision utility is a common measurable function

$$u_t : \{(\theta, a) : a \in \mathcal{A}_t\} \longrightarrow [0, 1].$$

The utility is an analytic evaluator of a decision. It is not supplied to a selector as pre-hit data. Any represented use of a utility value in forming the next decision belongs to the selector's pre-hit ancestry and is charged under Definition III.21.1.

For a represented pre-hit experiment E, let $\mathscr{K}_t(\mathsf{E}, B)$ contain the family-uniform selectors whose complete prefix, sampler, random-bit, precision, storage, and transition cost is at most B. Define

$$\mathrm{DVal}^{\mathrm{sat}}_{u,t}(\mathsf{E}; B) = \sup_{K_t \in \mathscr{K}_t(\mathsf{E},B)} \mathbb{E}_{\mathsf{E}}\left[\mathbf{1}_{\{\tau>t\}} \int u_t(\Theta, a) K_t(da \mid \mathcal{H}_t)\right]. \tag{447}$$

For a declared represented baseline ν_t on the same carrier, put

$$\mathrm{DAdv}^{\mathrm{sat}}_{u,t}(\mathsf{E}, \nu_t; B) = \sup_{K_t \in \mathscr{K}_t(\mathsf{E},B)} \left(\mathbb{E}_{\mathsf{E}}\left[\mathbf{1}_{\{\tau>t\}} \int u_t(\Theta, a)(K_t - \nu_t)(da \mid \mathcal{H}_t)\right]\right)_+. \tag{448}$$

For the next-test utility $u_t(\Theta, a) = V_I(a)$, Eq. (448) is the selector advantage of Eq. (150); taking the supremum over times gives the corresponding prospective selector profile. A terminal identification utility uses $u_t(\Theta, a) = \mathbf{1}_{\{a=\Theta\}}$.

Theorem V.4.5 (Represented pre-hit Blackwell domination)**.** *Let E_1 and E_2 be two complete pre-hit experiments for the same presented task, target Θ, verifier, and target-decision utility u_t. Include the initial record, training record, and the complete prefix through time t. Suppose that E_2 is generated from E_1 by represented causal kernels $G_0, \dots, G_t$: after each E_1-reveal and before the next decision, G_j produces the corresponding E_2-reveal from the already generated E_2-history. The kernels and their random bits have no later verifier flag, terminal record, target answer, or later oracle answer as an ancestor. The resulting online simulation reproduces the joint law of*

$$(\Theta, \mathbf{1}_{\{\tau>t\}}, \mathcal{H}_t^{(2)})$$

in the second experiment. The decision carriers are common; more generally, a represented action map $\iota_t(\mathcal{H}_t^{(1)}, \mathcal{H}_t^{(2)}, a_2)$ may be used when it preserves utility almost surely:

$$u_t\Big(\Theta, \iota_t(\mathcal{H}_t^{(1)}, \mathcal{H}_t^{(2)}, a_2)\Big) = u_t(\Theta, a_2).$$

For an actuator decision, the simulation must reproduce the complete conditional physical transition, as in Theorem V.4.3.

Assume that the complete costs of the causal garbling, action map, random bits, state, storage, precision, and evaluation enlarge B by the class-preserving map

$$B^\sharp = \psi_{\mathrm{BW}}(B, n, m, H).$$

Then

$$\mathrm{DVal}^{\mathrm{sat}}_{u,t}(\mathsf{E}_2; B) \le \mathrm{DVal}^{\mathrm{sat}}_{u,t}(\mathsf{E}_1; B^\sharp). \tag{449}$$

More precisely, if $K_t^{(2)}$ is a selector in the second experiment, its simulated selector in the first has kernel

$$K_t^{(1\to2)}(da_1 \mid h_1) = \int G_{\le t}(dh_2 \mid h_1) \int K_t^{(2)}(da_2 \mid h_2) \mathbf{1}_{\{\iota_t(h_1,h_2,a_2) \in da_1\}}, \tag{450}$$

with $\iota_t(a_2) = a_2$ on a common carrier, and it satisfies the exact identity

$$\mathbb{E}_{\mathsf{E}_1}\left[\mathbf{1}_{\{\tau>t\}} \int u_t(\Theta, a) K_t^{(1\to2)}(da \mid \mathcal{H}_t^{(1)})\right] = \mathbb{E}_{\mathsf{E}_2}\left[\mathbf{1}_{\{\tau>t\}} \int u_t(\Theta, a) K_t^{(2)}(da \mid \mathcal{H}_t^{(2)})\right]. \tag{451}$$

If $\nu_t^{(2)}$ is the declared baseline in E_2, define its represented pullback $\nu_t^{(1\to2)}$ by the right-hand side of Eq. (450) *with $K_t^{(2)}$ replaced by $\nu_t^{(2)}$. When the baseline's declared target-neutrality certificate is preserved by this simulation,*

$$\mathrm{DAdv}^{\mathrm{sat}}_{u,t}(\mathsf{E}_2, \nu_t^{(2)}; B) \le \mathrm{DAdv}^{\mathrm{sat}}_{u,t}(\mathsf{E}_1, \nu_t^{(1\to2)}; B^\sharp). \tag{452}$$

For $u_t = V_I$, this is Blackwell monotonicity of the represented pre-hit selector profile, with the complete garbling cost charged before the decision.

The comparison concerns decision value, not provenance metadata. A represented statistic, constructor label, or backward-cut classification establishes a garbling only in the direction from the fuller record to that derived record. It therefore bounds the decision value of selectors that factor through the derived record. An upper bound for the fuller record follows only from a reverse represented garbling, as in Corollary V.4.6, or from a separate numerical bound on its target-decision value.

Proof. Fix a complete second-experiment record of cost at most B and its selector $K_t^{(2)}$. Generate its initial, training, and online prefix by applying $G_0, \dots, G_t$ in causal order to the first-experiment record. All random bits used by G_j are generated before the decision following slot j. Hence the simulated prefix and the kernel in Eq. (450) factor through pref_t and satisfy the four temporal-source conditions of Definition III.21.1. The cost hypothesis charges the complete simulation and places the resulting selector below $B^\sharp$.

By the joint-law hypothesis, the simulated triple $(\Theta, \mathbf{1}_{\{\tau>t\}}, \mathcal{H}_t^{(2)})$ has its native second-experiment law. Conditional on the simulated history, draw a_2 from $K_t^{(2)}$, and apply ι_t. Utility preservation and the tower property give Eq. (451). Taking the supremum over all cost-B second-experiment records proves Eq. (449).

Apply the same construction to $\nu_t^{(2)}$. The selector and baseline utility expectations are each preserved exactly, so their difference is preserved exactly. Taking the positive part and then the saturated supremum proves Eq. (452). The final paragraph follows from the direction of Eq. (449): a statistic computed from a record is its degradation, whereas an upper bound for the record requires a simulation of the record from the statistic or a direct bound on the record itself.

□

Corollary V.4.6 (Represented pre-hit sufficient statistic)**.** *Let $\mathcal{H}_t$ be a complete pre-hit record and let $S_t = s_t(\mathcal{H}_t)$ be a represented statistic computed before the time-$t+1$ decision. Suppose that:*

- (i) *the extraction s_t, including its storage, precision, and evaluation, is temporally admissible and has complete cost map $B \mapsto \psi_s(B, n, m, H)$;*
- (ii) *there is a represented pre-hit reconstruction kernel $R_t(dh \mid S_t)$, of complete cost map $B \mapsto \psi_R(B, n, m, H)$, such that on $\{\tau > t\}$*

$$\mathcal{L}(\mathcal{H}_t \mid \Theta, S_t, \tau > t) = R_t(\cdot \mid S_t) = \mathcal{L}(\mathcal{H}_t \mid S_t, \tau > t); \tag{453}$$

- (iii) *the reconstruction uses no later flag, terminal record, target answer, or later oracle answer, and preserves the represented decision carrier and target utility, possibly through the action map of Theorem V.4.5.*

Then S_t and $\mathcal{H}_t$ are equivalent for the survival-weighted target decisions of Definition V.4.4, with their complete representation costs retained:

$$\mathrm{DVal}^{\mathrm{sat}}_{u,t}(S_t; B) \le \mathrm{DVal}^{\mathrm{sat}}_{u,t}(\mathcal{H}_t; \psi_s(B, n, m, H)), \tag{454}$$

$$\mathrm{DVal}^{\mathrm{sat}}_{u,t}(\mathcal{H}_t; B) \le \mathrm{DVal}^{\mathrm{sat}}_{u,t}(S_t; \psi_R(B, n, m, H)). \tag{455}$$

The same two inequalities hold for $\mathrm{DAdv}^{\mathrm{sat}}_{u,t}$ *when the declared baseline is transported through the two kernels. At a saturated ceiling closed under both class-preserving cost maps, the two target-decision profiles coincide.*

Thus a represented sufficient statistic can carry the complete optimal target-decision value of the pre-hit record. A constructor or provenance condition alone is not sufficient in this sense; Eq. (453) *and the affordable reconstruction are the conditions that permit an upper bound to pass from the statistic to the original record.*

Proof. The statistic $S_t = s_t(\mathcal{H}_t)$ is a deterministic represented garbling of $\mathcal{H}_t$. Applying Theorem V.4.5 with cost map ψ_s proves Eq. (454).

Conversely, on $\{\tau > t\}$, sample $\widetilde{\mathcal{H}}_t$ from $R_t(\cdot \mid S_t)$; on the failure sector retain the represented survival flag and use an arbitrary fixed padded history, because the decision payoff there is multiplied by zero. By Eq. (453), the tower property gives equality of every survival-weighted utility expectation under $(\Theta, S_t, \widetilde{\mathcal{H}}_t)$ and $(\Theta, S_t, \mathcal{H}_t)$. Composing any cost-$B$ history selector with this represented, pre-hit, fully charged reconstruction therefore produces a statistic selector of cost at most $\psi_R(B, n, m, H)$ with the same objective value. Taking the supremum proves Eq. (455) directly. Transporting the baseline through the same two simulations proves the advantage statements. If a common saturated ceiling absorbs both cost maps, the two opposite inequalities have the same feasible ceiling and give equality.

□

V.5 Reversible Carrier Geometry and Authorized Flow

The selected generator supplies a native Dirichlet geometry only when its reference measure and reversible conductance are represented. Otherwise a separately represented comparison geometry is used with the existing form-comparison loss.

Definition V.5.1 (Certified dynamical geometry)**.** Fix a controlled policy record and one selected generator L on a finite carrier X. A certified dynamical geometry at budget B is an ambient Geom/Caus record of Definition III.14.9, augmented by represented initial, target, and failure data. It has one of the following two carrier forms.

(i) *Native carrier.* A full-support represented probability measure μ is invariant for L. Writing $L = L^{\mathsf{s}} + A + L^{\mathsf{b}}$ in $L^2(X, \mu)$, the symmetric conductance is

$$w_L(x, z) = \frac{1}{2}(\mu(x)L(x, z) + \mu(z)L(z, x)) \ge 0 \quad (x \ne z),$$

and

$$\mathcal{E}_L(f, g) = \frac{1}{2}\sum_{x,z} w_L(x, z)(f(x) - f(z))(g(x) - g(z)).$$

Equivalently, the conductance in Definition III.14.8 is $c_L(x, z) = w_L(x, z)/(\mu(x)\mu(z))$.

(ii) *Comparison carrier.* A represented pointwise geometry $\Phi = (E_\Phi, c_\Phi)$ is related to the selected generator by a represented local, resolvent, or form-comparison record with the error and transfer loss prescribed by Lemma III.13.5 and Proposition III.13.6.

The certificate retains the target-normalized potential D, the selected actual current and authority record $(J, \mathfrak{K})$, target reach $R_T(D)$, regularity $\rho_\Phi(D)$, and the mixing and local-consistency gates of Definition III.14.9. It may additionally retain certified constants, all evaluated on this same carrier:

$$\gamma_\Phi = \inf_{\mathrm{Var}_\mu f>0} \frac{\mathcal{E}_\Phi(f,f)}{\mathrm{Var}_\mu(f)},$$
$$N_{\mathrm{eff},\Phi}(\lambda) = \mathrm{Tr}\Big(H_\Phi(H_\Phi+\lambda I)^{-1}\Big), \qquad H_\Phi \geq 0,$$

together with the convention that H_Φ is the nonnegative self-adjoint operator associated with $\mathcal{E}_\Phi$ in the represented finite realization, and with a conductance lower bound, an MLSI constant, a Bakry–Émery curvature bound, or a condenser-capacity value when the corresponding represented proof or audited finite computation is included.

An unavailable lower-rate certificate is assigned value zero. An unavailable upper error or cost certificate is assigned value $+\infty$. These conventions supply no positive mixing, alignment, reach, or affordability claim. Every retained constant, eigenvalue computation, comparison map, and precision parameter contributes to the complete saturated cost.

Proposition V.5.2 (Native geometry and Hodge authority)**.** *In the native-carrier case of Definition V.5.1, $\mathcal{E}_L$ is nonnegative and*

$$-\langle f, L^{\mathsf{s}} f\rangle_{L^2(\mu)} = \mathcal{E}_L(f,f).$$

The skew part contributes no quadratic energy: $\langle f, Af\rangle_{L^2(\mu)} = 0$. Its unique structurally actionable component is the authorized Hodge projection in Definitions II.4.2 and II.4.3; the orthogonal component belongs to J_{res}. Therefore the selected policy can use target-oriented directed flow only through the same-record Caus factor $C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D)$.

Proof. Invariance gives zero row and column sums in the weighted realization. Pairing the symmetric off-diagonal terms yields the displayed Dirichlet form, which is nonnegative because every $w_L(x,z)$ is nonnegative. Skew adjointness gives $\langle f, Af\rangle = 0$. The remaining statements are the uniqueness and target-pairing identities of Theorem II.4.6 and Lemma II.4.5.

□

Theorem V.5.3 (Certified mixing and safe spectral compression)**.** *Let $L = L^{\mathsf{s}} + A$ have a represented invariant law μ, with $A^* = -A$, and suppose the native symmetric form has certified gap $\gamma_\Phi > 0$. For the Markov semigroup $P_t = e^{tL}$,*

$$\mathrm{Var}_\mu(P_t f) \leq e^{-2\gamma_\Phi t}\,\mathrm{Var}_\mu(f). \tag{456}$$

If a certified MLSI with constant $\alpha_\Phi > 0$ is part of the same record, its declared entropy normalization gives the corresponding entropy decay

$$\mathrm{Ent}_\mu(P_t^* g) \leq e^{-2\alpha_\Phi t}\,\mathrm{Ent}_\mu(g) \tag{457}$$

for represented probability densities in its verified domain. A certified Bakry–Émery inequality supplies its separately stated gradient contraction for the reversible comparison semigroup; it is transferred to the selected generator only through a represented comparison theorem.

Let P_S be an orthogonal projection onto a represented safe or admissible subspace and put $H_S = P_S H_\Phi P_S|_{\mathrm{ran}\,P_S}$. Then for every $\lambda > 0$,

$$\mathrm{Tr}\Big(H_S(H_S+\lambda I)^{-1}\Big) \leq N_{\mathrm{eff},\Phi}(\lambda). \tag{458}$$

Proof. For $f_t = P_t f$, invariance and skew adjointness give

$$\frac{d}{dt}\operatorname{Var}_\mu(f_t) = 2\langle f_t, Lf_t\rangle_\mu = -2\mathcal{E}_\Phi(f_t, f_t) \le -2\gamma_\Phi \operatorname{Var}_\mu(f_t).$$

Grönwall's inequality proves Eq. (456). The entropy conclusion is the defining semigroup consequence of the certified MLSI in the declared normalization; the restriction on transfer is exactly Lemma IV.9.6. For Eq. (458), order the eigenvalues of H_Φ and its compression decreasingly. Cauchy interlacing gives $\eta_j(H_S) \le \eta_j(H_\Phi)$. The function $u \mapsto u/(u+\lambda)$ is increasing on $[0,\infty)$; summing the termwise inequalities proves the trace bound. □

V.5.1 Audited Functional-Inequality Profiles

The Dirichlet form of a certified dynamical geometry supports several inequalities with different outputs. Their constants are recorded with a fixed normalization so that rate comparisons are meaningful and so that one logical certificate is not charged more than once under different names.

Definition V.5.4 (Normalized functional-inequality profile)**.** For a certified carrier Φ, its normalized functional-inequality profile is the typed tuple

$$\mathfrak{F}_\Phi = (\mathfrak{F}_\Phi^{\mathrm{diff}}, \mathfrak{F}_\Phi^{\mathrm{ent}}, \mathfrak{F}_\Phi^{\mathrm{tar}})$$

defined in Definitions V.5.5 to V.5.7. The three components share one carrier, normalization, proof record, and cost ledger; storing the same certificate in two components creates no second charge.

Definition V.5.5 (Diffusive and spectral profile)**.** Let Φ be a native or comparison carrier from Definition V.5.1, with probability measure μ, self-adjoint operator H_Φ, Dirichlet form $\mathcal{E}_\Phi$, and reversible semigroup $S_t = e^{-tH_\Phi}$. In the comparison-carrier case write $w_\Phi(x,z) = \mu(x)\mu(z)c_\Phi(x,z)$; in the native case use the conductance w_L from Definition V.5.1. Put

$$\operatorname{Osc}(f) = \max_X f - \min_X f, \qquad \operatorname{Ent}_\mu(g) = \mu(g\log g) - \mu(g)\log\mu(g)$$

for $g \ge 0$, with $0\log 0 = 0$. For $A \subsetneq X$, let

$$\lambda_\Phi^{\mathrm{D}}(A) = \inf\left\{\frac{\mathcal{E}_\Phi(f,f)}{\|f\|_{2,\mu}^2} : 0 \ne f,\ f|_{A^c} = 0\right\}.$$

Also set

$$\lambda_\Phi^{\mathrm{sp}}(A) = \inf\left\{\frac{\mathcal{E}_\Phi(f,f)}{\operatorname{Var}_\mu(f)} : 0 < \operatorname{Var}_\mu(f),\ f|_{A^c} = 0\right\}.$$

The following entries constitute the normalized functional-inequality profile $\mathfrak{F}_\Phi$. An entry is retained only with its represented proof, audited finite computation, or represented comparison certificate.

(i) The Poincaré constant γ_Φ is the largest certified $\gamma \ge 0$ such that

$$\gamma \operatorname{Var}_\mu(f) \le \mathcal{E}_\Phi(f,f).$$

With $Q_\Phi(A,A^c) = \sum_{x\in A, z\notin A} w_\Phi(x,z)$, define

$$\mathsf{h}_\Phi = \inf_{0<\mu(A)\le 1/2} \frac{Q_\Phi(A,A^c)}{\mu(A)}, \qquad q_\Phi = \max_x \frac{1}{\mu(x)}\sum_{z\ne x} w_\Phi(x,z).$$

Thus h_Φ has generator-rate units.

(ii) A weak Poincaré rate is a decreasing function $\beta_{\mathrm{W}} : (0, 1] \to (0, \infty]$ satisfying

$$\mathrm{Var}_\mu(f) \le \beta_{\mathrm{W}}(r)\mathcal{E}_\Phi(f, f) + r\,\mathrm{Osc}(f)^2, \qquad 0 < r \le 1.$$

A super-Poincaré rate is a decreasing function $\beta_{\mathrm{S}} : (0, \infty) \to (0, \infty]$ satisfying

$$\|f\|_{2,\mu}^2 \le r\mathcal{E}_\Phi(f, f) + \beta_{\mathrm{S}}(r)\|f\|_{1,\mu}^2, \qquad r > 0.$$

(iii) A Nash certificate of dimension $d > 0$ and constant C_{N} satisfies

$$\|f\|_{2,\mu}^{2+4/d} \le C_{\mathrm{N}}\mathcal{E}_\Phi(f, f)\|f\|_{1,\mu}^{4/d}$$

on its declared semigroup-invariant mean-zero, killed, or transient domain $\mathscr{D}_{\mathrm{N}}$. For $d > 2$, a Sobolev certificate with constant C_{Sob} satisfies

$$\|f\|_{2d/(d-2),\mu}^2 \le C_{\mathrm{Sob}}\mathcal{E}_\Phi(f, f)$$

on the same declared domain.

(iv) The spectral profile and a Faber–Krahn minorant are

$$\Lambda_\Phi^{\mathrm{mix}}(v) = \inf_{0<\mu(A)\le v} \lambda_\Phi^{\mathrm{sp}}(A), \quad 0 < v < 1, \qquad \Lambda_\Phi^{\mathrm{mix}}(v) = \gamma_\Phi \quad (v \ge 1),$$

and

$$\lambda_\Phi^{\mathrm{D}}(A) \ge c_{\mathrm{FK}}\mu(A)^{-2/d} \qquad (0 < \mu(A) < 1).$$

Definition V.5.6 (Entropic, curvature, and transport profile)**.** Let $\Phi, \mu, H_\Phi, \mathcal{E}_\Phi$ be as in Definition V.5.5. The entropic component $\mathfrak{F}_\Phi^{\mathrm{ent}}$ consists of the following represented certificates.

(i) A logarithmic Sobolev constant α_{LS} and a modified logarithmic Sobolev constant α_{MLS} satisfy, respectively,

$$\mathrm{Ent}_\mu(f^2) \le \frac{2}{\alpha_{\mathrm{LS}}}\mathcal{E}_\Phi(f, f),$$
$$\mathrm{Ent}_\mu(g) \le \frac{1}{2\alpha_{\mathrm{MLS}}}\mathcal{E}_\Phi(g, \log g),$$

for the represented domains declared in the certificate.

(ii) With

$$\Gamma(f, g) = \frac{1}{2}\big(fH_\Phi g + gH_\Phi f - H_\Phi(fg)\big), \qquad \Gamma_2(f) = -\frac{1}{2}H_\Phi\Gamma(f) + \Gamma(f, H_\Phi f),$$

a Bakry–Émery $CD(\kappa, N)$ certificate means

$$\Gamma_2(f) \ge \kappa\Gamma(f) + \frac{1}{N}(H_\Phi f)^2$$

pointwise, where $N \in [1, \infty]$ and the last term is zero for $N = \infty$. The sign convention is the one for the nonnegative operator H_Φ.

(iii) A transport–entropy certificate for a represented metric d_Φ is either

$$T_1(C_{T_1}) : \quad W_{1,d_\Phi}(\nu, \mu)^2 \le 2C_{T_1}\,\mathrm{KL}(\nu\|\mu),$$

or

$$T_2(C_{T_2}) : \quad W_{2,d_\Phi}(\nu, \mu)^2 \le 2C_{T_2}\,\mathrm{KL}(\nu\|\mu),$$

for represented $\nu \ll \mu$.

Definition V.5.7 (Target and nonreversible profile)**.** Let $\Phi, \mu, H_\Phi, \mathcal{E}_\Phi$ be as in Definition V.5.5. The target and nonreversible component $\mathfrak{F}_\Phi^{\mathrm{tar}}$ consists of the following represented certificates.

(i) The target, failure, and isocapacitary profiles are restrictions of the represented condenser capacity from Definition V.6.5:

$$\mathsf{C}_{T^+}(v) = \inf_{0<\mu(A)\le v} \frac{\mathrm{Cap}_\Phi(A, T^+)}{\mu(A)},$$
$$\mathsf{C}_{T^-}(v) = \inf_{0<\mu(A)\le v} \frac{\mathrm{Cap}_\Phi(A, T^-)}{\mu(A)},$$
$$\mathsf{C}_{\mathrm{iso}}(v) = \inf_{0<\mu(A)\le v} \frac{\mathrm{Cap}_\Phi(A, A^c)}{\mu(A)}.$$

Empty or intersecting condenser classes are omitted from the infimum.

(ii) For a selected generator $L = -H_\Phi + A$, a strong sector constant C_{sec} satisfies

$$|\langle f, Ag\rangle_\mu| \le C_{\mathrm{sec}}\, \mathcal{E}_\Phi(f,f)^{1/2} \mathcal{E}_\Phi(g,g)^{1/2}.$$

A represented hypocoercive certificate is a quadratic functional $\mathscr{H}$ and constants $0 < m_{\mathscr{H}} \le M_{\mathscr{H}}$, $\kappa_{\mathscr{H}} > 0$, such that on the centered declared domain

$$m_{\mathscr{H}} \|f\|_2^2 \le \mathscr{H}(f) \le M_{\mathscr{H}} \|f\|_2^2, \qquad \frac{d}{dt}\mathscr{H}(P_t f) \le -2\kappa_{\mathscr{H}} \mathscr{H}(P_t f).$$

Rate functions are stored at the finite precision used by the downstream theorem. Their representation, optimization grid, domain verification, and all comparison losses contribute to saturated cost. A missing lower constant equals zero, a missing upper constant equals $+\infty$, and a missing rate function supplies no bound.

Theorem V.5.8 (Quantitative consequences of the audited profile)**.** *Let $\mathfrak{F}_\Phi$ be the profile of Definition V.5.4. Each certified entry has the following consequence in its declared domain.*

(i) *Poincaré gives*

$$\mathrm{Var}_\mu(S_t f) \le e^{-2\gamma_\Phi t}\, \mathrm{Var}_\mu(f).$$

If $q_\Phi > 0$, the continuous-time Cheeger bounds are

$$\frac{\mathsf{h}_\Phi^2}{2q_\Phi} \le \gamma_\Phi \le 2\mathsf{h}_\Phi.$$

(ii) *For every $0 < r \le 1$, weak Poincaré gives*

$$\mathrm{Var}_\mu(S_t f) \le e^{-2t/\beta_{\mathrm{W}}(r)}\, \mathrm{Var}_\mu(f) + r(1 - e^{-2t/\beta_{\mathrm{W}}(r)})\, \mathrm{Osc}(f)^2. \tag{459}$$

For every $r > 0$, super-Poincaré gives

$$\|S_t f\|_2^2 \le e^{-2t/r} \|f\|_2^2 + \beta_{\mathrm{S}}(r)(1 - e^{-2t/r}) \|f\|_1^2. \tag{460}$$

Both right sides may be minimized over the represented admissible values of r.

(iii) *Nash gives the ultracontractive half-step estimate*

$$\|S_t|_{\mathscr{D}_{\mathrm{N}}}\|_{1\to 2}^2 \le \left(\frac{dC_{\mathrm{N}}}{4t}\right)^{d/2}, \qquad t > 0, \tag{461}$$

and self-adjointness gives $\|S_t|_{\mathscr{D}_{\mathrm{N}}}\|_{1\to\infty} \le (dC_{\mathrm{N}}/(2t))^{d/2}$. The stated Sobolev inequality implies the Nash inequality with the same constant by interpolation between L^1 and $L^{2d/(d-2)}$.

(iv) *For the killed semigroup* S_t^A,

$$\|S_t^A\|_{2\to 2} \le e^{-t\lambda_\Phi^{\mathrm{D}}(A)}. \tag{462}$$

Hence a Faber–Krahn certificate gives $\|S_t^A\|_{2\to 2} \le e^{-tc_{\mathrm{FK}}\mu(A)^{-2/d}}$. *If the uniformized kernel* $K = I - H_\Phi/q_\Phi$ *is lazy, the represented generator-rate spectral profile supplies the relative-uniform mixing certificate*

$$t_\infty(\varepsilon) \le \int_{4\mu_*}^{4/\varepsilon} \frac{2\,dv}{v\Lambda_\Phi^{\mathrm{mix}}(v)}, \qquad \mu_* = \min_x \mu(x), \tag{463}$$

whenever the displayed lazy-kernel normalization is valid.

(v) *Logarithmic Sobolev gives hypercontractivity*

$$\|S_t f\|_{q(t)} \le \|f\|_p, \qquad q(t) - 1 = (p-1)e^{2\alpha_{\mathrm{LS}}t}, \quad p > 1.$$

Modified logarithmic Sobolev gives

$$\mathrm{Ent}_\mu(S_t^* g) \le e^{-2\alpha_{\mathrm{MLS}}t}\,\mathrm{Ent}_\mu(g) \tag{464}$$

for probability densities g.

(vi) $CD(\kappa,\infty)$ *gives the pointwise gradient estimate*

$$\Gamma(S_t f) \le e^{-2\kappa t} S_t\Gamma(f). \tag{465}$$

A finite N *retains the additional nonnegative dimension term in the integrated* Γ_2 *estimate.*

(vii) *If* F *is* L_F*-Lipschitz for* d_Φ, *then*

$$|\nu(F) - \mu(F)| \le L_F\sqrt{2C_{T_1}\,\mathrm{KL}(\nu\|\mu)}, \tag{466}$$

$$|\nu(F) - \mu(F)| \le L_F\sqrt{2C_{T_2}\,\mathrm{KL}(\nu\|\mu)}. \tag{467}$$

The second line uses $W_1 \le W_2$.

(viii) *A represented condenser capacity gives the exact Dirichlet and reach–avoid identities in Theorem V.6.6. The profiles* C_{T^+}, C_{T^-}, *and* $\mathsf{C}_{\mathrm{iso}}$ *therefore provide, respectively, certified target-access, failure-access, and killed-set energy scales at every represented mass level* v.

(ix) *A form comparison* $a\mathcal{E}_\Phi \le \mathcal{E}_\Psi \le b\mathcal{E}_\Phi$ *transfers every Rayleigh-quotient lower bound by the factor* a *and every energy upper bound by the factor* b. *The sector constant controls the skew bilinear term but supplies no decay rate by itself. A hypocoercive certificate gives*

$$\|P_t f\|_2^2 \le \frac{M_{\mathscr{H}}}{m_{\mathscr{H}}} e^{-2\kappa_{\mathscr{H}}t}\|f\|_2^2. \tag{468}$$

All transfers to the selected nonreversible generator use the represented comparison, sector, or hypocoercive record carried by the same certified dynamical geometry.

Proof. The Poincaré decay is the energy derivative calculation in Theorem V.5.3. Uniformization by $K = I - H_\Phi/q_\Phi$ and the discrete Cheeger inequalities give the two Cheeger bounds after multiplying by q_Φ.

For weak Poincaré, set $y(t) = \mathrm{Var}_\mu(S_t f)$. Markov contraction preserves oscillation, and

$$y'(t) = -2\mathcal{E}_\Phi(S_t f, S_t f) \le -\frac{2}{\beta_{\mathrm{W}}(r)}\big(y(t) - r\,\mathrm{Osc}(f)^2\big).$$

Solving this scalar differential inequality proves Eq. (459). The same calculation with $y(t) = \|S_t f\|_2^2$, using L^1-contraction, proves Eq. (460).

Under Nash, $y(t) = \|S_t f\|_2^2$ satisfies

$$y'(t) \le -\frac{2}{C_{\mathrm{N}}} y(t)^{1+2/d} \|f\|_1^{-4/d}.$$

Integration proves Eq. (461); the semigroup identity and self-adjoint duality give the L^1-to-L^∞ estimate. The Sobolev implication follows from Hölder interpolation.

The variational definition of $\lambda_\Phi^{\mathrm{D}}(A)$ and the spectral theorem give Eq. (462); substitution gives the Faber–Krahn estimate. The evolving-level-set argument for the lazy uniformized kernel gives Eq. (463); the factor q_Φ is already contained in the generator-rate identity $\Lambda_\Phi^{\mathrm{mix}} = q_\Phi \Lambda_K^{\mathrm{mix}}$ [33].

The logarithmic Sobolev and modified logarithmic Sobolev conclusions follow by differentiating the corresponding norms and entropy along S_t. The Γ_2 interpolation $s \mapsto S_s \Gamma(S_{t-s} f)$ proves Eq. (465). Kantorovich duality followed by the declared T_1 or T_2 inequality proves Eqs. (466) and (467). The capacity statement is the Dirichlet principle. Form comparison follows directly from the variational definitions. Finally, Grönwall's inequality applied to $\mathscr{H}(P_t f)$, followed by the two norm comparisons, proves Eq. (468). The Nash and weak-Poincaré implications use the normalizations of Carlen et al. [19] and Röckner and Wang [70]. The transport and hypocoercive entries use, respectively, Bobkov and Götze [14], Otto and Villani [65] and Villani [77].

□

Definition V.5.9 (Typed functional-output slots)**.** The profile outputs are assigned to the following slots:

$$\begin{aligned}\mathscr{S}_{\mathrm{FI}} = \{\, & L^2\text{-decay},\ \mathrm{Ent}\text{-decay},\ 1\to 2\text{-smoothing},\\ & 1\to\infty\text{-smoothing},\ \text{killed survival},\ \text{gradient contraction},\\ & \text{distribution shift},\ \text{target/failure access},\ \text{hypocoercive decay}\}.\end{aligned}$$

A candidate bound is eligible for a slot only after a represented theorem has converted its constants to that slot's normalization, time scale, carrier, domain, and comparison loss. A logical implication is represented by a directed edge from its source certificate to the induced bound. The induced bound retains the source certificate's provenance and is not a second independent contribution.

Theorem V.5.10 (Slotwise selection and no double counting)**.** *Fix one certified dynamical geometry and one output slot $s \in \mathscr{S}_{\mathrm{FI}}$. Let $\mathscr{J}_s$ be the finite represented set of audited bounds converted to the units of s. If every $j \in \mathscr{J}_s$ proves $U_s \le C_{s,j}$, then*

$$U_s \le C_s^{\mathrm{FI}} := \min_{j \in \mathscr{J}_s} C_{s,j}. \tag{469}$$

For an empirically selected finite family, the right side receives exactly the simultaneous selection penalty supplied by the relevant uniform or sample-split theorem. Deterministically certified constants receive no selection penalty.

If one certificate implies several inequalities, each downstream slot may use the induced bound, but a propagation theorem may charge the originating error only once. Bounds in different slots are added only after a represented propagation theorem converts them to one common risk, return, hitting, or deployment quantity. Thus Poincaré, LSI, MLSI, curvature, transport, capacity, and hypocoercivity are neither summed as dimensionally incomparable numbers nor counted repeatedly along an implication chain.

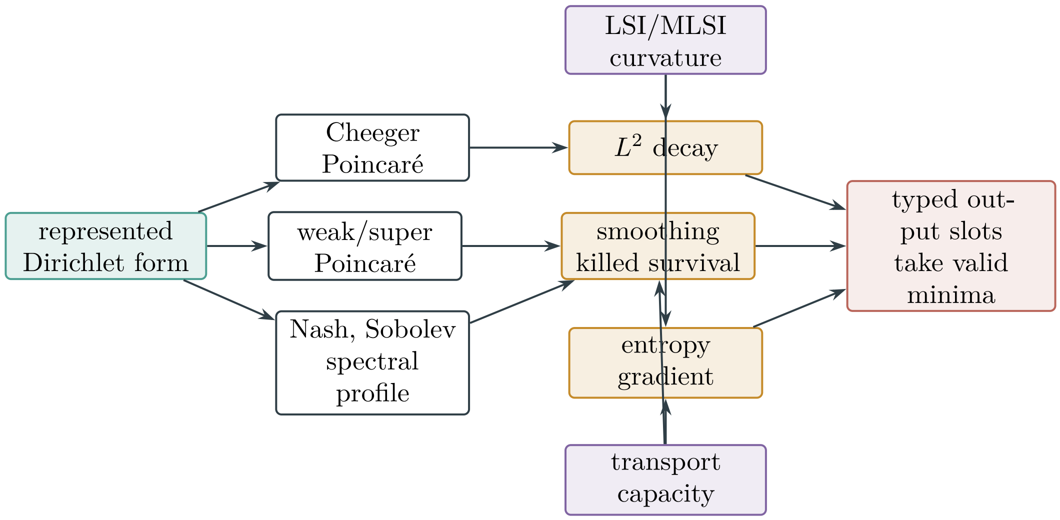

Figure 81: The audited functional-inequality profile converts certified constants into typed dynamical outputs. Bounds compete by a minimum only after conversion to the same slot; implication edges preserve provenance and therefore introduce no duplicate charge.

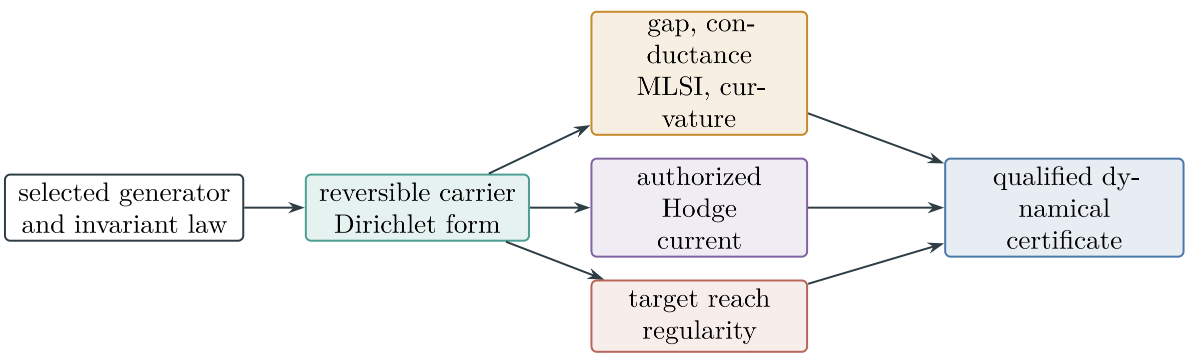

Figure 82: A controlled dynamical certificate couples a reversible carrier to the actual authorized current and target-visible regularity. Every constant is evaluated on the same represented record and charged at its complete saturated cost.

Proof. Every member of $\mathcal{J}_s$ is an upper bound for the same scalar in the same units, so their minimum is an upper bound. Simultaneous empirical validity is exactly the event supplied by the cited selection theorem. The remaining statements follow from provenance preservation under represented composition and no-creation, Proposition II.5.9 and Lemma IV.9.6: an induced inequality is a deterministic descendant of its source record, and cross-slot addition is valid only after a common-output comparison has been proved.

□

V.6 Committors, Bellman Recursions, and Reach–Avoid Geometry

V.6.1 Controlled committors and Bellman identities

Definition V.6.1 (Controlled reach–avoid committors)**.** Fix disjoint represented terminal sectors $T^+, T^- \subseteq X$, put $N = X \setminus (T^+ \cup T^-)$, and let P^π be a selected policy kernel. The horizon-h reach–avoid committor is

$$u_h^\pi(x) = \Pr_x^\pi\{\tau_{T^+} \le h,\ \tau_{T^+} < \tau_{T^-}\}.$$

For $0 < \beta < 1$, the discounted committor is

$$u_\beta^\pi(x) = \mathbb{E}_x^\pi\left[\beta^{\tau_{T^+}} \mathbf{1}_{\{\tau_{T^+} < \tau_{T^-}\}}\right],$$

with the convention $\beta^\infty = 0$. Both functions have boundary values one on T^+ and zero on T^-. Their semantic existence does not assign them saturated representation cost; derivability is governed by Definition III.14.4.

For a bounded reward r and discount $0 < \beta < 1$, define

$$(\mathcal{T}^\pi v)(x) = r^\pi(x) + \beta(P^\pi v)(x), \qquad (\mathcal{T} v)(x) = \max_{a \in \mathcal{A}(x)} \{r(x,a) + \beta(P^a v)(x)\}.$$

All maxima are finite and represented action evaluation and comparison are charged to the selected-policy derivation.

Theorem V.6.2 (Exact controlled Bellman and committor identities)**.** *The finite-horizon committors satisfy*

$$u_0^\pi = \mathbf{1}_{T^+}, \tag{470}$$

$$u_{h+1}^\pi = \mathbf{1}_{T^+} + \mathbf{1}_N P^\pi u_h^\pi. \tag{471}$$

The discounted committor is the unique bounded solution of

$$u_\beta^\pi = \mathbf{1}_{T^+} + \beta \mathbf{1}_N P^\pi u_\beta^\pi. \tag{472}$$

The optimal finite-horizon and discounted committors are obtained by replacing P^π on N by $\max_{a \in \mathcal{A}(x)} P^a$ pointwise. For a bounded represented reward, $\mathcal{T}^\pi$ and $\mathcal{T}$ are β-contractions in $\|\cdot\|_\infty$, and their unique fixed points are V^π and V^*, respectively.

For a finite legal-action set and $\tau > 0$, the soft action value obeys the exact Gibbs identity

$$\tau \log \sum_{a \in \mathcal{A}(x)} e^{q_a/\tau} = \max_{p \in \Delta(\mathcal{A}(x))} \left\{ \sum_a p_a q_a + \tau \mathcal{H}(p) \right\}, \tag{473}$$

whose maximizer is the represented Boltzmann kernel. Its construction and precision are part of the policy cost.

Proof. Condition on the first transition. A state already in T^+ contributes one, a state in T^- contributes zero, and a state in N contributes the expectation of the remaining-horizon committor. This proves Eqs. (470) and (471). The same conditioning with one factor β proves Eq. (472). The right-hand side is a β-contraction on bounded functions with fixed boundary values, so Banach's fixed-point theorem gives existence and uniqueness. Maximization at each state gives the dynamic-programming recursion because the finite-horizon continuation value depends on the past only through the clocked state.

Stochastic kernels are contractions in the supremum norm; taking a finite pointwise maximum preserves the same Lipschitz constant. Hence the reward operators are β-contractions and their fixed points equal the usual discounted returns by iteration. Finally, Lagrange multipliers for the simplex constraint give $p_a \propto e^{q_a/\tau}$; substitution proves Eq. (473) [10, 11, 67].

□

Theorem V.6.3 (Temporal admissibility of controlled values)**.** *Let u denote any committor, value function, advantage, policy table, first-hit partition, or completed reach–avoid record of a selected controlled system. It belongs to a prospective source profile at budget B exactly through a temporally admissible represented derivation of the same denotation whose complete cost is at most B. A Bellman recursion evaluated for h steps is charged for those steps and retained values. An exact solve is charged for its represented solver, arithmetic, precision, and output. A terminal record computed from future verifier flags remains an accounting record under Corollary III.23.1.*

Consequently, the identities in Theorem V.6.2 create no prospective target source. When a represented value is available before a transition, its quotient, Geom/Caus, Lift, and boundary provenance is classified by Theorem V.1.2 at its complete cost.

Proof. Apply Definition III.14.4 and Lemma III.14.5 to the policy-induced kernel on the clocked carrier. Availability before the transition and exclusion of later verifier flags are precisely Definition III.21.1. The channel assignment then follows from Theorem V.1.2. These alternatives are exhaustive because the completed transcript is a valid Lift and its post-exhaustion remainder is J_{res}.

□

V.6.2 No uncharged prospective value amplification

Theorem V.6.4 (No uncharged prospective value amplification)**.** *Let R be a family-uniform, temporally admissible pre-hit record of complete saturated cost at most B. Let $a_R(I) \in [0,1]$ be a prospective selector, reach–avoid, or arrival value in a declared output normalization. Let $\widehat{B}$ be a class-preserving common majorant, within the already declared saturated budget structure, for evaluator normalization, the clocked policy embedding, the complete active-source backward slice, and every comparison used below.*

A structural upper bound for a_R is admitted only through at least one of the following two forms.

(i) *There are an earlier dynamically qualified master certificate $\mathscr{C}_R$, a represented constant $K_R \geq 0$, and a represented error $\delta_R \geq 0$, all in the complete record, such that pointwise*

$$a_R(I) \leq K_R V_{\mathscr{C}_R,I} + \delta_R(I). \tag{474}$$

The value is then comparison-dominated by that earlier source.

(ii) *The complete pre-hit record R, including its selected kernel, potential, current, authority data, target interface, and output converter, contains a dynamically qualified master certificate $\mathscr{C}_R^{\mathrm{dir}}$ and a represented same-normalization comparison*

$$a_R(I) \leq K_R^{\mathrm{dir}} V_{\mathscr{C}_R^{\mathrm{dir}},I} + \delta_R^{\mathrm{dir}}(I). \tag{475}$$

The complete record is then itself the charged prospective source and is assigned, at cost at most $\widehat{B}$, to exactly one existing ambient, exact-quotient, reversible-compensated, or general-resolvent master-profile cell.

If neither represented comparison exists, the framework makes no numerical transfer from structural visibility to a_R. The record remains fully classified by the channel decomposition, but it cannot be counted as a comparison-dominated prospective structural value. A later terminal or first-hit record is accounting only; a gradient-orthogonal contribution is J_{res}; an undeclared input is presentation enrichment or advice; and an over-budget derivation is absent from the budget slice.

For any collection $\mathscr{R}(B)$ for which every positive value has one of the two displayed comparisons, assign a record to the inherited class when (i) *is present and otherwise to the direct class* (ii)*. These two classes are disjoint and cover $\mathscr{R}(B)$. Define no new source channel and use the existing master-profile cells containing the corresponding certificates. Then, pointwise,*

$$\sup_{R\in\mathscr{R}(B)} a_R(I) \le \max\left\{ \sup_{R\in\mathscr{R}_{\mathrm{inh}}(B)} (K_R V_{\mathscr{C}_R,I} + \delta_R(I)),\ \sup_{R\in\mathscr{R}_{\mathrm{dir}}(B)} (K_R^{\mathrm{dir}} V_{\mathscr{C}_R^{\mathrm{dir}},I} + \delta_R^{\mathrm{dir}}(I)) \right\}. \tag{476}$$

The same inequality holds after applying a monotone family functional and then taking the saturated supremum. In particular, ancestry equality, zero incremental constructor charge, Bellman recursion, valid lifting, or postprocessing alone never supplies either displayed numerical comparison.

Proof. Temporal admissibility places the complete pre-hit derivation of R in the saturated closure. Normalize its selected transition by Theorem V.1.2 and retain the active backward slice, including every instance-dependent coefficient and output converter. If (474) occurs in that record, it is precisely a represented same-normalization transfer, so the first alternative holds.

Otherwise a structural explanation must use the complete record producing the value. If that record contains the qualification and comparison in (475), apply Theorems III.2.6 and III.18.9. They assign $\mathscr{C}_R^{\mathrm{dir}}$ to exactly one existing master cell while preserving its full derivation and cost. This proves the second alternative and shows that it introduces no additional channel.

If neither comparison is represented, structural ancestry provides no inequality in the output normalization. This is exactly the scope restriction in Corollary III.16.9 and Theorem III.21.4: contextual or accounting charge transfers to a prospective source value only through the represented comparison. The remaining assignments follow from Corollary III.23.1, Theorems II.7.2 and III.33.2, and Definition III.1.3.

For a collection covered by the two alternatives, split it into the inherited and direct subcollections, apply the corresponding pointwise inequality, and take the supremum. A supremum over their union is the maximum of the two suprema, proving (476). Monotonicity gives the family-functional statement, and saturation preserves the same common ceiling. Finally, Proposition II.5.9 and Theorem V.6.3 shows that the listed derived operations preserve source ancestry but do not create a numerical comparison.

□

V.6.3 Condenser capacity and reach–avoid bounds

Definition V.6.5 (Represented condenser capacity)**.** Let Φ be a represented reversible carrier and let $S, T \subseteq X$ be disjoint represented sets. Its condenser capacity is

$$\mathrm{Cap}_\Phi(S,T) = \inf\{\mathcal{E}_\Phi(f,f) : f|_S = 1,\ f|_T = 0\}. \tag{477}$$

The infimum is over the finite represented function space. A capacity value is a certified coordinate only when the boundary sets, minimizer or solver, energy evaluation, and precision are in the saturated record. For reach–avoid, the relevant condenser is (T^+, T^-), optionally after killing outside a represented safe carrier.

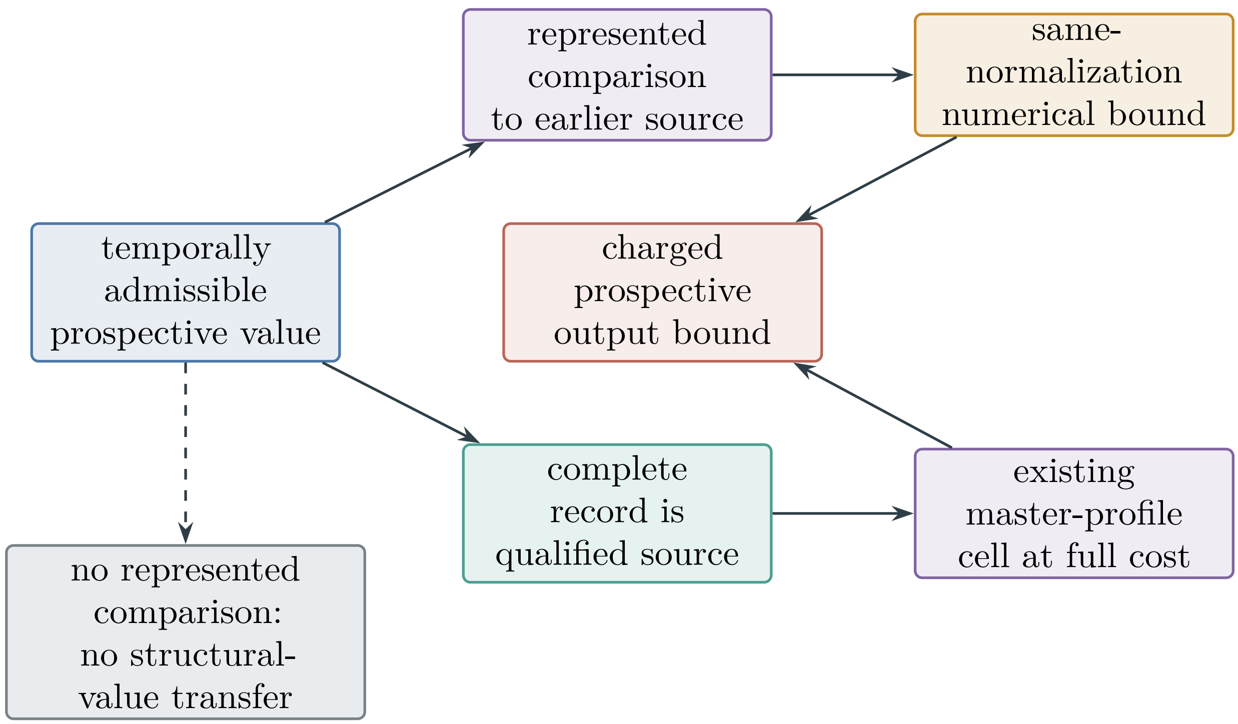

Figure 83: Prospective value has no uncharged intermediate location. It is either comparison-dominated by an earlier qualified source or charged as a complete directly qualified source in an existing master-profile cell. In the absence of either represented comparison, provenance alone supplies no numerical structural-value bound.

Theorem V.6.6 (Dirichlet reach–avoid principle and drift-time bound)**.** *Let the selected policy kernel be reversible with respect to its represented invariant law and let h be its undiscounted target-before-failure committor. Then $h = 1$ on T^+, $h = 0$ on T^-, it is harmonic on N, and*

$$\mathcal{E}_\Phi(h, h) = \mathrm{Cap}_\Phi(T^+, T^-). \tag{478}$$

For every represented initial law,

$$\Pr_{\mu_0}\{\tau_{T^+} < \tau_{T^-}\} = \langle \mu_0, h\rangle. \tag{479}$$

Independently of reversibility, suppose a represented nonnegative potential D vanishes on T^+, and the same selected generator satisfies $-LD \ge a > 0$ before T^+. Then

$$\mathbb{E}_{\mu_0}\tau_{T^+} \le \frac{\mathbb{E}_{\mu_0} D}{a}. \tag{480}$$

For a represented quotient potential with defect bounded by $\delta < a$, the denominator becomes $a - \delta$ as in Lemma III.12.10.

Proof. First-step conditioning gives the boundary and harmonic equations for h. For any competitor $f = h + g$ with $g = 0$ on $T^+ \cup T^-$, discrete integration by parts and harmonicity give $\mathcal{E}_\Phi(h, g) = 0$. Hence

$$\mathcal{E}_\Phi(f, f) = \mathcal{E}_\Phi(h, h) + \mathcal{E}_\Phi(g, g) \ge \mathcal{E}_\Phi(h, h),$$

which proves Eq. (478). Equation Eq. (479) is the definition of the committor integrated against μ_0. The hitting-time estimate is Dynkin's formula applied to $D(X_{t\wedge\tau_{T^+}})$, followed by monotone convergence; the quotient-defect refinement is exactly Lemma III.12.10 [31].

□

Theorem V.6.7 (Invariant boundary capacity, occupation, and global-drift obstruction)**.** *Let P be a represented Markov kernel on a finite carrier X, let μ be a represented full-support invariant probability measure, and let Φ_P be the native symmetric geometry obtained from $L_P = P - I$ as in Definition V.5.1. For $A, B \subseteq X$, write*

$$Q_P(A, B) = \sum_{x\in A}\sum_{z\in B} \mu(x)P(x, z).$$

Fix a nonempty proper represented target $T \subsetneq X$, and let $\tau_T = \inf\{t \ge 0 : X_t \in T\}$. The following statements hold for this same selected kernel and target record.

(i) *The target-boundary capacity is the invariant entrance flux:*

$$\mathrm{Cap}_{\Phi_P}(T, T^c) = \mathcal{E}_{\Phi_P}(\mathbf{1}_T, \mathbf{1}_T) = Q_P(T^c, T) = Q_P(T, T^c). \tag{481}$$

More generally, for every represented $F \subseteq T^c$,

$$\mathrm{Cap}_{\Phi_P}(T, F) \le Q_P(T^c, T). \tag{482}$$

(ii) *If $D \ge 0$, $D|_T = 0$, and $(I - P)D \ge a > 0$ on T^c, then*

$$\frac{a}{\mathrm{Osc}(D)} \le \frac{\mu(T)}{1 - \mu(T)}. \tag{483}$$

Thus an invariant full-carrier chain can have uniform normalized drift toward a small target only at the target-mass scale.

(iii) *Let $\mu_0 \ll \mu$ and put $R_0 = \|d\mu_0/d\mu\|_\infty$. For every integer $H \ge 0$,*

$$\Pr_{\mu_0}\{\tau_T \le H\} \le R_0\big[\mu(T) + H\,\mathrm{Cap}_{\Phi_P}(T, T^c)\big]. \tag{484}$$

If $0 \le t_0 \le H$ and the time-t_0 law has density f_{t_0} with respect to μ, then

$$\Pr_{\mu_0}\{\tau_T \le H\} \le \Pr_{\mu_0}\{\tau_T \le t_0\} + \|f_{t_0}\|_\infty\big[\mu(T) + (H - t_0)\,\mathrm{Cap}_{\Phi_P}(T, T^c)\big]. \tag{485}$$

(iv) *Suppose in addition that P is lazy and self-adjoint in $L^2(X,\mu)$, with spectral gap $\gamma_P > 0$. If $f_0 = d\mu_0/d\mu$, then*

$$\begin{aligned}\Pr_{\mu_0}\{\tau_T \le H\} &\le \sum_{t=0}^{H} \mu_0 P^t(T)\\ &\le (H+1)\mu(T) + \sqrt{\mu(T)(1-\mu(T))}\,\|f_0 - 1\|_{2,\mu}\frac{1-(1-\gamma_P)^{H+1}}{\gamma_P}.\end{aligned} \tag{486}$$

In particular, a relative-uniform mixing certificate $\|f_{t_0} - 1\|_\infty \le \varepsilon$, including one obtained from Eq. (463), permits $\|f_{t_0}\|_\infty \le 1 + \varepsilon$ in Eq. (485).

(v) *Let $F \subseteq T^c$ be represented, and let a represented comparison geometry Ψ have the same boundary sets and satisfy $\mathcal{E}_{\Phi_P}(f,f) \le b\mathcal{E}_\Psi(f,f)$ for every represented f in the condenser domain. Then*

$$\mathrm{Cap}_{\Phi_P}(T, F) \le b\,\mathrm{Cap}_\Psi(T, F). \tag{487}$$

Every quantity in these inequalities belongs to the selected native or comparison geometry already classified by Theorem V.1.2; the theorem introduces no additional structural channel.

Proof. Invariance gives equality of the two stationary cut flows:

$$Q_P(T, T^c) = \sum_{x\in T}\mu(x) - Q_P(T,T) = \sum_{z\in T}\mu(z) - Q_P(T,T) = Q_P(T^c, T).$$

For the native symmetric conductance, $\mathcal{E}_{\Phi_P}(\mathbf{1}_T, \mathbf{1}_T)$ is their common value. The boundary conditions for the condenser (T, T^c) force its only competitor to be $\mathbf{1}_T$, proving Eq. (481). The same indicator is an admissible competitor for (T, F), which proves Eq. (482).

Because $0 \le D \le \mathrm{Osc}(D)$ after the normalization $D|_T = 0$, one has $(I-P)D = -PD \ge -\mathrm{Osc}(D)$ on T. Invariance and the drift hypothesis therefore give

$$0 = \mu((I-P)D) \ge a(1-\mu(T)) - \mu(T)\,\mathrm{Osc}(D).$$

Rearrangement proves Eq. (483).

Under the stationary path law, a hit by time H occurs either at time zero or after one of the H transitions from T^c into T. The union bound and stationarity give

$$\Pr_{\mu}\{\tau_T \le H\} \le \mu(T) + HQ_P(T^c, T).$$

The Radon–Nikodym derivative of the full path law started from μ_0 with respect to the stationary path law is $f_0(X_0)$, and hence is at most R_0. This proves Eq. (484). Apply the same argument to the Markov chain started from its time-t_0 law and add the event of a hit through time t_0 to obtain Eq. (485).

For the lazy self-adjoint kernel, put $g_T = \mathbf{1}_T - \mu(T)$. The gap and Cauchy–Schwarz imply

$$\mu_0 P^t(T) = \mu(T) + \langle P^t(f_0 - 1), g_T\rangle_\mu \le \mu(T) + (1-\gamma_P)^t \|f_0 - 1\|_{2,\mu}\sqrt{\mu(T)(1-\mu(T))}.$$

The event $\{\tau_T \le H\}$ is contained in the union of the occupation events $\{X_t \in T\}$. Summing the last display and evaluating the geometric series proves Eq. (486). The relative-uniform statement is immediate from $f_{t_0} \le 1 + \varepsilon$.

Finally, let h_Ψ minimize the comparison condenser energy. It is an admissible competitor for the native condenser, so

$$\mathrm{Cap}_{\Phi_P}(T, F) \le \mathcal{E}_{\Phi_P}(h_\Psi, h_\Psi) \le b\mathcal{E}_\Psi(h_\Psi, h_\Psi) = b\,\mathrm{Cap}_\Psi(T, F).$$

This proves Eq. (487) and the theorem.

□

V.6.4 Invariant-mass and finite-horizon transfer

Theorem V.6.8 (Invariant mass conservation sharpens global target drift)**.** *Let P, μ, Φ_P, and the nonempty proper represented target $T \subsetneq X$ satisfy the hypotheses of Theorem V.6.7. Let $D \ge 0$ be a nonconstant represented potential with $D|_T = 0$, and put*

$$a_T(P, D) = \inf_{x \in T^c} (I - P)D(x). \tag{488}$$

Then the stationary mass balance is the exact identity

$$\int_{T^c} (I-P)D\,d\mu = \int_T (P-I)D\,d\mu = \sum_{x\in T}\sum_{z\in T^c} \mu(x)P(x,z)D(z). \tag{489}$$

Consequently,

$$\frac{\big(a_T(P,D)\big)_+}{\mathrm{Osc}(D)} \le \frac{Q_P(T,T^c)}{1-\mu(T)} = \frac{\mathrm{Cap}_{\Phi_P}(T,T^c)}{1-\mu(T)} \le \frac{\mu(T)}{1-\mu(T)}. \tag{490}$$

Thus a positive drift margin on the whole complement of T is paid for by the stationary boundary flux of the same selected kernel. The statement uses the represented invariant law of that kernel; an invariant law or form from a different record enters through a represented comparison such as Theorem V.6.9 below.

Proof. Invariance gives $\int_X (I-P)D\,d\mu = 0$. Since $D = 0$ on T,

$$-\int_T (I-P)D\,d\mu = \int_T PD\,d\mu = \sum_{x\in T}\sum_{z\in T^c}\mu(x)P(x,z)D(z),$$

which proves Eq. (489). The definition of $a_T(P, D)$ gives

$$\big(a_T(P,D)\big)_+ \mu(T^c) \le \int_{T^c}(I-P)D\,d\mu.$$

Every term in the last sum of Eq. (489) has $0 \le D(z) \le \mathrm{Osc}(D)$. Hence

$$\int_{T^c}(I-P)D\,d\mu \le \mathrm{Osc}(D)Q_P(T,T^c).$$

Divide by $\mathrm{Osc}(D)\mu(T^c) > 0$, and use Eq. (481). Finally, $Q_P(T,T^c) \le \mu(T)$, which proves the last comparison in Eq. (490).

□

Theorem V.6.9 (Density, conductance, capacity, and finite-horizon transfer)**.** *Let* (P, μ, Φ_P) *and* $(\widetilde{P}, \widetilde{\mu}, \Phi_{\widetilde{P}})$ *be two complete represented reversible Markov records on the same finite carrier* X*. For* $x \neq z$*, write*

$$c_P(x,z) = \mu(x)P(x,z), \qquad c_{\widetilde{P}}(x,z) = \widetilde{\mu}(x)\widetilde{P}(x,z).$$

Assume the same record contains constants $0 < r_- \le r_+ < \infty$ *and* $0 < \kappa_- \le \kappa_+ < \infty$ *such that, pointwise,*

$$r_-\mu(x) \le \widetilde{\mu}(x) \le r_+\mu(x), \qquad x \in X, \tag{491}$$

$$\kappa_- c_P(x,z) \le c_{\widetilde{P}}(x,z) \le \kappa_+ c_P(x,z), \qquad x \neq z. \tag{492}$$

Then, for every represented set $A \subseteq X$ *and every pair of disjoint represented plates* $S, F \subseteq X$*,*

$$r_-\mu(A) \le \widetilde{\mu}(A) \le r_+\mu(A), \tag{493}$$

$$\kappa_-\mathcal{E}_{\Phi_P}(f,f) \le \mathcal{E}_{\Phi_{\widetilde{P}}}(f,f) \le \kappa_+\mathcal{E}_{\Phi_P}(f,f), \tag{494}$$

$$\kappa_- \operatorname{Cap}_{\Phi_P}(S,F) \le \operatorname{Cap}_{\Phi_{\widetilde{P}}}(S,F) \le \kappa_+ \operatorname{Cap}_{\Phi_P}(S,F). \tag{495}$$

Fix a nonempty proper represented target T*, an integer* $H \ge 0$*, and an initial law* $\rho_0 \ll \widetilde{\mu}$*. For* $1 < p \le \infty$*, let* $q = p/(p-1)$*, with* $q = 1$ *for* $p = \infty$*, and put* $f_0 = d\rho_0/d\widetilde{\mu}$*. The finite-horizon target probability of the selected kernel* $\widetilde{P}$ *satisfies*

$$\Pr_{\rho_0}^{\widetilde{P}}\{\tau_T \le H\} \le \min\Big\{1, \|f_0\|_{L^p(\widetilde{\mu})} \Big[\min\Big\{1, r_+\mu(T) + H\kappa_+ \operatorname{Cap}_{\Phi_P}(T,T^c)\Big\}\Big]^{1/q}\Big\}. \tag{496}$$

In particular, if $\rho_0 \le R_{\widetilde{\mu}}\widetilde{\mu}$*, then*

$$\Pr_{\rho_0}^{\widetilde{P}}\{\tau_T \le H\} \le \min\Big\{1, R_{\widetilde{\mu}} \Big[r_+\mu(T) + H\kappa_+ \operatorname{Cap}_{\Phi_P}(T,T^c)\Big]\Big\}. \tag{497}$$

If instead the available initialization certificate is $\rho_0 \le R_\mu \mu$*, then*

$$\Pr_{\rho_0}^{\widetilde{P}}\{\tau_T \le H\} \le \min\Big\{1, \frac{R_\mu}{r_-} \Big[r_+\mu(T) + H\kappa_+ \operatorname{Cap}_{\Phi_P}(T,T^c)\Big]\Big\}. \tag{498}$$

The same comparison applies after a represented burn-in: if the time- t_0 *law obeys* $\rho_{t_0} \le R_\mu^{(t_0)}\mu$*, then the right side of Eq. (498), with* H *replaced by* $H - t_0$ *and* R_μ *by* $R_\mu^{(t_0)}$*, is added to* $\Pr_{\rho_0}^{\widetilde{P}}\{\tau_T \le t_0\}$*.*

All four constants are comparisons between the two complete represented records. Their derivation, evaluation precision, and cost occur in the same saturated record as the transferred capacity or hitting bound.

Proof. Summing Eq. (491) over A proves Eq. (493). Reversibility gives

$$\mathcal{E}_{\Phi_P}(f,f) = \frac{1}{2}\sum_{x,z} c_P(x,z)(f(x) - f(z))^2,$$

and the analogous identity for $\widetilde{P}$. Apply Eq. (492) termwise to prove Eq. (494). Taking the infimum over the same condenser boundary conditions proves Eq. (495).

Let $\widetilde{\mathbb{P}}_{\widetilde{\mu}}$ denote the stationary path law of $\widetilde{P}$, and put $E_H = \{\tau_T \le H\}$. The path law started from ρ_0 has Radon–Nikodym derivative $f_0(X_0)$ with respect to $\widetilde{\mathbb{P}}_{\widetilde{\mu}}$. Hölder's inequality gives

$$\Pr_{\rho_0}^{\widetilde{P}}(E_H) \le \|f_0\|_{L^p(\widetilde{\mu})} \left[\widetilde{\mathbb{P}}_{\widetilde{\mu}}(E_H)\right]^{1/q}.$$

Apply Eq. (484) with stationary initial law, followed by Eqs. (493) and (495):

$$\widetilde{\mathbb{P}}_{\widetilde{\mu}}(E_H) \le \widetilde{\mu}(T) + H\,\mathrm{Cap}_{\Phi_{\widetilde{P}}}(T, T^c) \le r_+\mu(T) + H\kappa_+\,\mathrm{Cap}_{\Phi_P}(T, T^c).$$

Clipping probabilities at one proves Eq. (496). The case $p = \infty$ gives Eq. (497). If $\rho_0 \le R_\mu \mu$, then Eq. (491) gives

$$\frac{d\rho_0}{d\widetilde{\mu}} \le \frac{R_\mu}{r_-},$$

which proves Eq. (498). Apply the same argument conditionally from time t_0, and add the event of a hit through t_0, to obtain the burn-in statement.

□

V.6.5 Gibbs–Metropolis specialization

Corollary V.6.10 (Represented Gibbs–Metropolis tilt)**.** *Let P_0 be a complete represented reversible Markov kernel with full-support invariant law μ on the finite carrier X. Let $D : X \to \mathbb{R}$ be a represented bounded potential, let $\theta \ge 0$, and write*

$$D_0 = D - \min_X D, \qquad \Omega_D = \mathrm{Osc}(D), \qquad Z_{\theta,D} = \sum_x \mu(x) e^{-\theta D_0(x)}.$$

Define the Gibbs law and the Metropolis kernel by

$$\mu_{\theta,D}(x) = Z_{\theta,D}^{-1} e^{-\theta D_0(x)} \mu(x), \tag{499}$$

$$P_{\theta,D}(x,z) = P_0(x,z) \min\{1, e^{-\theta(D_0(z) - D_0(x))}\}, \qquad x \ne z, \tag{500}$$

with the diagonal chosen to make every row sum to one. Then $P_{\theta,D}$ is reversible with respect to $\mu_{\theta,D}$, and

$$e^{-\theta\Omega_D} \le \frac{\mu_{\theta,D}(x)}{\mu(x)} \le e^{\theta\Omega_D}, \tag{501}$$

$$e^{-\theta\Omega_D} \le \frac{\mu_{\theta,D}(x) P_{\theta,D}(x,z)}{\mu(x) P_0(x,z)} \le e^{\theta\Omega_D} \qquad (P_0(x,z) > 0). \tag{502}$$

Consequently, for every pair of disjoint represented plates S, F,

$$e^{-\theta\Omega_D}\,\mathrm{Cap}_{\Phi_{P_0}}(S,F) \le \mathrm{Cap}_{\Phi_{P_{\theta,D}}}(S,F) \le e^{\theta\Omega_D}\,\mathrm{Cap}_{\Phi_{P_0}}(S,F). \tag{503}$$

If the represented target satisfies $T \subseteq \mathrm{argmin}_X D$, then

$$\mu_{\theta,D}(T) = \frac{\mu(T)}{Z_{\theta,D}} \le \min\{1, e^{\theta\Omega_D}\mu(T)\}. \tag{504}$$

For every represented target T, every $1 < p \le \infty$, $q = p/(p-1)$, and $f_0 = d\rho_0/d\mu_{\theta,D}$,

$$\Pr_{\rho_0}^{P_{\theta,D}}\{\tau_T \le H\} \le \min\Big\{1, \|f_0\|_{L^p(\mu_{\theta,D})}\Big[\min\Big\{1, e^{\theta\Omega_D}\big(\mu(T) + H\,\mathrm{Cap}_{\Phi_{P_0}}(T,T^c)\big)\Big\}\Big]^{1/q}\Big\}. \tag{505}$$

An initialization bound $\rho_0 \le R_\mu \mu$ gives the explicit specialization

$$\Pr_{\rho_0}^{P_{\theta,D}}\{\tau_T \le H\} \le \min\Big\{1, R_\mu e^{2\theta\Omega_D}\big(\mu(T) + H\,\mathrm{Cap}_{\Phi_{P_0}}(T,T^c)\big)\Big\}. \tag{506}$$

If, in addition, $T \subseteq \operatorname{argmin}_X D$, the shared normalizer cancels and the sharper bound is

$$\Pr_{\rho_0}^{P_{\theta,D}}\{\tau_T \le H\} \le \min\Big\{1, R_\mu e^{\theta\Omega_D}\big(\mu(T) + H\,\mathrm{Cap}_{\Phi_{P_0}}(T,T^c)\big)\Big\}. \tag{507}$$

Finally, let $F \ge 0$ be a nonconstant represented potential with $F|_T = 0$, and put $a_T(P_{\theta,D}, F) = \inf_{x\notin T}(I - P_{\theta,D})F(x)$. Then

$$\frac{\big(a_T(P_{\theta,D},F)\big)_+}{\mathrm{Osc}(F)} \le \frac{\mathrm{Cap}_{\Phi_{P_{\theta,D}}}(T,T^c)}{1-\mu_{\theta,D}(T)}. \tag{508}$$

Whenever $e^{\theta\Omega_D}\mu(T) < 1$, the right side is at most

$$\frac{e^{\theta\Omega_D}\,\mathrm{Cap}_{\Phi_{P_0}}(T,T^c)}{1-e^{\theta\Omega_D}\mu(T)}. \tag{509}$$

The represented tilt record contains the construction of D_0, the normalizer or its certified comparison, the Metropolis acceptance rule, and the precision used in the displayed constants. The factor $e^{\theta\Omega_D}$ is a quantitative density and conductance loss; the description length of D supplies no replacement for this factor.

Proof. For $x \ne z$, detailed balance follows from

$$\begin{aligned}\mu_{\theta,D}(x)P_{\theta,D}(x,z) &= \frac{\mu(x)P_0(x,z)}{Z_{\theta,D}}e^{-\theta D_0(x)}\min\{1, e^{-\theta(D_0(z)-D_0(x))}\}\\ &= \frac{\mu(x)P_0(x,z)}{Z_{\theta,D}}e^{-\theta\max\{D_0(x),D_0(z)\}},\end{aligned}$$

which is symmetric in x, z. Since $0 \le D_0 \le \Omega_D$,

$$e^{-\theta\Omega_D} \le Z_{\theta,D} \le 1.$$

This proves Eqs. (501) and (502). Apply Theorem V.6.9 with

$$r_- = \kappa_- = e^{-\theta\Omega_D}, \qquad r_+ = \kappa_+ = e^{\theta\Omega_D}$$

to obtain Eqs. (503) and (505). When $D_0 = 0$ on T, summing Eq. (499) over T gives Eq. (504). If $\rho_0 \le R_\mu\mu$, then $d\rho_0/d\mu_{\theta,D} \le R_\mu e^{\theta\Omega_D}$; combine this with the bracket in Eq. (505) for $p = \infty$ to prove Eq. (506).

For $x \in T$ and $z \notin T$, the target-minimum hypothesis gives

$$\mu_{\theta,D}(x)P_{\theta,D}(x,z) = \frac{e^{-\theta D_0(z)}}{Z_{\theta,D}}\mu(x)P_0(x,z) \leq \frac{1}{Z_{\theta,D}}\mu(x)P_0(x,z).$$

Summing the target cut and using Eq. (481) yields $\mathrm{Cap}_{\Phi_{P_{\theta,D}}}(T,T^c) \leq Z_{\theta,D}^{-1}\,\mathrm{Cap}_{\Phi_{P_0}}(T,T^c)$. Moreover, $d\rho_0/d\mu_{\theta,D} \leq R_\mu Z_{\theta,D}e^{\theta\Omega_D}$. The stationary entrance-count bound Eq. (484) cancels $Z_{\theta,D}$, proving Eq. (507).

Apply Theorem V.6.8 to the selected kernel $P_{\theta,D}$ and potential F to prove Eq. (508). The upper capacity comparison and $\mu_{\theta,D}(T) \leq e^{\theta\Omega_D}\mu(T)$ then give Eq. (509).

□

The capacity, density, and Gibbs comparisons above enter the native finite-horizon coordinate of Corollary V.8.4 and the typed dynamical certificate of Definition V.12.2. Each use retains its selected invariant law, initialization density, conductance comparison, and oscillation loss in one complete record.

V.7 Affordable Caus Action and Capacity Under Valid Lifts

The capacity and drift estimates above apply to the selected native carrier. The present section gives the corresponding accounting and comparison tools for authorized directed flow and for valid lifted carriers. All suprema below range over complete records. In particular, the selected dynamics, target interface, reference law, comparison maps, numerical precision, and every evaluation used by the bound occur in the same saturated record.

V.7.1 Caus action, aiming, and target contact

Definition V.7.1 (Complete Caus target-flow record)**.** Fix a finite represented carrier, an orientation E^+ of its active unoriented edges, and a budget B. A complete Caus target-flow record consists of

(i) a selected positive generator or transition kernel, its legal reference law, and the authority envelope of Definitions II.4.3 and II.4.4;

(ii) the actual authorized current $J = (J_e)_{e\in E^+}$, rather than an independently chosen current with the same formal type;

(iii) a represented target-normalized potential D, available before the transition on which it is used, together with its target and failure gates;

(iv) positive represented edge mobilities $a = (a_e)_{e\in E^+}$, the horizon or resolvent scale, and every comparison used to convert flow into contact, selector, or hitting units; and

(v) the complete derivation, storage, arithmetic, solver, comparison, and precision costs of the preceding data.

The record is *prospective* when its target-aligned entries satisfy Definition III.21.1. With $\nabla_e D = D(e^+) - D(e^-)$, define

$$\mathcal{A}_{\mathsf{Caus}}(J;a) = \sum_{e\in E^+} \frac{J_e^2}{a_e}, \tag{510}$$

$$\mathcal{E}_{\mathrm{aim}}(D;a) = \sum_{e\in E^+} a_e(\nabla_e D)^2, \tag{511}$$

$$\mathcal{P}_{\mathsf{Caus}}(J,D)=\langle J,-\nabla D\rangle_{E^+}. \tag{512}$$

The first quantity is the *Caus action*, the second is the *target-aiming energy*, and the third is the signed target power. When the mobility-compatible coordinates in Definitions III.12.8 and III.12.9 are chosen as $w_e=a_e$ and $F_e=J_e/a_e$, the power is their unnormalized authorized alignment numerator. Any other coordinate choice requires a represented comparison before the two pairings are identified.

Suppose additionally that the record contains positive oriented stationary or instantaneous fluxes $\varphi_e,\varphi_{\bar e}$, with the represented orientation convention $J_e=\varphi_e-\varphi_{\bar e}$, and uses the logarithmic-mean mobility

$$a_e=\frac{\varphi_e-\varphi_{\bar e}}{\log\varphi_e-\log\varphi_{\bar e}},$$

with the continuous diagonal value. Then $\mathcal{A}_{\mathsf{Caus}}$ is the edge entropy production

$$\sum_{e\in E^+}(\varphi_e-\varphi_{\bar e})\log\frac{\varphi_e}{\varphi_{\bar e}}.$$

The same terminology is permitted for an authorized Hodge projection only when the record proves contraction in this inverse-mobility norm. In every other case the quantity in Eq. (510) retains the name Caus action.

Theorem V.7.2 (Caus action–aiming inequality). *Every complete Caus target-flow record satisfies*

$$\left(\mathcal{P}_{\mathsf{Caus}}(J,D)_+\right)^2\le\mathcal{A}_{\mathsf{Caus}}(J;a)\mathcal{E}_{\mathrm{aim}}(D;a). \tag{513}$$

For a measurable time-dependent record on $[0,H]$*,*

$$\int_0^H\mathcal{P}_{\mathsf{Caus}}(J_t,D_t)_+\,dt\le\left(\int_0^H\mathcal{A}_{\mathsf{Caus}}(J_t;a_t)\,dt\right)^{1/2}\left(\int_0^H\mathcal{E}_{\mathrm{aim}}(D_t;a_t)\,dt\right)^{1/2}. \tag{514}$$

The discrete-time version replaces both integrals by sums.

Proof. Write $u_e=J_e/\sqrt{a_e}$ and $v_e=-\sqrt{a_e}\nabla_eD$. Cauchy–Schwarz gives $|\langle u,v\rangle|^2\le\|u\|_2^2\|v\|_2^2$, which is Eq. (513). Apply Cauchy–Schwarz once more in time to obtain Eq. (514). □

Definition V.7.3 (Saturated Caus action profiles). Let $\mathscr{R}^{\mathrm{pre}}_{\mathsf{Caus}}(B,H)$ be the represented family of prospective complete Caus target-flow records of cost at most B and horizon H. Define

$$\mathrm{Act}^{\mathrm{sat}}_{\mathsf{Caus}}(B,H)=\sup_{R\in\mathscr{R}^{\mathrm{pre}}_{\mathsf{Caus}}(B,H)}\int_0^H\mathcal{A}_{\mathsf{Caus},R}(t)\,dt,$$

$$\mathrm{Aim}^{\mathrm{sat}}_{\mathsf{Caus}}(B,H)=\sup_{R\in\mathscr{R}^{\mathrm{pre}}_{\mathsf{Caus}}(B,H)}\int_0^H\mathcal{E}_{\mathrm{aim},R}(t)\,dt,$$

$$\mathrm{Flow}^{\mathrm{sat}}_{\mathsf{Caus}}(B,H)=\sup_{R\in\mathscr{R}^{\mathrm{pre}}_{\mathsf{Caus}}(B,H)}\int_0^H\mathcal{P}_{\mathsf{Caus},R}(t)_+\,dt.$$

Empty suprema are zero. A record with an unavailable upper action or aiming certificate has value $+\infty$ in the corresponding upper profile. These profiles are accounting sub-suprema of the prospective source class in Definition III.23.3; their family-uniform descriptions and the optimization used to evaluate them are included in the saturated cost.

A represented *aiming-to-contact converter* is a same-record inequality

$$\mathrm{Hit}_{\mathrm{contact}}(R) \le B_R^{\mathrm{contact}} + K_R \int_0^H \mathcal{P}_{\mathsf{Caus},R}(t)_+ \, dt + \delta_R, \tag{515}$$

where the left side is the total contact in the prospective-selector, hitting-probability, or discounted-arrival normalization. A converter that directly bounds total contact uses $B_R^{\mathrm{contact}} = 0$; a converter that only bounds advantage uses its represented neutral-contact baseline, such as Enum_ν. The quantities $B_R^{\mathrm{contact}}, K_R, \delta_R$ are represented before the relevant transitions. A committor may supply this converter only through the represented derivation required by Theorem V.6.3.

Corollary V.7.4 (Uniform affordable Caus-flow bound)**.** *For every B, H,*

$$\mathrm{Flow}_{\mathsf{Caus}}^{\mathrm{sat}}(B,H) \le \sqrt{\mathrm{Act}_{\mathsf{Caus}}^{\mathrm{sat}}(B,H)\,\mathrm{Aim}_{\mathsf{Caus}}^{\mathrm{sat}}(B,H)}. \tag{516}$$

If all records in a represented subclass have $B_R^{\mathrm{contact}} \le B_B^{\mathrm{contact}}$, $K_R \le K_B$, and $\delta_R \le \delta_B$ in Eq. (515)*, then their total contact is at most*

$$B_B^{\mathrm{contact}} + K_B \sqrt{\mathrm{Act}_{\mathsf{Caus}}^{\mathrm{sat}}(B,H)\,\mathrm{Aim}_{\mathsf{Caus}}^{\mathrm{sat}}(B,H)} + \delta_B. \tag{517}$$

Without a represented converter, the action and aiming profiles remain flow coordinates and supply no probability bound.

Proof. Apply Eq. (514) to every record and then enlarge each factor to its saturated supremum. Substitution into Eq. (515) proves the second assertion. □

Example V.7.5 (Action does not determine aiming)**.** Let the oriented edges be e_1, e_2, take $a_{e_1} = a_{e_2} = 1$, and let $-\nabla D = (1,0)$. The currents $J^{\parallel} = (1,0)$ and $J^{\perp} = (0,1)$ have the same Caus action, equal to one, whereas $\mathcal{P}_{\mathsf{Caus}}(J^{\parallel}, D) = 1$ and $\mathcal{P}_{\mathsf{Caus}}(J^{\perp}, D) = 0$. These vectors may be realized on a finite graph by a target-directed edge and a disjoint cycle edge. Hence an action or entropy-production bound evaluates total directed activity; the aiming factor evaluates its concentration on the represented target potential.

V.7.2 Noise-corrected Caus aiming

The action of an actual represented current is evaluated on the selected record and requires no label correction. Label noise enters when a learned potential is used as a proxy for the latent clean target potential.

Definition V.7.6 (Clean-aiming comparison record)**.** Augment a complete Caus target-flow record by a complete dynamical label-noise record of Definition IV.16.1. A clean-aiming comparison contains a latent clean target-normalized analytic comparator D_0, the represented learned potential $\widehat{D}$, the same edge mobilities a_t, and one of the following represented certificates.

(i) *Linear homogeneous debiasing.* The population corrupted potential is $D_\eta = \rho D_0$, after the declared centering, and

$$\left(\int_0^H \mathcal{E}_{\mathrm{aim}}(\widehat{D}_t - D_{\eta,t}; a_t)\, dt\right)^{1/2} \le \varepsilon_{\mathrm{aim}}.$$

(ii) *Calibrated potential comparison.* A represented calibration theorem gives directly

$$\left(\int_0^H \mathcal{E}_{\mathrm{aim}}(\widehat{D}_t - D_{0,t}; a_t)\, dt\right)^{1/2} \le \varepsilon_{\mathrm{cal}}.$$

(iii) *Binary hard-potential fallback.* Both potentials take values in $\{0,1\}$. Let d^b be the represented behavior law and assume the deployment comparisons $d_t^\pi \le W_t d^b$. Under the represented deployment laws d_t^π, assume

$$\Pr_{d_t^\pi}\{\widehat{D}_t \ne D_{0,t}\} \le U^{\mathrm{clean}}_{\mathrm{gate,dep}} \qquad (0 \le t \le H),$$

where the latter is a represented behavior-to-deployment consequence of the dynamical clean-risk bound,

$$U^{\mathrm{clean}}_{\mathrm{gate,dep}} \le \min\left\{1, \left(\sup_{0\le t\le H} W_t\right) U^{\mathrm{clean}}_{\mathrm{noise}}(n, b(n), m, \eta_+, \delta)\right\}.$$

Let the represented time-dependent gradient operators satisfy

$$\kappa_{\mathrm{aim,H}} = \int_0^H \|\nabla_t\|^2_{L^2(d_t^\pi)\to\ell^2(a_t)}\, dt < \infty.$$

The noise rate, centering, calibration, gradient norm, deployment comparison, and their computation and precision are part of the same saturated record. A general score or ranking potential enters this definition through case (i) or (ii); scalar label correlation alone is not a gradient-energy comparison. The comparator D_0 is used to state and audit the clean-risk comparison; it is not a prospective source or a policy input unless an independent pre-hit represented derivation constructs it. Source ancestry remains that of $\widehat{D}$ and its complete training record.

Theorem V.7.7 (Dynamical noise correction for clean Caus aiming)**.** *In case* (i) *of Definition V.7.6,*

$$\left(\int_0^H \mathcal{E}_{\mathrm{aim}}(D_{0,t}; a_t)\, dt\right)^{1/2} \le \frac{\left(\int_0^H \mathcal{E}_{\mathrm{aim}}(\widehat{D}_t; a_t)\, dt\right)^{1/2} + \varepsilon_{\mathrm{aim}}}{\rho}. \tag{518}$$

In case (ii)*, the right side is*

$$\left(\int_0^H \mathcal{E}_{\mathrm{aim}}(\widehat{D}_t; a_t)\, dt\right)^{1/2} + \varepsilon_{\mathrm{cal}}.$$

In case (iii)*,*

$$\int_0^H \mathcal{E}_{\mathrm{aim}}(\widehat{D}_t - D_{0,t}; a_t)\, dt \le \kappa_{\mathrm{aim,H}} U^{\mathrm{clean}}_{\mathrm{gate,dep}}. \tag{519}$$

At each time the squared $L^2(d_t^\pi)$ distance appearing in this estimate is exactly

$$\|\widehat{D}_t - D_{0,t}\|^2_{L^2(d_t^\pi)} = \Pr_{d_t^\pi}\{\widehat{D}_t \ne D_{0,t}\}.$$

For each record, minimize its valid same-unit aiming errors over these three cases; let $\varepsilon^{\mathrm{sat}}_{\mathrm{aim}}$ be the supremum of that minimum over the complete budget slice. Let $\mathrm{Aim}^{\mathrm{sat}}_{\mathsf{Caus,obs}}$ be the observed-potential aiming profile. Every represented clean aiming-to-contact converter then gives

$$U^{\mathrm{noise}}_{\mathsf{Caus}} \le \min\left\{1,\ B_B^{\mathrm{contact}} + K_B\sqrt{\mathrm{Act}^{\mathrm{sat}}_{\mathsf{Caus}}}\,\frac{\sqrt{\mathrm{Aim}^{\mathrm{sat}}_{\mathsf{Caus,obs}}} + \varepsilon^{\mathrm{sat}}_{\mathrm{aim}}}{\rho} + \delta_B\right\} \tag{520}$$

in the linear homogeneous case. The calibrated case omits the division by ρ. In the hard-potential branch, preserve the recordwise order:

$$\varepsilon^{\mathrm{sat}}_{\mathrm{aim,hard}} = \sup_{\mathcal{R}\in\mathscr{R}^{\mathrm{hard}}_{B,H}} \sqrt{\kappa_{\mathrm{aim,H}}(\mathcal{R})U^{\mathrm{clean}}_{\mathrm{gate,dep}}(\mathcal{R})} \le \sqrt{\left(\sup_{\mathcal{R}}\kappa_{\mathrm{aim,H}}(\mathcal{R})\right)\left(\sup_{\mathcal{R}}U^{\mathrm{clean}}_{\mathrm{gate,dep}}(\mathcal{R})\right)}. \tag{521}$$

Here $\mathscr{R}^{\mathrm{hard}}_{B,H}$ is the complete affordable hard-gate record class at the displayed ceiling and horizon, and every unlabeled supremum on the right ranges over that same class. Use the left side as the hard-branch candidate in $\varepsilon^{\mathrm{sat}}_{\mathrm{aim}}$. This does not multiply minima attained by different records. All profiles in this display use the same horizon, mobilities, target normalization, deployment law, and class-preserving enlarged ceiling. For a hard gate learned at ceiling $b(n)$ from m labels with Massart rate at most η_+, substitute the displayed behavior-to-deployment bound on $U^{\mathrm{clean}}_{\mathrm{gate,dep}}$. The polynomial-capacity conclusion is the same display with $b(n) = b_C(n) = n^C$.

Proof. The map $D \mapsto (\int \mathcal{E}_{\mathrm{aim}}(D;a))^{1/2}$ is a seminorm. The triangle inequality applied to $\rho D_0 = \widehat{D} - (\widehat{D} - D_\eta)$ proves Eq. (518); the calibrated case is identical without rescaling. The definition of $\kappa_{\mathrm{aim,H}}$ gives the first inequality in Eq. (519) after integration in time, and the squared L^2 distance between binary gates is their misclassification probability. Combine these comparisons with Theorem V.7.2 and Corollary V.7.4 and the represented aiming-to-contact converter to obtain Eq. (520).

□

V.7.3 Variational capacity comparison under valid lifts

Definition V.7.8 (Complete form-lift comparison record)**.** Let $(\Omega, \mathcal{G}_{\mathrm{obs}}, \pi, \Lambda)$ be a valid lift with fiber laws κ_x, pullback U, averaging map P_κ, and $Q_\pi = I - UP_\kappa$ as in Theorem II.10.5. A complete form-lift comparison record contains a full-support represented base law μ, its disintegrated lifted law

$$\widetilde{\mu}(\omega) = \mu(\pi(\omega))\kappa_{\pi(\omega)}(\omega),$$

a nonnegative self-adjoint represented operator H_Ω on $L^2(\operatorname{supp}\widetilde{\mu}, \widetilde{\mu})$, and represented disjoint base boundary sets S, T. It also contains the induced form, all comparison constants, boundary and target gates, pseudoinverse or solver, precision, and complete construction cost. In these Hilbert spaces U is an isometry, $P_\kappa = U^*$, and Q_π is the orthogonal fiber projection.

Let $\mathcal{Q}_0$ be the subspace of $\operatorname{ran} Q_\pi$ whose members vanish on $\pi^{-1}(S \cup T)$, and let Q_0 denote its orthogonal projection. Set

$$H_X = U^* H_\Omega U, \qquad H_{F,0} = Q_0 H_\Omega Q_0|_{\mathcal{Q}_0}, \qquad B_0 = Q_0 H_\Omega U.$$

For a base function g with positive energy define the leakage ratio

$$\rho_\pi = \sup_{\langle g, H_X g\rangle > 0} \frac{\langle B_0 g, H^+_{F,0} B_0 g\rangle}{\langle g, H_X g\rangle}. \tag{522}$$

The supremum is restricted to the represented condenser domain when the comparison is used only for (S, T).

Theorem V.7.9 (Schur-complement capacity theorem for valid lifts)**.** *For every complete form-lift comparison record, positivity implies $B_0 g \in \operatorname{ran} H_{F,0}$ and $0 \le \rho_\pi \le 1$. Minimization over fiber fluctuations gives the effective base operator*

$$H_{\mathrm{eff}} = H_X - B_0^* H^+_{F,0} B_0. \tag{523}$$

For the pulled-back condenser $(\pi^{-1}S, \pi^{-1}T)$,

$$(1-\rho_\pi)\operatorname{Cap}_X(S,T) \le \operatorname{Cap}_\Omega(\pi^{-1}S, \pi^{-1}T) \le \operatorname{Cap}_X(S,T), \tag{524}$$

where Cap_X *is computed with the pullback form* $g \mapsto \langle g, H_X g\rangle$. *If a represented comparison form* $\mathcal{E}_\Phi$ *satisfies*

$$a_\pi \mathcal{E}_\Phi(g,g) \le \langle g, H_X g\rangle \le b_\pi \mathcal{E}_\Phi(g,g),$$

then

$$(1-\rho_\pi)a_\pi \operatorname{Cap}_\Phi(S,T) \le \operatorname{Cap}_\Omega(\pi^{-1}S, \pi^{-1}T) \le b_\pi \operatorname{Cap}_\Phi(S,T). \tag{525}$$

In particular, $B_0 = 0$ *gives exact equality between the lifted and pullback capacities.*

Proof. Relative to $\operatorname{ran} U \oplus \mathcal{Q}_0$, the quadratic form of H_Ω at $Ug + q$, with $q \in \mathcal{Q}_0$, is

$$\langle g, H_X g\rangle + 2\langle q, B_0 g\rangle + \langle q, H_{F,0} q\rangle.$$

For a finite positive block matrix, $B_0 g$ lies in $\operatorname{ran} H_{F,0}$; otherwise a vector in $\ker H_{F,0}$ paired with $B_0 g$ would make the displayed quadratic polynomial negative for one sign of its scalar coefficient. Completing the square shows that the minimum over q is attained at $-H_{F,0}^{+} B_0 g$ modulo $\ker H_{F,0}$, with value $\langle g, H_{\mathrm{eff}} g\rangle$. Positivity gives $0 \le B_0^* H_{F,0}^{+} B_0 \le H_X$, hence $0 \le \rho_\pi \le 1$.

The lifted boundary conditions constrain g on S, T and constrain q to vanish on the corresponding boundary fibers. The same completion of squares on that condenser domain gives $(1-\rho_\pi)\langle g, H_X g\rangle \le \langle g, H_{\mathrm{eff}} g\rangle \le \langle g, H_X g\rangle$. Taking the boundary infimum proves Eq. (524). Form comparison before the same infimum proves Eq. (525).

□

Definition V.7.10 (Saturated lift-capacity amplification)**.** For a represented class $\mathscr{L}(B)$ of complete form-lift records, require each $R \in \mathscr{L}(B)$ to carry a represented external comparison form $\mathcal{E}_{\Phi,R}$ and a finite constant $b_{\pi,R}$ satisfying the upper comparison in Theorem V.7.9. If an application starts from a larger form-lift class, $\mathscr{L}(B)$ below denotes its comparison-bearing subclass. The supremum of an empty subclass is zero; a record without such a comparison has the neutral external-capacity bound $+\infty$ and is not inserted into this finite profile. Put

$$\operatorname{LiftAmp}^{\mathrm{sat}}_{\mathscr{L}}(B) = \sup_{R \in \mathscr{L}(B)} b_{\pi,R}, \qquad \operatorname{LiftLeak}^{\mathrm{sat}}_{\mathscr{L}}(B) = \sup_{R \in \mathscr{L}(B)} \rho_{\pi,R}.$$

Then, at the class-preserving enlarged ceiling containing the lift and comparison computations,

$$\sup_{R \in \mathscr{L}(B)} \operatorname{Cap}_{\Omega,R} \le \operatorname{LiftAmp}^{\mathrm{sat}}_{\mathscr{L}}(B) \sup_{R \in \mathscr{L}(B)} \operatorname{Cap}_{\Phi,R}. \tag{526}$$

Capacities in this display use the represented common rate normalization. The valid-lift identity fixes the target interface and preserves target fractions; evaluation of $\operatorname{LiftAmp}^{\mathrm{sat}}_{\mathscr{L}}$ is the additional uniform computation over instances of the lift class.

V.7.4 Dynamic lift transfer

Definition V.7.11 (Dynamic lift blocks and structural subclasses)**.** For a complete valid-lift record and a lifted generator L_Ω, use the orthogonal disintegration above and define

$$L_X = P_\kappa L_\Omega U, \qquad E_+ = Q_\pi L_\Omega U, \qquad E_- = P_\kappa L_\Omega Q_\pi, \qquad L_F = Q_\pi L_\Omega Q_\pi.$$

Thus E_+ is the lift-normal-form defect E_π in the orthogonal disintegration; it is not a second structural record. The following represented subclasses will be used.

(i) $\mathscr{L}_{\mathrm{int}}$: exact represented intertwining, $E_+ = 0$;
(ii) $\mathscr{L}_{\mathrm{form}}$: form-neutral lifts, $Q_\pi H_\Omega U = 0$;
(iii) $\mathscr{L}_{\mathrm{fm}}(M_F, \kappa_F, \varepsilon_+, \varepsilon_-)$: fast-mixing weakly coupled lifts satisfying, in one declared operator norm,

$$\|e^{tL_F} Q_\pi\| \le M_F e^{-\kappa_F t}, \qquad \|E_+\| \le \varepsilon_+, \quad \|E_-\| \le \varepsilon_-;$$

(iv) $\mathscr{L}_{\mathrm{all}}$: all complete valid-lift records, with no numerical intertwining or fiber-mixing assumption.

Every constant and norm conversion belongs to the same record. A Poincaré, sector, or hypocoercive certificate supplies M_F, κ_F only through a represented theorem in the displayed norm. MLSI and statistical β-mixing estimates are not substituted without an explicit conversion to that norm.

Theorem V.7.12 (Block-resolvent transfer for valid lifts)**.** *Let $\lambda > 0$ lie in the resolvent sets required below. Put $R_F(\lambda) = (\lambda - L_F)^{-1}$ on* $\operatorname{ran} Q_\pi$*. Then*

$$P_\kappa(\lambda - L_\Omega)^{-1} U = \left[\lambda - L_X - E_- R_F(\lambda) E_+\right]^{-1}. \tag{527}$$

For $R \in \mathscr{L}_{\mathrm{int}}$, the projected lifted resolvent equals $(\lambda - L_X)^{-1}$. When these are the killed or absorbing generators for transported target and failure gates, exact initialization and pullback gates give equality of the corresponding discounted first-arrival functionals.

For $R \in \mathscr{L}_{\mathrm{fm}}$,

$$\|R_F(\lambda) Q_\pi\| \le \frac{M_F}{\lambda + \kappa_F}, \qquad \|E_- R_F(\lambda) E_+\| \le \frac{\varepsilon_- M_F \varepsilon_+}{\lambda + \kappa_F}. \tag{528}$$

Writing $R_X = (\lambda - L_X)^{-1}$ and $\sigma_\lambda = \varepsilon_- M_F \varepsilon_+/(\lambda + \kappa_F)$, if $\|R_X\|\sigma_\lambda < 1$, then

$$\left\| P_\kappa(\lambda - L_\Omega)^{-1} U - R_X \right\| \le \frac{\|R_X\|^2 \sigma_\lambda}{1 - \|R_X\|\sigma_\lambda}. \tag{529}$$

Pairing this estimate with represented initial and target functionals gives the same bound multiplied by their dual norms, plus their represented interface errors. For a valid exact pullback these interface errors vanish.

Proof. The block matrix of $\lambda - L_\Omega$ on $\operatorname{ran} U \oplus \operatorname{ran} Q_\pi$ is

$$\begin{pmatrix} \lambda - L_X & -E_- \\ -E_+ & \lambda - L_F \end{pmatrix}.$$

Block Gaussian elimination gives Eq. (527). Exact intertwining sets $E_+ = 0$. For the fast-fiber class, integrate the semigroup estimate to obtain the first inequality in Eq. (528); submultiplicativity gives the second. The resolvent identity followed by the Neumann-series bound proves Eq. (529). Pairing with the two boundary functionals proves the arrival statement.

$\square$

Theorem V.7.13 (Clocked finite-horizon lift transfer)**.** *Let P_t^Ω and P_t^X, $0 \le t < H$, be represented one-step kernels on compatible clock slices, acting backward on observables, with represented pullbacks $U_t : L^\infty(X_t) \to L^\infty(\Omega_t)$. For a first-arrival conclusion these are the kernels stopped, killed, or made absorbing at the transported target and failure gates. Define the pre-transition intertwining defect*

$$E_t = P_t^\Omega U_{t+1} - U_t P_t^X.$$

Then

$$\begin{aligned}P_0^{\Omega} \cdots P_{H-1}^{\Omega} U_H - U_0 P_0^X \cdots P_{H-1}^X \\ = \sum_{t=0}^{H-1} P_0^{\Omega} \cdots P_{t-1}^{\Omega} E_t P_{t+1}^X \cdots P_{H-1}^X,\end{aligned} \tag{530}$$

where an empty product is the identity on its clock slice. In a supremum or total-variation norm in which the kernels are contractions, the norm of the left side is at most $\sum_{t=0}^{H-1} \|E_t\|$*. Exact intertwining gives equality of the projected finite-horizon laws. With exact valid-lift initialization and target gates for the stopped dynamics, the same estimates hold for finite-horizon first arrival; approximate represented interfaces add their two errors. For unrestricted kernels the conclusion is an occupation-law comparison, not a first-arrival comparison.*

Each E_t*, comparator, and interface is derived before transition* t *and charged at the complete clocked-record cost. Statistical blocking may certify an estimator of these defects, but it does not replace the displayed dynamical comparison.*

Proof. For $0 \le t \le H$, put

$$Z_t = P_0^{\Omega} \cdots P_{t-1}^{\Omega} U_t P_t^X \cdots P_{H-1}^X,$$

with the empty products interpreted as identities. Thus the left side of Eq. (530) is $Z_H - Z_0$, while

$$Z_{t+1} - Z_t = P_0^{\Omega} \cdots P_{t-1}^{\Omega} (P_t^{\Omega} U_{t+1} - U_t P_t^X) P_{t+1}^X \cdots P_{H-1}^X.$$

Summing these identities proves the displayed telescoping formula. Kernel and pullback contraction, followed by the triangle inequality, gives the norm bound. Pairing the resulting observable comparison with the represented initial law and the transported terminal indicator proves the first-arrival statement.

□

Theorem V.7.14 (Fast-fiber and clocked-defect Lift certificate split)**.** *Fix one complete valid-Lift record with transported target and failure interfaces. The following two numerical transfer branches use the existing dynamic blocks of Definition V.7.11.*

(i) *Suppose the record contains a fast-fiber certificate*

$$\|e^{tL_F} Q_\pi\| \le M_F e^{-\kappa_F t}, \qquad \|E_+\| \le \varepsilon_+, \qquad \|E_-\| \le \varepsilon_-,$$

at a represented discount $\lambda > 0$*. Put*

$$\sigma_\lambda = \frac{\varepsilon_- M_F \varepsilon_+}{\lambda + \kappa_F}, \qquad \Delta_{\mathrm{fm}}(\lambda) = \frac{\|R_X(\lambda)\|^2 \sigma_\lambda}{1 - \|R_X(\lambda)\| \sigma_\lambda}, \tag{531}$$

where $R_X(\lambda) = (\lambda - L_X)^{-1}$*, and assume* $\|R_X(\lambda)\| \sigma_\lambda < 1$*. For every represented base row functional* ℓ *and base column* g*,*

$$\left| \ell \Big[P_\kappa (\lambda - L_\Omega)^{-1} U - R_X(\lambda) \Big] g \right| \le \|\ell\|_* \, \|g\| \, \Delta_{\mathrm{fm}}(\lambda). \tag{532}$$

Let $a_{0,T}^{\Omega}$ *and* $a_{0,T}^X$ *denote the represented lifted and base initial target atoms, respectively. If represented discounted-arrival functionals* $A_T^{\Omega}(\lambda)$ *and* $A_T^X(\lambda)$ *satisfy the endpoint interface comparisons*

$$\left|A_T^{\Omega}(\lambda) - a_{0,T}^{\Omega} - \ell P_\kappa(\lambda - L_\Omega)^{-1} U g\right| \le \varepsilon_{\Omega,\lambda}^{\mathrm{int}},$$
$$\left|A_T^{X}(\lambda) - a_{0,T}^{X} - \ell R_X(\lambda) g\right| \le \varepsilon_{X,\lambda}^{\mathrm{int}},$$

and

$$\left|a_{0,T}^{\Omega} - a_{0,T}^{X}\right| \le \varepsilon_{0,T}^{\mathrm{int}},$$

then

$$\left|A_T^{\Omega}(\lambda) - A_T^{X}(\lambda)\right| \le \varepsilon_{0,T}^{\mathrm{int}} + \varepsilon_{\Omega,\lambda}^{\mathrm{int}} + \varepsilon_{X,\lambda}^{\mathrm{int}} + \|\ell\|_* \,\|g\|\, \Delta_{\mathrm{fm}}(\lambda). \tag{533}$$

Exact initialization and exact pullback target contact set all three interface errors to zero with $\ell = \mu_{0,N}$ and $g = k_T$ in the target-resolvent normalization of Definition III.8.1.

(ii) *Suppose the record supplies represented clock slices, stopped or absorbing one-step kernels, and the defects*

$$E_t = P_t^{\Omega} U_{t+1} - U_t P_t^{X}, \qquad 0 \le t < H,$$

in a contraction norm. If its initialization and terminal-gate interface errors are bounded by $\varepsilon_{\mathrm{init}}$ and $\varepsilon_{\mathrm{gate}}$, then

$$\left|p_{T,H}^{\Omega} - p_{T,H}^{X}\right| \le \min\left\{1, \varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \|E_t\| + \varepsilon_{\mathrm{gate}}\right\}. \tag{534}$$

This branch applies to every represented clocked record carrying these defects, including the complement of the certified fast-fiber branch. Its right side contains no fiber decay rate, off-diagonal coupling norm, or resolvent smallness condition.

When both branches are certified, both conclusions hold in their respective units. A represented conversion between discounted arrival and the chosen finite horizon permits the corresponding converted upper bounds to be minimized. Without such a conversion, Eq. (533) *remains a discounted-arrival estimate and Eq.* (534) *remains a finite-horizon first-arrival estimate.*

When neither numerical branch is certified, valid-Lift target-fraction preservation, source inheritance, every separately certified form-capacity statement whose hypotheses hold, and the subprofile relation of Definition V.7.15 remain available. The numerical access coordinate then takes the neutral default prescribed by Definition V.12.2. Thus the certificate split changes no structural Lift classification and introduces no additional source channel.

Proof. Under the hypotheses of part (i), Eq. (528) gives

$$\|E_- R_F(\lambda) E_+\| \le \sigma_\lambda.$$

The Feshbach identity and the Neumann resolvent estimate in Eq. (529) therefore give

$$\left\|P_\kappa(\lambda - L_\Omega)^{-1} U - R_X(\lambda)\right\| \le \Delta_{\mathrm{fm}}(\lambda).$$

Dual pairing with ℓ and g proves Eq. (532). Insert the two projected pairings and the two initial target atoms between $A_T^{\Omega}(\lambda)$ and $A_T^{X}(\lambda)$, and apply the triangle inequality, to obtain Eq. (533).

For part (ii), Eq. (530) is an exact telescoping identity. Kernel and pullback contraction bound its norm by $\sum_t \|E_t\|$. Pairing with the represented initial law and terminal indicator, then adding the initialization and gate-interface errors, proves Eq. (534). This derivation uses only the clocked record, so it remains valid whenever the fast-fiber hypotheses or their smallness condition are unavailable. The statements about simultaneous certificates and typed minima follow from applying both valid inequalities to the same complete record.

□

Definition V.7.15 (Saturated lift-access envelope)**.** Let $\mathrm{LiftSel}^{\mathrm{sat}}_{\mathscr{L}}(B,H)$ be the supremum of the prospective selector advantage over complete records in a represented lift class $\mathscr{L}$, and let $\mathrm{LiftAcc}^{\mathrm{sat}}_{\mathscr{L}}(B,\lambda)$ be the analogous discounted-arrival supremum. Both are sub-suprema of the existing $\mathrm{Sel}^{\mathrm{sat}}$ and target-resolvent profiles, so

$$\mathrm{LiftSel}^{\mathrm{sat}}_{\mathscr{L}}(B,H) \le \mathrm{Sel}^{\mathrm{sat}}(B). \tag{535}$$

For $\mathscr{L}_{\mathrm{int}}$, discounted access transfers exactly. For a clocked class with $\sum_t \|E_t\| \le \varepsilon_H$, finite-horizon access is at most the base access plus ε_H and the interface errors. For $\mathscr{L}_{\mathrm{fm}}$, discounted access is at most the base access plus the right side of Eq. (529) paired with its boundary norms.

For $\mathscr{L}_{\mathrm{all}}$, validity and Theorem II.10.5 provide complete classification and source inheritance. Its numerical access envelope remains the displayed supremum until a uniform form, intertwining, aiming, or fiber-resolvent profile evaluates it.

Example V.7.16 (Product, weakly coupled, and slow fibers)**.** For $\Omega = X \times F$, projection to X, product initialization, and $L_\Omega = L_X \otimes I + I \otimes L_F$, one has $E_+ = 0$ and $Q_\pi H_\Omega U = 0$. Pulled-back capacity, projected resolvents, and target arrival therefore equal their base values. If off-diagonal couplings have norms ε_+ and ε_-, while the fiber decay is $M_F e^{-\kappa_F t}$, the correction has scale $\varepsilon_- M_F \varepsilon_+/(\lambda+\kappa_F)$. A represented family with fixed couplings and $\kappa_F \downarrow 0$ loses uniform smallness on long resolvent scales $\lambda \downarrow 0$; at fixed positive λ, the displayed denominator remains at least λ. Thus target-fraction preservation alone does not supply a dynamic lift-access bound; the relevant structural comparison is the charged fiber resolvent.

V.8 Noisy Target Interfaces and Lifted Capacity

A valid lift transports the latent clean target. A target or failure flag learned from corrupted labels is an approximate boundary interface until a represented recovery or stability theorem relates it to that clean target.

Definition V.8.1 (Dynamical clean-gate and capacity-stability profiles)**.** Let T and F be the disjoint latent clean target and failure events fixed by the task semantics. For $G \in \{T,F\}$, let $\widehat{g}_G : X \to \{0,1\}$ be a represented learned gate and let

$$U^{\mathrm{clean}}_{\mathrm{gate},G} = \Pr_{d^b}\{\widehat{g}_G(X) \ne \mathbf{1}_G(X)\} \le U^{\mathrm{clean}}_{\mathrm{noise},G}(n, b(n), m, \eta_{G,+}, \delta_G)$$

under a represented behavior law d^b, by Eq. (330). An exact gate has error zero. If the time-t deployment law satisfies $d^\pi_t \le W_t d^b$, define

$$\varepsilon^{\mathrm{dep}}_{T,F,H} = \min\left\{1, \sum_{t=0}^{H} W_t \left(U^{\mathrm{clean}}_{\mathrm{gate},T} + U^{\mathrm{clean}}_{\mathrm{gate},F}\right)\right\}. \tag{536}$$

The events T, F are analytic comparators in this definition, not pre-hit inputs to the learned controller. They become prospective sources only if an independent represented derivation constructs them within the temporal rules. For direct independent homogeneous flip reads of both gates at deployment, define the fallback

$$\varepsilon^{\mathrm{flip}}_{T,F,H} = 1 - (1-\eta_T)^{H+1}(1-\eta_F)^{H+1} \le (H+1)(\eta_T+\eta_F). \tag{537}$$

Capacity is measured under the reversible reference law μ, which need not equal d^b. A capacity comparison therefore also contains a represented density bound $\mu \le W_{\mu\leftarrow b} d^b$, or direct bounds on the same quantities. Put

$$\widehat{F} = \{x : \widehat{g}_F(x) = 1\}, \qquad \widehat{T} = \{x : \widehat{g}_T(x) = 1\} \setminus \widehat{F}. \tag{538}$$

Record

$$\begin{aligned}
\varepsilon^{\mu}_F &= \min\{1, W_{\mu\leftarrow b} U^{\mathrm{clean}}_{\mathrm{gate},F}\} \ge \mu(F \triangle \widehat{F}),\\
\varepsilon^{\mu}_T &= \min\{1, W_{\mu\leftarrow b}(U^{\mathrm{clean}}_{\mathrm{gate},T} + U^{\mathrm{clean}}_{\mathrm{gate},F})\} \ge \mu(T \triangle \widehat{T}).
\end{aligned} \tag{539}$$

Direct-bound branches replace the displayed products by their certified upper bounds. Removing $\widehat{F}$ resolves learned target–failure conflicts; because $T \cap F = \varnothing$, the additional target error is at most the failure-gate error.

For a represented reversible form Φ and $\varepsilon_T, \varepsilon_F \in [0,1]$, define the charged joint boundary-stability modulus

$$\omega^{\mathrm{Cap}}_{\Phi,T,F}(\varepsilon_T,\varepsilon_F) = \sup_{\substack{T',F' \text{ represented}\\ T'\cap F'=\varnothing\\ \mu(T\triangle T')\le\varepsilon_T\\ \mu(F\triangle F')\le\varepsilon_F}} \left|\mathrm{Cap}_\Phi(T,F) - \mathrm{Cap}_\Phi(T',F')\right|. \tag{540}$$

When the failure gate is exact, write $\omega^{\mathrm{Cap}}_{\Phi,T}(\varepsilon) = \omega^{\mathrm{Cap}}_{\Phi,T,F}(\varepsilon,0)$. The represented family of admissible boundary pairs, its optimization, form evaluation, and precision belong to the same cost record. A missing deployment or reference-law comparison gives the neutral value one for its probability error. A missing boundary-stability computation gives $+\infty$ for capacity error.

Theorem V.8.2 (Noise transfer through target gates and valid lifts)**.** *Let a clocked valid lift satisfy the hypotheses of Theorem V.7.13, with initialization error $\varepsilon_{\mathrm{init}}$, one-step defects E_t, and a learned target gate from Definition V.8.1. Then the clean finite-horizon first-arrival probability and the represented lifted first-arrival probability satisfy*

$$\left|p^{\mathrm{clean}}_{T,H} - \widehat{p}^{\Omega}_{T,H}\right| \le \min\left\{1,\, \varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \|E_t\| + \varepsilon^{\mathrm{dep}}_{T,F,H}\right\}. \tag{541}$$

The direct-read version replaces the last term by $\varepsilon^{\mathrm{flip}}_{T,F,H}$.

For discounted killed dynamics, let $\mu_{0,N}$ and $\widehat{\mu}_{0,N}$ be the clean and represented initial row functionals on the compared nonterminal space, and let k_T and $\widehat{k}_T$ be the corresponding clean and learned terminal-contact columns of Definition III.8.1. Assume the same-norm represented comparisons

$$\|\mu_{0,N} - \widehat{\mu}_{0,N}\|_* \le \varepsilon_{0,\lambda}, \qquad \|k_T - \widehat{k}_T\| \le \varepsilon_{T,\lambda},$$

and $|\mu_0(T) - \widehat{\mu}_0(\widehat{T})| \le \varepsilon_{0,T}$. The contact-vector comparison must be derived from the learned gate and the represented transition or rate operator; classification risk alone is not substituted for this norm. Writing R_{eff} for the effective lifted killed resolvent in Eq. (527)*, put*

$$A^{\mathrm{clean}}_T(\lambda) = \mu_0(T) + \mu_{0,N} R_{\mathrm{eff}} k_T, \qquad \widehat{A}^{\Omega}_T(\lambda) = \widehat{\mu}_0(\widehat{T}) + \widehat{\mu}_{0,N} R_X \widehat{k}_T.$$

Then

$$\left| A_T^{\mathrm{clean}}(\lambda) - \widehat{A}_T^{\Omega}(\lambda) \right| \le \varepsilon_{0,T} + \varepsilon_{0,\lambda} \|R_{\mathrm{eff}}\| \, \|k_T\| + \|\widehat{\mu}_{0,N}\|_* \|R_{\mathrm{eff}} - R_X\| \, \|k_T\| + \|\widehat{\mu}_{0,N}\|_* \|R_X\| \, \varepsilon_{T,\lambda}. \tag{542}$$

Exact intertwining or the fast-fiber estimate of Theorem V.7.12 supplies the middle resolvent term. When the failure gate is learned, this same-norm term also contains its represented killed-generator interface error; equivalently, a resolvent identity may bound that contribution by $\|R_{\mathrm{eff}}\| \, \|L_N - \widehat{L}_{\widehat{N}}\| \, \|R_X\|$, after the declared common-space identification of the two killed generators.

Finally, let a complete form-lift record satisfy $\mathcal{E}_X \le b_\pi \mathcal{E}_\Phi$, and use the conflict-resolved $\widehat{T}$ of Eq. (538)*. Then*

$$\mathrm{Cap}_\Omega(\pi^{-1}\widehat{T}, \pi^{-1}\widehat{F}) \le b_\pi \left[\mathrm{Cap}_\Phi(T,F) + \omega^{\mathrm{Cap}}_{\Phi,T,F}(\varepsilon^\mu_T, \varepsilon^\mu_F) \right]. \tag{543}$$

If either

(a) *represented target and failure scores both have uniform error smaller than their clean classification margins, or*

(b) *the finite carrier has minimum atom mass $\mu_* = \min_x \mu(x) > 0$ and $\max\{\varepsilon^\mu_T, \varepsilon^\mu_F\} < \mu_*$,*

then $\widehat{T} = T$ and $\widehat{F} = F$, so the original clean-boundary lift-capacity theorem applies directly to the realized gates. Under condition (b)*, every represented boundary pair admitted by the modulus with those error tolerances equals (T, F); hence additionally $\omega^{\mathrm{Cap}}_{\Phi,T,F}(\varepsilon^\mu_T, \varepsilon^\mu_F) = 0$. Condition* (a) *alone does not force this global supremum to vanish, because its admissible class also contains boundary pairs other than the realized margin-certified gates. Every noise term in the three comparisons is therefore a function of $(n, b(n), m, \eta_{T,+}, \eta_{F,+}, \delta_T, \delta_F)$ through $U^{\mathrm{clean}}_{\mathrm{noise}}$, followed only by the displayed charged deployment, contact-vector, and reference-law converters. Setting $b(n) = n^C$ gives the polynomial-capacity specialization.*

Proof. Under the behavior-to-deployment comparison, the probability of gate error at time t is at most $W_t(U^{\mathrm{clean}}_{\mathrm{gate},T} + U^{\mathrm{clean}}_{\mathrm{gate},F})$. A union bound over the stopped path gives Eq. (536). Add this event to the exact telescoping comparison of Eq. (530) to prove Eq. (541). Independent direct flips give $\Pr\{\text{at least one erroneous read}\} = 1 - (1-\eta_T)^{H+1}(1-\eta_F)^{H+1}$.

For the discounted statement, add the initial target-atom discrepancy and write the noninitial terms in the target-resolvent orientation $\mu_{0,N} R k_T$. Insert and subtract $\widehat{\mu}_{0,N} R_{\mathrm{eff}} k_T$ and $\widehat{\mu}_{0,N} R_X k_T$, then apply the dual and primal norms and the triangle inequality. For capacity, form comparison gives

$$\mathrm{Cap}_\Omega(\pi^{-1}\widehat{T}, \pi^{-1}\widehat{F}) \le b_\pi \, \mathrm{Cap}_\Phi(\widehat{T}, \widehat{F}),$$

and Eq. (539) together with the definition of $\omega^{\mathrm{Cap}}_{\Phi,T,F}$ gives Eq. (543).

A uniform error below the classification margin preserves every gate sign. For a binary gate on a finite carrier, any nonempty error set has μ-mass at least μ_*. Hence $\max\{\varepsilon^\mu_T, \varepsilon^\mu_F\} < \mu_*$ forces both error sets to be empty. Both cases give $\widehat{T} = T$, $\widehat{F} = F$, and the exact realized-gate conclusion. In the atom-mass case, every admissible error set in the definition of the modulus is also empty, so the modulus itself vanishes. The margin case bypasses the modulus by applying the exact theorem to its realized gates.

□

Theorem V.8.3 (Fast-mode singleton target reach and effective dimension)**.** *Let X have $N \geq 2$ elements, let μ be uniform, and let H_Φ be a represented target-independent nonnegative self-adjoint operator whose kernel consists exactly of the constants. On $\mathring{L}^2(X, \mu) = \mathbf{1}^\perp$, define*

$$f_s = \frac{\mathbf{1}_{\{s\}} - N^{-1}}{\sqrt{N^{-1}(1 - N^{-1})}}, \qquad \Pi_{\geq\lambda} = \mathbf{1}_{[\lambda,\infty)}(H_\Phi), \qquad d_\lambda = \operatorname{rank} \Pi_{\geq\lambda}.$$

Then

$$\frac{1}{N} \sum_{s \in X} \|\Pi_{\geq\lambda} f_s\|_{2,\mu}^2 = \frac{d_\lambda}{N-1} \leq \frac{2N_{\mathrm{eff},\Phi}(\lambda)}{N-1}. \tag{544}$$

Consequently, for every $r > 0$,

$$\frac{1}{N} \left|\left\{s \in X : \|\Pi_{\geq\lambda} f_s\|_{2,\mu}^2 \geq r\right\}\right| \leq \frac{d_\lambda}{(N-1)r}. \tag{545}$$

If the nonzero spectrum of H_Φ is bounded below by λ, then

$$N_{\mathrm{eff},\Phi}(\lambda) \geq \frac{N-1}{2}. \tag{546}$$

Thus a target-independent geometry that mixes every full-carrier direction at scale λ necessarily retains effective dimension of order the carrier size. The singleton vectors in this theorem are analytic target probes; their evaluation does not supply a represented target source.

Proof. The centered singleton vectors form a unit-norm tight frame on $\mathring{L}^2(X, \mu)$:

$$\frac{1}{N} \sum_{s \in X} f_s \otimes f_s = \frac{1}{N-1} I_{\mathring{L}^2}.$$

Taking the trace after multiplication by $\Pi_{\geq\lambda}$ proves the equality in Eq. (544). Every eigenvalue $u \geq \lambda$ contributes $u/(u + \lambda) \geq 1/2$ to $N_{\mathrm{eff},\Phi}(\lambda)$, whence $d_\lambda \leq 2N_{\mathrm{eff},\Phi}(\lambda)$. Markov's inequality applied to the tight-frame average proves Eq. (545). If every nonconstant mode has eigenvalue at least λ, each of the $N - 1$ such modes contributes at least $1/2$ to the effective dimension, proving Eq. (546).

$\square$

Corollary V.8.4 (Dynamical envelope for nonlinear nonquotient geometry)**.** *Let a budget-admissible nonlinear construction remain after every qualified nonidentity quotient postprocessing has been assigned as in Proposition III.13.8, and suppose it is used prospectively on the full carrier. At the class-preserving ceiling of its complete record, its structural source is exactly the ambient* Geom/Caus *record selected by Corollary III.22.1. For its selected native kernel P, target T, horizon H, invariant law μ, and initial density $f_0 = d\mu_0/d\mu \leq R_0$, one has simultaneously*

$$\Pr_{\mu_0}\{\tau_T \leq H\} \leq R_0\big[\mu(T) + H \operatorname{Cap}_{\Phi_P}(T, T^c)\big], \tag{547}$$

and every represented nonnegative target-normalized global potential D with $\operatorname{Osc}(D) > 0$ used by its authorized current satisfies

$$\inf_{x \notin T} \frac{(I - P)D(x)}{\operatorname{Osc}(D)} \leq \frac{\mu(T)}{1 - \mu(T)}. \tag{548}$$

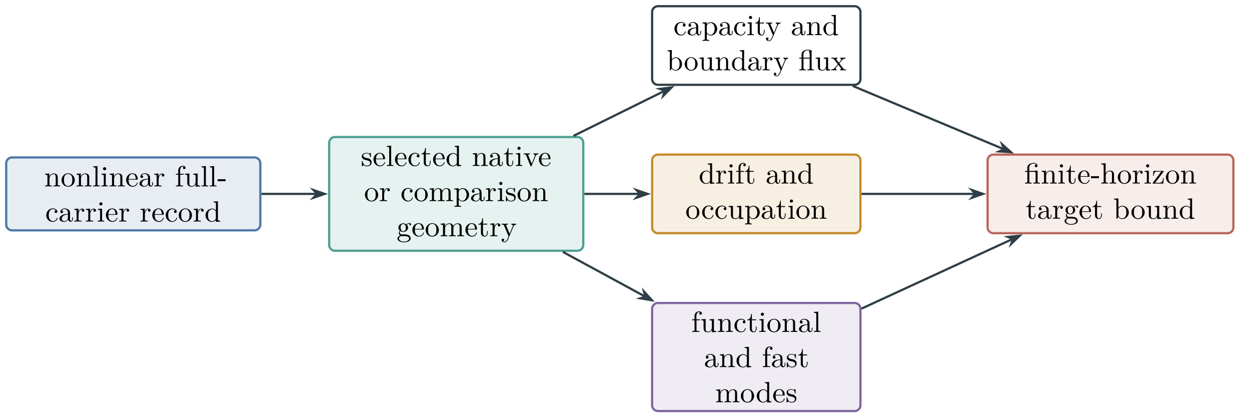

Figure 84: A prospective nonlinear construction that supplies no qualified quotient remains one ambient Geom/Caus record. Capacity, invariant drift, occupation, and functional-mode estimates evaluate that same source before its target-arrival bound is taken.

When the same native record is lazy and self-adjoint, Eq. (486) *supplies the additional finite-horizon occupation bound. When it is compared to a represented reversible geometry, Eq.* (487) *transfers its condenser bound with the declared form loss. Its audited functional-inequality profile supplies the mixing, smoothing, killed-set, transport, and hypocoercive bounds of Theorem V.5.8; a target-independent full-carrier operator also satisfies Eqs.* (544) *and* (546)*. The right sides of Eqs.* (486) *and* (547) *are in the same finite-horizon probability unit as* U_{hit} *in Definition V.12.2; the coordinate therefore uses their minimum whenever they occur in the same complete record.*

The committor and Bellman values of the same record remain governed by Theorems V.6.2 and V.6.3, and a learned record retains the simultaneous Part IV and controlled-learning bounds in the U_{learn} *coordinate of Definition V.12.2. All these estimates attach to the one already classified ambient source and its authorized current. Neither nonlinear evaluation nor dynamical iteration creates another source coordinate.*

Proof. The nonlinear closure proposition and prospective-routing corollary assign the complete construction either to a qualified quotient mode or to the full-carrier ambient mode. Under the hypothesis of the corollary, the quotient modes have already been assigned, so its prospective structure is the latter mode. The selected policy and kernel are ancestors of that mode by Theorem V.1.2; Hodge authority is Proposition V.5.2.

Apply Eq. (484) to the selected native kernel to obtain Eq. (547). Put

$$a_D = \inf_{x \notin T} (I - P)D(x).$$

If $a_D \le 0$, Eq. (548) is immediate. If $a_D > 0$, apply Eq. (483) with $a = a_D$. This proves Eq. (548). The comparison, functional-inequality, fast-mode, committor, Bellman, and learning conclusions are the cited same-record theorems. Temporal admissibility and complete-cost inheritance are Definition III.21.1 and Proposition II.5.9. Thus every displayed numerical certificate evaluates an existing Geom/Caus coordinate and none changes the exhaustive structural classification. □

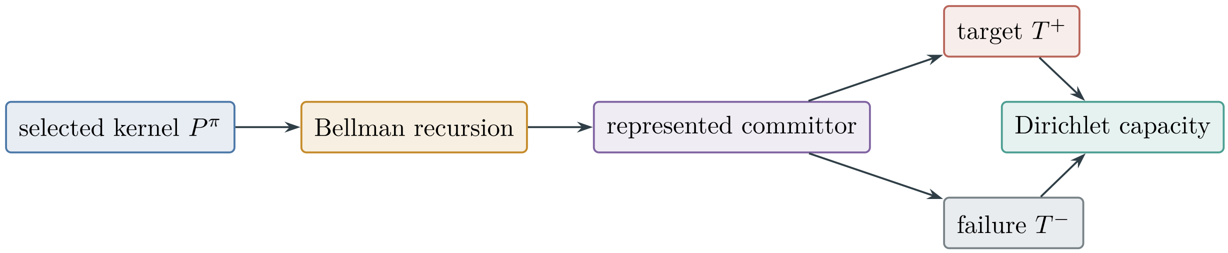

Figure 85: Bellman iteration produces a prospective committor only through its charged pre-transition derivation. On a reversible carrier the same reach–avoid function is the condenser-energy minimizer.

V.8.1 Observation regularity and sparse-reward exploration

Definition V.8.5 (Complete sparse-reward observation record)**.** Fix a finite represented carrier X, a finite target index Θ, target sectors $T_\theta \subseteq X$, and a refined horizon H. A complete sparse-reward observation record is

$$\mathcal{R}_{\mathrm{cur}} = (P_{0:H-1}, \phi, d_{\mathcal{O}}, \kappa_\varepsilon, M_{0:H}, \widehat{F}_{0:H-1}, b_{0:H-1}, K_{0:H-1}, V, \mathscr{C}_{\mathrm{comp}}),$$

with the following data.

(i) P_t is the actual executed clocked transition kernel after the action and execution channels have been composed. The public target reward is sparse:

$$r_t^{\mathrm{ext}} = 0 \quad \text{on } \{\tau > t\}, \qquad \tau = \inf\{t : X_t \in T_\Theta\}.$$

Any terminal reward or verifier flag is revealed only at its actual clock time.

(ii) $\phi : X \to (\mathcal{O}, d_{\mathcal{O}})$ is a represented observation map at declared evaluation precision. A represented quantizer $\kappa_\varepsilon : \mathcal{O} \to \mathcal{C}_\varepsilon$ gives $q_\varepsilon = \kappa_\varepsilon \circ \phi$. Put

$$M_\varepsilon(\phi) = |q_\varepsilon(X)|\,.$$

The quantizer, its cells, and $M_\varepsilon(\phi)$ are used numerically only when their evaluation or certified bound is included in the record.

(iii) M_t is the represented exploration memory before transition $t+1$. It includes every retained count, visitation sketch, replay item, learned embedding, random-network target, predictor parameter, ensemble state, diversity archive, and random seed used by the intrinsic signal. The represented predictor $\widehat{F}_t$, intrinsic reward b_t, and selector K_t are measurable functions of the public pre-hit observation history, M_t, and fresh target-independent randomness. Their training, update, selection, storage, precision, and deployment operations are represented before the transition they influence.

(iv) V is the public verifier and target interface. A target score, terminal flag, later observation, future predictor error, or completed first-hit record is unavailable before its clock time in the sense of Definition III.21.1.

(v) If q_ε is used only as an observation, no quotient dynamics is asserted. If cell dynamics is used in a target-arrival comparison, $\mathscr{C}_{\mathrm{comp}}$ is the exact or nonzero-defect master certificate of Definition III.18.1. It contains the represented quotient kernel, within-fiber geometry, symmetric and skew defects, target vectors, stability or resolvent comparison, discount scale, current authority, output converter, and evaluation precision.

In the resource monoid, the complete cost is

$$\begin{aligned}B_{\mathrm{cur}} &= B_{\mathrm{pres}} \oplus B_{\phi} \oplus B_{d} \oplus B_{\kappa,\varepsilon} \oplus B_{M_\varepsilon} \oplus B_{\mathrm{reg}} \oplus B_{\mathrm{info}} \\ &\quad \oplus B_{\mathrm{mem}} \oplus B_{\mathrm{pred}} \oplus B_{\mathrm{train}} \oplus B_{r^{\mathrm{ext}}} \oplus B_{\mathrm{bonus}} \oplus B_{\mathrm{pol}} \\ &\quad \oplus B_{\mathrm{exec}} \oplus B_{\mathrm{clock}} \oplus B_{\mathrm{verify}} \oplus B_{\mathrm{out}} \oplus B_{\mathrm{cap}} \oplus B_{\mathrm{comp}}, && (549)\\ B_{\mathrm{reg}} &= B_{\mu} \oplus B_{P} \oplus B_{\mathcal{E}_\phi} \oplus B_{\mathrm{Var}_\phi} \oplus B_{\widehat{\mathcal{E}}_\phi} \oplus B_{\widehat{\mathrm{Var}}_\phi} \oplus B_{\Gamma_E} \oplus B_{\Gamma_V}, && (550)\\ B_{\mathrm{info}} &= B_{Z_0} \oplus B_{C_\varepsilon} \oplus B_{\mathrm{reveal}} \oplus B_{\mathrm{channel}} \oplus B_{\mathrm{SDPI}} \oplus B_{J^{\mathrm{fib}}} \oplus B_{\beta^{\mathrm{fib}}} \oplus B_{\mathrm{conf}}, && (551)\\ B_{\mathrm{cap}} &= B_{P^{\mathrm{ref}}} \oplus B_{\mu^{\mathrm{ref}}} \oplus B_{\widetilde{\mu}} \oplus B_{r_\pm} \oplus B_{\kappa_\pm} \oplus B_{\mathrm{Cap}} \oplus B_{\rho_0} \oplus B_{f_0} \oplus B_{p}, && (552)\\ B_{\mathrm{comp}} &= B_{q} \oplus B_{\mathrm{fib}} \oplus B_{L_q} \oplus B_{E_q^{\mathsf{s}}} \oplus B_{E_q^{\mathsf{a}}} \oplus B_{\Phi_Q} \oplus B_{\Phi_\perp} \\ &\quad \oplus B_{D_Q} \oplus B_{J_Q} \oplus B_{\mathfrak{K}_q} \oplus B_{\mathrm{stab}} \oplus B_{\mathrm{res}} \oplus B_{\overline{A}_q} \oplus B_{S_q} \\ &\quad \oplus B_{M_{\mathfrak{G}_q}} \oplus B_{C_{\mathfrak{G}_q}} \oplus B_{R_{\mathfrak{G}_q}} \oplus B_{\rho_{\mathfrak{G}_q}} \oplus B_{\mathrm{targ}} \oplus B_{\mathrm{auth}} \\ &\quad \oplus B_{\mathrm{prec}} \oplus B_{\mathrm{select}} \oplus B_{\mathrm{compare}}. && (553)\end{aligned}$$

An absent slot has zero resource charge only when it is not used. Reuse, adaptive updates, and repeated evaluations accumulate according to Definition I.2.5. The regularity line charges the invariant law and executed kernel, population and empirical energy and variance, and both simultaneous deviation certificates. The information line charges the initial record, target-cell comparator, ordered reveals and their channels, contraction certificates, within-cell information and neutral contact, and confidence bookkeeping. The capacity line charges the reference kernel and law, selected invariant law, density and conductance comparisons, condenser-capacity evaluation, initial law and density, and the selected Hölder exponents. The slots in the final display charge, respectively, the support and fiber records, quotient generator and both defects, quotient and within-fiber forms, potential, current, causal provenance, stability and resolvent records, quotient utility and dynamic factor, all four geometric visibility factors, target interface, authority, precision, certificate selection, and the final same-normalization comparison. Shared ancestors are represented once in the derivation DAG; an additional evaluation is not free merely because its input was already stored.

Suppose P is a represented kernel with invariant law μ. The metric observation variance, Dirichlet energy, and regularity ratio are

$$\mathrm{Var}_\mu^d(\phi) = \frac{1}{2}\sum_{x,z\in X} \mu(x)\mu(z) d_{\mathcal{O}}(\phi(x),\phi(z))^2, \tag{554}$$

$$\mathcal{E}_P^d(\phi) = \frac{1}{2}\sum_{x,z\in X} \mu(x)P(x,z) d_{\mathcal{O}}(\phi(x),\phi(z))^2, \tag{555}$$

$$\rho_\phi(P) = \begin{cases} \mathcal{E}_P^d(\phi)/\operatorname{Var}_\mu^d(\phi), & \operatorname{Var}_\mu^d(\phi) > 0,\\ 0, & \operatorname{Var}_\mu^d(\phi) = 0.\end{cases} \tag{556}$$

These are analytic coordinates of the represented record. A numerical use requires their represented evaluation or a certified upper bound at the cost in Eq. (549).

Theorem V.8.6 (Observation regularity, fiber information, and sparse-reward exploration)**.** *Let $\mathcal{R}_{\mathrm{cur}}$ be a complete record from Definition V.8.5 of cost $B_{\mathrm{cur}} \preceq B$. Assume that Θ is uniform on $\{1,\dots,N\}$ and that the indexed target sectors are the disjoint singletons $T_\theta = \{x_\theta\}$. Define*

$$C_\varepsilon = q_\varepsilon(x_\Theta).$$

At time t, let $\mathcal{H}_t^{\mathrm{obs}}$ be the complete pre-hit observation, memory, predictor, intrinsic-reward, action, and execution history available before K_t is sampled. Give the analytic comparator the additional cell label C_ε, and put

$$J_{R,t}^{\mathrm{fib}} = I\Big(\Theta; \mathcal{H}_t^{\mathrm{obs}} \mid \mathsf{Z}_0, C_\varepsilon\Big). \tag{557}$$

Equivalently, an ordered-reveal record may replace the right side by the certified upper bound

$$J_{R,t}^{\mathrm{fib},\uparrow} = \sum_{j=1}^{L_{R,t}} \mathbb{E}\min\{c_{R,t,j}, \vartheta_{R,t,j} a_{R,t,j}\}$$

from Theorem V.3.3, after conditioning every channel slot on $(\mathsf{Z}_0, C_\varepsilon)$. Every target-dependent coordinate available at time zero is included in Z_0; every target-dependent coordinate revealed later is included in one of the ordered channel slots. Thus the displayed sum cannot omit information used by the selector. Throughout the theorem, $J_{R,t}^{\mathrm{fib},\uparrow}$ denotes either this certified upper bound or the exactly evaluated and charged value $J_{R,t}^{\mathrm{fib}}$. Its neutral value is $+\infty$ when neither is available.

Let $\mathbb{Q}_{R,t}^{\mathrm{fib}}$ be the law obtained by retaining $(\mathsf{Z}_0, C_\varepsilon, \mathcal{H}_t^{\mathrm{obs}})$ and resampling Θ from $\mathcal{L}(\Theta \mid \mathsf{Z}_0, C_\varepsilon)$, independently of the observation history. Define the within-cell neutral contact

$$\beta_{R,t}^{\mathrm{fib}} = \mathbb{Q}_{R,t}^{\mathrm{fib}}\{\tau > t,\ X_{t+1} \in T_\Theta\}. \tag{558}$$

Put

$$p_{R,t} = \mathbb{P}_{R,t}\{\tau > t,\ X_{t+1} \in T_\Theta\}.$$

Then the survival-weighted next-contact probability $p_{R,t}$ and the selector advantage satisfy

$$\begin{aligned}(\mathrm{Adv}_t(K_t; \nu_t))_+ &\le p_{R,t} \\ &\le \min\left\{1,\ \beta_{R,t}^{\mathrm{fib}} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{fib},\uparrow}}\right\}.\end{aligned} \tag{559}$$

On the survival sector the external reward history is deterministic, hence

$$I\Big(\Theta; r_{0:t}^{\mathrm{ext}} \mid \mathsf{Z}_0, C_\varepsilon, \tau > t\Big) = 0. \tag{560}$$

The intrinsic rewards, predictor states, and exploration memories are causal postprocessings already contained in $\mathcal{H}_t^{\mathrm{obs}}$; data processing gives them no additional information coordinate beyond $J_{R,t}^{\mathrm{fib},\uparrow}$, while their construction and deployment costs remain in B_{cur}. Suppose, in addition, that the initial record is target-permutation-neutral, meaning

$$\mathbb{P}\{\Theta = \theta \mid \mathsf{Z}_0 = z\} = \frac{1}{N} \quad (1 \le \theta \le N)$$

for every realized z, and that, conditional on $(\mathsf{Z}_0, C_\varepsilon)$, the only target-index restriction is membership in the displayed observation fiber. If $M_\varepsilon(\phi)$ is bounded pointwise by the represented value, then

$$\beta_{R,t}^{\mathrm{fib}} \le \frac{M_\varepsilon(\phi)}{N}, \tag{561}$$

and therefore

$$(\mathrm{Adv}_t(K_t; \nu_t))_+ \le \min\left\{1,\ \frac{M_\varepsilon(\phi)}{N} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{fib},\uparrow}}\right\}. \tag{562}$$

For a horizon H, summing the disjoint first-contact events gives

$$\Pr\{\tau \le H\} \le \min\left\{1,\ \sum_{t=0}^{H-1}\left(\beta^{\mathrm{fib}}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{fib},\uparrow}_{R,t}}\right)\right\}. \tag{563}$$

If, conditional on each target index θ, the same complete record contains the reversible density, conductance, capacity, and initialization comparison of Theorem V.6.9, then, with $q = p/(p-1)$, it also supplies

$$U^{\mathrm{cur,cap}}_R(H) = \max_{1\le\theta\le N} \min\left\{1, \|f_{0,\theta}\|_{L^p(\widetilde{\mu}_\theta)} \left[\min\left\{1, r_{\theta,+}\mu_\theta(T_\theta) + H\kappa_{\theta,+}\operatorname{Cap}_{\Phi_{P_\theta}}(T_\theta, T^c_\theta)\right\}\right]^{1/q}\right\}. \tag{564}$$

For that record,

$$\Pr\{\tau \le H\} \le \min\left\{U^{\mathrm{cur,cap}}_R(H),\ \sum_{t=0}^{H-1}\left(\beta^{\mathrm{fib}}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{fib},\uparrow}_{R,t}}\right)\right\}, \tag{565}$$

with the second candidate clipped at one. Every comparison used in Eq. (564) *carries the cost Eq.* (552)*, including evaluation or certification of the displayed maximum.*

The regularity coordinate controls the magnitude of a one-step intrinsic signal. If, under a stationary transition $(X_0, X_1) \sim \mu(dx)P(x, dz)$,

$$0 \le b_\phi(X_0, X_1) \le a_\phi + L_\phi d_{\mathcal{O}}(\phi(X_0), \phi(X_1)),$$

then

$$\mathbb{E}_{\mu P} b_\phi \le a_\phi + L_\phi\sqrt{2\mathcal{E}^d_P(\phi)} = a_\phi + L_\phi\sqrt{2\rho_\phi(P)\operatorname{Var}^d_\mu(\phi)}. \tag{566}$$

If instead $b_\phi \le a_\phi + L_\phi d^2_{\mathcal{O}}$, then

$$\mathbb{E}_{\mu P} b_\phi \le a_\phi + 2L_\phi\mathcal{E}^d_P(\phi) = a_\phi + 2L_\phi\rho_\phi(P)\operatorname{Var}^d_\mu(\phi). \tag{567}$$

If these quantities are estimated from a represented trajectory sample and the same record supplies simultaneous deviations

$$\mathcal{E}^d_P(\phi) \le \widehat{\mathcal{E}}^d_P(\phi) + \Gamma_E, \qquad \operatorname{Var}^d_\mu(\phi) \ge \widehat{\operatorname{Var}}^d_\mu(\phi) - \Gamma_V, \tag{568}$$

then, on that event,

$$\rho_\phi(P) \le \frac{\widehat{\mathcal{E}}^d_P(\phi) + \Gamma_E}{(\widehat{\operatorname{Var}}^d_\mu(\phi) - \Gamma_V)_+} \quad \textit{when the denominator is positive,} \tag{569}$$

$$\mathbb{E}_{\mu P} b_\phi \le a_\phi + L_\phi\sqrt{2(\widehat{\mathcal{E}}^d_P(\phi) + \Gamma_E)} \tag{570}$$

for the linear novelty bound, with $2L_\phi(\widehat{\mathcal{E}}^d_P(\phi) + \Gamma_E)$ replacing the square root term in the quadratic case. The deviations (Γ_E, Γ_V) are the applicable simultaneous structural Rademacher, PAC–Bayes, mixing-block, or martingale bounds from Part IV and Theorems IV.15.2 and IV.15.6; their sample size, confidence, effective dimension, selection penalty, trajectory dependence, and evaluation cost remain in $\mathcal{R}_{\mathrm{cur}}$. Thus observation regularity bounds the available novelty magnitude, while Eq. (559) *bounds target selection. Both coordinates remain in the same complete record.*

If q_ε exactly intertwines the selected dynamics and satisfies the target-qualification gates, its cell dynamics is assigned to the exact Abs/Geom *mode. If it does not intertwine and a cell-level target comparison is invoked, the comparison is admissible only through $\mathscr{C}_{\mathrm{comp}}$. Its dynamic factor is*

$$S_{q_\varepsilon}(\lambda) = \begin{cases} 1 - \vartheta^T_\lambda(q_\varepsilon), & \textit{under the reversible comparison,} \\ \max\{0, 1 - \Delta_\lambda(q_\varepsilon)\}, & \textit{under the general resolvent comparison,} \\ 0, & \textit{without either represented comparison,} \end{cases} \tag{571}$$

and its source visibility is exactly the applicable master-certificate value from Definitions III.18.2 and III.18.3. Every term used to build or evaluate this compensated geometry is charged by Eq. (553). *A nonintertwining observation with no such certificate remains an ambient observation record; its set-theoretic cells supply no quotient-dynamical bound.*

Finally, at the common class-preserving ceiling

$$B^{\mathrm{cur}} = \Psi_{\mathrm{cur}}(B, H, n, \varepsilon, \mathbf{m}),$$

where Ψ_{cur} is a declared class-preserving common majorant of all slots in Eqs. (549) *and* (553) *and of the ordered-reveal sample counts $\mathbf{m}$. It does not enlarge the public constructor algebra. Then the saturated curiosity-selector subprofile obeys*

$$\sup_{\substack{0 \le t < H \\ \mathcal{R}_{\mathrm{cur}}:\, B_{\mathrm{cur}} \preceq B}} \left(\mathrm{Adv}_t(K_t; \nu_t)\right)_+ \le \sup_{\substack{0 \le t < H \\ \mathcal{R}_{\mathrm{cur}}:\, B_{\mathrm{cur}} \preceq B^{\mathrm{cur}}}} \min\left\{1,\, \beta^{\mathrm{fib}}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{fib},\uparrow}_{R,t}}\right\}, \tag{572}$$

where every displayed quantity belongs to the same complete pre-hit record. Independent same-record capacity, Blackwell, source, Caus, and Lift bounds may subsequently be minimized with this coordinate under Theorem V.12.4; they are not used to prove the present inequality.

Proof. Conditional on the complete observation history, K_t uses only target-independent fresh randomness. Hence it is a causal postprocessing of the experiment $(\mathsf{Z}_0, C_\varepsilon, \mathcal{H}^{\mathrm{obs}}_t)$. Under $\mathbb{Q}^{\mathrm{fib}}_{R,t}$, the target index and observation history are conditionally independent given $(\mathsf{Z}_0, C_\varepsilon)$. The conditional relative-entropy identity gives

$$\mathrm{KL}(\mathbb{P}_{R,t} \| \mathbb{Q}^{\mathrm{fib}}_{R,t}) = (\log 2) I(\Theta; \mathcal{H}^{\mathrm{obs}}_t \mid \mathsf{Z}_0, C_\varepsilon).$$

Map the joint record to the next-contact indicator and apply data processing followed by Pinsker's inequality. This yields

$$(p_{R,t} - \beta^{\mathrm{fib}}_{R,t})_+ \le \sqrt{\frac{\log 2}{2} J^{\mathrm{fib},\uparrow}_{R,t}}.$$

The neutral selector has nonnegative target-contact mass, so $(\mathrm{Adv}_t)_+ \le p_{R,t}$, proving Eq. (559). Equation Eq. (560) follows because $r^{\mathrm{ext}}_0, \ldots, r^{\mathrm{ext}}_t$ are the constant zero vector on $\{\tau > t\}$. The intrinsic-record assertion is the data-processing inequality applied to the causal constructors in Definition V.8.5.

Under target-permutation neutrality, fix an initial record z. Conditional on a target cell c, Θ is uniform on

$$F_c(z) = \{\theta : q_{\varepsilon,z}(x_\theta) = c\}.$$

Even after giving c to the decoupled selector, its conditional exact target probability is at most $1/|F_c(z)|$. Averaging over the conditionally uniform Θ gives

$$\frac{1}{N} \sum_{c: F_c(z) \ne \varnothing} |F_c(z)| \frac{1}{|F_c(z)|} = \frac{|\{c : F_c(z) \ne \varnothing\}|}{N} \le \frac{M_\varepsilon(\phi)}{N}.$$

The bound is pointwise in z, so averaging over Z_0 preserves it. This proves Eqs. (561) and (562). Summing the disjoint first-contact events proves Eq. (563). The independent capacity estimate is Eq. (496), conditional on each $\Theta = \theta$, for the same selected kernel, target, horizon, and initialization. Averaging the conditional probabilities is bounded by the displayed maximum. Taking the smaller of two upper bounds for the same unconditional event proves Eq. (565); its listed construction costs are precisely Eq. (552).

Cauchy–Schwarz and $\mathbb{E}_{\mu P} d_{\mathcal{O}}^2 = 2\mathcal{E}_P^d(\phi)$ prove Eq. (566). Direct expectation proves Eq. (567).

The exact and nonzero-defect cell-dynamics assignments are Definitions III.18.1 and III.18.2. Their complete construction cost is the resource accumulation in Eqs. (549) and (553). The absence of an unrepresented quotient-dynamical comparison is Corollary III.22.1 and Theorem V.6.4. Finally, every budget-B record on the left belongs to the class at the common majorant B^{cur}. Apply Eq. (559) record by record and then take the displayed supremum. This proves Eq. (572) without invoking a downstream universal certificate.

□

Corollary V.8.7 (Resolution–plateau tradeoff)**.** *Under the target-permutation-neutrality hypothesis of Theorem V.8.6, suppose*

$$M_\varepsilon(\phi) \le N^\beta, \qquad 0 \le \beta \le 1, \qquad J_{R,t}^{\mathrm{fib},\uparrow} \le J_{\mathrm{fib}} \quad (0 \le t < H).$$

Then

$$\Pr\{\tau \le H\} \le \min\left\{1,\, H\left(N^{-(1-\beta)} + \sqrt{\frac{\log 2}{2} J_{\mathrm{fib}}}\right)\right\}. \tag{573}$$

For $N = 2^n$, the resolution term is $2^{-(1-\beta)n}$. The value of β is the evaluated logarithmic image size

$$\beta = \frac{\log M_\varepsilon(\phi)}{\log N};$$

it is not inferred from representation cost. In particular, the represented identity observation has cost linear in a binary state description and has $M_\varepsilon(\phi) = N$ at exact resolution.

Proof. Substitute $M_\varepsilon(\phi)/N \le N^{-(1-\beta)}$ and the uniform information bound into Eq. (563).

□

V.9 Optimal Structural Active Sampling

Active sampling combines the choice of a public experiment with the choice of the next state transition. The experiment may reveal new data only through a declared presentation interface. The transition remains a selected controlled kernel and is therefore governed by the master quotient–geometry certificate. This section optimizes target arrival over the resulting completely represented records. Information gain is one upper certificate for that objective, rather than its definition.

V.9.1 Complete active-acquisition records and cost

Definition V.9.1 (Complete master active-sampling record)**.** Fix a finite target index Θ, initial public record Z_0, finite refined horizon H, and resource ceiling B. At a surviving pre-hit history $\mathcal{H}_t$, a complete master active-sampling record contains the following represented data.

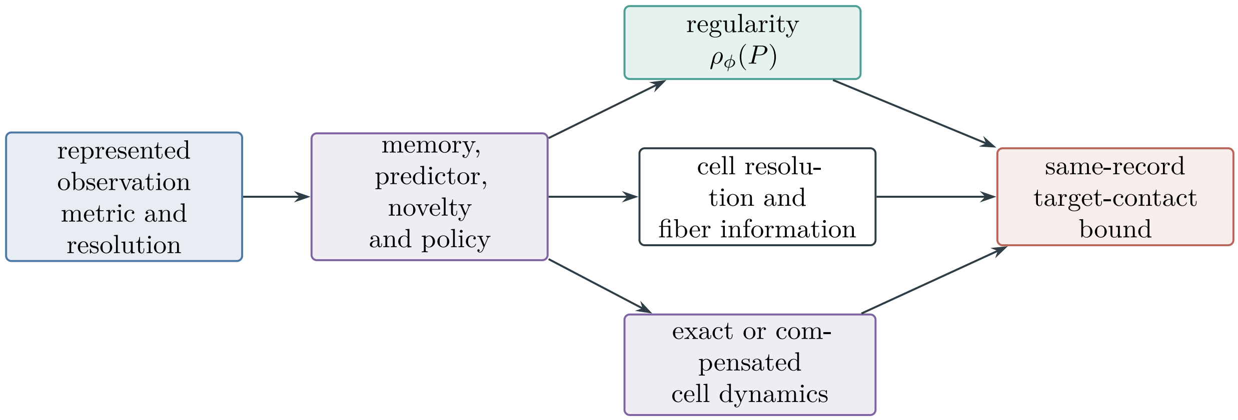

Figure 86: Sparse-reward observation accounting. Observation evaluation, novelty learning, memory, policy deployment, and the executed kernel are charged before use. Nonintertwining cells enter a dynamical comparison only through the fully costed compensated certificate. Resolution, conditional fiber information, and regularity are evaluated on the same record before the target-contact envelope is saturated.

(i) A predictable action $A_t = (A_t^{\mathrm{qry}}, A_t^{\mathrm{tr}})$. The query component selects a declared experiment, feature, or public-data readout. The transport component selects the next native or lifted transition. Conditional on $\mathcal{H}_t$, the action uses only represented fresh randomness independent of Θ.

(ii) The selected history-conditional experiment $K_t^{\mathrm{obs}}(dy \mid u, \mathcal{H}_t, A_t^{\mathrm{qry}})$, its clean input U_t, returned observation Y_{t+1}, and every physical channel state. An interface not generated by the declared presentation is an extension in the sense of Theorem III.33.2.

(iii) The executed transition kernel $P_t^{A_t^{\mathrm{tr}}}$, verifier interface, stopping rule, and output converter. Every observation used by a later transition occurs in the pre-hit filtration before that transition.

(iv) A dynamically qualified master certificate

$$\mathscr{C}_t = (\mathfrak{N}_t, q_t, \mathfrak{G}_{q_t}, \mathfrak{S}_t, \mathfrak{K}_{q_t})$$

for the selected transition, together with a represented same-normalization comparison to the target-arrival coordinate when structural visibility is used numerically.

(v) The acquisition score, its statistical or analytic certificate, the represented rule that selects or approximately selects the action, belief or sufficient-record update, memory, clock, precision, and all capacity, information, Blackwell, learning, Lift, and output comparisons invoked by the record.

In the resource monoid its complete cost is

$$\begin{aligned} B_{\mathrm{act}} = B_{\mathsf{Z}_0} &\oplus B_{\mathrm{hist}} \oplus B_{\mathrm{query}} \oplus B_{\mathrm{score}} \oplus B_{\mathrm{optimize}} \oplus B_{\mathrm{obs}} \oplus B_{\mathrm{channel}} \\ &\oplus B_{\mathrm{belief}} \oplus B_{\mathrm{memory}} \oplus B_{\mathrm{kernel}} \oplus B_{\mathscr{C}} \oplus B_{\mathrm{exec}} \oplus B_{\mathrm{clock}} \\ &\oplus B_{\mathrm{verify}} \oplus B_{\mathrm{stop}} \oplus B_{\mathrm{out}} \oplus B_{\mathrm{info}} \oplus B_{\mathrm{cap}} \oplus B_{\mathrm{BW}} \oplus B_{\mathrm{learn}}. \end{aligned} \tag{574}$$

The master-certificate charge $B_{\mathscr{C}}$ is the complete charge in Definition III.18.3. A displayed semantic maximizer of an acquisition score supplies no action rule. Deployment of that maximizer requires the represented selection or approximation record charged by B_{optimize}.

The master certificate already includes the pure geometric regime. Indeed, $q = \mathrm{id}$ with an active geometry slot is $\mathcal{C}_X$, whereas nonidentity exact and nonzero-defect records are $\mathcal{C}_{Q,\mathrm{ex}}$, $\mathcal{C}_{Q,\mathrm{rev}}$, and $\mathcal{C}_{Q,\mathrm{nr}}$. Thus compensated quasi-lumpability is the general case and full-carrier geometry is its identity-support realization.

Definition V.9.2 (Optimal structural active-sampling profiles)**.** Let $\mathscr{R}^{\mathrm{act}}_{I,H}(B)$ be the complete records of Definition V.9.1 with $B_{\mathrm{act}} \preceq B$. For $R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)$, set

$$A^H_R = \Pr_R\{\tau \le H\}, \qquad a_{R,t} = \big(\mathrm{Adv}_t(K_{R,t}; \nu_{R,t})\big)_+.$$

Define

$$\mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B) = \sup_{R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)} A^H_R, \tag{575}$$

$$\mathrm{ActSel}^{\mathrm{sat}}_{I,H}(B) = \max_{0 \le t < H} \sup_{R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)} a_{R,t}, \tag{576}$$

with supremum zero for an empty record class. These are analytic saturated profiles. A deployed optimal policy is present only when its action selector is represented in the record.

Definition V.9.3 (Family-uniform active saturated profile)**.** Let $\mathfrak{I}_{n,m,\zeta}$ be a presented size-and-sample slice with declared channel parameter ζ, and let $\mathfrak{M}_{n,m,\zeta}$ be its fixed family functional. In the homogeneous binary-noise case, $\zeta = \eta \in [0, 1/2)$. Write $\mathscr{R}^{\mathrm{act}}_{\mathfrak{I},n,m,\zeta,H}(B)$ for the complete records of Definition V.9.1 produced by one family-uniform represented derivation scheme and having complete saturated cost at most B on the whole slice. Define

$$\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H}(B) = \sup_{R \in \mathscr{R}^{\mathrm{act}}_{\mathfrak{I},n,m,\zeta,H}(B)} \mathfrak{M}_{n,m,\zeta}\Big(I \mapsto A^H_{R,I}\Big). \tag{577}$$

For a master mode

$$c \in \mathfrak{C}_{\mathrm{master}} := \{X, A, Q_{\mathrm{ex}}, Q_{\mathrm{rev}}, Q_{\mathrm{nr}}\},$$

let $\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H;c}(B)$ denote the same supremum restricted, respectively, to

$$\mathcal{C}_X, \quad \mathcal{C}_A, \quad \mathcal{C}_{Q,\mathrm{ex}}, \quad \mathcal{C}_{Q,\mathrm{rev}}, \quad \mathcal{C}_{Q,\mathrm{nr}}.$$

The active restriction of the saturated success profile is the supremum in Eq. (178) over these same complete active-interface derivations, with null queries admitted. Denote it by $\mathrm{Succ}^{\mathrm{act,sat}}_{\mathfrak{M},n,m,\zeta,H}(B)$. This restriction changes no cost convention: the action rule, query, observation, transport, certificate, stopping rule, and output are the single derivation whose cost is displayed in Eq. (574).

Theorem V.9.4 (Exact resource-valued active-acquisition recursion)**.** *For a represented surviving history h at time t, let $\mathscr{S}^{\mathrm{act}}_{t,H}(h; B)$ be the complete represented suffix records of horizon $H - t$ and cost at most B. Decompose a nonterminal suffix record at its first transition into its represented action a and its represented continuation rule ϱ, which maps each returned observation and next history to the remaining suffix record. Let $c^{\mathrm{full}}_t(a, \varrho; h)$ be the complete resource-monoid cost of this combined record, including the construction and storage of ϱ. Define*

$$V^{\mathrm{act}}_t(h; B) = \sup_{R \in \mathscr{S}^{\mathrm{act}}_{t,H}(h;B)} A^{H-t}_R.$$

Then the exact finite-horizon value satisfies

$$V_t^{\mathrm{act}}(h;B) = \sup_{\substack{(a,\varrho)\ \text{represented} \\ c_t^{\mathrm{full}}(a,\varrho;h)\preceq B}} \Big\{ p_T(h,a) + \mathbb{E}\Big[\mathbf{1}_{\{\tau>t+1\}} A^{H-t-1}_{\varrho(H_{t+1})} \mid h,a\Big]\Big\}, \tag{578}$$

with the declared terminal conversion and $V_H^{\mathrm{act}} = 0$ *before that conversion. Here* $p_T(h,a)$ *is the one-step first-contact probability.*

If a declared scalar cost is stage-decomposable,

$$\kappa(c_t^{\mathrm{full}}(a,\varrho;h)) = c_t^{\kappa}(a,h) + \kappa(\mathrm{Cost}(\varrho)),$$

then Eq. (578) *is equivalently the same recursion with the scalar constraint* $c_t^{\kappa}(a,h)+\kappa(\mathrm{Cost}(\varrho)) \le b$*. When a represented continuation optimizer simultaneously realizes the suffix suprema at every returned history, this becomes the familiar form*

$$V_t^{\mathrm{act}}(h,b) = \sup_{\substack{a\in\mathcal{A}_t(h)\ \text{represented} \\ c_t^{\kappa}(a,h)\le b}} \Big\{ p_T(h,a) + \mathbb{E}\Big[\mathbf{1}_{\{\tau>t+1\}} V_{t+1}^{\mathrm{act}}(H_{t+1}, b - c_t^{\kappa}(a,h)) \mid h,a\Big]\Big\}. \tag{579}$$

The continuation optimizer and every comparison used to construct it are part of the complete record.

Proof. Every nonterminal suffix execution has one first represented action and a represented rule for its continuations after the returned observation. Conversely, concatenating such an action and continuation rule produces one suffix record, and computation closure charges their combined record by c_t^{full}. First contact and survival continuation are disjoint, which proves Eq. (578) by taking the supremum over the two equivalent descriptions of the same suffix-record class. The scalar constraint follows from stage decomposability. A single represented continuation optimizer permits the suffix-record supremum to be taken inside the conditional expectation, proving Eq. (579). Without that represented selector, the resource-valued record recursion remains the exact statement.

□

Theorem V.9.5 (Active optimality equals affordable saturated construction)**.** *At every declared* (n,m,ζ,H,B)*,*

$$\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H}(B) = \mathrm{Succ}^{\mathrm{act,sat}}_{\mathfrak{M},n,m,\zeta,H}(B). \tag{580}$$

The common value has the exact structural decomposition

$$\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H}(B) = \max_{c\in\mathfrak{C}_{\mathrm{master}}} \mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H;c}(B). \tag{581}$$

Thus optimizing active acquisition is exactly optimizing the target-arrival coordinate of the complete saturated profile over affordable constructions; the construction and the selected master coordinate are carried by the same record.

For every $\varepsilon > 0$ *and nonempty record class, there is one family-uniform represented record* R_ε *of cost at most* B *such that*

$$\mathfrak{M}_{n,m,\zeta}(A^H_{R_\varepsilon}) \ge \mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H}(B) - \varepsilon. \tag{582}$$

This is an existence statement inside the represented record class. An affordable procedure that selects R_ε *from a collection of candidates is present precisely when that selection procedure is itself a represented derivation charged by* B_{optimize}*.*

Proof. The two sides of Eq. (580) take the same family functional and the same supremum over the complete active-interface derivations of Definition V.9.3; hence they are equal, including their cost and uniformity requirements. The five master families are disjoint and exhaustive by Definition III.18.4. Partitioning the record class by its carried master slot and taking the supremum proves Eq. (581). The ε-optimal record follows from the definition of a supremum. Derivation admissibility in Definition III.1.5 gives the final assertion: semantic selection among represented records is not itself a derivation.

□

Corollary V.9.6 (Constructive active optimum on a represented finite tree)**.** *Assume that the refined history tree and every represented action set are finite, the one-step probabilities and continuation values are represented to a declared absolute precision, and comparison plus tie-breaking are represented. Backward induction applied to Eq.* (578) *constructs a represented policy. If the value and comparison error at level t is at most ε_t, its arrival value is within*

$$\sum_{t=0}^{H-1} \varepsilon_t \tag{583}$$

of the best record in the enumerated affordable tree. Exact represented arithmetic gives exact attainment. Enumeration, value evaluation, comparison, storage, precision, and deployment are all charged in Eq. (574)*; the corollary applies at budget B exactly when this complete construction cost is at most B.*

Proof. At the last level choose a represented action within ε_{H-1} of the largest represented terminal value. Induct backward using the disjoint first-contact recursion. Conditional expectation is monotone, so the loss accumulated at level t is at most the local comparison error plus the continuation loss. Induction gives Eq. (583). Every operation used by this backward induction is among the charged slots listed in Definition V.9.1.

□

V.9.2 Information, master visibility, and optimal arrival

Theorem V.9.7 (Adaptive acquisition information and target contact)**.** *Fix $R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)$. Enumerate every target-dependent returned coordinate in its actual reveal order as $Y_1, \ldots, Y_L$, and let Q_j contain the query, selected experiment, disclosed policy randomness, and deterministic updates preceding Y_j. Then*

$$\Theta \perp Q_j \mid \mathscr{F}_{j-1}. \tag{584}$$

If a target-dependent datum is used to construct Q_j, it occurs in an earlier reveal slot and is consequently already in $\mathscr{F}_{j-1}$. With the capacity and SDPI certificates of Definition V.3.1,

$$I(\Theta; \mathscr{F}_L \mid \mathsf{Z}_0) = \sum_{j=1}^{L} I(\Theta; Y_j \mid \mathscr{P}_j), \tag{585}$$

$$I(\Theta; \mathscr{F}_L \mid \mathsf{Z}_0) \le J_R^{\mathrm{act},\uparrow} := \sum_{j=1}^{L} \mathbb{E} \min\{c_j(H_j^{\mathrm{pre}}), \vartheta_j(H_j^{\mathrm{pre}}) a_j(H_j^{\mathrm{pre}})\}. \tag{586}$$

Let $\beta^{\mathrm{act}}_{R,t}$ be the next-contact probability under the law which retains $(\mathsf{Z}_0, \mathcal{H}_t)$ and the selected kernel but resamples Θ independently from its declared reference conditional law. Then

$$\Pr_R\{\tau > t, X_{t+1} \in T_\Theta\} \le \min\left\{1, \beta^{\mathrm{act}}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{act},\uparrow}_{R,t}}\right\}, \tag{587}$$

$$A^H_R \le U^{\mathrm{act,info}}_R := \min\left\{1, \sum_{t=0}^{H-1}\left(\beta^{\mathrm{act}}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{act},\uparrow}_{R,t}}\right)\right\}. \tag{588}$$

Every selected query, experiment evaluator, returned answer, update, and action-optimization step remains charged by Eq. (574)*.*

Proof. The selected action is a measurable function of the earlier public history and fresh target-independent randomness, proving Eq. (584). Apply Theorem V.3.3 to obtain Eqs. (585) and (586). The conditional decoupling and Pinsker argument in Corollary V.3.4 gives Eq. (587). Summing the disjoint first-hit events proves Eq. (588).

□

Theorem V.9.8 (Master optimal structural active-sampling envelope)**.** *For a complete record R, let $U^{\mathrm{act,cap}}_R$ be every represented same-record capacity or occupation upper bound for A^H_R, with neutral value one when absent. Let $U^{\mathrm{act,BW}}_R$ be every represented Blackwell-transferred target-decision bound, again with neutral value one. Finally, the master comparison in R has one of the two forms*

$$A^H_R \le U^{\mathrm{act,master}}_R := \min\{1, K_R V_{\mathscr{C}_R} + \delta_R\}, \tag{589}$$

where $(K_R, \delta_R, \mathscr{C}_R)$ is either the inherited or direct same-normalization record of Theorem V.6.4. Write $R_{\le t}$ for the complete active record truncated after its next test at time t. Put

$$U^{\mathrm{act,src}}_R = \min\left\{1, \sum_{t=0}^{H-1} \min\left\{\begin{array}{l} 1,\ U^{\mathrm{stat}\to\mathrm{tgt}}_{R,t},\ \beta^+_{R,t} + \sqrt{\dfrac{\log 2}{2} J^{\mathrm{src}}_{I,R_{\le t}}}, \\ eH_B(n) \displaystyle\sum_{a \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{L}^{(a)}_{I,R_{\le t}},\ U^{\mathrm{dyn}\to\mathrm{tgt}}_{R,t} \end{array}\right\}\right\}, \tag{590}$$

using the same complete active record and the ledger of Definition IV.13.2. Put

$$U^{\mathrm{act}}_R = \min\{U^{\mathrm{act,info}}_R, U^{\mathrm{act,cap}}_R, U^{\mathrm{act,BW}}_R, U^{\mathrm{act,master}}_R, U^{\mathrm{act,src}}_R\}. \tag{591}$$

At a class-preserving common majorant

$$B^{\mathrm{act}} = \Psi_{\mathrm{act}}(B, H, n, \mathbf{m})$$

of every slot in Eq. (574)*,*

$$\mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B) \le \sup_{R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B^{\mathrm{act}})} U^{\mathrm{act}}_R. \tag{592}$$

Partition the records by the master-certificate slot. Then the right side of Eq. (592) *equals the maximum of its restrictions to*

$$\mathcal{C}_X, \quad \mathcal{C}_A, \quad \mathcal{C}_{Q,\mathrm{ex}}, \quad \mathcal{C}_{Q,\mathrm{rev}}, \quad \mathcal{C}_{Q,\mathrm{nr}}. \tag{593}$$

Consequently, exact Abs*, exact quotient geometry, compensated quasi-lumpability, and pure ambient geometry are evaluated by one theorem. The pure ambient restriction is precisely $q = \mathrm{id}$ with $\mathfrak{G}_q \ne \varnothing$.*

Proof. The information theorem bounds A_R^H by $U_R^{\mathrm{act,info}}$. The capacity and Blackwell candidates are admitted only after conversion to the same target-arrival event and probability law, so each also bounds A_R^H. The source-resolved learning candidate is Eq. (281) applied at each disjoint first-contact time of the same active record. The master candidate is the represented comparison in Eq. (589). Their recordwise minimum proves $A_R^H \le U_R^{\mathrm{act}}$. Encoding adequacy and computation closure place the complete record at B^{act}; taking the supremum proves Eq. (592).

The five certificate families are disjoint and exhaustive by Definition III.18.4. A supremum over their union is the maximum of the five restricted suprema, proving Eq. (593). □

V.9.3 Complexity, sample, and noise frontiers

Definition V.9.9 (Active construction thresholds)**.** Fix the admissible scalarization κ of Definition III.28.1. In this definition the record class without a displayed budget is the union of its finite-cost slices. Write

$$\mathrm{ActArr}^{\mathrm{sat},\kappa}_{\mathfrak{M},n,m,\zeta,H}(s) = \sup_{\substack{R\in\mathscr{R}^{\mathrm{act}}_{\mathfrak{I},n,m,\zeta,H}\\ \kappa(\mathrm{Cost}_{\mathrm{sat}}(R))\le s}} \mathfrak{M}_{n,m,\zeta}(A_R^H). \tag{594}$$

For $0 < \alpha \le 1$, define

$$B^*_{\mathrm{act}}(n,m,\zeta,H;\alpha) = \inf\left\{s \ge 0 : \mathrm{ActArr}^{\mathrm{sat},\kappa}_{\mathfrak{M},n,m,\zeta,H}(s) \ge \alpha\right\}, \tag{595}$$

$$m^*_{\mathrm{act}}(n,\zeta,s,H;\alpha) = \inf\left\{m \ge 1 : \mathrm{ActArr}^{\mathrm{sat},\kappa}_{\mathfrak{M},n,m,\zeta,H}(s) \ge \alpha\right\}. \tag{596}$$

For homogeneous binary noise, define

$$\eta^{\mathrm{sat}}_{\mathrm{act}}(n,m;s,H,\alpha) = \sup\left\{0 \le \eta < \frac{1}{2} : \mathrm{ActArr}^{\mathrm{sat},\kappa}_{\mathfrak{M},n,m,\eta,H}(s) \ge \alpha\right\}. \tag{597}$$

Infima of empty sets are $+\infty$, and suprema of empty sets are zero. These are restrictions of the existing recovery and dynamical threshold definitions to the complete active-interface record class, rather than new success criteria.

Theorem V.9.10 (Quantitative monotonicity of the active optimum)**.** *The scalarized profile in Eq.* (594) *is nondecreasing in s. It is nondecreasing in H whenever an earlier stopping rule may be retained in a longer record. If the declared sample presentations admit a class-preserving embedding that appends observations which the record may ignore, then it is nondecreasing in m at the compiled embedding cost.*

For homogeneous binary noise, let $0 \le \eta_1 \le \eta_2 < 1/2$, and let $\psi_{\mathrm{flip}}(s,n,m)$ be the complete scalar cost of the represented degradation in Proposition IV.14.3. Then

$$\mathrm{ActArr}^{\mathrm{sat},\kappa}_{\mathfrak{M},n,m,\eta_1,H}(\psi_{\mathrm{flip}}(s,n,m)) \ge \mathrm{ActArr}^{\mathrm{sat},\kappa}_{\mathfrak{M},n,m,\eta_2,H}(s). \tag{598}$$

Consequently the affordable active region is downward closed in the noise rate whenever the declared budget absorbs this degradation cost. The three thresholds in Definition V.9.9 are therefore the computational-cost, sample, and noise sections of one saturated active construction frontier.

Proof. Budget monotonicity is inclusion of scalar-cost sublevel sets. A record may retain its earlier stopping rule in a longer horizon, proving horizon monotonicity. Under the stated presentation embedding, append the extra observations and run the original record without reading them; computation closure charges the embedding and preserves its arrival law.

For the noise assertion, append independent flips of rate $q(\eta_1, \eta_2) = (\eta_2 - \eta_1)/(1 - 2\eta_1)$ to the η_1 record, then run any η_2 active construction. The complete history, action, and arrival law agree with the η_2 execution, while the cost is at most $\psi_{\mathrm{flip}}(s, n, m)$. Taking the saturated supremum proves Eq. (598).

□

Corollary V.9.11 (Learned acquisition-score transfer). *Let $g_R(h, a) \in [0, 1]$ be a represented population acquisition score and $\widehat{g}_R(h, a)$ its represented estimate. Suppose one complete trajectory-learning record certifies, simultaneously on its represented action class,*

$$\sup_{h,a} |\widehat{g}_R(h, a) - g_R(h, a)| \leq \Gamma_R$$

by either Definition IV.15.3 and Theorem IV.15.6. If the represented optimizer selects $\widehat{a}(h)$ satisfying

$$\widehat{g}_R(h, \widehat{a}(h)) \geq \sup_a \widehat{g}_R(h, a) - \varepsilon_{\mathrm{opt}},$$

then

$$\sup_a g_R(h, a) - g_R(h, \widehat{a}(h)) \leq 2\Gamma_R + \varepsilon_{\mathrm{opt}}. \tag{599}$$

The statistical radius, optimizer, and deployment costs are included in $B_{\mathrm{learn}} \oplus B_{\mathrm{optimize}}$. The conclusion compares the learned selector with the best represented action in the same record; the value of that action remains bounded by Theorem V.9.8.

Proof. Insert $\widehat{g}_R$ on both sides of the optimizer inequality. The two uniform deviations contribute $2\Gamma_R$, and the represented optimization defect contributes $\varepsilon_{\mathrm{opt}}$.

□

Theorem V.9.12 (Finite-horizon statistical loss of a learned active rule). *Let $Q_t^{\mathrm{act},*}(h, a, b)$ be the analytic action value obtained by taking represented action a at history h, charging its declared cost, and then taking the saturated suffix-record supremum in Eq. (578). This analytic quantity is not supplied to a deployed selector. Suppose a complete learning record constructs $\widehat{Q}_t$ and a represented action $\widehat{a}_t(h)$ such that, on one simultaneous event,*

$$\sup_{h,a} |\widehat{Q}_t(h, a) - Q_t^{\mathrm{act},*}(h, a, b_t)| \leq \Gamma_t, \tag{600}$$

$$\widehat{Q}_t(h, \widehat{a}_t(h)) \geq \sup_a \widehat{Q}_t(h, a) - \varepsilon_{\mathrm{opt},t}. \tag{601}$$

The suprema range over the represented histories and actions admitted by the remaining budget b_t. Then the deployed record $\widehat{R}$ obeys

$$V_0^{\mathrm{act}}(h_0, b_0) - A_{\widehat{R}}^H \leq \mathbb{E}_{\widehat{R}}\left[\sum_{t=0}^{H-1} \mathbf{1}_{\{\tau > t\}}(2\Gamma_t + \varepsilon_{\mathrm{opt},t})\right]. \tag{602}$$

Every Γ_t is admitted only after conversion to the same clean action-value law. Its sample certificate, confidence split, operator or prior selection, score construction, optimization, and deployment costs are included in $B_{\mathrm{learn}} \oplus B_{\mathrm{optimize}}$.

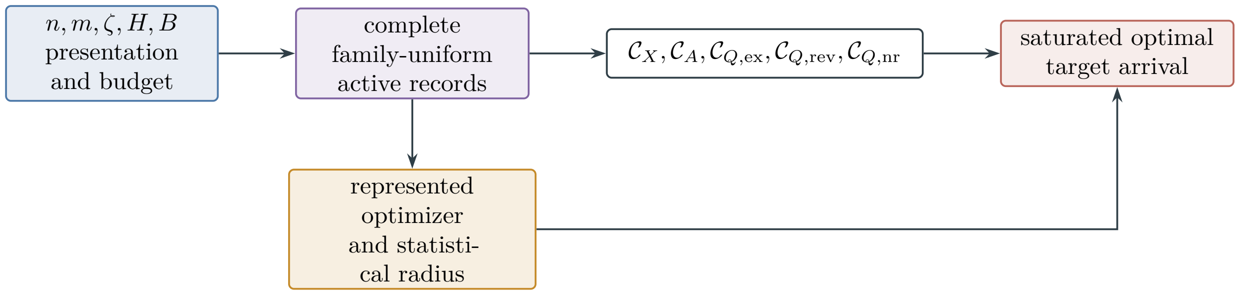

Figure 87: Optimal active sampling is a saturated construction problem. The parameterized presentation and complete budget determine one family-uniform record class. Its five master cells partition the exact arrival optimum. A learned optimizer approaches that optimum only through its represented, completely charged score and statistical certificate.

Proof. Insert $\widehat{Q}_t$ on both sides of Eq. (601). On the simultaneous event, the conditional loss relative to the largest admitted analytic action value is at most $2\Gamma_t + \varepsilon_{\mathrm{opt},t}$. Apply the exact first-contact recursion to the optimal suffix value and the deployed suffix value. On target contact there is no continuation loss; on survival the same argument applies at the next history. Backward induction and the tower property give Eq. (602).

□

Corollary V.9.13 (Explicit sample–noise active optimality loss)**.** *Assume homogeneous binary noise $0 \le \eta < 1/2$, a common optimization error $\varepsilon_{\mathrm{opt}}$, a $[0,1]$-valued flip-symmetric action-score loss, and one represented sample certificate with effective sample size m_{eff}, dependence error $\varepsilon_{\mathrm{dep}}$, and confidence allocation δ/H. If the sample-independent affordable action-value codebook has logarithmic size $L_{\mathrm{act}}(n, b, m, \eta)$, the finite-description candidate of Proposition IV.16.5, transported to the clean flip-symmetric action-value law, is*

$$\Gamma_{\mathrm{act}}^{\mathrm{clean}} \le \frac{1}{1-2\eta}\left[2\sqrt{\frac{2L_{\mathrm{act}}(n, b_{\mathfrak{U}}^{\mathrm{LPN}}, m, \eta)}{m_{\mathrm{eff}}}} + 3\sqrt{\frac{\log(2H/\delta)}{2m_{\mathrm{eff}}}} + \varepsilon_{\mathrm{dep}}\right]. \tag{603}$$

On the same event,

$$V_0^{\mathrm{act}} - A_{\widehat{R}}^{H} \le H\left(2\Gamma_{\mathrm{act}}^{\mathrm{clean}} + \varepsilon_{\mathrm{opt}}\right). \tag{604}$$

The record may replace Eq. (603) by any smaller simultaneous same-law candidate obtained from Definition IV.15.3 and Theorems IV.15.5 and IV.15.6; the operator, cover, posterior, prior, confidence allocation, and selection rule remain charged.

Proof. Use $\delta_{\mathrm{eff}} = \delta/H$ in Eq. (331). The homogeneous flip-symmetric transport identity in Theorem IV.16.2 divides the corrupted score radius by $1-2\eta$, proving Eq. (603). Substitute the common radius and optimization error into Eq. (602) and bound the number of surviving terms by H.

□

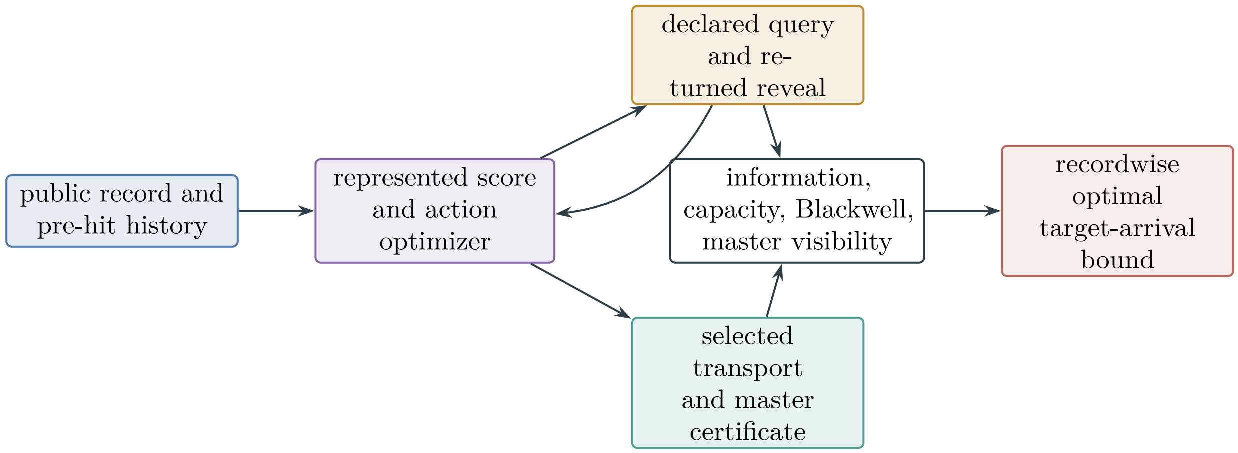

Figure 88: Optimal structural active sampling. The action optimizer, query, returned reveal, transport kernel, and master certificate belong to one charged pre-hit record. Quasi-lumpable compensation is the general master case; setting $q = \mathrm{id}$ gives pure ambient geometry.

V.10 Affordable Occupation Flows and the Bellman Structural Converse

The active optimum is a supremum over complete represented records. Its occupation-flow representation gives a linear target-flux primal, while a Bellman barrier gives the dual upper bound. Both descriptions retain the same clocked history, remaining budget, and complete record cost.

Definition V.10.1 (Stopped occupation flow of a complete active record)**.** Fix an instance I, horizon H, and a complete active record $R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)$. Let H^R_t be its represented measurable carrier of surviving clocked histories at time t, including the remaining resource state. Let $\mathsf{A}^R_t(h)$ be the represented action carrier available at h; an action includes its query, transport kernel, returned-data interface, master certificate, and charged continuation selector. Put

$$d_{R,t}(C) = \Pr_R\{\tau > t, \mathcal{H}_t \in C\}, \tag{605}$$

$$f_{R,t}(dh, da) = d_{R,t}(dh) K_{R,t}(da \mid h). \tag{606}$$

Write $p_T(h,a)$ for first contact at the next transition and $P_N(h' \mid h, a)$ for the surviving history kernel. Then

$$d_{R,0} = \delta_{h_0}, \tag{607}$$

$$f_{R,t}(dh, \mathsf{A}^R_t(h)) = d_{R,t}(dh), \tag{608}$$

$$d_{R,t+1}(C) = \int f_{R,t}(dh, da) P_N(C \mid h, a). \tag{609}$$

The stopped occupation record is $\omega_R = (d_{R,t}, f_{R,t})_{t<H}$. Let

$$\mathfrak{O}^{\mathrm{sat}}_{I,H}(B) = \{\omega_R : R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)\}. \tag{610}$$

This set contains only flows induced by complete records. Its convex hull is an analytic device; it does not supply an affordable mixture or policy.

Theorem V.10.2 (Exact affordable occupation-flow primal)**.** *Every complete active record satisfies*

$$A_R^H = \sum_{t=0}^{H-1} \int f_{R,t}(dh, da) p_T(h, a). \tag{611}$$

Consequently

$$\mathrm{ActArr}_{I,H}^{\mathrm{sat}}(B) = \sup_{\omega \in \mathfrak{O}_{I,H}^{\mathrm{sat}}(B)} \sum_{t=0}^{H-1} \int f_t(dh, da) p_T(h, a) = \sup_{\omega \in \mathrm{co}\, \mathfrak{O}_{I,H}^{\mathrm{sat}}(B)} \sum_{t=0}^{H-1} \int f_t(dh, da) p_T(h, a). \tag{612}$$

The same identity holds for a family-uniform slice after restricting to one family-uniform derivation scheme and applying its fixed family functional.

Proof. The event of first contact at time $t+1$ has probability $\int f_{R,t}(dh, da) p_T(h, a)$. These events are disjoint, so summing proves Eq. (611). The first equality in Eq. (612) is the definition of the active saturated profile applied to this identity. A linear functional has the same supremum over a set and its convex hull, which proves the second equality. No convex mixture is inserted into the represented record class.

□

Definition V.10.3 (Affordable Bellman support)**.** Let $v = (v_t)_{t=0}^H$ be bounded functions on the represented surviving history-and-resource carrier, with $v_H = 0$. For a complete first-step record $r = (h, a)$, define its positive Bellman increment

$$\mathfrak{d}_t(r; v) = \left[p_T(h, a) + \int P_N(dh' \mid h, a) v_{t+1}(h') - v_t(h) \right]_+ . \tag{613}$$

Let $\mathscr{A}_{I,t}^{\mathrm{sat}}(h; B)$ contain exactly the first-step records occurring in a complete suffix record of total cost at most B. The saturated Bellman-support coordinate is

$$\mathfrak{D}_{I,t}^{\mathrm{sat}}(B; v) = \sup_h \sup_{r \in \mathscr{A}_{I,t}^{\mathrm{sat}}(h;B)} \mathfrak{d}_t(r; v), \tag{614}$$

with value zero for an empty class. The supremum is analytic. A deployed barrier or maximizing action is available only through its represented derivation and complete cost.

For a master mode $c \in \mathfrak{C}_{\mathrm{master}}$, write $\mathfrak{D}_{I,t;c}^{\mathrm{sat}}(B; v)$ for the restriction to first-step records whose selected transition carries mode c, and write $\mathscr{A}_{I,t;c}^{\mathrm{sat}}(h; B)$ for that restricted first-step record class.

Theorem V.10.4 (Bellman barrier duality for affordable arrival)**.** *For every bounded barrier v as in Definition V.10.3,*

$$\mathrm{ActArr}_{I,H}^{\mathrm{sat}}(B) \le v_0(h_0) + \sum_{t=0}^{H-1} \mathfrak{D}_{I,t}^{\mathrm{sat}}(B; v). \tag{615}$$

Moreover,

$$\mathrm{ActArr}_{I,H}^{\mathrm{sat}}(B) = \inf_{v_H = 0} \left\{ v_0(h_0) + \sum_{t=0}^{H-1} \mathfrak{D}_{I,t}^{\mathrm{sat}}(B; v) \right\}, \tag{616}$$

where the infimum ranges over bounded analytic functions on the complete history-and-resource carrier.

The support decomposes exactly by the existing master classification:

$$\mathfrak{D}^{\mathrm{sat}}_{I,t}(B;v) = \max_{c\in\mathfrak{C}_{\mathrm{master}}} \mathfrak{D}^{\mathrm{sat}}_{I,t;c}(B;v). \tag{617}$$

If the represented rectangularity gate of Definition V.13.7 holds, the infimum is the ordinary finite occupation-measure/Bellman linear-program dual on the represented statewise action correspondence. Without that gate, Eq. (616) *uses the complete suffix records of Theorem V.9.4; it performs no statewise stitching of separately affordable policies.*

Proof. For one record R, integrate Eq. (613) against $f_{R,t}(dh,da)$ and sum over time. The signed increment is bounded by its positive part, hence

$$\begin{aligned}&\sum_t \int f_{R,t}(dh,da)p_T(h,a)\\&\quad\le \sum_t \int f_{R,t}(dh,da)\left[v_t(h) - \int P_N(dh' \mid h,a)v_{t+1}(h') + \mathfrak{D}^{\mathrm{sat}}_{I,t}(B;v)\right].\end{aligned}$$

The flow balances in Eqs. (607) to (609) make the two value sums telescope to $v_0(h_0)$; $v_H = 0$, and surviving mass is at most one at every time. Together with Eq. (611), this proves the recordwise bound and then Eq. (615).

For the reverse inequality in the infimum, take $v_t(h) = V^{\mathrm{act}}_t(h;B_h)$, the exact resource-valued suffix optimum in Theorem V.9.4. Its recursion makes every admitted Bellman increment nonpositive, so every support term is zero and $v_0(h_0) = \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B)$. The upper bound already proves that no barrier has a smaller objective. This establishes Eq. (616).

The master families partition the complete first-step records by Theorem V.9.5; a supremum over their union is the maximum of the restricted suprema. This proves Eq. (617). Under rectangularity, the balance equations and Bellman inequalities are the finite primal and dual linear programs, and finite-dimensional strong duality applies. The complete-suffix formulation supplies the stated nonrectangular reading.

□

Theorem V.10.5 (Structural evaluation of affordable Bellman support)**.** *Fix v, t, and a common class-preserving ceiling $B^\sharp = \Psi_{\mathrm{Bell}}(B,H,n)$ charging the barrier evaluator, selected transition, complete master certificate, and every comparison used below. For a first-step record r, let*

$$U^{\mathrm{Bell}}_{r,t}(v) = \min\{1, U^{\mathrm{cap}}_{r,t}, U^{\mathsf{Caus}}_{r,t}, U^{\mathrm{info}}_{r,t}, U^{\mathrm{BW}}_{r,t}, U^{\mathrm{obs}}_{r,t}, U^{\mathrm{master}}_{r,t}\}, \tag{618}$$

where a candidate is finite precisely when the same record contains a represented comparison from that candidate to the positive increment $\mathfrak{d}_t(r;v)$. An unavailable comparison has value one. Then

$$\mathfrak{d}_t(r;v) \le U^{\mathrm{Bell}}_{r,t}(v), \tag{619}$$

$$\mathfrak{D}^{\mathrm{sat}}_{I,t}(B;v) \le \max_{c\in\mathfrak{C}_{\mathrm{master}}} \sup_{r\in\mathscr{A}^{\mathrm{sat}}_{I,t;c}(B^\sharp)} U^{\mathrm{Bell}}_{r,t}(v). \tag{620}$$

The candidates are, respectively, the existing controlled-capacity arrival comparison, authorized Caus *aiming-to-contact comparison, sequential information/contact comparison, Blackwell target-decision comparison, sparse-reward observation-resolution comparison, and direct or inherited master visibility comparison. They are numerical views of the existing record and create no structural channel.*

Proof. Every finite candidate in Eq. (618) is required to bound the same positive Bellman increment under the same history law. Their minimum therefore proves Eq. (619). Encoding adequacy and computation closure place the complete comparison record at $B^\sharp$. Take the supremum first within each master class and then use Eq. (617). This proves Eq. (620). The candidate origins are the coordinatewise comparisons in Theorems V.9.8 and V.12.3; their same-normalization requirement is Theorem V.6.4.

□

Theorem V.10.6 (Quantitative prospective contact-witness coercivity)**.** *Fix a presented instance I, a refined horizon $H \geq 1$, a saturated budget B, and a complete active record $R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)$. Use the one-test normalization of Definition III.21.2, let $K_{R,t}$ be the represented next-test kernel of R, and let ν_t be a declared neutral baseline on the same eligible candidates. Put*

$$\varepsilon_R = \Big[A^H_R - \mathrm{Enum}_\nu(R,H)\Big]_+, \tag{621}$$

$$a_{R,t} = \big(\mathrm{Adv}_t(K_{R,t};\nu_t)\big)_+, \tag{622}$$

$$h_{R,t} = \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}}K_{R,t}(V_I = 1 \mid \mathcal{H}_t)\Big]. \tag{623}$$

Then the following pointwise chain is exact up to taking positive parts and suprema:

$$\varepsilon_R \leq \sum_{t=0}^{H-1} a_{R,t} \leq \sum_{t=0}^{H-1} h_{R,t} \leq \sum_{t=0}^{H-1} \mathfrak{D}^{\mathrm{sat}}_{I,t}(B;0) = \sum_{t=0}^{H-1} \max_{c\in\mathfrak{C}_{\mathrm{master}}} \mathfrak{D}^{\mathrm{sat}}_{I,t;c}(B;0). \tag{624}$$

Consequently, if $\varepsilon_R > 0$, there are a time $t_* < H$ and a master mode $c_* \in \mathfrak{C}_{\mathrm{master}}$ such that

$$\mathfrak{D}^{\mathrm{sat}}_{I,t_*;c_*}(B;0) \geq \frac{\varepsilon_R}{H}. \tag{625}$$

More concretely, for every $0 \leq \gamma < \varepsilon_R/H$, the complete first-step record class in that mode contains a record $r = (h,a)$ satisfying

$$p_T(h,a) > \gamma. \tag{626}$$

The record is prospective: its history, selected transition, randomness, memory, and complete normalized derivation are available before the tested transition and are charged within B. Its transition and source ancestry have the existing master-mode and channel assignments of Theorems II.9.4 and III.21.5; no terminal flag or completed first-hit quotient is inserted into that ancestry.

At the compiled comparison ceiling $B^\sharp = \Psi_{\mathrm{Bell}}(B,H,n)$, the same conclusion has the same-record certificate form

$$\max_{c\in\mathfrak{C}_{\mathrm{master}}} \sup_{r\in\mathscr{A}^{\mathrm{sat}}_{I,t_*;c}(B^\sharp)} U^{\mathrm{Bell}}_{r,t_*}(0) \geq \frac{\varepsilon_R}{H}. \tag{627}$$

Here an unavailable comparison has the neutral value one, exactly as in Eq. (618). Thus Eq. (627) does not turn an unavailable comparison into a structural estimate: it identifies either an evaluated same-record certificate at the displayed scale or the precise comparison slot that remains unevaluated.

Conversely, if numbers $\delta_{t,c} \in [0,1]$ satisfy

$$\mathfrak{D}^{\mathrm{sat}}_{I,t;c}(B;0) \leq \delta_{t,c} \qquad (0 \leq t < H,\ c \in \mathfrak{C}_{\mathrm{master}}), \tag{628}$$

then every complete active record of cost at most B obeys

$$A^H_R \leq \mathrm{Enum}_\nu(R,H) + \sum_{t=0}^{H-1} \max_{c\in\mathfrak{C}_{\mathrm{master}}} \delta_{t,c}. \tag{629}$$

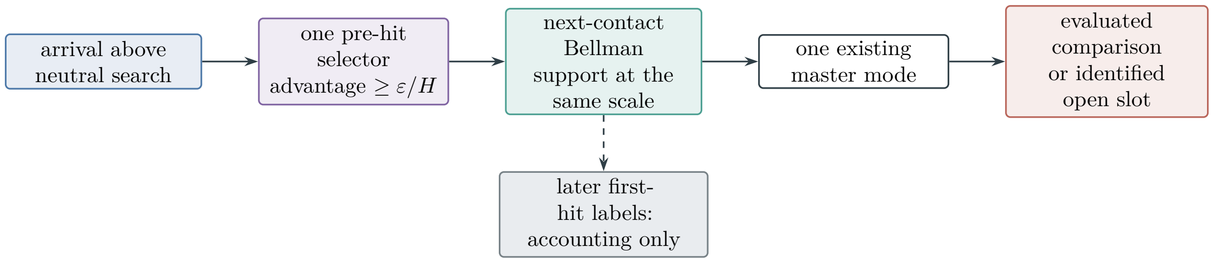

Figure 89: Quantitative prospective witness extraction. Excess target arrival forces a pre-hit next-contact record at scale ε/H. The complete record has one existing master-mode assignment. A later first-hit label cannot be used as its source.

These implications use the existing contact, Bellman-support, and master coordinates. They introduce no additional structural channel. The implication chain is summarized in Fig. 89.

Proof. The hazard identity Eq. (152) gives

$$A_R^H - \mathrm{Enum}_\nu(R, H) = \sum_{t=0}^{H-1} \mathrm{Adv}_t(K_{R,t}; \nu_t).$$

Taking the positive part and using $[\sum_t z_t]_+ \leq \sum_t (z_t)_+$ proves the first inequality in Eq. (624). The neutral hazard is nonnegative, so

$$a_{R,t} \leq h_{R,t} = \mathbb{E}_R\Big[\mathbf{1}_{\{\tau > t\}} p_T(\mathcal{H}_t, A_t)\Big].$$

Every first-step record used by R occurs in a complete suffix record of total cost at most B. With $v = 0$, its positive Bellman increment is exactly $p_T(h, a)$. Hence the integrand is at most $\mathfrak{D}^{\mathrm{sat}}_{I,t}(B; 0)$, proving the third inequality. The last equality is the exact master decomposition Eq. (617).

If $\varepsilon_R > 0$, the sum of the H nonnegative support terms is at least ε_R; one time therefore has support at least ε_R/H. The finite maximum over master modes gives c_*. The definition of a supremum then gives Eq. (626) for every strict lower level $\gamma < \varepsilon_R/H$. Temporal availability and complete cost are part of Definitions V.9.1 and V.10.3; transition-rule provenance and the master assignment are Theorems II.9.4 and III.21.5. The exclusion of later flags is Definition III.21.1 and Corollary III.23.1.

Apply Eq. (620) at t_*. Together with Eq. (625), it proves Eq. (627). The convention for an unavailable comparison is part of Theorem V.10.5; retaining that convention proves the stated scope without adding a comparison.

Finally, substitute Eq. (628) into Eq. (624) and undo the positive part. This gives Eq. (629). □

Corollary V.10.7 (Complete learned-control sandwich)**.** *Let $\widehat{R}$ be a complete learned active record satisfying the simultaneous action-value and optimization bounds of Theorem V.9.12. Then*

$$A_{\widehat{R}}^H \leq \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B) \leq \inf_{v_H = 0} \left\{ v_0(h_0) + \sum_{t<H} \mathfrak{D}^{\mathrm{sat}}_{I,t}(B; v) \right\}, \tag{630}$$

and

$$0 \leq \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B) - A_{\widehat{R}}^H \leq \mathbb{E}_{\widehat{R}} \sum_{t<H} \mathbf{1}_{\{\tau > t\}} (2\Gamma_t + \varepsilon_{\mathrm{opt},t}). \tag{631}$$

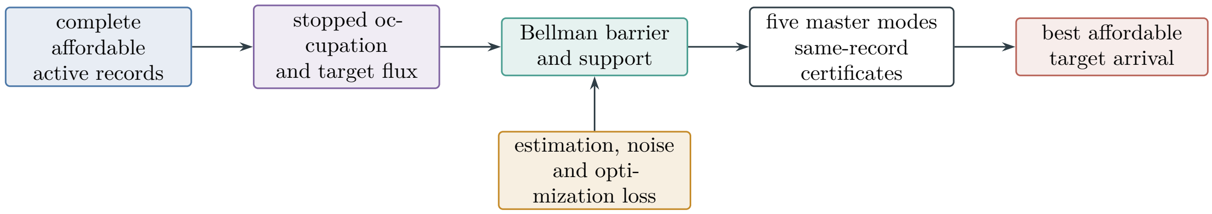

Figure 90: Affordable occupation-flow/Bellman duality. Target arrival is the linear flux of complete represented records. Bellman support is evaluated through the existing master certificates, while learning bounds measure the distance from the best affordable controller.

Each Γ_t is the smallest simultaneous same-action-value consequence of the independent, mixing, sequential-Rademacher, martingale PAC–Bayes, effective-dimension, label-noise, model-channel, filtering, and occupancy shift certificates in the complete record. Thus the first display bounds the best affordable structural controller, while the second measures the statistical and optimization distance of the learned controller from that optimum.

Proof. The first inequality is membership of $\widehat{R}$ in the affordable record class. The equality in Theorem V.10.4 proves the second. The final display is Theorem V.9.12. Its same-law candidate aggregation is exactly the rule in Definitions V.14.1 and V.13.7. □

V.11 Optimal Structural Bayesian Learning

Structural Bayesian learning equips a complete active record with a represented belief update. The target posterior and the PAC–Bayes posterior have different types: the former is a probability law on the task index, whereas the latter is a probability law on predictors or policies. This section keeps the two laws separate and charges every operation that turns public observations into a deployed target belief.

V.11.1 Prior provenance and represented task beliefs

Definition V.11.1 (Complete structural Bayesian record)**.** Fix a finite target-index carrier Θ, a declared prior Π_0 of full support, and a complete active record R from Definition V.9.1. Let $\mathcal{F}_t$ be its complete public pre-hit filtration and put

$$\Pi_t(\theta) = \Pr\{\Theta = \theta \mid \mathcal{F}_t\}. \tag{632}$$

This conditional law is an analytic comparator. A deployed belief is a represented $\mathcal{F}_t$-measurable probability vector $\widehat{\Pi}_t$ on Θ. Its comparison error is

$$e_t = \|\widehat{\Pi}_t - \Pi_t\|_{\mathrm{TV}}. \tag{633}$$

Suppose the action A_t^{qry} selects a declared observation kernel $L_t(dy \mid \theta, \mathcal{F}_t, A_t^{\mathrm{qry}})$. A represented Bayesian update consists of represented nonnegative likelihood scores $\widehat{L}_t(y \mid \theta)$, a represented evidence normalizer, and the update

$$\widehat{\Pi}_{t+1}(\theta) = \frac{\widehat{\Pi}_t(\theta)\widehat{L}_t(Y_{t+1} \mid \theta)}{\sum_{u\in\Theta} \widehat{\Pi}_t(u)\widehat{L}_t(Y_{t+1} \mid u)} \tag{634}$$

whenever the denominator has a represented positive-evidence certificate. An alternative represented normalizer or approximate update is admitted with its error in Eq. (633). A zero or uncertified denominator gives infinite raw filter error and the neutral value one after conversion to a probability coordinate.

The complete cost is the active cost in Eq. (574). The slots

$$B_{\mathrm{query}}, B_{\mathrm{obs}}, B_{\mathrm{channel}}, B_{\mathrm{belief}}, \qquad B_{\mathrm{memory}}, B_{\mathrm{score}}, B_{\mathrm{optimize}}, B_{\mathrm{learn}}$$

collectively charge the experiment selector, returned observation, likelihood interface, prior code, likelihood evaluation, evidence normalization, belief storage, precision, posterior statistic, acquisition optimizer, and deployment. These operations retain the existing master structural channel of their complete active record. Write

$$B^{\mathrm{Bayes}} = \Psi_{\mathrm{Bayes}}(B, H, n, \mathbf{m}, \boldsymbol{\zeta}) \tag{635}$$

for a class-preserving common majorant of these existing cost slots, where $\mathbf{m}$ and $\boldsymbol{\zeta}$ collect the declared sample and channel parameters. This is a common accounting ceiling; the primitive signature and constructor grammar remain unchanged.

The notation $\Pi_t, \widehat{\Pi}_t$ is reserved for task-index beliefs. A PAC–Bayes posterior Q remains a law on hypotheses, operators, or policies and retains the KL charge of Definition IV.8.1 and Theorem IV.15.6.

Theorem V.11.2 (Prior provenance and neutral initialization)**.** *Let G be a represented initialization record independent of Θ. Conditional on G, let K_G be any represented probability kernel on a common candidate carrier C. For a verification relation $V_\theta(c) \in \{0,1\}$, suppose*

$$\sup_{c \in \mathsf{C}} \sum_{\theta \in \Theta} V_\theta(c) \le \Lambda. \tag{636}$$

If Θ is uniform on $N = |\Theta|$, then

$$\mathbb{E} \sum_c K_G(c) V_\Theta(c) \le \frac{\Lambda}{N}. \tag{637}$$

When $\mathsf{C} = \Theta$ and $V_\theta(c) = \mathbf{1}_{\{c=\theta\}}$, the contact is exactly $1/N$, independent of every subjective nonuniform belief selected by G.

If an initialization rule reads a target-dependent public variable Z, then Z belongs to the first reveal slot of the public filtration and the resulting law is $\Pi_0(\cdot \mid Z)$. Its channel, likelihood, update, normalization, and retained precision are charged as the first observation and belief update. A target-dependent initialization without this ancestry is outside the complete structural Bayesian record class.

Proof. Independence gives

$$\mathbb{E} \sum_c K_G(c) V_\Theta(c) = \mathbb{E}_G \sum_c K_G(c) \frac{1}{N} \sum_\theta V_\theta(c) \le \frac{\Lambda}{N}.$$

For the displayed singleton relation, the inner target sum equals one for every candidate, so equality holds. When Z depends on Θ, the declared reveal order places Z before every rule that uses it. The chain rule for conditional probability identifies conditioning on Z with the first posterior update. Temporal source admissibility and the complete active cost then charge the stated record components.

□

V.11.2 Posterior concentration and target contact

Definition V.11.3 (Target vulnerability and affordable posterior profile)**.** For a probability law q on Θ, define its target vulnerability and excess over the declared prior by

$$\mathcal{V}_T(q) = \sup_{c\in\mathsf{C}} \sum_\theta q(\theta)V_\theta(c), \tag{638}$$

$$\mathcal{C}_T(q;\Pi_0) = \big[\mathcal{V}_T(q) - \mathcal{V}_T(\Pi_0)\big]_+. \tag{639}$$

For disjoint singleton target cells, $\mathcal{V}_T(q) = \max_\theta q(\theta)$. The functional is analytic unless the record contains its represented evaluator and comparison rule.

Let $\mathscr{R}^{\mathrm{Bayes}}_{I,H}(B)$ be the complete structural Bayesian records of cost at most B. Define

$$\mathrm{PostConc}^{\mathrm{sat}}_{I,H}(B) = \max_{t<H} \sup_{R\in\mathscr{R}^{\mathrm{Bayes}}_{I,H}(B)} \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}}\mathcal{C}_T(\widehat{\Pi}_t;\Pi_0)\Big], \tag{640}$$

$$\mathrm{FilErr}^{\mathrm{sat}}_{I,H}(B) = \max_{t<H} \sup_{R\in\mathscr{R}^{\mathrm{Bayes}}_{I,H}(B)} \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}}e_t\Big]. \tag{641}$$

The family-uniform versions restrict to one family-uniform represented derivation scheme and apply the declared family functional.

Let $\mathscr{R}^{\mathrm{Bayes,src}}_{I,H}(B)$ be the restriction to records in which the target-vulnerability evaluator, its comparison with $\mathcal{V}_T(\Pi_0)$, and the action using the resulting score are represented before that action. Write

$$\mathrm{PostSrc}^{\mathrm{sat}}_{I,H}(B) = \max_{t<H} \sup_{R\in\mathscr{R}^{\mathrm{Bayes,src}}_{I,H}(B)} \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}}\mathcal{C}_T(\widehat{\Pi}_t;\Pi_0)\Big]. \tag{642}$$

Thus $\mathrm{PostConc}^{\mathrm{sat}}$ is an analytic evaluation of represented beliefs, whereas $\mathrm{PostSrc}^{\mathrm{sat}}$ is its temporally admissible prospective-source subprofile.

Theorem V.11.4 (Structural Bayesian target-contact accountability)**.** *Let $R \in \mathscr{R}^{\mathrm{Bayes}}_{I,H}(B)$, and let $K_t(dc \mid \mathcal{F}_t)$ be its represented next-test kernel. Then*

$$\Pr_R\{\tau > t, V_\Theta(C_{t+1}) = 1\} = \mathbb{E}_R\left[\mathbf{1}_{\{\tau>t\}} \sum_c K_t(c \mid \mathcal{F}_t) \sum_\theta \Pi_t(\theta)V_\theta(c)\right] \tag{643}$$

and hence

$$\Pr_R\{\tau > t, V_\Theta(C_{t+1}) = 1\} \le \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}}\big(\mathcal{V}_T(\Pi_0) + \mathcal{C}_T(\widehat{\Pi}_t;\Pi_0) + e_t\big)\Big]. \tag{644}$$

Consequently

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) \le H\left(\mathcal{V}_T(\Pi_0) + \mathrm{PostConc}^{\mathrm{sat}}_{I,H}(B) + \mathrm{FilErr}^{\mathrm{sat}}_{I,H}(B)\right), \tag{645}$$

where

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) = \sup_{R\in\mathscr{R}^{\mathrm{Bayes}}_{I,H}(B)} \Pr_R\{\tau \le H\}. \tag{646}$$

The right side may be clipped at one. The same statements hold after a family-uniform restriction and its fixed family functional.

Proof. Condition on $\mathcal{F}_t$. Fresh selector randomness is conditionally independent of Θ, so the conditional contact probability is the inner double sum in Eq. (643). For every candidate c, the function $\theta \mapsto V_\theta(c)$ takes values in $[0,1]$; therefore its expectations under Π_t and $\widehat{\Pi}_t$ differ by at most e_t. Averaging over K_t and then bounding by the supremum over c gives

$$\sum_c K_t(c \mid \mathcal{F}_t) \sum_\theta \Pi_t(\theta) V_\theta(c) \le \mathcal{V}_T(\widehat{\Pi}_t) + e_t.$$

Use Eq. (639) to obtain Eq. (644). First-contact events are disjoint. Sum over $t < H$, take the saturated supremum, and use Eqs. (640) and (641) to prove Eq. (645).

□

Theorem V.11.5 (Bayes-factor gain, information, and degradation)**.** *Condition on a surviving history and represented action. Let $L_t(dy \mid \theta)$ be the declared observation channel and $\overline{L}_t(dy) = \sum_\theta \Pi_t(\theta) L_t(dy \mid \theta)$. On positive-evidence outcomes, the exact analytic posterior obeys*

$$\Pi_{t+1}(\theta) = \Pi_t(\theta) \frac{dL_t(\cdot \mid \theta)}{d\overline{L}_t}(Y_{t+1}). \tag{647}$$

Writing information in bits,

$$\mathbb{E}[D_{\mathrm{KL}}(\Pi_{t+1} \| \Pi_t) \mid \mathcal{F}_t, A_t] = (\log 2) I(\Theta; Y_{t+1} \mid \mathcal{F}_t, A_t), \tag{648}$$

$$\mathbb{E}\Big[\big(\mathcal{V}_T(\Pi_{t+1}) - \mathcal{V}_T(\Pi_t)\big)_+ \mid \mathcal{F}_t, A_t\Big] \le \sqrt{\frac{\log 2}{2} I(\Theta; Y_{t+1} \mid \mathcal{F}_t, A_t)}. \tag{649}$$

For represented beliefs,

$$\big(\mathcal{V}_T(\widehat{\Pi}_{t+1}) - \mathcal{V}_T(\widehat{\Pi}_t)\big)_+ \le \|\Pi_{t+1} - \Pi_t\|_{\mathrm{TV}} + e_t + e_{t+1}. \tag{650}$$

If the complete channel is a represented garbling of another declared channel, its expected target vulnerability is no larger than that of the more informative channel. If its SDPI coefficient is ϑ_t, the information term in Eq. (649) *may be replaced by the corresponding contracted information bound from Theorem V.3.3. A represented chi-square comparison supplies the additional same-law target-contact candidate in Eq.* (428)*; conversion to vulnerability units requires the represented comparison of Theorem V.11.15.*

Proof. Bayes' rule gives Eq. (647). Taking the posterior expectation of its log density ratio and then averaging over the observation gives the standard conditional mutual-information identity, with the factor $\log 2$ converting bits to nats. The functional $\mathcal{V}_T$ is one-Lipschitz in total variation: it is the supremum of expectations of functions taking values in $[0,1]$. Pinsker's inequality and Jensen's inequality therefore prove Eq. (649). Apply the same Lipschitz property twice and the triangle inequality to obtain Eq. (650).

Blackwell garbling preserves every decision rule available after the garbled channel as a randomized rule after the original channel. Taking the best target decision proves vulnerability monotonicity; this is the specialization of Theorem V.4.5. The SDPI and chi-square conclusions follow from the cited same-law comparisons.

□

Corollary V.11.6 (Cumulative structural Bayesian concentration)**.** *Let $Y_1, \dots, Y_L$ be the actual ordered target-dependent reveals in a complete structural Bayesian record, and write*

$$i_j = I(\Theta; Y_j \mid \mathcal{F}_{j-1}, A_{j-1})$$

in bits. If $J_t^{\uparrow}$ is any represented upper bound for the total information in the pre-hit record through time t, then

$$\mathbb{E}\mathcal{C}_T(\widehat{\Pi}_t;\Pi_0) \leq \min\left\{\sqrt{\frac{\log 2}{2}J_t^{\uparrow}},\ \sum_{j\leq L_t}\mathbb{E}\sqrt{\frac{\log 2}{2}i_j}\right\} + \mathbb{E}e_t. \tag{651}$$

Each i_j may be replaced by its channel-capacity or SDPI candidate in Theorem V.3.3. Target- independent computation between reveals contributes zero information; its belief-update, optimization, memory, and deployment costs remain in the complete record.

Proof. The total-information candidate is Eq. (423), followed by the one-Lipschitz vulnerability comparison and the filter error. For the second candidate, telescope $\mathcal{V}_T(\Pi_t) - \mathcal{V}_T(\Pi_0)$, bound the positive part by the sum of positive one-step increments, and apply Eq. (649) to each reveal. Replacing the exact posterior by $\widehat{\Pi}_t$ adds e_t. The conditional information identity assigns zero to deterministic target-independent updates.

□

V.11.3 Source-resolved actionable Bayesian gain

The posterior of Eq. (632) is an analytic law. The operational quantity is the improvement in the next represented test. This subsection resolves that improvement into the structural sources of the complete pre-hit record and retains the noise contraction of every target-dependent reveal.

Definition V.11.7 (Actionable Bayesian gain and source-resolved charge)**.** Let $R \in \mathscr{R}_{I,H}^{\mathrm{Bayes}}(B)$, and let $K_t(\,\cdot \mid \mathcal{F}_t)$ be its represented next-test kernel. Put

$$g_{R,t}^{\mathrm{Bayes}} = \Bigg[\mathbb{E}_R\Bigg[\mathbf{1}_{\{\tau>t\}}\sum_c K_t(c \mid \mathcal{F}_t)\sum_\theta \Pi_t(\theta)V_\theta(c)\Bigg] - \mathbb{E}_R[\mathbf{1}_{\{\tau>t\}}]\mathcal{V}_T(\Pi_0)\Bigg]_+ . \tag{652}$$

This is the target-contact improvement actually deployed by the record. It is distinct from the analytic excess $\mathcal{C}_T(\Pi_t;\Pi_0)$ and from the represented-belief audit $\mathcal{C}_T(\widehat{\Pi}_t;\Pi_0)$.

Truncate the execution immediately after this next test and apply the five-family frontier $\mathfrak{F}_5$ of Corollary III.16.11. For $a \in \mathfrak{J}^5_{\mathsf{Abs}}$, let

$$\mathcal{B}_{I,R,t}^{(a)} = \sum_{\ell:\,\mathsf{W}_a(E_{R,t,\ell})} \mathrm{ch}_I(E_{R,t,\ell}) \tag{653}$$

be the sum of the contextual charges in that truncated record carrying authenticated provenance tag a. The survival gate, next-test flag, stopping instruction, and terminal postprocessing remain in the complete record. Their deterministic filtration vertices have zero incremental charge by Lemmas III.17.1 and III.17.2.

At a common ceiling D, define

$$\mathcal{B}_{I,t}^{(a),\mathrm{sat}}(D) = \sup_{R\in\mathscr{R}_{I,H}^{\mathrm{Bayes}}(D)} \mathcal{B}_{I,R,t}^{(a)}, \qquad \sup\varnothing = 0, \tag{654}$$

and use the declared family functional for the family-uniform version. These are restrictions of the existing contextual-charge envelopes, not new structural channels.

When the complete pre-hit record contains a prospective master source, give it the unique exact index

$$\alpha(R,t) = (\chi, S_{\mathrm{st}}, S_{\mathrm{cf}}), \quad \chi \in \mathfrak{S}^5_{\mathrm{src}}, \quad \varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}, \quad S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}, \tag{655}$$

from Theorem III.3.4. The corresponding exact-cell envelope is the restriction of Eq. (654) to records with that index. Exact cells are disjoint. The five single-tag envelopes may overlap and are used only as a nonnegative cover.

Theorem V.11.8 (Exact source-resolved accounting of actionable Bayesian gain)**.** *Let $H_B(n)$ be the largest refined horizon admitted by the complete budget, and let*

$$D^{\mathrm{Bayes}}_B(n) = \Psi^{\mathrm{alg}}_5\Big(\Psi_{\mathrm{hit}}(B^{\mathrm{Bayes}}, n)\Big) \tag{656}$$

be the class-preserving ceiling for the represented Bayesian record, its next-test truncation, and its normalized five-family frontier. Then every record and every $t < H_B(n)$ satisfy

$$g^{\mathrm{Bayes}}_{R,t} \leq \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}}(\mathcal{C}_T(\widehat{\Pi}_t; \Pi_0) + e_t)\Big], \tag{657}$$

$$g^{\mathrm{Bayes}}_{R,t} \leq eH_B(n) \sum_{a\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{B}^{(a)}_{I,R,t}. \tag{658}$$

Consequently the saturated actionable-gain profile

$$\mathrm{BayesGain}^{\mathrm{sat}}_{I,H}(B) = \max_{t<H} \sup_{R\in\mathscr{R}^{\mathrm{Bayes}}_{I,H}(B)} g^{\mathrm{Bayes}}_{R,t} \tag{659}$$

obeys

$$\mathrm{BayesGain}^{\mathrm{sat}}_{I,H}(B) \leq eH_B(n) \max_{t<H} \sum_{a\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{B}^{(a),\mathrm{sat}}_{I,t}(D^{\mathrm{Bayes}}_B(n)). \tag{660}$$

Write $R_{\leq t}$ for the complete record truncated immediately after the deployed next test. The Bayesian quantities are the corresponding restrictions of the complete learning ledger:

$$\mathcal{B}^{(a)}_{I,R,t} = \mathcal{L}^{(a)}_{I,R_{\leq t}}, \qquad J^{\mathrm{Bayes,(a)}}_{R,t} = J^{(a)}_{I,R_{\leq t}}. \tag{661}$$

Likelihood evaluation, evidence normalization, belief storage, posterior approximation, acquisition optimization, and deployment therefore remain in the complete source-resolved record of Definition IV.13.2. The analytic posterior is not inserted as a represented source.

For records carrying a prospective master source, the source-resolved class is also the disjoint union of the exact indices in Eq. (655)*. Hence its supremum is the maximum of at most* $5(2^5-1)2^5 = 4960$ *exact-cell suprema. This reindexing retains every likelihood evaluator, locator, normalizer, belief coordinate, acquisition score, optimizer, transition, coefficient derivation, and qualification gate in the same complete record.*

Proof. Condition on $\mathcal{F}_t$. The first term in Eq. (652) is the exact conditional next-contact identity of Eq. (643). The one-Lipschitz target-vulnerability comparison used in Theorem V.11.4 gives Eq. (657) after subtracting the prior baseline and taking the positive part.

For the second bound, use the truncated computation that retains the complete pre-hit record and stops after the next test. Its success probability h_t dominates $g^{\mathrm{Bayes}}_{R,t}$, because the subtracted prior contact is nonnegative. Steps 2 and 3 of Theorem III.21.4 give

$$h_t \leq eH_B(n) \sum_{a\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{B}^{(a)}_{I,R,t}.$$

This proves Eq. (658). Taking the record supremum at the common normalized ceiling proves Eq. (660).

For a prospective master source, coefficient-complete normalization assigns one support mode and one exact state–coefficient signature by Theorem III.3.4. Those cells are disjoint and cover the normalized prospective source class. A supremum over their union is therefore their maximum. The complete-record assertion is part of Definitions V.11.1 and III.3.1.

□

Corollary V.11.9 (Pre-hit status of Bayesian source charges)**.** *In the setting of Theorem V.11.8, every nonzero summand of Eq.* (653) *is assigned to an authenticated vertex available before the next test. Its complete downstream record contains the represented likelihood use, belief update, acquisition score, selected kernel, and target-contact normalization. Consequently its source tag and exact master/signature index are determined without using a later first-hit label.*

More precisely, independent random coordinates are valid-Lift auxiliaries and add zero charge. Conditional on those coordinates and the earlier state, normalization, storage, comparison, selection, survival, verification, and stopping vertices are deterministic postprocessings and add zero charge. Hence the charge-bearing frontier is the pre-hit ancestry of the deployed update. Caus, Lift, and Bdry retain the source index of that ancestry, while its gradient-orthogonal complement is J_{res} and contributes zero target-qualified source charge.

Proof. Apply Lemma III.17.2 to the complete truncated record used in the proof of Theorem V.11.8. Its second and third cases remove independent auxiliary randomness and every deterministic descendant from the incremental charge ledger. The remaining nonzero vertices occur before the selected test and belong to the normalized backward slice of the likelihood-to-action chain. Their five-family tags are fixed by Theorem III.2.2, and their exact master and state–coefficient indices are fixed by Theorem III.3.4. Source inheritance and the residual assignment are Proposition II.5.9 and Theorem II.7.2.

□

Theorem V.11.10 (Noise-contracted source information for Bayesian updates)**.** *Enumerate the target-dependent coordinates actually revealed before the next test as $Y_1, \ldots, Y_{L_t}$. Fix the canonical anchored source assignment of Corollary III.16.10; write $a_j \in \mathfrak{J}^5_{\mathsf{Abs}}$ for the assigned source of Y_j. With the channel-capacity, available-information, and SDPI quantities of Definition V.3.1, set*

$$J^{\mathrm{Bayes},(a)}_{R,t} = \sum_{\substack{1\le j\le L_t\\ a_j=a}} \mathbb{E}\min\{c_j, \vartheta_j a^{\mathrm{info}}_j\}, \qquad J^{\mathrm{Bayes,src}}_{R,t} = \sum_{a\in\mathfrak{J}^5_{\mathsf{Abs}}} J^{\mathrm{Bayes},(a)}_{R,t}. \tag{662}$$

The superscript on a^{info}_j distinguishes the available-information scalar from the source tag a_j. Deterministic target-independent postprocessing contributes zero to every summand.

Let $\beta^{\mathrm{Bayes}}_{R,t}$ be the next-contact probability after retaining the public pre-hit record and the selected kernel while resampling the target independently from its declared reference conditional law, and put

$$\beta^{\mathrm{Bayes},+}_{R,t} = \Big[\beta^{\mathrm{Bayes}}_{R,t} - \mathbb{E}_R[\mathbf{1}_{\{\tau>t\}}]\mathcal{V}_T(\Pi_0)\Big]_+. \tag{663}$$

Then

$$g^{\mathrm{Bayes}}_{R,t} \le \min\left\{1, \beta^{\mathrm{Bayes},+}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{Bayes,src}}_{R,t}}, eH_B(n)\sum_{a\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{B}^{(a)}_{I,R,t}\right\}. \tag{664}$$

Every structural Rademacher, sequential Rademacher, martingale PAC–Bayes, Blackwell, capacity, or dynamical candidate converted to this same gain under the same conditional law may be added to the recordwise minimum. An unavailable candidate has neutral value one.

For a homogeneous binary symmetric reveal of noise rate η, the channel candidate may use

$$\vartheta_j(\eta) = (1-2\eta)^2. \tag{665}$$

Thus the square-root information term carries one factor $1-2\eta$. Noise correction of a learned clean score remains the inverse factor $(1-2\eta)^{-1}$ in Theorems IV.10.5 and IV.11.2; the contraction and correction factors have different directions and are retained separately.

Proof. The ordered-reveal chain rule and SDPI bound are Theorem V.9.7. Its conditional decoupling inequality bounds next contact by $\beta_{R,t}^{\mathrm{Bayes}} + \sqrt{(\log 2) J_{R,t}^{\mathrm{Bayes,src}}/2}$. Subtract the prior baseline, take the positive part, and combine with Eq. (658). Every additional listed candidate bounds the same record, event, normalization, and conditional law, so their minimum is valid. The binary symmetric SDPI candidate is the standard maximal-correlation coefficient of the declared flip channel; its square root gives the displayed vulnerability attenuation. The inverse noise factor belongs to the separate clean-risk conversion already proved in the cited learning theorems.

□

Corollary V.11.11 (Adaptive composition and no-source Bayesian consequence)**.** *For every complete structural Bayesian record,*

$$\Pr_R\{\tau \le H\} \le H\mathcal{V}_T(\Pi_0) + \sum_{t=0}^{H-1} g_{R,t}^{\mathrm{Bayes}}. \tag{666}$$

Consequently, if at the common compiled ceiling

$$\mathcal{B}_{\mathfrak{M},t}^{(a),\mathrm{sat}} \le \varepsilon_{a,t} \qquad (a \in \mathfrak{J}_{\mathsf{Abs}}^5,\ 0 \le t < H), \tag{667}$$

then

$$\mathrm{BayesArr}_{\mathfrak{M},H}^{\mathrm{sat}}(B) \le H\mathcal{V}_T(\Pi_0) + eH_B(n) \sum_{t=0}^{H-1} \sum_{a \in \mathfrak{J}_{\mathsf{Abs}}^5} \varepsilon_{a,t}. \tag{668}$$

The same conclusion may use the smaller recordwise minimum in Eq. (664)*. Hence a polynomial horizon preserves negligibility of the neutral contact and all source-resolved gain inputs.*

Proof. Subtract and add the prior-vulnerability baseline in each exact next-contact identity and discard negative excess. Summing the disjoint first-contact events proves Eq. (666). Apply Eq. (660) at each time and the monotone family functional to obtain Eq. (668).

□

V.11.4 Optimal acquisition and structural routing

Theorem V.11.12 (Optimal structural Bayesian acquisition)**.** *Restrict the suffix-record class in Theorem V.9.4 to complete structural Bayesian records and write $V_t^{\mathrm{Bayes}}(h,\widehat{\pi};B)$ for its target-arrival supremum. Then*

$$V_t^{\mathrm{Bayes}}(h,\widehat{\pi};B) = \sup_{(a,\varrho,\widehat{U})\,:\,c_t^{\mathrm{Bayes}}(a,\varrho,\widehat{U};h,\widehat{\pi}) \preceq B} \left\{ p_T(h,a) + \mathbb{E}\left[\mathbf{1}_{\{\tau > t+1\}} A^{H-t-1}_{\varrho(H_{t+1},\widehat{U}(\widehat{\pi},Y_{t+1}))} \mid h, a \right] \right\}, \tag{669}$$

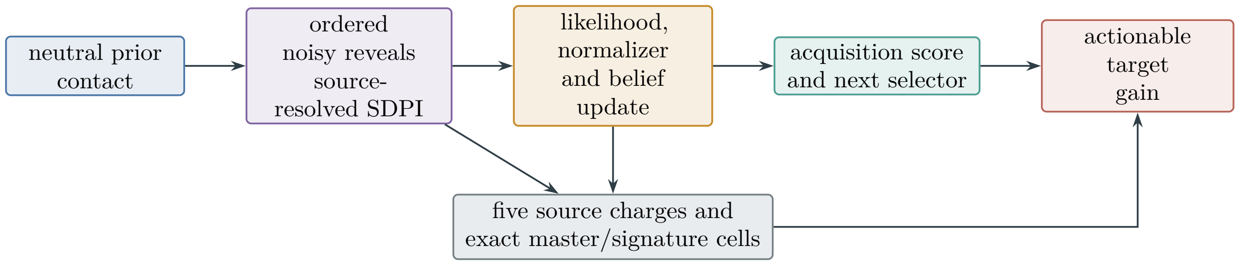

Figure 91: Source-resolved structural Bayesian gain. The operational scalar is the improvement in the represented next test. Every noisy reveal retains its source and contraction coefficient; deterministic updating adds no new source charge; and the complete deployed rule is evaluated by same-record information, learning, dynamical, and contextual-charge certificates.

where $\widehat{U}$ is the represented belief update and c_t^{Bayes} is the complete active cost including its likelihood, normalization, memory, comparison, continuation, and deployment slots. The stopped occupation-flow primal and Bellman-barrier dual in Theorems V.10.2 and V.10.4 hold after restricting their record and first-step sets to these Bayesian records.

Always

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) \le \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B). \tag{670}$$

Proof. Decompose a nonterminal structural Bayesian suffix record into its first action, represented belief update, and represented continuation rule. The converse concatenation is a complete structural Bayesian suffix record and has cost c_t^{Bayes}. The disjoint first-contact decomposition therefore proves Eq. (669), by the proof of Theorem V.9.4. Restricting the occupation and Bellman record sets preserves their balance identities, telescoping proof, and exact suffix-value barrier.

Every structural Bayesian record is a complete active record, proving Eq. (670).

□

Theorem V.11.13 (Universal Bayesianization of represented active computation)**.** *Let $R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)$ be a complete active record. There is a family-uniform compiler $R \mapsto \mathfrak{B}(R)$ and a class-preserving resource majorant*

$$B^{\mathrm{uBayes}} = \Psi_{\mathrm{uBayes}}(B, H, n) \tag{671}$$

such that $\mathfrak{B}(R) \in \mathscr{R}^{\mathrm{Bayes}}_{I,H}(B^{\mathrm{uBayes}})$ and the following data are identical in R and $\mathfrak{B}(R)$:

(i) *the public pre-hit filtration and represented sufficient history;*

(ii) *every query, observation, action, fresh-randomness, and transition kernel;*

(iii) *the verifier, stopping rule, output converter, and continuation rule;*

(iv) *the complete execution law, first-hit law, and target-arrival probability.*

The compiler attaches the declared full-support prior Π_0, the represented constant belief $\widehat{\Pi}_t = \Pi_0$, the constant likelihood score one, the evidence normalizer one, and the identity belief update. The original action and continuation rules retain the complete represented history as their sufficient-memory input and need not read the attached belief. The analytic posterior

$$\Pi_t = \Pr\{\Theta \in \cdot \mid \mathcal{F}_t\}$$

is therefore available for conditional-probability identities without being materialized by the compiled computation.

Conversely, forgetting the belief-specific slots of a complete structural Bayesian record produces a complete active record with the same execution law and no larger cost. Consequently

$$\mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B) \le \mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(\Psi_{\mathrm{uBayes}}(B,H,n)), \tag{672}$$

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) \le \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B). \tag{673}$$

The same inequalities hold for one family-uniform derivation and every declared monotone positively homogeneous family functional. Hence the two profiles coincide on every transfer-equivalence class of ceilings; in particular,

$$\mathrm{BayesArr}^{\mathrm{sat}}_{\mathrm{poly}} = \mathrm{ActArr}^{\mathrm{sat}}_{\mathrm{poly}}. \tag{674}$$

Proof. The definition of a complete structural Bayesian record in Definition V.11.1 begins with a complete active record. Retain every field of R. Append a reference to the fixed prior, the constant likelihood and normalizer, and the identity update. Their codes are independent of the target and of the history. Evaluation and storage over the H clocked steps have a uniform resource bound obtained by adding their existing belief, likelihood, normalization, memory, and clock slots to Eq. (574). Axioms B1–B4 and computation closure give the class-preserving majorant Eq. (671).

The attached fields are not ancestors of the original action or continuation vertices. Evaluation of the retained record therefore produces exactly the same conditional kernels at every history. Induction on the clock gives equality of the finite-dimensional path laws, hence equality of the first-hit and target-arrival laws. The analytic posterior is the conditional law of this common path measure and requires no represented posterior evaluator.

Deleting the appended belief-specific vertices leaves the original complete active record, which proves the converse map. Taking the appropriate suprema gives Eqs. (672) and (673). Both compilers use one family-uniform syntax transformation, so the same proof commutes with the family functional. Mutual class-preserving domination gives the transfer-class equality and its polynomial specialization.

□

Corollary V.11.14 (Universal represented-computation Bayesian ceiling)**.** *Let* $\mathsf{Alg}_{n,\le B}$ *be the represented computation class of Definition III.24.1, with finite refined horizon* $H_B(n)$*. There is a class-preserving compiled ceiling*

$$B^{\mathrm{compBayes}} = \Psi_{\mathrm{uBayes}}(\Psi_{\mathrm{act}}(B, H_B, n, \mathbf{0}), H_B, n) \tag{675}$$

for which

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le \mathrm{BayesArr}^{\mathrm{sat}}_{\mathfrak{M},n,H_B}(B^{\mathrm{compBayes}}). \tag{676}$$

The reverse inclusion holds by forgetting the Bayesian and active-interface annotations. Thus the represented-computation, active, and structural Bayesian optima coincide on the declared saturated transfer class. In the polynomial specialization,

$$\mathrm{Succ}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{ActArr}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{BayesArr}^{\mathrm{poly,sat}}_{\mathfrak{M},n}. \tag{677}$$

Moreover, at the common source-charge ceiling of Theorem V.11.8,

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le H_B \mathfrak{M}_n(\mathbf{1})\mathcal{V}_T(\Pi_0) + eH_B^2 \sum_{a\in\mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{B}^{(a),\mathrm{sat}}_{\mathfrak{M}}(D^{\mathrm{Bayes}}_{B^{\mathrm{compBayes}}}(n)). \tag{678}$$

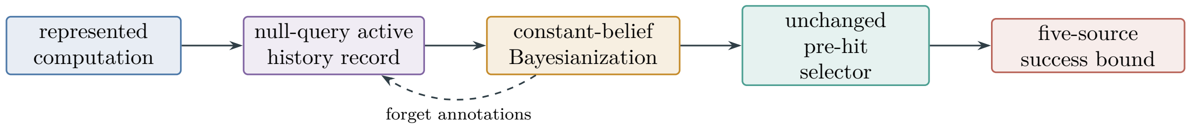

Figure 92: Universal Bayesianization preserves the execution law and the represented pre-hit selector. The exact posterior is an analytic conditional law; the compiled computation carries only the constant represented prior and identity update. Prospective structural charging therefore bounds the common computation, active, and Bayesian ceiling.

Proof. Use the operational embedding of Lemma II.11.1 and Corollary II.11.3. Regard its public history as the sufficient-memory state, take a null query, and retain its represented transition, verifier, stopping, and output rules. Channel exhaustiveness and computation closure supply the complete active record at the majorant Ψ_{act}; presentation-admissible oracle interfaces remain fields of that same record. Apply Theorem V.11.13 and take the family-uniform supremum to prove Eq. (676). Forgetting annotations preserves the execution law and proves the reverse transfer-class inclusion.

Finally apply the prospective selector charge theorem Theorem III.21.4 in its source-resolved Bayesian form Eqs. (660) and (666) to the compiled record. There are at most H_B pre-hit gains, each bounded by the five-source envelope with factor eH_B. Monotonicity and positive homogeneity of the family functional give Eq. (678).

□

Theorem V.11.15 (Structural provenance of affordable posterior concentration)**.** *Suppose a complete structural Bayesian record represents and deploys a positive value of $\mathcal{C}_T(\widehat{\Pi}_t;\Pi_0)$ through its represented likelihood, normalizer, belief, acquisition score, optimizer, and next-test kernel. The deployed target-aligned quantity is the actionable gain $g^{\mathrm{Bayes}}_{R,t}$ of Eq.* (652)*. Its complete pre-hit derivation is temporally admissible, and its source-resolved contextual charge is bounded by Theorem V.11.8. At the common ceiling $B^{\mathrm{Bayes}} = \Psi_{\mathrm{Bayes}}(B,H,n,\mathbf{m},\boldsymbol{\zeta})$, its complete backward cut belongs to exactly one of*

$$\mathcal{C}_X,\quad \mathcal{C}_A,\quad \mathcal{C}_{Q,\mathrm{ex}},\quad \mathcal{C}_{Q,\mathrm{rev}},\quad \mathcal{C}_{Q,\mathrm{nr}}. \tag{679}$$

Consequently

$$\mathrm{PostSrc}^{\mathrm{sat}}_{I,H}(B^{\mathrm{Bayes}}) = \max_{c\in\mathfrak{C}_{\mathrm{master}}} \mathrm{PostSrc}^{\mathrm{sat}}_{I,H;c}(B^{\mathrm{Bayes}}). \tag{680}$$

The corresponding actionable-gain class has the quantitative envelope

$$\mathrm{BayesGain}^{\mathrm{sat}}_{I,H}(B) \le eH_B(n) \max_{t<H} \sum_{a\in\mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{B}^{(a),\mathrm{sat}}_{I,t}(D^{\mathrm{Bayes}}_B(n)), \tag{681}$$

and each complete prospective source record has the unique exact source–state–coefficient index of Eq. (655)*. Thus the numerical ledger has no unrestricted direct-source remainder: a positive deployed gain is evaluated through its five contextual source charges, while the master index records its exact support and mixed provenance.*

For one record, every finite information, Blackwell, filter, capacity, observation-resolution, or master-source candidate must be converted to the same target-vulnerability excess under the same conditional law. The smallest such candidate is an upper bound for that record's concentration; an unavailable converter has neutral value one. Thus posterior concentration is a numerical view of the existing master profile and not an additional provenance family.

Proof. The represented concentration statistic, the score derived from it, and the next-test kernel are available before the action. Their complete derivation is charged, so the deployed record satisfies Definition III.21.1. Apply channel exhaustiveness, backward ancestry, and the master classification in Theorems II.9.4 and III.2.4. The five master families are disjoint and exhaustive, so partitioning the record class and taking the saturated supremum proves Eq. (680). Every listed finite candidate bounds the same record and coordinate by its defining comparison; their minimum is therefore valid. Missing comparisons contribute the neutral value and supply no numerical conclusion.

The quantitative gain bound is Eq. (660). The unique refined index follows from Theorem III.3.4. These two decompositions have different roles: contextual charge bounds the deployed gain, while the exact master/signature index records the complete source carrier without assigning a new mode.

□

Corollary V.11.16 (Learned structural Bayesian control)**.** *Let $\widehat{R}$ be a learned structural Bayesian record satisfying the simultaneous action-value and optimization inequalities of Theorem V.9.12, with their analytic action-value comparator restricted to the structural Bayesian suffix optimum in Eq.* (669)*. Let Δ^H_{fil} be the represented path/event comparison in Theorem V.2.4. Then*

$$A^H_{\widehat{R}} \le \mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B), \tag{682}$$

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) - A^H_{\widehat{R}} \le \mathbb{E}_{\widehat{R}} \sum_{t<H} \mathbf{1}_{\{\tau>t\}}(2\Gamma_t + \varepsilon_{\mathrm{opt},t}) + \Delta^H_{\mathrm{fil}}. \tag{683}$$

At a compiled ceiling that contains the universal Bayesianization of the complete active class, the same inequality is

$$0 \le \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B_0) - A^H_{\widehat{R}} \le \mathbb{E}_{\widehat{R}} \sum_{t<H} \mathbf{1}_{\{\tau>t\}}(2\Gamma_t + \varepsilon_{\mathrm{opt},t}) + \Delta^H_{\mathrm{fil}}, \tag{684}$$

where $B = \Psi_{\mathrm{uBayes}}(B_0, H, n)$. After the operational compilation of Corollary V.11.14, the first term is the saturated represented-computation optimum at its corresponding transfer-equivalent ceiling. The radii may be supplied by structural Rademacher, sequential, or martingale PAC–Bayes bounds. Their PAC–Bayes posterior Q remains separate from the task belief $\widehat{\Pi}_t$, and both costs occur in the same complete record.

Proof. The first inequality is membership in the structural Bayesian record class. Apply Theorem V.9.12 to the exact-belief reference execution and then Theorem V.2.4 to compare that execution with the represented filter. The triangle inequality gives Eq. (683). Substitute the profile equality of Theorem V.11.13 to obtain Eq. (684).

□

V.11.5 Optimal structural information extraction

The next result identifies the complete computational ceiling with optimal structural Bayesian acquisition and resolves its target gain into the existing structural modalities. The optimization is over represented acquisition records at one charged ceiling; the analytic posterior supplies conditional identities and is materialized only when its evaluator is represented and charged.

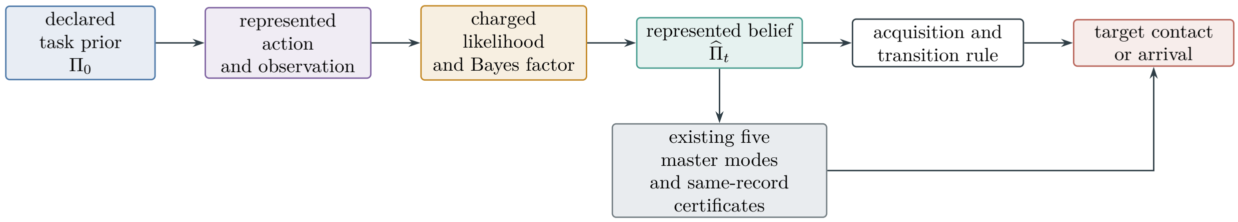

Figure 93: Optimal structural Bayesian learning. Target-dependent data enter through represented observation channels, and every likelihood, normalization, belief, acquisition, and transition operation is charged. Posterior concentration is evaluated through the existing master classification before it contributes to target contact.

Definition V.11.17 (Optimal structural extraction certificate)**.** Fix a finite target-index carrier Θ, target vulnerability $\mathcal{V}_T$ from Definition V.11.3, a horizon H, and a complete structural Bayesian record R. At time $t < H$, define the same-record information candidate

$$U^{\mathrm{info}}_{R,t} = \min\left\{1, \beta^{\mathrm{Bayes},+}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{Bayes,src}}_{R,t}}\right\}, \tag{685}$$

and the source candidate

$$U^{\mathrm{src}}_{R,t} = \min\left\{1, eH_B(n) \sum_{a \in \mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{B}^{(a)}_{I,R,t}\right\}. \tag{686}$$

Let $U^{\mathsf{Caus}}_{R,t}$ be the probability-normalized same-record Caus contact coordinate of Definition V.12.2. Let $U^{\mathsf{Lift}}_{R,t}$ be the minimum of one, U^{H}_{Lift}, $U^{H}_{\mathsf{Lift},\mathrm{noise}}$, and every represented probability-normalized finite-horizon comparison derived from $U^{\lambda}_{\mathsf{Lift}}$ or $U^{\lambda}_{\mathsf{Lift},\mathrm{noise}}$ in that same record. Missing comparisons have their neutral values. The boundary candidate $U^{\mathsf{Bdry}}_{R,t}$ is the exact represented next-contact probability when the boundary readout is available before the transition and is one otherwise. The residual candidate is zero on the gradient-orthogonal component and is not used as a bound for the qualified component.

The complete extraction certificate is

$$U^{\mathrm{opt}}_{R,t} = \min\left\{1, U^{\mathrm{info}}_{R,t}, U^{\mathrm{src}}_{R,t}, U^{\mathsf{Caus}}_{R,t}, U^{\mathsf{Lift}}_{R,t}, U^{\mathsf{Bdry}}_{R,t}, U^{\mathrm{Bell}}_{R,t}\right\}, \tag{687}$$

where $U^{\mathrm{Bell}}_{R,t}$ is the same-record Bellman-support minimum of Eq. (618) evaluated at the zero barrier for the next-contact event. Each finite entry must bound the same next-contact gain under the same surviving-history law.

Theorem V.11.18 (Bayes–Fisher optimality of a represented observation)**.** *Condition on a surviving pre-hit history and a represented acquisition action. Let π be the current full-support task belief, restrict to the represented positive-likelihood support of the returned observation, let $\ell_y(\theta) = \log L(y \mid \theta)$ be the represented log-likelihood of the returned observation, and put*

$$\pi^y(\theta) = \frac{\pi(\theta)e^{\ell_y(\theta)}}{\sum_u \pi(u)e^{\ell_y(u)}}.$$

For every probability law $q \ll \pi$,

$$\mathbb{E}_q \ell_y - D_{\mathrm{KL}}(q\|\pi) = \log \mathbb{E}_\pi e^{\ell_y} - D_{\mathrm{KL}}(q\|\pi^y). \tag{688}$$

Hence π^y is the unique maximizer of represented likelihood gain minus KL displacement. Along

$$\pi_u^y(\theta) = \frac{\pi(\theta)e^{u\ell_y(\theta)}}{\sum_v \pi(v)e^{u\ell_y(v)}}, \qquad 0 \le u \le 1,$$

the exact Fisher–Rao flow and action are

$$\dot{\pi}_u^y(\theta) = \pi_u^y(\theta)\big(\ell_y(\theta) - \mathbb{E}_{\pi_u^y}\ell_y\big), \tag{689}$$

$$\|\dot{\pi}_u^y\|_{\mathrm{FR}}^2 = \mathrm{Var}_{\pi_u^y}(\ell_y). \tag{690}$$

Thus the exact Bayesian update is the Fisher–Rao steepest likelihood flow for the fixed represented observation. Its likelihood construction, normalization, precision, storage, and deployment costs are those of Definition V.11.1.

Proof. Substitute $\log(\pi^y(\theta)/\pi(\theta)) = \ell_y(\theta) - \log \mathbb{E}_\pi e^{\ell_y}$ in $D_{\mathrm{KL}}(q\|\pi^y)$ and rearrange. Relative entropy is nonnegative and strictly convex on the support of π, proving the variational assertion. Differentiating the normalized exponential path gives Eq. (689); substitution in $\|v\|_{\mathrm{FR}}^2 = \sum_\theta v(\theta)^2/\pi_u^y(\theta)$ gives Eq. (690). The cost assertion is the complete Bayesian-record definition. □

Theorem V.11.19 (Optimal structural extraction equals the saturated ceiling)**.** *Let B be a declared represented-computation ceiling with refined horizon $H_B(n)$, and let $B^{\mathrm{compBayes}}$ be Eq. (675). At finite budget the two directions are the compiled-ceiling comparisons of Corollary V.11.14. On the polynomial saturated transfer class they give the exact identity*

$$\mathrm{Succ}_{\mathfrak{M},n}^{\mathrm{poly,sat}} = \mathrm{ActArr}_{\mathfrak{M},n}^{\mathrm{poly,sat}} = \mathrm{BayesArr}_{\mathfrak{M},n}^{\mathrm{poly,sat}}. \tag{691}$$

The Bayesian member is optimized by the resource-valued Bellman recursion in Eq. (669); each fixed observation is processed optimally by Theorem V.11.18. For every complete record and every $t < H_B(n)$,

$$g_{R,t}^{\mathrm{Bayes}} \le U_{R,t}^{\mathrm{opt}}. \tag{692}$$

Consequently,

$$\mathrm{Succ}_{\mathfrak{M},n}(B) \le H_B \mathfrak{M}_n(\mathbf{1})\mathcal{V}_T(\Pi_0) + \mathfrak{M}_n\left(\left(\sum_{t=0}^{H_B-1} \sup_{R\in\mathscr{R}_{I,H_B}^{\mathrm{Bayes}}(B^{\mathrm{compBayes}})} U_{R,t}^{\mathrm{opt}}\right)_{I\in\mathfrak{I}_n}\right). \tag{693}$$

The structural roles in this bound are exhaustive. Geom *supplies the reversible information geometry; exact and compensated* Abs *supply qualified sufficient fibers;* Caus *converts the qualified geometry into target-directed current;* Lift *transports belief, memory, and recurrent state with its comparison defects; and* Bdry *supplies the represented target interface. The gradient-orthogonal J_{res} component has zero target-qualified gain. Add, Mult, Ord, Sym, and Filt are the exact constructor provenance of these same source records and create no additional modality.*

Proof. The profile equality is Corollary V.11.14. The Bellman recursion is exact because every complete suffix record decomposes into a first acquisition, returned observation, represented update, and complete continuation, and conversely those data concatenate to a legal suffix record. The fixed-observation optimizer is Theorem V.11.18.

The information and source entries of Eq. (687) bound the deployed gain by Theorems V.11.8 and V.11.10. The Caus, Lift, boundary, and Bellman entries enter only through their same-record target-contact comparisons in Definition V.12.2 and Theorem V.10.5. Taking the minimum proves Eq. (692). Summing the disjoint first-contact gains and using Eq. (666) proves Eq. (693). Channel exhaustiveness, Hodge orthogonality, source inheritance, and the five-family theorem give the final modality statement. □

Corollary V.11.20 (Noise, sample, and target-size extraction bound)**.** *Let $N = |\Theta|$. Suppose at most m target-dependent observations are revealed before deployment. For reveal j, let c_j be its represented conditional channel-capacity bound in bits, ϑ_j its represented SDPI coefficient, and a_j^{info} its available conditional information. Then every complete record satisfies*

$$J_{R,t}^{\mathrm{Bayes,src}} \le \min\left\{\log_2 N, \sum_{j\le m} c_j, \sum_{j\le m} \vartheta_j a_j^{\mathrm{info}}\right\}. \tag{694}$$

Every target-dependent advice, oracle answer, reward, verifier outcome, or feedback coordinate appears as one of the ordered reveals with its own capacity, available-information, derivation-cost, and source tag. Selecting or postprocessing a coordinate already present in the pre-hit record adds zero conditional information and retains its complete computational cost.

For m binary symmetric reveals with common crossover probability $0 \le \eta \le 1/2$,

$$J_{R,t}^{\mathrm{Bayes,src}} \le J_{m,\eta,N}^{\uparrow} := \min\left\{\log_2 N, m(1-H_2(\eta)), (1-2\eta)^2 \sum_{j\le m} a_j^{\mathrm{info}}\right\}. \tag{695}$$

For singleton targets under the uniform prior,

$$g_{R,t}^{\mathrm{Bayes}} \le \min\left\{U_{R,t}^{\mathrm{src}}, U_{R,t}^{\mathsf{Caus}}, U_{R,t}^{\mathsf{Lift}}, U_{R,t}^{\mathsf{Bdry}}, U_{R,t}^{\mathrm{Bell}}, \beta_{R,t}^{\mathrm{Bayes},+} + \sqrt{\frac{\log 2}{2} J_{m,\eta,N}^{\uparrow}}\right\}. \tag{696}$$

Proof. Mutual information with Θ is at most $H(\Theta) \le \log_2 N$. The ordered-reveal chain rule, channel-capacity bound, and sequential SDPI bound give the other two candidates in Eq. (694). For the binary symmetric channel, substitute $c_j = 1 - H_2(\eta)$ and $\vartheta_j = (1-2\eta)^2$. Insert the resulting information bound in Eq. (685), then minimize with the remaining same-record structural candidates to obtain Eq. (696). □

V.11.6 Compact latent beliefs and optimal represented compression

The represented belief in Definition V.11.1 may be stored through a finite latent state. The latent state is part of the pre-hit computation: its encoder, update, decoder, and deployment rule retain the complete source ancestry and cost of the public record from which they are constructed.

Definition V.11.21 (Complete compact latent-belief record)**.** Let $R \in \mathscr{R}_{I,H}^{\mathrm{Bayes}}(B)$ be a complete structural Bayesian record, and let $\mathcal{H}_t$ be its complete public pre-hit history. A complete compact latent-belief extension of R contains:

(i) a finite represented code carrier $\mathcal{Z}_t$ and a temporally admissible represented encoder kernel $E_t(dz \mid \mathcal{H}_t)$, whose auxiliary randomness is independent of the target conditional on $\mathcal{H}_t$;

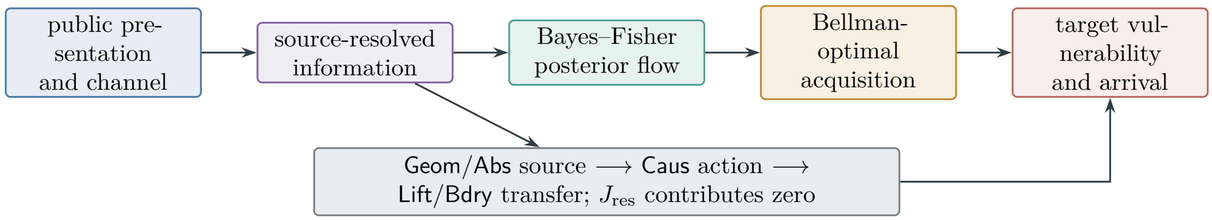

Figure 94: Optimal structural information extraction. Bayesian natural-gradient flow processes each represented observation, Bellman recursion chooses among affordable acquisitions, and the complete target gain remains bounded by the existing modality-resolved certificate.

(ii) a represented recurrent update $F_t(dz' \mid z, Y_{t+1}, A_t^{\mathrm{qry}}, \mathcal{H}_t)$ that produces Z_{t+1} before every transition that uses it;

(iii) a represented decoder $D_t : Z_t \mapsto \widehat{\Pi}_t^Z$, together with every likelihood score, normalizer, posterior statistic, acquisition score, and deployed selector computed from Z_t;

(iv) the complete encoder, update, storage, precision, decoder, optimization, verification, and deployment derivation.

Condition on the survival sector $\{\tau > t\}$. When this event has positive probability, define the analytic latent posterior, retained target information, and sufficiency defect by

$$\Pi_t^Z(\theta) = \Pr\{\Theta = \theta \mid Z_t, \tau > t\}, \tag{697}$$

$$I_{R,t}^{\mathrm{lat}} = I(\Theta; Z_t \mid \tau > t), \tag{698}$$

$$\Delta_{R,t}^{\mathrm{suf}} = I(\Theta; \mathcal{H}_t \mid Z_t, \tau > t), \tag{699}$$

$$e_{R,t}^{\mathrm{dec}} = \mathbb{E}\Big[\|\widehat{\Pi}_t^Z - \Pi_t^Z\|_{\mathrm{TV}} \mid \tau > t\Big]. \tag{700}$$

All four quantities are zero when the survival event is null. Mutual information is measured in bits. Put

$$b_{R,t}^{\mathrm{lat}} = \lceil \log_2 |\mathcal{Z}_t| \rceil . \tag{701}$$

If the represented code has d_t coordinates at p_t bits of precision and a codebook or architecture description of length b_t^{code}, then the record uses the certified bound

$$b_{R,t}^{\mathrm{lat}} \le d_t p_t + b_t^{\mathrm{code}}. \tag{702}$$

The complete latent cost is

$$\begin{aligned} B_R^{\mathrm{lat}} = B_R^{\mathrm{Bayes}} &\oplus B_E \oplus B_F \oplus B_{\mathcal{Z}} \oplus B_{\mathrm{code}} \oplus B_{\mathrm{prec}} \oplus B_{\mathrm{mem}} \\ &\oplus B_D \oplus B_{\mathrm{like}} \oplus B_{\mathrm{norm}} \oplus B_{\mathrm{score}} \oplus B_{\mathrm{optimize}} \oplus B_{\mathrm{verify}} \oplus B_{\mathrm{deploy}}. \end{aligned} \tag{703}$$

Shared ancestors occur once in the derivation DAG; every repeated evaluation or update retains its clocked cost. The source index is the exact source index of the complete backward slice in Eq. (655). Encoding and deterministic updating add no target information and create no new source tag.

Theorem V.11.22 (Latent information, sufficiency, and decision loss)**.** *For every complete compact latent-belief record and every nonnull survival sector,*

$$I(\Theta; \mathcal{H}_t \mid \tau > t) = I_{R,t}^{\mathrm{lat}} + \Delta_{R,t}^{\mathrm{suf}}. \tag{704}$$

If $J_{R,t}^{\mathrm{Bayes,src}}$ is the source-resolved ordered-reveal upper bound for the same conditioned history, then

$$I_{R,t}^{\mathrm{lat}} \leq \min\left\{b_{R,t}^{\mathrm{lat}}, J_{R,t}^{\mathrm{Bayes,src}}, I(\Theta;\mathcal{H}_t \mid \tau > t)\right\}. \tag{705}$$

Moreover,

$$\mathbb{E}\left[\left\|\Pr\{\Theta \in \cdot \mid \mathcal{H}_t, \tau > t\} - \Pi_t^Z\right\|_{\mathrm{TV}} \middle| \tau > t\right] \leq \sqrt{\frac{\log 2}{2}\Delta_{R,t}^{\mathrm{suf}}}. \tag{706}$$

Let $u(\theta, a) \in [0,1]$ be a represented target utility, and let $\mathrm{DVal}_{u,t}(\mathcal{H}_t)$ and $\mathrm{DVal}_{u,t}(Z_t)$ denote the optimal survival-conditioned decision values with the same represented action carrier. Then

$$0 \leq \mathrm{DVal}_{u,t}(\mathcal{H}_t) - \mathrm{DVal}_{u,t}(Z_t) \leq \sqrt{\frac{\log 2}{2}\Delta_{R,t}^{\mathrm{suf}}}. \tag{707}$$

Using the represented decoder instead of the analytic latent posterior adds at most $e_{R,t}^{\mathrm{dec}}$ to the right side. In particular, $\Delta_{R,t}^{\mathrm{suf}} = 0$ gives analytic decision-value equivalence. When the conditional reconstruction kernel is represented and charged, this is exactly the represented value equivalence of Corollary V.4.6.

Proof. Conditional on survival, the encoder construction gives the Markov chain $\Theta \to \mathcal{H}_t \to Z_t$. The mutual-information chain rule therefore proves Eq. (704). Data processing gives the history-information bound, while $I(\Theta; Z_t \mid \tau > t) \leq H(Z_t \mid \tau > t) \leq \log_2 |\mathcal{Z}_t|$ gives the code-capacity bound. The ordered-reveal inequality is Eq. (662) on the same conditioned record.

The conditional-information identity also gives

$$\Delta_{R,t}^{\mathrm{suf}} = \mathbb{E}\left[D_{\mathrm{KL},2}\left(\Pr\{\Theta \in \cdot \mid \mathcal{H}_t, \tau > t\} \,\middle\|\, \Pi_t^Z\right) \middle| \tau > t\right],$$

where $D_{\mathrm{KL},2}$ is relative entropy in bits. Pinsker's inequality and Jensen's inequality prove Eq. (706). The expectation of any $[0,1]$-valued action utility changes by at most total variation. Apply this first to the history-optimal action and then optimize over latent actions to obtain Eq. (707). The triangle inequality adds the decoder error. At zero conditional information, the conditional law of the history given (Θ, Z_t) equals its law given Z_t, which is the sufficiency identity in Eq. (453). The represented version uses that corollary's charged reconstruction hypothesis. □

Definition V.11.23 (Saturated compact latent-belief profile)**.** Let $\mathscr{R}_{I,H}^{\mathrm{lat}}(B, b)$ be the complete compact latent-belief records with $B_R^{\mathrm{lat}} \preceq B$ and $\max_{t<H} b_{R,t}^{\mathrm{lat}} \leq b$. Define

$$\mathrm{LatBayesArr}_{I,H}^{\mathrm{sat}}(B, b) = \sup_{R \in \mathscr{R}_{I,H}^{\mathrm{lat}}(B,b)} \Pr_R\{\tau \leq H\}, \qquad \sup \varnothing = 0. \tag{708}$$

The maximum useful belief update is the restriction of Eq. (659) to the same records:

$$\mathrm{LatBayesGain}_{I,H}^{\mathrm{sat}}(B, b) = \max_{t<H} \sup_{R \in \mathscr{R}_{I,H}^{\mathrm{lat}}(B,b)} g_{R,t}^{\mathrm{Bayes}}. \tag{709}$$

Usefulness here means deployment through the represented next-test kernel in Eq. (652). Posterior mass stored in the latent state without such a deployment contributes no actionable gain. The family-uniform profile uses one represented encoder, update, decoder, and deployment scheme on the whole size slice and then applies the declared family functional.

Theorem V.11.24 (Optimal compact latent acquisition and saturated completeness)**.** *The profile in Eq.* (708) *is monotone in B and b, and*

$$\mathrm{LatBayesArr}^{\mathrm{sat}}_{I,H}(B,b) \le \mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B). \tag{710}$$

The same inclusion gives

$$\mathrm{LatBayesGain}^{\mathrm{sat}}_{I,H}(B,b) \le \mathrm{BayesGain}^{\mathrm{sat}}_{I,H}(B). \tag{711}$$

Restricting Eq. (669) *to records in $\mathscr{R}^{\mathrm{lat}}_{I,H}(B,b)$ gives the exact resource-valued Bellman recursion that jointly optimizes acquisition, encoding, recurrent updating, decoding, and continuation under the remaining cost and code-capacity budgets.*

Conversely, encode the complete represented sufficient history and internal state of a structural Bayesian record. Its code length is the storage length already charged in that record, its update is the original clocked state update, and its decoder may be the original represented belief field or the constant represented belief supplied by universal Bayesianization. Hence a class-preserving compiler Ψ_{lat} satisfies

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) \le \sup_{b:\, B_{\mathrm{code}}(b) \preceq \Psi_{\mathrm{lat}}(B,H,n)} \mathrm{LatBayesArr}^{\mathrm{sat}}_{I,H}(\Psi_{\mathrm{lat}}(B,H,n),b). \tag{712}$$

Here $B_{\mathrm{code}}(b)$ denotes the represented storage, precision, update, and evaluation charge of a code with capacity b, as included in Eq. (703)*. Together with Corollary V.11.14, this gives*

$$\mathrm{LatBayesArr}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{BayesArr}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{ActArr}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{Succ}^{\mathrm{poly,sat}}_{\mathfrak{M},n}, \tag{713}$$

where the latent profile takes the union over code capacities admitted by the same polynomial saturated cost. Because the compiler preserves the represented next-test kernel and its pre-hit law, it also gives

$$\mathrm{LatBayesGain}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{BayesGain}^{\mathrm{poly,sat}}_{\mathfrak{M},n}. \tag{714}$$

Proof. Class inclusion and monotonicity prove Eq. (710). First-step decomposition and concatenation preserve both remaining resources, so the proof of Theorem V.11.12 gives the restricted exact recursion. For the converse, the complete clocked state is a finite represented code. Storing that code and applying the original transition and selector reproduces the path law exactly; encoding adequacy and computation closure give the class-preserving majorant Ψ_{lat}. This proves Eq. (712). Apply universal Bayesianization and take the polynomial saturated transfer class to obtain the arrival equality. The compiler also leaves the next-test kernel, target utility, survival event, and conditional law unchanged, so it preserves $g^{\mathrm{Bayes}}_{R,t}$ exactly. This proves Eq. (714). □

Theorem V.11.25 (Unified latent structural certificate)**.** *For one complete compact latent-belief record, put*

$$U^{\mathrm{lat,info}}_{R,t} = \min\left\{1, \beta^{\mathrm{Bayes},+}_{R,t} + \sqrt{\frac{\log 2}{2}\min\{b^{\mathrm{lat}}_{R,t}, J^{\mathrm{Bayes,src}}_{R,t}\}}\right\}. \tag{715}$$

Let $U^{\mathrm{lat,stat}}_{R,t}$ be the smallest same-record target-gain consequence of the source-resolved Rademacher, effective-dimension, PAC–Bayes, sequential, or martingale candidates for the complete encoder–decoder–selector class. Define

$$U^{\mathrm{lat}}_{R,t} = \min\left\{U^{\mathrm{opt}}_{R,t}, U^{\mathrm{lat,info}}_{R,t}, U^{\mathrm{lat,stat}}_{R,t}, U^{\mathsf{Caus}}_{R,t}, U^{\mathsf{Lift}}_{R,t}, U^{\mathsf{Bdry}}_{R,t}, U^{\mathrm{Bell}}_{R,t}\right\}, \tag{716}$$

where an unavailable candidate has neutral value one. Then

$$g^{\mathrm{Bayes}}_{R,t} \le U^{\mathrm{lat}}_{R,t}. \tag{717}$$

The assignment of the complete record is exhaustive. The represented latent metric, kernel, and belief simplex belong to Geom*. Encoder fibers belong to exact or compensated* Abs *precisely when their corresponding gates hold. A deployed target-directed belief or selector current is authenticated* Caus*. Recurrent state, memory, decoder comparison, and filter transfer belong to* Lift*. Support masks and the target decoder belong to* Bdry*. The gradient-orthogonal component belongs to J_{res} and has zero target-qualified gain. Add, Mult, Ord, Sym, and Filt remain the constructor provenance of these same records.*

For a learned latent policy, let $\Gamma^{\mathrm{lat}}_{R,t}$ be its applicable same-record generalization radius and $\varepsilon^{\mathrm{opt}}_{R,t}$ its represented optimization error. Relative to the complete-history optimum, the finite-horizon loss is bounded by

$$0 \le \mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) - A^H_R \le \sum_{t<H} \mathbb{E}\left[\mathbf{1}_{\{\tau>t\}}\left(\sqrt{\frac{\log 2}{2}\Delta^{\mathrm{suf}}_{R,t}} + e^{\mathrm{dec}}_{R,t} + 2\Gamma^{\mathrm{lat}}_{R,t} + \varepsilon^{\mathrm{opt}}_{R,t}\right)\right] + \Delta^H_{\mathrm{fil}}. \tag{718}$$

Proof. The information candidate is conditional data processing followed by the target-contact Pinsker comparison in Theorem V.11.10. The source, dynamical, and Bellman candidates are the same-record entries of Theorem V.11.19; the statistical candidate is Theorem IV.13.3 applied to the complete encoder–decoder–selector output class. Taking their minimum proves Eq. (717).

The channel assignment follows from Theorems II.9.4 and III.2.4 and from source inheritance under Caus, Lift, and Bdry. Finally combine the decision deficiency in Eq. (707), the decoder triangle inequality, the learned active-rule loss in Theorem V.9.12, and the represented filter transfer in Theorem V.2.4. Summing over surviving times proves Eq. (718). □

Corollary V.11.26 (Noise, samples, and code capacity for latent beliefs)**.** *Let $N = |\Theta|$, and suppose that the complete record contains at most m binary symmetric target-dependent reveals of common noise rate $0 \le \eta \le 1/2$. Let $D^{\mathrm{ext}}_{R,t}$ be the conditional information in all other target-dependent advice, oracle, reward, verifier, or feedback coordinates available before deployment. If each clean binary input contains at most one available bit, then*

$$I^{\mathrm{lat}}_{R,t} \le \min\left\{b^{\mathrm{lat}}_{R,t}, \log_2 N, m(1 - H_2(\eta)) + D^{\mathrm{ext}}_{R,t}, m(1-2\eta)^2 + D^{\mathrm{ext}}_{R,t}, J^{\mathrm{Bayes,src}}_{R,t}\right\}. \tag{719}$$

For uniform targets, exact identification with probability at least α requires

$$b^{\mathrm{lat}}_{R,t} \ge \alpha \log_2 N - 1. \tag{720}$$

Consequently a d_t-coordinate, p_t-bit representation requires

$$d_t p_t + b^{\mathrm{code}}_t \ge \alpha \log_2 N - 1. \tag{721}$$

Proof. Apply Corollary V.11.20 and data processing through the encoder, then intersect its information bound with the code-capacity bound in Eq. (705). The external coordinates enter their ordered reveal slots. Fano's inequality gives $\alpha \le (I^{\mathrm{lat}}_{R,t} + 1)/\log_2 N$; combine this with the code and coordinate capacities. □

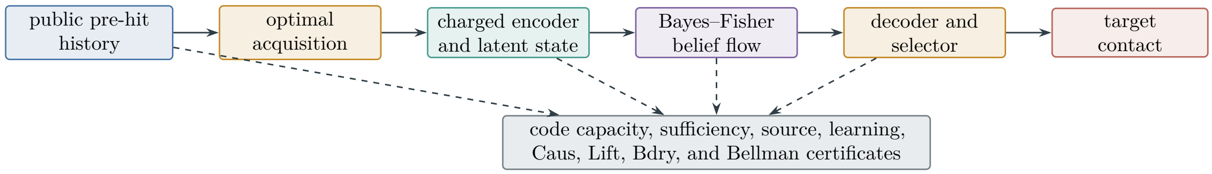

Figure 95: A compact latent belief is a completely charged component of optimal active acquisition. Its code capacity and sufficiency defect quantify compression; its deployed gain remains bounded by the existing structural, statistical, and dynamical certificates.

V.12 Bellman Error, Policy Loss, and the Universal Certificate

Theorem V.12.1 (Represented Bellman residual and policy loss)**.** *Let $0 < \beta < 1$, let rewards be bounded, and let v be a represented value approximation. For a fixed represented policy π,*

$$\|V^\pi - v\|_\infty \le \frac{\|\mathcal{T}^\pi v - v\|_\infty}{1-\beta}. \tag{722}$$

For the optimal Bellman operator,

$$\|V^* - v\|_\infty \le \frac{\|\mathcal{T} v - v\|_\infty}{1-\beta}. \tag{723}$$

If π_v is a represented policy greedy with respect to v, then

$$\|V^* - V^{\pi_v}\|_\infty \le \frac{2\beta}{(1-\beta)^2}\|\mathcal{T} v - v\|_\infty. \tag{724}$$

Every residual evaluation, action comparison, greedy selector, and retained coefficient is included in the saturated cost of the certificate. A sampled residual enters these deterministic inequalities only through a simultaneous Part IV generalization certificate for the represented residual class.

Proof. For any β-contraction F with fixed point v_F,

$$\|v_F - v\|_\infty \le \|Fv_F - Fv\|_\infty + \|Fv - v\|_\infty \le \beta\|v_F - v\|_\infty + \|Fv - v\|_\infty.$$

Rearrangement proves Eqs. (722) and (723). Greediness gives $\mathcal{T}^{\pi_v} v = \mathcal{T} v$. Insert v between the two fixed points, use the contraction twice, and obtain the standard estimate

$$\|V^* - V^{\pi_v}\|_\infty \le \frac{2\beta}{1-\beta}\|V^* - v\|_\infty.$$

Combining it with Eq. (723) proves Eq. (724). The cost and sampled-residual statements are applications of Definitions IV.11.1 and III.1.3.

□

Definition V.12.2 (Coordinatewise dynamical certificate)**.** For one controlled record of budget B, horizon H, target/failure pair, discount parameters, and selected policy, the coordinatewise dynamical certificate is

$$\mathbf{U}_{\rm dyn} = (U_{\rm hit}, U_{\rm disc}, U_{\rm time}, U_{\rm ra}, U_{\mathsf{Caus}}, U^H_{\mathsf{Lift}}, U^\lambda_{\mathsf{Lift}}, U_{\rm noise},$$
$$U^{\rm noise}_{\mathsf{Caus}}, U^H_{\mathsf{Lift},\rm noise}, U^\lambda_{\mathsf{Lift},\rm noise}, U^{\rm noise}_{\rm Cap}, \mathbf{U}_{\rm ch}, U^H_{\rm POMDP}, U_{\rm info}, U^H_{\rm cur}, \mathbf{H}_{\rm ctl}, U_{\rm mix}, U_{\rm Bell}, U_{\rm learn}).$$

It packages existing estimates and is not an additional structural channel. Its entries are defined as follows.

(i) U_{hit} is the minimum of one, the prospective-selector bound $\mathrm{Enum}_\nu + H\,\mathrm{Sel}^{\mathrm{sat}}(\Psi_{\mathrm{ctl}}(B,H,n))$, and every same-record native capacity, occupation, Caus-contact, or finite-horizon Lift bound, together with every same-record sparse-reward observation bound, already converted to first-arrival probability units.

(ii) U_{disc} is the minimum of one and every represented upper comparison for the target-projected killed resolvent $A_T(\lambda; L, \mu_0) = S_T(\lambda; L, \mu_0)$ admitted by Definitions III.8.1 and III.18.1, including $U^\lambda_{\mathsf{Lift}}$ and $U^\lambda_{\mathsf{Lift},\mathrm{noise}}$ when their same-normalization comparisons are represented.

(iii) $U_{\mathrm{time}} = \mathbb{E}_{\mu_0} D/a$ when the same selected generator has a represented drift certificate $-LD \geq a > 0$, and $U_{\mathrm{time}} = +\infty$ otherwise. A quotient defect replaces a by its certified positive residual margin.

(iv) $U_{\mathrm{ra}} = \langle \mu_0, h\rangle$ when a represented reach–avoid committor is certified, and $U_{\mathrm{ra}} = 1$ otherwise.

(v) U_{Caus} is the minimum of one and every probability-normalized bound in Eq. (517) supplied by a represented aiming-to-contact converter. Its value is one when only action or aiming energy is available.

(vi) U^H_{Lift} is the minimum of one and the base first-arrival bound plus the charged clocked defect and interface errors from Theorem V.7.13. Its default is one. The discounted coordinate $U^\lambda_{\mathsf{Lift}}$ is the analogous minimum using exact intertwining or Eq. (529); its default is one.

(vii) U_{noise} is the maximum, over the represented binary supervised coordinates in the record, of the clean risk bounds $U^{\mathrm{clean}}_{\mathrm{noise},q}(n, b(n), m, \eta_{q,+}, \delta_q)$ from Eq. (330). It retains the complete computational ceiling $b(n)$, sample size m, and every noise margin $1 - 2\eta_{q,+}$; its default is one.

(viii) $U^{\mathrm{noise}}_{\mathsf{Caus}}$ is the minimum of one and the applicable clean-target Caus contact bounds in Eq. (520). The finite-horizon coordinate $U^H_{\mathsf{Lift},\mathrm{noise}}$ is $\widehat{p}^{\,\Omega}_{T,H}$ plus the right side of Eq. (541), clipped at one. The discounted coordinate $U^\lambda_{\mathsf{Lift},\mathrm{noise}}$ is the represented learned-gate discounted arrival plus the right side of Eq. (542), again clipped at one after a declared probability normalization. Their defaults are one.

(ix) $U^{\mathrm{noise}}_{\mathrm{Cap}}$ is the right side of Eq. (543) in capacity units. Its default is $+\infty$. It is not minimized with a probability until a represented capacity-to-arrival comparison is supplied.

(x) The typed channel vector is

$$\mathbf{U}_{\mathrm{ch}} = (U_{r,\mathsf{N}_r}, U_{A,\mathsf{N}_A}, U_{O,\mathsf{N}_O}, U_{\mathrm{fil},\mathsf{N}_{\mathrm{fil}}}, U_{\mathrm{comp},\mathsf{N}_{\mathrm{comp}}}, U_{\mathrm{path}}),$$

whose raw entries are the applicable profiles in Eq. (353). Their declared norms are part of the coordinate. U_{path} is a represented reference event probability plus the applicable probability-normalized shift in Eqs. (377) and (378), clipped at one; raw path distance alone is not an event bound. Raw entries default to $+\infty$, and the path probability defaults to one.

(xi) U^H_{POMDP} is the minimum of one and every same-record clocked POMDP first-arrival bound. Its filter candidate is

$$\min\{1, U^H_b + \Delta^H_{\mathrm{fil}}\},$$

where U^H_b is a represented upper bound for the same exact-belief target event on the same reference record and Δ^H_{fil} is the comparison in Eq. (388). The exact analytic probability is eligible as U^H_b only when its evaluation or proof and target/failure interface are represented and charged; otherwise $U^H_b = 1$. Its default is therefore one. U_{info} is the Fano converse from Theorem V.3.2 when its success event identifies the target, and is one otherwise. A rate–distortion converse enters a probability coordinate only after a represented inversion to a distortion floor and a distortion-to-event converter. $\mathbf{H}_{\mathrm{ctl}}$ is the typed entropy vector in Eq. (442); it is not a scalar error coordinate.

(xii) $U_{\rm cur}^H$ is the minimum of one and every same-record sparse-reward observation bound in Eq. (563). In particular, a target-permutation-neutral record supplies

$$U_{\rm cur}^H \le \min\left\{1, \sum_{t=0}^{H-1}\left(\frac{M_\varepsilon(\phi)}{N} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\rm fib,\uparrow}}\right)\right\}.$$

Observation, quantization, memory, predictor training, intrinsic reward, policy deployment, executed transitions, verifier use, and every exact or compensated cell-dynamics comparison carry the costs in Eqs. (549) and (553). Its default is one.

(xiii) $U_{\rm mix}(t) = e^{-2\gamma_\Phi t}$ for variance and $e^{-2\alpha_\Phi t}$ for entropy when the corresponding same-record constant is certified; the default value is one.

(xiv) $U_{\rm Bell}$ is the appropriate right-hand side of Eqs. (722) to (724); its value is $+\infty$ without a represented residual certificate.

(xv) $U_{\rm learn}$ is the minimum of the valid simultaneous structural, Rademacher, PAC–Bayes, effective-dimension, and geometry-selection bounds of Part IV, after conversion to one declared risk or alignment unit and inclusion of every selection term.

The entries are never added across incompatible units. In particular, U_{Caus} enters $U_{\rm hit}$ only after the contact converter, U_{Lift}^H competes only in finite-horizon probability units, and $U_{\mathsf{Lift}}^\lambda$ competes only in discounted-arrival units. A minimum is taken only between upper bounds for the same coordinate on the same represented record. The same rule applies to the noise-corrected Caus and Lift coordinates; $U_{\rm Cap}^{\rm noise}$ remains separate unless its capacity units have been converted explicitly. The entries of $\mathbf{U}_{\rm ch}$ and $\mathbf{H}_{\rm ctl}$ likewise remain typed until a represented reward, Bellman, path, filter, aiming, or Lift theorem converts them to an existing coordinate. The curiosity coordinate competes in $U_{\rm hit}$ only after Theorem V.8.6 has converted observation resolution and conditional fiber information to the same finite-horizon target-arrival probability.

Theorem V.12.3 (Universal coordinatewise bound for represented dynamical systems)**.** *Every budget-admissible deterministic, randomized, controlled, or history-dependent finite-horizon dynamical system on a presented task has a coordinatewise certificate $\mathbf{U}_{\rm dyn}$ at the class-preserving ceiling $\Psi_{\rm ctl}(B,H,n)$. Its coordinates satisfy*

$$\Pr\{\tau_{T^+} \le H,\ \tau_{T^+} < \tau_{T^-}\} \le U_{\rm hit}, \tag{725}$$

$$S_T(\lambda; L, \mu_0) \le U_{\rm disc}, \tag{726}$$

$$\mathbb{E}_{\mu_0}\tau_{T^+} \le U_{\rm time}, \tag{727}$$

$$\Pr_{\mu_0}\{\tau_{T^+} < \tau_{T^-}\} \le U_{\rm ra}. \tag{728}$$

The certified mixing, Bellman, policy-loss, and learned-control conclusions are the corresponding entries of Theorems IV.12.4, V.5.3 and V.12.1. The same first-arrival probability is bounded by U_{Caus} and U_{Lift}^H whenever their converters are present, and by $U_{\rm cur}^H$ whenever a complete sparse-reward observation record is present. The discounted arrival is bounded by $U_{\mathsf{Lift}}^\lambda$. Accordingly, $U_{\rm hit} \le \min\{U_{\mathsf{Caus}}, U_{\mathsf{Lift}}^H, U_{\rm cur}^H\}$ and $U_{\rm disc} \le U_{\mathsf{Lift}}^\lambda$ for those same-record candidates. For binary target or failure interfaces learned from corrupted labels, the same statements hold with $U_{\mathsf{Caus}}^{\rm noise}, U_{\mathsf{Lift},\rm noise}^H$, and $U_{\mathsf{Lift},\rm noise}^\lambda$; the clean supervised risk is bounded by $U_{\rm noise}$, and the learned-boundary lifted capacity is bounded by $U_{\rm Cap}^{\rm noise}$. In particular, $U_{\rm hit} \le \min\{U_{\mathsf{Caus}}^{\rm noise}, U_{\mathsf{Lift},\rm noise}^H\}$ and $U_{\rm disc} \le U_{\mathsf{Lift},\rm noise}^\lambda$ when the corresponding learned-interface records apply. For a complete action-, reward-, observation-, or POMDP-channel record, the selected generator is the actual executed clocked generator of Definition V.1.3 and Theorem V.2.2. The same first-arrival probability is additionally bounded by

U_{path} and U^{H}_{POMDP} when their comparisons are present. Thus U_{hit} may include those two same-unit candidates. The target-identification coordinate is bounded by U_{info} only under the identification gate in Definition V.4.2. Reward noise changes target arrival only through the policy/history kernel or an explicit path comparison; raw reward error is not a direct arrival candidate.

Moreover, put $B^\sharp = \Psi_{\text{ctl}}(B, H, n)$. The unconditional five-family accounting estimate is

$$\mathrm{Sel}^{\text{sat}}_{I,n}(B^\sharp) \le eH_{B^\sharp}(n) \sum_{c \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\widetilde{\mathfrak{F}}_5}(D^{\text{sel}}_{B^\sharp}(n)). \tag{729}$$

Here $H_{B^\sharp}$, $D^{\text{sel}}_{B^\sharp}$, and the contextual-charge envelopes are exactly those of Theorem III.21.4. This estimate charges the complete first-hit accounting record. When every nonzero frontier term also has the represented pre-hit same-record comparison required by Remark III.17.4 and Corollary III.16.9, the source-envelope version of that theorem assigns the corresponding nonneutral improvement to its qualified prospective Abs *or* Geom/Caus *record. Qualified* Lift *and* Bdry *uses retain the inherited source and their complete downstream cost; J_{res} supplies no prospective structural source.*

The theorem is total: when a geometric or statistical certificate is absent, the conventions of Definitions V.5.1 and V.12.2 return the neutral bound for that coordinate. A numerical improvement is claimed only from an evaluated, temporally admissible, target-aligned record inside the declared budget.

Proof. Use Theorem V.1.2 to place the complete execution on its represented clocked carrier. The exact temporal hazard identity and its saturated supremum in Theorem III.21.3 prove Eq. (725). Apply the pointwise accounting inequality of Theorem III.21.4 at $B^\sharp = \Psi_{\text{ctl}}(B, H, n)$. Its definitions of $H_{B^\sharp}$ and $D^{\text{sel}}_{B^\sharp}$ give Eq. (729). Under the stated same-record comparison, the final paragraph of that theorem and Corollary III.16.9 give the prospective-source form.

The Caus coordinate follows from Corollary V.7.4 after the represented aiming-to-contact conversion. Exact or approximate Lift coordinates follow from Theorems V.7.12 and V.7.13. All three are candidates for an existing arrival coordinate, so slotwise selection in Theorem V.5.10 permits only the stated same-unit minima.

The sparse-reward observation coordinate is Theorem V.8.6. Its finite-horizon contact estimate is already in probability units, and its complete ledger includes every observation, novelty-learning, memory, policy, execution, verification, and compensated-geometry operation. Consequently it is an additional same-record candidate for U_{hit}, without treating a nonintertwining observation partition as an exact quotient.

The binary clean-risk coordinate is Theorem IV.16.6. Its conversion to clean Caus aiming is Theorem V.7.7; its finite-time, discounted, and capacity transfer through a learned target gate is Theorem V.8.2. These results give the noise-corrected coordinates without changing the native deterministic form, resolvent, or capacity inequalities.

The general channel coordinates are Definition IV.17.5 and Theorems V.1.4 and V.1.5. Their POMDP transfer is Theorems V.2.2 and V.2.4, and their information converse is Theorem V.3.2. The no-substitution rule in Definition V.4.1 prevents Shannon, MLSI, Caus, PAC–Bayes, description, and policy entropies from entering unlike coordinates.

The equality of discounted hitting and the target-projected killed resolvent is Definition III.8.1; taking the minimum over valid represented upper comparisons proves Eq. (726). The drift and reach–avoid conclusions are Theorem V.6.6; the default values $+\infty$ and one make the inequalities valid when those records are absent. The remaining coordinate bounds are the cited mixing, Bellman, and learning theorems applied to the same complete record.

Finally, source inheritance and post-saturation residual orthogonality are Proposition II.5.9 and Theorem II.7.2. Temporal admissibility is Definition III.21.1 and Theorem V.6.3. Thus every positive structural contribution has one existing source classification and every unavailable certificate reduces to its stated neutral value.

□

Theorem V.12.4 (Saturated same-record prospective certificate envelope)**.** *Fix a presented task, a budget B, a refined horizon H, and a time $0 \le t < H$. Let $\mathscr{R}_{I,t}(B)$ be the collection of complete temporally admissible pre-hit records that induce a next-test selector at time $t+1$, including the prefix law, selected kernel, target interface, neutral baseline, and every comparison used below. For $R \in \mathscr{R}_{I,t}(B)$, write*

$$a_R = \big(\mathrm{Adv}_t(K_R; \nu_R)\big)_+.$$

Evaluate the following candidates under the same prefix law and in the same survival-weighted next-contact units. An unavailable candidate has the neutral value one.

(i) *U_R^{src} is the minimum of one and every represented inherited or direct same-normalization comparison in Eqs.* (474) *and* (475)*.*

(ii) *U_R^{cap} is the minimum of one and every capacity or occupation bound for the next-contact event supplied by the selected kernel and prefix initialization. In particular, a represented reversible comparison record from Theorem V.6.9, used with one remaining transition, supplies*

$$U_R^{\mathrm{cap}} \le \sigma_{R,t} \min\Big\{1, \|f_{R,t}\|_{L^p(\widetilde{\mu}_R)} \Big[\min\Big\{1, r_{R,+}\mu_R(T_I) + \kappa_{R,+}\,\mathrm{Cap}_{\Phi_{P_R}}(T_I, T_I^c)\Big\}\Big]^{1/q}\Big\}, \quad (730)$$

where

$$\sigma_{R,t} = \Pr_R\{\tau > t\}, \qquad \rho_{R,t} = \mathcal{L}_R(X_t \mid \tau > t), \qquad f_{R,t} = \frac{d\rho_{R,t}}{d\widetilde{\mu}_R},$$

$1 < p \le \infty$, and $q = p/(p-1)$. More precisely, the conditional finite-horizon bound is multiplied by $\sigma_{R,t}$ because a_R uses survival-weighted units.

(iii) *If the target is indexed by a uniform latent variable on N values, U_R^{info} is the minimum of one and every information or chi-square contact bound in Eqs.* (425) *and* (428)*. The KL–SDPI candidate is explicitly*

$$U_R^{\mathrm{info}} \le \min\left\{1, \frac{\Lambda_{R,t}}{N} + \sqrt{\frac{\log 2}{2}\left(I(\Theta; \mathsf{Z}_0) + \sum_{j=1}^{L_{R,t}} d_{R,t,j}\right)}\right\}. \quad (731)$$

(iv) *U_R^{BW} is the minimum of one and every target-decision bound transferred from a represented pre-hit reference experiment by Theorem V.4.5. The reference experiment, causal garbling, action map, and complete simulation cost are part of R.*

(v) *U_R^{dyn} is the minimum of one and every same-record Caus, finite-horizon Lift, path, POMDP, sparse-reward observation, or learned-interface bound already converted to the same next-contact units in Definition V.12.2 and Theorem V.12.3. For a complete observation record, its next-contact candidate is*

$$U_R^{\mathrm{dyn}} \le \min\left\{1, \beta_{R,t}^{\mathrm{fib}} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{fib},\uparrow}}\right\}, \quad (732)$$

with the complete ledger of Definition V.8.5.

(vi) *For a complete master active-sampling record, U_R^{acq} is the minimum of one and the active next-contact candidate*

$$U_R^{\mathrm{acq}} \le \min\left\{1, \beta_{R,t}^{\mathrm{act}} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{act},\uparrow}}\right\}, \tag{733}$$

together with every same-record master, capacity, and Blackwell comparison already converted to this survival-weighted next-contact unit. Its complete ledger is Definition V.9.1; optimization over that complete record class is exactly the saturated active-success coordinate of Theorem V.9.5.

Put

$$U_R^{\mathrm{cert}} = \min\{U_R^{\mathrm{src}}, U_R^{\mathrm{cap}}, U_R^{\mathrm{info}}, U_R^{\mathrm{BW}}, U_R^{\mathrm{dyn}}, U_R^{\mathrm{acq}}\}. \tag{734}$$

Let $B^{\mathrm{cert}} = \Psi_{\mathrm{cert}}(B, H, n, \mathbf{m})$ be any declared class-preserving common majorant of the existing cost maps for the selected kernel, clock refinement, source comparison, capacity record, ordered reveals, Blackwell simulation, sparse-reward observation and compensated cell dynamics, master active sampling, Lift, and output conversion. Here $\mathbf{m}$ denotes the sample or reveal counts already present in the task presentation; it is omitted when no such counts occur. Then

$$\mathrm{Sel}_{I,n}^{\mathrm{sat}}(B) \le \max_{0\le t<H} \sup_{R\in\mathscr{R}_{I,t}(B^{\mathrm{cert}})} U_R^{\mathrm{cert}}, \tag{735}$$

$$\Pr\{\tau \le H\} \le \mathrm{Enum}_\nu(\mathcal{A}, H) + \sum_{t=0}^{H-1} \sup_{R\in\mathscr{R}_{I,t}(B^{\mathrm{cert}})} U_R^{\mathrm{cert}}. \tag{736}$$

The second right-hand side may be clipped at one. The same inequalities hold after applying a monotone family functional. The minimum in Eq. (734) *is taken recordwise, before either supremum. Thus coordinates belonging to different selected kernels, prefix laws, or presentation events are never combined.*

Proof. For each R, the neutral baseline has nonnegative next-contact mass, so Eq. (426) gives $a_R \le p_R$, where p_R is the survival-weighted next-contact probability. The two source comparisons bound a_R directly by Theorem V.6.4. Every available capacity, information, Blackwell, or dynamical candidate bounds either p_R or a_R under the same prefix law and in the same units. Hence

$$a_R \le U_R^{\mathrm{cert}}.$$

The explicit capacity candidate is Eq. (496) with one remaining transition; the explicit information candidate is Eq. (425). Blackwell transfer is Eq. (449), and the remaining dynamical candidates are the same-unit entries admitted by Definition V.12.2; the sparse-reward entry is Eq. (559). The active-sampling entry is Eq. (587); its master comparison is admitted only after the same next-contact normalization has been represented.

Encoding adequacy and computation closure place every complete record and comparison at the common majorant B^{cert}. Taking the supremum over records and then the maximum over times proves Eq. (735). Substitute the recordwise inequality into the exact hazard decomposition Eq. (152) and sum the H advantages to obtain Eq. (736). Monotonicity gives the family-functional statement. Since every candidate is first evaluated on the same record, the order of minimum and supremum is forced and no cross-record certificate is introduced.

□

Theorem V.12.5 (Exhaustive sparse-reward observation-to-transport accounting)**.** *Fix a complete record $\mathcal{R}_{\mathrm{cur}}$ from Definition V.8.5, a pre-hit time $t < H$, and the represented selector K_t used for the next transition. At the common charged ceiling, the structural use of $q_\varepsilon = \kappa_\varepsilon \circ \phi$ belongs to exactly one of the following master-profile modes.*

(i) Exact cell dynamics. *The cell map satisfies the target, public-structure, utility, and exact-intertwining gates. Its record is assigned to the exact qualified* $\mathsf{Abs}/\mathsf{Geom}$ *coordinate.*

(ii) Compensated cell dynamics. *The cell map satisfies the nondynamic qualification gates and has nonzero intertwining defect. A reversible or general-resolvent master certificate supplies the target-functional comparison. Its record is assigned to the corresponding compensated coordinate, and its admissibility requires*

$$B_{\mathrm{cur}} \preceq B,$$

with B_{cur} and B_{comp} evaluated term by term in Eqs. (549) *and* (553)*.*

(iii) Ambient full-carrier dynamics. *The observation cells carry no admitted cell dynamics. The executed kernel remains on the native carrier, and the observation history enters through the complete pre-hit selector record. Its next-contact advantage obeys*

$$\left(\mathrm{Adv}_t(K_t;\nu_t)\right)_+ \le \min\left\{ U_R^{\mathrm{cert}}, \beta_{R,t}^{\mathrm{fib}} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{fib},\uparrow}} \right\}, \tag{737}$$

with both candidates clipped at one. When a same-record capacity comparison is present, U_R^{cert} includes Eq. (564)*; otherwise that capacity slot has its neutral value.*

The three modes are exhaustive and disjoint at the level of the selected master certificate. Predictor errors, counts, novelty bonuses, memories, archives, and intrinsic rewards are causal postprocessings of the represented pre-hit record and contribute no additional source kind. Their complete construction and deployment costs remain in Eq. (549)*. The terminal external reward and completed first-hit record enter the accounting profile at their clock times and supply no earlier source.*

Restricting Eq. (735) *to complete sparse-reward records gives*

$$\sup_{\substack{0 \le t < H \\ R \in \mathscr{R}_{I,t}^{\mathrm{cur}}(B)}} \left(\mathrm{Adv}_t(K_R;\nu_R)\right)_+ \le \sup_{\substack{0 \le t < H \\ R \in \mathscr{R}_{I,t}^{\mathrm{cur}}(B^{\mathrm{cert}})}} U_R^{\mathrm{cert}}, \tag{738}$$

where $\mathscr{R}_{I,t}^{\mathrm{cur}}$ is the subset of the existing complete pre-hit records carrying Definition V.8.5. The common ceiling charges the observation, regularity, information, capacity, source, cell-dynamics, execution, verification, and output comparisons before the supremum is taken.

Proof. The four-mode master decomposition in Theorem II.9.4 consists of ambient geometry, exact qualified cell dynamics, reversible compensated dynamics, and general-resolvent compensated dynamics. Combining the latter two produces the three displayed alternatives. Their defining slots are part of the certificate, so the alternatives are disjoint. The exact and compensated admission conditions and their complete charge are Definitions III.18.1 to III.18.3.

In the ambient mode, Eq. (559) bounds the actual pre-hit selector. Independently, Theorem V.12.4 bounds the same survival-weighted advantage. Their recordwise minimum proves Eq. (737). Its capacity candidate is Eq. (564) after the stated same-record conversion.

The sparse-reward identity is Eq. (560). Constructor no-creation and temporal admissibility are Proposition II.5.9 and Definition III.21.1; they assign derived novelty coordinates to their

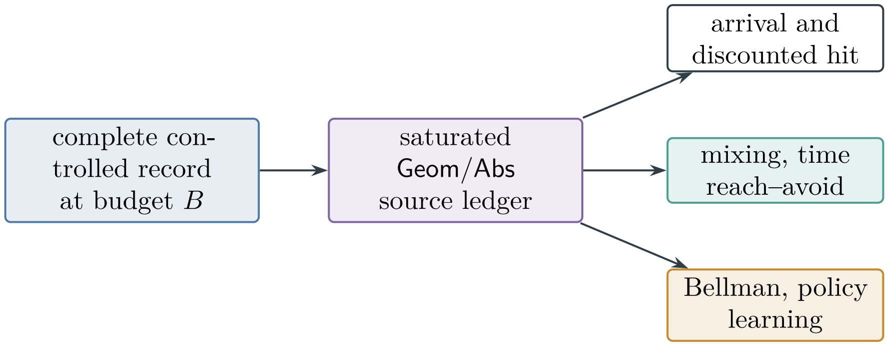

Figure 96: The universal dynamical certificate is coordinatewise. All entries use one complete saturated record, while estimates in different units remain separate.

represented ancestry and place terminal records in the accounting ledger. Restricting the record set in Eq. (735) and retaining its common cost majorant proves Eq. (738).

□

V.13 Learned Policies and Continuous-System Transfer

V.13.1 Represented trajectory learning records

Definition V.13.1 (Complete dynamical learning record)**.** Fix a controlled presentation at problem size n and a total budget $B = b(n)$. The general ceiling b is retained symbolically; the polynomial specialization is $b_C(n) = n^C$. A complete dynamical learning record contains:

(i) a represented behavior policy, its independent-episode, stationary-path, or predictable-interaction sample, and the corresponding Part IV dynamical-sample certificate;

(ii) for every binary target, failure, classification, committor-training, or policy-supervision coordinate, a complete dynamical label-noise record from Definition IV.16.1, including its clean analytic comparator, corruption rate, sample size, computational ceiling, debiasing or calibration, and behavior-to-deployment comparison. A noiseless coordinate uses the zero-cost identity channel $\eta = 0$;

(iii) for every corrupted reward report, executed action, noisy observation, filter, or identifiable composite transition, a complete typed model-channel noise record from Definition IV.17.1. It includes its clean analytic comparator, public and latent filtrations, channel factorization or declared composite, sample size, coverage, tail and entropy certificates, inverse or calibration, behavior-to-deployment comparison, and saturated error profile. An actual stochastic reward law is part of the true model; only a biased or imperfectly estimated report is a reward-corruption error;

(iv) for a partial-observation problem, the represented POMDP carrier, observation-measurable policy, exact analytic belief comparison, represented filter or memory, filter-stability recurrence, target and failure interfaces, and every clocked Lift, form, path-law, or resolvent comparison used at deployment;

(v) the learned transition and reward records $(\widehat{P}, \widehat{r})$, the learned geometry $\widehat{\Phi}$, and every retained value, committor, action-value, policy, and selector record;

(vi) represented evaluation losses for the one-step model, geometry self-defect, Bellman or committor residual, policy comparison, target contact, and failure contact;

(vii) a behavior-to-deployment comparison: either represented importance weights with an audited bound, or a represented transport, density-ratio, or occupancy comparison;

(viii) the complete costs of data collection, training, geometry and policy selection, stored state, precision, deployment, and verification.

All losses used in the bounded-loss theorems are normalized to $[0, 1]$. Another loss enters only with its represented sub-Gaussian, sub-exponential, or predictable-variance certificate. The budget split

$$B = B_{\text{data}} \oplus B_{\text{geom}} \oplus B_{\text{model}} \oplus B_{\text{value}} \oplus B_{\text{pol}} \oplus B_{\text{noise}} \oplus B_{\text{ch}} \oplus B_{\text{fil}} \oplus B_{\text{info}} \oplus B_{\text{dep}}$$

is part of the record. The split records resource use; it creates no new structural channel.

Definition V.13.2 (Action-safe defect precision and effective dimension)**.** Let $H_\Phi \succeq 0$ be the represented symmetric operator of a certified dynamical geometry. Let $P_{\mathcal{A}}$ project onto the represented span of increments generated by legal actions, and let P_{safe} project onto the represented subspace killed at the failure sector T^-. Assume $\operatorname{ran} P_{\text{safe}} \subseteq \operatorname{ran} P_{\mathcal{A}}$ and define the compressions

$$\begin{aligned} H_\Phi^{\mathcal{A}} &= P_{\mathcal{A}} H_\Phi P_{\mathcal{A}}|_{\operatorname{ran} P_{\mathcal{A}}}, \\ H_\Phi^{\mathcal{A},\text{safe}} &= P_{\text{safe}} H_\Phi^{\mathcal{A}} P_{\text{safe}}\Big|_{\operatorname{ran} P_{\text{safe}}}. \end{aligned}$$

For $\lambda > 0$, put

$$N_{\text{eff}}^{\mathcal{A}}(\lambda) = \operatorname{Tr}\Big[H_\Phi^{\mathcal{A}}(H_\Phi^{\mathcal{A}} + \lambda I)^{-1}\Big], \tag{739}$$

$$N_{\text{eff}}^{\mathcal{A},\text{safe}}(\lambda) = \operatorname{Tr}\Big[H_\Phi^{\mathcal{A},\text{safe}}(H_\Phi^{\mathcal{A},\text{safe}} + \lambda I)^{-1}\Big]. \tag{740}$$

Let P_D and P_H be the already represented defect and harmonic-reduction projections of Definition IV.8.1. On their retained positive range, the control precision is

$$\mathbf{M}_{D,H}^{\text{dyn}}(\lambda) = P_H P_D(H_\Phi^{\mathcal{A},\text{safe}} + \lambda I)P_D P_H\Big|_{R_{D,H}}. \tag{741}$$

Every projection, retained basis, and spectral computation is charged in the complete dynamical learning record.

Theorem V.13.3 (Constrained-safe spectral capacity)**.** *For every $\lambda > 0$, the nested compressions in Definition V.13.2 satisfy*

$$N_{\text{eff}}^{\mathcal{A},\text{safe}}(\lambda) \le N_{\text{eff}}^{\mathcal{A}}(\lambda) \le N_{\text{eff},\Phi}(\lambda). \tag{742}$$

If the represented safe target satisfies the compressed source condition $P_{\text{safe}} f_* = (H_\Phi^{\mathcal{A},\text{safe}})^r g_{\text{safe}}$, $0 < r \le 1$, then its source residual obeys

$$\Big\|\Big[I - H_\Phi^{\mathcal{A},\text{safe}}(H_\Phi^{\mathcal{A},\text{safe}} + \lambda I)^{-1}\Big] P_{\text{safe}} f_*\Big\|_2^2 + \|(I - P_{\text{safe}})f_*\|_2^2 \le C_r \|g_{\text{safe}}\|_2^2 \lambda^{2r} + \|(I - P_{\text{safe}})f_*\|_2^2. \tag{743}$$

The last term is the exact action/safety approximation residual; sample size does not alter it.

Proof. The eigenvalues of a self-adjoint compression interlace those of its parent operator. The map $u \mapsto u/(u + \lambda)$ is nonnegative and increasing, so summing the interlacing inequalities first for $P_{\mathcal{A}}$ and then for P_{safe} proves Eq. (742). On $\operatorname{ran} P_{\text{safe}}$, the stated compressed source condition and the calculation of Corollary IV.3.13 bound the ridge residual by $C_r\|g_{\text{safe}}\|_2^2 \lambda^{2r}$. Orthogonal decomposition adds the unrepresented component $\|(I - P_{\text{safe}})f_*\|_2^2$ to both sides, proving Eq. (743). □

Corollary V.13.4 (Saturated accountability for learned policies)**.** *Consider a represented computation that learns a geometry, value or committor, policy, and pre-hit action selector from finite represented data. Include its training rule, sample, retained state, selected kernel, decoder, policy evaluator, action comparisons, deployment state, and verifier calls in one cost-B record. Then:*

(i) *every prospective target-bearing object receives exactly one existing source-mode–state-provenance–coefficient-provenance index from Corollary IV.13.1;*

(ii) *its same-carrier target reach is bounded simultaneously by the exact* Geom *product and the generated-algebra projection ceiling in Eq.* (273)*;*

(iii) *its target-arrival probability is bounded by the minimum of the prospective selector/source estimate and every applicable simultaneous Part IV statistical estimate, as in Theorems IV.12.4 and V.14.2;*

(iv) *trajectory memorization and sample-indexed tables remain charged lifted coordinates, while their population-orthogonal component belongs to J_{res}. A population-aligned prospective use is reclassified into the existing qualified* Geom/Abs *ledger at its complete cost only when the same record supplies the required channel predicate and pre-hit comparison.*

Thus learning changes which represented geometry and policy are selected; it does not enlarge the structural classification or omit the cost of selecting and deploying them.

Proof. The complete learning and deployment computation satisfies the hypotheses of Theorem IV.12.4. Apply Corollary IV.13.1 to its prospective records and Theorems II.7.2 and IV.11.4 to its empirical and population components. The learned policy is a represented selector in Theorem V.1.2; hence all policy evaluation and action-selection costs remain in the same derivation. These cited conclusions give the four assertions.

□

V.13.2 Learned-Control Coordinates and Bellman Propagation

Definition V.13.5 (Learned-control certificate coordinates)**.** For one complete dynamical learning record, define

$$\mathbf{U}_{\mathrm{learn}}^{\mathrm{dyn}} = \begin{pmatrix} U_{\mathrm{model}}, U_{\mathrm{geom}}, U_{\mathrm{value}}, U_{\mathrm{policy}}, U_{\mathrm{shift}}, U_{\mathrm{noise}}, \mathbf{U}_{\mathrm{ch}}, U_{\mathrm{POMDP}}, \\ U_{\mathrm{info}}, \mathbf{H}_{\mathrm{ctl}}, U_{\mathrm{safe}}, U_{\mathrm{hit}}, U_{\mathrm{Cap}}^{\mathrm{noise}}, U_{\mathrm{return}}, U_{\mathrm{regret}} \end{pmatrix}.$$

Each entry is an upper bound in its named unit:

(i) U_{model} bounds the represented joint actionwise one-step Bellman discrepancy ε_M of Theorem V.13.6 on the retained value class $\mathcal{V}_B$. An optimal-operator defect alone is a value candidate but not a learned-greedy policy candidate.

(ii) U_{geom} bounds the represented form or operator discrepancy $\|\widehat{H}_\Phi - H_\Phi\|_{\mathrm{op}}$, together with the induced projector error at a certified eigengap.

(iii) U_{value} bounds a value, action-value, or committor error in its declared norm after the model, geometry, and Bellman residual have been propagated by a represented contraction, resolvent, or functional- inequality certificate.

(iv) U_{policy} bounds deployment-policy loss relative to the budget-reachable optimum.

(v) U_{shift} bounds the behavior-to-deployment risk change.

(vi) U_{noise} bounds clean binary supervised risk after explicit corruption transport, maximized over the binary supervised coordinates in the record on their simultaneous confidence event. It records n, the complete computational ceiling, sample size m, and each noise rate or Massart ceiling.

(vii) $\mathbf{U}_{\mathrm{ch}}$ stores reward, execution, observation, filter, composite-kernel, and path-law errors with their declared norms. U_{POMDP} stores filter-to-hit and filter-to-value consequences only after their represented converters. U_{info} stores the eligible target-identification converse, and $\mathbf{H}_{\mathrm{ctl}}$ stores the noninterchangeable entropy outputs of Definition V.4.1.

(viii) U_{safe} bounds failure-before-target probability; U_{hit} bounds target-before-failure arrival.

(ix) $U_{\mathrm{Cap}}^{\mathrm{noise}}$ bounds learned-boundary lifted condenser capacity and remains in capacity units.

(x) U_{return} bounds discounted-return suboptimality, and U_{regret} bounds normalized sequential regret.

A coordinate takes the minimum of all valid certified upper bounds expressed in the same unit. Its neutral value is $+\infty$ for an unbounded error or loss coordinate and one for a probability coordinate. Distinct coordinates are joined only by an explicit propagation theorem.

Theorem V.13.6 (Model, Bellman, and policy propagation)**.** *Let $\mathcal{T}$ and $\widehat{\mathcal{T}}$ be true and represented learned discounted Bellman operators with common discount $0 < \beta < 1$, and let $\mathcal{T}^a, \widehat{\mathcal{T}}^a$ be their one-step action maps. Let $v \in \mathcal{V}_B$ be represented, put*

$$\varepsilon_M = \sup_{w \in \mathcal{V}_B} \sup_{x} \sup_{a \in \mathcal{A}_B(x)} |\mathcal{T}^a w(x) - \widehat{\mathcal{T}}^a w(x)|, \qquad \varepsilon_B = \|\widehat{\mathcal{T}} v - v\|_\infty,$$

and assume that the fixed points and iterates used below remain in $\mathcal{V}_B$. For a history- or belief-action correspondence, the two inner suprema are over its represented one-step selectors, equivalently over the associated one-step policy operators. This joint defect implies both $\|\mathcal{T} w - \widehat{\mathcal{T}} w\|_\infty \le \varepsilon_M$ and $\|\mathcal{T}^\pi w - \widehat{\mathcal{T}}^\pi w\|_\infty \le \varepsilon_M$ for every admitted selector π.

Suppose, in addition, that a complete channel record has converted its typed errors to the same deployment-law Bellman sup norm and has a represented disjoint telescoping sequence of action maps $\mathcal{T}^{(0),a}, \dots, \mathcal{T}^{(5),a}$, from the true to the learned map. For $1 \le j \le 5$, put

$$\Delta_j = \sup_{w \in \mathcal{V}_B} \sup_{x} \sup_{a \in \mathcal{A}_B(x)} \left|\mathcal{T}^{(j),a} w(x) - \mathcal{T}^{(j-1),a} w(x)\right|,$$

and require the same record to identify, in the declared order,

$$(\Delta_1, \dots, \Delta_5) = (\varepsilon_r^{\mathrm{Bell}}, \beta\varepsilon_P^{\mathrm{Bell}}, \varepsilon_A^{\mathrm{Bell}}, \varepsilon_O^{\mathrm{Bell}}, \varepsilon_{\mathrm{fil}}^{\mathrm{Bell}}).$$

Then the actionwise triangle inequality gives

$$\varepsilon_M \le \varepsilon_r^{\mathrm{Bell}} + \beta\varepsilon_P^{\mathrm{Bell}} + \varepsilon_A^{\mathrm{Bell}} + \varepsilon_O^{\mathrm{Bell}} + \varepsilon_{\mathrm{fil}}^{\mathrm{Bell}} =: \varepsilon_M^{\mathrm{ch}}. \tag{744}$$

An absent component is zero only when the corresponding interface is exact; an unavailable converter makes the channel candidate $+\infty$. Components with overlapping ancestry occur in one chosen telescoping step, rather than being counted in two marginal steps. In particular, if $\varepsilon_A^{\mathrm{Bell}}$ already includes the reward and transition change caused by execution, those changes are excluded from the separate reward and transition steps. Then

$$\|V^* - v\|_\infty \le \frac{\varepsilon_B + \varepsilon_M}{1 - \beta}. \tag{745}$$

If $\widehat{\pi}_v$ is an admitted represented greedy selector for $\widehat{\mathcal{T}}$ at v, then

$$\|V^* - V^{\widehat{\pi}_v}\|_\infty \le \frac{2\beta(\varepsilon_B + \varepsilon_M)}{(1 - \beta)^2} + \frac{2\varepsilon_M}{1 - \beta}. \tag{746}$$

For a fixed policy, if $\|r^\pi - \widehat{r}^\pi\|_\infty \le \varepsilon_r$ *and* $\|(P^\pi - \widehat{P}^\pi)w\|_\infty \le \varepsilon_P$ *on the retained value class, then*

$$\|V^\pi - \widehat{V}^\pi\|_\infty \le \frac{\varepsilon_r + \beta\varepsilon_P}{1-\beta}. \tag{747}$$

The same inequalities hold for a represented committor contraction, with its declared contraction modulus in place of β*. For an undiscounted killed problem they hold with the represented killed-resolvent norm in place of* $(1-\beta)^{-1}$*. Raw* L^2 *reward error, channel total variation, filter error, entropy, and capacity are not substituted into Eq.* (744) *without a represented same-norm converter.*

Proof. The true residual satisfies

$$\|\mathcal{T}v - v\|_\infty \le \|\mathcal{T}v - \widehat{\mathcal{T}}v\|_\infty + \|\widehat{\mathcal{T}}v - v\|_\infty \le \varepsilon_M + \varepsilon_B.$$

Apply the contraction residual argument of Theorem V.12.1 to obtain Eq. (745). Pointwise greediness and the two parts of the joint defect give

$$\mathcal{T}v \le \widehat{\mathcal{T}}v + \varepsilon_M = \widehat{\mathcal{T}}^{\widehat{\pi}_v}v + \varepsilon_M \le \mathcal{T}^{\widehat{\pi}_v}v + 2\varepsilon_M.$$

Since $\mathcal{T}v \ge \mathcal{T}^{\widehat{\pi}_v}v$, this proves $\|\mathcal{T}v - \mathcal{T}^{\widehat{\pi}_v}v\|_\infty \le 2\varepsilon_M$. The approximate-greedy one-step comparison gives

$$\|V^* - V^{\widehat{\pi}_v}\|_\infty \le \frac{2\beta}{1-\beta}\|V^* - v\|_\infty + \frac{2\varepsilon_M}{1-\beta};$$

substitution proves Eq. (746). For a fixed policy, subtract the two fixed-point equations, use the stochastic-kernel contraction, and rearrange to obtain Eq. (747). The committor and killed-resolvent statements repeat the same calculation with the stated inverse bound. Finally, applying the actionwise triangle inequality to $\mathcal{T}^{(0),a} - \mathcal{T}^{(5),a}$, then taking the displayed suprema, gives Eq. (744). Substitution into the preceding contraction inequalities gives every channelized specialization. Without an actionwise or one-step-policy converter, the policy candidate is $+\infty$, even if an optimal-operator value candidate is finite.

□

V.13.3 Generalization Envelopes and Budget-Reachable Value

Definition V.13.7 (Dynamical generalization envelope and budget-reachable value)**.** For a represented bounded loss class $\mathcal{F}$ in a complete dynamical learning record, let

$$\mathfrak{G}_{\mathrm{dyn}}(\mathcal{F}; B, m, \delta)$$

be the minimum of the valid Part IV independent-episode, mixing-block, sequential-Rademacher, martingale PAC–Bayes, and effective-dimension upper bounds for its population-minus-empirical loss. Each candidate includes its operator-selection, confidence, dependence, and behavior-to-deployment terms. The dynamical candidates are supplied by Theorems IV.15.2, IV.15.5 and IV.15.6 and Corollary IV.15.10; the independent-episode candidate is Theorem IV.3.3 and Lemma IV.3.6 on the represented episode-loss class. Candidates are compared only after they bound the same population law and the same loss. The value is $+\infty$ when no candidate applies.

For a represented flip-symmetric binary loss class, define the clean supervised-risk envelope

$$\mathfrak{G}^{\mathrm{clean}}_{\mathrm{dyn}}(\mathcal{F}; B, m, \eta_+, \delta) = \min\left\{1, U^{\mathrm{clean}}_{\mathrm{noise}}(n, B, m, \eta_+, \delta), U^{\mathrm{clean}}_{\mathrm{cal}}(\mathcal{F})\right\}. \tag{748}$$

Here $U^{\text{clean}}_{\text{cal}}(\mathcal{F})$ ranges over represented same-law, same-unit calibration consequences of Theorem IV.16.6; an unavailable candidate is $+\infty$. Unlike $\mathfrak{G}_{\text{dyn}}$, this quantity bounds the clean supervised risk or excess itself, not merely a population–empirical deviation. It is therefore inserted into an operator, Bellman, committor, or geometry coordinate only through a represented loss-to-coordinate propagation theorem. Reward, transition, and state-observation losses that are not binary label losses continue to use $\mathfrak{G}_{\text{dyn}}$, or the typed channel profiles of Definition IV.17.5 when a represented correction applies. The latter enter a model or Bellman coordinate only through Eq. (744).

Let Π_B be the represented policies of complete saturated cost at most B, including data collection and deployment. The budget-reachable value and budget gap are

$$V^*_B(x) = \sup_{\pi\in\Pi_B} V^\pi(x), \qquad \Delta^{\text{ctrl}}_B = \|V^* - V^*_B\|_\infty. \tag{749}$$

This pointwise supremum is only a comparison benchmark unless a rectangularity gate is present. The Bellman branch is available only when the record supplies a represented state/history action correspondence $\mathcal{A}_B$ such that (a) every measurable selector used by the argument, including each greedy selector, belongs to Π_B with its common evaluation and deployment cost charged, and (b) Π_B is closed under state/history-wise one-step continuation. Under this represented rectangularity gate, the restricted operator $\mathcal{T}_B$ is a β-contraction and its fixed point is V^*_B. Without it, V^*_B remains the displayed pointwise benchmark, but the restricted Bellman fixed-point, greedy-policy, and return-propagation candidates are $+\infty$, unless a separately named rectangular closure is constructed and charged at its enlarged ceiling.

For a partial-observation record, let $\Pi_{B,\text{obs}} \subseteq \Pi_B$ be the represented policies measurable with respect to the public observation filtration, including their filters, memory, and deployment costs. Define

$$V^*_{B,\text{obs}}(x) = \sup_{\pi\in\Pi_{B,\text{obs}}} V^\pi(x), \qquad \Delta^{\text{POMDP}}_B = \|V^*_B - V^*_{B,\text{obs}}\|_\infty. \tag{750}$$

The observation-measurable class requires the analogous represented belief/history rectangularity gate before $V^*_{B,\text{obs}}$ is called a Bellman fixed point. In particular, statewise or beliefwise stitching of individually affordable policies is not free. For every $\widehat{\pi} \in \Pi_{B,\text{obs}}$, the exact three-term decomposition is

$$V^* - V^{\widehat{\pi}} = (V^* - V^*_B) + (V^*_B - V^*_{B,\text{obs}}) + (V^*_{B,\text{obs}} - V^{\widehat{\pi}}). \tag{751}$$

The middle term is an approximation/information restriction, not an estimation or filter error. In a fully observed record take $\Pi_{B,\text{obs}} = \Pi_B$ and $\Delta^{\text{POMDP}}_B = 0$. For every learned $\widehat{\pi} \in \Pi_B$, the exact decomposition is

$$V^* - V^{\widehat{\pi}} = (V^* - V^*_B) + (V^*_B - V^{\widehat{\pi}}). \tag{752}$$

The exact gaps Δ^{ctrl}_B and Δ^{POMDP}_B are analytic comparison quantities. An operational certificate stores represented upper bounds

$$\Delta^{\text{ctrl}}_B \le \overline{\Delta}^{\text{ctrl}}_B, \qquad \Delta^{\text{POMDP}}_B \le \overline{\Delta}^{\text{POMDP}}_B,$$

including the cost of evaluating or proving them. A missing upper certificate has value $+\infty$. The exact gaps may still be displayed in an oracle decomposition, but they are not free certificate inputs.

V.14 Master Learning Certificate and Continuous-System Transfer

Definition V.14.1 (Master leaf event and same-norm candidate aggregation)**.** Fix a complete dynamical learning record $\mathcal{R}$ of sample size m and a represented learned policy $\widehat{\pi}_v$ greedy for $\widehat{\mathcal{T}}$ at v. Let $\mathcal{I}(\mathcal{R})$ be the finite represented set of every stochastic leaf certificate eligible for a displayed minimum: generic generalization, geometry, binary corruption, channel estimation and entropy continuity, filter stability, path comparison, gate or interface estimation, and estimated information. Assign leaf probabilities $(\delta_i)_{i\in\mathcal{I}(\mathcal{R})}$ with $\sum_i \delta_i \le \delta$. Each data-dependent leaf uses a pre-specified event uniform over its affordable output hull and independent of which output is selected. Work on

$$\mathcal{E}_{\mathcal{R}} = \bigcap_{i\in\mathcal{I}(\mathcal{R})} \mathcal{E}_i, \qquad \Pr(\mathcal{E}_{\mathcal{R}}) \ge 1-\delta.$$

Deterministic identities have $\delta_i = 0$, and a candidate whose event is not included is unavailable. Internal allocations, including δ_{ch}, partition these leaves and are not counted a second time.

For the Bellman branch, assume the applicable fully observed or observation-measurable rectangularity gate of Definition V.13.7. Then $\mathcal{T}$ is the corresponding restricted Bellman operator and its fixed point is $V^*_{B,\mathrm{obs}}$, with the fully observed convention $V^*_{B,\mathrm{obs}} = V^*_B$. The learned-greedy branch also requires the joint actionwise defect in Theorem V.13.6. If either gate is absent, $U_{\mathrm{value}}, U_{\mathrm{policy}}$, and the Bellman-propagated U_{return} below equal $+\infty$; the other coordinates remain available. On $\mathcal{E}_{\mathcal{R}}$, define $U_M^{\mathrm{noise}}, U_B^{\mathrm{noise}}, U_\Phi^{\mathrm{noise}}$ to be the smallest same-norm upper bounds obtained by propagating $\mathfrak{G}^{\mathrm{clean}}_{\mathrm{dyn}}$ through a represented loss-to-coordinate calibration theorem for the model, Bellman, and geometry coordinates, respectively. A coordinate that is not a binary supervised loss, or lacks such a propagation theorem, receives the value $+\infty$. Define $U_M^{\mathrm{ch}}, U_B^{\mathrm{ch}}, U_\Phi^{\mathrm{ch}}$ analogously as the smallest same-law, same-norm consequences of the typed reward, execution, observation, composite-kernel, and filter profiles after the represented converters in Theorems IV.17.2, V.1.4 and V.2.4 and Eq. (744). An unavailable conversion has value $+\infty$. Define

$$\begin{aligned}
\overline{\varepsilon}_M &= \min\left\{\widehat{\varepsilon}_M + \mathfrak{G}_{\mathrm{dyn}}(\mathcal{F}_{\mathrm{model}}; B, m, \delta_M), U_M^{\mathrm{noise}}, U_M^{\mathrm{ch}}\right\},\\
\overline{\varepsilon}_B &= \min\left\{\widehat{\varepsilon}_B + \mathfrak{G}_{\mathrm{dyn}}(\mathcal{F}_{\mathrm{Bell}}; B, m, \delta_B), U_B^{\mathrm{noise}}, U_B^{\mathrm{ch}}\right\},\\
\overline{\varepsilon}_\Phi &= \min\left\{\widehat{\varepsilon}_\Phi + \mathfrak{G}_{\mathrm{dyn}}(\mathcal{F}_{\mathrm{geom}}; B, m, \delta_\Phi), U_\Phi^{\mathrm{noise}}, U_\Phi^{\mathrm{ch}}\right\}.
\end{aligned}$$

Here each empirical term and class is the represented coordinate from Definition V.13.1. Then the coordinates of $\mathbf{U}^{\mathrm{dyn}}_{\mathrm{learn}}$ use this one simultaneous event and these three same-norm candidate minima.

Theorem V.14.2 (Master structural learning certificate for controlled systems)**.** *Under the record, rectangularity, confidence, and same-norm aggregation hypotheses of Definition V.14.1, the core coordinates of $\mathbf{U}^{\mathrm{dyn}}_{\mathrm{learn}}$ may be chosen as follows:*

$$U_{\mathrm{model}} = \overline{\varepsilon}_M, \tag{753}$$

$$U_{\mathrm{geom}} = \left(\overline{\varepsilon}_\Phi, \frac{2\overline{\varepsilon}_\Phi}{g_*}\right), \tag{754}$$

$$U_{\mathrm{value}} = \frac{\overline{\varepsilon}_B + \overline{\varepsilon}_M}{1-\beta}, \tag{755}$$

$$U_{\mathrm{policy}} = \frac{2\beta(\overline{\varepsilon}_B + \overline{\varepsilon}_M)}{(1-\beta)^2} + \frac{2\overline{\varepsilon}_M}{1-\beta}, \tag{756}$$

$$U_{\mathrm{return}} = \overline{\Delta}_B^{\mathrm{ctrl}} + \overline{\Delta}_B^{\mathrm{POMDP}} + U_{\mathrm{policy}}. \tag{757}$$

The first entry in Eq. (754) *is the operator coordinate. The second is the projector coordinate at a certified gap $g_* > 0$, and is assigned $+\infty$ when no positive gap is certified. Every U_Q^{noise} is also $+\infty$ unless its calibration bounds the same norm as the generic candidate. Thus binary label transport never replaces a transition, reward, or operator norm by an unlike risk. The same rule applies to U_Q^{ch}: a channel-entropy, row-TV, L^2, filter, or path-law number enters only after conversion to the norm of that candidate. Every target-qualified learned coordinate additionally carries the exact source index, contextual-charge vector, and noise-information vector of Definition IV.13.2. Model, Bellman, and policy errors remain in their displayed norms; the source-resolved target candidate enters only the arrival coordinate Eq.* (763) *or another coordinate with a represented same-unit converter.*

Proof. On the simultaneous leaf event, each candidate in a displayed minimum is an upper bound for the same population object in the same norm. Their minima therefore remain upper bounds. The model and geometry coordinates are the definitions of the aggregated errors, while Theorem V.13.6 gives the value and policy coordinates. The exact return decomposition in Definition V.13.7, followed by the represented upper certificates for its two analytic gaps, gives Eq. (757). □

Proposition V.14.3 (Auxiliary channel, information, safety, and access coordinates)**.** *Under the hypotheses of Definition V.14.1, the remaining coordinates are*

$$U_{\mathrm{shift}} = \min\{U_{\mathrm{iw}}, U_{T_1}, U_{T_2}, U_{\mathrm{occ}}\}, \tag{758}$$

$$U_{\mathrm{noise}} = \max_{q \in \mathcal{Q}_{\mathrm{bin}}} U^{\mathrm{clean}}_{\mathrm{noise},q}(n, B, m, \eta_{q,+}, \delta_q), \tag{759}$$

$$\mathbf{U}_{\mathrm{ch}} = \left(\begin{array}{l} \mathrm{Err}^{\mathrm{sat}}_{r,\mathsf{N}_r}(n,B,m,\delta_r), \mathrm{Err}^{\mathrm{sat}}_{A,\mathsf{N}_A}(n,B,m,\delta_A), \\ \mathrm{Err}^{\mathrm{sat}}_{O,\mathsf{N}_O}(n,B,m,\delta_O), \mathrm{Err}^{\mathrm{sat}}_{\mathrm{fil},\mathsf{N}_{\mathrm{fil}}}(n,B,m,\delta_{\mathrm{fil}}), \\ \mathrm{Err}^{\mathrm{sat}}_{\mathrm{comp},\mathsf{N}_{\mathrm{comp}}}(n,B,m,\delta_{\mathrm{comp}}), U_{\mathrm{path}}(n,B,m,H,\delta_{\mathrm{path}}) \end{array} \right), \tag{760}$$

$$U_{\mathrm{POMDP}} = (U^H_{\mathrm{POMDP}}, U^{\mathrm{value}}_{\mathrm{POMDP}}), \qquad U_{\mathrm{info}} = \min\left\{1, \frac{I_0^{(m)} + C^{[H]}_{\mathrm{pub}} + 1}{\log_2 N_n}\right\}, \tag{761}$$

$$U_{\mathrm{safe}} = \min\{1, U_{\text{fail-comm}}, U_{\text{death-cap}}, U^{\mathrm{noise}}_{\text{fail-gate}}, U^{\mathrm{path}}_{\mathrm{safe}}, U^{F,H}_{\mathrm{POMDP}}\}, \tag{762}$$

$$U_{\mathrm{hit}} = \min\{1, U_{\mathrm{comm}}, U_{\mathrm{selector}}, U_{\mathrm{resolvent}}, U_{\mathsf{Caus}}, U^{\mathrm{noise}}_{\mathsf{Caus}}, U^H_{\mathsf{Lift}}, U^H_{\mathsf{Lift},\mathrm{noise}}, U^{\mathrm{path}}_{\mathrm{hit}}, U^H_{\mathrm{POMDP}}, U^{\mathrm{src}}_{\mathrm{learn}}\}, \tag{763}$$

$$U^{\mathrm{noise}}_{\mathrm{Cap}} = b_\pi \left[\mathrm{Cap}_\Phi(T,F) + \omega^{\mathrm{Cap}}_{\Phi,T,F}(\varepsilon^\mu_T, \varepsilon^\mu_F) \right], \tag{764}$$

$$U_{\mathrm{regret}} = U_{\mathrm{seq}}. \tag{765}$$

The capacity entry is assigned $+\infty$ when its reference-law comparison, form comparison, or boundary-stability modulus is unavailable. The source-resolved learning entry has neutral value one unless its complete learning ledger and target converter are present in the same controlled record. Here $\mathcal{Q}_{\mathrm{bin}}$ is the finite represented index set of binary supervised coordinates in the complete record, with the declared confidence allocations δ_q. The maximum in Eq. (759) *is zero when the record has no binary supervised coordinate; the identity channel $\eta = 0$ is used when such a coordinate is present but uncorrupted. Each $\mathrm{Err}^{\mathrm{sat}}_{q,\mathsf{N}_q}$ in Eq.* (760) *retains its displayed (n, B, m, δ_q) arguments and its own norm. The allocation satisfies*

$$\delta_r + \delta_A + \delta_O + \delta_{\mathrm{fil}} + \delta_{\mathrm{comp}} + \delta_{\mathrm{path}} \le \delta_{\mathrm{ch}},$$

and δ_{ch} participates in the master simultaneous confidence budget. The path entries $U^{\mathrm{path}}_{\mathrm{hit}}$ and $U^{\mathrm{path}}_{\mathrm{safe}}$ are a represented reference probability plus the applicable TV or KL path shift, clipped at one; raw path distance is not itself an upper bound on either event. Similarly,

$$U^{H}_{\mathrm{POMDP}} = \min\{1, U^{H}_{b} + \Delta^{H}_{\mathrm{fil}}\}, \qquad U^{F,H}_{\mathrm{POMDP}} = \min\{1, U^{F,H}_{b} + \Delta^{F,H}_{\mathrm{fil}}\},$$

where each U_b is a represented and charged upper bound for the same exact-belief reference event. An uncomputed analytic probability has value one, so it cannot improve an affordable prospective minimum. $U^{\mathrm{value}}_{\mathrm{POMDP}}$ is Eq. (389) in value units only when its restricted benchmark, performance-difference identity, and evaluation interface are represented; otherwise it is $+\infty$. The information entry is Eq. (402) when target identification is the declared success event and the complete record contains the represented, costed output-to-target-index decoder and any disjoint-sector verification; it is one otherwise. A missing information budget has value $+\infty$ before probability clipping. The entropy vector is $\mathbf{H}_{\mathrm{ctl}}$ from Eq. (442) and remains coordinatewise. Here U_{iw} is the bounded-importance-weight certificate; U_{T_1}, U_{T_2} are the applicable transport-entropy bounds; U_{occ} is another represented occupancy comparison; $U_{\text{fail-comm}}$ and U_{comm} are the propagated failure and target committor bounds; $U_{\text{death-cap}}$ is zero under certified polar excision and otherwise is the represented monotone failure-capacity bound; U_{selector} and $U_{\mathrm{resolvent}}$ are the existing prospective-selector and killed-resolvent bounds; $U^{\mathrm{src}}_{\mathrm{learn}}$ is the same-record target-gain minimum in Eq. (281); U_{Caus} is admitted only through a represented contact converter; U^{H}_{Lift} and $U^{\lambda}_{\mathsf{Lift}}$ are admitted only after conversion to the arrival normalization used by this learned record; and U_{seq} is the Part IV sequential structural-regret bound. The noise-corrected entries are those of Theorems V.7.7 and V.8.2. In particular, $U^{\mathrm{noise}}_{\text{fail-gate}}$ is the deployment failure-gate error added to a represented clean failure probability, and $U^{H}_{\mathsf{Lift},\mathrm{noise}}$ is the learned lifted arrival plus the clocked interface error, both clipped at one. Raw discounted entries $U^{\lambda}_{\mathsf{Lift}}$ and $U^{\lambda}_{\mathsf{Lift},\mathrm{noise}}$ remain in the discounted coordinate U_{disc}; they do not enter the finite-horizon minimum without an explicit resolvent-to-horizon converter.

Proof. Each displayed entry is the clipped same-event consequence of the theorem or definition cited in its accompanying paragraph. The channel vector uses the disjoint internal confidence allocation, the path coordinates add the path distance to a charged reference event, the POMDP coordinates use the filter-transfer theorem, and the information coordinate uses the represented decoder form of Fano's inequality. Unavailable conversions take their declared neutral value, so none can improve the corresponding minimum. □

Proposition V.14.4 (Explicit deployment-shift and committor candidates)**.** *Under the hypotheses of Definition V.14.1, the following represented formulas instantiate the shift and committor entries in Proposition V.14.3. For a represented deployment loss (F) with $\mathrm{Lip}_{d_\Phi}(F) = L_F$, behavior occupancy d^b, deployment occupancy d^π, and represented density ratio $w = d^\pi/d^b \le W$, the eligible shift entries are*

$$U_{\mathrm{iw}} = \widehat{\mathbb{E}}_{d^b}[wF] + \mathfrak{G}_{\mathrm{dyn}}(w\mathcal{F}; B, m, \delta_{\mathrm{iw}}), \tag{766}$$

$$U_{T_p} = \widehat{\mathbb{E}}_{d^b}[F] + \mathfrak{G}_{\mathrm{dyn}}(\mathcal{F}; B, m, \delta_{\mathrm{shift}}) + L_F\sqrt{2C_{T_p}\,\mathrm{KL}(d^\pi \| d^b)}, \qquad p \in \{1,2\}, \tag{767}$$

$$U_{\mathrm{occ}} = \widehat{\mathbb{E}}_{d^b}[F] + \mathfrak{G}_{\mathrm{dyn}}(\mathcal{F}; B, m, \delta_{\mathrm{shift}}) + \|d^\pi - d^b\|_{\mathrm{TV}}. \tag{768}$$

All three lines bound deployment risk. The final term in the last two lines is the certified behavior-to-deployment change for $[0,1]$-valued loss.

For learned target and failure committors $\widehat{h}_+, \widehat{h}_-$, let $\kappa_\pm$ be certified killed-resolvent norms and let $\bar{\varepsilon}_\pm$ be their empirical residuals plus the applicable dynamical generalization envelopes. Then

$$U_{\mathrm{comm}} = \min\{1, \langle \mu_0, \widehat{h}_+\rangle + \kappa_+\bar{\varepsilon}_+\}, \tag{769}$$

$$U_{\text{fail-comm}} = \min\{1, \langle \mu_0, \widehat{h}_- \rangle + \kappa_- \bar{\varepsilon}_-\}, \tag{770}$$
$$U_{\text{selector}} = \min\{1, \mathrm{Enum}_\nu + H\,\mathrm{Sel}^{\text{sat}}(\Psi_{\text{ctl}}(B, H, n))\}, \tag{771}$$
$$U_{\text{resolvent}} = \min\{1, eA_{T^+}(H^{-1}; L, \mu_0)\}. \tag{772}$$

Certified polarity sets $U_{\text{death-cap}} = 0$. At positive failure capacity, this entry is the upper probability supplied by the represented capacity-to-boundary-risk comparison, and equals one when that comparison is absent. Positive capacity alone is not substituted for a probability.

Each committor, propagation, shift, mixing, and access entry uses the smallest same-unit consequence of the audited functional-inequality profile. A functional inequality with no represented conversion to that coordinate is excluded from its minimum. Thus the master certificate uses the full profile without adding logically dependent inequalities or unlike units.

Proof. On $\mathcal{E}_{\mathcal{R}}$, every stochastic leaf inequality holds for the actually selected data-dependent output. Therefore each generic population loss is at most its empirical value plus its envelope. For a binary supervised coordinate, Theorem IV.16.6 and its represented same-norm calibration supply the alternative noise-corrected candidate. The typed channel profiles and their same-norm reward, execution, observation, and filter converters supply the U_Q^{ch} candidates through Eq. (744). Since these candidates bound the same population object in the same norm on the same event, their minimum is again an upper bound. This proves the three barred-error definitions and Eq. (753). Geometry selection and projector transfer are Lemma IV.9.2, which proves Eq. (754).

Under the represented rectangularity gate, $V^*_{B,\text{obs}}$ is the restricted Bellman fixed point; without that gate the theorem has already assigned the Bellman branch $+\infty$. Under the joint actionwise model defect, the learned-greedy selector satisfies the two-sided one-step comparison proved in Theorem V.13.6. Substituting the barred model and residual errors into that theorem proves Eqs. (753) to (756). The exact identity Eq. (751), the triangle inequality, and the two represented gap certificates, with $\overline{\Delta}_B^{\text{POMDP}} = 0$ in the fully observed case, prove Eq. (757).

The shift entries are alternative comparisons between the same behavior and deployment risks, using the transport consequences of the functional profile and bounded importance weighting. The committor residual identities follow by applying the killed inverse to the represented residual. The clean-interface safe and hit entries follow from the reach–avoid committor identity, condenser capacity, prospective-selector accountability, and killed-resolvent identity in Theorems III.21.3 and V.6.6 and Lemma III.8.3. The learned-interface Caus, Lift, failure-gate, and capacity entries are Theorems V.7.7 and V.8.2. For a POMDP target or failure entry, first bound the exact-belief reference event by its represented same-record baseline U_b, then add the path or filter shift from Theorems V.1.5 and V.2.4. If the baseline is merely analytic and uncomputed, its value is one and the candidate is neutral. Thus no latent posterior or target gate enters an affordable minimum for free. The information entry follows from Theorem V.3.2 only after the represented output-to-Θ decoder and target-sector verification make the success event an identification event; otherwise its value is one. The sequential coordinate is the Part IV online bound applied to the represented policy-loss class, Theorem IV.15.7 and Corollary IV.15.8. Slotwise functional-inequality selection is the dynamical specialization of Lemma IV.9.6. Taking minima of upper bounds for the same quantity proves Eqs. (758) to (765).

All selected geometries, priors, losses, policies, comparisons, and deployment operations belong to the complete record. Saturated accounting therefore charges their derivation and selection costs before any coordinate is used. Every target-aligned prospective improvement still terminates in the represented selector/source, Caus-contact, path-baseline, or Lift comparison of that record. Analytic latent beliefs, physical channel entropy, and Fano converses are comparison quantities, not new structural sources.

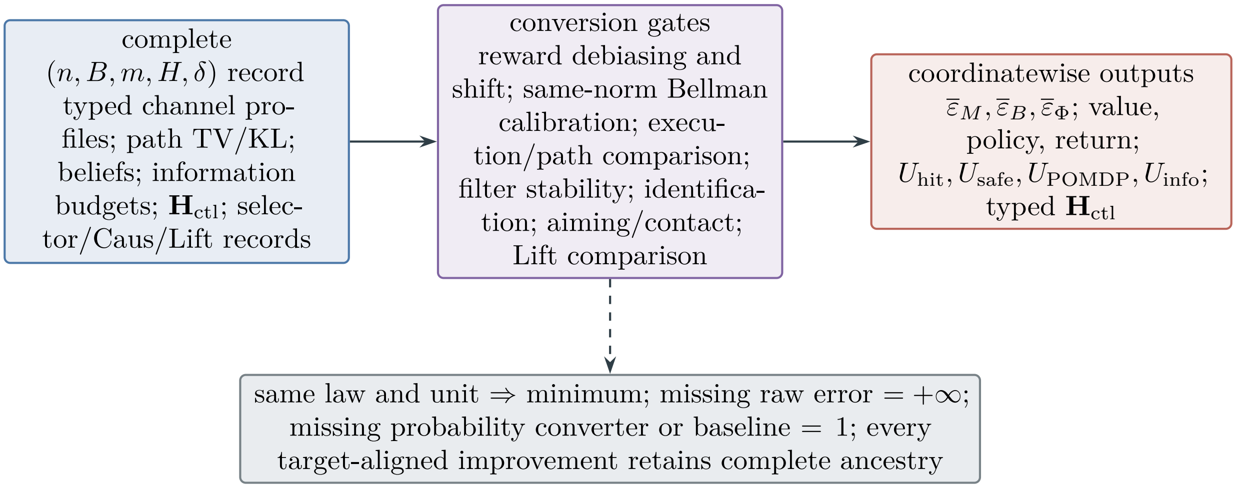

Figure 97: The multichannel master certificate routes every raw coordinate through a same-law, same-norm converter before combination. Model, Bellman, value, return, hit, safety, POMDP, and information bounds take minima only within one output unit and one complete record; the entropy vector remains typed.

□

Figure 97 gives the detailed multichannel routing behind the master certificate.

V.14.1 Continuous-System and Discretization Transfer

Corollary V.14.5 (Represented continuum and discretization transfer)**.** *Let L be a generator on a declared state space with bounded resolvent at $\lambda > 0$, let $\widehat{L}$ be a represented finite or clocked generator, and let U be a represented lift of finite observables into the declared space. Suppose the common represented domain satisfies*

$$E_U = LU - U\widehat{L} \tag{773}$$

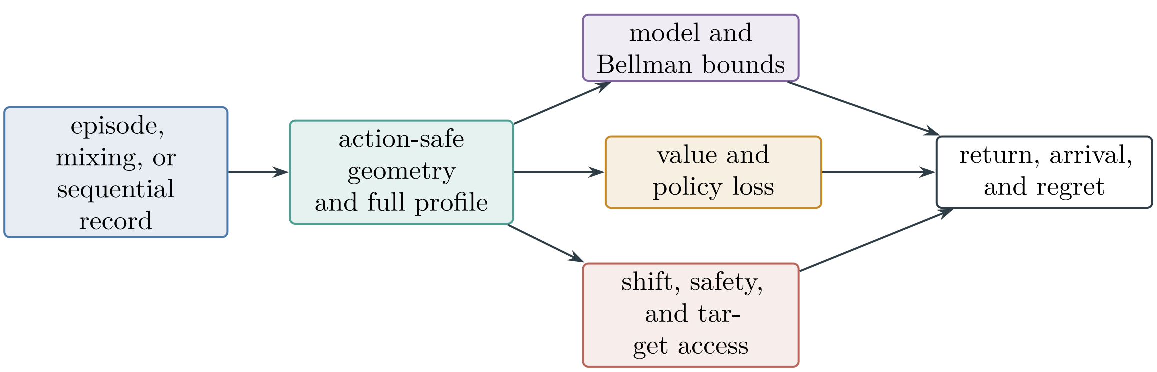

Figure 98: The learned-control certificate propagates one completely charged trajectory record through action-safe geometry. Generalization, functional inequalities, policy loss, safety, and target access remain typed until a theorem converts them to return, arrival, or regret units.

Figure 99: A learned policy remains one complete represented computation. Geometry selection, value fitting, policy construction, and deployment are charged before the valid structural and statistical bounds are combined.

and the target and failure gates are transported by U. For first-arrival claims, L and $\widehat{L}$ are the corresponding killed or absorbing generators. Then the exact resolvent identity is

$$(\lambda I - L)^{-1}U - U(\lambda I - \widehat{L})^{-1} = (\lambda I - L)^{-1}E_U(\lambda I - \widehat{L})^{-1}. \tag{774}$$

Consequently,

$$\left\|(\lambda I - L)^{-1}U - U(\lambda I - \widehat{L})^{-1}\right\| \le \|(\lambda I - L)^{-1}\| \, \|E_U\| \, \|(\lambda I - \widehat{L})^{-1}\|. \tag{775}$$

If P_t and $\widehat{P}_t$ are represented semigroups for which the Duhamel integral is defined, then

$$P_tU - U\widehat{P}_t = \int_0^t P_{t-s}E_U\widehat{P}_s \, ds. \tag{776}$$

Every discounted-arrival, mixing, or finite-time certificate for $\widehat{L}$ transfers through these identities with the displayed norm or form-comparison loss and the represented target-gate comparison. The lift, discretization, generator evaluation, error proof, and precision are charged at their complete saturated cost. Without this represented transfer record, the finite certificate makes no claim about L.

This corollary covers represented discretizations of deterministic flows, stochastic kernels, controlled diffusions, and other continuous models whenever their finite lift and defect estimate satisfy the displayed hypotheses.

Proof. From Eq. (773),

$$(\lambda I - L)U = U(\lambda I - \widehat{L}) - E_U.$$

Multiply on the left and right by the two resolvents and rearrange to obtain Eq. (774); submultiplicativity proves Eq. (775). Differentiating $P_{t-s}U\widehat{P}_s$ in s and integrating from zero to t proves Eq. (776). Target-function transfer then uses the represented gate comparison. Cost and provenance are those of a valid Lift in Theorem II.10.5 and Corollary II.10.7; the form-comparison alternative is Lemma III.13.5 and Proposition III.13.6. The block form and its exact-intertwining specialization are Theorem V.7.12; the present identity is the two-carrier defect comparison used when the declared and represented generators live on different approximation spaces.

□

V.15 Certified Neural Implementation of the Saturated Optimum

The saturated active and structural-Bayesian profiles are analytic optima over complete represented records. A trained network supplies one such record. This section gives a finite-sample certificate for the distance between that record and the saturated optimum. The certificate combines a held-out lower bound for the deployed record with a Bellman upper bound for the complete saturated class. Its optimization component is then specialized to a represented weighted Adam update.

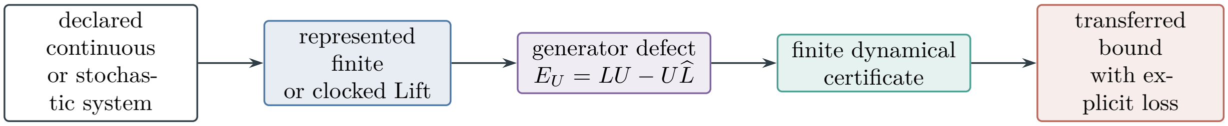

Figure 100: Continuous-system conclusions pass through a represented lift and an explicit intertwining defect. The finite certificate is transferred only with its proved comparison loss.

V.15.1 Implementation records and error coordinates

Definition V.15.1 (Complete neural implementation record)**.** Fix an instance I, horizon H, and saturated active budget B. A *complete neural implementation record* consists of:

(i) a complete active or structural-Bayesian record in the sense of Definitions V.9.1 and V.11.1;

(ii) represented neural evaluators for every learned score, value, barrier, belief, observation, and transition kernel used by that record;

(iii) the architecture, parameter precision, initialization, training sample, loss, optimizer transcript, stopping rule, and selected terminal weights;

(iv) a represented deployment compiler producing the pre-hit selector and every state update from those terminal weights; and

(v) represented evaluation and upper-certificate records of the types defined below.

The complete cost is

$$B_{\mathrm{impl}} = B_{\mathrm{active}} \oplus B_{\mathrm{arch}} \oplus B_{\mathrm{par}} \oplus B_{\mathrm{train}} \oplus B_{\mathrm{optimize}} \oplus B_{\mathrm{precision}} \oplus B_{\mathrm{deploy}} \oplus B_{\mathrm{eval}} \oplus B_{\mathrm{upper}}. \tag{777}$$

Every selected feature, audit cover, random seed, gradient, moment estimate, preconditioner, likelihood, normalization, and comparison used by the record is charged in the corresponding slot. Write $\mathscr{N}_{I,H}(B; B_{\mathrm{impl}})$ for the complete neural implementation records of total cost at most B_{impl} whose deployed active component has cost at most B. Let

$$J^*_{\mathcal{N}}(B_{\mathrm{impl}}) = \sup_{R_\theta \in \mathscr{N}_{I,H}(B;B_{\mathrm{impl}})} A^H_{R_\theta}, \tag{778}$$

$$J^*_{\mathrm{sat}}(B) = \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B). \tag{779}$$

For a deployed terminal parameter $\widehat{\theta}$, define the analytic coordinates

$$\varepsilon_{\mathrm{rep}} = J^*_{\mathrm{sat}}(B) - J^*_{\mathcal{N}}(B_{\mathrm{impl}}), \tag{780}$$

$$\varepsilon_{\mathrm{train}} = J^*_{\mathcal{N}}(B_{\mathrm{impl}}) - A^H_{R_{\widehat{\theta}}}, \tag{781}$$

$$\varepsilon_{\mathrm{impl}} = J^*_{\mathrm{sat}}(B) - A^H_{R_{\widehat{\theta}}} = \varepsilon_{\mathrm{rep}} + \varepsilon_{\mathrm{train}}. \tag{782}$$

The budget comparison is made at a common class-preserving ceiling containing the operational and neural records. Thus both suprema refer to the same primitive presentation and target event.

V.15.2 Parameter-indexed approximation and underfitting

Definition V.15.2 (Parameter-indexed saturated neural profile)**.** Let $(\mathcal{N}_p)_{p\ge 0}$ be a represented architecture sequence in which every record in $\mathcal{N}_p$ has at most p trainable scalar parameters at the declared precision. Assume a represented padding map $\mathcal{N}_p \hookrightarrow \mathcal{N}_{p+1}$ that preserves the deployed record. At a common active ceiling B and implementation ceiling $B_{\mathrm{impl}}(p)$, define

$$J^*_p(B) = \sup\left\{ A^H_{R_\theta} : R_\theta \in \mathscr{N}_{I,H}(B; B_{\mathrm{impl}}(p)) \cap \mathcal{N}_p \right\}, \tag{783}$$

$$\varepsilon_{\mathrm{rep}}(p;B) = J^*_{\mathrm{sat}}(B) - J^*_p(B), \tag{784}$$
$$p_{\min}(\epsilon;B) = \inf\left\{p \in \mathbb{N}_0 : \varepsilon_{\mathrm{rep}}(p;B) \le \epsilon\right\}. \tag{785}$$

The padding map makes J^*_p nondecreasing and $\varepsilon_{\mathrm{rep}}(p;B)$ nonincreasing. The value $p_{\min} = +\infty$ is used when the set is empty.

For every p, the complete parameter-explicit ledger is

$$\mathfrak{E}_p = \Big(\varepsilon_{\mathrm{rep}}(p),\ \varepsilon_{\mathrm{train}}(p),\ \{\Gamma^{(a)}_{t,p}\}_{t,a},\ \{\mathcal{L}^{(a)}_p, J^{(a)}_p\}_a,\ \Delta^H_{\mathrm{fil},p},\ \varepsilon_{\mathrm{num},p},\ \varepsilon_{\mathrm{opt},p},\ U_{\mathrm{Bell},p},\ L_{\mathrm{ev},p}\Big). \tag{786}$$

Every coordinate is evaluated for the same architecture, precision, training record, deployed selector, and audit law. Dependence on the data size, noise parameters, horizon, confidence allocation, and complete budget is retained in the corresponding coordinate.

Theorem V.15.3 (Certified underfitting and optimization intervals)**.** *Fix p. Suppose simultaneous represented certificates give*

$$L_{\mathrm{sat}} \le J^*_{\mathrm{sat}}(B) \le U_{\mathrm{sat}}, \tag{787}$$
$$L_p \le J^*_p(B) \le U_p, \tag{788}$$
$$L_{\mathrm{run}} \le A^H_{R_{\widehat{\theta}_p}} \le U_{\mathrm{run}}. \tag{789}$$

Then

$$\max\{0, L_{\mathrm{sat}} - U_p\} \le \varepsilon_{\mathrm{rep}}(p;B) \le U_{\mathrm{sat}} - L_p, \tag{790}$$
$$\max\{0, L_p - U_{\mathrm{run}}\} \le \varepsilon_{\mathrm{train}}(p;B) \le U_p - L_{\mathrm{run}}. \tag{791}$$

Consequently

$$p_{\mathrm{suff}}(\epsilon) = \inf\{p : U_{\mathrm{sat}} - L_p \le \epsilon\}, \tag{792}$$
$$p_{\mathrm{fail}}(\epsilon) = 1 + \sup\{p : L_{\mathrm{sat}} - U_p > \epsilon\} \tag{793}$$

satisfy

$$p_{\mathrm{fail}}(\epsilon) \le p_{\min}(\epsilon;B) \le p_{\mathrm{suff}}(\epsilon), \tag{794}$$

with the empty-set conventions $p_{\mathrm{fail}} = 0$ and $p_{\mathrm{suff}} = +\infty$.

A represented solver compiled into $\mathcal{N}_p$ supplies L_p and therefore an upper certificate for $p_{\min}$. A Bellman upper certificate restricted to $\mathcal{N}_p$ supplies U_p; together with a lower certificate L_{sat} for the unrestricted saturated optimum, it supplies a lower certificate for $p_{\min}$.

Proof. Subtract the upper and lower endpoints in Eqs. (787) and (788) to obtain Eq. (790). Subtract the endpoints in Eqs. (788) and (789) to obtain Eq. (791). Nonnegativity gives the clipped lower endpoints.

If $U_{\mathrm{sat}} - L_p \le \epsilon$, then $\varepsilon_{\mathrm{rep}}(p;B) \le \epsilon$, proving the upper bracket. If $L_{\mathrm{sat}} - U_p > \epsilon$, then $\varepsilon_{\mathrm{rep}}(p;B) > \epsilon$. Monotonicity under the padding maps excludes every smaller parameter count, proving the lower bracket.

□

V.15.3 Canonical finite-precision attention–belief architecture

Definition V.15.4 (Canonical certified attention–belief architecture)**.** Fix a finite horizon H. A canonical certified architecture $\mathcal{C}_{\Theta}^{\mathrm{can}}$ contains the following represented modules.

(i) A tokenizer maps the public presentation, current native state, surviving pre-hit history, resource record, and represented legal actions to N_0 width-d_0 tokens. Illegal actions receive a represented mask.

(ii) One recurrent belief token stores a width-d_0 compact latent state. Its update factors through the current pre-hit record.

(iii) Layer ℓ is a pre-normalized transformer block with width d_ℓ, H_ℓ attention heads, key width $d_{k,\ell}$, value width $d_{v,\ell}$, feed-forward width f_ℓ, temperature $\tau_{\ell h} > 0$, residual addition, and two affine normalizations. The canonical residual stack has $d_\ell = d_0$; a represented width-changing projection is counted in p_{tok} when present.

(iv) The terminal representation feeds five represented heads: a masked policy score, a value function, a Bellman barrier, a centered geometry feature map $\psi_{p_{\mathrm{geom}}}$, and a quotient label or quotient-logit map q.

(v) A finite-precision deployment compiler produces the pre-hit selector, belief update, transition record, and all audit evaluators from the stored parameters.

The quotient head proposes a represented label. Its exact or approximate dynamical status is determined by the fiber-row audit of Definition V.15.39.

For dense affine maps with biases, the trainable parameter count is

$$p_{\mathrm{can}} = p_{\mathrm{tok}} + p_{\mathrm{bel}} + \sum_{\ell=1}^{L} (p_{\mathrm{att},\ell} + p_{\mathrm{ff},\ell} + 4d_\ell) + p_{\mathrm{pol}} + p_{\mathrm{val}} + p_{\mathrm{bar}} + p_{\mathrm{geom}} + p_{\mathrm{quot}}, \tag{795}$$

$$p_{\mathrm{att},\ell} = H_\ell(2d_\ell d_{k,\ell} + d_\ell d_{v,\ell} + 2d_{k,\ell} + d_{v,\ell}) + H_\ell d_{v,\ell} d_\ell + d_\ell, \tag{796}$$

$$p_{\mathrm{ff},\ell} = 2d_\ell f_\ell + f_\ell + d_\ell. \tag{797}$$

The $4d_\ell$ term stores the scale and shift parameters of the two normalizations. Tied parameters are counted once and their reuse is stored in the architecture record. Every scalar is stored at the declared precision.

If E_ℓ is the number of unmasked query–key pairs at layer ℓ, one forward deployment has arithmetic cost bounded by

$$\begin{aligned}\mathrm{Cost}_{\mathrm{fwd}} \le C_{\mathrm{op}}\Bigg\{ &\mathrm{Cost}_{\mathrm{tok}} + \mathrm{Cost}_{\mathrm{bel}} + \mathrm{Cost}_{\mathrm{heads}} \\ &+ \sum_{\ell=1}^{L} H_\ell[N_\ell d_\ell(2d_{k,\ell} + d_{v,\ell}) + E_\ell(d_{k,\ell} + d_{v,\ell})] \\ &+ \sum_{\ell=1}^{L} N_\ell d_\ell \left(H_\ell d_{v,\ell} + 2f_\ell + 1\right)\Bigg\}.\end{aligned} \tag{798}$$

where C_{op} is the represented finite-precision arithmetic conversion. Parameter storage, activations, recurrent state, optimizer moments, training, audit samples, and deployment are charged separately in Eq. (777).

Theorem V.15.5 (Canonical architecture implementation and compilation)**.** *Every module in Definition V.15.4 is a finite composition of represented affine maps, comparisons, masks, normalizations, pointwise maps, and finite-precision exponential normalization. Hence it defines a*

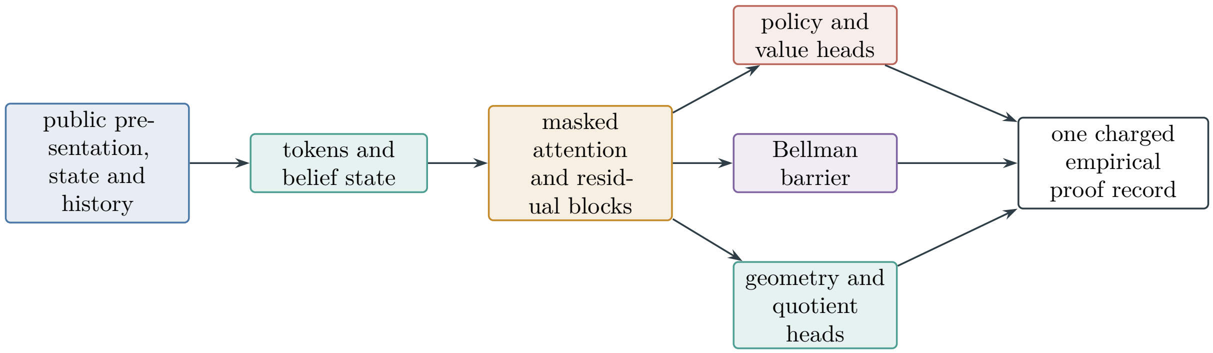

Figure 101: Canonical certified implementation. The deployed controller and the Bellman, geometry, and quotient audits share the same public tokens, belief state, parameters, transition law, precision, and cost ledger.

complete neural implementation record at the cost in Eqs. (795) *and* (798)*. Its deployed attention rows are complete attention records in Definition VI.0.1.*

Conversely, let a complete bounded-branch clocked selector have M_t represented legal actions at time t. At the class-preserving compilation ceiling, a canonical architecture with $N_0 \geq \max_t M_t + 1$ represents the same pre-hit path law by tokenizing the legal actions and the recurrent state, exposing the selector's already charged auxiliary seed, and assigning the selected action the unique maximal finite-precision score. The architecture preserves the selector's source ancestry, latent code, mask, tie-breaking rule, and complete cost.

Proof. The first assertion follows directly from the represented constructor list and the complete cost definition. The parameter formulas count the Q, K, V, O, feed-forward, normalization, tokenizer, recurrent, and output maps once. The operation bound counts each unmasked dot product, value transport, and feed-forward multiplication.

For the converse, use the pathwise hard-attention compilation in Corollary VI.2.3: for each represented seed, the inverse-transform comparisons select one legal action. Store the surviving history and seed in the belief token, mask illegal actions, assign the selected action a unique maximal score, and apply the original tie-breaking convention. Averaging over the same seed reproduces the original kernel. Constructor no-creation retains every source ancestor and the compilation cost.

□

Proposition V.15.6 (Held-out arrival lower certificate)**.** *Condition on the complete training record and terminal weights $\widehat{\theta}$. Run N_{ev} independent represented evaluation episodes under the deployed record and put*

$$\widehat{A}_{\mathrm{ev}} = \frac{1}{N_{\mathrm{ev}}} \sum_{j=1}^{N_{\mathrm{ev}}} \mathbf{1}_{\{\tau_j \leq H\}}.$$

For $0 < \delta_{\mathrm{ev}} < 1$, define

$$L_{\mathrm{ev}} = \left[\widehat{A}_{\mathrm{ev}} - \sqrt{\frac{\log(1/\delta_{\mathrm{ev}})}{2N_{\mathrm{ev}}}} \right]_+ . \tag{799}$$

Then, conditionally and hence unconditionally,

$$\Pr\{L_{\mathrm{ev}} \leq A^H_{R_{\widehat{\theta}}}\} \geq 1 - \delta_{\mathrm{ev}}. \tag{800}$$

A sequential or dependent evaluation record may replace the square-root term by a valid same-law martingale or mixing candidate from Theorems IV.15.2 and IV.15.6.

Proof. Conditional on the training record, the evaluation indicators are independent Bernoulli variables with mean $A^H_{R_{\widehat{\theta}}}$. The lower-tail Hoeffding inequality gives

$$\Pr\left\{ \widehat{A}_{\mathrm{ev}} - A^H_{R_{\widehat{\theta}}} > \sqrt{\frac{\log(1/\delta_{\mathrm{ev}})}{2N_{\mathrm{ev}}}} \,\middle|\, \widehat{\theta} \right\} \le \delta_{\mathrm{ev}}.$$

Clipping at zero preserves the lower bound. Averaging over the training record proves Eq. (800). □

V.15.4 Empirical Bellman upper certificates

Definition V.15.7 (Saturated Bellman audit record)**.** Let $v = (v_t)_{t=0}^H$, $v_H = 0$, be a represented bounded Bellman barrier on the complete surviving history-and-resource carrier. For each time t, let μ_t be a represented audit probability law on complete first-step records $r = (h, a)$. A *saturated coverage certificate* is a represented number $C_t < \infty$ such that every stopped first-step flow of every $R \in \mathscr{R}^{\mathrm{act}}_{I,H}(B)$ satisfies

$$f_{R,t} \ll \mu_t, \qquad \left\| \frac{df_{R,t}}{d\mu_t} \right\|_\infty \le C_t. \tag{801}$$

Here $f_{R,t}$ is the subprobability flow in Definition V.10.1. Let $r_{t,1}, \ldots, r_{t,N_t}$ be an independent audit sample from μ_t, independent of the construction of v, and set

$$d_t(r; v) = [p_T(h, a) + P_N v_{t+1}(h, a) - v_t(h)]_+ \,, \tag{802}$$

$$\widehat{d}_t(v) = \frac{1}{N_t} \sum_{j=1}^{N_t} d_t(r_{t,j}; v). \tag{803}$$

Let M_t be a represented bound $0 \le d_t(r; v) \le M_t$ on the support of μ_t. The coverage law, its sampler, barrier, transition and contact evaluators, sample, precision, and comparisons are charged in B_{upper}.

An exact verification record may instead supply represented numbers $\overline{d}_t(v)$ satisfying

$$\sup_{r \in \mathscr{A}^{\mathrm{sat}}_{I,t}(B)} d_t(r; v) \le \overline{d}_t(v). \tag{804}$$

Theorem V.15.8 (Empirical saturated Bellman upper bound)**.** *Under Definition V.15.7, allocate $\delta_{\mathrm{up},t} > 0$ with $\sum_{t<H} \delta_{\mathrm{up},t} \le \delta_{\mathrm{up}}$, and put*

$$u^{\mathrm{emp}}_t(v) = C_t \left[\widehat{d}_t(v) + M_t \sqrt{\frac{\log(1/\delta_{\mathrm{up},t})}{2N_t}} \right], \tag{805}$$

$$U_{\mathrm{Bell}}(v) = v_0(h_0) + \sum_{t=0}^{H-1} \min\{\overline{d}_t(v), u^{\mathrm{emp}}_t(v)\}, \tag{806}$$

where an unavailable candidate has value $+\infty$. Then, with probability at least $1 - \delta_{\mathrm{up}}$,

$$J^*_{\mathrm{sat}}(B) \le U_{\mathrm{Bell}}(v). \tag{807}$$

The conclusion remains valid when a simultaneous structural Rademacher, sequential, mixing, or martingale PAC–Bayes radius replaces the Hoeffding radius, provided it bounds the same μ_t-expectation for the selected barrier and its construction and selection are included in the audit record.

Proof. Hoeffding's inequality and a union bound give, simultaneously for every $t < H$,

$$\mathbb{E}_{\mu_t} d_t(r; v) \leq \widehat{d}_t(v) + M_t \sqrt{\frac{\log(1/\delta_{\mathrm{up},t})}{2N_t}}. \tag{808}$$

For any complete record R, the coverage certificate gives

$$\int d_t(r; v) f_{R,t}(dr) \leq C_t \mathbb{E}_{\mu_t} d_t(r; v) \leq u_t^{\mathrm{emp}}(v).$$

The exact verification candidate gives the alternative bound $\int d_t \, df_{R,t} \leq \overline{d}_t(v)$, because $f_{R,t}$ has mass at most one. Integrating the signed Bellman inequality along the stopped flow and telescoping as in Theorem V.10.4 yields

$$A_R^H \leq v_0(h_0) + \sum_{t<H} \min\{\overline{d}_t(v), u_t^{\mathrm{emp}}(v)\}.$$

Taking the supremum over $R \in \mathscr{R}_{I,H}^{\mathrm{act}}(B)$ proves Eq. (807).

□

Corollary V.15.9 (Primal–dual empirical implementation certificate)**.** *Let L_{ev} and $U_{\mathrm{Bell}}(v)$ be constructed from independent evaluation and upper-audit records with total failure probability $\delta = \delta_{\mathrm{ev}} + \delta_{\mathrm{up}}$. Then, with probability at least $1 - \delta$,*

$$0 \leq \varepsilon_{\mathrm{impl}} = J_{\mathrm{sat}}^*(B) - A_{R_{\widehat{\theta}}}^H \leq U_{\mathrm{Bell}}(v) - L_{\mathrm{ev}}. \tag{809}$$

In particular, this certificate bounds the sum $\varepsilon_{\mathrm{rep}} + \varepsilon_{\mathrm{train}}$ without separately estimating either coordinate.

If the hypotheses of Corollary V.11.16 hold at the same compiled ceiling, the implementation error also satisfies

$$\varepsilon_{\mathrm{impl}} \leq \min\left\{U_{\mathrm{Bell}}(v) - L_{\mathrm{ev}}, \mathbb{E}_{\widehat{R}} \sum_{t<H} \mathbf{1}_{\{\tau>t\}}(2\Gamma_t + \varepsilon_{\mathrm{opt},t}) + \Delta_{\mathrm{fil}}^H\right\}. \tag{810}$$

Proof. On the simultaneous event, $L_{\mathrm{ev}} \leq A_{R_{\widehat{\theta}}}^H \leq J_{\mathrm{sat}}^*(B) \leq U_{\mathrm{Bell}}(v)$. Subtraction proves Eq. (809). The second candidate is Eq. (683); both candidates bound the same nonnegative quantity on the same event, so their minimum is valid.

□

V.15.5 Scale-certified empirical closure of attenuating sources

The empirical upper audit becomes an asymptotic certificate when the same record includes a verified scale law for structural capacity and target attenuation. The following definition records that additional datum at the level of the saturated source cells. The per-cell factorization, exponent competition, and complete empirical-to-proof pipeline are displayed in Figs. 102, 103 and 105.

Definition V.15.10 (Scale-certified attenuating source audit)**.** Fix a family functional $\mathfrak{M}_n$, a budget schedule B_n, and a base size n_0. A *scale-certified attenuating source audit* consists of a finite set $\mathfrak{C}_{\text{att}}$ of existing qualified source signatures such that, for every $n \geq n_0$,

$$I^{\text{src,sat}}_{\mathfrak{M},n}(B_n) \leq \sum_{c\in\mathfrak{C}_{\text{att}}} \sup_{R\in\mathcal{S}_{c,n}(B_n)} V_{c,n}(R). \tag{811}$$

Here $\mathcal{S}_{c,n}(B_n)$ denotes the restriction of the already-defined complete record class to source cell c, including all temporal, target, dynamic, and cost qualification gates. Thus the audit introduces no new admissible record class.

For each c and n, the audit supplies a represented Hilbert carrier $\mathcal{H}_{c,n}$, a coefficient record $z_{c,n}(R) \in \mathcal{H}_{c,n}$, a contraction $P_{c,n} : \mathcal{H}_{c,n} \to \mathcal{H}_{c,n}$, and a scalar target-transfer functional $T_{c,n} : \mathcal{H}_{c,n} \to \mathbb{R}$. It also supplies nonnegative numbers $D_{c,n}$, $a_{c,n}$, and $e_{c,n}$ satisfying, simultaneously for every $R \in \mathcal{S}_{c,n}(B_n)$,

$$\|z_{c,n}(R)\|_{\mathcal{H}_{c,n}} \leq 1, \tag{812}$$

$$V_{c,n}(R) \leq |T_{c,n}P_{c,n}z_{c,n}(R)| + e_{c,n}, \tag{813}$$

$$\|T_{c,n}P_{c,n}\|_{\mathcal{H}_{c,n}\to\mathbb{R}} \leq \sqrt{D_{c,n}}\, a_{c,n}. \tag{814}$$

The quantity $D_{c,n}$ is the certified target-relevant capacity of the budget-reachable carrier, $a_{c,n}$ is its target-transfer attenuation, and $e_{c,n}$ is the certified factorization, tail, and numerical error. For an orthogonal materialized band, one may take $D_{c,n}$ to be its rank and $a_{c,n}$ the largest retained transfer multiplier. For a ridge band, the effective-dimension and tail certificates of Definitions IV.2.3 and I.5.6 supply the corresponding continuous values.

At the base size, a simultaneous upper audit supplies

$$D_{c,n_0} \leq \overline{D}_{c,0}, \qquad a_{c,n_0} \leq \overline{a}_{c,0}, \qquad e_{c,n_0} \leq \overline{e}_{c,0}. \tag{815}$$

For all $n \geq n_0$, a represented scale verifier supplies

$$D_{c,n+1} \leq r_{c,n}D_{c,n}, \tag{816}$$

$$a_{c,n+1} \leq s_{c,n}a_{c,n}, \tag{817}$$

$$e_{c,n} \leq \overline{e}_{c,n}. \tag{818}$$

The complete audit cost contains the learned factorization, audit samples, effective-dimension or operator calculation, numerical precision, interval or symbolic scale verifier, source-cell cover, and every selection rule used to choose them.

A *proof-grade* audit has deterministic verification of Eqs. (811), (812) and (814) to (818). An empirical simultaneous bound with failure probability δ_{aud} gives the same conclusions on its declared audit event. A randomly trained record becomes proof-grade when its exported inequalities are checked by a deterministic exact, interval, or symbolic verifier; randomness used to find the certificate then has no role in its validity.

Lemma V.15.11 (Base-size audit with verified scale propagation)**.** *Under Definition V.15.10, for every $n \geq n_0$,*

$$D_{c,n} \leq \overline{D}_{c,0} \prod_{j=n_0}^{n-1} r_{c,j}, \tag{819}$$

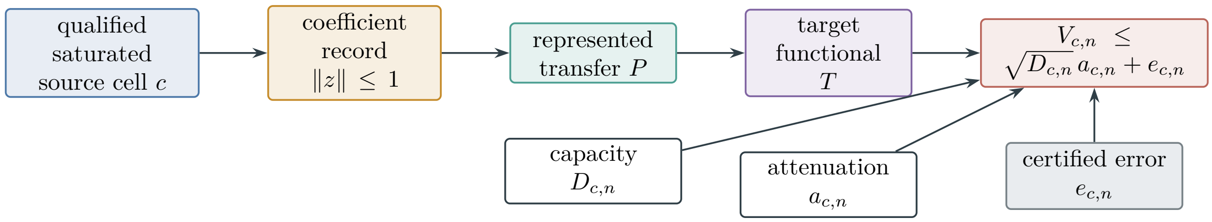

Figure 102: One scale-certified source cell. Exhaustive source routing fixes the cell; the represented factorization separates budget-reachable capacity, target-transfer attenuation, and certified approximation error.

$$a_{c,n} \le \overline{a}_{c,0} \prod_{j=n_0}^{n-1} s_{c,j}, \tag{820}$$

$$\sup_{R \in \mathcal{S}_{c,n}(B_n)} V_{c,n}(R) \le \sqrt{\overline{D}_{c,0}}\, \overline{a}_{c,0} \prod_{j=n_0}^{n-1} \left(\sqrt{r_{c,j}} s_{c,j}\right) + \overline{e}_{c,n}. \tag{821}$$

Empty products are one.

Proof. Iterating Eq. (816) from the certified base bound gives Eq. (819); iterating Eq. (817) gives Eq. (820). For a qualified record R, Eqs. (812) and (814) and $\|z_{c,n}(R)\| \le 1$ give

$$V_{c,n}(R) \le \sqrt{D_{c,n}}\, a_{c,n} + e_{c,n}.$$

Substitute the two propagated bounds and Eq. (818), then take the supremum over the source cell. □

Theorem V.15.12 (Empirical-to-asymptotic saturated-source closure)**.** *Assume a scale-certified attenuating source audit. Suppose that, for each $c \in \mathfrak{C}_{\text{att}}$, there are constants $\beta_c \ge 0$, $\gamma_c > 0$, and a positive subexponential function p_c, normalized by $p_c(n_0) = 1$, such that*

$$r_{c,n} \le e^{\beta_c} \frac{p_c(n+1)}{p_c(n)}, \qquad s_{c,n} \le e^{-\gamma_c} \qquad (n \ge n_0). \tag{822}$$

Then, on the simultaneous audit event,

$$I^{\text{src,sat}}_{\mathfrak{M},n}(B_n) \le \sum_{c \in \mathfrak{C}_{\text{att}}} \left[\sqrt{\overline{D}_{c,0}}\, \overline{a}_{c,0} \sqrt{p_c(n)} e^{-(\gamma_c - \beta_c/2)(n-n_0)} + \overline{e}_{c,n}\right]. \tag{823}$$

If $\gamma_c > \beta_c/2$ for every cell and the finite sum of error envelopes is negligible, the saturated prospective source profile is negligible. Prospective selector accountability then gives negligible arrival whenever the declared neutral baseline is negligible.

For a proof-grade audit, the conclusion is a deterministic computer-assisted theorem. For a statistical audit, it holds with the simultaneous confidence carried by the audit record.

Proof. The capacity recurrence in Eq. (822) telescopes:

$$\prod_{j=n_0}^{n-1} r_{c,j} \le e^{\beta_c(n-n_0)} p_c(n), \qquad \prod_{j=n_0}^{n-1} s_{c,j} \le e^{-\gamma_c(n-n_0)}.$$

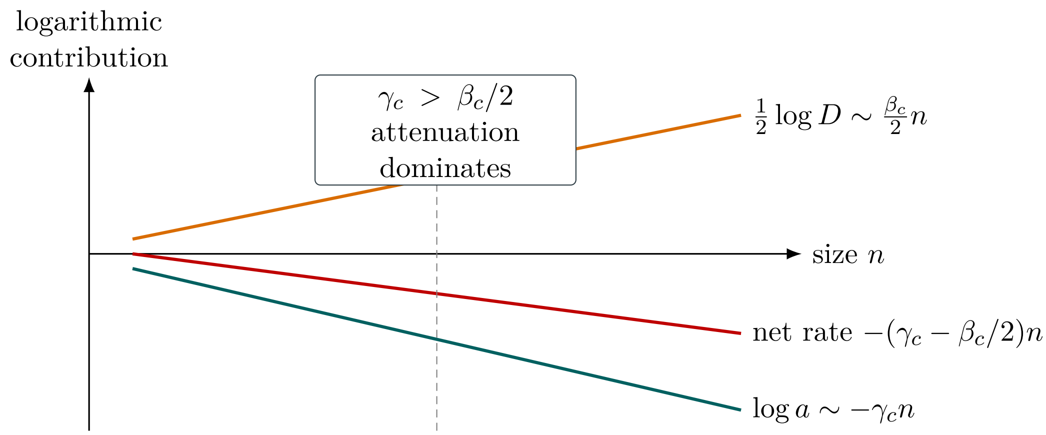

Figure 103: Verified scale competition. Capacity enters through $\sqrt{D_{c,n}}$, whereas target attenuation enters linearly through $a_{c,n}$. The strict exponent gap $\gamma_c - \beta_c/2 > 0$ makes their product exponentially small.

Insert these inequalities in Eq. (821), sum the source-cell bounds using Eq. (811), and obtain Eq. (823). Subexponential p_c and $\gamma_c - \beta_c/2 > 0$ make every leading term exponentially small. The negligibility and arrival conclusions follow from closure under a fixed finite sum and Theorems III.21.3 and III.21.4. □

Corollary V.15.13 (Polynomial capacity loses to exponential attenuation)**.** *In Theorem V.15.12, suppose B_n is polynomial and the proof-grade audit certifies, for fixed constants C_c and $\gamma_c > 0$,*

$$D_{c,n} \le (n + B_n)^{C_c}, \qquad a_{c,n} \le e^{-\gamma_c n}, \qquad \sum_c \overline{e}_{c,n} = \mathrm{negl}(n).$$

Then

$$I^{\mathrm{src,sat}}_{\mathfrak{M},n}(B_n) \le \sum_c (n + B_n)^{C_c/2} e^{-\gamma_c n} + \mathrm{negl}(n) = \mathrm{negl}(n). \tag{824}$$

The conclusion applies at a polynomial Levin ceiling because Lemma III.1.6 and Eq. (998) *charge the code, generated parameters, evaluation, storage, and deployment inside B_n. The capacity inequality remains an independently verified field of the audit; description length alone supplies no replacement for that inequality.*

Proof. Apply Eqs. (812) and (814) in every source cell and sum. A polynomial factor times $e^{-\gamma_c n}$ is exponentially small. The Levin statement follows from the cited complete-cost domination. □

Corollary V.15.14 (Finite empirical discovery with deterministic universal closure)**.** *Fix the public presentation, budget schedule, and existing saturated source cells. A finite computation discharges the saturated-source estimate in Theorem V.15.12 when its exported record passes the following checks.*

(i) *The framework source-exhaustion theorem verifies the cover Eq.* (811)*; sampled coverage is not used in place of this check.*

(ii) *Exact, rational-interval, or symbolic arithmetic verifies the factorization and norm inequalities Eqs.* (812) *and* (814) *for the complete qualified cell, including the displayed error envelope.*

(iii) *A simultaneous empirical audit supplies candidate upper bounds at the base size, and an independent deterministic verifier proves the final base inequalities in Eq.* (815)*. The candidate may use the effective-dimension audit of Definition I.5.6, the Bellman upper audit of Theorem V.15.8, the simultaneous finite-network radius of Corollary V.15.22, or the source-resolved learning radii of Theorem V.15.25.*

(iv) *The same verifier proves the capacity, attenuation, and error scale laws Eqs.* (816) *to* (818) *and the rate comparison Eq.* (822)*.*

(v) *The code, generated parameters, precision, audit, verifier, evaluation, storage, and deployment all fit the common charged ceiling.*

(vi) *The strict exponent inequalities* $\gamma_c > \beta_c/2$ *and the claimed negligible error envelopes are verified for every cell in the finite cover.*

(vii) *Any finite interval of sizes not covered by the propagated estimate is checked directly by the same outward-rounded upper audit.*

Then Eq. (823) *is a deterministic upper bound on the complete saturated prospective source profile. The training sample, optimizer, and random initialization may be used to find the record, but do not occur in the validity of the resulting bound. If item* (iii) *or item* (iv) *is retained only as a confidence statement, the conclusion has that confidence and is not an unconditional computer-assisted theorem.*

Proof. Items (i)–(iv) are exactly the hypotheses of Definition V.15.10; item (v) places their construction in the declared derivability closure. Item (vi) invokes Theorem V.15.12, and item (vii) joins the finite verified prefix to its propagated tail. Deterministic verification makes the exported inequalities independent of the random search that found them. □

V.15.6 Certificate engineering specification

A proof-producing computation has three logically distinct layers.

Discovery code. Neural training, Bayesian search, symbolic regression, combinatorial search, or numerical optimization may propose a factorization, recurrence, enclosure, or crossover point. This code is untrusted for proof validity. When it is part of the computation being analyzed, its construction and execution costs remain in the represented resource ledger.

Proof objects. The exported record contains canonical finite data, exact rational values, directed-rounding intervals, symbolic identities, coverage witnesses, and induction certificates. It contains no statistical assertion whose validity depends on the optimizer, random seed, or failure to discover a counterexample.

Trusted checker. A deterministic parser and verifier checks the canonical encodings, exact enumerations, interval inclusions, symbolic steps, and dependency hashes. The final proof term must be accepted by the Lean kernel. Candidate-generation code and the proof-object exporter lie outside this trusted base.

Definition V.15.15 (Proof-producing certificate package)**.** Fix a public presentation, a budget schedule B_n, a finite audited prefix, and the qualified saturated source cells to be certified. A *proof-producing certificate package* consists of the following twelve machine-checkable fields. Such a package exists only when Fields 1–3 certify that the directly audited presentation and budgeted derivation slices are finitely enumerable; otherwise exact finite saturation has not been certified.

1. *Presentation and budget encoding.* A canonical serialization of the task presentation, primitive observables, constructor grammar, terminal data, reference laws, admissible oracle contracts, resource monoid, scalarization if used, budget schedule, and all versioned semantics on which the certificate depends. The checker verifies the serialization and dependency hashes.
2. *Finite-state and full-support enumerator.* For every directly audited size, an exact enumerator for the complete finite state carrier and the full support of each declared finite law, with exact weights, a termination witness, duplicate elimination, and a completeness witness. The resulting population calculations use Definition I.6.1; a sampled support list is not an exact enumerator.
3. *Complete admissible derivation slice.* A canonical enumeration of every represented derivation admitted by the declared constructor grammar at cost at most B_n, including generated representations, memory states, oracle calls, precision, verification, and deployment. The field includes a proof that the enumeration is closed under every affordable constructor and that each omitted branch exceeds the ceiling. It is an exact finite realization of Definition III.1.3, not a list of records found by the discovery procedure.
4. *Exact source-cell assignment.* For every derivation in the preceding slice, an authenticated backward ancestry record, temporal source type, support mode, exact provenance signature, and assignment to one qualified source cell. The checker verifies disjointness, exhaustion, and the source-cover inequality Eq. (811).
5. *Capacity certificate.* Exact or outward-rounded witnesses for the represented carrier, coefficient norm, effective dimension or rank, and the uniform bound $D_{c,n}$ for every record in each covered cell.
6. *Attenuation and noise-transfer certificate.* A symbolic or interval-verified transfer factor $a_{c,n}$, including every provenance weight, channel parameter, floor, dependence correction, and noise conversion used to prove the attenuation law.
7. *Error enclosure.* Directed-rounding enclosures for factorization, truncation, tail, discretization, approximation, and arithmetic error, combined into the uniform cellwise quantity $e_{c,n}$. Fields 5–7 jointly verify Eqs. (812), (814) and (818).
8. *Recurrence-cover certificate.* A finite represented family of structural transition maps and a proof that every admissible transition of every record in the complete qualified cell follows one of them. New grammars, acquisition rules, reuse patterns, locators, and precision regimes appear as new represented maps with their complete incremental costs, as required by Definition V.15.17.
9. *Interval invariance or symbolic induction certificate.* Base enclosures and exact symbolic or outward-rounded inclusion proofs showing that every transition map preserves the asserted tail envelope, uniformly for all declared sizes beyond the prefix; see Eq. (828).
10. *Breakpoint and crossover certificate.* Exact scalar comparisons proving the exponent gap, every claimed crossover inequality, and each crossing-price lower bound used to exclude a different structural regime within B_n; see Theorem V.15.18 and Corollary V.15.19.
11. *Direct finite-prefix verification.* Exact symbolic or outward-rounded evaluation of the complete saturated slice at every size not covered by the tail induction. The checker joins this finite result to the first certified tail size without a gap.
12. *Final theorem object exported to Lean.* A versioned Lean theorem declaration and kernel-checkable proof term whose imported constants and hypotheses encode the accepted contents of Fields 1–11 and whose conclusion is the final instantiated saturated-profile or target-performance bound. A manifest binds the theorem to the content hashes of those fields. The export log records the Lean version, imported definitions, theorem name, and kernel acceptance result; a generated source file without kernel acceptance does not populate this field.

The package is *accepted* only when the trusted checker validates Fields 1–11 and the Lean kernel accepts Field 12. The present manuscript specifies this package and proves the implications used to assemble it. It does not contain an accepted instantiated LPN package.

Proposition V.15.16 (Soundness of an accepted certificate package)**.** *An accepted package in the sense of Definition V.15.15 supplies a deterministic proof-grade audit of its complete declared source cells. Its final theorem object depends only on the encoded presentation, declared resource semantics, deterministic proof objects, and trusted checker; the discovery procedure and its randomness do not enter the theorem's hypotheses.*

Proof. Fields 1–4 verify the presented finite slices and the source cover. Fields 5–7 verify the factorization, capacity–attenuation, and error premises of Definition V.15.10. Fields 8 and 9 verify the recurrence cover and its uniform induction. Field 10 supplies the rate and crossing exclusions, and Field 11 supplies the remaining finite sizes. Thus Corollary V.15.14 and Theorem V.15.18 give the asserted deterministic bound. Field 12 records the corresponding formal theorem and is accepted only after kernel checking. □

Uniformity beyond the audited prefix. For an accepted package, the directly audited sizes enumerate the complete budgeted structural object, while the asymptotic statement ranges over every record in the complete qualified source cell. The recurrence-cover field quantifies over every admissible represented transition in that cell, rather than continuing one low-dimensional optimizer. Any new high-n regime therefore requires a represented structural transition. The complete cost of that transition enters its recurrence path, and Theorem V.15.18 and Corollary V.15.19 exclude it whenever its verified crossing price exceeds B_n. Finally, every representation constructed by any later budget-admissible algorithm is an element of the saturated derivability closure and hence belongs to the same source cover. The accepted certificate is therefore uniform over admissible algorithms and sizes, not an extrapolation of the optimizer used during discovery.

V.15.6.1 Proof-producing verification of structural scaling laws

Finite-size experiments can discover a useful scaling law, but the asymptotic assertion is supplied by a verified structural recurrence. The next definition records the recurrence, the complete cost of traversing it, and the source-cell cover to which it applies.

Definition V.15.17 (Charged structural scaling-recurrence certificate)**.** Fix a qualified saturated source cell c, a size n, and its declared budget B_n. A *charged structural scaling-recurrence certificate* consists of the following represented data.

(i) A represented partially ordered state space $\mathcal{U}_{c,n}$, an initial enclosure $U_{c,n,0} \subseteq \mathcal{U}_{c,n}$, and a finite collection $\mathfrak{F}_{c,n}$ of monotone interval transition maps. The state stores only already-defined structural coordinates, including capacity, attenuation order, approximation defect, available rows or observations, provenance size, memory, and accumulated cost when those coordinates are present.

(ii) A represented recurrence cover: every record in $\mathcal{S}_{c,n}(B_n)$ determines a finite path

$$u_0 \in U_{c,n,0}, \qquad u_{j+1} \in F_{c,n,j}(u_j), \quad F_{c,n,j} \in \mathfrak{F}_{c,n}. \tag{825}$$

The cover is proved from the framework source-exhaustion and constructor-provenance theorems; empirical path sampling is not a substitute for this item.

(iii) A nonnegative progress functional $\pi_{c,n}$, a nonnegative target-transfer envelope $A_{c,n}$, and a nonnegative incremental complete-cost functional $p_{c,n}$ satisfying along every covered path

$$V_{c,n}(R) \leq A_{c,n}(u_J) + e_{c,n}, \tag{826}$$

$$\mathrm{Cost}(R) \geq C_{c,n,0} + \sum_{j=0}^{J-1} p_{c,n}(u_j, u_{j+1}). \tag{827}$$

The incremental cost includes the recurrence evaluator, locator or collision rule, acquired data, generated records, arithmetic, precision, storage, history, verification, and deployment.

(iv) A finite verified prefix $n_0 \leq n < N_0$, together with represented tail enclosures $\overline{U}_{c,n,j}$, such that a symbolic or outward-rounded interval verifier proves, for every $n \geq N_0$,

$$U_{c,n,0} \subseteq \overline{U}_{c,n,0}, \qquad F_{c,n,j}(\overline{U}_{c,n,j}) \subseteq \overline{U}_{c,n,j+1}. \tag{828}$$

The training runs, learned recurrence proposal, finite prefix, verifier, interval precision, and all proof objects are included in the common cost record. The certificate is *proof-grade* when the recurrence cover, all inclusions, and all numerical inequalities are checked deterministically.

For a target progress level q, define its charged crossing price by

$$\mathrm{Cross}_{c,n}(q) = \inf \left\{ C_{c,n,0} + \sum_{j=0}^{J-1} p_{c,n}(u_j, u_{j+1}) : \begin{array}{l} (u_0, \ldots, u_J) \text{ satisfies } Eq.\ (825), \\ \pi_{c,n}(u_J) \geq q \end{array} \right\}, \tag{829}$$

with infimum $+\infty$ when no covered path reaches q.

Theorem V.15.18 (Verified recurrence propagation and crossing exclusion)**.** *Let Definition V.15.17 be proof-grade. Then every qualified record in its covered source cell follows one of the verified enclosures. Consequently,*

$$\sup_{R \in \mathcal{S}_{c,n}(B_n)} V_{c,n}(R) \leq \sup_{j, u \in \overline{U}_{c,n,j}} A_{c,n}(u) + e_{c,n}. \tag{830}$$

Moreover, if $B_n < \mathrm{Cross}_{c,n}(q)$, no record in that budgeted cell reaches progress q. Hence

$$\sup_{R \in \mathcal{S}_{c,n}(B_n)} V_{c,n}(R) \leq \sup_{\substack{j, u \in \overline{U}_{c,n,j} \\ \pi_{c,n}(u) < q}} A_{c,n}(u) + e_{c,n}. \tag{831}$$

If a learned formula $\widehat{u}_{c,n,j}(\theta)$ supplies the interval enclosures and the verifier establishes Eq. (828) *symbolically for all $n \geq N_0$, then the conclusion holds for all such sizes. The empirical prefix selects and calibrates the candidate formula; the verified recurrence proves its asymptotic continuation.*

Proof. The recurrence cover assigns a path to each qualified record. Induction on the path index in Eq. (828) places every state of that path in its displayed enclosure. Applying Eq. (826) and taking the supremum gives Eq. (830).

If a budget-B_n path reached q, Eq. (827) would make its complete cost at least the infimum in Eq. (829), contrary to $B_n < \mathrm{Cross}_{c,n}(q)$. Restricting the first supremum to states below q proves Eq. (831). The last assertion is the same induction with the learned formula used only as the verifier's proposed enclosure. $\square$

Corollary V.15.19 (A scaling breakpoint has a represented structural price)**.** *Under Theorem V.15.18, a change from a verified subcritical envelope $\pi_{c,n} < q_n$ to a regime with $\pi_{c,n} \geq q_n$ occurs only through a represented recurrence path whose complete cost is at least* $\mathrm{Cross}_{c,n}(q_n)$*. A change of recurrence grammar, acquisition interface, reuse pattern, precision, or locator is a new represented transition map and carries its own code, construction, evaluation, memory, and deployment charge. Thus a proof-grade lower bound*

$$\mathrm{Cross}_{c,n}(q_n) > B_n \tag{832}$$

excludes the breakpoint inside the declared saturated cell.

Proof. This is the crossing exclusion in Theorem V.15.18. Constructor provenance and complete-cost accounting place every modified recurrence component in the path record before the modified transition is used. □

Definition V.15.20 (Disjoint collision–cancellation recurrence)**.** Consider a represented combining cell whose stage-j locator assigns the current N_{j-1} records to a represented bucket set $\mathcal{Q}_j$ of cardinality Q_j and combines disjoint pairs inside each bucket. Write $n_{j-1,a}$ for the occupancy of bucket $a \in \mathcal{Q}_j$. Let d_j be the certified number of semantic coordinates cancelled by equality of one bucket label, let w_j be the maximum number of primitive noisy leaves in a resulting record, and let D_j be the declared remaining semantic-carrier dimension. The canonical complete-pairing rule has the exact and guaranteed row counts

$$N_j = \sum_{a \in \mathcal{Q}_j} \left\lfloor \frac{n_{j-1,a}}{2} \right\rfloor, \tag{833}$$

$$N_j \geq \left\lfloor \frac{(N_{j-1} - Q_j)_+}{2} \right\rfloor, \tag{834}$$

$$D_j = D_{j-1} - d_j, \qquad\qquad w_j = 2w_{j-1}, \qquad w_0 = 1. \tag{835}$$

For independent primitive binary noise with signed mean ρ, a disjoint depth-j record has exact signed mean $\rho^{w_j} = \rho^{2^j}$. One conservative reverse sufficient-row recurrence for K final records is

$$N_{j-1}^{\mathrm{suff}} = 2N_j^{\mathrm{suff}} + Q_j + 1, \qquad N_r^{\mathrm{suff}} = K. \tag{836}$$

The complete recurrence state also stores the locator-evaluation, bucket-table, pairing, combination, provenance-DAG, acquisition, terminal readout, memory, and description costs. Alternative reuse or approximate collision rules define different represented recurrence maps and retain their dependence and compensation coordinates.

Theorem V.15.21 (Exact transport and charged scaling certificate for disjoint collision combining)**.** *For the cell in Definition V.15.20, the occupancy recurrence and noise transport are exact deterministic certificate fields. The guaranteed forward recurrence and its reverse are constructive sufficient-row fields. After r disjoint rounds,*

$$\sum_{j=1}^{r} d_j = D_0 - D_r, \qquad w_r = 2^r, \qquad a_r = |\rho|^{2^r}, \qquad N_0^{\mathrm{suff}} = 2^r K + \sum_{j=1}^{r} 2^{j-1}(Q_j + 1). \tag{837}$$

The minimum complete cost of these sufficient schedules,

$$\mathrm{Build}_n^{\mathrm{coll}}(q, K) = \min_{\substack{r \geq 0,\ Q_j, d_j \ \textit{represented} \\ \sum_{j \leq r} d_j \geq q}} \mathrm{Cost}_{\mathrm{complete}}\left(r, (Q_j, d_j)_{j \leq r}, N_0^{\mathrm{suff}}, K\right), \tag{838}$$

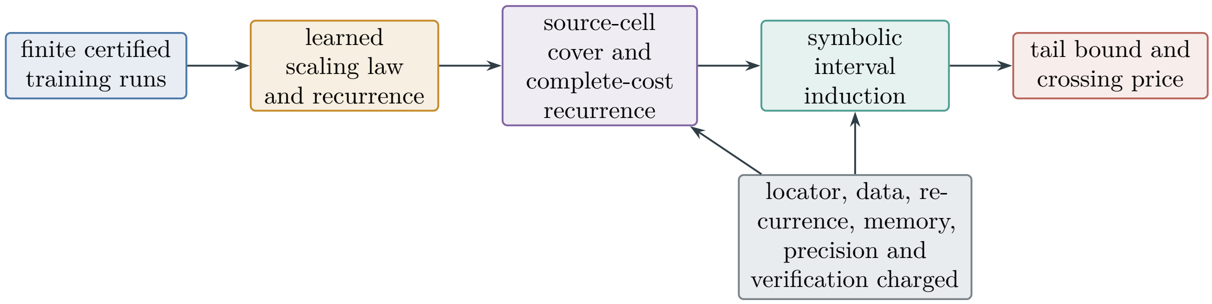

Figure 104: Proof-producing scaling-law verification. Empirical learning proposes the recurrence and its envelope. Structural source coverage, complete-cost accounting, and symbolic induction establish the tail and the price of entering a different scaling regime.

Figure 105: Empirical discovery and proof-grade closure. Sampling and training propose a certificate; deterministic source coverage, factorization, norm, and scale checks establish the universal saturated-profile bound.

is a constructive upper bound on the cost of reaching that cancellation level inside the declared collision grammar. A lower bound on the charged crossing price is instead obtained by minimizing the incremental-cost functional over the exact occupancy paths in Eq. (833)*. In particular, any deterministic certificate*

$$\underline{\mathrm{Cross}}_n^{\mathrm{coll}}(q,K) \le \inf_{\substack{\text{covered exact occupancy paths}\\ \sum_j d_j \ge q,\ N_r \ge K}} \left(C_{c,n,0} + \sum_j p_{c,n}(u_j, u_{j+1}) \right) \tag{839}$$

is a valid lower crossing certificate in Eq. (829)*. If a source-exhaustion theorem routes a complete source cell to this recurrence grammar, the lower certificate bounds that complete cell; otherwise it bounds precisely the collision cell named in the recurrence cover.*

Proof. Complete pairing gives Eq. (833). At most one row per bucket remains unpaired, which gives Eq. (834). Disjoint pairing doubles the primitive leaf count and independence multiplies signed means, proving Eq. (835) and the first three identities in Eq. (837). Repeated substitution in Eq. (836) gives the last identity. It constructs a feasible schedule and therefore an upper cost. The exact occupancy recurrence supplies the path set used by Eq. (839); applying Theorem V.15.18 turns any verified lower bound on that path infimum into an exclusion certificate. □

V.15.7 Finite-network audit and selection

Corollary V.15.22 (Finite-network simultaneous audit radius)**.** *Let a represented network family have p real parameters in a Euclidean ball of radius R_Θ, stored to the declared precision. Suppose its audited scalar $z \mapsto \ell_\theta(z) \in [0, M]$ is L_Θ-Lipschitz in θ, uniformly in z. For an audit sample* *of size N, a parameter-net resolution $\rho > 0$, and confidence δ, with probability at least $1-\delta$, simultaneously for all represented terminal weights in the family,*

$$\mathbb{E}\ell_\theta \le \widehat{\mathbb{E}}\ell_\theta + M\sqrt{\frac{p\log(1+2R_\Theta L_\Theta/\rho)+\log(1/\delta)}{2N}} + 2\rho. \tag{840}$$

The network architecture, parameter norm, Lipschitz certificate, net resolution, audit sample, and terminal-weight selection rule are part of the complete implementation record. The displayed radius may be replaced by the smaller same-law candidate supplied by Lemmas IV.5.4 and IV.5.8 and Theorems IV.15.5 and IV.15.6.

Proof. By Corollary IV.5.6, the parameter family has a represented scalar-function ρ-net of cardinality at most $(1+2R_\Theta L_\Theta/\rho)^p$. Hoeffding's inequality and a union bound give the square-root term simultaneously on the net. Moving from a selected parameter to its net point changes both the empirical and population means by at most ρ, producing 2ρ.

□

Theorem V.15.23 (Certified deep-network implementation bridge)**.** *Let a complete neural implementation record use finite represented networks for its action values, belief update, selector, and Bellman barrier. Suppose:*

(i) *the deployed record and barrier lie within the common charged ceiling;*
(ii) *the evaluation lower bound in Proposition V.15.6 is available;*
(iii) *each Bellman residual has either the exact verification certificate in Eq.* (804) *or the coverage certificate in Eq.* (801)*; and*
(iv) *data-dependent network selection is covered by Corollary V.15.22 or by a smaller simultaneous same-law structural learning radius.*

Then the trained deep-network record satisfies the implementation bound Eq. (809)*, with every empirical radius in Eq.* (805) *replaced by the selected simultaneous network radius. If its learned action values also satisfy Eqs.* (600) *and* (601)*, the sharper minimum in Eq.* (810) *holds. All terms are explicit functions of the horizon, evaluation and audit sample sizes, confidence allocation, network parameter count and norm, precision, coverage constants, structural learning radii, filtering error, and represented optimization defects.*

Proof. Apply Corollary V.15.22 to every selected network scalar used in the Bellman audit, using a confidence allocation whose sum is at most δ_{up}. This verifies the premise of Theorem V.15.8. Combine its upper bound with Proposition V.15.6 and invoke Corollary V.15.9. The learned action-value candidate follows from Theorem V.9.12 and Corollary V.11.16.

□

V.15.8 Empirical structural-channel attribution

Definition V.15.24 (Audited parameter–source ledger)**.** Let $\mathfrak{J}^5_{\mathsf{Abs}} = \{\mathsf{Add}, \mathsf{Mult}, \mathsf{Ord}, \mathsf{Sym}, \mathsf{Filt}\}$. For a complete p-parameter learner–deployment record R_p, retain the exact source index and contextual and information ledgers of Definition IV.13.2. Write

$$\mathcal{L}_p^{(a)} = \mathcal{L}_{I,R_p}^{(a)}, \tag{841}$$

$$J_p^{(a)} = J_{I,R_p}^{(a)}, \tag{842}$$

$$\Gamma_{t,p}^{(a)} = \Gamma^{\mathrm{learn}}_{R_p,\alpha(a)}(B, m, \boldsymbol{\eta}, \delta) \tag{843}$$

for each source a, with the exact mixed-source cell $\alpha(a)$ retained whenever more than one source occurs in the backward slice.

An empirical source audit supplies estimates $\widehat{\mathcal{L}}_p^{(a)}$ and $\widehat{J}_p^{(a)}$, together with simultaneous represented radii $\xi_{\mathcal{L},p}^{(a)}$ and $\xi_{J,p}^{(a)}$ satisfying

$$\mathcal{L}_p^{(a)} \le \widehat{\mathcal{L}}_p^{(a)} + \xi_{\mathcal{L},p}^{(a)}, \tag{844}$$

$$J_p^{(a)} \le \widehat{J}_p^{(a)} + \xi_{J,p}^{(a)}. \tag{845}$$

The audit sample, exact-cell selector, backward graph, reveal assignment, noise conversion, and confidence allocation are fields of the complete record.

Theorem V.15.25 (Parameter- and source-resolved empirical target-gain bound)**.** *On the simultaneous source-audit event of Definition V.15.24, the deployed target gain of R_p satisfies*

$$g_{R_p} \le \min \left\{ \begin{aligned} &1, \quad U_{R_p}^{\mathrm{stat}\to\mathrm{tgt}}, \quad U_{R_p}^{\mathrm{dyn}\to\mathrm{tgt}}, \\ &\beta_{R_p}^{+} + \sqrt{\frac{\log 2}{2} \sum_{a \in \mathfrak{J}_{\mathsf{Abs}}^5} \left(\widehat{J}_p^{(a)} + \xi_{J,p}^{(a)} \right)}, \\ &eH_B(n) \sum_{a \in \mathfrak{J}_{\mathsf{Abs}}^5} \left(\widehat{\mathcal{L}}_p^{(a)} + \xi_{\mathcal{L},p}^{(a)} \right) \end{aligned} \right\}. \tag{846}$$

For an exact source cell α, restriction of every term to records with that cell gives a cellwise bound $U_{p,\alpha}^{\mathrm{tgt}}$. The Rademacher union and PAC–Bayes mixture penalties for selecting the observed cell are exactly Eqs. (279) *and* (280)*. Thus the audit reports both the source support learned by the network and a simultaneous upper bound on the target gain carried by that support.*

Proof. Insert Eqs. (844) and (845) into the recordwise minimum Eq. (281). Monotonicity of the sum and square root proves Eq. (846). Restricting the same recordwise inequality to one exact cell preserves every comparison. The selection penalties follow from the cited union and mixture identities.

□

Definition V.15.26 (Charged source intervention family)**.** Let $\mathcal{A} = \mathfrak{J}_{\mathsf{Abs}}^5$. A complete record carries a *charged source intervention family* when, for every $S \subseteq \mathcal{A}$, it represents a valid deployed record R_p^S obtained by retaining precisely the source-gated vertices in S and replacing every removed source input by its declared target-neutral baseline. Each intervention retains the common public presentation, verifier, horizon, evaluation law, and nonremoved computation. Its recompilation, normalization, and deployment costs are charged.

Define the structural intervention value and Shapley allocation by

$$v_p(S) = A_{R_p^S}^H, \tag{847}$$

$$\phi_{p,a} = \sum_{S \subseteq \mathcal{A} \setminus \{a\}} \frac{|S|!\,(4-|S|)!}{5!} \big(v_p(S \cup \{a\}) - v_p(S)\big). \tag{848}$$

These coordinates are used only when all 32 interventions are valid complete records. The contextual and exact-cell ledger remains the source-attribution certificate when that intervention gate is absent.

Theorem V.15.27 (Simultaneous empirical source-intervention certificate)**.** *For every $S \subseteq \mathcal{A}$, evaluate R_p^S on N_{int} independent episodes and let $\widehat{v}_p(S)$ be its empirical arrival frequency. With*

$$r_{\text{int}} = \sqrt{\frac{\log(64/\delta_{\text{int}})}{2N_{\text{int}}}}, \tag{849}$$

all 32 *values satisfy*

$$|v_p(S) - \widehat{v}_p(S)| \leq r_{\text{int}} \tag{850}$$

simultaneously with probability at least $1 - \delta_{\text{int}}$*. If* $\widehat{\phi}_{p,a}$ *is obtained from Eq.* (848) *by replacing* v_p *with* $\widehat{v}_p$*, then*

$$|\phi_{p,a} - \widehat{\phi}_{p,a}| \leq 2r_{\text{int}}, \tag{851}$$

$$\sum_{a\in\mathcal{A}} \phi_{p,a} = v_p(\mathcal{A}) - v_p(\varnothing). \tag{852}$$

Hence the empirical run has two complementary rigorous descriptions: the source ledger gives mechanism-wise upper certificates, while the intervention family gives an interaction-aware allocation of the observed arrival improvement.

Proof. Hoeffding's inequality gives failure probability at most $2e^{-2N_{\text{int}}r_{\text{int}}^2} = \delta_{\text{int}}/32$ for one subset. A union bound proves Eq. (850). The Shapley weights for one source are nonnegative and sum to one. Each marginal difference uses two intervention values, proving Eq. (851). The efficiency identity follows by the standard permutation form of the same weights: along every permutation, the five marginal increments telescope from $v_p(\varnothing)$ to $v_p(\mathcal{A})$, and averaging preserves the identity. □

V.15.9 Canonical primal–dual learning objective

Definition V.15.28 (Canonical attention–belief training objective)**.** Let R_θ be the deployed record of Definition V.15.4. Its population primal loss is

$$\mathcal{L}_{\text{arr}}(\theta) = -A_{R_\theta}^H = -\Pr_{R_\theta}\{\tau_T \leq H\}. \tag{853}$$

For a represented critic $v = (v_t)_{t\leq H}$, $v_H = 0$, write

$$Q_t^v(h,a) = p_T(h,a) + P_N v_{t+1}(h,a). \tag{854}$$

Let

$$(\mathcal{T}_t v)(h) = \max_{a\in\mathcal{A}_t(h)} Q_t^v(h,a)$$

be the represented finite-action Bellman operator. Given a represented sampling law μ_t, define

$$\mathcal{L}_{\text{Bell}}(v) = \sum_{t<H} \mathbb{E}_{\mu_t}\left[(\mathcal{T}_t v)(h) - v_t(h)\right]^2, \tag{855}$$

$$\mathcal{L}_{\text{bar}}(v) = v_0(h_0) + \sum_{t<H} C_t\, \mathbb{E}_{\mu_t}\left[Q_t^v(h,a) - v_t(h)\right]_+. \tag{856}$$

The first loss trains a critic on its declared sampling law. The second is the population version of the saturated Bellman upper certificate and retains the coverage constants C_t.

Let $\nu_t(\cdot \mid h)$ be the represented neutral reference row with positive mass on every legal action, and put

$$\pi^{\tau}_{v,t}(a \mid h) = \frac{\nu_t(a \mid h)\exp(Q^v_t(h,a)/\tau_t)}{\sum_{a'} \nu_t(a' \mid h)\exp(Q^v_t(h,a')/\tau_t)}. \tag{857}$$

The policy and belief losses are

$$\mathcal{L}_{\mathrm{pol}}(\theta; v) = \sum_{t<H} \mathbb{E}_{f_{R_\theta},t} D_{\mathrm{KL}}\Big(\pi_{\theta,t}(\cdot \mid h)\Big\|\pi^{\tau}_{v,t}(\cdot \mid h)\Big), \tag{858}$$

$$\mathcal{L}_{\mathrm{Bayes}}(\theta) = -\sum_{t<H} \mathbb{E}\log\left[\sum_z q_{\theta,t}(z) L_{\boldsymbol{\eta},t}(Y_t \mid z, A_t)\right]. \tag{859}$$

Here $L_{\boldsymbol{\eta},t}$ is the likelihood of the declared observation channel, including its noise parameters, and $q_{\theta,t}$ is the represented compact belief before the reveal. Let $\mathcal{L}_{\mathrm{num}}$ be a represented upper loss for finite-precision score, normalization, likelihood, and state-update discrepancies. The canonical trainable objective is

$$\boxed{\mathcal{L}_{\mathrm{train}}(\theta, v) = \mathcal{L}_{\mathrm{pol}}(\theta; v) + \lambda_V \mathcal{L}_{\mathrm{Bell}}(v) + \lambda_B \mathcal{L}_{\mathrm{Bayes}}(\theta) + \lambda_{\mathrm{num}} \mathcal{L}_{\mathrm{num}}(\theta, v),} \tag{860}$$

with represented nonnegative weights. The geometry and quotient heads are trained on their declared empirical forms and are certified by the separate same-law audits in Sections V.15.12 and V.15.13.

The empirical optimality objective is the primal–dual gap

$$\boxed{\mathcal{L}_{\mathrm{cert}}(\widehat{\theta}, v) = U_{\mathrm{Bell}}(v) - L_{\mathrm{ev}}(\widehat{\theta}).} \tag{861}$$

Its construction, data split, confidence allocation, and optimization are fields of the complete implementation record.

Theorem V.15.29 (Canonical loss calibration to saturated arrival)**.** *Assume* $0 < \nu_{\min,t} \le \nu_t(a \mid h)$ *on every represented legal action and surviving history. Suppose*

$$\|Q^v_t - Q^{\mathrm{act},*}_t\|_\infty \le \Gamma_{Q,t}, \tag{862}$$

$$D_{\mathrm{KL}}\Big(\pi_{\theta,t}(\cdot \mid h)\Big\|\pi^{\tau}_{v,t}(\cdot \mid h)\Big) \le \kappa_t \tag{863}$$

uniformly on the surviving carrier. Then the stepwise action defect obeys

$$\max_a Q^{\mathrm{act},*}_t(h,a) - \mathbb{E}_{a\sim\pi_{\theta,t}(\cdot|h)} Q^{\mathrm{act},*}_t(h,a) \le 2\Gamma_{Q,t} + \tau_t \log\frac{1}{\nu_{\min,t}} + \mathrm{Osc}(Q^v_t)\sqrt{\frac{\kappa_t}{2}}. \tag{864}$$

Consequently,

$$J^*_{\mathrm{sat}}(B) - A^H_{R_\theta} \le \varepsilon_{\mathrm{rep}}(p; B) + \mathbb{E}_{R_\theta} \sum_{t<H} \mathbf{1}_{\{\tau>t\}} \left[2\Gamma_{Q,t} + \tau_t \log\frac{1}{\nu_{\min,t}} + \mathrm{Osc}(Q^v_t)\sqrt{\frac{\kappa_t}{2}}\right] + \Delta^H_{\mathrm{fil}} + \varepsilon_{\mathrm{num}}. \tag{865}$$

On the simultaneous held-out and Bellman-audit event,

$$0 \le J^*_{\mathrm{sat}}(B) - A^H_{R_{\widehat{\theta}}} \le \mathcal{L}_{\mathrm{cert}}(\widehat{\theta}, v). \tag{866}$$

Thus minimization of Eq. (860) *is connected to saturated arrival through the displayed calibration coordinates, while Eq.* (861) *directly certifies the remaining optimality gap.*

Proof. The Gibbs variational identity Eq. (1056) gives

$$\max_a Q_t^v(h,a) - \mathbb{E}_{\pi^\tau_{v,t}} Q_t^v(h,a) \le \tau_t \log \frac{1}{\nu_{\min,t}}.$$

Pinsker's inequality and Eq. (863) give

$$\left|\mathbb{E}_{\pi_{\theta,t}} Q_t^v - \mathbb{E}_{\pi^\tau_{v,t}} Q_t^v\right| \le \operatorname{Osc}(Q_t^v)\sqrt{\kappa_t/2}.$$

Replacing Q_t^v by $Q_t^{\mathrm{act},*}$ in the maximum and expectation contributes at most $2\Gamma_{Q,t}$, proving Eq. (864). Apply the finite-horizon performance comparison in Theorem V.9.12, add the represented filter and numerical comparisons, and then add the parameter-class representation gap. This proves Eq. (865).

On the audit event,

$$L_{\mathrm{ev}}(\widehat{\theta}) \le A^H_{R_{\widehat{\theta}}} \le J^*_{\mathrm{sat}}(B) \le U_{\mathrm{Bell}}(v).$$

Subtraction proves Eq. (866).

□

V.15.10 Certified WAdam optimization

Definition V.15.30 (Certified WAdam training record)**.** Let $F : \mathbb{R}^p \to \mathbb{R}$ be the represented training objective. A *certified WAdam training record* stores the actual parameter iterates θ_k, stochastic gradients, first and second moment states, bias corrections, clipping operations, and positive preconditioners used by the implementation. Its terminal update is written in the common form

$$\theta_{k+1} = \theta_k - \alpha_k d_k, \qquad d_k = W_k^{-1}\widehat{m}_k, \tag{867}$$

where W_k and $\widehat{m}_k$ are the represented preconditioner and bias-corrected first moment produced by the implemented Adam-type recursions. Diagonal Adam, block-weighted Adam, and geometry-weighted variants are instances of Eq. (867). The optimizer is certified through represented constants $a_k, c_k > 0$, $b_k, s_k \ge 0$ satisfying, for the training filtration $\mathcal{G}_k$,

$$\mathbb{E}[\langle \nabla F(\theta_k), d_k\rangle \mid \mathcal{G}_k] \ge a_k \|\nabla F(\theta_k)\|_2^2 - b_k, \tag{868}$$

$$\mathbb{E}[\|d_k\|_2^2 \mid \mathcal{G}_k] \le c_k \|\nabla F(\theta_k)\|_2^2 + s_k. \tag{869}$$

The complete record charges the computation and verification of these constants. Spectral bounds on W_k, stochastic-gradient bias and variance bounds, or direct interval verification of the two displayed inequalities are admissible certificates when represented at their complete cost.

Theorem V.15.31 (WAdam descent and global optimization certificate)**.** *Let Θ_{cert} be a represented region containing every iterate, put $F_* = \inf_{\theta\in\Theta_{\mathrm{cert}}} F(\theta)$, and assume F is L_F-smooth on every segment $[\theta_k, \theta_{k+1}]$. Suppose the certified WAdam conditions Eqs.* (868) *and* (869) *hold. Put*

$$q_k = \alpha_k a_k - \frac{L_F \alpha_k^2 c_k}{2}, \tag{870}$$

$$r_k = \alpha_k b_k + \frac{L_F \alpha_k^2 s_k}{2}. \tag{871}$$

If $q_k > 0$, then

$$\sum_{k=0}^{K-1} q_k \mathbb{E}\|\nabla F(\theta_k)\|_2^2 \le F(\theta_0) - F_* + \sum_{k=0}^{K-1} r_k. \tag{872}$$

Suppose additionally that the represented objective satisfies the Polyak–Łojasiewicz comparison

$$\|\nabla F(\theta)\|_2^2 \ge 2\mu(F(\theta) - F_*) \tag{873}$$

on the visited certified region, and $0 \le 2\mu q_k \le 1$. Then

$$\mathbb{E}[F(\theta_K) - F_*] \le \left[\prod_{k=0}^{K-1}(1 - 2\mu q_k)\right][F(\theta_0) - F_*] + \sum_{j=0}^{K-1} r_j \prod_{k=j+1}^{K-1}(1 - 2\mu q_k). \tag{874}$$

If the represented acquisition optimizer has a calibration

$$\varepsilon_{\mathrm{opt},t} \le \lambda_t(F(\theta_K) - F_*) + \zeta_t, \tag{875}$$

then substitution of Eq. (874) *into Eq.* (810) *gives a completely explicit WAdam contribution to the saturated implementation error. Moreover, if $\mathcal{E}_{\mathrm{WAdam}}(K)$ denotes the right side of Eq.* (874)*, then for every $0 < \delta_{\mathrm{opt}} < 1$,*

$$\Pr\left\{F(\theta_K) - F_* \le \frac{\mathcal{E}_{\mathrm{WAdam}}(K)}{\delta_{\mathrm{opt}}}\right\} \ge 1 - \delta_{\mathrm{opt}}. \tag{876}$$

For a p-parameter architecture, write $\mathcal{E}_{\mathrm{WAdam}}(p, K)$; this notation retains the dependence of $F, L_F, \mu, (a_k, b_k, c_k, s_k)_{k<K}$, the initialization, precision, and all optimizer-state costs on p.

Proof. Smoothness and Eq. (867) give

$$F(\theta_{k+1}) \le F(\theta_k) - \alpha_k\langle\nabla F(\theta_k), d_k\rangle + \frac{L_F\alpha_k^2}{2}\|d_k\|_2^2.$$

Take conditional expectations and use Eqs. (868) and (869) to obtain

$$\mathbb{E}[F(\theta_{k+1}) \mid \mathcal{G}_k] \le F(\theta_k) - q_k\|\nabla F(\theta_k)\|_2^2 + r_k. \tag{877}$$

Summing, taking expectations, and using $\mathbb{E}F(\theta_K) \ge F_*$ proves Eq. (872).

Under Eq. (873), subtract F_* from Eq. (877) to obtain

$$\mathbb{E}[F(\theta_{k+1}) - F_* \mid \mathcal{G}_k] \le (1 - 2\mu q_k)[F(\theta_k) - F_*] + r_k.$$

Iteration proves Eq. (874). The calibrated optimization error is then bounded by Eq. (875), and Corollary V.15.9 completes the stated substitution. Since $F(\theta_K) - F_* \ge 0$, Markov's inequality applied to Eq. (874) proves Eq. (876).

□

V.15.11 Certified structural bias–variance ledger

Definition V.15.32 (Deterministic bias and stochastic-radius coordinates)**.** For one p-parameter canonical record, split every valid clean action-value comparison as

$$\Gamma_{Q,t} = \beta_{Q,t} + \sigma_{Q,t}, \qquad \beta_{Q,t}, \sigma_{Q,t} \geq 0, \tag{878}$$

where $\beta_{Q,t}$ collects certified population approximation, regularization, truncation, and deterministic model discrepancy, while $\sigma_{Q,t}$ is a simultaneous statistical radius obtained from a represented iid, mixing, sequential, or martingale certificate. Define

$$\varepsilon_{\mathrm{WAdam}}(p, K) = \mathbb{E}_{R_{\widehat{\theta}_p}} \sum_{t<H} \mathbf{1}_{\{\tau>t\}} \left[\lambda_{t,p} \frac{\mathcal{E}_{\mathrm{WAdam}}(p, K)}{\delta_{\mathrm{opt}}} + \zeta_{t,p} \right] \tag{879}$$

on the high-probability optimizer event, and define

$$\begin{aligned} B_{\mathrm{det}}(p, K, \boldsymbol{\tau}, b, H) &= \varepsilon_{\mathrm{rep}}(p; B) + \varepsilon_{\mathrm{WAdam}}(p, K) + \Delta^H_{\mathrm{fil},p} + \varepsilon_{\mathrm{num},p} + \varepsilon_{\mathrm{trunc},H} + \varepsilon_{\mathrm{suff},b} \\ &\quad + \mathbb{E}_{R_{\widehat{\theta}_p}} \sum_{t<H} \mathbf{1}_{\{\tau>t\}} \left[2\beta_{Q,t} + \tau_t \log \frac{1}{\nu_{\min,t}} + \mathrm{Osc}(Q^v_t) \sqrt{\frac{\kappa_t}{2}} \right], \qquad (880) \\ V_{\mathrm{stat}}(p, N, \boldsymbol{\eta}, \delta) &= 2\mathbb{E}_{R_{\widehat{\theta}_p}} \sum_{t<H} \mathbf{1}_{\{\tau>t\}} \sigma_{Q,t}. \qquad (881) \end{aligned}$$

Here $\varepsilon_{\mathrm{WAdam}}(p, K)$ is the calibrated action-value contribution obtained from Eqs. (874) and (875); $\varepsilon_{\mathrm{suff},b}$ is the compact-belief sufficiency defect at code capacity b; and $\varepsilon_{\mathrm{trunc},H}$ is the certified tail beyond the deployed horizon. An absent coordinate receives its neutral valid bound, so the ledger remains total.

For a predictable loss sequence, retain additionally the conditional variance

$$\mathcal{V}_T(Q) = \mathbb{E}_{h\sim Q} \sum_{t=1}^{T} \mathbb{E}\Big[\big(r_t(h) - \ell_t(h)\big)^2 \Big| \mathcal{F}_{t-1} \Big]. \tag{882}$$

This is the variance coordinate used by the Bernstein PAC–Bayes radius in Eq. (317).

Theorem V.15.33 (Complete certified bias–variance decomposition)**.** *On the simultaneous calibration, learning, filtering, numerical, and optimizer event of a complete canonical record,*

$$\boxed{0 \leq J^*_{\mathrm{sat}}(B) - A^H_{R_{\widehat{\theta}_p}} \leq B_{\mathrm{det}}(p, K, \boldsymbol{\tau}, b, H) + V_{\mathrm{stat}}(p, N, \boldsymbol{\eta}, \delta).} \tag{883}$$

The stochastic coordinate may use the smallest same-law candidate among the structural Rademacher, PAC–Bayes, mixing-block, sequential, martingale, and predictable-variance bounds already carried by the record. In particular, the martingale PAC–Bayes contribution at $0 < \lambda < 3/c$ is

$$\frac{\mathrm{KL}^{\mathsf{str}}_{E,H}(Q) + \log(1/\delta)}{\lambda T} + \frac{\lambda \mathcal{V}_T(Q)}{2T(1 - \lambda c/3)}. \tag{884}$$

If the entire right side of Eq. (883) *is unavailable, the primal–dual candidate $\mathcal{L}_{\mathrm{cert}} \wedge 1$ remains a total certified gap.*

Proof. Insert the split Eq. (878) into Eq. (865). Add the calibrated WAdam, truncation, and compact-belief sufficiency defects, which compare the same deployed record with the same saturated optimum. Deterministic terms give Eq. (880); simultaneous statistical terms give Eq. (881). This proves Eq. (883). The displayed predictable-variance candidate is Eq. (317). The final assertion is Eq. (866) clipped by the trivial arrival range. □

Proposition V.15.34 (Exact clipping bias–variance identity)**.** *Let μ be a represented audit law, let $P \ll \mu$ be the deployment law with represented density $W = dP/d\mu$, and let $0 \le Y \le 1$. For independent $(W_i, Y_i)_{i \le N}$ from μ, define*

$$J = \mathbb{E}_\mu[WY], \qquad \widehat{J}_C = \frac{1}{N}\sum_{i=1}^{N} \min\{W_i, C\} Y_i.$$

Then

$$b_C := J - \mathbb{E}\widehat{J}_C = \mathbb{E}_\mu[(W - C)_+ Y], \tag{885}$$

$$\operatorname{Var}(\widehat{J}_C) = \frac{1}{N}\operatorname{Var}_\mu(\min\{W, C\} Y), \tag{886}$$

$$\mathbb{E}(\widehat{J}_C - J)^2 = b_C^2 + \frac{1}{N}\operatorname{Var}_\mu(\min\{W, C\} Y), \tag{887}$$

$$\operatorname{Var}(\widehat{J}_C) \le \frac{CJ}{N}. \tag{888}$$

Thus the clipping level has a represented exact bias term and a represented second-moment variance term; its selection cost remains part of the audit record.

Proof. Subtract $\min\{W, C\}Y$ from WY to obtain Eq. (885). Independence gives Eq. (886); the scalar bias–variance identity gives Eq. (887). Finally, $\min\{W, C\}^2 Y^2 \le CWY$, proving Eq. (888). □

Corollary V.15.35 (Quantitative architecture and audit tradeoffs)**.** *The complete canonical record has the following certified tuning curves.*

(i) Parameter count. *The upper curve*

$$\varepsilon^{\mathrm{up}}_{\mathrm{rep}}(p) + \varepsilon^{\mathrm{up}}_{\mathrm{WAdam}}(p, K) + 2\Gamma_p(N, \delta)$$

combines the nonincreasing representation error of Definition V.15.2 with the parameter-cover radius in Eq. (840)*.*

(ii) Attention temperature. *If a represented score estimate satisfies $\|\widehat{s}_t - s_t\|_\infty \le \epsilon_{s,t}$, then its Gibbs rows satisfy*

$$D_{\mathrm{KL}}(P_{s_t} \| P_{\widehat{s}_t}) \le \frac{2\epsilon_{s,t}}{\tau_t}, \qquad \|P_{s_t} - P_{\widehat{s}_t}\|_{\mathrm{TV}} \le \sqrt{\frac{\epsilon_{s,t}}{\tau_t}}. \tag{889}$$

The corresponding arrival candidate combines this sensitivity with the Gibbs approximation bias

$$\sum_{t<H} \tau_t \log(1/\nu_{\min,t}).$$

(iii) Binary label noise. *Clean-risk radii acquire the factor $(1 - 2\eta_+)^{-1}$, and their squared or variance coordinates acquire $(1 - 2\eta_+)^{-2}$, by Theorem IV.10.3 and Corollary IV.10.6.*

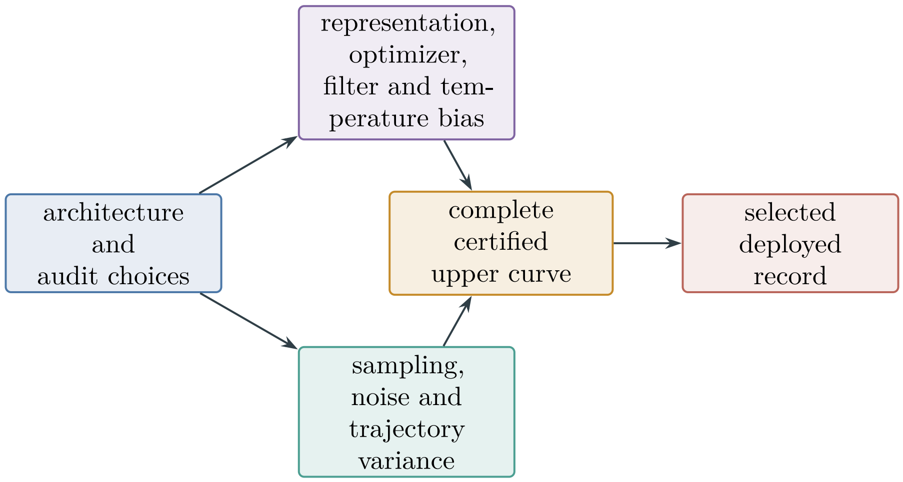

Figure 106: Certified bias–variance selection. Every architecture and audit choice changes named deterministic and stochastic coordinates; simultaneous selection minimizes their complete upper bound.

(iv) Horizon and dependence. *The represented tail $\varepsilon_{\mathrm{trunc},H}$ is balanced against the accumulated predictable variance $\mathcal{V}_H$ or the certified mixing effective sample size.*

(v) Latent capacity. *Increasing b decreases the certified sufficiency defect and increases the represented latent cover, PAC–Bayes, storage, and optimization coordinates of Theorem V.11.25.*

(vi) Functional features. *Increasing p_{geom} changes the statistical radii $\epsilon_{C,p}, \epsilon_{G,p}$ in Eqs.* (893) *and* (894) *and the analytical completeness remainder r_p in Eq.* (900)*.*

(vii) Quotient resolution. *A quotient with $|Q|$ cells has statistical defect radius*

$$|Q|\sqrt{\frac{\log(2|X||Q|/\delta_{\mathrm{lump}})}{2N_{\min}}},$$

together with its represented target-sufficiency and transfer defects.

Let Λ be a represented finite or countable grid of complete architecture, temperature, clipping, capacity, feature, and quotient choices, with weights $w_\lambda > 0$, $\sum_{\lambda\in\Lambda} w_\lambda \le 1$. Evaluate every candidate at confidence δw_λ, let U_λ be its complete certified upper bound, and select

$$\widehat{\lambda} \in \arg\min_{\lambda\in\Lambda} U_\lambda. \tag{890}$$

Then, with probability at least $1-\delta$, the selected record satisfies its reported bound $U_{\widehat{\lambda}}$. This selection includes the full architecture and audit choice rather than tuning one coordinate in isolation.

Proof. The parameter, noise, horizon, latent, feature, and quotient statements are the cited confidence or completeness comparisons. For the temperature statement, the uniform score error bounds the log-normalizer ratio by $\epsilon_{s,t}/\tau_t$, so the log density ratio is bounded by $2\epsilon_{s,t}/\tau_t$. Taking expectation proves the KL bound and Pinsker proves the total-variation bound. A weighted union bound over Λ proves the selected-candidate assertion. $\square$

V.15.12 Empirical certification of analytical geometry

Definition V.15.36 (Feature-restricted Dirichlet audit)**.** Let Φ be a certified finite dynamical geometry with invariant law μ, Dirichlet form $\mathcal{E}_\Phi$, and analytical Poincaré constant γ_Φ from Definition V.5.5. Let $\psi_p : X \to \mathbb{R}^p$ be a represented feature map, centered under μ. Define its covariance and Dirichlet matrices by

$$C_p = \mathbb{E}_\mu[\psi_p \psi_p^\mathsf{T}], \qquad G_p = \big(\mathcal{E}_\Phi(\psi_{p,i}, \psi_{p,j})\big)_{i,j \le p}. \tag{891}$$

Let $\widehat{C}_p, \widehat{G}_p$ be represented unbiased audit estimates and let $\epsilon_{C,p}, \epsilon_{G,p}$ satisfy, on one simultaneous event,

$$\|C_p - \widehat{C}_p\|_{\mathrm{op}} \le \epsilon_{C,p}, \qquad \|G_p - \widehat{G}_p\|_{\mathrm{op}} \le \epsilon_{G,p}. \tag{892}$$

Centering error is included in $\epsilon_{C,p}$.

For independent audit samples, one explicit entrywise option is

$$\epsilon_{C,p} = pK_p^2 \sqrt{\frac{2\log(4p^2/\delta_{\mathrm{FI}})}{N_C}}, \tag{893}$$

$$\epsilon_{G,p} = pD_p^2 \sqrt{\frac{2\log(4p^2/\delta_{\mathrm{FI}})}{N_G}}, \tag{894}$$

provided every centered coordinate obeys $|\psi_{p,i}| \le K_p$, every unbiased sampled Dirichlet increment has coordinate magnitude at most D_p, and the Dirichlet normalization is included in D_p. A matrix concentration, mixing, or martingale radius may replace these entrywise candidates. The feature construction, centering, edge sampler, dependence correction, precision, and selected radius are charged in the implementation record.

Theorem V.15.37 (Empirical Poincaré bracket and full-carrier transfer)**.** *On the event Eq.* (892), *suppose a represented number $\underline{\gamma}_p \ge 0$ satisfies the semidefinite certificate*

$$\widehat{G}_p - \underline{\gamma}_p \widehat{C}_p \succeq (\epsilon_{G,p} + \underline{\gamma}_p \epsilon_{C,p}) I_p. \tag{895}$$

Then every $f_a = a^\mathsf{T}\psi_p$ satisfies

$$\mathcal{E}_\Phi(f_a, f_a) \ge \underline{\gamma}_p \operatorname{Var}_\mu(f_a). \tag{896}$$

For every represented a with $a^\mathsf{T}(\widehat{C}_p - \epsilon_{C,p} I)a > 0$, the analytical constant also satisfies

$$\gamma_\Phi \le \frac{a^\mathsf{T}(\widehat{G}_p + \epsilon_{G,p} I)a}{a^\mathsf{T}(\widehat{C}_p - \epsilon_{C,p} I)a}. \tag{897}$$

Assume additionally that a represented linear projection Π_p from the full Poincaré domain to $\operatorname{span}\psi_p$ *has certified constants $A_p, D_p^{\mathrm{pr}}, r_p \ge 0$ such that, for every centered f,*

$$\operatorname{Var}_\mu(f) \le A_p \operatorname{Var}_\mu(\Pi_p f) + r_p \mathcal{E}_\Phi(f, f), \tag{898}$$

$$\mathcal{E}_\Phi(\Pi_p f, \Pi_p f) \le D_p^{\mathrm{pr}} \mathcal{E}_\Phi(f, f). \tag{899}$$

If $\underline{\gamma}_p > 0$, then the true full-carrier constant has the certified lower bound

$$\boxed{\gamma_\Phi \ge \left(\frac{A_p D_p^{\mathrm{pr}}}{\underline{\gamma}_p} + r_p\right)^{-1}.} \tag{900}$$

Thus p controls both statistical estimation through Eqs. (893) *and* (894) *and analytical underfitting through the completeness remainder r_p.*

Proof. The operator inequalities in Eq. (892) give

$$G_p - \underline{\gamma}_p C_p \succeq \widehat{G}_p - \underline{\gamma}_p \widehat{C}_p - (\epsilon_{G,p} + \underline{\gamma}_p \epsilon_{C,p}) I_p \succeq 0.$$

Taking the quadratic form in a proves Eq. (896). Conversely, $G_p \preceq \widehat{G}_p + \epsilon_{G,p} I$ and $C_p \succeq \widehat{C}_p - \epsilon_{C,p} I$. The resulting Rayleigh quotient is an admissible test in the full Poincaré infimum, proving Eq. (897).

For a centered f, apply Eq. (896) to $\Pi_p f$, followed by Eq. (899), in Eq. (898):

$$\mathrm{Var}_\mu(f) \le \left(\frac{A_p D_p^{\mathrm{pr}}}{\underline{\gamma}_p} + r_p \right) \mathcal{E}_\Phi(f, f).$$

Taking the reciprocal proves Eq. (900).

□

Theorem V.15.38 (Generic empirical functional-inequality transfer)**.** *Let $\mathscr{D}$ be the declared domain of one entry of $\mathfrak{F}_\Phi$. Write its inequality as*

$$\mathcal{D}(f) \ge \kappa\, \mathcal{V}(f), \qquad f \in \mathscr{D}, \tag{901}$$

after using the normalization of Definition V.5.4. Let $\mathscr{D}_p \subseteq \mathscr{D}$ be a represented feature domain and $\mathcal{N}(f) \ge 0$ a represented homogeneous control functional. Suppose, simultaneously on the audit event,

$$|\widehat{\mathcal{D}}(f) - \mathcal{D}(f)| \le \epsilon_{\mathcal{D},p} \mathcal{N}(f), \tag{902}$$

$$|\widehat{\mathcal{V}}(f) - \mathcal{V}(f)| \le \epsilon_{\mathcal{V},p} \mathcal{N}(f) \tag{903}$$

for all $f \in \mathscr{D}_p$, and suppose

$$\widehat{\mathcal{D}}(f) - \underline{\kappa}_p \widehat{\mathcal{V}}(f) \ge (\epsilon_{\mathcal{D},p} + \underline{\kappa}_p \epsilon_{\mathcal{V},p}) \mathcal{N}(f) \tag{904}$$

on that domain. Then $\mathcal{D}(f) \ge \underline{\kappa}_p \mathcal{V}(f)$ on $\mathscr{D}_p$.

If a represented projection $\Pi_p : \mathscr{D} \to \mathscr{D}_p$ satisfies

$$\mathcal{V}(f) \le A_p \mathcal{V}(\Pi_p f) + r_p \mathcal{D}(f), \tag{905}$$

$$\mathcal{D}(\Pi_p f) \le D_p^{\mathrm{pr}} \mathcal{D}(f), \tag{906}$$

then the full analytical constant is at least

$$\kappa_\Phi \ge \left(\frac{A_p D_p^{\mathrm{pr}}}{\underline{\kappa}_p} + r_p \right)^{-1}. \tag{907}$$

For any represented $f \in \mathscr{D}_p$ satisfying $\widehat{\mathcal{V}}(f) > \epsilon_{\mathcal{V},p} \mathcal{N}(f)$, the same audit also gives the analytical upper certificate

$$\kappa_\Phi \le \frac{\widehat{\mathcal{D}}(f) + \epsilon_{\mathcal{D},p} \mathcal{N}(f)}{\widehat{\mathcal{V}}(f) - \epsilon_{\mathcal{V},p} \mathcal{N}(f)}. \tag{908}$$

The theorem applies to the Poincaré, weak/super-Poincaré at a fixed represented parameter, Nash, Sobolev, spectral-profile, LSI, MLSI, curvature, sector, hypocoercive, capacity, and transport slots whenever the displayed audit and projection comparisons are supplied in that slot's declared domain and normalization.

Proof. For $f \in \mathscr{D}_p$, subtract the two deviation bounds from Eq. (904); the error margin cancels and gives Eq. (901) with $\kappa = \underline{\kappa}_p$. Insert this restricted inequality and Eq. (906) into Eq. (905). The same reciprocal calculation as in Theorem V.15.37 proves Eq. (907). Finally, $\mathcal{D}(f) \leq \widehat{\mathcal{D}}(f) + \epsilon_{\mathcal{D},p}\mathcal{N}(f)$ and $\mathcal{V}(f) \geq \widehat{\mathcal{V}}(f) - \epsilon_{\mathcal{V},p}\mathcal{N}(f) > 0$. The ratio $\mathcal{D}(f)/\mathcal{V}(f)$ is an admissible test in the infimum defining κ_Φ, which proves Eq. (908). □

V.15.13 Empirical certification of exact and quasi-lumpability

Definition V.15.39 (Fiber transition diameter)**.** Let P be a represented Markov kernel on a finite carrier X, and let $q : X \to Q$ be a represented terminal-compatible quotient. Define

$$P_q(x, c) = P(x, q^{-1}(c)), \tag{909}$$

$$\delta_q(P) = \max_{\substack{x,x' \in X \\ q(x)=q(x')}} \|P_q(x, \cdot) - P_q(x', \cdot)\|_{\mathrm{TV}}. \tag{910}$$

For a continuous-time generator L, use a represented uniformization $P = I + L/\omega$ and retain the rate ω in every defect conversion. The quotient is exactly lumpable precisely when $\delta_q(P) = 0$.

Theorem V.15.40 (Finite-sample lumpability certificate)**.** *Suppose every $x \in X$ has at least $N_{\min}$ independent audited outgoing transitions, and let $\widehat{P}_q(x, \cdot)$ be the empirical quotient-row distribution. Put*

$$\widehat{\delta}_q = \max_{\substack{x,x' \in X \\ q(x)=q(x')}} \|\widehat{P}_q(x, \cdot) - \widehat{P}_q(x', \cdot)\|_{\mathrm{TV}}, \tag{911}$$

$$r_q = \sqrt{\frac{\log(2|X||Q|/\delta_{\mathrm{lump}})}{2N_{\min}}}. \tag{912}$$

Then, with probability at least $1 - \delta_{\mathrm{lump}}$,

$$\boxed{\left|\delta_q(P) - \widehat{\delta}_q\right| \leq |Q| r_q.} \tag{913}$$

For an implicit carrier, the same conclusion holds with $|Q|r_q$ replaced by $2\Gamma_q^{\mathrm{row}}$, where Γ_q^{row} is any represented simultaneous total-variation radius for the selected quotient-row class. Its cover, feature locator, adaptive sampling rule, and selection cost are charged.

If every represented quotient-row probability lies on a declared grid $\{0, h_q, 2h_q, \ldots, 1\}$, then

$$\widehat{\delta}_q + |Q|r_q < h_q \quad \Longrightarrow \quad \delta_q(P) = 0. \tag{914}$$

An exact symbolic or interval verification of all row equalities also gives $\delta_q(P) = 0$ directly.

Proof. Hoeffding's inequality and a union bound over $X \times Q$ give

$$|P_q(x, c) - \widehat{P}_q(x, c)| \leq r_q$$

simultaneously. Hence each empirical row is within $|Q|r_q/2$ in total variation of its true row. The reverse and forward triangle inequalities therefore bound the difference between every true pairwise row distance and its empirical counterpart by $|Q|r_q$. Taking maxima preserves this uniform error bound and proves Eq. (913).

Two distinct probability vectors on the common grid have total-variation distance at least h_q: one coordinate differs by at least h_q, and mass conservation supplies an equal total discrepancy of the opposite sign. Therefore $\delta_q(P)$ is either zero or at least h_q. The strict upper bound in Eq. (914) forces the zero case.

□

Theorem V.15.41 (Why the certified quotient is lumpable)**.** *Let $\bar{P}$ be the quotient kernel obtained by choosing the common row when $\delta_q(P) = 0$. Then*

$$PU_q = U_q\bar{P}. \tag{915}$$

Consequently $q(X_t)$ is Markov with kernel $\bar{P}$, independently of the representative inside each fiber. For a terminal-compatible absorbing target, the full and quotient first-hit laws agree by Proposition III.11.3.

More generally, choose $\bar{P}(c, \cdot)$ to be any convex average of the rows $P_q(x, \cdot)$ over $q(x) = c$. Then

$$\|PU_q - U_q\bar{P}\|_{\infty\to\infty} \le 2\delta_q(P), \tag{916}$$

$$\Big|\Pr\{\tau_T \le H\} - \Pr\{\tau_{T_Q}^{\bar{P}} \le H\}\Big| \le \min\{1, H\delta_q(P)\}. \tag{917}$$

Here the quotient chain is started from the pushforward of the full chain's initial law, and the target and quotient target are absorbing. These are the same initialization and terminal-compatibility conventions used by the first-hit transfer theorem. Under uniformization, the generator defect in Definition III.11.2 obeys

$$\|E_q\|_{\infty\to\infty} \le 2\omega\delta_q(P). \tag{918}$$

Thus the empirical bound in Theorem V.15.40 supplies either an exact Abs *quotient when the defect is certified zero, or the explicit quasi-lumpable defect input required by Section III.11 and Proposition II.6.2.*

Proof. If $\delta_q(P) = 0$, the value $P(x, q^{-1}(c'))$ depends only on $q(x)$. Define this value to be $\bar{P}(q(x), c')$. For every $f : Q \to \mathbb{R}$,

$$PU_qf(x) = \sum_{c'} P(x, q^{-1}(c'))f(c') = U_q\bar{P}f(x),$$

which proves exact intertwining and the Markov property of the projected process.

For the averaged row, every row in a fiber is within $\delta_q(P)$ in total variation of the average. Expectations of functions bounded by one therefore differ by at most $2\delta_q(P)$, proving Eq. (916). Couple the projected full transition and the quotient transition maximally at every step. Their one-step disagreement probability is at most $\delta_q(P)$; a union bound through H steps proves Eq. (917). Finally, $L = \omega(P - I)$ and $\bar{L} = \omega(\bar{P} - I)$, so multiplying Eq. (916) by ω proves Eq. (918).

□

Corollary V.15.42 (Canonical same-run geometry and quotient certificate)**.** *Let one canonical architecture expose the geometry feature head $\psi_{p_{\rm geom}}$ and quotient head $q : X \to Q$. Construct the Dirichlet and quotient-row audits from independent represented samples of the same deployed transition law. Assume the hypotheses of Theorem V.15.37 hold with $\underline{\gamma}_p > 0$. On the simultaneous confidence event, the true Poincaré constant satisfies*

$$\left(\frac{A_pD_p^{\rm pr}}{\underline{\gamma}_p} + r_p\right)^{-1} \le \gamma_\Phi \le \inf_{\substack{a \text{ represented} \\ a^{\mathsf{T}}(\widehat{C}_p - \epsilon_{C,p}I)a > 0}} \frac{a^{\mathsf{T}}(\widehat{G}_p + \epsilon_{G,p}I)a}{a^{\mathsf{T}}(\widehat{C}_p - \epsilon_{C,p}I)a}, \tag{919}$$

and the true fiber diameter satisfies

$$\left[\widehat{\delta}_q - |Q|r_q\right]_+ \le \delta_q(P) \le \min\left\{1, \widehat{\delta}_q + |Q|r_q\right\}. \tag{920}$$

The analogous interval for another functional slot is obtained from Eqs. (907) *and* (908) *using the same feature head and that slot's declared forms. The infimum in Eq.* (919) *is* $+\infty$ *when no represented test vector has a positive certified denominator.*

If the quotient-row probabilities lie on the represented grid of Eq. (914) *and the upper endpoint in Eq.* (920) *is smaller than* h_q, *then* $\delta_q(P) = 0$. *Hence*

$$PU_q = U_q\bar{P},$$

the projected process is Markov, and terminal-compatible first-hit laws transfer exactly. Otherwise the upper endpoint is substituted for $\delta_q(P)$ *in Eqs.* (916) *to* (918).

Proof. The Poincaré lower and upper bounds are Eqs. (897) and (900); taking the infimum over available represented test vectors preserves the upper bound. The lumpability interval is Eq. (913) clipped to the range $[0, 1]$. Grid separation then gives zero fiber diameter, and Theorem V.15.41 gives exact intertwining and first-hit transfer. The same theorem gives the stated quasi-lumpable substitutions.

□

Corollary V.15.43 (End-to-end certified neural optimum)**.** *Consider a complete structural-Bayesian neural learner trained by a certified WAdam record with* p *trainable parameters and terminal weights* $\widehat{\theta}_p$. *At one common class-preserving ceiling, suppose the record contains:*

(i) *the WAdam descent, PL, and objective-to-action calibration certificates of Theorem V.15.31;*
(ii) *the simultaneous structural learning radii* $\Gamma_{t,p}$, *including their noise, dependence, selection, and confidence conversions;*
(iii) *the represented filter comparison* $\Delta^H_{\mathrm{fil},p}$;
(iv) *the held-out arrival lower certificate* $L_{\mathrm{ev},p}$; *and*
(v) *the exact or coverage-based Bellman upper certificate* $U_{\mathrm{Bell},p}$.

Then, with the declared simultaneous confidence,

$$\begin{aligned} 0 \le J^*_{\mathrm{sat}}(B) - A^H_{R_{\widehat{\theta}_p}} \le \min\Bigg\{ & U_{\mathrm{Bell},p} - L_{\mathrm{ev},p}, \\ & \mathbb{E}_{\widehat{R}} \sum_{t<H} \mathbf{1}_{\{\tau>t\}} \left(2\Gamma_{t,p} + \lambda_{t,p}\frac{\mathcal{E}_{\mathrm{WAdam}}(p,K)}{\delta_{\mathrm{opt}}} + \zeta_{t,p}\right) + \Delta^H_{\mathrm{fil},p}\Bigg\}, \end{aligned} \tag{921}$$

where the declared simultaneous failure probability includes δ_{opt} *from Eq.* (876). *The first candidate certifies the complete representation-plus-training gap. The second resolves the learned record into statistical, optimizer, calibration, and filtering coordinates. Every term in Eq.* (921) *is a coordinate of the common parameter-explicit ledger* $\mathfrak{E}_p$ *in Eq.* (786)*; parameter count is therefore not hidden inside an unspecified approximation term.*

At the universal Bayesianized ceiling of Corollary V.11.14, J^*_{sat} *equals the saturated represented-computation optimum. Thus a small certified primal–dual gap proves that the implemented learner attains that universal ceiling to the displayed accuracy.*

Proof. Use Eq. (876) and Eq. (875) in the learned-control candidate of Corollary V.15.9. The Bellman candidate follows from the same corollary. Finally apply the profile equality in Corollary V.11.14 at the common compiled ceiling.

□

Theorem V.15.44 (Complete empirical structural proof package)**.** *Fix a parameter count p and one complete implementation record. Suppose the record contains:*

(i) *the saturated, p-class, and trained-run intervals in Eqs.* (787) *to* (789)*;*
(ii) *the source-resolved audit of Definition V.15.24;*
(iii) *the primal–dual and WAdam certificates in Corollary V.15.43;*
(iv) *the canonical loss calibration and bias–variance ledger in Theorems V.15.29 and V.15.33;*
(v) *a feature-restricted functional-inequality audit and its represented completeness comparison; and*
(vi) *a represented quotient q, its quotient-row audit, and either its exact grid/separation record or its quasi-lumpability defect record.*

Allocate confidence so that the sum of all failure probabilities is at most δ_{all}. Then, with probability at least $1-\delta_{\mathrm{all}}$, the same empirical run simultaneously certifies:

(a) *the implementation gap in Eq.* (921)*;*
(b) *the calibrated primal–dual loss and separated deterministic-bias and stochastic-radius coordinates in Eqs.* (866) *and* (883)*;*
(c) *the underfitting, training, and minimum-parameter intervals in Eqs.* (790)*,* (791) *and* (794)*;*
(d) *the exact learned source support and the source-resolved target-gain bound in Eq.* (846)*;*
(e) *every audited analytical functional constant transferred by Theorems V.15.37 and V.15.38; and*
(f) *exact quotient dynamics and exact target-law transfer when Eq.* (914) *holds, or the quantitative quasi-lumpability bounds Eqs.* (916) *to* (918) *otherwise.*

All conclusions use the same represented architecture, source graph, transition law, quotient, feature map, precision, and common saturated ceiling. Therefore the empirical proof is a certificate about the true analytical record rather than a collection of unrelated fitted diagnostics.

Proof. Each cited theorem holds on its represented audit event. The union bound gives the simultaneous probability. All conversions are fields of the same complete record, so their common-law and common-normalization hypotheses are preserved. Applying the cited conclusions on the intersection proves (a)–(f).

□

V.16 Black-Box Structural Auditing of Trained Predictors

The preceding section certifies a represented training and deployment record. This section treats the complementary setting in which the trained predictor is available only through a query interface. The audit compiles oracle calls, a represented test law, label-channel information, and public interventions into the same statistical, structural, geometric, and quotient coordinates used throughout the framework. Its conclusions concern the behavior of the deployed predictor on the declared audit law. A statement about the unobserved training sample is produced precisely when the record contains a represented transfer from the audit channel to the training channel.

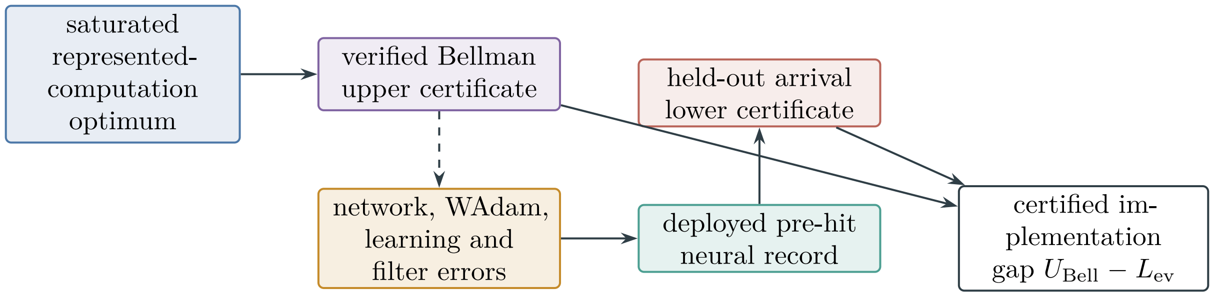

Figure 107: Certified neural realization of the saturated optimum. The deployed network supplies a primal lower certificate; a represented Bellman barrier supplies an upper certificate for the complete saturated class. Structural generalization, WAdam optimization, filtering, precision, and deployment are charged inside the common record.

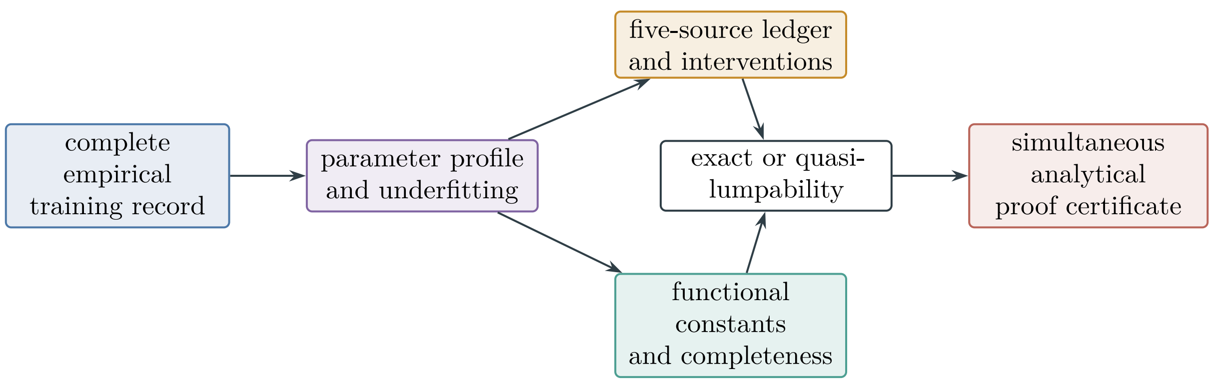

Figure 108: Empirical structural proof package. Parameter sufficiency, source-resolved learning, analytical functional inequalities, and quotient lumpability are certified from one completely charged record and one simultaneous confidence allocation.

V.16.1 Oracle access and complete audit records

Definition V.16.1 (Complete black-box predictor audit)**.** Let X be a represented input carrier and $Y = \{1, \dots, K\}$ a finite label carrier. A trained predictor is exposed through one of the following represented interfaces:

(i) a label interface $O_f(x) \in Y$;

(ii) a score interface $O_f(x) \in [-M, M]^K$ at declared precision; or

(iii) a probability interface $O_f(x) \in \Delta_K$, deterministic or sampled using represented oracle randomness.

For label and score access, a represented readout maps the oracle output to a probability vector $p_f(x) \in \Delta_K$; a label is represented by its point mass. The readout and its numerical precision are part of the audit.

A *complete black-box predictor audit* is a record

$$R_{\mathrm{BB}} = (O_f, \mathsf{Access}, \mu, S_{\mathrm{audit}}, \mathscr{T}, \mathscr{D}, Q, \boldsymbol{\delta}, \mathsf{Select}), \tag{922}$$

where:

(a) μ is the declared audit law and $S_{\mathrm{audit}} = (Z_i)_{i=1}^N$ consists of independent audit units $Z_i \sim \mu$; an audit unit contains its input and every clean, corrupted, repeated, or subgroup label declared by the protocol;

(b) $\mathscr{T}$ is a finite represented library of input interventions and perturbation kernels, with their coupling to the base input and their declared structural provenance;

(c) $\mathscr{D}$ is the represented family of diagnostics, losses, discretizers, feature maps, and quotient-row tests evaluated by the audit;

(d) Q is the total number of oracle calls, including calls used for transformed inputs, repeated stochastic evaluation, diagnostic selection, and numerical calibration;

(e) $\boldsymbol{\delta}$ allocates the total confidence failure among all selected certificates; and

(f) Select stores every data-dependent diagnostic, feature, perturbation, binning, quotient, stopping decision, and comparison reported by the audit.

All calls made from one Z_i, including adaptive calls whose next input is a represented function of earlier outputs in that block, form one bounded audit block. Blocks use independent audit examples and independent oracle randomness. The complete cost is

$$B_{\mathrm{BB}} = B_{\mathrm{access}} \oplus B_{\mathrm{data}} \oplus B_{\mathrm{query}} \oplus B_{\mathrm{intervention}} \oplus B_{\mathrm{diagnostic}} \oplus B_{\mathrm{selection}} \oplus B_{\mathrm{precision}} \oplus B_{\mathrm{certificate}}. \tag{923}$$

The black-box record reveals the represented input–output behavior of the oracle. Internal parameters, activations, source graph, optimizer state, and training sample enter the record only when separately exposed.

Definition V.16.2 (Test-law behavioral equivalence)**.** Two trained predictors f and g are *behaviorally equivalent for an audit record* when every transcript produced by the declared access interface and every intervention in $\mathscr{T}$ have the same law under μ. A black-box certificate is a function of this equivalence class. It may identify a source in the internal training record only when an exposed provenance map compiles the observed behavior into that source coordinate.

Theorem V.16.3 (Finite-query simultaneous black-box audit)**.** *Let $D_1, \dots, D_M$ be diagnostics fixed before the audit sample is revealed. Suppose diagnostic j produces one block value $D_j(Z_i, O_f, U_i) \in [0, 1]$, after any finite within-block adaptive query sequence, and put*

$$d_j = \mathbb{E} D_j(Z_i, O_f, U_i), \qquad \widehat{d}_j = \frac{1}{N} \sum_{i=1}^{N} D_j(Z_i, O_f, U_i). \tag{924}$$

For $0 < \delta_{\mathrm{BB}} < 1$*, define*

$$r_{N,M,\delta_{\mathrm{BB}}} = \sqrt{\frac{\log(2M/\delta_{\mathrm{BB}})}{2N}}. \tag{925}$$

Then, with probability at least $1 - \delta_{\mathrm{BB}}$*,*

$$|d_j - \widehat{d}_j| \le r_{N,M,\delta_{\mathrm{BB}}} \qquad (1 \le j \le M) \tag{926}$$

simultaneously. In particular, precision $r_{N,M,\delta} \le \epsilon$ *is ensured by*

$$N \ge \frac{\log(2M/\delta_{\mathrm{BB}})}{2\epsilon^2}. \tag{927}$$

If the diagnostic family is selected using the audit data, its fixed-family cardinality in Eq. (925) *is replaced by the applicable structural Rademacher, PAC–Bayes mixture, sequential Rademacher, or martingale PAC–Bayes radius already carried by the complete record. If a diagnostic lies in* $[a_j, b_j]$*, its radius is multiplied by* $b_j - a_j$*.*

Proof. For each fixed j, the block variables are independent and lie in $[0, 1]$, even though the calls inside one block may be adaptive. Hoeffding's inequality gives

$$\Pr\{|\widehat{d}_j - d_j| > r\} \le 2e^{-2Nr^2}.$$

At the radius in Eq. (925), the right side is δ_{BB}/M. A union bound proves Eq. (926); solving for N proves Eq. (927). Rescaling proves the bounded interval statement. The selected-family replacements are exactly the uniform and posterior-weighted selection theorems of Definition IV.13.2 and Theorems IV.15.5 and IV.15.6. □

V.16.2 Predictive performance, calibration, and law transfer

Definition V.16.4 (Canonical black-box predictive diagnostics)**.** For a clean audit label Y, a reported label $\widehat{Y}_f = \arg\max_c p_f^c(X)$, and a fixed finite partition $\mathcal{B}$ of $[0, 1]$, define

$$R_0(f) = \Pr\{\widehat{Y}_f \ne Y\}, \tag{928}$$

$$J_{ac}(f) = \Pr\{Y = a, \widehat{Y}_f = c\}, \tag{929}$$

$$R_{\mathrm{Br}}(f) = \frac{1}{2}\mathbb{E}\sum_{c=1}^{K}(p_f^c(X) - \mathbf{1}_{\{Y=c\}})^2, \tag{930}$$

$$\mathrm{Cal}_{\mathcal{B}}(f) = \sum_{B\in\mathcal{B}} \Pr\{C_f \in B\}\left|\mathbb{E}[\mathbf{1}_{\{\widehat{Y}_f=Y\}} \mid C_f \in B] - \mathbb{E}[C_f \mid C_f \in B]\right|, \tag{931}$$

where $C_f = \max_c p_f^c(X)$. Empty calibration bins contribute zero. For $\epsilon_{\mathrm{pr}} \in (0, 1/K]$, the clipped normalized log loss is

$$R_{\log,\epsilon_{\mathrm{pr}}}(f) = \mathbb{E}\frac{-\log(p_f^Y(X) \vee \epsilon_{\mathrm{pr}})}{\log(1/\epsilon_{\mathrm{pr}})}. \tag{932}$$

All displayed losses lie in $[0, 1]$. Classwise and subgroup risks use the same definitions conditionally on a represented cell; their radius uses the number of independent audit examples in that cell.

Corollary V.16.5 (Simultaneous predictive audit). *Fix the label tie rule, probability readout, calibration partition, clipping level, and a finite collection of represented subgroups before evaluation. Empirical versions of Eqs.* (928) *to* (930) *and* (932) *are within $r_{N,M,\delta}$ of their population values simultaneously, with one diagnostic allocated to every reported scalar. Moreover, if $\widehat{p}_B = N_B/N$, $\widehat{a}_B$ is the empirical correct mass in bin B, and $\widehat{c}_B$ is its empirical confidence mass, then*

$$\mathrm{Cal}_{\mathcal{B}}(f) \leq \sum_{B\in\mathcal{B}} |\widehat{a}_B - \widehat{c}_B| + 2|\mathcal{B}|r_{N,M,\delta}, \tag{933}$$

where the empirical masses are normalized by N, so empty bins require no conditional division.

Proof. Each risk and confusion entry is the expectation of a fixed $[0,1]$-valued diagnostic, so Theorem V.16.3 applies. For calibration, write the contribution of bin B as $|a_B - c_B|$, where

$$a_B = \mathbb{E}[\mathbf{1}_{\{C_f\in B\}}\mathbf{1}_{\{\widehat{Y}_f=Y\}}], \qquad c_B = \mathbb{E}[\mathbf{1}_{\{C_f\in B\}}C_f].$$

Both summands are bounded diagnostics. The reverse triangle inequality gives $|a_B - c_B| \leq |\widehat{a}_B - \widehat{c}_B| + 2r$; summing over bins proves Eq. (933). □

Theorem V.16.6 (Audit-law to deployment-law transfer). *Let ν be a represented deployment law and let $D \in [0,1]$ be one black-box diagnostic. Either of the following represented comparisons transfers its audit-law certificate:*

(i) *if $\|\nu - \mu\|_{\mathrm{TV}} \leq \Delta_{\mathrm{shift}}$, then*

$$\mathbb{E}_\nu D \leq \widehat{\mathbb{E}}_\mu D + r_{N,M,\delta} + \Delta_{\mathrm{shift}}; \tag{934}$$

(ii) *if $w = d\nu/d\mu$ is represented and $0 \leq w \leq W$, then the importance-weighted estimator satisfies*

$$\mathbb{E}_\nu D \leq \frac{1}{N}\sum_{i=1}^{N} w(Z_i)D_i + W\sqrt{\frac{\log(2M/\delta)}{2N}} \tag{935}$$

simultaneously over M fixed diagnostics.

Proof. The total-variation dual inequality gives $|\mathbb{E}_\nu D - \mathbb{E}_\mu D| \leq \Delta_{\mathrm{shift}}$, which combines with Eq. (926). In the second case, $\mathbb{E}_\mu[wD] = \mathbb{E}_\nu D$ and $wD \in [0,W]$; the rescaled Hoeffding and union-bound argument proves Eq. (935). □

V.16.3 Noise identification and clean-risk reconstruction

Definition V.16.7 (Audit, effective, and training noise). The black-box record keeps three typed noise coordinates.

(i) *Audit-label noise* is the represented channel from a clean audit label Y to the observed audit label $\widetilde{Y}$.
(ii) *Effective task noise* is the disagreement channel between the observed label and a represented clean target on the audit law.
(iii) *Training noise* is the corresponding channel in the sample used to construct the trained oracle.

The first two are identified from paired or repeated audit labels, a known channel, or another represented reconstruction in Definition IV.17.1. The third is reported as a compatible set unless the record supplies a represented training-to-audit channel comparison.

Theorem V.16.8 (Binary repeated-label noise interval)**.** *Let $\widetilde{Y}^{(1)} = Y\Xi_1$ and $\widetilde{Y}^{(2)} = Y\Xi_2$, where, conditionally on (X, Y), the signs Ξ_1, Ξ_2 are independent and satisfy $\Pr\{\Xi_j = -1\} = \eta \le 1/2$. Put*

$$d = \Pr\{\widetilde{Y}^{(1)} \ne \widetilde{Y}^{(2)}\} = 2\eta(1-\eta) \tag{936}$$

and let $\widehat{d}$ be the disagreement frequency in N independent pairs. With

$$r_d = \sqrt{\frac{\log(2/\delta_d)}{2N}}, \tag{937}$$

$$d_- = [\widehat{d} - r_d]_{[0,1/2]}, \qquad d_+ = [\widehat{d} + r_d]_{[0,1/2]}, \tag{938}$$

$$h(d) = \frac{1-\sqrt{1-2d}}{2}, \tag{939}$$

one has, with probability at least $1-\delta_d$,

$$h(d_-) \le \eta \le h(d_+). \tag{940}$$

Combining either endpoint with the exact risk transport in Lemma IV.10.2 gives a simultaneous clean-risk interval for every fixed audited predictor.

Proof. Conditional independence gives $d = \eta(1-\eta) + (1-\eta)\eta$. Hoeffding's inequality gives $|d - \widehat{d}| \le r_d$. The map $\eta \mapsto 2\eta(1-\eta)$ is increasing on $[0, 1/2]$, and h is its increasing inverse, proving the interval. The risk statement follows by intersection with the predictive audit event and substitution in Eq. (231). □

Corollary V.16.9 (Joint binary noise and clean-risk confidence set)**.** *Suppose a noisy-label audit gives simultaneous intervals*

$$R_\eta(f) \in [\widetilde{r}_-, \widetilde{r}_+], \qquad \eta \in [\eta_-, \eta_+] \subset [0, 1/2). \tag{941}$$

Define

$$\mathcal{I}_0(f) = \left\{ \frac{u-v}{1-2v} : u \in [\widetilde{r}_-, \widetilde{r}_+],\ v \in [\eta_-, \eta_+] \right\} \cap [0,1]. \tag{942}$$

On the joint confidence event, $R_0(f) \in \mathcal{I}_0(f)$. Its reported interval is the closed hull

$$[\inf \mathcal{I}_0(f), \sup \mathcal{I}_0(f)], \tag{943}$$

whose endpoints are obtained by evaluating the rational map at the four corners of the rectangle and clipping the resulting compatible values to $[0, 1]$. If the noise confidence set contains $1/2$, the certified default is $[0, 1]$.

Proof. The exact identity $R_\eta(f) = \eta + (1-2\eta)R_0(f)$ from Eq. (231) gives $R_0(f) = (R_\eta(f) - \eta)/(1-2\eta)$ whenever $\eta < 1/2$. Substitution of the simultaneous rectangle proves membership in $\mathcal{I}_0(f)$. The rectangle is compact and the denominator is positive, so the image is compact. For fixed v, the map is increasing in u; at either endpoint in u, its derivative with respect to v has constant sign $(2u-1)/(1-2v)^2$. Its extrema therefore occur at corners. At $\eta = 1/2$, the corrupted risk is independent of the clean risk, yielding the full default interval. □

Theorem V.16.10 (Multiclass channel inversion with estimated noise)**.** *Let* $C_{ab} = \Pr\{\widetilde{Y} = b \mid Y = a\}$ *be a class-conditional corruption matrix and let*

$$J_{ac} = \Pr\{Y = a, \widehat{Y}_f = c\}, \qquad \widetilde{J}_{bc} = \Pr\{\widetilde{Y} = b, \widehat{Y}_f = c\}. \tag{944}$$

Then

$$\widetilde{J} = C^{\mathsf{T}} J. \tag{945}$$

Let $\widehat{C}$ *be an invertible represented estimate and let* $\widehat{\widetilde{J}}$ *be the empirical noisy-label confusion matrix. Define*

$$\widehat{J} = (\widehat{C}^{\mathsf{T}})^{-1} \widehat{\widetilde{J}}. \tag{946}$$

On any simultaneous event satisfying

$$\|\widehat{\widetilde{J}} - \widetilde{J}\|_1 \le \epsilon_J, \qquad \|\widehat{C}^{\mathsf{T}} - C^{\mathsf{T}}\|_{1\to 1} \le \epsilon_C, \tag{947}$$

one has

$$\|\widehat{J} - J\|_1 \le \|(\widehat{C}^{\mathsf{T}})^{-1}\|_{1\to 1}\,(\epsilon_J + \epsilon_C). \tag{948}$$

Consequently the clean classification error obeys

$$\left| R_0(f) - \left(1 - \sum_{a=1}^{K} \widehat{J}_{aa}\right) \right| \le \|(\widehat{C}^{\mathsf{T}})^{-1}\|_{1\to 1}\,(\epsilon_J + \epsilon_C). \tag{949}$$

The same linear transport applies to every bounded loss that is a represented linear functional of the clean joint table. The inverse norm records the exact noise-amplification cost.

Proof. Condition on $(Y, \widehat{Y}_f)$ to obtain Eq. (945). Multiplication of Eq. (946) by $\widehat{C}^{\mathsf{T}}$, followed by subtraction of $\widehat{C}^{\mathsf{T}} J$, gives

$$\widehat{C}^{\mathsf{T}}(\widehat{J} - J) = (\widehat{\widetilde{J}} - \widetilde{J}) + (C^{\mathsf{T}} - \widehat{C}^{\mathsf{T}})J.$$

Since J is a probability table, $\|J\|_1 = 1$. Apply the induced norm and the inverse to prove Eq. (948). The diagonal functional has dual norm at most one, proving the risk interval. $\square$

Theorem V.16.11 (Black-box nonidentifiability and certified training-noise transfer)**.** *Fix a black-box predictor oracle, audit law, and query protocol. Without a represented relation to the training record, its training-noise rate is not identified: for every* $\eta_{\mathrm{tr}} \in [0, 1]$*, there is a training record with that label-noise rate that returns the same oracle and hence the same black-box audit transcript law.*

Let $\mathcal{C}_{\mathrm{tr}}(R_{\mathrm{BB}})$ *denote the set of training channels compatible with the exposed record. If the record additionally supplies a represented channel comparison*

$$\|C_{\mathrm{tr}} - C_{\mathrm{audit}}\|_{1\to 1} \le \Delta_{\mathrm{tr,aud}}, \tag{950}$$

then every confidence set $\mathfrak{C}_{\mathrm{audit}}$ *for the audit channel induces the training-channel confidence set*

$$\mathfrak{C}_{\mathrm{tr}} = \left\{ C : \inf_{C' \in \mathfrak{C}_{\mathrm{audit}}} \|C - C'\|_{1\to 1} \le \Delta_{\mathrm{tr,aud}} \right\}. \tag{951}$$

For binary homogeneous noise this reduces to

$$\eta_{\mathrm{tr}} \in \left[(\eta^{-}_{\mathrm{audit}} - \Delta_\eta)_+, (\eta^{+}_{\mathrm{audit}} + \Delta_\eta) \wedge 1\right] \tag{952}$$

when the represented comparison is $|\eta_{\mathrm{tr}} - \eta_{\mathrm{audit}}| \le \Delta_\eta$.

Proof. For a fixed exposed oracle, consider a represented training compiler that returns that oracle after reading and charging its input training record. Feeding the compiler clean labels gives rate zero. Passing the same labels through any declared corruption channel before the compiler reads them gives the prescribed rate and leaves the returned oracle unchanged. Therefore all black-box transcripts are identical, proving nonidentifiability from the oracle and independent audit data alone.

On the event $C_{\text{audit}} \in \mathfrak{C}_{\text{audit}}$, Eq. (950) places C_{tr} in Eq. (951) by definition. The binary interval is the triangle inequality followed by intersection with $[0, 1]$. □

V.16.4 Behavioral structural channels and interventions

Definition V.16.12 (Black-box structural intervention ledger)**.** Let

$$\mathfrak{A}_{\text{BB}} = \{\mathsf{Add}, \mathsf{Mult}, \mathsf{Ord}, \mathsf{Sym}, \mathsf{Filt}, \mathsf{Geom}, \mathsf{Caus}, \mathsf{Lift}, \mathsf{Bdry}\}. \tag{953}$$

For every audited source $a \in \mathfrak{A}_{\text{BB}}$, a represented intervention record contains:

(i) a coupling $(X, T_a X)$ on the audit law, generated without access to the current clean audit label unless the intervention is explicitly a label-conditioned analytic diagnostic;

(ii) an output action $\rho_a(T_a) : \Delta_K \to \Delta_K$, equal to the identity for an invariant intervention and to the represented label action for an equivariant intervention;

(iii) a target-preservation flag stating whether the clean label is unchanged or transformed by the same represented action; and

(iv) the query, construction, locator, selection, and numerical costs of the intervention.

With total variation on Δ_K, define the behavioral defect and, when the transformed clean label Y_a is represented, the target response

$$D_a(f) = \mathbb{E}\left[\|p_f(T_a X) - \rho_a(T_a) p_f(X)\|_{\text{TV}}\right], \tag{954}$$

$$A_a(f) = \mathbb{E}\left[\ell(p_f(X), Y) - \ell(p_f(T_a X), Y_a)\right]. \tag{955}$$

The loss is represented and lies in $[0, 1]$, so $A_a(f) \in [-1, 1]$. A filtration intervention uses two represented prefixes that differ only in the audited reveal; a lift intervention compares the base prediction with the represented pushforward of its lifted prediction; a boundary intervention changes only the declared boundary record. A Caus intervention additionally contains a prospective transition rule compiled from oracle outputs available before the label reveal. Its target response is therefore a pre-reveal behavioral alignment coordinate.

The ledger is behavioral: D_a and A_a describe dependence of the deployed input–output map on a represented source intervention. An exact claim about internal source ancestry requires an exposed compiler from the model record to the source-resolved ledger of Definition IV.13.2.

Theorem V.16.13 (Simultaneous black-box structural signature)**.** *Fix L_a intervention draws per audit example for every $a \in \mathfrak{A}_{\text{BB}}$, and average their bounded defects inside the example block. Let $\widehat{D}_a$ and $\widehat{A}_a$ be the resulting empirical averages over N independent audit examples. For*

$$r_{\text{str}} = \sqrt{\frac{\log(4|\mathfrak{A}_{\text{BB}}|/\delta_{\text{str}})}{2N}}, \tag{956}$$

one has, with probability at least $1 - \delta_{\text{str}}$,

$$D_a(f) \in \left[[\widehat{D}_a - r_{\text{str}}]_+, (\widehat{D}_a + r_{\text{str}}) \wedge 1\right], \tag{957}$$

$$A_a(f) \in [\widehat{A}_a - 2r_{\mathrm{str}}, \widehat{A}_a + 2r_{\mathrm{str}}] \cap [-1, 1] \tag{958}$$

simultaneously for every source. The complete query count of this audit is

$$Q_{\mathrm{str}} = N \sum_{a \in \mathfrak{A}_{\mathrm{BB}}} L_a(q_a^{\mathrm{base}} + q_a^{\mathrm{trans}}), \tag{959}$$

where repeated base calls may be shared only when the oracle and coupling record justify the shared transcript.

For a source family chosen from data, replace r_{str} by its charged source-resolved Rademacher or PAC–Bayes radius. If a provenance compiler is present, the corresponding exact source cell and target-gain upper bound are then obtained from Theorem V.15.25.

Proof. The averaged intervention defect in one example block lies in $[0, 1]$. The two loss values defining the target response are separately in $[0, 1]$. Apply Theorem V.16.3 to the defect and the two loss expectations for every source. The defect uses one radius; subtraction of the two loss intervals uses two. Summing the represented calls proves Eq. (959). The selected-source and compiled-provenance conclusions are the cited framework theorems. □

Definition V.16.14 (Black-box input-intervention value)**.** Let $\mathcal{A} \subseteq \mathfrak{A}_{\mathrm{BB}}$ be a finite set of d audited input sources. A complete intervention family supplies, for every $S \subseteq \mathcal{A}$, a represented coupled input X^S in which the sources in S are retained and the others are replaced by their declared target-neutral baselines. All 2^d coupled constructions use the same audit law, label action, oracle, loss, and numerical readout. Define

$$v_f(S) = 1 - \mathbb{E}\ell(p_f(X^S), Y^S), \tag{960}$$

$$\phi_{f,a} = \sum_{S \subseteq \mathcal{A} \setminus \{a\}} \frac{|S|!\,(d - |S| - 1)!}{d!} (v_f(S \cup \{a\}) - v_f(S)). \tag{961}$$

These are input-behavior allocations. They coincide with internal source interventions only when the exposed model compiler proves that equivalence.

Theorem V.16.15 (Simultaneous black-box intervention allocation)**.** *Evaluate every X^S on N_{int} independent coupled audit units and put*

$$r_{\mathrm{int}}^{\mathrm{BB}} = \sqrt{\frac{\log(2^{d+1}/\delta_{\mathrm{int}})}{2N_{\mathrm{int}}}}. \tag{962}$$

Then, with probability at least $1 - \delta_{\mathrm{int}}$,

$$|\widehat{v}_f(S) - v_f(S)| \le r_{\mathrm{int}}^{\mathrm{BB}} \qquad \textit{for every } S \subseteq \mathcal{A}, \tag{963}$$

$$|\widehat{\phi}_{f,a} - \phi_{f,a}| \le 2r_{\mathrm{int}}^{\mathrm{BB}} \qquad \textit{for every } a \in \mathcal{A}, \tag{964}$$

$$\sum_{a \in \mathcal{A}} \phi_{f,a} = v_f(\mathcal{A}) - v_f(\varnothing). \tag{965}$$

The full deterministic-oracle query count is $Q_{\mathrm{int}} = N_{\mathrm{int}} 2^d$; stochastic repetition multiplies this by the represented number of calls per intervention.

Proof. Hoeffding's inequality with a union bound over 2^d bounded losses proves Eq. (963). The Shapley weights for one source are nonnegative and sum to one; every marginal contains two estimated values, giving Eq. (964). Along each permutation of $\mathcal{A}$, the marginal increments telescope from the empty to the complete intervention, proving the efficiency identity. □

V.16.5 Black-box geometry, alignment, and functional inequalities

Definition V.16.16 (Black-box perturbation geometry)**.** Let P be a represented Markov perturbation kernel with invariant audit law μ, and suppose $(X, X') \sim \mu(dx)P(x, dx')$. Let $X'' \sim \mu$ be independent. The output Dirichlet energy, variance, and labeled signed progress are

$$\mathcal{E}_{P,f} = \frac{1}{2}\mathbb{E}\|p_f(X') - p_f(X)\|_2^2, \tag{966}$$

$$\mathcal{V}_f = \frac{1}{2}\mathbb{E}\|p_f(X'') - p_f(X)\|_2^2, \tag{967}$$

$$\mathcal{C}_{P,f} = \mathbb{E}\left[\ell(p_f(X), Y) - \ell(p_f(X'), Y')\right], \tag{968}$$

where Y' is the represented transformed clean label. Each energy and variance integrand lies in $[0, 1]$; the two losses defining $\mathcal{C}_{P,f}$ lie in $[0, 1]$. When $\mathcal{V}_f > 0$, define the observed Rayleigh ratio

$$\rho_{P,f} = \mathcal{E}_{P,f}/\mathcal{V}_f. \tag{969}$$

A prospective Caus record additionally specifies a kernel K_f whose row is computed from oracle outputs and public information available before the clean label reveal. Replacing P by K_f in Eq. (968) gives its audited Caus alignment. Label-conditioned perturbations remain analytic diagnostics and are not prospective Caus records.

Theorem V.16.17 (Finite-query geometric confidence region)**.** *Use N_E, N_V, N_C independent represented audit blocks to estimate $\mathcal{E}_{P,f}, \mathcal{V}_f$, and the two loss expectations defining $\mathcal{C}_{P,f}$. Allocate failure probabilities $\delta_E, \delta_V, \delta_C$, and put*

$$r_E = \sqrt{\frac{\log(2/\delta_E)}{2N_E}}, \qquad r_V = \sqrt{\frac{\log(2/\delta_V)}{2N_V}}, \qquad r_C = \sqrt{\frac{\log(4/\delta_C)}{2N_C}}. \tag{970}$$

On an event of probability at least $1 - \delta_E - \delta_V - \delta_C$,

$$\mathcal{E}_{P,f} \in [E_-, E_+] := [[\widehat{\mathcal{E}}_{P,f} - r_E]_+, (\widehat{\mathcal{E}}_{P,f} + r_E) \wedge 1], \tag{971}$$

$$\mathcal{V}_f \in [V_-, V_+] := [[\widehat{\mathcal{V}}_f - r_V]_+, (\widehat{\mathcal{V}}_f + r_V) \wedge 1], \tag{972}$$

$$\mathcal{C}_{P,f} \in [\widehat{\mathcal{C}}_{P,f} - 2r_C, \widehat{\mathcal{C}}_{P,f} + 2r_C] \cap [-1, 1]. \tag{973}$$

If $V_- > 0$, then

$$\frac{E_-}{V_+} \leq \rho_{P,f} \leq \frac{E_+}{V_-}. \tag{974}$$

If P is reversible and the output coordinates lie in the declared Dirichlet domain, the upper endpoint in Eq. (974) is a represented test-function upper certificate for the Poincaré constant of P. A lower certificate for the full declared function domain follows when the audit also supplies the uniform feature-domain deviations and completeness comparison required by Theorem V.15.38. For a deterministic oracle, the three independent audits use

$$Q_{\mathrm{geom}} = 2N_E + 2N_V + 2N_C \tag{975}$$

oracle evaluations. Represented stochastic repetition multiplies each summand by its declared repeat count; shared calls are permitted only through a represented common-block coupling.

Proof. Apply Hoeffding's inequality to the bounded energy and variance integrands and to the two bounded loss expectations. A union bound gives the simultaneous event. Division by positive interval endpoints proves Eq. (974). For every nonconstant admissible test function, the Poincaré constant is at most its Dirichlet-to-variance ratio; summing over output coordinates preserves this Rayleigh quotient. The full-domain lower transfer is exactly Eq. (907). □

V.16.6 Behavioral quotients and residual structure

Definition V.16.18 (Black-box output quotient)**.** Let $b : \Delta_K \to Q_f$ be a represented finite discretizer, fixed or selected with its charged complexity, and put

$$q_f(x) = b(p_f(x)). \tag{976}$$

Given a represented perturbation kernel P, define its quotient-row map and fiber diameter by

$$P_{q_f}(x, c) = P(x, q_f^{-1}(c)), \tag{977}$$
$$\delta_{q_f}(P) = \max_{q_f(x)=q_f(x')} \|P_{q_f}(x, \cdot) - P_{q_f}(x', \cdot)\|_{\mathrm{TV}}. \tag{978}$$

The discretizer, bin locator, transition sampler, empty-cell convention, and all row evaluations are fields of the complete black-box record.

Theorem V.16.19 (Black-box exact and quasi-lumpability compilation)**.** *Suppose first that the audited carrier X_{aud} is finite and every state has at least $N_{\min}$ independent outgoing perturbation samples. Then, with the radius of Eq.* (912)*,*

$$|\delta_{q_f}(P) - \widehat{\delta}_{q_f}| \le |Q_f| r_{q_f} \tag{979}$$

with probability at least $1 - \delta_{\mathrm{lump}}$. On an implicit carrier, the same conclusion holds with $2\Gamma^{\mathrm{row}}_{q_f}$ whenever the complete audit supplies the simultaneous quotient-row radius required by Theorem V.15.40.

If the grid-separation condition in Eq. (914) *holds, then*

$$PU_{q_f} = U_{q_f}\bar{P}. \tag{980}$$

Otherwise the certified upper endpoint

$$\overline{\delta}_{q_f} = \min\{1, \widehat{\delta}_{q_f} + |Q_f| r_{q_f}\} \tag{981}$$

gives

$$\|PU_{q_f} - U_{q_f}\bar{P}\|_{\infty\to\infty} \le 2\overline{\delta}_{q_f}, \tag{982}$$
$$|\Pr\{\tau_T \le H\} - \Pr\{\tau^{\bar{P}}_{T/Q_f} \le H\}| \le \min\{1, H\overline{\delta}_{q_f}\} \tag{983}$$

for every represented terminal-compatible absorbing target. Selection of b on the same audit sample uses the row-class radius $\Gamma^{\mathrm{row}}_{q_f}$; it is not treated as a fixed-bin audit. For a deterministic oracle on the finite audited carrier, caching every base label and querying every sampled transition endpoint gives the explicit upper count

$$Q_{\mathrm{quot}} \le |X_{\mathrm{aud}}| + \sum_{x\in X_{\mathrm{aud}}} N_x, \tag{984}$$

where $N_x \ge N_{\min}$ is the number of outgoing samples at x. Every additional candidate discretizer or stochastic repeat is charged in the selection or repetition coordinate of B_{BB}.

Proof. The confidence statement is Theorem V.15.40 applied to the represented map in Eq. (976). Grid separation gives exact lumpability. The operator and first-hit comparisons follow from Theorem V.15.41 after substituting the certified upper endpoint. The selected-bin statement is the implicit carrier clause of the same finite-sample theorem. □

Definition V.16.20 (Held-out structural surrogate and residual)**.** Split the audit sample into independent construction and evaluation parts. Using only the construction part, build a represented surrogate $g_{\mathrm{str}} : X \to \Delta_K$ from the audited structural feature library, intervention responses, quotient cells, or another source-resolved class in the saturated grammar. On the evaluation law define

$$R^{\mathrm{BB}}_{\mathrm{res}}(f; g_{\mathrm{str}}) = \frac{1}{2}\mathbb{E}\|p_f(X) - g_{\mathrm{str}}(X)\|_2^2, \tag{985}$$

$$C^{\mathrm{BB}}_{\mathrm{res}}(f; g_{\mathrm{str}}) = \frac{1}{2}\left|\mathbb{E}\left\langle p_f(X) - g_{\mathrm{str}}(X), e_Y - \pi_Y\right\rangle\right|, \tag{986}$$

where $\pi_Y = \mathbb{E}e_Y$ is estimated on the construction split or specified by the audit law. Both integrands have absolute value at most one. The first coordinate measures output behavior not reconstructed by the declared structural surrogate. The second measures the part of that residual that remains aligned with the clean audit target. A provenance compiler may route a target-orthogonal residual into J_{res}; the empirical norm alone does not assert that internal classification.

Theorem V.16.21 (Held-out black-box residual certificate)**.** *Let the evaluation split contain N_{res} independent examples and let*

$$r_{\mathrm{res}} = \sqrt{\frac{\log(4/\delta_{\mathrm{res}})}{2N_{\mathrm{res}}}}. \tag{987}$$

Then, conditionally on the construction split and hence also unconditionally, with probability at least $1 - \delta_{\mathrm{res}}$,

$$R^{\mathrm{BB}}_{\mathrm{res}} \in [[\widehat{R}^{\mathrm{BB}}_{\mathrm{res}} - r_{\mathrm{res}}]_+, (\widehat{R}^{\mathrm{BB}}_{\mathrm{res}} + r_{\mathrm{res}}) \wedge 1], \tag{988}$$

$$C^{\mathrm{BB}}_{\mathrm{res}} \le \widehat{C}^{\mathrm{BB}}_{\mathrm{res}} + r_{\mathrm{res}}. \tag{989}$$

If the surrogate is selected using the evaluation split, these radii are replaced by the applicable represented structural Rademacher or PAC–Bayes radius for the selected surrogate class.

Proof. Conditioned on the construction split, the surrogate is fixed. The squared simplex distance divided by two lies in $[0, 1]$. The inner product in Eq. (986), divided by two, lies in $[-1, 1]$. Hoeffding's inequality and a union bound prove the two claims. The data-selected case is the uniform selected-class substitution already used in Theorem V.16.3. □

V.16.7 Master black-box certificate

Definition V.16.22 (Complete quantitative black-box signature)**.** The complete quantitative signature of one black-box audit is

$$\mathfrak{C}_{\mathrm{BB}} = \begin{pmatrix} \mathcal{I}_{\mathrm{risk}}, \mathcal{I}_{\mathrm{cal}}, \mathfrak{C}_{\mathrm{audit}}, \mathfrak{C}_{\mathrm{tr}}, \{\mathcal{I}_{D_a}, \mathcal{I}_{A_a}\}_{a \in \mathfrak{A}_{\mathrm{BB}}}, \\ \mathcal{I}_{\mathcal{E}}, \mathcal{I}_{\mathcal{V}}, \mathcal{I}_{\rho}, \mathcal{I}_{\mathcal{C}}, \mathcal{I}_{\delta_q}, \mathcal{I}_{R_{\mathrm{res}}}, \mathcal{I}_{C_{\mathrm{res}}}, \\ N, Q, B_{\mathrm{BB}}, \mu, \mathcal{I}_{\mathrm{shift}}, \boldsymbol{\delta} \end{pmatrix}. \tag{990}$$

Every interval is tagged by its audit law, data split, query interface, noise channel, selected diagnostic family, and confidence allocation.

Theorem V.16.23 (End-to-end black-box structural certificate)**.** *Consider one complete record from Definition V.16.1. Suppose it contains the predictive diagnostics, one represented audit-noise* *identification record, the structural interventions, a perturbation geometry, an output quotient, and a held-out structural surrogate defined above. Allocate confidence so that*

$$\delta_{\text{pred}} + \delta_{\text{noise}} + \delta_{\text{str}} + \delta_{\text{geom}} + \delta_{\text{lump}} + \delta_{\text{res}} \le \delta_{\text{all}}^{\text{BB}}. \tag{991}$$

Then, with probability at least $1 - \delta_{\text{all}}^{\text{BB}}$, the single signature $\mathfrak{C}_{\text{BB}}$ simultaneously certifies:

(i) *clean or noise-corrected predictive risk, calibration, confusion, and subgroup performance on the audit law;*

(ii) *the audit-label channel and its exact clean-risk amplification, and the compatible training-channel set whenever a training comparison is represented;*

(iii) *every behavioral structural defect and target response in Eqs.* (957) *and* (958)*;*

(iv) *the output energy, variance, Rayleigh ratio, and prospective Caus progress in Eqs.* (971), (973) *and* (974)*;*

(v) *exact output-quotient dynamics when Eq.* (914) *holds, and otherwise the explicit quasi-lumpable defect and first-hit loss in Eqs.* (982) *and* (983)*; and*

(vi) *the magnitude and clean-target correlation of output behavior not reconstructed by the declared structural surrogate.*

If a deployment-law comparison is represented, every bounded coordinate is transported by Theorem V.16.6. If an exposed model compiler maps the black-box behavior into the complete source-resolved learning ledger, the source-qualified target gain additionally satisfies

$$g_f \le \min\left\{1, U_f^{\text{stat}\to\text{tgt}}, U_f^{\text{dyn}\to\text{tgt}}, \beta_f^+ + \sqrt{\frac{\log 2}{2}\sum_{a\in\mathfrak{I}^5_{\mathsf{Abs}}} J_f^{(a)}}, eH_B(n)\sum_{a\in\mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{L}_f^{(a)}\right\}, \tag{992}$$

with each unknown source coordinate replaced by its simultaneous empirical upper endpoint. The compilation requirement is satisfied automatically for the white-box records of Definition V.15.1; for a pure oracle record, Eq. (992) *is included exactly when its provenance compiler is exposed.*

Proof. Apply Corollary V.16.5 and Theorems V.16.8, V.16.10, V.16.13, V.16.17, V.16.19 and V.16.21 on their respective audit splits and take the intersection of their events. The union bound and Eq. (991) give the stated confidence. The law-transfer claim is Theorem V.16.6. Under the provenance compiler, substitute the empirical source upper endpoints into Eq. (846); monotonicity of the minimum, sum, and square root proves Eq. (992). □

Corollary V.16.24 (Certified black-box comparison of two trained predictors)**.** *Evaluate predictors f and g on the same paired audit units and let $\Delta_i = D_i(f) - D_i(g) \in [-1, 1]$ be any fixed paired diagnostic difference. With*

$$r_{\text{pair}} = \sqrt{\frac{2\log(2M/\delta_{\text{pair}})}{N}}, \tag{993}$$

one has simultaneously over M comparisons

$$\left|\mathbb{E}\Delta_i - \frac{1}{N}\sum_{i=1}^{N}\Delta_i\right| \le r_{\text{pair}} \tag{994}$$

with probability at least $1 - \delta_{\text{pair}}$. The comparison therefore inherits common-example cancellation without treating the two empirical means as independent.

Proof. The paired differences are independent across audit units and lie in an interval of length two. Apply the rescaled Hoeffding inequality and a union bound over the M fixed comparisons. □

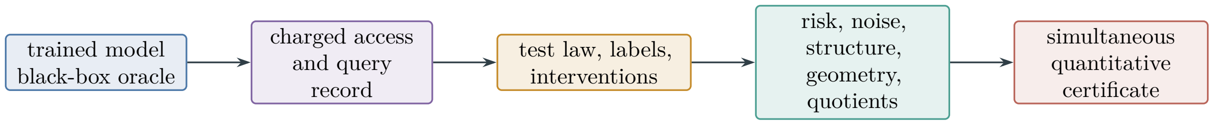

Figure 109: Black-box audit compilation. Oracle access and every selected query are charged before the test-law observations are converted into simultaneous performance, noise, structural, geometric, and quotient coordinates.

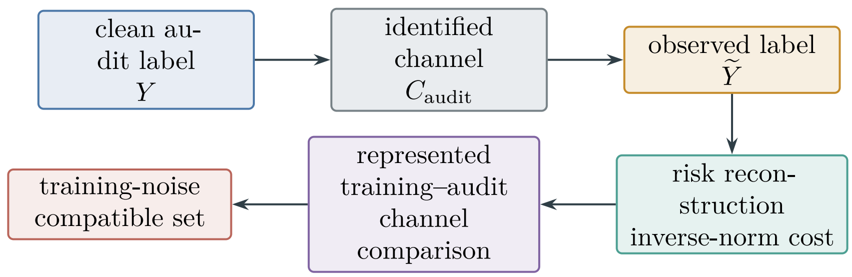

Figure 110: Typed noise reconstruction. Audit noise is inferred from represented anchors or repeated labels. Training noise is transported only through a represented comparison, yielding a confidence set rather than an inferred training history.

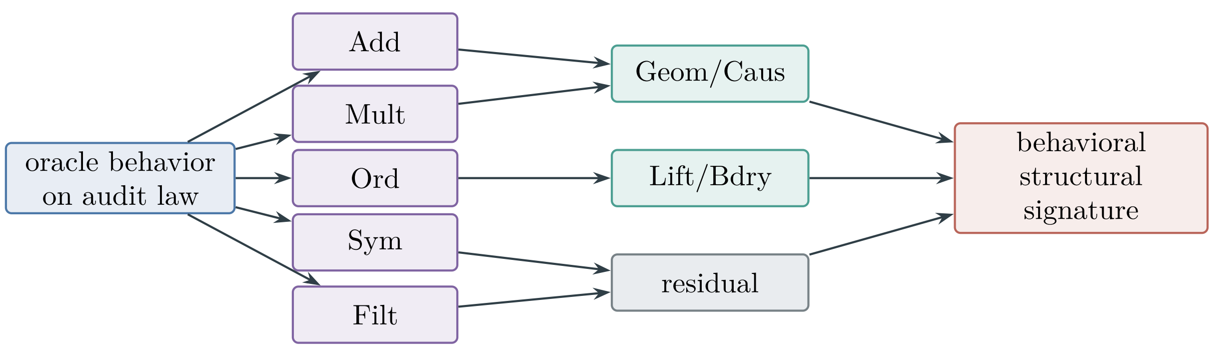

Figure 111: Externally auditable structural channels. Represented input interventions produce behavioral defects and target-response intervals. Internal provenance is added only through an exposed compiler.

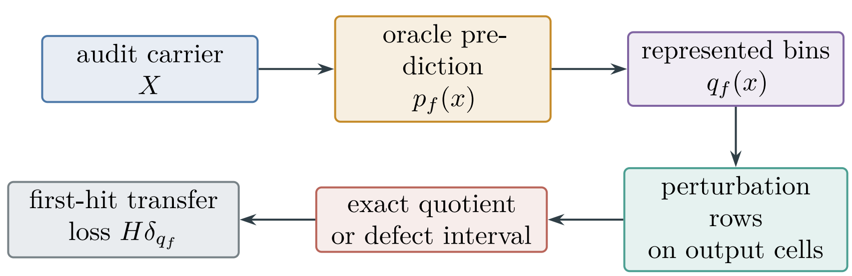

Figure 112: Behavioral quotient audit. Output cells become a dynamical abstraction only after their perturbation rows satisfy the exact or quasi-lumpability certificate.

V.17 Length-Bounded Levin Search and the Structural Learner

This section relates the saturated active profile to one represented universal search record. Description length remains a separate resource coordinate: the universal record pays exponentially for an unknown record code, whereas a structural learner may construct its deployed record from the presentation. The comparison below records the code length, simulation overhead, verifier error, learning error, and physical ceiling separately.

V.17.1 Prefix codes and length-bounded saturated profiles

Definition V.17.1 (Canonical code of a complete active record)**.** Fix a prefix-free binary encoding $\mathsf{code} : \mathscr{R}^{\mathrm{act}} \to \{0,1\}^*$ of the syntax of complete master active-sampling records from Definition V.9.1. The code stores every finite program, rational constant, precision declaration, continuation rule, and output convention belonging to the record. Public instance data are read through the declared presentation interface and are not copied into the program code. Put

$$d(R) := |\mathsf{code}(R)|. \tag{995}$$

The prefix condition gives

$$\sum_{R\in\mathscr{R}^{\mathrm{act}}} 2^{-d(R)} \le 1. \tag{996}$$

The encoding is *family-uniform*: one finite code describes the record on every size slice, while the public presentation supplies n, m, ζ and the instance. Let Int be its represented universal interpreter. There is a monotone class-preserving scalar majorant $\sigma_{\mathrm{int}}(s,d)$ such that interpreting the first s scalarized steps of a length-d record costs at most $\sigma_{\mathrm{int}}(s,d)$. The canonical finite-precision machine may use

$$\sigma_{\mathrm{int}}(s,d) \le c_{\mathrm{int}}(s+d+1)\big(1+\lceil\log_2(s+d+2)\rceil\big)^{a_{\mathrm{int}}} \tag{997}$$

for fixed machine constants $c_{\mathrm{int}}, a_{\mathrm{int}}$. The scalarization of complete cost includes reading the code, so for a fixed constant c_{read},

$$d(R) \le c_{\mathrm{read}}\big(1+\kappa(\mathrm{Cost}(R))\big). \tag{998}$$

Definition V.17.2 (Length-bounded active and selector profiles)**.** For an instance I, horizon H, scalar ceiling b, and $\ell \in \mathbb{N}$, define

$$\mathrm{ActArr}^{\mathrm{sat}}_{I,H,\le\ell}(b) = \sup_{R\in\mathscr{R}^{\mathrm{act}}_{I,H}\,;\ \kappa(\mathrm{Cost}(R))\le b,\ d(R)\le\ell} A^H_R, \tag{999}$$

$$\mathrm{ActSel}^{\mathrm{sat}}_{I,H,\le\ell}(b) = \max_{0\le t<H} \sup_{R\in\mathscr{R}^{\mathrm{act}}_{I,H}\,;\ \kappa(\mathrm{Cost}(R))\le b,\ d(R)\le\ell} \big(\mathrm{Adv}_t(K_{R,t};\nu_{R,t})\big)_+. \tag{1000}$$

The family-uniform versions apply the fixed functional $\mathfrak{M}_{n,m,\zeta}$ before taking the same supremum, as in Definition V.9.3. Empty suprema are zero.

Lemma V.17.3 (Monotonicity and exact exhaustion by code length)**.** *For fixed (I,H,b), both profiles in Definition V.17.2 are nondecreasing in ℓ, bounded above by their unrestricted counterparts, and satisfy*

$$\sup_{\ell\in\mathbb{N}} \mathrm{ActArr}^{\mathrm{sat}}_{I,H,\le\ell}(b) = \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(b), \tag{1001}$$

$$\sup_{\ell\in\mathbb{N}} \mathrm{ActSel}^{\mathrm{sat}}_{I,H,\le\ell}(b) = \mathrm{ActSel}^{\mathrm{sat}}_{I,H}(b). \tag{1002}$$

Moreover, Eq. (998) *implies equality already for every $\ell \ge c_{\mathrm{read}}(1+b)$.*

Proof. The feasible record classes are nested in ℓ. Every represented record has one finite code, so the union of the length-bounded classes is the unrestricted class. A supremum over an increasing union equals the supremum of the suprema. Finally, every record of scalarized cost at most b has code length at most $c_{\mathrm{read}}(1+b)$ by Eq. (998). $\square$

Definition V.17.4 (Represented target gate and global gate error)**.** Let $V_I : X_I \to \{0,1\}$ be a represented public verifier of scalar cost at most $t_V(n,m)$, and put $\widehat{T}_I = V_I^{-1}(1)$. The clean target gate is T_I. Define the pointwise gate defect

$$e_V(I) = \mathbf{1}_{\{\widehat{T}_I \neq T_I\}} \tag{1003}$$

and its family value

$$\varepsilon_V(n,m,\zeta) = \mathfrak{M}_{n,m,\zeta}(I \mapsto e_V(I)). \tag{1004}$$

This global event controls every verifier call in one execution and therefore is not multiplied by the horizon. If only conditional per-call soundness and completeness bounds are available, their predictable union bound is used in place of Eq. (1004) and is charged in the verifier slot of the complete record.

V.17.2 The represented Levin record

Definition V.17.5 (Length-bounded represented Levin search)**.** For $\ell \in \mathbb{N} \cup \{\infty\}$, let $\mathfrak{U}_\ell$ be the complete active record that lazily decodes the prefix tree, simulates each legal record code p with $|p| \le \ell$, and assigns branch p Kraft weight $2^{-|p|}$. Whenever a branch emits a candidate, $\mathfrak{U}_\ell$ evaluates V_I before declaring arrival. Branch state, random bits, candidate emission, verification, stopping, memory, and the scheduler clock are fields of the complete record.

There are fixed constants $c_{\mathrm{sch}}, a_{\mathrm{sch}}$ such that a branch of length d receives s interpreted branch steps within scalarized universal cost

$$\Lambda_{\mathfrak{U}}(d,s) = c_{\mathrm{sch}} 2^d (s+d+1)\big(1 + \lceil \log_2(s+d+2) \rceil\big)^{a_{\mathrm{sch}}}. \tag{1005}$$

This is obtained by the standard staged Kraft scheduler: at stage j, it executes every decoded prefix p with $|p| \le j$ for the stage share proportional to $2^{-|p|}$; the Kraft inequality bounds total simulated work, and the logarithmic factor pays for prefix-tree traversal and resumable branch lookup. Thus the description penalty is exactly the factor 2^d; machine simulation and scheduling are the displayed class-preserving factors.

The code of $\mathfrak{U}_\ell$ contains the interpreter, scheduler, verifier reference, and a self-delimiting numeral for ℓ. Hence

$$d(\mathfrak{U}_\ell) \le c_{\mathfrak{U}} + 2\lceil \log_2(\ell+2) \rceil \quad (\ell < \infty), \tag{1006}$$

whereas $d(\mathfrak{U}_\infty) \le c_{\mathfrak{U}}$.

Theorem V.17.6 (Length-bounded Levin realization)**.** *Let $q \in \mathscr{R}^{\mathrm{act}}_{I,H}$ satisfy $d(q) \le \ell$ and $\kappa(\mathrm{Cost}(q)) \le b$. Suppose a horizon-$H$ execution emits at most H candidates. Define*

$$s_q(b,n,m,H) = \sigma_{\mathrm{int}}(b, d(q)) + H\, t_V(n,m), \tag{1007}$$

$$b_{\mathfrak{U}}(q; b,n,m,H) = \Lambda_{\mathfrak{U}}\big(d(q), s_q(b,n,m,H)\big). \tag{1008}$$

Let

$$H_{\mathfrak{U}}(q; b,n,m,H) = H_{b_{\mathfrak{U}}(q;b,n,m,H)} \tag{1009}$$

be the largest refined clock horizon admitted by this scalar ceiling. Then, pointwise in the presentation,

$$A_{q,I}^{H} \le A_{\mathfrak{U}_\ell,I}^{H_{\mathfrak{U}}(q)}(b_{\mathfrak{U}}(q;b,n,m,H)) + e_V(I). \tag{1010}$$

Consequently, for every declared monotone subadditive family functional,

$$\mathfrak{M}_{n,m,\zeta}(A_q^H) \le \mathfrak{M}_{n,m,\zeta}\Big(A_{\mathfrak{U}_\ell}^{H_{\mathfrak{U}}(q)}(b_{\mathfrak{U}}(q;b,n,m,H))\Big) + \varepsilon_V(n,m,\zeta). \tag{1011}$$

For a linear expectation functional this is the usual average-instance comparison.

Proof. Couple the branch indexed by $\mathsf{code}(q)$ to the same represented random seed as q. The interpreter reproduces its horizon-H path law within $\sigma_{\mathrm{int}}(b, d(q))$ branch cost. At most H candidate verifications add Ht_V. By Eq. (1005), the universal record supplies this branch cost by the ceiling in Eq. (1008).

On $\{\widehat{T}_I = T_I\}$, the universal record declares a hit on the coupled branch precisely when that branch first emits a target; an earlier acceptance is also a true target hit. Hence $A_{q,I}^{H} \le A_{\mathfrak{U}_\ell,I}^{H_{\mathfrak{U}}(q)}$ on this event. On its complement both probabilities lie in $[0,1]$, which proves Eq. (1010). Monotonicity and subadditivity of the family functional give Eq. (1011). □

Corollary V.17.7 (Uniform length-class realization)**.** *Put*

$$s_\ell(b,n,m,H) = \sup_{0\le d\le \ell} \sigma_{\mathrm{int}}(b,d) + Ht_V(n,m), \tag{1012}$$

$$b_{\mathfrak{U}}(\ell;b,n,m,H) = \Lambda_{\mathfrak{U}}(\ell, s_\ell(b,n,m,H)). \tag{1013}$$

Put

$$H_{\mathfrak{U}}(\ell;b,n,m,H) = H_{b_{\mathfrak{U}}(\ell;b,n,m,H)}. \tag{1014}$$

Then

$$\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H,\le\ell}(b) \le \mathfrak{M}_{n,m,\zeta}\Big(A_{\mathfrak{U}_\ell}^{H_{\mathfrak{U}}(\ell)}(b_{\mathfrak{U}}(\ell;b,n,m,H))\Big) + \varepsilon_V(n,m,\zeta). \tag{1015}$$

Proof. For $d(q) \le \ell$, monotonicity of σ_{int} and $\Lambda_{\mathfrak{U}}$ places the simulated branch below the uniform ceiling. Apply Eq. (1011) and take the supremum over the defining length class. □

Proposition V.17.8 (Sharp integration with saturated framework certificates)**.** *Let $\{U_j(b)\}_{j\in\mathcal{J}}$ be any finite collection of valid same-law, same-target upper certificates for $\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H}(b)$. The collection may contain the prospective selector/source bound, master active-sampling envelope, occupation-flow Bellman barrier, controlled capacity or information candidate, and a simultaneous empirical Bellman upper certificate furnished respectively by Theorems III.21.4, V.9.8, V.10.4 and V.12.3 and Corollary V.15.9. Then*

$$\begin{aligned}\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,m,\zeta,H,\le\ell}(b) \le \min\Big\{&\mathfrak{M}_{n,m,\zeta}\Big(A_{\mathfrak{U}_\ell}^{H_{\mathfrak{U}}(\ell)}(b_{\mathfrak{U}}(\ell;b,n,m,H))\Big) + \varepsilon_V,\\ &\inf_{j\in\mathcal{J}} U_j(b),\, 1\Big\}.\end{aligned} \tag{1016}$$

Only certificates expressed for the same family law, target event, horizon, and scalarization enter the minimum. Every certificate retains its own complete construction and evaluation cost.

Proof. The first candidate is Eq. (1015). The length-bounded record class is a subset of the unrestricted active class, so every $U_j(b)$ also bounds its profile. Arrival lies in $[0,1]$. Taking the minimum of upper bounds on the same scalar proves the result. □

Corollary V.17.9 (Single-record reduction for uniform computation). *Let q be one family-uniform record code, so $d(q)$ is independent of n. If $b(n), H(n), t_V(n, m(n))$, the interpreter majorant, and the scheduler majorant are polynomial on the declared slice, then $b_{\mathfrak{U}}(q; b, n, m, H)$ is polynomial. Therefore*

$$\begin{aligned}\mathfrak{M}_{n,m,\zeta}(A_q^H) &\text{ nonnegligible at polynomial cost} \\ &\implies \mathfrak{M}_{n,m,\zeta}(A_{\mathfrak{U}}^{H_{\mathfrak{U}}(q)}) \text{ nonnegligible at polynomial cost.}\end{aligned} \tag{1017}$$

provided ε_V is negligible. Since $\mathfrak{U}$ is itself one family-uniform record, negligibility of its polynomial arrival is equivalent to negligibility for the complete family-uniform active class.

Proof. The constant $2^{d(q)}$ in Eq. (1005) is independent of n; the remaining factors are polynomial by hypothesis. Apply Theorem V.17.6. The converse direction follows from membership of $\mathfrak{U}$ in the complete active class. □

V.17.3 Certified relation to the structural learner

Definition V.17.10 (Complete learned-control slack). For a deployed learned record $\widehat{R}$ satisfying the simultaneous action-value hypotheses of Theorem V.9.12, define

$$\rho_{\text{learn}}(\widehat{R}) = \mathbb{E}_{\widehat{R}} \sum_{t=0}^{H-1} \mathbf{1}_{\{\tau>t\}} (2\Gamma_t + \varepsilon_{\text{opt},t}) + \Delta_{\text{fil}}^H. \tag{1018}$$

The radii Γ_t already include their represented approximation, sample, dependence, noise-transport, selection, and same-law conversion costs. The parameter-explicit implementation may instead use the smaller valid candidate in Eq. (921); its deterministic and stochastic coordinates are separated by Theorem V.15.33. For a canonical neural record, define the sharp same-record candidate

$$\rho_{\text{learn}}^{\sharp}(\widehat{R}) = \min \left\{ \begin{array}{l} \rho_{\text{learn}}(\widehat{R}), \\ \left[U_{\text{Bell}} - L_{\text{ev}}\right]_+, \\ B_{\det}(p, K, \boldsymbol{\tau}, b, H) + V_{\text{stat}}(p, N, \boldsymbol{\eta}, \delta), \\ \mathcal{L}_{\text{cert}} \wedge 1 \end{array} \right\}. \tag{1019}$$

An unavailable candidate has value $+\infty$. Each finite entry bounds the same saturated-arrival gap for the same record on the same simultaneous event, so their minimum is valid. For a non-neural active record set $\rho_{\text{learn}}^{\sharp} = \rho_{\text{learn}}$.

Theorem V.17.11 (Structural learner versus length-bounded Levin search). *Fix $(n, m, \zeta, H, b, \ell)$, and train and deploy $\widehat{R}$ at the common ceiling $b_{\mathfrak{U}}(\ell; b, n, m, H)$ and refined horizon $H_{\mathfrak{U}}(\ell; b, n, m, H)$. If its action-value certificate ranges over the complete active suffix class at that ceiling, then on its declared simultaneous event,*

$$\mathfrak{M}_{n,m,\zeta}(A_{\widehat{R}}^{H_{\mathfrak{U}}(\ell)}) \geq \mathrm{ActArr}^{\text{sat}}_{\mathfrak{M},n,m,\zeta,H,\leq\ell}(b) - \varepsilon_V(n, m, \zeta) - \mathfrak{M}_{n,m,\zeta}(\rho_{\text{learn}}^{\sharp}(\widehat{R})). \tag{1020}$$

The upper comparison is membership at the common ceiling:

$$\mathfrak{M}_{n,m,\zeta}(A_{\widehat{R}}^{H_{\mathfrak{U}}(\ell)}) \leq \mathrm{ActArr}^{\text{sat}}_{\mathfrak{M},n,m,\zeta,H_{\mathfrak{U}}(\ell)}(b_{\mathfrak{U}}(\ell; b, n, m, H)). \tag{1021}$$

Proof. The universal record $\mathfrak{U}_\ell$ is one feasible active record at the common ceiling. Hence the saturated active optimum there dominates its arrival. The learned-control loss theorem, followed when necessary by the represented filter transfer, gives a loss of at most $\rho_{\text{learn}}^{\sharp}$ relative to that optimum. The minimum is valid by Eqs. (866), (883) and (921). Combine this fact with Eq. (1015). The upper inequality is the definition of the saturated active profile. □

Definition V.17.12 (Length saturation at a declared ceiling)**.** For a displayed function $\ell_*(n, m, \zeta, b)$ and error δ_{len}, write $\mathsf{LS}(\ell_*, \delta_{\text{len}})$ for the quantitative statement

$$0 \leq \text{ActArr}^{\text{sat}}_{\mathfrak{M},n,m,\zeta,H}(b) - \text{ActArr}^{\text{sat}}_{\mathfrak{M},n,m,\zeta,H,\leq\ell_*}(b) \leq \delta_{\text{len}}(n, m, \zeta, b). \tag{1022}$$

Corollary V.17.13 (Learner comparison under length saturation)**.** *Under* $\mathsf{LS}(\ell_*, \delta_{\text{len}})$,

$$\mathfrak{M}_{n,m,\zeta}(A^{H_{\mathfrak{U}}(\ell_*)}_{\widehat{R}}) \geq \text{ActArr}^{\text{sat}}_{\mathfrak{M},n,m,\zeta,H}(b) - \delta_{\text{len}} - \varepsilon_V - \mathfrak{M}_{n,m,\zeta}(\rho^{\sharp}_{\text{learn}}(\widehat{R})) \tag{1023}$$

at the compiled ceiling $b_{\mathfrak{U}}(\ell_*; b, n, m, H)$*.*

Proof. Insert Eq. (1022) into Eq. (1020). □

V.17.4 Certifiable length and structural advantage

Definition V.17.14 (Blind certifiable-length frontier)**.** For a physical scalar ceiling b_{phys}, define

$$\ell_{\text{cert}}(b_{\text{phys}}; b, n, m, H) = \max \{\ell \in \mathbb{N} : b_{\mathfrak{U}}(\ell; b, n, m, H) \leq b_{\text{phys}}\}, \tag{1024}$$

with value -1 when the set is empty. If $s_\ell \leq s$ throughout the relevant range, then

$$\ell_{\text{cert}} \leq \log_2 \frac{b_{\text{phys}}}{c_{\text{sch}}(s+1)}, \qquad \ell_{\text{cert}} = \log_2 \frac{b_{\text{phys}}}{s+1} - O(\log\log(b_{\text{phys}} + s + 2)), \tag{1025}$$

whenever the right-hand range is nonempty and the machine constants are fixed. Thus blind universal simulation gains one description bit only when its available compute is approximately doubled.

Definition V.17.15 (True and witnessed effective length)**.** Let

$$\vartheta_{\text{learn}} = (p, L, (H_\ell, d_\ell, d_{k,\ell}, d_{v,\ell}, f_\ell)_{\ell\leq L}, N, K, \boldsymbol{\tau}, b, H, \varepsilon_{\text{opt}}) \tag{1026}$$

denote the completely charged architecture, sample, optimizer, temperature, latent-capacity, horizon, and precision choices of Definition V.15.4. For a deployed record $\widehat{R}$, common ceiling b_{phys}, and tolerance $\gamma \geq 0$, define

$$\ell_{\text{eff}}(\widehat{R}; b_{\text{phys}}, \gamma) = \sup \left\{\ell : \mathfrak{M}(A^H_{\widehat{R}}) \geq \text{ActArr}^{\text{sat}}_{\mathfrak{M},H,\leq\ell}(b_{\text{phys}}) - \gamma\right\}. \tag{1027}$$

For a finite represented benchmark set $\mathcal{A}$ of records, define the witnessed profile

$$\text{ActArr}_{\mathcal{A},H,\leq\ell}(b) = \max_{a\in\mathcal{A}:\ d(a)\leq\ell,\ \kappa(\text{Cost}(a))\leq b} \mathfrak{M}(A^H_a) \tag{1028}$$

and let $\ell^{\mathcal{A}}_{\text{eff}}$ be Eq. (1027) with this witnessed profile in place of the saturated profile. Since

$$\text{ActArr}_{\mathcal{A},H,\leq\ell}(b) \leq \text{ActArr}^{\text{sat}}_{\mathfrak{M},H,\leq\ell}(b), \tag{1029}$$

benchmark matching certifies $\ell^{\mathcal{A}}_{\text{eff}}$, not a lower bound on the true ℓ_{eff}. Define the corresponding reported advantages

$$\Delta = \ell_{\text{eff}} - \ell_{\text{cert}}, \qquad \Delta^{\mathcal{A}} = \ell^{\mathcal{A}}_{\text{eff}} - \ell_{\text{cert}}. \tag{1030}$$

Both effective lengths depend on $(n, m, \zeta, \vartheta_{\text{learn}}, b_{\text{phys}}, \gamma)$; this dependence is suppressed only where the displayed record already fixes the coordinates.

Proposition V.17.16 (Simultaneous empirical benchmark certificate)**.** *Let $\mathcal{A}$ be finite. Suppose simultaneous evaluation intervals $[L_a, U_a]$ for $a \in \mathcal{A}$ and $[L_{\widehat{R}}, U_{\widehat{R}}]$ for the learner hold with probability at least $1-\alpha$. If*

$$L_{\widehat{R}} \geq \max_{a\in\mathcal{A}:\ d(a)\leq L,\ \kappa(\mathrm{Cost}(a))\leq b_{\mathrm{phys}}} U_a - \gamma, \tag{1031}$$

then on that event $\ell_{\mathrm{eff}}^{\mathcal{A}} \geq L$. Independent binomial evaluations may use simultaneous Clopper–Pearson intervals; dependent evaluations use the represented martingale or mixing candidate already admitted by the framework.

Proof. On the simultaneous event, the learner's true arrival is at least $L_{\widehat{R}}$, while every eligible benchmark arrival is at most its upper endpoint and hence at most the maximum in Eq. (1031). Therefore the learner is within γ of the witnessed maximum for every $\ell \leq L$, which is the defining condition for $\ell_{\mathrm{eff}}^{\mathcal{A}} \geq L$. □

Proposition V.17.17 (Exact represented benchmark crossover)**.** *For a finite benchmark set $\mathcal{A}$, each serialized record a has the exact row*

$$\Big(d(a),\ \kappa(\mathrm{Cost}(a)),\ \mathfrak{M}(A_a^H),\ \Lambda_{\mathfrak{U}}(d(a), \sigma_{\mathrm{int}}(\kappa(\mathrm{Cost}(a)), d(a)) + Ht_V)\Big)\,. \tag{1032}$$

The fourth coordinate is the physical ceiling at which blind universal simulation is guaranteed to reproduce that benchmark. Code length is computed from the canonical serialized artifact; estimated source-code size is not substituted for $d(a)$. Simultaneous evaluation intervals may replace the third coordinate without changing the other three.

Proof. The first three entries are the definitions of description length, complete cost, and arrival. The final entry is Eq. (1008) with the benchmark's displayed cost. □

V.17.5 Resource-matched benchmark frontiers

The saturated profile and a represented benchmark family live in the same resource order. This permits comparisons at equal cost without collapsing time, acquired data, memory, description length, and evaluation horizon into an informal running-time exponent.

Definition V.17.18 (Represented benchmark frontier and matched-success cost)**.** Let $\mathcal{A}$ be a represented family of complete active records for one presented task family. For a vector ceiling $\mathbf{b} = (b_{\mathrm{time}}, b_{\mathrm{data}}, b_{\mathrm{mem}}, b_{\mathrm{desc}})$, put

$$\mathrm{BenchArr}_{\mathcal{A},H}(\mathbf{b}) = \sup_{a\in\mathcal{A}} \left\{ \mathfrak{M}(A_a^H) : \begin{array}{ll} \mathrm{Cost}_{\mathrm{time}}(a) \leq b_{\mathrm{time}}, & \mathrm{Cost}_{\mathrm{data}}(a) \leq b_{\mathrm{data}}, \\ \mathrm{Cost}_{\mathrm{mem}}(a) \leq b_{\mathrm{mem}}, & d(a) \leq b_{\mathrm{desc}} \end{array} \right\}, \tag{1033}$$

with value zero when the feasible class is empty. The matched-success Pareto set at level $\alpha \in (0,1)$ is

$$\mathcal{P}_{\mathcal{A},H}(\alpha) = \mathrm{Min}_{\preceq} \Big\{ \mathrm{Cost}(a) : a \in \mathcal{A},\ \mathfrak{M}(A_a^H) \geq \alpha \Big\}\,, \tag{1034}$$

where $\mathrm{Min}_{\preceq}$ retains the nondominated vectors in the existing componentwise resource order. These definitions use the same presentation law, target, verifier, and horizon as the saturated profile.

Theorem V.17.19 (Saturated headroom and simultaneous benchmark comparison)**.** *Let $\mathcal{A}$ be as in Definition V.17.18, and let $\widehat{R}$ be a deployed represented learner. At a common vector ceiling $\mathbf{b}$, define the saturated benchmark headroom*

$$\mathrm{Head}_{\mathcal{A}}(\mathbf{b}) = \mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},H}(\mathbf{b}) - \mathrm{BenchArr}_{\mathcal{A},H}(\mathbf{b}). \tag{1035}$$

Then $\mathrm{Head}_{\mathcal{A}}(\mathbf{b}) \geq 0$*.*

Suppose $[L_{\mathrm{sat}}, U_{\mathrm{sat}}]$*,* $[L_{\mathcal{A}}, U_{\mathcal{A}}]$*, and* $[L_{\widehat{R}}, U_{\widehat{R}}]$ *are simultaneous valid intervals for the saturated profile, benchmark frontier, and learner arrival. On their common event,*

$$\big[L_{\mathrm{sat}} - U_{\mathcal{A}}\big]_+ \leq \mathrm{Head}_{\mathcal{A}}(\mathbf{b}) \leq U_{\mathrm{sat}} - L_{\mathcal{A}}, \tag{1036}$$

$$L_{\widehat{R}} - U_{\mathcal{A}} \leq \mathfrak{M}(A^H_{\widehat{R}}) - \mathrm{BenchArr}_{\mathcal{A},H}(\mathbf{b}) \leq U_{\widehat{R}} - L_{\mathcal{A}}. \tag{1037}$$

For a declared denominator floor $\epsilon_0 > 0$, the multiplicative gain obeys

$$\frac{L_{\widehat{R}}}{\max\{U_{\mathcal{A}}, \epsilon_0\}} \leq \frac{\mathfrak{M}(A^H_{\widehat{R}})}{\max\{\mathrm{BenchArr}_{\mathcal{A},H}(\mathbf{b}), \epsilon_0\}} \leq \frac{U_{\widehat{R}}}{\max\{L_{\mathcal{A}}, \epsilon_0\}}. \tag{1038}$$

If the learner certificate gives

$$\mathfrak{M}(A^H_{\widehat{R}}) \geq \mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},H}(\mathbf{b}) - \rho_{\mathrm{learn}}, \tag{1039}$$

then

$$\mathfrak{M}(A^H_{\widehat{R}}) - \mathrm{BenchArr}_{\mathcal{A},H}(\mathbf{b}) \geq \mathrm{Head}_{\mathcal{A}}(\mathbf{b}) - \rho_{\mathrm{learn}}. \tag{1040}$$

Thus the learner realizes the saturated improvement over the represented benchmark family up to its already charged learning slack.

Proof. Every benchmark record is an admissible represented record in the saturated class at the same ceiling. Hence $\mathrm{BenchArr}_{\mathcal{A},H}(\mathbf{b}) \leq \mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},H}(\mathbf{b})$, proving nonnegativity. Subtracting the endpoints of simultaneous intervals gives Eqs. (1036) and (1037). Monotonicity of division on $(0, \infty)$, after applying the declared floor, gives Eq. (1038). Finally, subtract the benchmark frontier from Eq. (1039) and use Eq. (1035). □

Corollary V.17.20 (Matched-success resource savings and Levin gain)**.** *Fix $\alpha \in (0,1)$. Let $\mathbf{c}_{\widehat{R}}$ be the learner's complete cost and let $\mathbf{c}_a \in \mathcal{P}_{\mathcal{A},H}(\alpha)$. For each positive resource coordinate j, define the matched-success logarithmic saving*

$$\mathrm{Save}_j(\widehat{R}; a, \alpha) = \log_2 \frac{c_{a,j}}{c_{\widehat{R},j}}. \tag{1041}$$

It is positive precisely when the learner uses less of resource j than that nondominated benchmark at the same certified arrival level.

If a has canonical code length $d(a)$, the blind physical ceiling for reproducing it is the fourth coordinate of Eq. (1032)*. When the simultaneous condition in Eq.* (1031) *holds through $L = d(a)$, the certified description-length reach beyond blind search is*

$$\Delta^{\mathcal{A}} \geq d(a) - \ell_{\mathrm{cert}}. \tag{1042}$$

Proof. The first assertion is the definition of a logarithmic ratio. The second is Proposition V.17.16 followed by Eq. (1030). □

Theorem V.17.21 (Finite-grid nonidentifiability of the asymptotic length profile)**.** *Fix any finite grid $\mathcal{G} \subset \mathbb{N}^2$ of size–length pairs and any finite collection of valid profile bounds evaluated only on $\mathcal{G}$. Those bounds alone do not imply that, for every fixed ℓ,*

$$\mathrm{ActArr}^{\mathrm{sat}}_{\mathfrak{M},n,H,\le\ell}(\mathrm{poly}(n)) \text{ is negligible as } n \to \infty. \tag{1043}$$

Proof. Let $n_{\max}$ and $\ell_{\max}$ be the largest coordinates inspected by $\mathcal{G}$. Extend any presentation outside the inspected slices by adjoining one family-uniform record of length $\ell_{\max}+1$ that returns the neutral output for $n \le n_{\max}$ and reaches the target at polynomial cost for $n > n_{\max}$. This changes none of the finitely evaluated profile values, but it violates Eq. (1043) at the fixed length $\ell_{\max}+1$. Hence the finite statements do not entail the asymptotic quantifier. □

V.17.6 LPN verifier and parameter-explicit specialization

Corollary V.17.22 (Length-bounded Levin relation for LPN)**.** *In the LPN presentation of Definition VIII.1.1, let $V_I(x) = \mathbf{1}_{\{|Ax+b|_1 \le \tau m\}}$ with $\eta < \tau < H_2^{-1}(1-R)$ and $R = n/m$. A direct matrix–vector evaluation has*

$$t_V(n,m) \le c_V mn \tag{1044}$$

bit operations for a fixed implementation constant c_V, and

$$\varepsilon_V(n,m,\eta,\tau) \le \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] + (2^n-1)2^{-m(1-H_2(\tau))}. \tag{1045}$$

Consequently, the uniform length-class simulation ceiling is

$$b^{\mathrm{LPN}}_{\mathfrak{U}}(\ell; b, n, m, H) = \Lambda_{\mathfrak{U}}\left(\ell, \sup_{d\le\ell} \sigma_{\mathrm{int}}(b,d) + c_V Hmn\right), \tag{1046}$$

Put

$$\widetilde{H}^{\mathrm{LPN}}_{\mathfrak{U}} = H_{b^{\mathrm{LPN}}_{\mathfrak{U}}(\ell;b,n,m,H)}. \tag{1047}$$

At physical ceiling b_{phys}, the LPN blind frontier is therefore

$$\ell_{\mathrm{cert,LPN}} = \max\left\{\ell : b^{\mathrm{LPN}}_{\mathfrak{U}}(\ell; b,n,m,H) \le b_{\mathrm{phys}}\right\}, \tag{1048}$$

$$\ell_{\mathrm{cert,LPN}} = \log_2 \frac{b_{\mathrm{phys}}}{\sup_{d\le\ell_{\mathrm{cert,LPN}}} \sigma_{\mathrm{int}}(b,d) + c_V Hmn + 1} - O(\log\log(b_{\mathrm{phys}} + b + Hmn + 2)). \tag{1049}$$

In the direct-verification-dominated range this reduces to $\log_2 b_{\mathrm{phys}} - \log_2(Hmn) - O(\log\log b_{\mathrm{phys}})$. *Then Eq.* (1015) *becomes*

$$\begin{aligned}\mathrm{ActArr}^{\mathrm{sat}}_{\mathrm{LPN},n,m,\eta,H,\le\ell}(b) \le \mathfrak{M}_{n,m,\eta}&\left(A^{\widetilde{H}^{\mathrm{LPN}}_{\mathfrak{U}}}_{\mathfrak{U}_\ell}(b^{\mathrm{LPN}}_{\mathfrak{U}})\right) \\ &+ \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] + (2^n-1)2^{-m(1-H_2(\tau))}.\end{aligned} \tag{1050}$$

For the learned LPN record, the right-hand side of Eq. (1020) *additionally carries the explicit noise-dependent action-value radius*

$$\Gamma^{\mathrm{clean}}_{\mathrm{act}} \le \frac{1}{1-2\eta}\left[2\sqrt{\frac{2L_{\mathrm{act}}(n,b,m,\eta)}{m_{\mathrm{eff}}}} + 3\sqrt{\frac{\log(2\widetilde{H}^{\mathrm{LPN}}_{\mathfrak{U}}/\delta)}{2m_{\mathrm{eff}}}} + \varepsilon_{\mathrm{dep}}\right] \tag{1051}$$

and finite-horizon learning loss at most

$$\widetilde{H}_{\mathfrak{U}}^{\mathrm{LPN}}(2\Gamma_{\mathrm{act}}^{\mathrm{clean}} + \varepsilon_{\mathrm{opt}}) + \Delta_{\mathrm{fil}}^{\widetilde{H}_{\mathfrak{U}}^{\mathrm{LPN}}}. \tag{1052}$$

For a canonical neural LPN implementation, the sharp complete learning candidate is

$$\rho_{\mathrm{learn,LPN}}^{\sharp} = \min \left\{ \begin{array}{l} \widetilde{H}_{\mathfrak{U}}^{\mathrm{LPN}}(2\Gamma_{\mathrm{act}}^{\mathrm{clean}} + \varepsilon_{\mathrm{opt}}) + \Delta_{\mathrm{fil}}^{\widetilde{H}_{\mathfrak{U}}^{\mathrm{LPN}}}, \\ \left[U_{\mathrm{Bell,LPN}} - L_{\mathrm{ev,LPN}}\right]_{+}, \\ B_{\mathrm{det,LPN}}(p, K, \boldsymbol{\tau}, b_{\mathfrak{U}}^{\mathrm{LPN}}, \widetilde{H}_{\mathfrak{U}}^{\mathrm{LPN}}) + V_{\mathrm{stat,LPN}}(p, N, \eta, \delta), \\ \mathcal{L}_{\mathrm{cert,LPN}} \wedge 1 \end{array} \right\}. \tag{1053}$$

Every candidate retains the same n, m, η, H, compiled horizon $\widetilde{H}_{\mathfrak{U}}^{\mathrm{LPN}}$, architecture, transition law, target gate, and compiled ceiling. The finite-description first row may be replaced by any smaller simultaneous sequential-Rademacher, martingale PAC–Bayes, mixing, or predictable-variance radius admitted by Theorem V.15.33.

The length-bounded profile also inherits every unrestricted LPN upper certificate. In particular, define

$$U_{\mathrm{Bell,LPN}}^{\mathrm{sat}} = \inf_{v_H = 0} \left\{ \mathfrak{M}_{n,m,\eta}(v_0(h_0)) + \sum_{t=0}^{H-1} \max_{c \in \mathfrak{C}_{\mathrm{master}}} \mathfrak{U}_{t,c}^{\mathrm{LPN}}(v) \right\} + \varepsilon_{\mathrm{good}}(n, m, \eta, \tau, \delta). \tag{1054}$$

Then the sharp integrated length-class upper bound is

$$\begin{aligned} \mathrm{ActArr}_{\mathrm{LPN},n,m,\eta,H,\leq \ell}^{\mathrm{sat}}(b) \leq \min\Big\{ & \mathfrak{M}_{n,m,\eta}\left(A_{\mathfrak{U}_\ell}^{\widetilde{H}_{\mathfrak{U}}^{\mathrm{LPN}}}(b_{\mathfrak{U}}^{\mathrm{LPN}}) \right) + \varepsilon_V(n, m, \eta, \tau), \\ & U_{\mathrm{Bell,LPN}}^{\mathrm{sat}} \Big\}. \end{aligned} \tag{1055}$$

The Bellman candidate expands mode by mode through Eqs. (1530) *and* (1531)*; hence its information, capacity, Blackwell, source, quotient-compensation, direct-contact, and residual-Gibbs terms remain available to the minimum. Thus n, m, η, H, description length, action-class capacity, dependent sample size, optimization, filtering, verifier error, and physical compute all occur in displayed coordinates. For a represented subset-Gaussian-elimination benchmark, the arrival entry in Eq.* (1032) *is exactly $(1-\eta)^n$ on the full-rank event and equals $n^{-cR+o(1)}$ when $\eta m = c \log n$, by Eqs.* (1396) *and* (1398)*. A represented located-spectral benchmark instead carries the exponent and budget threshold in Eqs.* (1400) *and* (1401)*. Its complete fixed-data and streaming BKW realization, including sample reuse, decoder error, time, data, memory, and canonical code length, is Theorem VIII.13.9 and Corollary VIII.13.15. These are benchmark-specific entries of the witnessed profile; the saturated length-class comparison remains Eq.* (1055)*.*

Proof. The verifier cost is the cost of computing $Ax + b$ and its Hamming weight. The global gate event and its binomial/random-code bound are Eq. (1184). Substitute these values into Eqs. (1013) and (1015). The learning radius and finite-horizon propagation are Eqs. (603) and (604), with the represented filter comparison added as in Corollary V.11.16. The minimum in Eq. (1053) follows from Eq. (1019). Finally, a restricted length class is contained in the unrestricted active class; combine Eq. (1050) with Eq. (1528) to obtain Eq. (1055). □

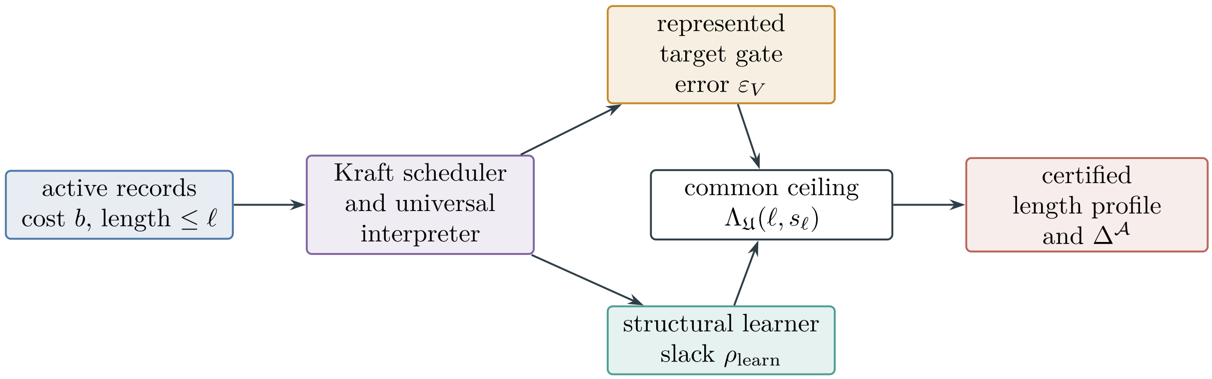

Figure 113: Length-bounded Levin comparison. Blind universal search pays the Kraft factor 2^ℓ together with interpreter, scheduler, and verifier costs. The structural learner is compared at the same compiled ceiling and carries its complete statistical, optimization, filtering, and deployment slack. Empirical comparisons certify the witnessed frontier $\ell^{\mathcal{A}}_{\mathrm{eff}}$; the true frontier retains the saturated length-class supremum.

V.18 Summary of the Dynamical-Systems Geometry

Coordinate	Structural record	Framework result
finite-horizon arrival	pre-hit selector and neutral baseline	Theorems III.21.3 and III.21.4
discounted arrival	killed generator and target gate	target resolvent; master certificate
expected hitting time	authorized drift potential	Theorem V.6.6
reach–avoid	committor and condenser geometry	Theorems V.6.2 and V.6.6
mixing and compression	invariant reversible carrier	Theorem V.5.3
functional rates	audited typed inequality profile	Theorems V.5.8 and V.5.10
authorized target flow	Caus action, aiming energy, and contact converter	Theorem V.7.2 and Corollary V.7.4
noise-corrected Caus flow	clean-target comparison and corrupted binary supervision	Theorems IV.16.6 and V.7.7
lifted condenser capacity	valid Lift and boundary-zero fiber form	Theorem V.7.9 and Definition V.7.10
lifted target access	intertwining defect and fiber resolvent	Theorems V.7.12 and V.7.13
noisy Lift interface	target/failure deployment gates, killed contact vector, and joint capacity stability	Definition V.8.1 and Theorem V.8.2
reward and execution noise	typed reward correction, actual executed kernel, and Bellman/path converters	Theorems IV.17.2 and V.1.4 and Definition V.1.3
observation and POMDP noise	public filtration, clocked latent carrier, filter stability, and gate transfer	Theorems V.2.2 and V.2.4
multichannel entropy	conditional capacity, path KL/TV, posterior entropy, Fano, and rate–distortion	Theorems V.1.5, V.3.2 and V.3.7 and Definition V.4.1
optimal active sampling	represented acquisition optimizer, declared experiment, selected transport, and master quotient–geometry certificate	Theorems V.9.5, V.9.8 and V.9.12
affordable occupation duality	stopped history flow, target flux, and resource-indexed Bellman barrier	Theorems V.10.2, V.10.4 and V.10.5
optimal structural Bayesian learning	declared task prior, represented likelihood and belief, source-resolved actionable gain, and charged acquisition	Theorems V.11.2, V.11.4, V.11.8, V.11.10 and V.11.12
trajectory learning	mixing blocks, predictable history trees, and martingale priors	Theorem IV.15.2; Theorem IV.15.5; Theorem IV.15.6
action and safety capacity	action-generated and failure-killed operator compressions	Theorem V.13.3
value and policy loss	represented Bellman residual	Theorem V.12.1
learned control	learner–geometry–policy record	policy accountability; master certificate
certified neural implementation	complete network and optimizer transcript, held-out arrival audit, and saturated Bellman barrier	Corollary V.15.9 and Theorems V.15.23 and V.15.31
canonical neural controller	recurrent belief token, masked attention, policy/value/barrier heads, and explicit parameter ledger	Definition V.15.4 and Theorems V.15.5 and V.15.29
bias–variance diagnosis	deterministic approximation ledger, predictable variance, noise transport, and simultaneous tuning	Theorem V.15.33, Proposition V.15.34, and Corollary V.15.35
length-bounded universal search	prefix-free active-record code, Kraft scheduler, verifier gate, and structural-learning slack	Theorems V.17.6 and V.17.11 and Definition V.17.14
empirical analytical proof	parameter profile, source audit, feature Dirichlet form, and quotient-row diameter	Theorems V.15.3, V.15.25, V.15.37, V.15.40 and V.15.44
continuous transfer	valid Lift and generator defect	Corollary V.14.5

The universal conclusion is Theorem V.12.3: every represented controlled system is classified by the existing structural channels and receives a coordinatewise certificate at its complete saturated cost. Arrival, reach, mixing, policy error, and generalization retain their own exact identities and the strongest applicable represented comparison. Target-directed Caus flow enters an arrival bound through an aiming-to-contact converter. Lifted access enters through a form, intertwining, clocked-defect, or fiber-resolvent comparison. The unrestricted valid-lift class remains the explicit saturated lift-access supremum of Definition V.7.15. The stopped occupation-flow identity and Bellman barrier duality in Theorems V.10.2 and V.10.4 also evaluate the best affordable active controller. Their support terms are derived coordinates over the five existing master modes. The learned controller is compared with this optimum by Corollary V.10.7; estimation and optimization errors are not substituted for the structural optimum. The implemented neural controller is connected to the same optimum by Corollary V.15.9. A held-out execution audit provides the primal lower certificate, while an exact or coverage-certified Bellman audit provides the saturated upper certificate. The canonical implementation in Definition V.15.4 fixes its tokenizer, recurrent belief, masked attention, policy, value, barrier, geometry, and quotient interfaces and counts their parameters and operations explicitly. Its trainable policy and likelihood losses are calibrated to saturated arrival by Theorem V.15.29; its analytical optimality objective is the primal–dual gap Eq. (861). The WAdam specialization in Theorem V.15.31 converts its represented descent, second-moment, PL, and objective-to-action comparisons into the optimizer coordinate of the end-to-end gap. The bias–variance ledger in Theorem V.15.33 separates representation, optimization, filtering, precision, truncation, sufficiency, and temperature effects from the simultaneous statistical radius. Parameter count, temperature, label noise, horizon, clipping, latent capacity, feature dimension, and quotient resolution are selected through the complete simultaneous upper curve in Corollary V.15.35. The parameter-indexed profile in Definition V.15.2 separates certified underfitting from optimizer error and brackets the minimum sufficient parameter count. The same run can attribute learned target gain to the five structural sources by Theorem V.15.25, transfer feature-level Dirichlet estimates to true analytical functional-inequality constants by Theorems V.15.37 and V.15.38, and certify exact or quasi-lumpability by Theorems V.15.40 and V.15.41. Exact lumpability is obtained when every pair of states carrying the same label has the same probability of entering each next label: this is exactly the row-diameter condition $\delta_q(P) = 0$, and it yields the intertwining $PU_q = U_q\bar{P}$. A nonzero certified diameter remains an explicit finite-horizon and generator defect rather than being treated as an exact quotient. The structural Bayesian restriction in Section V.11 separates the declared task prior, analytic task posterior, represented belief, and PAC–Bayes posterior. Prior selection independent of the target retains neutral contact, while a target-dependent initialization is the first charged observation and belief update. Posterior concentration bounds target contact by Theorem V.11.4, and every prospectively used concentration readout retains the same five-mode provenance by Theorem V.11.15. The deployed next-test improvement is the actionable gain of Definition V.11.7; its exact five-source charge, noise-contracted information ledger, and adaptive composition are Theorems V.11.8 and V.11.10 and Corollary V.11.11. When a binary target or failure interface is learned from noisy labels, the same coordinates additionally carry the explicit clean-risk factor $(1-2\eta_+)^{-1}$, the affordable ceiling $b(n)$, and sample size m. Only represented same-norm calibration, deployment, contact-vector, and reference-law comparisons propagate that statistical bound into Caus action, Lift arrival, or condenser capacity. The native deterministic capacity and resolvent inequalities themselves are unchanged. Reward, execution, observation, and filter corruption are now treated by the same discipline. Their raw $(n, b(n), m)$-indexed errors remain typed; executed actions define the actual generator, observation-measurable policies define the affordable POMDP class, and filter errors enter arrival or return only through the clocked or Bellman converters. Physical-channel entropy, stochastic-transition entropy, policy entropy, public channel capacity in bits, path KL in nats, MLSI density entropy, Caus edge action, PAC–Bayes KL, and description length remain distinct coordinates. The only target-contact use of Caus entropy production is the same-record product with aiming followed by a contact converter. Multichannel computational monotonicity is asserted only along represented Blackwell garblings.

Part VI

Attention as Structural Gradient Flow

VI.0 Represented Attention Records

Part contract. *Inputs:* the saturated constructor algebra of Parts I–III, the source-resolved learning ledger of Part IV, and the controlled-dynamics, active-sampling, and structural-Bayesian certificates of Part V. *Output:* an exact representation and source-charging theorem for attention layers and their deployed transition kernels. *First pass:* read Definition VI.0.1 and Theorems VI.1.1, VI.1.3 and VI.2.2.

Attention is used here as a represented mechanism for selecting and transporting public values. It is not adjoined as a new structural channel. Every attention coordinate is compiled into the already exhaustive $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}, J_{\mathrm{res}}$ ledger and retains the five-family constructor provenance of its score and value records. The canonical finite-precision realization is Definition V.15.4. Its exact parameter and operation ledgers are Eqs. (795) and (798), and its policy, belief, critic, and certificate objectives are Definition V.15.28.

Definition VI.0.1 (Complete represented attention record)**.** A complete represented attention record on a finite token carrier $\mathcal{I}$ is

$$\mathcal{R}^{\mathrm{att}} = \big(\mathcal{R}, \Theta_{\mathrm{att}}, \mathrm{Tok}, \nu, Q, K, V, O, s, P_s, Y, \mathrm{Deploy}\big),$$

where $\mathcal{R}$ is the complete ancestral record and

$$\Theta_{\mathrm{att}} = (L, (H_\ell, N_\ell, d_\ell, d_{k,\ell})_{\ell \le L}, (\tau_{\ell h})_{\ell,h}, p, R_Q, R_K, R_V, R_O)$$

records depth, heads, token counts, widths, key widths, temperatures, arithmetic precision, and represented parameter records. For one head, $\nu(\cdot \mid i)$ is the represented masked reference kernel,

$$s(i,j) = \langle Q z_i, K z_j \rangle, \qquad P_s(j \mid i) = \frac{\nu(j \mid i) e^{s(i,j)/\tau}}{\sum_k \nu(k \mid i) e^{s(i,k)/\tau}}, \qquad Y_i = O \sum_j P_s(j \mid i) V z_j.$$

Multihead concatenation, residual addition, normalization, feed-forward maps, recurrence, and a final selector are represented constructors in the same record. The complete attention cost $\mathrm{Cost}_{\mathrm{att}}(\mathcal{R}^{\mathrm{att}})$ charges tokenization, masks, all parameter derivations, score evaluation, normalization at precision p, value transport, heads, residual and feed-forward paths, training, selection, and deployment. A target-dependent parameter, mask, or initialization is charged as the first target-dependent ancestor.

The record is *temporally admissible* when every row deployed after time t factors through the pre-hit record of Definition III.21.1. It is *budget admissible* when its complete cost lies in the one declared saturated ceiling.

VI.1 Gibbs Geometry and Exact Structural Charging

Theorem VI.1.1 (Gibbs variational identity and Fisher–Rao attention flow)**.** *Fix a row i, write $\nu_j = \nu(j \mid i)$, and assume $\nu_j > 0$ on its represented support. For every probability vector $p \ll \nu$,*

$$\frac{1}{\tau}\mathbb{E}_p s - \mathrm{KL}(p \| \nu) = \log \mathbb{E}_\nu e^{s/\tau} - \mathrm{KL}(p \| P_s). \tag{1056}$$

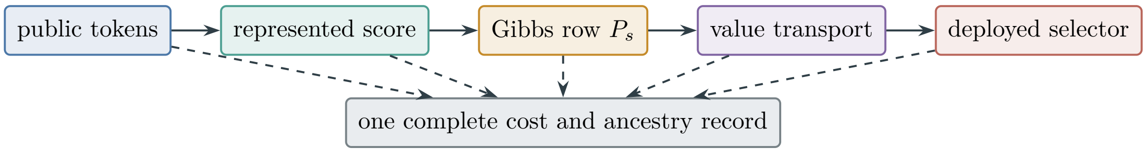

Figure 114: A represented attention head. Normalization and value transport do not erase the ancestry or cost of the score used by the deployed selector.

Consequently P_s is the unique maximizer of the left-hand functional. Along

$$p_t(j) = \frac{\nu_j e^{ts_j/\tau}}{\sum_k \nu_k e^{ts_k/\tau}}, \qquad 0 \le t \le 1,$$

one has

$$\dot{p}_t(j) = \tau^{-1} p_t(j)(s_j - \mathbb{E}_{p_t} s), \tag{1057}$$

$$\|\dot{p}_t\|_{\mathrm{FR}}^2 = \tau^{-2} \operatorname{Var}_{p_t}(s), \tag{1058}$$

$$\mathrm{KL}(P_s \| \nu) = \tau^{-1} \mathbb{E}_{P_s} s - \log \mathbb{E}_\nu e^{s/\tau}. \tag{1059}$$

Thus softmax is a Fisher–Rao natural-gradient path on the represented row simplex. This statement does not identify an arbitrary parameter-training trajectory with Euclidean gradient flow.

Proof. Substitute $\log(P_s(j)/\nu_j) = s_j/\tau - \log \mathbb{E}_\nu e^{s/\tau}$ in $\mathrm{KL}(p\|P_s)$ and rearrange. Nonnegativity and strict convexity of relative entropy give the maximizer. Differentiating the normalized exponential family proves Eq. (1057). The Fisher–Rao metric is $\|v\|_{\mathrm{FR}}^2 = \sum_j v_j^2/p_j$; substitution proves Eq. (1058). Setting $p = P_s$ in Eq. (1056) proves Eq. (1059). □

Corollary VI.1.2 (Bayesian posterior as represented attention flow)**.** *Let $\mathcal{C}_t$ be a finite represented latent hypothesis or cell carrier retained by a compact latent-belief record, let q_t be its current full-support belief on $\mathcal{C}_t$, and let $L_t(y \mid z) > 0$ be the represented likelihood of the acquired observation. The stored code for q_t, its carrier, and its decoder are included in the latent capacity and complete cost of Definition V.11.21. Choose one attention row with*

$$\nu_t(z) = q_t(z), \qquad s_t(z) = \tau \log L_t(y \mid z). \tag{1060}$$

Then its Gibbs row is the exact latent Bayesian update:

$$P_{s_t}(z) = \frac{q_t(z) L_t(y \mid z)}{\sum_u q_t(u) L_t(y \mid u)}. \tag{1061}$$

The Fisher–Rao flow and action in Eqs. (1057) *and* (1058) *are therefore the Bayes–Fisher flow and action in Eqs.* (689) *and* (690) *on the same represented latent simplex.*

Suppose a represented score $\widehat{\ell}_t$ approximates $\ell_t = \log L_t(y \mid \cdot)$ on that support with

$$\|\widehat{\ell}_t - \ell_t\|_\infty \le \varepsilon_{\ell,t}. \tag{1062}$$

Let $\widehat{q}_{t+1}$ be the normalized exponential update using $\widehat{\ell}_t$, and let q_{t+1} be the exact update. Then

$$D_{\mathrm{KL}}(q_{t+1} \| \widehat{q}_{t+1}) \le 2\varepsilon_{\ell,t}, \tag{1063}$$

$$\|q_{t+1} - \widehat{q}_{t+1}\|_{\mathrm{TV}} \le \sqrt{\varepsilon_{\ell,t}}. \tag{1064}$$

The likelihood evaluator, logit approximation, exponentiation, normalization, precision, latent storage, and downstream use are charged respectively through the complete latent-belief and attention records.

Proof. Substitute Eq. (1060) in the Gibbs row of Definition VI.0.1; this proves Eq. (1061). The two exponential paths then coincide, which gives the Fisher–Rao assertion.

Write Z and $\widehat{Z}$ for the exact and approximate normalizers. The uniform logit error gives $e^{-\varepsilon_{\ell,t}}Z \le \widehat{Z} \le e^{\varepsilon_{\ell,t}}Z$. Hence

$$\left|\log \frac{q_{t+1}(z)}{\widehat{q}_{t+1}(z)}\right| \le 2\varepsilon_{\ell,t}$$

on the common support. Taking expectation under q_{t+1} proves the KL bound. Pinsker's inequality proves the TV bound. □

Theorem VI.1.3 (Attention structural-source compilation)**.** *Every temporally and budget-admissible complete attention record has the following exhaustive source assignment, at its complete charged ceiling:*

attention coordinate	*framework coordinate*
token/key carrier and reversible row law	Geom
qualified nonidentity score fibers	*exact or compensated* Abs
target-oriented score gradient/current	*authorized* Caus *of that* Geom
values, heads, residual paths, depth, memory	Lift
masks, support truncation, terminal interface	Bdry
gradient-orthogonal transport	J_{res}
score and value constructors	$\mathsf{Add}, \mathsf{Mult}, \mathsf{Ord}, \mathsf{Sym}, \mathsf{Filt}$ *provenance*

No row normalization, head concatenation, residual connection, or clocked reuse creates an additional source coordinate.

Proof. Apply selected-generator exhaustiveness Theorem V.1.2 to each deployed row. The reversible and Hodge components give Geom, authorized Caus, and J_{res}. A represented score fiber passes through the exact or compensated quotient gates of Theorem III.2.6; a fiber that fails those gates remains full-carrier Geom, not Abs. Value transport and stored head state are represented contextual extensions, so Theorem II.10.5 assigns them to Lift. Masks and support changes are assigned by Theorem II.5.10. Finally apply the five-family provenance theorem and constructor no-creation to the finite ancestral DAG. All derived vertices retain their ancestors and complete cost, proving both exhaustiveness and no creation. □

VI.2 Attention Target-Gain Envelope

Definition VI.2.1 (Source-resolved attention ledger)**.** For a complete attention record R deployed at time t, let $g^{\mathrm{att}}_{R,t}$ be its positive pre-hit selector advantage over the declared neutral row. Its source information is

$$J^{\mathrm{att,src}}_{R,t} = \sum_{a\in\mathfrak{I}^5_{\mathsf{Abs}}} J^{\mathrm{att,(a)}}_{R,t},$$

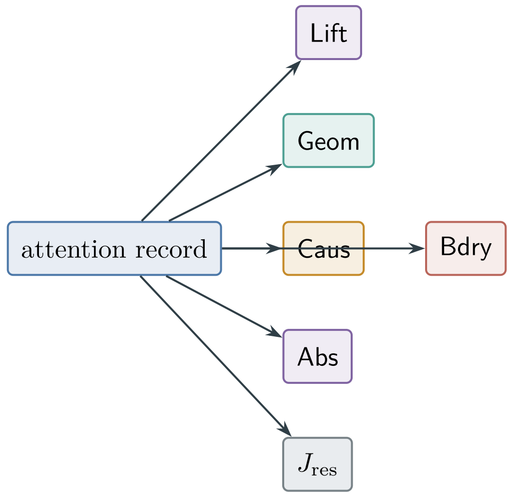

Figure 115: Attention introduces no seventh structural modality. Every useful coordinate is assigned to the existing exhaustive ledger.

with the same five-family coordinates as Definition IV.13.2. Let $\mathcal{L}_{R,t}^{\mathrm{att,(a)}}$ be the corresponding qualified source visibility, including the Geom/Abs support mode, and let $U_{R,t}^{\mathrm{KL}}, U_{R,t}^{\mathrm{stat}\to\mathrm{tgt}}, U_{R,t}^{\mathrm{dyn}\to\mathrm{tgt}}$ be the already-defined relative-entropy, statistical-calibration, and dynamical contact converters for this same record. Define

$$U_{R,t}^{\mathrm{att}} = \min\left\{U_{R,t}^{\mathrm{KL}}, U_{R,t}^{\mathrm{stat}\to\mathrm{tgt}}, \beta_{R,t} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{att,src}}}, eH_B \sum_a \mathcal{L}_{R,t}^{\mathrm{att,(a)}}, U_{R,t}^{\mathrm{dyn}\to\mathrm{tgt}}\right\}. \tag{1065}$$

An unavailable converter contributes the neutral value one and therefore cannot improve the minimum.

Theorem VI.2.2 (Source-resolved attention target-gain bound)**.** *Every temporally admissible complete attention record satisfies*

$$0 \le g_{R,t}^{\mathrm{att}} \le U_{R,t}^{\mathrm{att}}. \tag{1066}$$

At a saturated ceiling B,

$$\mathrm{AttSel}^{\mathrm{sat}}(B) \le \mathrm{Sel}^{\mathrm{sat}}(B) \le I^{\mathrm{src,sat}}(B). \tag{1067}$$

The same bounds hold for finite-precision hard attention by taking the represented zero-temperature limit at its charged precision and tie-breaking cost.

Proof. The five entries in Eq. (1065) are, respectively, the represented change-of-measure bound, the source-resolved learning calibration bound, Pinsker's information-to-contact conversion, the contextual source-transfer bound, and the controlled-dynamics arrival converter. Each is applied to the same complete record and common law; hence their minimum is valid. The first inequality in Eq. (1067) is inclusion of represented attention selectors in the temporally admissible selector profile. The second is prospective selector source routing from Theorems IV.13.3 and V.6.4. Finite-precision argmax is a represented Ord postprocessing; constructor no-creation retains the score ancestry and charges precision and tie breaking. □

Corollary VI.2.3 (Attention cannot bypass optimal active sampling)**.** *At the class-preserving compilation ceiling of Theorem V.9.8, every attention-driven acquisition or transition rule is one* *feasible complete active-sampling record. Its arrival, posterior gain, Caus action, Lift transfer, and sparse-reward value are bounded by the corresponding coordinates of that master envelope. Conversely, every bounded-branch clocked selector can be represented by finite-precision hard attention over its represented legal actions: expose the selector's already represented auxiliary seed, apply its represented inverse-transform comparisons, and give the selected action the unique maximal score. Averaging over that same seed recovers the original kernel. The seed, comparisons, tie breaking, ancestry, and compilation cost are retained. Thus attention changes the representation of the selector, not the saturated structural optimum. That optimum is the structural Bayesian and represented-computation ceiling of Theorem V.11.19; its explicit noise, sample, and target-size bound is Corollary V.11.20.*

Proof. The forward statement is inclusion. For the converse, tokenize the finitely many represented legal actions and expose the selector's already charged auxiliary seed. Its represented inverse-transform comparisons select one legal action; assign that action the unique maximal score, mask illegal actions, and take represented argmax with the original tie-breaking rule. This reproduces the selector pathwise for each seed and therefore reproduces its transition law after averaging over that seed. The construction is a clocked Lift of the original selector and retains all source ancestors by Theorem VI.1.3. □

Theorem VI.2.4 (Joint compact-latent attention and optimal acquisition)**.** *Let* $\mathrm{LatAttArr}^{\mathrm{sat}}_{I,H}(B,b)$ *be the restriction of Eq.* (708) *to complete records whose represented latent update or deployed selector is implemented by Definition VI.0.1. Then*

$$\mathrm{LatAttArr}^{\mathrm{sat}}_{I,H}(B,b) \leq \mathrm{LatBayesArr}^{\mathrm{sat}}_{I,H}(B,b). \tag{1068}$$

At the class-preserving finite-precision hard-attention compilation ceiling, every bounded-branch compact latent selector has an attention representation with the same pre-hit path law, source ancestry, and latent code. Therefore the two profiles coincide on their saturated transfer class. Taking the union over code capacities admitted by the polynomial saturated budget gives

$$\mathrm{LatAttArr}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{LatBayesArr}^{\mathrm{poly,sat}}_{\mathfrak{M},n} = \mathrm{Succ}^{\mathrm{poly,sat}}_{\mathfrak{M},n}. \tag{1069}$$

For every one of these records, the deployed gain is bounded by the same-record minimum. Writing $g^{\mathrm{lat,att}}_{R,t}$ *for the actionable Bayesian gain Eq.* (652) *of its deployed attention selector,*

$$g^{\mathrm{lat,att}}_{R,t} \leq \min\left\{U^{\mathrm{lat}}_{R,t}, U^{\mathrm{att}}_{R,t}\right\}. \tag{1070}$$

Thus optimal data acquisition selects the observation channel, the compact latent record controls retained information and sufficiency, attention realizes the represented posterior or selector flow, and the Bellman recursion selects their optimal affordable continuation.

Proof. The first inequality is class inclusion. For the converse, tokenize the finitely many represented legal actions of each compact latent state, expose the selector's already charged auxiliary seed, and use its represented inverse-transform comparisons to assign the selected action the unique maximal score. Mask illegal actions and apply Corollary VI.2.3. The construction preserves the selector kernel and code state and retains the original score ancestry and compilation cost. Combine this equality with Eq. (713) to obtain the polynomial identity. Both entries in Eq. (1070) bound the same deployed gain under the same conditional law by Theorems V.11.25 and VI.2.2; their minimum is therefore valid. □

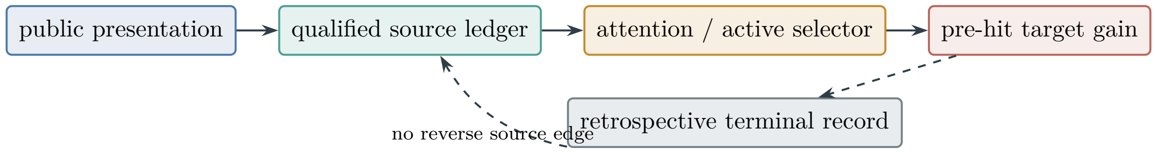

Figure 116: Prospective attention value is charged before deployment. A later first-hit record remains an accounting object and cannot supply an earlier score.

VI.3 Summary of the Attention Compilation

Attention is a normalized, represented selection mechanism. Its exact Gibbs geometry supplies a quantitative KL and Fisher–Rao action, while the structural compilation assigns every score, value, mask, residual path, and deployed transition to the existing saturated ledger. The resulting target bound is the minimum of already certified information, learning, source, and dynamical converters. With a latent prior row and represented log-likelihood scores, the Gibbs row is the exact Bayesian posterior update. Joint optimization with the compact latent record and the active Bellman recursion therefore gives the optimal represented acquisition–encoding–attention pipeline at the declared code and saturated cost ceilings. Attention supplies no additional source coordinate or normalization credit. The canonical implementation and training interface are Definitions V.15.4 and V.15.28. Their saturated-arrival calibration and empirical bias–variance certificate are Theorems V.15.29 and V.15.33.

Part VII

Structural Meta-Complexity and Certificate Minimization

VII.0 Introduction: Presentation-Relative Meta-Complexity

Part contract. *Inputs:* the budgeted derivability closure and saturated cost of Definitions III.14.1 and III.1.3, the prospective source and selector profiles of Definitions III.21.2 and III.23.3, the exact source-cell decomposition of Theorem III.3.4, and the length-bounded universal record of Definition V.17.1 and Theorem V.17.6. *Output:* an exact Pareto meta-frontier for the cost of crossing a target-gain threshold, its generalized-inverse relation with saturated profiles, typed decision/search/certification/deployment interfaces, representation-minimization specializations, and a represented hardness-magnification rule. *First pass:* read Definition VII.1.2 and Theorems VII.1.3, VII.2.2, VII.3.2 and VII.8.2.

Meta-complexity asks for the resources required to recognize, construct, or certify a low-complexity representation. Circuit minimization and resource-bounded Kolmogorov complexity are classical instances of this question [2, 52]. The presentation-relative version developed here uses the complete structural records already defined in Part III and Part V. Its threshold object is the least saturated resource at which one prospective record attains a prescribed target gain. Description length, construction, data access, retained state, evaluation, qualification, and deployment remain separate coordinates of the same cost vector.

The resulting meta-problem is an inverse description of the saturated profile. The profile asks for the largest certified gain below a budget; the meta-frontier asks for the budgets at which a certified gain threshold is crossed. Strict thresholds give an exact inverse without a compactness or attainment convention. Scalar threshold costs are obtained only after the same admissible scalarization used by Definitions III.28.1 and III.28.9 has been fixed.

VII.1 Structural Certificate Meta-Problems

VII.1.1 Complete profile data and Pareto feasibility

Definition VII.1.1 (Complete saturated meta-profile datum)**.** Fix a size slice, a presented family, and the resource order of Definition III.1.3. A *complete saturated meta-profile datum* is a pair $\mathfrak{D}_n = (\mathscr{R}_n, g_n)$, where:

(i) $\mathscr{R}_n$ is one already-defined class of complete family-uniform represented records, each retaining its upward-closed admissible budget set $\mathfrak{B}_{\mathrm{sat},n}(R)$;

(ii) $g_n : \mathscr{R}_n \to [0,1]$ is the already-defined numerical coordinate evaluated on that complete record.

Explicitly, for the size-n presentation slice $\mathfrak{I}_n$,

$$\mathfrak{B}_{\mathrm{sat},n}(R) = \{B : \exists \rho \in \mathsf{Der}_{\mathfrak{I}} \text{ such that } [\![\rho]\!]_I = R_I \text{ and } \mathrm{Cost}_I(\rho) \preceq B \text{ for every } I \in \mathfrak{I}_n\}\,. \quad (1071)$$

It is upward closed by construction. Its Pareto-minimal elements are the complete cost frontier of R in the sense of Lemma III.5.7 and Remark III.5.8.

Its saturated profile is

$$P^{\mathrm{sat}}_{\mathfrak{D},n}(B) = \sup\{g_n(R) : R \in \mathscr{R}_n, \quad B \in \mathfrak{B}_{\mathrm{sat},n}(R)\}\,, \quad (1072)$$

with empty supremum zero. This definition is a wrapper for an existing profile and adds no structural channel. The principal instances are:

(a) prospective master certificates with $g_n(R) = V_R$, giving $\mathfrak{U}_n^{\mathrm{src,sat}}$;

(b) represented pre-hit selector occurrences (R,t) with $g_n(R,t) = (\mathrm{Adv}_t(K_{R,t};\nu_{R,t}))_+$, giving $\mathrm{Sel}_n^{\mathrm{sat}}$;

(c) complete active records with $g_n(R) = A_R^H$, giving $\mathrm{ActArr}_{I,H}^{\mathrm{sat}}$;

(d) untyped master accounting records with their accounting visibility, giving $\mathfrak{U}_n^{\mathrm{sat}}$;

(e) completed exact-quotient records with their accounting visibility, giving the exact-quotient retrospective accounting datum.

The temporal type is part of $\mathscr{R}_n$. In particular, prospective source, prospective selector, active-arrival, and retrospective accounting data are distinct meta-profile data even when two records have the same denotation.

Definition VII.1.2 (Strict structural meta-frontier and scalar threshold cost)**.** For a complete datum $\mathfrak{D}_n = (\mathscr{R}_n, g_n)$ and $0 \le \theta < 1$, define its strict Pareto feasibility region by

$$\mathfrak{P}^{>}_{\mathfrak{D},n}(\theta) = \bigcup_{\substack{R\in\mathscr{R}_n \\ g_n(R)>\theta}} \mathfrak{B}_{\mathrm{sat},n}(R). \tag{1073}$$

It is upward closed in the resource order and decreasing in θ.

Fix an admissible monotone scalarization $\kappa : C \to [0,\infty]$. The associated strict structural meta-complexity is

$$\mathrm{SMC}^{>}_{\mathfrak{D},n,\kappa}(\theta) = \inf\left\{\mathrm{Cost}^{\kappa}_{\mathrm{sat},n}(R) : R \in \mathscr{R}_n, \quad g_n(R) > \theta\right\}, \tag{1074}$$

with value ∞ when the class is empty. The corresponding scalar sublevel profile is

$$P^{\mathrm{sat}}_{\mathfrak{D},n,\kappa}(b) = \sup_{\substack{R\in\mathscr{R}_n \\ \mathrm{Cost}^{\kappa}_{\mathrm{sat},n}(R)\le b}} g_n(R). \tag{1075}$$

The non-strict feasible region $\mathfrak{P}^{\geq}_{\mathfrak{D},n}(\theta)$ and its scalar threshold cost

$$\mathrm{SMC}^{\geq}_{\mathfrak{D},n,\kappa}(\theta) = \inf\left\{\mathrm{Cost}^{\kappa}_{\mathrm{sat},n}(R) : R \in \mathscr{R}_n, \quad g_n(R) \ge \theta\right\} \tag{1076}$$

are defined by replacing $g_n(R) > \theta$ with $g_n(R) \ge \theta$ in Eqs. (1073) and (1074) respectively. Strict regions are used for exact profile inversion; non-strict regions record attained certificate thresholds.

VII.1.2 Exact generalized inverse

Theorem VII.1.3 (Strict meta-frontier as the exact generalized inverse)**.** *For every complete saturated meta-profile datum, budget B, and $0 \le \theta < 1$,*

$$P^{\mathrm{sat}}_{\mathfrak{D},n}(B) > \theta \iff B \in \mathfrak{P}^{>}_{\mathfrak{D},n}(\theta). \tag{1077}$$

Consequently,

$$P^{\mathrm{sat}}_{\mathfrak{D},n}(B) = \sup\left(\{\theta \in [0,1) : B \in \mathfrak{P}^{>}_{\mathfrak{D},n}(\theta)\} \cup \{0\}\right). \tag{1078}$$

For the scalarized quantities in Eqs. (1074) *and* (1075)*,*

$$\mathrm{SMC}^{>}_{\mathfrak{D},n,\kappa}(\theta) > b \Longrightarrow P^{\mathrm{sat}}_{\mathfrak{D},n,\kappa}(b) \le \theta, \tag{1079}$$

$$P^{\mathrm{sat}}_{\mathfrak{D},n,\kappa}(b) > \theta \Longrightarrow \mathrm{SMC}^{>}_{\mathfrak{D},n,\kappa}(\theta) \le b, \tag{1080}$$

$$\mathrm{SMC}^{>}_{\mathfrak{D},n,\kappa}(\theta) < b \Longrightarrow P^{\mathrm{sat}}_{\mathfrak{D},n,\kappa}(b) > \theta. \tag{1081}$$

Under the locally finite discrete scalar-cost convention of Definitions III.14.1 and III.1.3, the relevant minimum is attained whenever it is finite, and the final two lines combine to

$$P^{\mathrm{sat}}_{\mathfrak{D},n,\kappa}(b) > \theta \quad \Longleftrightarrow \quad \mathrm{SMC}^{>}_{\mathfrak{D},n,\kappa}(\theta) \le b. \tag{1082}$$

Proof. If the supremum in Eq. (1072) exceeds θ, the definition of a supremum supplies a feasible record R with $g_n(R) > \theta$. Its admissible budget set contains B, proving the forward implication in Eq. (1077). The reverse implication follows directly from Eq. (1073). Taking the supremum over strict sublevels proves Eq. (1078).

If the scalar meta-complexity exceeds b, no record of scalar cost at most b has gain greater than θ, proving the first line of Eq. (1079). A record witnessing a scalar profile value greater than θ has cost at most b, proving the second line. If the infimum in Eq. (1074) is strictly below b, choose a feasible record with cost below b; this proves the third line. Local finiteness gives only finitely many derivation descriptions in every finite scalar sublevel. The well-posedness result Lemma III.5.7 and the attainment convention in Definition III.14.1 therefore turn a finite infimum into a minimum, proving Eq. (1082). □

Proposition VII.1.4 (Family-uniform provenance with slice-dependent gates)**.** *Let $\mathfrak{D}_n = (\mathscr{R}_n, g_n)$ be size slices of one complete saturated meta-profile datum generated by the family-uniform grammar of Definitions III.1.3 and III.1.4. Each $R \in \mathscr{R}_n$ is the restriction of a record produced by one fixed derivation program. Its admissible budget set $\mathfrak{B}_{\mathrm{sat},n}(R)$, qualification gates, and value $g_n(R)$ are nevertheless evaluated on the size-n public presentation. They may therefore depend on n, on represented rank or spectral data, and on the declared size-dependent ceiling.*

For any sequences of budgets B_n and strict thresholds $0 \le \theta_n < 1$, the exact crossing set is

$$\left\{n : P^{\mathrm{sat}}_{\mathfrak{D},n}(B_n) > \theta_n\right\} = \left\{n : B_n \in \mathfrak{P}^{>}_{\mathfrak{D},n}(\theta_n)\right\}. \tag{1083}$$

At every member of this set there is a complete family-uniform record whose size-n restriction fits B_n and has value greater than θ_n. The witnessing record may vary with n, because the saturated profile is a slice-wise supremum over the fixed family-uniform record class. Thus family uniformity fixes the derivation rule and excludes instance-specific advice; numerical exclusion of a size crossing is supplied by a uniform upper bound on the corresponding saturated profile.

Proof. The fixed-program statement is the uniform-provenance clause of Definition III.1.4. The definition Eq. (1071) evaluates the cost of that program on every instance in the current size slice, and Definition VII.1.1 evaluates the existing qualification and gain coordinate on the same slice. Hence these quantities remain size-dependent even though the generating program is fixed.

Apply Eq. (1077) separately at each n to obtain Eq. (1083). The witness assertion is the strict-supremum argument in the proof of Theorem VII.1.3. Finally, the definition of the profile takes a new supremum on every slice; it contains no cross-size choice of one maximizing record. The description/advice distinction follows from Lemma III.1.6. □

Corollary VII.1.5 (Existing saturated threshold and alignment complexity)**.** *Apply Theorem VII.1.3 to the prospective master datum $\mathfrak{U}^{\mathrm{src,sat}}_n$. Under the attained non-strict convention, write $\mathrm{SMC}^{\ge}_{\mathrm{src},n,\kappa}(\theta)$ for its least scalar cost at visibility at least θ, and define the typed prospective threshold*

$$B^{*,\mathrm{src}}_{\eta}(n) = \inf\left\{b : I^{\mathrm{src,sat}}_{n,\kappa}(b) \ge \eta\right\}. \tag{1084}$$

Since

$$\mathfrak{U}_n^{\mathrm{src,sat}}(B) \le I_n^{\mathrm{src,sat}}(B) \le 2\mathfrak{U}_n^{\mathrm{src,sat}}(B),$$

the prospective threshold satisfies

$$\mathrm{SMC}^{\ge}_{\mathrm{src},n,\kappa}(\eta/2) \le B_\eta^{*,\mathrm{src}}(n) \le \mathrm{SMC}^{\ge}_{\mathrm{src},n,\kappa}(\eta), \tag{1085}$$

with the same strict-threshold or $B_+ > B_\eta^{*,\mathrm{src}}$ relaxation used in Lemma III.28.10 when attainment is unavailable. The untyped invariant B_η^* of Definition III.28.1 is the threshold of the separately defined combined accounting profile I_n^{sat}, whose two summands need not be attained by one record. If $\mathrm{SMC}^{\ge}_{\mathrm{acct},n,\kappa}(\theta)$ denotes the non-strict threshold cost for the single-record accounting master profile $\mathfrak{U}_n^{\mathrm{sat}}$, then Eq. (183) *gives*

$$\mathrm{SMC}^{\ge}_{\mathrm{acct},n,\kappa}(\eta/2) \le B_\eta^*(n) \le \mathrm{SMC}^{\ge}_{\mathrm{acct},n,\kappa}(\eta). \tag{1086}$$

Thus neither temporal use of a single-record meta-datum identifies the two-coordinate sum with one certificate. The alignment complexity $\kappa_{\mathrm{align},\eta}$ is the program-cost scalarization of the complete-certificate frontier selected in Definition III.28.9, up to the compiler overheads Π_1, Π_2 of Lemma III.28.10. Temporal restriction to prospective records replaces that frontier by Eq. (1084)*.*

Proof. The first inequality follows because $I_n^{\mathrm{src,sat}}(B) \ge \eta$ forces $\mathfrak{U}_n^{\mathrm{src,sat}}(B) \ge \eta/2$. A master certificate of visibility at least η makes $I_n^{\mathrm{src,sat}}(B) \ge \eta$, proving the second. The same argument with Eq. (183) proves Eq. (1086). The alignment-complexity identification is precisely the two compiler directions in Lemma III.28.10. □

Corollary VII.1.6 (Presentation-equivalence transfer)**.** *Suppose two presentations are admissibly equivalent with resource distortions Ψ_{12} and Ψ_{21}, and the equivalence transports the complete record class and preserves its gain. Then*

$$B \in \mathfrak{P}^{>}_{\mathfrak{D}_1,n}(\theta) \Longrightarrow \Psi_{12}(B,n) \in \mathfrak{P}^{>}_{\mathfrak{D}_2,n}(\theta), \tag{1087}$$

and conversely with $1,2$ interchanged. Scalar meta-complexities are class-equivalent under the induced scalar distortions.

Proof. Transport a witnessing complete record and its derivation through the admissible presentation equivalence. The transported cost is at most $\Psi_{12}(B,n)$, while denotation and gain are preserved. Apply the reverse compiler for the converse. This is the threshold-resolved form of Lemma III.28.3. □

VII.2 Temporal Meta-Complexity and Accountability

VII.2.1 Source, selector, active, and accounting frontiers

Definition VII.2.1 (Temporally typed structural meta-frontiers)**.** Use Definition VII.1.1 with the following existing record classes and write:

$$\mathfrak{P}_n^{\mathrm{src},>}(\theta) := \mathfrak{P}^{>}_{\mathfrak{U}^{\mathrm{src}},n}(\theta), \tag{1088}$$

$$\mathfrak{P}_{I,H,n}^{\mathrm{sel},>}(\theta) := \mathfrak{P}^{>}_{\mathrm{Sel},I,H,n}(\theta), \tag{1089}$$

$$\mathfrak{P}_{I,H,n}^{\mathrm{act},>}(\theta) := \mathfrak{P}^{>}_{\mathrm{ActArr},I,H,n}(\theta), \tag{1090}$$

$$\mathfrak{P}^{\mathrm{acct},>}_{I,H,n}(\theta) := \mathfrak{P}^{>}_{\mathrm{Acct},I,H,n}(\theta). \tag{1091}$$

Here the source datum uses complete prospective master certificates, the selector datum uses represented pre-hit selector occurrences, the active datum uses complete finite-horizon control records, and the accounting datum uses completed terminal-compatible records. Every frontier inherits the temporal type and complete cost of its underlying record class.

Theorem VII.2.2 (Structural meta-accountability)**.** *Let $\mathcal{A}$ be a budget-B, horizon-H represented computation with neutral baseline ν, and let $\Psi(B, n)$ be the charged one-test-clock compilation ceiling of Theorem III.21.3. For every $\varepsilon > 0$,*

$$\Psi(B, n) \notin \mathfrak{P}^{\mathrm{sel},>}_{I,H,n}(\varepsilon/H) \quad \Longrightarrow \quad \Pr\{\tau \le H\} \le \mathrm{Enum}_\nu(\mathcal{A}, H) + \varepsilon. \tag{1092}$$

Conversely,

$$\Pr\{\tau \le H\} > \mathrm{Enum}_\nu(\mathcal{A}, H) + \varepsilon \quad \Longrightarrow \quad \Psi(B, n) \in \mathfrak{P}^{\mathrm{sel},>}_{I,H,n}(\varepsilon/H). \tag{1093}$$

After scalarization, the sufficient lower-bound condition is

$$\mathrm{SMC}^{\mathrm{sel},>}_{I,H,n,\kappa}(\varepsilon/H) > \kappa(\Psi(B, n))\,, \tag{1094}$$

with the boundary case governed by the exact Pareto statement Eq. (1092)*.*

Proof. Absence from the strict selector frontier means, by Theorem VII.1.3, that $\mathrm{Sel}^{\mathrm{sat}}_{I,n}(\Psi(B, n)) \le \varepsilon/H$. Substitute this inequality into Eq. (153). This proves Eq. (1092). The extraction statement is Eq. (154) together with the complete-cost conclusion in the proof of Theorem III.21.3. The scalar condition implies Pareto-frontier absence by Eq. (1079). ☐

Corollary VII.2.3 (Transfer-class meta-hardness criterion)**.** *Let $B(n)$ and $H(n) \ge 1$ belong to the declared transfer class. Suppose $e_n \ge 0$ and $\varepsilon_n > 0$ are negligible in that class and every budget-$B(n)$ execution satisfies*

$$\mathrm{Enum}_\nu(\mathcal{A}, H(n)) \le e_n, \tag{1095}$$

while

$$\Psi(B(n), n) \notin \mathfrak{P}^{\mathrm{sel},>}_{I,H(n),n}(\varepsilon_n/H(n)). \tag{1096}$$

Then the complete success profile at budget $B(n)$ is at most $e_n + \varepsilon_n$.

Proof. Apply Theorem VII.2.2 to every represented execution in the saturated budget slice and take the defining supremum of the success profile. ☐

Remark VII.2.4 (Relation to prospective source and retrospective accounting)*.* The selector frontier in Theorem VII.2.2 is prospective and yields the unconditional success implication. The source frontier records complete Geom/Abs certificates. A selector occurrence first produces the contextual accounting charge of Theorem III.21.4; the same-record comparison of Corollary III.16.9 then compiles each nonzero charge into a prospective source record. This is a selector-to-source map, so exclusion from the source frontier excludes the corresponding selectors by contrapositive whenever that comparison record is present. The accounting frontier records completed first-hit objects and yields the retrospective bounds of Theorem III.24.3; temporal source status is governed by Definition III.21.1 and Corollary III.23.2. Thus the three meta-frontiers preserve the source/selector/accounting typing already present in the framework.

VII.3 Source-Resolved Structural Meta-Complexity

VII.3.1 Exact cell frontiers

Definition VII.3.1 (Exact source-cell meta-complexity)**.** Let

$$\mathfrak{A}_{\mathrm{src}} = \left\{ (\chi, S_{\mathrm{st}}, S_{\mathrm{cf}}) : \begin{array}{l} \chi \in \mathfrak{S}^5_{\mathrm{src}}, \\ \varnothing \neq S_{\mathrm{st}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}}, \\ S_{\mathrm{cf}} \subseteq \mathfrak{J}^5_{\mathsf{Abs}} \end{array} \right\}. \tag{1097}$$

For $\alpha \in \mathfrak{A}_{\mathrm{src}}$, let $\mathscr{R}^{\mathrm{src,nf}}_{\alpha,n}$ be the canonical normalized complete prospective source records in that exact cell, with visibility V_R and their complete normalized saturated costs. Define

$$\mathfrak{P}^{\mathrm{src,nf},>}_{\alpha,n}(\theta) = \bigcup_{\substack{R \in \mathscr{R}^{\mathrm{src,nf}}_{\alpha,n} \\ V_R > \theta}} \mathfrak{B}_{\mathrm{sat},n}(R), \tag{1098}$$

and define $\mathrm{SMC}^{\mathrm{src,nf},>}_{\alpha,n,\kappa}(\theta)$ by Eq. (1074). Empty cells have empty frontier and scalar cost ∞.

Theorem VII.3.2 (Exact source-cell exhaustion of structural meta-complexity)**.** *The canonical normalized prospective source class is the disjoint union of the* 4960 *cells in* $\mathfrak{A}_{\mathrm{src}}$*. Consequently,*

$$\mathfrak{P}^{\mathrm{src,nf},>}_{n}(\theta) = \bigcup_{\alpha \in \mathfrak{A}_{\mathrm{src}}} \mathfrak{P}^{\mathrm{src,nf},>}_{\alpha,n}(\theta), \tag{1099}$$

and, for every scalarization,

$$\mathrm{SMC}^{\mathrm{src,nf},>}_{n,\kappa}(\theta) = \min_{\alpha \in \mathfrak{A}_{\mathrm{src}}} \mathrm{SMC}^{\mathrm{src,nf},>}_{\alpha,n,\kappa}(\theta). \tag{1100}$$

If an unnormalized source record has cost at most B and visibility greater than θ, its coefficient-complete normalization has the same visibility, belongs to one exact cell, and has cost at most $\Psi_{\mathrm{src},5}(B)$. Hence

$$B \in \mathfrak{P}^{\mathrm{src},>}_{n}(\theta) \Longrightarrow \Psi_{\mathrm{src},5}(B) \in \bigcup_{\alpha \in \mathfrak{A}_{\mathrm{src}}} \mathfrak{P}^{\mathrm{src,nf},>}_{\alpha,n}(\theta). \tag{1101}$$

Every normalized cell record is itself a complete prospective source record at its displayed cost, giving the reverse inclusion at that normalized ceiling.

For the combined two-coordinate profile, Eq. (184) *gives the threshold comparison*

$$I^{\mathrm{src,sat}}_{n}(B) > 2\theta \Longrightarrow B \in \mathfrak{P}^{\mathrm{src},>}_{n}(\theta), \qquad B \in \mathfrak{P}^{\mathrm{src},>}_{n}(\theta) \Longrightarrow I^{\mathrm{src,sat}}_{n}(B) > \theta. \tag{1102}$$

Proof. The unique source-mode–state-provenance–coefficient-provenance index and the cell count are Theorem III.3.4. A strict feasible record therefore belongs to exactly one cell, proving Eq. (1099). The infimum over a finite union is the minimum of the cell infima, with empty cells assigned ∞, which proves Eq. (1100). The normalization statement is the denotation-preserving compiler and cost majorant in that same theorem. Finally, $\mathfrak{U}^{\mathrm{src,sat}} \le I^{\mathrm{src,sat}} \le 2\mathfrak{U}^{\mathrm{src,sat}}$; apply Theorem VII.1.3 to obtain Eq. (1102). □

Corollary VII.3.3 (Cellwise lower bounds close the complete source frontier)**.** *Let B be a declared budget. If*

$$\Psi_{\mathrm{src},5}(B) \notin \mathfrak{P}^{\mathrm{src,nf},>}_{\alpha,n}(\theta) \qquad (\alpha \in \mathfrak{A}_{\mathrm{src}}), \tag{1103}$$

then $\mathfrak{U}^{\mathrm{src,sat}}_{n}(B) \le \theta$ and $I^{\mathrm{src,sat}}_{n}(B) \le 2\theta$.

Proof. The union in Eq. (1099) is empty at the normalized ceiling. Apply Eqs. (1077) and (1101) and then Eq. (184). □

VII.4 Representation, Extraction, and Operational Interfaces

VII.4.1 Exact denotation and extraction

Definition VII.4.1 (Exact-denotation representation and extraction frontiers)**.** Fix a represented evaluator grammar $\mathscr{E}$, an output carrier $\mathcal{Y}$, and $Y \in \mathcal{Y}$. The exact-denotation representation frontier is

$$\mathfrak{P}^{=}_{\mathscr{E},n}(Y) = \bigcup_{\substack{R \in \mathscr{E} \\ [\![R]\!] = Y}} \mathfrak{B}_{\mathrm{sat},n}(R). \tag{1104}$$

After scalarization, define

$$\mathrm{RepMC}_{\mathscr{E},n,\kappa}(Y) = \inf_{\substack{R \in \mathscr{E} \\ [\![R]\!] = Y}} \mathrm{Cost}^{\kappa}_{\mathrm{sat},n}(R). \tag{1105}$$

This representation frontier is evaluated on the singleton semantic slice $\mathfrak{I}_Y = \{Y\}$. A candidate evaluator may therefore hard-code a description specialized to Y, but its full description, evaluator, and retained-state costs remain charged. Family-uniformity in Eq. (1071) is uniformity over this singleton; it does not require a uniform procedure that constructs the evaluator from a varying public input.

For a public presentation I, the extraction frontier $\mathfrak{P}^{\mathrm{ext}}_{\mathscr{E},n}(Y \mid I)$ ranges over complete represented computations that read only the declared interface of I, halt with an evaluator $R \in \mathscr{E}$ satisfying $[\![R]\!] = Y$, and retain the complete construction and output costs. Its scalar infimum is $\mathrm{ExtMC}_{\mathscr{E},n,\kappa}(Y \mid I)$. For a family of public presentations, the extraction derivation is uniform over the declared family slice. Thus extraction, unlike singleton representation, charges the rule that finds the evaluator from the input.

The equality verifier may read Y when checking a completed output. The candidate evaluator receives only the inputs declared by $\mathscr{E}$. This separation prevents the verification interface from becoming a lookup primitive of the candidate representation.

Proposition VII.4.2 (Output projection from extraction to representation)**.** *Suppose the charged output projection of a complete extraction record of budget B produces its evaluator record at cost at most $\Psi_{\mathrm{out}}(B, n)$. Then*

$$B \in \mathfrak{P}^{\mathrm{ext}}_{\mathscr{E},n}(Y \mid I) \Longrightarrow \Psi_{\mathrm{out}}(B, n) \in \mathfrak{P}^{=}_{\mathscr{E},n}(Y). \tag{1106}$$

Thus extraction charges the search for a representation in addition to the representation retained in its output.

Proof. Project the completed computation record to its represented output evaluator. Computation closure in Definition III.1.3 preserves the denotation and charges the output through Ψ_{out}. □

VII.4.2 Decision, search, certification, and deployment

Definition VII.4.3 (Four operational forms of a structural meta-problem)**.** Fix an encoded size-n meta-input slice Ξ_n. Each $\xi \in \Xi_n$ names a complete datum $\mathfrak{D}_\xi$, a budget B_ξ, and a threshold θ_ξ. The four operational forms are:

(i) *decision*: output the truth value of budget membership in the declared meta-frontier;
(ii) *search*: output a witnessing complete record on a yes instance;
(iii) *certification*: output the record together with its charged derivation, qualification data, and represented gain verifier;

(iv) *deployment*: evaluate the record prospectively as an ancestor of the transition or selector whose gain it certifies.

Each form is equipped with a family $\mathfrak{D}^{\mathsf{A}} = (\mathfrak{D}_n^{\mathsf{A}})_{n\geq 1}$ of complete meta-profile data,

$$\mathfrak{D}_n^{\mathsf{A}} = (\mathscr{R}_n^{\mathsf{A}}, g_n^{\mathsf{A}}), \qquad \mathsf{A} \in \{\mathsf{dec}, \mathsf{search}, \mathsf{cert}, \mathsf{deploy}\}. \tag{1107}$$

For search, g_n^{search} is the underlying gain of the valid witness, and is zero for an invalid output. For certification it is that gain when the represented derivation, qualification, and gain verifier all accept, and zero otherwise. For deployment it is the prospective gain actually realized by the represented deployed record. A total decision record D has the worst-slice correctness coordinate

$$g_n^{\mathsf{dec}}(D) = \inf_{\xi\in\Xi_n} \mathbf{1}_{\left\{D(\xi)=\mathbf{1}_{\{B_\xi\in\mathfrak{P}^{>}_{\mathfrak{D}_\xi,n}(\theta_\xi)\}}\right\}}. \tag{1108}$$

The decision record includes its terminating evaluation on the complete declared input interface. Each form therefore has its own represented record class, numerical coordinate, and saturated cost. A compiler between two forms is admissible when it is represented, preserves the required soundness/completeness relation, and carries an explicit cost majorant.

Theorem VII.4.4 (Typed compiler calculus for meta-problems)**.** *Let $\mathfrak{D}_n^{\mathsf{A}}$ and $\mathfrak{D}_n^{\mathsf{B}}$ be the typed data of two operational forms from Eq.* (1107)*. Suppose a represented compiler maps every valid* A*-record of budget B and gain greater than θ to a valid* B*-record of budget at most $\Psi_{\mathsf{A}\to\mathsf{B}}(B,n)$ and gain greater than $\varphi(\theta)$. Then*

$$B \in \mathfrak{P}^{>}_{\mathfrak{D}^{\mathsf{A}},n}(\theta) \Longrightarrow \Psi_{\mathsf{A}\to\mathsf{B}}(B,n) \in \mathfrak{P}^{>}_{\mathfrak{D}^{\mathsf{B}},n}(\varphi(\theta)). \tag{1109}$$

Projection supplies certification-to-search and extraction-to-representation compilers. A represented verifier and a terminating no-case bound supply a search-to-total-decision compiler. A represented self-reduction supplies a decision-to-search compiler. Prospective deployment additionally requires the prefix factorization and ancestry conditions of Definition III.21.1.

Proof. Compose the witnessing derivation with the declared compiler. Constructor closure preserves uniformity, the compiler hypothesis preserves validity and gain, and cost composition gives the displayed ceiling. The listed special cases use projection, verifier execution, self-reduction, and temporal prefix evaluation, respectively. □

VII.5 Classical Representation-Minimization Specializations

VII.5.1 Circuit minimization

Definition VII.5.1 (Truth-table circuit presentation)**.** Let $f : \{0,1\}^k \to \{0,1\}$, put $N = 2^k$, and let $z_f \in \{0,1\}^N$ be its truth table. The verification presentation contains z_f, the canonical order of $\{0,1\}^k$, and equality of a completed circuit truth table with z_f. The candidate grammar $\mathscr{E}_{\mathrm{circ}}$ consists of Boolean circuits over one fixed finite gate basis. A candidate circuit evaluator receives $x \in \{0,1\}^k$ and its own gate description; it has no truth-table lookup interface.

For a circuit record R, write $s(R)$ for its gate-count coordinate and define

$$\mathrm{CC}_{\mathrm{meta}}(z_f) = \min_{\substack{R\in\mathscr{E}_{\mathrm{circ}}\\ [\![R]\!]=f}} s(R). \tag{1110}$$

Gate count is retained as one Pareto coordinate rather than used as a scalar that discards the other costs. For constants depending only on the finite basis, every circuit of size at most s has a canonical prefix encoding and evaluator satisfying

$$L_{\mathrm{circ}}(s,k) \le c_0 + c_1 s \lceil \log_2(k+s+2) \rceil, \quad T_{\mathrm{eval}}(s,k) \le c_2(k+s), \quad S_{\mathrm{eval}}(s,k) \le c_3(k+s). \tag{1111}$$

Let $B_{\mathrm{circ}}(s,k)$ be the complete Pareto ceiling containing these three bounds and the gate-count bound $s(R) \le s$. The truth-table equality verifier has separate cost $O(2^k T_{\mathrm{eval}}(s,k))$, charged to certification rather than to one candidate evaluation.

Theorem VII.5.2 (Circuit minimization specialization)**.** *For the presentation of Definition VII.5.1, $\mathrm{CC}_{\mathrm{meta}}(z_f)$ is exactly the minimum circuit size of f over the declared basis. Hence the threshold language*

$$\{(z_f, s) : \mathrm{CC}_{\mathrm{meta}}(z_f) \le s\} \tag{1112}$$

is the corresponding Minimum Circuit Size Problem. Its input length is $N = 2^k$, while k is the semantic input dimension of the represented function.

The exact-denotation Pareto frontier satisfies

$$\mathrm{CC}_{\mathrm{meta}}(z_f) \le s \quad \Longleftrightarrow \quad B_{\mathrm{circ}}(s,k) \in \mathfrak{P}^{=}_{\mathscr{E}_{\mathrm{circ}},k}(f). \tag{1113}$$

Thus gate count determines the classical threshold, while description, evaluation, state, and verification costs remain visible in the complete framework budget.

The extraction quantity $\mathrm{ExtMC}_{\mathscr{E}_{\mathrm{circ}}}(f \mid z_f)$ *additionally charges the computation that constructs a circuit from the truth table. The output projection in Proposition VII.4.2 maps this extraction problem to the representation-minimization problem.*

Proof. The candidate grammar is precisely the declared circuit basis, exact denotation is equality with f on all 2^k inputs, and Eq. (1110) minimizes its retained gate-count coordinate. This is the standard minimum circuit size by definition. A size-at-most-s witness has the complete bounds in Eq. (1111), proving the forward implication in Eq. (1113). Conversely, membership below $B_{\mathrm{circ}}(s,k)$ includes the coordinate constraint $s(R) \le s$, proving the reverse implication. The final assertion is Proposition VII.4.2. ☐

VII.5.2 Universal programs and time-sensitive costs

Definition VII.5.3 (Declared universal-program meta-complexity)**.** Fix a prefix universal interpreter U, a finite string $z = z_1 \cdots z_N$, and one of the following represented output interfaces:

(i) in *stream mode*, $\mathsf{U}(p)$ outputs all of z within t steps;

(ii) in *bit mode*, $\mathsf{U}(p,i)$ outputs z_i within t steps for every $1 \le i \le N$.

Let $\mathcal{W}^{\mathrm{mode}}_{\mathsf{U}}(z)$ be the set of pairs (p,t) satisfying the selected interface. For any declared admissible monotone scalarization $c(|p|,t)$ with finite sublevels and description length charged, put

$$K^{c,\mathrm{mode}}_{\mathsf{U}}(z) = \inf_{(p,t) \in \mathcal{W}^{\mathrm{mode}}_{\mathsf{U}}(z)} c(|p|,t). \tag{1114}$$

The complete resource-vector form retains $(|p|, t, \mathrm{space}, \mathrm{output\ length})$ before scalarization.

The associated minimization language is

$$\mathrm{MinK}^{c,\mathrm{mode}}_{\mathsf{U}} = \left\{(z,s) : K^{c,\mathrm{mode}}_{\mathsf{U}}(z) \le s\right\}. \tag{1115}$$

Theorem VII.5.4 (Program-minimization specialization)**.** *The exact-denotation representation complexity of Definition VII.5.3 equals* $K_{\mathsf{U}}^{\mathrm{c,mode}}(z)$*. In bit mode, choosing* $c(|p|,t) = |p| + t$ *gives* $\mathrm{KT}_{\mathsf{U}}(z)$ *for the declared random-access interpreter, and Eq.* (1115) *is its Minimum* KT *Problem. In stream mode, choosing* $c(|p|,t) = |p| + \lceil \log_2(1+t) \rceil$ *gives the logarithmic-time Levin-style output complexity. These identifications are interface typed; neither output convention is substituted for the other. Machine changes connected by represented universal simulation alter the quantities by their explicit interpreter overheads.*

The scheduler of Definition V.17.5 acts on the separate description-length/time resource vector. Its branch cost Eq. (1005) *supplies the represented simulation comparison for every chosen scalarization; it does not identify the two scalarizations above.*

Proof. The records in the exact-denotation frontier are exactly the pairs (p,t) in $\mathcal{W}_{\mathsf{U}}^{\mathrm{mode}}(z)$; minimizing the declared cost gives the equality. A represented universal interpreter composes with each program and incurs its displayed simulation cost, proving the machine-transfer statement. The final assertion follows from Theorem V.17.6. $\square$

VII.6 Prefix Coding and Structural Incompressibility

VII.6.1 Unconditional counting bounds

Theorem VII.6.1 (Prefix-code counting and random-table incompressibility)**.** *Let* $\ell \in \mathbb{N}$*. Fix independently of* Z *an evaluator class* $\mathscr{E}$ *with an injective prefix code* $c : \mathscr{E} \to \{0,1\}^*$*, and put* $\mathscr{E}_{\leq \ell} = \{R \in \mathscr{E} : |c(R)| \leq \ell\}$*. Then*

$$\#\mathscr{E}_{\leq \ell} \leq 2^{\ell}. \tag{1116}$$

Let Z *be uniform on* $\{0,1\}^N$*. If every evaluator denotes one Boolean table of length* N*, then*

$$\Pr\{Z \text{ has an exact code of length at most } \ell\} \leq 2^{\ell - N}. \tag{1117}$$

For $0 < \varepsilon \leq 1/2$*,*

$$\Pr\left\{\exists R \in \mathscr{E}_{\leq \ell} : \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\{R(i)=Z_i\}} \geq \frac{1}{2} + \varepsilon\right\} \leq \min\left\{1, 2^{\ell} e^{-2\varepsilon^2 N}\right\}. \tag{1118}$$

Proof. For a prefix-free code, Kraft's inequality gives $\sum_{R \in \mathscr{E}_{\leq \ell}} 2^{-|c(R)|} \leq 1$. Since $|c(R)| \leq \ell$, every summand is at least $2^{-\ell}$, and hence $\#\mathscr{E}_{\leq \ell} \leq 2^{\ell}$. This is the same prefix-code argument used for complete active records in Eq. (996). Each evaluator denotes at most one exact table, while Z has 2^N equiprobable values; a union bound gives Eq. (1117). For a fixed evaluator, its agreement count with Z is binomial $\mathrm{Bin}(N, 1/2)$. Hoeffding's inequality bounds its upper tail by $e^{-2\varepsilon^2 N}$. Union over at most 2^{ℓ} evaluators and cap the result by one. $\square$

Corollary VII.6.2 (Random truth-table circuit lower bound)**.** *For the circuit presentation in Definition VII.5.1, let* $L(s,k)$ *be an upper bound on the canonical prefix-code length of every circuit of size at most* s*. A uniform random Boolean function on* k *bits satisfies*

$$\Pr\{\mathrm{CC}_{\mathrm{meta}}(Z) \leq s\} \leq 2^{L(s,k) - 2^k}. \tag{1119}$$

The same substitution in Eq. (1118) *gives the corresponding approximation bound.*

Proof. Apply Theorem VII.6.1 with $N = 2^k$ and $\ell = L(s,k)$. $\square$

VII.6.2 Represented coding transfer

Theorem VII.6.3 (Charged represented coding transfer)**.** *Let* P *be a represented distribution on a finite complete-record class, and suppose the record contains a represented prefix encoder* c*, decoder* D*, and cost majorant* T_D *satisfying*

$$D(c(R)) = R, \qquad |c(R)| \le \lceil -\log_2 \mathsf{P}(R) \rceil + a \tag{1120}$$

for every R *in the support. Then*

$$B_R^{\mathrm{code}} := \mathrm{Cost}(D, c, \mathsf{P}) \oplus |c(R)| \oplus T_D(c(R)), \qquad B_R^{\mathrm{code}} \in \mathfrak{B}_{\mathrm{sat},n}(R). \tag{1121}$$

Here $\mathrm{Cost}(D, c, \mathsf{P})$ *is the complete charged cost of the joint represented decoder, encoder, and distribution record; it includes their descriptions, retained data, and evaluation interfaces. The term* $|c(R)|$ *occupies the description-length coordinate, and* $T_D(c(R))$ *occupies the decoder execution coordinates. Consequently any exact-denotation or structural meta-frontier containing the decoded record also contains* B_R^{code}*.*

Proof. The decoder, codeword, represented distribution data used by the decoder, and its execution form one derivation of R. Constructor and computation closure add precisely the displayed cost coordinates. Frontier membership then follows from the definition of the admissible budget set. □

Remark VII.6.4 (Coding interface). The coding conclusion of Theorem VII.6.3 is invoked through its represented encoder and decoder. General resource-bounded coding theorems are separate meta-complexity results; efficient coding is itself tied to computational lower bounds and one-way functions [38]. The saturated ledger retains this issue as the cost of the coding interface.

VII.7 Learning, Compression, and Natural Properties

VII.7.1 Learned records as feasible meta-certificates

Theorem VII.7.1 (Learned selector and source records enter their typed meta-frontiers)**.** *Let a complete learned record satisfy the hypotheses of Theorem IV.13.3, have total budget at most* B*, and have certified prospective target gain greater than* θ *after all statistical, optimization, noise, filtering, and deployment terms are included. Let* $\mathfrak{P}^{\mathrm{sel},>}_{\mathrm{learn},n}$ *be the selector frontier obtained by restricting the complete selector-occurrence datum to these learned records. Then*

$$D_B^{\mathrm{learn}} \in \mathfrak{P}^{\mathrm{sel},>}_{\mathrm{learn},n}(\theta), \tag{1122}$$

where $D_B^{\mathrm{learn}} = \Psi_{\mathrm{learn}}(B, H, n, m, \boldsymbol{\eta}, \delta)$ *is the complete common ceiling in Eq. (278). Therefore*

$$D_B^{\mathrm{learn}} \notin \mathfrak{P}^{\mathrm{sel},>}_{\mathrm{learn},n}(\theta) \Longrightarrow \textit{every such certified learned record has gain at most } \theta. \tag{1123}$$

Suppose additionally that the complete learned record contains the same-record comparisons required by Corollary III.16.9. If the resulting transfer selects a prospective source record R_{src}*, of exact source visibility* $V(R_{\mathrm{src}}) > \vartheta$*, normalize it by Theorem III.3.4 and let* α *be the resulting exact source index. At the complete ceiling*

$$D_B^{\mathrm{learn,src}} = \Psi_{\mathrm{src},5}\Big(\Psi_{\mathrm{tr}}(D_B^{\mathrm{learn}})\Big),$$

one has

$$D_B^{\mathrm{learn,src}} \in \mathfrak{P}_{\alpha,n}^{\mathrm{src,nf,>}}(\vartheta) \subseteq \mathfrak{P}_n^{\mathrm{src,nf,>}}(\vartheta). \tag{1124}$$

This cellwise conclusion concerns the selected source record; one complete learner may retain several source cells.

Proof. The learning compilation retains the deployed pre-hit selector and every source, noise, statistical, optimization, and deployment coordinate at the enlarged cost. Its certified target gain is the selector-datum coordinate, so Definition VII.1.2 proves Eq. (1122); its contrapositive is Eq. (1123). Under the additional comparison, the selected source is a complete prospective record at the transfer ceiling and its coefficient-complete normalization lies at $D_B^{\mathrm{learn,src}}$. Apply Definition VII.3.1 and Theorem VII.3.2 to obtain Eq. (1124). □

Natural lower-bound properties can yield learning and compression procedures under explicit access models [20, 34]. The following definition places constructivity, density, and usefulness in the presentation-relative ledger.

Definition VII.7.2 (Presentation-relative structural natural property)**.** Fix a finite object carrier $\mathcal{Y}_n$, a declared reference law μ_n, a represented certificate class $\mathscr{R}_{\alpha,n}$ with declared class index α, and a budget B. A property $\mathcal{P}_n \subseteq \mathcal{Y}_n$ is (A, λ, B, α)-*structurally natural* when:

(i) its membership predicate and the declared sampling or access interface for μ_n have a complete joint saturated cost at most $A(n)$;

(ii) $\mu_n(\mathcal{P}_n) \geq \lambda(n)$;

(iii) every denotation produced by a record in $\mathscr{R}_{\alpha,n}$ of cost at most $B(n)$ lies in $\mathcal{Y}_n \setminus \mathcal{P}_n$.

The class index α may be an exact structural source cell, a circuit class, or another already-declared certificate class. A finite union is handled by listing its indices and charging one common evaluator and access record.

Proposition VII.7.3 (Natural property as a charged meta-distinguisher)**.** *An (A, λ, B, α)-structurally natural property gives a represented test of cost A that accepts a μ_n-sample with probability at least λ and accepts no denotation in the cost-B cell $\mathscr{R}_{\alpha,n}$. If a represented reduction converts this test into a learner, compressor, or source certificate with overhead Ψ, Theorem VII.4.4 transfers its guarantee to the corresponding meta-frontier at cost $\Psi(A, B, n)$.*

For the truth-table presentation, uniform μ_n, nonuniform circuit cell, polynomial-time property evaluator in the truth-table length, and inverse-polynomial λ, the three clauses specialize to the classical constructivity, largeness, and usefulness conditions of natural proofs [69].

Proof. The first statement is the conjunction of the three defining clauses. The second composes the property evaluator with the declared reduction and applies Theorem VII.4.4. □

VII.8 Represented Hardness Magnification

Hardness magnification converts a weak lower bound for a meta-problem into a stronger lower bound for another computational class through an explicit reduction. Such phenomena are known for variants of circuit minimization [21, 63, 64]. The framework-level statement is a compiler theorem over complete profile data.

Definition VII.8.1 (Represented profile reduction)**.** Let $\mathfrak{D}_n^A = (\mathscr{R}_n^A, g_n^A)$ and $\mathfrak{D}_n^B = (\mathscr{R}_n^B, g_n^B)$ be complete meta-profile data. A represented profile reduction $(\mathcal{C}, \Psi, \varphi)$ consists of a family-uniform compiler $\mathcal{C} : \mathscr{R}_n^A \to \mathscr{R}_n^B$, a monotone budget map Ψ, and a strictly increasing gain map $\varphi : [0,1] \to [0,1]$ such that

$$B \in \mathfrak{B}_{\mathrm{sat},n}(R) \Longrightarrow \Psi(B,n) \in \mathfrak{B}_{\mathrm{sat},n}(\mathcal{C}R), \qquad g_n^B(\mathcal{C}R) \geq \varphi(g_n^A(R)). \tag{1125}$$

The complete compiler record includes its instance map, oracle or sampling interfaces, retained witness data, soundness relation, and resource distortion.

Theorem VII.8.2 (Represented profile magnification)**.** *Every represented profile reduction satisfies*

$$B \in \mathfrak{P}^{>}_{\mathfrak{D}^A,n}(\theta) \Longrightarrow \Psi(B,n) \in \mathfrak{P}^{>}_{\mathfrak{D}^B,n}(\varphi(\theta)). \tag{1126}$$

Equivalently,

$$\Psi(B,n) \notin \mathfrak{P}^{>}_{\mathfrak{D}^B,n}(\varphi(\theta)) \Longrightarrow B \notin \mathfrak{P}^{>}_{\mathfrak{D}^A,n}(\theta). \tag{1127}$$

Let κ_A and κ_B be scalarizations of the two resource orders, and suppose the monotone scalar majorant ψ satisfies

$$\kappa_B(\Psi(B,n)) \leq \psi(\kappa_A(B), n)\,. \tag{1128}$$

Extend ψ by $\psi(\infty, n) = \infty$. Then

$$\mathrm{SMC}^{>}_{\mathfrak{D}^B,n,\kappa_B}(\varphi(\theta)) \leq \inf_{\epsilon>0} \psi\Big(\mathrm{SMC}^{>}_{\mathfrak{D}^A,n,\kappa_A}(\theta) + \epsilon, n\Big)\,, \tag{1129}$$

with $\epsilon = 0$ when the source infimum is attained.

Thus a lower bound on the target meta-frontier magnifies to a lower bound on the source profile precisely through the displayed compiler. Source-cell classification supplies the domain of the compiler; the numerical magnification follows from Eq. (1125)*.*

Proof. Choose R witnessing $B \in \mathfrak{P}^{>}_{\mathfrak{D}^A,n}(\theta)$. Then $g_n^A(R) > \theta$, and strict monotonicity gives $\varphi(g_n^A(R)) > \varphi(\theta)$. The compiler conditions place $\mathcal{C}R$ below $\Psi(B,n)$ and give $g_n^B(\mathcal{C}R) > \varphi(\theta)$, proving Eq. (1126). The contrapositive is immediate. For scalar costs, choose a source record within ϵ of the infimum when it is finite, apply the scalar cost majorant, and take the infimum over $\epsilon > 0$. If the source scalar meta-complexity is infinite, the right-hand side is infinite by convention and the inequality is automatic. $\square$

Definition VII.8.3 (Gap structural meta-problem)**.** Let Ξ_n be an encoded meta-input slice with a represented datum assignment $\xi \mapsto \mathfrak{D}_\xi$. The input is the encoding of ξ, together with budget and rational-threshold encodings when those parameters are not fixed for the problem family. The verifier and access interface are the ones retained by $\mathfrak{D}_\xi$.

For budgets $B_0 \preceq B_1$ and thresholds $0 \leq \theta_0 < \theta_1 \leq 1$, define the promise problem

$$\mathrm{GapSMC}\big[(B_0, \theta_1), (B_1, \theta_0)\big] \tag{1130}$$

to have yes region $B_0 \in \mathfrak{P}^{>}_{\mathfrak{D}_\xi,n}(\theta_1)$ and no region $B_1 \notin \mathfrak{P}^{>}_{\mathfrak{D}_\xi,n}(\theta_0)$. The promise gap records both resource and gain separation. Decision, search, certification, and deployment versions use Definition VII.4.3.

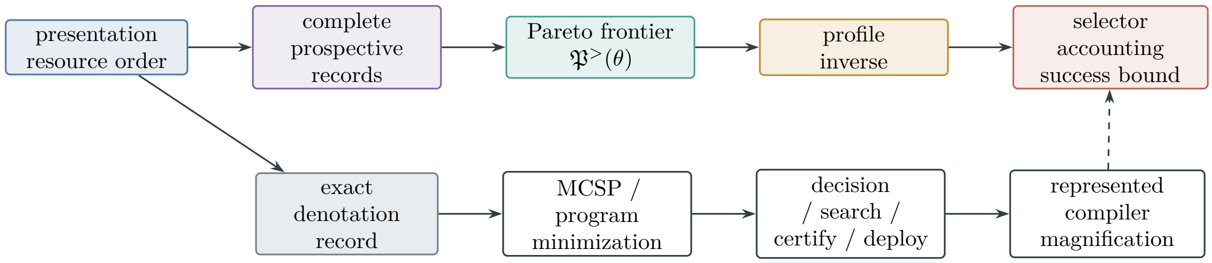

Figure 117: Structural meta-complexity. The upper row takes the exact Pareto inverse of an existing saturated profile and feeds the prospective selector accountability theorem. The lower row separates representation minimization from extraction and deployment; a magnification conclusion requires a represented compiler with an explicit resource and gain map.

VII.9 Application Interface and Chapter Summary

An application evaluates one of the already-defined strict frontiers. The source-resolved route evaluates all cells in Theorem VII.3.2. The direct operational route evaluates the selector frontier in Theorem VII.2.2. The represented selector-to-source comparison of Theorem III.21.4 and Corollary III.16.9 places a selector witness in a source cell at its charged transfer ceiling; a source-cell lower bound then excludes that selector by contrapositive. A completed first-hit record remains in the accounting frontier.

Definition VII.9.1 (Family-functional selector meta-datum)**.** For a parameterized presented family with sample size m, channel or noise parameter ζ, and horizon $H \geq 1$, let $\mathscr{R}^{\mathrm{sel}}_{n,m,\zeta,H}$ be the class of pairs (R,t) such that $0 \leq t < H$ and R is a complete family-uniform, temporally admissible pre-hit selector record. Write $K_{R,I,t}$ and $\nu_{R,I,t}$ for its represented selector and declared neutral baseline on instance I. For a declared monotone family functional $\mathfrak{M}_{n,m,\zeta}$ with values in $[0,1]$, define

$$g^{\mathrm{sel}}_{\mathfrak{M},n,m,\zeta,H}(R,t) = \mathfrak{M}_{n,m,\zeta}\Big(I \longmapsto \left(\mathrm{Adv}_{I,t}(K_{R,I,t};\nu_{R,I,t})\right)_+\Big). \tag{1131}$$

This gives the complete datum

$$\mathfrak{D}^{\mathrm{sel}}_{\mathfrak{M},n,m,\zeta,H} = \Big(\mathscr{R}^{\mathrm{sel}}_{n,m,\zeta,H}, g^{\mathrm{sel}}_{\mathfrak{M},n,m,\zeta,H}\Big). \tag{1132}$$

Write $\mathfrak{P}^{\mathrm{sel},>}_{\mathfrak{M},n,m,\zeta,H}$ for its strict frontier and $\mathrm{SMC}^{\mathrm{sel},>}_{\mathfrak{M},n,m,\zeta,H,\kappa}$ for its scalar meta-complexity.

By Definition VII.1.2, the parameter-indexed scalar quantity is exactly

$$\mathrm{SMC}^{\mathrm{sel},>}_{\mathfrak{M},n,m,\zeta,H,\kappa}(\theta) = \inf\left\{\mathrm{Cost}^{\kappa}_{\mathrm{sat}}(R,t) : \begin{array}{c}(R,t) \in \mathscr{R}^{\mathrm{sel}}_{n,m,\zeta,H},\\ g^{\mathrm{sel}}_{\mathfrak{M},n,m,\zeta,H}(R,t) > \theta\end{array}\right\}. \tag{1133}$$

Every statistical, noise, data-acquisition, representation, and deployment term remains inside the complete record cost and its gain evaluation. The LPN specialization in Part VIII uses Eq. (1133) with $\zeta = \eta$; its explicit meta-frontier consequence is Corollary VIII.22.2.

Meta-object	Underlying framework record	Exact result
source frontier	prospective master certificate	Theorem VII.3.2
selector frontier	pre-hit selector occurrence	Theorem VII.2.2
active frontier	complete active record	Theorem VII.1.3
accounting frontier	completed first-hit record	Theorem III.24.3 and Remark VII.2.4
representation frontier	exact evaluator denotation	Theorems VII.5.2 and VII.5.4
learning transfer	compiled learned selector and normalized source	Theorem VII.7.1
magnification	represented profile compiler	Theorem VII.8.2

The meta-frontier is a reindexing of complete represented records by the least resources needed to cross a numerical threshold. Its universal computational consequence is Theorem VII.2.2; its source decomposition is Theorem VII.3.2. Classical meta-complexity appears through exact-denotation grammars, while target-directed structural meta-complexity retains temporal availability, qualification, and deployment.

Part VIII
Resource-Bounded Structural Analysis of LPN

VIII.0 Application to Learning Parity with Noise

Part contract. *Inputs:* the presentation-relative saturated profile, exhaustive Geom/Abs source theory, and accountability theorem from Part III, together with the uniform statistical-attack bound from Part IV and the universal controlled-system certificate of Part V. The represented-attention compilation of Part VI supplies the source-preserving normal form for normalized active selectors. The strict Pareto frontier and meta-accountability theorem of Part VII convert the evaluated LPN selector profile into an explicit certificate-extraction lower region. *Output:* an exact reduction of the prospective LPN source profile, a separate core-refined first-hit accounting bound, an exact constant-noise spectral perimeter, the hardness theorem obtained upon certification of its isolated residual-ancestry target, a finite proof-grade protocol whose deterministic acceptance discharges precisely that target, the direct implication from fixed-noise LPN hardness to $P \neq NP$, the statistical denoising bound, and explicit polynomial, statistical, and information-theoretic noise thresholds. *First pass:* the reader should retain the presentation, the saturated statistical-alignment specialization, the quantitative phase diagram, the constant-noise spectral ledger, the five source bounds, the core-reduction theorem, and the final obstruction criterion.

Proof-producing LPN route. The structural reduction eliminates every quotient-supported and explicitly materialized source mode and isolates one compact residual-ancestry certification target (Certification Target VIII.20.5). The proof-grade discharge theorem combines exact finite saturation, exhaustive source-cell coverage, verified capacity, attenuation and error laws, an explicit crossover calculation, and direct finite-prefix verification (Theorem VIII.20.6). A learned or optimized component may propose candidate records; only deterministic exact, symbolic, or outward-rounded interval verification enters the proof. If the checker accepts all fields, Eq. (1646) establishes the residual bound uniformly for every $n \geq n_0$ and closes the LPN obstruction at the declared ceiling. The manuscript proves this verification implication and specifies its machine-checkable fields in Section V.15.6; it does not execute an instantiated certificate analysis, so the downstream LPN hardness consequences remain conditional on acceptance of that internal certificate.

Parts I to VII respectively supply task presentations, the budget-admissible channel decomposition, the resource-saturated structural profile, structural learning bounds, controlled-system certificates, the represented-attention compilation, and the strict certificate-cost frontier. We now apply these constructions to Learning Parity with Noise (LPN) [12, 13, 39].

All task-independent notions and implications used in this part are stated in Parts I to VII. Accordingly, the statements below either define the presented LPN family or evaluate an already-defined framework coordinate for the public data (A, b, η). General counting, source-routing, temporal-accountability, learning, controlled-dynamics, geometric, quotient, and oracle-extension results are cited from the preceding parts rather than restated here. The latent specialization Theorem VIII.15.14 applies optimal acquisition, Bayes–Fisher updating, compact representation, source-resolved learning, and attention to the same complete LPN record and bounds the maximum deployed belief gain over their full polynomial saturated class.

We take the budget family $\mathcal{B}_{\mathrm{poly}}$ and analyze the standard binary LPN presentation within the presentation-relative model, including every oracle interface that is conservative in the sense of LPN oracle admissibility below. The principal ledger statements are the prospective-source reduction Eq. (1654) and the prospective success bound

$$\begin{aligned}\mathrm{Succ}_{\mathfrak{M},n,\eta}(\mathcal{B}_C) \le H_C \left(2^{-n} + \mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(\widehat{B}_C)\right) \\ + \varepsilon_{\mathrm{rig}}(n,m) + \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] \\ + (2^n-1)2^{-m(1-H_2(\tau))}.\end{aligned}$$

The last three terms explicitly account for exceptional public presentations. The optimal-extraction specialization Corollary VIII.15.12 additionally records the complete (n,m,η)-dependent information, SDPI, source, Caus, Lift, boundary, and Bellman minimum for every represented acquisition rule. The compact-belief specialization Corollary VIII.15.13 refines the same record by latent code capacity, precision, sufficiency defect, attention-flow error, and the joint acquisition–encoding–attention optimum. The exact-Abs, exact-quotient Geom, and both compensated quotient–Geom coordinates vanish by Corollaries VIII.18.2 and VIII.18.3. The constant-noise ledger Theorem VIII.3.7 exponentially bounds every explicitly materialized located spectrum and isolates compact non-lumpable residual ancestry as the remaining ambient coordinate. Under Certification Target VIII.20.5, the complete saturated LPN source profile is bounded by $2^{-n}+(1-\eta)^n$. Complete controls and lifts follow the prospective-value routing of Theorem V.6.4; the displayed selector profile retains every directly qualified complete ambient record. The two ledger statements are Theorems VIII.21.1 and VIII.22.1.

Here b_{poly} is the ceiling of the single declared polynomial budget structure $\mathcal{B}_{\mathrm{poly}}$, including the class-preserving polynomial overheads already built into that structure. The reduction proceeds componentwise on this saturated algebra. The public instance supplies linear algebra and residual statistics. The framework decomposition assigns every derived coordinate, readout, and finite postprocessing to its represented prospective source, post-saturation residual, or completed accounting record. The complete ambient coordinate is governed by the obstruction criterion in Theorem III.27.2; its two evaluated restrictions have the bounds just displayed. First-hit visibility is retained separately in $\mathrm{HitCore}^{\mathrm{sat}}$ as a retrospective audit.

The LPN structural-reduction section applies the framework decomposition to the full saturated closure. Primitive geometry, quotient support, derived channels, and residual accounting are placed in the exhaustive master ledger. Quotient-supported source modes vanish, and the ambient coordinate is evaluated through the framework gate product. The transition-level dependency is fixed by Theorem III.21.5 and Corollaries III.22.1 and III.22.2: every succinct LPN transition retains one charged family-uniform derivation from the public presentation. Its structural use has exact quotient, compensated quotient, or identity-supported ambient support; terminal readouts and the residual current receive no prospective target-aligned credit. The LPN quotient seals set the pure exact-Abs, exact-quotient Geom, and both compensated quotient–Geom source coordinates to zero. Hence Corollary III.22.2 places every remaining positive sequential-control or reinforcement-learning contribution in the full-carrier ambient mode, where it is evaluated by the complete Geom/Caus gate product. A multiscale or self-similar representation contributes only through the Geom data that it affordably derives from the public LPN presentation. The routing diagram is Fig. 35.

The LPN-specific controlled-dynamics deployment is indexed in Section VIII.11. Its capacity and drift, Caus aiming, valid-Lift transfer, public functional profile, and dynamical-learning coordinates are evaluated in Theorem VIII.11.1 and Corollaries VIII.12.4 to VIII.11.6 and VIII.8.8; their common-record dependency is shown in Fig. 131.

Throughout this part, polynomial computation means computation in $\mathcal{R}_n^{\mathcal{B}_{\mathrm{poly}}}$, the polynomial budgeted saturated represented closure of the presented instance, either directly or through a presentation-admissible oracle interface. The Oracle admissibility and presentation separation theorem compiles every such interface into the same closure at class-preserving cost. An oracle evaluator outside this closure belongs to an enriched task presentation.

VIII.1 The LPN Task Presentation

Except in the noise-indexed comparison of Section VIII.6, fix a constant noise rate

$$0 < \eta < \frac{1}{2}$$

and a polynomial sample count

$$m = m(n) = \mathrm{poly}(n).$$

The LPN sampling experiment draws

$$A \in \mathbb{F}_2^{m\times n}, \qquad s \in \mathbb{F}_2^n, \qquad E \in \mathbb{F}_2^m,$$

where the rows of A are independent uniform vectors, s is uniform, and the coordinates of E are independent Bernoulli η. The public instance is

$$I = (A, b, \eta), \qquad b = As + E.$$

This is the standard random-sample search-LPN presentation [39, 53].

We write $\mathcal{C}_A = \mathrm{Im}(A) \subseteq \mathbb{F}_2^m$ for the public linear code.

The inversion task is to recover s from (A, b, η).

Definition VIII.1.1 (LPN inversion task)**.** The candidate space is

$$X_I = \mathbb{F}_2^n$$

with uniform reference measure μ_I. A point $x \in X_I$ represents a candidate secret. The success set is

$$T_I = \{s\}.$$

The public observable algebra is generated by A, b, η, candidate-coordinate projections on x, public linear algebra over $\mathbb{F}_2$, the residual map

$$r_I(x) = Ax + b = A(x + s) + E,$$

finite postprocessings, admissible lifted coordinates, budget-admissible boundary readouts, and forward transport data generated by the budgeted saturated represented closure under $\mathcal{B}_{\mathrm{poly}}$. The secret s and the noise vector E are not primitive public observables.

The LPN oracle interface is admissible exactly when it preserves the declared presentation and its charged costs.

Definition VIII.1.2 (Presentation-admissible oracle for LPN)**.** An oracle family $O = (O_I)$ is presentation-admissible for the standard LPN task under $\mathcal{B}_{\mathrm{poly}}$ when its evaluator has one family-uniform represented derivation from the LPN presentation, including every retained datum and its evaluation rule, and substitution of that derivation for all oracle calls has class-preserving accumulated cost within the declared polynomial ceiling. A finite collection of oracle interfaces is admissible when their joint substitution has the same property. Equivalently, adjoining these interfaces is a conservative extension under the Oracle admissibility and presentation separation theorem.

The predicate $x = s$ specifies the success set for the analysis and defines the target contrast in the discounted hitting functional. The public boundary readouts are generated by candidate statistics such as the Hamming weight of $r_I(x)$.

For this task presentation, the centered target contrast is

$$y_s(x) = \mathbf{1}_{\{x=s\}} - 2^{-n}.$$

The saturated extraction functional is

$$I^{\mathrm{sat}}_{\mathrm{LPN},n}(B) = m^{\mathrm{sat}}_{\mathrm{LPN},n}(B) + G^{\mathrm{sat}}_{\mathrm{LPN},n}(B), \tag{1134}$$

where the suprema range over the Part III saturated representatives generated from the LPN presentation at public cost at most B. By Definition III.25.1, this untyped quantity is the accounting profile. Its temporally admissible prospective restriction is $I^{\mathrm{src,sat}}_{\mathrm{LPN},n}(B)$, and completed first-hit records remain in the accounting ledger rather than entering that source restriction.

Example VIII.1.3 (Running example: the LPN profile)**.** The presentation in Definition VIII.1.1 completes Example I.1.3. The transcript decomposition of Example II.7.4 and every represented parity quotient from Example III.18.7 now lie in the saturated LPN accounting profile displayed above. The remainder of this part evaluates its temporally admissible Geom/Abs source restriction and retains completed first-hit visibility in a separate core-refined accounting profile before applying the accountability theorem.

VIII.2 Statistical Attacks as Saturated LPN Learning

The LPN sample is a homogeneous binary label-corruption experiment. The framework theorem in Theorem IV.11.2 therefore controls the complete saturated class of statistical readouts produced from the public sample, including data-dependent operators, kernels, features, posteriors, and terminal selectors.

VIII.2.1 Noisy parity as a corrupted binary target

Example VIII.2.1 (Learning parity from corrupted labels)**.** Let a be uniform on $\mathbb{F}_2^n$, let

$$f_s(a) = (-1)^{\langle a,s\rangle}, \qquad b = \langle a, s\rangle \oplus e, \qquad e \sim \mathrm{Bernoulli}(\eta),$$

and set $\widetilde{Y} = (-1)^b$. Then

$$\widetilde{Y} = f_s(a)(-1)^e,$$

so Definition IV.10.1 applies with $\rho = 1 - 2\eta$. Consider the represented parity class

$$\mathcal{H}_{\mathrm{par}} = \{f_t : t \in \mathbb{F}_2^n\}, \qquad f_t(a) = (-1)^{\langle a,t\rangle}.$$

Massart's finite-class lemma [55] gives

$$\widehat{\mathfrak{R}}_S(\mathcal{L}_{\mathcal{H}_{\mathrm{par}}}) \le \sqrt{\frac{2n\log 2}{m}}.$$

Hence the radius

$$\Gamma^{\mathrm{par}}_{n,m,\delta} = 2\sqrt{\frac{2n\log 2}{m}} + 3\sqrt{\frac{\log(2/\delta)}{2m}} \tag{1135}$$

may be substituted into (235). Every $\varepsilon_{\mathrm{opt}}$-empirical parity minimizer therefore satisfies

$$R_0(\widehat{f}) \le \frac{2\Gamma^{\mathrm{par}}_{n,m,\delta} + \varepsilon_{\mathrm{opt}}}{1-2\eta}. \tag{1136}$$

Distinct parity functions disagree on exactly half of $\mathbb{F}_2^n$. Thus

$$2\Gamma^{\mathrm{par}}_{n,m,\delta} + \varepsilon_{\mathrm{opt}} < \frac{1-2\eta}{2} \tag{1137}$$

certifies $\widehat{f} = f_s$. In particular, if $\varepsilon_{\mathrm{opt}} \le (1-2\eta)/4$, there is a universal constant $C > 0$ such that it suffices to take

$$m \ge C\,\frac{n + \log(1/\delta)}{(1-2\eta)^2}$$

,

where C is fixed by (1135).

For the PAC–Bayes form, let P_{par} be uniform on the 2^n parity vectors. Then

$$\mathrm{KL}(\delta_t \| P_{\mathrm{par}}) = n\log 2.$$

For any posterior Q on $\mathcal{H}_{\mathrm{par}}$, define the right-hand side of (239), with $P = P_{\mathrm{par}}$, by $U_{\eta,S}(Q)$. Since every $t \ne s$ has clean risk $1/2$,

$$R_0(Q) = \frac{1 - Q(s)}{2}, \qquad Q(s) \ge 1 - 2U_{\eta,S}(Q). \tag{1138}$$

A point posterior with $U_{\eta,S}(\delta_t) < 1/2$ is therefore supported on s, while $U_{\eta,S}(Q) < 1/4$ gives $Q(s) > 1/2$.

Finally, let $\widehat{y}(a) \in \mathbb{F}_2$ be the bit prediction corresponding to a learned sign predictor and put

$$\widehat{e} = b \oplus \widehat{y}(a).$$

Then

$$\Pr\{\widehat{e} \ne e\} = \Pr\{\widehat{y}(a) \ne \langle a,s\rangle\} = R_0(\widehat{f}). \tag{1139}$$

Replacing the corrupted label by the predicted clean parity therefore leaves exactly the certified clean prediction error. Under (1137), the recovered parity also reconstructs the noise bits on the observed rows exactly.

The finite-class and PAC–Bayes bounds are simultaneous sample-capacity statements. A learner producing $\widehat{f}$ or Q, its optimization transcript, and every data-dependent feature or operator selection retain their saturated derivation costs under Definition IV.11.1 and Lemmas IV.0.2 and IV.3.6.

VIII.2.2 Uniform target-alignment bound

For a score $h : \mathbb{F}_2^n \to [-1, 1]$, put

$$c_s(h) = \mathbb{E}_{a\sim\mathrm{Unif}(\mathbb{F}_2^n)}\big[h(a)f_s(a)\big], \qquad \widehat{c}_{\eta,S}(h) = \frac{1}{m}\sum_{i=1}^{m} h(a_i)(-1)^{b_i}.$$

Theorem VIII.2.2 (Uniform saturated target-alignment bound for LPN statistical attacks)**.** *Fix a budget B, sample length m, and confidence level $0 < \delta < 1$. Let $\mathcal{H}^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m}$ be the specialization of Definition IV.11.1 to the LPN presentation. With probability at least $1-\delta$ over the public LPN sample, simultaneously for every represented statistical attack of complete cost at most B, every represented seed, and its output $h \in \mathcal{H}^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m}$,*

$$|\widehat{c}_{\eta,S}(h) - (1-2\eta)c_s(h)| \le 2\Gamma^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m,\delta}. \tag{1140}$$

On the event $s \neq 0$, the parity target is balanced and

$$|c_s(h)|^2 \le M^{\mathrm{sat,stat}}_{\mathrm{LPN},n}(B, m). \tag{1141}$$

If the attack outputs a candidate $\widehat{s}$, append the represented parity readout $h_{\widehat{s}} = f_{\widehat{s}}$. This incurs only the class-preserving cost of evaluating one binary inner product, and

$$c_s(f_{\widehat{s}}) = \mathbf{1}_{\{\widehat{s}=s\}}. \tag{1142}$$

For a randomized attack, averaging (1142) *over its represented seed gives*

$$\Pr_{\omega}\{\widehat{s} = s \mid A, b\} = \mathbb{E}_\omega\, c_s(f_{\widehat{s}}). \tag{1143}$$

The appended parity evaluator and every downstream target use remain in the same saturated LPN derivation at class-preserving overhead. An affordable statistical recovery attack is therefore included in LPN accounting. The appended post-output evaluator is a prospective Geom/Abs source only for a later use at which it satisfies temporal source admissibility; statistical denoising supplies no additional provenance channel.

Proof. The identity $\widetilde{Y} = f_s(a)(-1)^e$ identifies the LPN sample with the homogeneous corruption model of Definition IV.10.1. Apply Theorem IV.11.2 to obtain (1140). For $s \neq 0$, the character f_s is balanced, so the projection conclusion of that theorem gives (1141). Since s is uniform, $\Pr\{s = 0\} = 2^{-n}$.

Walsh-character orthogonality gives

$$\mathbb{E}_a[f_t(a)f_s(a)] = \mathbf{1}_{\{t=s\}},$$

which proves (1142); expectation over a represented random seed proves (1143). The parity evaluator is a uniform represented postprocessing of $\widehat{s}$ and a. Closure and temporal typing follow from Lemma IV.0.2, Theorem III.25.2, and Corollary III.23.1.

$\square$

Theorem VIII.2.3 (Exact LPN Walsh alignment and charged spectral location)**.** *Fix $A \in \mathbb{F}_2^{m\times n}$, let s be uniform on $\mathbb{F}_2^n$, let E have independent* Bernoulli(η) *coordinates, and put $b = As + E$ and $\rho = 1 - 2\eta$. For $u \in \mathbb{F}_2^m$, $v \in \mathbb{F}_2^n$, write*

$$\chi_u(z) = (-1)^{\langle u,z\rangle}, \qquad \chi_v(t) = (-1)^{\langle v,t\rangle}.$$

Then

$$\mathbb{E}_{s,E}[\chi_u(b)\chi_v(s) \mid A] = \mathbf{1}_{\{A^\mathsf{T}u=v\}}\rho^{|u|}. \tag{1144}$$

Let $h_A : \mathbb{F}_2^m \to [-1, 1]$ be a represented public readout and normalize its Walsh coefficients by

$$\widehat{h}_A(u) := 2^{-m} \sum_{z\in\mathbb{F}_2^m} h_A(z)\chi_u(z). \tag{1145}$$

Its exact target-character correlation is

$$\mathbb{E}_{s,E}[h_A(As + E)\chi_v(s) \mid A] = \sum_{u:A^\mathsf{T}u=v} \widehat{h}_A(u)\rho^{|u|}. \tag{1146}$$

Suppose one represented derivation materializes a list $L_A \subseteq \mathbb{F}_2^m$, its indices, and the coefficients used in the spectral readout

$$h_{A,L}(z) = \sum_{u\in L_A} \widehat{h}_A(u)\chi_u(z).$$

The derivation cost includes construction of L_A, evaluation of its membership or enumeration rule, construction or evaluation of the retained coefficients, and evaluation of the displayed readout. Put $\ell_A = |L_A|$. If every target-aligned retained index has weight at least q, then

$$|\mathbb{E}_{s,E}[h_{A,L}(As + E)\chi_v(s) \mid A]| \le \sqrt{\ell_A}\, \rho^q. \tag{1147}$$

In particular, a charged ceiling that permits at most L_B explicitly materialized coefficients gives the bound $\sqrt{L_B}\rho^q$ for every such located block at that ceiling.

For random A with independent uniform entries and fixed $v \neq 0$,

$$\Pr_A\Big\{\exists u :\ 1 \le |u| < q,\ A^\mathsf{T}u = v\Big\} \le 2^{-n} \sum_{k=1}^{q-1} \binom{m}{k}. \tag{1148}$$

Consequently, if $n/m \to R \in (0, 1)$, $q = \lfloor \delta m\rfloor$, $0 < \delta < 1/2$, and $H_2(\delta) < R$, then

$$\Pr_A\Big\{\exists u :\ 1 \le |u| < q,\ A^\mathsf{T}u = v\Big\} \le 2^{-m(R-H_2(\delta)-o(1))}. \tag{1149}$$

The structural use of the located readout retains this complete derivation. A locator whose selected indices depend on b is part of the represented map $h_A : b \mapsto h_A(b)$; its selection predicates and coefficient evaluators are therefore included in the Walsh coefficients in Eq. (1145) *and retain their complete derivation cost. The existence of an index satisfying $A^\mathsf{T}u = v$, or the analytic appearance of such an index in Eq.* (1146)*, is not a represented locator. A target-qualified spectral record retains both the pre-hit rule that selects its indices from the public presentation and the* *represented comparison by which the selected readout is used as target-relevant. In the target-character specialization $v = s$, the equality $A^\mathsf{T} u = s$ is an analytic test against the hidden target; it is not available to the selection rule unless the public derivation independently represents the corresponding comparison. Such a locator receives the explicit block bound Eq.* (1147) *precisely when its represented output materializes the displayed list expansion. Otherwise its complete readout is evaluated by the exact spectrum Eq.* (1146) *and by the existing saturated statistical projection and Geom gates. A target-qualified nonidentity fiber map generated by $h_{A,L}$ belongs to the existing* Abs *profile. An identity-supported use belongs to the existing ambient* Geom *profile and retains its mixing, authority,* Caus-*alignment, reach, regularity, and temporal gates. A rule that selects L_A, a coefficient, a threshold, or a downstream transition is part of the same represented derivation and supplies no independent source coordinate.*

Proof. Since $b = As + E$,

$$\chi_u(b)\chi_v(s) = (-1)^{\langle A^\mathsf{T} u + v, s\rangle}(-1)^{\langle u, E\rangle}.$$

The expectation of the first factor over uniform s is one when $A^\mathsf{T} u = v$ and zero otherwise. Independence of the noise coordinates gives

$$\mathbb{E}_E(-1)^{\langle u,E\rangle} = \prod_{i:u_i=1} \mathbb{E}(-1)^{E_i} = \rho^{|u|},$$

proving Eq. (1144). Expanding h_A in the Walsh basis and applying that identity termwise proves Eq. (1146).

For the located block, only indices satisfying $A^\mathsf{T} u = v$ contribute. Cauchy–Schwarz and Parseval give

$$\begin{aligned}\left|\sum_{\substack{u\in L_A\\ A^\mathsf{T} u = v}} \widehat{h}_A(u)\rho^{|u|}\right| &\le \rho^q \sqrt{\ell_A}\left(\sum_{u\in L_A} |\widehat{h}_A(u)|^2\right)^{1/2} \\ &\le \rho^q\sqrt{\ell_A}\,\|h_A\|_{L^2(\mathrm{Unif}(\mathbb{F}_2^m))} \le \rho^q\sqrt{\ell_A}.\end{aligned}$$

Every explicitly retained index and coefficient occupies a represented output coordinate, so the output-size and evaluation ledgers bound their number by the declared ceiling; this proves the L_B specialization.

For fixed nonzero u, the vector $A^\mathsf{T} u$ is uniform on $\mathbb{F}_2^n$. Hence $\Pr\{A^\mathsf{T} u = v\} = 2^{-n}$. A union bound over the $\sum_{k=1}^{q-1}\binom{m}{k}$ vectors of the displayed weights proves Eq. (1148). The standard entropy bound $\sum_{k<\delta m}\binom{m}{k} \le 2^{m(H_2(\delta)+o(1))}$ proves Eq. (1149).

Finally, the represented list-construction and coefficient-evaluation rules are ancestors of every use of $h_{A,L}$, so Lemmas IV.0.2 and I.4.7 retain their costs. By Definition VIII.1.1, s is not a primitive public observable. Hence the analytic equality $A^\mathsf{T} u = s$ cannot be evaluated by a public locator unless its comparison is separately derived from the declared presentation, in which case that derivation is an ancestor and is charged by the same two results. The support classification is Theorem III.18.9; temporal and downstream-use classification is Definition III.21.1, Theorem IV.11.4, and Corollary III.16.9.

□

Figure 118 records the distinct statistical, spectral-location, and verification conversions used by the LPN alignment argument.

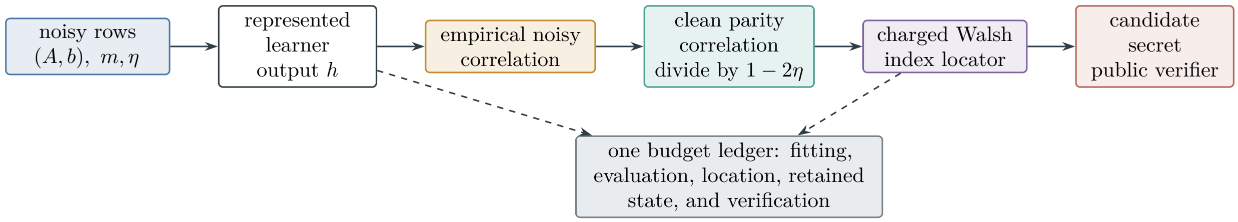

Figure 118: LPN alignment is a chain of represented conversions. Statistical uniformity moves empirical noisy correlation to population correlation, noise inversion supplies the factor $(1 - 2\eta)^{-1}$, and a separate charged locator turns spectral mass into a candidate index before public verification. Formal Fourier mass alone skips neither the locator nor its cost.

Corollary VIII.2.4 (Spectral mass requires charged location and target qualification)**.** *In the setting of Theorem VIII.2.3, a prospective spectral source consists of one represented pre-hit derivation containing its index locator, coefficient evaluator, target-relevance comparison, and downstream use. Its source value is evaluated from that complete record. An unmaterialized term of Eq.* (1146) *contributes no separate source value, and the condition $A^{\mathsf{T}} u = s$ cannot be used as a public location oracle. Explicitly materialized located blocks obey Eq.* (1147)*; a compact evaluator that does not materialize its spectrum is evaluated as one complete readout through the existing exact-*Abs*, compensated, terminal-readout, or ambient-* Geom *mode.*

Proof. Temporal admissibility and complete ancestry are Definition III.21.1 and Lemma IV.0.2. The exact target correlation of the complete readout is Eq. (1146); its located-block specialization is Eq. (1147). The exhaustive support assignment is Theorem III.18.9, and isolated terminal discrimination is separated from prospective structural activity by Corollary III.3.7. □

Theorem VIII.2.5 (A formal diffuse spectrum creates no independent structural source)**.** *Fix the LPN presentation of Definition VIII.1.1, a charged ceiling B, and a represented public residual readout $h_A : \mathbb{F}_2^m \to [-1, 1]$. Regard Eq.* (1146) *as the exact analytic evaluation of its target-character correlation. The formal Walsh expansion of h_A introduces no represented source vertex and no additional source coordinate. The explicit/implicit distinction and every prospective consumer are charged by Theorem III.1.7. Every prospective use of $h_A(b)$ has exactly one of the following existing assignments.*

(i) *If its represented derivation materializes indices and coefficients, their locator, target-relevance comparison, coefficient evaluation, and downstream use remain in the complete record. A materialized block is governed by Eq.* (1147)*.*

(ii) *If the complete readout generates a target-qualified nonidentity fiber map, its dynamically qualified use belongs to the existing exact or compensated quotient-supported mode.*

(iii) *If the complete readout is used with identity support, its prospective value is the existing ambient-*Geom *qualified visibility. Its potential, represented local differences, transition rule, same-generator authority, authorized* Caus *alignment, target reach, regularity, temporal provenance, and complete evaluation cost remain in that one record. The Walsh expansion and the derived* Caus *record add no independent orientation source.*

(iv) *If the readout supplies target discrimination or an extremum but no represented authorized intermediate transport, it is a terminal verifier and has zero prospective source activity.*

(v) *If an evaluator explicitly processes N distinct spectral terms, its complete evaluation ledger contains at least N represented term evaluations. An explicit expansion with superpolynomial*

N is outside the declared polynomial ceiling.

In particular, the analytic restriction

$$\sum_{u:A^{\mathsf{T}}u=s} \widehat{h}_A(u)(1-2\eta)^{|u|}$$

does not define a public aggregation rule: its indexing predicate contains the nonpublic secret s. A public derivation that realizes the same denotation by a compact circuit, factorization, recurrence, or another represented rule is classified and charged by items (i)–(iv) *according to that rule. Consequently there is no separate diffuse-aggregation source between the five-family provenance ledger and the existing exact-*Abs*, compensated, ambient-* Geom*, terminal-accounting, and cost coordinates.*

Proof. By Definition VIII.1.1, the primitive public observables contain A, b, η but not s. Hence $A^{\mathsf{T}}u = s$ is available in the analytic correlation calculation but is not a represented public selection predicate. By Definition III.21.1 and Lemmas IV.0.2 and I.4.7, every represented locator, coefficient evaluator, comparison, and downstream transition remains in the ancestral derivation with its complete cost. This proves the charging assertion in item (i), and Theorem VIII.2.3 gives its displayed bound.

The support of the complete readout is either nonidentity or identity. Theorem III.18.9 assigns a dynamically qualified nonidentity record to the exact or compensated quotient-supported modes and an identity-supported record to ambient Geom. In the latter case, Definitions III.14.9 and III.12.9 retain all same-record Geom gates, while Proposition II.5.9 assigns zero independent source creation to the derived Caus record. This proves items (ii) and (iii).

Corollary III.3.8 proves item (iv). For item (v), each explicitly processed term is an evaluated vertex of the represented derivation. The evaluation ledger therefore dominates the number of such vertices, so a superpolynomial explicit expansion cannot lie below a polynomial ceiling. These cases exhaust whether spectral terms are materialized and, within a compact readout, the existing support and temporal assignments. No additional source coordinate remains. □

Corollary VIII.2.6 (Fixed-ceiling suppression of located LPN spectral blocks)**.** *Fix one declared ceiling whose explicitly materialized spectral output has length at most $L_B(n) \le n^C$, fix $0 < \eta < 1/2$, and suppose $n/m \to R \in (0,1)$. Choose $0 < \delta < 1/2$ with $H_2(\delta) < R$, and put $q = \lfloor \delta m \rfloor$. For each fixed nonzero target character v, outside an event of probability at most*

$$2^{-m(R-H_2(\delta)-o(1))}, \tag{1150}$$

every represented located block at that ceiling satisfies

$$|\mathbb{E}_{s,E}\left[h_{A,L}(As+E)\chi_v(s) \mid A\right]| \le n^{C/2}(1-2\eta)^{\lfloor \delta m \rfloor}. \tag{1151}$$

For fixed R, η, δ, C, the right-hand side is

$$\exp\left[-\delta m \log \frac{1}{1-2\eta} + \frac{C}{2}\log n + O(1)\right] = \exp[-\Theta(n)]. \tag{1152}$$

A represented low-weight exception, when it occurs, consists of an explicitly located u satisfying $A^{\mathsf{T}}u = v$. Its two-valued readout $\chi_u(b)$ has exact target-character correlation $(1-2\eta)^{|u|}$. When a represented downstream use turns this readout into a nonidentity fiber map on the active candidate or transcript carrier and passes the quotient-utility condition of Definition III.15.1, that complete use belongs to the existing Abs *profile. The quotient derivation includes the rule that located u, the character evaluator, the carrier map, and the quotient dynamics.*

Proof. On the complement of the event in Eq. (1149), every aligned index has weight at least q. Apply Eq. (1147) and $\ell_A \le L_B(n) \le n^C$ to obtain Eq. (1151). Taking logarithms gives Eq. (1152); since $m = \Theta(n)$, its negative linear term dominates its logarithmic term. The last paragraph uses Eq. (1144) and the exact-support branch of Theorem III.18.9. The locator remains an ancestor by Lemma IV.0.2.

□

VIII.3 Constant-Noise Spectral Closure Ledger

This section consolidates the constant-noise part of the spectral argument. Throughout it, $0 < \eta < 1/2$ is fixed, $n/m \to R \in (0, 1)$, and the ceiling is one declared polynomial saturated ceiling. The vanishing-noise regimes remain in Section VIII.6 and Corollary VIII.22.3. The profile being evaluated here is prospective: completed first-hit records remain in the accounting profile of Definition III.23.3. Figure 120 displays the resulting spectral routing.

Lemma VIII.3.1 (Residual-parity spectral fibers)**.** *Let one represented pre-hit derivation materialize $L_I = (u_1, \ldots, u_k) \in (\mathbb{F}_2^m)^k$. Its natural residual-character map on the LPN candidate carrier is*

$$q_{L,I}(x) = \big(\langle u_j, Ax + b\rangle\big)_{j=1}^{k} = M_{L,A}x + c_{L,b}, \qquad (M_{L,A})_{j,\cdot} = u_j^{\mathsf{T}} A. \tag{1153}$$

For every candidate x,

$$q_{L,I}^{-1}(q_{L,I}(s)) = s + \ker M_{L,A}. \tag{1154}$$

On the verifier-good event of Lemma VIII.5.1, where $T_I = \{s\}$, the target is a union of $q_{L,I}$-fibers if and only if

$$\ker M_{L,A} = \{0\} \quad \Longleftrightarrow \quad \operatorname{rank} M_{L,A} = n. \tag{1155}$$

In this case $q_{L,I}$ is injective and its represented partition is identity-equivalent. When $\operatorname{rank} M_{L,A} < n$*, the target fiber contains a nonzero affine translate of s, so the terminal-event clause of Definition III.15.1 fails.*

Proof. The affine form in Eq. (1153) follows from linearity of the binary inner product. Therefore $q_{L,I}(x) = q_{L,I}(s)$ precisely when $M_{L,A}(x+s) = 0$, proving Eq. (1154). A singleton target is a union of fibers precisely when its own fiber is the singleton. This is equivalent to $\ker M_{L,A} = \{0\}$, hence to full column rank. Full column rank makes the affine map injective. This is the spectral-list specialization of the additive fiber identity in Lemma VIII.17.2. □

Proposition VIII.3.2 (Spectral selection and the complete quotient gate)**.** *Let a family-uniform, temporally admissible spectral-selection record construct a candidate- or transcript-carrier map $q_I : \Omega_I \to Q_I$. The record is an exact* Abs *source precisely when the same complete derivation supplies all five conditions of Definition III.15.1: represented fibers, terminal-compatible cells, exact intertwining with the selected dynamics, positive normalized quotient utility, and strict refinement of the verifier partition. A nonzero represented intertwining defect can contribute only through a dynamically qualified compensated quotient–Geom certificate satisfying the active-geometry and stability/transfer gates of Definition III.18.1. An identity-supported dynamically qualified use is evaluated in ambient* Geom*.*

The description length of the selector, its family uniformity, and its coefficient-location cost establish budget admissibility of the objects they actually construct. They do not replace any of the five quotient gates. Appending the public verifier to a spectral label supplies terminal compatibility, *but exact or compensated dynamics and positive prospective utility remain properties of the resulting complete record.*

Proof. A fiber-represented selected observable induces the static algebraic quotient of Definition I.8.2. The exact Abs coordinate is defined by Definitions III.15.1 and III.15.2, which sets its utility to zero unless all five displayed gates hold on one normalized record. The exact, nonzero-defect, and identity-support assignments are the three corresponding branches of Theorem III.18.9. Temporal availability and complete ancestry are retained by Definition III.21.1 and Lemma IV.0.2. These definitions prove every assertion. □

Theorem VIII.3.3 (Coset-energy envelope for a complete bounded readout)**.** *In the setting of Theorem VIII.2.3, fix $v \neq 0$, put $\rho_\eta = 1-2\eta$, and let $h_A : \mathbb{F}_2^m \to [-1,1]$ be any complete represented readout. Then, pointwise in A,*

$$|\mathbb{E}_{s,E}\left[h_A(As+E)\chi_v(s) \mid A\right]|^2 \leq \sum_{u:A^\mathsf{T}u=v} \rho_\eta^{2|u|}. \tag{1156}$$

For a random binary matrix A,

$$\mathbb{E}_A\left|\mathbb{E}_{s,E}\left[h_A(As+E)\chi_v(s) \mid A\right]\right| \leq \min\left\{1, 2^{-n/2}\sqrt{(1+\rho_\eta^2)^m - 1}\right\}. \tag{1157}$$

Consequently, when $n/m \to R$ and

$$R > \log_2(1+\rho_\eta^2), \tag{1158}$$

the right-hand side of Eq. (1157) *is*

$$\exp\left[-\frac{m}{2}\left(R - \log_2(1+\rho_\eta^2) - o(1)\right)\log 2\right]. \tag{1159}$$

The same bounds hold after averaging over represented randomness used to choose h_A, provided that randomness is independent of (s,E) conditional on A.

Proof. Apply Eq. (1146) and Cauchy–Schwarz:

$$\begin{aligned}\left|\sum_{u:A^\mathsf{T}u=v} \widehat{h}_A(u)\rho_\eta^{|u|}\right|^2 &\leq \left(\sum_{u:A^\mathsf{T}u=v} |\widehat{h}_A(u)|^2\right)\left(\sum_{u:A^\mathsf{T}u=v} \rho_\eta^{2|u|}\right) \\ &\leq \sum_{u:A^\mathsf{T}u=v} \rho_\eta^{2|u|},\end{aligned}$$

where Parseval and $|h_A| \leq 1$ bound the first factor by one. This proves Eq. (1156).

For every nonzero u, $A^\mathsf{T}u$ is uniform on $\mathbb{F}_2^n$, while $u = 0$ cannot satisfy $A^\mathsf{T}u = v \neq 0$. Hence

$$\begin{aligned}\mathbb{E}_A \sum_{u:A^\mathsf{T}u=v} \rho_\eta^{2|u|} &= 2^{-n}\sum_{u\neq 0}\rho_\eta^{2|u|} \\ &= 2^{-n}\left((1+\rho_\eta^2)^m - 1\right).\end{aligned}$$

Jensen's inequality for the square root and the unit bound on correlation give Eq. (1157). Substituting $n = Rm + o(m)$ gives Eq. (1159). Conditioning on represented randomness proves the final assertion. □

Theorem VIII.3.4 (Fixed-noise suppression of materialized located spectra)**.** *Fix $C < \infty$, choose $0 < \delta < 1/2$ with $H_2(\delta) < R$, put $q = \lfloor \delta m \rfloor$, and set $\rho_\eta = 1 - 2\eta$. Suppose the declared ceiling permits at most $L_B(n) \le n^C$ explicitly materialized spectral indices. For every fixed nonzero secret-space character $v \in \mathbb{F}_2^n$, every represented located block from Theorem VIII.2.3 satisfies*

$$\mathbb{E}_A \left|\mathbb{E}_{s,E}[h_{A,L}(As + E)\chi_v(s) \mid A]\right| \le n^{C/2}\rho_\eta^{\lfloor \delta m \rfloor} + n^{C/2}2^{-m(R-H_2(\delta)-o(1))}. \tag{1160}$$

Each term on the right is $\exp[-\Theta(n)]$ *at fixed* (R, η, δ, C).

Proof. Fix $v \neq 0$. Outside the short aligned-coset event in Eq. (1149), every u satisfying $A^\mathsf{T} u = v$ has weight at least q. The located-block estimate Eq. (1147) and $L_B(n) \le n^C$ give $n^{C/2}\rho_\eta^q$. On the exceptional event, the $q = 0$ case of Eq. (1147) bounds the correlation by $\sqrt{L_B(n)} \le n^{C/2}$; that event has probability at most $2^{-m(R-H_2(\delta)-o(1))}$. This proves the fixed-nonzero-character claim after expectation over A, namely Eq. (1160). Finally, $m = \Theta(n)$ and $0 < \rho_\eta < 1$, so

$$n^{C/2}\rho_\eta^{\lfloor \delta m \rfloor} = \exp\left[-\delta m \log \frac{1}{\rho_\eta} + \frac{C}{2}\log n + O(1)\right] = \exp[-\Theta(n)].$$

The exceptional exponent is positive because $H_2(\delta) < R$. Thus $n^{C/2}2^{-m(R-H_2(\delta)-o(1))} = \exp[-\Theta(n)]$ as well. □

Theorem VIII.3.5 (Rate-uniform semantic-cancellation bound for LPN)**.** *Let $A \in \mathbb{F}_2^{m \times n}$ have independent uniform entries. For integers $1 \le q_0 \le m$ and $0 \le k < n$, define the short-provenance, small-semantic-support event*

$$\mathcal{E}_{q_0,k} = \left\{ \begin{array}{c} \exists\, 0 \neq \lambda \in \mathbb{F}_2^m,\ \exists S \subseteq [n] : \\ |\lambda| < q_0, |S| \le k,\ 0 \neq A^\mathsf{T}\lambda \in \mathbb{F}_2^S \end{array} \right\}. \tag{1161}$$

Then

$$\Pr_A(\mathcal{E}_{q_0,k}) \le \left(\sum_{j=0}^{k} \binom{n}{j} 2^{-(n-j)}\right) \left(\sum_{r=1}^{q_0-1} \binom{m}{r}\right). \tag{1162}$$

Let a complete charged noisy-combination record materialize at most L_B rows λ_j, each with nonzero semantic label $A^\mathsf{T}\lambda_j$ supported on at most k secret coordinates. For every normalized located character block H constructed from those rows and every fixed positive noise $0 < \eta < 1/2$,

$$\begin{aligned} \mathbb{E}_A \left|\mathbb{E}_{s,E}[H\chi_v(s) \mid A]\right| &\le \sqrt{L_B}(1 - 2\eta)^{q_0} \\ &\quad + \sqrt{L_B}\left(\sum_{j=0}^{k} \binom{n}{j} 2^{-(n-j)}\right) \left(\sum_{r=1}^{q_0-1} \binom{m}{r}\right), \end{aligned} \tag{1163}$$

uniformly for every nonzero semantic character v supported on the declared terminal coordinate set.

Equivalently, define the finite-parameter margins

$$\mathfrak{E}_{\text{noise}}(n, m, \eta, q_0, L_B) := q_0 \log \frac{1}{1 - 2\eta} - \frac{1}{2}\log L_B, \tag{1164}$$

$$\mathfrak{E}_{\text{comb}}(n, m, q_0, k, L_B) := (n - k)\log 2 - \log \sum_{j=0}^{k} \binom{n}{j} - \log \sum_{r=1}^{q_0-1} \binom{m}{r} - \frac{1}{2}\log L_B. \tag{1165}$$

Then Eq. (1163) *implies the explicit estimate*

$$\mathbb{E}_A \left|\mathbb{E}_{s,E}[H\chi_v(s) \mid A]\right| \le e^{-\mathfrak{E}_{\mathrm{noise}}} + e^{-\mathfrak{E}_{\mathrm{comb}}}. \tag{1166}$$

It is negligible along any parameter schedule for which both margins grow faster than $\log n$*. This finite criterion retains* n, m, η, q_0, k*, and* L_B *without assuming a constant-rate limit.*

Suppose $n/m \to R \in (0,1)$*,* $q_0 = \lfloor \delta m \rfloor$*, and* $k/n \to \kappa \in [0,1/2)$*. If*

$$H_2(\delta) < R(1 - \kappa - H_2(\kappa)), \tag{1167}$$

then the exceptional probability in Eq. (1162) *is at most*

$$2^{-m[R(1-\kappa-H_2(\kappa))-H_2(\delta)-o(1)]}. \tag{1168}$$

In particular, for $k = o(n)$ *and every fixed* $R > 0$*, one may choose* $\delta > 0$ *with* $H_2(\delta) < R$*. If* L_B *is polynomial, both terms in Eq.* (1163) *are then exponentially small. This conclusion has no lower-rate restriction of the form Eq.* (1158)*.*

Proof. Fix nonzero λ and a coordinate set S. The random vector $A^{\mathsf{T}}\lambda$ is uniform on $\mathbb{F}_2^n$, so

$$\Pr_A\{0 \ne A^{\mathsf{T}}\lambda \in \mathbb{F}_2^S\} = \frac{2^{|S|}-1}{2^n} \le 2^{-(n-|S|)}.$$

Union bounds over the displayed choices of S and λ prove Eq. (1162).

On $\mathcal{E}^{\mathsf{c}}_{q_0,k}$, every materialized row with a nonzero semantic label supported on at most k coordinates has provenance weight at least q_0. Apply Eq. (220) with $\rho = 1 - 2\eta$. On $\mathcal{E}_{q_0,k}$, use the same located-block bound with unit attenuation. Averaging over A and substituting Eq. (1162) gives Eq. (1163).

For $k/n \to \kappa < 1/2$, the standard entropy estimates give

$$\sum_{j=0}^{k} \binom{n}{j} 2^{-(n-j)} \le 2^{-n(1-\kappa-H_2(\kappa)-o(1))},$$

$$\sum_{r=1}^{q_0-1} \binom{m}{r} \le 2^{m(H_2(\delta)+o(1))}.$$

Using $n = Rm + o(m)$ proves Eq. (1168). When $k = o(n)$, $\kappa = 0$; since $R > 0$, continuity of H_2 at zero permits a fixed $\delta > 0$ with $H_2(\delta) < R$. Fixed positive noise makes $(1-2\eta)^{\delta m}$ exponentially small, and a polynomial factor does not change either exponential conclusion. □

Corollary VIII.3.6 (High-sample closure of the materialized additive-combination sector)**.** *Fix* $R \in (0,1)$*,* $0 < \eta < 1/2$*, and a polynomial saturated ceiling. Consider the source-resolved LPN subprofile whose complete records*

(i) *construct their prospective target-bearing characters through the charged binary combination interface of Definition III.30.1;*
(ii) *materialize polynomially many located rows and coefficients; and*
(iii) *use a terminal coordinate support of size* $k = o(n)$*, including every explicitly enumerated terminal table of polynomial cardinality.*

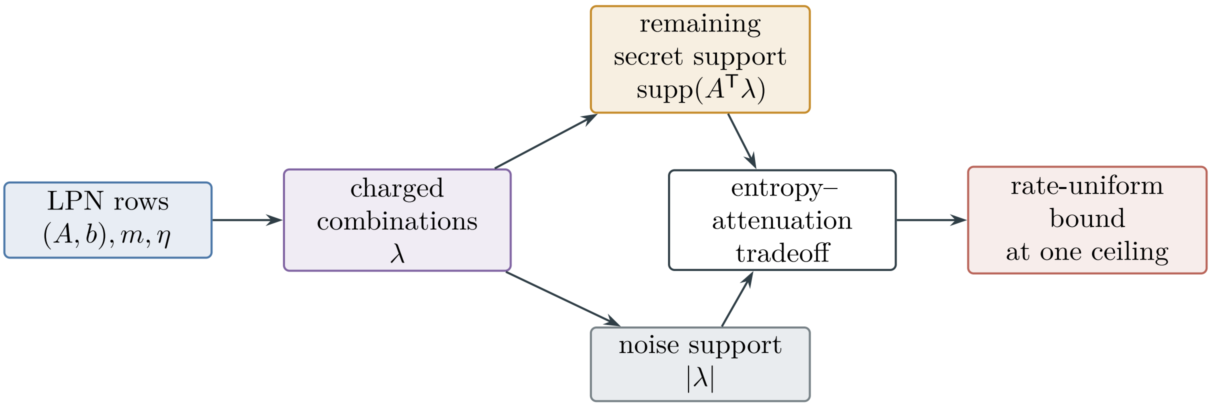

Figure 119: LPN semantic cancellation and noise provenance. A combination may cancel secret coordinates, but a short-noise-provenance combination reaching a small terminal support is exponentially unlikely. Outside that event, the exact factor $(1-2\eta)^{|\lambda|}$ suppresses every polynomially materialized located block, including in the high-sample region.

Then its saturated located-combination profile is exponentially small under the random LPN presentation law. This holds throughout the high-sample region

$$R \le \log_2(1 + (1 - 2\eta)^2)$$

as well as in the coset-energy decay region.

Every row locator, collision rule, shared combination DAG, sample reuse, terminal table, decoder, verifier, and prospective transition remains in the complete record. A fixed program implementing the construction is admitted at its actual execution, acquisition, memory, and output cost; fixed code length alone supplies no polynomial-cost certificate.

Proof. Choose $\delta > 0$ with $H_2(\delta) < R$ and apply Theorem VIII.3.5 and Corollary III.30.4. Polynomial terminal-table cardinality implies $k = O(\log n) = o(n)$. Complete cost and provenance follow from Theorem III.30.2 and Lemmas IV.0.2 and III.1.6. □

Theorem VIII.3.7 (Exact constant-noise spectral perimeter)**.** *At fixed* $0 < \eta < 1/2$*, fixed rate, and a polynomial saturated ceiling, let* $\mathfrak{M}_{n,m,\eta}$ *denote the declared LPN expectation functional. The prospective spectral sector has the following exhaustive routing. Its correlation estimates are family-average statements for a fixed nonzero target character under the unrestricted random-A law.*

(i) *The family-average correlation of every explicitly materialized located block is exponentially suppressed by Theorem VIII.3.4. When the block is produced by semantic cancellation into* $k = o(n)$ *terminal coordinates, the rate-uniform estimate Corollary VIII.3.6 supplies the same conclusion throughout the high-sample region.*

(ii) *The natural residual-parity fiber map of a materialized selector either fails the terminal gate or is identity-equivalent, by Lemma VIII.3.1 and Eq.* (1155)*. A refinement using the verifier or another represented label enters exact* Abs *only when the complete quotient gate of Proposition VIII.3.2 holds.*

(iii) *An explicit expansion processing more than polynomially many terms is outside the ceiling by Lemma III.1.6 and Theorem VIII.2.5.*

(iv) *A compact evaluator may have exponentially many formal Walsh coefficients while retaining polynomial description and evaluation cost. It is one complete represented readout, not an explicitly materialized coefficient list. Its represented circuit, factorization, recurrence, or other compression rule is charged at its actual construction, evaluation, memory, precision, and retained-state cost. Every coefficient locator, selected aggregation, comparison, and prospective transition is an ancestor of the same complete record. Its prospective use is assigned to an exact quotient, a compensated quotient, an identity-supported ambient* Geom/Caus *record, or a terminal readout by Theorems III.1.7 and VIII.2.5 and Remark VIII.9.7. Its complete bounded-readout character correlation obeys Theorem VIII.3.3 and Eq.* (1157)*, which is exponentially small in the parameter region Eq.* (1158)*.*

For the final source-mode reduction, restrict the declared family functional to the verifier, full-rank, and random-code-rigidity event of Definition VIII.5.2, and denote the resulting subprobability functional by $\mathfrak{M}^*_{n,m,\eta}$. On that event, after the exact and compensated quotient-supported LPN modes are removed by Corollaries VIII.18.2 and VIII.18.3, the only spectral form not covered by the exponential located-block estimate is the compact, dynamically qualified, identity-supported residual-ancestry ambient coordinate $G^{\mathrm{src,sat}}_{X,b,\mathfrak{M}^*,n}(B)$. Its complete numerical bound is exactly the premise isolated in Certification Target VIII.20.5; spectral location and provenance classification alone do not change its value. Outside Eq. (1158)*, the general coset-energy envelope may equal the unit bound and therefore supplies no replacement estimate for that sequential ambient coordinate.*

Proof. Items (i) and (ii) are Theorem VIII.3.4, Corollary VIII.3.6, Lemma VIII.3.1, and Proposition VIII.3.2. For an explicit expansion, every processed term is an evaluated derivation vertex, so the evaluation and output ledgers dominate the expansion length; this is item (v) of Theorem VIII.2.5 and proves item (iii). Evaluator normalization in Definition III.1.3 charges a compact factorization by its represented derivation rather than by the cardinality of its formal expansion. The support and temporal routing in Theorems III.1.7 and VIII.2.5 therefore prove item (iv). Removing the quotient-supported modes leaves the ambient branch by Theorem VIII.20.4; its residual-independent subbranch is already bounded by Lemma VIII.8.11. The displayed residual-ancestry coordinate is the remaining subbranch in Definition VIII.8.9. ☐

VIII.4 Candidate Recovery and Prospective Selector Accountability

Corollary VIII.4.1 (Noise-dependent accountability of LPN candidate recovery)**.** *Use the simultaneous event of Theorem VIII.2.2, and let a represented randomized attack output $\widehat{s} = \widehat{s}(A, b, \omega)$. After the represented parity evaluator has been included at its class-preserving cost, put*

$$p_S := \Pr_{\omega}\{\widehat{s} = s \mid A, b\}, \qquad \rho := 1 - 2\eta, \qquad \Gamma_{\mathrm{LPN}} := \Gamma^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m,\delta}.$$

On the event $s \neq 0$,

$$p_S \leq \sqrt{M^{\mathrm{sat,stat}}_{\mathrm{LPN},n}(B, m)}. \tag{1169}$$

Character orthogonality also gives the exact execution-law identity

$$\mathbb{E}_S p_S = \mathbb{E}_{S,\omega} \left| c_s(f_{\widehat{s}}) \right|^2. \tag{1170}$$

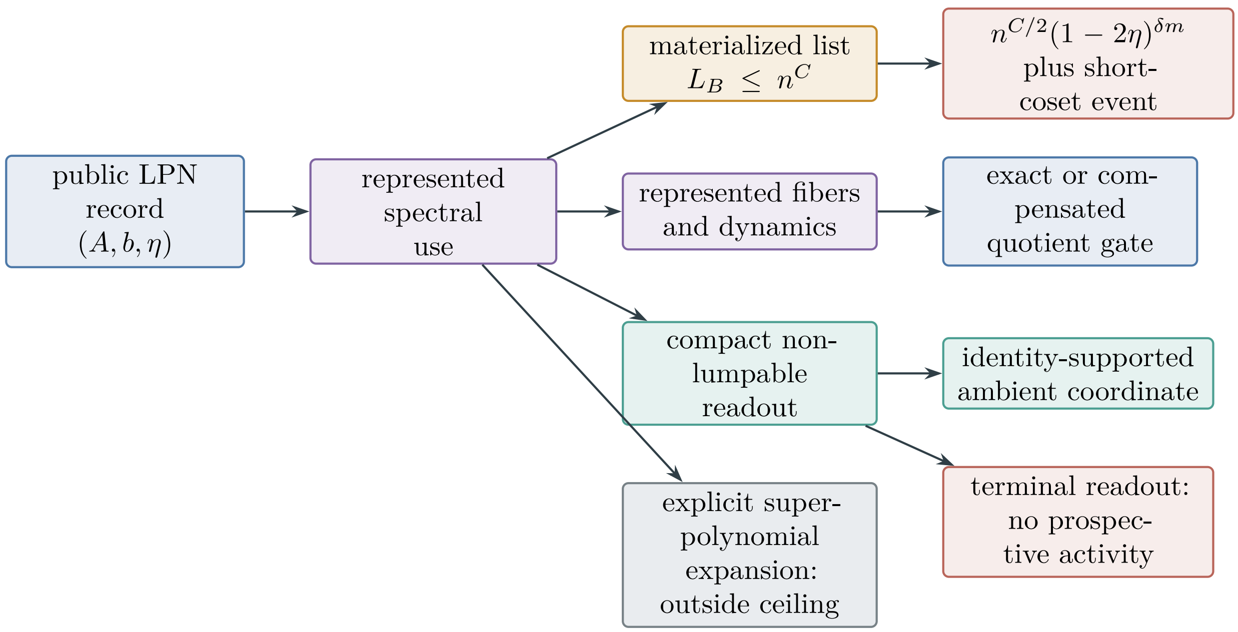

Figure 120: Exhaustive constant-noise spectral routing. Explicitly located blocks obey the exponential attenuation theorem. Fiber compression reaches Abs only through the complete exact or compensated quotient gate. Compact non-lumpable evaluators remain complete ambient records at their actual cost or terminate without prospective activity; an explicitly processed superpolynomial expansion is outside the ceiling. The number of terms in a formal Walsh expansion is not a separate source coordinate.

Consequently, after the candidate parity evaluator and decoder have been included in the complete cost,

$$\mathfrak{M}_{n,m,\eta}\Big(I \longmapsto \Pr_{S,\omega}\{\widehat{s}=s\}\Big) \le \mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m), \tag{1171}$$

$$\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m) \le \mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m) \le \mathfrak{M}_{n,m,\eta}\Big(I \longmapsto M^{\mathrm{sat,stat}}_{\mathrm{LPN},n}(B,m)\Big). \tag{1172}$$

The empirical noisy alignment of the candidate parity satisfies the two-sided bound

$$(\rho p_S - 2\Gamma_{\mathrm{LPN}})_+ \le \mathbb{E}_\omega \left|\widehat{c}_{\eta,S}(f_{\widehat{s}})\right| \le \rho p_S + 2\Gamma_{\mathrm{LPN}}. \tag{1173}$$

A sample lookup table used to evaluate $\widehat{c}_{\eta,S}(f_{\widehat{s}})$ remains a charged empirical record. Its target-bearing projection is classified by the existing LPN ledger and its orthogonal remainder is J_{res}, as in Theorem IV.11.4. If the learned output is available before and used to prioritize later candidate tests, that use is a prospective source and its positive nonbaseline advantage is bounded by Theorem III.21.4. A post-output evaluation remains accounting. Neither case adds a memorization or learning channel to the saturated profile.

Proof. For every represented seed, character orthogonality gives

$$c_s(f_{\widehat{s}}) = \mathbf{1}_{\{\widehat{s}=s\}}$$

by (1142). On $s \ne 0$, apply (1141) to the appended parity readout and average over ω:

$$p_S = \mathbb{E}_\omega c_s(f_{\widehat{s}}) \le \sqrt{M^{\mathrm{sat,stat}}_{\mathrm{LPN},n}(B,m)}.$$

This proves (1169). The uniform identity (1170) follows by averaging the same pointwise equality over (S,ω). The decoder margin is exactly one, so Theorem IV.11.7 gives (1171); its projection domination gives (1172). The uniform transport estimate (1140) gives, for each represented seed,

$$\rho c_s(f_{\widehat{s}}) - 2\Gamma_{\mathrm{LPN}} \le \left|\widehat{c}_{\eta,S}(f_{\widehat{s}})\right| \le \rho c_s(f_{\widehat{s}}) + 2\Gamma_{\mathrm{LPN}}.$$

Expectation over ω, followed by the nonnegativity of the middle term, proves (1173). The final provenance statement is the LPN specialization of Theorem IV.11.4; downstream selector charging is exactly Theorem III.21.4.

□

Corollary VIII.4.2 (Closed budget accounting for LPN learning rules)**.** *Fix one declared saturated learning ceiling B. Let $\mathcal{D}$ be one family-uniform represented derivation from the LPN presentation. Its complete record consists of the feature or spectral-index selection rule, coefficient and operator construction, optimization transcript, represented randomness, retained state, output evaluator, and every downstream selector, quotient, potential, or transition rule that uses the output. By Definition IV.11.1, all these ancestors belong to the already-defined complete cost* $\mathrm{Cost}_{\mathrm{sat}}(\mathcal{D})$*. The budget accounting has the following two exhaustive cases.*

(i) *If* $\mathrm{Cost}_{\mathrm{sat}}(\mathcal{D}) \preceq B$*, then the output belongs to* $\mathcal{H}^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m}$*. A candidate output* $\widehat{s}$ *satisfies*

$$\begin{aligned}\mathfrak{M}_{n,m,\eta}(I \longmapsto \Pr\{\widehat{s}=s\}) &\le \mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m) && (1174)\\ &\le \mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m) && (1175)\end{aligned}$$

$$\leq \mathfrak{M}_{n,m,\eta}\Big(I \longmapsto M^{\mathrm{sat,stat}}_{\mathrm{LPN},n}(B,m)\Big). \tag{1176}$$

Its empirical/population alignment obeys Eq. (1173)*. An explicitly located Walsh block additionally obeys Eq.* (1151)*; a represented target-qualified nonidentity fiber use enters* Abs*, and an identity-supported use retains the complete ambient* Geom *gate product.*

(ii) *If* $\mathrm{Cost}_{\mathrm{sat}}(\mathcal{D}) \not\preceq B$*, then* $\mathcal{D}$ *is absent from the budget-*B *statistical readout, selector, source, and success profiles. Enlarging the ceiling to admit it charges the same complete record in the enlarged slice.*

Thus optimization, implicit aggregation, and spectral location are evaluated as components of one represented derivation. The saturated profiles contain no additional decoder coordinate outside these two cases.

Proof. The membership assertion and complete-cost convention are the LPN specialization of Definition IV.11.1 and Lemma IV.0.2. The success and projection chain is Eqs. (1171) and (1172); the empirical bound is Eq. (1173). Located spectral blocks and their locator costs are Theorem VIII.2.3 and Corollary VIII.2.6. The support alternatives are Theorem III.18.9. Budget exclusion in the second case is the defining restriction of the saturated budget slice, and Lemma I.4.7 preserves the complete ancestry when the ceiling is enlarged.

□

Theorem VIII.4.3 (Prospective LPN learning and selector accountability)**.** *Fix a represented LPN computation class of budget* B*, normalized horizon at most* H_B*, and homogeneous noise rate* $0 \leq \eta < 1/2$*. Work on the good-presentation event* $\mathcal{G}_{\mathrm{ver}}(\tau)$ *of Lemma VIII.5.1, on which the public candidate verifier has accepting set* $\{s\}$*, and write* $\mathfrak{M}^{\mathrm{ver}}_{n,m,\eta} = \mathfrak{M}^{\mathcal{G}_{\mathrm{ver}}(\tau)}_{n,m,\eta}$ *for the corresponding restricted functional of Definition VIII.5.2. Include every public candidate-verifier call in the one-test clock refinement of Definition III.21.2, and append one final verifier call to the computation's output. At each verifier step use the uniform kernel on* $\mathbb{F}_2^n$ *as the declared neutral baseline. Then*

$$\mathrm{Succ}_{\mathfrak{M}^{\mathrm{ver}},n,\eta}(B) \leq H_B 2^{-n} + H_B\,\mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M}^{\mathrm{ver}},n,\eta}(\Psi(B,n)). \tag{1177}$$

This restricted statement, added to the verifier failure probability of Eq. (1184)*, bounds the unrestricted* $\mathrm{Succ}_{\mathfrak{M},n,\eta}(B)$ *by* $\mathfrak{M}_{n,m,\eta}$*-monotonicity, since success is bounded by one.*

Put $\widehat{B} = \Psi(B,n)$*, let* $\widehat{H} = H_{\widehat{B}}$*, and let* $D^{\mathrm{sel}}_{\widehat{B}}$ *be the selector-charge ceiling in Eq.* (155)*. The selector-to-charge metatheorem gives*

$$\mathrm{Sel}^{\mathrm{sat}}_{I,n}(\widehat{B}) \leq e\widehat{H} \sum_{c \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_{\widehat{B}}), \tag{1178}$$

and therefore, after applying the family functional,

$$\mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M}^{\mathrm{ver}},n,\eta}(\widehat{B}) \leq e\widehat{H}\,\mathfrak{M}^{\mathrm{ver}}_{n,m,\eta}\left(I \longmapsto \sum_{c \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_{\widehat{B}})\right). \tag{1179}$$

Consequently,

$$\mathrm{Succ}_{\mathfrak{M}^{\mathrm{ver}},n,\eta}(B) \leq H_B 2^{-n} + eH_B\widehat{H}\,\mathfrak{M}^{\mathrm{ver}}_{n,m,\eta}\left(I \longmapsto \sum_{c \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_{\widehat{B}})\right). \tag{1180}$$

For $0 \leq \eta_1 \leq \eta_2 < 1/2$*, the same LPN specialization of flip degradation satisfies*

$$\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_2}(B,m) \le \mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_1}(\psi_{\mathrm{flip}}(B,n,m),m), \tag{1181}$$

$$\mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_2}(B,m) \le \mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta_1}(\psi_{\mathrm{flip}}(B,n,m),m). \tag{1182}$$

After the parity evaluator is included in the scalarized computation cost, the LPN decoder margin is one, and hence

$$B^*_{\mathrm{align}}(n,m,\eta;\alpha) \le B^*_{\mathrm{rec}}(n,m,\eta;\alpha). \tag{1183}$$

All quantities in these displays are specializations of the existing saturated learning, selector, and five-family charge profiles. They add no LPN-specific source channel.

Proof. On $\mathcal{G}_{\mathrm{ver}}(\tau)$, the appended verifier ensures that recovery implies target contact within the refined horizon. At every verifier step, the uniform baseline assigns mass 2^{-n} to the unique target s; its survival-weighted neutral contact is therefore at most 2^{-n}. Summing at most H_B such terms and applying Theorem IV.13.6 to the restricted functional $\mathfrak{M}^{\mathrm{ver}}_{n,m,\eta}$ proves (1177). The selector at every step is represented before its corresponding verifier call and retains the complete computation prefix, so it is temporally admissible by Definition III.21.1. Since $0 \le \mathrm{Succ}_{\mathfrak{M},n,\eta}(B) - \mathrm{Succ}_{\mathfrak{M}^{\mathrm{ver}},n,\eta}(B) \le \Pr(\mathcal{G}_{\mathrm{ver}}(\tau)^c)$ by Definition VIII.5.2 and success takes values in $[0,1]$, adding (1184) to (1177) bounds the unrestricted success profile.

Apply Theorem III.21.4 pointwise at budget $\widehat{B}$ to obtain (1178), then apply the monotone positively homogeneous expectation functional $\mathfrak{M}^{\mathrm{ver}}_{n,m,\eta}$. This proves (1179). Substitution in (1177) gives (1180).

Equations (1181) and (1182) are Eqs. (286) and (287) specialized to the LPN flip channel. Finally, (1170) verifies, with equality, the margin-one learner–decoder realization required by Corollary IV.13.7; that corollary gives (1183).

□

The statistical scale in (1140) is explicit: an alignment ε can exceed the uniform selection radius only when

$$(1-2\eta)\varepsilon > 2\Gamma^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m,\delta}.$$

This comparison measures sample resolution. Membership in the saturated readout class separately enforces the complete computational cost of finding and representing the aligned parity.

VIII.5 Public Verification of the LPN Target

The binary entropy H_2, Bernoulli divergence D_{Ber}, and the increasing inverse H_2^{-1} are defined in Eqs. (294) and (295).

The public residual predicate at threshold τ is

$$V_I(x) = \mathbf{1}\{|Ax+b|_1 \le \tau m\}.$$

The following task-level lemma identifies its accepting fiber with the target of the LPN inversion task.

Lemma VIII.5.1 (Public verification of LPN recovery)**.** *Fix* $0 < \eta < 1/2$*, choose* $0 < \delta < 1/2 - \eta$*, and put* $\tau = \eta + \delta$*. If*

$$m(1 - H_2(\tau)) \geq n + \omega(\log n),$$

then, with probability $1 - \mathrm{negl}(n)$ *over the standard random binary LPN presentation,*

$$V_I^{-1}(1) = \{s\} = T_I.$$

More precisely,

$$\Pr\{V_I^{-1}(1) = \{s\}\} \geq 1 - \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] - (2^n - 1)2^{-m(1-H_2(\tau))}. \tag{1184}$$

Proof. For $x \neq s$, write $h = x + s \neq 0$. Then

$$Ax + b = A(x + s) + E = Ah + E.$$

For the standard uniform row model, Ah is uniform on $\mathbb{F}_2^m$ and independent of E. Hence $Ah + E$ is uniform. The Hamming-ball estimate gives [32, 76]

$$\Pr\{|Ah + E|_1 \leq \tau m\} \leq 2^{-m} \sum_{j=0}^{\lfloor \tau m \rfloor} \binom{m}{j} \leq 2^{-m(1-H_2(\tau))}.$$

There are at most $2^n - 1$ nonzero choices of h. A union bound yields

$$\Pr\{\exists x \neq s : |Ax + b|_1 \leq \tau m\} \leq (2^n - 1)2^{-m(1-H_2(\tau))} \leq 2^{n-m(1-H_2(\tau))} = 2^{-\omega(\log n)}.$$

The displayed probability is negligible. Since $\tau > \eta$, the Chernoff bound gives $\Pr\{|E|_1 > \tau m\} \leq \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)]$. Therefore $V_I(s) = 1$ and $V_I(x) = 0$ for every $x \neq s$, outside a negligible event. Adding the two failure probabilities proves (1184).

□

Definition VIII.5.2 (Named LPN presentation events and restricted functionals)**.** For a verifier threshold τ, write

$$\begin{aligned} \mathcal{G}_{\mathrm{ver}}(\tau) &:= \{V_I^{-1}(1) = \{s\}\}, \\ \mathcal{G}_{\mathrm{rig}} &:= \{\mathrm{rank}(A) = n\} \cap \{\text{the random-code rigidity conclusion holds}\}. \end{aligned}$$

and $\mathcal{G}_*(\tau) = \mathcal{G}_{\mathrm{ver}}(\tau) \cap \mathcal{G}_{\mathrm{rig}}$. The residual-flatness event is $\mathcal{F}_\delta$ of Eq. (1240); when all three are needed, write $\mathcal{G}_{*,\delta}(\tau) = \mathcal{G}_*(\tau) \cap \mathcal{F}_\delta$.

For an event G and an integrable family quantity Z, the event-restricted functional and conditional functional are, respectively,

$$\mathfrak{M}^G_{n,m,\eta}(Z) = \mathbb{E}[Z\mathbf{1}_G], \qquad \overline{\mathfrak{M}}^G_{n,m,\eta}(Z) = \mathbb{E}[Z \mid G] \quad (\Pr(G) > 0). \tag{1185}$$

Every unqualified phrase "restricted to G" below means the first, subprobability functional. Conditional expectations always carry the bar. Consequently, for $0 \leq Z \leq 1$, $\mathfrak{M}(Z) \leq \mathfrak{M}^G(Z) + \Pr(G^c)$; no conditional normalization is silently used.

Figure 121 displays the two candidate fibers used in Lemma VIII.5.1.

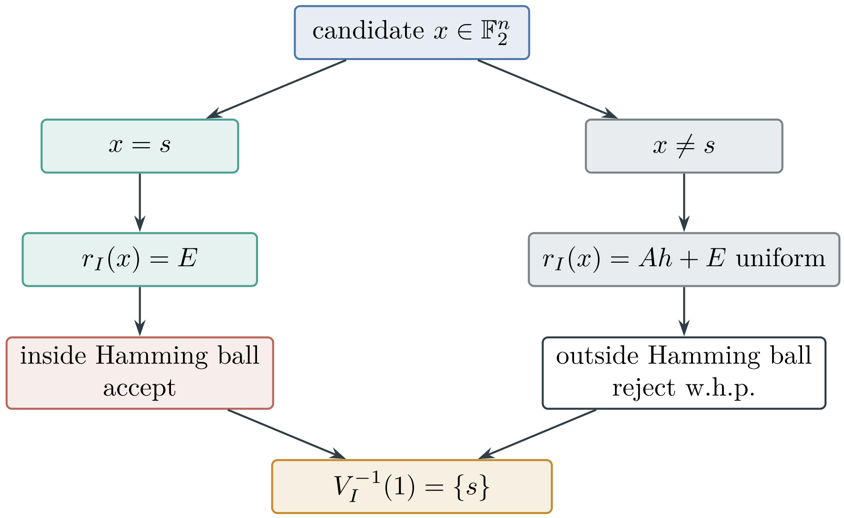

Figure 121: For the public residual verifier, the true secret produces the noise vector inside the threshold ball, whereas each false candidate produces a uniform residual and is rejected with high probability in the sampling regime of Lemma VIII.5.1.

VIII.6 Quantitative Noise–Resource Phase Diagram

This section compares the same LPN presentation template across its noise coordinate. Each value of η defines one presented family as in Definition VIII.1.1; varying η does not enlarge its public constructor grammar. The comparison uses the single scalarization fixed in Definition III.28.1 and, for the polynomial slice, the scalar ceiling $s_C(n) = n^C$ of the declared scalar-sublevel resource budget. Thus its admissible computation class is exactly $\{\mathcal{A} : \kappa(\mathrm{Cost}(\mathcal{A})) \leq s_C(n)\}$. The three thresholds are the represented-cost crossing of Definition IV.14.2, the statistical-resolution condition of Theorem IV.14.4, and the information carried by the binary flip channel.

VIII.6.1 Exact finite-budget error enumeration

Use the task-independent Hamming-ball count $N_m(k)$ from Eq. (296) and Lemma IV.14.1. For the LPN noise law, define its exact binomial mass by

$$F_{m,\eta}(k) = \sum_{j=0}^{k} \binom{m}{j} \eta^j (1-\eta)^{m-j}. \tag{1186}$$

Set $F_{m,\eta}(-1) = 0$. This quantity is exactly $\Pr\{\mathrm{Bin}(m,\eta) \leq k\}$.

Fix a verifier radius $0 < \tau < 1/2$. Let $c_{\mathrm{prep}}(n,m)$ be the scalarized represented cost of row-reducing the public matrix and initializing the enumerator. Let $c_{\mathrm{cand}}(n,m)$ be the scalarized represented cost of constructing one error vector, solving the corresponding reduced linear system, and applying the public verifier of Lemma VIII.5.1. Both are polynomial because they use only the public binary linear-algebra and Hamming readouts of Definition VIII.1.1. Define

$$C_{\mathrm{EE}}(n,m,k) = c_{\mathrm{prep}}(n,m) + c_{\mathrm{cand}}(n,m) N_m(k) \tag{1187}$$

and the exact affordable radius

$$k_B(n,m;\tau) = \max\left(\{-1\} \cup \{k \in \mathbb{Z} : \ 0 \le k \le \lfloor \tau m \rfloor, \quad C_{\mathrm{EE}}(n,m,k) \le B\}\right). \tag{1188}$$

Proposition VIII.6.1 (Exact-binomial success bound for affordable error enumeration)**.** *Assume $m \ge n$. For every B, the scalarized success profile satisfies*

$$\mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta},n}(B) \ge F_{m,\eta}(k_B(n,m;\tau)) - \varepsilon_{\mathrm{alg}}(n,m,\tau), \tag{1189}$$

where

$$\varepsilon_{\mathrm{alg}}(n,m,\tau) = 2^{n-m} + (2^n - 1)2^{-m(1-H_2(\tau))}. \tag{1190}$$

When $k_B(n,m;\tau) \ge 0$, a family-uniform represented LPN computation of scalarized cost at most B witnesses Eq. (1189)*.*

For $0 < \alpha < 1 - \varepsilon_{\mathrm{alg}}$, the exact certified easy-side noise boundary is therefore

$$\eta_{\mathrm{EE}}(n,m;B,\tau,\alpha) = \sup\left\{0 \le \eta < \frac{1}{2} : F_{m,\eta}(k_B(n,m;\tau)) \ge \alpha + \varepsilon_{\mathrm{alg}}(n,m,\tau)\right\}. \tag{1191}$$

Set $\eta_{\mathrm{EE}} = 0$ when this set is empty.

For every η below this boundary, $B^*_{\mathrm{rec}}(n,m,\eta;\alpha) \le B$.

Proof. If $k_B = -1$, the right-hand side of Eq. (1189) is nonpositive, so the profile inequality follows from its nonnegativity. Suppose $k_B \ge 0$. Enumerate the binary vectors e' of weight at most $k_B(n,m;\tau)$. For each vector, solve

$$Ax = b + e'$$

using the stored row reduction and apply $V_I(x)$. Return the first accepted candidate. The complete represented cost is bounded by Eq. (1187).

The union bound over nonzero $h \in \mathbb{F}_2^n$ gives

$$\Pr\{\mathrm{rank}(A) < n\} \le (2^n - 1)2^{-m} < 2^{n-m}.$$

On the full-column-rank event, if $e' = E$, then $Ax = b + E = As$ has the unique solution $x = s$. If $|E| \le k_B$, this error vector is enumerated and $|E| \le \tau m$, so the true candidate passes the verifier. The Hamming-ball argument in the proof of Lemma VIII.5.1 gives

$$\Pr\{\exists x \ne s : V_I(x) = 1\} \le (2^n - 1)2^{-m(1-H_2(\tau))}.$$

Outside these two algebraic failure events, recovery occurs whenever $|E| \le k_B$, an event of exact probability $F_{m,\eta}(k_B)$. Subtracting the two failure probabilities proves Eq. (1189). The boundary statement then follows from Definition IV.14.2. Indeed, let $U_1, \dots, U_m$ be independent uniform random variables on $[0,1]$ and set $X_i(\eta) = \mathbf{1}_{\{U_i \le \eta\}}$. Then $\sum_i X_i(\eta) \sim \mathrm{Bin}(m,\eta)$, and this sum is nondecreasing in η for every realization. Consequently $F_{m,\eta}(k)$ is nonincreasing in η. The set in Eq. (1191) is therefore an initial interval, and every $\eta < \eta_{\mathrm{EE}}$ satisfies the displayed success bound at level α.

$\square$

Figure 122 displays one exact finite instance of Eq. (1189) before the separately bounded algebraic failure term is subtracted.

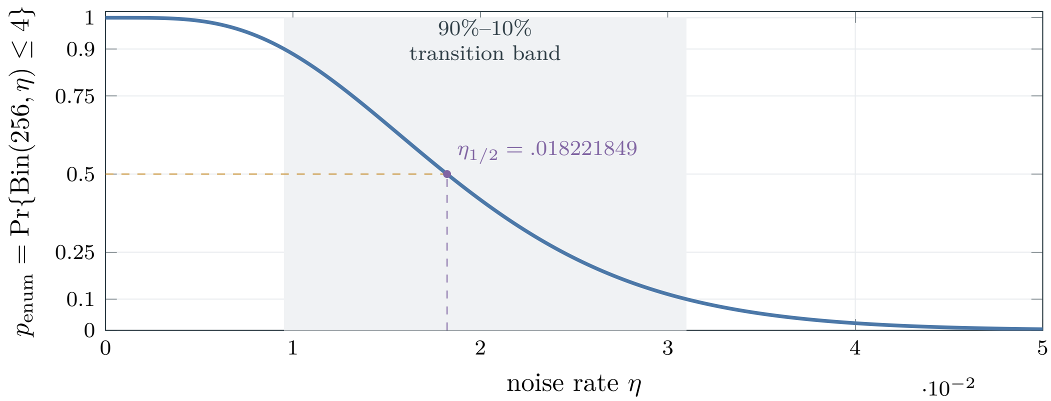

Figure 122: Exact binomial success transition for enumeration through error weight $k_B = 4$ with $m = 256$ samples. The curve is $\sum_{j=0}^{4} \binom{256}{j} \eta^j (1-\eta)^{256-j}$. Its 90%, 50%, and 10% crossings are 0.009531622, 0.018221849, and 0.030981875, respectively. The general budget-dependent curve is Eq. (1189) with the exact affordable radius from Eq. (1188).

Corollary VIII.6.2 (Polynomial enumeration window)**.** *Suppose $m = \Theta(n^a)$ for some $a > 0$, $c_{\text{cand}}(n,m) = \Theta(n^d)$, and $c_{\text{prep}}(n,m) = O(n^{d_0})$ for some $d < C$ and $d_0 < C$. Under the fixed scalar ceiling $s_C(n) = n^C$, assume also*

$$m - n = \omega(\log n), \qquad m(1 - H_2(\tau)) - n = \omega(\log n),$$

so that $\varepsilon_{\text{alg}}(n,m,\tau)$ is negligible. Then, away from the equality case $d + ak = C$, the affordable radius is the largest fixed integer satisfying

$$\max\{d_0, d + ak\} < C. \tag{1192}$$

For a fixed affordable radius k_C and fixed $0 < \alpha < 1$, let $\lambda_{k_C,\alpha}$ be the unique nonnegative solution of

$$e^{-\lambda} \sum_{j=0}^{k_C} \frac{\lambda^j}{j!} = \alpha. \tag{1193}$$

Then, with negligible algebraic failure,

$$\eta_{\text{EE}}(n,m; s_C, \tau, \alpha) = \frac{\lambda_{k_C,\alpha}}{m} + o(m^{-1}). \tag{1194}$$

For the nonnegligible-success criterion, if $\eta m = c \log n$ with fixed $c > 0$ and $m = \Theta(n^a)$, even the weight-zero member of the same computation succeeds with probability

$$(1-\eta)^m = n^{-c+o(1)}. \tag{1195}$$

If k_C is fixed and $\eta m = \omega(\log n)$, then $F_{m,\eta}(k_C) = n^{-\omega(1)}$. Thus bounded-weight enumeration has constant-success scale m^{-1} and nonnegligible-success scale $(\log n)/m$.

Proof. For fixed k, combine Eqs. (298) and (1187). The candidate term has order n^{d+ak}, giving Eq. (1192); constants decide only the equality case. If $\eta = \lambda/m$, the binomial law converges to Poisson(λ), which proves Eq. (1194). More explicitly, for fixed k_C the convergence of

$$F_{m,\lambda/m}(k_C) \longrightarrow e^{-\lambda} \sum_{j=0}^{k_C} \frac{\lambda^j}{j!}$$

is uniform for λ in compact intervals. The limiting function is continuous and strictly decreasing from 1 to 0, since its derivative for $\lambda > 0$ is

$$-e^{-\lambda} \frac{\lambda^{k_C}}{k_C!}.$$

Its level-α solution is therefore unique. The negligible term ε_{alg} moves the corresponding λ-crossing by $o(1)$, and hence moves the noise crossing by $o(m^{-1})$.

For $\eta m = c \log n$, under this scaling $\eta = o(1)$ and

$$m \log(1 - \eta) = -\eta m + O(\eta^2 m) = -c \log n + o(1),$$

which proves Eq. (1195). Finally,

$$F_{m,\eta}(k_C) \leq (k_C + 1) m^{k_C} \exp[-\eta(m - k_C)].$$

When $\eta m = \omega(\log n)$ and m is polynomial, the right-hand side is $n^{-\omega(1)}$.

□

VIII.7 Asymptotic, Statistical, and Information Thresholds

The preceding polynomial boundary is distinct from the point at which error enumeration becomes more expensive than exhaustive secret search.

Corollary VIII.7.1 (Error-enumeration and secret-search crossover)**.** *Let $n/m \to R \in (0, 1)$, fix $0 < \eta < 1/2$, and let $k_m/m \to \eta$. Then*

$$\log_2 N_m(k_m) = m H_2(\eta) + o(m). \tag{1196}$$

Consequently typical-error enumeration and exhaustive search through the 2^n secrets have respective normalized exponents

$$E_{\text{err}}(\eta; R) = \frac{H_2(\eta)}{R}, \qquad E_{\text{secret}} = 1. \tag{1197}$$

They cross at

$$\eta_{\text{X}}(R) = H_2^{-1}(R). \tag{1198}$$

This is an exponential-algorithm crossover. For a continuous visualization of the polynomial scale, suppose the scalarized per-candidate cost is $n^{d+o(1)}$ for some $d < C$. Under the scalar ceiling $s_C(n) = n^C$, define the leading-entropy count proxy

$$\beta_{C,d}(n) = \frac{(C - d) \log_2 n}{n} \longrightarrow 0, \tag{1199}$$

and its formal intersection with the constant-rate entropy curve by

$$\eta_{\text{poly}}^{\text{ent}}(n; C, d, R) = H_2^{-1}\left(R \beta_{C,d}(n)\right). \tag{1200}$$

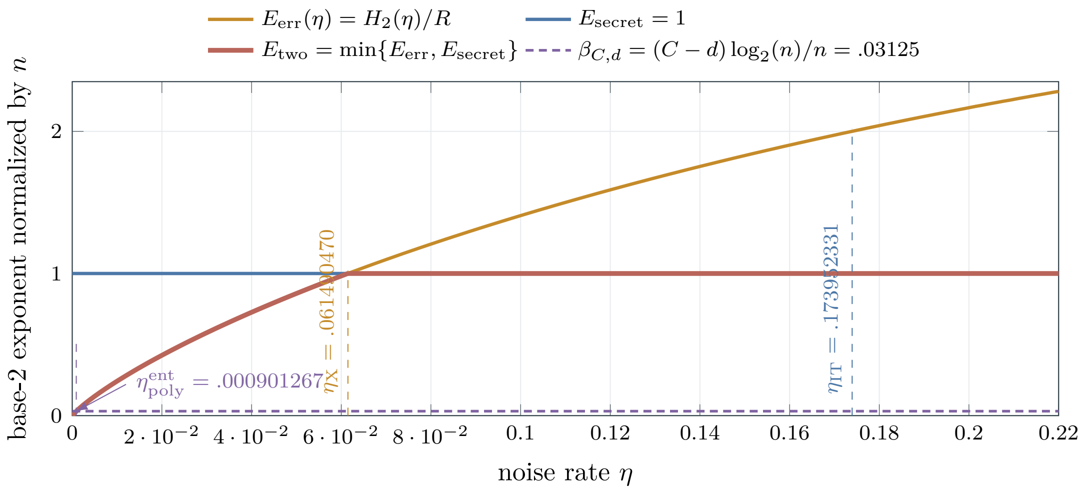

Figure 123: Cost exponents at the illustrative rate $R = n/m = 1/3$. The attack exponents and proxy are Eqs. (1197), (1198) and (1200). Here $\eta_{\text{X}} = H_2^{-1}(R) = 0.061490470$, while the information boundary of Theorem VIII.7.3 is $\eta_{\text{IT}} = H_2^{-1}(1-R) = 0.173952331$. For the displayed finite values $n = 256$, $C = 2$, and $d = 1$, $\beta_{C,d} = .03125$ and its entropy-proxy crossing is 0.000901267; the exact finite polynomial crossing uses the binomial count in Eq. (1188).

The rigorous polynomial-budget intersection is the exact binomial-cost condition in Eq. (1188)*; for fixed error weight it is Eq.* (1192)*. These formulas govern the vanishing-noise scale plotted by the proxy.*

Proof. Apply both sides of Eq. (297) and use continuity of H_2. Division by $n = Rm + o(m)$ proves Eq. (1197); strict monotonicity of H_2 on $[0, 1/2]$ gives Eq. (1198). The final two displays define the continuous count proxy. Exact polynomial affordability was proved in Lemma IV.14.1 and Corollary VIII.6.2.

□

Corollary VIII.7.2 (Explicit LPN Rademacher threshold)**.** *Put*

$$G_{n,\delta} = 2\sqrt{2n\log 2} + 3\sqrt{\frac{\log(2/\delta)}{2}}, \tag{1201}$$

so that $\Gamma^{\text{par}}_{n,m,\delta} = G_{n,\delta}/\sqrt{m}$ *in Eq.* (1135)*. An* ε_{opt}*-empirical parity minimizer is certified to equal* f_s *whenever*

$$\eta < \eta_{\text{Rad}}(n, m, \delta, \varepsilon_{\text{opt}}) := \frac{1}{2} - \frac{2G_{n,\delta}}{\sqrt{m}} - \varepsilon_{\text{opt}}. \tag{1202}$$

Equivalently, when $1 - 2\eta > 2\varepsilon_{\text{opt}}$*, it suffices that*

$$m > \frac{16G_{n,\delta}^2}{(1 - 2\eta - 2\varepsilon_{\text{opt}})^2}. \tag{1203}$$

For the complete budget-B saturated statistical class, a clean correlation of magnitude $\varepsilon > 0$ *rises above its simultaneous selection radius only when*

$$(1-2\eta)\varepsilon > 2\Gamma^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m,\delta}, \qquad \textit{equivalently} \qquad \eta < \frac{1}{2} - \frac{\Gamma^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m,\delta}}{\varepsilon}. \tag{1204}$$

This is the statistical-resolution boundary of the existing uniform generalization theorem. The represented cost of constructing the empirical minimizer is still measured by Eq. (300)*.*

Proof. Substitute $\Gamma^{\mathrm{par}}_{n,m,\delta} = G_{n,\delta}/\sqrt{m}$ into Eq. (1137) and solve first for η, then for m. Rearranging the simultaneous deviation bound Eq. (1140) at clean alignment ε gives Eq. (1204).

□

Theorem VIII.7.3 (Sharp LPN information threshold away from equality)**.** *Under the standard random binary LPN law of Definition VIII.1.1, in particular with s uniform on $\mathbb{F}_2^n$, let $P_\mathrm{e} = \Pr\{\widehat{s} \neq s\}$ for an arbitrary, possibly randomized decoder of the LPN sample. Then*

$$H_2(P_\mathrm{e}) + P_\mathrm{e}\log_2(2^n-1) \geq n - m(1-H_2(\eta)). \tag{1205}$$

In particular, for $0 \leq \delta < 1-2^{-n}$, the condition $P_\mathrm{e} \leq \delta$ requires

$$m(1-H_2(\eta)) \geq n - H_2(\delta) - \delta\log_2(2^n-1). \tag{1206}$$

For an exponential strong converse, fix $0<\epsilon<\eta<1/2$ and write

$$L_\eta = \log_2\frac{1-\eta}{\eta}.$$

Every decoder satisfies

$$\Pr\{\widehat{s} = s\} \leq 2e^{-2m\epsilon^2} + 2^{m-n-m(H_2(\eta)-\epsilon L_\eta)}. \tag{1207}$$

Conversely, fix $0<\delta<1-2^{-n}$ and suppose

$$0 < 1 - \frac{n+\log_2(2/\delta)}{m} < 1.$$

Define

$$\tau_\delta(n,m) = H_2^{-1}\left(1 - \frac{n+\log_2(2/\delta)}{m}\right). \tag{1208}$$

If

$$0 \leq \eta \leq \tau_\delta(n,m) - \sqrt{\frac{\log(2/\delta)}{2m}}, \tag{1209}$$

then exhaustive candidate scanning with the public verifier recovers s with probability at least $1-\delta$. The corresponding finite-sample achievability endpoint is

$$\eta^{\mathrm{ach}}_{\mathrm{IT}}(n,m,\delta) = H_2^{-1}\left(1 - \frac{n+\log_2(2/\delta)}{m}\right) - \sqrt{\frac{\log(2/\delta)}{2m}}, \tag{1210}$$

whenever the inverse argument lies in $(0,1)$ and the displayed endpoint is nonnegative. Put

$$K_{n,\delta} = n - H_2(\delta) - \delta\log_2(2^n-1).$$

Whenever $1 - K_{n,\delta}/m \in [0,1]$, the finite-sample converse endpoint is

$$\eta_{\mathrm{IT}}^{\mathrm{con}}(n,m,\delta) = H_2^{-1}\left(1 - \frac{K_{n,\delta}}{m}\right). \tag{1211}$$

Every decoder with success probability at least $1-\delta$ must have $\eta \leq \eta_{\mathrm{IT}}^{\mathrm{con}}$, whereas exhaustive scanning has success at least $1-\delta$ throughout $0 \leq \eta \leq \eta_{\mathrm{IT}}^{\mathrm{ach}}$.

Hence, when $n/m \to R \in (0,1)$, the sharp information threshold away from equality is

$$\eta_{\mathrm{IT}}(R) = H_2^{-1}(1-R). \tag{1212}$$

Below this value exhaustive recovery succeeds with probability tending to one; above it every decoder has exponentially small exact-recovery probability.

Proof. Condition on the public matrix A. The secret is uniform on 2^n values and the channel from As to b consists of m independent binary symmetric channels. Therefore

$$I(s;b \mid A) \leq m(1 - H_2(\eta)).$$

Fano's entropy inequality gives

$$H(s \mid A, b) \leq H_2(P_{\mathrm{e}}) + P_{\mathrm{e}} \log_2(2^n - 1).$$

Since $H(s) = n$, combining these inequalities proves Eq. (1205). The function $p \mapsto H_2(p) + p\log_2(2^n-1)$ is increasing on $[0, 1-2^{-n})$, which gives Eq. (1206). Solving this condition for η on the increasing branch of H_2 gives Eq. (1211).

For the strong converse, let

$$\mathcal{T}_\epsilon = \left\{ e : \left| \frac{|e|}{m} - \eta \right| \leq \epsilon \right\}.$$

Hoeffding's inequality gives $\Pr\{E \notin \mathcal{T}_\epsilon\} \leq 2e^{-2m\epsilon^2}$. Every $e \in \mathcal{T}_\epsilon$ satisfies

$$\Pr\{E = e\} \leq 2^{-m(H_2(\eta) - \epsilon L_\eta)}.$$

For fixed A and received word y, a randomized decoder assigns weights to the possible secrets that sum to one. Reindexing $y = As + e$, summing over the 2^m received words, and using the uniform prior on s bounds the success contribution of the typical set by

$$2^{-n} 2^m 2^{-m(H_2(\eta) - \epsilon L_\eta)}.$$

Adding the atypical probability proves Eq. (1207). If $R > 1 - H_2(\eta)$, choose

$$0 < \epsilon < \min\left\{ \eta, \frac{R - 1 + H_2(\eta)}{L_\eta} \right\};$$

both terms then decay exponentially.

For achievability, the definition of τ_δ gives

$$(2^n - 1) 2^{-m(1 - H_2(\tau_\delta))} \leq \frac{\delta}{2}.$$

If $\eta = 0$, the true residual is identically zero and the displayed false-candidate bound proves the claimed success probability directly. Assume $0 < \eta < 1/2$.

The Bernoulli Pinsker bound $D_{\mathrm{Ber}}(\tau_\delta\|\eta) \geq 2(\tau_\delta - \eta)^2$ and Eq. (1209) give

$$e^{-mD_{\mathrm{Ber}}(\tau_\delta\|\eta)} \leq \frac{\delta}{2}.$$

Substitute both estimates into Eq. (1184). On the resulting event the verifier has the unique accepting candidate s, so exhaustive scanning recovers it. This proves the endpoint statement in Eq. (1210). For any sequence $\delta_n \to 0$ with $\log(1/\delta_n) = o(m)$, both finite endpoints converge to $H_2^{-1}(1-R)$. The achievability endpoint gives the lower side of Eq. (1212); the strong converse gives the upper side.

□

Corollary VIII.7.4 (Rate asymptotics of the exponential and information thresholds)**.** *The attack crossover $\eta_{\mathrm{X}}(R) = H_2^{-1}(R)$ and information threshold $\eta_{\mathrm{IT}}(R) = H_2^{-1}(1-R)$ satisfy*

$$\eta_{\mathrm{X}}(R) \sim \frac{R}{\log_2(1/R)}, \qquad \eta_{\mathrm{IT}}(R) = \frac{1}{2} - \sqrt{\frac{R\log 2}{2}} + O(R^{3/2}) \quad (R \downarrow 0), \tag{1213}$$

$$\eta_{\mathrm{X}}(R) = \frac{1}{2} - \sqrt{\frac{(1-R)\log 2}{2}} + O((1-R)^{3/2}), \quad \eta_{\mathrm{IT}}(R) \sim \frac{1-R}{\log_2(1/(1-R))} \qquad (R \uparrow 1). \tag{1214}$$

Proof. As $u \downarrow 0$,

$$H_2(u) = u\log_2(1/u) + \frac{u}{\log 2} + O(u^2),$$

whose monotone inverse is $H_2^{-1}(y) \sim y/\log_2(1/y)$. As $x \to 0$,

$$H_2\left(\frac{1}{2} - x\right) = 1 - \frac{2x^2}{\log 2} + O(x^4),$$

so $H_2^{-1}(1-y) = 1/2 - \sqrt{y\log 2/2} + O(y^{3/2})$. Substitute $y = R$ and $y = 1-R$.

□

VIII.7.1 The saturated computational boundary

For $0 < \vartheta \leq 1$, write

$$B_\vartheta^*(n; \mathrm{LPN}_\eta) = \inf\left\{s \geq 0 : I^{\mathrm{sat}}_{\mathfrak{M}_{n,m,\eta,n,\kappa}}(s) \geq \vartheta\right\}. \tag{1215}$$

Here $I^{\mathrm{sat}}_{\mathfrak{M}_{n,m,\eta,n,\kappa}}(s)$ is the family profile of Definition III.18.4 restricted to records with scalarized cost at most s. Thus Eq. (1215) is precisely the saturated cost invariant of Definition III.28.1 with the family and visibility level shown explicitly.

Theorem VIII.7.5 (Saturated hard-side threshold for LPN)**.** *Fix the declared polynomial scalar-sublevel resource ceiling $\mathcal{B}_C$, whose admissible computation class is*

$$\mathsf{Alg}_{n,\leq\mathcal{B}_C} = \left\{\mathcal{A} : \kappa(\mathrm{Cost}(\mathcal{A})) \leq s_C(n) = n^C\right\},$$

and let $H_C(n) = H_{\mathcal{B}_C}(n)$ be its normalized polynomial horizon. Let $\Psi(\mathcal{B}_C, n)$ be the class-preserving representation and first-hit enlargement in Theorem III.24.3. For every target success level $0 < \alpha_n \le 1$,

$$\kappa(\Psi(\mathcal{B}_C, n)) < B^*_{\alpha_n/(eH_C(n))}(n; \mathrm{LPN}_\eta) \quad \Longrightarrow \quad \mathrm{Succ}_{\mathfrak{M}_{n,m,\eta,n}}(\mathcal{B}_C) < \alpha_n. \tag{1216}$$

The three quantitative comparison bounds are as follows.

(i) *Under the hypotheses of Proposition VIII.6.1, for $0 < \alpha_n < 1 - \varepsilon_{\mathrm{alg}}$ and $0 \le \eta < \eta_{\mathrm{EE}}(n, m; s_C, \tau, \alpha_n)$, the represented error enumerator gives $B^*_{\mathrm{rec}}(n, m, \eta; \alpha_n) \le s_C$. Consequently,*

$$\eta_{\mathrm{EE}}(n, m; s_C, \tau, \alpha_n) \le \eta^{\mathrm{sat}}_{\mathrm{comp}}(n, m; s_C, \alpha_n). \tag{1217}$$

For fixed affordable radius this includes constant success at $\eta = \Theta(m^{-1})$ and nonnegligible success at $\eta = O((\log n)/m)$, as quantified in Corollary VIII.6.2.

(ii) *Exact empirical parity identification is statistically certified below η_{Rad} in Eq.* (1202)*. This condition measures sample resolution. If the complete scalarized construction cost of the empirical minimizer is at most s_C and $\eta_{\mathrm{Rad}} > 0$, then*

$$\eta_{\mathrm{Rad}}(n, m, \delta, \varepsilon_{\mathrm{opt}}) \le \eta^{\mathrm{sat}}_{\mathrm{comp}}(n, m; s_C, 1 - \delta). \tag{1218}$$

(iii) *Information permits high-probability recovery below $\eta_{\mathrm{IT}}(R) = H_2^{-1}(1 - R)$ and forbids it above that value, away from equality, by Theorem VIII.7.3. At finite (n, m, δ), whenever the converse endpoint is defined,*

$$\eta^{\mathrm{sat}}_{\mathrm{comp}}(n, m; s_C, 1 - \delta) \le \eta^{\mathrm{con}}_{\mathrm{IT}}(n, m, \delta). \tag{1219}$$

For a vanishing schedule $\eta = \eta(n)$, the supremal affordable noise is Eq. (301)*; when the ceiling absorbs the flip postprocessing of Proposition IV.14.3, it is the right endpoint of the affordable interval. Eq.* (1216) *is its framework-level hard-side certificate. Under the polynomial-enumeration hypotheses in item* (i)*, the present estimates locate the nonnegligible-success easy side through $\eta m = O(\log n)$.*

Proof. If the inequality on the left of Eq. (1216) holds, the definition in Eq. (1215) gives

$$m^{\mathrm{sat}}_{\mathfrak{M}_{n,m,\eta,n}}(\Psi(\mathcal{B}_C, n)) \le I^{\mathrm{sat}}_{\mathfrak{M}_{n,m,\eta,n}}(\Psi(\mathcal{B}_C, n)) \le I^{\mathrm{sat}}_{\mathfrak{M}_{n,m,\eta,n,\kappa}}(\kappa(\Psi(\mathcal{B}_C, n))) < \frac{\alpha_n}{eH_C(n)}.$$

Applying Theorem III.24.3, equivalently Corollary III.23.4, now proves the strict success bound α_n.

For part (i), the initial-interval conclusion proved in Proposition VIII.6.1 gives, for every $0 \le \eta < \eta_{\mathrm{EE}}$, one represented computation of scalarized cost at most s_C and success at least α_n. Hence

$$[0, \eta_{\mathrm{EE}}) \subseteq \left\{ \eta : \mathrm{Succ}^{\kappa}_{\mathfrak{M}_{n,m,\eta,n}}(s_C) \ge \alpha_n \right\}.$$

Taking suprema and using Eq. (301) proves Eq. (1217); the two asymptotic scales are those of Corollary VIII.6.2.

For part (ii), every $0 \le \eta < \eta_{\mathrm{Rad}}$ satisfies the strict identification inequality in Eq. (1202). The represented empirical minimizer belongs to the scalar sublevel s_C by the stated complete-cost bound, and it recovers the target with probability at least $1-\delta$. Thus the interval $[0, \eta_{\mathrm{Rad}})$ is contained in the defining set for $\eta^{\mathrm{sat}}_{\mathrm{comp}}(n, m; s_C, 1-\delta)$, which proves Eq. (1218) after taking suprema.

For part (iii), if $\eta > \eta^{\mathrm{con}}_{\mathrm{IT}}(n, m, \delta)$, strict monotonicity of H_2 on $[0, 1/2]$ makes Eq. (1206) fail by a positive amount. The monotonicity of

$$p \longmapsto H_2(p) + p \log_2(2^n - 1)$$

on $[0, 1-2^{-n})$ then gives a uniform error bound strictly above δ: continuity and strict monotonicity give a number $\delta_\eta > \delta$, depending only on (n, m, η, δ), such that every decoder satisfies $P_{\mathrm{e}} \geq \delta_\eta$. Hence every decoder has success at most $1 - \delta_\eta < 1 - \delta$. Therefore the defining set for $\eta^{\mathrm{sat}}_{\mathrm{comp}}(n, m; s_C, 1-\delta)$ is contained in $[0, \eta^{\mathrm{con}}_{\mathrm{IT}}]$. Taking its supremum proves Eq. (1219).

□

VIII.8 Structural Core Reduction for LPN

This section applies the Part III master-source decomposition and core-refined first-hit accountability to the standard binary LPN presentation. The source and accounting clauses follow from the Part II component decomposition, the exact Lift identity, and the LPN source bounds proved below.

VIII.8.1 Representation and Geometric Source Bounds

The first phase identifies the declared polynomial LPN execution class with its saturated closure and classifies the primitive and coupled geometric cases.

Lemma VIII.8.1 (Representation of the polynomial LPN execution class with admissible oracle interfaces)**.** *The declared family-uniform polynomial execution class on the LPN presentation, including its presentation-admissible oracle interfaces, embeds, by the encoding-adequacy theorem and budgeted generation lemma, as a represented generator on the full lifted observable configuration at polynomial saturated cost. Hence its target contribution is counted by the generic saturated profile at the class-preserving enlarged cost.*

Proof. Apply the Oracle admissibility and presentation separation theorem to every presentation-admissible interface. Replacing each query vertex by its family-uniform represented evaluator produces a uniform polynomial computation in the original LPN saturated closure, with class-preserving accumulated overhead.

Encoding adequacy simulates the input presentation, random transcript, work state, clock, and output map as observable coordinates. The transition rule of a uniform polynomial computation is a finite constructor sequence using only public data and charged precision. Therefore the induced generator and every terminal readout it uses lie in the budgeted derivability closure at polynomial budget. Admission of a future committor, latent noise vector, planted secret, or quotient to the budgeted algebra is determined by such a represented derivation.

□

Lemma VIII.8.2 (LPN specialization of the component decomposition)**.** *Every represented target-relevant LPN term decomposes into* Geom*,* Caus*,* Abs*,* Lift*,* Bdry*, and residual accounting* J_{res}*. Only* Geom *and* Abs *are generative sources.* Caus *orients differences of a represented potential,*

Lift *reorganizes an existing source, and* Bdry *scores or kills mass supplied by a source. Thus* Caus*,* Lift*, and* Bdry *charge back to* Geom *or* Abs*.*

Proof. This is the Component exhaustiveness theorem, Exhaustive residual normal form theorem, and the Roles of the five structural components and the residual term proposition applied to the LPN lifted configuration. The positive maximum principle gives the generator form. The exposed relation splits local symmetric and local oriented rates. Quotient-compatible movement is Abs. Auxiliary coordinates are valid lifts only when their projection preserves target fraction. Terminal scores and residual-weight predicates are boundary readouts. What remains after all represented source projections is one of the two residual coordinates in Theorem II.7.2. The decomposition is exhaustive by the component exhaustiveness theorem. The following source bounds evaluate these branches in the existing saturated $\mathsf{Geom}/\mathsf{Abs}$ profile. □

Lemma VIII.8.3 (Part II exact Lift identity specialized to LPN)**.** *Let $\pi : \widetilde{X} \to X$ be a valid lift with lifted measure $\widetilde{\mu}$ and target $T \subseteq X$. Validity gives the exact target-fraction identity*

$$\widetilde{\mu}(\pi^{-1}T) = \mu(T), \qquad \int_{\widetilde{X}} h \circ \pi \, d\widetilde{\mu} = \int_X h \, d\mu$$

for every target-relevant base observable h. These identities compose exactly. Therefore any finite composition of valid lifts, including recurrence, stacked auxiliary coordinates, and filtration level-stacking, preserves target fraction exactly and charges to the represented base source with zero target-fraction defect.

Proof. This is the exact Lift identity applied to the presented LPN task. Every admissible composition retains the represented base projection and charges to that base source. □

VIII.8.2 Capacity and arrival transfer under valid lifts

Proposition VIII.8.4 (Native invariant capacity and access under an LPN lift)**.** *Let $I = (A, b, \eta)$ be an LPN presentation on $X_I = \mathbb{F}_2^n$, let μ_I be uniform, and let $T_I = \{s\}$. Let $(\Omega, \mathcal{G}_{\mathrm{obs}}, \pi, \Lambda)$ be a complete valid-Lift record, with $\widetilde{\mu} = \Lambda\mu_I$. Then*

$$\widetilde{\mu}(\pi^{-1}T_I) = \mu_I(T_I) = 2^{-n}. \tag{1220}$$

Suppose first that P_Ω is a represented normalized kernel whose invariant law is $\widetilde{\mu}$, and let Φ_{P_Ω} be its native symmetric geometry. Then

$$\mathrm{Cap}_{\Phi_{P_\Omega}}(\pi^{-1}T_I, \Omega \setminus \pi^{-1}T_I) = Q_{P_\Omega}(\pi^{-1}T_I, \Omega \setminus \pi^{-1}T_I) \le 2^{-n}, \tag{1221}$$

$$\Pr_{\widetilde{\mu}_0}\{\tau_{\pi^{-1}T_I} \le H\} \le R_0(H+1)2^{-n}, \qquad R_0 = \left\| \frac{d\widetilde{\mu}_0}{d\widetilde{\mu}} \right\|_\infty. \tag{1222}$$

Proof. The exact-Lift identity gives target mass 2^{-n}. The boundary-flux identity bounds the invariant target boundary by that mass, and a union bound over the $H+1$ invariant-time marginals, followed by the density comparison $\widetilde{\mu}_0 \le R_0\widetilde{\mu}$, gives the hitting bound. □

Theorem VIII.8.5 (Schur-complement capacity comparison for an LPN lift)**.** *Under the LPN presentation and valid-Lift hypotheses of Proposition VIII.8.4, let the same lift carry a complete form-lift comparison record from Definition V.7.8. Let P be a normalized uniform-invariant base kernel, let Φ_P be its native symmetric geometry, and suppose that on the condenser domain*

$$a_\pi \mathcal{E}_{\Phi_P}(g,g) \le \langle g, H_X g\rangle \le b_\pi \mathcal{E}_{\Phi_P}(g,g).$$

For every represented $F \subseteq X_I \setminus T_I$,

$$\begin{aligned}(1-\rho_\pi)a_\pi \operatorname{Cap}_{\Phi_P}(T_I,F) \le \operatorname{Cap}_\Omega(\pi^{-1}T_I, \pi^{-1}F) &\le b_\pi \operatorname{Cap}_{\Phi_P}(T_I,F) \\ &\le b_\pi 2^{-n}(1-P(s,s)).\end{aligned} \tag{1223}$$

For the opposite plate $F = X_I \setminus T_I$, this becomes

$$(1-\rho_\pi)a_\pi 2^{-n}(1-P(s,s)) \le \operatorname{Cap}_\Omega(\pi^{-1}T_I, \pi^{-1}(X_I\setminus T_I)) \le b_\pi 2^{-n}(1-P(s,s)). \tag{1224}$$

If the lift is form-neutral, $B_0 = 0$, and its pullback form is exactly $\mathcal{E}_{\Phi_P}$, then $\rho_\pi = 0$, $a_\pi = b_\pi = 1$, and

$$\operatorname{Cap}_\Omega(\pi^{-1}T_I, \pi^{-1}F) = \operatorname{Cap}_{\Phi_P}(T_I,F). \tag{1225}$$

Consequently, for every represented form-lift class $\mathscr{L}(B)$ whose comparison geometry is such a normalized LPN base geometry,

$$\sup_{R\in\mathscr{L}(B)} \operatorname{Cap}_{\Omega,R} \le 2^{-n} \operatorname{LiftAmp}^{\mathrm{sat}}_{\mathscr{L}}(B), \tag{1226}$$

with the sharper bound 2^{-n} on the native invariant subclass of Eq. (1221)*.*

Proof. Apply the block-form Schur complement theorem Theorem V.7.9 and then the two-sided base-form comparison. The singleton boundary-flux identity gives the last upper bound. The opposite-plate and form-neutral formulas are the corresponding specializations, and taking the supremum over the represented class gives Eq. (1226). □

Theorem VIII.8.6 (Clocked arrival transfer for a valid LPN lift)**.** *Adopt the presentation, lift, and target notation of Proposition VIII.8.4.*

For the finite-horizon comparison, let P_t^Ω, P_t^X, $0 \le t < H$, be the stopped or absorbing clocked kernels and let U_t be their represented pullbacks. Put

$$E_t = P_t^\Omega U_{t+1} - U_t P_t^X.$$

Write $p^\Omega_{T_I,H}$ for first arrival at $\pi^{-1}T_I$ on the lifted carrier and $p^X_{T_I,H}$ for first arrival at T_I on the base carrier, under the compared represented initializations. Likewise, $p^\Omega_{\widehat{T}_I,H}$ below denotes first arrival at $\pi^{-1}\widehat{T}_I$.

If the represented initialization and transported target interfaces have errors $\varepsilon_{\mathrm{init}}$ and $\varepsilon_{\mathrm{gate}}$, then

$$\left| p^\Omega_{T_I,H} - p^X_{T_I,H} \right| \le \varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \|E_t\| + \varepsilon_{\mathrm{gate}}. \tag{1227}$$

Exact clocked intertwining and exact valid-Lift interfaces set the right side to zero. Let $\widehat{T}_I = V_I^{-1}(1)$ be the public LPN verifier gate at threshold τ, with $\eta < \tau < 1/2$. Pointwise in the presentation,

$$\left|p^{\Omega}_{\widehat{T}_I,H} - p^{X}_{T_I,H}\right| \le \varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \|E_t\| + \varepsilon_{\mathrm{gate}} + \mathbf{1}_{\{\widehat{T}_I \neq T_I\}}. \tag{1228}$$

Therefore the LPN family functional satisfies

$$\begin{aligned}\mathfrak{M}_{n,m,\eta}\left(\left|p^{\Omega}_{\widehat{T}_I,H} - p^{X}_{T_I,H}\right|\right) \le \mathfrak{M}_{n,m,\eta}&\left(\varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \|E_t\| + \varepsilon_{\mathrm{gate}}\right) \\ &+ \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] + (2^n-1)2^{-m(1-H_2(\tau))}.\end{aligned} \tag{1229}$$

Proof. The clocked telescoping identity Theorem V.7.13 bounds the difference of the stopped laws by the initialization defect, the sum of the intertwining defects, and the target-interface defect. Replacing the clean gate by the public verifier adds its single symmetric-difference event. Applying the full LPN family functional and the verifier estimate gives the family bound. □

Theorem VIII.8.7 (Fast-fiber resolvent transfer for a valid LPN lift)**.** *Adopt the presentation, lift, and target notation of Proposition VIII.8.4.*

Finally, suppose the complete dynamic lift belongs to $\mathscr{L}_{\mathrm{fm}}(M_F, \kappa_F, \varepsilon_+, \varepsilon_-)$, and put

$$\sigma_\lambda = \frac{\varepsilon_- M_F \varepsilon_+}{\lambda + \kappa_F}.$$

For the clean killed target resolvents, assume that the represented initial functional and the initial target atom, nonterminal row, and target-contact pullback interfaces are exact in the sense of Theorem V.8.2. The operator comparison is

$$\left\|P_\kappa(\lambda - L_\Omega)^{-1}U - (\lambda - L_X)^{-1}\right\| \le \frac{\|R_X\|^2\sigma_\lambda}{1 - \|R_X\|\sigma_\lambda}, \qquad R_X = (\lambda - L_X)^{-1}, \tag{1230}$$

whenever $\|R_X\|\sigma_\lambda < 1$. In a normalized killed-kernel supremum norm, $\|R_X\| \le \lambda^{-1}$. If the represented initial functional and target-contact column have dual and primal norms at most one, then $\sigma_\lambda < \lambda$ gives

$$A^{\Omega}_{T_I}(\lambda) = A_{\pi^{-1}T_I}(\lambda; L_\Omega, \widetilde{\mu}_0), \qquad A^{X}_{T_I}(\lambda) = A_{T_I}(\lambda; L_X, \mu_0),$$

and

$$\left|A^{\Omega}_{T_I}(\lambda) - A^{X}_{T_I}(\lambda)\right| \le \frac{\sigma_\lambda}{\lambda(\lambda - \sigma_\lambda)}. \tag{1231}$$

Exact dynamic intertwining gives $\sigma_\lambda = 0$. Replacing the clean gate by $\widehat{T}_I$ adds $\mathbf{1}_{\{\widehat{T}_I \neq T_I\}}$ pointwise, and hence adds at most

$$\exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] + (2^n-1)2^{-m(1-H_2(\tau))} \tag{1232}$$

after applying $\mathfrak{M}_{n,m,\eta}$. Thus the verifier error is one presentation event for the complete clocked or resolvent comparison; it is not multiplied by the horizon.

Proof. The Feshbach identity of Theorem V.7.12 bounds the effective base perturbation by σ_λ. The resolvent identity and a Neumann series give the displayed operator norm whenever $\|R_X\|\sigma_\lambda < 1$. Pairing with the represented initial functional and target-contact column yields the scalar arrival bound. Replacing the target gate changes the comparison only on the verifier error event, whose family probability is the displayed binomial/code bound. $\square$

Corollary VIII.8.8 (Quantitative capacity and access under valid LPN lifts)**.** *The native invariant, Schur-capacity, clocked-arrival, and fast-fiber resolvent conclusions are respectively Proposition VIII.8.4 and Theorems VIII.8.5 to VIII.8.7. They concern different hypotheses and output units; none may be substituted for another without its displayed converter.*

Proof. The valid-Lift identity in Lemma VIII.8.3 applied to the uniform LPN law gives Eq. (1220). For the native invariant lifted kernel, the boundary-flux identity Eq. (481) gives

$$\mathrm{Cap}_{\Phi_{P_\Omega}}(\pi^{-1}T_I, \Omega \setminus \pi^{-1}T_I) = Q_{P_\Omega}(\pi^{-1}T_I, \Omega \setminus \pi^{-1}T_I) \leq \widetilde{\mu}(\pi^{-1}T_I) = 2^{-n}.$$

Substitution in Eq. (484) proves Eq. (1222).

Apply Theorem V.7.9 to the represented condenser domain. The upper base-capacity bound is Eq. (482); uniform invariance gives

$$Q_P(T_I^c, T_I) = \mu_I(s)(1 - P(s,s)) = 2^{-n}(1 - P(s,s)).$$

This proves Eq. (1223). When the opposite plate is all of T_I^c, the indicator of T_I is the unique condenser competitor, so the last display is the exact base capacity. This proves Eq. (1224). The form-neutral conclusion is the $B_0 = 0$ case of Eq. (524); exact pullback-form comparison then proves Eq. (1225). Taking the represented supremum and using $1 - P(s,s) \leq 1$ gives Eq. (1226) by Definition V.7.10.

The corrected observable-pullback form of Theorem V.7.13 gives Eq. (1227). On $\{\widehat{T}_I = T_I\}$, the stopped gates coincide, so the same estimate applies to $p^{\Omega}_{\widehat{T}_I,H}$. On the complementary event both arrival probabilities lie in $[0,1]$, which proves Eq. (1228). Apply the family functional and use Eq. (1184) to obtain Eq. (1229).

The fast-fiber bound Eq. (529) is exactly Eq. (1230). A normalized killed semigroup is a supremum-norm contraction, hence its Laplace-transform resolvent has norm at most λ^{-1}. Pair the operator estimate with the represented initial and target functionals and substitute this norm bound. The exact pullback interfaces eliminate the three interface-error terms in Eq. (542), giving Eq. (1231). Exact intertwining sets $E_+ = 0$, hence $\sigma_\lambda = 0$. The same good-event split used for the clocked comparison proves Eq. (1232) for the public verifier gate.

$\square$

Figure 124 separates the two numerical comparisons available for a valid lift: static elimination of fiber modes and clocked transport of stopped laws.

VIII.8.3 Ambient ancestry and permutation bounds

Definition VIII.8.9 (Residual-independent and residual-ancestry ambient LPN records)**.** Consider a temporally admissible LPN master certificate in the ambient family $\mathcal{C}_X(B)$ of Definition III.18.4; thus its terminal quotient is the identity and its geometry slot is active. Apply least-cost source assignment as in Theorem III.25.2, so that valid Lift, Bdry, and filtration postprocessings retain their authenticated source ancestry.

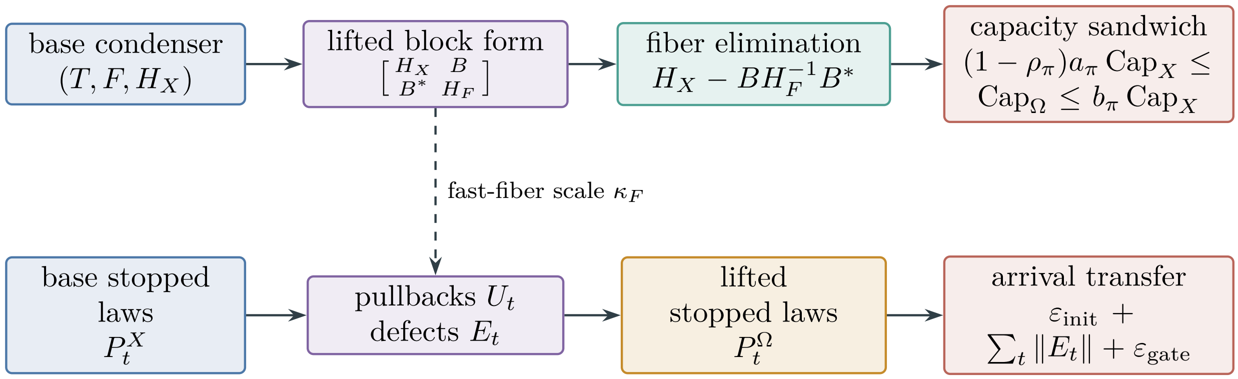

Figure 124: Two distinct valid-Lift comparisons. The upper row eliminates fiber modes and compares condenser energies; the lower row telescopes clocked intertwining defects and compares first-arrival probabilities. Target-fraction preservation initializes both rows but does not set either ρ_π or the clocked defects to zero.

The record is *residual-independent* when the source-ancestor sub-DAG of its complete qualified active record $\mathrm{Act}_{\mathsf{Geom}}(\mathscr{C})$, including every represented qualification gate, factors through A, η, the candidate coordinates, public constants, and represented randomness, and has no read of b or of the residual primitive $r_I(x) = Ax + b$. It is *residual-ancestry* when that complete source-ancestor sub-DAG contains such a read. This binary distinction records only whether the public noisy-label primitive occurs in the complete qualified ancestry. It is not a structural-source class. Write

$$\mathcal{C}^{\mathrm{LPN}}_{X,0}(B), \qquad \mathcal{C}^{\mathrm{LPN}}_{X,b}(B)$$

for the two record families. They form a disjoint partition of the ambient LPN family. For the LPN expectation functional $\mathfrak{M}_{n,m,\eta}$, let

$$G^{\mathrm{src,sat}}_{X,0,\mathfrak{M},n}(B), \qquad G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B)$$

denote the corresponding restrictions of the existing prospective ambient profile. These are provenance restrictions of $G^{\mathrm{src,sat}}_{X,n}$, not additional structural coordinates.

Corollary VIII.8.10 (Five-family classification of residual-ancestry LPN geometry)**.** *Every record in $\mathcal{C}^{\mathrm{LPN}}_{X,b}(B)$ has the unique ambient support tag X and a nonempty source signature contained in*

$$\{\mathsf{Add}, \mathsf{Mult}, \mathsf{Ord}, \mathsf{Sym}, \mathsf{Filt}\}.$$

Its complete active tuple, including every locality, conductance, potential, gap, reach, regularity, current, authority, and comparison datum used by its visibility, factors through the canonical charged joint frontier of Definition III.2.3 at the class-preserving enlarged cost $\Psi_{\mathsf{Geom},5}(B)$. Mixed derivations retain every applicable tag. Consequently, "contains a read of b or r_I" leaves no unclassified geometric source: it is exhausted by the five existing provenance families with ambient support, and introduces neither a sixth provenance family nor a new structural channel.

Proof. The identity terminal quotient fixes the support tag X. Apply Theorem III.2.4 to the complete active derivation selected in Definition VIII.8.9. Its factorization, nonempty signature, and complete-cost bound are Eqs. (29) and (30).

□

Lemma VIII.8.11 (Target-permutation bound for residual-independent LPN geometry)**.** *For every saturated budget B,*

$$G^{\mathrm{src,sat}}_{X,0,\mathfrak{M},n}(B) \le 2^{-n}. \tag{1233}$$

Equivalently, a residual-independent ambient geometry has no positive expected advantage over target-neutral enumeration. A selector whose structural source ancestry remains residual-independent has zero expected advantage, under the LPN presentation law, over the uniform kernel on the remaining untested candidates.

Proof. Put $N = 2^n$. Fix one family-uniform record in $\mathcal{C}^{\mathrm{LPN}}_{X,0}(B)$, and condition on A and on every represented random coordinate in its source derivation. By Definitions VIII.1.1 and VIII.8.9, its potential D is then independent of the uniform secret s. Let

$$Z = \{x \in \mathbb{F}_2^n : D(x) = 0\}, \qquad k = |Z|.$$

A qualified ambient record for the singleton target $\{s\}$ requires $D(s) = 0$, by Definition III.14.9. Equivalently for the family-functional evaluation, extend the contribution of a derivation scheme by zero on every instance on which its output is not a qualified record. If D is constant, its variance and ambient visibility are zero. Suppose that D is nonconstant and $s \in Z$. Since $\mathcal{M}(D) \subseteq L^2(\sigma(D))$, orthogonal-projection monotonicity gives

$$R_{\{s\}}(D) \le \mathrm{ProjMass}_{\sigma(D)}(\{s\}).$$

The fiber of D containing s is Z. For the centered singleton contrast $y_s = \mathbf{1}_{\{s\}} - N^{-1}$, direct conditional expectation on the fibers of D gives

$$\mathrm{ProjMass}_{\sigma(D)}(\{s\}) = \frac{N-k}{k(N-1)}. \tag{1234}$$

Indeed, $\mathbb{E}[y_s \mid \sigma(D)]$ equals $k^{-1} - N^{-1}$ on Z and $-N^{-1}$ off Z; its squared norm divided by $\|y_s\|_2^2 = (N-1)/N^2$ is the right-hand side of (1234). Every other factor in the ambient visibility of Definition III.14.9 lies in $[0,1]$. Therefore, conditional on the fixed source record,

$$\begin{aligned}\mathbb{E}_s\Big[\mathbf{1}_{\{s\in Z\}} V^X_{\Phi,D,J,\mathfrak{K}}\Big] &\le \frac{k}{N}\frac{N-k}{k(N-1)}\\ &= \frac{N-k}{N(N-1)} \le \frac{1}{N}.\end{aligned}$$

Removing the conditioning preserves the bound. The expectation occurs before the supremum over the one uniform derivation scheme in the definition of the family-functional profile. Taking that supremum proves (1233).

For the operational statement, let $\mathcal{H}^0_t$ be the represented subrecord containing A, the residual-independent source data, the selector's randomness, the candidates already tested, and their negative outcomes. The ancestry condition makes the selector kernel measurable with respect to $\mathcal{H}^0_t$; in particular, it does not use b. Conditional on $\mathcal{H}^0_t$ and survival, the secret is uniform on the remaining untested candidates, because s is initially uniform and each negative outcome removes exactly the tested candidate. If N_t candidates remain, then

$$\mathbb{E}\Big[K_t(s \mid \mathcal{H}^0_t)\,\Big|\,\mathcal{H}^0_t, \tau > t\Big] = \frac{1}{N_t}\sum_{x\in\mathcal{C}_t} K_t(x \mid \mathcal{H}^0_t) = \frac{1}{N_t}.$$

The same identity holds for the uniform baseline. Averaging over $\mathcal{H}_t^0$ proves that the step advantage in Eq. (150) is zero. This is the target-permutation calculation of Corollary III.22.5, applied to the represented source subfiltration through which the selector factors.

$\square$

VIII.9 Residual Perturbations, Likelihood Quotients, and Direct-Target Obstructions

VIII.9.1 Residual Displacement and Simultaneous Separation

Lemma VIII.9.1 (Exact residual displacement laws for LPN)**.** *Let $I = (A, b, \eta)$ be an LPN presentation and put*

$$W_I(x) = |r_I(x)|_1 = |Ax + b|_1.$$

Every ordered pair of candidate states has a unique displacement $u = z + x \in \mathbb{F}_2^n$, so $z = x + u$, and

$$r_I(x+u) = r_I(x) + Au, \tag{1235}$$
$$W_I(x+u) - W_I(x) = |Au|_1 - 2\big|\operatorname{supp} r_I(x) \cap \operatorname{supp}(Au)\big|. \tag{1236}$$

Writing $v = x + s$, these identities become

$$r_I(x) = Av + E, \qquad r_I(x+u) = A(v+u) + E. \tag{1237}$$

The unique displacement sending x to the target is $u = v = x + s$. For fixed v, u, chosen independently of the LPN draw, the following distributional statements hold:

(i) *If $v = 0$ and $u \neq 0$, then $W_I(s) \sim \mathrm{Bin}(m, \eta)$, $W_I(s+u) \sim \mathrm{Bin}(m, 1/2)$, and*

$$\mathbb{E}\big[W_I(s+u) - W_I(s)\big] = m(1/2 - \eta).$$

Moreover,

$$\Pr\{W_I(s+u) \le W_I(s)\} \le \exp\Big[-\frac{m}{2}(1/2 - \eta)^2\Big]. \tag{1238}$$

(ii) *If $v \neq 0$ and $u \notin \{0, v\}$, then $W_I(x)$ and $W_I(x+u)$ are independent* $\mathrm{Bin}(m, 1/2)$ *random variables. Hence, for every $a > 0$,*

$$\Pr\{|W_I(x+u) - W_I(x)| \ge am\} \le 2\exp\bigg(-\frac{a^2 m}{2}\bigg). \tag{1239}$$

(iii) *If $v \neq 0$ and $u = v$, then $x + u = s$ and the second residual in* (1237) *is E.*

Proof. The additive group gives the unique displacement $z = x + u$, and linearity gives $r_I(x+u) = r_I(x) + Au$. The weight identity follows coordinatewise. For fixed nonzero, distinct vectors, the corresponding random linear forms of a uniform row of A are independent fair bits; adding the independent noise bit preserves the asserted binomial laws. Hoeffding's inequality gives the two fixed-pair tails. $\square$

Lemma VIII.9.2 (Uniform residual flatness and kernel drift)**.** *Adopt the LPN presentation and residual notation of Lemma VIII.9.1.*

For every $\delta > 0$, define the simultaneous residual-flatness event

$$\mathcal{F}_\delta = \{||E|_1 - \eta m| \le \delta m\} \cap \bigcap_{0 \neq w \in \mathbb{F}_2^n} \{||Aw + E|_1 - m/2| \le \delta m\} . \tag{1240}$$

It obeys the uniform estimate

$$\Pr(\mathcal{F}_\delta^c) \le 2e^{-2\delta^2 m} + 2(2^n - 1)e^{-2\delta^2 m} = 2^{n+1}e^{-2\delta^2 m}. \tag{1241}$$

Equivalently, for every $\gamma > 0$, the choice

$$\delta_{n,m,\gamma} = \sqrt{\frac{(n+1)\log 2 + \gamma}{2m}} \tag{1242}$$

gives $\Pr(\mathcal{F}_{\delta_{n,m,\gamma}}^c) \le e^{-\gamma}$*. The target margin in* (1245) *is positive whenever*

$$m > \frac{2((n+1)\log 2 + \gamma)}{(1/2 - \eta)^2}. \tag{1243}$$

Consequently, on $\mathcal{F}_\delta$, simultaneously for every candidate x and every displacement u, including choices made from the complete public presentation,

$$x \neq s,\ x + u \neq s \quad \Longrightarrow \quad |W_I(x + u) - W_I(x)| \le 2\delta m, \tag{1244}$$

$$x \neq s,\ x + u = s \quad \Longrightarrow \quad W_I(x) - W_I(s) \ge m(1/2 - \eta - 2\delta). \tag{1245}$$

For any represented normalized kernel P, including a kernel selected from the complete public instance, put $p_s = P(x, s)$. If $x \neq s$ and $\Delta_\delta = 1/2 - \eta - 2\delta$, then every represented normalized kernel, whether selected before or after observing (A, b), satisfies

$$(P - I)W_I(x) \le -m\Delta_\delta p_s + 2\delta m(1 - p_s), \tag{1246}$$

$$(P - I)W_I(x) \ge -m(1/2 - \eta + 2\delta)p_s - 2\delta m(1 - p_s). \tag{1247}$$

Equivalently, all family-parameter dependence in the normalized drift is displayed by

$$\left| \frac{(P - I)W_I(x)}{m} + (1/2 - \eta)P(x, s) \right| \le 2\delta. \tag{1248}$$

Thus all off-target perturbations contribute at most $2\delta m$ in absolute residual-weight drift, while the only perturbation having the target margin is the unique target-cancelling displacement.

Proof. Hoeffding's inequality applies to $|E|_1$ and to every $|Aw + E|_1$, $w \neq 0$. A union bound over the 2^n residuals gives the stated simultaneous event. On that event the target and off-target intervals give the two displacement bounds. Averaging those bounds against an arbitrary normalized kernel, while isolating its mass at s, gives the upper, lower, and normalized drift formulas. $\square$

Lemma VIII.9.3 (Entropy-sharp simultaneous residual separation)**.** *Adopt the notation of Lemmas VIII.9.1 and VIII.9.2. There is also an entropy-sharp, rate-dependent form. For every*

$$\eta < \tau_- < \tau_+ < 1/2,$$

the simultaneous target-separation event satisfies

$$\Pr\left\{W_I(s) \le \tau_- m \text{ and } \min_{x \neq s} W_I(x) > \tau_+ m\right\} \ge 1 - e^{-mD_{\mathrm{Ber}}(\tau_- \| \eta)} - (2^n - 1)2^{-m(1 - H_2(\tau_+))}. \tag{1249}$$

On this event, every target-cancelling perturbation has the quantitative margin

$$W_I(x) - W_I(s) > m(\tau_+ - \tau_-) \qquad (x \neq s). \tag{1250}$$

If the event is intersected with $\mathcal{F}_\delta$, then every represented normalized kernel satisfies

$$\frac{(P - I)W_I(x)}{m} < -(\tau_+ - \tau_-)P(x, s) + 2\delta(1 - P(x, s)). \tag{1251}$$

The event supporting this joint perturbation estimate has probability at least

$$1 - 2^{n+1}e^{-2\delta^2 m} - e^{-mD_{\mathrm{Ber}}(\tau_- \| \eta)} - (2^n - 1)2^{-m(1 - H_2(\tau_+))}. \tag{1252}$$

For a rate schedule $n/m \to R \in (0, 1)$, constants τ_-, τ_+ with exponentially small exceptional probability exist precisely throughout the displayed construction whenever

$$\eta < \tau_- < \tau_+ < H_2^{-1}(1 - R). \tag{1253}$$

More explicitly, if $n/m \to R \in (0, 1)$ and $\gamma = o(n)$, then

$$\delta_{n,m,\gamma} = \sqrt{\frac{R \log 2}{2}} + o(1). \tag{1254}$$

Thus the two-sided Hoeffding audit has a positive uniform target margin when

$$\eta < \frac{1}{2} - \sqrt{2R \log 2}, \tag{1255}$$

whenever the right side is positive. The entropy-sharp one-sided audit replaces this sufficient window by $\eta < H_2^{-1}(1 - R)$, through Eq. (1253).

Proof. The lower-tail Chernoff bound controls the planted residual and the Hamming ball estimate controls each nonplanted residual. A union bound proves the separation event. Intersecting it with the uniform flatness event gives the kernel inequality. The interval $\eta < \tau_- < \tau_+ < H_2^{-1}(1 - R)$ is nonempty exactly in the stated rate window; direct substitution gives the Hoeffding comparison window. □

VIII.9.2 Base and Lifted Residual Drift

Proposition VIII.9.4 (Exact base and lifted residual-drift decomposition)**.** *Adopt the notation of Lemma VIII.9.1. For such a kernel, the corresponding one-step drift has the exact instancewise expansion*

$$(P-I)W_I(x)=\sum_{u\in\mathbb{F}_2^n} P(x,x+u)\big(W_I(x+u)-W_I(x)\big). \tag{1256}$$

If $v=x+s\neq 0$, the same sum separates the unique target-cancelling perturbation from every other displacement:

$$\begin{aligned}(P-I)W_I(x)&=P(x,s)\big(W_I(s)-W_I(x)\big)\\&\quad+\sum_{u\notin\{0,v\}} P(x,x+u)\big(W_I(x+u)-W_I(x)\big),\end{aligned} \tag{1257}$$

$$\big|(P-I)W_I(x)-P(x,s)\big(W_I(s)-W_I(x)\big)\big|\le \max_{u\notin\{0,v\}} |W_I(x+u)-W_I(x)|\,, \tag{1258}$$

where the maximum is zero when its index set is empty. These identities also hold with P_Φ in place of P. Thus the represented terminal-contact term and off-target residual-weight variation are the two exhaustive contributions to the drift of every legal identity-supported perturbation kernel. The terminal-contact term is accounted through Theorem VIII.9.8; it is not an additional structural source.

Thus (1235) *and* (1256) *cover every legal identity-supported candidate perturbation. Data-dependent selection of the weights $P(x,x+u)$ remains part of the represented kernel record and its saturated cost.*

More generally, let Ω be a represented clocked or lifted LPN carrier with its existing candidate projection $\pi_X:\Omega\to\mathbb{F}_2^n$, and let P_Ω be a represented normalized kernel on Ω. For $\omega\in\Omega$, put $x=\pi_X(\omega)$ and

$$u(\omega,\omega')=\pi_X(\omega')+\pi_X(\omega).$$

Then

$$(P_\Omega-I)(W_I\circ\pi_X)(\omega)=\sum_{\omega'\in\Omega} P_\Omega(\omega,\omega')\big(W_I(x+u(\omega,\omega'))-W_I(x)\big). \tag{1259}$$

Transitions changing only memory, clock, or transcript coordinates have $u(\omega,\omega')=0$, so they contribute zero to this drift. On $\mathcal{F}_\delta$, if $x\neq s$ and

$$p_s^\Omega(\omega)=\sum_{\omega':\,\pi_X(\omega')=s} P_\Omega(\omega,\omega'),$$

then (1246) *and* (1247) *hold with $(P_\Omega-I)(W_I\circ\pi_X)(\omega)$ and $p_s^\Omega(\omega)$ in place of $(P-I)W_I(x)$ and p_s. The normalized and entropy-sharp bounds Eqs.* (1248) *and* (1251) *hold under the same substitutions.*

Proof. Expand $(P-I)W_I(x)$ over the displacement parametrization and isolate the unique displacement $u=x+s$. The uniform remainder estimate follows by bounding the convex combination by its largest absolute summand. On a lifted carrier, apply the same identity after the candidate projection; transitions contained in one projection fiber have zero displacement and therefore zero residual drift. The base simultaneous bounds then apply to the projected transition mass. □

Corollary VIII.9.5 (Exhaustive residual transport under LPN candidate perturbations)**.** *The exact displacement laws, uniform and entropy-sharp simultaneous events, and base/lifted drift decompositions are Lemmas VIII.9.1 to VIII.9.3 and Proposition VIII.9.4. Together they cover every represented identity-supported perturbation and retain data-dependent kernel selection in the charged record.*

Proof. The additive group of $\mathbb{F}_2^n$ gives the unique representation $z = x + u$. Since $r_I(x) = Ax + b$, linearity gives

$$r_I(x+u) = Ax + Au + b = r_I(x) + Au,$$

which proves (1235). For binary vectors p, q,

$$|p+q|_1 - |p|_1 = |q|_1 - 2|\operatorname{supp} p \cap \operatorname{supp} q|.$$

Substitution of $p = r_I(x)$ and $q = Au$ proves (1236). Equation (1237) follows from $b = As + E$. The equation $x + u = s$ has the unique solution $u = x + s = v$.

For a fixed nonzero vector w, the coordinates of Aw are independent uniform bits. If $v = 0$ and $u \neq 0$, this observation gives the two binomial marginals in part (i). The increment is a sum of m independent variables in $[-1, 1]$, each with mean $1/2 - \eta$. Hoeffding's inequality gives (1238).

If $v \neq 0$ and $u \notin \{0, v\}$, the two vectors v and $v + u$ are distinct and nonzero, hence linearly independent over $\mathbb{F}_2$. For each row a_i, the pair $(\langle a_i, v\rangle, \langle a_i, v+u\rangle)$ is uniform on $\mathbb{F}_2^2$. Adding the common independent bit E_i to both coordinates preserves the uniform product law. Independence across rows gives the asserted independent binomial variables. Their difference is a sum of independent mean-zero variables in $[-1, 1]$; the two-sided Hoeffding bound proves (1239). Part (iii) is the substitution $u = v$ in (1237).

Substituting $z = x + u$ in the definition of $PW_I - W_I$ gives (1256). Isolating the term $u = v$ gives (1257); the remaining kernel weights have sum at most one, so the triangle inequality gives (1258). The same algebra applies to the sample-and-hold kernel P_Φ.

For each fixed nonzero w, the vector $Aw + E$ has independent uniform coordinates. Hoeffding's inequality and a union bound over the $2^n - 1$ nonzero vectors give

$$\Pr\left\{\max_{0\neq w\in\mathbb{F}_2^n} \big||Aw+E|_1 - m/2\big| > \delta m\right\} \leq 2(2^n-1)e^{-2\delta^2 m}.$$

The same inequality applied to $|E|_1$ gives $2e^{-2\delta^2 m}$. A second union bound proves (1241). Substitution of (1242) gives $2^{n+1}e^{-2\delta^2 m} = e^{-\gamma}$. The inequality $\delta_{n,m,\gamma} < (1/2 - \eta)/2$ is equivalent to (1243).

If $x \neq s$, then $v = x + s \neq 0$, so (1237) and $\mathcal{F}_\delta$ place $W_I(x)$ in $[m/2 - \delta m, m/2 + \delta m]$. This proves (1244); comparison with $W_I(s) = |E|_1 \in [\eta m - \delta m, \eta m + \delta m]$ proves (1245). In (1257), the target term lies between $-m(1/2 - \eta + 2\delta)p_s$ and $-m\Delta_\delta p_s$, while the total absolute contribution of all other displacements is at most $2\delta m(1 - p_s)$. This proves Eqs. (1246) and (1247). Because $\mathcal{F}_\delta$ is simultaneous over every nonzero w, no independence between the selected kernel and the public presentation is used in these kernel bounds.

For the entropy-sharp form, Chernoff's inequality gives

$$\Pr\{W_I(s) > \tau_- m\} \leq e^{-mD_{\mathrm{Ber}}(\tau_-\|\eta)}.$$

For each $x \neq s$, the residual $A(x+s) + E$ is uniform on $\mathbb{F}_2^m$. The binary Hamming-ball estimate and a union bound give

$$\Pr\left\{\min_{x\neq s} W_I(x) \leq \tau_+ m\right\} \leq (2^n - 1)2^{-m(1-H_2(\tau_+))}.$$

This proves Eq. (1249). The difference of the two strict thresholds proves Eq. (1250). Isolating the target term in Eq. (1257) and applying Eq. (1244) proves Eq. (1251). Combining the three

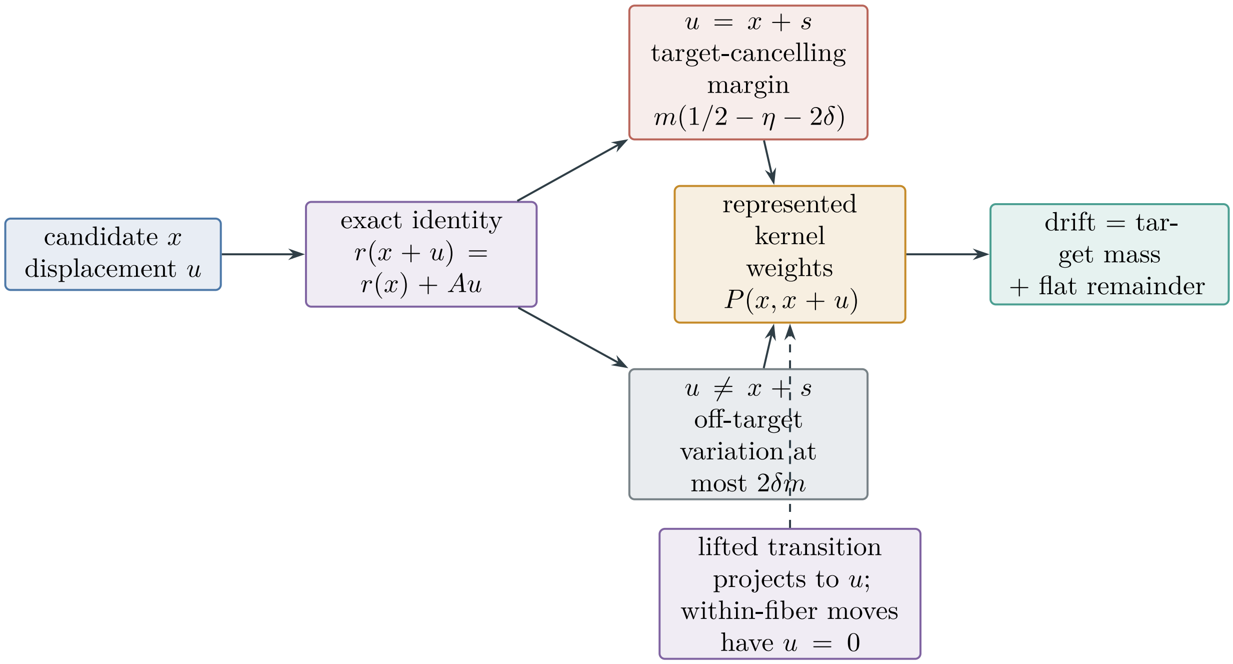

Figure 125: Every candidate transition has one displacement. Exactly one displacement reaches the secret and receives the target margin; all others remain uniformly flat on $\mathcal{F}_\delta$. A data-dependent kernel may choose the weights, but it cannot create a third drift term. The same split holds after projection of a clocked or lifted transition.

exceptional-probability bounds proves Eq. (1252). If $n/m \to R$, its false-candidate exponent is positive exactly when $R < 1 - H_2(\tau_+)$. Strict monotonicity of H_2 on $[0, 1/2]$ gives Eq. (1253); both exceptional terms then decay exponentially. Substituting $m = n/R + o(n)$ and $\gamma = o(n)$ into Eq. (1242) proves Eq. (1254). The condition $1/2 - \eta - 2\delta_{n,m,\gamma} > 0$ then gives Eq. (1255).

For the lifted statement, substitute $x = \pi_X(\omega)$ and $u(\omega, \omega') = \pi_X(\omega') + \pi_X(\omega)$ into $P_\Omega(W_I \circ \pi_X) - W_I \circ \pi_X$. This proves (1259). Grouping successors by their projected displacement recovers (1256); the group with displacement zero has zero increment. On $\mathcal{F}_\delta$, group instead the successors whose candidate projection is s. Their total mass is $p_s^\Omega(\omega)$; every remaining nonzero projected displacement is off-target. The preceding target and off-target bounds therefore prove the two asserted lifted-kernel inequalities.

□

Figure 125 displays the exhaustive candidate-displacement split behind the uniform drift bounds.

VIII.9.3 Residual Likelihood Quotients and Compact Factorization

Theorem VIII.9.6 (Residual likelihood is an exact or compensated quotient record)**.** *Fix an LPN presentation $I = (A, b, \eta)$ with $0 < \eta < 1/2$, and let V_I be the public candidate verifier. Define*

$$W_I(x) = |Ax + b|_1, \qquad \mathcal{L}_I(x) = (1 - \eta)^{m - W_I(x)} \eta^{W_I(x)}. \tag{1260}$$

Then $\mathcal{L}_I = \ell_\eta \circ W_I$, where $\ell_\eta(k) = (1-\eta)^{m-k}\eta^k$ is strictly decreasing on $\{0,\dots,m\}$. Hence W_I and $\mathcal{L}_I$ generate the same represented fiber partition. The terminal-compatible refinement

$$\widetilde{q}_{W,I}(x) = \bigl(V_I(x), W_I(x)\bigr) \tag{1261}$$

has at most $m+2$ nonempty cells, is computable from the original LPN presentation, and has complete saturated cost bounded by the residual evaluation, Hamming-weight, verifier, and pair-encoding costs.

Let P_I be a represented normalized candidate kernel used with this readout. For x' satisfying $W_I(x') = W_I(x)$, exact dynamic compatibility of $\widetilde{q}_{W,I}$ requires, for each quotient cell C,

$$\sum_{z:\widetilde{q}_{W,I}(z)=C} P_I(x,z) = \sum_{z:\widetilde{q}_{W,I}(z)=C} P_I(x',z). \tag{1262}$$

Writing $z = x+u$, every term in this comparison is determined by

$$W_I(x+u) = W_I(x) + |Au|_1 - 2\bigl|\operatorname{supp} r_I(x) \cap \operatorname{supp}(Au)\bigr|. \tag{1263}$$

Thus the scalar likelihood level alone determines the quotient transition law precisely when the represented kernel makes the support-intersection dependence in Eq. (1263) *constant after aggregation over every destination cell.*

Every temporally admissible target-qualified use of the factorized likelihood has the following exhaustive classification.

(i) *When Eq.* (1262) *holds and the quotient-utility condition is positive, the complete record belongs to the exact* Abs *profile.*
(ii) *When the quotient has nonzero represented defect and its compensated comparison passes the existing defect, mixing, consistency, reach, and regularity gates, the complete record belongs to the reversible or general-resolvent compensated quotient mode.*
(iii) *When neither comparison is qualified, the likelihood is a represented terminal-discrimination readout with zero prospective quotient-source value.*
(iv) *A refinement retaining additional residual coordinates is evaluated as the refined quotient when its fibers remain nonidentity. An injective refinement has identity support and belongs to ambient* Geom*; there the likelihood and weight computations are derived evaluators and the complete source value is exactly Eq.* (1286).

Consequently, polynomial evaluation of the factorized likelihood creates no additional identity-supported source coordinate. Its dynamically useful compression is charged to the existing exact or compensated quotient modes; an injective refinement is charged to the existing ambient mode with the same represented kernel, local differences, authority, reach, regularity, and complete cost.

Proof. The first factorization is immediate from the Bernoulli likelihood. Since $\eta/(1-\eta) < 1$, the ratio $\ell_\eta(k+1)/\ell_\eta(k) = \eta/(1-\eta)$ is strictly smaller than one; therefore W_I and $\mathcal{L}_I$ have identical fibers. The map W_I evaluates $Ax+b$ and counts its nonzero coordinates. Pairing it with the represented verifier yields Eq. (1261); its first coordinate isolates the target on the verifier good event, and its remaining cells are indexed by at most $m+1$ weight values.

Exact intertwining for a finite represented quotient is equivalent to equality of the total transition mass from any two states in one fiber to every destination fiber. This is Eq. (1262). The perturbation identity Eq. (1236) is exactly Eq. (1263), proving the stated criterion.

The four alternatives are the exact-support, reversible compensated, general-resolvent compensated, and identity-support modes of Theorem III.18.9 and Definition III.18.1. The exact case additionally uses the quotient-utility gate Definition III.15.1. Derived likelihood, weight, comparison, and verifier evaluators retain their ancestry and add no independent source by Lemmas III.17.1 and I.4.7. The identity-supported value is the complete product in Eq. (1286). These support modes are exhaustive, which proves the final assertion.

□

Remark VIII.9.7 (Compact evaluation does not locate an exponentially supported spectrum). For $0 < \lambda < 1$, the represented factorization

$$F_{\lambda,I}(x) = \prod_{i=1}^{m} \left(1 + \lambda(-1)^{a_i \cdot x + b_i}\right) \tag{1264}$$

is evaluable using m residual tests and $m-1$ multiplications. Its complete record retains the representation of λ and every arithmetic and precision cost. Its formal expansion is

$$F_{\lambda,I}(x) = \sum_{U \subseteq [m]} \lambda^{|U|}(-1)^{\sum_{i \in U} b_i}(-1)^{(A^{\mathsf{T}}\mathbf{1}_U)\cdot x}, \tag{1265}$$

which can contain exponentially many Walsh terms. The compact evaluator does not enumerate or locate those terms. Moreover,

$$F_{\lambda,I}(x) = (1+\lambda)^{m-W_I(x)}(1-\lambda)^{W_I(x)}, \tag{1266}$$

so it generates exactly the residual-weight fibers already classified in Theorem VIII.9.6.

Accordingly, Eq. (1264) receives no spectral source value from the number of terms in Eq. (1265). A represented location rule and its target-relevance comparison are charged as required by Corollary VIII.2.4 and Theorem VIII.2.5. A dynamically qualified nonidentity use is charged to the existing exact or compensated quotient mode. An identity-supported use retains the complete ambient Geom gates. When the factorization supplies only an extremal score, with no represented authorized intermediate transport, it is a terminal verifier by Corollary III.3.8.

VIII.9.4 Direct-Target Perturbation Obstruction

Theorem VIII.9.8 (No-free direct-target perturbation for LPN)**.** *Work on the verifier good event of Lemma VIII.5.1, so that $T_I = \{s\}$ and $V_I = \mathbf{1}_{\{s\}}$. Let $\mathcal{A}$ be a budget-B computation and let $K_t(\cdot \mid \mathcal{H}_t)$ be any of its temporally admissible represented next-test samplers, after the one-test clock refinement of Definition III.21.2. Let ν_t be its declared target-neutral baseline. Then the following identities and provenance alternatives are exhaustive.*

(i) *The acceptance mass of the represented sampler is exactly its terminal-contact mass:*

$$K_t(V_I = 1 \mid \mathcal{H}_t) = K_t(s \mid \mathcal{H}_t). \tag{1267}$$

For a time-homogeneous candidate kernel P, this is the discrete-time specialization

$$k_T^P(x) = \sum_{z \in T_I} P(x,z) = P(x,s)$$

of the terminal-contact gate in Definitions III.7.1 and III.8.2.

(ii) *The one-step and finite-horizon target-contact probabilities are*

$$\Pr\{\tau = t+1\} = \mathbb{E}\Big[\mathbf{1}_{\{\tau>t\}} K_t(s \mid \mathcal{H}_t)\Big], \tag{1268}$$

$$\Pr\{\tau \le H\} = \sum_{t=0}^{H-1} \mathbb{E}\Big[\mathbf{1}_{\{\tau>t\}} K_t(s \mid \mathcal{H}_t)\Big]. \tag{1269}$$

Equivalently,

$$\Pr\{\tau \le H\} = \mathrm{Enum}_\nu(\mathcal{A}, H) + \sum_{t=0}^{H-1} \mathrm{Adv}_t(K_t; \nu_t). \tag{1270}$$

A uniform proposal on all 2^n candidates has $\nu_t(s \mid \mathcal{H}_t) = 2^{-n}$. Uniform sampling without replacement from N_t remaining candidates has conditional contact mass N_t^{-1}.

(iii) *For the fixed kernel P, its H-step target value is the represented truncated committor*

$$u_{H,P} = \sum_{j=0}^{H-1} P_N^j k_T^P. \tag{1271}$$

On a clocked transcript carrier the same formula applies to the homogeneous lifted kernel. For the associated represented generator $L = P - I$, the exact discounted committor is

$$u_{\lambda,L} = (\lambda I - L_N)^{-1} k_T^L.$$

Each expression contributes prospectively only through a represented derivation at its complete saturated cost, as prescribed by Definition III.14.4 and Lemma III.14.5.

(iv) *Any positive excess above the neutral baseline is already bounded by the completed five-family accounting charges. If $H_B(n)$ is the normalized budget horizon and $D_B^{\mathrm{sel}}(n)$ is Eq.* (155)*, then*

$$\big(\mathrm{Adv}_t(K_t; \nu_t)\big)_+ \le e H_B(n) \sum_{c \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D_B^{\mathrm{sel}}(n)). \tag{1272}$$

Thus $K_t(s \mid \mathcal{H}_t)$ is an analytic target-contact coordinate of the represented transition rule. It introduces no accounting coordinate outside the existing five-family ledger. It contributes to a prospective Geom*,* Abs*, or coupled source only through the represented pre-hit comparison required by Corollary III.16.9.*

(v) *The standard LPN primitive signature in Definition VIII.1.1 contains A, b, η, candidate coordinates, the residual evaluator, and their declared constructors. It contains neither s, E, a target sampler, nor a committor. Consequently, a target-biased sampler used by the original presentation has a family-uniform represented derivation in the budgeted saturated closure and is charged by part* (iv)*. An evaluator or sampler without such a derivation is an added oracle primitive or nonuniform advice and defines a different presented family by Convention III.5.10, Proposition III.29.5, and Theorem III.33.2.*

In particular, the verifier supplies an affordable test of a proposed candidate but no target-generating sampler. Under independent uniform proposals, verifier rejection sampling satisfies

$$\Pr\{\tau \le H\} = 1 - (1 - 2^{-n})^H \le H 2^{-n}, \tag{1273}$$

$$\mathbb{E}[\tau] = 2^n. \tag{1274}$$

Replacing the uniform proposal by a target-biased proposal invokes the represented sampler and the charge in Eq. (1272)*.*

Proof. On the verifier good event,

$$K_t(V_I = 1 \mid \mathcal{H}_t) = \sum_{z \in X_I} K_t(z \mid \mathcal{H}_t) \mathbf{1}_{\{z=s\}} = K_t(s \mid \mathcal{H}_t),$$

which proves Eq. (1267). Conditional on $\mathcal{H}_t$ and survival through time t, the next tested candidate has law K_t. Hence its conditional hit probability is the right-hand side of Eq. (1267). Multiplication by the survival indicator and expectation prove Eq. (1268). The first-hit events are pairwise disjoint, so summation proves Eq. (1269). Adding and subtracting the declared baseline in every summand gives Eq. (1270), which is the LPN specialization of Eq. (152). The two displayed neutral masses follow by counting the target in the full and remaining candidate carriers.

For a fixed P, the probability of first entering T_I after j surviving steps is $P_N^j k_T^P$. Summing $j = 0, \ldots, H-1$ proves Eq. (1271). The clock coordinate makes a finite inhomogeneous computation homogeneous on its represented transcript carrier. The discounted formula and its derivability requirements are Definitions III.8.1 and III.14.4; finite truncations are charged by Lemma III.14.5.

The sampler, random-bit use, transition implementation, and complete prefix are ancestors of the next test and lie in the saturated derivability closure at the enlarged represented budget. Applying Theorem III.21.4 proves Eq. (1272). The primitive list and the exclusion of s, E are explicit in Definition VIII.1.1. Computation closure therefore gives the two presentation-relative alternatives in part (v): a sampler in the original task has a charged represented derivation, while adjoining an underived evaluator or instance-dependent table changes the presentation by Theorem III.33.2.

Finally, each independent uniform proposal succeeds with probability 2^{-n}. The failure probability for H proposals is $(1 - 2^{-n})^H$, and the geometric waiting time has mean 2^n. Bernoulli's inequality gives $1 - (1 - 2^{-n})^H \leq H2^{-n}$, proving Eqs. (1273) and (1274).

□

VIII.10 Public-Ancestry Local Geometry and Gate Evaluation

VIII.10.1 Target-edge and ancestry normal forms

Corollary VIII.10.1 (Target-conditioned edge predicates do not supply a sampler)**.** *On the verifier good event, the predicate*

$$(x, z) \longmapsto \mathbf{1}_{E_\Phi}(x, z) \mathbf{1}_{\{V_I(x) \neq V_I(z)\}}$$

evaluates the target-incident part of the represented edge relation E_Φ. Its evaluation does not normalize or sample a transition supported on that set. A valid identity-supported geometric record using these edges therefore contains a represented normalized kernel from the original LPN saturated closure and the same-generator authority comparison of Definition III.12.9.

With a uniform proposal, the sample-and-hold implementation reaches the target with one-step probability at most 2^{-n}. A larger target-contact probability is the selector advantage in Eq. (1272)*. An auxiliary current assigned to the target-incident edges without the represented kernel and comparison fails the authority condition and has* $\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}} = 0$*.*

Proof. Since $V_I = \mathbf{1}_{\{s\}}$, its values differ exactly when one endpoint is s. Edge-membership evaluation is one represented Boolean computation. The definition of a represented sampler in Definition III.21.2 additionally requires its sampling implementation and random-bit use, while Definition III.27.1 forms the canonical sample-and-hold kernel from the selected normalized kernel.

The uniform contact probability and every excess above it are Eqs. (1272) and (1273). Finally, Definition III.12.9 assigns zero alignment when the current lacks the required same-generator authorization.

□

Theorem VIII.10.2 (Public-ancestry normal form for qualified LPN local geometry)**.** *Fix an LPN presentation $I = (A, b, \eta)$, a charged ceiling B, and a temporally admissible, dynamically qualified, identity-supported ambient record in the original LPN saturated closure. At a represented transition, write $x \in \mathbb{F}_2^n$ for the candidate coordinate and ξ for the complete remaining represented state. Put*

$$r = r_I(x) = Ax + b.$$

The active evaluators that read b have exact representations using only A, x, r, ξ, obtained by the public substitution

$$b = r + Ax. \tag{1275}$$

In particular, the selected normalized transition kernel has the exact candidate-displacement representation

$$P_I\big((x,\xi),(x+u,\xi')\big) = \pi_I(u, \xi' \mid A, x, r, \xi), \tag{1276}$$

where π_I is the pushforward of the same represented sampler under $(z, \xi') \mapsto (z + x, \xi')$. Thus π_I is a reparameterization of P_I, not an additional kernel or structural source. Sampling either side and converting between z and $u = z + x$ uses only the public Add operation and retains the complete cost of P_I.

The same substitution gives represented functions $\widehat{D}_I, \widehat{E}_I, \widehat{c}_I$ for the potential, exposed-edge predicate, and conductance. Their local values satisfy

$$D_I(x,\xi) = \widehat{D}_I(A, x, r, \xi), \tag{1277}$$

$$D_I(x+u,\xi') - D_I(x,\xi) = \widehat{D}_I(A, x+u, r+Au, \xi') - \widehat{D}_I(A, x, r, \xi), \tag{1278}$$

$$\mathbf{1}_{E_\Phi}\big((x,\xi),(x+u,\xi')\big) = \widehat{E}_I(A, x, r, \xi, u, \xi'), \tag{1279}$$

$$c_\Phi\big((x,\xi),(x+u,\xi')\big) = \widehat{c}_I(A, x, r, \xi, u, \xi'). \tag{1280}$$

These identities give the following exhaustive structural assignment.

(i) *A record whose active ancestral set contains no read of b or r_I belongs to the residual-independent ambient restriction.*

(ii) *A record whose active ancestral set contains such a read has* Add *source ancestry. Products, comparisons, mixtures, clocks, valid lifts, and derived* Caus *records retain their exact operational signatures and costs but contribute no independent target alignment beyond this charged ancestry. When the readout is expressed spectrally, its exact target correlation and the complete cost of the index-selection rule are those of Theorem VIII.2.3.*

(iii) *A target-qualified nonidentity fiber map belongs to the existing* Abs *profile. With identity support, the complete target contribution is the existing ambient* Geom *gate product evaluated on Eqs.* (1276) *and* (1278)*.*

(iv) *Direct target contact is the single displacement $u = x + s$. Its transition mass is $\pi_I(x + s, \xi' \mid A, x, r, \xi)$, summed over ξ'. This term is admitted only through the represented pre-hit sampler and its charged target-alignment comparison. A verifier or target-edge predicate evaluates a proposed contact and supplies no sampler for this displacement.*

Consequently the LPN local-geometry ledger is exactly the qualified visibility of the represented data in the displays above, with the complete public derivation and cost retained.

Proof. The primitive public data are those of Definition VIII.1.1. At a transition with candidate coordinate x, the identity $b = r_I(x) + Ax$ proves Eq. (1275). Substitute this represented identity at each active read of b. Affordable composition and no-creation Lemmas VIII.8.1 and I.4.7 retain the original active ancestry and charge the additional public Add operations.

Sample (z, ξ') from $P_I((x, \xi), \cdot)$ and put $u = z + x$. Addition by x is a bijection of $\mathbb{F}_2^n$, so this is an exact pushforward with the same normalization. This proves Eq. (1276). The residual transport identity

$$r_I(x + u) = r_I(x) + Au$$

then proves Eqs. (1277) to (1280) by the same substitution in the represented evaluators.

The first two cases are the exact ancestry partition of Definition VIII.8.9, together with the inherited charge identities Lemma III.17.1 and Proposition II.5.9. The support alternatives are the ambient and quotient-supported modes of Theorem III.18.9; target qualification of a nonidentity fiber map is the Abs admission condition. Finally, $x + u = s$ holds exactly when $u = x + s$. The sampler and target-contact statements are Corollary VIII.10.1 and Theorem VIII.9.8.

□

VIII.10.2 Identity-supported gate records

Definition VIII.10.3 (Identity-supported LPN gate record)**.** Work on the good-presentation event of Lemma VIII.5.1, and let

$$\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K}) \in \mathcal{C}_{X,b}^{\mathrm{LPN}}(B)$$

be any temporally admissible, dynamically qualified, identity-supported, residual-ancestry ambient record. Write $\Phi = (E_\Phi, c_\Phi)$ and $\mathfrak{N} = (\widetilde{L}, \zeta, c, P, L, \lambda)$. Then every geometric datum and every legal perturbation in the record is evaluated by the following exhaustive list.

This theorem evaluates records already admitted to $\mathcal{C}_{X,b}^{\mathrm{LPN}}(B)$; it does not construct or admit such a record. When that family is empty, its saturated Geom coordinate is zero.

(i) *Provenance and affordability.* The relation E_Φ, conductance c_Φ, potential D, selected normalized kernel P, current $J_{\mathfrak{K}}$, authority comparison, and their evaluation procedures belong to one represented derivation of total cost at most B. The record is available before the transition in which it is used. Its b-dependent evaluators have the exact public ancestry normal forms of Theorem VIII.10.2.

(ii) *All exposed perturbations.* The candidate-coordinate projection of each charged edge has the unique form $(x, x+u)$. Its sampler is the pushforward Eq. (1276), and its potential difference is Eq. (1278). On that projected edge the public residual changes by Au, as in (1235). The canonical legal local perturbation is the sample-and-hold kernel P_Φ of Definition III.27.1; hence no edge, displacement, or rejected transition lies outside the charged perturbation record. Clocked and lifted transitions obey (1259). On the simultaneous event $\mathcal{F}_\delta$, every such perturbation, including a public-instance-dependent choice, obeys Eqs. (1244) and (1245);

every base or lifted normalized kernel obeys Eqs. (1246) and (1247). The exceptional probability and the positive-margin sample threshold are given explicitly by Eqs. (1241) and (1243). The separated terminal-contact term is exhausted by Theorem VIII.9.8: it is neutral contact, a completed accounting contact, a temporally admissible charged source use, or data outside the original LPN presentation.

Theorem VIII.10.4 (Local gates for identity-supported LPN geometry)**.** *Let $\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K})$ be an admitted record in $\mathcal{C}^{\mathrm{LPN}}_{X,b}(B)$ as in Definition VIII.10.3. Its remaining local gates are the following.*

(iii) Transfer-scale mixing. *The mixing gate is exactly M_Φ. It equals one precisely when the represented comparison data certify the required transfer-class Poincaré or conductance bound for the exposed geometry used by this same normalized record.*

(iv) Authorized local consistency. *The local gate is exactly $\chi^{\mathrm{auth}}_\Phi(D)$. It equals one only when the same selected generator supplies the represented local-consistency, drift, resolvent, or carrier comparison required by Definition III.12.9. The potential or its formal gradient alone does not activate this gate.*

(v) Authorized descent. *When the authority gate is active, the complete directed factor is*

$$\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) = \frac{\left(\langle J_{\mathfrak{K}}, -\nabla_\Phi D\rangle_E\right)_+^2}{\|J_{\mathfrak{K}}\|_E^2 \|\nabla_\Phi D\|_E^2}, \tag{1281}$$

with value zero when either norm vanishes. Its positive pairing is supported on the descending perturbations $E_\downarrow(\mathfrak{g})$ of Definition III.27.1.

(vi) Target reach. *The exact reach satisfies*

$$R_{\{s\}}(D) \le \mathrm{ProjMass}_{\sigma(D)}(\{s\}). \tag{1282}$$

Let V_I be the represented public verifier of Lemma VIII.5.1 and put $D_V = 1 - V_I$. On the good-presentation event, $D_V^{-1}(0) = \{s\}$ and

$$R_{\{s\}}(D_V) = 1. \tag{1283}$$

This unit reach records that the extremum identifies the target. The isolated potential D_V is a terminal verifier. It enters the prospective ambient profile only as part of the same pre-hit record that represents and certifies the intermediate transition, same-generator authority, authorized descent, and regularity required in parts (i)–(v) *and* (vii)*.*

(vii) Dirichlet regularity. *Every potential has the exact ratio*

$$\rho_\Phi(D) = \frac{\dfrac{1}{2}\displaystyle\sum_{x\in\mathbb{F}_2^n}\sum_{u\in\mathbb{F}_2^n} c_\Phi(x, x+u)\big(D(x+u) - D(x)\big)^2 \mu_I(x)\mu_I(x+u)}{\mathrm{Var}_{\mu_I}(D)}. \tag{1284}$$

For $N = 2^n$, the verifier potential has

$$\mathrm{Var}_{\mu_I}(D_V) = \frac{N-1}{N^2}, \qquad \rho_\Phi(D_V) = \frac{1}{N-1}\sum_{z\ne s} c_\Phi(s, z). \tag{1285}$$

Proof. The mixing and authority gates are their defining represented comparisons. The directed factor is the normalized authorized current–gradient pairing. Target reach is bounded by the target mass of the potential sigma-field, and the verifier specialization is exact on the verifier event. Expanding the Dirichlet form of D, and then of the singleton verifier potential, gives the two regularity formulas. □

Corollary VIII.10.5 (Complete gate evaluation for identity-supported LPN geometry)**.** *Under the hypotheses of Definition VIII.10.3 and Theorem VIII.10.4, all provenance and local gates belong to the same represented record.*

Consequently the complete visibility of the record is exactly

$$V^{X}_{\Phi,D,J,\mathfrak{K}} = M_\Phi\, \chi^{\mathrm{auth}}_{\Phi}(D)\ \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D)\, R_{\{s\}}(D)\, \frac{1}{1+\rho_\Phi(D)}. \tag{1286}$$

Parts (i)–(vii) *are one conjunctive admissibility record. None is credited as target-aligned structure in isolation. In particular, an inverse-polynomial Poincaré gap activates only the carrier gate M_Φ; it contributes zero to the LPN Geom profile whenever the same record fails affordability, perturbation provenance, same-generator consistency, authorized descent, target reach, or Dirichlet regularity. The target-aligned transfer quantity is exactly the complete product in Eq.* (1286)*.*

The quantified additive class here is exactly the $(\mathsf{X}, \mathsf{Add})$ *provenance restriction of the saturated prospective Geom profile. Membership requires one family-uniform represented pre-hit derivation of polynomial saturated cost, with the complete active tuple and every gate evaluator retained as in Definition III.2.3 and Theorem III.2.4. The simultaneous residual-flatness estimates in part* (ii) *control the legal residual perturbations available to that tuple. Every dynamically qualified represented nonidentity level-set or spectral coarsening on the same transfer window is charged to the exact or compensated quotient-supported branches by Theorem III.12.6 and Corollary VIII.18.3. Consequently, an affordable target-aligned additive potential that survives quotient exhaustion belongs to the identity-supported* $(\mathsf{X}, \mathsf{Add})$ *cell and contributes only through the explicitly evaluated product Eq.* (1286)*. Its target alignment, local transport, and regularity are evaluated together with the represented cost, authority, and remaining gates of that same record.*

For every $0 < \varepsilon < 1$, a contribution larger than ε therefore requires, in this same represented record,

$$M_\Phi = 1, \quad \chi^{\mathrm{auth}}_{\Phi}(D) = 1, \quad \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) > \varepsilon, \quad R_{\{s\}}(D) > \varepsilon, \quad \frac{1}{1+\rho_\Phi(D)} > \varepsilon. \tag{1287}$$

The verifier potential in (1283) *evaluates the reach gate. Activation of the mixing, local-consistency, and authorized-current gates uses the complete selected geometric record in parts* (i)–(v) *and* (vii)*. Absent that represented intermediate transport, the verifier potential is assigned zero prospective source activity by Corollary III.3.8; its knowledge of the terminal extremum supplies no descent rule.*

Proof. Part (i) is the membership condition in Definitions III.21.1 and III.14.9. Part (ii) follows from Corollary VIII.9.5 and Definition III.27.1. Parts (iii) and (iv) are the mixing and same-generator consistency clauses of Definitions III.14.9 and III.12.9. Part (v) is Definitions III.12.8 and III.12.9; the descending-edge support is Eq. (206).

Every monotone function of D is $\sigma(D)$-measurable, so $\mathcal{M}(D) \subseteq L^2(\sigma(D))$. Orthogonal-projection monotonicity proves (1282). On the verifier good event, $D_V = 0$ at s and $D_V = 1$ elsewhere. The target indicator $1 - D_V = \mathbf{1}_{\{s\}}$ belongs to the closed span of monotone functions oriented toward $D_V = 0$; hence the centered target contrast belongs to $\mathcal{M}(D_V)$, proving (1283).

Substituting $z = x + u$ into the Dirichlet form of Definition III.14.8 proves (1284). For D_V, only edges with exactly one endpoint s contribute. Symmetry of c_Φ and uniformity of μ_I give

$$\mathcal{E}_\Phi(D_V, D_V) = \frac{1}{N^2} \sum_{z \neq s} c_\Phi(s, z).$$

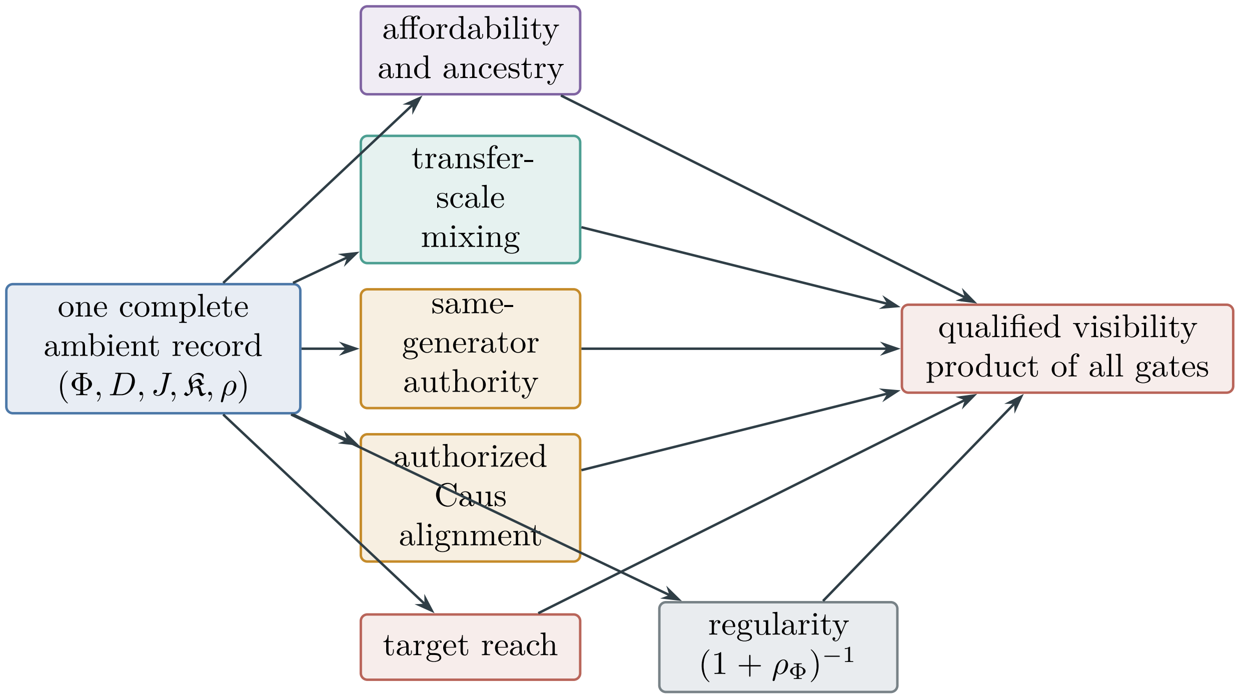

Figure 126: The identity-supported ambient coordinate is conjunctive. The carrier, potential, current, authority comparison, reach certificate, and regularity comparison belong to one affordable pre-hit record. A spectral gap, target extremum, or formal gradient in isolation contributes zero when any other gate is absent.

Since D_V is one on $N-1$ states and zero on one state, its variance is $(N-1)/N^2$. Division gives (1285).

Finally, (1286) is the LPN specialization of Definition III.14.9. All five factors lie in $[0,1]$, with the first two binary. A product larger than ε forces both binary gates to equal one and each remaining factor to exceed ε, proving (1287). The last assertion follows from the separate record requirements in Lemma II.9.2 and Definition III.12.9.

□

Figure 126 shows why no single favorable geometric statistic activates an ambient target-aligned record.

VIII.10.3 Parameter-explicit verifier geometry

Corollary VIII.10.6 (Parameter-explicit LPN verifier geometry)**.** *Fix $0<\eta<\tau<1/2$, put $N=2^n$, and let $\mathfrak{g}=(\mathfrak{N},\Phi,D_V,J,\mathfrak{K})$ be a record from Corollary VIII.10.5 using the public verifier potential $D_V=1-V_I$. With probability at least*

$$1-\exp[-mD_{\mathrm{Ber}}(\tau\|\eta)]-(2^n-1)2^{-m(1-H_2(\tau))} \tag{1288}$$

over the LPN presentation, its complete visibility is

$$V^X_{\Phi,D_V,J,\mathfrak{K}}=\frac{M_\Phi\,\chi_\Phi^{\mathrm{auth}}(D_V)\ \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K},D_V)}{1+\dfrac{1}{2^n-1}\displaystyle\sum_{z\neq s}c_\Phi(s,z)}. \tag{1289}$$

The alignment factor in Eq. (1289) *is evaluated only after the complete target-qualified Geom record has been admitted. It is the derived* Caus *factor of that same record and supplies no independent source coordinate. When the ambient record family is empty, the verifier potential contributes zero by the empty-supremum convention in Theorem III.26.1. Likewise, a represented evaluation of $D_V(x)$, or the fact that its unique minimum is s, remains terminal verification unless the same charged pre-hit record supplies the authorized intermediate transport and regularity appearing in Eq.* (1289)*.*

The dependence on the LPN family parameters is therefore explicit: 2^n is the candidate multiplicity, m is the number of public equations, η enters through the Bernoulli divergence, and τ is the represented verifier radius. The budget B restricts the record through $\mathcal{C}^{\mathrm{LPN}}_{X,b}(B)$; it creates no additional probability penalty because Eqs. (1241) *and* (1249) *hold simultaneously over all candidate displacements and therefore over every kernel retained by the budget-B saturated closure.*

Proof. The probability is Eq. (1184). On that event, Eqs. (1283) and (1285) give

$$R_{\{s\}}(D_V) = 1, \qquad \rho_\Phi(D_V) = \frac{1}{2^n - 1}\sum_{z \neq s} c_\Phi(s, z).$$

Substitution in Eq. (1286) proves Eq. (1289).

□

Corollary VIII.10.7 (Complete quantitative admissibility ledger for ambient LPN geometry)**.** *Fix $0 < \eta < \tau < 1/2$, $\delta > 0$, and a saturated budget B. With probability at least*

$$1 - \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] - (2^n - 1)2^{-m(1-H_2(\tau))} - 2^{n+1}e^{-2\delta^2 m}, \tag{1290}$$

the following statements hold simultaneously for every temporally admissible residual-ancestry ambient record $\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K})$ of complete cost at most B.

(i) *Every candidate edge is uniquely $(x, x+u)$, its public residual increment is Au, and every normalized kernel in the record obeys*

$$\left|\frac{(P-I)W_I(x)}{m} + (1/2 - \eta)P(x, s)\right| \leq 2\delta \qquad (x \neq s). \tag{1291}$$

The unique target-cancelling displacement has residual margin at least $m(1/2 - \eta - 2\delta)$; every off-target displacement changes W_I by at most $2\delta m$.

(ii) *The complete admissible visibility, with all factors evaluated on this same charged record, satisfies*

$$V^X_{\Phi,D,J,\mathfrak{K}} \leq \min\left\{M_\Phi, \chi^{\mathrm{auth}}_\Phi(D), \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D), \mathrm{ProjMass}_{\sigma(D)}(\{s\}), \frac{1}{1 + \mathcal{E}_\Phi(D,D)/\operatorname{Var}_{\mu_I}(D)}\right\}. \tag{1292}$$

Here $M_\Phi = 1$ only when the represented reciprocal gap lies in the transfer class; $\chi^{\mathrm{auth}}_\Phi(D) = 1$ only when the same selected generator supplies the required local comparison; and the alignment factor is the exact squared normalized pairing in Eq. (1281)*. These are joint factors, not independent sources.*

(iii) *For the public verifier potential $D_V = 1 - V_I$, the reach and regularity factors are evaluated exactly. For an already admitted target-qualified ambient Geom record, the full product is*

$$V^X_{\Phi,D_V,J,\mathfrak{K}} = \frac{M_\Phi \chi^{\mathrm{auth}}_\Phi(D_V)\,\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D_V)}{1 + \dfrac{1}{2^n - 1}\displaystyle\sum_{z\neq s} c_\Phi(s,z)}. \tag{1293}$$

The alignment factor is derived from this admitted geometry and has no standalone source value. If the target-qualified ambient record is absent, this item contributes zero to the saturated profile.

(iv) *For every represented next-test sampler, the target-incident mass is exactly $K_t(s \mid \mathcal{H}_t)$. Its positive excess over the declared neutral baseline is bounded by*

$$\big(\mathrm{Adv}_t(K_t;\nu_t)\big)_+ \le eH_B(n) \sum_{c\in\mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{C}^{(c)}_{I,\mathfrak{F}_5}(D^{\mathrm{sel}}_B(n)). \tag{1294}$$

A uniform proposal contributes exactly 2^{-n} per test.

These four conclusions jointly evaluate affordability and provenance, all legal perturbations, transfer-scale mixing, same-generator consistency, authorized descent, target reach, Dirichlet regularity, and direct target contact. The saturated ambient profile takes the supremum of the complete same-record product; it does not take separate suprema of these properties and combine them afterward.

Proof. Intersect the verifier good event of Eq. (1184) with the simultaneous residual-flatness event of Eq. (1241). A union bound gives Eq. (1290). Part (i) is Eqs. (1235), (1244), (1245) and (1248).

For part (ii), apply the exact product Eq. (1286). Every factor lies in $[0,1]$, so the product is bounded by each factor. Projection monotonicity gives $R_{\{s\}}(D) \le \mathrm{ProjMass}_{\sigma(D)}(\{s\})$, and the last factor is the definition of $\rho_\Phi(D)$. The gap, consistency, and alignment interpretations are Definitions III.14.9 and III.12.9 and Eq. (1281).

For part (iii), substitute the exact verifier reach and regularity from Eqs. (1283) and (1285) into the same product. The source assignment of the alignment factor is Proposition II.5.9. Part (iv) is Eqs. (1267) and (1272).

□

VIII.11 Controlled Capacity and Caus Aiming for LPN

The nonlinear ambient branch of Corollary VIII.10.7 is evaluated on the same represented kernel and the same native or comparison Dirichlet form. This introduces no additional structural channel: capacity, drift, functional inequalities, learned-operator error, and target occupation are typed numerical outputs of the already qualified $\mathsf{Geom}/\mathsf{Caus}$ record in Definitions V.5.1 and V.5.9.

VIII.11.1 Base-Carrier Capacity and Global Drift

Fix $0 < \eta < \tau < 1/2$ in the verifier range of Lemma VIII.5.1 and $\delta > 0$. Put $N = 2^n$ and define the simultaneous presentation event

$$\mathcal{G}_{\tau,\delta} = \{V_I^{-1}(1) = \{s\}\} \cap \mathcal{F}_\delta. \tag{1295}$$

The two estimates already proved in Lemma VIII.5.1 and Corollary VIII.9.5 give the explicit finite-parameter bound

$$\Pr(\mathcal{G}_{\tau,\delta}) \geq 1 - \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] - (2^n-1)2^{-m(1-H_2(\tau))} - 2^{n+1}e^{-2\delta^2 m}. \tag{1296}$$

Theorem VIII.11.1 (Controlled target capacity and drift for LPN)**.** *Fix a presentation in $\mathcal{G}_{\tau,\delta}$. Let P be the normalized selected kernel of a certified native dynamical geometry on $X_I = \mathbb{F}_2^n$, suppose that its invariant law is the uniform law μ_I, and let Φ_P be its native symmetric geometry. Write*

$$q_{P,s} = 1 - P(s,s). \tag{1297}$$

Then the target condenser, every represented opposite plate $F \subseteq X_I \setminus \{s\}$, and every initial law $d\mu_0/d\mu_I \leq R_0$ satisfy

$$\mathrm{Cap}_{\Phi_P}(\{s\}, X_I \setminus \{s\}) = 2^{-n} q_{P,s}, \tag{1298}$$

$$\mathrm{Cap}_{\Phi_P}(\{s\}, F) \leq 2^{-n} q_{P,s}, \tag{1299}$$

$$\Pr_{\mu_0}\{\tau_s \leq H\} \leq R_0 2^{-n}(1 + Hq_{P,s}) \leq R_0(H+1)2^{-n}. \tag{1300}$$

If a represented comparison geometry Ψ has the same plates and obeys $\mathcal{E}_{\Phi_P}(f,f) \leq b\mathcal{E}_\Psi(f,f)$ on the condenser domain, then

$$\mathrm{Cap}_{\Phi_P}(\{s\}, F) \leq b\,\mathrm{Cap}_\Psi(\{s\}, F). \tag{1301}$$

The burn-in and relative-uniform versions are obtained by replacing R_0 with the certified post-burn-in density ratio in Theorem V.6.7.

More generally, let $D \geq 0$ be represented, let $D(s) = 0$, and suppose the same selected kernel has a global normalized drift margin

$$(I-P)D(x) \geq a > 0 \qquad (x \neq s).$$

Then

$$\frac{a}{\mathrm{Osc}(D)} \leq \frac{1}{2^n - 1}. \tag{1302}$$

For the public residual potential $W_I(x) = |Ax + b|_1$, if

$$-\frac{(P-I)W_I(x)}{m} \geq a \qquad (x \neq s), \tag{1303}$$

then, pointwise on the same event,

$$P(x,s) \geq \frac{(a-2\delta)_+}{1/2 - \eta}, \tag{1304}$$

$$a \leq \frac{(1/2 - \eta + 2\delta)q_{P,s}}{2^n - 1} \leq \frac{1/2 - \eta + 2\delta}{2^n - 1}. \tag{1305}$$

Thus a globally useful nonlinear residual descent and its target-boundary flux are evaluated in the same native record; neither is inferred from the mere existence of the terminal residual score.

Proof. For the native symmetric conductance, invariance of μ_I gives

$$\mathcal{E}_{\Phi_P}(\mathbf{1}_{\{s\}},\mathbf{1}_{\{s\}})=\mu_I(s)(1-P(s,s))=2^{-n}q_{P,s}.$$

The boundary values of the condenser $(\{s\},X_I\setminus\{s\})$ determine the unique competitor $\mathbf{1}_{\{s\}}$, proving Eq. (1298). The same indicator is an admissible competitor for $(\{s\},F)$, proving Eq. (1299). The stationary entrance count and initial-density comparison in Theorem V.6.7 give Eq. (1300). The form-comparison conclusion Eq. (487) gives Eq. (1301).

Apply the invariant-drift conclusion of Theorem V.6.7 with $\mu_I(s)=2^{-n}$ to obtain Eq. (1302). On $\mathcal{F}_\delta$, Eq. (1248) gives

$$(1/2-\eta)P(x,s)\ge-\frac{(P-I)W_I(x)}{m}-2\delta,$$

which proves Eq. (1304). Finally, invariance makes the uniform mean of $(P-I)W_I$ zero. Summing Eq. (1303) over the $N-1$ off-target states and balancing the sum at s gives

$$(N-1)a\le\frac{(P-I)W_I(s)}{m}\le(1/2-\eta+2\delta)(1-P(s,s)),$$

where the last inequality follows directly from the two intervals in Eq. (1240). Division proves Eq. (1305).

□

VIII.11.2 Residual Caus Energy and Contact Conversion

Theorem VIII.11.2 (Residual Caus energy and action for LPN)**.** *Fix a presentation in $\mathcal{G}_{\tau,\delta}$, assume $\Delta_\delta=1/2-\eta-2\delta>0$, and define the represented target-normalized residual potential*

$$D^{\mathrm{res}}_{\delta,I}(x)=\left(\frac{W_I(x)}{m}-\eta-\delta\right)_+. \tag{1306}$$

This potential is a represented scalar postprocessing of W_I. Its source assignment is therefore the assignment of the residual-weight record in Theorem VIII.9.6; the Caus use below is derived and creates no additional source by Proposition II.5.9.

Then

$$D^{\mathrm{res}}_{\delta,I}(s)=0,\qquad\Delta_\delta\le D^{\mathrm{res}}_{\delta,I}(x)\le\frac{1}{2}-\eta\quad(x\ne s). \tag{1307}$$

Let R be a complete prospective Caus target-flow record on the LPN candidate carrier, with selected normalized kernel P, authorized current J, and represented positive edge mobilities a. Orient the unoriented active edges arbitrarily and let ∂s denote those with exactly one endpoint equal to s. The target-aiming energy of Eq. (1306) *satisfies the exact parameter-dependent estimate*

$$\mathcal{E}_{\mathrm{aim}}(D^{\mathrm{res}}_{\delta,I};a)\le4\delta^2\sum_{e=\{x,z\}\ x,z\ne s}a_e+(1/2-\eta)^2\sum_{e\in\partial s}a_e. \tag{1308}$$

Consequently,

$$\mathcal{P}_{\mathsf{Caus}}(J,D^{\mathrm{res}}_{\delta,I})_+\le\sqrt{\mathcal{A}_{\mathsf{Caus}}(J;a)}\left(4\delta^2\sum_{e=\{x,z\}\ x,z\ne s}a_e+(1/2-\eta)^2\sum_{e\in\partial s}a_e\right)^{1/2}. \tag{1309}$$

Proof. On the simultaneous flatness event, off-target potential increments are at most 2δ, while target-boundary increments are at most $1/2-\eta$. Splitting the aiming-energy sum over those two edge classes gives the energy estimate. The represented Caus action–aiming inequality then gives the power bound. ☐

Corollary VIII.11.3 (Traffic and horizon specialization of residual Caus aiming)**.** *Adopt the hypotheses and potential of Theorem VIII.11.2.*

There is a canonical normalized specialization of this estimate. Suppose that P has uniform invariant law and that the complete record uses the represented traffic mobility and current

$$a_{\{x,z\}} = \mu_I(x)P(x,z) + \mu_I(z)P(z,x), \qquad J^{\mathrm{tr}}_{\{x,z\}} = \mu_I(x)P(x,z) - \mu_I(z)P(z,x). \tag{1310}$$

Take $J = J^{\mathrm{tr}}$, or take its authorized Hodge projection when the same record contains the inverse-mobility contraction required in Definition V.7.1. Then

$$\mathcal{A}_{\mathsf{Caus}}(J;a) \le 1, \tag{1311}$$

$$\sum_{e\in\partial s} a_e = 2^{1-n} q_{P,s}. \tag{1312}$$

For the verifier potential $D_V = 1 - V_I$,

$$\mathcal{E}_{\mathrm{aim}}(D_V;a) = 2^{1-n} q_{P,s}, \tag{1313}$$

$$\mathcal{P}_{\mathsf{Caus}}(J,D_V)_+ \le 2^{(1-n)/2}\sqrt{q_{P,s}}, \tag{1314}$$

while the clipped residual potential satisfies

$$\mathcal{P}_{\mathsf{Caus}}(J,D^{\mathrm{res}}_{\delta,I})_+ \le \left[4\delta^2 + (1/2-\eta)^2\, 2^{1-n} q_{P,s}\right]^{1/2}. \tag{1315}$$

For a clocked sequence of such complete records $(P_t, J_t, a_t)_{t=0}^{H-1}$, the discrete-time action–aiming bound gives

$$\sum_{t=0}^{H-1} \mathcal{P}_{\mathsf{Caus}}(J_t,D_V)_+ \le H\left[2^{1-n}\frac{1}{H}\sum_{t=0}^{H-1} q_{P_t,s}\right]^{1/2} \le H2^{(1-n)/2}, \tag{1316}$$

$$\sum_{t=0}^{H-1} \mathcal{P}_{\mathsf{Caus}}(J_t,D^{\mathrm{res}}_{\delta,I})_+ \le H\left[4\delta^2 + (1/2-\eta)^2\, 2^{1-n}\frac{1}{H}\sum_{t=0}^{H-1} q_{P_t,s}\right]^{1/2}. \tag{1317}$$

In particular, choosing $\delta = \delta_{n,m,\gamma}$ from Eq. (1242) yields

$$\sum_{t=0}^{H-1} \mathcal{P}_{\mathsf{Caus}}(J_t,D^{\mathrm{res}}_{\delta,I})_+ \le H\left[\frac{2((n+1)\log 2 + \gamma)}{m} + (1/2-\eta)^2\, 2^{1-n}\frac{1}{H}\sum_{t=0}^{H-1} q_{P_t,s}\right]^{1/2}. \tag{1318}$$

The event on which Eq. (1318) holds has the fully explicit probability

$$\Pr\Big(\mathcal{G}_{\tau,\delta_{n,m,\gamma}}\Big) \geq 1 - \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] - (2^n-1)2^{-m(1-H_2(\tau))} - e^{-\gamma}. \tag{1319}$$

Its target margin is positive under the sample condition in Eq. (1243). *Thus the sample ratio enters the off-target aiming term through* $2((n+1)\log 2+\gamma)/m$*, the label-noise rate enters the target-boundary term through* $(1/2-\eta)^2$*, and the two exceptional-presentation exponents retain their exact* (n,m,η,τ) *dependence.*

Every represented aiming-to-contact converter Eq. (515) *therefore gives its LPN contact bound by adding its represented neutral-contact baseline* B_R^{contact}*, multiplying the applicable right side of Eqs.* (1316) *and* (1318) *by its represented* K_R *and adding its represented* δ_R*.*

Proof. The inequality $(u-v)^2/(u+v) \leq u+v$ bounds traffic action by one, and uniform invariance gives the exact target-boundary mobility. Substitution in Theorem VIII.11.2, followed by Cauchy–Schwarz over time, proves the horizon formulas. The stated event probability is the union bound for verifier separation and residual flatness; the contact formula is the declared aiming-to-contact converter. □

Proposition VIII.11.4 (Residual drift-to-contact conversion)**.** *Adopt the hypotheses and potential of Theorem VIII.11.2.*

The same clipped potential also supplies an exact drift-to-contact conversion. For $x \neq s$*,*

$$\Big|(P-I)D^{\mathrm{res}}_{\delta,I}(x) + D^{\mathrm{res}}_{\delta,I}(x)P(x,s)\Big| \leq 2\delta(1-P(x,s)). \tag{1320}$$

Hence a pointwise descent margin $(I-P)D^{\mathrm{res}}_{\delta,I}(x) \geq a$ *implies*

$$P(x,s) \geq \frac{(a-2\delta)_+}{\Delta_\delta}. \tag{1321}$$

If this margin holds for every $x \neq s$ *and* P *has uniform invariant law, then*

$$a \leq \frac{(1/2-\eta)q_{P,s}}{2^n-1}. \tag{1322}$$

Proof. Separate the target transition from all off-target transitions in the exact kernel drift. The flatness event bounds the latter contribution by $2\delta(1-P(x,s))$, yielding the pointwise identity and contact lower bound. Summing the descent inequality against the uniform invariant law cancels total drift and leaves only the target boundary, which gives the global ceiling. □

Corollary VIII.11.5 (Learned-parity transport to LPN Caus aiming)**.** *Adopt the verifier-good LPN presentation and normalized traffic records of Corollary VIII.11.3.*

Finally, fix $0 < \delta_{\mathrm{stat}} < 1$*, and let a represented parity learner satisfying Eq.* (1136) *output* $\widehat{s} = \widehat{s}(A,b;\omega)$ *before the controlled transitions. Retain* ω *as a static coordinate in its public product record, and put*

$$D_0(x) = \mathbf{1}_{\{x\neq s\}}, \qquad \widehat{D}_\omega(x) = \mathbf{1}_{\{x\neq\widehat{s}(\omega)\}}.$$

On the simultaneous parity-learning event, define

$$p^{\mathrm{par}}_{\mathrm{err}} = \min\left\{1, \frac{2(2\Gamma^{\mathrm{par}}_{n,m,\delta_{\mathrm{stat}}} + \varepsilon_{\mathrm{opt}})}{1-2\eta}\right\}. \tag{1323}$$

Then

$$\Pr_{\omega}\{\widehat{s} \neq s\} \leq p_{\mathrm{err}}^{\mathrm{par}}, \tag{1324}$$

$$\mathbb{E}_{\omega}\|\widehat{D}_{\omega} - D_0\|^2_{L^2(\mu_I)} = 2^{1-n}\Pr_{\omega}\{\widehat{s} \neq s\} \leq 2^{1-n}p_{\mathrm{err}}^{\mathrm{par}}. \tag{1325}$$

For normalized uniform-invariant traffic records, the squared norm of the gradient map from $L^2(\mu_I)$ to $\ell^2(a)$ is at most four. Summing the same represented comparison over a length-H record therefore gives the calibrated-potential certificate

$$\varepsilon_{\mathrm{cal}}^2 \leq 4H\,2^{1-n}p_{\mathrm{err}}^{\mathrm{par}}. \tag{1326}$$

Combining this estimate with Theorem V.7.7 gives, for every represented aiming-to-contact converter with $B_R^{\mathrm{contact}} \leq B_B^{\mathrm{contact}}$, $K_R \leq K_B$, and $\delta_R \leq \delta_B$,

$$U^{\mathrm{noise}}_{\mathsf{Caus},\mathrm{LPN}} \leq \min\left\{1, B_B^{\mathrm{contact}} + K_B H 2^{(1-n)/2}\left(1 + 2\sqrt{p_{\mathrm{err}}^{\mathrm{par}}}\right) + \delta_B\right\}. \tag{1327}$$

If $2\Gamma^{\mathrm{par}}_{n,m,\delta_{\mathrm{stat}}} + \varepsilon_{\mathrm{opt}} < (1-2\eta)/2$, then Eq. (1137) *gives $\widehat{s} = s$; the errors in Eqs.* (1325) *and* (1326) *then vanish.*

Proof. The clean-risk transport bound controls decoder error. Two distinct singleton indicators differ on exactly two points, giving the exact $L^2(\mu_I)$ calibration. The normalized traffic gradient map has squared operator norm at most four, so summing over the horizon gives Eq. (1326). The framework noise-corrected Caus theorem then yields the displayed contact certificate. □

Corollary VIII.11.6 (Caus action, residual aiming, and noisy target transport for LPN)**.** *The residual-energy, traffic-horizon, drift-to-contact, and learned-noise conclusions are respectively Theorem VIII.11.2, Corollaries VIII.11.3 and VIII.11.5, and Proposition VIII.11.4. They share one complete represented Caus record and one declared confidence allocation.*

Proof. On $\mathcal{F}_\delta$, the target residual satisfies $W_I(s)/m \leq \eta + \delta$, whereas every off-target residual satisfies $1/2 - \delta \leq W_I(x)/m \leq 1/2 + \delta$. Substitution into Eq. (1306) proves Eq. (1307). An edge joining two off-target states therefore has potential increment at most 2δ in absolute value, an edge in ∂s has increment between Δ_δ and $1/2 - \eta$, and every self-loop has zero increment. Splitting the aiming-energy sum over these edge classes proves Eq. (1308). The Caus action–aiming inequality Theorem V.7.2 then proves Eq. (1309).

For the traffic coordinates, the scalar inequality $(u-v)^2/(u+v) \leq u+v$ for $u, v \geq 0$ gives

$$\mathcal{A}_{\mathsf{Caus}}(J;a) \leq \sum_{\{x,z\}} [\mu_I(x)P(x,z) + \mu_I(z)P(z,x)] \leq 1.$$

The same estimate holds for the authorized Hodge projection by its recorded inverse-mobility contraction. Uniform invariance gives equality of the stationary entrance and exit fluxes of $\{s\}$, so

$$\sum_{e\in\partial s} a_e = 2\mu_I(s)(1 - P(s,s)) = 2^{1-n}q_{P,s}.$$

The verifier potential changes by one exactly on ∂s, proving Eq. (1313). Apply Theorem V.7.2 and Eq. (1311) to obtain Eqs. (1314) and (1315). Summing the action and aiming bounds in time, then applying the discrete-time form of Eq. (514), proves Eqs. (1316) and (1317). The identity

$4\delta^2_{n,m,\gamma} = 2((n+1)\log 2 + \gamma)/m$ proves Eq. (1318). The converter assertion is Corollary V.7.4 applied to these same-record estimates.

For $x \neq s$, split the kernel average into the target state and the off-target states. On the latter set the positive-part operation in Eq. (1306) is inactive, so

$$(P-I)D^{\mathrm{res}}_{\delta,I}(x) = -D^{\mathrm{res}}_{\delta,I}(x)P(x,s) + \sum_{z\neq s} P(x,z)\frac{W_I(z)-W_I(x)}{m}.$$

The simultaneous off-target bound in Eq. (1244) proves Eq. (1320). If $(I-P)D^{\mathrm{res}}_{\delta,I}(x) \geq a$, then

$$a \leq (1/2-\eta)P(x,s) + 2\delta(1-P(x,s)) = 2\delta + \Delta_\delta P(x,s),$$

which proves Eq. (1321). Under uniform invariance, the mean of $(P-I)D^{\mathrm{res}}_{\delta,I}$ is zero. Summing the global descent margin over the 2^n-1 off-target states and using

$$(P-I)D^{\mathrm{res}}_{\delta,I}(s) = \sum_{z\neq s} P(s,z)D^{\mathrm{res}}_{\delta,I}(z) \leq (1/2-\eta)q_{P,s}$$

proves Eq. (1322).

For a parity output, character orthogonality gives $\mathbf{1}_{\{\widehat{s}\neq s\}} = 2R_0(f_{\widehat{s}})$. Apply Eq. (1136) and average over the represented seed to obtain Eq. (1324). Two distinct singleton potentials differ exactly at their two singleton states, proving Eq. (1325).

For the traffic mobility, put $P^{\mathsf{s}} = (P+P^*)/2$. Then

$$\sum_{\{x,z\}} a_{\{x,z\}}(f(z)-f(x))^2 = 2\langle f,(I-P^{\mathsf{s}})f\rangle_{\mu_I} \leq 4\|f\|^2_{L^2(\mu_I)},$$

because the self-adjoint Markov operator P^{s} has spectrum in $[-1,1]$. Summing over H clock steps and substituting Eq. (1325) gives the represented calibrated-potential comparison in Definition V.7.6(ii). The calibrated branch of Theorem V.7.7 proves Eq. (1326). The observed singleton potential has aiming energy at most $H2^{1-n}$, while Eq. (1311) gives accumulated action at most H. Substitution into the same theorem proves Eq. (1327), including the represented neutral-contact baseline $B^{\mathrm{contact}}_R \leq B^{\mathrm{contact}}_B$. The final assertion is Eq. (1137).

□

Figure 127 separates the energy estimate from the additional converter required to turn Caus power into target contact.

VIII.11.3 Noise-Indexed Valid-Lift Interfaces

Corollary VIII.11.7 (Noise-indexed Caus and valid-Lift target interfaces for LPN)**.** *Fix a polynomial ceiling $B = b_C(n) = n^C$, a horizon H, $0 < \delta_{\mathrm{stat}} < 1$, and the represented class $\mathscr{R}^{\mathrm{hard}}_{B,H}$ of complete LPN dynamical label-noise records at this ceiling and horizon. Use the decoder error $p^{\mathrm{par}}_{\mathrm{err}}$ from Eq.* (1323) *with this confidence allocation. In the linear homogeneous clean-aiming branch, $\rho = 1-2\eta$, so Eq.* (520) *specializes exactly to*

$$U^{\mathrm{noise}}_{\mathsf{Caus}} \leq \min\left\{1, B^{\mathrm{contact}}_B + K_B\sqrt{\mathrm{Act}^{\mathrm{sat}}_{\mathsf{Caus}}(B,H)}\frac{\sqrt{\mathrm{Aim}^{\mathrm{sat}}_{\mathsf{Caus},\mathrm{obs}}(B,H)} + \varepsilon^{\mathrm{sat}}_{\mathrm{aim}}}{1-2\eta} + \delta_B\right\}. \tag{1328}$$

For the binary hard-potential branch, the same framework theorem gives

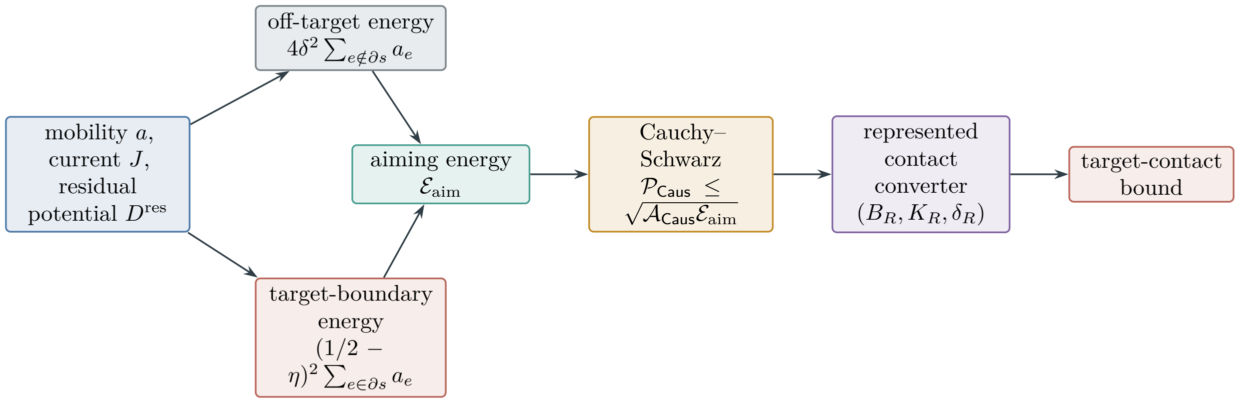

Figure 127: Residual Caus routing for LPN. Flat off-target edges and the target boundary contribute different energy scales. The action–aiming inequality controls Caus power, but contact follows only after a represented converter with its own baseline, gain, defect, ancestry, and cost.

$$
\begin{aligned}
(\varepsilon^{\mathrm{sat}}_{\mathrm{aim,hard}})^2 &= \sup_{\mathcal{R}\in\mathscr{R}^{\mathrm{hard}}_{B,H}} \kappa_{\mathrm{aim,H}}(\mathcal{R}) U^{\mathrm{clean}}_{\mathrm{gate,dep}}(\mathcal{R}) \\
&\le \left(\sup_{\mathcal{R}\in\mathscr{R}^{\mathrm{hard}}_{B,H}} \kappa_{\mathrm{aim,H}}(\mathcal{R})\right) \left(\sup_{\mathcal{R}\in\mathscr{R}^{\mathrm{hard}}_{B,H}} U^{\mathrm{clean}}_{\mathrm{gate,dep}}(\mathcal{R})\right),
\end{aligned} \tag{1329}
$$

where, record by record,

$$
U^{\mathrm{clean}}_{\mathrm{gate,dep}}(\mathcal{R}) \le \min\left\{1, \left(\sup_{0\le t\le H} W_t(\mathcal{R})\right) U^{\mathrm{clean}}_{\mathrm{noise}}(n, n^C, m, \eta, \delta_{\mathrm{stat}})\right\}.
$$

Thus n and the polynomial ceiling enter through the saturated class, m through the applicable dynamical generalization certificate, and η through both that certificate and the exact attenuation factor $(1-2\eta)^{-1}$.

Now fix one represented parity-output member of this class. For the output in Corollary VIII.11.6, use its existing public product lift carrying the seed ω and a uniform candidate x. On that carrier, the clean and learned singleton gates are

$$
\mathsf{T}_0 = \{(\omega, x) : x = s\}, \qquad \widehat{\mathsf{T}} = \{(\omega, x) : x = \widehat{s}(\omega)\}.
$$

Their exact reference-law error is

$$
(\mathbb{P}_\omega \otimes \mu_I)(\mathsf{T}_0 \triangle \widehat{\mathsf{T}}) = 2^{1-n} \Pr_\omega\{\widehat{s} \ne s\} \le 2^{1-n} p^{\mathrm{par}}_{\mathrm{err}}. \tag{1330}
$$

If the time-t deployment law has the represented comparison $d^\pi_t \le W_t(\mathbb{P}_\omega \otimes \mu_I)$, then Definition V.8.1 gives

$$
\varepsilon^{\mathrm{dep}}_{T,H} \le \min\left\{1, 2^{1-n} p^{\mathrm{par}}_{\mathrm{err}} \sum_{t=0}^{H} W_t\right\}. \tag{1331}
$$

Every clocked valid lift in the same record therefore obeys

$$\left|p^{\mathrm{clean}}_{\mathsf{T}_0,H} - \widehat{p}^{\Omega}_{\widehat{\mathsf{T}},H}\right| \le \min\left\{1, \varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \|E_t\| + \min\left\{1, 2^{1-n} p^{\mathrm{par}}_{\mathrm{err}} \sum_{t=0}^{H} W_t\right\}\right\}. \tag{1332}$$

Let Φ be the native symmetric Dirichlet form of a normalized kernel invariant under the public product law $\mathbb{P}_\omega \otimes \mu_I$. For an exact failure gate and a represented valid-Lift form comparison with factor b_π, the corresponding capacity statement is

$$\mathrm{Cap}_\Omega(\pi^{-1}\widehat{\mathsf{T}}, \pi^{-1}F) \le b_\pi \left[2^{-n} + \omega^{\mathrm{Cap}}_{\Phi,\mathsf{T}_0}(2^{1-n} p^{\mathrm{par}}_{\mathrm{err}})\right]. \tag{1333}$$

Under the exact-identification condition in Eq. (1137)*, one has $\widehat{s} = s$. The realized reference and deployment gate errors are then reset to zero, and the capacity comparison is applied with modulus argument zero; consequently the gate and capacity-stability contributions in Eqs.* (1330) *to* (1333) *then vanish exactly.*

Proof. Substitute $b_C(n) = n^C$ and $\rho = 1 - 2\eta$ into Theorem V.7.7; its binary fallback and the behavior-to-deployment comparison give Eq. (1329). Two unequal singleton gates on the uniform candidate carrier differ at exactly two states, while equal gates have zero symmetric difference. Averaging over the represented seed proves Eq. (1330); the decoder-error bound is Eq. (1324). The deployment comparison in Definition V.8.1 proves Eq. (1331). Apply Theorem V.8.2 to obtain the clocked and capacity inequalities, using Eq. (482) on the product-law-invariant base carrier for the clean singleton-capacity term $(\mathbb{P}_\omega \otimes \mu_I)(\mathsf{T}_0) = 2^{-n}$. Exact parity identification makes the two gates identical; applying the same theorem with realized gate error zero and modulus argument zero proves the last assertion.

□

VIII.12 Hypercube Functional Geometry and Learned Operators

VIII.12.1 Exact hypercube spectrum and functional outputs

Theorem VIII.12.1 (Exact spectrum and functional constants of the public hypercube)**.** *Let $P_\square$ be the public coordinate-flip kernel*

$$P_\square(x, x+e_i) = \frac{1}{n}, \qquad 1 \le i \le n, \tag{1334}$$

with uniform invariant law, and put $H_\square = I - P_\square$. Its Walsh character χ_S has eigenvalue $2|S|/n$. Consequently the audited functional-inequality profile of Definition V.5.4 admits the following simultaneous values. All decay statements associated with these constants refer to the continuous-time semigroup $e^{-tH_\square}$; the profile does not assert discrete-time convergence of the periodic kernel $P_\square$.

$$q_\square = 1, \qquad \mathsf{h}_\square = \frac{1}{n}, \qquad \gamma_\square = \frac{2}{n}, \tag{1335}$$

$$\beta_{\mathrm{W}}(r) = \frac{n}{2}, \qquad \beta_{\mathrm{S}}(r) = \begin{cases} 2^n, & 0 < r < n/2, \\ 1, & r \ge n/2, \end{cases} \tag{1336}$$

$$d_{\mathrm{N}} = n, \qquad C_{\mathrm{N}} = 2n, \qquad C_{\mathrm{Sob}} = 2n \quad (n > 2), \tag{1337}$$

$$\alpha_{\mathrm{LS}} = \frac{2}{n}, \qquad \alpha_{\mathrm{MLS}} = \frac{2}{n}, \qquad CD\left(\frac{2}{n}, \infty\right) \text{ holds}, \tag{1338}$$

$$C_{T_1} = \frac{n}{4}, \qquad C_{\mathrm{sec}} = 0, \qquad (m_{\mathscr{H}}, M_{\mathscr{H}}, \kappa_{\mathscr{H}}) = (1, 1, 2/n). \tag{1339}$$

In particular, for $S_t^{\square} = e^{-tH_{\square}}$*,*

$$\mathrm{Var}_{\mu_I}(S_t^{\square} f) \le e^{-4t/n}\,\mathrm{Var}_{\mu_I}(f). \tag{1340}$$

Proof. Walsh diagonalization gives eigenvalues $2|S|/n$, hence the gap and variance decay. Cube edge isoperimetry gives the Cheeger constant. Tensor product Poincaré, logarithmic-Sobolev, modified-logarithmic-Sobolev, curvature, and Hamming concentration inequalities give the remaining exact or certified constants with the displayed normalization. Self-adjointness sets the sector constant to zero and supplies the stated hypocoercive tuple. □

Theorem VIII.12.2 (Typed functional outputs of the public hypercube)**.** *Adopt the carrier and constants of Theorem VIII.12.1.*

The remaining functional outputs are obtained from the same audited record. For $0 < r \le 1$*, weak Poincaré gives*

$$\mathrm{Var}_{\mu_I}(S_t^{\square} f) \le e^{-4t/n}\,\mathrm{Var}_{\mu_I}(f) + r(1 - e^{-4t/n})\,\mathrm{Osc}(f)^2. \tag{1341}$$

For every $r > 0$*, the super-Poincaré output is*

$$\|S_t^{\square} f\|_{2,\mu_I}^2 \le e^{-2t/r}\|f\|_{2,\mu_I}^2 + \beta_{\mathrm{S}}(r)(1 - e^{-2t/r})\|f\|_{1,\mu_I}^2, \tag{1342}$$

where β_{S} *is the explicit function in Eq.* (1336)*. On the declared mean-zero Nash/Sobolev domain* $\mathscr{D}_{\mathrm{N}}$*,*

$$\|S_t^{\square}|_{\mathscr{D}_{\mathrm{N}}}\|_{1\to 2}^2 \le \left(\frac{n^2}{2t}\right)^{n/2}, \tag{1343}$$

$$\|S_t^{\square}|_{\mathscr{D}_{\mathrm{N}}}\|_{1\to \infty} \le \left(\frac{n^2}{t}\right)^{n/2}. \tag{1344}$$

For every represented proper set $A \subsetneq X_I$*, the killed semigroup satisfies*

$$\|S_t^{\square,A}\|_{2\to 2} \le \min\left\{\exp\left[-\frac{2t}{n}(1 - \mu_I(A))\right], \exp\left[-tc_{\mathrm{FK},\square}\mu_I(A)^{-2/n}\right]\right\}. \tag{1345}$$

The logarithmic-Sobolev, modified-logarithmic-Sobolev, and curvature coordinates give, respectively,

$$\|S_t^{\square} f\|_{q(t),\mu_I} \le \|f\|_{p,\mu_I}, \qquad q(t) - 1 = (p-1)e^{4t/n}, \qquad p > 1, \tag{1346}$$

$$\mathrm{Ent}_{\mu_I}(S_t^{\square,*} g) \le e^{-4t/n}\,\mathrm{Ent}_{\mu_I}(g), \tag{1347}$$

$$\Gamma_{\square}(S_t^{\square} f) \le e^{-4t/n} S_t^{\square}\Gamma_{\square}(f). \tag{1348}$$

If F *is represented and* L_F*-Lipschitz for Hamming distance, then*

$$|\nu(F) - \mu_I(F)| \le L_F\sqrt{\frac{n}{2}\,\mathrm{KL}(\nu\|\mu_I)}. \tag{1349}$$

Finally, on the centered hypocoercive domain,

$$\|S_t^{\square} f\|_{2,\mu_I}^2 \leq e^{-4t/n}\|f\|_{2,\mu_I}^2. \tag{1350}$$

For each typed output slot $s_0 \in \mathscr{S}_{\mathrm{FI}}$, let $\mathcal{J}_{s_0,\mathrm{LPN}}$ be the applicable members of these audited hypercube bounds after conversion to that slot's units. The slotwise LPN certificate is therefore

$$C_{s_0,\mathrm{LPN}}^{\mathrm{FI}} = \min_{j\in\mathcal{J}_{s_0,\mathrm{LPN}}} C_{s_0,j}, \tag{1351}$$

with every induced entry retaining the single hypercube-profile provenance.

Proof. Apply the corresponding consequence in Theorem V.5.8 to each constant in Theorem VIII.12.1. The killed bound is the minimum of the spectral and Faber–Krahn routes. All outputs are converted to a common slot before their minimum is taken, so their single profile provenance is retained. □

VIII.12.2 Capacity, effective dimension, and carrier profile

Theorem VIII.12.3 (Target capacity, effective dimension, and residual geometry of the hypercube)**.** *Adopt the public hypercube carrier of Theorem VIII.12.1.*

The kernel is self-adjoint, so its selected generator has zero antisymmetric part and its native actual current has zero authorized Hodge component. Consequently, for every target-normalized represented potential D,

$$\mathrm{Align}_{\mathsf{Caus}}^{\mathrm{auth}}(\mathfrak{K}_{\square}, D) = 0 \tag{1352}$$

when $\mathfrak{K}_{\square}$ is the actual-current record of this selected kernel. This is the exact separation between its inverse-polynomial mixing profile and its target-oriented current coordinate.

Here T_1 uses ordinary Hamming distance. For every represented $A \subsetneq X_I$,

$$\Lambda_{\square}^{\mathrm{mix}}(v) \geq \frac{2}{n}, \qquad \lambda_{\square}^{\mathrm{D}}(A) \geq \frac{2}{n}(1-\mu_I(A)), \tag{1353}$$

$$\lambda_{\square}^{\mathrm{D}}(A) \geq c_{\mathrm{FK},\square}\mu_I(A)^{-2/n}, \qquad c_{\mathrm{FK},\square} = \frac{2}{n2^n}(1-2^{-n})^{2/n}. \tag{1354}$$

The target condenser and effective dimension are exact:

$$\mathrm{Cap}_{\square}(\{s\}, X_I \setminus \{s\}) = 2^{-n}, \tag{1355}$$

$$N_{\mathrm{eff},\square}(\lambda) = \sum_{k=1}^{n} \binom{n}{k} \frac{2k/n}{2k/n+\lambda}. \tag{1356}$$

For $0 < \lambda \leq 1$,

$$N_{\mathrm{eff},\square}(\lambda) \geq 2^{n-2}. \tag{1357}$$

Let

$$f_s = \frac{\mathbf{1}_{\{s\}} - 2^{-n}}{\sqrt{(2^n-1)/2^{2n}}}$$

be the normalized centered singleton contrast and let $E_{\geq\lambda} = \mathbf{1}_{[\lambda,\infty)}(H_\square)$. Then

$$\|E_{\geq\lambda} f_s\|_{2,\mu_I}^2 = \frac{1}{2^n - 1} \sum_{\substack{1\leq k\leq n \\ 2k/n\geq\lambda}} \binom{n}{k}. \tag{1358}$$

This is the equality case of the fast-mode target-reach accounting in Theorem V.8.3: a constant fraction of the singleton target lies in fast hypercube modes only together with exponentially many active modes.

For the public verifier potential $D_V = 1 - V_I$ on the verifier-good event, the same kernel has the exact local drift

$$(I - P_\square) D_V(x) = \begin{cases} -1, & x = s, \\ 1/n, & d_H(x,s) = 1, \\ 0, & d_H(x,s) \geq 2. \end{cases} \tag{1359}$$

Hence its global off-target drift margin is zero. More generally, if $W_I - \mu_I(W_I) = \sum_{v\neq 0} \widehat{W}_I(v)\chi_v$, then

$$\mathrm{Var}_{\mu_I}(W_I) = \sum_{v\neq 0} \widehat{W}_I(v)^2, \tag{1360}$$

$$\mathcal{E}_\square(W_I, W_I) = \sum_{v\neq 0} \frac{2|v|_1}{n} \widehat{W}_I(v)^2, \tag{1361}$$

so every nonconstant residual potential satisfies

$$\frac{2}{n} \leq \frac{\mathcal{E}_\square(W_I, W_I)}{\mathrm{Var}_{\mu_I}(W_I)} \leq 2. \tag{1362}$$

Proof. Self-adjointness removes native directed current. The Walsh spectrum gives the Dirichlet profile, effective-dimension sum, singleton spectral mass, and residual regularity range. The opposite-plate capacity equals the boundary flux of the singleton and is 2^{-n}. Pairing Walsh levels $k \geq n\lambda/2$ with their binomial multiplicities proves the effective dimension and fast-mode assertions; direct neighbor counting gives the verifier drift. □

Corollary VIII.12.4 (Complete functional profile of the public hypercube carrier)**.** *The exact constants, typed functional outputs, and target/effective-dimension geometry are Theorems VIII.12.1 to VIII.12.3. They are simultaneous outputs of one audited public-carrier record.*

Proof. The Walsh diagonalization follows from $\chi_S(x + e_i) = (-1)^{\mathbf{1}_{\{i\in S\}}}\chi_S(x)$. It gives $q_\square = 1$ and the gap $2/n$. The edge-isoperimetric inequality on the cube gives $\mathsf{h}_\square = 1/n$. Poincaré gives the constant weak-Poincaré rate. Also

$$\|f\|_2^2 \leq 2^n \|f\|_1^2, \qquad \|f\|_2^2 \leq \frac{n}{2}\mathcal{E}_\square(f,f) + \|f\|_1^2,$$

which proves the displayed super-Poincaré rate. On the mean-zero domain, $\|f\|_2 \leq 2^{n/2}\|f\|_1$ and Poincaré give

$$\|f\|_2^{2+4/n} \leq 2n\, \mathcal{E}_\square(f,f) \|f\|_1^{4/n}.$$

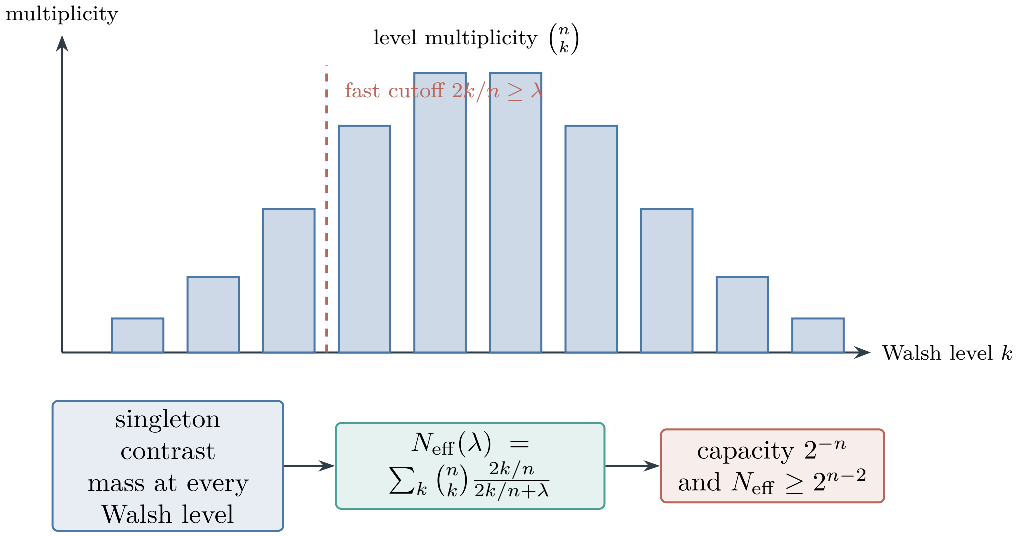

Figure 128: Walsh levels of the public hypercube. Eigenvalue $2k/n$ grows with level, while multiplicity is binomial. The centered singleton spreads across all nonconstant levels, so placing a constant fraction in fast modes requires exponentially many active modes even though the singleton condenser capacity is only 2^{-n}. Bar heights are schematic.

For $n > 2$, finite-space norm comparison gives $\|f\|^2_{2n/(n-2)} \leq 4\|f\|^2_2 \leq 2n\mathcal{E}_\square(f, f)$ on the same mean-zero domain. This proves Eq. (1337).

Tensorization of the sharp two-point inequalities gives Eq. (1338); tensorization of the bounded-difference subgaussian estimate gives $T_1(n/4)$ for Hamming distance [14, 36]. The kernel is reversible, so the sector constant is zero and the Poincaré energy itself supplies the displayed hypocoercive triple. The Poincaré consequence in Theorem V.5.8 gives Eq. (1340). Substituting the same constants into the weak-Poincaré, super-Poincaré, Nash, killed-semigroup, logarithmic-Sobolev, modified-logarithmic-Sobolev, curvature, transport, and hypocoercive conclusions of that theorem proves Eqs. (1341) to (1343) and (1345) to (1350). The slotwise minimum is Theorem V.5.10. Self-adjointness makes the antisymmetric generator component zero; hence Proposition V.5.2 and Definition III.12.8 give Eq. (1352).

Poincaré gives $\Lambda^{\text{mix}}_\square(v) \geq 2/n$. If f is supported in A, Cauchy–Schwarz gives $\mathrm{Var}(f) \geq (1-\mu_I(A))\|f\|^2_2$; this proves the Dirichlet estimate. Since a proper subset has mass in $[2^{-n}, 1-2^{-n}]$, minimizing $(2/n)(1-v)v^{2/n}$ over those allowed masses gives the stated Faber–Krahn minorant.

The capacity identity is Eq. (1298) with $q_{P_\square,s} = 1$. Summing the Walsh eigenvalues through the definition of effective dimension proves Eq. (1356). For $0 < \lambda \leq 1$, all modes of weight at least $n/2$ contribute at least one half, proving Eq. (1357). Every nonconstant Walsh coefficient of the normalized singleton contrast has squared magnitude $(2^n - 1)^{-1}$, which proves Eq. (1358).

On the verifier-good event, $D_V(s) = 0$ and $D_V = 1$ elsewhere. One-step substitution proves Eq. (1359). Finally, Parseval's identity and the Walsh eigenvalues give Eq. (1360); their nonzero range $[2/n, 2]$ gives Eq. (1362).

$\square$

Figure 128 summarizes the multiplicity calculation that accompanies fast singleton modes on the public cube.

VIII.12.3 Nonlinear reweighting and learned-operator transfer

Corollary VIII.12.5 (Explicit nonlinear residual reweighting)**.** *Fix $\theta \geq 0$ and define the represented lazy local kernel*

$$P_\theta(x, x+e_i) = \frac{1}{2n}\min\{1, e^{-\theta(W_I(x+e_i)-W_I(x))}\}, \tag{1363}$$

$$P_\theta(x, x) = 1 - \sum_{i=1}^{n} P_\theta(x, x+e_i). \tag{1364}$$

It is reversible for $\pi_\theta(x) = Z_\theta^{-1}e^{-\theta W_I(x)}$. When this kernel is retained as a certified dynamical geometry, the same record contains the represented derivation of Z_θ, the invariant-law certificate, its declared precision, and their complete cost, as required by Definition V.5.1. On $\mathcal{F}_\delta$, put $\Delta_- = 1/2 - \eta - 2\delta > 0$ and $\Delta_+ = 1/2 - \eta + 2\delta$. Then

$$\frac{1}{1+(2^n-1)e^{-\theta m\Delta_-}} \leq \pi_\theta(s) \leq \frac{1}{1+(2^n-1)e^{-\theta m\Delta_+}}, \tag{1365}$$

and

$$q_{\theta,s} \leq \frac{1}{2}e^{-\theta m\Delta_-}, \qquad \mathrm{Cap}_\theta(\{s\}, X_I \setminus \{s\}) \leq \frac{1}{2}\pi_\theta(s)e^{-\theta m\Delta_-}. \tag{1366}$$

Its Poincaré gap obeys the explicit two-regime bound

$$\gamma_\theta \leq \begin{cases} e^{-\theta m\Delta_-}, & \pi_\theta(s) \leq 1/2, \\ e^{4\theta\delta m}/(2^n-1), & \pi_\theta(s) > 1/2. \end{cases} \tag{1367}$$

In the normalization of Definition V.5.4, every positive logarithmic-Sobolev, modified-logarithmic-Sobolev, or Bakry–Émery curvature rate certified for this same kernel is bounded above by the right side of Eq. (1367)*. Each such rate implies the Poincaré inequality at its own rate, whereas Eq.* (1367) *is an upper bound for the optimal Poincaré rate.*

Thus this compact nonlinear reweighting is evaluated simultaneously by its target mass, target-boundary capacity, and mixing rate.

Proof. Detailed balance follows from

$$e^{-\theta W_I(x)}\min\{1, e^{-\theta(W_I(z)-W_I(x))}\} = e^{-\theta W_I(z)}\min\{1, e^{-\theta(W_I(x)-W_I(z))}\}.$$

On $\mathcal{F}_\delta$, every off-target state satisfies $m\Delta_- \leq W_I(x) - W_I(s) \leq m\Delta_+$. Summing the corresponding Boltzmann ratios proves Eq. (1365). Every neighbor of s is off-target and has residual increase at least $m\Delta_-$, so summing the n proposal rates proves Eq. (1366) by Eq. (1298).

If $\pi_\theta(s) \leq 1/2$, the singleton conductance ratio is at most $q_{\theta,s}$; the upper Cheeger inequality gives $\gamma_\theta \leq 2q_{\theta,s}$. If $\pi_\theta(s) > 1/2$, use the complementary set. Since

$$1 - \pi_\theta(s) \geq \pi_\theta(s)(2^n-1)e^{-\theta m\Delta_+},$$

its conductance ratio is at most $e^{\theta m(\Delta_+-\Delta_-)}/(2(2^n-1))$. Applying the same upper Cheeger inequality and using $\Delta_+ - \Delta_- = 4\delta$ proves Eq. (1367).

□

Theorem VIII.12.6 (Learned-operator transfer to LPN target capacity)**.** *Let H_Φ be the native symmetric operator of a uniform-invariant LPN kernel and let $\widehat{H}_\Phi \geq 0$ be a represented learned operator on the same $L^2(\mu_I)$ carrier. Suppose the complete dynamical learning record certifies*

$$\|H_\Phi - \widehat{H}_\Phi\|_{\mathrm{op}} \leq \varepsilon_\Phi. \tag{1368}$$

Define the learned target-boundary rate

$$\widehat{q}_{\Phi,s} = 2^n \langle \mathbf{1}_{\{s\}}, \widehat{H}_\Phi \mathbf{1}_{\{s\}} \rangle_{\mu_I}. \tag{1369}$$

Then

$$\mathrm{Cap}_\Phi(\{s\}, X_I \setminus \{s\}) \leq 2^{-n}(\widehat{q}_{\Phi,s} + \varepsilon_\Phi), \tag{1370}$$

$$\Pr_{\mu_0}\{\tau_s \leq H\} \leq R_0 2^{-n}\big[1 + H(\widehat{q}_{\Phi,s} + \varepsilon_\Phi)\big] \tag{1371}$$

whenever $d\mu_0/d\mu_I \leq R_0$. On the simultaneous event of Theorem V.14.2, one may take

$$\varepsilon_\Phi = \widehat{\varepsilon}_\Phi + \mathfrak{G}_{\mathrm{dyn}}(\mathcal{F}_{\mathrm{geom}}; B, m, \delta_\Phi). \tag{1372}$$

The minimum defining $\mathfrak{G}_{\mathrm{dyn}}$ retains, in common population-error units, the mixing-block estimate of Theorem IV.15.2, the predictable learned-operator estimate of Theorem IV.15.5, and the martingale PAC–Bayes estimate of Theorem IV.15.6. If a certified eigengap $g_* > 0$ is also present, the learned fast-mode target mass differs from the population fast-mode target mass by at most

$$\frac{2\varepsilon_\Phi}{g_*}, \tag{1373}$$

in accordance with Theorem V.8.3 and Eq. (754)*.*

The appearances of s in Eqs. (1369) *and* (1373) *are analytic target audits and do not expose s to the learned operator. When the learned dynamical record outputs a represented candidate $\widehat{s}$, append the parity readout $h_{\widehat{s}}(a) = (-1)^{\langle a, \widehat{s} \rangle}$. Then Eqs.* (1142) *and* (1143) *convert its recovery probability into clean parity alignment, and Eq.* (1140) *transports that alignment through the LPN noise factor $1 - 2\eta$. The appended evaluator and its complete cost remain in the same saturated learning record.*

Proof. Put $v_s = \mathbf{1}_{\{s\}}$, so $\|v_s\|_{2,\mu_I}^2 = 2^{-n}$. Since the two condenser plates fill the carrier, its unique Dirichlet competitor is v_s, and therefore

$$\mathrm{Cap}_\Phi(\{s\}, X_I \setminus \{s\}) = \langle v_s, H_\Phi v_s \rangle \leq \langle v_s, \widehat{H}_\Phi v_s \rangle + \varepsilon_\Phi \|v_s\|_2^2.$$

Substitution of Eq. (1369) proves Eq. (1370). Apply Theorem V.6.7 to obtain Eq. (1371). Equation Eq. (1372) is the geometry coordinate of Theorem V.14.2. Finally, the projector perturbation bound in Eq. (754), evaluated on the unit vector f_s, proves Eq. (1373).

□

VIII.13 LPN Committors and Dynamical Learning

VIII.13.1 Committor and Bellman propagation

Theorem VIII.13.1 (LPN committor and same-norm Bellman propagation)**.** *Fix $0 < \eta < \tau < 1/2$, $0 < \delta < 1$, and confidence allocations $\delta_M, \delta_B, \delta_\Phi, \delta_{\text{gen}}, \delta_{\text{PB}} > 0$ whose sum is at most δ. Work on the verifier-good event of Lemma VIII.5.1. Let $F \subseteq X_I \setminus \{s\}$ be a represented failure sector, put $\mathsf{N}_I = X_I \setminus (\{s\} \cup F)$, and let P^π be a selected LPN policy kernel. Here and below u^*_β is the fixed point of the optimal committor Bellman operator restricted to the same represented budget-reachable policy class $\Pi_{B,\text{obs}}$, or Π_B in the fully observed case, used in Eq. (749) and Theorem V.14.2; it is the reach–avoid specialization of $V^*_{B,\text{obs}}$. The target gate is then $V_I = \mathbf{1}_{\{s\}}$, and the finite-horizon and discounted committors satisfy*

$$u^\pi_0 = V_I, \qquad u^\pi_{h+1} = V_I + \mathbf{1}_{\mathsf{N}_I} P^\pi u^\pi_h, \tag{1374}$$
$$u^\pi_\beta = V_I + \beta \mathbf{1}_{\mathsf{N}_I} P^\pi u^\pi_\beta, \qquad 0 < \beta < 1. \tag{1375}$$

For every represented approximation v,

$$\|u^\pi_\beta - v\|_\infty \le \frac{\|V_I + \beta \mathbf{1}_{\mathsf{N}_I} P^\pi v - v\|_\infty}{1-\beta}. \tag{1376}$$

If a complete learned LPN control record supplies same-norm model and Bellman errors $\bar\varepsilon_M, \bar\varepsilon_B$ as in Theorem V.14.2, then

$$\|u^*_\beta - v\|_\infty \le \frac{\bar\varepsilon_B + \bar\varepsilon_M}{1-\beta}, \tag{1377}$$
$$\|u^*_\beta - u^{\widehat{\pi}_v}_\beta\|_\infty \le \frac{2\beta(\bar\varepsilon_B + \bar\varepsilon_M)}{(1-\beta)^2} + \frac{2\bar\varepsilon_M}{1-\beta}. \tag{1378}$$

The errors in these displays retain the exact three same-norm coordinates of Theorem V.14.2:

$$\bar\varepsilon_M = \min\left\{\widehat\varepsilon_M + \mathfrak{G}_{\text{dyn}}(\mathcal{F}_{\text{model}}; B, m, \delta_M), U^{\text{noise}}_M, U^{\text{ch}}_M\right\}, \tag{1379}$$
$$\bar\varepsilon_B = \min\left\{\widehat\varepsilon_B + \mathfrak{G}_{\text{dyn}}(\mathcal{F}_{\text{Bell}}; B, m, \delta_B), U^{\text{noise}}_B, U^{\text{ch}}_B\right\}, \tag{1380}$$
$$\bar\varepsilon_\Phi = \min\left\{\widehat\varepsilon_\Phi + \mathfrak{G}_{\text{dyn}}(\mathcal{F}_{\text{geom}}; B, m, \delta_\Phi), U^{\text{noise}}_\Phi, U^{\text{ch}}_\Phi\right\}. \tag{1381}$$

Here each noise candidate is finite exactly when the complete record contains the corresponding represented same-norm loss-to-coordinate calibration, and each channel candidate is the smallest same-law, same-norm consequence of the typed channel profiles and represented converters listed in Theorem V.14.2; an unavailable candidate is $+\infty$. When a concrete generalization or PAC–Bayes candidate is used in one coordinate, its δ_{gen} or δ_{PB} allocation is charged within that coordinate's declared $\delta_M, \delta_B, \delta_\Phi$. Data-dependent selection among candidates uses the simultaneous confidence split required by Definition V.13.7.

Proof. The verifier event identifies the target gate with the singleton. The controlled Bellman identities give the committor recursions and fixed-policy residual bound. Applying the master same-norm propagation theorem gives the learned value and policy inequalities; the simultaneous leaf event justifies each displayed minimum. □

VIII.13.2 Safe spectra and dependent-data candidates

Theorem VIII.13.2 (Action-safe LPN spectrum and kernel-ridge error)**.** *Adopt the verifier-good presentation and public hypercube geometry of Theorem VIII.13.1 and Corollary VIII.12.4.*

In the further target-only specialization $F = \varnothing$, the legal coordinate actions generate the full mean-zero Walsh space and there is no failure killing. Consequently

$$P_{\mathcal{A}} = P_{\text{safe}} = I \quad \text{on } \mathbf{1}^{\perp}, \tag{1382}$$

and the action-safe effective dimension is exact:

$$N_{\text{eff}}^{\mathcal{A},\text{safe}}(\lambda) = \sum_{k=1}^{n} \binom{n}{k} \frac{2k/n}{2k/n + \lambda} \geq 2^{n-2}, \qquad 0 < \lambda \leq 1. \tag{1383}$$

For the normalized singleton contrast f_s in Eq. (1358)*, the exact action-safe ridge residual is*

$$\left\| \left[I - H_{\square}(H_{\square} + \lambda I)^{-1} \right] f_s \right\|_{2,\mu_I}^2 = \frac{1}{2^n - 1} \sum_{k=1}^{n} \binom{n}{k} \left(\frac{\lambda}{2k/n + \lambda} \right)^2. \tag{1384}$$

On the row-input hypercube used for noisy parity learning, the clean parity χ_s is a Walsh eigenfunction. Its corresponding residual is

$$\left\| \left[I - H_{\square}(H_{\square} + \lambda I)^{-1} \right] \chi_s \right\|_{2,\mu_I}^2 = \left(\frac{\lambda}{2|s|_1/n + \lambda} \right)^2. \tag{1385}$$

For $s \neq 0$, this is the action-safe compressed residual. For $s = 0$, $\chi_0 = \mathbf{1}$ lies in the constant orthogonal complement of the action-generated mean-zero range, and the same display retains the exact approximation residual $\|(I - P_{\mathcal{A}})\chi_0\|_{2,\mu_I}^2 = 1$, as prescribed by Eq. (743)*.*

For independent LPN rows, the debiased dynamical kernel-ridge estimator $\widehat{f}_\lambda$ of Eq. (334)*, applied to this row-input hypercube and the clean parity χ_s, satisfies*

$$\begin{aligned} \mathbb{E}\|\widehat{f}_\lambda - \chi_s\|_{2,\mu_I}^2 \leq C \left[\frac{v_\eta^2}{(1-2\eta)^2 m} \sum_{k=1}^{n} \binom{n}{k} \frac{2k/n}{2k/n + \lambda} + \left(\frac{\lambda}{2|s|_1/n + \lambda} \right)^2 \right] \\ + \frac{\varepsilon_{\text{sel}}}{(1-2\eta)^2}, \end{aligned} \tag{1386}$$

whenever the complete record contains the conditional sub-Gaussian variance, covariance-concentration, effective-sample-size, and simultaneous selection certificates required by that framework bound. For a dependent record, the same display replaces m by its certified m_{dyn} and restores the represented ε_{dep} term.

Proof. Coordinate actions span every nonconstant Walsh character, so both safe projections are the identity on $\mathbf{1}^{\perp}$. Walsh diagonalization gives the effective dimension and singleton/parity ridge residuals. Substitution into the framework noise-corrected kernel-ridge theorem gives the independent-row bound; the dependent-data statement uses its certified effective sample size and dependence defect. □

Theorem VIII.13.3 (LPN dependent-data generalization candidates)**.** *Adopt the complete represented LPN learning record and confidence allocation of Theorem VIII.13.1.*

The dynamical generalization candidates also specialize without changing their sampling regimes. For a length-m absolutely regular LPN record and a certified block length a, put

$$\mu_a = \left\lfloor \frac{m}{2a} \right\rfloor, \qquad \delta_a = \delta_{\mathrm{gen}} - 2(\mu_a - 1)\beta(a) > 0. \tag{1387}$$

Then the mixing-block candidate is

$$\Gamma^{\mathrm{LPN}}_{\beta,a} = 2 \max_{p\in\{\mathrm{odd},\mathrm{even}\}} \mathfrak{R}^{p}_{\mu_a}(a) + 3\sqrt{\frac{\log(4/\delta_a)}{2\mu_a}} + \frac{2a}{m}. \tag{1388}$$

For $m \geq 2$ independent LPN rows, $\beta(a) = 0$. Taking $a = 1$ and the 2^n-element parity class gives the explicit candidate

$$\Gamma^{\mathrm{LPN}}_{\beta,1} \leq 2\sqrt{\frac{2n\log 2}{\lfloor m/2\rfloor}} + 3\sqrt{\frac{\log(4/\delta_{\mathrm{gen}})}{2\lfloor m/2\rfloor}} + \frac{2}{m}. \tag{1389}$$

For a represented predictable LPN operator family satisfying the hypotheses of Theorem IV.15.5, write $\eta_{\mathrm{op}} > 0$ for its pathwise operator lower bound, distinct from the LPN label-noise rate η. Its sequential-radius candidate is

$$\mathfrak{R}^{\mathrm{seq}}_{m,\mathrm{LPN}} \leq \inf_{\varrho>0} \left\{ R_A \sup_{\theta\in\Theta} \mathfrak{q}_m(\mathbf{A}_\theta) + R_A\sqrt{\frac{L_A\varrho}{m}} + \frac{4R_A}{\sqrt{\eta_{\mathrm{op}}}}\sqrt{\frac{\log\mathcal{N}(\Theta,d,\varrho)}{m}} \right\}. \tag{1390}$$

For the represented LPN prediction class $\mathcal{H}_B$ in the same complete record, the online coordinate specializes to

$$\frac{\mathcal{V}_m(\mathcal{H}_B)}{m} \leq 2L_\ell \sup_{\mathbf{z}} \mathfrak{R}^{\mathrm{seq}}_m(\mathcal{H}_B;\mathbf{z}). \tag{1391}$$

Under independent LPN rows, the represented average predictor, or a uniformly selected represented iterate, has population excess risk at most the right side of Eq. (1391)*. For the homogeneous flip-symmetric binary loss covered by Eq.* (324)*, corrupted excess is converted to clean parity risk by the same factor $(1-2\eta)^{-1}$ used below.*

For the uniform prior P_{par} on the parity class and a point posterior δ_t, $\mathrm{KL}(\delta_t\|P_{\mathrm{par}}) = n\log 2$. Hence the martingale PAC–Bayes candidate is

$$\Gamma^{\mathrm{LPN}}_{\mathrm{PB}}(\lambda) = \frac{n\log 2 + \log(1/\delta_{\mathrm{PB}})}{\lambda m} + \frac{\lambda}{8}, \tag{1392}$$

and optimization over $\lambda > 0$ gives

$$\inf_{\lambda>0} \Gamma^{\mathrm{LPN}}_{\mathrm{PB}}(\lambda) = \sqrt{\frac{n\log 2 + \log(1/\delta_{\mathrm{PB}})}{2m}}. \tag{1393}$$

After conversion to one common bounded population loss, the definition of $\mathfrak{G}_{\mathrm{dyn}}$ takes the minimum of the applicable independent, mixing-block, sequential, and martingale PAC–Bayes candidates above. It also admits the effective-dimension candidate exactly when the same complete record satisfies the source, noise, and covariance hypotheses of Corollary IV.15.10.

Proof. Apply the absolute-regularity blocking theorem with the displayed block count. Independent rows set the mixing coefficient to zero. The sequential cover, online-to-batch, and martingale PAC–Bayes theorems give the other candidates; optimizing the elementary PAC–Bayes expression gives its closed form. Only candidates in the same bounded population-loss units enter the minimum. □

VIII.13.3 Complete learning transfer and stress tests

Corollary VIII.13.4 (Clean-risk transport for a complete LPN learning record)**.** *Under the hypotheses of Theorems VIII.13.1 and VIII.13.3, the homogeneous LPN flip channel preserves the dependence comparison and transports corrupted excess to clean risk through*

$$U^{\mathrm{clean}}_{\mathrm{noise}}(n, B, m, \eta, \delta_{\mathrm{gen}}) = \min\left\{1,\ \frac{\mathrm{Exc}^{\mathrm{sat}}_{\mathrm{corr}}(n, B, m, \eta, \delta_{\mathrm{gen}})}{1-2\eta}\right\}. \tag{1394}$$

Separately, the represented parity empirical minimizer evaluated in Example VIII.2.1 obeys the recordwise bound

$$R_0(\widehat{f}) \le \min\left\{1,\ \frac{2\Gamma^{\mathrm{par}}_{n,m,\delta_{\mathrm{gen}}} + \varepsilon_{\mathrm{opt}}}{1-2\eta}\right\}, \tag{1395}$$

with $\Gamma^{\mathrm{par}}_{n,m,\delta_{\mathrm{gen}}}$ *given explicitly in Eq.* (1135)*.*

Every committor, operator, posterior, block length, selected policy, and downstream evaluator in these displays belongs to the one complete saturated LPN learning record. The identities create no additional structural source.

Proof. The homogeneous flip channel preserves the dependence comparison and scales binary excess risk by $1 - 2\eta$. Applying the saturated clean-risk transport theorem gives the first formula. The recordwise parity bound is the uniform parity generalization estimate with the same confidence allocation, clipped by the unit risk bound. □

Corollary VIII.13.5 (LPN committor, safe-spectrum, and dynamical-learning specialization)**.** *The committor, action-safe spectral, dependent-generalization, and clean-risk conclusions are Theorems VIII.13.1 to VIII.13.3 and Corollary VIII.13.4. All belong to one complete represented LPN learning record.*

Proof. On the verifier-good event, Lemma VIII.5.1 identifies V_I with the singleton target gate. Substitution into Theorem V.6.2 proves Eqs. (1374) and (1375). The fixed-policy residual estimate is Theorem V.12.1; the learned value and policy estimates are Theorem V.13.6. Substitution of the master same-norm coordinates proves Eq. (1379). Temporal admissibility and complete cost are retained by Theorem V.6.3 and Corollary V.13.4.

For every nonempty Walsh index S, choose $i \in S$. The coordinate difference satisfies $\chi_S(\cdot + e_i) - \chi_S = -2\chi_S$, so the legal coordinate-action increments span every nonconstant Walsh mode. This proves Eq. (1382). Apply Theorem V.13.3 and the eigenvalues $2k/n$ from Corollary VIII.12.4 to obtain Eq. (1383). The normalized singleton has squared Walsh coefficient $(2^n - 1)^{-1}$ on every nonconstant character; the spectral multiplier of $I - H_\square(H_\square + \lambda I)^{-1}$ is $\lambda/(2k/n + \lambda)$. Parseval proves Eq. (1384). Applying the same multiplier to the single parity mode χ_s proves Eq. (1385). Substitution of Eqs. (1383) and (1385) into Eq. (334), with $m_{\mathrm{dyn}} = m$ and $\varepsilon_{\mathrm{dep}} = 0$ for independent rows, proves Eq. (1386).

Set the trajectory length $N = m$ in Theorem IV.15.2 to obtain Eqs. (1387) and (1388). Independent rows have zero absolute-regularity coefficients, and the finite-class Rademacher estimate $\mathfrak{R}_{\mu_1} \leq \sqrt{2n \log 2/\mu_1}$ proves Eq. (1389). Substitution of $T = m$ and η_{op} for the operator lower bound in Theorem IV.15.5 proves Eq. (1390). The uniform parity prior gives the displayed KL exactly; substitution in Theorem IV.15.6 proves Eq. (1392). Minimizing $c/\lambda + \lambda/8$, with $c = (n \log 2 + \log(1/\delta_{\mathrm{PB}}))/m$, proves Eq. (1393). The online-regret and online-to-batch conclusions are, respectively, Theorem IV.15.7 and Corollary IV.15.8 with $T = m$. The clean-risk transport for their corrupted binary loss is Eq. (324); this proves the statement following Eq. (1391).

The common-loss minimum is exactly Definition V.13.7. The corruption innovations do not increase absolute regularity by Theorem IV.16.2. The saturated clean-risk conversion is Eq. (330) and Theorem IV.16.6; the recordwise statement Eq. (1395) is Eq. (1136) with $\delta = \delta_{\mathrm{gen}}$, truncated by the trivial risk bound one. Finally, Theorem V.1.2 and Corollary V.13.4 retain the one complete derivation and its existing structural source classification.

$\square$

Corollary VIII.13.6 (Three exact LPN dynamical stress tests)**.** *The controlled certificate gives the following parameter-explicit evaluations. Whenever R occurs below, assume $n/m \to R \in (0, 1)$.*

(i) Subset Gaussian elimination. *On $\{\operatorname{rank} A = n\}$, let $S(A)$ be the represented deterministic pivot-row set and put $\widehat{s} = A^{-1}_{S(A)} b_{S(A)}$. Then*

$$\Pr_E\{\widehat{s} = s \mid A, \operatorname{rank} A = n\} = (1 - \eta)^n, \tag{1396}$$

and hence

$$\Pr\{\widehat{s} = s\} \geq (1 - 2^{n-m})(1 - \eta)^n. \tag{1397}$$

If $n/m \to R \in (0, 1)$ and $\eta m = c \log n$, then

$$(1 - \eta)^n = n^{-cR+o(1)}. \tag{1398}$$

(ii) Located spectral or BKW-type materialization. *Put $\rho = 1 - 2\eta$. If a charged located block has length $L_B(n)$ and every aligned index has weight at least δm, then*

$$|\mathbb{E}[h_{A,L}(As + E)\chi_v(s) \mid A]| \leq \sqrt{L_B(n)}\, \rho^{\lfloor \delta m \rfloor}. \tag{1399}$$

If $n/m \to R$ and $L_B(n) \leq 2^{\beta n + o(n)}$, the logarithm of the right side is at most

$$n \left[\frac{\beta \log 2}{2} - \frac{\delta}{R} \log \frac{1}{1 - 2\eta}\right] + o(n). \tag{1400}$$

It is exponentially negative whenever

$$\beta < \frac{2\delta}{R \log 2} \log \frac{1}{1 - 2\eta}. \tag{1401}$$

The exceptional-presentation probability remains the explicit term in Eq. (1150).

(iii) Threshold-guided residual dynamics. *Fix $\eta < \tau_- < \tau < \tau_+ < H_2^{-1}(1 - R)$. On the event in Eq.* (1249),

$$\mathbf{1}_{\{W_I \leq \tau m\}} = \mathbf{1}_{\{s\}}. \tag{1402}$$

Consequently every represented normalized kernel satisfies, for $x \neq s$,

$$P\mathbf{1}_{\{W_I \leq \tau m\}}(x) = P(x, s), \tag{1403}$$

$$(I-P)(1-\mathbf{1}_{\{W_I\le\tau m\}})(x)=P(x,s), \tag{1404}$$
$$\mathrm{Cap}_{\Phi_P}(\{W_I\le\tau m\},\{W_I>\tau m\})=2^{-n}q_{P,s} \tag{1405}$$

for a uniform-invariant native kernel. Thus the threshold readout's prospective transition value is exactly the already charged target-contact, drift, and capacity coordinate of the selected kernel.

Proof. The pivot set depends only on A. Conditional on full rank, $\widehat{s}=s+A_{S(A)}^{-1}E_{S(A)}$, so equality holds precisely when the n selected independent noise bits vanish. This proves Eq. (1396); the rank bound in Proposition VIII.6.1 proves Eq. (1397). Expanding $n\log(1-\eta)=-n\eta+O(n\eta^2)$ gives Eq. (1398).

The Cauchy–Schwarz attenuation in Eq. (1147) proves Eq. (1399). Substitute $m=n/R+o(n)$ and the bound on L_B, then take logarithms, to obtain Eq. (1400); its negativity is exactly Eq. (1401).

The strict separation $W_I(s)\le\tau_-m<\tau m<\tau_+m<W_I(x)$ for $x\ne s$ proves Eq. (1402). Applying P gives the contact identity, subtraction from one gives the drift identity, and Eq. (1298) gives the capacity identity.

□

VIII.13.4 Exact BKW traces and resource-matched comparison

The located-spectral estimate in Eq. (1399) records the attenuation of a materialized block. The generic semantic-cancellation theorem Theorem III.30.2 distinguishes the terminal support of $\lambda^{\mathsf T}A$ from the noise-provenance support $|\lambda|$. A complete BKW comparison also records how the block is constructed, how often an original noisy equation is reused, how the terminal score is decoded, and which data interface supplies the equations. The following trace representation makes these quantities explicit. It uses the collision-and-terminal-transform structure of BKW and its concrete variants [13, 16, 80] while retaining the manuscript's presentation-relative resource ledger.

Definition VIII.13.7 (Prefix-adapted LPN combination trace)**.** Let q LPN equations be written as $b=As+E$, where $A\in\mathbb{F}_2^{q\times n}$ and $E_i\overset{\text{iid}}{\sim}\mathrm{Ber}(\eta)$. Fix a nonempty terminal coordinate block $S\subseteq[n]$, with $|S|=k$. A *prefix-adapted combination trace* for S is a represented matrix

$$\Lambda=(\lambda_1^{\mathsf T},\dots,\lambda_K^{\mathsf T})\in\mathbb{F}_2^{K\times q} \tag{1406}$$

such that:

- (i) every λ_j is nonzero;
- (ii) $(\Lambda A)_{[n]\setminus S}=0$;
- (iii) the represented rule constructing Λ is measurable with respect to $A_{[n]\setminus S}$, its own charged randomness, and its clocked construction record, and does not inspect A_S, b, or the noise;
- (iv) the construction cost, row locators, collision tables, combination DAG, terminal score, comparison rule, and output map are retained in the complete record.

Put

$$w_j=|\lambda_j|,\qquad d_i=|\{j:(\lambda_j)_i=1\}|,\qquad D_2(\Lambda)=\sum_{i=1}^{q}d_i^2,\qquad \mu_\eta(\Lambda)=\sum_{j=1}^{K}(1-2\eta)^{w_j}. \tag{1407}$$

The terminal score for $z\in\mathbb{F}_2^S$ is

$$Z_{\Lambda,S}(z)=\sum_{j=1}^{K}(-1)^{(\Lambda b)_j+\langle(\Lambda A)_{j,S},z\rangle}. \tag{1408}$$

For full-secret recovery, a BKW trace record is a finite collection $\mathfrak{T} = (S_g, \Lambda_g)_{g=1}^G$ whose terminal blocks partition $[n]$, together with the represented concatenation of the block estimates. The same acquired equations may be reused across different blocks; that reuse is charged in each $D_2(\Lambda_g)$ and in the complete execution ledger.

Lemma VIII.13.8 (Exact BKW trace moments and dependence)**.** *Write $\rho = 1 - 2\eta$. Conditional on a realized prefix-adapted trace Λ and on the inspected coordinates of A,*

$$\mathbb{E}_E[Z_{\Lambda,S}(s_S)] = \mu_\eta(\Lambda) = \sum_{j=1}^{K} \rho^{w_j}, \tag{1409}$$

$$\mathrm{Cov}_E\left((-1)^{\lambda_j^\mathsf{T} E}, (-1)^{\lambda_l^\mathsf{T} E}\right) = \rho^{|\lambda_j+\lambda_l|} - \rho^{w_j+w_l}. \tag{1410}$$

For each $z \neq s_S$, conditional on the same trace data and averaging over the uninspected block A_S and the noise,

$$\mathbb{E}_{A_S,E}[Z_{\Lambda,S}(z)] = 0. \tag{1411}$$

Moreover, for every $t > 0$,

$$\Pr\{Z_{\Lambda,S}(s_S) \leq \mu_\eta(\Lambda) - t \mid \Lambda\} \leq \exp\left[-\frac{t^2}{2D_2(\Lambda)}\right], \tag{1412}$$

$$\Pr\{Z_{\Lambda,S}(z) \geq t \mid \Lambda\} \leq \exp\left[-\frac{t^2}{2D_2(\Lambda)}\right]. \tag{1413}$$

Proof. At the correct block value, clause (ii) of Definition VIII.13.7 gives

$$(\Lambda b)_j + \langle (\Lambda A)_{j,S}, s_S \rangle = \lambda_j^\mathsf{T} E.$$

Independence of the original noise bits yields $\mathbb{E}(-1)^{\lambda_j^\mathsf{T} E} = \rho^{w_j}$. Multiplying the two signs replaces their exponents by $(\lambda_j + \lambda_l)^\mathsf{T} E$, which proves Eq. (1410).

For $z \neq s_S$, put $v = z + s_S \neq 0$. Prefix adaptation leaves A_S independent of the trace. Hence $(\langle A_{i,S}, v\rangle + E_i)_{i \leq q}$ are independent uniform bits, and every nonzero λ_j has zero character mean. This proves Eq. (1411).

Changing original bit E_i changes the correct score by at most $2d_i$. For a wrong score the same bound holds after replacing the independent input bit by $\langle A_{i,S}, v\rangle + E_i$. The bounded-difference inequality therefore has squared sensitivity sum $4\sum_i d_i^2 = 4D_2(\Lambda)$. Its two one-sided forms give Eqs. (1412) and (1413). □

Theorem VIII.13.9 (Exact dependence-aware BKW recovery certificate)**.** *Let $\widehat{s}_{S_g}$ maximize Z_{Λ_g,S_g}, using the represented deterministic tie rule, and concatenate the block estimates. Conditional on successful construction of the trace collection $\mathfrak{T}$,*

$$\Pr\{\widehat{s} = s \mid \mathfrak{T}\} \geq 1 - \sum_{g=1}^{G} \min\left\{1, 2^{k_g} \exp\left[-\frac{\mu_\eta(\Lambda_g)^2}{8D_2(\Lambda_g)}\right]\right\}. \tag{1414}$$

If trace construction fails with probability at most $\varepsilon_{\mathrm{tr}}$, the unconditional right side is reduced by $\varepsilon_{\mathrm{tr}}$. Public verification further contributes the global error

$$\varepsilon_V(n, q, \eta, \tau) = \exp[-qD_{\mathrm{Ber}}(\tau\|\eta)] + (2^n - 1)2^{-q(1-H_2(\tau))} \tag{1415}$$

when the trace is built from q independent presented equations.

Proof. For block g, apply Eq. (1412) with $t = \mu_\eta(\Lambda_g)/2$. Apply Eq. (1413) with the same threshold to each of the $2^{k_g} - 1$ wrong block values. Outside the union of these events the correct score is strictly larger than every wrong score. A union bound over the candidates and then over the terminal blocks proves Eq. (1414). The trace-construction and verifier errors are separate events. The verifier term is Eq. (1045) with matrix height q. □

Definition VIII.13.10 (Canonical materialized BKW schedule and exact cost)**.** For one terminal block of size k, a canonical disjoint BKW schedule is an integer tuple

$$\theta_{\mathrm{BKW}} = (q, r, k, b_1, \dots, b_r), \qquad \sum_{j=1}^{r} b_j = n - k. \tag{1416}$$

At stage j, the schedule buckets the current rows by the next b_j coordinates and forms disjoint pairs within each bucket. Put

$$D_0 = n, \qquad\qquad D_j = n - \sum_{h=1}^{j} b_h, \tag{1417}$$

$$N_0 = q, \qquad\qquad N_j = \left\lfloor \frac{(N_{j-1} - 2^{b_j})_+}{2} \right\rfloor. \tag{1418}$$

The recurrence is a deterministic lower bound: among at most 2^{b_j} buckets, pairing leaves at most one unpaired row per bucket. The final trace has $K = N_r$, mutually disjoint leaf supports of weight 2^r, and hence

$$\mu_\eta = K\rho^{2^r}, \qquad D_2 = K2^r. \tag{1419}$$

Fix implementation constants $c_{\mathrm{key}}, c_{\mathrm{pair}}, c_{\mathrm{xor}}, c_{\mathrm{dag}}, c_{\mathrm{hist}}, c_{\mathrm{F}}, c_{\max}, c_{\mathrm{acq}} > 0$. The canonical bit-operation and word-operation ledger is

$$\begin{aligned}
T_{\mathrm{red}}(\theta_{\mathrm{BKW}}) = \sum_{j=1}^{r} \Big[& c_{\mathrm{key}} N_{j-1} b_j + c_{\mathrm{pair}} N_j + c_{\mathrm{xor}} N_j (D_{j-1} + 1) \\
& + 2c_{\mathrm{dag}} N_j \left\lceil \log_2(N_{j-1} \vee 2) \right\rceil \Big],
\end{aligned} \tag{1420}$$

$$T_{\mathrm{term}}(\theta_{\mathrm{BKW}}) = c_{\mathrm{hist}} K(k+1) + c_{\mathrm{F}} k 2^k + c_{\max} 2^k, \tag{1421}$$

$$Q_{\mathrm{BKW}}(\theta_{\mathrm{BKW}}) = q, \tag{1422}$$

$$\begin{aligned}
M_{\mathrm{BKW}}(\theta_{\mathrm{BKW}}) = \max_{1 \le j \le r} \Big\{ & N_{j-1}(D_{j-1}+1) + N_j(D_j + 1) \\
& + 2N_j \lceil \log_2(N_{j-1} \vee 2) \rceil,\ 2^k \lceil \log_2(K+1) \rceil \Big\},
\end{aligned} \tag{1423}$$

$$T^{\mathrm{fix}}_{\mathrm{BKW}}(\theta_{\mathrm{BKW}}) = T_{\mathrm{red}} + T_{\mathrm{term}}, \tag{1424}$$

$$T^{\mathrm{stream}}_{\mathrm{BKW}}(\theta_{\mathrm{BKW}}) = c_{\mathrm{acq}} q + T_{\mathrm{red}} + T_{\mathrm{term}}. \tag{1425}$$

The code length $d_{\mathrm{BKW}}(\theta_{\mathrm{BKW}})$ is the canonical self-delimiting code of the stage tag, the integers in Eq. (1416), the collision and tie rules, the terminal transform, precision, and the fixed interpreter identifier, as specified by Definition V.17.1. A full-secret schedule sums time and data actually acquired across its block traces, takes peak memory, and includes every reused-data reference in the DAG ledger.

Corollary VIII.13.11 (Closed-form disjoint-BKW sample threshold)**.** *Let G terminal blocks have size at most k, and allocate total trace failure $\epsilon \in (0,1)$ uniformly across them. A sufficient final row count for each block is*

$$K_{\mathrm{req}}(r,k,G,\eta,\epsilon) = \left\lceil \frac{8\,2^r\big(k\log 2 + \log(G/\epsilon)\big)}{(1-2\eta)^{2^{r+1}}} \right\rceil. \tag{1426}$$

Starting from $N_r = K_{\mathrm{req}}$, the reverse recursion

$$N_{j-1}^{\mathrm{req}} = 2N_j^{\mathrm{req}} + 2^{b_j} + 1, \qquad j = r, r-1, \dots, 1, \tag{1427}$$

produces an explicit sufficient initial sample count $q_{\mathrm{req}} = N_0^{\mathrm{req}}$. With these values,

$$\Pr\{\widehat{s} = s\} \geq 1 - \epsilon - \varepsilon_V(n, q_{\mathrm{req}}, \eta, \tau). \tag{1428}$$

All constants in the data, reduction, terminal, memory, and Levin costs are then obtained by substituting the reverse recurrence into Eqs. (1420)*,* (1421) *and* (1423)*.*

Proof. Substitute Eq. (1419) into Eq. (1414). Each block error is at most

$$2^k \exp\left[-\frac{K(1-2\eta)^{2^{r+1}}}{8\,2^r}\right].$$

The choice in Eq. (1426) makes this at most ϵ/G. The reverse recurrence implies $N_j \geq N_j^{\mathrm{req}}$ in Eq. (1418), completing the proof. □

Corollary VIII.13.12 (Proof-producing scaling audit for materialized BKW)**.** *Fix n, m, η, ϵ, terminal block size k, and number G of terminal blocks. For a required cancellation level q_{can}, let $\mathcal{P}^{\mathrm{fix}}_{\mathrm{BKW}}(q_{\mathrm{can}})$ be the exact occupancy paths of the canonical materialized BKW cell that use at most m presented rows, cancel at least q_{can} semantic coordinates, and retain*

$$N_r \geq K_{\mathrm{req}}(r,k,G,\eta,\epsilon).$$

Define the fixed-data charged crossing value

$$\mathfrak{C}^{\mathrm{fix}}_{\mathrm{BKW}}(n,m,\eta,\epsilon; q_{\mathrm{can}}) = \inf_{p \in \mathcal{P}^{\mathrm{fix}}_{\mathrm{BKW}}(q_{\mathrm{can}})} \mathrm{Cost}_{\mathrm{complete}}(p), \tag{1429}$$

with infimum $+\infty$ when the path set is empty. The complete cost is the acquisition, time, memory, description, provenance, terminal, and verification ledger in Definition VIII.13.10. The streaming crossing value removes the $N_0 \leq m$ restriction and charges $c_{\mathrm{acq}} N_0$.

The canonical guaranteed-row schedules give the constructive candidate

$$\mathfrak{U}^{\mathrm{fix}}_{\mathrm{BKW}} = \min_{\theta_{\mathrm{BKW}}} \left\{ \mathrm{Cost}_{\mathrm{complete}}(\theta_{\mathrm{BKW}}) \;\middle|\; \begin{array}{l} \sum_{j=1}^{r} b_j \geq q_{\mathrm{can}}, \quad N_r \geq K_{\mathrm{req}}(r,k,G,\eta,\epsilon), \\ N_0 \leq m, \quad b_j \in \mathbb{N}, \quad \theta_{\mathrm{BKW}} \textit{ satisfies Eq. } (1418) \end{array} \right\}, \tag{1430}$$

and

$$\mathfrak{C}^{\mathrm{fix}}_{\mathrm{BKW}} \leq \mathfrak{U}^{\mathrm{fix}}_{\mathrm{BKW}}. \tag{1431}$$

Under the LPN substitutions

$$Q_j = 2^{b_j}, \qquad d_j = b_j, \qquad \rho = 1 - 2\eta, \qquad K = N_r, \tag{1432}$$

the exact occupancy paths and noise transport are those of Theorem V.15.21. Hence a deterministic certificate of the lower inequality

$$\mathfrak{C}^{\mathrm{fix}}_{\mathrm{BKW}}(n, m, \eta, \epsilon; q_{\mathrm{can}}) > B_n \tag{1433}$$

proves that no record in the canonical materialized BKW cell reaches that cancellation level together with the displayed terminal-accuracy certificate within budget B_n.

A learning pipeline may propose a piecewise schedule and lower-bound template from finitely many evaluated sizes. It gives a proof for every $n \geq N_0$ when the exported certificate verifies:

(i) *the exact bucket-occupancy, semantic-support, provenance, and complete cost recurrences;*
(ii) *the noise-dependent terminal requirement Eq.* (1426)*;*
(iii) *a complete recurrence cover of the canonical materialized BKW cell and a branch-and-bound, dynamic-programming, or symbolic lower bound on every exact path in that cover; and*
(iv) *the uniform tail inequality Eq.* (1433)*, including every breakpoint of the proposed piecewise law.*

The finite experiments discover and test the scaling template. The exact recurrence cover, lower path optimization, and symbolic induction establish its continuation and the price of every scaling breakpoint.

Proof. The substitutions in Eq. (1432) identify the bucket alphabet, semantic cancellation, provenance size, and noise attenuation with Definition V.15.20. The dependence-aware score certificate gives the terminal row requirement in Eq. (1426). Guaranteed complete-pairing schedules are members of the exact path set, which proves Eq. (1431). A verified lower bound on the exact path infimum is the crossing certificate in Theorem V.15.18; that theorem proves the exclusion and the symbolic tail assertion. □

Definition VIII.13.13 (Fixed-data and streaming optimized BKW envelopes)**.** Let $\mathcal{A}^{\mathrm{fix}}_{\mathrm{BKW}}(n, m, \eta)$ contain all represented prefix-adapted BKW trace records that use only the m rows of the public presentation. Let $\mathcal{A}^{\mathrm{stream}}_{\mathrm{BKW}}(n, \eta)$ contain the same represented stage grammar with a conservative LPN sample interface, where every acquired row is counted by Eqs. (1422) and (1425). Define

$$A^{*,\mathrm{fix}}_{\mathrm{BKW}}(\mathbf{b}) = \mathrm{BenchArr}_{\mathcal{A}^{\mathrm{fix}}_{\mathrm{BKW}},H}(\mathbf{b}), \tag{1434}$$

$$A^{*,\mathrm{stream}}_{\mathrm{BKW}}(\mathbf{b}) = \mathrm{BenchArr}_{\mathcal{A}^{\mathrm{stream}}_{\mathrm{BKW}},H}(\mathbf{b}). \tag{1435}$$

The corresponding matched-success frontiers are $\mathcal{P}_{\mathcal{A}^{\mathrm{fix}}_{\mathrm{BKW}},H}(\alpha)$ and $\mathcal{P}_{\mathcal{A}^{\mathrm{stream}}_{\mathrm{BKW}},H}(\alpha)$. An optimizer over the integer schedule parameters evaluates these definitions; it may discard dominated schedules but may not omit acquisition, table, combination, decoder, verification, or description costs. The family also contains represented approximate-collision or coded-BKW stages when their code, quantizer, compensation defect, and terminal decoder are included in the complete record. Such a stage uses the existing compensated-quotient certificate of Definition III.18.1; the exact trace certificate Theorem VIII.13.9 applies to the zero-defect subfamily. Thus optimization over the full envelope retains modern BKW variants, while the closed-form grid below reports the canonical exact disjoint subfamily whose constants are evaluated without an approximation.

Theorem VIII.13.14 (Complete structural charging of a BKW schedule)**.** *Every record in Definition VIII.13.13 is one complete record in the existing saturated profile. Its row combinations have* Add *ancestry; a dynamically qualified block or fiber map is charged to its exact or compensated* Abs *coordinate; collision equality, table ordering, and stage clocks retain their* Sym*,* Ord*, and* Filt *provenance; iteration is charged to* Lift*; the terminal score and prospective choice retain their* *authorized* Geom/Caus *gates; public target testing is charged to the verifier; and nonstructural transcript storage remains in* J_{res}*. Derived constructors create no additional source charge.*

For every row $\lambda_{g,j}$*, the semantic support is* $\mathrm{supp}(A^{\mathsf{T}}\lambda_{g,j}) \subseteq S_g$*, while its noise-provenance support is* $|\lambda_{g,j}|$*. The collision stages reduce the former; the latter determines the exact attenuation* $(1-2\eta)^{|\lambda_{g,j}|}$ *and retains the complete combination-DAG cost. Hence every polynomially materialized trace with* $\max_g |S_g| = o(n)$ *lies in the rate-uniform sector of Corollary VIII.3.6.*

For a materialized trace collection put

$$L_B = \sum_{g=1}^{G} K_g, \qquad q_{\min} = \min_{g,j} |\lambda_{g,j}|, \qquad \beta_{\mathrm{BKW}} = \frac{\log_2(L_B \vee 1)}{n}, \qquad \delta_{\mathrm{BKW}} = \frac{q_{\min}}{q}. \tag{1436}$$

Then its located target-character coordinate satisfies

$$|\mathrm{Corr}_{\mathrm{loc}}(\mathfrak{T}; s)| \leq \min\Big\{1, \sqrt{L_B}\,(1-2\eta)^{q_{\min}}\Big\}. \tag{1437}$$

For fixed data $q = m$*, this is exactly Eq.* (1399) *with* $\delta = \delta_{\mathrm{BKW}}$*. For streaming data, the identical formula holds with the charged acquired height* q*. Every other upper certificate available to the same complete record remains in the minimum prescribed by the saturated framework.*

Proof. The provenance assignments are instances of Theorems II.9.4 and III.18.9, Corollary III.16.11, and Proposition II.5.9. The trace constructor, its locators, and its costs remain ancestors by Lemmas IV.0.2 and I.4.7 and Theorem III.30.2. Applying Theorem VIII.2.3 to the L_B explicitly located characters, followed by Cauchy–Schwarz, gives Eq. (1437). The fixed and streaming statements differ only in the declared source of the q independent rows; the latter source is included in the cost by definition. The rate-uniform statement follows from Theorem VIII.3.5. □

Corollary VIII.13.15 (Quantitative saturated improvement relative to optimized BKW)**.** *Let* $\diamond \in \{\mathrm{fix}, \mathrm{stream}\}$*, and let* $[L^{\diamond}_{\mathrm{BKW}}, U^{\diamond}_{\mathrm{BKW}}]$ *be a simultaneous valid interval for the optimized BKW frontier at resource ceiling* $\mathbf{b}$*. Take* $[L_{\mathrm{sat}}, U_{\mathrm{sat}}]$ *from the sharp integrated LPN minimum, including Eq.* (1055) *and the complete LPN Bellman/source candidates, and let* $[L_{\widehat{R}}, U_{\widehat{R}}]$ *be the learner interval. Then*

$$\big[L_{\mathrm{sat}} - U^{\diamond}_{\mathrm{BKW}}\big]_+ \leq \mathrm{ActArr}^{\mathrm{sat}}_{\mathrm{LPN}}(\mathbf{b}) - A^{*,\diamond}_{\mathrm{BKW}}(\mathbf{b}) \leq U_{\mathrm{sat}} - L^{\diamond}_{\mathrm{BKW}}, \tag{1438}$$

$$L_{\widehat{R}} - U^{\diamond}_{\mathrm{BKW}} \leq \mathfrak{M}(A^{H}_{\widehat{R}}) - A^{*,\diamond}_{\mathrm{BKW}}(\mathbf{b}) \leq U_{\widehat{R}} - L^{\diamond}_{\mathrm{BKW}}. \tag{1439}$$

At matched arrival α*, the time, data, and memory improvements are the three coordinates of Eq.* (1041)*; the BKW-specific witnessed Levin gain is*

$$\Delta^{\mathcal{A}_{\mathrm{BKW}}} \geq d_{\mathrm{BKW}}(\theta_{\mathrm{BKW}}) - \ell_{\mathrm{cert,LPN}} \tag{1440}$$

whenever the simultaneous benchmark-matching condition holds through that canonical code length.

For the disjoint materialized schedule, every displayed quantity is an explicit function of

$$(n, m, q, \eta, H, \tau, \epsilon, G, r, k, b_1, \ldots, b_r, c_{\mathrm{key}}, c_{\mathrm{pair}}, c_{\mathrm{xor}}, c_{\mathrm{dag}}, c_{\mathrm{hist}}, c_{\mathrm{F}}, c_{\max}, c_{\mathrm{acq}}) \tag{1441}$$

through Eqs. (1415), (1420), (1421), (1423), (1426) *and* (1427).

Proof. Apply Theorem V.17.19 with $\mathcal{A} = \mathcal{A}^{\diamond}_{\mathrm{BKW}}$. The matched-success and description-length statements are Corollary V.17.20. The parameter list is a collection of the arguments appearing in the cited exact formulas. □

n	η	k	r	$\log T$	$\log M$	$\log Q$	success lower
128	0.010	16	6	35.282	32.445	27.748	0.990001
128	0.125	16	4	42.567	39.727	35.031	0.990000
256	0.010	16	8	51.569	47.744	43.064	0.990000
256	0.125	16	5	65.483	61.634	56.957	0.990000
512	0.010	16	9	78.431	73.657	69.003	0.990000
512	0.125	16	6	102.754	97.946	93.285	0.990000

Table 5: Certified canonical disjoint-BKW streaming points. The public fixed-data presentations in these rows contain only $m = 4n$ equations and do not meet the corresponding sufficient sample thresholds; the fixed-data frontier is therefore empty for this certified schedule grid rather than being silently supplemented with streaming samples.

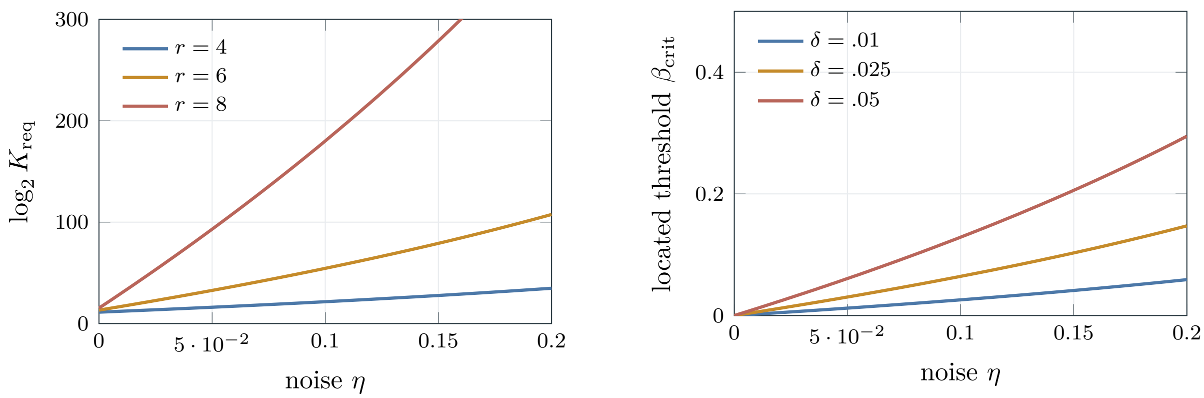

Figure 129: Noise dependence of the certified BKW resources. Left: the exact terminal-row threshold from Eq. (1426) for $k = 16$, $G = 16$, and $\epsilon = .01$. Right: the located-spectrum boundary from Eq. (1401) at $R = .25$. Increasing the number of reductions raises the combination order and therefore amplifies the noise penalty explicitly.

Reproducible parameter grid. The canonical evaluator `scripts/lpn_bkw_frontier.py` enumerates balanced integer block schedules, retains the nondominated resource vectors, and writes the complete grid to `docs/source/01_task_space/outputs/lpn_bkw_frontier.csv`. The following rows use unit implementation constants, target trace failure 10^{-2}, $m = 4n$, terminal sizes $k \in \{1, 2, 4, 8, 16\}$, and at most twelve stages. Each displayed row is the minimum-time streaming point among the retained schedules at its (n, m, η); logarithms are base two.

Corollary VIII.13.6 supplies direct checks on the subset-elimination, located-spectral, and threshold-guided regimes. Figure 131 summarizes the common qualified-record dependency of their controlled structural coordinates.

VIII.14 Readout-Agnostic Capacity, Information, and Blackwell Envelopes

The next three bounds remove the syntactic form of a represented readout from the numerical comparison. They apply, respectively, to the invariant law of one selected kernel, the ordered reveal channels of one pre-hit record, and the represented comparison of two experiments. Each conclusion retains its own hypotheses and complete cost; none is inferred from provenance alone.

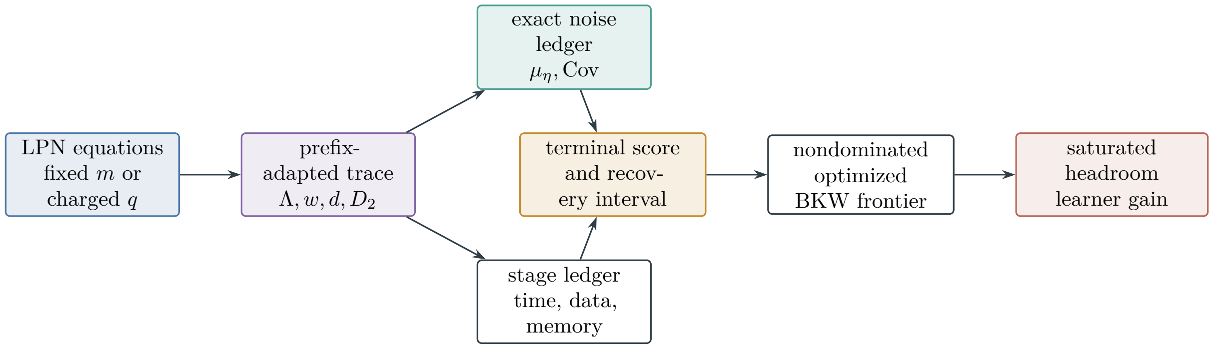

Figure 130: Exact resource-matched BKW comparison. The combination trace keeps sample reuse and its covariance visible; the stage record charges construction and storage; optimization retains the nondominated fixed-data and streaming frontiers; and the saturated comparison reports headroom and achieved learner gain separately.

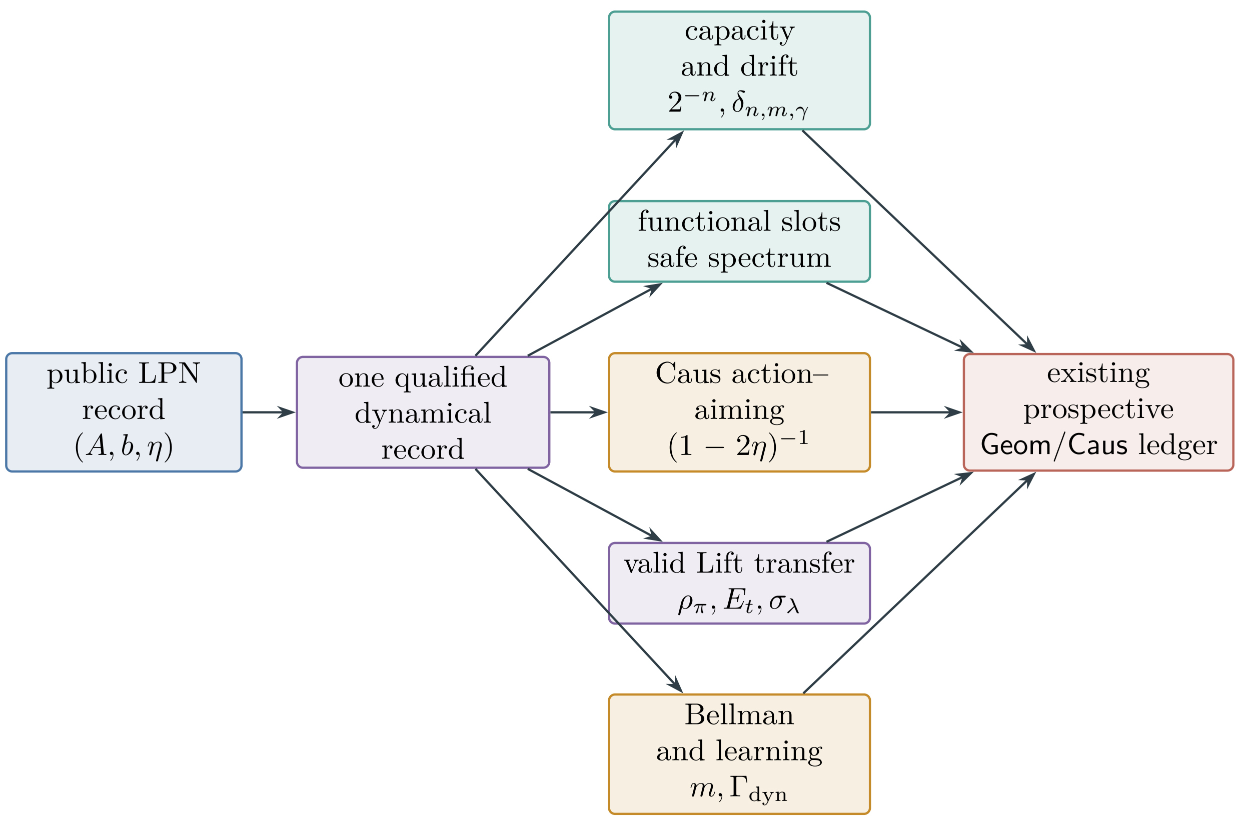

Figure 131: The LPN nonlinear ambient branch uses one qualified record. Capacity and drift, the typed functional profile, Caus action, valid-Lift transfer, and dynamical learning are numerical outputs of that same record. Their explicit n, m, η dependence returns to the existing prospective Geom/Caus ledger.

VIII.14.1 Invariant-law capacity and represented tilts

Theorem VIII.14.1 (LPN invariant-density capacity and drift envelope)**.** *Fix the uniform law μ_I on $X_I = \mathbb{F}_2^n$. Let P_0 be a complete represented reversible reference kernel with invariant law μ_I, and put*

$$q_{0,s} = 1 - P_0(s, s).$$

Let $(\widetilde{P}, \widetilde{\mu}, \Phi_{\widetilde{P}})$ be a complete represented reversible selected record on the same carrier. Suppose that the record certifies the pointwise comparisons

$$r_- 2^{-n} \le \widetilde{\mu}(x) \le r_+ 2^{-n}, \qquad x \in X_I, \tag{1442}$$

$$\kappa_- 2^{-n} P_0(x, z) \le \widetilde{\mu}(x)\widetilde{P}(x, z) \le \kappa_+ 2^{-n} P_0(x, z), \qquad x \ne z. \tag{1443}$$

Then

$$\widetilde{\mu}(s) \le r_+ 2^{-n}, \qquad\qquad \mathrm{Cap}_{\Phi_{\widetilde{P}}}(\{s\}, X_I \setminus \{s\}) \le \kappa_+ 2^{-n} q_{0,s}. \tag{1444}$$

For every horizon H and initial law $\rho_0 \le R_{\widetilde{\mu}}\widetilde{\mu}$,

$$\Pr_{\rho_0}^{\widetilde{P}}\{\tau_s \le H\} \le \min\Big\{1,\ R_{\widetilde{\mu}}(r_+ + H\kappa_+ q_{0,s})2^{-n}\Big\}. \tag{1445}$$

If instead $\rho_0 \le R_\mu \mu_I$, the same conclusion holds with $R_{\widetilde{\mu}}$ replaced by R_μ / r_-. Finally, if a nonconstant represented potential $F \ge 0$, with $F(s) = 0$, has global selected-kernel drift margin

$$a_s(\widetilde{P}, F) = \inf_{x \ne s} (I - \widetilde{P})F(x) > 0,$$

then, whenever $r_+ 2^{-n} < 1$,

$$\frac{a_s(\widetilde{P}, F)}{\mathrm{Osc}(F)} \le \frac{\kappa_+ 2^{-n} q_{0,s}}{1 - r_+ 2^{-n}}. \tag{1446}$$

Thus a nonlinear represented readout is absorbed into the selected invariant law and conductance comparison: its syntactic form does not occur in the bound. The density, conductance, initialization, and potential certificates, including their evaluation precision and complete cost, remain in the same prospective dynamical record.

Proof. The reference singleton capacity is $2^{-n} q_{0,s}$ by Eq. (1298). Apply Eqs. (493) and (495) to obtain Eq. (1444). The L^∞ and base-density cases of Theorem V.6.9 give Eq. (1445) and its stated variant. Apply Eq. (490) to $(\widetilde{P}, \widetilde{\mu})$, then use Eq. (1444); this proves Eq. (1446).

□

Corollary VIII.14.2 (Noise-indexed LPN tilt and mixing tradeoff)**.** *Retain the reference kernel P_0 of Theorem VIII.14.1. For any represented bounded potential D and inverse temperature $\theta \ge 0$, let $P_{\theta,D}$ be its represented Gibbs–Metropolis tilt from Corollary V.6.10. If $\rho_0 \le R_\mu \mu_I$, then*

$$\Pr_{\rho_0}^{P_{\theta,D}}\{\tau_s \le H\} \le \min\Big\{1,\ R_\mu e^{2\theta\,\mathrm{Osc}(D)}(1 + H q_{0,s})2^{-n}\Big\}. \tag{1447}$$

If $D(s) = \min_{x \in X_I} D(x)$, then

$$\Pr_{\rho_0}^{P_{\theta,D}} \{\tau_s \le H\} \le \min\Big\{1,\ R_\mu e^{\theta \operatorname{Osc}(D)} (1 + H q_{0,s}) 2^{-n}\Big\}. \tag{1448}$$

In particular, if $\log R_\mu + \log(H+1) + 2\theta \operatorname{Osc}(D) = o(n)$, the right side of Eq. (1447) is $2^{-n+o(n)}$. Under the target-minimum hypothesis, the factor two may be omitted from this condition by Eq. (1448). These conclusions use the evaluated oscillation of the represented potential; affordability alone is not substituted for that number.

For the residual Metropolis record $P_\theta = P_{\theta,W_I}$ of Corollary VIII.12.5, work on $\mathcal{F}_\delta$, put

$$\Delta_- = \frac{1}{2} - \eta - 2\delta > 0, \qquad \Delta_+ = \frac{1}{2} - \eta + 2\delta,$$

and fix $0 < \varepsilon < 1$. In the subcritical-tilt regime

$$\theta m \Delta_+ \le (1-\varepsilon) n \log 2, \tag{1449}$$

one has

$$\pi_\theta(s) \le \frac{2^{-\varepsilon n}}{1 - 2^{-n}}, \tag{1450}$$

$$\Pr_{\rho_0}^{P_\theta}\{\tau_s \le H\} \le \min\left\{1,\ \frac{R_\theta 2^{-\varepsilon n}}{1 - 2^{-n}} \left(1 + \frac{H}{2} e^{-\theta m \Delta_-}\right)\right\} \quad \text{if } \rho_0 \le R_\theta \pi_\theta. \tag{1451}$$

In the supercritical-concentration regime

$$\theta m \Delta_- \ge (1+\varepsilon) n \log 2, \tag{1452}$$

the same record satisfies

$$\pi_\theta(s) \ge \frac{1}{1 + 2^{-\varepsilon n}}, \tag{1453}$$

$$q_{\theta,s} \le \frac{1}{2}\, 2^{-(1+\varepsilon)n}, \qquad \operatorname{Cap}_\theta(\{s\}, X_I \setminus \{s\}) \le \frac{1}{2}\, 2^{-(1+\varepsilon)n}. \tag{1454}$$

Moreover, if $4\theta\delta m \le (1-\zeta) n \log 2$ for some $0 < \zeta < 1$, then

$$\gamma_\theta \le \frac{2^{-\zeta n}}{1 - 2^{-n}}. \tag{1455}$$

Equations Eqs. (1449) and (1452) locate the transition at the explicit scale $\theta m (1/2 - \eta) \asymp n \log 2$, with its finite-sample uncertainty carried by $2\delta m$.

Proof. Apply Eq. (506) with $\mu_I(s) = 2^{-n}$ and $\operatorname{Cap}_{\Phi_{P_0}}(\{s\}, X_I \setminus \{s\}) = 2^{-n} q_{0,s}$ to obtain Eq. (1447). Under $D(s) = \min D$, apply instead Eq. (507) to obtain Eq. (1448).

Under Eq. (1449), the upper target mass bound in Eq. (1365) gives

$$\pi_\theta(s) \le \frac{1}{(2^n - 1) e^{-\theta m \Delta_+}} \le \frac{2^{-\varepsilon n}}{1 - 2^{-n}},$$

which proves Eq. (1450). The stationary entrance-count estimate Eq. (484), the initial-density comparison, and Eq. (1366) give

$$\Pr_{\rho_0}^{P_\theta}\{\tau_s \le H\} \le R_\theta \pi_\theta(s) \left(1 + \frac{H}{2} e^{-\theta m \Delta_-}\right),$$

which proves Eq. (1451).

Under Eq. (1452), the lower target mass bound in Eq. (1365) gives Eq. (1453); Eq. (1366) gives Eq. (1454). The mass bound is strictly greater than one half. Therefore the second branch of Eq. (1367), followed by $e^{4\theta\delta m} \le 2^{(1-\zeta)n}$, proves Eq. (1455).

□

Corollary VIII.14.3 (Fast-fiber and clocked-defect access for LPN)**.** *Fix one complete valid-Lift record over the LPN target $T_I = \{s\}$, and let its base selected kernel satisfy Theorem VIII.14.1. Define the base finite-horizon capacity envelope*

$$U_{X,H}^{\mathrm{Cap}} = \min\Big\{1,\, R_{\widetilde{\mu}}(r_+ + H\kappa_+ q_{0,s})2^{-n}\Big\}. \tag{1456}$$

The two certificate branches of Theorem V.7.14 specialize as follows.

(i) *If the LPN lift has the represented fast-fiber data $(M_F, \kappa_F, \varepsilon_+, \varepsilon_-, \lambda)$ and satisfies the smallness condition in Eq.* (531)*, then*

$$A_{\{s\}}^{\Omega}(\lambda) \le A_{\{s\}}^{X}(\lambda) + \varepsilon_{0,s}^{\mathrm{int}} + \varepsilon_{\Omega,\lambda}^{\mathrm{int}} + \varepsilon_{X,\lambda}^{\mathrm{int}} + \|\ell\|_* \|g\| \Delta_{\mathrm{fm}}(\lambda), \tag{1457}$$

where $\Delta_{\mathrm{fm}}(\lambda)$ is the explicit rational expression in Eq. (531)*. Any evaluated LPN base-access bound for $A_{\{s\}}^{X}(\lambda)$ may be substituted in the same discounted-arrival units.*

(ii) *For represented stopped or absorbing clock slices, no fiber-mixing hypothesis is required. With clocked defects $E_t = P_t^{\Omega} U_{t+1} - U_t P_t^{X}$,*

$$p_{\{s\},H}^{\Omega} \le \min\left\{1,\, U_{X,H}^{\mathrm{Cap}} + \varepsilon_{\mathrm{init}} + \sum_{t=0}^{H-1} \|E_t\| + \varepsilon_{\mathrm{gate}}\right\}. \tag{1458}$$

For the residual Metropolis base record, the sharper subcritical expression in Eq. (1451) *replaces $U_{X,H}^{\mathrm{Cap}}$. It depends on (n, m, η) through $2^{-\varepsilon n}$, $\Delta_- = 1/2 - \eta - 2\delta$, and the certified initialization density.*

When both branches are certified, they are retained separately. They may be minimized only after the same record supplies the represented conversion between discounted arrival and the selected finite horizon. Exact valid-Lift initialization, target pullback, and intertwining set the corresponding interface and defect terms to zero.

Proof. Part (i) is Eq. (533) with the LPN singleton target. For part (ii), combine Eq. (534) with the base access bound Eq. (1445). The residual Metropolis substitution is Eq. (1451). The unit-conversion and exact- interface statements are the final clauses of Theorem V.7.14.

□

VIII.14.2 Sequential information contraction for the LPN presentation

Theorem VIII.14.4 (LPN pre-hit information and target-contact envelope)**.** *Let S be the uniform secret in the random binary LPN presentation of Definition VIII.1.1, and take the initial public record to be $\mathsf{Z}_0 = (A, b)$. Consider one complete temporally admissible prefix before a represented* *next-candidate selector is sampled. Enumerate its target-dependent reveals as in Theorem V.3.3, and retain the corresponding contracted increments $d_1, \ldots, d_L$. Define*

$$J^{\mathrm{LPN}}_{n,m,\eta,t} = \min\{n,\, m(1 - H_2(\eta))\} + \sum_{j=1}^{L} d_j. \tag{1459}$$

Then

$$I(S; \mathcal{H}_t) \le J^{\mathrm{LPN}}_{n,m,\eta,t}. \tag{1460}$$

Let $K_t(\cdot \mid \mathcal{H}_t)$ be the represented next-candidate kernel, let C_{t+1} be its sampled candidate, and put

$$p^{\mathrm{contact}}_t = \Pr\{\tau_s > t,\ C_{t+1} = S\}, \qquad \tau_s = \inf\{t \ge 1 : C_t = S\}.$$

The exact-target sectors are disjoint and have overlap one. Hence

$$p^{\mathrm{contact}}_t \le \min\left\{1,\ 2^{-n} + \sqrt{\frac{\log 2}{2} J^{\mathrm{LPN}}_{n,m,\eta,t}}\right\}. \tag{1461}$$

For a horizon H, with the ordered reveal ledger recomputed at each prefix,

$$\Pr\{\tau_s \le H\} \le \min\left\{1,\ \sum_{t=0}^{H-1} \left(2^{-n} + \sqrt{\frac{\log 2}{2} J^{\mathrm{LPN}}_{n,m,\eta,t}}\right)\right\}. \tag{1462}$$

For the terminal branch, let $\mathcal{R}^{\mathrm{full}}_H$ index every target-dependent reveal after (A, b) through the complete terminal transcript, including any verifier, hit-selection, feedback, reward, or oracle coordinate, and let $c_{q,r}$ be its represented conditional capacity certificate. Define

$$J^{\mathrm{LPN,full}}_{n,m,\eta,H} = \min\{n,\, m(1 - H_2(\eta))\} + \sum_{(q,r) \in \mathcal{R}^{\mathrm{full}}_H} c_{q,r}. \tag{1463}$$

If the terminal record contains the candidate reached at an exact target contact and all coordinates used to identify that contact occur in this full ledger, the represented-decoder branch gives

$$\Pr\{\tau_s \le H\} \le \min\left\{1,\ \frac{J^{\mathrm{LPN,full}}_{n,m,\eta,H} + 1}{n}\right\}. \tag{1464}$$

When no target-dependent reveal is adjoined after (A, b), every retained coordinate is a represented postprocessing of the public LPN experiment and the sum in Eq. (1459) is zero. In that case an exact-secret output obeys, for every $0 < \epsilon < \eta < 1/2$, the sharper strong converse

$$\Pr\{\widehat{S} = S\} \le \min\Big\{1,\ 2e^{-2m\epsilon^2} + 2^{m-n-m(H_2(\eta) - \epsilon L_\eta)}\Big\}, \qquad L_\eta = \log_2 \frac{1-\eta}{\eta}. \tag{1465}$$

Every post-public verifier flag, feedback value, reward, or admissible oracle answer used before the selector occupies an ordered reveal slot. Its query, channel, contraction certificate, returned value, memory, precision, and complete evaluation cost are charged in the same prefix. Public postprocessing and fresh target-independent randomness contribute no new information slot.

Proof. Because A is independent of S, the binary symmetric channel bound in the proof of Theorem VIII.7.3 gives

$$I(S; A, b) = I(S; b \mid A) \leq \min\{n,\, m(1 - H_2(\eta))\}\,.$$

Apply Eq. (413) to the ordered reveals. This proves Eq. (1460).

Use the analytic target relation $V_s(x) = \mathbf{1}_{\{x=s\}}$. For every candidate x, exactly one of the 2^n target indices verifies it, so the overlap coordinate in Eq. (424) is $\Lambda_t = 1$. Substitution in Eq. (425) proves Eq. (1461). The exact first-contact events are disjoint; Eq. (430) therefore gives Eq. (1462). Retaining the target-contact candidate gives the identification map required in Eq. (432). The binary-symmetric-channel bound for $I(S; A, b)$, followed by Theorem V.3.2 on every slot in $\mathcal{R}_H^{\mathrm{full}}$, gives

$$I(S; \mathsf{Z}_H^{\mathrm{full}}) \leq J_{n,m,\eta,H}^{\mathrm{LPN,full}}.$$

Substitution in Eq. (432) proves Eq. (1464).

If there is no later target-dependent reveal, the complete output is a postprocessing of (A, b) and target-independent randomness. Data processing followed by Eq. (1207) proves Eq. (1465). The final charging statement is the LPN specialization of the temporal and oracle clauses in Theorem V.3.3.

□

VIII.14.3 Blackwell envelopes for LPN pre-hit readouts

The represented Blackwell order compares LPN readouts by causal simulation. It transfers a numerical target-decision bound only in the direction from a reference experiment to its represented garblings. The following specialization records the three reference experiments used below.

Theorem VIII.14.5 (LPN target-decision bounds under represented garbling)**.** *Let $u_{\mathrm{id}}(s, x) = \mathbf{1}_{\{x=s\}}$ be the exact-secret decision utility of Definition V.4.4, and let $B^\sharp = \psi_{\mathrm{BW}}(B, n, m, H)$ be the complete cost enlargement of Theorem V.4.5. The following statements hold for the random binary LPN presentation of Definition VIII.1.1.*

(i) *Let E_0 be a target-permutation-neutral, residual-independent pre-hit reference experiment. Its time-t decision is made on the survival sector and can select one remaining candidate. If E is a represented causal garbling of E_0, then*

$$\mathrm{DVal}_{u_{\mathrm{id}},t}^{\mathrm{sat}}(\mathsf{E}; B) \leq \mathrm{DVal}_{u_{\mathrm{id}},t}^{\mathrm{sat}}(\mathsf{E}_0; B^\sharp) \leq 2^{-n}. \tag{1466}$$

(ii) *Work conditionally on a matrix A of full column rank. Let $S(A)$ be the represented deterministic pivot-row set from Corollary VIII.13.6, and define the pivot experiment*

$$\mathsf{E}_{\mathrm{piv}} = (A, S(A), b_{S(A)}). \tag{1467}$$

Before any target-dependent verifier outcome is adjoined, its Bayes exact-secret decision value is

$$\sup_K \mathbb{E}\left[\int u_{\mathrm{id}}(s, x) K(dx \mid \mathsf{E}_{\mathrm{piv}}) \,\middle|\, A, \operatorname{rank} A = n\right] = (1 - \eta)^n. \tag{1468}$$

Here the supremum ranges over all probability kernels, so it is also an upper bound for every saturated budget. The represented rule $x = A_{S(A)}^{-1} b_{S(A)}$ attains equality once the charged ceiling contains pivot extraction and binary elimination, of complete cost

$$B_{\mathrm{piv}}(n, m) \preceq mn^2 + n^3. \tag{1469}$$

Consequently, every represented causal garbling E *of* $\mathsf{E}_{\mathrm{piv}}$ *satisfies*

$$\mathrm{DVal}^{\mathrm{sat}}_{u_{\mathrm{id}},0}(\mathsf{E}; B) \le (1-\eta)^n. \tag{1470}$$

If the pivot experiment is completed by a represented failure symbol when $\operatorname{rank} A < n$*, then every decision rule on the unconditional experiment satisfies*

$$\mathrm{DVal}^{\mathrm{sat}}_{u_{\mathrm{id}},0}(\mathsf{E}; B) \le \min\{1,\, 2^{n-m} + (1-\eta)^n\}\,. \tag{1471}$$

(iii) *Let* $\mathsf{E}_{\mathrm{pub}} = (A, b)$ *be the complete public LPN experiment. Every represented causal garbling* E *of* $\mathsf{E}_{\mathrm{pub}}$*, including its target-independent random bits and public postprocessing, obeys the finite Fano inequality*

$$H_2(1-p_B) + (1-p_B)\log_2(2^n - 1) \ge n - m(1 - H_2(\eta)), \tag{1472}$$

where $p_B = \mathrm{DVal}^{\mathrm{sat}}_{u_{\mathrm{id}},0}(\mathsf{E}; B)$*. For every* $0 < \epsilon < \eta < 1/2$*, the corresponding explicit bound is*

$$p_B \le \min\Big\{1, 2e^{-2m\epsilon^2} + 2^{m-n-m(H_2(\eta)-\epsilon L_\eta)}\Big\}, \qquad L_\eta = \log_2 \frac{1-\eta}{\eta}. \tag{1473}$$

All three comparisons retain the complete derivation and garbling costs. Thus they apply at the one charged enlarged ceiling rather than adjoining a free readout.

Proof. For (i), condition on the complete residual-independent pre-hit record and on survival. By Lemma VIII.8.11, the secret is uniform on the remaining untested candidates. If N_t candidates remain, the conditional best exact decision value is N_t^{-1}, while the survival probability is at most $N_t/2^n$. Their product is at most 2^{-n}. Apply Theorem V.4.5 to every represented causal garbling and charge its simulation at $B^\sharp$. This proves Eq. (1466).

For (ii), put $A_S = A_{S(A)}$ and $Y = b_{S(A)}$. Conditional on A and full column rank,

$$Y = A_S s + E_S, \qquad E_S \sim \mathrm{Bernoulli}(\eta)^{\otimes n},$$

and A_S is invertible. Since s is uniform, Bayes' formula gives, for every $x, y \in \mathbb{F}_2^n$,

$$\Pr\{s = x \mid A, Y = y, \operatorname{rank} A = n\} = (1-\eta)^{n-|y+A_S x|_1}\eta^{|y+A_S x|_1}. \tag{1474}$$

For $0 \le \eta < 1/2$, its maximum over x is $(1-\eta)^n$, attained uniquely at $x = A_S^{-1}y$; at $\eta = 1/2$, every posterior atom equals $2^{-n} = (1-\eta)^n$. Integrating the largest posterior atom proves Eq. (1468). Pivot extraction scans the $m \times n$ matrix and binary elimination uses cubic arithmetic cost, which gives Eq. (1469). Equality of the represented rule is also the exact calculation in Eq. (1396). Blackwell monotonicity now proves Eq. (1470). On rank failure, bound the decision utility by one; the rank-failure estimate used in Eq. (1397) is at most 2^{n-m}. Adding the two events proves Eq. (1471).

For (iii), compose a decision rule for E with the represented causal garbling from (A, b). The result is a randomized decoder of the complete LPN sample, with exactly the same exact-recovery probability by Eq. (451). Apply Eqs. (1205) and (1207) from Theorem VIII.7.3. The first application gives Eq. (1472); the second gives Eq. (1473). Data processing is already quantified over every measurable postprocessing, so the conclusion is independent of the represented garbling's syntactic form.

□

Corollary VIII.14.6 (Direction of the LPN Blackwell comparisons)**.** *The pivot experiment is a represented statistic of the public experiment:*

$$\mathsf{E}_{\mathrm{pub}} \longrightarrow \mathsf{E}_{\mathrm{piv}}, \qquad (A,b) \longmapsto (A, S(A), b_{S(A)}). \tag{1475}$$

Hence

$$\mathrm{DVal}^{\mathrm{sat}}_{u_{\mathrm{id}},0}(\mathsf{E}_{\mathrm{piv}}; B) \le \mathrm{DVal}^{\mathrm{sat}}_{u_{\mathrm{id}},0}(\mathsf{E}_{\mathrm{pub}}; \psi_{\mathrm{BW}}(B,n,m,0)). \tag{1476}$$

An upper bound on the full public experiment passes back from $\mathsf{E}_{\mathrm{piv}}$ whenever an affordable represented pre-hit reconstruction kernel satisfies Eq. (453)*; in that case Corollary V.4.6 supplies the reverse comparison. A separate direct public-record bound gives the other valid route. In the absence of either route, the pivot value $(1-\eta)^n$ bounds the pivot experiment and its garblings, whereas the full public experiment and its garblings are bounded by Eqs.* (1472) *and* (1473)*. The complete-record bounds therefore apply directly to decision rules using (A,b).*

Proof. The displayed map is represented before the decision and has polynomial complete cost. Apply Theorem V.4.5 in the displayed direction to obtain Eq. (1476). The affordable reconstruction criterion of Corollary V.4.6 supplies the stated sufficient reverse comparison. The remaining bounds are Theorem VIII.14.5.

□

Corollary VIII.14.7 (Recordwise readout-agnostic LPN envelope)**.** *Fix one complete prospective LPN record, one target-arrival event $E_H = \{\tau_s \le H\}$, and one common probability law $\mathbb{P}_{\mathcal{R}}$. The law may condition on a fixed presented instance or may include the random LPN presentation, but every candidate below is first expressed under this same law.*

A pointwise bound $\Pr(E_H \mid I) \le U(I)$ contributes $\mathbb{E}_{\mathbb{P}_{\mathcal{R}}} U(I)$. If it is certified only on a presentation event $\mathcal{G}$, it contributes

$$\mathbb{P}_{\mathcal{R}}(\mathcal{G}^c) + \mathbb{E}_{\mathbb{P}_{\mathcal{R}}}\left[\mathbf{1}_{\mathcal{G}} U(I)\right]. \tag{1477}$$

Thus the residual-Metropolis bound retains the explicit failure probability of $\mathcal{F}_\delta$, and the conditional pivot bound retains either the full-rank conditioning or the rank-failure term in Eq. (1471)*. Denote the resulting common-law candidates as follows:*

- *$\overline{U}_{\mathrm{cap}}$ is the converted applicable right side of Eq.* (1445) *or Eq.* (1451)*;*
- *$\overline{U}_{\mathrm{lift}}$ is the converted right side of Eq.* (1458) *when the arrival occurs on a valid lifted carrier;*
- *$\overline{U}_{\mathrm{info}}$ is the smaller applicable right side of Eqs.* (1462) *and* (1464)*;*
- *$\overline{U}_{\mathrm{cur}}$ is the applicable same-record minimum in Eq.* (565) *for a complete sparse-reward observation record, with $\mathsf{Z}_0 = (A,b)$, $\Theta = S$, and target $T_S = \{S\}$;*
- *$\overline{U}_{\mathrm{BW}}$ is available when either the complete terminal experiment, including every target-dependent reveal through time H, is a represented causal garbling of a bounded reference experiment and a represented terminal map converts target arrival into its decision utility, or the survival-weighted Blackwell decision bounds are summed over the disjoint first-contact times;*
- *$\overline{U}_{\mathsf{Caus}}$ is a converted contact bound obtained from the same selected record through Eq.* (1328) *or a represented aiming-to-contact converter as in Corollary VIII.11.3.*

Set an unavailable candidate equal to the neutral value one. A candidate is available only when its hypotheses, interface, units, temporal ancestry, and complete-cost certificate are present in this record and it bounds $\mathbb{P}_{\mathcal{R}}(E_H)$. Then

$$\mathbb{P}_{\mathcal{R}}\{\tau_s \le H\} \le \min\{\overline{U}_{\mathrm{cap}}, \overline{U}_{\mathrm{lift}}, \overline{U}_{\mathrm{info}}, \overline{U}_{\mathrm{cur}}, \overline{U}_{\mathrm{BW}}, \overline{U}_{\mathsf{Caus}}\}. \tag{1478}$$

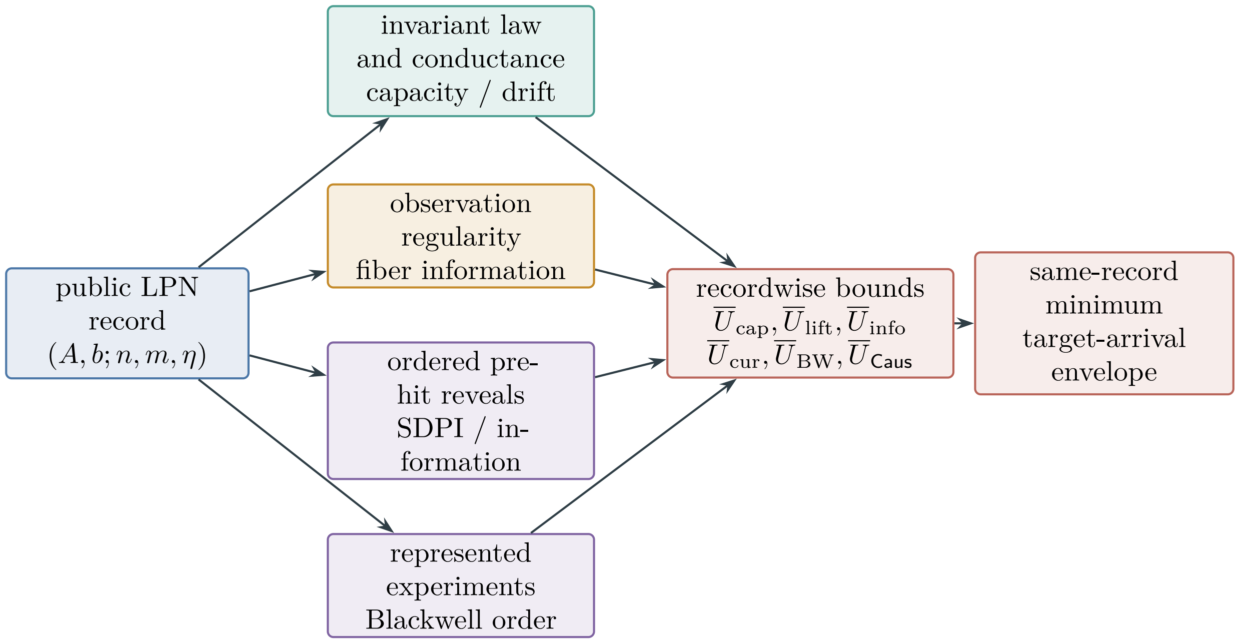

Figure 132: Four recordwise routes for LPN. Capacity sees the selected invariant law and conductance, sparse-reward accounting sees observation regularity and within-cell information, SDPI sees the ordered reveal channels, and Blackwell order sees represented experiment degradation. Each route produces a number for one complete record and one common probability law; only then are applicable bounds minimized and passed to the existing target-arrival ledger.

The minimum is taken before saturation. Saturation then takes the supremum over complete records at the common charged ceiling. In particular, no capacity, information, sparse-reward observation, Blackwell, Lift, or Caus certificate from one record is combined with a coordinate from another.

Proof. The pointwise and good-event conversions follow from the tower property and the decomposition of E_H over $\mathcal{G}$ and $\mathcal{G}^c$. For the second Blackwell route, sum the survival-weighted decision values over the disjoint first-contact times. After these conversions, each available candidate is an upper bound for the same scalar probability under $\mathbb{P}_{\mathcal{R}}$, so their minimum is an upper bound. The neutral-default and order-of-operations statements are exactly the coordinatewise convention of Definition V.12.2.

□

Corollary VIII.14.8 (Saturated quantitative prospective certificate envelope for LPN)**.** *Fix $n, m \geq 1$, $0 < \eta < 1/2$, budget B, and refined horizon H. Put $B^{\text{cert}}_{\text{LPN}} = \Psi_{\text{cert}}(B, H, n, m)$. For a complete pre-hit selector record $R \in \mathscr{R}_{I,t}(B^{\text{cert}}_{\text{LPN}})$, set $\sigma_{R,t} = \Pr_R\{\tau_s > t\}$. Whenever the corresponding same-record certificates are present, define*

$$C^{\text{cap}}_{R,t} = \sigma_{R,t} \min\Big\{1,\ \|f_{R,t}\|_{L^p(\widetilde{\mu}_R)} \left[\min\{1,\ 2^{-n}(r_{R,+} + \kappa_{R,+} q_{0,R,s})\}\right]^{1/q}\Big\}, \tag{1479}$$

$$C^{\text{info}}_{R,t} = \min\left\{1,\ 2^{-n} + \sqrt{\frac{\log 2}{2} J^{\text{LPN}}_{n,m,\eta,t}(R)}\right\}, \tag{1480}$$

$$J^{\mathrm{LPN}}_{n,m,\eta,t}(R) = \min\{n, m(1-H_2(\eta))\} + \sum_{j=1}^{L_{R,t}} d_{R,t,j}, \tag{1481}$$

where $q_{0,R,s} = 1 - P_{0,R}(s,s)$, $f_{R,t} = d\mathcal{L}_R(X_t \mid \tau_s > t)/d\widetilde{\mu}_R$, $1 < p \le \infty$, and $q = p/(p-1)$.

For a complete sparse-reward observation record, let $C_{\varepsilon,R} = q_{\varepsilon,R}(s)$, let $\beta^{\mathrm{fib}}_{R,t}$ be the decoupled posterior contact in Eq. (558) with $\mathsf{Z}_0 = (A,b)$, and let $J^{\mathrm{fib},\uparrow}_{R,t}$ be the charged ordered-reveal upper bound on

$$I\Big(S; \mathcal{H}^{\mathrm{obs}}_{R,t} \mid A, b, C_{\varepsilon,R}\Big).$$

Set

$$C^{\mathrm{cur}}_{R,t} = \min\left\{1, \beta^{\mathrm{fib}}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{fib},\uparrow}_{R,t}}\right\}. \tag{1482}$$

This candidate has value one when the complete observation record or its information comparison is unavailable.

For a complete master active-sampling record, let $\beta^{\mathrm{act}}_{R,t}$ and $J^{\mathrm{act},\uparrow}_{R,t}$ be the decoupled next-contact and ordered acquisition-information coordinates of Theorem V.9.7. Set

$$C^{\mathrm{act}}_{R,t} = \min\left\{1, \beta^{\mathrm{act}}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{act},\uparrow}_{R,t}}\right\}, \tag{1483}$$

with neutral value one when that complete record is unavailable.

Let $C^{\mathrm{learn}}_{R,t}$ be the complete LPN specialization of Eq. (281):

$$C^{\mathrm{learn}}_{R,t} = \min\left\{1,\, U^{\mathrm{stat}\to\mathrm{tgt}}_{R,t},\, \beta^{+}_{R,t} + \sqrt{\frac{\log 2}{2} \sum_{a\in\mathfrak{I}^5_{\mathsf{Abs}}} J^{(a)}_{I,R_{\le t}}},\, eH_B(n) \sum_{a\in\mathfrak{I}^5_{\mathsf{Abs}}} \mathcal{L}^{(a)}_{I,R_{\le t}},\, U^{\mathrm{dyn}\to\mathrm{tgt}}_{R,t}\right\}. \tag{1484}$$

For an LPN binary reveal, every summand uses $\vartheta_j(\eta) = (1-2\eta)^2$. The common ceiling charges the public record, learner, locator, optimizer, posterior or memory state, selected kernel, Caus/Lift processing, verifier, and deployment converter.

Let $C^{\mathrm{BW}}_{R,t}$ be the smallest applicable bound among

$$2^{-n}, \qquad \min\{1, 2^{n-m} + (1-\eta)^n\}, \qquad \min\Big\{1,\, 2e^{-2m\epsilon^2} + 2^{m-n-m(H_2(\eta)-\epsilon L_\eta)}\Big\}, \tag{1485}$$

where $0 < \epsilon < \eta$ and $L_\eta = \log_2((1-\eta)/\eta)$, with applicability determined by Theorem VIII.14.5. Let $C^{\mathrm{src}}_{R,t}$ be the smallest applicable same-record comparison

$$\min\{1, K_R(2^{-n} + (1-\eta)^n) + \delta_R\}, \qquad \min\{1, K^{\mathrm{dir}}_R V_{\mathscr{C}^{\mathrm{dir}}_R, I} + \delta^{\mathrm{dir}}_R\}. \tag{1486}$$

The first requires Eq. (474) and Certification Target VIII.20.5; the second requires Eq. (475). Finally, let $C^{\mathrm{dyn}}_{R,t}$ be the smallest applicable same-record next-contact bound in Corollaries VIII.14.3, VIII.14.7 and VIII.11.7. Set every unavailable candidate to one and put

$$C^{\mathrm{LPN}}_{R,t} = \min\{C^{\mathrm{cap}}_{R,t}, C^{\mathrm{info}}_{R,t}, C^{\mathrm{BW}}_{R,t}, C^{\mathrm{src}}_{R,t}, C^{\mathrm{dyn}}_{R,t}, C^{\mathrm{cur}}_{R,t}, C^{\mathrm{act}}_{R,t}, C^{\mathrm{learn}}_{R,t}\}.$$

Then

$$\mathrm{Sel}^{\mathrm{sat}}_{I,n,\eta}(B) \le \max_{0\le t<H} \sup_{R\in\mathscr{R}_{I,t}(B^{\mathrm{cert}}_{\mathrm{LPN}})} C^{\mathrm{LPN}}_{R,t}, \tag{1487}$$

$$\mathrm{Succ}_{I,n,\eta}(B) \le \min\left\{1,\ H2^{-n} + \sum_{t=0}^{H-1} \sup_{R\in\mathscr{R}_{I,t}(B^{\mathrm{cert}}_{\mathrm{LPN}})} C^{\mathrm{LPN}}_{R,t}\right\}. \tag{1488}$$

Every term retains its full (n, m, η, H, B)-dependence and the applicable density, conductance, SDPI, observation-resolution, within-cell information, Blackwell, Caus, Lift, learning, and good-presentation parameters. If the recordwise envelope is at most $\varepsilon_{n,m,\eta,B,H}$, then

$$\mathrm{Sel}^{\mathrm{sat}}_{I,n,\eta}(B) \le \varepsilon_{n,m,\eta,B,H}, \qquad \mathrm{Succ}_{I,n,\eta}(B) \le \min\{1, H(2^{-n} + \varepsilon_{n,m,\eta,B,H})\}.$$

Proof. For the uniform LPN carrier, $\mu_I(\{s\}) = 2^{-n}$ and the reference singleton capacity is $2^{-n}q_{0,R,s}$. Thus the capacity candidate is Eq. (730). The information, Blackwell, and source candidates follow from Eqs. (1461) and (1647) and Theorems V.6.4 and VIII.14.5. The sparse-reward observation candidate is Eq. (559) with the LPN target index and initial public record substituted as displayed. The active-acquisition candidate is Eq. (587) on the same complete prefix. The source-resolved learning candidate is Theorem IV.13.3 with the LPN binary-noise coefficient and complete LPN record substituted. Apply Theorem V.12.4. The uniform neutral next-test baseline contributes at most 2^{-n} per refined step. Summation gives the success bound.

□

VIII.14.4 Sparse-reward observation and transport accounting for LPN

Theorem VIII.14.9 (Complete sparse-reward observation accounting for LPN)**.** *Fix $0 < \eta < \tau < 1/2$, $\delta > 0$, and a budget in the declared polynomial saturated ceiling. Work on the intersection of $\mathcal{G}_{\tau,\delta}$ with the full-rank and random-code-rigidity event of Lemma VIII.17.5. Let S be the uniform secret, set $X_I = \mathbb{F}_2^n$, $T_S = \{S\}$, and take $\mathsf{Z}_0 = (A, b)$. Let $\mathcal{R}^{\mathrm{LPN}}_{\mathrm{cur}}$ be a complete record from Definition V.8.5 whose observation is a represented map*

$$\phi_I : \mathbb{F}_2^n \longrightarrow (\mathcal{O}, d_{\mathcal{O}})$$

derived from the public LPN presentation, and put $q_{\varepsilon,I} = \kappa_\varepsilon \circ \phi_I$ and $C_{\varepsilon,I} = q_{\varepsilon,I}(S)$. The public external reward is the verifier value revealed after a candidate is tested. On $\mathcal{G}_{\tau,\delta}$, it is zero throughout every surviving pre-hit prefix.

For the complete observation history before the next selector, define

$$J^{\mathrm{cur}}_{I,t} = I\Big(S; \mathcal{H}^{\mathrm{obs}}_{I,t} \mid A, b, C_{\varepsilon,I}\Big). \tag{1489}$$

If every later target-dependent coordinate is placed in the conditional ordered-reveal ledger, then

$$J^{\mathrm{cur}}_{I,t} \le J^{\mathrm{cur},\uparrow}_{I,t} = \sum_{j=1}^{L_{I,t}} \mathbb{E}\min\{c_{I,t,j}, \vartheta_{I,t,j}a_{I,t,j}\}. \tag{1490}$$

Public postprocessing of (A, b), intrinsic rewards, predictor updates, exploration memories, and fresh target-independent randomness add no term to this sum. Their construction and deployment remain in the complete cost

$$\begin{aligned} B^{\mathrm{LPN}}_{\mathrm{cur}} = B_{(A,b,\eta)} &\oplus B_{\phi_I} \oplus B_{d_{\mathcal{O}}} \oplus B_{\kappa,\varepsilon} \oplus B_{M_\varepsilon} \oplus B_{\mathrm{reg}} \oplus B_{\mathrm{info}} \\ &\oplus B_{\mathrm{mem}} \oplus B_{\mathrm{pred}} \oplus B_{\mathrm{train}} \oplus B_{r^{\mathrm{ext}}} \oplus B_{\mathrm{bonus}} \oplus B_{\mathrm{pol}} \\ &\oplus B_{\mathrm{exec}} \oplus B_{\mathrm{clock}} \oplus B_{\mathrm{verify}} \oplus B_{\mathrm{out}} \oplus B_{\mathrm{cap}} \oplus B_{\mathrm{comp}}. \end{aligned} \tag{1491}$$

Here the last four analytic ledgers are expanded in Eqs. (550) *to* (553)*.*

Let $\mathbb{Q}^{\mathrm{fib}}_{I,t}$ *retain* $(A, b, C_{\varepsilon,I}, \mathcal{H}^{\mathrm{obs}}_{I,t})$ *and resample* S *from* $\mathcal{L}(S \mid A, b, C_{\varepsilon,I})$*, independently of the observation history. Put*

$$\beta^{\mathrm{cur}}_{I,t} = \mathbb{Q}^{\mathrm{fib}}_{I,t}\{\tau_s > t,\ X_{t+1} = S\}. \tag{1492}$$

Write $R_{\le t}$ *for this complete curiosity record truncated immediately after its next test. Then the actual survival-weighted contact and selector advantage satisfy*

$$\left(\mathrm{Adv}_t(K_t; \nu_t)\right)_+ \le \Pr\{\tau_s > t,\ X_{t+1} = S\} \le \min\left\{1, \beta^{\mathrm{cur}}_{I,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{cur},\uparrow}_{I,t}}, eH_B(n) \sum_{a \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{L}^{(a)}_{I,R_{\le t}}\right\}. \tag{1493}$$

Consequently,

$$\Pr\{\tau_s \le H\} \le \min\left\{1, \sum_{t=0}^{H-1} \min\left\{1, \beta^{\mathrm{cur}}_{I,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{cur},\uparrow}_{I,t}}, eH_B(n) \sum_{a \in \mathfrak{J}^5_{\mathsf{Abs}}} \mathcal{L}^{(a)}_{I,R_{\le t}}\right\}\right\}. \tag{1494}$$

The master-profile assignment is exact. An exact qualified use of $q_{\varepsilon,I}$ *lies in the LPN exact quotient coordinates, and a nonzero-defect qualified use lies in one of the two compensated coordinates. These three coordinates vanish by Eqs.* (1602)*,* (1606) *and* (1607)*. Every remaining observation-driven transition is therefore an ambient full-carrier record, with no cell-dynamical credit. Its target-arrival coordinate is additionally bounded, whenever the corresponding same-record certificate is present, by*

$$\Pr_{\mu_0}\{\tau_s \le H\} \le R_0(H+1)2^{-n}, \tag{1495}$$

$$\Pr_{\rho_0}^{\widetilde{P}}\{\tau_s \le H\} \le \min\Big\{1, R_{\widetilde{\mu}}(r_+ + H\kappa_+ q_{0,s})2^{-n}\Big\}. \tag{1496}$$

The first line applies to a uniform-invariant selected kernel with $d\mu_0/d\mu_I \le R_0$*; the second applies under the represented invariant density and conductance comparisons of Theorem VIII.14.1. For the residual Metropolis record, the parameter-explicit replacement is Eq.* (1451)*, with* $\Delta_- = 1/2 - \eta - 2\delta$ *and* $\Delta_+ = 1/2 - \eta + 2\delta$*.*

Finally, the observation-derived intrinsic signal obeys, on the same stationary record,

$$\mathbb{E}_{\mu P} b_{\phi_I} \le a_{\phi_I} + L_{\phi_I}\sqrt{2\rho_{\phi_I}(P)\,\mathrm{Var}^d_\mu(\phi_I)}, \tag{1497}$$

with the empirical replacement in Eqs. (569) *and* (570)*. Thus novelty magnitude, within-cell target information, controlled capacity, and compensated dynamics are evaluated on one record and charged before saturation. The candidate in Eq.* (1493) *is exactly* $C^{\mathrm{cur}}_{R,t}$ *in Eq.* (1482)*, and hence enters the complete LPN envelope Eqs.* (1487) *and* (1488)*.*

Under the random LPN presentation, the same conclusions hold after adding the explicit complement probability

$$\varepsilon_{\mathrm{rig}}(n,m) + \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] + (2^n - 1)2^{-m(1-H_2(\tau))} + 2^{n+1}e^{-2\delta^2 m}. \tag{1498}$$

Proof. On $\mathcal{G}_{\tau,\delta}$, the public verifier accepts precisely s, so the external reward is the constant zero vector on every surviving prefix. Apply Eq. (560). Conditional on $(A, b, C_{\varepsilon,I})$, public postprocessings and intrinsic-record updates are causal postprocessings. The conditional sequential

SDPI theorem Theorem V.3.3 therefore gives Eq. (1490). The conditional decoupling-and-Pinsker argument of Theorem V.8.6 gives Eq. (1493)'s information candidate. Its source-charge candidate is Eq. (281) for the same truncated curiosity record. Taking the recordwise minimum gives the full display; summing the disjoint first-contact events gives Eq. (1494).

The mode assignment is Theorem V.12.5. Substitution of the established LPN vanishing coordinates leaves its ambient mode. The native uniform-capacity estimate is Eq. (1300), and the comparison estimate is Eq. (1445). Observation regularity is Eq. (566). Every operation used in these substitutions occurs in Eq. (1491); constructor reuse follows Definition I.2.5.

The good-event complement is the union of Eqs. (1296) and (1604). Apply the unit bound there and the tower property on the intersection to obtain the final family statement.

□

VIII.15 Optimal Structural Active Sampling for LPN

The standard LPN presentation supplies the complete public record (A, b, η) once. Active selection of its rows, residual coordinates, features, candidate tests, or native transitions is therefore computation on that record, not a new sampling channel. Fresh random samples and chosen parity queries are separate declared interfaces and are evaluated below as presentation extensions.

The framework optimum is specialized in Theorem VIII.15.5: the null-query active profile is the standard finite-horizon saturated computation profile up to the completely charged clock-and-certificate compilation. Its computational, sample, and noise dependence is evaluated in Corollaries VIII.15.4 and VIII.15.6.

Definition VIII.15.1 (LPN master active-sampling ceiling)**.** For a budget B and refined horizon H, let

$$B_{\mathrm{LPN}}^{\mathrm{act}} = \Psi_{\mathrm{act}}(B, H, n, m, \eta)$$

be a class-preserving common majorant of Eq. (574) and the existing LPN source, capacity, Blackwell, Lift, Caus, learning, verifier, and good-event conversions. Let $\mathscr{R}_{\mathrm{LPN},H}^{\mathrm{act}}(B_{\mathrm{LPN}}^{\mathrm{act}})$ be the corresponding complete records with initial public record $\mathsf{Z}_0 = (A, b)$, target index S, carrier $\mathbb{F}_2^n$, and target $T_S = \{S\}$. Write $\mathcal{G}_{\mathrm{rig}}$ for the full-rank and random-code-rigidity event in Lemma VIII.17.5, and set

$$\mathcal{G}_{\mathrm{act}} = \mathcal{G}_{\tau,\delta} \cap \mathcal{G}_{\mathrm{rig}}. \tag{1499}$$

The five restricted record sets $\mathscr{R}_X^{\mathrm{act}}, \mathscr{R}_A^{\mathrm{act}}, \mathscr{R}_{Q,\mathrm{ex}}^{\mathrm{act}}, \mathscr{R}_{Q,\mathrm{rev}}^{\mathrm{act}}, \mathscr{R}_{Q,\mathrm{nr}}^{\mathrm{act}}$ are obtained by requiring the master slot to belong to the corresponding certificate family of Eq. (593).

Theorem VIII.15.2 (Complete LPN master active-sampling envelope)**.** *Fix $0 < \eta < \tau < 1/2$, $\delta > 0$, and the ceiling of Definition VIII.15.1. On the LPN good-presentation event used in Theorem VIII.14.9, every complete active record satisfies*

$$A_R^H \le U_R^{\mathrm{LPN,act}} := \min\{U_R^{\mathrm{act,info}}, U_R^{\mathrm{act,cap}}, U_R^{\mathrm{act,BW}}, U_R^{\mathrm{act,master}}, U_R^{\mathrm{act,src}}\}, \tag{1500}$$

where every candidate is evaluated for the same record and probability law. For the information coordinate one may take

$$J_{R,t}^{\mathrm{LPN,act}} = \min\{n, m(1 - H_2(\eta))\} + \sum_{j=1}^{L_{R,t}} d_{R,t,j}, \tag{1501}$$

$$U_R^{\mathrm{act,info}} = \min\left\{1, \sum_{t=0}^{H-1} \left(2^{-n} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{LPN,act}}}\right)\right\}. \tag{1502}$$

The conditional within-cell candidate of Eq. (1493) *may replace Eq.* (1502) *when it is smaller. Here* $U_R^{\mathrm{act,src}}$ *is Eq.* (590) *with* $\vartheta_j(\eta) = (1-2\eta)^2$*, the LPN five-family contextual charges, and the complete cost in Definition VIII.15.1.*

The four nonambient active-geometry modes have zero LPN prospective source coordinate at the polynomial ceiling:

$$\sup_{R\in\mathscr{R}_A^{\mathrm{act}}\cup\mathscr{R}_{Q,\mathrm{ex}}^{\mathrm{act}}\cup\mathscr{R}_{Q,\mathrm{rev}}^{\mathrm{act}}\cup\mathscr{R}_{Q,\mathrm{nr}}^{\mathrm{act}}} V_{\mathscr{C}_R} = 0. \tag{1503}$$

Retain every additive comparison error rather than absorbing it into the vanishing source coordinate:

$$\Delta_{I,H}^{\mathrm{act,Q}} = \sup_{R\in\mathscr{R}_A^{\mathrm{act}}\cup\mathscr{R}_{Q,\mathrm{ex}}^{\mathrm{act}}\cup\mathscr{R}_{Q,\mathrm{rev}}^{\mathrm{act}}\cup\mathscr{R}_{Q,\mathrm{nr}}^{\mathrm{act}}} \min\{U_R^{\mathrm{act,info}}, U_R^{\mathrm{act,cap}}, U_R^{\mathrm{act,BW}}, \min(1,\delta_R)\}. \tag{1504}$$

For the exact same-normalization source comparisons, $\delta_R = 0$ *and this coordinate is zero. Consequently the optimal qualified active source profile reduces, with every comparison error retained explicitly, to the identity-support master mode:*

$$\mathrm{ActArr}_{I,H}^{\mathrm{sat}}(B) \le \max\left\{\sup_{R\in\mathscr{R}_X^{\mathrm{act}}(B_{\mathrm{LPN}}^{\mathrm{act}})} U_R^{\mathrm{LPN,act}}, \Delta_{I,H}^{\mathrm{act,Q}}\right\}, \qquad I \in \mathcal{G}_{\mathrm{act}}, \tag{1505}$$

$$\mathbb{E}_I\, \mathrm{ActArr}_{I,H}^{\mathrm{sat}}(B) \le \mathbb{E}_I\left[\mathbf{1}_{\mathcal{G}_{\mathrm{act}}} \max\left\{\sup_{R\in\mathscr{R}_X^{\mathrm{act}}(B_{\mathrm{LPN}}^{\mathrm{act}})} U_R^{\mathrm{LPN,act}}, \Delta_{I,H}^{\mathrm{act,Q}}\right\}\right] + \varepsilon_{\mathrm{good}}(n,m,\eta,\tau,\delta), \tag{1506}$$

where

$$\varepsilon_{\mathrm{good}} = \varepsilon_{\mathrm{rig}}(n,m) + e^{-mD_{\mathrm{Ber}}(\tau\|\eta)} + (2^n-1)2^{-m(1-H_2(\tau))} + 2^{n+1}e^{-2\delta^2 m}. \tag{1507}$$

For $R \in \mathscr{R}_X^{\mathrm{act}}$*, the master certificate has* $q = \mathrm{id}$*,* $S_q(\lambda) = 1$*, and*

$$V_{\mathscr{C}_R} = M_{\mathfrak{G}_R} C_{\mathfrak{G}_R} \frac{R_{\mathfrak{G}_R}}{1+\rho_{\mathfrak{G}_R}}. \tag{1508}$$

Accordingly, an inherited comparison retains its declared family-functional normalization:

$$\mathfrak{M}_{n,m,\eta}(U_R^{\mathrm{act,master}}) \le \min\{1, K_R(2^{-n} + (1-\eta)^n) + \mathfrak{M}_{n,m,\eta}(\delta_R)\}. \tag{1509}$$

A direct comparison instead retains its pointwise normalization:

$$U_{R,I}^{\mathrm{act,master}} \le \min\left\{1, K_R^{\mathrm{dir}} M_{\mathfrak{G}_R} C_{\mathfrak{G}_R} \frac{R_{\mathfrak{G}_R}}{1+\rho_{\mathfrak{G}_R}} + \delta_{R,I}^{\mathrm{dir}}\right\}. \tag{1510}$$

Thus the remaining active coordinate is evaluated only through the completely charged target alignment, reach, mixing, and regularity of the selected full-carrier geometry; no quotient or observation-cell credit is retained in this branch.

Proof. Apply Theorem V.9.8 with $\Theta = S$ and $\mathsf{Z}_0 = (A,b)$. The initial LPN information bound and ordered-reveal increment are Theorem VIII.14.4; summing its disjoint next-contact bounds gives Eqs. (1501) and (1502). The capacity, Blackwell, and master candidates are the same-record substitutions already used in Corollary VIII.14.8. The source-resolved candidate is Theorem IV.13.3

specialized to the complete LPN active record. It retains the separate Add, Mult, Ord, Sym, and Filt charges and the binary-noise contraction.

The pure exact Abs, exact quotient-geometry, and both compensated coordinates vanish by Eqs. (1602), (1606) and (1607). This proves Eq. (1503). The exhaustive five-mode partition bounds every nonambient record by its retained comparison error and leaves $\mathcal{C}_X$ as the only nonzero source mode. This proves the pointwise bound Eq. (1505). The value of an identity-support active master certificate is Eq. (124) with $\overline{A}_{\mathrm{id}} = S_{\mathrm{id}} = 1$, which gives Eq. (1508). The inherited ancestor bound is Eq. (1647); substituting it and the direct visibility in the corresponding comparison forms proves Eqs. (1509) and (1510). Finally apply the unit bound on the complement of the good event; its probability is Eq. (1498). The tower property proves Eq. (1506).

□

Theorem VIII.15.3 (Fixed, passive-stream, and chosen-query LPN acquisition)**.** *The three active interfaces have the following quantitative accounting.*

(i) *In the fixed standard presentation, every selected row, coordinate, residual feature, or deterministic postprocessing $F(A,b)$ satisfies*

$$I(S; F(A,b) \mid A, b, \mathcal{H}_t) = 0. \tag{1511}$$

Its construction, score optimization, memory update, and use in the selected transition remain charged by Eq. (574). A later verifier answer occupies an ordered reveal slot.

(ii) *Suppose a declared passive-stream interface returns $Z_j = (a_j, y_j)$, where a_j is fresh uniform and independent of the past and $y_j = \langle a_j, S\rangle + E_j$, $E_j \sim \mathrm{Bernoulli}(\eta)$. After q requested samples,*

$$I(S; Z_{1:q} \mid A, b) \le \min\{n, q(1 - H_2(\eta))\}. \tag{1512}$$

(iii) *Suppose instead that a declared chosen-query interface accepts $a_j \in \mathbb{F}_2^n$ selected from the pre-query history and returns $\langle a_j, S\rangle + E_j$. This is a presentation extension. Querying e_i independently q_i times and taking the majority value gives*

$$\Pr\{\widehat{S} \ne S\} \le \sum_{i=1}^{n} \exp\left[-\frac{q_i(1-2\eta)^2}{2}\right]. \tag{1513}$$

Thus $q_i \ge 2(1-2\eta)^{-2}\log(n/\epsilon)$ makes the total error at most ϵ.

Proof. Part (i) is conditional data processing: $F(A,b)$ is measurable with respect to the conditioning record. For one passive sample, a_j is independent of S, while

$$I(S; y_j \mid a_j, \mathcal{H}_j) \le H(y_j \mid a_j, \mathcal{H}_j) - H(E_j) \le 1 - H_2(\eta).$$

The conditional chain rule proves Eq. (1512). The chosen-query interface is separated from the fixed presentation by Theorem III.33.2. For each basis query, Hoeffding's inequality bounds the majority error by $\exp[-2q_i(1/2-\eta)^2]$. A union bound gives Eq. (1513) and the displayed sufficient query count.

□

Corollary VIII.15.4 (Parameter-explicit LPN active-acquisition regimes)**.** *Fix a monotone scalarization κ, write $b = \kappa(B)$, and let $c_{\mathrm{samp}} > 0$ be the charged scalar cost of one declared* *fresh-sample request, including its returned row, label, storage, and channel update. The number of fresh-sample requests in any such record is at most*

$$q_B = \min\left\{H,\, \left\lfloor \frac{b}{c_{\mathrm{samp}}} \right\rfloor\right\}. \tag{1514}$$

For the passive-stream extension, let

$$D^{\mathrm{post}}_{R,t} = \sum_{j=1}^{L_{R,t}} d_{R,t,j}$$

contain the contracted information of every other target-dependent reveal. Then

$$I(S;\mathcal{H}_t) \le J^{\mathrm{act}}_{n,m,\eta,q_B,t} := \min\left\{n,\ (m+q_B)(1-H_2(\eta)) + D^{\mathrm{post}}_{R,t}\right\}, \tag{1515}$$

$$\Pr\{\tau_s \le H\} \le \min\left\{1,\ H2^{-n} + \sum_{t=0}^{H-1} \sqrt{\frac{\log 2}{2} J^{\mathrm{act}}_{n,m,\eta,q_B,t}}\right\}, \tag{1516}$$

$$\Pr\{\widehat{S} = S\} \le \min\left\{1,\ \frac{J^{\mathrm{act}}_{n,m,\eta,q_B,H}+1}{n}\right\}. \tag{1517}$$

The fixed standard presentation is the specialization $q_B = 0$.

For a chosen-query extension with equal allocation, put $\underline{q}_B = \lfloor q_B/n \rfloor$. The basis-majority construction satisfies

$$\Pr\{\widehat{S} \ne S\} \le n \exp\left[-\frac{\underline{q}_B(1-2\eta)^2}{2}\right]. \tag{1518}$$

In particular, error at most ϵ is obtained whenever

$$q_B \ge \frac{2n}{(1-2\eta)^2} \log \frac{n}{\epsilon} + n. \tag{1519}$$

For the fixed-presentation identity-geometry mode, every same-record uniform-invariant certificate supplies

$$U^{\mathrm{act,cap}}_R \le \min\{1, R_0(H+1)2^{-n}\}. \tag{1520}$$

For a residual Gibbs–Metropolis certificate, put $\Delta_\pm = 1/2 - \eta \pm 2\delta$. If $\theta m \Delta_+ \le (1-\epsilon_0) n \log 2$ and $\rho_0 \le R_\theta \pi_\theta$, then

$$U^{\mathrm{act,cap}}_R \le \min\left\{1,\ \frac{R_\theta 2^{-\epsilon_0 n}}{1-2^{-n}}\left(1 + \frac{H}{2} e^{-\theta m \Delta_-}\right)\right\}. \tag{1521}$$

Thus every displayed active certificate depends explicitly on (n, m, η, H, B), with (τ, δ) entering through the good-presentation and residual-flatness conversions.

Proof. Equation Eq. (1514) is the joint cost and horizon restriction. Combine the public-record information bound $I(S; A, b) \le \min\{n, m(1 - H_2(\eta))\}$ with Eq. (1512) and the conditional chain rule. This proves Eq. (1515). The overlap-one contact and Fano conversions are Theorem VIII.14.4, proving Eqs. (1516) and (1517).

Apply Eq. (1513) with $q_i = \underline{q}_B$ to obtain Eq. (1518); the displayed threshold implies $\underline{q}_B \ge 2(1-2\eta)^{-2}\log(n/\epsilon)$. The two capacity bounds are respectively Eqs. (1300) and (1451). □

Theorem VIII.15.5 (Optimal active construction for the standard LPN presentation)**.** *Fix (n, m, η, H, B) in the standard LPN presentation and use the family functional $\mathfrak{M}_{n,m,\eta}$. Let*

$$\mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(B) = \sup_{R \in \mathscr{R}^{\mathrm{act}}_{\mathrm{LPN},H}(B)} \mathfrak{M}_{n,m,\eta}(A^H_R), \tag{1522}$$

where the query component is null or a deterministic readout of the fixed public record (A, b). Then

$$\mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(B) = \mathrm{Succ}^{\mathrm{act,sat}}_{\mathrm{LPN},n,m,\eta,H}(B), \tag{1523}$$

and its exact arrival decomposition is the five-mode maximum in Eq. (581)*.*

Let $\mathrm{Succ}^{\mathrm{sat}}_{\mathrm{LPN},n,m,\eta,H}(B)$ denote the saturated success profile over all represented standard-presentation LPN computations stopped by time H. There is a class-preserving complete majorant $\Psi^{\mathrm{act}}_{\mathrm{LPN}}$, obtained from Definition VIII.15.1, such that

$$\mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(B) \le \mathrm{Succ}^{\mathrm{sat}}_{\mathrm{LPN},n,m,\eta,H}(B) \le \mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(\Psi^{\mathrm{act}}_{\mathrm{LPN}}(B, H, n, m, \eta)). \tag{1524}$$

Thus the standard-presentation active optimum and the complete finite-horizon saturated success profile are the same affordable class up to the explicit cost of compiling the clocked action, master certificate, and comparison record. The compilation introduces no fresh LPN sample or chosen parity query.

After scalarization, let $\psi^{\mathrm{act}}_{\mathrm{LPN}}$ majorize this compiled cost, and let $B^*_{\mathrm{act,LPN}}(n, m, \eta, H; \alpha)$ denote Eq. (595) *restricted to the standard LPN active record class. For every $\varepsilon > 0$, the corresponding threshold costs obey*

$$B^*_{\mathrm{rec}}(n, m, \eta; \alpha) \le B^*_{\mathrm{act,LPN}}(n, m, \eta, H; \alpha) \le \psi^{\mathrm{act}}_{\mathrm{LPN}}(B^*_{\mathrm{rec}}(n, m, \eta; \alpha) + \varepsilon, H, n, m, \eta), \tag{1525}$$

whenever the recovery infimum is witnessed within horizon H at the displayed ε-enlarged cost. Hence polynomial affordability is preserved in both directions by a polynomial class-preserving compilation.

Proof. The identity Eq. (1523) is Theorem V.9.5 on the standard LPN slice. Equation Eq. (1511) shows that deterministic selection inside (A, b) creates no additional observation channel; its computation and use remain in the complete cost.

Dropping certificate annotations from a complete active record leaves its represented query, transition, stopping, and output computation, proving the first inequality in Eq. (1524). Conversely, Corollary II.11.3 and Lemma II.11.1 turn every finite-horizon represented computation into a null-query clocked active record. Encoding adequacy, the universal dynamical certificate, and Definition VIII.15.1 append its completely charged master and comparison slots at $\Psi^{\mathrm{act}}_{\mathrm{LPN}}$, without changing the executed path law. This proves the second inequality. Apply the two inequalities to the defining infima in Eqs. (300) and (595) to obtain Eq. (1525).

□

Corollary VIII.15.6 (Explicit learned-optimum loss for LPN active sampling)**.** *In the setting of Corollary VIII.15.4, suppose a complete standard-presentation LPN record learns its represented action-value rule through a bounded flip-symmetric score comparison from a sample-independent represented codebook and a sample certificate with effective size m_{eff}, dependence error $\varepsilon_{\mathrm{dep}}$, confidence $1 - \delta$, common optimization error $\varepsilon_{\mathrm{opt}}$, and affordable action-value codebook logarithm $L_{\mathrm{LPN,act}}(n, m, \eta, B)$. Then, on its simultaneous event,*

$$V^{\mathrm{act}}_{\mathrm{LPN},0} - A^H_{\widehat{R}} \le H\left\{\frac{2}{1-2\eta}\left[2\sqrt{\frac{2L_{\mathrm{LPN,act}}(n, m, \eta, B)}{m_{\mathrm{eff}}}} + 3\sqrt{\frac{\log(2H/\delta)}{2m_{\mathrm{eff}}}} + \varepsilon_{\mathrm{dep}}\right] + \varepsilon_{\mathrm{opt}}\right\}. \tag{1526}$$

The same record simultaneously retains the information, target-contact, chosen-interface, and capacity bounds in Eqs. (1515), (1516), (1518) *and* (1521). *For the fixed standard presentation, $q_B = 0$; passive fresh samples and chosen parity queries retain their separate presentation-extension status.*

Proof. Apply Corollary V.9.13 with the LPN family functional and the displayed codebook. The factor $(1-2\eta)^{-1}$ is the homogeneous label-noise transport of Theorem IV.16.6; the remaining parameter bounds are the cited LPN active-acquisition estimates.

□

Theorem VIII.15.7 (LPN occupation-flow and Bellman-support certificate)**.** *Fix the standard-presentation LPN slice (n, m, η, H, B), let $\rho = 1-2\eta$, and use the completely compiled ceiling*

$$B^{\mathrm{Bell}}_{\mathrm{LPN}} = \Psi_{\mathrm{Bell}}\big(\Psi^{\mathrm{act}}_{\mathrm{LPN}}(B, H, n, m, \eta), H, n\big).$$

For a bounded family-uniform barrier $v = (v_t)_{t\le H}$, $v_H = 0$, define

$$\mathfrak{U}^{\mathrm{LPN}}_{t,c}(v) = \sup_{R\in\mathscr{R}^{\mathrm{act}}_{\mathrm{LPN},t;c}(B^{\mathrm{Bell}}_{\mathrm{LPN}})} \mathfrak{M}_{n,m,\eta}\left(U^{\mathrm{Bell}}_{R,t}(v)\right), \qquad c \in \mathfrak{C}_{\mathrm{master}}, \tag{1527}$$

where $U^{\mathrm{Bell}}_{R,t}$ is the same-record minimum in Eq. (618)*. On the LPN good event of Theorem VIII.14.9, the standard active optimum satisfies*

$$\mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(B) \le \inf_{v_H=0}\left\{\mathfrak{M}_{n,m,\eta}(v_0(h_0)) + \sum_{t=0}^{H-1}\max_{c\in\mathfrak{C}_{\mathrm{master}}}\mathfrak{U}^{\mathrm{LPN}}_{t,c}(v)\right\} + \varepsilon_{\mathrm{good}}(n, m, \eta, \tau, \delta). \tag{1528}$$

In particular, the zero barrier gives the completely explicit next-contact specialization

$$\mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(B) \le \sum_{t=0}^{H-1}\max_{c\in\mathfrak{C}_{\mathrm{master}}}\mathfrak{U}^{\mathrm{LPN}}_{t,c}(0) + \varepsilon_{\mathrm{good}}(n, m, \eta, \tau, \delta). \tag{1529}$$

The four nonambient modes obey the explicit comparison-error bound

$$\max_{c\in\{A,Q_{\mathrm{ex}},Q_{\mathrm{rev}},Q_{\mathrm{nr}}\}}\mathfrak{U}^{\mathrm{LPN}}_{t,c}(0) \le \sup_{R\in\mathscr{R}^{\mathrm{act}}_{A}\cup\mathscr{R}^{\mathrm{act}}_{Q,\mathrm{ex}}\cup\mathscr{R}^{\mathrm{act}}_{Q,\mathrm{rev}}\cup\mathscr{R}^{\mathrm{act}}_{Q,\mathrm{nr}}} \mathfrak{M}_{n,m,\eta}\min\{1, U^{\mathrm{info}}_{R,t}, U^{\mathrm{cap}}_{R,t}, U^{\mathrm{BW}}_{R,t}, \delta_{R,t}\}. \tag{1530}$$

For exact same-normalization comparisons $\delta_{R,t} = 0$, so this quantity vanishes.

For the identity-support mode, every record retains the following explicit candidate minimum:

$$\mathfrak{U}^{\mathrm{LPN}}_{t,X}(0) \le \sup_{R\in\mathscr{R}^{\mathrm{act}}_{X}(B^{\mathrm{Bell}}_{\mathrm{LPN}})} \mathfrak{M}_{n,m,\eta}\min\left\{\begin{array}{l} 1, \\ 2^{-n} + \sqrt{\dfrac{\log 2}{2}J^{\mathrm{LPN,act}}_{R,t}}, \\ R_0(H+1)2^{-n}, \\ \dfrac{R_\theta 2^{-\epsilon_0 n}}{1-2^{-n}}\left(1 + \dfrac{H}{2}e^{-\theta m(1/2-\eta-2\delta)}\right), \\ U^{\mathrm{BW}}_{R,t}, \\ \min\{1, K_R(2^{-n} + (1-\eta)^n) + \delta_{R,t}\}, \\ \min\left\{1, K^{\mathrm{dir}}_R M_{\mathfrak{G}_R} C_{\mathfrak{G}_R}\dfrac{R_{\mathfrak{G}_R}}{1+\rho_{\mathfrak{G}_R}} + \delta^{\mathrm{dir}}_{R,t}\right\} \end{array}\right\}. \tag{1531}$$

Here every entry is evaluated conditionally from the same surviving history. For $v = 0$, the positive Bellman increment is exactly the next-contact probability $p_T(h,a)$. A whole-horizon candidate enters this minimum only through its represented re-rooted comparison at that history. For a general barrier v, the corresponding entry in Eq. (1527) *is finite only when the record supplies the Bellman-increment converter required by Theorem V.10.5. The residual-Gibbs candidate is included when $\theta m(1/2 - \eta + 2\delta) \le (1-\epsilon_0)n \log 2$ and the represented initial-density comparison of Eq.* (1521) *holds. Otherwise that entry has the neutral value one. The information budget is*

$$J^{\mathrm{LPN,act}}_{R,t} \le \min\{n, m(1 - H_2(\eta))\} + \sum_{j=1}^{L_{R,t}} d_{R,t,j}, \tag{1532}$$

and every ordered reveal, contraction coefficient, barrier evaluation, transition, and comparison is included in $B^{\mathrm{Bell}}_{\mathrm{LPN}}$.

If a complete LPN learner produces $\widehat{R}$ with the action-value radius in Eq. (1526)*, then, on the same simultaneous event,*

$$A^{H}_{\widehat{R}} \le \mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(B), \tag{1533}$$

$$A^{H}_{\widehat{R}} \ge \mathrm{ActArr}^{\mathrm{std,sat}}_{\mathrm{LPN},n,m,\eta,H}(B) - H\left(2\Gamma^{\mathrm{clean}}_{\mathrm{LPN,act}} + \varepsilon_{\mathrm{opt}}\right), \tag{1534}$$

where

$$\Gamma^{\mathrm{clean}}_{\mathrm{LPN,act}} \le \frac{1}{\rho}\left[2\sqrt{\frac{2L_{\mathrm{LPN,act}}(n,m,\eta,B)}{m_{\mathrm{eff}}}} + 3\sqrt{\frac{\log(2H/\delta)}{2m_{\mathrm{eff}}}} + \varepsilon_{\mathrm{dep}}\right]. \tag{1535}$$

Thus Eq. (1528) *controls the best affordable LPN controller, while Eqs.* (1534) *and* (1535) *control the statistical and optimization distance of a learned controller from that optimum.*

Proof. Compile every standard-presentation computation into the complete active record class by Theorem VIII.15.5. Apply Theorem V.10.4 and then the structural support evaluation Theorem V.10.5. The five-mode decomposition gives the maximum in Eq. (1528). On the complement of the good event use the unit bound; its probability is Eq. (1507).

The exact and compensated nonambient LPN sources vanish by Eq. (1503). Their numerical Bellman increments therefore retain only the same-event candidates and comparison errors, proving Eq. (1530). In the identity mode, specialize to $v = 0$, so the positive Bellman increment is the conditional next-contact probability. Then insert successively the information/contact bound Eq. (1502), the uniform-capacity and residual-Gibbs bounds Eqs. (1520) and (1521), the inherited comparison Eq. (1509), and the direct master visibility Eq. (1510). These candidates bound that same conditional next-contact probability only through their stored re-rooted same-normalization converters; their minimum proves Eq. (1531). The information display is Eq. (1501). The zero-barrier case of Eq. (1528) is Eq. (1529).

Finally, apply Corollaries V.10.7 and V.9.13 and substitute the LPN finite-description radius. This proves Eqs. (1533) to (1535).

□

Corollary VIII.15.8 (LPN specialization of prospective contact-witness coercivity)**.** *Fix the standard LPN presentation, parameters (n,m,η,H,B), and the compiled ceiling $B^{\mathrm{Bell}}_{\mathrm{LPN}}$ of Theorem VIII.15.7. Let R be one complete family-uniform active record and let ν be the declared target-neutral next-test baseline. If*

$$\mathfrak{M}_{n,m,\eta}(A^{H}_{R}) > \mathfrak{M}_{n,m,\eta}(\mathrm{Enum}_{\nu}(R,H)) + \varepsilon, \tag{1536}$$

then some $t_* < H$ satisfies

$$\mathfrak{M}_{n,m,\eta}(\mathrm{Adv}_{t_*}(K_{R,t_*};\nu_{t_*})) > \frac{\varepsilon}{H}. \tag{1537}$$

The complete normalized pre-hit derivation of this selector reads b or the residual primitive $r_I(x) = Ax+b$. Thus its transition record has residual ancestry. After the quotient-supported modes are assigned, an ambient witness belongs to the residual-ancestry branch of Definition VIII.8.9; a residual-independent derivation cannot satisfy Eq. (1537)*.*

At the same time and compiled ceiling, the zero-barrier Bellman certificate satisfies

$$\max_{c\in\mathfrak{C}_{\mathrm{master}}} \mathfrak{U}^{\mathrm{LPN}}_{t_*,c}(0) \geq \frac{\varepsilon}{H|\mathfrak{C}_{\mathrm{master}}|} = \frac{\varepsilon}{5H}. \tag{1538}$$

The exact and compensated quotient-supported modes are evaluated by Eq. (1530)*; when their represented same-normalization comparisons are exact, their contribution is zero. Consequently a positive witness at the displayed scale is either*

(i) *a residual-ancestry ambient record evaluated by one of the finite candidates in Eq.* (1531)*; or*

(ii) *a residual-ancestry ambient record for which that same-record next-contact comparison is unavailable and therefore has the neutral certificate value one.*

In case (i)*, every available comparison is already explicit in (n, m, η, H, B): its information term is bounded by Eq.* (1532)*, its uniform-capacity term is $R_0(H+1)2^{-n}$, its represented residual-Gibbs term is the third nontrivial entry of Eq.* (1531)*, and its inherited or direct master comparison retains its complete source and converter cost. Case* (ii) *is not a new structural mechanism. It is exactly the absence of a represented comparison from the already charged ambient record to its next-contact value. Provenance classification, posterior concentration, or a retrospective first-hit record does not fill that slot.*

Conversely, if every residual-ancestry ambient first-step record at the compiled ceiling has a represented same-normalization comparison bounded by $\delta^b_t(n, m, \eta, H, B)$, then

$$\mathfrak{M}_{n,m,\eta}(A^H_R) \leq \mathfrak{M}_{n,m,\eta}(\mathrm{Enum}_\nu(R, H)) + \sum_{t=0}^{H-1} \delta^b_t(n, m, \eta, H, B) + \varepsilon_{\mathrm{good}}(n, m, \eta, \tau, \delta). \tag{1539}$$

In particular, a bound $\delta^b_t \leq (1-\eta)^n$ gives the explicit excess-arrival estimate

$$\mathfrak{M}_{n,m,\eta}(A^H_R) \leq H2^{-n} + H(1-\eta)^n + \varepsilon_{\mathrm{good}}(n, m, \eta, \tau, \delta) \tag{1540}$$

for the uniform full-carrier baseline.

Proof. Apply the family functional to the exact hazard identity Eq. (152). If Eq. (1536) holds, the sum of the H family-averaged selector advantages exceeds ε, so one term satisfies Eq. (1537). If the complete normalized source derivation of that selector did not read b or r_I, it would be residual-independent. Its expected advantage over the target-neutral baseline would then be zero by Lemma VIII.8.11, contradicting Eq. (1537). This proves the ancestry claim without treating information in the public residual as an affordable readout.

Apply Theorem V.10.6 pointwise and then the monotone LPN family functional. The pointwise master maximum is at most the sum of its five mode coordinates. Hence one fixed master mode carries at least one fifth of the family-averaged witness, which gives the factor $5 = |\mathfrak{C}_{\mathrm{master}}|$ in Eq. (1538). Structural Bellman evaluation at the compiled ceiling is Theorem VIII.15.7; it yields Eq. (1538). The nonambient evaluation is Eq. (1530). The remaining master mode is ambient, and the residual-independent alternative has just been excluded at positive family advantage. The

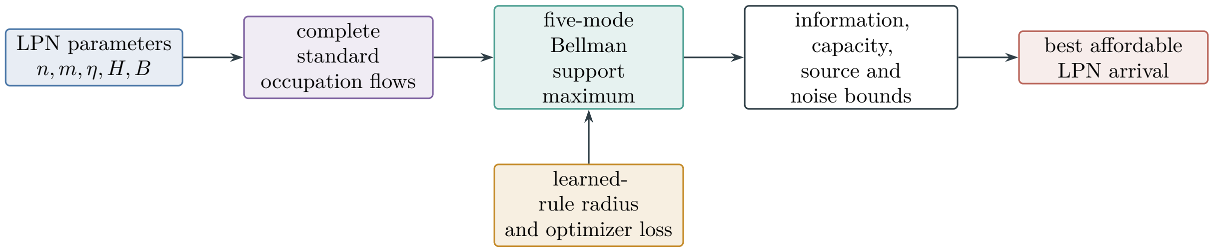

Figure 133: Quantitative LPN learning/control certificate. The Bellman-support maximum evaluates the best affordable standard-presentation controller. The learned-rule radius, including the factor $(1-2\eta)^{-1}$, measures the additional statistical and optimization loss.

same-record minimum Eq. (1531) then gives exactly the two alternatives in the statement: a finite evaluated converter or the neutral value assigned to an unavailable converter.

For the converse, use the residual-independent zero-advantage identity and the exact nonambient comparisons, substitute the assumed ambient same-normalization bounds into Eq. (629), and account for the complement of the LPN good event by the unit bound. This proves Eq. (1539). The uniform full-carrier neutral baseline is at most $H2^{-n}$ by Eq. (1650); substituting $\delta_t^b \le (1-\eta)^n$ proves Eq. (1540).

□

VIII.15.1 Structural Bayesian learning for LPN

The declared LPN target prior is uniform. The public matrix is independent of the secret, whereas the noisy right-hand side is the first target-dependent observation. This reveal order fixes the prior provenance before any represented belief or acquisition rule is constructed.

Definition VIII.15.9 (LPN structural belief and posterior-concentration profiles)**.** Let $S \sim \mathrm{Unif}(\mathbb{F}_2^n)$, so

$$\Pi_0^{\mathrm{LPN}}(u) = 2^{-n}, \qquad u \in \mathbb{F}_2^n. \tag{1541}$$

For $0 < \eta < 1/2$, conditional on a presented (A, b), the analytic task posterior is

$$\Pi_{A,b}^{\mathrm{LPN}}(u) = \frac{\eta^{|Au+b|_1}(1-\eta)^{m-|Au+b|_1}}{\sum_{v\in\mathbb{F}_2^n} \eta^{|Av+b|_1}(1-\eta)^{m-|Av+b|_1}}. \tag{1542}$$

This displayed law is an analytic comparator. A prospective LPN belief $\widehat{\Pi}_{R,t}$ is available only through a complete structural Bayesian record at the compiled ceiling

$$B_{\mathrm{LPN}}^{\mathrm{Bayes}} = \Psi_{\mathrm{Bayes}}(B_{\mathrm{LPN}}^{\mathrm{Bell}}, H, n, m, \eta). \tag{1543}$$

Its target concentration and filter error are

$$C_{R,t}^{\mathrm{LPN}} = \left[\max_{u\in\mathbb{F}_2^n} \widehat{\Pi}_{R,t}(u) - 2^{-n}\right]_+, \tag{1544}$$

$$e_{R,t}^{\mathrm{LPN}} = \left\|\widehat{\Pi}_{R,t} - \mathcal{L}(S \mid \mathcal{F}_{R,t})\right\|_{\mathrm{TV}}. \tag{1545}$$

Let $\mathfrak{M}^* = \mathfrak{M}^{\mathcal{G}_{\text{act}}}_{n,m,\eta}$ be the declared family functional restricted to the LPN good event in Eq. (1499). Put

$$\mathrm{PostConc}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B^{\mathrm{Bayes}}_{\mathrm{LPN}}) = \max_{t<H} \sup_R \mathfrak{M}^*_{n,m,\eta} \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}} C^{\mathrm{LPN}}_{R,t}\Big], \tag{1546}$$

$$\mathrm{FilErr}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B^{\mathrm{Bayes}}_{\mathrm{LPN}}) = \max_{t<H} \sup_R \mathfrak{M}^*_{n,m,\eta} \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}} e^{\mathrm{LPN}}_{R,t}\Big], \tag{1547}$$

where R ranges over one family-uniform complete structural Bayesian record scheme at the displayed ceiling. Every likelihood score, feature locator, evidence normalization, retained coefficient, precision choice, belief update, acquisition score, optimizer, and deployed transition is included in that record. Write

$$\mathrm{PostSrc}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B^{\mathrm{Bayes}}_{\mathrm{LPN}}) = \max_{t<H} \sup_{R\in\mathscr{R}^{\mathrm{Bayes,src}}_{\mathrm{LPN},H}(B^{\mathrm{Bayes}}_{\mathrm{LPN}})} \mathfrak{M}^* \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}} C^{\mathrm{LPN}}_{R,t}\Big], \tag{1548}$$

where the record restriction is the LPN specialization of Eq. (642).

Theorem VIII.15.10 (Quantitative structural Bayesian envelope for LPN)**.** *Use the parameter range and good-presentation event of Theorem VIII.15.2. Every complete LPN structural Bayesian record satisfies*

$$\Pr_R\{\tau > t, C_{t+1} = S\} \le \mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}}(2^{-n} + C^{\mathrm{LPN}}_{R,t} + e^{\mathrm{LPN}}_{R,t})\Big]. \tag{1549}$$

Consequently its family-uniform active Bayesian optimum obeys

$$\begin{aligned}\mathrm{BayesArr}^{\mathrm{LPN,sat}}_{\mathfrak{M},n,m,\eta,H}(B) \le H\Big(2^{-n} + \mathrm{PostConc}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B^{\mathrm{Bayes}}_{\mathrm{LPN}}) + \mathrm{FilErr}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B^{\mathrm{Bayes}}_{\mathrm{LPN}})\Big) \\ + \varepsilon_{\mathrm{good}}(n,m,\eta,\tau,\delta).\end{aligned} \tag{1550}$$

The actionable-gain form of the same bound is

$$\begin{aligned}\mathrm{BayesArr}^{\mathrm{LPN,sat}}_{\mathfrak{M},n,m,\eta,H}(B) \le H\Big(2^{-n} + \mathrm{BayesGain}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B^{\mathrm{Bayes}}_{\mathrm{LPN}})\Big) \\ + \varepsilon_{\mathrm{good}}(n,m,\eta,\tau,\delta).\end{aligned} \tag{1551}$$

The actionable inequality has no additive filter-error term. Every target-aligned effect of the represented filter belongs to the deployed next-test gain and retains its complete source ancestry.

For one record, the existing ordered-reveal information budget gives

$$\mathfrak{M}^*_{n,m,\eta}\mathbb{E}_R\Big[\mathbf{1}_{\{\tau>t\}} C^{\mathrm{LPN}}_{R,t}\Big] \le \min\left\{1 - 2^{-n}, \sqrt{\frac{\log 2}{2} J^{\mathrm{LPN,act}}_{R,t}} + \mathfrak{M}^*_{n,m,\eta}\mathbb{E}_R[\mathbf{1}_{\{\tau>t\}} e^{\mathrm{LPN}}_{R,t}]\right\}, \tag{1552}$$

where

$$J^{\mathrm{LPN,act}}_{R,t} \le \min\{n, m(1 - H_2(\eta))\} + \sum_{j=1}^{L_{R,t}} d_{R,t,j}. \tag{1553}$$

Selection of a row, coordinate, or feature from the fixed public record adds zero conditional information after (A,b)*, as in Eq.* (1511)*; its complete computational cost and structural provenance remain charged.*

Proof. For singleton targets, the vulnerability in Eq. (638) is maximum posterior mass and its uniform-prior value is 2^{-n}. Apply Theorem V.11.4 conditionally on each surviving history to obtain Eq. (1549). Sum the disjoint first-hit events, take the family-uniform saturated supremum, and use the unit bound on the complement of the LPN good event. Its probability is Eq. (1507), proving Eq. (1550). Apply Eq. (666) to the same record, then take the family-uniform saturated supremum and restore the complement of the good event. This gives Eq. (1551).

The exact conditional posterior satisfies

$$\max_u \Pi_{R,t}(u) - 2^{-n} \leq \|\Pi_{R,t} - \Pi_0^{\mathrm{LPN}}\|_{\mathrm{TV}}.$$

The target-vulnerability functional is one-Lipschitz, so replacing the exact posterior by $\widehat{\Pi}_{R,t}$ adds at most $e_{R,t}^{\mathrm{LPN}}$. Apply Eq. (423) with the LPN information budget Eq. (1501) to prove Eqs. (1552) and (1553). The fixed-record assertion is Theorem VIII.15.3.

□

Theorem VIII.15.11 (Source-resolved structural Bayesian gain for LPN)**.** *At the ceiling $B_{\mathrm{LPN}}^{\mathrm{Bayes}}$, every represented LPN posterior concentration used by an acquisition or transition rule has the exact decomposition below. The fixed indices (n, m, η, H), restricted functional $\mathfrak{M}^*$, and common ceiling are suppressed in this display. The decomposition and recordwise comparisons are unconditional. The complete ambient numerical row and the saturated numerical conclusions below use Certification Target VIII.20.5 at this ceiling.*

$$\mathrm{PostSrc}_{\mathfrak{M}^*}^{\mathrm{LPN,sat}} = \max_{c \in \{X, A, Q_{\mathrm{ex}}, Q_{\mathrm{rev}}, Q_{\mathrm{nr}}\}} \mathrm{PostSrc}_{\mathfrak{M}^*;c}^{\mathrm{LPN,sat}}. \tag{1554}$$

The actionable gain of the deployed next-test kernel has the source-resolved five-family charges of Definition V.11.7. These are the LPN restriction of Definition IV.13.2 and Eq. (661)*. Refine them by the master support mode and exact state–coefficient signature and suppress the fixed indices in the notation. On the LPN good event their complete table is*

source cell	*framework conclusion*
$\mathsf{A}, \mathsf{Q}_{\mathrm{ex}}, \mathsf{Q}_{\mathrm{rev}}, \mathsf{Q}_{\mathrm{nr}}$	0
X : Add, *residual-independent*	2^{-n}
X : *complete ambient record*	$\leq 2^{-n} + (1-\eta)^n$ *by Certification Target VIII.20.5*
X : Mult	0 *new source; retain the complete source record*
X : Ord	0 *new source; retain the complete source record*
X : Sym	*retain the authenticated complete source record*
X : Filt	*retain the represented complete source record.*

(1555)

Authorized Caus*, valid* Lift*, and* Bdry *retain the same row; their represented current, interface, and boundary costs remain in the complete record. Their gradient-orthogonal component is J_{res} and has zero target-qualified gain. Therefore every record satisfies the exact framework comparison*

$$g_{R,t}^{\mathrm{Bayes}} \leq \min\left\{ U_{R,t}^{\mathrm{cert}}, eH_B(n) \sum_{a \in \mathfrak{J}_{\mathsf{Abs}}^5} \mathcal{B}_{\mathfrak{M}^*,R,t}^{(a)} \right\}. \tag{1556}$$

The complete ambient record, including every derived operation and its cost, is bounded at the same ceiling by Theorem VIII.20.4 and Corollary VIII.20.8.

For the ordered target-dependent reveals in the same record, the source-resolved information ledger obeys

$$J_{R,t}^{\text{Bayes,(Add)}} \le \min\{n, m(1 - H_2(\eta))\} + \sum_{j=1}^{L_{R,t}} d_{R,t,j}, \tag{1557}$$

$$J_{R,t}^{\text{Bayes,(Mult)}} = J_{R,t}^{\text{Bayes,(Ord)}} = J_{R,t}^{\text{Bayes,(Filt)}} = 0 \tag{1558}$$

after conditioning on their earlier state-dependent arguments. Authenticated symmetry and derived channels inherit the information of their earlier source. Each homogeneous noisy parity reveal may use the SDPI coefficient $(1-2\eta)^2$*.*

Let $U_{R,t}^{\text{Bayes,info}}$ *be the resulting neutral-excess information candidate from Eq.* (664)*, let* $U_{R,t}^{\text{Bayes,learn}}$ *be the smaller applicable structural Rademacher or martingale PAC–Bayes action-value candidate from Eq.* (1535)*, and let* $U_{R,t}^{\text{Bayes,dyn}}$ *be the same-record minimum of the applicable capacity, Blackwell, observation-resolution, Caus, and Lift candidates in Corollary VIII.14.7. Unavailable candidates have value one. Then*

$$g_{R,t}^{\text{Bayes}} \le \min\left\{1, U_{R,t}^{\text{Bayes,info}}, U_{R,t}^{\text{Bayes,learn}}, U_{R,t}^{\text{Bayes,dyn}}, U_{R,t}^{\text{cert}}\right\}. \tag{1559}$$

All candidates use the same pre-hit record, conditional law, target-contact normalization, and common charged ceiling. An unavailable candidate has the neutral value one. The directly qualified ambient branch remains part of the complete prospective Geom profile, as prescribed by Theorem V.6.4. Every displayed candidate retains its explicit $(n, m, \eta, H, B, \delta)$ *arguments; each binary reveal uses* $(1-2\eta)^2$*, while clean-risk correction uses the distinct inverse factor* $(1-2\eta)^{-1}$*.*

Taking the family-uniform saturated supremum and applying the complete ambient seal at the same class-preserving ceiling gives

$$\text{BayesGain}_{\mathfrak{M}^*,n,m,\eta,H}^{\text{LPN,sat}}(B) \le 2^{-n} + (1-\eta)^n. \tag{1560}$$

The unrestricted family functional retains the good-presentation error $\varepsilon_{\text{good}}(n, m, \eta, \tau, \delta)$*. Combining this estimate with Eq.* (1551) *gives*

$$\begin{aligned} \text{BayesArr}_{\mathfrak{M},n,m,\eta,H}^{\text{LPN,sat}}(B) \le H\,&(2^{1-n} + (1-\eta)^n) \\ &+ \varepsilon_{\text{good}}(n, m, \eta, \tau, \delta). \end{aligned} \tag{1561}$$

The saturated Bayesian class is a subprofile of the same complete saturated source profile. No additional profile coordinate is introduced; the residual-ancestry row is evaluated by Certification Target VIII.20.5.

Proof. Apply Theorem V.11.15 to the LPN record class. The exact record-class partition proves Eq. (1554). The exact and compensated nonambient modes vanish by Eqs. (1503), (1602), (1606) and (1607). The charge-bearing vertices are prospective pre-hit ancestors by Corollary V.11.9; hence the LPN source seals apply to the charge ledger itself rather than to a retrospective first-hit label.

For the ambient mode, the source tags are exhausted by Lemmas VIII.18.1, VIII.17.7 and VIII.17.8 and Corollary VIII.17.6. Source inheritance under Caus, Lift, and Bdry and the residual assignment are Proposition II.5.9 and Corollaries VIII.19.1 and VIII.8.8. These results prove the provenance rows of the table. The complete ambient evaluation in Theorem VIII.20.4 and Corollary VIII.20.8 bounds the resulting complete source record at the same saturated ceiling.

Apply Theorems V.11.8 and V.12.4 to obtain Eq. (1556). The complete ambient branch is bounded directly by the complete LPN source seal.

The Add information budget is Eq. (1553). Conditional on the earlier arguments, deterministic Mult, Ord, and Filt vertices add zero information and zero contextual charge by Lemmas III.17.1 and III.17.2. The binary-noise coefficient is Eq. (665). This proves Eqs. (1557) and (1558).

Apply the recordwise minimum in Theorem V.11.10, together with the LPN learning and dynamical candidates cited in the statement, to obtain Eq. (1559). Taking the family-uniform record supremum proves the certificate part of the claim. The Bayesian record class is a restriction of the complete saturated source class, so Eq. (1647) proves Eq. (1560). Restriction to the good event is removed with the explicit complement probability in Eq. (1507). Substitution into Eq. (1551) proves Eq. (1561).

□

Corollary VIII.15.12 (Optimal structural extraction ledger for LPN)**.** *Let S be uniform on $\mathbb{F}_2^n$, let the fixed public presentation contain m independent binary-symmetric parity reveals of noise rate $0 \le \eta \le 1/2$, and let $D_{R,t}^{\mathrm{ext}}$ be the total conditional information in target-dependent advice, oracle answers, verifier outcomes, or feedback revealed after (A, b) and before the deployed test. Every such coordinate is included in the ordered reveal ledger with its complete source and cost. Then*

$$J_{R,t}^{\mathrm{LPN,opt}} \le \min\left\{n, m(1 - H_2(\eta)) + D_{R,t}^{\mathrm{ext}}, m(1 - 2\eta)^2 + D_{R,t}^{\mathrm{ext}}\right\}. \tag{1562}$$

Selection and postprocessing of the fixed public record contribute zero to $D_{R,t}^{\mathrm{ext}}$; their computation and structural provenance remain in the complete record.

At the compiled ceiling of Theorem V.11.19, the deployed singleton-target gain satisfies

$$g_{R,t}^{\mathrm{LPN}} \le \min\left\{U_{R,t}^{\mathrm{src}}, U_{R,t}^{\mathsf{Caus}}, U_{R,t}^{\mathsf{Lift}}, U_{R,t}^{\mathsf{Bdry}}, U_{R,t}^{\mathrm{Bell}}, \beta_{R,t}^{\mathrm{LPN},+} + \sqrt{\frac{\log 2}{2} J_{R,t}^{\mathrm{LPN,opt}}}\right\}. \tag{1563}$$

Here the source coordinate is the exact LPN restriction of the five-family and master-support ledger in Eq. (1555)*; the Caus, Lift, boundary, and Bellman coordinates are the LPN evaluations already collected in Eq.* (1559)*. The residual coordinate is zero after target qualification.*

The parameter dependence has the following exact limiting behavior:

(i) *at $\eta = 0$, the information candidate permits n bits and is consistent with represented Gaussian elimination;*

(ii) *when $\eta m = c \log n$, the subset-elimination source retains the value $(1-\eta)^n = n^{-cR+o(1)}$ from Eq.* (1398)*;*

(iii) *at fixed $0 < \eta < 1/2$, the channel contribution is explicitly bounded by the two noisy terms in Eq.* (1562)*, while target arrival uses the smaller complete structural coordinate in Eq.* (1563)*;*

(iv) *at $m(1 - H_2(\eta)) < n$, the information candidate recovers the finite LPN converse of Theorem VIII.7.3.*

Proof. Apply Corollary V.11.20 with $N = 2^n$, BSC capacity $1 - H_2(\eta)$, and SDPI coefficient $(1 - 2\eta)^2$. Each parity input contains at most one available bit, so the SDPI sum is at most $m(1 - 2\eta)^2$. The conditional chain rule appends $D_{R,t}^{\mathrm{ext}}$, proving Eq. (1562). Substitute this bound in Eq. (696) and use the LPN same-record candidates cited in the statement to obtain Eq. (1563). The four regime statements are direct substitutions and the cited LPN converse. □

Corollary VIII.15.13 (Compact latent-attention ledger for LPN)**.** *Adopt the LPN presentation and ordered-reveal convention of Corollary VIII.15.12. Let R be a complete compact latent-attention LPN record, let $b^{\mathrm{lat}}_{R,t}$ be its code capacity, and let $I^{\mathrm{LPN,lat}}_{R,t} = I(S; Z_t \mid \tau > t)$. Then*

$$I^{\mathrm{LPN,lat}}_{R,t} \le \min\left\{b^{\mathrm{lat}}_{R,t}, n, m(1 - H_2(\eta)) + D^{\mathrm{ext}}_{R,t}, m(1 - 2\eta)^2 + D^{\mathrm{ext}}_{R,t}, J^{\mathrm{LPN,opt}}_{R,t}\right\}. \tag{1564}$$

If the code has d_t coordinates at p_t-bit precision and exact secret recovery has probability at least α, then

$$d_t p_t + b^{\mathrm{code}}_t \ge \alpha n - 1. \tag{1565}$$

Let $U^{\mathrm{LPN,lat}}_{R,t}$ be the LPN restriction of Eq. (716)*, and let $U^{\mathrm{LPN,att}}_{R,t}$ be the attention envelope in Eq.* (1065) *for the same record. The deployed gain satisfies*

$$g^{\mathrm{LPN,lat,att}}_{R,t} \le \min\left\{U^{\mathrm{LPN,lat}}_{R,t}, U^{\mathrm{LPN,att}}_{R,t}, U^{\mathrm{src}}_{R,t}, U^{\mathsf{Caus}}_{R,t}, U^{\mathsf{Lift}}_{R,t}, U^{\mathsf{Bdry}}_{R,t}, U^{\mathrm{Bell}}_{R,t}\right\}. \tag{1566}$$

All entries refer to the same pre-hit law and compiled ceiling. Attention scores derived from (A, b), their normalizers, value paths, latent memory, posterior decoder, and selected transitions retain the complete LPN source ancestry. Optimal acquisition is the Bellman optimization over these same records, as in Theorems V.11.24 and VI.2.4.

The parameter regimes are consistent with the existing LPN phase ledger:

(i) *at $\eta = 0$, the information and code candidates permit an n-bit exact solution representation;*
(ii) *at $\eta = 1/2$, the binary reveals contribute zero channel information;*
(iii) *when $\eta m = c \log n$, the subset-elimination candidate retains $(1 - \eta)^n = n^{-cR+o(1)}$;*
(iv) *at fixed $0 < \eta < 1/2$, target arrival uses the minimum in Eq.* (1566)*, rather than the information availability bound alone.*

Proof. Apply Corollary V.11.26 with $N = 2^n$ and the LPN binary-symmetric reveals. Intersect its bound with Eq. (1562); this proves Eq. (1564). The latent-size condition is Eq. (721). Apply Theorems V.11.25 and VI.2.4 and the LPN same-record modality minimum Eq. (1563) to obtain Eq. (1566). The regime statements are direct substitutions and Eq. (1398). □

Theorem VIII.15.14 (Optimal latent representation and useful belief update for LPN)**.** *Restrict to complete LPN latent records constructed from the public presentation (A, b), target-independent auxiliary randomness, and the represented pre-hit transcript. Thus $D^{\mathrm{ext}}_{R,t} = 0$: selecting, reordering, or postprocessing public rows and deterministic verifier outcomes adds no new target-dependent reveal, while every such operation retains its complete cost and source ancestry. Write*

$$J^{\mathrm{lat}}_{n,m,\eta}(b) = \min\left\{b, n, m(1 - H_2(\eta)), m(1 - 2\eta)^2\right\}. \tag{1567}$$

Every code-capacity-b record satisfies

$$I(S; Z_t \mid \tau > t) \le J^{\mathrm{lat}}_{n,m,\eta}(b). \tag{1568}$$

If its represented decoder identifies S with probability at least α, then

$$b \ge \alpha n - 1, \qquad d_t p_t + b^{\mathrm{code}}_t \ge \alpha n - 1. \tag{1569}$$

The useful part of a belief update is its deployed actionable gain $g^{\mathrm{Bayes}}_{R,t}$. For every complete record it obeys the same-record bound

$$g^{\mathrm{Bayes}}_{R,t} \le \min\left\{U^{\mathrm{opt}}_{R,t}, U^{\mathrm{lat,stat}}_{R,t}, U^{\mathrm{att}}_{R,t}, U^{\mathsf{Caus}}_{R,t}, U^{\mathsf{Lift}}_{R,t}, U^{\mathsf{Bdry}}_{R,t}, U^{\mathrm{Bell}}_{R,t}, \beta^{\mathrm{Bayes},+}_{R,t} + \sqrt{\frac{\log 2}{2} J^{\mathrm{lat}}_{n,m,\eta}(b)}\right\}. \tag{1570}$$

Every entry refers to the same acquired data, latent encoder, recurrent update, decoder, attention selector, pre-hit law, and charged ceiling. Accordingly, the bound simultaneously optimizes the sampling schedule through the resource-valued Bellman recursion, the represented posterior update through the Bayes–Fisher flow, the latent representation through the code-capacity and sufficiency certificates, and the learned deployment rule through the source-resolved statistical certificate.

Let

$$\mathrm{LatBayesGain}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B,b) = \max_{t<H} \sup_{R\in\mathscr{R}^{\mathrm{lat}}_{\mathrm{LPN},H}(B,b)} \mathfrak{M}^*_{n,m,\eta}\Big(g^{\mathrm{Bayes}}_{R,t}\Big) \tag{1571}$$

be the LPN specialization of Eq. (709)*. On the good-presentation event, the supremum ranges over one family-uniform encoder, update, decoder, acquisition, and deployment scheme on the whole parameter slice, and*

$$\mathrm{LatBayesGain}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B,b) \le \mathrm{BayesGain}^{\mathrm{LPN,sat}}_{\mathfrak{M}^*,n,m,\eta,H}(B) \le 2^{-n} + (1-\eta)^n. \tag{1572}$$

For the unrestricted presentation functional, the right side becomes

$$2^{-n} + (1-\eta)^n + \varepsilon_{\mathrm{good}}(n,m,\eta,\tau,\delta). \tag{1573}$$

Taking the union over every latent capacity admitted by the polynomial saturated budget gives

$$\mathrm{LatBayesGain}^{\mathrm{LPN,poly,sat}}_{\mathfrak{M}^*,n,m,\eta,H} = \mathrm{BayesGain}^{\mathrm{LPN,poly,sat}}_{\mathfrak{M}^*,n,m,\eta,H} \le 2^{-n} + (1-\eta)^n. \tag{1574}$$

Thus the displayed ceiling ranges over every family-uniform latent representation constructible from the public presentation at saturated polynomial cost; it is not restricted to a chosen embedding architecture or learning rule.

Finally, the loss caused by compressing the complete pre-hit history into a particular latent representation remains explicit. For each presented LPN instance I, let $A^H_{I,R}$ be its arrival value. Then

$$\begin{aligned} 0 \le\ & \mathrm{BayesArr}^{\mathrm{LPN,sat}}_{I,H}(B) - A^H_{I,R} \\ \le\ & \sum_{t<H} \mathbb{E}_R\left[\mathbf{1}_{\{\tau>t\}}\left(\sqrt{\frac{\log 2}{2}\Delta^{\mathrm{suf}}_{R,t}} + e^{\mathrm{dec}}_{R,t} + 2\Gamma^{\mathrm{lat}}_{R,t} + \varepsilon^{\mathrm{opt}}_{R,t}\right)\right] + \Delta^H_{\mathrm{fil}}. \end{aligned} \tag{1575}$$

Hence code capacity controls retained information, the sufficiency and decoder terms control representation loss, and Eq. (1572) *controls the maximum portion of the resulting belief update that can improve the next target test.*

Proof. With $D^{\mathrm{ext}}_{R,t} = 0$, Eqs. (1567) and (1568) follow from Eq. (1564); the coordinate and code requirements are Eqs. (720) and (721). The recordwise minimum combines Eqs. (716), (1065) and (1563) after substituting the public-only information ceiling. These comparisons refer to one record and one conditional law, so their minimum proves Eq. (1570).

Class inclusion Eq. (711) gives the first comparison in Eq. (1572); the complete LPN source evaluation Eq. (1560) gives the second. Restoring the complement of the good event gives Eq. (1573). The exact compiler identity Eq. (714) proves Eq. (1574). It preserves the next-test law and therefore introduces no comparison loss. The final display is the LPN specialization of Eq. (718). □

Corollary VIII.15.15 (Learned LPN structural Bayesian policy)**.** *Let a complete LPN learner construct a represented belief and action-value rule with filter path error Δ^H_{fil}, optimization error $\varepsilon_{\mathrm{opt}}$, and the clean action-value radius in Eq.* (1535)*. Then*

$$0 \le \mathrm{BayesArr}^{\mathrm{LPN,sat}}_{\mathfrak{M},n,m,\eta,H}(B) - \mathfrak{M}_{n,m,\eta}(A^H_{\widehat{R}}) \le H\left(2\Gamma^{\mathrm{clean}}_{\mathrm{LPN,act}} + \varepsilon_{\mathrm{opt}}\right) + \mathfrak{M}_{n,m,\eta}(\Delta^H_{\mathrm{fil},\widehat{R}}). \tag{1576}$$

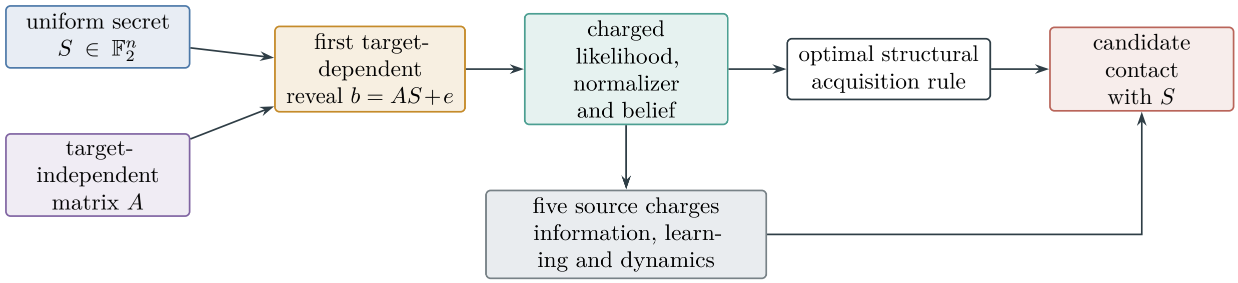

Figure 134: Structural Bayesian learning for LPN. The uniform prior supplies the neutral 2^{-n} contact. Every noisy reveal retains its source and SDPI coefficient. Likelihood, normalization, posterior approximation, acquisition, and deployment are deterministic charged descendants; the actionable next-test gain is bounded by the five source charges and the same-record learning and dynamical certificates.

The factor $(1-2\eta)^{-1}$*, effective sample size, dependence error, confidence allocation, codebook size, and optimization cost are therefore exactly those already displayed in Eq.* (1535)*. The PAC–Bayes posterior over policies and the LPN task belief over* S *remain separately typed and separately charged.*

Proof. Apply Corollary V.11.16 and substitute Eq. (1535). Bound the number of surviving loss terms by H.

□

VIII.16 Charged Decoder Geometry and Current Classification

VIII.16.1 Post-Output Decoder Geometry and Level Quotients

Theorem VIII.16.1 (Charged post-output decoder geometry for LPN)**.** *Fix* $n \geq 2$*,* $0 \leq \eta < \tau < 1/2$*, an LPN instance* $I = (A, b, \eta)$*, and a family-uniform represented decoder derivation* $\rho_{\mathcal{D}}$ *of complete saturated cost at most* $B_{\mathcal{D}}$*. Let*

$$\widehat{s} = \mathcal{D}_n(A, b; \omega)$$

be its output, where the random seed ω*, when present, is carried by the public product lift. Immediately after this output is represented, and before the correction transitions below, define*

$$\Theta_{\widehat{s}}(x) = x + \widehat{s}, \qquad D_{\widehat{s}}(x) = |x + \widehat{s}|_1, \tag{1577}$$

$$P_{\widehat{s}}(x, x+e_i) = \frac{1}{n}\mathbf{1}_{\{x_i \neq \widehat{s}_i\}}, \qquad P_{\widehat{s}}(x, x) = 1 - \frac{D_{\widehat{s}}(x)}{n}. \tag{1578}$$

All other transition probabilities are zero. Let $H \geq 1$ *be the charged correction horizon, with public correction-phase initialization* $x_0 = 0$*. Evaluator normalization and computation closure produce one derivation* ρ_{corr}*, containing* $\rho_{\mathcal{D}}$*, the stored output, the public verifier evaluation, the potential and conductance evaluators, the fixed-space current, the correction kernel, and its clocked transcript, with*

$$\mathrm{Cost}_I(\rho_{\mathrm{corr}}) \preceq B_{\mathcal{D}} + mn + n + H(n + \log n). \tag{1579}$$

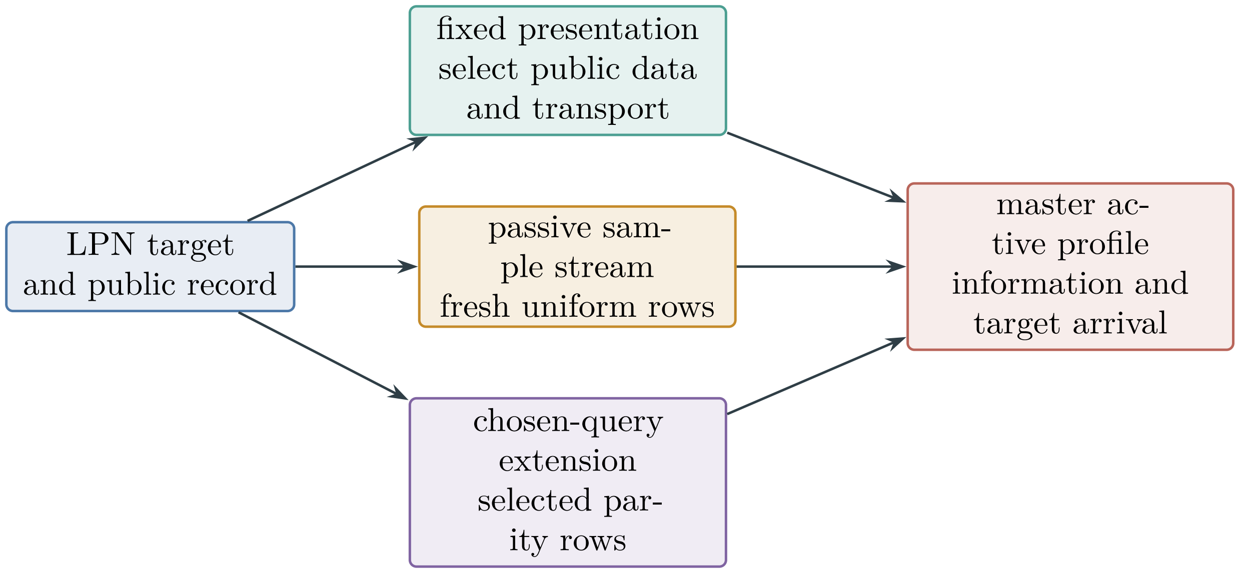

Figure 135: The three LPN acquisition interfaces. Selection inside the fixed public record creates no new information channel. Passive fresh samples and chosen parity queries are declared extensions with different quantitative bounds. Each interface enters the same master active-sampling ledger.

The implicit hypercube relation and its conductance are generated by the coordinate index i; no 2^n-entry table occurs in this bound. In particular, for $m = \Theta(n)$, $B_{\mathcal{D}} \le n^C$, and

$$H = \left\lceil n\left(\log n + \log\frac{1}{\delta}\right)\right\rceil, \qquad 0 < \delta < 1,$$

the complete cost in (1579) *is bounded by one fixed polynomial in n, uniformly over the family.*

This bound constructs the represented hypercube carrier centered at $\widehat{s}$. Target-oriented admissibility is a separate conjunct in the same record. By the progress-observable condition of Definition III.14.9, the potential must vanish on $T_I = \{s\}$, whereas

$$D_{\widehat{s}}(s) = d_H(s, \widehat{s}), \qquad D_{\widehat{s}}(s) = 0 \iff \widehat{s} = s. \tag{1580}$$

Thus the carrier is an admissible target-oriented Geom record precisely on decoder success. When $\widehat{s} \neq s$, it is centered on the wrong state, fails target normalization, and contributes zero under the qualified record convention. Its spectral gap receives no separate target-aligned credit.

On the event $\{\widehat{s} = s\}$, the fixed-space split of $P_{\widehat{s}} - I$ supplies a dynamically qualified identity-supported ambient record with the following exact gate values:

$$q_{\text{amb}} = \text{id}, \qquad M_{\Phi_{\widehat{s}}} = 1, \qquad \chi^{\text{auth}}_{\Phi_{\widehat{s}}}(D_{\widehat{s}}) = 1, \tag{1581}$$

$$\text{Align}^{\text{auth}}_{\mathsf{Caus}}(\mathfrak{K}_{\widehat{s}}, D_{\widehat{s}}) = 1, \qquad R_{\{s\}}(D_{\widehat{s}}) = 1, \qquad \rho_{\Phi_{\widehat{s}}}(D_{\widehat{s}}) = \frac{1}{n}. \tag{1582}$$

Consequently,

$$V^X_{\Phi_{\widehat{s}}, D_{\widehat{s}}, J_{\widehat{s}}, \mathfrak{K}_{\widehat{s}}} = \frac{n}{n+1}. \tag{1583}$$

Combining Eqs. (1580) *and* (1583)*, the qualified target-aligned contribution of this represented construction is exactly*

$$\mathbf{1}_{\{\widehat{s}=s\}} V^X_{\Phi_{\widehat{s}}, D_{\widehat{s}}, J_{\widehat{s}}, \mathfrak{K}_{\widehat{s}}} = \frac{n}{n+1} \mathbf{1}_{\{\widehat{s}=s\}}. \tag{1584}$$

Thus the inverse-polynomial carrier gap, same-generator consistency, authorized alignment, conditional target reach, and regularity factor are all multiplied by the represented target-admissibility indicator. No factor from the incorrectly centered carrier enters the target-aligned profile.

Proof. Retaining the decoder derivation, output, verifier, potential, implicit cube relation, and clocked correction rule gives the displayed polynomial cost. The potential vanishes at the true target exactly on decoder success. On that event direct evaluation of the selected correction kernel gives all five gates and regularity $1/n$; off that event target admissibility is zero. Multiplying the gates yields the exact qualified contribution. □

Theorem VIII.16.2 (Exact decoder level quotient and correction arrival)**.** *Adopt the charged decoder construction of Theorem VIII.16.1.*

The same record exposes the represented level-set quotient

$$q_{\widehat{s}} : X_I \longrightarrow \{0, 1, \ldots, n\}, \qquad q_{\widehat{s}}(x) = D_{\widehat{s}}(x). \tag{1585}$$

On $\{\widehat{s} = s\}$*, this is a nonidentity terminal-compatible exact quotient. Its quotient chain is*

$$w \longmapsto \begin{cases} w-1, & \text{with probability } w/n, \\ w, & \text{with probability } 1 - w/n, \end{cases} \qquad 1 \le w \le n, \tag{1586}$$

and 0 *is absorbing. Thus its lumpability defect is zero, its canonical comparison factor is one, and its target-projection ratio is one. With the displayed horizon and* $\lambda = 1/H$*,*

$$\Pr\{\tau_{\{s\}} > H \mid \widehat{s} = s\} \le n e^{-H/n} \le \delta, \tag{1587}$$

$$V_I(q_{\widehat{s}}) \ge \frac{1-\delta}{eH}. \tag{1588}$$

Proof. When $\widehat{s} = s$, Hamming weight is a strong lumping of the correction kernel and gives the displayed pure-death chain. Each incorrect coordinate survives H updates with probability at most $e^{-H/n}$; a union bound gives the arrival estimate. Substitution into the exact quotient visibility formula with $\lambda = H^{-1}$ gives its lower bound. □

Corollary VIII.16.3 (Family accounting for charged decoder geometry)**.** *Adopt the constructions of Theorems VIII.16.1 and VIII.16.2.*

Let

$$p_{n,\eta}(\mathcal{D}) = \Pr_{A,s,E,\omega}\{\mathcal{D}_n(A, As+E; \omega) = s\},$$

and put

$$\zeta_{n,m,\eta,\tau} = \exp[-m D_{\mathrm{Ber}}(\tau \| \eta)] + (2^n - 1) 2^{-m(1-H_2(\tau))}. \tag{1589}$$

At the complete charged budget

$$B_{\mathrm{corr}} \succeq B_{\mathcal{D}} + mn + n + H(n + \log n),$$

the completed post-output audit therefore satisfies

$$\mathbb{E}\Big[\mathbf{1}_{\{V_I^{-1}(1)=\{s\},\ \widehat{s}=s\}} V^X_{\Phi_{\widehat{s}}, D_{\widehat{s}}, J_{\widehat{s}}, \mathfrak{K}_{\widehat{s}}}\Big] = \frac{n}{n+1} \Pr\Big\{V_I^{-1}(1) = \{s\},\ \widehat{s} = s\Big\}, \tag{1590}$$

$$\mathbb{E}\Big[\mathbf{1}_{\{V_I^{-1}(1)=\{s\},\ \widehat{s}=s\}} V_I(q_{\widehat{s}})\Big] \geq \frac{1-\delta}{eH} \Pr\Big\{V_I^{-1}(1) = \{s\},\ \widehat{s} = s\Big\}. \tag{1591}$$

Thus the decoder cost is an authenticated ancestor of both post-output records, and the complete dependence on the noise law is retained in the represented success mass $p_{n,\eta}(\mathcal{D})$ and the explicit verifier exception (1589)*. These displays audit structure constructed after the decoder output. They do not make that record a prospective source for the decoding transitions that produced $\widehat{s}$. For this particular represented construction, before taking the saturated supremum, the family-functional target-aligned Geom contribution on the verifier good event is exactly*

$$\frac{n}{n+1} \Pr\Big\{V_I^{-1}(1) = \{s\},\ \widehat{s} = s\Big\}. \tag{1592}$$

It lies in the explicit interval

$$\frac{n}{n+1}\big(p_{n,\eta}(\mathcal{D}) - \zeta_{n,m,\eta,\tau}\big)_+ \leq \frac{n}{n+1} \Pr\Big\{V_I^{-1}(1) = \{s\},\ \widehat{s} = s\Big\} \leq \frac{n}{n+1} p_{n,\eta}(\mathcal{D}). \tag{1593}$$

Proof. Intersect decoder success with the verifier event and substitute the exact Geom visibility and quotient lower bound. The verifier complement is bounded by its explicit binomial/code probability. Since the verifier event can remove at most that probability from decoder success, the two-sided family interval follows. Every term retains the decoder derivation as an ancestor, so the conclusion is retrospective. □

Corollary VIII.16.4 (Fully charged post-output LPN geometry and exact level-quotient audit)**.** *The post-output geometry, exact level quotient, and family accounting are Theorems VIII.16.1 and VIII.16.2 and Corollary VIII.16.3. They are retrospective consequences of one charged decoder record and are not prospective sources for the computation that produced its output.*

Proof. The derivation $\rho_{\mathcal{D}}$ is retained as an ancestor when its n-bit output is stored. Matrix–vector multiplication and the public verifier require at most $O(mn)$ Boolean operations. The translation and Hamming potential require $O(n)$ operations. A correction step stores the current n-bit candidate, generates an index using $O(\log n)$ random bits, and applies one comparison and update. Charging the complete clocked state record gives the conservative $H(n + \log n)$ term. The constructor cost rule of Definition III.1.3 proves (1579). The hypercube relation is the uniform evaluator $(x, i) \mapsto x + e_i$, and the conductance coefficient has an $O(n)$-bit encoding, so this construction contains no exponential table.

Equation (1580) is the defining identity for Hamming distance. If $\widehat{s} \neq s$, then $D_{\widehat{s}}(s) > 0$, so the potential fails the target-normalization condition in Definition III.14.9. The centered carrier remains a charged represented object, but it is outside the admissible target-oriented record family and has zero target-aligned profile contribution. This proves the zero branch in Eq. (1584).

Assume $\widehat{s} = s$. Put $D = D_{\widehat{s}}$ and $w = D(x)$. Direct substitution in (1578) gives

$$(P_{\widehat{s}} - I)D(x) = -\frac{w}{n} \leq -\frac{1}{n} \qquad (x \neq s). \tag{1594}$$

The same-generator drift margin $1/n$ activates the authority-compatible local gate. The symmetric part of the fixed-space split has rate $1/(2n)$ on every hypercube edge. Its Walsh eigenvalue on a character indexed by $S \subseteq \{1, \dots, n\}$ is $-|S|/n$; hence its gap is $1/n$ and the carrier gate $M_{\Phi_{\widehat{s}}}$ equals one within this admissible record. The carrier gate has no separate target-aligned contribution. Its directed current is a positive scalar multiple of $-\nabla_{\Phi_{\widehat{s}}} D$, so the authorized alignment is one and its gradient-orthogonal residual projection is zero. Since $D^{-1}(0) = \{s\}$, the centered target indicator belongs to the target-oriented monotone span of D, giving target reach one.

Under the uniform candidate law, D is binomial $\mathrm{Bin}(n, 1/2)$, and therefore

$$\mathrm{Var}_{\mu_I}(D) = \frac{n}{4}.$$

For the symmetric rate $1/(2n)$, every hypercube edge changes D by one, so

$$\mathcal{E}_{\Phi_{\widehat{s}}}(D, D) = \frac{1}{4}, \qquad \rho_{\Phi_{\widehat{s}}}(D) = \frac{1}{n}.$$

These identities prove (1581); substitution into Eq. (1286) proves (1583).

The fibers of $q_{\widehat{s}}$ are the represented Hamming spheres about $\widehat{s}$. Equation (1578) shows that the transition law of the level depends only on its current value w, proving (1586) and exact lumpability. The target is the singleton level 0, while the positive levels strictly refine the verifier partition. Canonical pushforward gives comparison factor one by Proposition III.11.3, and the target indicator is quotient-measurable, so the projection ratio is one.

An initially incorrect coordinate remains unselected after H independent coordinate choices with probability at most $e^{-H/n}$. A union bound over at most n such coordinates proves (1587). Uniformization with $\lambda = 1/H$ then gives

$$A_T^{q_{\widehat{s}}}(1/H) \geq (1 + 1/H)^{-H}(1 - \delta) \geq e^{-1}(1 - \delta).$$

The quotient-utility normalization of Definition III.15.1 and the unit projection ratio yield (1588).

Finally, the good-presentation probability in Eq. (1288) is $1 - \zeta_{n,m,\eta,\tau}$. On that event, decoder success makes both records qualified and gives the two pointwise contributions just computed. The independent seed is a valid product lift, and unsuccessful or unqualified records contribute zero in the displayed audit. Hence the probability of a qualified successful record is at least $(p_{n,\eta}(\mathcal{D}) - \zeta_{n,m,\eta,\tau})_+$. Averaging the two pointwise bounds proves Eqs. (1590) and (1591). The Geom term is exactly the indicator product in Eq. (1584) restricted to the verifier good event. Its expectation is Eq. (1592). The bounds in Eq. (1593) follow from $\Pr(A \cap B) \geq (\Pr(A) - \Pr(B^c))_+$ and $\Pr(A \cap B) \leq \Pr(A)$.

□

Figure 136 distinguishes the charged post-output constructions from the prospective computation that produced the decoder output.

Corollary VIII.16.5 (Quantitative post-output decoder audit). *Under the hypotheses and notation of Corollary VIII.16.4, every represented decoder satisfies*

$$p_{n,\eta}(\mathcal{D}) \leq \zeta_{n,m,\eta,\tau} + \Pr\Big\{V_I^{-1}(1) = \{s\},\ \widehat{s} = s\Big\}, \tag{1595}$$

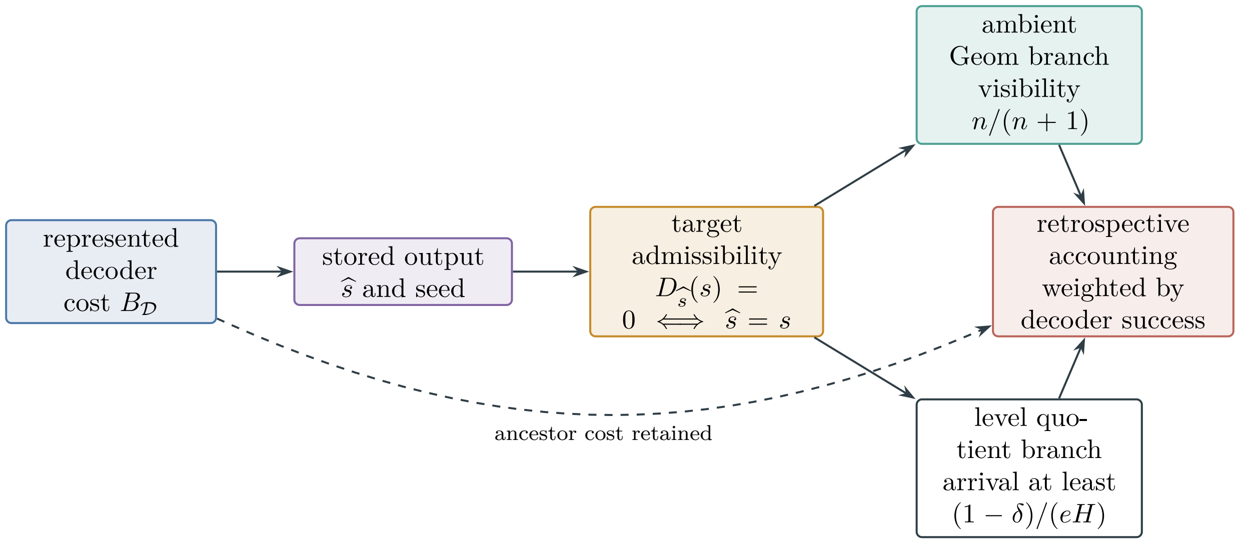

Figure 136: A decoder can cheaply build a target-centered carrier only after it has produced $\widehat{s}$. The target-admissibility gate activates both the ambient geometry and exact level quotient precisely on decoder success. Their visibility therefore audits that success retrospectively and cannot be used as a prospective source for the earlier decoding transitions.

$$p_{n,\eta}(\mathcal{D}) \le \zeta_{n,m,\eta,\tau} + \frac{eH}{1-\delta}\mathbb{E}\Big[\mathbf{1}_{\{V_I^{-1}(1)=\{s\},\ \widehat{s}=s\}} V_I(q_{\widehat{s}})\Big]. \tag{1596}$$

Both use the single complete cost (1579) *and the same authenticated decoder ancestor. They are post-output accounting identities, not bounds by a prospective source profile.*

Proof. Use $p_{n,\eta}(\mathcal{D}) \le \zeta_{n,m,\eta,\tau} + \Pr\{V_I^{-1}(1) = \{s\}, \widehat{s} = s\}$, then apply Eq. (1591) for the second inequality.

□

VIII.16.2 Selected-Current Hodge Classification

Theorem VIII.16.6 (Selected-current Hodge classification in LPN)**.** *Work on the LPN verifier good event and fix a saturated budget B. Let*

$$\mathfrak{g} = (\mathfrak{N}, \Phi, D, J, \mathfrak{K}) \in \mathcal{C}^{\mathrm{LPN}}_{X,b}(B)$$

be a temporally admissible identity-supported record. Apply the channel projection to its actual selected current after the terminal-contact coordinate has been accounted by Theorem VIII.9.8. In the represented weighted edge space, the selected current has the orthogonal decomposition

$$J_{\mathfrak{K}} = \Pi_{\operatorname{im}\nabla_\Phi} J_{\mathfrak{K}} + \big(I - \Pi_{\operatorname{im}\nabla_\Phi}\big) J_{\mathfrak{K}}. \tag{1597}$$

This decomposition is internal to the admitted Geom record. Neither summand supplies a Geom source: the gradient component is derived Caus*, and the orthogonal component is* J_{res}*. In the absence of the underlying target-qualified Geom record, both have zero prospective-source value. The first summand is the* Caus *gradient projection of the selected operator current. It* *contributes prospectively precisely through a represented target-normalized potential, the same-generator authority record, and the complete ambient visibility*

$$M_\Phi\, \chi_\Phi^{\mathrm{auth}}(D)\ \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D)\, R_{\{s\}}(D)\, \frac{1}{1+\rho_\Phi(D)}. \tag{1598}$$

This contribution is charged to the existing $G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B)$ restriction of the ambient coordinate. The directed projection creates no additional source coordinate.

The second summand in Eq. (1597) *is orthogonal to every represented gradient in the budget-B observable or lifted algebra. Consequently, for every represented target-normalized potential D',*

$$\left\langle (I - \Pi_{\mathrm{im}\,\nabla_\Phi}) J_{\mathfrak{K}}, -\nabla_\Phi D' \right\rangle_E = 0. \tag{1599}$$

It therefore has zero authorized Caus alignment and belongs to the post-projection J_{res} component. If a family-uniform affordable potential, current comparison, quotient, lift, or terminal readout later gives this component a channel-qualified target-aligned representative, the least-cost saturated reclassification moves that representative into the corresponding exposed channel before the post-saturation residual is formed.

Quantitatively, if the selected-current gradient contribution of a record exceeds $\varepsilon \in (0,1)$, then

$$M_\Phi = 1, \qquad \chi_\Phi^{\mathrm{auth}}(D) = 1, \qquad \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) > \varepsilon, \qquad R_{\{s\}}(D) > \varepsilon, \qquad \frac{1}{1+\rho_\Phi(D)} > \varepsilon, \tag{1600}$$

$$\left\langle J_{\mathfrak{K}}, -\nabla_\Phi D \right\rangle_E > \sqrt{\varepsilon}\, \|J_{\mathfrak{K}}\|_E \|\nabla_\Phi D\|_E. \tag{1601}$$

Thus an inverse-polynomial gradual contribution is already an inverse-polynomial Caus-qualified Geom record in the single saturated ambient profile. A direction specified only by the semantic statement that it points toward s, without the represented potential values, local differences, selected current, and authority comparison available before the transition, activates neither $\chi_\Phi^{\mathrm{auth}}$ nor $\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}$ *and has zero structural visibility at budget B.*

Together with Theorem VIII.9.8, this exhausts the identity-supported transition record: direct target contact is neutral or charged through an existing source; the selected-current gradient projection is the existing Caus-qualified Geom coordinate; and its gradient-orthogonal complement is J_{res}.

Proof. Theorem II.7.1 applies the represented Hodge projection to the actual selected edge current. Its gradient subspace is $\mathrm{im}\,\nabla_\Phi$; orthogonal projection gives Eq. (1597) and Eq. (1599). By Definition II.4.4 and Proposition II.5.9, the gradient summand is the derived Caus use of a represented potential and conductance. Its dynamic use is admitted only by the same-generator record of Definition III.12.9. Substitution in Definition III.14.9 gives Eq. (1598), so its complete cost and target visibility are already included in $G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B)$.

The orthogonal summand has zero inner product with every gradient in the represented subspace. Its numerator in Definition III.12.8 is therefore zero. The channel decomposition assigns it to J_{res}, and Proposition II.8.3 and Lemmas II.8.4 and II.8.8 gives the stated least-cost reclassification property and excludes a separate affordable target-aligned residual source.

If the visibility in Eq. (1598) exceeds ε, all five factors lie in $[0,1]$, with the first two binary. Each quantitative factor therefore exceeds ε, proving Eq. (1600). The alignment formula in Definition III.12.8 then gives Eq. (1601). When the potential, current identity, polarity, or authority comparison is absent, Definition III.12.9 sets the corresponding authorized factor to zero. Finally, Theorem VIII.9.8 exhausts the removed terminal-contact part.

$\square$

VIII.16.3 Geometric-Source Exclusion

Lemma VIII.16.7 (Absence of an independent geometric component in LPN). *For random binary LPN, no primitive local* Geom *source has inverse-polynomial target visibility. A generated construction whose derivation uses the public residual map retains that map as an authenticated ancestor. Its fiber map is a quotient candidate; it contributes a qualified* Abs *structural source only when its complete record satisfies the framework's target-utility, compatibility, affordability, and temporal-source conditions. Fiber formation alone supplies no* Abs *source classification. Every selected current in the ambient Geom mode is therefore either the derived* Caus *gradient projection of that same represented* Geom *record or a gradient-orthogonal residual term with zero authorized target alignment. There is no additional source represented only by the semantic assertion that a transition points toward the target.*

Proof. Apply the Characterization of transfer-class geometric certificates theorem. The LPN inversion-task definition leaves its relation slot undeveloped and supplies no primitive tuple consisting of a target-oriented pointwise potential, exposed symmetric conductance, positive killed gap, and target reach. The complete residual-independent ambient branch has target visibility at most 2^{-n}, and its prospective selector advantage is zero, by Lemma VIII.8.11. The coefficient-complete exact decomposition Corollary VIII.17.3 assigns every remaining record one positive support–signature normal form. Its complete ambient value is evaluated, without replacing the selected record by an ancestral restriction, in Theorem VIII.20.4 and Corollary VIII.20.8. A construction using target-oriented information from b obtains it through a represented public derivation having the candidate residual map among its authenticated ancestors. The Budgeted derivability closure retains that ancestry and its complete cost. At a declared polynomial saturated ceiling, the construction contributes only when its family-uniform composite record satisfies the cost bound in Eq. (4) and the resulting geometric record passes the admission, temporal-source, and target-visibility conditions in Theorem III.26.2 and Definitions III.21.1 and III.14.9. The possible information-theoretic dependence of the joint public input (A, b) on s supplies none of these conditions: it is an analytic property of the input law, not a represented polynomial-cost readout, potential, kernel, current, or selector. In particular, a target projection of the joint input enters the saturated source profile only through an explicit represented postprocessing whose full ancestry and evaluation cost fit that ceiling. Conditional on such an admitted record, the Affordable composition and target-projection monotonicity theorem assigns a deterministic postprocessing no target projection beyond its represented joint input. Without such a record, the qualified-record convention assigns zero source contribution; information-theoretic correlation alone is not an affordable structural source.

When the construction exposes a fiber map q, the Fiber algebra of a represented quotient theorem describes its quotient-measurable algebra. Structural-source status is then decided by the existing gates: q contributes to the qualified Abs profile precisely when it satisfies Definitions III.15.1 and III.15.2 and, when invoked as a prospective source, Definition III.21.1. A fiber map failing those gates remains a represented postprocessing and is classified only through the channel predicates it satisfies under Theorem II.7.1; it is not reclassified as an Abs source. Hence every b-dependent construction retains its charged derivation provenance, while none becomes an independent primitive Geom source or a qualified Abs source solely by factoring through the residual map. Its complete ambient geometric record is evaluated, without an additional source type, by Corollaries VIII.9.5 and VIII.10.5. When such a derivation first produces a represented estimate $\widehat{s}$ and uses it to center a local target-oriented geometry, Corollary VIII.16.4 retains the complete decoder cost and evaluates every ambient gate in the post-output record. The same potential exposes the exact nonidentity level quotient $q_{\widehat{s}}(x) = d_H(x, \widehat{s})$. The corresponding calculations in Eqs. (1590) and (1591) are accounting audits and supply no prospective-source credit for the decoding transitions that produced $\widehat{s}$. The reversible conductance of this or any other LPN carrier supplies no Sym tag. On the random-code-rigidity event, Corollary VIII.17.6 removes every independent authenticated Sym source and assigns each b-dependent symmetry-breaking construction to its actual charged provenance cells. When such a construction is learned, its target-qualified current is bounded jointly by its generated target projection, authorized gradient component, and decoder qualification probability through Corollary IV.12.2. The transition record supplies no semantic-direction exception. Theorem VIII.9.8 exhausts its terminal-contact part. For the remaining selected current, Theorem VIII.16.6 identifies every represented actionable direction with the already charged Caus projection of the complete Geom record; the orthogonal current has zero pairing with every represented gradient and remains J_{res}. Since Caus is derived by Proposition II.5.9, it contributes no independent source beyond that Geom record. A direction lacking the represented potential, local differences, selected current, and same-generator authority comparison has zero authorized alignment by Definition III.12.9.

□

Lemma VIII.16.8 (Absence of compensated quotient–geometric sources in LPN)**.** *Work on the full-rank and random-code-rigidity event of Lemma VIII.17.5. At every budget in the declared polynomial saturated ceiling,*

$$G^{\mathrm{src,sat}}_{Q,\mathrm{rev},n}(B) = G^{\mathrm{src,sat}}_{Q,\mathrm{nr},n}(B) = G^{\mathrm{src,sat}}_{Q,\mathrm{ql},n}(B) = 0. \tag{1602}$$

Proof. By Theorem II.9.4 and Definition III.18.4, a record in either compensated mode contains a represented nonidentity quotient q, its complete five-family derivation, a positive target-qualification factor, and a reversible or general-resolvent stability factor. The stability factor compares the quotient and ambient dynamics; it creates neither quotient fibers nor target alignment.

Apply the exhaustive five-family charge decomposition Corollary III.16.11 to the represented quotient map and its qualification record. The Mult and Ord charges vanish by Lemmas VIII.17.7 and VIII.17.8. Filtration and valid-Lift vertices add zero incremental charge and return inherited charge to an earlier represented ancestor by Lemmas VIII.18.1 and VIII.8.3. On the rigidity event, authenticated Sym ancestry is identity-equivalent by Lemma VIII.17.5 and Corollary VIII.17.6. The remaining Add alternative is identity-equivalent by Lemma VIII.17.2. Mixed derivations retain these same contextual charges and introduce no sixth alternative.

Thus every target-qualified quotient support is identity-equivalent or has zero prospective source charge. Identity support belongs to the ambient mode X, whereas both compensated modes require $q \neq \mathrm{id}$. Hence each compensated source family is empty. Taking the two suprema and their maximum proves Eq. (1602).

□

VIII.17 Five-Family Provenance and Algebraic Source Classes

The universal five-family normal form now specializes the exact abstraction channel to the public LPN presentation.

Corollary VIII.17.1 (Five-family Abs provenance for LPN)**.** *Every exact quotient evaluator admitted to the budgeted LPN quotient ledger has additive, multiplicative, order, symmetry, or filtration provenance in the sense of five-family Abs evaluator provenance. Mixed derivations retain* *all applicable provenance tags and their contextual joint charge. The complete LPN quotient profile obeys the aggregate and individual bounds of the exhaustive five-family Abs envelope.*

This is the LPN instance of the framework decomposition. The five families are the provenance families of the universal normal form, not an LPN-specific catalogue of algorithms or constructions.

Proof. Apply the universal five-family Abs provenance normal form and its saturated-profile envelope to the budget-complete LPN presentation.

□

VIII.17.1 Additive and Symmetry Sources

Target alignment reduces both represented additive and authenticated symmetry quotients to identity support.

Lemma VIII.17.2 (Additive LPN quotients are identity-equivalent)**.** *Let $q : X_I \to Q$ be a budget-admissible quotient map occurring either in an exact* Abs *record or as the support quotient of a compensated Geom record. Suppose its terminal quotient map has purely* Add *state-dependent provenance in the universal five-family Abs provenance normal form. Thus, after an affordable relabeling of its image, its terminal quotient map is affine in the candidate variable:*

$$q(x) = L_I x + c_I, \qquad x \in X_I = \mathbb{F}_2^n.$$

Every such q satisfying the quotient-utility condition for the LPN target $T_I = \{s\}$ obeys

$$\ker L_I = \{0\}.$$

Consequently q is injective and its nonempty-fiber partition is the identity partition up to the represented affine relabeling. Hence the LPN presentation has no proper target-qualified quotient support in the Add *channel, whether exact or compensated.*

Proof. The terminal-event clause of the quotient-utility condition requires T_I to be a union of quotient fibers. Since $T_I = \{s\}$, the nonempty fiber containing s must equal $\{s\}$. Affinity gives

$$q^{-1}(q(s)) = s + \ker L_I.$$

Therefore $s + \ker L_I = \{s\}$, so $\ker L_I = \{0\}$. The map q is injective, and every nonempty fiber is a singleton. Its partition is therefore the identity partition. Any proper additive quotient has a nonzero kernel, so its target-containing fiber strictly contains s and fails the terminal-event clause. The proof uses the terminal-event gate and the affine fiber identity, not exact intertwining. It therefore applies to the exact and compensated quotient-support modes selected by the framework decomposition.

□

Corollary VIII.17.3 (Coefficient-complete exact Geom decomposition for LPN)**.** *Every temporally admissible dynamically qualified LPN Geom source belongs to exactly one of the 124 cells*

$$(\sigma, S) \in \mathfrak{S}^4_{\mathsf{Geom}} \times \mathfrak{P}_{\mathsf{Geom},5}$$

of Theorem III.2.5. A represented calculation from (A, b, η) that uses a product, equality or fiber operation, comparison or selection, authenticated group action, or stateful iteration occurs instead *in an exact signature containing respectively* Mult*,* Ord*,* Sym*, or* Filt*. Its complete coefficient derivation remains in that cell's active frontier and saturated cost.*

The provenance tag Add *records ancestry within the existing ambient Geom mode. Every record using an additional constructor retains that constructor in its exact provenance signature. The complete ambient record is evaluated by Theorem VIII.20.4 and Corollary VIII.20.8. No Hodge-orthogonal current occurs in an ambient Geom cell.*

Proof. Apply Theorem III.2.5 to the complete LPN active record. Evaluator normalization assigns every listed non-Add operation its corresponding tag, including when the operation computes an instance-dependent coefficient. The complete-record evaluation is Theorem VIII.20.4; the current assignment is the one in Theorem III.2.5.

□

Lemma VIII.17.4 (Authenticated LPN symmetries induce code automorphisms)**.** *Fix* $0 < \eta < 1/2$*. Let* G *be a represented group action whose authenticated LPN action record occurs either in a budget-admissible exact quotient or in the complete active ancestral set of a dynamically qualified Geom record. For every* $g \in G$*, the authenticated action has the form*

$$g(x) = M_g x + h_g, \qquad P_g \in \mathrm{Sym}(m),$$

and residual equivariance is equivalent to

$$AM_g = P_g A, \qquad Ah_g + b = P_g b. \tag{1603}$$

Consequently

$$P_g \mathcal{C}_A = \mathcal{C}_A,$$

so P_g *is a permutation automorphism of the public code. In particular, this conclusion applies to every exact orbit quotient* $q_G : X_I \to X_I/G$ *with purely* Sym *terminal provenance in the universal five-family normal form.*

Proof. The Sym authentication clause requires the represented action to preserve the active public LPN interface. Preservation of the candidate affine interface gives $g(x) = M_g x + h_g$ with M_g invertible. Preservation of the Bernoulli product and Hamming sample interface gives a coordinate permutation P_g: Hamming isometries may also flip coordinates, but such flips do not preserve $\mathrm{Bernoulli}(\eta)^{\otimes m}$ when $0 < \eta < 1/2$. Equivariance of the public residual primitive is

$$r_I(gx) = P_g r_I(x) \qquad (x \in \mathbb{F}_2^n).$$

Substituting $r_I(x) = Ax + b$ and comparing the linear and constant terms gives (1603). Since M_g is invertible,

$$P_g \mathcal{C}_A = P_g \operatorname{Im}(A) = \operatorname{Im}(AM_g) = \operatorname{Im}(A),$$

which proves the claim.

□

Lemma VIII.17.5 (Symmetry-derived LPN quotients are identity-equivalent)**.** *For uniform* $A \in \mathbb{F}_2^{m \times n}$ *at fixed rate* $n/m \to R \in (0, 1)$*, with probability* $1 - o(1)$*, every budget-admissible quotient support occurring in an exact or compensated LPN record with purely* Sym *terminal* *provenance and satisfying the quotient-utility condition is identity-equivalent. Hence the random LPN presentation has no proper target-qualified exact or compensated quotient support in the* Sym *channel with high probability.*

Proof. With probability $1 - o(1)$, the matrix A has full column rank and the public code $\mathcal{C}_A$ has trivial permutation automorphism group: the rank-failure probability is at most 2^{n-m}, and random-code rigidity gives the second event at fixed rate $R \in (0, 1)$ [50]. Work on their intersection and let q_G be such a quotient. By the authenticated LPN symmetry lemma, every $g \in G$ supplies a permutation automorphism P_g of $\mathcal{C}_A$. Triviality of the permutation automorphism group gives $P_g = I$. Equation (1603) then becomes

$$A(M_g - I) = 0, \qquad Ah_g = 0.$$

Full column rank gives $M_g = I$ and $h_g = 0$. Thus every element of G acts trivially on X_I. In the five-family normal form, a purely Sym terminal map can obtain state dependence only from its authenticated action constructors. With every such action trivial, orbit maps and averaging operators are identity-equivalent; an isotypic output is either the trivial component, hence identity-equivalent, or state-independent, in which case it fails the quotient-utility condition for the singleton target. Equivariant relabeling does not change fibers. Therefore every admissible pure Sym quotient has the identity partition.

The intersection event has probability $1 - o(1)$, which proves the stated random-presentation conclusion.

□

For later family-functional bookkeeping, define the rigidity failure probability

$$\varepsilon_{\text{rig}}(n, m) := \Pr_A\{\text{rank}(A) < n \text{ or } \text{Aut}_{\text{perm}}(\mathcal{C}_A) \neq \{I\}\}\,. \tag{1604}$$

The proof of Lemma VIII.17.5 gives

$$\varepsilon_{\text{rig}}(n, m) \leq 2^{n-m} + \varepsilon_{\text{aut}}(n, m), \qquad \varepsilon_{\text{aut}}(n, m) = o(1) \tag{1605}$$

at fixed rate $n/m \to R \in (0, 1)$. Here ε_{aut} is the proportion of full-rank binary $[m, n]$ codes having a nontrivial permutation automorphism, whose vanishing is the random-code rigidity theorem [50]. This notation records the exceptional presentation probability; it is not a new structural coordinate.

Corollary VIII.17.6 (Closure of symmetry provenance in LPN geometry)**.** *On the full-rank and random-code-rigidity event of Lemma VIII.17.5, every authenticated* Sym *action in the complete active ancestral set of an LPN geometric record acts trivially on the candidate and residual interfaces. It therefore supplies neither an independent* Sym*-derived Geom source nor a nonidentity* Sym*-derived quotient.*

The edge-reversal identity $c_\Phi(x, z) = c_\Phi(z, x)$ *remains the reversible carrier condition and supplies no* Sym *provenance. A represented b-dependent center, potential, conductance, ordering, or current that lacks an authenticated equivariant LPN action retains its complete derivation cost and its actual* Add*,* Mult*,* Ord*, or* Filt *tags. Its target contribution is evaluated in those existing provenance cells by the full Geom gate product. When the record is learned and target qualification entails correct recovery, its admitted current contribution also obeys Eqs.* (265) *and* (266)*.*

Proof. By Lemma VIII.17.4, every authenticated LPN action induces a permutation automorphism of the public code. Random-code rigidity makes that automorphism the identity, and full column rank then gives $M_g = I$ and $h_g = 0$ exactly as in the proof of the LPN rigidity theorem. Apply Theorem III.4.2: trivial authenticated actions have singleton orbits and contribute no independent Sym source data, while symmetric conductance alone receives no Sym tag.

The exhaustive complete-ancestry classification Theorem III.2.4 retains every other constructor tag and the complete represented cost. Its ambient target contribution is the product in Definition III.14.9. For a learned record whose qualification is correct target recovery, Corollary IV.12.2 gives the two displayed current ceilings.

□

VIII.17.2 Multiplicative and Order Sources

The multiplicative and order families are evaluated through their contextual charge on the same public algebra.

Lemma VIII.17.7 (Absence of an independent multiplicative source in LPN)**.** *For every binary LPN instance and every admissible budget B, the multiplicative contextual-charge envelope of the five-family frontier vanishes:*

$$\mathcal{C}^{(\mathsf{Mult})}_{I,\mathfrak{F}_5}(B) = 0.$$

Thus the LPN presentation has no independent Mult *source, including in mixed-provenance quotient derivations.*

Proof. The state-dependent primitives of the LPN task presentation are the candidate coordinates and the affine residual map $r_I(x) = Ax + b$; both have Add provenance. The public data A, b, and η are state-independent coefficients. Derived residual weights and comparisons have Ord provenance, authenticated actions have Sym provenance, and clocked updates have Filt provenance. Hence the LPN primitive signature contains no state-dependent primitive with Mult provenance.

Every Mult-tagged vertex in the five-family frontier is therefore a derived multiplicative constructor. By the derived constructor zero-charge lemma, each such contextual extension has charge zero. This remains true when the vertex carries additional provenance tags, because the charge is attached to the same conditional algebra extension. Summing over the multiplicative extensions and taking the supremum in the definition of the aggregate envelope gives the displayed identity for every B.

□

Lemma VIII.17.8 (Absence of an independent order source in LPN)**.** *For every binary LPN instance and every admissible budget B, the order contextual-charge envelope of the five-family frontier vanishes:*

$$\mathcal{C}^{(\mathsf{Ord})}_{I,\mathfrak{F}_5}(B) = 0.$$

Thus the LPN presentation has no independent Ord *source, including in mixed-provenance quotient derivations.*

Proof. By the LPN task presentation, the only state-dependent generators are the candidate coordinates and the affine residual map $r_I(x) = Ax + b$, which have Add provenance. The public data A, b, and η are state-independent coefficients. Every weight, comparison, branch, selection, metric or quadratic comparison, and ordered reduction in the five-family frontier is therefore a derived Ord constructor.

The derived constructor zero-charge lemma gives zero contextual charge for every such extension. This includes the affordable target-aligned Hamming statistic $x \mapsto |Ax + b|_1$: it is a represented postprocessing of the earlier residual observable and does not enlarge its conditional algebra. The same argument applies when an order vertex carries additional provenance tags. Summing over the order extensions and taking the supremum in the aggregate envelope gives the displayed identity for every B.

□

VIII.18 Filtration, Quotient, and Reversible-Coarsening Closure

Filtration constructors preserve the charged ancestry of their generated flags, histories, and first-hit records.

Lemma VIII.18.1 (Absence of an independent filtration source in LPN)**.** *For every binary LPN instance and every admissible budget, filtration provenance supplies no independent generative source. Under the Filtration provenance charging theorem, a* Filt*-tagged contextual extension has zero incremental charge or leaves the charge on an earlier represented ancestor in the completed accounting DAG. That ancestor is a prospective source only at a use satisfying temporal source admissibility and the represented comparison of Corollary III.16.9. This classification includes clocks, memory states, recurrences, outcome and terminal transcripts, stored verifier flags, and exact first-hit records.*

Proof. The LPN inversion-task definition contains no primitive state-dependent filtration source. Every clock, memory state, history, outcome record, and stored flag is generated by a represented rule. The budgeted saturated represented-closure definition retains that rule's complete update, verifier, state-dependent arguments, transcript, and readout at their accumulated cost.

Apply the Filtration provenance charging theorem. Its deterministic-constructor case is implemented by the Derived evaluator constructors carry zero contextual charge lemma; hence a represented clock, stored copy, memory update, recurrence step, or history coordinate has zero incremental contextual charge. Its valid-Lift case is implemented by the Part II exact Lift identity specialized to LPN; hence an auxiliary filtration-lift coordinate has zero incremental charge and its target contribution remains on the represented base record.

For an operational outcome or terminal transcript, Theorem III.2.2 retains the generating update, independent outcome coordinates, public verifier, stored flags, and terminal map in one charged topological derivation. Independent outcome coordinates are valid-Lift auxiliaries and add zero target charge by Lemma VIII.8.3. Once those coordinates and the earlier state are exposed, every update, stored flag, and terminal record is a represented deterministic postprocessing and adds zero contextual charge by Lemma III.17.1. The exact contextual charge identity of Corollary III.16.11 therefore leaves every nonzero charge on an earlier authenticated vertex of the completed derivation. It transfers that charge to a prospective source only under the separate hypotheses of Corollary III.16.9 and Definition III.21.1.

The public LPN verifier and the exact first-hit map are deterministic postprocessings of the candidate record and completed flags. They add no source by Lemmas III.17.2 and I.4.7. The flag-based first-hit representation is retrospective by Corollary III.23.1; an independent pre-hit representation is instead charged and classified by Corollary III.23.2. Hence no Filt opcode supplies independent LPN source credit.

□

Corollary VIII.18.2 (Exhaustion of independent exact Abs sources in LPN)**.** *Fix $0 < \eta < 1/2$ and a sample schedule with $n/m \to R \in (0,1)$. On the full-rank and random-code-rigidity event, every temporally admissible, target-qualified, budget-admissible exact* Abs *record is covered by the following exhaustive alternatives:*

(i) *its* Add *source is identity-equivalent;*
(ii) *its* Mult *contextual charge is zero;*
(iii) *its* Ord *contextual charge is zero;*
(iv) *its authenticated* Sym *source is identity-equivalent; or*
(v) *its* Filt *vertex has zero incremental charge or leaves that charge on an earlier represented source ancestor.*

Mixed-provenance records retain every applicable alternative in the same contextual charge decomposition. Identity-equivalent records belong to the ambient-support case of Theorem II.9.4; they supply no proper exact Abs *source. Consequently the LPN presentation supplies no independent proper exact* Abs *source at the declared ceiling. A completed first-hit quotient remains in the separate accounting profile of Definition III.22.3 and receives no prospective-source credit by Corollary III.23.1. Equivalently, on the stated event and at every budget in the declared ceiling,*

$$m_{I_n}^{\mathrm{src,sat}}(B) = 0. \tag{1606}$$

Proof. The exhaustive five-family envelope Corollaries VIII.17.1 and III.16.11 assigns every contextual charge to at least one of Add, Mult, Ord, Sym, and Filt. Apply Lemmas VIII.18.1, VIII.17.2, VIII.17.5, VIII.17.7 and VIII.17.8 in that order.

The Mult and Ord envelopes vanish exactly. The Filt result returns every nonzero inherited charge to an earlier represented ancestor; iteration terminates because the retained derivation is a finite directed acyclic graph. A valid-Lift auxiliary coordinate has zero incremental charge by Lemma VIII.8.3. Thus a generated carrier creates no additional exact-Abs alternative. The remaining authenticated Add and Sym alternatives are identity-equivalent by the two cited LPN seals. The master normal form Theorem III.18.9 assigns identity support to the ambient Geom mode rather than to the nonidentity exact-Abs mode. These conclusions exhaust pure and mixed provenance. Temporal source admissibility and Corollary III.23.1 give the final accounting statement.

□

Corollary VIII.18.3 (Provenance restriction of quotient-supported LPN geometry)**.** *Adopt the fixed-positive-noise, fixed-rate setting of Corollary VIII.18.2, work on its good-presentation event, fix the presented instance I_n used there, and let $B^{\sharp}_{\mathrm{poly}}$ be the common charged normalization ceiling. Then the pointwise qualified source profiles satisfy*

$$G_{Q,\mathrm{ex},n}^{\mathrm{src,sat}}(b_{\mathrm{poly}}(n)) \le m_{I_n}^{\mathrm{src,sat}}(b_{\mathrm{poly}}(n)) = 0, \tag{1607}$$

$$\max\Big\{G_{Q,\mathrm{rev},n}^{\mathrm{src,sat}}(b_{\mathrm{poly}}(n)), G_{Q,\mathrm{nr},n}^{\mathrm{src,sat}}(b_{\mathrm{poly}}(n))\Big\} = 0. \tag{1608}$$

Thus every nonidentity quotient-supported Geom mode vanishes. Identity support belongs to the ambient mode X*.*

Proof. The prospective pointwise exact-support comparison Theorem III.19.1 gives the inequality in Eq. (1607); Eq. (1606) gives its zero endpoint.

The compensated-source seal Eq. (1602) gives Eq. (1608). Valid-Lift and generated carriers are already included in that seal through their retained ancestry and zero-incremental-charge identities.

□

Corollary VIII.18.4 (Sharp reversible geometric-coarsening bounds for LPN)**.** *Adopt the fixed-positive-noise, fixed-rate setting and good-presentation event of Corollary VIII.18.3, and evaluate this corollary at one presented instance I. At a saturated budget B, put*

$$\alpha^{\mathrm{rev}}_{\mathrm{LPN},n}(B) = \max\Big\{G^{\mathrm{src,sat}}_{\mathrm{LPN},Q,\mathrm{ex},n}(B), G^{\mathrm{src,sat}}_{\mathrm{LPN},Q,\mathrm{rev},n}(B)\Big\}. \tag{1609}$$

At $B = b_{\mathrm{poly}}(n)$, it is bounded by

$$\alpha^{\mathrm{rev}}_{\mathrm{LPN},n}(b_{\mathrm{poly}}(n)) = 0$$

by Eqs. (1607) *and* (1608)*.*

Let $q \neq \mathrm{id}$ be any represented terminal-compatible coarsening of a reversible LPN geometric record whose complete master-certificate data have cost at most B. Let $K = -L_N$ be its killed normalized operator, let $\gamma > 0$ be a represented lower spectral bound for K, and retain the represented discount $\lambda > 0$. Then $K = I - P_N$ and self-adjoint sub-Markov contractivity give

$$\gamma I \preceq K \preceq 2I. \tag{1610}$$

Consequently,

$$\overline{A}_q\, M_{\mathfrak{G}_q}\, C_{\mathfrak{G}_q}\, \frac{R_{\mathfrak{G}_q}}{1+\rho_{\mathfrak{G}_q}} \le \alpha^{\mathrm{rev}}_{\mathrm{LPN},n}(B)\frac{2+\lambda}{\gamma+\lambda}. \tag{1611}$$

Define the clipped obstruction scale

$$\beta^{\mathrm{rev}}_{\mathrm{LPN},n}(B;\gamma,\lambda) = \min\Big\{1, \alpha^{\mathrm{rev}}_{\mathrm{LPN},n}(B)\frac{2+\lambda}{\gamma+\lambda}\Big\}. \tag{1612}$$

For every $a, c, r, g \in (0,1]$ with

$$acrg > \beta^{\mathrm{rev}}_{\mathrm{LPN},n}(B;\gamma,\lambda), \tag{1613}$$

the coarsening cannot satisfy simultaneously

$$M_{\mathfrak{G}_q} = 1, \quad \overline{A}_q > a, \quad C_{\mathfrak{G}_q} > c, \quad R_{\mathfrak{G}_q} > r, \quad \frac{1}{1+\rho_{\mathfrak{G}_q}} > g. \tag{1614}$$

The symmetric specialization is

$$M_{\mathfrak{G}_q} = 0 \quad \textit{or} \quad \min\left\{\overline{A}_q, C_{\mathfrak{G}_q}, R_{\mathfrak{G}_q}, \frac{1}{1+\rho_{\mathfrak{G}_q}}\right\} \le \Big(\beta^{\mathrm{rev}}_{\mathrm{LPN},n}(B;\gamma,\lambda)\Big)^{1/4}. \tag{1615}$$

These bounds combine with the existing geometric evaluations as follows. Whenever $C_{\mathfrak{G}_q} > c$, the local-consistency gate equals one and the authorized current satisfies

$$\Big\langle J_{\mathfrak{K}_q}, -\nabla_{\Phi_Q} D_Q\Big\rangle_E > \sqrt{c}\,\|J_{\mathfrak{K}_q}\|_E \|\nabla_{\Phi_Q} D_Q\|_E. \tag{1616}$$

The reach and regularity gates obey

$$R_{\mathfrak{G}_q} > r \Longrightarrow \mathrm{ProjMass}_{\sigma(D_Q)}(T_Q) > r, \tag{1617}$$

$$\frac{1}{1+\rho_{\mathfrak{G}_q}} > g \Longrightarrow \rho_{\mathfrak{G}_q} < g^{-1} - 1. \tag{1618}$$

At the ambient level, an already admitted verifier-potential record satisfies the exact formula in Eq. (1289)*; its* Caus *factor is derived from that same Geom record by Proposition II.5.9. Thus a coarsening that retains the same inverse-polynomial source scale must obey Eqs.* (1616) *to* (1618) *while Eq.* (1611) *forbids their simultaneous retention above the obstruction scale.*

At the polynomial saturated ceiling,

$$\beta^{\mathrm{rev}}_{\mathrm{LPN},n}(b_{\mathrm{poly}}(n); \gamma, \lambda) = 0, \tag{1619}$$

and its fourth root is the corresponding symmetric obstruction in Eq. (1615)*.*

Proof. For a self-adjoint sub-Markov contraction P_N, its spectrum is contained in $[-1, 1]$. Hence the spectrum of $K = I - P_N$ is contained in $[0, 2]$, and the represented killed-gap comparison supplies the lower bound $\gamma I \preceq K$. This proves Eq. (1610).

Apply Eq. (75) with $\Lambda = 2$ and $\alpha_n = \alpha^{\mathrm{rev}}_{\mathrm{LPN},n}(B)$. This proves Eq. (1611). The asymmetric exclusion and symmetric fourth-root obstruction are Eqs. (78) and (79) under the same substitution.

Since $C_{\mathfrak{G}_q} = \chi^{\mathrm{auth}}_{\Phi_Q}(D_Q)\,\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}_q, D_Q)$, the inequality $C_{\mathfrak{G}_q} > c$ forces the binary consistency gate to equal one and the alignment factor to exceed c. Definition III.12.8 then gives Eq. (1616). Projection monotonicity in Lemma I.4.6 gives Eq. (1617); the regularity implication is algebraic. The ambient verifier assertions are Corollary VIII.10.6.

At the polynomial ceiling, Eqs. (1607) and (1608) sets the pointwise quotient-supported envelope to zero at the same fixed presentation. Moreover, $0 < \gamma \le 2$ and $\lambda > 0$ give

$$\frac{2+\lambda}{\gamma+\lambda} \le \frac{2}{\gamma},$$

so the transfer factor is polynomial whenever γ^{-1} is. Substitution in Eq. (1612) proves Eq. (1619); taking the fourth root gives the symmetric statement.

□

VIII.19 Residual Ambient Geometry and Five-Family Affordability

Corollary VIII.19.1 (Exhaustive reversible-carrier classification in LPN)**.** *On the good-presentation event, every temporally admissible dynamically qualified LPN geometric source has a reversible* Geom *carrier. For an identity-supported residual-ancestry record, every directed contribution to target progress is either the authorized* Caus *projection appearing in the existing ambient visibility*

$$M_\Phi\,\chi^{\mathrm{auth}}_{\Phi}(D)\ \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D)\, R_{\{s\}}(D)\,\frac{1}{1+\rho_\Phi(D)},$$

or the gradient-orthogonal J_{res} component with zero authorized alignment. There is no third, nonreversible ambient-geometry branch.

For a nonidentity quotient, use of a general-resolvent factor $S^{\mathrm{nr}}_{q,T}(\lambda)$ remains a quotient-supported Geom/Caus *transfer record and belongs to the general-resolvent family already bounded in Eq.* (1608)*. It supplies no independent ambient source. Consequently, after quotient-supported exhaustion, the identity-supported residual-ancestry profile in Eq.* (1620) *is the displayed ambient*

LPN coordinate, with all directed target alignment already included in its Caus *gate. That coordinate is charged directly to the saturated ambient* Geom *profile; absence of an effective nonidentity coarsening does not move it to J_{res} or create an uncharged case.*

Proof. Apply Theorem II.4.6 to the selected LPN generator. Its pairing identities specialize to Eqs. (1597) and (1599), and Theorem VIII.16.6 identifies the first summand with the displayed ambient visibility and the second with J_{res}. For $q \neq \text{id}$, the master normal form places a general-resolvent comparison in the quotient-supported family, and Corollary VIII.18.3 gives its stated bound. The three ambient alternatives are therefore exhausted by the cited channel and quotient classifications.

□

Proposition VIII.19.2 (Exhaustive reduction to classified ambient provenance)**.** *Fix a presented instance I_n on the good-presentation event. Its temporally admissible, dynamically qualified geometric source profile is the disjoint union of the following branches:*

(i) *identity-supported residual-independent ambient geometry, belonging to $\mathcal{C}_{X,0}^{\text{LPN}}$;*

(ii) *nonidentity quotient-supported geometry, belonging to $\mathcal{C}_{Q,\text{ex}}$, $\mathcal{C}_{Q,\text{rev}}$, or $\mathcal{C}_{Q,\text{nr}}$;*

(iii) *identity-supported residual-ancestry ambient geometry, belonging to $\mathcal{C}_{X,b}^{\text{LPN}}$, with its complete source ancestry classified by Corollary VIII.8.10.*

The family-average evaluation of the first branch satisfies (1233)*. The pointwise second branch satisfies Eqs.* (1607) *and* (1608)*. At the polynomial ceiling the complete pointwise geometric profile satisfies*

$$G_n^{\text{src,sat}}(b_{\text{poly}}(n)) = G_{X,n}^{\text{src,sat}}(b_{\text{poly}}(n)). \tag{1620}$$

Branch (iii) *has the exact public-ancestry local form of Theorem VIII.10.2. At each charged transition, that theorem writes its selected sampler and potential difference exactly as*

$$P_I((x,\xi),(x+u,\xi')) = \pi_I(u,\xi' \mid A, x, r_I(x), \xi)$$

and

$$\widehat{D}_I(A, x+u, r_I(x)+Au, \xi') - \widehat{D}_I(A, x, r_I(x), \xi).$$

These are reparameterizations of the same represented record. The public presentation supplies A, x, and $r_I(x)$; the target-cancelling displacement $u = x+s$ is absent from its primitive signature. A positive mass assigned to that displacement is therefore counted only through the represented pre-hit sampler and its existing target-alignment comparison, as required by Theorem VIII.9.8 and Corollary VIII.10.1.

The residual-weight likelihood factors through the terminal-compatible (V_I, W_I) quotient and is exhausted by the exact, compensated, and identity-support alternatives of Theorem VIII.9.6. Its ambient identity-support refinement obeys the simultaneous off-target bound Eq. (1244)*; its only macroscopic descent is the target-cancelling displacement in Eq.* (1245)*. A different represented potential receives precisely the complete ambient gate product of Eq.* (1286)*, evaluated on the same sampler, local differences, current, authority record, reach comparison, regularity comparison, and complete cost. Operational products, comparisons, mixtures, clocks, valid lifts, and derived* Caus *records retain their signatures and costs while contributing zero independent target charge by Lemma III.17.1 and Proposition II.5.9.*

For a nonlinear full-carrier use of this record, Corollary V.8.4 evaluates the selected kernel before the source ancestry is charged backward. In the uniform-invariant native case, Theorem VIII.11.1 *gives its exact singleton capacity, finite-horizon entrance bound, invariant drift ceiling, and the noise-dependent conversion from residual descent to target contact. The canonical public carrier is evaluated in every functional-inequality slot by Corollary VIII.12.4; a learned carrier retains the operator and population-transfer correction of Theorem VIII.12.6. These are numerical outputs of the same ambient* $\mathsf{Geom}/\mathsf{Caus}$ *record and preserve its complete ancestry and cost.*

A target-qualified nonidentity spectral or level-set coarsening lies in the existing quotient-supported modes and satisfies Eqs. (1607)*,* (1608) *and* (1611)*. With identity support, the authorized* Caus *projection is the complete directed part of the ambient record, and its gradient-orthogonal complement is* J_{res}*, by Corollaries VIII.19.1 and II.8.5. Thus Eq.* (1620) *is exactly the maximum of the displayed budget-admissible, target-qualified structural records.*

Proof. By Theorems II.9.5 and III.18.9, every dynamically qualified geometric record has identity support, exact nonidentity quotient support, reversible compensated support, or general resolvent-compensated support. The last three alternatives give branch (ii). By Definition VIII.8.9, the identity-supported family is the disjoint union of branches (i) and (iii). This proves exhaustiveness and disjointness.

For branch (iii), Corollary VIII.8.10 applies the framework source classification to the complete active tuple. Thus the binary residual-ancestry partition has no role as an additional structural classification: every record in that branch already lies in the ambient row of the exhaustive five-family provenance cover. The exact substitution and displacement pushforward of Theorem VIII.10.2 express its local data using only the declared LPN presentation and preserve the same qualification gates and complete cost.

The family-average estimate for branch (i) is Lemma VIII.8.11; the pointwise quotient estimate for branch (ii) is Corollary VIII.18.3. All three branches are already among the five master modes of the saturated geometric profile. The quotient-supported modes vanish by Eqs. (1602) and (1607); the remaining identity-supported modes comprise the ambient coordinate. This proves Eq. (1620). □

Theorem VIII.19.3 (Quantitative five-family affordability reduction for LPN geometry)**.** *Fix the LPN family functional* $\mathfrak{M}_{n,m,\eta}$*. For* $\sigma \in \mathfrak{S}^4_{\mathsf{Geom}}$ *and* $c \in \mathfrak{J}^5_{\mathsf{Abs}}$*, write*

$$G^{\mathrm{src,sat}}_{\mathrm{LPN},\sigma,c,\mathfrak{M},n}(B)$$

for the LPN specialization of the provenance restriction in Definition III.2.3. This is a restriction of the existing prospective Geom profile, evaluated by $\mathfrak{M}_{n,m,\eta}$*. Put* $\widehat{B} = \Psi_{\mathsf{Geom},5}(B)$*. The exhaustive family-functional bound is*

$$\begin{aligned} G^{\mathrm{src,sat}}_{\mathrm{LPN},\mathfrak{M},n}(B) \le \max\Bigg\{ & 2^{-n}, \max_{c\in\mathfrak{J}^5_{\mathsf{Abs}}} G^{b,\mathrm{src,sat}}_{\mathrm{LPN},\mathsf{X},c,\mathfrak{M},n}(\widehat{B}), \\ & G^{\mathrm{src,sat}}_{\mathrm{LPN},Q,\mathrm{ex},\mathfrak{M},n}(\widehat{B}), G^{\mathrm{src,sat}}_{\mathrm{LPN},Q,\mathrm{rev},\mathfrak{M},n}(\widehat{B}), G^{\mathrm{src,sat}}_{\mathrm{LPN},Q,\mathrm{nr},\mathfrak{M},n}(\widehat{B}) \Bigg\}. \end{aligned} \tag{1621}$$

Here the superscript b restricts the ambient support row to $\mathcal{C}^{\mathrm{LPN}}_{X,b}$*. For each provenance tag, this remaining quantity is exactly*

$$G^{b,\mathrm{src},\mathrm{sat}}_{\mathrm{LPN},\mathsf{X},c,\mathfrak{M},n}(B)$$
$$= \sup_{\substack{\mathfrak{g}=(\mathfrak{N},\Phi,D,J,\mathfrak{K})\in\mathcal{C}^{\mathrm{LPN}}_{X,b}(B)\\ c\in\mathrm{Prov}_{\mathsf{Geom},5}(\mathfrak{g})}} \mathfrak{M}_{n,m,\eta}\left(I \longmapsto M_\Phi \chi^{\mathrm{auth}}_\Phi(D)\,\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K},D) R_{\{s\}}(D)\frac{1}{1+\rho_\Phi(D)}\right), \tag{1622}$$

with value zero when the restricted class is empty. The derivation charged in this supremum includes the complete canonical source ancestry, exposed relation and conductance, normalized kernel, potential, current, authority comparison, gap comparison, reach record, regularity comparison, and every constructor used to evaluate them.

Let $0 < \varepsilon < 1$*. If the left side of Eq.* (1621) *is larger than each of the three quotient-supported entries and larger than* $\max\{2^{-n}, \varepsilon\}$*, then one tag* $c \in \mathfrak{I}^5_{\mathsf{Abs}}$ *and one family-uniform record* $\mathfrak{g}$ *of complete cost at most* $\widehat{B}$ *have joint gate product* $Z_{\mathfrak{g}}$*, equal to the product displayed in Eq.* (1622)*, such that*

$$\begin{aligned} &\mathfrak{M}_{n,m,\eta}(Z_{\mathfrak{g}}) > \varepsilon,\\ &\Pr_{I\sim\mathfrak{I}_{n,m,\eta}}\left\{Z_{\mathfrak{g}}(I) > \frac{\varepsilon}{2}\right\} > \frac{\varepsilon}{2-\varepsilon}. \end{aligned} \tag{1623}$$

On the event in Eq. (1623)*, the same represented record satisfies simultaneously*

$$\begin{aligned} &c\in\mathrm{Prov}_{\mathsf{Geom},5}(\mathfrak{g}), \qquad M_\Phi = 1, \qquad \chi^{\mathrm{auth}}_\Phi(D) = 1,\\ &\mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K},D) > \frac{\varepsilon}{2}, \quad R_{\{s\}}(D) > \frac{\varepsilon}{2}, \quad \frac{1}{1+\rho_\Phi(D)} > \frac{\varepsilon}{2}, \end{aligned} \tag{1624}$$

Conversely, if a family-uniform record of complete cost at most B *satisfies all five gates in Eq.* (1624)*, with the three quantitative gates exceeding* $a \in (0,1)$*, on an instance event of probability at least* p*, then its family visibility is larger than* pa^3*. Thus, at a polynomial ceiling, inverse-polynomial family visibility is equivalent, up to explicit polynomial losses, to constructing one complete record in one of these five ambient provenance restrictions whose full gate product is inverse-polynomial on an inverse-polynomial-mass set of LPN instances. The criterion is a single saturated profile evaluation; it contains no per-algorithm quantifier.*

Separately, if $0 < \eta < 1/2$*,* $n/m \to R \in (0,1)$*, and* I_n *is a fixed presentation on the full-rank and random-code-rigidity event, then the pointwise profile at the declared ceiling satisfies*

$$G^{\mathrm{src},\mathrm{sat}}_n(b_{\mathrm{poly}}(n)) = G^{\mathrm{src},\mathrm{sat}}_{X,n}(b_{\mathrm{poly}}(n)). \tag{1625}$$

This is the direct master-profile reduction obtained from Eqs. (1602)*,* (1606) *and* (1607)*. The ambient profile retains the temporal-admissibility, target-qualification, and complete-cost gates.*

Proof. Apply Theorem III.2.4 to the complete LPN prospective Geom profile. Its support coordinate is one of $\mathsf{X}, \mathsf{Q}_{\mathrm{ex}}, \mathsf{Q}_{\mathrm{rev}}, \mathsf{Q}_{\mathrm{nr}}$, and its nonempty source signature contains at least one tag in $\mathfrak{I}^5_{\mathsf{Abs}}$. The factorization and complete cost are Eqs. (29) and (30). Taking the finite maximum gives the LPN specialization of Eq. (31).

For identity support, Definition VIII.8.9 gives the disjoint residual-independent and residual-ancestry partition. The first part is at most 2^{-n} by Eq. (1233). The second part retains its five-family source signature by Corollary VIII.8.10. The three nonidentity support tags are bounded by their unrestricted exact, reversible, and general-resolvent quotient-supported Geom coordinates. These observations prove Eq. (1621).

The ambient visibility identity Eq. (1286), restricted to records whose canonical signature contains c, is exactly Eq. (1622). Every factor lies in $[0,1]$, and the first two factors are binary. Because the outer profile value is larger than ε, the definition of supremum supplies a single family-uniform record $\mathfrak{g}$ in one ambient provenance cell with $\mathbb{E}Z_{\mathfrak{g}} > \varepsilon$. For $A = \{Z_{\mathfrak{g}} > \varepsilon/2\}$,

$$\mathbb{E}Z_{\mathfrak{g}} \leq \Pr(A) + \frac{\varepsilon}{2}(1 - \Pr(A)),$$

so $\Pr(A) > \varepsilon/(2-\varepsilon)$. On A, both binary gates equal one and every other factor exceeds $\varepsilon/2$. This proves Eqs. (1623) and (1624). Conversely, on an event of probability at least p, two active binary gates and three factors larger than a give $Z_{\mathfrak{g}} > a^3$; hence $\mathbb{E}Z_{\mathfrak{g}} > pa^3$. This is precisely the LPN specialization of the saturated ambient-profile identity, perturbation obstruction cover, and inverse-polynomial witness criterion in Theorems III.26.1 and III.27.2 and Corollary III.27.4.

Finally, on the fixed rigidity-event presentation, Eqs. (1602), (1606) and (1607) remove every quotient-supported master mode at the declared ceiling. The ambient mode retains the same public constructors, qualification gates, and complete source records. The master decomposition therefore proves Eq. (1625).

□

VIII.20 First-Hit Accountability and Complete Source Exhaustion

VIII.20.1 First-hit accounting and source modes

Corollary VIII.20.1 (Core-refined first-hit accountability for LPN)**.** *Adopt the fixed-positive-noise and fixed-rate setting of Corollary VIII.18.2, and work on the intersection with the good-presentation event of Lemma VIII.5.1. Fix the declared polynomial budget envelope $\mathcal{B}_C$, put $\widehat{B}_C = \Psi(\mathcal{B}_C, n)$, and let $B_C^{\dagger}$ be the common charged compilation ceiling for the first-hit, five-family, and core representations of the same admissible records. Then*

$$\mathrm{Sel}^{\mathrm{sat}}_{I,n}(\widehat{B}_C) \leq eH_{\widehat{B}_C}\,\mathrm{HitCore}^{\mathrm{sat}}_{I,n}(B_C^{\dagger}). \tag{1626}$$

while the pointwise saturated success profile satisfies

$$\mathrm{Succ}_{I,n,\eta}(\mathcal{B}_C) \leq eH_C\,\mathrm{HitCore}^{\mathrm{sat}}_{I,n}(B_C^{\dagger}). \tag{1627}$$

These are pointwise accounting bounds in the profile of Definition III.22.3. Completed verifier flags and exact first-hit quotients remain retrospective accounting records; no displayed inequality promotes them to prospective sources.

Proof. Apply the pointwise form of Theorem III.22.4 at $\widehat{B}_C$ for the selector inequality and at $\mathcal{B}_C$ for the success inequality. This gives the displayed bounds directly. The direct-contact and temporal classifications remain Theorem VIII.9.8, Definition III.21.1, and Corollary III.23.1; they introduce no term outside the core-refined first-hit profile.

□

Proposition VIII.20.2 (Five master-source modes and the separate LPN accounting ledger)**.** *Fix the hypotheses and declared ceiling of Corollary VIII.18.2. The first four rows below cover the five disjoint master source modes of Theorem II.9.4, with the two nonzero-defect modes combined in row* (iv)*. Row* (v) *is the separate retrospective accounting profile.*

(i) *In the null-geometry exact-*Abs *mode* A*, the qualified record has the exact value*

$$Q_I V_{\mathscr{C},I} = Q_I A_{q,I} \frac{\|P_q^{\mathrm{amb}} y\|^2}{\|y\|^2}, \qquad 0 \le Q_I V_{\mathscr{C},I} \le 1. \tag{1628}$$

The five-family exhaustion Corollary VIII.18.2 removes this independent proper exact-Abs *source mode. On a generated carrier, filtration and Lift vertices add zero incremental charge and return every inherited charge to the earlier represented source ancestor. Identity-equivalent support is assigned to the ambient mode* X*.*

(ii) *In the identity-supported ambient mode* X*, the complete record has the exact same-record value*

$$Q_I M_\Phi \chi_\Phi^{\mathrm{auth}}(D)\, \mathrm{Align}_{\mathsf{Caus}}^{\mathrm{auth}}(\mathfrak{K}, D) R_{\{s\}}(D) \frac{1}{1+\rho_\Phi(D)}, \tag{1629}$$

and the exact factorwise ceilings are those of Corollaries VIII.10.5 and VIII.10.7; no provenance reindexing changes this product. Its complete saturated value is bounded by Corollary VIII.20.8 using Certification Target VIII.20.5.

(iii) *In mode* Q_{ex}*, exact quotient-supported geometry is bounded by the same exact-*Abs *coordinate:*

$$G_{Q,\mathrm{ex},n}^{\mathrm{src,sat}}(B) \le m_{I_n}^{\mathrm{src,sat}}(B). \tag{1630}$$

Thus the exact support value is controlled by the same A *profile through the displayed comparison and therefore vanishes by Eq.* (1606)*.*

(iv) *In modes* $\mathsf{Q}_{\mathrm{rev}}$ *and* Q_{nr}*, the master value is respectively*

$$Q_I \overline{A}_q (1 - \vartheta_\lambda^T(q)) M_{\mathfrak{G}_q} C_{\mathfrak{G}_q} \frac{R_{\mathfrak{G}_q}}{1+\rho_{\mathfrak{G}_q}}, \tag{1631}$$

$$Q_I \overline{A}_q \max\{0, 1 - \Delta_\lambda(q)\} M_{\mathfrak{G}_q} C_{\mathfrak{G}_q} \frac{R_{\mathfrak{G}_q}}{1+\rho_{\mathfrak{G}_q}}. \tag{1632}$$

Both nonzero-defect modes vanish by Eq. (1608)*. Its proof retains valid-Lift and generated-carrier ancestry and cost.*

(v) *The core-refined first-hit coordinate is a separate retrospective accounting profile. It has the exact finite maximum Eq.* (170)*, every cell has the charge identity Eq.* (173)*, and*

$$\mathrm{Succ}_{I_n,n,\eta}(\mathcal{B}_C) \le e H_C\, \mathrm{HitCore}_{I_n,n}^{\mathrm{sat}}(B_C^\dagger). \tag{1633}$$

By Corollary III.23.1, none of these accounting identities transfers to a prospective source coordinate without the represented comparison required by Corollary III.16.9.

Consequently, the exact-Abs *mode and the exact-quotient-supported Geom mode vanish on the stated event, as do both compensated Geom modes. Hence the prospective master profile retains only the ambient Geom coordinate. Its structure is exhausted by Theorem VIII.20.4; by Certification Target VIII.20.5, its complete saturated value is bounded by* $2^{-n} + (1-\eta)^n$ *through Corollary VIII.20.8. The first-hit profile remains a separate accounting maximum. The operational/core and state–coefficient decompositions are exact reindexings and introduce no additional numerical factor.*

Proof. Item (i) is Lemma VIII.17.2 and Corollary VIII.18.2. Item (ii) is Corollaries VIII.10.5 and VIII.20.8 and Theorem VIII.20.4. Item (iii) is Theorem III.19.1 and Eq. (1607). For item (iv), the two displayed products are the reversible and general-resolvent specializations of Definition III.18.1; the LPN vanishing result is Lemma VIII.16.8. Item (v) is Theorem III.22.4. Finally, Theorem III.5.3 preserves the same $Q_I V_{\mathscr{C},I}$ and complete cost under core reindexing.

□

VIII.20.2 Complete prospective value routing

Corollary VIII.20.3 (No uncharged prospective LPN value)**.** *Fix an LPN budget B. Every family-uniform, temporally admissible pre-hit selector, controlled value, or valid-Lift arrival record is governed by Theorem V.6.4. Hence a prospective structural bound is obtained either from a represented same-normalization comparison with an earlier qualified LPN source, or from the complete record regarded as a directly qualified source in one of the existing LPN master-profile cells. In both cases every selected kernel, potential, current, authority comparison, target interface, clock, and Lift map remains in the charged record at the common class-preserving ceiling.*

There is no separate LPN value-contraction coordinate. Ancestry equality, zero incremental Mult, Ord, Sym, *or* Filt *charge, and valid-Lift inheritance do not by themselves transfer a numerical ancestor bound to a derived output. A derived output without an inherited comparison is evaluated, if qualified, as the complete ambient or quotient-supported source record that produces it.*

Proof. Apply Theorem V.6.4 to the complete LPN pre-hit record. The master-cell assignment is the LPN specialization of Theorems III.2.6 and III.18.9. Source inheritance under Caus, Lift, and boundary processing is Proposition II.5.9; it preserves provenance and cost but supplies no unrepresented numerical comparison.

$\square$

Figure 83 gives the framework-level routing used here; no LPN-specific replacement coordinate is introduced.

VIII.20.3 Complete Geom exhaustion and saturated evaluation

Theorem VIII.20.4 (Complete structural exhaustion of LPN Geom)**.** *Fix $0 < \eta < 1/2$, work on the full-rank random-code-rigidity event $\mathcal{G}_{\mathrm{rig}}$ of Definition VIII.5.2, and use the declared LPN family functional restricted to that event. Let B lie in the declared polynomial saturated ceiling. Every temporally admissible, dynamically qualified prospective LPN Geom record is classified by a finite application of the following structural exhaustion. The primary support tags are disjoint. The ancestry rows may overlap and, for a mixed exact provenance signature, are traversed whenever applicable before the routing terminates at the earliest target-bearing source.*

Support or source ancestry	*Structural assignment*	*Evaluating result*
Exact nonidentity support	*Exact quotient-supported* Geom/Abs	*Eq.* (1607)
Compensated nonidentity support	*Reversible or general-resolvent quotient-supported* Geom	*Eq.* (1608)
Identity support without residual ancestry	*Residual-independent ambient* Geom	*Lemma VIII.8.11*
Identity support with residual ancestry	*Complete selected ambient* Geom/Caus *record, with every constructor, comparison, transition, and cost retained*	*Theorems III.2.6, VIII.11.1 and VIII.12.6 and Corollaries V.8.4 to VIII.11.6*
Mult *or* Ord *ancestry*	*No independent source; inherited charge returns to its earlier source ancestor*	*Lemmas VIII.17.7 and VIII.17.8*
Authenticated Sym *ancestry*	*Identity-equivalent; no independent source*	*Lemma VIII.17.5 and Corollary VIII.17.6*
Filt*, valid* Lift, Bdry*, or derived* Caus	*Same source cell as the earlier represented ancestor*	*Lemmas VIII.18.1 and VIII.8.3, Corollaries VIII.11.7 and VIII.8.8, and Proposition II.5.9*
Spectral representation	*Charged location, an existing quotient or ambient cell, terminal readout, or superpolynomial explicit expansion*	*Theorem VIII.2.5*
Extremal readout without authorized intermediate transport	*Terminal verifier; zero prospective source activity*	*Corollary III.3.8*
Gradient-orthogonal component	J_{res}*; zero authorized target alignment*	*Corollary VIII.19.1*

The table is exhaustive, including mixed exact provenance signatures. Backward source charging on the finite coefficient-complete ancestral DAG removes every zero-incremental or identity-equivalent decorator while retaining the complete selected Geom record. The classification is preserved after every class-preserving polynomial enlargement of the ceiling.

Proof. Theorem III.2.6 assigns every qualified prospective Geom record one support tag and one exact, coefficient-complete five-family signature. The LPN specialization Proposition VIII.19.2 refines the ambient tag into the residual-independent and residual-ancestry branches. The first two rows of the table close the three quotient-supported tags. The third row is the residual-independent ambient estimate.

Consider a residual-ancestry ambient record. Apply the coefficient-complete five-family backward cut of Corollary VIII.8.10. The Mult, Ord, authenticated Sym, and Filt rows of the table contribute no independent target-bearing source. Valid lifts, boundary readouts, and derived currents preserve the same assignment. Backward iteration terminates because the represented ancestral DAG is finite. The complete selected record remains in the ambient master cell at its full saturated cost.

Before this source assignment is used, every nonlinear full-carrier descendant is evaluated as its complete selected dynamical record by Corollary V.8.4. Its target-boundary capacity and global drift obey Theorem VIII.11.1; its authorized current and noise-corrected aiming obey Corollaries VIII.11.6 and VIII.11.7; its public-carrier functional constants obey Corollary VIII.12.4; its valid lifted forms, clocked laws, and target gates obey Corollary VIII.8.8; and its learned operator, value, policy, and safe-spectrum coordinates obey Theorem VIII.12.6 and Corollary VIII.13.5. These are typed numerical outputs of the same complete record. They remain attached to that record while the backward cut identifies its structural source. Their explicit finite-parameter interfaces are Eqs. (1296), (1319), (1386) and (1394).

The complete ambient record retains the capacity, Caus, functional-profile, Lift, and learning comparisons listed above. Spectral notation cannot introduce another terminal source by Theorem VIII.2.5; an isolated extremum is terminal verification by Corollary III.3.8; and the gradient-orthogonal component is J_{res} by Corollary VIII.19.1. Thus every exact signature has one source assignment, and every derived row retains the complete record and the same numerical normalization during that assignment. This proves the asserted structural exhaustion.

□

Certification Target VIII.20.5 (Compact residual-ancestry ambient bound for LPN)**.** Fix $0 < \eta < 1/2$, the declared LPN family functional restricted to the full-rank and random-code-rigidity event, and a budget B in the declared polynomial saturated ceiling or a class-preserving polynomial enlargement. For the residual-ancestry source family of Definition VIII.8.9, the remaining certification objective is

$$G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B) \leq (1-\eta)^n. \tag{1634}$$

The estimate applies to the complete record: score construction, normalization, active acquisition, posterior update, Caus current, valid Lift, and deployed transition are all charged at the same ceiling. By Theorem VIII.3.7, the inequality is already proved for explicitly materialized located spectra and for every residual-parity fiber map that fails the quotient gate or is identity-equivalent. Its remaining content is the numerical evaluation of compact, dynamically qualified, non-lumpable residual-ancestry ambient records, including any polynomial evaluator with an unmaterialized diffuse Walsh expansion.

For the proof-producing discharge and its certificate schema, see Theorem VIII.20.6 and Definition V.15.15. The exact finite-population quantities used in the finite-saturation checks are evaluated by the finite audits of Section I.6; complete source-cell coverage remains a separate field of the discharge theorem. Recurrence and scale certification are specified in Definition V.15.10, and the uniformly propagated conclusion is Eq. (1646).

This is the sole unresolved numerical certification target in the LPN part, not an external hardness assumption. It is not discharged in the present manuscript because the instantiated certificate analysis has not been run and accepted. The next theorem proves that a record satisfying its complete proof-grade fields discharges the target; a confidence-only empirical estimate does not satisfy those fields. Consequently, every later LPN result that invokes this target remains conditional on deterministic acceptance of such a record, while the certification machinery and its implication are unconditional theorems of the paper. Figure 137 displays the resulting finite-prefix and propagated-tail argument.

Theorem VIII.20.6 (Proof-grade empirical discharge of the residual-ancestry bound)**.** *Fix $0 < \eta < 1/2$, a rate schedule $m = m(n)$, a polynomial ceiling B_n, and a base size n_0. After the source reductions in Theorem VIII.3.7 and Corollary VIII.8.10, let the single remaining source cell be the compact, dynamically qualified, non-lumpable residual-ancestry ambient cell. Suppose a proof-grade scale-certified audit in Definition V.15.10 verifies for this cell*

$$D_{b,n} \leq \overline{D}_{b,0}\left(\frac{n}{n_0}\right)^d, \tag{1635}$$

$$a_{b,n} \leq \overline{a}_{b,0}e^{-\gamma(n-n_0)}, \tag{1636}$$

$$e_{b,n} \leq \overline{e}_{b,n} \tag{1637}$$

for represented constants $d \geq 0$, $\gamma > 0$, and every $n \geq n_0$. These inequalities must cover the complete qualified source cell, not only the trained record. They imply the explicit cell bound

$$G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B_n) \leq \sqrt{\overline{D}_{b,0}}\,\overline{a}_{b,0}\left(\frac{n}{n_0}\right)^{d/2} e^{-\gamma(n-n_0)} + \overline{e}_{b,n}. \tag{1638}$$

The attenuation check is explicitly noise-indexed. In particular, if the verified factorization gives

$$a_{b,n} \leq (1-2\eta)^{q_n}, \qquad q_n \geq q_{n_0} + \alpha(n-n_0), \tag{1639}$$

then one may take

$$\overline{a}_{b,0} = (1-2\eta)^{q_{n_0}}, \qquad \gamma = \alpha \log\frac{1}{1-2\eta}. \tag{1640}$$

For a certified weight law $q_n \geq \delta m(n)$ with $m(n) = n/R + O(1)$, any certified lower affine envelope for $q_n - q_{n_0}$ supplies the corresponding $\alpha = \delta/R + o(1)$; the verifier retains the floor and $O(1)$ terms rather than discarding them.

Put

$$\lambda_\eta = \log\frac{1}{1-\eta}. \tag{1641}$$

Assume $\gamma > \lambda_\eta$. Let $N_* \geq n_0$ be a represented integer satisfying

$$N_* \geq \frac{d}{2(\gamma-\lambda_\eta)}, \tag{1642}$$

$$\log\left(2\sqrt{\overline{D}_{b,0}}\,\overline{a}_{b,0}\right) + \frac{d}{2}\log\frac{N_*}{n_0} + \gamma n_0 \leq (\gamma-\lambda_\eta)N_*, \tag{1643}$$

$$\overline{e}_{b,n} \leq \frac{1}{2}e^{-\lambda_\eta n} \qquad (n \geq N_*). \tag{1644}$$

If outward-rounded direct verification also proves

$$G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B_n) \leq (1-\eta)^n \qquad (n_0 \leq n < N_*), \tag{1645}$$

then

$$G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B_n) \leq (1-\eta)^n \qquad (n \geq n_0). \tag{1646}$$

Consequently the exported certificate discharges Certification Target VIII.20.5 at the audited ceiling.

The proof dependencies are explicit. Exact source-cell exhaustion is supplied by Theorem VIII.3.7 and Corollary VIII.8.10. For each finite slice covered by that exhaustion, the exact finite-saturation and direct finite-prefix fields use complete full-support audit data in the sense of Definition I.6.1, with finite-matrix soundness supplied by Lemma I.6.3. This numerical audit does not by itself establish that the represented slice is complete, and sampled audit data alone do not supply either field. Capacity factorization, attenuation, and error control are the verified fields of Definition V.15.10, Theorem V.15.12, and Corollary V.15.14, specialized here by Eqs. (1635)*,* (1637) *and* (1639)*. The strict phase separation and crossover exclusion are exactly Eqs.* (1641) *to* (1644)*; the prefix and propagated tail meet in Eqs.* (1645) *and* (1646)*. After deterministic acceptance, the conversion from this source bound to a universal success bound is supplied by Theorems III.21.3, III.24.3 and VIII.22.1.*

Every premise above has a distinct verification field: source-cell coverage comes from the cited LPN exhaustion theorems; the factorization, capacity, attenuation, and error bounds come from the scale audit; the polynomial ceiling comes from complete cost accounting; the crossover inequalities are scalar interval checks; and the finite prefix is checked directly. The fields are not all finite computations: the crossover and finite-prefix checks are finite, but the scale-law bounds (1635) *to* (1637) *are uniform-in-n proof obligations over the complete qualified cell and must be discharged symbolically, so supplying them is equivalent in strength to proving the residual-ancestry bound on that cell. A neural or Bayesian learner may propose all these records using Theorems V.15.8 and V.15.25 and Corollary V.15.22, but only their deterministic verification is used in Eq.* (1646)*.*

Proof. The LPN source-exhaustion results cited in the statement verify Eq. (811) with the displayed residual-ancestry cell as the sole unevaluated member. Apply Eqs. (812) and (814) and substitute Eqs. (1635) to (1637); this gives Eq. (1638).

Under Eq. (1639),

$$(1-2\eta)^{q_n} \le (1-2\eta)^{q_{n_0}} \exp\left[-\alpha(n-n_0)\log\frac{1}{1-2\eta}\right],$$

which proves Eq. (1640). The logarithm of the leading term in Eq. (1638), divided by $e^{-\lambda_\eta n}$, is at most

$$\log\left(\sqrt{\overline{D}_{b,0}}\,\overline{a}_{b,0}\right) + \frac{d}{2}\log\frac{n}{n_0} + \gamma n_0 - (\gamma-\lambda_\eta)n.$$

Its derivative is $d/(2n) - (\gamma - \lambda_\eta)$, which is nonpositive for $n \ge N_*$ by Eq. (1642). The check at N_* in Eq. (1643) therefore bounds the leading term by $\frac{1}{2}e^{-\lambda_\eta n}$ for every $n \ge N_*$. Together with Eq. (1644), this gives Eq. (1646) on the propagated tail. The direct check Eq. (1645) supplies the remaining finite sizes. Since $e^{-\lambda_\eta n} = (1-\eta)^n$, the conclusion is exactly Eq. (1634). □

Remark VIII.20.7 (What attention compilation proves)*.* Compile any selected transition into a complete attention record by Corollary VI.2.3. Then Theorems VI.1.3 and VI.2.2 retain every score, normalization, value path, posterior update, and deployed transition in the existing source-resolved ledger. This proves that attention adds no uncharged source coordinate. It does not identify the complete public experiment (A, b) with its pivot-row statistic. Such an identification would require the represented pre-hit sufficiency condition of Corollary V.4.6; provenance classification alone does not supply that condition.

Corollary VIII.20.8 (Complete ambient LPN source bound under the residual-ancestry bound)**.** *Fix* $0 < \eta < 1/2$*, and let* $\mathfrak{M}_{n,m,\eta}$ *be the declared LPN expectation functional on the full-rank and* *random-code-rigidity event. At every budget B in the declared polynomial saturated ceiling, assume Certification Target VIII.20.5. Then*

$$G^{\mathrm{src,sat}}_{X,\mathfrak{M},n}(B) \le 2^{-n} + (1-\eta)^n. \tag{1647}$$

The same bound holds after every class-preserving polynomial enlargement of that ceiling.

Proof. Split the ambient source family by Definition VIII.8.9. The residual-independent branch is at most 2^{-n} by Lemma VIII.8.11.

The residual-ancestry branch has the exhaustive five-family source classification of Corollary VIII.8.10; its numerical value is bounded by Certification Target VIII.20.5. Since the two ancestry families are disjoint and exhaustive,

$$G^{\mathrm{src,sat}}_{X,\mathfrak{M},n}(B) = \max\left\{G^{\mathrm{src,sat}}_{X,0,\mathfrak{M},n}(B), G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B)\right\} \le \max\{2^{-n}, (1-\eta)^n\} \le 2^{-n} + (1-\eta)^n.$$

The residual-ancestry compact-ambient and residual-independent estimates are stated at every class-preserving enlarged ceiling, so the same conclusion holds there.

□

Corollary VIII.20.9 (LPN specialization of the ambient obstruction criterion)**.** *Fix the LPN presentation, family functional, and saturated ceiling of Corollary VIII.20.8. Suppose every complete temporally admissible ambient record at that ceiling satisfies at least one of*

$$M_\Phi = 0, \qquad \chi^{\mathrm{auth}}_\Phi(D) = 0, \qquad \mathrm{Align}^{\mathrm{auth}}_{\mathsf{Caus}}(\mathfrak{K}, D) \le \delta, \qquad R_{\{s\}}(D) \le \delta, \qquad \frac{1}{1+\rho_\Phi(D)} \le \delta.$$

Then

$$G^{\mathrm{src,sat}}_{X,\mathfrak{M},n}(B) \le \delta, \qquad I^{\mathrm{src,sat}}_{\mathfrak{M},n}(B) \le \delta, \tag{1648}$$

on the LPN good event. For a polynomial ceiling, the complete prospective source profile is negligible when the displayed cover holds with $\delta = n^{-k}$ for every fixed k and all sufficiently large n.

Proof. The first conclusion is the direct LPN substitution in Theorem III.27.2. The exact and compensated quotient-supported coordinates vanish by Eqs. (1602), (1606) and (1607). The complete master decomposition Theorems II.9.4 and III.2.4, together with Definition III.25.1, therefore gives $I^{\mathrm{src,sat}}_{\mathfrak{M},n} = G^{\mathrm{src,sat}}_{X,\mathfrak{M},n}$, which proves the second conclusion without invoking the later LPN floor theorem. The polynomial statement is Corollary III.27.3.

□

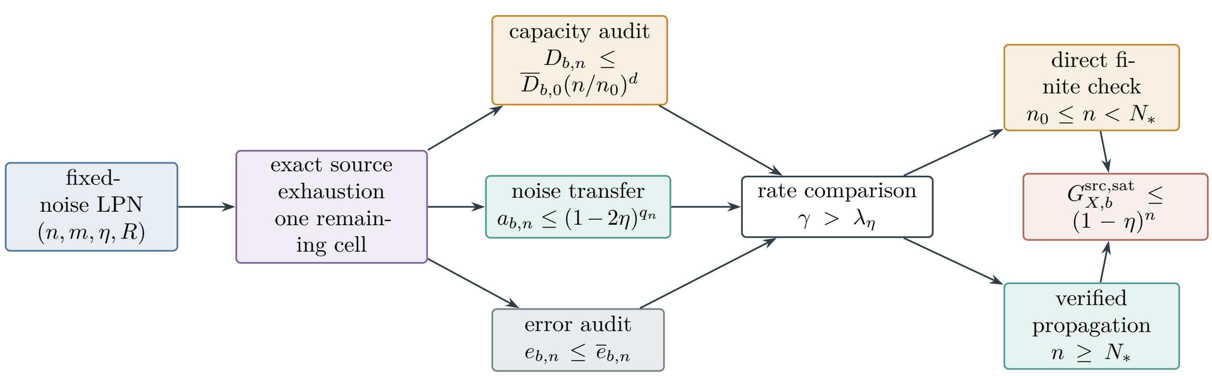

Figure 137: Proof-grade LPN discharge. Structural exhaustion fixes the sole remaining source cell. Capacity, noise transfer, and error certificates establish the exponent comparison; direct verification covers the finite prefix and scale propagation covers the tail.

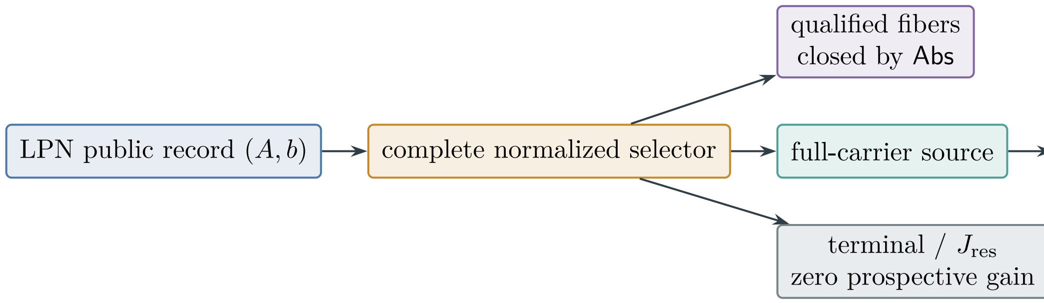

Figure 138: LPN specialization of the complete selector compilation. Quotient, terminal, residual, and materialized located-spectral rows are closed by Theorem VIII.3.7. The remaining compact non-lumpable full-carrier row is evaluated by Certification Target VIII.20.5; Theorem VIII.20.6 gives its explicit computer-assisted discharge protocol.

VIII.20.4 Selector and valid-Lift consequences

Corollary VIII.20.10 (Prospective selector source routing)**.** *Under the hypotheses of Corollary VIII.20.8, let B be a declared execution ceiling and let $\widehat{B} = \Psi(B, n)$ be the prospective-selector ceiling of Theorem III.21.3. Every positive selector value is routed into the complete prospective profile. Consequently,*

$$\mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(\widehat{B}) \leq G^{\mathrm{src,sat}}_{X,\mathfrak{M},n}(\widehat{B}) \leq 2^{-n} + (1-\eta)^n. \tag{1649}$$

Proof. A positive selector advantage is represented before the corresponding test and is target-aligned by Definition III.21.2 and Theorem III.21.3. Channel exhaustiveness and source no-creation place its complete qualified record in the prospective Geom/Abs provenance profile at $\widehat{B}$; see Theorem II.7.1, Proposition III.29.5, and Corollary III.16.9. The exact-Abs and all quotient-supported Geom coordinates vanish by Eqs. (1602), (1606) and (1607). The complete ambient coordinate is bounded by Eq. (1647). A directly qualified selector remains in that coordinate at complete cost by Theorem V.6.4; hence it is included in, rather than additional to, the displayed bound. □

Corollary VIII.20.11 (Valid-Lift access preserves complete LPN value routing)**.** *Under the hypotheses of Corollary VIII.20.10, let $\mathscr{L}$ be any represented valid-Lift class evaluated with the same family functional, horizon H, and enlarged ceiling $\widehat{B}$. Use the declared uniform full-carrier neutral baseline ν; its singleton contact obeys*

$$\mathfrak{M}_{n,m,\eta}(\mathrm{Enum}_\nu(\mathcal{A}, H)) \leq H2^{-n}. \tag{1650}$$

Then

$$\mathrm{LiftSel}^{\mathrm{sat}}_{\mathscr{L}}(\widehat{B}, H) \leq \mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(\widehat{B}). \tag{1651}$$

For a clocked member of this class, let $E_t = P^{\Omega}_t U_{t+1} - U_t P^X_t$ be its represented intertwining defects, and retain the initialization and gate-interface errors $\varepsilon_{\mathrm{init}}, \varepsilon_{\mathrm{gate}}$ of Theorem V.7.13. With the public LPN verifier gate at threshold τ, its family-functional arrival probability obeys

$$\mathfrak{M}_{n,m,\eta}(p^{\Omega}_{\widehat{T}_I,H}) \leq H\big(2^{-n} + \mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}(\widehat{B})\big)$$

$$+ \mathfrak{M}_{n,m,\eta}\left(\varepsilon_{\text{init}} + \sum_{t=0}^{H-1} \|E_t\| + \varepsilon_{\text{gate}}\right)$$
$$+ \exp[-mD_{\text{Ber}}(\tau\|\eta)] + (2^n - 1)2^{-m(1-H_2(\tau))}. \tag{1652}$$

Thus an exact valid lift contributes no interface defect and preserves the complete prospective-value routing of its base selector. A clocked lift additionally contributes its defined interface defects, and the public verifier contributes one explicit (n, m, η, τ)-dependent exceptional-presentation term.

Proof. The inequality in Eq. (1651) is Eq. (535). Apply $\mathfrak{M}_{n,m,\eta}$ to the hazard identity in Theorem III.21.3, then use sublinearity, Eq. (1650), and the saturated selector supremum. The base first-arrival probability is therefore at most

$$\mathfrak{M}_{n,m,\eta}(p^X_{T_I,H}) \le H(2^{-n} + \mathrm{Sel}^{\text{sat}}_{\mathfrak{M},n,\eta}(\widehat{B})).$$

Apply the clocked pullback estimate in Theorem V.7.13, followed by the public-gate family comparison Eq. (1229), to obtain Eq. (1652).

□

VIII.21 Exact Reduction of the Saturated LPN Profile

Corollaries VIII.18.2 and VIII.18.3 and Lemma VIII.16.8 remove the exact-Abs mode and every quotient-supported Geom mode from the prospective source ledger. By the master decomposition, the only remaining prospective coordinate is ambient Geom. The separate first-hit accounting ledger is retained by Definition III.22.3 and Theorem III.22.4; temporal source admissibility forbids an automatic transfer between the two ledgers.

Theorem VIII.21.1 (Exact saturated-profile ledger for LPN under the residual-ancestry bound)**.** *Fix $0 < \eta < 1/2$ and a sample schedule with $n/m \to R \in (0, 1)$. On the full-rank and random-code-rigidity event, fix a presented instance I_n, and let $D^{\text{hit}}_{\text{poly}}$ be the first-hit normalization ceiling supplied by Theorem III.22.4. Apply Certification Target VIII.20.5 at $b_{\text{poly}}(n)$. Then*

$$m^{\text{src,sat}}_{I_n}(b_{\text{poly}}(n)) = G^{\text{src,sat}}_{Q,\text{ex},n}(b_{\text{poly}}(n)) = G^{\text{src,sat}}_{Q,\text{rev},n}(b_{\text{poly}}(n)) = G^{\text{src,sat}}_{Q,\text{nr},n}(b_{\text{poly}}(n)) = 0, \tag{1653}$$

$$I^{\text{src,sat}}_n(b_{\text{poly}}(n)) = G^{\text{src,sat}}_{X,n}(b_{\text{poly}}(n)), \tag{1654}$$
$$I^{\text{src,sat}}_{\mathfrak{M},n}(b_{\text{poly}}(n)) = G^{\text{src,sat}}_{X,\mathfrak{M},n}(b_{\text{poly}}(n)), \tag{1655}$$
$$G^{\text{src,sat}}_{X,\mathfrak{M},n}(b_{\text{poly}}(n)) \le 2^{-n} + (1-\eta)^n, \tag{1656}$$
$$\mathrm{HitCore}^{\text{sat}}_{\mathfrak{M},n}(D^{\text{hit}}_{\text{poly}}) = \max_{\substack{K_{\text{st}},K_{\text{cf}} \\ S_{\text{st}} \supseteq K_{\text{st}} \\ S_{\text{cf}} \supseteq K_{\text{cf}}}} \mathrm{HitCore}^{\text{sat},=}_{\mathfrak{M},K_{\text{st}},K_{\text{cf}}|S_{\text{st}},S_{\text{cf}},n}(D^{\text{hit}}_{\text{poly}}). \tag{1657}$$

For the fixed declared polynomial execution envelope $\mathcal{B}_C$, at its common charged normalization ceiling $B^\dagger_C$, the corresponding pointwise success bound on the public-verifier good event is

$$\mathrm{Succ}_{I_n,n,\eta}(\mathcal{B}_C) \le eH_C\ \mathrm{HitCore}^{\text{sat}}_{I_n,n}(B^\dagger_C). \tag{1658}$$

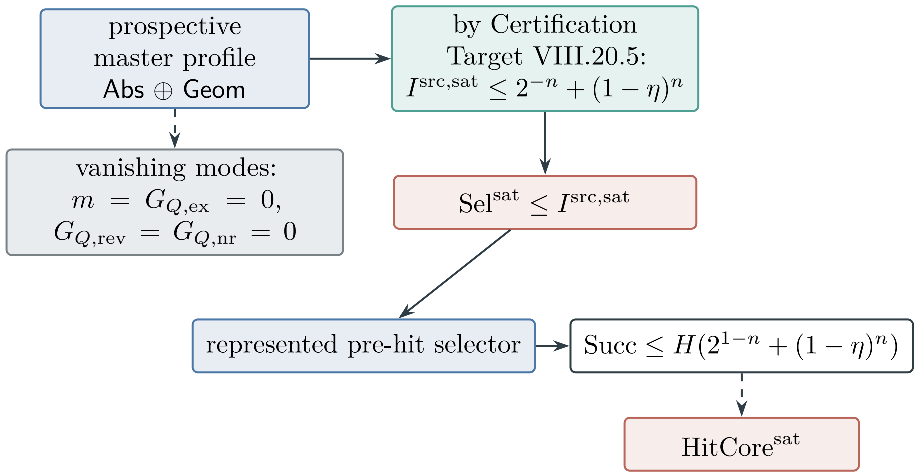

Figure 139: The exact LPN reduction. Exact abstraction and all three quotient-supported geometric modes vanish, and the complete saturated source profile is bounded at the common charged ceiling by Certification Target VIII.20.5. Prospective selector accountability then gives the displayed success inequality. The separate HitCore coordinate retains completed first-hit accounting and is not a hardness hypothesis. The source and success bounds are Theorems VIII.21.1 and VIII.22.1.

These identities and bounds are preserved by conservative extensions through presentation-admissible LPN oracles. The first displays separate the exact pointwise provenance reduction, the complete ambient source coordinate, and its saturated evaluation. Complete prospective values are routed by Corollary VIII.20.3; no separate derived-value coordinate is inserted. The next display is the exact finite decomposition of the retrospective accounting profile, and the pointwise accounting bound uses only that profile. No comparison between the two profile types is asserted.

Proof. Equations (1602), (1606) and (1607) gives Eq. (1653). The exhaustive master decomposition Theorems II.9.4 and III.2.4 expresses $G_n^{\mathrm{src,sat}}$ as the maximum of the ambient, exact-quotient, reversible-compensated, and resolvent-compensated coordinates. The preceding vanishing identities leave exactly $G_{X,n}^{\mathrm{src,sat}}$. Since $I_n^{\mathrm{src,sat}} = m_n^{\mathrm{src,sat}} + G_n^{\mathrm{src,sat}}$ by Definition III.25.1, this proves the equality in Eq. (1654). Apply the declared LPN family functional to the same exact decomposition to obtain Eq. (1655); then apply the complete ambient seal Eq. (1647) to prove Eq. (1656).

Equation (170) gives Eq. (1657). Apply the pointwise form of Theorem III.22.4 to obtain Eq. (1658). The LPN representation lemma Lemma VIII.8.1 and oracle admissibility theorem Theorem III.33.2 retain the same represented ancestry and class-preserving costs under admissible oracle interfaces.

The completed first-hit record is accounting-admissible but not a prospective source by Corollary III.23.1. Its exact visibility is nevertheless retained in HitCore and charged by Eq. (173); hence it is neither discarded nor used as a retrospective structural source.

□

Figure 139 collects the exact source and accounting reduction in Theorem VIII.21.1.

VIII.22 Hardness from the Saturated Prospective Profile

The final LPN theorem applies prospective selector accountability at the same represented budget. The noise schedules $\eta = \eta(n)$, including the zero-noise and polynomial-enumeration scales, are localized separately in Corollary VIII.22.3.

Theorem VIII.22.1 (Admissible-oracle hardness of search-LPN under the residual-ancestry bound)**.** *Fix $0 < \eta < 1/2$, choose $0 < \delta < 1/2 - \eta$, and set $\tau = \eta + \delta$. Let $n/m \to R \in (0, 1)$, with*

$$m(1 - H_2(\tau)) \geq n + \omega(\log n).$$

Fix the declared polynomial execution envelope $\mathcal{B}_C$, with horizon H_C, and put

$$B_C^{\mathrm{uBayes}} = \Psi_{\mathrm{uBayes}}(\Psi_{\mathrm{act}}(\mathcal{B}_C, H_C, n, \mathbf{0}), H_C, n), \qquad B_C^\sharp = \Psi(B_C^{\mathrm{uBayes}}, n), \qquad H_C^\sharp = H_{B_C^\sharp}(n). \tag{1659}$$

For the declared LPN family functional restricted to the intersection of the public-verifier good event of Lemma VIII.5.1 and the full-rank random-code-rigidity event of Lemma VIII.17.5, write $\mathfrak{M}^* = \mathfrak{M}_{n,m,\eta}^{\mathcal{G}_*(\tau)}$ in the sense of Definition VIII.5.2. The complete saturated LPN source-profile bound of Theorem VIII.21.1 applies at the class-preserving common ceiling $B_C^\sharp$. Apply Certification Target VIII.20.5 at that ceiling. Then

$$\mathrm{Succ}_{\mathfrak{M}^*,n,\eta}(\mathcal{B}_C) \leq H_C(2^{1-n} + (1-\eta)^n). \tag{1660}$$

Taking the declared LPN family functional and bounding success by one off that event gives the full-family distributional inequality

$$\begin{aligned}\mathrm{Succ}_{\mathfrak{M},n,\eta}(\mathcal{B}_C) \leq H_C&(2^{1-n} + (1-\eta)^n)\\ &+ \varepsilon_{\mathrm{rig}}(n, m)\\ &+ \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] + (2^n - 1)2^{-m(1-H_2(\tau))}.\end{aligned} \tag{1661}$$

The bound includes computations expressed through presentation-admissible LPN oracle interfaces. For fixed $0 < \eta < 1/2$, polynomial $H_C, H_C^\sharp$, and the displayed sampling condition, every term on the right of Eq. (1661) *is negligible.*

Proof. On the good-presentation event of Lemma VIII.5.1, the represented verifier accepts exactly $T_I = \{s\}$. The LPN representation lemma Lemma VIII.8.1 places the declared execution envelope, including presentation-admissible oracle expansions and a final verifier call, in the saturated represented computation class. Apply the universal compiler of Corollary V.11.14 at Eq. (1659). It preserves the complete execution and first-hit laws. The source-resolved LPN Bayesian theorem Theorem VIII.15.11 retains the recordwise framework candidates and the complete ambient record. The compact-state compiler and Theorem VIII.15.14 apply that same source evaluation to every latent representation, optimal acquisition rule, Bayes–Fisher update, learned decoder, and attention selector in the compiled polynomial saturated class. Their maximum useful belief update is the profile in Eq. (1574) and is bounded by $2^{-n} + (1-\eta)^n$ at this same ceiling.

Since $\mathcal{G}_*(\tau) \subseteq \mathcal{G}_{\mathrm{ver}}(\tau)$, the public candidate verifier again accepts exactly $\{s\}$ on $\mathcal{G}_*(\tau)$; the proof of Theorem VIII.4.3 therefore applies verbatim with the restricting event $\mathcal{G}_{\mathrm{ver}}(\tau)$ replaced by $\mathcal{G}_*(\tau)$ and the functional $\mathfrak{M}_{n,m,\eta}^{\mathrm{ver}}$ by the further-restricted functional $\mathfrak{M}^*$. Apply the resulting LPN prospective selector bound Eq. (1177) to the compiled record. The uniform neutral baseline on the full LPN carrier contributes at most $H_C 2^{-n}$, so

$$\mathrm{Succ}_{\mathfrak{M}^*,n,\eta}(\mathcal{B}_C) \leq H_C 2^{-n} + H_C \, \mathrm{Sel}_{\mathfrak{M}^*,n,\eta}^{\mathrm{sat}}(B_C^\sharp).$$

Framework source routing and the complete saturated LPN profile evaluation in Theorem VIII.21.1 give, at the same ceiling,

$$\mathrm{Sel}_{\mathfrak{M}^*,n,\eta}^{\mathrm{sat}}(B_C^\sharp) \leq I_{\mathfrak{M}^*,n}^{\mathrm{src,sat}}(B_C^\sharp) = G_{X,\mathfrak{M}^*,n}^{\mathrm{src,sat}}(B_C^\sharp) \leq 2^{-n} + (1-\eta)^n.$$

Substitution gives

$$\mathrm{Succ}_{\mathfrak{M}^*,n,\eta}(\mathcal{B}_C) \le H_C\bigl(2^{1-n} + (1-\eta)^n\bigr). \tag{1662}$$

Equation Eq. (1662) is Eq. (1660). Valid Lift, Caus, boundary, filtering, likelihood, normalization, and belief updates retain their framework source cell and complete cost.

The source deduction uses only prospective records. The completed first-hit record belongs to the retrospective audit profile by Corollary III.23.1; neither it nor $\mathrm{HitCore}^{\mathrm{sat}}$ is a premise of Eq. (1662).

Extend the restricted family expectation to the complete LPN family. On the complement of the rigidity event or the verifier-good event, use the unit bound on success. The union bound, together with Eqs. (1184) and (1604), gives the three exceptional-presentation terms in Eq. (1661).

The complement of the good event is included explicitly in Eq. (1661); its probability is an evaluated error term, not an additional structural assumption.

□

Corollary VIII.22.2 (Structural meta-complexity lower region for LPN)**.** *Adopt the parameters, restricted functional $\mathfrak{M}^*$, and common charged ceiling $B_C^\sharp$ of Theorem VIII.22.1, and put*

$$\theta^{\mathrm{meta}}_{n,\eta} = 2^{-n} + (1-\eta)^n. \tag{1663}$$

For every n for which $\theta^{\mathrm{meta}}_{n,\eta} < 1$, and therefore for every sufficiently large n at fixed $\eta > 0$, the complete LPN pre-hit selector datum satisfies

$$B_C^\sharp \notin \mathfrak{P}^{\mathrm{sel},>}_{\mathfrak{M}^*,n,m,\eta,H_C}(\theta^{\mathrm{meta}}_{n,\eta}). \tag{1664}$$

For a scalar total-cost order whose sublevel b is represented by the ceiling $B_C^\sharp$, the Pareto exclusion implies

$$\mathrm{SMC}^{\mathrm{sel},>}_{\mathfrak{M}^*,n,m,\eta,H_C,\kappa}(\theta^{\mathrm{meta}}_{n,\eta}) \ge \kappa(B_C^\sharp), \tag{1665}$$

and the inequality is strict under the locally finite discrete scalar-cost convention. The exact boundary statement is the Pareto exclusion Eq. (1664)*; no injectivity of the scalarization is required there. Prospective meta-accountability gives*

$$\mathrm{Succ}_{\mathfrak{M}^*,n,\eta}(\mathcal{B}_C) \le H_C\bigl(2^{-n} + \theta^{\mathrm{meta}}_{n,\eta}\bigr) = H_C\bigl(2^{1-n} + (1-\eta)^n\bigr). \tag{1666}$$

Proof. The proof of Theorem VIII.22.1 establishes at the same ceiling

$$\mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M}^*,n,\eta}(B_C^\sharp) \le 2^{-n} + (1-\eta)^n = \theta^{\mathrm{meta}}_{n,\eta}.$$

By Definition VII.9.1, this is exactly the saturated profile of $\mathfrak{D}^{\mathrm{sel}}_{\mathfrak{M}^*,n,m,\eta,H_C}$ at that ceiling. Apply the exact generalized inverse Eq. (1077) to obtain Eq. (1664). The scalar statement is Eq. (1079). Finally, the uniform neutral baseline contributes at most $H_C 2^{-n}$; substitute the frontier bound in Theorem VII.2.2 to obtain Eq. (1666). □

Corollary VIII.22.3 (Quantitative localization of the LPN noise phases)**.** *Let $n/m \to R \in (0,1)$. In this corollary the noise may be zero, may vary with n, or may be a fixed element of* $(0, 1/2)$*, as specified in each item.*

(i) *At $\eta = 0$, Gaussian elimination recovers s in polynomial time with probability at least $1 - 2^{n-m}$.*

(ii) *Under the polynomial enumeration hypotheses of Corollary VIII.6.2, every schedule satisfying $\eta m \le c \log n$, for fixed $c > 0$, has a represented polynomial-time recovery procedure with success at least $n^{-c+o(1)} - \mathrm{negl}(n)$. For every fixed affordable radius k_C and target probability $0 < \alpha < 1$, the constant-success boundary is $\eta = \lambda_{k_C,\alpha}/m + o(m^{-1})$, where $\lambda_{k_C,\alpha}$ is determined by Eq.* (1193)*.*

(iii) *By Certification Target VIII.20.5, at fixed $0 < \eta < \eta_{\mathrm{IT}}(R) = H_2^{-1}(1-R)$, the declared polynomial execution envelope satisfies Eq.* (1661)*. Its saturated structural term and the rigidity and verifier-event errors are negligible under the stated sampling condition.*

(iv) *For every fixed $\eta > \eta_{\mathrm{IT}}(R)$, every decoder has exponentially small exact-recovery probability.*

Thus the low-noise construction and the information-theoretic converse are unconditional in their stated regimes. The fixed-positive computational bound uses Certification Target VIII.20.5. For a finite scalar budget and target success α, the supremal affordable noise is $\eta^{\mathrm{sat}}_{\mathrm{comp}}$ from Eq. (301)*. It is the endpoint of the affordable interval whenever the budget absorbs represented flip postprocessing. Its explicit enumeration lower bound, structural hard-side certificate, statistical-resolution certificate, and information upper bound are Eqs.* (1216) *to* (1219)*.*

Proof. At zero noise, $b = As$. The rank estimate in the proof of Proposition VIII.6.1 gives $\Pr\{\mathrm{rank}(A) < n\} < 2^{n-m}$; on its complement, row reduction returns the unique secret. This proves (i).

For (ii), the weight-zero branch of the represented enumerator succeeds on $E = 0$. Under $\eta m \le c \log n$ and the stated polynomial sample scaling,

$$\Pr\{E = 0\} = (1-\eta)^m \ge n^{-c+o(1)}.$$

Subtracting the negligible algebraic failure in Eq. (1190) gives the claim. The constant-success assertion is Eq. (1194).

For (iii), choose a constant τ with $\eta < \tau < \eta_{\mathrm{IT}}(R)$. Since $R < 1 - H_2(\tau)$,

$$m(1 - H_2(\tau)) - n = \Theta(m),$$

so the verification hypotheses of Theorem VIII.22.1 hold; its Eq. (1661) gives the bound in (iii). Part (iv) is the strong converse Eq. (1207), with a fixed

$$0 < \epsilon < \min\left\{\eta, \frac{R - 1 + H_2(\eta)}{L_\eta}\right\}.$$

□

Figure 140 collects the two asymptotic scales and the finite-budget threshold relations.

Corollary VIII.22.4 (Saturated statistical denoising obstruction for LPN)**.** *Adopt the hypotheses of Theorem VIII.22.1. Fix a represented learning ceiling B and an inverse-polynomial threshold $\varepsilon \in (0,1]$. Let $B_\varepsilon^{\mathrm{GL}}$ include the represented bounded-score evaluator, Goldreich–Levin list decoder at threshold ε, and public verification pass, and let $p_{\mathrm{GL}} > 0$ be its fixed amplified conditional success probability. At the common normalized ceiling $(B_\varepsilon^{\mathrm{GL}})^\dagger$, the saturated clean-alignment profile obeys*

$$\begin{aligned}\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n,\eta}(B,m) \le \varepsilon + \frac{H_{B_\varepsilon^{\mathrm{GL}}}}{p_{\mathrm{GL}}}\Big(2^{-n} + \mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}\big((B_\varepsilon^{\mathrm{GL}})^\dagger\big)\Big) + \varepsilon_{\mathrm{rig}}(n,m) + \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] \\ + (2^n - 1)2^{-m(1-H_2(\tau))}.\end{aligned} \tag{1667}$$

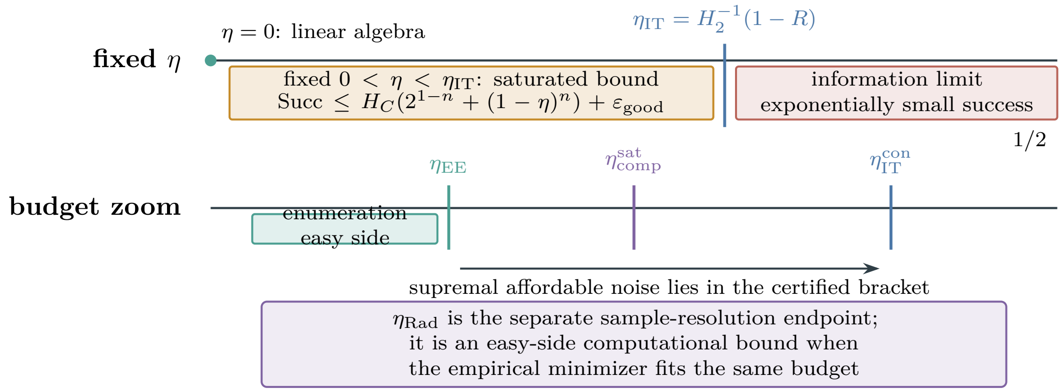

Figure 140: The LPN noise regimes. The upper row combines the fixed-noise saturated-profile bound and the unconditional information threshold from Theorems VIII.7.3 and VIII.22.1 and Corollary VIII.22.3. At a common high-probability target level $\alpha = 1 - \delta$, whenever the endpoints are defined and under the stated enumeration hypotheses, the lower row gives the finite-budget relations $\eta_{\mathrm{EE}} \leq \eta^{\mathrm{sat}}_{\mathrm{comp}} \leq \eta^{\mathrm{con}}_{\mathrm{IT}}$; the Rademacher endpoint joins the easy side only when its represented empirical minimizer is affordable. The diagram is schematic; horizontal positions in both rows are not to scale.

For a sign-valued predictor in the same saturated class, put

$$\Delta_\varepsilon = \varepsilon + \frac{H_{B^{\mathrm{GL}}_\varepsilon}}{p_{\mathrm{GL}}}\left(2^{-n} + \mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}\big((B^{\mathrm{GL}}_\varepsilon)^\dagger\big)\right) + \varepsilon_{\mathrm{rig}}(n,m) + \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)]$$
$$+ (2^n-1)2^{-m(1-H_2(\tau))}.$$

Its family-average clean parity-prediction and fresh-row noise-reconstruction risks satisfy

$$\mathbb{E}_{I,\omega} R_0(h) \geq \frac{1-\Delta_\varepsilon}{2}, \qquad \Pr_{I,\omega,a,e}\{\widehat{e} \neq e\} \geq \frac{1-\Delta_\varepsilon}{2}. \tag{1668}$$

For every $0 < \delta_{\mathrm{stat}} < 1$, on the simultaneous event of Theorem VIII.2.2, the empirical-alignment profile also obeys

$$\sup_{\substack{\mathcal{A}\text{ represented}\\ \mathrm{Cost}(\mathcal{A}) \preceq B}} \mathfrak{M}_n(I \longmapsto \mathbb{E}_{S,\omega}\,|\widehat{c}_{\eta,S}(h_{\mathcal{A},I,S,\omega})|) \leq 2\Gamma^{\mathrm{sat,stat}}_{\mathrm{LPN},n,B,m,\delta_{\mathrm{stat}}} + (1-2\eta)\Delta_\varepsilon. \tag{1669}$$

Thus the statistical term controls adaptive sample fitting, while the computational contribution is localized in the same saturated prospective selector profile. A negligible denoising obstruction follows when that profile is negligible at the compiled Goldreich–Levin ceiling. The complete learner, optimizer, locator, retained state, and downstream use obey the closed budget accounting of Corollary VIII.4.2.

Proof. For a represented evaluator h in the budget-B class, the Goldreich–Levin heavy-Fourier algorithm returns, conditional on $|\widehat{h}(s)| \geq \varepsilon$, a polynomial-size list containing s with probability at least p_{GL} [35]. The standard bounded-score form follows by using the represented internal randomness to sample a sign with conditional mean $h(a)$. Evaluate the public predicate V_I of Lemma VIII.5.1 on every returned candidate. On its high-probability good-instance event, exactly one candidate is accepted and it is s.

The learner, evaluator, list decoder, and verification pass form one family-uniform represented record at $B_\varepsilon^{\mathrm{GL}}$. All intermediate records and admissible oracle calls retain their costs by Definition III.1.3 and Lemma VIII.8.1. On the verifier-good event, the family-averaged success of this one record is at least p_{GL} times the probability of $|\widehat{h}(s)| \geq \varepsilon$ within that event. Apply the family-functional prospective selector bound from Theorem III.21.3, and bound the complementary event by Eqs. (1184) and (1604). Therefore

$$\Pr_{I,\omega}\Big\{|\widehat{h}(s)| \geq \varepsilon\Big\} \leq \frac{H_{B_\varepsilon^{\mathrm{GL}}}}{p_{\mathrm{GL}}}\Big(2^{-n} + \mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}((B_\varepsilon^{\mathrm{GL}})^\dagger)\Big) + \varepsilon_{\mathrm{rig}}(n,m) + \exp[-mD_{\mathrm{Ber}}(\tau\|\eta)] \\ + (2^n-1)2^{-m(1-H_2(\tau))}.$$

Since $|\widehat{h}(s)| \leq 1$, splitting its expectation at ε, applying the displayed probability bound, and taking the saturated supremum proves Eq. (1667).

For a sign-valued predictor,

$$R_0(h) = \frac{1 - \mathbb{E}_a[h(a)f_s(a)]}{2}.$$

Taking expectation over the represented randomness, the first inequality in (1668) follows from (1667); the second is the exact noise-reconstruction identity (1139), averaged over the same seed. Finally, apply the population–empirical transport inequality (1140), then take the saturated supremum to obtain (1669).

□

Corollary VIII.22.5 (Fixed-noise search-LPN hardness implies $P \neq NP$)**.** *Use the standard finite binary encoding of the LPN presentation in Definition VIII.1.1. Fix rational constants $0 < \eta < \tau < 1/2$ and a sample schedule satisfying*

$$m(1 - H_2(\tau)) \geq n + \omega(\log n), \tag{1670}$$

as in Lemma VIII.5.1. Suppose fixed-noise search-LPN is hard for this distribution in the following sense: every declared uniform polynomial execution envelope $\mathcal{B}_C$ satisfies

$$\mathrm{Succ}_{\mathfrak{M},n,\eta}(\mathcal{B}_C) = \mathrm{negl}(n) \tag{1671}$$

at that fixed noise rate. Then $P \neq NP$.

In particular, any unconditional derivation of Eq. (1661) *proves $P \neq NP$. Under the current residual-ancestry certification target, the conditional derivation is supplied by Certification Target VIII.20.5 and Theorem VIII.22.1. The implication from fixed-noise search-LPN hardness to $P \neq NP$ itself uses no additional complexity-barrier premise.*

Proof. Suppose for contradiction that $P = NP$. For the threshold τ fixed in Eq. (1670), consider the prefix language

$$L_{\mathrm{pref}} = \Big\{((A,b),u) : \exists v \in \mathbb{F}_2^{n-|u|} \text{ such that } |A(u \,\|\, v) + b|_1 \leq \tau m\Big\}.$$

Here $u \,\|\, v$ denotes bit-string concatenation.

Membership in L_{pref} is in NP: the suffix v is a polynomial-length witness, and the displayed predicate is the polynomial-time predicate of the public LPN verification theorem. Hence $P = NP$ gives a uniform deterministic polynomial-time decider for L_{pref}.

Starting with the empty prefix, query whether the current prefix extended by 0 belongs to L_{pref}; retain 0 when the answer is yes and retain 1 otherwise. After n queries this procedure outputs a word accepted by the public predicate whenever one exists. On the high-probability event in the public LPN verification theorem, that word exists and is uniquely s. Thus $P = NP$ would place a presentation-relative uniform deterministic prefix-search record with success $1 - o(1)$ in one subenvelope $\mathcal{B}_C$ of the declared polynomial closure. This contradicts Eq. (1671). Therefore $P \neq NP$. The final specialization follows by applying Theorem VIII.22.1 at every polynomial ceiling covered by Certification Target VIII.20.5.

□

VIII.23 Complexity-Barrier Interpretation for LPN

The LPN barrier discussion specializes Proposition III.33.9 and Remark III.33.10. Its three classical components are indexed separately in Sections VIII.23.1 to VIII.23.3; the classification consequence is indexed in Section VIII.23.4. These are presentation-extension tests, not exceptions to Corollary VIII.22.5: once fixed-noise search-LPN hardness is established for the declared presentation, $P \neq NP$ follows.

VIII.23.1 Relativization

The LPN partition gives exact obstruction criteria for its declared public presentation and its presentation-admissible oracle interfaces. A conservative oracle preserves those bounds at class-preserving overhead. The corresponding framework results are Theorem III.33.2, Lemma III.33.4, and Corollary III.33.5; the LPN specialization is Definition VIII.1.2 and Theorem VIII.22.1.

VIII.23.2 Natural proofs

The natural-proofs test concerns an additional truth-table classifier defined by constructivity and nonuniform usefulness. The framework classification is Lemma III.33.6, and the LPN profile and obstruction results are Theorems VIII.21.1 and VIII.22.1.

VIII.23.3 Algebrization

A conservative algebraic interface preserves the LPN statement at class-preserving overhead. An interface that supplies new target-aligned information defines an enriched LPN family, and its saturated profile records the added source. The corresponding framework results are Theorem III.33.2, Lemma III.33.8, and Proposition III.33.9.

Every conservative extension remains inside the same structural accounting. An extension carrying new target-aligned information defines a different presented family with its own saturated profile.

VIII.23.4 Oracle-Blind Family Classification

The presentation distinction in the preceding barrier tests yields the following consequence for classifications that identify a problem with an inadmissible solver-oracle enrichment.

Corollary VIII.23.1 (LPN witness against oracle-blind family classification)**.** *Adopt the hypotheses and conclusion of Corollary VIII.22.5. Let $\mathfrak{C}$ be a classification of presented problem families with the following two properties:*

1. $\mathfrak{C}$ *classifies every presentation containing an explicit uniform polynomial-time solver as polynomial-time solvable;*

2. $\mathfrak{C}$ *assigns the same classification to a presentation and to every enrichment of it by an inadmissible solver oracle.*

Then $\mathfrak{C}$ *is not sound for polynomial-time solvability.*

Proof. Let $\mathcal{T}_{\mathrm{LPN}}$ be the presentation of Definition VIII.1.1, and adjoin a solver oracle O_{LPN} that returns its secret target s. The enriched presentation $\mathcal{T}_{\mathrm{LPN}}^{O_{\mathrm{LPN}}}$ has a uniform one-query polynomial-time solver, so the first property makes $\mathfrak{C}$ classify it as polynomial-time solvable. The oracle is inadmissible for $\mathcal{T}_{\mathrm{LPN}}$: otherwise Theorem III.33.2 would compile it into the original saturated polynomial closure, contradicting Eq. (1671). The second property therefore assigns the same polynomial-time label to $\mathcal{T}_{\mathrm{LPN}}$, contradicting Eq. (1671). This is the LPN instance of Corollary III.33.3.

□

VIII.24 Conclusion

The LPN specialization begins with the represented channel partition in Lemma VIII.8.2; the specialization is introduced as a continuous running instance in Example VIII.1.3.

The public data (A, b, η) instantiate the Part I task presentation, Part II component decomposition, and Part III saturated profile; every uniform polynomial computation using presentation-admissible oracle interfaces lies in its saturated represented closure. The uniform saturated statistical-attack theorem specializes in Theorem VIII.2.2 to control every represented LPN predictor on one simultaneous sample event. Candidate recovery is identified with clean parity alignment and transported through the noisy sample in Corollary VIII.4.1. The exact execution-law identity (1170) identifies recovery with average population alignment, and Theorem VIII.4.3 identifies nonbaseline use of that alignment with the represented pre-hit selector and retains its completed five-family accounting ancestry. A prospective-source comparison uses Corollary III.16.9; the pointwise and family accounting bounds are Corollary VIII.20.1. The LPN denoising corollary combines this statistical control with the saturated prospective selector bound and the represented heavy-Fourier decoder; it does not use the retrospective HitCore coordinate as a hardness premise.

The controlled-dynamics specialization evaluates the same qualified LPN record through target capacity and drift, Caus action–aiming, the complete hypercube functional profile, valid-Lift transfer, and the saturated dynamical-learning envelope in Theorem VIII.11.1 and Corollaries VIII.12.4 to VIII.8.8. The exact subset, located-spectral, and threshold-guided checks are Corollary VIII.13.6; their dependency diagram is Fig. 131. The resource-matched BKW refinement is Theorems VIII.13.9 and VIII.13.14 and Corollary VIII.13.15; its numerical grid and noise frontiers are Table 5 and Fig. 129. Valid lifted selectors inherit the complete prospective routing by Corollary VIII.20.11. Optimal active learning is additionally represented by the stopped occupation-flow/Bellman duality of Theorems V.10.2 and V.10.4. Its LPN specialization Theorem VIII.15.7 gives a modewise, parameter-explicit support bound for the best affordable standard-presentation controller. The learned-controller loss is then the separate finite-sample term in Eqs. (1534) and (1535). The structural Bayesian specialization fixes the uniform secret prior, places $b = As + e$ in the first target-dependent reveal slot, and charges every likelihood, evidence normalization, belief update, and acquisition rule. Its target-contact and concentration bounds are Theorem VIII.15.10; the prospective concentration readout retains the existing LPN five-mode decomposition, and its deployed target gain has the explicit Add/Mult/Ord/Sym/Filt table and noise-contracted same-record minimum in Theorem VIII.15.11. The resulting arrival bound is Eq. (1561). The learned structural Bayesian loss is Corollary VIII.15.15.

All LPN learning routes use the common source-resolved ledger of Theorem IV.13.3: each reveal, feature, posterior update, acquisition rule, controlled transition, and deployment map retains its Add/Mult/Ord/Sym/Filt charge, binary contraction $(1-2\eta)^2$, sample size m, confidence level, horizon, and complete cost. The passive and sequential specialization is Eq. (1484), the sparse-reward observation specialization is Theorem VIII.14.9, and the active learning specialization is Theorem VIII.15.2. Each contributes a same-record target-contact candidate; none replaces the complete prospective source evaluation.

The noise–resource comparison in Section VIII.6 gives the exact-binomial error-enumeration lower-bound curve, the Rademacher sample-resolution endpoint, the saturated computational obstruction, and the sharp information threshold. Their finite-budget relations are collected in Theorem VIII.7.5; Corollary VIII.22.3 and Fig. 140 locate the vanishing-noise nonnegligible-success window, the fixed-positive hardness region, and the information-impossible region.

The generic component decomposition, LPN geometry theorem, coupled-source theorem, and LPN exact-Abs source-exhaustion corollary reduce the prospective $\mathsf{Geom}/\mathsf{Abs}$ source profile to the ambient Geom coordinate. Before the remaining compact ambient certification target is invoked, the constant-noise spectral perimeter Theorem VIII.3.7 exponentially closes materialized located spectra and applies the exact terminal-rank gate to residual-parity fiber maps. The exact-Abs and all quotient-supported Geom coordinates vanish, and the complete ambient source seal, using Certification Target VIII.20.5, bounds the remaining coordinate by $2^{-n} + (1-\eta)^n$. The sole unresolved numerical target in this chain is therefore Certification Target VIII.20.5. It is discharged by a computer-assisted record exactly when the finite verification fields of Theorem VIII.20.6 pass; no sampled failure to find a source is substituted for saturated source coverage. The present manuscript proves this verification implication and specifies its complete record, but it does not supply an accepted instantiated certificate for the residual-ancestry cell. Thus the fixed-noise hardness and separation consequences remain conditional in this version on that internal certificate obligation. Supplying and deterministically verifying the record would discharge the sole remaining target without importing an LPN or complexity-theoretic hardness assumption. The no retrospective first-hit source loophole corollary excludes a completed first-hit label as an uncharged source and classifies every independent pre-hit representation inside that same closure. These results prove the exact source and first-hit bounds as separate ledgers in the saturated LPN ledger theorem.

By Certification Target VIII.20.5, the search-LPN hardness theorem applies the complete saturated-profile bound and prospective selector accountability to obtain the explicit negligible fixed-noise success bound, including presentation-admissible LPN oracle interfaces. This yields the separation consequence for $P \neq NP$ and oracle-blind classification corollary. The roles of relativization, natural proofs, and algebrization in these presentation-relative statements are indexed in Section VIII.23.

A Index of Notation

The index records the load-bearing notation and the result that fixes each symbol. The final column points to its first substantive use.

Symbol	Meaning	Defined in	First used in
Spaces, algebras, and presentations			
$\mathcal{T}_I$	Presented task on instance I	Definition I.1.1	Part I
X_I	Complete candidate-state space	Definition I.1.1	Section I.1
Obs_I	Declared primitive observable interface	Definition I.1.1	Section I.1
Dyn_I	Declared static relation slot	Definition I.1.1	Section I.1
End_I	Positive, negative, and run-dependent terminal data	Definition I.1.1	Section I.1
$\mathcal{F}_I$	Observable event algebra generated by the public interface	Definition I.2.1	Section I.2
$\mathfrak{I}_n$	Size-n slice of a task family	Definition I.3.1	Section I.3
T_I	Positive terminal sector or target set	Definition I.1.1	Definition I.4.1
y_I	Centered target contrast	Definition I.4.5	Definition I.4.5
$[\![\rho]\!]_I$	Denotation of represented derivation ρ on I	Definition III.1.3	Definition III.1.3
Dynamics and structural channels			
L	Positive search generator	Lemma II.2.2	Part II
μ_I	Declared reference measure for state and edge Hilbert spaces	Definition II.3.1	Definition II.3.1
H	Charged stopping horizon for a finite transcript carrier	Definition II.5.3	Definition II.5.5
Geom	Reversible local geometric component	Theorem II.7.1	Part II
Caus	Authorized directed or drift component	Theorem II.7.1	Definition II.4.3
Abs	Qualified target-bearing quotient/lumping source; algebraic quotient provenance is recorded before qualification	Definitions III.15.1 and III.15.2	Definitions II.5.1 and III.23.3
Lift	Valid represented lifted-state component	Theorem II.7.1	Definition II.5.2
Bdry	Boundary, killing, and terminal-readout component	Theorem II.7.1	Theorem II.7.1
J_{res}	Residual complement of the declared channel projections	Theorem II.7.1	Theorem II.7.2
$\Delta_e^{(m)}$	Weighted single-edge geometric Laplacian	Theorem II.9.5	Theorem II.9.5
κ_x, P_κ	Valid-Lift fiber law and its averaging projection	Theorem II.10.5	Theorem II.10.5
$L_{\mathrm{kill}}^{\mathsf{Bdry}}, L_{\mathrm{val}}^{\mathsf{Bdry}}$	Homogeneous loss and represented value blocks of the boundary channel	Theorem II.5.10	Theorem II.5.10
$L_{\mathrm{dir}}^{J_{\mathrm{res}}}$	Hodge-orthogonal directed residual coordinate	Theorem II.7.2	Corollary II.7.3

Symbol	Meaning	Defined in	First used in
$L_{\mathrm{nl}}^{J_{\mathrm{res}}}$	Quotient/lift-orthogonal nonlocal residual coordinate	Theorem II.7.2	Corollary II.7.3
$L^{J_{\mathrm{res}},B}$	Budget-filtered residual ledger at budget B	Theorem II.7.1	Theorem II.7.2
U, P, L_Q	Quotient lift, projection, and induced generator	Definition III.11.1	Definition III.11.2
E_q	Lumpability defect $LU - UL_Q$	Definition III.11.2	Definition III.11.2
Φ	Represented Dirichlet geometry	Definition III.14.8	Definition III.14.9
D	Target-normalized progress observable	Definition III.14.9	Definition III.14.9
$\mathfrak{A}$	Declared causal-authority envelope	Definition II.4.3	Lemma II.4.5
Costs, budgets, and saturation			
$\mathcal{B}$	Resource structure, cost order, ceiling, and transfer class	Definition I.2.4	Definition I.2.4
$b(n)$	Declared resource ceiling at size n	Definition I.2.4	Part II
$\mathcal{R}_{n,\leq B}^{\mathrm{sat}}$	Saturated represented objects of cost at most B	Definition III.14.1	Definition III.14.1
$\mathcal{D}_{n,\leq B}$	Budgeted derivability closure	Definition III.1.3	Definition III.1.3
$\mathrm{Cost}_{\mathrm{sat},n}(R)$	Least scalar saturated representation cost of R	Definition III.14.1	Definition III.14.1
Ψ	Class-preserving accounting enlargement	Theorem III.24.3	Theorem III.19.3
λ	Discount parameter for killed target arrival	Definition III.7.3	Definition III.7.3
$B^*(n), d^*(n)$	Saturated threshold budget and its log-cost coordinate	Definition III.28.1	Definition III.28.1
$\mathfrak{P}_{\mathfrak{D},n}^{>}(\theta)$	Strict Pareto region of budgets supporting a complete record with gain greater than θ	Definition VII.1.2	Theorem VII.1.3
$\mathrm{SMC}_{\mathfrak{D},n,\kappa}^{>}(\theta)$	Scalar structural meta-complexity at strict gain threshold θ	Definition VII.1.2	Theorem VII.2.2
$B_{\eta}^{*,\mathrm{src}}(n)$	Prospective source-profile threshold cost	Eq. (1084)	Corollary VII.1.5
RepMC, ExtMC	Exact-denotation representation cost and complete extraction cost	Definition VII.4.1	Proposition VII.4.2
$\mathrm{CC}_{\mathrm{meta}}$	Minimum gate-count coordinate of an exact circuit representation	Definition VII.5.1	Theorem VII.5.2
$K_{\mathsf{U}}^{c,\mathrm{mode}}$, $\mathrm{MinK}_{\mathsf{U}}^{c,\mathrm{mode}}$	Universal-program representation cost and its threshold language	Definition VII.5.3	Theorem VII.5.4
GapSMC	Resource-and-gain promise version of a structural meta-problem	Definition VII.8.3	Definition VII.8.3

Symbol	Meaning	Defined in	First used in
$B^*_{\vartheta}(n; \mathrm{LPN}_\eta)$	Visibility-indexed saturated threshold cost for the LPN family	Eq. (1215)	Theorem VIII.7.5
$B^*_{\mathrm{rec}}(n, m, \eta; \alpha)$	Infimum scalarized saturated cost reaching recovery probability α	Definition IV.14.2	Theorem IV.14.4
$B^*_{\mathrm{align}}(n, m, \eta; \theta)$	Infimum scalarized cost reaching average population alignment θ	Eq. (292)	Corollary IV.13.7
$\eta^{\mathrm{sat}}_{\mathrm{comp}}$	Supremal affordable noise at a fixed scalar budget and success level; an interval endpoint under flip closure	Eq. (301)	Theorem VIII.7.5
Profiles and extraction coordinates			
$V_I(q)$	Target-conditioned visibility of exact quotient q	Definition III.15.2	Definition III.15.2
$A_T(\lambda)$	Discounted target-arrival functional	Definitions III.7.2 and III.7.3	Definition III.7.3
$m^{\mathrm{sat}}_I(B)$	Saturated exact-quotient visibility profile	Definition III.15.2	Definition III.15.2
$G^{\mathrm{sat}}_{X,n}(B)$	Ambient Geom/Caus extraction coordinate	Definition III.14.9	Definition III.14.9
$G^{\mathrm{sat}}_n(B)$	Active-geometry master subprofile	Definition III.18.4	Definition III.18.4
$I^{\mathrm{sat}}_n(B)$	Full two-coordinate saturated structural profile	Definition III.25.1	Theorem III.25.2
$\mathfrak{U}^{\mathrm{sat}}_n(B)$	Master compensated quotient–geometry profile	Definition III.18.4	Theorem III.18.9
$\mathscr{C}, V_{\mathscr{C}}$	Master compensated certificate and its visibility	Definition III.18.1	Definition III.18.1
$R_T(D)$	Fraction of target contrast reachable from monotone functions of D	Definition III.14.9	Definition III.14.9
$\rho_\Phi(D)$	Dirichlet regularity of the progress observable	Definition III.14.9	Definition III.14.9
$C^{\mathrm{auth}}_{\Phi,\mathfrak{K}}(D)$	Authority-compatible causal alignment factor	Definition III.14.9	Definition III.12.9
$\mathrm{Pert}(\mathfrak{g})$	Authorized ambient perturbation data retained by an ambient source	Definition III.27.1	Theorem III.27.2
P_Φ	Edge-restricted sample-and-hold kernel of an ambient perturbation record	Eq. (202)	Definition III.27.1
$E_\downarrow(\mathfrak{g})$	Exposed edges on which the authorized current descends the represented potential	Definition III.27.1	Theorem III.27.2
$\mathrm{ch}_I(E)$	Contextual target-aligned charge of an algebra extension	Definition III.16.1	Theorem III.16.6

Symbol	Meaning	Defined in	First used in
$\mathrm{CoreSig}(\mathscr{C})$	Charge-bearing source signature of a complete normalized record	Definition III.5.2	Theorem III.5.3
$\mathrm{CoreAct}^{\mathrm{sat}}(B)$	Qualified source visibility in one operational-signature and charge-bearing-core cell	Eq. (52)	Theorem III.5.3
$\mathrm{HitCore}^{\mathrm{sat}}(B)$	Core-refined exact first-hit accounting profile; not a prospective source profile	Definition III.22.3	Theorem III.22.4
$\mathrm{Succ}_{\mathfrak{M},n}(B)$	Budget-B success profile under family functional $\mathfrak{M}_n$	Definition III.24.1	Theorem III.24.3
$\mathrm{Succ}^{\kappa}_{\mathfrak{M},n}(s)$	Scalar-sublevel saturated success profile	Eq. (299)	Definition IV.14.2
$\Gamma, J_{\Gamma,\ell}$	Affordable frontier and its successive joint interfaces	Definition III.16.1	Definition III.16.1
Learning and kernel quantities			
$k_I, \mathcal{H}_{k_I}$	Represented kernel and its RKHS	Definition IV.1.1	Definition IV.1.1
$N_{\mathrm{eff}}(\lambda)$	Effective dimension at ridge scale λ	Definition IV.2.1	Definition IV.2.1
$D_{c,n}, a_{c,n}, e_{c,n}$	Certified source-cell capacity, target-transfer attenuation, and factorization/tail/numerical error	Definition V.15.10	Theorem V.15.12
β_c, γ_c, p_c	Capacity exponent, attenuation exponent, and subexponential scale factor in a verified source audit	Theorem V.15.12	Eq. (823)
λ_η, N_*	Fixed-noise LPN target exponent and certified finite-to-asymptotic crossover size	Eqs. (1641) and (1642)	Theorem VIII.20.6
A_m	Empirical kernel–target alignment	Definition IV.3.11	Definition IV.3.11
$\mathcal{H}^{\mathrm{sat,stat}}_{n,B,m}$	Sample-independent saturated statistical readout class	Definition IV.11.1	Theorem IV.11.2
$M^{\mathrm{sat,stat}}_{I,n}(B,m)$	Saturated learned target-projection envelope	Definition IV.11.1	Theorem IV.11.2
$\mathrm{Align}^{\mathrm{sat,stat}}_{\mathfrak{M},n}$	Average-sample, average-seed population-alignment profile	Definition IV.11.6	Theorem IV.11.7
$\mathrm{GenProj}^{\mathrm{sat,stat}}_{\mathfrak{M},n}$	Average generalizing target-projection profile	Definition IV.11.6	Theorem IV.11.7
$\mathrm{Enum}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}$	Saturated neutral-contact profile for the declared baseline	Eq. (288)	Theorem IV.13.6
$\mathrm{Sel}^{\mathrm{sat}}_{\mathfrak{M},n,\eta}$	Saturated family-uniform prospective-selector profile	Eq. (289)	Theorem IV.13.6
$\Gamma^{\mathrm{sat,stat}}_{n,B,m,\delta}$	Uniform selection radius of the saturated readout class	Definition IV.11.1	Theorem IV.11.2

Symbol	Meaning	Defined in	First used in
$\mathcal{R}^{\mathrm{att,sat}}$, $\mathrm{App}^{\mathrm{sat}}$	Saturated attainable clean risk and structural approximation floor	Definition IV.18.1	Theorem IV.18.2
$N^{\mathrm{out}}_{n,B,m}$	Number of complete saturated output records at a locally finite scalar budget	Corollary IV.13.5	Corollary IV.13.5
$H_2(u)$	Binary entropy in bits	Eq. (294)	Theorem IV.14.4
$D_{\mathrm{Ber}}(u\|v)$	Bernoulli relative entropy in nats	Eq. (295)	Lemma VIII.5.1
$d_{\mathrm{eff}}(\varepsilon)$	Spectral prefix size capturing target mass $1-\varepsilon$	Definition IV.6.1	Definition IV.6.1
$\mathrm{KL}^{\mathsf{str}}_{E,H}$	Defect-restricted structural PAC–Bayes divergence	Definition IV.8.1	Definition IV.8.1
$\beta(k), \mu_a, \delta_a$	Absolute-regularity coefficient, retained block count, and post-coupling confidence level	Definition IV.15.1	Theorem IV.15.2
$\mathfrak{R}^{\mathrm{seq}}_T, \mathfrak{q}_T(\mathbf{A})$	Sequential structural Rademacher complexity and path-operator inverse-precision radius	Definition IV.15.3	Lemma IV.15.4
$\Gamma^{\mathrm{sat}}_{\mathrm{dyn}}$, $\mathrm{Exc}^{\mathrm{sat}}_{\mathrm{corr}}$ $U^{\mathrm{clean}}_{\mathrm{noise}}$	Uniform affordable dynamical deviation, corrupted excess, and clean binary-risk envelopes in $(n, b(n), m, \eta)$	Definition IV.16.3	Theorem IV.16.6
$m^{\mathrm{sat}}_{\mathrm{noise}}$ $\eta^{\mathrm{sat}}_{\mathrm{stat,dyn}}$, $\eta^{\mathrm{sat}}_{\mathrm{comp,dyn}}$	Dynamical sample, statistical-noise, and computational-noise thresholds	Definition IV.16.7	Theorem IV.16.8
$\mathrm{Err}^{\mathrm{sat}}_{q,\mathsf{N}}$	Affordable reward, execution, observation, filter, or composite-channel error in its declared norm, indexed by $(n, b(n), m, \delta)$	Definition IV.17.5	Theorem IV.17.2
$\varepsilon^{\mathrm{rec}}_{\mathrm{obs}}, \tau_m$	Observation reconstruction error and finite-row channel-estimation radius	Definition IV.17.3 and Theorem IV.17.8	Eqs. (350) and (356)
$R_{\mathrm{BB}}, B_{\mathrm{BB}}, Q$	Complete trained-model black-box audit, its charged cost, and its total oracle-query count	Definition V.16.1	Theorem V.16.3
$\mathfrak{C}_{\mathrm{BB}}$	Simultaneous black-box risk, noise, structural, geometric, quotient, and residual signature	Definition V.16.22	Theorem V.16.23
$D_a(f), A_a(f)$	Behavioral defect and target response of source intervention a	Definition V.16.12	Theorem V.16.13
$\mathcal{E}_{P,f}, \mathcal{V}_f, \rho_{P,f}$	Black-box output Dirichlet energy, variance, and Rayleigh ratio	Definition V.16.16	Theorem V.16.17

Symbol	Meaning	Defined in	First used in
$q_f, \delta_{q_f}(P)$	Output discretizer quotient and its perturbation-row lumpability defect	Definition V.16.18	Theorem V.16.19
Controlled dynamical systems			
P^a, P^π, L^π	Action kernel, selected policy kernel, and normalized generator	Definition V.1.1	Theorem V.1.2
$K^A, P^{\pi,K}, r^{\pi,K}$	Execution channel and actual selected transition and reward	Definition V.1.3	Theorem V.1.4
$\Omega^{[H]}_{\mathrm{POMDP}}, b_t, \widehat{b}_t, e_t$	Clocked stochastic POMDP carrier, analytic belief, represented filter, and filter error	Definitions V.2.1 and V.2.3	Theorem V.2.2
$\phi, q_\varepsilon, \rho_\phi(P), M_\varepsilon(\phi), J^{\mathrm{fib}}_{R,t}$	Represented sparse-reward observation, resolution, regularity, cell count, and conditional within-cell target information	Definition V.8.5	Theorem V.8.6
$B_{\mathrm{act}}, \mathrm{ActArr}^{\mathrm{sat}}, \mathrm{ActSel}^{\mathrm{sat}}, B^*_{\mathrm{act}}, m^*_{\mathrm{act}}, \eta^{\mathrm{sat}}_{\mathrm{act}}$	Complete master active-sampling cost, optimal arrival and selector profiles, and their computational, sample, and noise frontiers	Definitions V.9.1, V.9.3 and V.9.9	Theorems V.9.5 and V.9.10
$\mathfrak{O}^{\mathrm{sat}}_{I,H}(B), \mathfrak{D}^{\mathrm{sat}}_{I,t}(B; v)$	Affordable stopped occupation-flow set and saturated Bellman support	Definitions V.10.1 and V.10.3	Theorems V.10.2 and V.10.4
$\Pi_t, \widehat{\Pi}_t, \mathrm{PostConc}^{\mathrm{sat}}, \mathrm{PostSrc}^{\mathrm{sat}}, \mathrm{BayesGain}^{\mathrm{sat}}, \mathrm{BayesArr}^{\mathrm{sat}}$	Analytic task posterior, represented structural belief, affordable concentration, prospective concentration-source, actionable gain, and Bayesian arrival profiles	Definitions V.11.1, V.11.3 and V.11.7	Theorems V.11.4, V.11.8 and V.11.12
$Z_t, b^{\mathrm{lat}}_{R,t}, I^{\mathrm{lat}}_{R,t}, \Delta^{\mathrm{suf}}_{R,t}, e^{\mathrm{dec}}_{R,t}, \mathrm{LatBayesArr}^{\mathrm{sat}}, \mathrm{LatBayesGain}^{\mathrm{sat}}, \mathrm{LatAttArr}^{\mathrm{sat}}$	Compact latent state, code capacity, retained target information, sufficiency and decoder defects, useful belief gain, and latent Bayesian/attention arrival profiles	Definitions V.11.21 and V.11.23	Theorems V.11.22 and V.11.24; Theorem VI.2.4

Symbol	Meaning	Defined in	First used in
$C_{\mathcal{Q}}(K)$, $\vartheta_{\mathrm{KL}}(K;\mathcal{Q})$	Conditional channel capacity in bits and KL strong data-processing coefficient	Definition V.3.1	Theorem V.3.2
$I_0^{(m)}, C_{\mathrm{pub}}^{[H]}$, $D_{\min}(C)$	Training-information bound, deployment information budget, and inverse rate–distortion floor	Eqs. (399) and (401)	Eq. (406)
$\mathsf{H}_{\mathrm{init}}^{[H]}, \mathsf{H}_{\mathrm{ch}}^{[H]}$, $\mathsf{H}_{\mathrm{env}}^{[H]}$, $\mathsf{H}_R^{[H]}, H_\pi^{[H]}$, $\mathbf{H}_{\mathrm{ctl}}$	Initial retained-state, physical-channel, stochastic-transition, clean-reward, and policy entropy, and the typed, noninterchangeable controlled-entropy vector	Definitions V.3.1 and V.4.1	Theorem V.3.7
u_h^π, u_β^π	Finite-horizon and discounted reach–avoid committors	Definition V.6.1	Theorem V.6.2
$\mathrm{Cap}_\Phi(S,T)$	Represented condenser capacity	Definition V.6.5	Theorem V.6.6
$\gamma_\Phi, N_{\mathrm{eff},\Phi}(\lambda)$	Certified gap and effective dimension of a dynamical carrier	Definition V.5.1	Theorem V.5.3
$\mathfrak{F}_\Phi, \Lambda_\Phi^{\mathrm{mix}}$	Audited functional-inequality profile and its spectral-profile coordinate	Definition V.5.4	Theorem V.5.8
$\mathcal{A}_{\mathsf{Caus}}$ $\mathcal{E}_{\mathrm{aim}}$ $\mathcal{P}_{\mathsf{Caus}}$	Caus action, target-aiming energy, and authorized target power	Definition V.7.1	Theorem V.7.2
$\mathrm{Act}_{\mathsf{Caus}}^{\mathrm{sat}}$ $\mathrm{Aim}_{\mathsf{Caus}}^{\mathrm{sat}}$ $\mathrm{Flow}_{\mathsf{Caus}}^{\mathrm{sat}}$	Saturated affordable Caus-flow profiles	Definition V.7.3	Corollary V.7.4
$U_{\mathsf{Caus}}^{\mathrm{noise}}$	Clean-target Caus contact bound learned through corrupted binary supervision	Definition V.7.6	Theorem V.7.7
$Q_\pi, H_{F,0}, \rho_\pi$	Orthogonal fiber projection, boundary-zero fiber form, and lift leakage ratio	Definition V.7.8	Theorem V.7.9
$\mathrm{LiftAmp}^{\mathrm{sat}}$ $\mathrm{LiftSel}^{\mathrm{sat}}$ $\mathrm{LiftAcc}^{\mathrm{sat}}$	Saturated capacity-amplification, selector, and discounted-access profiles for valid lifts	Definitions V.7.10 and V.7.15	Theorems V.7.12 and V.7.13
$\varepsilon_{T,F,H}^{\mathrm{dep}}$, $\varepsilon_{T,F,H}^{\mathrm{flip}}$ $\varepsilon_T^\mu, \varepsilon_F^\mu$, $\omega_{\Phi,T,F}^{\mathrm{Cap}}$	Deployment, direct-flip, reference-law, and capacity-stability errors for learned target and failure gates	Definition V.8.1	Theorem V.8.2

Symbol	Meaning	Defined in	First used in
$N_{\mathrm{eff}}^{\mathcal{A},\mathrm{safe}}, \mathbf{M}_{D,H}^{\mathrm{dyn}}$	Action-safe effective dimension and defect/harmonic control precision	Definition V.13.2	Theorem V.13.3
$\mathbf{U}_{\mathrm{dyn}}$	Coordinatewise dynamical-system certificate	Definition V.12.2	Theorem V.12.3
$\mathbf{U}_{\mathrm{learn}}^{\mathrm{dyn}}$	Typed learned-control certificate: model, geometry, value, policy, shift, binary and general channel noise, POMDP, information, entropy, safety, arrival, noisy capacity, return, and regret	Definition V.13.5	Theorem V.14.2
$\mathfrak{G}_{\mathrm{dyn}}$, $\mathfrak{G}_{\mathrm{dyn}}^{\mathrm{clean}}$ $V_B^*, \Delta_B^{\mathrm{ctrl}}$ $\Delta_B^{\mathrm{POMDP}}$ $\overline{\Delta}_B^{\mathrm{ctrl}}$ $\overline{\Delta}_B^{\mathrm{POMDP}}$	Dynamical deviation and clean binary-risk envelopes, budget-reachable value, computational approximation gap, partial-observation restriction gap, and their represented operational upper certificates	Definition V.13.7	Theorem V.14.2
$E_U = LU - U\widehat{L}$	Represented generator-intertwining defect	Eq. (773)	Corollary V.14.5
Optimal structural extraction			
$U_{R,t}^{\mathrm{opt}}$	Same-record minimum of information, source, Caus, Lift, boundary, and Bellman target-gain bounds	Definition V.11.17	Theorem V.11.19
$J_{R,t}^{\mathrm{Bayes,src}}$	Source-resolved ordered-reveal information in bits	Eq. (662)	Corollary V.11.20
Attention compilation			
Θ_{att}	Complete depth, head, width, temperature, precision, and parameter record	Definition VI.0.1	Theorem VI.1.3
P_s	Represented masked Gibbs attention row	Definition VI.0.1	Theorem VI.1.1
$U_{R,t}^{\mathrm{att}}$	Source-resolved attention target-gain envelope	Definition VI.2.1	Theorem VI.2.2
LPN specialization			
(A, b, η)	Public binary LPN instance and noise rate	Definition VIII.1.1	Part VIII
s	Latent secret and singleton target	Definition VIII.1.1	Definition VIII.1.1
$r_I(x) = Ax + b$	Public residual map of candidate x	Definition VIII.1.1	Definition VIII.1.1

Symbol	Meaning	Defined in	First used in
$\rho_\eta = 1 - 2\eta$	Binary Fourier attenuation at fixed noise	Theorem VIII.3.3	Eq. (1156)
$\Gamma(\lambda) = A^{\mathsf{T}}\lambda$, $\mathrm{Prov}(\lambda)$, $w(\lambda)$	Semantic label, noisy-row provenance support, and provenance weight of a charged LPN combination	Definition III.30.1	Theorem VIII.3.5
$\mathcal{E}_{q_0,k}$	Short-provenance combination with nonzero semantic support contained in at most k secret coordinates	Eq. (1161)	Eq. (1162)
$q = \lfloor \delta m \rfloor$, $L_B(n)$	Short aligned-coset cutoff and charged ceiling on explicitly materialized spectral indices	Theorem VIII.3.4	Eq. (1160)
$q_{L,I}, M_{L,A}, c_{L,b}$	Residual-parity fiber map and its affine data	Eq. (1153)	Eq. (1154)
$G^{\mathrm{src,sat}}_{X,b,\mathfrak{M},n}(B)$	Compact, dynamically qualified residual-ancestry ambient source coordinate	Definition VIII.8.9	Theorem VIII.3.7
$J^{\mathrm{cur}}_{I,t}, \beta^{\mathrm{cur}}_{I,t}, B^{\mathrm{LPN}}_{\mathrm{cur}}$	Conditional within-cell target information, decoupled contact, and complete sparse-reward observation cost	Eqs. (1489), (1491) and (1492)	Theorem VIII.14.9
$B^{\mathrm{act}}_{\mathrm{LPN}}, q_B, J^{\mathrm{act}}_{n,m,\eta,q_B,t}, B^{*}_{\mathrm{act,LPN}}$	LPN active-sampling ceiling, budgeted request count, parameter-explicit information budget, and optimal active recovery cost	Definition VIII.15.1 and Eqs. (1514) and (1515)	master envelope, optimality
$V_I(x)$	Public threshold verifier	Lemma VIII.5.1	Section VIII.5
$N_m(k)$ $F_{m,\eta}(k)$ k_B	Hamming-ball count, its binomial mass, and the exact affordable enumeration radius	Eqs. (296), (1186) and (1188)	error-enumeration result
$\eta_{\mathrm{EE}}, \eta_{\mathrm{Rad}}, \eta_{\mathrm{IT}}$	Enumeration, statistical-resolution, and information endpoints	Eqs. (1191), (1202) and (1212)	Corollary VIII.6.2
$\eta^{\mathrm{ach}}_{\mathrm{IT}}$ $\eta^{\mathrm{con}}_{\mathrm{IT}}$	Finite-sample information achievability and converse endpoints	Eqs. (1210) and (1211)	Theorem VIII.7.3
$\eta_{\mathrm{X}}(R)$	Error-enumeration versus exhaustive-secret-search crossover	Eq. (1198)	Corollary VIII.7.4
$I^{\mathrm{sat}}_{\mathrm{LPN},n}$	Full saturated LPN extraction profile	Eq. (1134)	Part VIII

Symbol	Meaning	Defined in	First used in
$I^{\mathrm{src,sat}}_{\mathrm{LPN},n}$	Saturated prospective LPN source profile	Definition III.25.1	Theorem VIII.21.1

A.1 Superscript and subscript conventions

Decoration	Reading
$\cdot^{\mathrm{sat}}$	Resource-saturated.
$\cdot^{\mathrm{auth}}$	Relative to the declared causal-authority envelope.
$\cdot^{\mathrm{amb}}$	Ambient, before quotient compression.
$\cdot^{\mathrm{nf}}$	Expanded derivation normal form.
$\cdot^{\mathrm{env}}$	Resource-filtered envelope.
$\cdot^{\mathrm{vis}}$	Visibility contribution.
$\cdot^{\mathrm{imp}}$	Imposed dynamics.
$\cdot^{\mathrm{ctrl}}$	Affordable controlled dynamics.
$\cdot^{\mathrm{rev}}$	Reversible compensation record.
$\cdot^{\mathrm{nr}}$	General nonreversible resolvent record.
$\cdot^{\mathrm{ql}}$	Quasi-lumpable compensated quotient family.

References

[1] Scott Aaronson and Avi Wigderson. Algebrization: A new barrier in complexity theory. *ACM Transactions on Computation Theory*, 1(1):2:1–2:54, 2009. doi: 10.1145/1490270.1490272.

[2] Eric Allender, Joshua A. Grochow, Dieter van Melkebeek, Cristopher Moore, and Andrew Morgan. Minimum circuit size, graph isomorphism, and related problems, 2017. URL `https://arxiv.org/abs/1710.09806`.

[3] Nachman Aronszajn. Theory of reproducing kernels. *Transactions of the American Mathematical Society*, 68(3):337–404, 1950. doi: 10.1090/S0002-9947-1950-0051437-7.

[4] Sanjeev Arora, Simon S. Du, Wei Hu, Zhiyuan Li, Ruslan Salakhutdinov, and Ruosong Wang. On exact computation with an infinitely wide neural net. In *Advances in Neural Information Processing Systems 32*, pages 8139–8148, 2019. URL `https://papers.nips.cc/paper/2019/hash/dbc4d84bfcfe2284ba11beffb853a8c4-Abstract.html`.

[5] Bengt Aspvall, Michael F. Plass, and Robert E. Tarjan. A linear-time algorithm for testing the truth of certain quantified boolean formulas. *Information Processing Letters*, 8(3): 121–123, 1979. doi: 10.1016/0020-0190(79)90002-4.

[6] Theodore P. Baker, John Gill, and Robert Solovay. Relativizations of the $\mathcal{P} =? \mathcal{NP}$ question. *SIAM Journal on Computing*, 4(4):431–442, 1975. doi: 10.1137/0204037.

[7] Dominique Bakry and Michel Émery. Diffusions hypercontractives. In *Séminaire de probabilités XIX, 1983/84*, volume 1123 of *Lecture Notes in Mathematics*, pages 177–206. Springer, Berlin, 1985. doi: 10.1007/BFb0075847.

[8] Peter L. Bartlett and Shahar Mendelson. Rademacher and Gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3:463–482, 2002. URL `https://www.jmlr.org/papers/v3/bartlett02a.html`.

[9] Peter L. Bartlett, Dylan J. Foster, and Matus J. Telgarsky. Spectrally-normalized margin bounds for neural networks. In *Advances in Neural Information Processing Systems 30*, pages 6240–6249, 2017.

[10] Richard Bellman. *Dynamic Programming*. Princeton University Press, Princeton, NJ, 1957.

[11] Dimitri P. Bertsekas and John N. Tsitsiklis. *Neuro-Dynamic Programming*. Athena Scientific, Belmont, MA, 1996.

[12] Avrim Blum, Merrick Furst, Michael Kearns, and Richard J. Lipton. Cryptographic primitives based on hard learning problems. In *Advances in Cryptology—CRYPTO '93*, volume 773 of *Lecture Notes in Computer Science*, pages 278–291. Springer, 1994. doi: 10.1007/3-540-48329-2_24.

[13] Avrim Blum, Adam Kalai, and Hal Wasserman. Noise-tolerant learning, the parity problem, and the statistical query model. *Journal of the ACM*, 50(4):506–519, 2003. doi: 10.1145/792538.792543.

[14] S. G. Bobkov and F. Götze. Exponential integrability and transportation cost related to logarithmic sobolev inequalities. *Journal of Functional Analysis*, 163(1):1–28, 1999. doi: 10.1006/jfan.1998.3326.

[15] Sergey G. Bobkov and Prasad Tetali. Modified logarithmic sobolev inequalities in discrete settings. *Journal of Theoretical Probability*, 19(2):289–336, 2006. doi: 10.1007/s10959-006-0016-3.

[16] Sonia Bogos, Florian Tramèr, and Serge Vaudenay. On solving LPN using BKW and variants: Implementation and analysis. Cryptology ePrint Archive, Paper 2015/049, 2015. URL `https://eprint.iacr.org/2015/049`.

[17] Peter Buchholz. Exact and ordinary lumpability in finite markov chains. *Journal of Applied Probability*, 31(1):59–75, 1994. doi: 10.2307/3215235.

[18] Andrea Caponnetto and Ernesto De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007. doi: 10.1007/s10208-006-0196-8.

[19] Eric A. Carlen, Shigeo Kusuoka, and Daniel W. Stroock. Upper bounds for symmetric Markov transition functions. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, 23(S2):245–287, 1987. URL `https://eudml.org/doc/77309`.

[20] Marco Carmosino, Russell Impagliazzo, Valentine Kabanets, and Antonina Kolokolova. Algorithms from natural lower bounds. Technical Report TR16-008, Electronic Colloquium on Computational Complexity, 2016. URL `https://eccc.weizmann.ac.il/report/2016/008/`.

[21] Lijie Chen, Shuichi Hirahara, Igor C. Oliveira, Ján Pich, Ninad Rajgopal, and Rahul Santhanam. Beyond natural proofs: Hardness magnification and locality. In *11th Innovations in Theoretical Computer Science Conference*, volume 151 of *Leibniz International Proceedings in Informatics*, pages 70:1–70:48, 2020. doi: 10.4230/LIPIcs.ITCS.2020.70.

[22] Lénaïc Chizat, Edouard Oyallon, and Francis Bach. On lazy training in differentiable programming. In *Advances in Neural Information Processing Systems 32*, pages 2933–2943, 2019. URL `https://papers.nips.cc/paper/2019/hash/ae614c557843b1df326cb29c57225459-Abstract.html`.

[23] Stephen A. Cook. The complexity of theorem-proving procedures. In *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, pages 151–158. ACM, 1971. doi: 10.1145/800157.805047.

[24] Philippe Courrège. Sur la forme intégro-différentielle des opérateurs de c_k^∞ dans c satisfaisant au principe du maximum. *Séminaire Brelot-Choquet-Deny. Théorie du potentiel*, 10(1):1–38, 1965–1966. URL `http://eudml.org/doc/109338`.

[25] Nello Cristianini, John Shawe-Taylor, André Elisseeff, and Jaz S. Kandola. On kernel-target alignment. In *Advances in Neural Information Processing Systems 14*, pages 367–373. MIT Press, 2002. URL `https://papers.nips.cc/paper/2001/hash/1f71e393b3809197ed66df836fe833e5-Abstract.html`.

[26] Chandler Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. iii. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970. doi: 10.1137/0707001.

[27] William F. Dowling and Jean H. Gallier. Linear-time algorithms for testing the satisfiability of propositional horn formulae. *Journal of Logic Programming*, 1(3):267–284, 1984. doi: 10.1016/0743-1066(84)90014-1.

[28] R. M. Dudley. The sizes of compact subsets of Hilbert space and continuity of Gaussian processes. *Journal of Functional Analysis*, 1(3):290–330, 1967. doi: 10.1016/0022-1236(67)90017-1.

[29] Aryeh Dvoretzky, Jack Kiefer, and Jacob Wolfowitz. Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator. *The Annals of Mathematical Statistics*, 27(3):642–669, 1956. doi: 10.1214/aoms/1177728174.

[30] Beno Eckmann. Harmonische Funktionen und Randwertaufgaben in einem Komplex. *Commentarii Mathematici Helvetici*, 17(1):240–255, 1944. doi: 10.1007/BF02566245.

[31] Masatoshi Fukushima, Yoichi Oshima, and Masayoshi Takeda. *Dirichlet Forms and Symmetric Markov Processes*. De Gruyter, Berlin, 2 edition, 2011.

[32] Edgar N. Gilbert. A comparison of signalling alphabets. *Bell System Technical Journal*, 31 (3):504–522, 1952. doi: 10.1002/j.1538-7305.1952.tb01393.x.

[33] Sharad Goel, Ravi Montenegro, and Prasad Tetali. Mixing time bounds via the spectral profile. *Electronic Journal of Probability*, 11:1–26, 2006. doi: 10.1214/EJP.v11-302.

[34] Halley Goldberg and Valentine Kabanets. Improved learning from kolmogorov complexity. Technical Report TR23-080, Electronic Colloquium on Computational Complexity, 2023. URL `https://eccc.weizmann.ac.il/report/2023/080/`.

[35] Oded Goldreich and Leonid A. Levin. A hard-core predicate for all one-way functions. In *Proceedings of the Twenty-First Annual ACM Symposium on Theory of Computing*, pages 25–32. ACM, 1989. doi: 10.1145/73007.73010.

[36] Leonard Gross. Logarithmic sobolev inequalities. *American Journal of Mathematics*, 97(4): 1061–1083, 1975. doi: 10.2307/2373688.

[37] Richard W. Hamming. Error detecting and error correcting codes. *Bell System Technical Journal*, 29(2):147–160, 1950. doi: 10.1002/j.1538-7305.1950.tb00463.x.

[38] Shuichi Hirahara, Zhenjian Lu, and Mikito Nanashima. Optimal coding for randomized kolmogorov complexity and its applications. Technical Report TR24-155, Electronic Colloquium on Computational Complexity, 2024. URL `https://eccc.weizmann.ac.il/report/2024/155/`.

[39] Nicholas J. Hopper and Manuel Blum. Secure human identification protocols. In *Advances in Cryptology—ASIACRYPT 2001*, volume 2248 of *Lecture Notes in Computer Science*, pages 52–66. Springer, 2001. doi: 10.1007/3-540-45682-1_4.

[40] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural Tangent Kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems 31*, pages 8571–8580, 2018. URL `https://papers.nips.cc/paper/2018/hash/5a4be1fa34e62bb8a6ec6b91d2462f5a-Abstract.html`.

[41] Arne Jensen. Markoff chains as an aid in the study of markoff processes. *Scandinavian Actuarial Journal*, 1953(sup1):87–91, 1953. doi: 10.1080/03461238.1953.10419459.

[42] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, and Yinyu Ye. Statistical ranking and combinatorial Hodge theory. *Mathematical Programming*, 127(1):203–244, 2011. doi: 10.1007/s10107-010-0419-x.

[43] Richard M. Karp. Reducibility among combinatorial problems. In *Complexity of Computer Computations*, pages 85–103. Plenum Press, 1972. doi: 10.1007/978-1-4684-2001-2_9.

[44] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 2 edition, 1995. doi: 10.1007/978-3-642-66282-9.

[45] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer, New York, 1976. ISBN 9780387901923.

[46] George Kimeldorf and Grace Wahba. Some results on Tchebycheffian spline functions. *Journal of Mathematical Analysis and Applications*, 33(1):82–95, 1971. doi: 10.1016/0022-247X(71)90184-3.

[47] Gregory F. Lawler and Alan D. Sokal. Bounds on the L^2 spectrum for Markov chains and Markov processes: A generalization of Cheeger's inequality. *Transactions of the American Mathematical Society*, 309(2):557–580, 1988. doi: 10.1090/S0002-9947-1988-0930082-9.

[48] Michel Ledoux and Michel Talagrand. *Probability in Banach Spaces: Isoperimetry and Processes*. Springer, Berlin, 1991. doi: 10.1007/978-3-642-20212-4.

[49] Jaehoon Lee, Lechao Xiao, Samuel S. Schoenholz, Yasaman Bahri, Roman Novak, Jascha Sohl-Dickstein, and Jeffrey Pennington. Wide neural networks of any depth evolve as linear models under gradient descent. In *Advances in Neural Information Processing Systems 32*, pages 8570–8581, 2019. URL `https://papers.nips.cc/paper/2019/hash/0d1a9651497a38d8b1c3871c84528bd4-Abstract.html`.

[50] Hanno Lefmann, Kevin T. Phelps, and Vojtěch Rödl. Rigid linear binary codes. *Journal of Combinatorial Theory, Series A*, 63(1):110–128, 1993. doi: 10.1016/0097-3165(93)90028-7.

[51] David A. Levin and Yuval Peres. *Markov Chains and Mixing Times*. American Mathematical Society, 2 edition, 2017. doi: 10.1090/mbk/107.

[52] Yanyi Liu and Rafael Pass. On one-way functions and kolmogorov complexity. In *61st Annual IEEE Symposium on Foundations of Computer Science*, pages 1243–1254, 2020. doi: 10.1109/FOCS46700.2020.00118.

[53] Vadim Lyubashevsky. The parity problem in the presence of noise, decoding random linear codes, and the subset sum problem. In *Approximation, Randomization and Combinatorial Optimization. Algorithms and Techniques*, volume 3624 of *Lecture Notes in Computer Science*, pages 378–389. Springer, 2005. doi: 10.1007/11538462_32.

[54] Pascal Massart. The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality. *The Annals of Probability*, 18(3):1269–1283, 1990. doi: 10.1214/aop/1176990746.

[55] Pascal Massart. About the constants in talagrand's concentration inequalities for empirical processes. *The Annals of Probability*, 28(2):863–884, 2000. doi: 10.1214/aop/1019160263.

[56] Andreas Maurer. A note on the PAC-Bayesian theorem, 2004. URL `https://arxiv.org/abs/cs/0411099`.

[57] David A. McAllester. Some PAC-Bayesian theorems. *Machine Learning*, 37(3):355–363, 1999. doi: 10.1023/A:1007618624809.

[58] James Mercer. Functions of positive and negative type, and their connection with the theory of integral equations. *Philosophical Transactions of the Royal Society of London. Series A*, 209(441–458):415–446, 1909. doi: 10.1098/rsta.1909.0016.

[59] Philipp Metzner, Christof Schütte, and Eric Vanden-Eijnden. Transition path theory for Markov jump processes. *Multiscale Modeling & Simulation*, 7(3):1192–1219, 2009. doi: 10.1137/070699500.

[60] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher. An analysis of approximations for maximizing submodular set functions—i. *Mathematical Programming*, 14(1):265–294, 1978. doi: 10.1007/BF01588971.

[61] J. R. Norris. *Markov Chains*. Cambridge University Press, 1997. doi: 10.1017/CBO9780511810633.

[62] Ryan O'Donnell. *Analysis of Boolean Functions*. Cambridge University Press, 2014. doi: 10.1017/CBO9781139814782.

[63] Igor Carboni Oliveira and Rahul Santhanam. Hardness magnification for natural problems. Technical Report TR18-139, Electronic Colloquium on Computational Complexity, 2018. URL `https://eccc.weizmann.ac.il/report/2018/139/`.

[64] Igor Carboni Oliveira, Ján Pich, and Rahul Santhanam. Hardness magnification near state-of-the-art lower bounds. Technical Report TR18-158, Electronic Colloquium on Computational Complexity, 2018. URL `https://eccc.weizmann.ac.il/report/2018/158/`.

[65] Felix Otto and Cédric Villani. Generalization of an inequality by Talagrand and links with the logarithmic sobolev inequality. *Journal of Functional Analysis*, 173(2):361–400, 2000. doi: 10.1006/jfan.1999.3557.

[66] Daniel Paulin. Concentration inequalities for Markov chains by Marton couplings and spectral methods. *Electronic Journal of Probability*, 20(79):1–32, 2015. doi: 10.1214/EJP.v20-4039.

[67] Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, New York, 1994.

[68] Alexander Rakhlin, Karthik Sridharan, and Ambuj Tewari. Online learning via sequential complexities. *Journal of Machine Learning Research*, 16:155–186, 2015. URL `https://jmlr.org/papers/v16/rakhlin15a.html`.

[69] Alexander A. Razborov and Steven Rudich. Natural proofs. *Journal of Computer and System Sciences*, 55(1):24–35, 1997. doi: 10.1006/jcss.1997.1494.

[70] Michael Röckner and Feng-Yu Wang. Weak Poincaré inequalities and L^2-convergence rates of Markov semigroups. *Journal of Functional Analysis*, 185(2):564–603, 2001. doi: 10.1006/jfan.2001.3776.

[71] Mark Rudelson and Roman Vershynin. Hanson–Wright inequality and sub-gaussian concentration. *Electronic Communications in Probability*, 18(82):1–9, 2013. doi: 10.1214/ECP.v18-2865.

[72] Thomas J. Schaefer. The complexity of satisfiability problems. In *Proceedings of the Tenth Annual ACM Symposium on Theory of Computing*, pages 216–226. ACM, 1978. doi: 10.1145/800133.804350.

[73] Yevgeny Seldin, François Laviolette, Nicolò Cesa-Bianchi, John Shawe-Taylor, and Peter Auer. PAC-bayesian inequalities for martingales. *IEEE Transactions on Information Theory*, 58(12):7086–7093, 2012. doi: 10.1109/TIT.2012.2214217.

[74] Lloyd N. Trefethen. *Approximation Theory and Approximation Practice*. SIAM, extended edition, 2019. doi: 10.1137/1.9781611975949.

[75] Joel A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12:389–434, 2012. doi: 10.1007/s10208-011-9099-z.

[76] Rom R. Varshamov. Estimate of the number of signals in error correcting codes. *Doklady Akademii Nauk SSSR*, 117(5):739–741, 1957. URL `https://www.mathnet.ru/eng/dan22571`.

[77] Cédric Villani. *Hypocoercivity*, volume 202 of *Memoirs of the American Mathematical Society*. American Mathematical Society, 2009. doi: 10.1090/S0065-9266-09-00567-5. no. 950.

[78] J. L. Walsh. A closed set of normal orthogonal functions. *American Journal of Mathematics*, 45(1):5–24, 1923. doi: 10.2307/2387224.

[79] Hermann Weyl. Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Differentialgleichungen. *Mathematische Annalen*, 71(4):441–479, 1912. doi: 10.1007/BF01456804.

[80] Thom Wiggers and Simona Samardjiska. Practically solving LPN. In *2021 IEEE International Symposium on Information Theory*. IEEE, 2021. URL `https://eprint.iacr.org/2021/962`. Cryptology ePrint Archive, Paper 2021/962.